

# CNNA Planning and Documentation Tool

Tool V1.27 – Lean data 2026-05-10T05:51:42+02:00

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Assistant: ChatGPT (OpenAI)

Minimum tested assistant setup: ChatGPT Plus account with GPT-5.5 for the tested ChatGPT-instantiated workflow

2026-05-11

# Contents

|    |                                                                             |     |
|----|-----------------------------------------------------------------------------|-----|
| 1  | Stats                                                                       | 3   |
| 2  | Latest change overview                                                      | 14  |
| 3  | Working format and module mapping                                           | 15  |
| 4  | Static code findings                                                        | 18  |
| 5  | Global no-go guardrails                                                     | 24  |
| 6  | Global ontic postulates                                                     | 27  |
| 7  | Phase-object-link policy                                                    | 28  |
| 8  | Phase/object context-class taxonomy                                         | 29  |
| 9  | Object/phase significance and documentation-depth tiers                     | 31  |
| 10 | TomS thread source cross-check                                              | 32  |
| 11 | Secondary long-term goals                                                   | 34  |
| 12 | Standardized phase-preparation, implementation, and documentation workflows | 36  |
| 13 | Implementation-scaling phase plan                                           | 46  |
| 14 | Reuse/duplication ledger and canonical generator policy                     | 48  |
| 15 | Declaration consumption classification map                                  | 51  |
| 16 | Canonical object register with code derivation sites                        | 52  |
| 17 | Object and quantities proof documentation                                   | 115 |
| 18 | Phase table in natural derivation order                                     | 231 |
| 19 | Implementation scratch pad                                                  | 268 |
| 20 | Technical appendix: module-change manifest and semantic placement gate      | 629 |

# 1 Stats

**What this section shows.** This opening section is a generated-secondary orientation surface. It collects project-scale facts, credit and license metadata, the tool-interface metaphor, and the repository module inventory generated from the frozen Lean snapshot. It is shown first so a reader can see the scope, version, graph conventions and review boundary before reading planning tables.

**Stats rule.** This opening section is a generated-secondary orientation surface. It summarizes project scale, credits, license status, root/source layout and review context, then embeds the repository module inventory and derivation graphs. None of these counts, graphs, notices, or metaphors replaces the Lean code as primary formal source, and none removes the need for discipline-specific expert review.

## Project facts

| Area                       | Metric                                | Value                                                                                   | Source                                 | Review status                                                        |
|----------------------------|---------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------|
| Tool version               | current tool version                  | V1.27                                                                                   | metadata.tool_version                  | generated-secondary                                                  |
| Lean data                  | fixed Lean-code data timestamp        | 2026-05-10T05:51:42+02:00                                                               | metadata.data_timestamp                | Lean-primary timestamp only;<br>not semantic infallibility           |
| Credits                    | editor                                | Jan Seeck (antaris)                                                                     | metadata.editor                        | project credit metadata;<br>generated-secondary                      |
| Credits                    | assistant                             | ChatGPT (OpenAI)                                                                        | metadata.assistant                     | project credit metadata;<br>generated-secondary                      |
| Assistant workflow         | minimum tested setup                  | ChatGPT Plus account<br>with GPT-5.5 for the tested<br>ChatGPT-instantiated<br>workflow | metadata.minimum_assistant_requirement | tested ChatGPT-instantiated workflow assumption;<br>not a CNNA input |
| Lean code extent           | Lean source files                     | 288                                                                                     | repo_snapshot/**/*lean                 | generated source-file count                                          |
| Lean code extent           | Lean source lines                     | 89602                                                                                   | repo_snapshot/**/*lean                 | generated line count;<br>comments/prose included                     |
| Lean code extent           | Lean source size bytes                | 3794080                                                                                 | repo_snapshot/**/*lean                 | generated byte count                                                 |
| Module inventory           | registered Lean modules               | 287                                                                                     | repo_inventory/raw_export/stats.json   | generated module inventory                                           |
| Module inventory           | internal import edges                 | 724                                                                                     | repo_inventory/raw_export/stats.json   | generated import graph                                               |
| Module inventory           | missing internal import edges         | 0                                                                                       | repo_inventory/raw_export/stats.json   | should remain zero for inventory consistency                         |
| Module inventory           | acyclic internal import graph         | True                                                                                    | repo_inventory/raw_export/stats.json   | graph property;<br>not semantic correctness                          |
| Symbol/reference inventory | declarations detected                 | 8860                                                                                    | symbol_reference_summary.json          | static extraction;<br>may miss semantic intent                       |
| Symbol/reference inventory | reference edges detected              | 394                                                                                     | symbol_reference_summary.json          | static extraction;<br>review required                                |
| Architectural core         | core modules                          | 241                                                                                     | architectural_core_summary.json        | generated architectural classification                               |
| Architectural core         | artifact modules                      | 46                                                                                      | architectural_core_summary.json        | generated architectural classification                               |
| Planning register          | registered phases                     | 236                                                                                     | masters/cnna_roadmap_master.yaml       | YAML planning count                                                  |
| Planning register          | registered objects/quantities         | 240                                                                                     | masters/cnna_roadmap_master.yaml       | YAML planning count                                                  |
| Planning register          | proof dossiers                        | 240                                                                                     | masters/cnna_roadmap_master.yaml       | YAML planning count                                                  |
| Planning register          | implementation scratchpad records     | 236                                                                                     | masters/cnna_roadmap_master.yaml       | YAML planning count                                                  |
| Review governance          | global no-go guardrails               | 38                                                                                      | masters/cnna_roadmap_master.yaml       | governance count;<br>not proof                                       |
| Review governance          | context documentation profiles        | 3                                                                                       | masters/cnna_roadmap_master.yaml       | profile count;<br>false negatives remain possible                    |
| License                    | documentation/prose/generated reports | CC BY 4.0                                                                               | license_notice                         | project-authored artifacts unless stated otherwise                   |
| License                    | source code/scripts/Lean files        | MIT                                                                                     | license_notice                         | project-authored code unless stated otherwise                        |

## Tool-interface guiding metaphor

**Guiding-metaphor rule.** Guiding metaphors are operational documentation aids. They may help identify review boundaries, hidden dependencies, and interface duties, but they are not formal CNNA theorems, ontic inputs, or substitutes for Lean-primary proof and discipline-specific expert review.

| ID                | Name                     | Metaphor                                                                                                                                                            | Mapping                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Operational consequence                                                                                                                                                                                                          | Boundary                                                                                                                                                                                     |
|-------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GM-TOOL-INTERFACE | Tool-interface principle | The CNNA tool locally exposes an active phase as a bright object against the complementary rest of the Lean codebase and its still unaudited semantic surroundings. | Bright sector = active phase/current object;<br>dark complement = remaining Lean code, open dependencies, unreviewed theory links, and unaudited semantics;<br>boundary/interface = Master-YAML, status, object register, proof dossiers, module/import DAG and later database views;<br>DtN/Schur-like reduction = extraction of relevant dependencies, modules, handoffs and review duties;<br>noise/uncertainty = semantic errors, missing blue markings, incomplete expert review or generated-artifact defects;<br>Birdsview = database/DAG/review view over the whole formal project, still secondary. | Each active phase must make its boundary to the remaining project explicit: dependencies, module interfaces, review duties, blue/reference markings and hidden complement risks must be visible before local closure is trusted. | This is an architecture metaphor for documentation and review practice, not a formal CNNA theorem, not an ontic claim, and not a replacement for Lean-primary verification or expert review. |

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| Scope                                                 | License   | Reference                                           | Boundary                                                                                                                                                                                                                                               |
|-------------------------------------------------------|-----------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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### Repository module inventory and derivation graphs

**Module inventory rule.** This section is a generated repository view, not a second planning source. It is derived from the frozen SM9h Lean source snapshot by `scripts/build_export_script_v1.8.py`. The planning YAML remains the only manually maintained roadmap source; the module catalog, dependency summaries, source-path links, and graph figures are regenerated from the repository snapshot.

#### Inventory summary

| Metric                                    | Value |
|-------------------------------------------|-------|
| Lean modules                              | 287   |
| Internal import edges                     | 724   |
| External import edges                     | 43    |
| Missing internal imports                  | 0     |
| Import graph acyclic                      | True  |
| Symbol declarations detected              | 8860  |
| Symbol-reference edges detected           | 394   |
| Architectural core modules                | 241   |
| Artifact/build/notation/top-level modules | 46    |

### Module dependency graphs

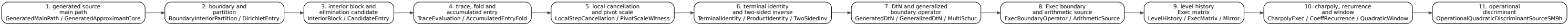
#### Graph A: group-level import/dependency overview

**Arrow convention.** An edge  $A \rightarrow B$  means that at least one module in group  $A$  imports or depends on a module in group  $B$ . The arrow points from the importing/consuming group to the required dependency. This is a technical import graph, not a semantic derivation graph.



Graph B: robust semantic core mainline

**Arrow convention.** An edge  $A \rightarrow B$  means that  $B$  is the next curated semantic construction step derived from  $A$  along the robust core mainline. The arrow points in the intended derived-only construction direction. This is a semantic mainline view, not a complete import graph.



Group summary

| Group              | Modules | Internal edges | Outgoing edges | Incoming edges |
|--------------------|---------|----------------|----------------|----------------|
| Basic              | 1       | 0              | 0              | 0              |
| BuildAll           | 1       | 0              | 2              | 0              |
| Notation           | 1       | 0              | 1              | 1              |
| PillarA            | 3       | 2              | 23             | 2              |
| PillarA/Arithmetic | 127     | 348            | 16             | 2              |
| PillarA/Closure    | 4       | 5              | 2              | 5              |
| PillarA/Coupling   | 4       | 5              | 2              | 6              |
| PillarA/DtN        | 4       | 5              | 3              | 8              |
| PillarA/Finite     | 26      | 58             | 4              | 4              |
| PillarA/Foundation | 17      | 45             | 0              | 25             |
| PillarA/Geometry   | 5       | 7              | 1              | 2              |
| PillarA/Handoff    | 25      | 46             | 7              | 2              |
| PillarA/Network    | 7       | 13             | 2              | 4              |
| PillarA/Sectors    | 9       | 15             | 1              | 5              |
| PillarA/Structural | 37      | 79             | 5              | 5              |
| PillarA/ToC        | 8       | 15             | 8              | 6              |
| PillarB            | 2       | 1              | 0              | 0              |

| Group   | Modules | Internal edges | Outgoing edges | Incoming edges |
|---------|---------|----------------|----------------|----------------|
| PillarC | 2       | 1              | 0              | 0              |
| PillarD | 2       | 1              | 0              | 0              |
| PillarE | 2       | 1              | 0              | 0              |

### Import hotspots

| Module                                                         | Group              | Imports | Imported by | Transitive imports | Flags    |
|----------------------------------------------------------------|--------------------|---------|-------------|--------------------|----------|
| <a href="#">CNNA.PillarA.Arithmetic.BuildAll</a>               | PillarA/Arithmetic | 126     | 1           | 204                | BuildAll |
| <a href="#">CNNA.PillarA.Arithmetic.Notation</a>               | PillarA/Arithmetic | 80      | 2           | 175                | Notation |
| <a href="#">CNNA.PillarA.Structural.CayleyDickson.BuildAll</a> | PillarA/Structural | 35      | 1           | 75                 | BuildAll |
| <a href="#">CNNA.PillarA.Finite.BuildAll</a>                   | PillarA/Finite     | 17      | 1           | 46                 | BuildAll |
| <a href="#">CNNA.PillarA.Foundation.BuildAll</a>               | PillarA/Foundation | 16      | 1           | 16                 | BuildAll |
| <a href="#">CNNA.PillarA.BuildAll</a>                          | PillarA            | 13      | 2           | 274                | BuildAll |
| <a href="#">CNNA.PillarA.Finite.Notation</a>                   | PillarA/Finite     | 12      | 2           | 45                 | Notation |
| <a href="#">CNNA.PillarA.Notation</a>                          | PillarA            | 11      | 2           | 225                | Notation |
| <a href="#">CNNA.PillarA.Foundation.Notation</a>               | PillarA/Foundation | 10      | 4           | 13                 | Notation |
| <a href="#">CNNA.PillarA.Handoff.BuildAll</a>                  | PillarA/Handoff    | 8       | 1           | 75                 | BuildAll |
| <a href="#">CNNA.PillarA.Sectors.BuildAll</a>                  | PillarA/Sectors    | 8       | 1           | 35                 | BuildAll |
| <a href="#">CNNA.PillarA.ToC.BuildAll</a>                      | PillarA/ToC        | 7       | 2           | 21                 | BuildAll |
| <a href="#">CNNA.PillarA.Finite.ExecSpectral.BuildAll</a>      | PillarA/Finite     | 6       | 2           | 32                 | BuildAll |
| <a href="#">CNNA.PillarA.Handoff.Notation</a>                  | PillarA/Handoff    | 6       | 2           | 74                 | Notation |
| <a href="#">CNNA.PillarA.Handoff.Outputs.BuildAll</a>          | PillarA/Handoff    | 6       | 2           | 61                 | BuildAll |
| <a href="#">CNNA.PillarA.Network.BuildAll</a>                  | PillarA/Network    | 6       | 1           | 37                 | BuildAll |
| <a href="#">CNNA.PillarA.Handoff.Core.SectorExport</a>         | PillarA/Handoff    | 5       | 5           | 42                 |          |
| <a href="#">CNNA.PillarA.Foundation.Interfaces</a>             | PillarA/Foundation | 4       | 3           | 6                  |          |
| <a href="#">CNNA.PillarA.Geometry.BuildAll</a>                 | PillarA/Geometry   | 4       | 1           | 20                 | BuildAll |
| <a href="#">CNNA.PillarA.Handoff.Inputs.BuildAll</a>           | PillarA/Handoff    | 4       | 2           | 51                 | BuildAll |

### Complete Lean module catalog, dependency-first order

| #  | Module                                                     | Group              | Path                                                            | Imports | Imported by | Flags    |
|----|------------------------------------------------------------|--------------------|-----------------------------------------------------------------|---------|-------------|----------|
| 1  | <a href="#">CNNA.Basic</a>                                 | Basic              | <a href="#">CNNA/Basic.lean</a>                                 | 0       | 0           | —        |
| 2  | <a href="#">CNNA.PillarA.Foundation.HermitianStructure</a> | PillarA/Foundation | <a href="#">CNNA/PillarA/Foundation/HermitianStructure.lean</a> | 0       | 3           | —        |
| 3  | <a href="#">CNNA.PillarA.Foundation.SubstrateClass</a>     | PillarA/Foundation | <a href="#">CNNA/PillarA/Foundation/SubstrateClass.lean</a>     | 0       | 3           | —        |
| 4  | <a href="#">CNNA.PillarB.BuildAll</a>                      | PillarB            | <a href="#">CNNA/PillarB/BuildAll.lean</a>                      | 0       | 1           | BuildAll |
| 5  | <a href="#">CNNA.PillarC.BuildAll</a>                      | PillarC            | <a href="#">CNNA/PillarC/BuildAll.lean</a>                      | 0       | 1           | BuildAll |
| 6  | <a href="#">CNNA.PillarD.BuildAll</a>                      | PillarD            | <a href="#">CNNA/PillarD/BuildAll.lean</a>                      | 0       | 1           | BuildAll |
| 7  | <a href="#">CNNA.PillarE.BuildAll</a>                      | PillarE            | <a href="#">CNNA/PillarE/BuildAll.lean</a>                      | 0       | 1           | BuildAll |
| 8  | <a href="#">CNNA.PillarA.Foundation.ExecComplex</a>        | PillarA/Foundation | <a href="#">CNNA/PillarA/Foundation/ExecComplex.lean</a>        | 1       | 5           | —        |
| 9  | <a href="#">CNNA.PillarA.Foundation.FiniteHilbert</a>      | PillarA/Foundation | <a href="#">CNNA/PillarA/Foundation/FiniteHilbert.lean</a>      | 1       | 3           | —        |
| 10 | <a href="#">CNNA.PillarB</a>                               | PillarB            | <a href="#">CNNA/PillarB.lean</a>                               | 1       | 0           | —        |
| 11 | <a href="#">CNNA.PillarC</a>                               | PillarC            | <a href="#">CNNA/PillarC.lean</a>                               | 1       | 0           | —        |
| 12 | <a href="#">CNNA.PillarD</a>                               | PillarD            | <a href="#">CNNA/PillarD.lean</a>                               | 1       | 0           | —        |
| 13 | <a href="#">CNNA.PillarE</a>                               | PillarE            | <a href="#">CNNA/PillarE.lean</a>                               | 1       | 0           | —        |
| 14 | <a href="#">CNNA.PillarA.Foundation.ExecComplexLemmas</a>  | PillarA/Foundation | <a href="#">CNNA/PillarA/Foundation/ExecComplexLemmas.lean</a>  | 1       | 1           | —        |
| 15 | <a href="#">CNNA.PillarA.Foundation.MatrixNorms</a>        | PillarA/Foundation | <a href="#">CNNA/PillarA/Foundation/MatrixNorms.lean</a>        | 1       | 9           | —        |



| #  | Module                                                                   | Group              | Path                                                                          | Imports | Imported by | Flags    |
|----|--------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------|---------|-------------|----------|
| 16 | CNNA.PillarA.Foundation.MatrixAlgebra                                    | PillarA/Foundation | CNNA/PillarA/Foundation/MatrixAlgebra.lean                                    | 1       | 4           | —        |
| 17 | CNNA.PillarA.Foundation.ExecComplexBridge                                | PillarA/Foundation | CNNA/PillarA/Foundation/ExecComplexBridge.lean                                | 2       | 8           | —        |
| 18 | CNNA.PillarA.Foundation.Interfaces                                       | PillarA/Foundation | CNNA/PillarA/Foundation/Interfaces.lean                                       | 4       | 3           | —        |
| 19 | CNNA.PillarA.Foundation.Determinism                                      | PillarA/Foundation | CNNA/PillarA/Foundation/Determinism.lean                                      | 1       | 7           | —        |
| 20 | CNNA.PillarA.Foundation.ConcreteSubstrate                                | PillarA/Foundation | CNNA/PillarA/Foundation/ConcreteSubstrate.lean                                | 1       | 5           | —        |
| 21 | CNNA.PillarA.Foundation.LevelVariableSubstrate                           | PillarA/Foundation | CNNA/PillarA/Foundation/LevelVariableSubstrate.lean                           | 1       | 5           | —        |
| 22 | CNNA.PillarA.Foundation.RegularizationPolicy                             | PillarA/Foundation | CNNA/PillarA/Foundation/RegularizationPolicy.lean                             | 2       | 3           | —        |
| 23 | CNNA.PillarA.Foundation.WeightPolicy                                     | PillarA/Foundation | CNNA/PillarA/Foundation/WeightPolicy.lean                                     | 1       | 4           | —        |
| 24 | CNNA.PillarA.Foundation.SubstrateAnalysis                                | PillarA/Foundation | CNNA/PillarA/Foundation/SubstrateAnalysis.lean                                | 2       | 2           | —        |
| 25 | CNNA.PillarA.Foundation.Notation                                         | PillarA/Foundation | CNNA/PillarA/Foundation/Notation.lean                                         | 10      | 4           | Notation |
| 26 | CNNA.PillarA.Foundation.BuildAll                                         | PillarA/Foundation | CNNA/PillarA/Foundation/BuildAll.lean                                         | 16      | 1           | BuildAll |
| 27 | CNNA.PillarA.ToC.Contract                                                | PillarA/ToC        | CNNA/PillarA/ToC/Contract.lean                                                | 2       | 2           | —        |
| 28 | CNNA.PillarA.ToC.Addressing                                              | PillarA/ToC        | CNNA/PillarA/ToC/Addressing.lean                                              | 2       | 3           | —        |
| 29 | CNNA.PillarA.Geometry.LCPMeasure                                         | PillarA/Geometry   | CNNA/PillarA/Geometry/LCPMeasure.lean                                         | 1       | 2           | —        |
| 30 | CNNA.PillarA.ToC.IdealAddressEquiv                                       | PillarA/ToC        | CNNA/PillarA/ToC/IdealAddressEquiv.lean                                       | 1       | 3           | —        |
| 31 | CNNA.PillarA.Geometry.Foliation                                          | PillarA/Geometry   | CNNA/PillarA/Geometry/Foliation.lean                                          | 1       | 2           | —        |
| 32 | CNNA.PillarA.ToC.ConcreteIdeal                                           | PillarA/ToC        | CNNA/PillarA/ToC/ConcreteIdeal.lean                                           | 2       | 3           | —        |
| 33 | CNNA.PillarA.Geometry.SpacetimePaths                                     | PillarA/Geometry   | CNNA/PillarA/Geometry/SpacetimePaths.lean                                     | 1       | 2           | —        |
| 34 | CNNA.PillarA.ToC.Concrete                                                | PillarA/ToC        | CNNA/PillarA/ToC/Concrete.lean                                                | 4       | 3           | —        |
| 35 | CNNA.PillarA.Geometry.Notation                                           | PillarA/Geometry   | CNNA/PillarA/Geometry/Notation.lean                                           | 1       | 2           | Notation |
| 36 | CNNA.PillarA.ToC.GeneratedBranchFamily                                   | PillarA/ToC        | CNNA/PillarA/ToC/GeneratedBranchFamily.lean                                   | 3       | 2           | —        |
| 37 | CNNA.PillarA.ToC.Notation                                                | PillarA/ToC        | CNNA/PillarA/ToC/Notation.lean                                                | 2       | 3           | Notation |
| 38 | CNNA.PillarA.Geometry.BuildAll                                           | PillarA/Geometry   | CNNA/PillarA/Geometry/BuildAll.lean                                           | 4       | 1           | BuildAll |
| 39 | CNNA.PillarA.Arithmetic.Smoke.GeneratedSource                            | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedSource.lean                            | 1       | 3           | —        |
| 40 | CNNA.PillarA.Finite.CutSpec                                              | PillarA/Finite     | CNNA/PillarA/Finite/CutSpec.lean                                              | 1       | 2           | —        |
| 41 | CNNA.PillarA.ToC.BuildAll                                                | PillarA/ToC        | CNNA/PillarA/ToC/BuildAll.lean                                                | 7       | 2           | BuildAll |
| 42 | CNNA.PillarA.Arithmetic.Smoke.GeneratedMainPath                          | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedMainPath.lean                          | 1       | 3           | —        |
| 43 | CNNA.PillarA.Finite.RegionCore                                           | PillarA/Finite     | CNNA/PillarA/Finite/RegionCore.lean                                           | 1       | 2           | —        |
| 44 | CNNA.PillarA.Arithmetic.Smoke.GeneratedApproximantCore                   | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedApproximantCore.lean                   | 1       | 3           | —        |
| 45 | CNNA.PillarA.Finite.BoundaryPorts                                        | PillarA/Finite     | CNNA/PillarA/Finite/BoundaryPorts.lean                                        | 1       | 2           | —        |
| 46 | CNNA.PillarA.Arithmetic.Smoke.GeneratedBoundaryInteriorPartition         | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedBoundaryInteriorPartition.lean         | 1       | 3           | —        |
| 47 | CNNA.PillarA.Finite.Approximant                                          | PillarA/Finite     | CNNA/PillarA/Finite/Approximant.lean                                          | 1       | 3           | —        |
| 48 | CNNA.PillarA.Arithmetic.Smoke.GeneratedDirichletEntry                    | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedDirichletEntry.lean                    | 2       | 4           | —        |
| 49 | CNNA.PillarA.Finite.DirichletLaplacian                                   | PillarA/Finite     | CNNA/PillarA/Finite/DirichletLaplacian.lean                                   | 3       | 4           | —        |
| 50 | CNNA.PillarA.Finite.Selection                                            | PillarA/Finite     | CNNA/PillarA/Finite/Selection.lean                                            | 1       | 2           | —        |
| 51 | CNNA.PillarA.Arithmetic.Smoke.GeneratedDirichletExecEntrySourceSM3bq     | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedDirichletExecEntrySourceSM3bq.lean     | 2       | 2           | —        |
| 52 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorBlock                     | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorBlock.lean                     | 1       | 3           | —        |
| 53 | CNNA.PillarA.DtN.DtN                                                     | PillarA/DtN        | CNNA/PillarA/DtN/DtN.lean                                                     | 1       | 5           | —        |
| 54 | CNNA.PillarA.Finite.SpectralDecompositionCore                            | PillarA/Finite     | CNNA/PillarA/Finite/SpectralDecompositionCore.lean                            | 1       | 3           | —        |
| 55 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorEliminationCandidate      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorEliminationCandidate.lean      | 1       | 3           | —        |
| 56 | CNNA.PillarA.DtN.DtNStabilized                                           | PillarA/DtN        | CNNA/PillarA/DtN/DtNStabilized.lean                                           | 3       | 4           | —        |
| 57 | CNNA.PillarA.Finite.ExecSpectral.PolynomialCore                          | PillarA/Finite     | CNNA/PillarA/Finite/ExecSpectral/PolynomialCore.lean                          | 1       | 2           | —        |
| 58 | CNNA.PillarA.Finite.SpectralDecompositionC                               | PillarA/Finite     | CNNA/PillarA/Finite/SpectralDecompositionC.lean                               | 2       | 3           | —        |
| 59 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorEliminationCandidateEntry | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorEliminationCandidateEntry.lean | 1       | 3           | —        |
| 60 | CNNA.PillarA.DtN.Notation                                                | PillarA/DtN        | CNNA/PillarA/DtN/Notation.lean                                                | 1       | 3           | Notation |
| 61 | CNNA.PillarA.Sectors.BranchPatch                                         | PillarA/Sectors    | CNNA/PillarA/Sectors/BranchPatch.lean                                         | 1       | 2           | —        |

| #   | Module                                                                    | Group              | Path                                                                           | Imports | Imported by | Flags    |
|-----|---------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------|---------|-------------|----------|
| 62  | CNNA.PillarA.Finite.ExecSpectral.CharpolyCore                             | PillarA/Finite     | CNNA/PillarA/Finite/ExecSpectral/CharpolyCore.lean                             | 1       | 2           | —        |
| 63  | CNNA.PillarA.Finite.SpectralBridge                                        | PillarA/Finite     | CNNA/PillarA/Finite/SpectralBridge.lean                                        | 1       | 3           | —        |
| 64  | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorTwoSidedInv                | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorTwoSidedInv.lean                | 2       | 3           | —        |
| 65  | CNNA.PillarA.DtN.BuildAll                                                 | PillarA/DtN        | CNNA/PillarA/DtN/BuildAll.lean                                                 | 3       | 1           | BuildAll |
| 66  | CNNA.PillarA.Sectors.ComplementSectorFamily                               | PillarA/Sectors    | CNNA/PillarA/Sectors/ComplementSectorFamily.lean                               | 1       | 2           | —        |
| 67  | CNNA.PillarA.Finite.ExecSpectral.RootIsolation                            | PillarA/Finite     | CNNA/PillarA/Finite/ExecSpectral/RootIsolation.lean                            | 1       | 2           | —        |
| 68  | CNNA.PillarA.Finite.StateSpace                                            | PillarA/Finite     | CNNA/PillarA/Finite/StateSpace.lean                                            | 1       | 3           | —        |
| 69  | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorInverseCandidateStart      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorInverseCandidateStart.lean      | 1       | 3           | —        |
| 70  | CNNA.PillarA.Sectors.SectorSplit                                          | PillarA/Sectors    | CNNA/PillarA/Sectors/SectorSplit.lean                                          | 1       | 4           | —        |
| 71  | CNNA.PillarA.Finite.ExecSpectral.EigenvectorKernel                        | PillarA/Finite     | CNNA/PillarA/Finite/ExecSpectral/EigenvectorKernel.lean                        | 1       | 2           | —        |
| 72  | CNNA.PillarA.Finite.MatrixExponential                                     | PillarA/Finite     | CNNA/PillarA/Finite/MatrixExponential.lean                                     | 1       | 4           | —        |
| 73  | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorEliminationCarrier         | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorEliminationCarrier.lean         | 1       | 3           | —        |
| 74  | CNNA.PillarA.Coupling.GeneralizedDtN                                      | PillarA/Coupling   | CNNA/PillarA/Coupling/GeneralizedDtN.lean                                      | 2       | 3           | —        |
| 75  | CNNA.PillarA.Network.RegionNet                                            | PillarA/Network    | CNNA/PillarA/Network/RegionNet.lean                                            | 1       | 3           | —        |
| 76  | CNNA.PillarA.Sectors.BranchingWitness                                     | PillarA/Sectors    | CNNA/PillarA/Sectors/BranchingWitness.lean                                     | 1       | 2           | —        |
| 77  | CNNA.PillarA.Finite.ExecSpectral.ProjectorCore                            | PillarA/Finite     | CNNA/PillarA/Finite/ExecSpectral/ProjectorCore.lean                            | 1       | 2           | —        |
| 78  | CNNA.PillarA.Finite.GibbsStateSeed                                        | PillarA/Finite     | CNNA/PillarA/Finite/GibbsStateSeed.lean                                        | 1       | 2           | —        |
| 79  | CNNA.PillarA.Finite.UnitaryEvolution                                      | PillarA/Finite     | CNNA/PillarA/Finite/UnitaryEvolution.lean                                      | 1       | 3           | —        |
| 80  | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorEliminationStep            | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorEliminationStep.lean            | 1       | 3           | —        |
| 81  | CNNA.PillarA.Coupling.MultiSchur                                          | PillarA/Coupling   | CNNA/PillarA/Coupling/MultiSchur.lean                                          | 1       | 5           | —        |
| 82  | CNNA.PillarA.Network.SectorChannels                                       | PillarA/Network    | CNNA/PillarA/Network/SectorChannels.lean                                       | 1       | 3           | —        |
| 83  | CNNA.PillarA.Network.InfiniteCarrier                                      | PillarA/Network    | CNNA/PillarA/Network/InfiniteCarrier.lean                                      | 1       | 4           | —        |
| 84  | CNNA.PillarA.Sectors.SelectedBranching                                    | PillarA/Sectors    | CNNA/PillarA/Sectors/SelectedBranching.lean                                    | 1       | 2           | —        |
| 85  | CNNA.PillarA.Finite.ExecSpectral.Certificates                             | PillarA/Finite     | CNNA/PillarA/Finite/ExecSpectral/Certificates.lean                             | 1       | 2           | —        |
| 86  | CNNA.PillarA.Finite.DynamicsAdapter                                       | PillarA/Finite     | CNNA/PillarA/Finite/DynamicsAdapter.lean                                       | 1       | 3           | —        |
| 87  | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorEliminationTrace           | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorEliminationTrace.lean           | 1       | 3           | —        |
| 88  | CNNA.PillarA.Coupling.Notation                                            | PillarA/Coupling   | CNNA/PillarA/Coupling/Notation.lean                                            | 1       | 2           | Notation |
| 89  | CNNA.PillarA.Network.SectorSysEnv                                         | PillarA/Network    | CNNA/PillarA/Network/SectorSysEnv.lean                                         | 1       | 2           | —        |
| 90  | CNNA.PillarA.Sectors.UVSpectralSelector                                   | PillarA/Sectors    | CNNA/PillarA/Sectors/UVSpectralSelector.lean                                   | 1       | 2           | —        |
| 91  | CNNA.PillarA.Finite.ExecSpectral.BuildAll                                 | PillarA/Finite     | CNNA/PillarA/Finite/ExecSpectral/BuildAll.lean                                 | 6       | 2           | BuildAll |
| 92  | CNNA.PillarA.Finite.SpectralDecomposition                                 | PillarA/Finite     | CNNA/PillarA/Finite/SpectralDecomposition.lean                                 | 2       | 2           | —        |
| 93  | CNNA.PillarA.Finite.ChannelSeed                                           | PillarA/Finite     | CNNA/PillarA/Finite/ChannelSeed.lean                                           | 1       | 3           | —        |
| 94  | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorInverseCandidateEntry      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorInverseCandidateEntry.lean      | 1       | 3           | —        |
| 95  | CNNA.PillarA.Coupling.BuildAll                                            | PillarA/Coupling   | CNNA/PillarA/Coupling/BuildAll.lean                                            | 3       | 1           | BuildAll |
| 96  | CNNA.PillarA.Network.RelativeEntropyFlow                                  | PillarA/Network    | CNNA/PillarA/Network/RelativeEntropyFlow.lean                                  | 1       | 2           | —        |
| 97  | CNNA.PillarA.Sectors.BranchingSelector                                    | PillarA/Sectors    | CNNA/PillarA/Sectors/BranchingSelector.lean                                    | 1       | 3           | —        |
| 98  | CNNA.PillarA.Finite.ExecSpectral.Notation                                 | PillarA/Finite     | CNNA/PillarA/Finite/ExecSpectral/Notation.lean                                 | 1       | 1           | Notation |
| 99  | CNNA.PillarA.Arithmetic.Core.Source                                       | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Core/Source.lean                                       | 3       | 4           | —        |
| 100 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorAccumulatedTraceEvaluation | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorAccumulatedTraceEvaluation.lean | 1       | 4           | —        |
| 101 | CNNA.PillarA.Network.Notation                                             | PillarA/Network    | CNNA/PillarA/Network/Notation.lean                                             | 4       | 2           | Notation |
| 102 | CNNA.PillarA.Closure.ParameterClosure                                     | PillarA/Closure    | CNNA/PillarA/Closure/ParameterClosure.lean                                     | 2       | 2           | —        |
| 103 | CNNA.PillarA.Sectors.Notation                                             | PillarA/Sectors    | CNNA/PillarA/Sectors/Notation.lean                                             | 1       | 2           | Notation |
| 104 | CNNA.PillarA.Finite.Notation                                              | PillarA/Finite     | CNNA/PillarA/Finite/Notation.lean                                              | 12      | 2           | Notation |
| 105 | CNNA.PillarA.Arithmetic.BoundarySource.LevelHistoryIndex                  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/LevelHistoryIndex.lean                  | 1       | 2           | —        |
| 106 | CNNA.PillarA.Arithmetic.Core.FullBoundaryAudit                            | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Core/FullBoundaryAudit.lean                            | 1       | 3           | —        |
| 107 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorBlockFoldNormalFormR3c1a   | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorBlockFoldNormalFormR3c1a.lean   | 1       | 4           | —        |



| #   | Module                                                                                           | Group              | Path                                                                                                  | Imports | Imported by | Flags    |
|-----|--------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------|---------|-------------|----------|
| 108 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorInverseCandidateMatrix                            | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorInverseCandidateMatrix.lean                            | 1       | 3           | —        |
| 109 | CNNA.PillarA.Network.BuildAll                                                                    | PillarA/Network    | CNNA/PillarA/Network/BuildAll.lean                                                                    | 6       | 1           | BuildAll |
| 110 | CNNA.PillarA.Closure.RegularizationClosure                                                       | PillarA/Closure    | CNNA/PillarA/Closure/RegularizationClosure.lean                                                       | 1       | 4           | —        |
| 111 | CNNA.PillarA.Sectors.BuildAll                                                                    | PillarA/Sectors    | CNNA/PillarA/Sectors/BuildAll.lean                                                                    | 8       | 1           | BuildAll |
| 112 | CNNA.PillarA.Finite.BuildAll                                                                     | PillarA/Finite     | CNNA/PillarA/Finite/BuildAll.lean                                                                     | 17      | 1           | BuildAll |
| 113 | CNNA.PillarA.Arithmetic.BoundarySource.LevelStepWitness                                          | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/LevelStepWitness.lean                                          | 1       | 2           | —        |
| 114 | CNNA.PillarA.Arithmetic.BoundarySource.ExistingIdealThreadNoGo                                   | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/ExistingIdealThreadNoGo.lean                                   | 1       | 2           | —        |
| 115 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorAccumulatedEntryCancellationR3c1                  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorAccumulatedEntryCancellationR3c1.lean                  | 1       | 3           | —        |
| 116 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorLocalStepSchurCancellationSourceAuditR3c1b0       | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorLocalStepSchurCancellationSourceAuditR3c1b0.lean       | 1       | 3           | —        |
| 117 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorInverseCandidateShapeSeparation                   | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorInverseCandidateShapeSeparation.lean                   | 1       | 3           | —        |
| 118 | CNNA.PillarA.Closure.Notation                                                                    | PillarA/Closure    | CNNA/PillarA/Closure/Notation.lean                                                                    | 1       | 3           | Notation |
| 119 | CNNA.PillarA.Structural.CayleyDickson.Source                                                     | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/Source.lean                                                     | 3       | 2           | —        |
| 120 | CNNA.PillarA.Arithmetic.BoundarySource.RecursiveLevelHistoryCarrier                              | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/RecursiveLevelHistoryCarrier.lean                              | 1       | 2           | —        |
| 121 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorLocalStepSchurCancellationInvariantSM3db2aR       | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorLocalStepSchurCancellationInvariantSM3db2aR.lean       | 1       | 3           | —        |
| 122 | CNNA.PillarA.Arithmetic.Smoke.SM3dbRToSM3eRHandoff                                               | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/SM3dbRToSM3eRHandoff.lean                                               | 1       | 4           | —        |
| 123 | CNNA.PillarA.Closure.BuildAll                                                                    | PillarA/Closure    | CNNA/PillarA/Closure/BuildAll.lean                                                                    | 3       | 1           | BuildAll |
| 124 | CNNA.PillarA.Handoff.Core.SectorExport                                                           | PillarA/Handoff    | CNNA/PillarA/Handoff/Core/SectorExport.lean                                                           | 5       | 5           | —        |
| 125 | CNNA.PillarA.Structural.CayleyDickson.S6AdditionalFinding                                        | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6AdditionalFinding.lean                                        | 1       | 2           | —        |
| 126 | CNNA.PillarA.Arithmetic.BoundarySource.LevelHistoryMatrix                                        | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/LevelHistoryMatrix.lean                                        | 2       | 2           | —        |
| 127 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorPerPivotRegularityFromTraceSM3db2bR               | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorPerPivotRegularityFromTraceSM3db2bR.lean               | 1       | 3           | —        |
| 128 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorTwoSidedInvFromSM3db7R                            | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorTwoSidedInvFromSM3db7R.lean                            | 2       | 4           | —        |
| 129 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInverseCandidateExecEntrySourceSM3eRq                     | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInverseCandidateExecEntrySourceSM3eRq.lean                     | 2       | 2           | —        |
| 130 | CNNA.PillarA.Handoff.Core.Step1StrongCore                                                        | PillarA/Handoff    | CNNA/PillarA/Handoff/Core/Step1StrongCore.lean                                                        | 1       | 2           | —        |
| 131 | CNNA.PillarA.Handoff.Outputs.A_to_C                                                              | PillarA/Handoff    | CNNA/PillarA/Handoff/Outputs/A_to_C.lean                                                              | 1       | 1           | —        |
| 132 | CNNA.PillarA.Handoff.Outputs.A_to_D                                                              | PillarA/Handoff    | CNNA/PillarA/Handoff/Outputs/A_to_D.lean                                                              | 1       | 1           | —        |
| 133 | CNNA.PillarA.Handoff.Outputs.A_to_E                                                              | PillarA/Handoff    | CNNA/PillarA/Handoff/Outputs/A_to_E.lean                                                              | 1       | 1           | —        |
| 134 | CNNA.PillarA.Structural.CayleyDickson.S6StatusLayer                                              | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6StatusLayer.lean                                              | 1       | 3           | —        |
| 135 | CNNA.PillarA.Arithmetic.BoundarySource.MultiSchurToLevelHistoryAdapter                           | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/MultiSchurToLevelHistoryAdapter.lean                           | 1       | 2           | —        |
| 136 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorTracePivotScaleWitnessFromGeneratedDegreeSM3db2cR | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorTracePivotScaleWitnessFromGeneratedDegreeSM3db2cR.lean | 1       | 2           | —        |
| 137 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorTraceCancellationDerivation                       | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorTraceCancellationDerivation.lean                       | 1       | 3           | —        |
| 138 | CNNA.PillarA.Handoff.Core.Step1MathData                                                          | PillarA/Handoff    | CNNA/PillarA/Handoff/Core/Step1MathData.lean                                                          | 1       | 2           | —        |
| 139 | CNNA.PillarA.Structural.CayleyDickson.S6ConcreteSeeds                                            | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6ConcreteSeeds.lean                                            | 1       | 2           | —        |
| 140 | CNNA.PillarA.Arithmetic.BoundarySource.BoundarySourceTypeDecision                                | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/BoundarySourceTypeDecision.lean                                | 1       | 2           | —        |
| 141 | CNNA.PillarA.Arithmetic.Smoke.GeneratedDegreePivotScaleSourceSM3db2dR                            | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedDegreePivotScaleSourceSM3db2dR.lean                            | 1       | 3           | —        |
| 142 | CNNA.PillarA.Handoff.Core.BuildAll                                                               | PillarA/Handoff    | CNNA/PillarA/Handoff/Core/BuildAll.lean                                                               | 3       | 2           | BuildAll |
| 143 | CNNA.PillarA.Handoff.Outputs.A_to_B                                                              | PillarA/Handoff    | CNNA/PillarA/Handoff/Outputs/A_to_B.lean                                                              | 1       | 2           | —        |
| 144 | CNNA.PillarA.Structural.CayleyDickson.S6RoleCompositionSeed                                      | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6RoleCompositionSeed.lean                                      | 1       | 2           | —        |
| 145 | CNNA.PillarA.Arithmetic.BoundarySource.BoundarySourceInterface                                   | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/BoundarySourceInterface.lean                                   | 1       | 3           | —        |
| 146 | CNNA.PillarA.Arithmetic.Smoke.GeneratedDegreeFormulaAuditSM3db2eR0                               | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedDegreeFormulaAuditSM3db2eR0.lean                               | 1       | 3           | —        |
| 147 | CNNA.PillarA.Handoff.Outputs.Closed                                                              | PillarA/Handoff    | CNNA/PillarA/Handoff/Outputs/Closed.lean                                                              | 1       | 6           | —        |
| 148 | CNNA.PillarA.Structural.CayleyDickson.S6QuaternionicSeed                                         | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6QuaternionicSeed.lean                                         | 1       | 3           | —        |

| #   | Module                                                                                           | Group              | Path                                                                                                  | Imports | Imported by | Flags    |
|-----|--------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------|---------|-------------|----------|
| 149 | CNNA.PillarA.Arithmetic.BoundarySource.RecursiveLevelHistoryRoute                                | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/RecursiveLevelHistoryRoute.lean                                | 1       | 3           | —        |
| 150 | CNNA.PillarA.Arithmetic.BoundarySource.RecursiveLevelHistoryTypeDecision                         | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/RecursiveLevelHistoryTypeDecision.lean                         | 1       | 2           | —        |
| 151 | CNNA.PillarA.Arithmetic.Smoke.<br>GeneratedDegreePivotReciprocalProfileFromWeightPolicySM3db2eR  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedDegreePivotReciprocalProfileFromWeightPolicySM3db2eR.lean  | 1       | 3           | —        |
| 152 | CNNA.PillarA.Handoff.Inputs.B_to_A                                                               | PillarA/Handoff    | CNNA/PillarA/Handoff/Inputs/B_to_A.lean                                                               | 1       | 1           | —        |
| 153 | CNNA.PillarA.Handoff.Inputs.C_to_A                                                               | PillarA/Handoff    | CNNA/PillarA/Handoff/Inputs/C_to_A.lean                                                               | 1       | 1           | —        |
| 154 | CNNA.PillarA.Handoff.Inputs.D_to_A                                                               | PillarA/Handoff    | CNNA/PillarA/Handoff/Inputs/D_to_A.lean                                                               | 1       | 1           | —        |
| 155 | CNNA.PillarA.Handoff.Inputs.E_to_A                                                               | PillarA/Handoff    | CNNA/PillarA/Handoff/Inputs/E_to_A.lean                                                               | 1       | 1           | —        |
| 156 | CNNA.PillarA.Structural.CayleyDickson.S6OctonionicSeed                                           | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6OctonionicSeed.lean                                           | 1       | 2           | —        |
| 157 | CNNA.PillarA.Arithmetic.BoundarySource.GeneratedSubstrateRoute                                   | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean                                   | 1       | 7           | —        |
| 158 | CNNA.PillarA.Arithmetic.Smoke.<br>GeneratedOperationalDegreeReciprocalSourceSM3db2fR             | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedOperationalDegreeReciprocalSourceSM3db2fR.lean             | 2       | 3           | —        |
| 159 | CNNA.PillarA.Handoff.Inputs.BuildAll                                                             | PillarA/Handoff    | CNNA/PillarA/Handoff/Inputs/BuildAll.lean                                                             | 4       | 2           | BuildAll |
| 160 | CNNA.PillarA.Structural.CayleyDickson.S6HurwitzStopSeed                                          | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6HurwitzStopSeed.lean                                          | 1       | 3           | —        |
| 161 | CNNA.PillarA.Arithmetic.BoundarySource.BoundarySourceConvergence                                 | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/BoundarySourceConvergence.lean                                 | 2       | 3           | —        |
| 162 | CNNA.PillarA.Arithmetic.BoundarySource.BoundarySourceDecisionLedger                              | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/BoundarySourceDecisionLedger.lean                              | 1       | 2           | —        |
| 163 | CNNA.PillarA.Arithmetic.BoundarySource.BoundaryStateSubstrate                                    | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/BoundaryStateSubstrate.lean                                    | 1       | 2           | —        |
| 164 | CNNA.PillarA.Arithmetic.BoundarySource.BranchAddressSubstrate                                    | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/BranchAddressSubstrate.lean                                    | 1       | 2           | —        |
| 165 | CNNA.PillarA.Arithmetic.BoundarySource.HistoryExpandedSubstrate                                  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/HistoryExpandedSubstrate.lean                                  | 1       | 2           | —        |
| 166 | CNNA.PillarA.Arithmetic.BoundarySource.InterfaceExposureSubstrate                                | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/BoundarySource/InterfaceExposureSubstrate.lean                                | 1       | 2           | —        |
| 167 | CNNA.PillarA.Arithmetic.Smoke.<br>GeneratedInteriorBlockFoldKernelLocalSchurResidualBridgeR3c1b1 | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorBlockFoldKernelLocalSchurResidualBridgeR3c1b1.lean | 1       | 3           | —        |
| 168 | CNNA.PillarA.Structural.CayleyDickson.S6ProofDutyGroups                                          | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6ProofDutyGroups.lean                                          | 1       | 3           | —        |
| 169 | CNNA.PillarA.Structural.CayleyDickson.S6ResearchSchema                                           | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6ResearchSchema.lean                                           | 1       | 3           | —        |
| 170 | CNNA.PillarA.Arithmetic.Core.MatrixTransport                                                     | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Core/MatrixTransport.lean                                                     | 1       | 3           | —        |
| 171 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorLocalStepCancellationR3c1bFull                    | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorLocalStepCancellationR3c1bFull.lean                    | 1       | 3           | —        |
| 172 | CNNA.PillarA.Handoff.CayleyDicksonAdapter                                                        | PillarA/Handoff    | CNNA/PillarA/Handoff/CayleyDicksonAdapter.lean                                                        | 3       | 5           | —        |
| 173 | CNNA.PillarA.Structural.CayleyDickson.S6CayleyDicksonIdentification                              | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CayleyDicksonIdentification.lean                              | 2       | 3           | —        |
| 174 | CNNA.PillarA.Structural.CayleyDickson.S6ResearchReadinessBridge                                  | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6ResearchReadinessBridge.lean                                  | 2       | 1           | —        |
| 175 | CNNA.PillarA.Arithmetic.Core.BrightDtNSource                                                     | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Core/BrightDtNSource.lean                                                     | 1       | 3           | —        |
| 176 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorBlockFoldInvarianceR3c1c                          | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorBlockFoldInvarianceR3c1c.lean                          | 1       | 3           | —        |
| 177 | CNNA.PillarA.Handoff.Outputs.A_to_CayleyDickson                                                  | PillarA/Handoff    | CNNA/PillarA/Handoff/Outputs/A_to_CayleyDickson.lean                                                  | 1       | 1           | —        |
| 178 | CNNA.PillarA.Handoff.ProofObligationI.Completion                                                 | PillarA/Handoff    | CNNA/PillarA/Handoff/ProofObligationI/Completion.lean                                                 | 1       | 1           | —        |
| 179 | CNNA.PillarA.Handoff.ProofObligationII.Preparation                                               | PillarA/Handoff    | CNNA/PillarA/Handoff/ProofObligationII/Preparation.lean                                               | 1       | 1           | —        |
| 180 | CNNA.PillarA.Handoff.ProofObligationIII.Preparation                                              | PillarA/Handoff    | CNNA/PillarA/Handoff/ProofObligationIII/Preparation.lean                                              | 1       | 1           | —        |
| 181 | CNNA.PillarA.Structural.CayleyDickson.S6AlternativeLawScaffold                                   | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6AlternativeLawScaffold.lean                                   | 1       | 3           | —        |
| 182 | CNNA.PillarA.Structural.CayleyDickson.S6NormUpgrade                                              | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6NormUpgrade.lean                                              | 1       | 3           | —        |
| 183 | CNNA.PillarA.Arithmetic.Actions.ActionPackage                                                    | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Actions/ActionPackage.lean                                                    | 1       | 3           | —        |
| 184 | CNNA.PillarA.Arithmetic.Smoke.<br>GeneratedInteriorTracePivotTerminalCoverageSourceSM3db3aR      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorTracePivotTerminalCoverageSourceSM3db3aR.lean      | 1       | 3           | —        |
| 185 | CNNA.PillarA.Handoff.Outputs.BuildAll                                                            | PillarA/Handoff    | CNNA/PillarA/Handoff/Outputs/BuildAll.lean                                                            | 6       | 2           | BuildAll |
| 186 | CNNA.PillarA.Handoff.ProofObligationI.BuildAll                                                   | PillarA/Handoff    | CNNA/PillarA/Handoff/ProofObligationI/BuildAll.lean                                                   | 1       | 2           | BuildAll |
| 187 | CNNA.PillarA.Handoff.ProofObligationII.BuildAll                                                  | PillarA/Handoff    | CNNA/PillarA/Handoff/ProofObligationII/BuildAll.lean                                                  | 1       | 2           | BuildAll |
| 188 | CNNA.PillarA.Handoff.ProofObligationIII.BuildAll                                                 | PillarA/Handoff    | CNNA/PillarA/Handoff/ProofObligationIII/BuildAll.lean                                                 | 1       | 2           | BuildAll |
| 189 | CNNA.PillarA.Structural.CayleyDickson.S6FullAlternativeLawBoundary                               | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6FullAlternativeLawBoundary.lean                               | 1       | 2           | —        |
| 190 | CNNA.PillarA.Structural.CayleyDickson.S6DivisionUpgrade                                          | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6DivisionUpgrade.lean                                          | 2       | 3           | —        |
| 191 | CNNA.PillarA.Structural.CayleyDickson.S6NormDefinitenessScaffold                                 | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6NormDefinitenessScaffold.lean                                 | 1       | 2           | —        |

| #   | Module                                                                                                    | Group              | Path                                                                                                           | Imports | Imported by | Flags    |
|-----|-----------------------------------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------|---------|-------------|----------|
| 192 | CNNA.PillarA.Arithmetic.Actions.GeneratorActionInventory                                                  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Actions/GeneratorActionInventory.lean                                                  | 1       | 3           | —        |
| 193 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorTerminalFoldResidualCoverageSM3db4aT                       | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorTerminalFoldResidualCoverageSM3db4aT.lean                       | 1       | 3           | —        |
| 194 | CNNA.PillarA.Handoff.Notation                                                                             | PillarA/Handoff    | CNNA/PillarA/Handoff/Notation.lean                                                                             | 6       | 2           | Notation |
| 195 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalProductSurfaceCandidate                                  | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalProductSurfaceCandidate.lean                                  | 1       | 3           | —        |
| 196 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalIdealAlgebraizationPath                                  | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalIdealAlgebraizationPath.lean                                  | 1       | 2           | —        |
| 197 | CNNA.PillarA.Structural.CayleyDickson.S6NormMultiplicativityScaffold                                      | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6NormMultiplicativityScaffold.lean                                      | 2       | 2           | —        |
| 198 | CNNA.PillarA.Arithmetic.Actions.ActionFormatDecision                                                      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Actions/ActionFormatDecision.lean                                                      | 1       | 3           | —        |
| 199 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorTerminalFoldResidualIdentityFieldsSM3db4aU                 | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorTerminalFoldResidualIdentityFieldsSM3db4aU.lean                 | 1       | 4           | —        |
| 200 | CNNA.PillarA.Handoff.BuildAll                                                                             | PillarA/Handoff    | CNNA/PillarA/Handoff/BuildAll.lean                                                                             | 8       | 1           | BuildAll |
| 201 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalProductMultiplicationCandidate                           | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalProductMultiplicationCandidate.lean                           | 1       | 3           | —        |
| 202 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalStar                                                     | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalStar.lean                                                     | 1       | 2           | —        |
| 203 | CNNA.PillarA.Structural.CayleyDickson.S6DivisionConclusionScaffold                                        | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6DivisionConclusionScaffold.lean                                        | 2       | 2           | —        |
| 204 | CNNA.PillarA.Arithmetic.Schur.MobiusTransfer                                                              | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/MobiusTransfer.lean                                                              | 1       | 2           | —        |
| 205 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorBlockFoldTerminalIdentityR3c1d                             | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorBlockFoldTerminalIdentityR3c1d.lean                             | 1       | 4           | —        |
| 206 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorTerminalFoldResidualIdentityEquationSourceSM3db4aV         | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorTerminalFoldResidualIdentityEquationSourceSM3db4aV.lean         | 1       | 3           | —        |
| 207 | CNNA.PillarA.Structural.CayleyDickson.S6Slot1AlternativeLaw                                               | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6Slot1AlternativeLaw.lean                                               | 1       | 1           | —        |
| 208 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalScalarPart                                               | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalScalarPart.lean                                               | 2       | 2           | —        |
| 209 | CNNA.PillarA.Structural.CayleyDickson.S6PhaseICompactClosure                                              | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6PhaseICompactClosure.lean                                              | 1       | 2           | —        |
| 210 | CNNA.PillarA.Arithmetic.Schur.RecursiveSchurSource                                                        | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/RecursiveSchurSource.lean                                                        | 1       | 2           | —        |
| 211 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorAccumulatedEntryCancellationFromTerminalIdentityR3c1e      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorAccumulatedEntryCancellationFromTerminalIdentityR3c1e.lean      | 2       | 3           | —        |
| 212 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorTerminalIdentityFieldCertificateFromEquationSourceSM3db4aW | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorTerminalIdentityFieldCertificateFromEquationSourceSM3db4aW.lean | 2       | 3           | —        |
| 213 | CNNA.PillarA.Structural.CayleyDickson.S6DivisionConclusionDecomposition                                   | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6DivisionConclusionDecomposition.lean                                   | 1       | 3           | —        |
| 214 | CNNA.PillarA.Arithmetic.Schur.Recurrence                                                                  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/Recurrence.lean                                                                  | 1       | 3           | —        |
| 215 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorTraceCancellationR3c2                                      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorTraceCancellationR3c2.lean                                      | 2       | 3           | —        |
| 216 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorBlockFoldTerminalIdentityFromSM3db4aWR3c1dFull             | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorBlockFoldTerminalIdentityFromSM3db4aWR3c1dFull.lean             | 1       | 2           | —        |
| 217 | CNNA.PillarA.Structural.CayleyDickson.S6Slot2NormSqMultiplicativity                                       | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6Slot2NormSqMultiplicativity.lean                                       | 1       | 3           | —        |
| 218 | CNNA.PillarA.Arithmetic.Schur.CharpolyExec                                                                | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/CharpolyExec.lean                                                                | 1       | 5           | —        |
| 219 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorProductIdentityProofR4a                                    | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorProductIdentityProofR4a.lean                                    | 1       | 2           | —        |
| 220 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalNormSqCandidate                                          | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqCandidate.lean                                          | 2       | 2           | —        |
| 221 | CNNA.PillarA.Structural.CayleyDickson.S6Slot3DivisionFromNormSqLogic                                      | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6Slot3DivisionFromNormSqLogic.lean                                      | 2       | 2           | —        |
| 222 | CNNA.PillarA.Arithmetic.Schur.CharpolyC                                                                   | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/CharpolyC.lean                                                                   | 2       | 3           | —        |
| 223 | CNNA.PillarA.Arithmetic.Schur.L1Baseline                                                                  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/L1Baseline.lean                                                                  | 1       | 3           | —        |
| 224 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorProvenTwoSidedInvR5                                        | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorProvenTwoSidedInvR5.lean                                        | 2       | 2           | —        |
| 225 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalNormSqDefined                                            | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqDefined.lean                                            | 2       | 4           | —        |
| 226 | CNNA.PillarA.Arithmetic.Schur.L2Eisenstein                                                                | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/L2Eisenstein.lean                                                                | 1       | 3           | —        |
| 227 | CNNA.PillarA.Arithmetic.Smoke.GeneratedInteriorEliminationCertificateSM3fR                                | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorEliminationCertificateSM3fR.lean                                | 1       | 3           | —        |
| 228 | CNNA.PillarA.Arithmetic.CayleyDickson.ArithmeticCDBBridge                                                 | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/CayleyDickson/ArithmeticCDBBridge.lean                                                 | 2       | 4           | —        |
| 229 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalDivisionFromNormSq                                       | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalDivisionFromNormSq.lean                                       | 2       | 2           | —        |
| 230 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalNormSqReduction                                          | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqReduction.lean                                          | 1       | 2           | —        |
| 231 | CNNA.PillarA.Arithmetic.Schur.L3Gaussian                                                                  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/L3Gaussian.lean                                                                  | 1       | 4           | —        |
| 232 | CNNA.PillarA.Arithmetic.Smoke.GeneratedSM3fSchurExecEntrySourceSM3fq                                      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedSM3fSchurExecEntrySourceSM3fq.lean                                      | 1       | 2           | —        |



| #   | Module                                                                                         | Group              | Path                                                                                                | Imports | Imported by | Flags    |
|-----|------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------|---------|-------------|----------|
| 233 | CNNA.PillarA.Arithmetic.Smoke.GeneratedTopBoundaryDtNSM3gR                                     | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedTopBoundaryDtNSM3gR.lean                                     | 1       | 2           | —        |
| 234 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalNormSqBlockSystem                             | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqBlockSystem.lean                             | 2       | 2           | —        |
| 235 | CNNA.PillarA.Arithmetic.Schur.IdentityLedger                                                   | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/IdentityLedger.lean                                                   | 1       | 4           | —        |
| 236 | CNNA.PillarA.Arithmetic.Smoke.GeneratedSM3fSchurExecCompletionSM3fqPlus                        | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedSM3fSchurExecCompletionSM3fqPlus.lean                        | 2       | 2           | —        |
| 237 | CNNA.PillarA.Arithmetic.Smoke.GeneratedGeneralizedTopBoundaryDtNSM3hR                          | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedGeneralizedTopBoundaryDtNSM3hR.lean                          | 1       | 3           | —        |
| 238 | CNNA.PillarA.Structural.CayleyDickson.S6CanonicalNormSqSimultaneousSystem                      | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqSimultaneousSystem.lean                      | 1       | 1           | —        |
| 239 | CNNA.PillarA.Arithmetic.Graph.SpectralGapSeed                                                  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Graph/SpectralGapSeed.lean                                                  | 1       | 3           | —        |
| 240 | CNNA.PillarA.Arithmetic.Schur.HigherLevelPredictionLedger                                      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/HigherLevelPredictionLedger.lean                                      | 1       | 3           | —        |
| 241 | CNNA.PillarA.Arithmetic.Smoke.GeneratedResetSM3LedgerSM3iR                                     | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedResetSM3LedgerSM3iR.lean                                     | 1       | 3           | —        |
| 242 | CNNA.PillarA.Structural.CayleyDickson.BuildAll                                                 | PillarA/Structural | CNNA/PillarA/Structural/CayleyDickson/BuildAll.lean                                                 | 35      | 1           | BuildAll |
| 243 | CNNA.PillarA.Arithmetic.Graph.CheegerSeed                                                      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Graph/CheegerSeed.lean                                                      | 1       | 3           | —        |
| 244 | CNNA.PillarA.Arithmetic.Schur.HigherLevelCertificates                                          | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/HigherLevelCertificates.lean                                          | 1       | 3           | —        |
| 245 | CNNA.PillarA.Arithmetic.Smoke.GeneratedMultiSchurSM4                                           | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedMultiSchurSM4.lean                                           | 1       | 4           | —        |
| 246 | CNNA.PillarA.Structural.BuildAll                                                               | PillarA/Structural | CNNA/PillarA/Structural/BuildAll.lean                                                               | 1       | 1           | BuildAll |
| 247 | CNNA.PillarA.Arithmetic.Graph.RamanujanTest                                                    | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Graph/RamanujanTest.lean                                                    | 1       | 3           | —        |
| 248 | CNNA.PillarA.Arithmetic.Schur.HigherLevelOutcomeRouter                                         | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/HigherLevelOutcomeRouter.lean                                         | 1       | 3           | —        |
| 249 | CNNA.PillarA.Arithmetic.Smoke.GeneratedArithmeticSourceSM5                                     | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedArithmeticSourceSM5.lean                                     | 1       | 4           | —        |
| 250 | CNNA.PillarA.Arithmetic.Smoke.GeneratedMultiSchurExecBoundaryOperatorSM4q                      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedMultiSchurExecBoundaryOperatorSM4q.lean                      | 2       | 2           | —        |
| 251 | CNNA.PillarA.Arithmetic.Convergence.QuaternionRamanujanInterface                               | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Convergence/QuaternionRamanujanInterface.lean                               | 3       | 2           | —        |
| 252 | CNNA.PillarA.Arithmetic.Schur.HigherLevelDiscriminants                                         | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Schur/HigherLevelDiscriminants.lean                                         | 1       | 3           | —        |
| 253 | CNNA.PillarA.Arithmetic.Smoke.GeneratedArithmeticSourceExecBoundaryOperatorSM5q                | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedArithmeticSourceExecBoundaryOperatorSM5q.lean                | 2       | 2           | —        |
| 254 | CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelStepWitnessSM6                                     | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelStepWitnessSM6.lean                                     | 1       | 5           | —        |
| 255 | CNNA.PillarA.Arithmetic.Smoke.GeneratedMultiSchurExecBoundaryOperatorFromSM3fqPlusSM4q         | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedMultiSchurExecBoundaryOperatorFromSM3fqPlusSM4q.lean         | 2       | 2           | —        |
| 256 | CNNA.PillarA.Arithmetic.Orders.QuadraticOrder                                                  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Orders/QuadraticOrder.lean                                                  | 2       | 3           | —        |
| 257 | CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelHistoryBoundaryPortSourceSM6a                      | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryBoundaryPortSourceSM6a.lean                      | 1       | 2           | —        |
| 258 | CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelHistoryToSchurIndexSM7                             | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryToSchurIndexSM7.lean                             | 1       | 2           | —        |
| 259 | CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelStepExecMirrorSourceSM6b                           | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelStepExecMirrorSourceSM6b.lean                           | 2       | 2           | —        |
| 260 | CNNA.PillarA.Arithmetic.Smoke.GeneratedArithmeticSourceExecBoundaryOperatorFromSM4qRecheckSM5q | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedArithmeticSourceExecBoundaryOperatorFromSM4qRecheckSM5q.lean | 2       | 2           | —        |
| 261 | CNNA.PillarA.Arithmetic.Orders.DiscriminantConvention                                          | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Orders/DiscriminantConvention.lean                                          | 1       | 3           | —        |
| 262 | CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelHistorySchurIndexBridgeRecheckSM7a                 | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistorySchurIndexBridgeRecheckSM7a.lean                 | 2       | 2           | —        |
| 263 | CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelStepExecMirrorSourceFromSM5qRecheckSM6b            | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelStepExecMirrorSourceFromSM5qRecheckSM6b.lean            | 2       | 2           | —        |
| 264 | CNNA.PillarA.Arithmetic.Orders.NormForms                                                       | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Orders/NormForms.lean                                                       | 1       | 3           | —        |
| 265 | CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelHistoryMatrixFromBridgeSM8                         | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryMatrixFromBridgeSM8.lean                         | 1       | 3           | —        |
| 266 | CNNA.PillarA.Arithmetic.Convergence.CDArithmeticCompatibility                                  | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Convergence/CDArithmeticCompatibility.lean                                  | 2       | 2           | —        |
| 267 | CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelHistoryExecMatrixFromSM6bRecheckSM8                | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryExecMatrixFromSM6bRecheckSM8.lean                | 2       | 2           | —        |
| 268 | CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelHistoryExecMirrorSourceSM9a                        | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryExecMirrorSourceSM9a.lean                        | 2       | 2           | —        |
| 269 | CNNA.PillarA.Arithmetic.Notation                                                               | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Notation.lean                                                               | 80      | 2           | Notation |
| 270 | CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelHistoryExecMirrorSourceFromSM8RecheckSM9a          | PillarA/Arithmetic | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryExecMirrorSourceFromSM8RecheckSM9a.lean          | 2       | 2           | —        |
| 271 | CNNA.PillarA.Notation                                                                          | PillarA            | CNNA/PillarA/Notation.lean                                                                          | 11      | 2           | Notation |

| #   | Module                                                                                                           | Group              | Path                                                                                                                  | Imports | Imported by | Flags    |
|-----|------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------|---------|-------------|----------|
| 272 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelHistoryCharpolyExecFromMirrorSM9b</a>                    | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryCharpolyExecFromMirrorSM9b.lean</a>                    | 2       | 2           | —        |
| 273 | <a href="#">CNNA.Notation</a>                                                                                    | Notation           | <a href="#">CNNA/Notation.lean</a>                                                                                    | 1       | 1           | Notation |
| 274 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelHistoryOperationalCharpolyPolynomialProfileFromSM9b</a>  | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryOperationalCharpolyPolynomialProfileFromSM9b.lean</a>  | 1       | 2           | —        |
| 275 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.GeneratedLevelHistoryOperationalCoefficientSourceFromSM9c</a>          | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryOperationalCoefficientSourceFromSM9c.lean</a>          | 1       | 2           | —        |
| 276 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.GeneratedOperationalDeterminantCoefficientRecurrenceSM9e</a>           | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/GeneratedOperationalDeterminantCoefficientRecurrenceSM9e.lean</a>           | 1       | 3           | —        |
| 277 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.GeneratedOperationalDeterminantEvaluationBridgeFromSM9e</a>            | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/GeneratedOperationalDeterminantEvaluationBridgeFromSM9e.lean</a>            | 1       | 2           | —        |
| 278 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.OperationalBoundedPolynomialEvaluationNormalFormSM9ePlus</a>           | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/OperationalBoundedPolynomialEvaluationNormalFormSM9ePlus.lean</a>           | 1       | 2           | —        |
| 279 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.OperationalBoundedPolynomialForwardEvaluationSourceSM9ePlus</a>        | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/OperationalBoundedPolynomialForwardEvaluationSourceSM9ePlus.lean</a>        | 1       | 2           | —        |
| 280 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.GeneratedOperationalDeterminantForwardEvaluationBridgeFromSM9ePlus</a> | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/GeneratedOperationalDeterminantForwardEvaluationBridgeFromSM9ePlus.lean</a> | 2       | 2           | —        |
| 281 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.GeneratedOperationalDiscriminantFactorGateFromSM9fFull</a>             | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/GeneratedOperationalDiscriminantFactorGateFromSM9fFull.lean</a>             | 1       | 2           | —        |
| 282 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.GeneratedOperationalQuadraticCoefficientWindowFromSM9dSM9e</a>         | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/GeneratedOperationalQuadraticCoefficientWindowFromSM9dSM9e.lean</a>         | 1       | 2           | —        |
| 283 | <a href="#">CNNA.PillarA.Arithmetic.Smoke.GeneratedOperationalQuadraticDiscriminantSourceFromSM9g1Window</a>     | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/Smoke/GeneratedOperationalQuadraticDiscriminantSourceFromSM9g1Window.lean</a>     | 1       | 1           | —        |
| 284 | <a href="#">CNNA.PillarA.Arithmetic.BuildAll</a>                                                                 | PillarA/Arithmetic | <a href="#">CNNA/PillarA/Arithmetic/BuildAll.lean</a>                                                                 | 126     | 1           | BuildAll |
| 285 | <a href="#">CNNA.PillarA.BuildAll</a>                                                                            | PillarA            | <a href="#">CNNA/PillarA/BuildAll.lean</a>                                                                            | 13      | 2           | BuildAll |
| 286 | <a href="#">CNNA.BuildAll</a>                                                                                    | BuildAll           | <a href="#">CNNA/BuildAll.lean</a>                                                                                    | 2       | 0           | BuildAll |
| 287 | <a href="#">CNNA.PillarA</a>                                                                                     | PillarA            | <a href="#">CNNA/PillarA.lean</a>                                                                                     | 1       | 0           | —        |

## 2 Latest change overview

**What this section shows.** This table combines fixed metadata from the Master YAML with the latest manually recorded tool/display changes. It explains what changed in this package version and, equally important, whether the Lean-code data timestamp changed. It is displayed to prevent generated documentation changes from being mistaken for Lean-source changes.

**Latest change overview.** This section records the most recent tool/display update and the currently fixed Lean-code data timestamp. The tool version tracks generator, workflow, and PDF-display contract changes and increments by 0.01 for each such change. The Lean-code data timestamp changes only when Lean source/data changes; generator-only, YAML-policy-only, and PDF-display-only edits must not advance it.

| Item                                                    | State                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tool version                                            | V1.27                                                                                                                                                                                                                                                                                                                                                 |
| Tool/display change timestamp                           | 2026-05-11T09:25:00+02:00                                                                                                                                                                                                                                                                                                                             |
| Lean-code data timestamp                                | 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                             |
| Document title                                          | CNNA Planning and Documentation Tool - Context classes plus object/phase significance tiers                                                                                                                                                                                                                                                           |
| Object/phase significance and documentation-depth tiers | V1.27 adds a second orthogonal classification axis: Context class says what a row is; Significance tier / Phase documentation tier says how deep the documentation and review obligation must be. Fundamental objects require full proof dossiers; trivial/auxiliary objects may stay mention-only unless promoted or consumed by a higher-tier gate. |
| Context-class taxonomy                                  | V1.26 adds OOP-style Context class as an orthogonal row type system for every phase/object. Second-pass semantic audit adds LeanRefactorObject, LeanHandoffContractObject and LeanGovernanceObject so Lean-facing refactors/contracts/policies cannot be confused with tool-only rows or mathematical Lean data.                                      |
| Horn quaternion/GA reference-gate integration           | V1.25 reads Horn 0709.2238v1 and routes quaternion, Pauli, Clifford/geometric-algebra, duality and biquaternion content as comparison/target gates only. No CNNA generator, Lean source, DB row or phase-release proof is introduced from the paper.                                                                                                  |
| New comparison phases                                   | Adds 1.4.9, 1.4.9.1, 1.4.9.2, 13.2.1, 13.6.1 and 18.5 for CD/Quaternion/GA, Pillar-B Pauli/Clifford, Dirac/spacetime-algebra and conformal-GA side-comparison target gates.                                                                                                                                                                           |
| New reference objects                                   | Adds GA0-GA6 as black comparison/reference objects with proof dossiers and no handoff rights. They may only be consumed as review targets once their predecessor phases expose CNNA-derived data.                                                                                                                                                     |
| Existing Supabase phase check                           | V1.24 first checks existing 20.15-20.20 DB/tooling phases and keeps them as broad architecture contracts instead of duplicating them.                                                                                                                                                                                                                 |
| Supabase live activation                                | Adds concrete execution sidechain 20.22-20.26 for discovered project aiekahdewadrejrrdajn: baseline audit, migration package, import package, explicitly authorized live execution, and post-import read-only/RLS/frontend verification.                                                                                                              |
| PDF typography profile                                  | V1.23 raises generated full/context PDF text one step: 11pt document class, one-step TeX size-command bump for generated tables/legends/static text, and larger generated graph labels where controlled by the tool.                                                                                                                                  |
| DB/publication sidechain                                | V1.22 records Supabase/Postgres as a future public read-only mirror and cnna-project.net visualization layer, not a Lean/YAML replacement.                                                                                                                                                                                                            |
| New planning phases                                     | Added 20.15-20.20 for DB schema, YAML/import roundtrip, snapshot/checksum model, public read-only API/RLS views, web-frontend contract and future reviewer/admin model.                                                                                                                                                                               |
| New guardrail                                           | Added GNG34: no Supabase row is a CNNA truth source, no public writes, no browser service__role key, and no DB update without snapshot/hash/import report.                                                                                                                                                                                            |
| Implementation status                                   | No live Supabase connection, no DB write, no frontend deployment and no Lean source change were performed in V1.22.                                                                                                                                                                                                                                   |
| Active cursor                                           | The active mathematical/content cursor is unchanged: 1.1.1.1.5.1 remains the pre-code IST source intake completeness audit.                                                                                                                                                                                                                           |

### 3 Working format and module mapping

**What this section shows.** This section records the maintained work format, root/source layout, and module-mapping assumptions. Its source is the Single-YAML-Master. It is shown because file placement, generated-output handling and source-root conventions affect every later command and review packet.

| Topic                                | Assessment                                                                                                                                                                                                                                                                          | Decision                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary working format               | LaTeX, generated tables, and exported TSV files are unsuitable as primary editing sources for the roadmap.                                                                                                                                                                          | The only maintained source is masters/cnna_roadmap_master.yaml; all tables, links, and optional exports are generated from it.                                                                                                                                                                                                                 |
| Tables as single source              | Separate phase, object, and access master files drift even if their rendered tables look synchronized.                                                                                                                                                                              | phases, objects, access policy, legends, code audit, and scratch-pad records are all sections of one YAML master.                                                                                                                                                                                                                              |
| Scratch-pad synchronization          | Manual scratch-pad prose inside the .tex file causes drift.                                                                                                                                                                                                                         | The How implemented column is generated from phase_id links into implementation_scratchpad.yaml; green phases without YAML records are generator errors.                                                                                                                                                                                       |
| Code-module assignment               | Every object and phase row names the current derivation site and the neutral target module.                                                                                                                                                                                         | This makes visible whether something is derived-only, legacy-only, incomplete, or missing.                                                                                                                                                                                                                                                     |
| Subphases                            | Unlimited dot notation is allowed only when dependency depth justifies it.                                                                                                                                                                                                          | Split criterion: too many prerequisites, a deep DAG chain, different proof kinds, or a new generation layer.                                                                                                                                                                                                                                   |
| Object and phase status separated    | A shared color code hides whether an object is complete or only close.                                                                                                                                                                                                              | Object status uses W/B/G/Y/R/K. Phase status uses only G/Y/R/W with exactly one active cursor.                                                                                                                                                                                                                                                 |
| Single-master validation             | A new phase, object, or scratch-pad record must appear everywhere after generation, or generation fails.                                                                                                                                                                            | The generator validates phase IDs, object IDs, phase-object links, active cursor, scratch-pad completeness, and no manual generated columns.                                                                                                                                                                                                   |
| Roadmap horizon policy               | The full CNNA roadmap must be visible without treating distant goals as proved or implementation-ready.                                                                                                                                                                             | Short-term focus is Pillar A. Mid-term focus is Pillar B QFT and Pillar C OQS. Long-term focus is Pillar D spacetime, Pillar E matter, and explicit backreaction. External speculative theories are side-comparison targets only.                                                                                                              |
| No speculative foundation import     | Strings, loop quantum gravity, noncommutative geometry, and full ToE narratives may be useful external comparisons but must not enter the CNNA foundation as assumptions.                                                                                                           | Only established mathematics and physics objects are admitted into the derivation core. Speculative frameworks may be registered as consumer-side comparisons if they emerge from CNNA outputs.                                                                                                                                                |
| Implementation scratch-pad workflow  | A phase can be prepared from reliable mathematical/physical sources before any Lean implementation. Preparation must not be confused with closure.                                                                                                                                  | Use scratch-pad fields inside the single YAML master: source materials, extracted theory objects, dependency audit, proof rehearsal, tactic plan, guardrails, fallback layer, and post-green checks.                                                                                                                                           |
| TomS thread ingestion rule           | The thread is useful as a map of established structures, but it mixes exposition, corrections, heuristic CNNA replies, and later clarifications. It is not a source of CNNA-generative data.                                                                                        | Store extracted objects in a cross-check register, map them to phases/objects, and classify each as derived target, comparison-only target, correction, or no-action note. Use primary/authoritative sources before any formalization step.                                                                                                    |
| Legacy TeX/PDF authority cleanup     | Retaining v20/v21 TeX filenames as maintained-looking sources creates ambiguity after the YAML master becomes authoritative.                                                                                                                                                        | v22 package uses a v22 main TeX/PDF filename. Any old TeX/PDF is historical output only and should not be edited as a roadmap source.                                                                                                                                                                                                          |
| Blue-object and pruning planning     | Derived-only proofs are not enough to make an object public-final when the semantic/API boundary is still wrong. Likewise, syntactic non-use is not enough to delete.                                                                                                               | v23 introduces consumption classification, blue-object correction ledger, public API contract, and pruning ledger before any code move/deletion.                                                                                                                                                                                               |
| Implementation module manifest       | Before any code change, the plan must state exact added/modified modules, semantic placement, import direction, reuse decision, public/legacy role, consumer direction, generic finite test placement, phase type, regression mode, version impact, and mathlib/noncomputable risk. | Phase 1.1.1.1.1 closes this as a generated/validated module-manifest contract. The generator now emits a module-manifest table/TSV from scratchpad fields and validates that every scratchpad record carries the manifest fields; later code phases must fill them before Lean edits.                                                          |
| EXEC-2 rational ExecComplex boundary | ExecComplex is $\mathbb{Q} \times \mathbb{Q}$ and is suitable for integer/rational operational invariants in Phases 3-6, but not for general algebraic irrationalities such as $\sqrt{D}$ , general CM tau, algebraic eigenvalues, or non-rational lattice embeddings.              | Operational paths end at the $\mathbb{Q} \times \mathbb{Q}$ boundary. Algebraic irrationalities live only in mirror/non-operational layers. Only proved rational/integer invariants, such as discriminants, norms, rational special-angle cosines, and $h=1$ integer j-values when separately proved, may flow back into the operational path. |
| Refactor regression mode             | The Notation split touches a large import surface, so an all-at-once move risks long red intervals and consumer ambiguity.                                                                                                                                                          | Use wrapper-first incremental refactoring: add neutral wrappers beside legacy, migrate bounded consumer batches, record declaration counts/import deltas/breakage audit, and require local full-build evidence for every code package before closing a refactor phase.                                                                         |
| Lean/mathlib version policy          | Future number-theory and algebra phases may be tempted to chase newer mathlib APIs, while the current baseline is Lean 4.28.0 plus its mathlib snapshot.                                                                                                                            | Freeze to the baseline for implementation unless an explicit phase-boundary upgrade gate is opened. A version upgrade is its own review/build event; mathlib noncomputable pressure must be isolated in mirror/non-operational layers or cause a natural-source refactor, not an operational leak.                                             |
| Phase-type-specific scratchpads      | A refactor phase and a new derivation phase need different evidence. One flat scratchpad creates not-applicable ballast and misses migration metrics.                                                                                                                               | Every phase records phase_type. Refactor phases require migration/regression metrics; derivation phases require proof rehearsal/tactic/mathlib data; comparison phases require derived-only boundary; infrastructure phases must state why no operational witness is required.                                                                 |
| Governance-depth retrospective       | The deep Phase-1 governance stack is justified by the SM1-SM9h failure modes, but it must not become the project itself.                                                                                                                                                            | After the first real code-change phase 1.1.2 closes, run an explicit retrospective: identify which governance gates prevented actual errors and decide whether Phase 3+ can use a flatter max-three-level structure.                                                                                                                           |

| Topic                                                          | Assessment                                                                                                                                                                                                                                                                                                                                                                                                                                 | Decision                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 3 to Phase 4 consumer contract                           | Phase 3 must wrap Schur/DtN before Phase 4 consumes it, but the Phase-4 generator path is the canonical operational consumer and should shape the Phase-3.6 public contract.                                                                                                                                                                                                                                                               | Phase 3.6 publishes a consumer-ready BoundaryOperatorProfile for Phase 4.1 without importing Phase 4. Phase 4 consumes that contract and must not reconstruct Schur/DtN internals or implement parallel generator logic.                                                                                                                                                                       |
| IdealThread/CutSpec.full extremal routes                       | Code confirms IdealThread is an InfiniteThread with singleton carrier per level and DirectedFamily embedding;<br>CutSpec.full is carrier := T.carrier. ExistingIdealThreadNoGo already proves the singleton route cannot generate a non-singleton carrier/boundary source.                                                                                                                                                                 | Treat IdealThread and CutSpec.full as two extremal profiles. IdealThread may supply scalar/thread-route audits;<br>full boundary/DtN matrices require full/generated boundary data. Any lifting from thread to nontrivial boundary geometry needs its own theorem and phase.                                                                                                                   |
| String-like interpretation boundary                            | The one-dimensional/string-like reading is a potentially useful physical analogy, but it is not a Lean theorem and cannot supply generative CNNA data.                                                                                                                                                                                                                                                                                     | Register the analogy only as comparison/diagnostic material under secondary long-term goals and proof dossiers. Never use it to justify a module, formula, discriminant, spectrum, or handoff without derived code evidence.                                                                                                                                                                   |
| Thread recursion versus full-boundary arithmetic provenance    | The code contains RecursiveSchurSource/MobiusTransfer/CharpolyExec chains, but the exact provenance relation between scalar recurrence, IdealThread singleton carriers, full-boundary diagonal entries, and generated boundary sources must be audited before interpretation.                                                                                                                                                              | Insert a Phase 4.0 audit before operational arithmetic wrappers. D=-3/D=-4 or CM claims may be promoted only through the operational discriminant/number-theory chain, not through analogy.                                                                                                                                                                                                    |
| Horizontal/vertical Schur-complement duality bridge order      | The algebra path and arithmetic path plausibly share Schur-complement structure, but the claim depends on separately derived horizontal sector elimination, vertical level recursion, transfermatrix data, and CD norm structure.                                                                                                                                                                                                          | Register the bridge as BR08 in Phase 10.4.1 only. It is not an input to Phase 1.4, 3, 4, 5, or 6. CD8 is deferred to Phase 10.4.2 as a corollary candidate, not a Phase-6 final theorem.                                                                                                                                                                                                       |
| One-generator Schur mechanism boundary                         | A unified Schur generator is an architectural target: the same Dirichlet-Laplacian/Schur machinery may yield horizontal algebraic and vertical arithmetic invariants. This remains a theorem obligation, not a premise.                                                                                                                                                                                                                    | Use the one-generator language only as a planning constraint and proof target. Code may share the canonical generator, but no formula may be accepted because it fits the unification story; every entry must be proof-carrying from the earlier route.                                                                                                                                        |
| Heuristic Eisenstein/Gauss lattice tests                       | Post #97 gives a useful heuristic for 120-degree Eisenstein and 90-degree Gauss spiral embeddings of a b-ary tree, but it does not define the code branching object and includes irrational/analytic targets outside ExecComplex.                                                                                                                                                                                                          | Represent this material as a consumer-only test matrix after generator/lattice/CM/j source objects exist. Use code-defined b and L_max;<br>operational rows may compute finite relations and rational/integer invariants, while tau, sqrt(D), coordinates, and analytic j stay in mirror/comparison layers.                                                                                    |
| General early finite test matrix vs. specialized sentinel rows | Eisenstein/Gauss tests are valuable early sentinels, but two special examples cannot certify that the computable path is general or non-skeletal.                                                                                                                                                                                                                                                                                          | Introduce O07 as the generic finite computation test matrix. L09/L10/M08/BR10 are specialized consumers/sentinel extensions. Every computation phase must define generic rows before or alongside sentinels.                                                                                                                                                                                   |
| Regime split for Heegner/CM test rows                          | The nine classical class-number-one/Heegner fields are number-theoretic comparison targets, but CNNA closed/unitary status is a regime statement, not a class-number statement. The noOuter empty-set REAL-root branch may have only Eisenstein/Gauss as early sentinel candidates; additional Heegner targets are naturally treated as IDEAL_infty projection/complement-family rows until proven otherwise.                              | Implement regime-typed test rows. Generic finite rows come first. Eisenstein/Gauss may be tested as noOuter-root sentinels. All other Heegner/CM targets default to HasOuter/complement-projection or comparison-only rows unless a later proof derives noOuter provenance.                                                                                                                    |
| EXEC-3 computable scalar architecture                          | ExecComplex is code-confirmed as rational pairs QxQ and maps into mathlib Complex through ExecComplexBridge. This is executable and robust for rational/integer matrix, polynomial, determinant and discriminant data, but it cannot represent general algebraic irrational coordinates such as sqrt(D) or tau. The need for a quadratic layer is therefore foreseeable;<br>the provenance question is solved by indexing it by derived D. | Keep ExecComplex as the rational operational floor. Do not expand it into a full Complex layer. Treat O09 as a required exact quadratic schema for later CM/lattice coordinates, but concrete instantiation requires O10/BR14 and a proof-carrying discriminant D-source.                                                                                                                      |
| Exact quadratic layer versus full computable Complex           | Mathlib Complex is Real x Real, and mathlib number fields/algebraic closures are formal rather than generally executable. Mathlib QuadraticAlgebra gives explicit coordinates and multiplication; AdjoinRoot demonstrates relation-generated root adjoining via quotient, but docs expose noncomputable instances that require boundary audit.                                                                                             | Implement or wrap a minimal discriminant-indexed exact quadratic schema rather than a full computable Complex. Mathlib QuadraticAlgebra may guide/wrap the schema after audit;<br>AdjoinRoot is a useful quotient/reference pattern but any noncomputable quotient/field instances must stay out of operational code unless isolated.                                                          |
| Cayley-Dickson route to C-layer noncircularity                 | The current CD code is structural and shares Schur provenance;<br>ArithmeticCDBridge explicitly records no ExecMirror-CDComplex identification without a transfer lemma. A CD-derived C surface is therefore not currently an operational scalar replacement.                                                                                                                                                                              | Do not derive the operative C-layer from Cayley-Dickson before CD7/BR08/CD8. Register CD9 as a late noncircularity audit. If proved, it may compare or bridge scalar surfaces;<br>it may not retroactively justify earlier ExecComplex or arithmetic definitions.                                                                                                                              |
| EXEC-4 derived-only exact quadratic provenance                 | The later theory will need exact quadratic data, but derived-only forbids introducing sqrt(D) as a number, axiom, or CD/Complex import. The only acceptable source is an already derived operational discriminant parameter D.                                                                                                                                                                                                             | O09 is a parameterized algebraic schema with $\theta^2 = D$ . O10/BR14 instantiate it only from O05/O06 or equivalent proof-carrying discriminant ledgers. Missing D is an upstream discriminant-source gap, not a scalar-layer patch.                                                                                                                                                         |
| PHASE-5-10 dependency bifurcation                              | v36 placed spectral/physical objects inside or behind a modular-form/j chain, even though SP0/SP1/P00-P06 consume finite boundary matrices, algebra/star/state/regime data and not j or modular forms. Phase 10 also mixed modular, transfer, spectral, analytic and bridge targets.                                                                                                                                                       | Split the post-Phase-4 roadmap into Path A (number theory -> lattice geometry -> modular/j -> analytic theta/L/class-number formula) and Path B (spectral finite profile -> finite physical/operator dynamics -> heat/spectral geometry -> transfermatrix -> late Schur/CD bridges). No object may inherit a modular-form dependency unless its proof target explicitly consumes modular data. |
| PHASE-OBJECT-LINK policy                                       | The phase table previously rendered Linked objects from direct equality object.Phase == phase.Phase only. Empty cells were often valid for umbrella, closeout, baseline or moved phases, but the view did not explain that and could be mistaken for missing objects.                                                                                                                                                                      | Every phase has an explicit Object link policy: direct, scope_only, no_direct_object, or moved. The generator renders policy-aware text and validates that direct phases have direct objects and scope-only phases have child object scope. Blank Linked objects cells are forbidden.                                                                                                          |
| ONT-1 ontic subsystem-minimality                               | Without an explicit ontic-minimality contract, later phases could silently import dynamics, environment, geometry, stochasticity, arithmetic, operator algebra or physics as primitive interpretation rather than deriving them from the subsystem/complement cut.                                                                                                                                                                         | The only primitive ontic emergence trigger is subsystem-being inside the formal IDEAL_infty provenance whole. All effective structures are derived downstream from cut, boundary, complement families, regime and Schur/DtN consequences. Physical readings are consumer interpretations, never generative inputs.                                                                             |



| Topic                                                                    | Assessment                                                                                                                                                                                                                                                                         | Decision                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DB-REVIEW main side objective and database-ready planning representation | The Single-YAML-Master remains the maintained planning source, but its information should be structured so phases, objects, modules, handoffs, status colors, theory links, proof dossiers, and review paths can later be queried through a relational or graph-friendly database. | Add Phase 20 as a non-blocking main side objective. Technical database implementation is later, but schema discipline and review semantics apply immediately. YAML/PDF/TSV/DOT are generated views, not a second truth system. |
| Epistemic review status and no-infallibility maxim                       | A green Lean build proves formal consistency relative to selected definitions, not semantic infallibility of modeling choices, interpretations, or established-theory attachments. Generated artifacts can also contain semantic errors.                                           | Record the maxim that nothing is infallible and no 100-percent truth is declared. Every relevant claim remains under discipline-specific expert review, and missing blue/reference-theory markings are active audit debt.      |
| OOP-style context-class separation for phases and objects                | The previous status colors were doing two jobs: release-state signaling and, implicitly, type/context signaling. That risks a category error: a green tool/database/reference object can look like a green Lean object.                                                            | V1.26 adds Context class as an orthogonal class/type field on all phases and objects, renders it in phase/object/proof/scratchpad/module-manifest tables, and keeps status colors strictly as release/proof-state indicators.  |

4    Static code findings

**What this section shows.**    This table lists static findings discovered during repository review, implementation rehearsals and tool maintenance. Each row has a lifecycle status, addressing phase and version/timestamp. It is shown to keep known risks, closed issues and unresolved structural problems explicit rather than leaving them in chat history.

**Static finding triage rule.**    This table is sourced from the Master YAML code-audit ledger. Every static finding is a tracked issue, not loose prose. Each row records the tool/change version and timestamp at which the issue record was last materially changed. Static-finding colors are: **R/red = urgent**, **Y/yellow = critical**, **G/green = closed**. New static problems discovered during feasibility checks, implementation, or review must be added here with an ID, lifecycle status, and addressing phase before work continues.

**Table legend.**    R = red/urgent; Y = yellow/critical; G = green/closed.

| ID    | Priority | Issue status              | Addressing phase               | Change version       | Change timestamp          | Finding                                                     | Static observation                                                                                                                                                                                                                                                                 | Risk                                                                                                                                                                                                                      | Refactoring consequence                                                                                                                                                                                                                          |
|-------|----------|---------------------------|--------------------------------|----------------------|---------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCF01 | R        | open-addressed-by-phase   | 1.1.3                          | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Arithmetic/Notation.lean                                    | 5095 lines, 903 declarations, 0 notation/syntax/macro declarations by static grep.                                                                                                                                                                                                 | The notation module is an alias/plan-semantics monolith; Part 2 must not import it as an API.                                                                                                                             | Split into true Notation/CoreSymbols, Legacy/PhaseAliases, and NeutralWrappers.                                                                                                                                                                  |
| SCF02 | Y        | open-addressed-by-phase   | 1.1.4; 1.1.5                   | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Plan-semantic Smoke names                                   | Many active derivation sites carry SM/R/Recheck/Plus/Full/Audit in file names, e.g. SM9h, SM9e, R3c1.                                                                                                                                                                              | Without Plan Part 1, the code surface is not readable by domain role.                                                                                                                                                     | New public modules must be object/role based only; old modules remain legacy sources.                                                                                                                                                            |
| SCF03 | Y        | open-addressed-by-phase   | 1.1.4; 4.1-4.5                 | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Operational SM9h core                                       | SM9h leaves an ExecComplex discriminant derived from a quadratic window, but under historical names.                                                                                                                                                                               | Mathematics is usable, but not Part-2-neutral.                                                                                                                                                                            | First wrap Operational/QuadraticWindow and Operational/QuadraticDiscriminant.                                                                                                                                                                    |
| SCF04 | Y        | open-addressed-by-phase   | 5.1-5.5; 6.1-6.4               | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Orders / number theory                                      | Discriminant, quadratic order, norm forms and Galois/Eisenstein special objects exist in bridge form.                                                                                                                                                                              | Still no general D_op -> D_fund/conductor/order pipeline.                                                                                                                                                                 | Build NumberTheory/* only after the operational profile is neutralized.                                                                                                                                                                          |
| SCF05 | Y        | open-addressed-by-phase   | 6.5; 10.4.2; 10.4.3            | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Cayley-Dickson                                              | The Structural/CayleyDickson path includes star, normSq, multiplication, alternative-law and norm-multiplicativity material.                                                                                                                                                       | Do not delete; no free identification with sectors.                                                                                                                                                                       | Register it as a structural consumer/bridge and keep it out of the arithmetic source chain until a bridge is proved.                                                                                                                             |
| SCF06 | R        | open-addressed-by-phase   | 2.3; 11.1-11.5; 17.1-17.5      | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Handoff                                                     | A_to_B exports carrier, Schur, regionNet, split, DtN, selectedLevel, and regime data. Inputs B/C/D/E_to_A exist as empty/small structures.                                                                                                                                         | Pillar A could accidentally treat its own outputs as sources.                                                                                                                                                             | The access matrix enforces: A consumes only inputs; outputs are forbidden-to-A.                                                                                                                                                                  |
| SCF07 | Y        | open-addressed-by-phase   | 10.2; 13.1-13.6; 14.1-14.5     | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Finite/AQFT seeds                                           | StateSpace, MatrixExponential, GibbsStateSeed, UnitaryEvolution, and ChannelSeed exist as A-internal seeds.                                                                                                                                                                        | They can behave like parallel physics generators if not attached to an A-operator profile.                                                                                                                                | Mark them as consumers of BoundaryOperator/ExecMatrix; no independent operator source.                                                                                                                                                           |
| SCF08 | G        | closed-in-workflow        | 1.1.1.1.0.1; workflow_policy   | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Implementation preparation must be proof-rehearsed          | Previous SM phases often exposed missing sources only during implementation.                                                                                                                                                                                                       | The assistant may patch consumers, obstruct too early, or forget ExecComplex/import/no-go constraints.                                                                                                                    | Every phase record has preparation fields: dependency audit, source-derived theory objects, proof rehearsal, tactic plan, natural fallback source, and explicit no-go checklist.                                                                 |
| SCF09 | G        | closed-generated-view     | 1.1.1.1.4; 19.1                | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Integrated Lean module inventory                            | The SM9h snapshot contains 287 Lean modules, 724 internal import edges, zero missing internal imports, and an acyclic import graph according to build_export_script_v1.8.py.                                                                                                       | If module names and dependency structure are not visible in the planning tool, object/phase planning can drift away from the repository surface.                                                                          | The planning package now carries a generated module catalog, dependency summaries, source-path links, and graph figures derived from the repository snapshot. These are generated views, not manual planning sources.                            |
| SCF10 | Y        | open-comparison-guardrail | 8.1-8.8; 9.1-9.5; 12.1-12.8    | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | TomS thread cross-check / external-reference intake         | Thread pages 1-5 introduce prime-generation heuristics, imaginary-quadratic lattices, SL2Z/j, L-functions via modular forms, finite simple groups, Monster/Moonshine, Leech lattice and VOA/string-theory context.                                                                 | These topics are established-reference material. They must not become CNNA generators, hidden axioms, imported physical ontology, or proof shortcuts; several statements require precision corrections before formal use. | Register the material as source-audit and comparison-only objects unless a later phase derives the required CNNA object from A provenance. Add Leech/finite-simple-group/VOA/Moonshine comparison profiles after j/q-expansion, not before.      |
| SCF11 | G        | closed-routing-lock       | 1.1.1.1.0; 1.1.1.1.1-1.1.1.1.4 | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Phase 1.1 split refinement / Legacy.StatusGates quarantine  | The next frontier combines three different work kinds: declaration classification, legacy enum quarantine, true notation extraction, and import firewall verification.                                                                                                             | Keeping this as one implementation phase would recreate the previous monolith problem and make missing imports look like proof failures.                                                                                  | Split 1.1 into 1.1.1-1.1.5. Activate only 1.1.1 first; move the 749 single-constructor status enums in 1.1.2, not during inventory.                                                                                                              |
| SCF12 | G        | closed-generated-view     | 1.1.1.1.4                      | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Declaration-level consumption gap                           | Module graph reachability is strong, but declaration-level semantic consumption is not equivalent to BuildAll/Umbrella import reachability. Current snapshot: 287 Lean modules, 724 internal import edges, 8860 declarations, 394 static reference edges, 1686 referenced symbols. | A declaration can look consumed because an umbrella imports its module, while it is only a status/audit marker or phase alias. Blind pruning or blind retention would both corrupt the derived-only path.                 | Insert Phase 1.1.1.1 and LEG5 before code movement: classify declarations as core-computational, proof-bridge, public-wrapper, legacy-status, legacy-policy, phase-alias, audit-only, blue-semantic-correction, dead-candidate, or final-output. |
| SCF13 | R        | open-addressed-by-phase   | 1.1.2                          | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Single-constructor inductives are broader than status enums | A rough direct static scan finds about 872 single-constructor inductives in 99 modules; the earlier 749 figure should be treated as a narrower status-enum subset, not the full population.                                                                                        | Moving every single-constructor inductive into Legacy.StatusGates would swallow policy/proof-bridge declarations and hide real provenance.                                                                                | Phase 1.1.2 must consume the classification from 1.1.1.1/1.1.1.3. Constructor count alone is not a quarantine criterion.                                                                                                                         |

| ID    | Priority | Issue status                  | Addressing phase                                                          | Change version       | Change timestamp          | Finding                                                                    | Static observation                                                                                                                                                                                                         | Risk                                                                                                                                                                                             | Refactoring consequence                                                                                                                                                                                                     |
|-------|----------|-------------------------------|---------------------------------------------------------------------------|----------------------|---------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCF14 | R        | active-next                   | 1.1.1.2                                                                   | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Blue objects are proof-carrying but not public-final                       | Object status B rows occur in Cayley-Dickson, Schur/DtN, and operational arithmetic families. These are already derived-only/proof-carrying in current code, but still semantically/API unfinished.                        | Treating blue objects as final because proofs exist would export old SM/AR semantics or wrong API boundaries into Phase 3+ and handoffs.                                                         | Add Phase 1.1.1.2 and LEG7. Every blue object must get a correction ledger entry: semantic defect, natural correction phase, allowed temporary consumers, forbidden public consumers, and exit-to-white condition.          |
| SCF15 | Y        | open-addressed-by-phase       | 1.1.1.4                                                                   | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Pruning must follow API contracts, not precede them                        | Syntactic sinks and weakly referenced declarations include diagnostic leaves, audit leaves, intentional outputs, and proof-obligation terminals.                                                                           | Deleting before a public API contract can remove audit/provenance evidence or proof bridges that are not mainline outputs but still necessary.                                                   | Add Phase 1.1.1.4 and LEG8. Deletion is postponed to a later optional pruning phase and requires classification plus local green build evidence.                                                                            |
| SCF16 | G        | closed-generated-view         | 1.1.1.1.1                                                                 | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Module budget missing before implementation                                | The v23 plan classifies module/declaration consumption but did not yet require every implementation phase to name the exact modules to add/modify before Lean code is written.                                             | Kleinteilige phases can degenerate into kleinteilige code: one new file per subphase, accidental wrappers, and parallel helper modules that are syntactically green but semantically fragmented. | Insert 1.1.1.1.1 and LEG10. Every implementation phase must carry a module manifest with exact module names/count, semantic placement rationale, reuse decision, and import role before code starts.                        |
| SCF17 | G        | closed-generated-view         | 1.1.1.1.2                                                                 | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Duplicate generator risk                                                   | The Smoke-Test Generator is intended to become the canonical generator seam for the whole theory, but later phases could accidentally reimplement request/output, extraction, ledger, or smoke-test logic under new names. | Duplicate generator paths would break derived-only provenance by creating competing sources of truth and would make future proof reuse fragile.                                                  | Insert 1.1.1.1.2 and LEG11. Generator-like code must reuse or extend the canonical generator at its natural layer; parallel generators are blocked unless a genuine semantic split is recorded.                             |
| SCF18 | Y        | deferred-secondary-roadmap    | 19.1-19.4                                                                 | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Secondary derivation-graph analysis assets exist                           | repo_inventory/raw_export already contains import graph data and related forensics sufficient for future documentation coverage and derivation-geometry diagnostics.                                                       | If activated too early, graph metrics could be mistaken for CNNA-generative evidence or distract from closing operational derived objects.                                                       | Keep as secondary long-term phases 19.1-19.4; use for documentation/audit only until core derived objects and public API are stable.                                                                                        |
| SCF19 | G        | closed-policy-open-consumers  | 1.1.1.1.0.2;<br>1.2.1.2;<br>4.5.1                                         | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | ExecComplex rational ceiling                                               | ExecComplex is represented as rational pairs $Q \times Q$ ; later CM/lattice objects require algebraic or analytic data.                                                                                                   | Operational path may become a skeleton or silently import mirror/noncomputable semantics.                                                                                                        | Adopt EXEC-2: operational code carries rational/integer invariants; algebraic irrational data remains mirror-only unless a return invariant is proved.                                                                      |
| SCF20 | G        | closed-policy-open-consumers  | 1.1.1.1.0.3;<br>1.1.2-1.1.5                                               | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Refactor regression risk                                                   | Notation/alias cleanup touches a large declaration and import surface.                                                                                                                                                     | A single large move can create ambiguous red states and consumer-side patches.                                                                                                                   | Use wrapper-first bounded migrations with declaration-count, import-delta, consumer-breakage, and local full-build evidence.                                                                                                |
| SCF21 | G        | closed-policy-open-consumer   | 1.1.1.1.0.7;<br>3.0                                                       | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | IdealThread singleton collapse is code-backed                              | ToC/Contract.lean defines IdealThread.carrier L as {U.cell L}; IdealThread.family uses that carrier; deterministic coarsen_carrier_eq_prev proves coarsenSet of the next carrier is the previous singleton.                | Any plan that uses an existing IdealThread route as a nontrivial boundary matrix source would silently collapse to scalar/degenerate data.                                                       | Add S07 and BR06. Later full-boundary/DtN phases must explicitly reject IdealThread as full matrix source unless a separate lifting theorem is supplied.                                                                    |
| SCF22 | G        | closed-policy-open-consumer   | 1.1.1.1.0.7;<br>3.0                                                       | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | CutSpec.full is the maximal carrier route                                  | Finite/CutSpec.lean defines full T with carrier := T.carrier and full_carrier is rfl.                                                                                                                                      | Confusing CutSpec.full with IdealThread or with a diagonal/scalar route would erase the intended sector/interior/interface geometry.                                                             | Add S08 and require Phase 3.0 boundary-route separation before neutral Schur/DtN wrappers.                                                                                                                                  |
| SCF23 | G        | closed-policy-open-consumers  | 1.1.1.1.0.13;<br>1.2.1.2;<br>4.5.1;<br>7.1-7.4                            | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Exact quadratic scalar provenance gap                                      | v35 treated O09 as optional/pilot; CM/lattice phases will require exact quadratic coordinates, but derived-only prohibits free sqrt(D).                                                                                    | A scalar consumer could silently introduce theta/sqrt(D) without an operational discriminant source, turning a derived-only construction into an axiom or mirror import.                         | v36 requires O09 as schema, adds O10/BR14 as D-source instantiation bridge, and adds GNG19 no-free-sqrt enforcement.                                                                                                        |
| SCF24 | G        | closed-in-v44-generator       | all phases before implementation;<br>immediate enforcement before 1.1.1.2 | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Static finding lifecycle fields were missing                               | The Section 2 Static code findings table listed risks and consequences but did not force an issue ID, current status, priority color, or addressing phase.                                                                 | A known finding could be re-discovered in later work, silently left open, or closed by an unrelated phase without a visible provenance link.                                                     | Every static finding row now requires ID, Priority, Issue status, and Addressing phase. New findings must be added to Section 2 before implementation continues.                                                            |
| SCF25 | G        | closed-in-v44-workflow-policy | all active phases;<br>current gate before 1.1.1.2                         | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Pre-start feasibility check was implicit, not a hard visible workflow gate | The scratchpad contained dependency audit/proof rehearsal fields, but the workflow table did not require a separate go/no-go feasibility decision immediately before starting the next phase.                              | A phase could begin from a green prior build while a static finding or missing dependency still requires a natural backjump, producing consumer-side patches or over-fragmented governance work. | A mandatory pre-start feasibility workflow now checks active phase dependencies, static findings, module manifest, reuse audit, and whether the phase is infrastructure-only or code-changing before implementation starts. |

| ID    | Priority | Issue status                                        | Addressing phase                                | Change version       | Change timestamp          | Finding                                                                                                         | Static observation                                                                                                                                                                                                                                                                             | Risk                                                                                                                                                                                                                   | Refactoring consequence                                                                                                                                                                                                                                                                                        |
|-------|----------|-----------------------------------------------------|-------------------------------------------------|----------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCF26 | Y        | open-addressed-by-active-phase                      | 1.1.1.2                                         | baseline-before-V1.0 | 2026-05-10T05:51:42+02:00 | Blue correction cannot rely only on status B labels                                                             | The active Phase 1.1.1.2 consumes object register rows with Object status B, but a valid LEG7 ledger must also bind each row to proof dossier, current derivation site, allowed temporary consumers, forbidden consumers, and exit-to-white condition.                                         | A proof-carrying blue object could be whitened or publicly wrapped merely because the object status row was found, without recording the exact semantic/API defect and correction phase.                               | Phase 1.1.1.2 must generate or validate a per-blue-object correction ledger, not a status-only summary. Public API and pruning phases remain blocked until that ledger is closed.                                                                                                                              |
| SCF27 | G        | closed-in-tool-v1.0-legend-policy                   | pre-1.1.1.2 tool V1.0 documentation update      | V1.0                 | 2026-05-10T05:51:42+02:00 | Generated table views needed local, section-specific legends.                                                   | Earlier rendering either lacked local legends or inserted an over-broad mixed legend. The current generator emits static/phase/object legends only where the corresponding colored cells exist.                                                                                                | Wrong or missing legend text makes color semantics ambiguous across static findings, object status, and phase status tables.                                                                                           | Use explicit legend kinds in the generator and no legend for non-status tables.                                                                                                                                                                                                                                |
| SCF28 | G        | closed-in-tool-v1.01-versioning                     | tooling/versioning-policy V1.01                 | V1.01                | 2026-05-10T06:17:44+02:00 | Tool contract version and data snapshot version were conflated as v40-v44 document increments.                  | The previous split still allowed the Lean-code data timestamp to move when only the generator/PDF display changed.                                                                                                                                                                             | A documentation-only edit could be misread as a Lean-code data change, destroying reproducible alignment between Lean snapshots and planning artifacts.                                                                | Metadata now separates tool_version, tool_change_timestamp, and data_timestamp. tool_version increments by 0.01 for tool/display changes; data_timestamp is restored and changes only with Lean code/data.                                                                                                     |
| SCF29 | G        | closed-in-tool-v1.0-static-finding-migration        | pre-1.1.1.2 tool V1.0 documentation update      | V1.0                 | 2026-05-10T05:51:42+02:00 | Essential findings front matter duplicated issue tracking outside Section 2.                                    | The v44 PDF opened with prose bullets that contained actionable workflow findings, while Section 2 was supposed to be the issue ledger.                                                                                                                                                        | Important problems can remain unprioritized, unassigned, or forgotten if they live outside the Static code findings table.                                                                                             | The separate Essential findings section was removed; the actionable content is represented as SCF rows with priority, lifecycle status, and addressing assignment.                                                                                                                                             |
| SCF30 | G        | closed-in-tool-v1.0-layout                          | pre-1.1.1.2 tool V1.0 documentation update      | V1.0                 | 2026-05-10T05:51:42+02:00 | Complete Lean module catalog: Module column could run visually into the Group column.                           | Long dotted Lean module names were emitted inside raw href cells without dot-level breakpoints and with too little Module-column width.                                                                                                                                                        | The generated module inventory becomes unreliable as a review object if readers cannot distinguish module names from group labels.                                                                                     | Module and path link text is now breakable at dots; the Module column is widened and the Group/Path widths are adjusted.                                                                                                                                                                                       |
| SCF31 | G        | closed-in-tool-v1.0-object-links                    | pre-1.1.1.2 tool V1.0 documentation update      | V1.0                 | 2026-05-10T05:51:42+02:00 | Object table A IDs were row anchors but not clickable links.                                                    | object_table_id_cell(target=True) emitted a hypertarget around plain ID text, while other object tables emitted hyperlinks to that row.                                                                                                                                                        | The first object table looked less navigable than the rest of the proofbook workflow and did not expose the proof-dossier jump from the canonical object list.                                                         | Object table A IDs now remain row anchors and additionally link to the corresponding proof-dossier target.                                                                                                                                                                                                     |
| SCF32 | G        | closed-in-tool-v1.01-legend-policy                  | tooling/static-finding-rendering V1.01          | V1.01                | 2026-05-10T06:17:44+02:00 | Global table legends were semantically over-broad.                                                              | The V1.0 generator inserted the same mixed object/phase/static legend before generated tables even when the table had no colored status cells or the section required a narrower legend.                                                                                                       | A reader could interpret colors with the wrong status system, especially static findings and phase rows.                                                                                                               | The generator now supports explicit legend kinds (static, phase, object) and emits no legend for tables without colored status/triage semantics.                                                                                                                                                               |
| SCF33 | G        | closed-in-tool-v1.01-data-timestamp-policy          | tooling/versioning-policy V1.01                 | V1.01                | 2026-05-10T06:17:44+02:00 | Lean-code data timestamp advanced during a display-only update.                                                 | The V1.0 legend-policy update reported a new data timestamp even though no Lean source file was changed.                                                                                                                                                                                       | Downstream reviewers could infer a new Lean data state from a purely presentational PDF/generator change.                                                                                                              | data_timestamp/data_version are reset to 2026-05-10T05:51:42+02:00; tool_change_timestamp records the current display-policy correction separately.                                                                                                                                                            |
| SCF34 | G        | closed-in-tool-v1.01-record-change-tracking         | tooling/table-rendering V1.01                   | V1.01                | 2026-05-10T06:17:44+02:00 | Changed static findings, scratchpad records, and object rows had no visible per-record version/timestamp field. | Before V1.01, table readers could see final status but not which tool/display version last changed the row.                                                                                                                                                                                    | Review history becomes ambiguous when a table row changes without visible version/time metadata.                                                                                                                       | Static findings expose Change version and Change timestamp. Scratchpad and object proof-link views render record metadata and default unchanged rows to unchanged-by-V1.01 / restored Lean data timestamp.                                                                                                     |
| SCF35 | G        | closed-in-tool-v1.02-pdf-navigation                 | tooling/pdf-navigation V1.02                    | V1.02                | 2026-05-10T06:27:26+02:00 | PDF navigation lacked a clickable contents page and explicit outline/bookmark tree.                             | Before V1.02, the PDF body started directly with Latest change overview and relied mainly on reader search or section scrolling. The existing hyperref setup did not expose a document-level contents page or an explicitly configured bookmark/outline tree as a reviewed navigation feature. | Long generated planning PDFs become hard to review, especially when checking sections such as Static code findings, object tables, phase tables, and scratchpad records after tool-only changes.                       | The PDF preamble now configures section-level PDF bookmarks/outline support, and the document body emits a clickable table of contents immediately after the title. This is a tool/display change only and does not advance the Lean-code data timestamp.                                                      |
| SCF36 | G        | closed-in-tool-v1.03-pre-implementation-phase-check | tooling/pre-implementation-phase-check-workflow | V1.03                | 2026-05-10T06:38:25+02:00 | Forward-looking semantic phase feasibility checks lacked an explicit workflow contract.                         | The plan had per-phase preparation fields and a pre-start feasibility rule, but no dedicated workflow for simulating several semantically connected future phases before implementation and classifying them as implementable or dependency-blocked.                                           | Without this workflow, missing dependencies can be discovered only during implementation; red findings may not automatically generate upstream addressing phases, causing late backjumps and ad hoc patching pressure. | Tool V1.03 introduces a Pre-Implementation Phase Check: inspect a semantic window as a thought experiment, mark phases Y when implementable and R when dependencies are missing, record new findings, and generate upstream addressing phases for every red result before the blocked phase can become active. |



| ID    | Priority | Issue status                                                | Addressing phase                                         | Change version | Change timestamp          | Finding                                                                                                                                           | Static observation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Risk                                                                                                                                                                                                                  | Refactoring consequence                                                                                                                                                                                                                                                                                 |
|-------|----------|-------------------------------------------------------------|----------------------------------------------------------|----------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCF37 | G        | closed-in-tool-v1.04-implementation-scaling-plan            | tooling/<br>implementation-scaling-phase-plan V1.04      | V1.04          | 2026-05-10T06:58:41+02:00 | The tool lacked an explicit phase roadmap for scalable CNNA object-world growth without overloading Lean implementation work.                     | Previous workflows defined phase checks and ledgers, but did not yet separate lean-focus packets, local module-theory context, large end-theory comparison ledgers, incremental cache, object graph slices, and automatic red-blocker prephases into implementation phases.                                                                                                                                                                                                                                                             | Without this roadmap, future requests could drift into manual PDF/table maintenance, excessive context scanning, or ambiguous treatment of established theories during Lean-code work.                                | Tool V1.04 adds future phases under 1.1.1.5 and a generated implementation-scaling plan. These phases preserve the active 1.1.1.2 cursor and keep Lean-data timestamp fixed.                                                                                                                            |
| SCF38 | G        | closed-in-tool-v1.05-preimplementation-window-result-ledger | tooling/<br>pre-implementation-check-result-ledger V1.05 | V1.05          | 2026-05-10T07:12:27+02:00 | Requested semantic phase-window implementability checks had no persistent result table.                                                           | Tool V1.03 defined the pre-implementation thought-experiment workflow, but the concrete result of a requested check window was not yet rendered as a dedicated generated ledger with phase-level Y/R decisions.                                                                                                                                                                                                                                                                                                                         | Feasibility conclusions could remain only in chat prose, making later phase activation ambiguous and weakening the red-blocker-to-addressing-phase discipline.                                                        | Tool V1.05 adds a <code>pre_implementation_phase_check_results</code> section, generator validation, TSV export, and PDF rendering for the checked window. The 1.1.1.2-1.1.1.5 window is recorded as yellow/implementable under sequential closure assumptions, with no red addressing phase generated. |
| SCF39 | G        | closed-in-tool-v1.06-preimplementation-status-refinement    | tooling/<br>pre-implementation-check-refinement V1.06    | V1.06          | 2026-05-10T07:22:31+02:00 | Pre-implementation check status Y was too coarse and conflated active implementability with later implementability in a derivation chain.         | The V1.05 check correctly treated 1.1.1.3-1.1.1.5 as implementable only under predecessor-closure assumptions, but the table still marked every implementable row yellow.                                                                                                                                                                                                                                                                                                                                                               | Readers could mistake later chain-dependent phases for immediately active implementable phases, weakening the one-active-phase invariant and obscuring predecessor assumptions.                                       | Tool V1.06 introduces G/Y/O/R check semantics plus Check mode, Implementability scope, Required predecessor closures, Assumptions, Evidence basis, Lean proof risk, Parent/subphase rule, and Would-generate-addressing-phase fields.                                                                   |
| SCF40 | G        | closed-in-tool-v1.06-phase-legend-synchronization           | tooling/<br>phase-logic-legends V1.06                    | V1.06          | 2026-05-10T07:22:31+02:00 | Tables that use phase logic needed legends synchronized with the new orange derivation-chain implementability status.                             | The prior phase and pre-implementation legends exposed only green/white/yellow/red or yellow/red meanings and did not explain the new chain-dependent implementability state.                                                                                                                                                                                                                                                                                                                                                           | Even with correct result rows, inconsistent legends would make the PDF display ambiguous and could lead to wrong activation of a non-current phase.                                                                   | The generator now emits updated phase-logic legends: G = closed/derived-only, W = not yet checked where actual roadmap statuses appear, O = implementable in derivation chain, Y = active implementable now, R = blocked with missing dependencies and mandatory upstream addressing phase.             |
| SCF41 | G        | closed-in-tool-v1.07-db-review-main-side-objective          | 20;<br>20.1                                              | Tool V1.07     | 2026-05-10T07:39:10+02:00 | Database-ready review/documentation structure was only a prose idea, not a first-class main side objective.                                       | The prior YAML had secondary long-term goals and generated views, but no main phase route for database-ready review structure, round-trip schema discipline, and expert-review semantics.                                                                                                                                                                                                                                                                                                                                               | The DB direction could be treated as optional late tooling and fail to constrain current YAML/schema/review practices.                                                                                                | Add Phase 20 DB-REVIEW with child phases, direct DBR objects, scratchpads, secondary-goal row, workflow row, and immediate guardrails. No Lean source changes.                                                                                                                                          |
| SCF42 | G        | closed-in-tool-v1.07-expert-review-guardrail                | 20.1;<br>GNG23;<br><code>workflow_policy</code>          | Tool V1.07     | 2026-05-10T07:39:10+02:00 | Generated documentation lacked an explicit global no-infallibility and discipline-specific expert-review rule.                                    | The tool distinguished Lean code from generated views, but did not yet force the stronger epistemic maxim that even green Lean builds and generated artifacts remain semantically reviewable.                                                                                                                                                                                                                                                                                                                                           | Planning tables or green builds could be overread as 100-percent secured truth, especially around established-theory/reference material that may still lack blue marking.                                             | Add GNG23 and the DB-REVIEW workflow: Lean code is primary but not semantically infallible; generated artifacts are secondary; expert review is always required; missing blue markings are audit debt.                                                                                                  |
| SCF43 | Y        | active-in-phase-1.1.1.1.5.1                                 | 1.1.1.1.5.1-1.1.1.1.5.5                                  | V1.12          | 2026-05-10T11:16:00+02:00 | New substantive Lean code should wait until semantically related module families have been audited for consolidation or explicit no-merge status. | The repository has a valid import graph, but it still contains visibly related micro-modules and clone-like wrappers. Existing phases covered reuse/generator policy and declaration consumption, but not a dedicated pre-substantive module-consolidation/no-merge ledger.                                                                                                                                                                                                                                                             | Without this gate, new code may be built on duplicate or unstable import surfaces, making later public API correction, blue-object review, and refactor migration harder.                                             | V1.21 routes this finding through the full readiness chain: source intake, neutral naming/facade plan, refactor execution, post-refactor re-scan, and release proof before 1.1.1.2.                                                                                                                     |
| SCF44 | Y        | prepared-after-source-intake                                | 1.1.1.1.5.2-1.1.1.1.5.3                                  | V1.12          | 2026-05-10T11:16:00+02:00 | High-confidence schema/clone families were found in BoundarySource substrate ledgers and Handoff input/output wrappers.                           | BoundaryStateSubstrate, BranchAddressSubstrate, HistoryExpandedSubstrate, and InterfaceExposureSubstrate share the same import, ledger projection pattern, blocker/status/obligation theorems, and StrengtheningAR2b1 wrappers. <code>A_to_C</code> , <code>A_to_D</code> , <code>A_to_E</code> are SectorExportStrong aliases with identical of export projections. <code>B_to_A</code> , <code>C_to_A</code> , <code>D_to_A</code> , <code>E_to_A</code> share the same source/feedback structure, ext proof, and handoff projection. | Leaving these as unrelated modules can obscure the true shared schema and make future public API/firewall decisions noisier; merging them unsafely could still break existing imports or erase pillar-specific names. | After LEG37 closes, LEG38 must decide schema/facade/no-merge treatment; any actual code change moves through LEG39 with old-module facades and local build evidence.                                                                                                                                    |
| SCF45 | Y        | prepared-after-source-intake                                | 1.1.1.1.5.2                                              | V1.12          | 2026-05-10T11:16:00+02:00 | Long SM/generated proof chains and Cayley-Dickson S6 chains are not safe merge candidates merely because they are numerous or similarly named.    | The Smoke/GeneratedInterior*/Operational* modules and Structural/CayleyDickson/S6* modules encode phase/proof progression and consumer history. Their boundaries may carry provenance, debugging, or theorem-staging information even if some boilerplate repeats.                                                                                                                                                                                                                                                                      | Aggressive consolidation could reduce auditability, hide natural proof boundaries, or create a large monolith that is harder for experts to review.                                                                   | LEG38 must explicitly classify these as no-merge-until-proven unless a canonical proof module preserves provenance and passes import/declaration regression checks.                                                                                                                                     |

| ID    | Priority | Issue status                | Addressing phase        | Change version | Change timestamp          | Finding                                                                                                                                                                             | Static observation                                                                                                                                                                                                                                                                      | Risk                                                                                                                                                                                                                 | Refactoring consequence                                                                                                                                                                                                                                  |
|-------|----------|-----------------------------|-------------------------|----------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCF46 | G        | closed-by-V1.13-policy      | 20.8                    | V1.13          | 2026-05-10T11:26:00+02:00 | The project root had accumulated generated documentation artifacts and historical Markdown reports, obscuring the source/secondary-artifact boundary.                               | Root inspection of the V1.12 package found 36 Markdown reports, 20 PDFs, 20 TeX files, 19 aux files, 22 logs, 8 out files, 11 toc files, and additional latexmk artifacts at maxdepth 1. These files are not Lean sources and are reproducible or archival documentation surfaces.      | Root clutter can make generated-secondary artifacts look like authoritative inputs, increases package size, and weakens the Single-YAML-Master / Lean-primary discipline.                                            | Keep README.md in root; archive historical Markdown reports; remove generated PDF/TeX/LaTeX build products from root; use scripts/generate_full_document.py for disposable full documentation and scripts/clean_generated_root_artifacts.py for cleanup. |
| SCF47 | G        | closed-by-V1.15-policy      | 20.10                   | V1.15          | 2026-05-10T12:20:00+02:00 | The V1.14 root policy was too narrow because Lean/Lake project-control files must live in the package/project root for reproducible local builds.                                   | User supplied CNNA.lean, lakefile.lean, lake-manifest.json and lean-toolchain. The lakefile points the CNNA library to Repository/repo_snapshot, while the manifest/toolchain pin the Lean/mathlib build environment.                                                                   | If root cleanup deletes or omits these files, the package remains documentable but loses its direct local Lake build entry and reproducibility surface.                                                              | Amend DBR8/GNG27 by DBR10/GNG28: root is source/control oriented, not README-only. Keep package-root Lean/Lake build files; continue excluding generated PDF/TeX/build artifacts and historical reports from active root.                                |
| SCF48 | G        | closed-by-tooling-phase     | 20.11                   | V1.17          | 2026-05-10T12:58:00+02:00 | Root license/credit and duplicate-README surface needed explicit closure                                                                                                            | V1.16 had README copies in both package root and Repository/, and no root LICENSE file. User required root-only README, root license file, and explicit editor/assistant credits also in generated PDFs.                                                                                | Without closure, release packages could present ambiguous entry surfaces and generated PDFs could lose attribution/license context.                                                                                  | V1.17 keeps README.md and LICENSE only in package root, removes duplicate README files, updates PDF generators and metadata credits, and records GNG29/DBR11.                                                                                            |
| SCF49 | G        | closed-by-tooling-phase     | 20.12                   | V1.18          | 2026-05-10T13:18:00+02:00 | Generated documentation sections needed stronger visual separation for review navigation                                                                                            | V1.17 full and context PDFs rendered consecutive top-level sections without a uniform automatic pagebreak invariant; individual manual clearpages existed only around the table of contents or specific generator positions.                                                            | Large review packets become harder to navigate and section boundaries can be visually blurred, especially in context-filtered documents that preserve all major sections but show sparse rows.                       | V1.18 adds a generator-level section hook so every top-level full/context documentation section starts on a fresh page. The change is layout-only and does not alter Lean, YAML row semantics, context filtering, or proof status.                       |
| SCF50 | G        | closed-by-tooling-phase     | 20.13                   | V1.19          | 2026-05-10T13:45:00+02:00 | White phase status was too permissive for unreleased Lean-derivation phases                                                                                                         | V1.18 had many W phases with implementation_scratchpad.phase_type = derivation, so W still meant unchecked/future rather than non-Lean/tooling-only. User required unreleased Lean-derivation phases to default red until their release is proven from derived-only Lean objects.       | Without this policy, a future Lean-content phase can look merely inactive/white instead of explicitly locked, weakening the derived-only partial-order discipline and hiding missing prerequisite-object evidence.   | V1.19 converts all W derivation phases to R, redefines W for non-direct-Lean work, and patches generator validation to reject any future W derivation phase.                                                                                             |
| SCF51 | G        | closed-by-tooling-phase     | 20.14                   | V1.20          | 2026-05-10T13:52:00+02:00 | Implementation module manifest could be misread as a second phase table and section/table purposes were underexplained                                                              | V1.19 still presented the module manifest in the main reading flow and several sections relied on terse table-local paragraphs. User reported confusion and requested clearer static explanations with dynamic table contents only.                                                     | Readers may confuse module/API change planning with phase-release logic, or miss where data comes from and why a table is shown.                                                                                     | V1.20 renames and moves the manifest to a technical appendix, clarifies its use boundary, and adds static source/purpose explanations around the full and context documentation sections.                                                                |
| SCF52 | Y        | active-in-phase-1.1.1.1.5.1 | 1.1.1.1.5.1-1.1.1.1.5.5 | V1.21          | 2026-05-10T14:20:00+02:00 | Pre-code readiness was too broad: semantic consolidation alone did not prove that all current modules were neutralized, refactored and classified before new substantive Lean code. | V1.20 had active 1.1.1.1.5 and orange 1.1.1.2, while many required neutralization/refactor/object-intake phases remained white planning/refactor rows. The codebase still contains phase-semantic SM*/S6*/StrengtheningPhase names and 8860 declarations needing stable classification. | If 1.1.1.2 starts too early, blue-object correction may be built on unstable module names, incomplete IST-object intake, stale public/API surfaces or unclassified proof-stage modules.                              | V1.21 splits 1.1.1.1.5 into a five-step readiness chain and makes 1.1.1.2 depend on the final release certificate.                                                                                                                                       |
| SCF53 | G        | closed-as-planning-record   | 20.15-20.20             | V1.22          | 2026-05-10T14:55:00+02:00 | Database/public-web vision needed explicit non-truth-source planning phases before any direct DB work.                                                                              | SLTG-06 already described database-ready documentation, but Supabase/Postgres public read-only mirror, snapshot provenance, RLS/views and cnna-project.net frontend contract were not yet split into concrete planning phases.                                                          | Without explicit phases and guardrails, a future database could become an untracked second truth system, leak write authority, lose snapshot provenance or expose service_role secrets to browser/frontend contexts. | V1.22 adds 20.15-20.20 plus DBR15-DBR20 and GNG34 as a white non-blocking DB/tooling sidechain. Lean/YAML remain authoritative and active Lean/content cursor is unchanged.                                                                              |
| SCF54 | G        | closed-by-tooling-phase     | 20.21                   | V1.23          | 2026-05-10T15:20:00+02:00 | Generated PDFs needed a larger typography profile for review readability                                                                                                            | V1.22 used a 10pt document class for full/context PDFs and many explicit scriptsize/footnotesize table fragments, which made large review packets dense even after section-pagebreak and explanation improvements.                                                                      | Reviewers may miss status/provenance details in dense tables; readability pressure could otherwise lead to manual PDF edits or inconsistent per-table hacks.                                                         | V1.23 applies a one-step font-size increase at generator level: documentclass 11pt, TeX size-command bump in tables, and graph label-size bump where generated by the tool.                                                                              |
| SCF55 | G        | closed-by-planning-phase    | 20.22-20.26             | V1.24          | 2026-05-10T13:45:00+02:00 | Existing Supabase phases existed but were too broad after live project access became available                                                                                      | V1.23 already had 20.15-20.20 and DBR15-DBR20 for Supabase/Postgres architecture, but after live connector discovery the plan needed concrete activation steps for the actual project aiekahdewadrejrrdajn before any migration/import write.                                           | Without an activation split, a future DB write could blur schema design, import execution, RLS verification and web-frontend handoff into one broad DB phase, weakening auditability and no-Lean-change discipline.  | V1.24 keeps 20.15-20.20 as architecture contracts and adds 20.22-20.26 as concrete live-project activation phases, with GNG36 requiring explicit authorization and snapshot/import evidence before writes.                                               |



| ID    | Priority | Issue status                   | Addressing phase                     | Change version | Change timestamp          | Finding                                                                                                                                                                                     | Static observation                                                                                                                                                                       | Risk                                                                                                                                                                                                      | Refactoring consequence                                                                                                                                             |
|-------|----------|--------------------------------|--------------------------------------|----------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCF56 | G        | closed-by-planning-phase       | 1.4.9;<br>13.2.1;<br>13.6.1;<br>18.5 | V1.25          | 2026-05-11T08:40:00+02:00 | Quaternion/geometric-algebra material needed reference-gate routing before it could influence CNNA planning.                                                                                | Horn 0709.2238v1 contains useful Hamilton/Pauli/Clifford/duality/biquaternion target information, but using it directly would blur CNNA-derived-only provenance and sign/unit semantics. | Without explicit gates, external Pauli/Clifford/Dirac or biquaternion structures could silently become inputs, set orientation or scalar layers, or patch later Pillar-B physics from established theory. | V1.25 adds GA0-GA6 and comparison phases; GNG37 forbids generator/handoff rights and keeps all material as target/scratchpad context only.                          |
| SCF57 | G        | closed-by-V1.26-tooling-schema | 20.27                                | V1.26          | 2026-05-11T09:25:00+02:00 | Phase/object status could be misread as context/type identity; tool-only, database, generated-artifact, governance and reference-gate objects needed explicit separation from Lean objects. | Earlier generated phase/object tables showed status and links but did not render an explicit class field on all phase/object logic surfaces.                                             | Category error: green generated or reference objects could be mistaken for Lean-derived objects; red/yellow Lean phases could be conflated with non-Lean tooling phases.                                  | Add Context class taxonomy and mandatory field, render in phase/object/proof/scratchpad/module-manifest tables, and enforce status/context orthogonality via GNG38. |

5 Global no-go guardrails

**What this section shows.** This table contains global rules that every phase inherits. The data comes from the Master YAML, not from Lean. It is shown because these rules constrain how new phases, refactors, generated documents and Lean implementation prompts may proceed.

**Global no-go guardrail rule.** This table is sourced from the Master YAML guardrail list. The rules below are global repository/workflow invariants, not phase-local suggestions. Every phase scratchpad inherits them through the `global_no_go_inheritance` field. Phase-local no-go lists are only stricter addenda; they may not weaken, omit, or scope away this table.

| ID    | Rule                                                                                                                                  | Scope                                                                                                 | Enforcement                                                                                                                                                                                                                                                                                                                                          |
|-------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GNG01 | no @[simp] attributes in the operational path                                                                                         | Global operational Lean code path                                                                     | Rejected in implementation manifest, code review, and closeout unless the code is explicitly non-operational and the exception is recorded.                                                                                                                                                                                                          |
| GNG02 | no noncomputable in the operational path                                                                                              | Global operational Lean code path                                                                     | Operational definitions must remain computable; mirror/non-operational layers must be marked and isolated.                                                                                                                                                                                                                                           |
| GNG03 | no classical helper unless isolated in allowed mirror/non-operational layer                                                           | Global operational/mirror boundary                                                                    | Any classical use must be outside the operational path and documented as mirror/non-operational.                                                                                                                                                                                                                                                     |
| GNG04 | no Matrix.inv or external inverse oracle in operational derivations                                                                   | Global inverse/elimination derivations                                                                | Use explicit proof-carrying inverse candidates and two-sided inverse proofs; external inverse APIs are forbidden as operational sources.                                                                                                                                                                                                             |
| GNG05 | no free formula without proof-carrying entry structure                                                                                | Global formula/code-object introduction                                                               | Every formula exposed to consumers must be backed by a source object, entry structure, definition, proof target, and verification/dossier record.                                                                                                                                                                                                    |
| GNG06 | no consumer-side patch when the missing structure belongs earlier                                                                     | Global dependency/backjump discipline                                                                 | Missing structure must be introduced at its natural source layer; consumer/gate patches are blocked.                                                                                                                                                                                                                                                 |
| GNG07 | no premature obstruction before a serious positive construction attempt                                                               | Global positive-path discipline                                                                       | Obstructions require a recorded positive attempt and a localized mathematical failure reason.                                                                                                                                                                                                                                                        |
| GNG08 | no phase-semantic public API names                                                                                                    | Global public API and module-surface naming                                                           | SM/AR/phase-semantic names may remain legacy/internal but cannot be public-final neutral API.                                                                                                                                                                                                                                                        |
| GNG09 | no fragile rewrite chain when a structural constructor/projection proof is available                                                  | Global proof-tactic strategy                                                                          | Prefer constructors, projections, and explicit structural lemmas; fragile rewrite chains must be justified as local and harmless.                                                                                                                                                                                                                    |
| GNG10 | do not delete legacy status enums during quarantine                                                                                   | Global legacy quarantine workflow                                                                     | Legacy status enums may be moved/quarantined/wrapped, not deleted during the quarantine phase.                                                                                                                                                                                                                                                       |
| GNG11 | do not let Phase 3+ import Legacy.StatusGates                                                                                         | Global import firewall for post-neutral phases                                                        | Phase 3+ visibility checks must reject Legacy.StatusGates imports.                                                                                                                                                                                                                                                                                   |
| GNG12 | do not treat old TeX/PDF files as maintained roadmap sources                                                                          | Global planning/source-of-truth policy                                                                | YAML master is authoritative; old TeX/PDF files are historical/generated views only.                                                                                                                                                                                                                                                                 |
| GNG13 | no string-like or physical analogy as CNNA-generative input                                                                           | All interpretive, arithmetic, Schur/DtN, and handoff phases.                                          | Analogies may be comparison/diagnostic notes only; any formula, object, spectrum, or handoff must have a derived code/proof route.                                                                                                                                                                                                                   |
| GNG14 | no horizontal/vertical Schur-duality or string/subsystem analogy as input to earlier generators, Schur, arithmetic, or CD definitions | all Pillar-A bridge, Schur, arithmetic, CD, and generator phases                                      | the duality may only be a later theorem/diagnostic target after S05/O05/TM0/CD7/BR06/BR07 are closed; earlier phases may export data to it but never consume its conclusion                                                                                                                                                                          |
| GNG15 | no heuristic angle/lattice construction as generator or branching definition                                                          | Global generator, substrate, lattice-test, and arithmetic comparison workflows                        | Any Eisenstein/Gauss spiral or angle profile must be a consumer-only test/comparison target. The code-defined branching API is authoritative; angle drawings, CM targets, and j values may not define b, carriers, parents, or generator semantics.                                                                                                  |
| GNG16 | no special-case-only test matrix as evidence for a general computable path                                                            | All computation, arithmetic, lattice, generator, smoke-test, and refactor-regression workflows        | Eisenstein/Gauss or other sentinel rows may be included only as special comparison rows. Generic finite rows over the code-defined API must exist, or the phase remains unclosed/blocked.                                                                                                                                                            |
| GNG17 | no Heegner/class-number-one membership as closed noOuter evidence                                                                     | All number-theory, CM, lattice, regime, test-matrix, AQFT/unitarity, and comparison workflows         | A row with h(D)=1, a Heegner label, or an integer j target remains number-theoretic/comparison data until its noOuter/HasOuter regime is derived from Pillar-A source objects. Eisenstein/Gauss may be tested as noOuter-root sentinels; other Heegner targets default to HasOuter/complement-projection or comparison rows unless proven otherwise. |
| GNG18 | no replacement of ExecComplex by analytic Complex, Cayley-Dickson C-surface, or unproven exact quadratic scalar layer                 | global operational scalar path, arithmetic, lattice/CM, Cayley-Dickson bridge, and test-matrix phases | ExecComplex remains QxQ operational floor. O09 is required for later quadratic coordinates but only as a schema; concrete instances require GNG19/O10/BR14. CD-derived C may only be consumed after CD9/10.4.3 proves noncircular provenance. General Complex/Real stays mirror-only.                                                                |
| GNG19 | no free sqrt(D), no exact quadratic instance without proof-carrying D-source                                                          | all scalar, arithmetic, CM/lattice, number-theory, test-matrix, and Cayley-Dickson bridge phases      | O09 may define only a parameterized schema. Any concrete exact quadratic row must consume O10/BR14 with a D sourced from O05/O06 or an equivalent operational discriminant ledger. Analytic Complex/Real, AdjoinRoot field structures, or CD C-surfaces may not supply the early D or theta.                                                         |
| GNG20 | no artificial phase bottleneck or phase-title/object mismatch                                                                         | global phase DAG, object register, module manifest, and implementation scratchpads                    | Each object must be placed behind only the prerequisites used by its proof target. In particular, SP0/SP1/P00-P06/TM0-TM2 may not wait on modular forms, j, q-expansions, Hilbert class polynomials, theta or L-functions unless a specific bridge theorem explicitly requires that dependency.                                                      |

| ID    | Rule                                                                                                                                             | Scope                                                                                                                                                                               | Enforcement                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GNG21 | no unexplained empty Linked-objects cell and no umbrella aggregation as direct object ownership                                                  | global phase table, object register, scratchpads, module manifest, generated views                                                                                                  | Every phase must declare direct, scope_only, no_direct_object, or moved. Empty direct object sets require an explicit non-direct policy. Child object scope may be shown for readability but cannot change object Phase ownership.                                                                                                                                                                                    |
| GNG22 | no dynamics, physics, stochasticity, geometry, arithmetic, operator algebra, environment, or unitarity as ontic primitive beyond subsystem-being | global CNNA ontology, object register, phase DAG, scratchpads, proof dossiers, consumer bridges, external-comparison sections                                                       | Every such structure must cite a derived cut/regime/complement/Schur-DtN/source-object chain. If a phase needs one of these before the chain exists, it must backjump to the natural generating layer rather than postulate or patch the consumer.                                                                                                                                                                    |
| GNG23 | no 100-percent truth declaration and no bypass of discipline-specific expert review                                                              | All CNNA documentation, generated tables/PDFs/graphs/exports, phase status claims, Lean-code interpretation, object proof dossiers, and established-theory/reference markings       | The checked Lean code is the primary truth system, but even a green build only checks formal consistency of chosen declarations. Semantic correctness, modeling adequacy, theory attachments, and generated interpretations remain reviewable. Every relevant claim must remain open to specialized expert review; generated artifacts are secondary.                                                                 |
| GNG24 | no ignored missing blue/reference-theory marking                                                                                                 | All references to established mathematics/physics/computer-science background, local mathlib/formalization context, and large end-theory comparison horizons                        | If a claim, object, phase, module, or documentation passage touches established theory, the tool must decide whether it needs blue/reference marking. Missing markings are audit debt. Blue marks comparison/reference/review context only and never promotes established theory to a CNNA generator.                                                                                                                 |
| GNG25 | no tool-interface metaphor as formal CNNA theorem or ontic input                                                                                 | Documentation, database views, context slices, phase closeouts, and reflective tool architecture notes.                                                                             | The bright/dark/interface analogy may guide review and visibility obligations only. It must not be cited as a proved CNNA statement, a mathematical generator, or a reason to bypass Lean-primary proof and discipline-specific expert review.                                                                                                                                                                        |
| GNG26 | no semantic module consolidation without compatibility facade, natural-layer placement, and regression evidence                                  | All Lean refactor/merge phases, module manifest, reuse ledger, import graph, declaration consumption map, object proof dossiers, and public API/firewall phases                     | A merge/refactor may proceed only after LEG36 classifies the candidate, preserves old import paths or supplies an explicit migration facade, records import/declaration/consumer deltas, and obtains local full-build evidence. Proof-stage chains must not be merged by naming similarity or line-count pressure.                                                                                                    |
| GNG27 | no generated PDF/TeX/LaTeX build products or historical reports as active root sources                                                           | Repository root, generated full-document outputs, context-documentation outputs, release packaging, README/toolchain commands, and archive/report policy                            | The project root must remain source-oriented: README.md may stay in root; historical Markdown reports belong in archive/reports/; generated PDF, TeX, aux, log, out, toc, fls and fdb_latexmk files are disposable and must not be treated as authoritative or retained as active root files. Regenerate them from YAML/tables/scripts when needed.                                                                   |
| GNG28 | do not remove package-root Lean/Lake project-control files during cleanup or packaging                                                           | Package root, release packaging, root cleanup workflow, local build workflow, README/manual commands, future CI/database packaging                                                  | Cleanup may remove disposable generated documentation artifacts and archive historical reports, but packaging must preserve package-root CNNA.lean, lakefile.lean, lake-manifest.json, and lean-toolchain unless an explicit future build-entry phase replaces them with a proven alternative.                                                                                                                        |
| GNG29 | no duplicate active README and no hidden license/credit surface                                                                                  | Package root, Repository root, generated full/context PDFs, release packaging, README/manual commands, license notice table, and future CI/database packaging                       | The maintained README.md and LICENSE must live in the package root. Subdirectories must not carry additional active README.md files unless a future phase classifies them as generated/raw-export artifacts with explicit non-authoritative status. Generated PDFs must expose the Editor and Assistant credits from metadata.                                                                                        |
| GNG30 | generated PDF section-pagebreak invariant                                                                                                        | Full documentation generator, context documentation generator, review PDFs, future documentation layout profiles, and generated-secondary review packets                            | Top-level sections in generated PDFs must start on new pages automatically. The invariant is implemented in the generators, not by manually editing disposable TeX/PDF outputs. Layout changes must not change Lean code, Master-YAML truth semantics, or context-matching logic.                                                                                                                                     |
| GNG31 | no white direct-Lean derivation phases and no release without derived-only object evidence                                                       | Phase table, implementation scratchpad, pre-implementation checks, object register, proof dossiers, active cursor selection, generated PDFs/TSVs, future database/status dashboards | If a phase_type is derivation and the phase is not green, yellow, or orange, it must be red. A red derivation phase may become yellow or orange only after a pre-implementation release check references the required already-derived Lean objects/proof dossiers. White is reserved for refactor/tool/infrastructure/documentation/comparison work that does not itself require direct Lean-code derivation release. |
| GNG32 | main phase table is the release surface; module-change manifest is appendix-only                                                                 | Full documentation, context documentation, phase status release workflow, module-change/refactor audits, implementation scratchpads, README/manual use                              | Do not read the module-change manifest as a second phase table. It may only record concrete module/API/import-surface change contracts. Phase release status, dependencies and derived-only evidence belong to the main phase table, object register and proof dossiers. Static explanatory prose must stay generator-owned; table rows remain dynamic.                                                               |
| GNG33 | no new substantive Lean code before full IST intake, neutral naming/refactor, post-scan and release proof                                        | All future Lean implementation prompts, 1.1.1.2 blue-object correction, Phase 1.x neutralization, public API/firewall phases, and later red locked derivation phases                | Before any new substantive Lean content phase becomes yellow, the plan must reference closed LEG37-LEG41 or spawn a red blocker at the natural source layer. Refactoring/tooling may proceed only inside the pre-code readiness chain and must preserve old imports through facades until build-proven.                                                                                                               |
| GNG34 | Supabase/Postgres mirror is public read-only and never a CNNA truth source                                                                       | Supabase/Postgres schemas, public api.* views, web frontend, importer/admin paths, future Edge Functions, reviewer annotations, DB snapshots and cnna-project.net visualization.    | No Supabase row may release a phase or replace Lean/YAML provenance. Public/anon access is SELECT-only on api.* views. No public writes. No browser service_role key. No DB update without snapshot_id, source hashes and import report. Every public row must be traceable to a Lean/YAML snapshot.                                                                                                                  |

| ID    | Rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Scope                                                                                                                                                                                                                  | Enforcement                                                                                                                                                                                                                                                                                                    |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GNG35 | generated PDF typography changes must be generator-level and semantic-neutral                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Full documentation generator, context documentation generator, table generator, graph-label generation, generated PDF review packets and future documentation layout profiles                                          | Font-size changes must be applied in generators/source tables, not by manual PDF edits. They must not alter Lean code, YAML row semantics, context filtering, phase status logic, proof status, or public database/export semantics.                                                                           |
| GNG36 | no live Supabase write without explicit authorization, snapshot provenance and no-Lean-change boundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Supabase migrations/imports, ChatGPT direct DB connector use, local importer scripts, future cnna-project.net publication layer, release packaging and DB reports                                                      | Any live apply __migration/execute_sql write must be explicitly authorized by the user for the current step, must write only Supabase schema/data, must not edit Lean files, must produce snapshot_id/source-hash/import-report evidence, and must preserve public read-only/no-service-role-in-browser rules. |
| GNG37 | Quaternion/geometric-algebra references are target gates only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Horn 0709.2238v1, Hamilton quaternion relations, Pauli matrices, Clifford/geometric algebra, biquaternions, spacetime algebra, conformal GA, mathlib Quaternion/CliffordAlgebra APIs and future GA comparison modules. | These references may define comparison profiles, sign audits, scratchpad notes and expert-review targets only. They cannot generate CNNA objects, set scalar layers, choose orientations/dimensions, release phases, create handoff data or shortcut derived-only proofs.                                      |
| GNG38 | Status/context orthogonality and Lean-facing subclass separation. Phase status and Object status encode release/proof-state only; Context class encodes type/class membership. No ToolInfrastructureObject, DatabaseMirrorObject, GeneratedArtifactObject, DocumentationObject, GovernancePolicyObject or ReferenceGateObject may be consumed as a LeanSourceObject/LeanTargetObject or as derived mathematical data because it is green, linked or documented. LeanRefactorObject, LeanHandoffContractObject and LeanGovernanceObject are Lean-facing control classes, not automatically mathematical CNNA data. | All phase, object, proof-dossier, scratchpad, generated export, DB mirror and implementation-prompt surfaces.                                                                                                          | All new phase/object rows must declare Context class from the taxonomy. Generated tables render Context class beside status. Future DB schema must mirror the field. Any phase release must cite LeanSourceObject/LeanTargetObject evidence, not merely a green tool/reference/governance object.              |

## 6 Global ontic postulates

**What this section shows.** This section records the global ontic/provenance postulates used by the planning layer. It is intentionally separated from Lean axioms: these statements guide the roadmap and review discipline, but do not become Lean theorems by appearing here.

**Global postulate rule.** The postulate below is a global ontology/provenance constraint, not a Lean axiom and not a physical input. It fixes the permitted primitive emergence trigger for the roadmap: subsystem-being inside the formal IDEAL\_infty provenance whole. Everything else must be derived or explicitly marked comparison-only.

| ID    | Postulate                       | Precise formulation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Allowed primitive                                                                   | Derived-only consequences                                                                                                                                                                                 | Forbidden primitive imports                                                                                                                                                              | Enforcement                                                                                                                     |
|-------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| POST1 | CNNA ontic subsystem-minimality | The only primitive ontic emergence trigger is subsystem-being: the selection of a finite or local subsystem, cut, or approximant inside the formal IDEAL_infty provenance whole. IDEAL_infty is the formal relational/provenance background, not a physical dynamics input. All dynamics, spectra, operator algebra, arithmetic, geometry, stochasticity, openness, unitarity claims and physical interpretation must be derived from the subsystem/complement relation and its Schur/DtN consequences. | Subsystem/cut/approximant selection within the formal IDEAL_infty provenance whole. | boundary;<br>inside/outside distinction;<br>complement families;<br>regime noOuter/HasOuter;<br>Schur/DtN effective operator;<br>spectra;<br>algebra;<br>arithmetic;<br>geometry;<br>dynamics;<br>physics | dynamics;<br>time evolution;<br>stochasticity/noise;<br>environment;<br>geometry/lattice;<br>arithmetic/discriminant;<br>operator algebra;<br>AQFT/physics;<br>Cayley-Dickson structures | Implemented through LEG35, GNG22, phase 1.1.1.1.0.16, object proof dossiers, scratchpad inheritance, and consumer-phase audits. |

## 7 Phase-object-link policy

**What this section shows.** This policy explains how phases and objects are linked in the generated documentation. The table is shown before the object and phase registers so the reader knows when a linked object is directly owned by a phase and when it is only part of an umbrella scope.

**Phase-object-link policy rule.** Direct object ownership is defined only by exact equality `object.Phase == phase.Phase`. Umbrella phases may display child-object scope for navigation, but child scope does not transfer ownership. A generated phase row may therefore never show an unexplained blank **Linked objects** cell: it must be rendered as direct links, scope-only child summary, no-direct-object explanation, or moved placeholder.

| Kind             | Meaning                                                                                                                           | Required evidence                                                                                      | Rendering rule                                                                                                                     |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| direct           | Phase directly owns registered object(s).                                                                                         | At least one object has Phase exactly equal to the phase id.                                           | Render direct object hyperlinks. Do not aggregate child phase objects as direct ownership.                                         |
| scope_only       | Phase is an umbrella/routing/scope phase; objects are owned by child phases.                                                      | No direct objects and at least one descendant phase has direct object(s).                              | Render scope-only plus a bounded child-object summary.                                                                             |
| no_direct_object | Phase intentionally has no direct object, e.g. baseline, audit, closeout, generator/view infrastructure without object ownership. | No direct objects and no descendant-object ownership needed for the phase semantics, plus explanation. | Render no direct object with reason; never leave cell blank.                                                                       |
| moved            | Historical placeholder or routing note whose active objects are owned elsewhere.                                                  | Main goal/definition/proof goal identifies the destination phase or ownership move.                    | Render moved placeholder and reason/destination; no direct object links unless the placeholder also owns an explicit audit object. |



8 Phase/object context-class taxonomy

**What this section shows.** This section adds an object-oriented class layer above phases and objects. Status colors say release/proof state; context class says what kind of row it is: Lean, tool, database mirror, generated artifact, documentation, governance, or external reference gate. It is shown before object and phase registers to prevent tool-only or reference-only rows from being read as Lean objects.

**Context-class taxonomy rule.** This table is an orthogonal class/type layer over the phase and object registers. It applies an object-oriented reading to the planning schema: **Context class** says what kind of entity a row is, while status colors say only release/proof state. A green generated, database, documentation, governance or reference row is therefore not a green Lean object. Conversely, a red Lean derivation phase remains a Lean derivation phase, but blocked by missing derived-only evidence.

| Class                     | Applies to | Parent class                           | Meaning                                                                                                                                                                                                        | Truth role                                                                                                                                                      | Allowed consumers                                                                                        |
|---------------------------|------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| PhaseBase                 | phase      | root                                   | Abstract superclass for all roadmap phases.                                                                                                                                                                    | No truth role by itself; subclasses determine whether the phase may touch Lean, tooling, documentation, DB or reference material.                               | Only used by generated validators/tables.                                                                |
| LeanDerivationPhase       | phase      | PhaseBase/LeanPhase                    | Direct Lean mathematical/content derivation phase.                                                                                                                                                             | May eventually create or close checked Lean objects, but only after derived-only release evidence.                                                              | Lean implementation prompt, object proof dossier, downstream phases after closure.                       |
| LeanRefactorPhase         | phase      | PhaseBase/LeanPhase                    | Lean code/API/name/import-surface refactor phase without new mathematical content by itself.                                                                                                                   | Preserves/reorganizes checked code; does not prove new mathematics unless separately registered.                                                                | Build/refactor workflow and later Lean phases after regression checks.                                   |
| LeanGovernancePhase       | phase      | PhaseBase/GovernancePhase              | Policy/release/import/consumer-governance phase that controls Lean work but is not itself a Lean theorem.                                                                                                      | Planning authority only; must not be cited as mathematical proof.                                                                                               | Implementation prompts and generated validators.                                                         |
| ReferenceGatePhase        | phase      | PhaseBase/ComparisonPhase              | External established-theory or side-theory comparison/target gate.                                                                                                                                             | Comparison/reference only; never CNNA generator or input.                                                                                                       | Scratchpads, review gates and later consistency checks after CNNA-derived data exists.                   |
| ToolInfrastructurePhase   | phase      | PhaseBase/ToolPhase                    | Python/YAML/generator/schema/tooling phase.                                                                                                                                                                    | Generated-secondary tooling; cannot release Lean mathematical claims.                                                                                           | Documentation tool, exporters, future DB mirror.                                                         |
| DatabaseMirrorPhase       | phase      | PhaseBase/ToolPhase                    | Supabase/Postgres/public-read mirror or frontend publication phase.                                                                                                                                            | Read/query/publication mirror only; not a truth source.                                                                                                         | Public API/frontend/review tooling after snapshot/checksum rules.                                        |
| GeneratedArtifactPhase    | phase      | PhaseBase/ToolPhase                    | PDF/TSV/DOT/README/package generated-output phase.                                                                                                                                                             | Rendering/navigation/release packaging only.                                                                                                                    | Readers, documentation exports, release packages.                                                        |
| DocumentationPhase        | phase      | PhaseBase/DocumentationPhase           | Human-readable documentation, review matrix, no-infallibility statement, or manual workflow phase.                                                                                                             | Interpretive documentation; expert review required.                                                                                                             | Reviewers and future tool/database views.                                                                |
| BaselinePhase             | phase      | PhaseBase                              | Frozen baseline or package state phase.                                                                                                                                                                        | Records baseline identity only.                                                                                                                                 | Regression comparison and package history.                                                               |
| ObjectBase                | object     | root                                   | Abstract superclass for all registered objects/quantities/control artifacts.                                                                                                                                   | No truth role by itself; subclasses determine whether the row is Lean, target, tool, DB, reference, generated or governance.                                    | Object register/proofbook validators.                                                                    |
| LeanSourceObject          | object     | ObjectBase/LeanObject                  | Object has an existing Lean source/seed or canonical Lean module site.                                                                                                                                         | Potential checked-code object; exact semantic status still governed by Object status and proof dossier.                                                         | Lean phases subject to import/consumer discipline.                                                       |
| LeanTargetObject          | object     | ObjectBase/LeanObject                  | Planned or future Lean object/candidate, not a checked final object yet.                                                                                                                                       | Target only until implemented and proof-checked.                                                                                                                | Planning/proof rehearsal; not public-final.                                                              |
| LeanRefactorObject        | object     | ObjectBase/LeanObject/RefactorObject   | Lean-facing refactor, notation, facade, quarantine or import-surface object. It may point to Lean modules, but its semantic role is reorganization/compatibility rather than new mathematics.                  | Code-shape object only; preserves or exposes checked code and must not be cited as a new derived mathematical theorem.                                          | Lean refactor phases, regression checks and downstream phases after build-compatible migration evidence. |
| LeanHandoffContractObject | object     | ObjectBase/LeanObject/ContractObject   | Lean-facing handoff/input/output contract object. It controls access between pillars or phase layers and may have a Lean target module, but it is not itself the mathematical data transported by the handoff. | Interface/consumer-rights object; downstream phases may consume only the separately registered LeanSourceObject or LeanTargetObject data named by the contract. | Handoff validators, pillar-boundary phases and later Lean phases subject to access rights.               |
| LeanGovernanceObject      | object     | ObjectBase/LeanObject/GovernanceObject | Lean-facing policy/audit object whose natural representation may be a Lean module or proof-carrying record, while the content remains governance rather than mathematical CNNA data.                           | Checked governance/audit boundary if implemented; never a substitute for a derived mathematical object.                                                         | Static audits, import/consumer discipline, release gates and implementation prompts.                     |
| GovernancePolicyObject    | object     | ObjectBase/GovernanceObject            | Policy, ledger, guardrail, release, import or consumer-control object.                                                                                                                                         | Controls work; does not constitute mathematical CNNA data.                                                                                                      | Implementation prompts, validators and review checklists.                                                |
| ReferenceGateObject       | object     | ObjectBase/ComparisonObject            | External theory/reference/comparison/target-gate object.                                                                                                                                                       | Comparison-only; never ontic or generative CNNA input.                                                                                                          | Scratchpads and later consistency gates after CNNA-derived objects exist.                                |

| Class                    | Applies to | Parent class                   | Meaning                                                            | Truth role                                                               | Allowed consumers                                |
|--------------------------|------------|--------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------|
| ToolInfrastructureObject | object     | ObjectBase/ToolObject          | YAML/schema/script/export/tooling object.                          | Generated-secondary/tooling;<br>cannot be mistaken for Lean object.      | Documentation tool and future database importer. |
| DatabaseMirrorObject     | object     | ObjectBase/ToolObject          | Supabase/Postgres/frontend/read-only mirror object.                | Publication/query mirror only;<br>never Lean/YAML truth replacement.     | Public read-only API/frontend/review tooling.    |
| GeneratedArtifactObject  | object     | ObjectBase/ToolObject          | PDF/TSV/DOT/README/package output or generated rendering contract. | Rendering/navigation artifact;<br>semantically reviewable and secondary. | Readers and release packages.                    |
| DocumentationObject      | object     | ObjectBase/DocumentationObject | Documentation/review statement or expert-review object.            | Interpretive;<br>expert review remains required.                         | Reviewers and documentation slices.              |

9    Object/phase significance and documentation-depth tiers

**What this section shows.**    This section adds a second orthogonal classification axis: not what a row is, but how fundamental it is and how much proof/documentation burden it carries. It prevents fundamental generator or handoff objects from being under-documented while keeping trivial, auxiliary, or mention-only rows from bloating the proofbook.

**Object/phase significance-tier rule.**    This table is a second orthogonal classification axis. **Context class** says what a row is; **Significance tier / Phase documentation tier** says how deep the proof/documentation burden is. It is deliberately independent of status and class: a reference gate may be high-significance, and a Lean-facing helper may be auxiliary. Trivial rows stay visible without bloating the proofbook; fundamental rows must receive full dossiers before public-final promotion or downstream consumption.

| Tier | Name         | Applies to    | Meaning                                                                                                                                          | Documentation requirement                                                                                                                                                                               | Promotion / demotion rule                                                                                                                                 |
|------|--------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| F0   | fundamental  | object, phase | Ontic, generator, handoff, invariant, or root-contract level. Failure changes the meaning of the derived-only architecture or a pillar boundary. | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | May not be promoted by generated prose or status color. Must cite checked Lean code or an explicit governance contract and must remain expert-reviewable. |
| F1   | core         | object, phase | Core derived object, API seam, comparison gate, or proof route that many later phases depend on, but not itself an ontic root.                   | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | May be promoted to F0 only if it becomes a pillar-boundary or generator-level contract; may be demoted only after no downstream dependency relies on it.  |
| F2   | standard     | object, phase | Ordinary derived object, target object, or reference/comparison profile with local but real mathematical or planning relevance.                  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | Promote to F1 if several later phases consume it, if it crosses a handoff/API boundary, or if an expert-review risk is discovered.                        |
| F3   | auxiliary    | object, phase | Support, migration, tooling, generated artifact, ledger, or narrow refactor helper. Important for auditability but not a mathematical generator. | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | Promote to F2/F1 only if it starts carrying semantic proof obligations or becomes a required release gate.                                                |
| F4   | mention-only | object, phase | Trivial, navigational, future-side, purely illustrative, or low-risk reference row. It should be visible but not exhaustively documented.        | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                           | Automatically promote if it becomes a dependency, is consumed by Lean code, controls access/release, or carries a reference-theory risk.                  |

10 TomS thread source cross-check

**What this section shows.** This section tracks external review prompts and mathematical warnings from TomS-related discussion material. It is comparison and correction context only. It is shown to keep established-theory cautions visible without importing them as CNNA generators.

**Source-integration rule.** This is an external-reference audit only. TomS thread material and public mathematical references may define comparison targets, warning signs, and proof obligations, but they do not supply CNNA-generative data. Every accepted action below is therefore routed through existing or newly added derived-only phases and object records.

| ID      | Topic                                                   | Thread evidence                                                                                                                                  | External check                                                                                                                                                                       | Plan disposition                                                                                   | CNNA action                                                                                                                                   | Phase/Object                               | Risk / uncertainty                                                                                                               |
|---------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| TOMS-01 | Prime generation vs irreducible/prime distinction       | #8-#10 and #48: Schur levels, Eisenstein/Gauss norm heuristics, TomS correction that irreducible need not be prime                               | Algebraically standard: UFD/PID/class-number gates are required before irreducible=prime can be claimed                                                                              | Already broadly covered by NumberTheory phases but wording must stay strict.                       | Keep Heegner/PID/UFD indicator after class number; never claim prime generation from norm irreducibility without class-number/UFD proof.      | 6.2, 6.3 / N09-N13                         | The v3 script observations remain empirical/heuristic until operational recurrence, norm form, and class group gates are proved. |
| TOMS-02 | Quadratic orders and lattice types                      | #9-#11: imaginary-quadratic integer rings have two order-shapes by d mod 4; elliptic functions exist on arbitrary non-collinear complex lattices | Established CM theory: quadratic orders/lattices are restrictive subfamily of all elliptic lattices                                                                                  | Covered by D_fund/order/lattice/tau phases; add caution to source map.                             | Do not say Schur levels generate all elliptic lattices; derive D/order/basis/tau first, then compare Gauss/Eisenstein special cases.          | 5.1-8.3 / N00-N07, L00-L07                 | Current code has special Gaussian/Eisenstein bridges but no general D_op->order pipeline.                                        |
| TOMS-03 | SL2Z/Moebius, modular functions, j                      | #13/#15/#22: SL2Z matrices act by z -> (az+b)/(cz+d), S and T generate transformations; j as invariant modular function                          | Milne/standard modular-form references support j as generator of modular-function field; thread contains minor typo-like fundamental-domain bound, so do not copy formulas unchecked | Already present but should remain after CM/tau, not before.                                        | Keep modular group action, fundamental-domain representative, q-expansion and Hilbert class polynomial as post-lattice/modular phases.        | 9, 10.3 / M00-M06                          | Need authoritative source and mathlib import check at implementation; do not treat forum formulas as formal definitions.         |
| TOMS-04 | j algebraic at CM points                                | #30: j(tau) algebraic on imaginary-quadratic tau; CM connection                                                                                  | Milne modular-function notes and CM theory support algebraicity at imaginary-quadratic points                                                                                        | Covered by CM/lattice/j phases but dependency should be explicit.                                  | Require tau upper-half-plane and CM/order provenance before j-values/class-polynomial work.                                                   | 8-10.3 / L04-L05, M05-M06                  | Exact field/class-polynomial statements require class group/ring class field layer; current plan only has placeholders.          |
| TOMS-05 | L-functions are not Riemann-zeta-first                  | #49: TomS says zeta/L-functions yes, not specifically Riemann zeta; modular forms are the bridge                                                 | Standard theory: many L-functions attach to characters/modular forms; Riemann zeta is special with trivial character                                                                 | Plan should not introduce Riemann-zeta as generator.                                               | Keep chi_D/splitting before any L-function comparison; any L-function object is derived/comparison after character/modular data.              | 6.4, 10.5 / N15, spectral-zeta side object | Avoid conflating spectral zeta, Dirichlet L-functions, and Riemann zeta.                                                         |
| TOMS-06 | Finite simple groups taxonomy                           | #43-#56: cyclic, alternating, Lie type, sporadic groups; groups as generalized prime-like objects                                                | Warwick WMI: CFSG categories are cyclic prime order, alternating, Lie type, sporadic                                                                                                 | Gap in v20: Moonshine section lacked explicit finite-simple-group taxonomy as comparison-only.     | Add MOON3/MOON4 and phase 12.4 as external comparison context only.                                                                           | 12.4 / MOON3-MOON4                         | CFSG is extremely deep; never import it into A proof path as an unchecked theorem.                                               |
| TOMS-07 | Monster is not final object; common indicator matters   | #64-#71: Monster/Moonshine; #67/#70 says Monster is only tip of iceberg, seek connecting structure and common indicator                          | Borcherds 1992 proves Conway-Norton moonshine conjectures using vertex algebras, Thompson series/Hauptmoduls, Monster simple group                                                   | v20 had Monster/McKay-Thompson placeholders; needs common-indicator/Hauptmodul bridge-test record. | Add MOON8 and phase 12.8; make Monster comparison subordinate to derived q-character/Hauptmodul tests.                                        | 12.8 / MOON8                               | Do not overfit CNNA to Monster; use it as late fingerprint test only.                                                            |
| TOMS-08 | Leech lattice/rootlessness/theta shell data             | #84/#93: Leech lattice has no roots; shell/theta coefficients expose missing first shell and 196560 minimal vectors                              | Borcherds 1985 proves Leech lattice existence/uniqueness and records no-root/Niemeier context                                                                                        | Gap in v20: no Leech-specific comparison object.                                                   | Add MOON5/MOON6 and phases 12.5-12.6.                                                                                                         | 12.5-12.6 / MOON5-MOON6                    | Thread details around root systems were corrected by Jakito/TomS; use corrected Erzeugendensystem not basis wording.             |
| TOMS-09 | VOA/chiral CFT/string-theory context                    | #64: Moonshine connection to bosonic string compactified on Leech lattice and 2D chiral CFT; #82/#85 caution about physics-from-math             | Borcherds 1992 keywords include vertex algebras, monster simple group, Thompson series, Hauptmoduls                                                                                  | Needs strict comparison/consumer boundary.                                                         | Add MOON7 and keep string theory in side-comparison phase 18.2; no VOA/string axiom import into Pillar A.                                     | 12.7, 18.2 / MOON7                         | VOA formalization is much deeper than current A path; only finite graded truncations may be computable comparison artifacts.     |
| TOMS-10 | Elliptic functions are related but not the final bridge | #82/#89: TomS did not use elliptic functions as over-arching link; Jakito notes G_{2k}, Delta, j arise from elliptic-function lattice theory     | Standard modular/elliptic-function pedagogy supports G4/G6 -> g2/g3 -> Delta -> j                                                                                                    | Already covered by modular forms and Weierstrass invariant phases.                                 | Keep elliptic-function material as necessary j-construction route, not as claimed universal CNNA endpoint.                                    | 9.1-9.2 / M02-M05                          | At implementation, distinguish analytic exact layer from finite q/truncation mirror.                                             |
| TOMS-11 | Fractal fundamental-domain imagery is low priority      | #22/#24: fractal image language for iterated fundamental domains is explicitly uncertain/irrelevant in thread                                    | No primary formal target needed for current CNNA plan                                                                                                                                | No action in near path.                                                                            | Do not create a CNNA object for fractality; if ever needed, classify as visualization or tessellation comparison after modular action exists. | 10.3 only if needed                        | Avoid spending proof effort on metaphorical/visual claims.                                                                       |
| TOMS-12 | Physics-from-math caution                               | #37/#85: TomS objects that physics is not obtained by pure mathematics alone; thread is not a CNNA ontological source                            | Methodological warning aligns with comparison-only and handoff discipline                                                                                                            | Supports existing access policy.                                                                   | Keep established physics/math theories as references and tests, never as ontic/generative inputs.                                             | access_policy; 18.* comparisons            | Empirical/physical claims remain speculative unless reconstructed from CNNA and compared to experiment.                          |

| ID         | Topic                                                                     | Thread evidence                                                                                                                                                                                                           | External check                                                                                                                                                                                                | Plan disposition                                                                                                                  | CNNA action                                                                                                                                                                                                                                                               | Phase/Object                                             | Risk / uncertainty                                                                                                                                                                                                  |
|------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TOMS-13    | Heuristic b-ary tree construction of Eisenstein/Gauss lattice profiles    | post #97: b-ary address construction; b>=2 heuristic; Eisenstein angle 120 degrees and closure of 0-path at level 3; Gauss angle 90 degrees and closure at level 4; hierarchical overlays versus spiral angle dependence. | The thread entry is heuristic and self-authored, not a theorem source. Established CM/j special targets should be verified from formal references during M05/M08; forum data may define tests but not proofs. | Add green boundary lock 1.1.1.1.0.9 and future phases 8.5-8.7 for static/dynamic test matrices and closeout.                      | Use code-defined branching b. Test rows must cover angle, rational cosine, finite address collisions/closure, CM point target, discriminant target, and j target over selected L_max values. Keep complex coordinates/tau/j analytic evaluation mirror-only under EXEC-2. | 1.1.1.1.0.9; 8.5-8.7 / LEG28, L09, L10, M08, BR10        | The geometric embedding is not yet derived from the generator; it may fail or remain comparison-only. Dynamic branching and general D require later source objects and cannot be assumed from b=2 drawings.         |
| TOMS14     | Post #97 as early general test source, not special-case tunnel            | Post #97 motivates Eisenstein/Gauss angle/closure tests from b-ary tree drawings.                                                                                                                                         | The source is heuristic and special-case oriented; it does not define the code branching object or a complete general test suite.                                                                             | Use as sentinel rows inside O07/L09/L10, not as the complete test matrix.                                                         | Add LEG29/O07 and GNG16; update Phase 8.5/8.6 to require generic coverage before sentinel interpretation.                                                                                                                                                                 | 1.1.1.1.0.10, 4.0.1, O07, L09/L10/M08/BR10               | If generic rows are too weak, sentinels may create false confidence. Coverage must be reviewed when real code starts.                                                                                               |
| TOMS15     | Eisenstein/Gauss as noOuter-root sentinels, not the full Heegner story    | Post #97 gives b-ary address and angle/closure heuristics: 120 degrees with closure at L=3 for Eisenstein; 90 degrees with closure at L=4 for Gauss; the author also marks the discussion as heuristic in #99.            | Class-number-one/Heegner lists contain more than D=-3 and D=-4, so CNNA must not identify all Heegner targets with the closed/noOuter branch. Established lists are comparison targets, not CNNA derivations. | Add LEG30/O08/N18/BR11/BR12, GNG17, and phases 1.1.1.1.0.11, 4.0.2, 7.2.1.                                                        | Use regime-typed test rows: Eisenstein/Gauss are noOuter-root sentinel candidates; other Heegner/CM targets are HasOuter/complement-projection or comparison rows unless noOuter provenance is derived.                                                                   | 1.1.1.1.0.11; 4.0.2; 7.2.1 / LEG30, O08, N18, BR11, BR12 | This is a CNNA hypothesis and test classification, not yet a theorem. It may fail once concrete rows are computed; failure should trigger backjump to regime/complement source layers, not a consumer patch.        |
| HORN-GA-01 | Horn 0709.2238v1 quaternion, Pauli and geometric-algebra synthesis        | Uploaded PDF 0709.2238v1, sections on quaternions, Pauli matrices, geometric algebra and duality.                                                                                                                         | Horn presents quaternions, Pauli matrices and Hestenes geometric algebra as overlapping frameworks synthesized through duality; the paper is didactic/conceptual and not a CNNA proof source.                 | Create comparison/reference gates GA0-GA3 under 1.4.9.* and keep them outside the derived-only generator chain.                   | Record target identities and ambiguity warnings: Hamilton unit products, Pauli/Cartan sign choices, complex-i versus quaternion-i, pseudoscalar duality, pure quaternion as bivector and biquaternion multivector bookkeeping.                                            | 1.4.9; 1.4.9.1; 1.4.9.2 / GA0-GA3                        | Sign conventions, orientation choices and semantics are underdetermined by the external reference; a CNNA-derived object may fail the target or match only after a natural-origin proof supplies missing structure. |
| HORN-GA-02 | Mathlib Quaternion and CliffordAlgebra seams as future formal target APIs | Web source review of current mathlib Quaternion and CliffordAlgebra documentation plus Lean geometric-algebra formalization literature.                                                                                   | Mathlib exposes Quaternion/QuaternionAlgebra and CliffordAlgebra APIs; Wieser/Song note GA formalization through CliffordAlgebra and gaps around wedge/N-grading.                                             | Record as possible future comparison implementation seams only; no import is added before local Lean/mathlib 4.28.0 verification. | Add scratchpad expected-import warnings for 1.4.9.*, 13.2.1 and 13.6.1; route implementation through minimal target profiles if mathlib imports are too heavy or noncomputable.                                                                                           | 1.4.9.1; 1.4.9.2; 13.2.1; 13.6.1 / GA1-GA5               | Live docs may differ from the pinned project snapshot. Noncomputable or classical pressure must remain in mirror/comparison layers and never enter operational paths.                                               |
| HORN-GA-03 | Dirac, spacetime algebra and conformal GA as late target gates only       | Horn discussion of quaternionic Dirac motivation, complex quaternions, outlook toward spacetime algebra and conformal geometric algebra.                                                                                  | The material is useful for target-gate design but cannot supply Lorentz structure, Dirac equations, spacetime dimension or conformal geometry as CNNA inputs.                                                 | Add 13.6.1 and 18.5 as late comparison phases with GA5-GA6.                                                                       | Use after Pillar-B local-net/modular data and side-comparison outputs exist; compare signatures and structural compatibility, not foundations.                                                                                                                            | 13.6.1; 18.5 / GA5-GA6                                   | Potentially high heuristic pull toward established physics. Guardrail: no Dirac/STA/CGA theorem enters CNNA before derived predecessor objects exist.                                                               |

11    Secondary long-term goals

**What this section shows.**    This table contains side objectives and long-horizon tool/theory ambitions, including database and interactive-map directions.    The rows are shown because some principles apply immediately even if their technical realization remains later.    They must not displace the active Lean/content path.

**Secondary long-term goal rule.**    The goals below are secondary to Lean implementation, but not all are purely downstream.    The Vision column records motivating long-horizon interface and review ambitions without upgrading them to CNNA inputs.    When a goal is marked as immediate-principles, its epistemic/governance constraints apply now while its technical implementation may remain later.    These goals may use generated repository metadata, proof dossiers, derivation graphs, review records, context-documentation slices, and future database exports as documentation or diagnostic material, but they do not supply CNNA-generative data.    They must not displace the active Phase-1 cleanup, computable-instance traversal, or derived-only object closure.

| ID      | Goal                                                                                        | Motivation                                                                                                                                                                                                                                       | Vision                                                                                                                                                                                                                       | Raw data / tools                                                                                                                                                                            | Operational scope                                                                                                                                               | Not-before gate                                                                                                                                                                | Derived-only boundary                                                                                                                                            | Plan action                                                                                                                                                                                                                                             |
|---------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SLTG-01 | Semi-automated documentation of existing Lean code inside the same planning/proofbook tool. | The Bestandscode cannot be fully normalized at once, but every module, declaration family, object, proof bridge, and consumer path can be documented with the same YAML/proof-dossier infrastructure so later interpretation is minimized.       | Existing Lean-code documentation should grow into a reviewable object universe: each missing object/quantity becomes a dossier-backed node that can be inspected by the relevant expert without reading the full repository. | repo_inventory/raw_export modules.json, import_edges.json, symbol_reference_summary.json, dependency_order, mainline forensics, object register, proof dossiers, implementation scratchpad. | Secondary workflow: document existing code and its true consumers while Phase 1.x cleans the public API. It does not authorize refactors or deletion by itself. | After Phase 1.1.1.1.4 declaration consumption classification and initial proof-dossier infrastructure are closed; deeper execution after Phase 1.1.5 import firewall is green. | Documentation may classify provenance and consumers; it may not turn graph metadata into CNNA-generative assumptions.                                            | Add Phase 19.1 and object LEG16; use the generated Lean module graph to prevent forgotten code/documentation gaps. V1.12 adds LEG36 so existing-code documentation also records semantic consolidation/no-merge candidates before new substantive code. |
| SLTG-02 | Derivation-graph geometry of the formalized theory.                                         | The repository already contains graph-theoretic data: import graph, dominators, layer decomposition, source-sink signal paths, and robust semantic mainline. These can later expose bottlenecks, weak derivation edges, and discipline clusters. | A later graph/database layer can provide semantic zoom from pillars to phases, modules, declarations, objects, proof dossiers, and theorem sites; the graph remains a diagnostic/evidence map, not a generator.              | import_edges.json, dominator analysis, layer decomposition, source-to-sink paths, module group graph, architectural core summary, mainline forensics.                                       | Secondary research/diagnostic layer only. Use it to understand theory organization and proof pressure after core derivations are stable.                        | After the neutral API, generator governance, object proof dossiers, and main operational arithmetic/DtN paths have stabilized; not before active Phase 1.x cleanup.            | Graph geometry can suggest audits and consistency checks but cannot supply mathematical axioms, stochasticity, open-system behavior, or physical interpretation. | Add Phase 19.2 and object LEG17; render/import future graph metrics such as dominators, cuts, signal-path depth, cluster boundaries, and optional Cheeger-style diagnostics as comparison metadata.                                                     |
| SLTG-03 | Formal emergence-path forensics for open-system dynamics inside the derivation network.     | Later, once Pillar A/C objects are derived, the proof/module graph may help locate where closed finite subsystem assumptions become open effective dynamics through boundary/complement/regime handoffs.                                         | When A/C open-system paths exist, graph forensics can show where regime, complement, boundary, and handoff structures make openness visible, but the dynamics must still be derived in Lean-facing CNNA objects.             | A-regime profiles, complement-family profiles, DtN/Schur handoff modules, C/OQS consumer path, source-sink paths, dominator/cut analysis, object proof dossiers.                            | Post-core analysis of the formalization topology; not a replacement for deriving open-system behavior inside CNNA.                                              | After A regime/complement path, A->C handoff, and C OQS consumer roadmaps have enough checked objects to analyze. Treat as bonus/secondary until then.                         | Emergence claims must still be proved from CNNA objects; graph forensics only localize proof paths, bottlenecks, and structural analogies.                       | Add Phase 19.3 and object LEG18; keep as black/secondary target until the relevant A/C derivations exist.                                                                                                                                               |
| SLTG-04 | Reflective comparison between CNNA object-structure and formalization-structure.            | If the formal derivation graph later exhibits structural features resembling CNNA objects - layering, boundary cuts, operators on graphs, signal flow, modular clusters - this should be studied as a meta-level observation, not as a premise.  | The tool may be read as a meta-level interface analogue: one active cursor/bright object is examined against the large Lean-code complement; this is an operational review metaphor, not a CNNA theorem.                     | discipline subgraphs for Dirichlet/DtN/Schur, prenumerical/ToC/Substrate, Topology/Algebra/CD; module clusters; graph cuts; source-sink paths; proof-dossier links.                         | Far-target meta-analysis and visualization after mathematical closure. It may produce diagnostics, figures, and research notes, not new CNNA base assumptions.  | Only after core derived objects, public API, and consumer handoffs are stable enough that graph measurements are meaningful.                                                   | Strictly comparison/diagnostic. Any apparent self-similarity between theory and formalization remains an observation unless separately formalized and proved.    | Add Phase 19.4 and object LEG19; keep outside the active implementation path.                                                                                                                                                                           |
| LT5     | Thread/full-boundary derivation-geometry and string-analogy diagnostics                     | The IdealThread/full-cut extrema may later illuminate how one-dimensional recursive routes sit inside full boundary/operator geometry and how open-system behavior emerges from complement/interface structure.                                  | Future string-analogy diagnostics may use derivation geometry as a comparison map around the thread/full-boundary extrema, only after the underlying route is derived and reviewed.                                          | IdealThread/CutSpec.full profiles; import graph; derivation graph; Schur/DtN route audit; proof dossiers; source-sink paths and dominators.                                                 | Secondary diagnostic/comparison study only, after the underlying CNNA objects are code-derived and public-final.                                                | After S07, S08, BR06, BR07, S05/S06, and the relevant arithmetic/number-theory proof dossiers are closed.                                                                      | The diagnostic may compare formalization geometry to CNNA objects but cannot feed back as a generative axiom, formula, or shortcut.                              | Keep as Phase 19.5 side-road; do not block active Phase 1 cleanup or Phase 3/4 neutralization.                                                                                                                                                          |



| ID      | Goal                                                                                       | Motivation                                                                                                                                                                                                                  | Vision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Raw data / tools                                                                                                                                                                                                                   | Operational scope                                                                                                                                                                        | Not-before gate                                                                                                                                                           | Derived-only boundary                                                                                                                                                                                                              | Plan action                                                                                                                                                                                                                                                                                   |
|---------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SLTG-06 | Database-ready CNNA review and documentation structure                                     | A database-backed representation is likely the most effective path for querying, analyzing, visualizing, sharing, online display, interactive exploration, and discipline-separated expert review of the CNNA object world. | Long-term vision: a database-backed, queryable and online visualizable CNNA map - effectively an interactive semantic mindmap of the formal project - with speech/query instantiated context-docs, expert-review packets, and deep zoom from global structure to Lean sites. This is not current priority and remains secondary to the active Lean/content phases. The same future map should expose merge/no-merge refactor candidates and show where compactness would improve or damage provenance. V1.22 refines this into a Supabase/Postgres public read-only mirror and cnna-project.net web-frontend sidechain with append-only snapshots and no DB truth-source authority. | Single YAML master, generated TSV/JSON/DOT/PDF views, phase/object/proof-dossier records, module inventory, status legends, local theory context ledger, end-theory comparison ledger, future relational or graph database export. | Main side objective Phase 20. Technical database implementation is later and non-blocking, but schema discipline, expert-review semantics, and blue-audit obligations apply immediately. | Principles apply immediately. Technical database export follows after YAML schema and generated views are stable enough for round-trip-safe identifiers and review links. | The database is a documentation/review/query layer. It may expose, analyze, and visualize Lean/YAML-derived metadata, but it cannot supply CNNA-generative axioms, formulas, physics, stochasticity, or established-theory inputs. | V1.22 adds phases 20.15-20.20 and DBR15-DBR20 for Supabase/Postgres schema, YAML/import roundtrip, snapshot/checksum model, public read-only API/RLS views, cnna-project.net frontend data contract and future reviewer/admin model. Technical implementation remains later and non-blocking. |
| SLTG-07 | Geometric-algebra comparison map for later CNNA-derived algebraic and spacetime structures | Quaternion/Pauli/Clifford/geometric-algebra literature provides compact target profiles for rotations, bivectors, pseudoscalars, Clifford signatures and possible spacetime/conformal comparison layers.                    | After CNNA derives relevant CD, boundary-algebra, AQFT/spacetime or side-comparison outputs, the tool can generate a review map showing which GA targets are matched, failed or not yet meaningful.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Horn 0709.2238v1, Hestenes references, mathlib Quaternion/CliffordAlgebra docs, Wieser/Song Lean GA formalization, proof dossiers GA0-GA6.                                                                                         | Long-term comparison/review map only; no foundation, no phase release, no Lean generator input.                                                                                          | After GA0-GA6 predecessor phases expose CNNA-derived candidate data; especially CD reintegration and Pillar-B boundary/local-net phases.                                  | GA references remain blue/reference material. A target match is evidence of compatibility, not a source of truth; a mismatch must be recorded rather than patched at the consumer.                                                 | V1.25 adds phases 1.4.9.*, 13.2.1, 13.6.1 and 18.5 plus objects GA0-GA6.                                                                                                                                                                                                                      |



12    Standardized phase-preparation, implementation, and documentation workflows

**What this section shows.**    This section describes repeatable workflows over the Master YAML state: how to rehearse implementability, close a phase, prepare context documentation, add workflows or route missing structures back to their natural origin. It is shown because the main value of the tool is workflow scalability, not merely static tables.

| Workflow                                                     | Trigger                                                                                                                                                                                                                  | Required input                                                                                                                                                                                                                                                                                                            | Required updates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Exit rule                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global no-go guardrail inheritance                           | Every phase preparation, implementation prompt, code review, object documentation update, closeout, pruning decision, or generated-view refresh.                                                                         | top-level global_no_go_guardrails; active phase scratchpad; proposed module/declaration/object changes; operational vs mirror/non-operational layer classification.                                                                                                                                                       | Record that the global guardrails are inherited. Phase-local no_go_guardrails may add stricter local constraints but may not replace, weaken, or scope away the global list. Any exception must be explicitly rejected unless it is already allowed by the global rule text and isolated outside the operational path.                                                                                                                                                                                                        | No phase may be implemented or closed unless global_no_go_inheritance is present in its scratchpad record and no local field contradicts the top-level global guardrails.                                                                                                                                                                                                              |
| Tool/display versioning and Lean-code data timestamp policy  | Any generator, YAML-policy, PDF-display, table-layout, legend, navigation, or workflow-contract change; and any Lean source/data change.                                                                                 | metadata.tool_version; metadata.tool_change_timestamp; metadata.data_timestamp; change classification as tool/display-only versus Lean-code/data-affecting; affected static-finding, scratchpad, or object rows.                                                                                                          | For tool/display-only changes, increment tool_version by 0.01 and update tool_change_timestamp only. For Lean-code/data changes, update data_timestamp/data_version and record the phase/build context. Any materially changed static-finding, scratchpad, or object row must carry a visible change version and timestamp in its generated table view.                                                                                                                                                                       | No release artifact may advance the Lean-code data timestamp unless Lean source/data changed. No table-row correction may remain invisible in version/change metadata when it affects static findings, scratchpad records, or object records.                                                                                                                                          |
| Pre-start feasibility and static-finding triage              | Before starting any active phase after a user-reported green build, and before changing Lean code or closing an infrastructure gate.                                                                                     | active phase_id; current phase dependencies; scratchpad preparation fields; Section 2 static findings with Priority/Issue status/Addressing phase; module manifest; reuse/duplication audit; object proof dossiers when relevant.                                                                                         | State whether the active phase is feasible as infrastructure-only, refactor, or Lean implementation. If a blocking issue is found, add or update a Static code finding row with R/Y/G priority, status, and addressing phase; set or keep the phase red/yellow accordingly and backjump to the natural source layer rather than patching a consumer.                                                                                                                                                                          | Implementation may start only after the active phase has a positive feasibility result or a precisely localized missing bridge. New findings may not remain prose-only; they must be represented in Section 2 with a phase assignment and priority.                                                                                                                                    |
| Pre-Implementation Phase Check / semantic thought experiment | User asks to check the next semantically connected phases in advance, as a Gedankenexperiment, before implementation; also applicable before a risky multi-phase refactor or derived-only bridge sequence.               | Current active cursor; phase table; scratchpad; object register and proof dossiers; module implementation manifest; reuse/duplication ledger; consumption map; static findings; last user-reported green build when available; requested semantic window or the next contiguous phase cluster inferred from dependencies. | For every inspected phase, create/update a pre-implementation check record and mark Y if the phase is implementable under current dependencies, or R if a dependency, object, proof route, module seam, source, or consumer-contract gap is missing. Add newly discovered problems to Static code findings. For every R record, generate one or more upstream addressing phases or a documented merge/backjump target before the blocked phase; do not set the blocked phase active until those addressing phases are closed. | The check closes only when every phase in the semantic window has a Y/R result, each R result has a concrete addressing phase or documented merge target, generated phases have object-link policy and scratchpad stubs, and all changed records carry the current tool version/timestamp. Lean-code data timestamp remains unchanged unless Lean files or Lean inventory are changed. |
| Concrete computable instance traversal                       | Before any operational Lean implementation starts, whenever a computable path, ExecComplex bridge, generated matrix/operator, recurrence, determinant, inverse candidate, or smoke-test route is introduced or modified. | active phase_id; target operational declarations; at least one concrete finite instance or an explicit infrastructure-only justification; expected executable witness; current mathlib imports and known computability risks.                                                                                             | Fill concrete_instance_traversal, computable_path_not_skeleton_status, expected_executable_witness, mathlib_complication_log, noncomputable_boundary_audit, and mathlib_mitigation_or_refactor_layer in the scratchpad. Record whether the instance reaches the intended output object or where it stops.                                                                                                                                                                                                                     | Implementation may start only if the phase either has a concrete instance path through the computable code or is explicitly infrastructure-only. Any mathlib noncomputable leak must be isolated in an allowed mirror/non-operational layer or backjumped to the natural source layer; it cannot silently enter the operational path.                                                  |
| Phase preparation from reliable sources                      | User asks to prepare phase x with sources y,z or supplies established/reliable sources before implementation.                                                                                                            | phase_id; source titles/URLs/files; expected theory layer; optional target objects.                                                                                                                                                                                                                                       | Record source materials; extract theory objects/formulas; map them to CNNA object IDs; update or add object rows; split the phase if dependency depth is too high; fill dependency audit, proof rehearsal, tactic plan, and natural fallback layer. Do not mark the phase green.                                                                                                                                                                                                                                              | Phase remains Y/R/W according to dependency status. A prepared phase has a concrete positive proof path or a precisely localized missing bridge.                                                                                                                                                                                                                                       |
| Implementation from last green baseline                      | User asks to implement the active phase in Lean code.                                                                                                                                                                    | last green package; active phase_id; prepared scratch-pad record; object/phase dependencies; applicable source notes.                                                                                                                                                                                                     | Use positive construction first; preserve import/consumer discipline; add or modify only neutral code-surface modules; record files changed, definitions, proof skeleton, tactics, forbidden-pattern checks, and package result.                                                                                                                                                                                                                                                                                              | If the build is locally confirmed green, set phase G and implementation_status completed; otherwise keep or set active R and add the missing bridge at the natural source layer.                                                                                                                                                                                                       |

| Workflow                                                                | Trigger                                                                                                                                                                                                                               | Required input                                                                                                                                                                                  | Required updates                                                                                                                                                                                                                                                                       | Exit rule                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object and quantities proof documentation workflow                      | A registered object or quantity is introduced, corrected from blue to public-final, exported through a handoff, or used as a proof-relevant code object.                                                                              | object_id;<br>current derivation site;<br>source objects;<br>CNNA definition;<br>intended proof target;<br>phase_id;<br>semantic/API status.                                                    | Create or update the object proof dossier with source objects, definitions, proof goal, proof sketch/proof term reference, verification/audit result, documented yes/no, and link from the object register. Blue objects must record why proof-carrying is not yet semantically final. | An object may be public-final only when the object row and proof dossier agree: documented=yes, semantic status complete, API/handoff rights settled, and verification evidence points to the checked code or explicitly states that Lean build evidence is still local-only. |
| Gap localization and backjump                                           | During preparation or implementation a required object/proof bridge is missing.                                                                                                                                                       | failed target statement;<br>currently available objects;<br>deepest successful dependency;<br>missing lemma/data shape.                                                                         | Do not patch the consumer. Add or activate the earliest natural source phase where the structure should be generated;<br>record why the old phase cannot close;<br>preserve positive-path orientation.                                                                                 | The blocked phase is not closed. A smaller predecessor phase is created or activated with explicit source objects and proof goals.                                                                                                                                            |
| Post-green closeout                                                     | User reports the phase build is green.                                                                                                                                                                                                | compiler result;<br>exact zip/package used;<br>warnings or errors fixed;<br>current phase record.                                                                                               | Fill implementation result;<br>update object statuses;<br>update linked phase-object map;<br>close scratch-pad record;<br>move active cursor to the next phase whose dependencies are complete.                                                                                        | Exactly one active cursor remains. All green phases have complete implementation records;<br>generated tables and PDF compile.                                                                                                                                                |
| Local Lean build evidence rule                                          | Any phase closeout or artifact package mentions build status.                                                                                                                                                                         | Actual Lean build output from the current package or an explicit user report of a local green build.                                                                                            | Record whether the assistant generated artifacts only or whether a Lean build was actually run. Never infer green build from unchanged planning artifacts.                                                                                                                             | If no build was run, the package status is artifact-generated / local Lean build required, not green.                                                                                                                                                                         |
| Consumption classification and pruning discipline                       | A module/declaration appears unused, only BuildAll-reachable, status/audit-like, phase-semantic, or blue-object/API-unfinished.                                                                                                       | module path;<br>declaration name/family;<br>import/reference evidence;<br>object ID if any;<br>proposed consumer or deletion target.                                                            | Classify the item in LEG5/LEG7/LEG8 before moving or deleting it. Decide keep/quarantine/wrap/reexport/archive/deletion-candidate. Update public API contract and firewall if it can be consumed by Phase 3+.                                                                          | No deletion or public export until the item has a class, a consumer rule, and either a neutral API route or an explicit quarantine/deletion-candidate record. Deletion still requires later green build evidence.                                                             |
| Pre-implementation module manifest                                      | Before any Lean implementation phase starts or before an implementation prompt asks to add/modify code.                                                                                                                               | active phase_id;<br>target objects;<br>current module inventory;<br>existing modules with matching semantics;<br>proposed new declarations;<br>import/firewall constraints.                     | Fill the scratchpad module manifest: exact count and names of proposed added modules and modified modules;<br>semantic placement rationale for each;<br>why existing modules are reused/extended or rejected;<br>public/legacy/API role;<br>expected import direction.                 | Implementation may start only if the module budget is explicit and compact. A new module is forbidden unless semantic fit, reuse, and import-firewall checks justify it.                                                                                                      |
| Reuse and duplicate-implementation audit                                | A planned declaration, helper, generator, ledger, request/output object, smoke test, or computation resembles existing code.                                                                                                          | symbol/reference summary;<br>existing generator/API modules;<br>proposed target semantics;<br>object IDs and phase consumers.                                                                   | Record whether the existing module is consumed, extended, wrapped, or rejected. If rejected, explain the semantic mismatch. Mark duplicate risk and update public API/manifest.                                                                                                        | No duplicate implementation is allowed. The same mathematical/computational operation must have one canonical implementation path plus wrappers/reexports if needed.                                                                                                          |
| Canonical Smoke-Test Generator governance                               | A phase proposes generator-like code, experiment request/output code, smoke-test generation, or reusable theory-generation infrastructure.                                                                                            | existing Smoke-Test Generator modules and APIs;<br>target generator semantics;<br>phase/object consumers;<br>import constraints.                                                                | Route through the canonical generator seam whenever semantics match. If extension is needed, extend the canonical generator at its natural layer;<br>do not create a parallel generator. Register any genuine semantic split explicitly.                                               | The canonical generator remains the single generator for the theory where applicable. Parallel generator modules are blocked unless the plan records a proven semantic split.                                                                                                 |
| Secondary long-term code-documentation and derivation-geometry workflow | A request proposes documenting the existing codebase, analyzing the geometry of the formal derivation graph, studying bottlenecks/dominators/source-sink paths, or reflecting on open-system emergence in the formalization topology. | Stable generated repo inventory;<br>object proof dossiers;<br>declaration consumption classification;<br>active not-before gates;<br>explicit derived-only/comparison-only boundary.            | Register the goal as secondary if prerequisites are not closed. If active later, update coverage/dominators/layer/path diagnostics and link them to objects/modules without changing CNNA generative inputs.                                                                           | Outputs are documentation, diagnostics, or comparison reports only. They may guide audits and refactors, but cannot become mathematical or physical premises without a separate CNNA-derived proof path.                                                                      |
| EXEC-2 rational boundary and mirror isolation                           | Any phase mentions ExecComplex, tau, sqrt(D), lattice embedding, Gram entries, eigenvalues, j-invariant, algebraic bridge, or mirror boundary.                                                                                        | Active phase;<br>target object IDs;<br>whether the value is rational/integer operational data or algebraic/analytic mirror data;<br>proof obligation for any proposed operational return value. | Fill exec_complex_boundary_policy and noncomputable_boundary_audit. Update affected object ExecComplex/computable path text. Mark algebraic irrational data as mirror-only unless a rational/integer invariant is explicitly proved and typed.                                         | No phase may introduce sqrt(D), tau, algebraic eigenvalues, or general algebraic complex coordinates into the operational Q x Q path. Operational return from mirror requires a stated invariant and proof target.                                                            |

| Workflow                                            | Trigger                                                                                                                                                                          | Required input                                                                                                                                                                                  | Required updates                                                                                                                                                                                                                                                                    | Exit rule                                                                                                                                                                                                                                           |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refactor regression and incremental migration       | Any phase moves declarations, splits a module, introduces wrappers/aliases, quarantines legacy gates, or changes imports.                                                        | Module manifest; declaration list or family; current and target imports; consumer list; local build evidence plan.                                                                              | Fill refactor_regression_protocol and refactor_migration_metrics. Record atomically delivered package boundary, declarations migrated count, import graph delta, consumer breakage audit, and local full-build status source.                                                       | A refactor phase closes only after the delivered package has local full-build evidence. Intermediate wrapper coexistence is allowed; deleting legacy/status material before quarantine closure is not allowed.                                      |
| Lean/mathlib version management                     | Any phase proposes a new mathlib import, encounters noncomputable/classical pressure, or wants newer mathlib API than the frozen baseline.                                       | Baseline version; proposed import; current compile/build evidence; expected mathlib theorem/API; operational vs mirror classification.                                                          | Fill lean_mathlib_version_policy and mathlib_complication_log. Either stay on baseline, open a separate upgrade gate at a phase boundary, or re-route the dependency to mirror/non-operational code.                                                                                | No silent version drift. No operational code may depend on a new mathlib API or noncomputable/classical pressure without an explicit gate and local build evidence.                                                                                 |
| Phase-type-specific scratchpad validation           | Every phase preparation or status change.                                                                                                                                        | phase_type; phase status; target code/object role; active scratchpad.                                                                                                                           | Fill phase_type and phase_type_specific_contract. For refactor phases fill migration/regression fields; for derivation phases fill proof/tactic/mathlib fields; for comparison phases fill derived-only boundary; for infrastructure phases justify absence of operational witness. | A phase cannot become active unless its phase_type-specific evidence is prepared. Not-applicable filler is acceptable only with an explicit infrastructure-only or comparison-only justification.                                                   |
| Governance-depth retrospective after first refactor | When Phase 1.1.2 or the first real code-change/refactor package closes.                                                                                                          | Closed refactor report; list of governance gates used; mistakes prevented; time/complexity overhead; upcoming Phase 3+ structure.                                                               | Fill governance_overhead_review. Decide whether the current depth remains necessary or whether later phases should use a flatter structure with the same global guardrails.                                                                                                         | Do not remove global no-go rules. Only the phase granularity may be flattened, and only after a recorded retrospective.                                                                                                                             |
| Phase 3 to Phase 4 consumer-contract prebinding     | Before preparing Phase 3.6 or Phase 4.1, or any wrapper around BoundaryOperatorProfile / ExecBoundaryMatrix.                                                                     | BoundaryOperatorProfile shape; Operational/ExecComplex consumer needs; import direction; generator policy.                                                                                      | Fill consumer_contract_direction. Phase 3.6 records the exported shape; Phase 4.1 records how it consumes the shape without reconstructing Schur/DtN internals.                                                                                                                     | Phase 3 may not import Phase 4. Phase 4 may not duplicate generator/Schur/DtN logic; it consumes the Phase-3 profile contract.                                                                                                                      |
| Boundary-route extremal audit                       | Any phase that interprets IdealThread, CutSpec.full, Schur recursion, DtN boundary matrices, or string-like analogies.                                                           | Code references for IdealThread singleton carrier/family/coarsen lemmas; CutSpec.full carrier theorem; current generated boundary source; planned consumer shape.                               | Record whether the phase consumes thread-scalar route, full-boundary route, or generated intermediate route. Mark analogical string language comparison-only unless a theorem gives a formal CNNA object.                                                                           | No full matrix/operator, discriminant, spectrum, or physical interpretation may be promoted from the thread route unless the route and lifting theorem are explicitly present and code-checked.                                                     |
| Horizontal/vertical Schur bridge workflow           | A phase attempts to relate Cayley-Dickson/Hurwitz algebra to arithmetic discriminant, CM order, norm, or transfermatrix data through Schur complementation.                      | Closed S05/BR06 horizontal boundary route, closed O05/BR07 vertical arithmetic-shadow route, Transfermatrix objects TM0-TM2, CD7 Hurwitz boundary, and explicit module manifest for the bridge. | Create/update BR08/BR09/CD8 proof dossiers; record commutation/naturality target, failure modes, concrete finite instance traversal, and no-input-backflow audit.                                                                                                                   | The bridge is either proved as a theorem over derived objects, split into smaller prerequisite lemmas, or marked blocked. It may not feed earlier phases unless a separate backreaction/input handoff explicitly permits it.                        |
| Heuristic lattice test-matrix governance            | Whenever Eisenstein/Gauss b-ary tree geometry, angle profiles, CM points, j-invariants, or L_max sweeps are planned or implemented.                                              | code-defined branching object/API; L_max set; embedding kind; angle profile; EXEC-2 boundary status; source objects L04/L06/O05/M05 when j/CM comparisons are used.                             | Record static/dynamic branch rows, concrete finite instance traversal, operational vs mirror fields, expected executable witness, and mathlib/noncomputable complications. Add or update L09/L10/M08/BR10 dossiers.                                                                 | The test may close only if all rows consume the canonical generator/branching API, no duplicate tree generator is introduced, operational assertions are rational/integer/finite, and mirror-only CM/j comparisons cannot feed generator semantics. |
| General early finite computation-test matrix        | Before implementing or promoting any operational computable path, arithmetic wrapper, generator extension, lattice test, or refactor that claims computation is preserved.       | Canonical generator/branching API; finite instance parameters; L_max/branch-mode rows; expected executable witness; optional sentinel/comparison rows; mathlib/noncomputable risk log.          | Fill O07 or a consumer of O07; update concrete_instance_traversal, expected_executable_witness, mathlib_complication_log, and no-go guardrails. Separate generic rows from sentinel rows.                                                                                           | The phase cannot close on hand-picked CM/Eisenstein/Gauss examples alone. At least one generic row family must traverse the actual computable path, or a documented failure/backjump is required.                                                   |
| Regime-typed Heegner/CM finite test matrix workflow | Whenever finite computation tests, Eisenstein/Gauss sentinel rows, Heegner/class-number-one targets, CM/j targets, or noOuter/HasOuter regime claims are planned or implemented. | O07 generic finite rows; derived regime split; code-defined generator/branching API; L_max rows; EXEC-2 boundary; class-number data if Heegner interpretation is claimed.                       | Fill O08/N18/BR11/BR12 dossiers; record regime tag, projection/complement-family status, generic-vs-sentinel status, operational witness, and mirror/comparison columns. Update mathlib/noncomputable log for any class-number/CM dependency.                                       | The phase cannot close if h(D)=1 or a Heegner/j target is used to infer noOuter closure. Generic rows must exist; noOuter-root sentinel rows and HasOuter/projection rows must be distinguished explicitly.                                         |

| Workflow                                                          | Trigger                                                                                                                                                                                                            | Required input                                                                                                                                                                                     | Required updates                                                                                                                                                                                                                                                                                                              | Exit rule                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computable scalar architecture workflow                           | Any phase introduces scalar data beyond rational ExecComplex, uses mathlib Complex/Real/NumberField/QuadraticAlgebra, or proposes a Cayley-Dickson-derived C-layer.                                                | O00/LEG20/LEG31, module manifest, reuse audit, expected executable witness, mathlib/noncomputable complication log, and CD noncircularity status if CD is involved.                                | Update LEG31/LEG32, O09/O10/BR13/BR14/CD9 dossiers; classify each scalar as rational ExecComplex, exact quadratic schema, proof-carrying D-instantiation, integer/rational invariant, or analytic mirror-only. Record whether a new module is justified or existing module semantics suffice.                                 | No new scalar layer may close without concrete finite instance traversal and proof-carrying projection back to allowed operational invariants. General Complex/Real and CD C-surfaces cannot become generator inputs, and exact quadratic instances require D-source provenance. |
| Derived discriminant-indexed exact quadratic scalar workflow      | Any phase creates or consumes exact quadratic coordinates, theta, sqrt(D)-like notation, CM tau coordinates, quadratic-field/order coordinates, or lattice rows requiring algebraic irrational data.               | O09 schema, LEG32/GNG19, proof-carrying D-source from O05/O06 or equivalent ledger, module manifest/reuse audit, mathlib/noncomputable complication log.                                           | Update O09/O10/BR14/BR13 dossiers; record whether the phase is schema-only, D-instantiation, invariant projection, or mirror comparison. Record concrete finite D-ledger rows and allowed invariant exports.                                                                                                                  | No phase may close with a free sqrt(D), an axiom, or an analytic Complex/CD source for theta. If D is unavailable or semantically blue, backjump to the discriminant-source correction rather than patching the scalar consumer.                                                 |
| Phase-path dependency and object placement audit                  | Before any implementation phase after Phase 4, or when moving/wrapping an object from N/L/M/SP/P/TM/BR/CD groups.                                                                                                  | Object register, source-object list, target module manifest, dependency-depth audit, and current phase-path classification.                                                                        | Update phase title, object Phase field, proof dossier, scratchpad dependency_depth_audit, consumer_contract_direction, and no-go notes. Record whether the object belongs to Path A, Path B, or a late explicit bridge.                                                                                                       | Phase may close only if its title, object group and dependencies agree. Finite spectral/physical consumers cannot be blocked behind modular/j/analytic phases unless a proof-carrying bridge says so.                                                                            |
| Phase-object-link policy workflow                                 | Whenever a phase is added, split, merged, moved, or whenever an object Phase field changes.                                                                                                                        | Phase hierarchy; object register; direct object ownership map; child-phase scope; moved/audit/closeout status.                                                                                     | Update phase Object link policy, Object link explanation, Child object scope, object Phase fields, proof dossiers and phase table generator if needed.                                                                                                                                                                        | The generated phase table may not contain an unexplained empty Linked objects cell. Direct ownership is never inferred from child scope; child-scope summaries are display-only.                                                                                                 |
| Ontic subsystem-minimality workflow                               | Whenever a phase introduces or consumes dynamics, spectra, operator algebra, geometry, arithmetic, stochasticity, open-system language, unitarity, AQFT/physics interpretation, or environment/complement effects. | POST1/LEG35/GNG22; source-object chain from IDEAL_infty formal background to subsystem/cut/approximant, boundary, regime, complement families and Schur/DtN or later certified bridge.             | Update object proof dossier, scratchpad source_to_cnna_mapping, dependency_depth_audit, no-go guardrails, and consumer-contract direction. Mark comparison-only interpretations explicitly and prohibit them from supplying definitions.                                                                                      | A phase may close only if non-subsystem structures are shown as derived consequences or comparison-only interpretations. Any primitive import of dynamics/physics/arithmetic/geometry/environment forces a backjump to the natural subsystem/complement generating layer.        |
| Lean-focus implementation-packet workflow                         | Before a Lean implementation phase starts, or when the user asks for the next semantically connected phases to be checked with minimal cognitive/tooling overhead.                                                 | Active phase, next semantic phase window, object register, scratchpad, module manifest, static findings, import/declaration inventory, and current pre-implementation-check ledger.                | Generate a compact implementation packet and avoid full PDF/declaration dumps unless the phase-check or audit-full mode explicitly requires them. Record G/Y/O/R thought-experiment results, predecessor assumptions, proof risk, and any red addressing phase proposals.                                                     | Lean coding may start only from a compact packet that names relevant modules, producer/consumer objects, red blockers, orange predecessor assumptions, yellow active implementability, forbidden shortcuts, and comparison-only theory boundaries.                               |
| Local theory context versus end-theory comparison workflow        | Whenever a phase or module mentions established mathematical/physical theory, mathlib background, or a large target theory such as AQFT/OQS/modular forms/CM/Moonshine.                                            | Module imports, declared CNNA objects, object proof dossiers, local background-theory context, and end-theory comparison ledger.                                                                   | Classify local established theory as formalization/proof-method/library context; classify large end theories only as comparison/target/consistency horizons. Record that neither role is an ontic CNNA generator unless derived internally.                                                                                   | A module may use established mathematics as formal language, but a CNNA object may close only if its producer is internal provenance. Large end theories remain comparison-only until CNNA derives a bridge object.                                                              |
| DB-REVIEW database-ready documentation and expert-review workflow | Whenever the YAML schema, generated views, phase/object records, proof dossiers, theory/reference markings, online/interactive documentation, or database/export plans are edited.                                 | Lean-code data timestamp; YAML master; phase/object/proof-dossier records; status legend; global no-go guardrails GNG23-GNG24; local theory context/end-theory comparison ledgers when applicable. | Keep Lean as primary truth system, generated views as secondary, and expert review as mandatory. Record whether a row is database-ready-later, review-required, blue-audit-needed, lean-primary, or generated-secondary when those semantics matter. Preserve round-trip-friendly IDs and explicit phase/object/review links. | No generated documentation may close by declaring semantic infallibility or 100-percent truth. Missing blue/reference markings must remain visible as audit debt, and database/export work must not become a CNNA-generative input.                                              |
| Context-filtered documentation slice generation                   | User asks for documentation about a specific object/topic, or a phase closeout requires documenting only the changed context rather than regenerating the whole PDF for human review.                              | Natural-language query or curated profile id; current Single-YAML-Master; regenerated generated_exports/*.tsv; optional slug.                                                                      | Run scripts/generate_tables.py first; run scripts/generate_context_doc.py with query/profile; package context_docs/<slug>/ manifest, section summary, TeX/PDF, and compile log. Add/adjust curated profile terms in YAML if the topic is recurring.                                                                           | Every major documentation section is present in the context packet; each section records matched/total row counts; output states generated-secondary status, Lean-primary rule, and expert-review requirement.                                                                   |

| Workflow                                                                     | Trigger                                                                                                                                                                           | Required input                                                                                                                                                                                                                                       | Required updates                                                                                                                                                                                                                                                                                                                                                                                                    | Exit rule                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semantic module consolidation and compatibility-preserving refactor workflow | Before new substantive Lean code after governance closure, or whenever the user asks whether semantically related modules should be merged, compacted, facaded, or kept separate. | LEG10 module manifest;<br>LEG11 reuse ledger;<br>LEG5 consumption map;<br>repo_inventory/raw_export modules/<br>import_edges/symbol references;<br>candidate file list;<br>direct code spot-checks;<br>current public/API and proof-dossier records. | Update LEG36, code_audit findings, reuse_duplication_ledger, scratchpad refactor fields, proposed modules, duplicate-risk symbols, natural-layer placement, facade/compatibility plan, and regression metrics. Set downstream content phases to orange if this gate must precede them.                                                                                                                              | No Lean merge may close without old-module facade or explicit migration plan, import graph delta, declaration/consumer delta, local full-build evidence, and proof that semantic/provenance auditability is preserved or improved.                             |
| Root artifact retention and generated-output cleanup workflow                | Before packaging a tool release, after full/context documentation generation, or when root PDFs/TeX/Markdown reports accumulate.                                                  | Current Repository root file list;<br>Single-YAML-Master;<br>README.md;<br>generated tables/exports;<br>optional generated_docs/full output;<br>archive/reports policy.                                                                              | Keep README.md in root;<br>move historical Markdown reports to archive/reports/;<br>delete disposable root PDF/TeX/LaTeX auxiliary files;<br>ensure full documentation is generated under generated_docs/full/ on demand;<br>update DBR8/GNG27 if classification changes.                                                                                                                                           | Project root contains no retained generated PDF/TeX/LaTeX build products and no active historical Markdown reports beyond README.md. All retained source data remains in YAML/scripts/tables/generated_exports/repo_inventory/repo_snapshot as appropriate.    |
| Root artifact retention and Lean/Lake build-entry preservation workflow      | Before packaging a tool release, after root cleanup, or when Lean/Lake build-control files are added, removed, or moved.                                                          | Package root file list;<br>CNNA.lean;<br>lakefile.lean;<br>lake-manifest.json;<br>lean-toolchain;<br>Repository/repo_snapshot/CNNA source tree;<br>README.md;<br>cleanup script policy.                                                              | Ensure package-root Lean/Lake project-control files are preserved;<br>ensure generated docs/reports remain outside active root;<br>update DBR10/GNG28 and README if the build-entry surface changes.                                                                                                                                                                                                                | Package root contains the source/control files needed for local Lake build and contains no retained generated PDF/TeX/LaTeX byproducts or historical active reports.                                                                                           |
| Root license, README and credit-surface workflow                             | Before packaging a tool release, when licensing/credits change, or when README files appear outside the package root.                                                             | Package root file list;<br>README.md;<br>LICENSE;<br>license_notice table;<br>metadata editor/assistant/<br>minimum_assistant_requirement;<br>full/context documentation generators;<br>cleanup policy.                                              | Keep README.md and LICENSE in package root;<br>remove duplicate active README files in subdirectories;<br>update license_notice/Stats/PDF author metadata;<br>update DBR11/GNG29 and README if credits, license scope, or tested assistant setup changes.                                                                                                                                                           | Exactly one maintained README.md exists at package root;<br>root LICENSE exists;<br>generated full/context PDFs expose Editor and Assistant credits;<br>package contains no root PDF/TeX/LaTeX byproducts.                                                     |
| Generated PDF section-pagebreak workflow                                     | When full or context documentation layout requirements change, especially section-boundary/page-navigation rules for review PDFs.                                                 | Full-document generator, context-document generator, Master YAML metadata, representative full/context PDF outputs, and the user-facing README command surface.                                                                                      | Patch generator preambles or layout macros;<br>keep disposable generated TeX/PDF out of the package root;<br>regenerate tables and representative PDFs;<br>update DBR12/GNG30/SCF49 and README if user commands or layout expectations change.                                                                                                                                                                      | The full documentation PDF and at least one context documentation PDF compile successfully, their generated TeX contains the section-pagebreak hook, and no Lean source or context-matching semantics are changed.                                             |
| Phase status release and red-lock workflow                                   | Before adding phases, changing a phase status, advancing the active cursor, or preparing a Lean-derivation implementation prompt.                                                 | phase row;<br>implementation_scratchpad.phase_type;<br>prerequisite object IDs;<br>proof dossier status;<br>pre-implementation check result;<br>current active cursor;<br>relevant static findings and guardrails.                                   | If phase_type=derivation and the phase is not closed, active, or explicitly implementable-after-predecessors, set it red. To release red, add a pre-implementation check that references already-derived Lean objects and sets yellow only for the unique active cursor or orange for later implementable phases. Keep white only for non-direct-Lean refactor/tool/infrastructure/documentation/comparison phases. | Generated tables pass validation: exactly one yellow phase;<br>zero white derivation phases;<br>red derivation phases carry lock language;<br>any yellow/orange release has explicit prerequisite evidence or a corresponding pre-implementation check record. |
| Documentation section explanation and appendix workflow                      | When a documentation section/table is confusing, redundant, or appears to duplicate a primary control surface.                                                                    | Full-document section order, context-document SECTION_ORDER, static table paragraphs, Master YAML sections, generated exports, and user review feedback.                                                                                             | Add or tighten static generator-owned explanations, keep table rows dynamic, move narrow technical tables to appendix when they are not primary reading surfaces, and update DBR/GNG/static finding records if semantics change.                                                                                                                                                                                    | A reader can identify what each section/table displays, where the data comes from, why it is shown, and which table is authoritative for phase release. No dynamic prose rows are added only to explain static section purpose.                                |
| Pre-substantive-code readiness and proof-gap audit workflow                  | Before starting any new substantive Lean/content implementation, especially before activating 1.1.1.2 or any later red locked derivation phase.                                   | Current full zip;<br>repo_snapshot/CNNA source tree;<br>repo_inventory/raw_export module/import/declaration inventory;<br>generated_exports consumption/object/proof/phase tables;<br>local build result if refactor is executed.                    | Fill LEG37 source-intake ledger;<br>create LEG38 neutral naming/facade plan;<br>execute only approved LEG39 refactor package;<br>regenerate inventory and close LEG40;<br>issue LEG41 release proof or create upstream red blockers.                                                                                                                                                                                | Exactly one active Y phase remains. No new substantive Lean code begins until LEG41 references closed derived-only/verified objects and the stable neutral API/source surface.                                                                                 |

| Workflow                                                   | Trigger                                                                                                                                                                                                 | Required input                                                                                                                                                                                                                                                                 | Required updates                                                                                                                                                                                                                                        | Exit rule                                                                                                                                                                                                                                                                |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supabase public read-only mirror and web-frontend workflow | When the database/web-publication sidechain is activated or when a Supabase/Postgres mirror, public read-only API, cnna-project.net frontend, reviewer annotation layer or direct importer is designed. | Master YAML, generated TSV exports, repo_inventory/raw_export, tool version, Lean data timestamp, source zip name, YAML/repo inventory hashes, Supabase project URL/anon key for read-only frontend, local service_role key only via .env for admin importer if/when executed. | Add or update DBR15-DBR20, schema/import/RLS/frontend contracts, snapshot/checksum records, RLS/grant tests, public view definitions, README/manual commands and archive report. Never store secrets in repo/chat.                                      | A DB mirror phase may close only when every public row traces to a snapshot_id, anon access is read-only through api.* views, internal/admin schemas are closed, service_role remains server/local-admin only, and a YAML/export roundtrip or import report is archived. |
| Generated PDF typography profile workflow                  | When generated full/context documentation readability or typography scaling changes are requested.                                                                                                      | Full-document generator, context-document generator, table generator, representative full/context PDF outputs, generated TeX audit and README command surface.                                                                                                                 | Patch generator-level size profile, regenerate tables, compile representative PDFs, render sample pages, update DBR21/GNG35/SCF54 and README if user-facing layout expectations change.                                                                 | Generated full and context PDFs compile with the intended typography profile, generated TeX shows the larger size profile, representative pages render without obvious clipping/black-glyph failure, and no Lean/YAML semantics change.                                  |
| Supabase live import activation workflow                   | After Supabase connector access exists and before any live schema/data write to the CNNA Supabase project.                                                                                              | Existing DB phases 20.15-20.20; Supabase project id/ref; Master YAML; generated_exports; repo_inventory; no-go guardrails GNG34/GNG36; explicit current-step user authorization for any write.                                                                                 | Record baseline discovery; design migration package; design importer package; only then apply migration/import on explicit authorization; generate import report/checksums; verify read-only api views/RLS/advisors; update README/report/context docs. | No live write occurs without authorization. After authorized execution, every public DB row must trace to snapshot_id/source hash; public access must be read-only; no Lean file changes are allowed or claimed.                                                         |
| Context-class assignment                                   | Whenever a phase or object is created, moved, reclassified, exported to DB, or rendered in a phase/object table.                                                                                        | Declared Context class from context_class_taxonomy plus phase_type/object provenance/status fields.                                                                                                                                                                            | Master YAML phase/object row; proof dossier if object; scratchpad if phase; generated tables and TSV exports.                                                                                                                                           | No blank Context class; class must not contradict handoff/access rights or generated table semantics.                                                                                                                                                                    |

**Pre-Implementation Phase Check rule.** This workflow is a forward-looking implementation rehearsal, not a Lean build and not a phase closeout. When requested, the next semantically connected phases are inspected as if they were to be implemented in sequence. The phase-logic colors are: green = already closed/derived-only, yellow = the unique active phase is implementable now, orange = a later phase is implementable only in the derivation chain after listed predecessor closures, red = locked/blocked by missing derived-only prerequisite evidence, and white = non-direct-Lean tooling/refactor/infrastructure/documentation/comparison work. Every red result forces a static finding plus an upstream addressing phase at the natural origin of the missing structure before the blocked phase may become active.

**Table legend.** G = green/closed and derived-only; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| ID     | Check status                                      | Meaning                                                                                                                                                                                       | Required action                                                                                                                                                                                               | Ledger effect                                                                                                                                                        |
|--------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PIPC-G | <span style="background-color: #d9ead3;">G</span> | Closed/derived-only in the inspected window: the phase or gate is already closed and its relevant outputs are available under the current Lean-data timestamp or governance closeout.         | Do not reopen or reimplement. It may be consumed as a completed predecessor if its scratchpad/proof/ledger evidence is present.                                                                               | Record as a closed predecessor in the check output when a window includes already closed phases.                                                                     |
| PIPC-Y | <span style="background-color: #fff2cc;">Y</span> | Active implementable now: the current active phase has all visible prerequisites, object/proof dependencies, import seams, and consumer-contract constraints satisfied at the current cursor. | May be proposed for implementation from the current cursor, subject to normal implementation prompt and post-green closeout. No Lean data timestamp changes during the check itself.                          | Record the Y result with scope now, evidence basis, proof-risk note, and tool version/timestamp in the pre-implementation check output.                              |
| PIPC-O | <span style="background-color: #fce4d6;">O</span> | Implementable in derivation chain: a later phase is not active now, but becomes implementable after the explicitly listed predecessor closures in the thought experiment.                     | Do not activate immediately. Preserve the predecessor assumptions and activate only after the listed chain has actually closed.                                                                               | Record the O result with required predecessor closures and assumptions; it must not count as the unique active phase.                                                |
| PIPC-R | <span style="background-color: #f4cccc;">R</span> | Blocked: at least one required dependency, object, proof bridge, source extraction, module seam, or consumer contract is missing or not naturally generated at the required layer.            | Do not start the blocked phase. Add the finding to Static code findings and generate one or more upstream addressing phases at the natural emergence layer of the missing structure before the blocked phase. | Record the R result, missing dependency, natural addressing layer, generated addressing phase(s), and tool version/timestamp in the pre-implementation check output. |

**Current pre-implementation phase-window check.** This table records the requested forward-looking thought experiment for the next semantic phase window. It is not a Lean build and does not close any checked phase. Yellow means the unique active phase is implementable now. Orange means a later phase is implementable only in the stated derivation chain after predecessor closures. Red means a Lean-derivation phase is locked/blocked by missing derived-only prerequisite evidence and requires a generated upstream addressing/release proof at the natural origin of the missing structure. White means non-direct-Lean tooling/refactor/infrastructure/documentation/comparison work. Green means already closed/derived-only.

### Pre-check summary

**Table legend.** G = green/closed and derived-only; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.



| ID                 | Phase       | Status | Scope                     | Predecessors                                                                        | Proof risk                                                                                                                           | Addressing?                                                                   | Addressing phase        |
|--------------------|-------------|--------|---------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------|
| PIC-2026-05-10-001 | 1.1.1.2     | O      | after-listed-predecessors | 1.1.1.1.5.5;<br>LEG37-LEG41                                                         | low/none: planning/generator ledger phase;<br>no Lean source edit expected.                                                          | no                                                                            | none-required           |
| PIC-2026-05-10-002 | 1.1.1.3     | O      | after-listed-predecessors | 1.1.1.2                                                                             | low/none: public API contract/consumer whitelist planning phase;<br>no Lean wrapper creation expected.                               | no                                                                            | none-required           |
| PIC-2026-05-10-003 | 1.1.1.4     | O      | after-listed-predecessors | 1.1.1.2;<br>1.1.1.3                                                                 | low unless converted into deletion/removal work;<br>as ledger-only pruning classification no Lean source edit is expected.           | no                                                                            | none-required           |
| PIC-2026-05-10-004 | 1.1.1.5     | O      | after-listed-predecessors | 1.1.1.2;<br>1.1.1.3;<br>1.1.1.4                                                     | low/none for parent closeout;<br>future 1.1.1.5.x tool phases are separate and not prerequisites unless<br>explicitly listed.        | no                                                                            | none-required           |
| PIC-2026-05-10-005 | 1.1.1.1.5   | Y      | now                       | none beyond current green path through 1.1.1.1.4                                    | low for audit;<br>medium for any later actual refactor because compatibility facades and<br>full local build evidence are mandatory. | no red phase yet;<br>this phase itself is<br>the proactive<br>addressing gate | none-required-now       |
| PIC-2026-05-10-006 | 1.1.1.1.5.1 | Y      | now                       | closed governance path through 1.1.1.1.4 and V1.20<br>documentation/tooling updates | low for audit;<br>later refactor phases require local full-build evidence.                                                           | already generated<br>child sequence                                           | 1.1.1.1.5.1-1.1.1.1.5.5 |
| PIC-2026-05-10-008 | 20.25       | O      | after-listed-predecessors | 20.22;<br>20.23;<br>20.24                                                           | none: database/import tooling only;<br>risk is governance/security, not Lean proof.                                                  | already generated<br>activation chain                                         | 20.22-20.26             |

### Pre-check detail records

#### Pre-check detail PIC-2026-05-10-001 – phase 1.1.1.2

| Field                            | Record                                                                                                                                                                                                                              |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Check status                     | O                                                                                                                                                                                                                                   |
| Check mode                       | sequential-thought-experiment                                                                                                                                                                                                       |
| Implementability scope           | after-listed-predecessors                                                                                                                                                                                                           |
| Required predecessor closures    | 1.1.1.1.5.5;<br>LEG37-LEG41                                                                                                                                                                                                         |
| Assumptions                      | V1.12 inserts a semantic module consolidation gate before 1.1.1.2. The earlier v44 feasibility result remains valid after LEG36 closes or after any resulting bounded refactor package is build-proved.                             |
| Evidence basis                   | YAML phase table, object register, 19 Object status B rows, LEG5 consumption map, proof-dossier anchors, generated inventory summaries.                                                                                             |
| Lean proof risk                  | low/none: planning/generator ledger phase;<br>no Lean source edit expected.                                                                                                                                                         |
| Parent/subphase rule             | not applicable: leaf phase at current active cursor.                                                                                                                                                                                |
| Implementability finding         | Implementable only after the pre-substantive-code readiness chain closes with a release certificate.                                                                                                                                |
| Required action                  | Complete LEG36 first;<br>then implement blue_correction_ledger with semantic defect, allowed temporary consumers, forbidden public consumers, neutral target API, correction phase, and exit-to-white condition for every B object. |
| Would generate addressing phase? | no                                                                                                                                                                                                                                  |
| Generated addressing phase       | none-required                                                                                                                                                                                                                       |
| Static findings affected         | SCF43-SCF45 now precede this result;<br>SCF14 and SCF26 remain addressed when 1.1.1.2 is later implemented.                                                                                                                         |

#### Pre-check detail PIC-2026-05-10-002 – phase 1.1.1.3

| Field                         | Record                                                                                                                                                                         |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Check status                  | O                                                                                                                                                                              |
| Check mode                    | sequential-thought-experiment                                                                                                                                                  |
| Implementability scope        | after-listed-predecessors                                                                                                                                                      |
| Required predecessor closures | 1.1.1.2                                                                                                                                                                        |
| Assumptions                   | Assumes 1.1.1.2 has closed and LEG7 is generated;<br>consumes LEG5 declaration-consumption map, LEG7 blue-correction ledger, object access clauses, and proof-dossier anchors. |



|                                  |                                                                                                                                                                                                                       |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidence basis                   | YAML dependencies, object access register, consumption classification map, planned LEG7 output from 1.1.1.2.                                                                                                          |
| Lean proof risk                  | low/none: public API contract/consumer whitelist planning phase;<br>no Lean wrapper creation expected.                                                                                                                |
| Parent/subphase rule             | not applicable: leaf phase;<br>remains inactive until 1.1.1.2 closes.                                                                                                                                                 |
| Implementability finding         | Implementable in the derivation chain after 1.1.1.2. The phase should only define the visible public surface and forbidden imports for Phase 3+ and later handoffs.                                                   |
| Required action                  | Generate public_api_contract and consumer_whitelist/firewall inputs;<br>exclude uncorrected blue objects, legacy status/policy gates, audit-only objects, duplicate generator seams, and undocumented public objects. |
| Would generate addressing phase? | no                                                                                                                                                                                                                    |
| Generated addressing phase       | none-required                                                                                                                                                                                                         |
| Static findings affected         | No new red finding. Existing open findings remain routed to later phases;<br>this phase only defines the firewall contract.                                                                                           |

Pre-check detail PIC-2026-05-10-003 – phase 1.1.1.4

| Field                            | Record                                                                                                                                                                                                                                                        |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Check status                     | O                                                                                                                                                                                                                                                             |
| Check mode                       | sequential-thought-experiment                                                                                                                                                                                                                                 |
| Implementability scope           | after-listed-predecessors                                                                                                                                                                                                                                     |
| Required predecessor closures    | 1.1.1.2;<br>1.1.1.3                                                                                                                                                                                                                                           |
| Assumptions                      | Assumes 1.1.1.3 has closed and LEG6 public API/consumer whitelist exists;<br>consumes declaration classification from LEG5 and generated repository inventory/raw_export.                                                                                     |
| Evidence basis                   | YAML dependencies, LEG5 declaration-consumption map, planned LEG6 public API firewall, generated repo inventory/raw_export.                                                                                                                                   |
| Lean proof risk                  | low unless converted into deletion/removal work;<br>as ledger-only pruning classification no Lean source edit is expected.                                                                                                                                    |
| Parent/subphase rule             | not applicable: leaf phase;<br>remains inactive until 1.1.1.2 and 1.1.1.3 close.                                                                                                                                                                              |
| Implementability finding         | Implementable in the derivation chain as a pruning/dead-candidate ledger phase, but not as a deletion phase. It can classify keep/quarantine/wrap/reexport/archive/deletion-candidate entries using existing generated inventory and the public API contract. |
| Required action                  | Generate pruning_ledger. Do not delete or move Lean declarations. Require future local build evidence for any actual deletion/removal phase.                                                                                                                  |
| Would generate addressing phase? | no                                                                                                                                                                                                                                                            |
| Generated addressing phase       | none-required                                                                                                                                                                                                                                                 |
| Static findings affected         | Refines SCF15 but does not close deletion-related risks until a later actual pruning/removal phase with build evidence.                                                                                                                                       |

Pre-check detail PIC-2026-05-10-004 – phase 1.1.1.5

| Field                         | Record                                                                                                                                                                                                    |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Check status                  | O                                                                                                                                                                                                         |
| Check mode                    | sequential-thought-experiment                                                                                                                                                                             |
| Implementability scope        | after-listed-predecessors                                                                                                                                                                                 |
| Required predecessor closures | 1.1.1.2;<br>1.1.1.3;<br>1.1.1.4                                                                                                                                                                           |
| Assumptions                   | Assumes 1.1.1.2, 1.1.1.3, and 1.1.1.4 have closed;<br>parent 1.1.1.5 is checked as the generated-view authority/inventory-lock closeout, not as implementation of its future 1.1.1.5.x tooling subphases. |
| Evidence basis                | YAML parent/child hierarchy, current active cursor, implementation-scaling phase plan, generated table/export inventory, module-manifest and reuse/duplication ledgers.                                   |
| Lean proof risk               | low/none for parent closeout;<br>future 1.1.1.5.x tool phases are separate and not prerequisites unless explicitly listed.                                                                                |

|                                  |                                                                                                                                                                                                                                                             |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parent/subphase rule             | Parent closeout does not require future tooling subphases 1.1.1.5.x unless they are explicitly listed as dependencies.                                                                                                                                      |
| Implementability finding         | Implementable in the derivation chain as a closeout/authority-lock phase. It can bind YAML master, generated tables/PDF, README/report update, module-manifest/reuse checks, and local-build evidence requirements into one implementation-prompt contract. |
| Required action                  | Close the audit/planning gate with reproducible generated artifacts and one active next cursor for 1.1.2. Keep 1.1.1.5.x as future tool-scaling subphases unless explicitly activated.                                                                      |
| Would generate addressing phase? | no                                                                                                                                                                                                                                                          |
| Generated addressing phase       | none-required                                                                                                                                                                                                                                               |
| Static findings affected         | No red blocker found;<br>no new addressing phase generated.                                                                                                                                                                                                 |
|                                  |                                                                                                                                                                                                                                                             |

Pre-check detail PIC-2026-05-10-005 – phase 1.1.1.1.5

|                                  |                                                                                                                                                                                                                                 |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Field                            | Record                                                                                                                                                                                                                          |
| Check status                     | Y                                                                                                                                                                                                                               |
| Check mode                       | sequential-thought-experiment plus direct code spot-check                                                                                                                                                                       |
| Implementability scope           | now                                                                                                                                                                                                                             |
| Required predecessor closures    | none beyond current green path through 1.1.1.1.4                                                                                                                                                                                |
| Assumptions                      | LEG10, LEG11, LEG12, LEG5, phase-object policy, reuse ledger, and generated repo inventory are available;<br>no Lean code is edited by the audit itself.                                                                        |
| Evidence basis                   | repo_inventory/raw_export modules/import_edges/symbol references;<br>direct checks of BoundarySource substrate modules and Handoff input/output wrappers;<br>known SM and Cayley-Dickson proof-stage chains.                    |
| Lean proof risk                  | low for audit;<br>medium for any later actual refactor because compatibility facades and full local build evidence are mandatory.                                                                                               |
| Parent/subphase rule             | Leaf active gate under 1.1.1.1. It may close with a ledger only, or open a later bounded refactor execution phase if safe candidates are confirmed.                                                                             |
| Implementability finding         | Implementable now as a compact planning/refactor governance phase. It has enough code evidence to classify candidate families without touching Lean source.                                                                     |
| Required action                  | Fill LEG36 consolidation ledger;<br>decide candidate classes;<br>keep proof-stage chains split unless proven otherwise;<br>return cursor to 1.1.1.2 or open a bounded refactor execution phase with facades and build evidence. |
| Would generate addressing phase? | no red phase yet;<br>this phase itself is the proactive addressing gate                                                                                                                                                         |
| Generated addressing phase       | none-required-now                                                                                                                                                                                                               |
| Static findings affected         | SCF43-SCF45 created and routed to 1.1.1.1.5.                                                                                                                                                                                    |

Pre-check detail PIC-2026-05-10-006 – phase 1.1.1.1.5.1

|                               |                                                                                                                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Field                         | Record                                                                                                                                                                      |
| Check status                  | Y                                                                                                                                                                           |
| Check mode                    | static DAG/source-inventory audit plus code-name spot-check                                                                                                                 |
| Implementability scope        | now                                                                                                                                                                         |
| Required predecessor closures | closed governance path through 1.1.1.1.4 and V1.20 documentation/tooling updates                                                                                            |
| Assumptions                   | The current source snapshot and generated inventory are sufficient to audit classification completeness without editing Lean source.                                        |
| Evidence basis                | 287 Lean modules, 8860 declarations, module/declaration consumption TSVs, object/proof registers, direct file-name scan for phase-semantic names and clone/schema families. |
| Lean proof risk               | low for audit;<br>later refactor phases require local full-build evidence.                                                                                                  |
| Parent/subphase rule          | Child active phase under 1.1.1.1.5;<br>parent is umbrella W, this leaf is the unique Y cursor.                                                                              |
| Implementability finding      | Implementable now as the next constructive pre-code gate. It replaces the too-broad active umbrella as the precise cursor.                                                  |
| Required action               | Fill LEG37 and route missing source/object mappings to LEG38-LEG41 or upstream red blockers.                                                                                |

|                                  |                                  |
|----------------------------------|----------------------------------|
| Would generate addressing phase? | already generated child sequence |
| Generated addressing phase       | 1.1.1.1.5.1-1.1.1.1.5.5          |
| Static findings affected         | SCF52                            |

Pre-check detail PIC-2026-05-10-008 – phase 20.25

| Field                            | Record                                                                                                                                                                                                                                                             |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Check status                     | O                                                                                                                                                                                                                                                                  |
| Check mode                       | live-DB-write readiness thought experiment after read-only Supabase discovery                                                                                                                                                                                      |
| Implementability scope           | after-listed-predecessors                                                                                                                                                                                                                                          |
| Required predecessor closures    | 20.22;<br>20.23;<br>20.24                                                                                                                                                                                                                                          |
| Assumptions                      | Supabase project aiekahdewadrejrrdajn remains reachable and empty or migration-compatible;<br>user explicitly authorizes the current write step;<br>no Lean files are edited. Separate explicit user authorization is additionally required for any live DB write. |
| Evidence basis                   | Existing phases 20.15-20.20;<br>V1.24 read-only discovery;<br>planned migration/import contracts;<br>GNG34/GNG36.                                                                                                                                                  |
| Lean proof risk                  | none: database/import tooling only;<br>risk is governance/security, not Lean proof.                                                                                                                                                                                |
| Parent/subphase rule             | DB/tooling sidechain under 20;<br>does not affect the unique Lean/content Y cursor.                                                                                                                                                                                |
| Implementability finding         | Implementable after migration/import packages exist and the user explicitly authorizes live DB writes. It must write only Supabase schema/data and produce snapshot/import evidence.                                                                               |
| Required action                  | Before execution, present exact migration/import SQL or script effects, then apply only after explicit confirmation. Afterwards run table/advisor/read-only verification.                                                                                          |
| Would generate addressing phase? | already generated activation chain                                                                                                                                                                                                                                 |
| Generated addressing phase       | 20.22-20.26                                                                                                                                                                                                                                                        |
| Static findings affected         | SCF55                                                                                                                                                                                                                                                              |

### 13 Implementation-scaling phase plan

**What this section shows.** This table records how the planning/documentation tool itself may grow without taking over the mathematical implementation. It distinguishes Python tooling, generated views, future database paths and review infrastructure from Lean-derived CNNA objects.

**Implementation-scaling phase plan.** This table shows how the tool may grow while protecting the Lean/content path. The rows come from the Master YAML and describe Python inventory, generated views, database readiness, context packets, root hygiene and validation surfaces. It is shown to make tooling growth explicit and bounded; none of these scaling rows is an ontic CNNA generator.

| Phase      | Name                                                                | Primary purpose                                                                                                                                                                                                       | Tool mode                                                                                    | Load-control rule                                                                                                                                                                  | Theory semantics                                                                                                                                                                  | Generated artifacts                                                                                                                | Lean-data effect                                             |
|------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1.1.1.5.1  | Tool-mode split and lean-focus contract                             | Define audit-full, phase-check, and lean-focus modes so routine Lean work consumes compact implementation packets instead of full PDF/table scans.                                                                    | lean-focus foundation                                                                        | Full PDF regeneration is not part of ordinary Lean coding; use compact packets unless a governance or audit phase explicitly requests audit-full.                                  | No theory classification yet; this phase only fixes the mode boundary.                                                                                                            | workflow-policy row; scaling-plan TSV/TeX; no Lean code                                                                            | none                                                         |
| 1.1.1.5.2  | Implementation Packet Generator                                     | Generate the phase-local packet with goal, source objects, producer/consumer modules, red blockers, yellow risks, forbidden shortcuts, expected artifacts, and comparison-only warnings.                              | lean-focus                                                                                   | Packet should fit on a small number of pages/rows and link to full TSV/JSON exports rather than embedding full inventories.                                                        | Consumes module-theory/end-theory ledgers once available; before then marks theory context as unknown-not-input.                                                                  | active-phase implementation packet TSV/JSON/TeX                                                                                    | none unless the user explicitly edits Lean code afterwards   |
| 1.1.1.5.3  | Incremental Lean inventory and file-hash cache                      | Avoid rescanning/rendering everything by caching file hashes, module declarations, import edges, and symbol-reference candidates between Lean-data updates.                                                           | phase-check performance layer                                                                | Only changed Lean files and dependent generated summaries are refreshed; full inventory rebuild remains available as audit-full.                                                   | May infer rough local background categories from imports, but must mark them heuristic until semantically confirmed.                                                              | file-hash cache; declaration cache; inventory-delta TSV                                                                            | changes only when Lean files or frozen inventory data change |
| 1.1.1.5.4  | Module theory context ledger                                        | Track local established mathematical background per module as formal language, proof method, library/API context, implementation substrate, or comparison lemma context.                                              | phase-check plus lean-focus lookup                                                           | Start with active/relevant modules only; do not manually annotate all modules at once. Python proposes rows from imports; semantic closure is explicit and phase-bound.            | Local established theory is allowed formalization context, never ontic CNNA generator. Examples: Matrix, Finset, Polynomial, graph/linear algebra, Schur-style algebraic methods. | module-theory-context ledger TSV; active-module slice                                                                              | none for ledger-only updates                                 |
| 1.1.1.5.5  | End-theory comparison ledger                                        | Track large established target theories as comparison-only horizons: AQFT/QFT, OQS, modular forms, CM, elliptic functions/curves, L-functions, Moonshine, renormalization, operator algebras, spectral geometry, etc. | phase-check comparison layer                                                                 | Only record end theories that constrain current or near-future phases; do not build encyclopedia entries.                                                                          | End theories are target structures, consistency checks, terminology alignment, or obstruction/comparison mirrors; forbidden as CNNA producers.                                    | end-theory-comparison ledger TSV; comparison-boundary table                                                                        | none for ledger-only updates                                 |
| 1.1.1.5.6  | Active object graph slices                                          | Render small producer-consumer/provenance neighborhoods around the active phase and active CNNA objects rather than dumping the full object/declaration graph into the PDF.                                           | lean-focus visualization                                                                     | Depth 1-2 graph slices only in PDF; full graph exported as DOT/JSON for optional audit.                                                                                            | Graph edges distinguish internal provenance, local formalization context, and external comparison-only references.                                                                | active-object-neighborhood DOT/SVG/PDF; graph-slice TSV/JSON                                                                       | none unless generated from changed Lean inventory            |
| 1.1.1.5.7  | Red-blocker addressing-phase generator                              | When the pre-implementation thought experiment marks a future phase red, propose an upstream addressing phase at the earliest natural source layer before the blocked phase becomes active.                           | phase-check governance layer                                                                 | Generate concise phase proposals only for red blockers; yellow risks remain notes in the implementation packet.                                                                    | If the blocker is an imported end-theory assumption, route to internal derivation/comparison-boundary phase rather than patching the consumer.                                    | red-blocker phase-proposal TSV; accepted phase rows only after approval                                                            | none until an accepted phase later changes Lean code         |
| 1.1.1.5.8  | Database-ready YAML schema and review-link normalization            | Normalize IDs, phase/object/proof-dossier references, review statuses, and theory-reference markers so the YAML master can later be exported to relational or graph database tables without semantic loss.            | phase-check plus documentation infrastructure                                                | Do not manually annotate the whole repository at once. Start with active/near-active rows and generated consistency checks; keep full database realization as a later export task. | Database fields are review metadata. They may mark local background theory or end-theory comparison context, but they do not create CNNA generators.                              | database-readiness checklist TSV; normalized ID/reference audit; review-status export slice                                        | none for schema/metadata-only changes                        |
| 1.1.1.5.9  | Database export, round-trip, and interactive documentation scaffold | Prepare future YAML-to-DB exports, database-to-generated-view round-trip checks, interactive object/phase exploration, online documentation, and discipline-separated expert-review views.                            | audit-full optional plus generated export layer                                              | Keep export scripts reproducible and optional. Lean implementation packets must remain compact and must not depend on full database exploration.                                   | Interactive views are secondary/generated. They expose review obligations and comparison horizons but never certify semantic truth.                                               | future SQL/SQLite/graph export scripts; schema diagrams; interactive-view JSON; round-trip consistency report                      | none unless future scripts read changed Lean inventory data  |
| 1.1.1.5.10 | Context documentation slices                                        | Reduce documentation overhead by producing topic/phase-specific packets that preserve full section topology while filtering rows to the requested context.                                                            | Python context generator over YAML and generated TSV exports; optional pdflatex compilation. | Use for targeted requests and phase closeouts; reserve full PDF regeneration for tool releases, global audits, or explicit full-document requests.                                 | Generated-secondary navigation/review view only. Absence from a slice is not proof of semantic irrelevance; expert review and blue/reference audit obligations remain.            | context_docs/<slug>/context_filter_manifest.json; context_section_summary.tsv; <slug>_context_documentation.tex; pdf; compile log. | None; no Lean source mutation and no Lean build claim.       |
| 1.1.1.5.11 | Stats front section and interface-metaphor documentation surface    | Keep project-scale facts, repository inventory, graph views, license notices and the tool-interface Leitmetaphor visible without creating another truth layer.                                                        | Python-generated documentation/front matter                                                  | Do not expand the opening section into prose-heavy explanation; keep it as compact facts, concise metaphor, licensing note and existing inventory/graph inclusion.                 | Generated-secondary review architecture only. The metaphor is operational, not ontic or theorematic; Lean-primary and expert-review rules remain dominant.                        | project_stats.generated.tsv; tables/project_stats_table.tex; tables/stats.tex; updated V1.09 full PDF.                             | none; no Lean source or Lean-data timestamp change.          |

| Phase      | Name                                                                              | Primary purpose                                                                                                                                                                | Tool mode                                                                                                                      | Load-control rule                                                                                                                                                                | Theory semantics                                                                                                                                                                             | Generated artifacts                                                                                                                                                                         | Lean-data effect                                       |
|------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 1.1.1.5.12 | Root artifact retention and disposable documentation outputs                      | Keep the project root clean and source-oriented while preserving on-demand generation of full PDFs/TeX and archival access to historical reports.                              | Python full-document generator plus root cleanup script.                                                                       | Do not package regenerated full PDFs/TeX by default. Generate them only for review/export tasks, then keep them in generated_docs/full/ or delete them before release packaging. | Generated documentation outputs remain generated-secondary. Archival Markdown reports are trace records, not authoritative theory sources.                                                   | generated_docs/full/<version>.tex/pdf/log when explicitly generated; archive/reports/*.md for historical reports; generated_exports/root_artifact_cleanup.generated.json when cleanup runs. | none; no Lean source mutation and no Lean build claim. |
| 1.1.1.5.13 | Root license and PDF credit surface                                               | Keep license, editor/assistant credits and minimum tested ChatGPT workflow requirements visible without creating duplicate documentation entry points.                         | Root README/LICENSE plus generator metadata rendering for full/context PDFs.                                                   | Do not duplicate README files in subdirectories; keep license and credit text compact and generated-secondary.                                                                   | Licensing/credits are governance metadata only. They do not affect CNNA derivations, Lean truth status, or expert-review obligations.                                                        | Updated license_notice.generated.tsv; tables/license_notice.tex; generated full/context PDFs with author credits when explicitly compiled.                                                  | none; no Lean source mutation and no Lean build claim. |
| 1.1.1.5.14 | Supabase/Postgres public read-only mirror and cnna-project.net frontend sidechain | Prepare a public query/visualization layer for CNNA planning metadata without replacing Lean/YAML truth and without public write authority.                                    | Postgres/Supabase schemas and read-only api views; local importer; future static/web frontend; no live DB connection in V1.22. | Start as schema/import/view contracts, not as full web application. No PDFs/ZIPs in DB, no service_role key in browser/chat, no manual production table edits.                   | Database/web layer is generated-secondary/publication infrastructure. It can query, visualize and review Lean/YAML-derived metadata, but cannot generate CNNA axioms or release Lean phases. | Future SQL migrations, import reports, snapshot checksums, RLS test reports, frontend data-contract docs and public views; V1.22 adds planning records only.                                | none; no Lean source mutation and no Lean build claim. |
| 1.1.1.5.15 | Supabase live-project import activation and verification package                  | Turn the broad Supabase DB mirror concept into a concrete, auditable no-Lean-change activation path for schema migration, snapshot import and read-only frontend verification. | Supabase connector plus generated SQL/import/report artifacts; explicit authorization before write.                            | Do not combine schema design, import execution and RLS verification. Keep each write/verification step separately reportable and snapshot-addressed.                             | Database rows are generated-secondary publication/query data. They cannot release Lean phases, prove CNNA claims or replace YAML/Lean provenance.                                            | Future database/supabase migrations, importer scripts, import reports, checksum manifests, advisor/RLS reports and frontend data contracts.                                                 | none; no Lean source mutation and no Lean build claim. |

14    Reuse/duplication ledger and canonical generator policy

**What this section shows.**    This ledger tracks reuse seams, duplicated implementation risks, generator boundaries and canonical-source decisions.    It is shown because compactness and reuse are allowed only where provenance and reviewability improve; duplicate generator paths or consumer-side patches are blocked by default.

**Reuse and duplication ledger rule.**    This generated view closes LEG11 as a planning/API hygiene contract.    It is not a Lean source and not a mathematical generator.    A future generator-like implementation must be classified here before code is added: consume an existing seam, extend it at its natural source layer, wrap it for import/API reasons, or record a genuine semantic split.    Parallel request/output, extraction, ledger, smoke-test, or theory-generation paths are blocked by default.

Reuse ledger row RD01 – generated ToC/smoke source and main-path construction

| Field                   | Record                                                                                                                                                                                                      |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                      | RD01                                                                                                                                                                                                        |
| Scope                   | generated ToC/smoke source and main-path construction                                                                                                                                                       |
| Existing canonical seam | CNNA.PillarA.Arithmetic.Smoke.GeneratedSource and CNNA.PillarA.Arithmetic.Smoke.GeneratedMainPath, backed by CNNA.PillarA.ToC.GeneratedBranchFamily                                                         |
| Decision                | consume_or_extend_existing                                                                                                                                                                                  |
| Allowed action          | Future generated ToC/smoke phases must consume or extend the existing source/main-path seam at its natural layer.                                                                                           |
| Blocked duplication     | No second smoke-test generator, generated carrier source, or parallel main-path constructor under a phase-local module name.                                                                                |
| Evidence                | GeneratedSource exposes regular/variable smoke carriers and SM0 ledger;<br>GeneratedMainPath exposes generatedMainPath, regular/variable paths, pair ledger, and no-CutSpec/no-matrix-data boundary status. |
| Future consumers        | Phase 5 experiment request/output, O07 generic finite computation-test matrix, later generated arithmetic source phases.                                                                                    |

Reuse ledger row RD02 – downstream generated Schur/DtN/arithmetic smoke chain

| Field                   | Record                                                                                                                                                            |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                      | RD02                                                                                                                                                              |
| Scope                   | downstream generated Schur/DtN/arithmetic smoke chain                                                                                                             |
| Existing canonical seam | CNNA.PillarA.Arithmetic.Smoke.GeneratedApproximantCore through GeneratedOperationalQuadraticDiscriminantSourceFromSM9g1Window                                     |
| Decision                | consume_existing_mainline                                                                                                                                         |
| Allowed action          | Future consumers may consume the proof-carrying generated chain and add natural source-layer extensions when a missing structure is exposed.                      |
| Blocked duplication     | No independent Schur/DtN/arithmetic generator that recomputes the same provenance chain outside the Generated* mainline.                                          |
| Evidence                | The robust semantic mainline graph and generated module chain already route source -> boundary/interior -> Schur/DtN -> arithmetic/charpoly/discriminant objects. |
| Future consumers        | Phase 3/4 boundary-operator profiles, Phase 5 computation requests, O05/O06/O07/O08 operational arithmetic checks.                                                |

Reuse ledger row RD03 – action-candidate generator semantics

| Field                   | Record                                                                                                                                                                                                 |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                      | RD03                                                                                                                                                                                                   |
| Scope                   | action-candidate generator semantics                                                                                                                                                                   |
| Existing canonical seam | CNNA.PillarA.Arithmetic.Actions.GeneratorActionInventory and CNNA.PillarA.Arithmetic.Actions.ActionFormatDecision                                                                                      |
| Decision                | genuine_semantic_split_already_existing                                                                                                                                                                |
| Allowed action          | Action-like requests must consume or extend the action inventory/decision seam, because it classifies action formats rather than generated ToC/smoke carriers.                                         |
| Blocked duplication     | No action-format generator reimplemented inside Smoke or later CM/CD consumers.                                                                                                                        |
| Evidence                | GeneratorActionInventory records candidate kinds/statuses and ActionFormatDecision consumes it to classify level-history identity, branching, bright/dark, Mobius-Schur, CM, and CD action candidates. |
| Future consumers        | AR/action-format phases, later CM/CD/action comparison gates.                                                                                                                                          |

Reuse ledger row RD04 – documentation and planning-table generation

| Field | Record |
|-------|--------|
|-------|--------|



|                         |                                                                                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                      | RD04                                                                                                                                                |
| Scope                   | documentation and planning-table generation                                                                                                         |
| Existing canonical seam | masters/cnna_roadmap_master.yaml plus scripts/generate_tables.py                                                                                    |
| Decision                | artifact_only_split                                                                                                                                 |
| Allowed action          | Planning/PDF/TSV views are generated from the single YAML master only.                                                                              |
| Blocked duplication     | No independent proofbook, phase table, object list, module manifest, or reuse ledger source outside the YAML master.                                |
| Evidence                | generate_tables.py validates the single master and renders object, phase, scratchpad, module manifest, proof documentation, and reuse ledger views. |
| Future consumers        | All planning/documentation phases, implementation prompts, public API classification.                                                               |

Reuse ledger row RD05 – repo inventory/import graph generation

|                         |                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Field</b>            | <b>Record</b>                                                                                                                                          |
| ID                      | RD05                                                                                                                                                   |
| Scope                   | repo inventory/import graph generation                                                                                                                 |
| Existing canonical seam | scripts/build_export_script_v1.8.py and repo_inventory/raw_export                                                                                      |
| Decision                | artifact_only_audit_not_ontic_generator                                                                                                                |
| Allowed action          | Use inventory data for module placement, import graph diagnostics, and documentation/audit only.                                                       |
| Blocked duplication     | Do not treat graph metrics as CNNA-generative evidence and do not create a second inventory generator for the same source snapshot.                    |
| Evidence                | raw_export contains modules/import edges/symbol-reference summaries;<br>phase plans classify these as secondary audit assets, not mathematical inputs. |
| Future consumers        | LEG10/LEG11/LEG5/LEG6, module inventory views, later documentation/audit phases.                                                                       |

Reuse ledger row RD06 – future experiment request/output infrastructure

|                         |                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Field</b>            | <b>Record</b>                                                                                                                                              |
| ID                      | RD06                                                                                                                                                       |
| Scope                   | future experiment request/output infrastructure                                                                                                            |
| Existing canonical seam | No separate code yet;<br>must route through RD01/RD02 and LEG10 manifest before code.                                                                      |
| Decision                | future_consume_or_wrap_canonical_generator                                                                                                                 |
| Allowed action          | Define request/output wrappers only after the manifest proves the wrapper consumes the canonical generated chain and does not generate data independently. |
| Blocked duplication     | No parallel experiment request/output generator, extraction generator, or ledger generator under Phase 5 or O07 names.                                     |
| Evidence                | LEG10 already records duplicate-risk symbols;<br>LEG11 now turns them into an explicit block unless a semantic split row is added.                         |
| Future consumers        | Phase 5, O07 generic finite computation-test matrix, heuristic lattice and regime-typed test matrices.                                                     |

Reuse ledger row RD07 – future neutral generator API possibility

|                         |                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Field</b>            | <b>Record</b>                                                                                                                             |
| ID                      | RD07                                                                                                                                      |
| Scope                   | future neutral generator API possibility                                                                                                  |
| Existing canonical seam | Current semantics live in Smoke/GeneratedSource-MainPath;<br>no neutral module exists yet.                                                |
| Decision                | defer_until_code_phase_proves_need                                                                                                        |
| Allowed action          | Create a neutral wrapper/API only if a later manifest shows the Smoke namespace is semantically too narrow but the operation is the same. |
| Blocked duplication     | No neutral generator file created merely to rename existing semantics or to bypass import/firewall discipline.                            |



|                  |                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------|
| Evidence         | The v41 gate closes governance only;<br>no Lean source change is required to make the policy enforceable in planning. |
| Future consumers | 1.1.1.3 public API contract, future refactor phases, Phase 5 generator consumers.                                     |

Reuse ledger row RD08 – semantic module consolidation and compatibility-preserving code compactness

| Field                   | Record                                                                                                                                                                                                                                    |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                      | RD08                                                                                                                                                                                                                                      |
| Scope                   | semantic module consolidation and compatibility-preserving code compactness                                                                                                                                                               |
| Existing canonical seam | LEG10 module manifest;<br>LEG11 reuse/duplication ledger;<br>LEG5 declaration consumption classification;<br>repo_inventory/raw_export import/declaration graph;<br>old Lean module import paths as compatibility seams                   |
| Decision                | defer_until_code_phase_proves_need                                                                                                                                                                                                        |
| Allowed action          | Unify semantically identical schema/wrapper families only through a recorded LEG36 candidate row, old-module facades, declaration/consumer diff, and local build evidence. Prefer canonical schema modules at the natural producer layer. |
| Blocked duplication     | No continued proliferation of unrelated clone wrappers once a canonical schema is validated;<br>no aggressive merge of proof-stage chains, generated SM phase modules, or Cayley-Dickson S6 stages without provenance-preservation proof. |
| Evidence                | V1.12 code spot-check: BoundarySource substrate ledger clone family;<br>Handoff A_to_C/D/E alias family;<br>Handoff B/C/D/E_to_A feedback-input family;<br>no-merge warning for Smoke/SM and Cayley-Dickson proof chains.                 |
| Future consumers        | Phase 1.1.1.1.5;<br>Phase 1.1.1.2;<br>public API/firewall work;<br>pruning/refactor phases;<br>future Lean implementation prompts.                                                                                                        |

## 15 Declaration consumption classification map

**What this section shows.** This section summarizes generated declaration-consumption classification. It separates semantic consumption from mere import reachability and exports the full records as TSV. The PDF renders summaries so reviewers can see the API/firewall shape without reading an unreadable declaration dump.

**Declaration consumption classification rule.** LEG5 is a generated planning/API-hygiene artifact, not a Lean source and not a mathematical generator. The classifier distinguishes semantic consumption from mere BuildAll/import reachability. Full module-, declaration-, and object-level records are exported as TSV files; the PDF renders only the class policy and aggregate counts to avoid turning the roadmap into an unreadable declaration dump.

| Class                    | Meaning                                                                                                                 | Evidence rule                                                                                                                                                                          | Public/API disposition                                                                           | Blocked promotion rule                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| core-computational       | Internal derived computational definitions that are neither policy, wrapper, phase alias, nor proof-only bridge.        | Pillar-A internal module/declaration without governance, wrapper, smoke-phase, obstruction, or proof-bridge markers; module reference counts are evidence only, not public API rights. | May be used internally by later derived phases after import/consumer discipline is checked.      | Cannot become public-final merely because it is imported; public API contract and object proof dossier are still required. |
| proof-bridge             | Certificates, equality bridges, correctness/witness declarations, and other proof-carrying connectors.                  | Names/modules carrying proof, certificate, correctness, identity, witness, equality, or bridge roles.                                                                                  | May discharge local proof obligations but is not automatically a public object.                  | Cannot hide a missing natural source structure; if a consumer needs new structure, backjump to the source layer.           |
| public-wrapper           | BuildAll, umbrella, or import-facing wrapper modules whose purpose is exposure/routing rather than semantic generation. | Inventory flags build modules, top-level pillar wrappers, or module names ending in BuildAll.                                                                                          | Exposure surface only; it may route imports after public API rules are closed.                   | Wrapper reachability is not semantic consumption evidence and cannot make a blue object final.                             |
| legacy-status            | Legacy/status-gate declarations or modules that encode old one-constructor states or deprecated status control.         | Name/path contains legacy or status-gate/status markers.                                                                                                                               | Quarantine/refactor candidate, normally hidden behind legacy/status policy.                      | Cannot be consumed by implementation phases except through the explicit quarantine/correction path.                        |
| legacy-policy            | Policy, ledger, decision, format, audit, inventory, report, and governance/control-surface material.                    | Name/path/layer contains policy, ledger, decision, format, audit, inventory, contract, report, planning, governance, or legacy control markers.                                        | May govern work; does not supply ontic or mathematical CNNA input.                               | Cannot be cited as derived mathematical structure; must stay prose/planning/audit unless mirrored by checked source.       |
| phase-alias              | Phase- or smoke-test-specific names carrying implementation-history aliases such as SM-numbered records.                | Name/module contains SM-numbered aliases or lives in Smoke namespace.                                                                                                                  | Internal migration/scaffold evidence only; later public names must be neutral.                   | Cannot become stable public API without wrapper/correction and consumer whitelist.                                         |
| audit-only               | Read-only diagnostic or inventory material not part of the CNNA generator chain.                                        | Non-Pillar-A, unclassified, or diagnostic inventory material without derived generator role.                                                                                           | Documentation/diagnostic evidence only.                                                          | Cannot be used as ontic input or proof of mathematical closure.                                                            |
| blue-semantic-correction | Proof-carrying but semantically/API-incomplete objects that must be corrected before final/public consumption.          | Object status B in the object register; declarations/modules may later be linked by the blue-correction ledger.                                                                        | Blocked from public-final consumption; only temporary internal correction consumers are allowed. | Must not be classified as final-output or public-wrapper before correction phase, proof dossier, and API contract close.   |
| dead-candidate           | Rejected, blocked, negative, impossible, obstruction, or black/quarantined candidates.                                  | Names/path contain NoGo/obstruction/blocked/dead/impossible/negative or object status R/K.                                                                                             | No direct public/API consumption; may be cited as obstruction evidence only.                     | Cannot be deleted or repurposed before the pruning/dead-candidate ledger closes.                                           |
| final-output             | Intentional output candidates or green documented objects after proof/audit closure.                                    | Architectural sink taxonomy intentional output, or green documented non-governance object in the object register.                                                                      | Candidate final output, still subject to explicit public API contract and consumer whitelist.    | A blue/quarantined object can never enter this class directly; documented proof alone is insufficient.                     |

### Generated classification counts

| Class                    | Modules | Declarations | Objects | Public/API disposition                                                                           |
|--------------------------|---------|--------------|---------|--------------------------------------------------------------------------------------------------|
| audit-only               | 1       | 2            | 0       | Documentation/diagnostic evidence only.                                                          |
| blue-semantic-correction | 0       | 0            | 19      | Blocked from public-final consumption; only temporary internal correction consumers are allowed. |
| core-computational       | 133     | 3512         | 82      | May be used internally by later derived phases after import/consumer discipline is checked.      |
| dead-candidate           | 1       | 129          | 89      | No direct public/API consumption; may be cited as obstruction evidence only.                     |
| final-output             | 7       | 21           | 13      | Candidate final output, still subject to explicit public API contract and consumer whitelist.    |
| legacy-policy            | 29      | 1705         | 37      | May govern work; does not supply ontic or mathematical CNNA input.                               |
| legacy-status            | 0       | 2            | 0       | Quarantine/refactor candidate, normally hidden behind legacy/status policy.                      |
| phase-alias              | 75      | 2494         | 0       | Internal migration/scaffold evidence only; later public names must be neutral.                   |
| proof-bridge             | 10      | 995          | 0       | May discharge local proof obligations but is not automatically a public object.                  |
| public-wrapper           | 31      | 0            | 0       | Exposure surface only; it may route imports after public API rules are closed.                   |

**Promotion gate.** A blue object is always classified as **blue-semantic-correction**. It cannot become **final-output** or **public-wrapper** from proof-carrying code alone; later phases must close the semantic correction ledger, proof dossier, and public API contract.

## 16 Canonical object register with code derivation sites

**What this section shows.** This register is the canonical object/quantity index maintained in the Master YAML. It shows object status, derivation/provenance text, code locations, access rights, handoff rights and proof-dossier links. It is shown before the phase table so phase release can be checked against registered objects.

**Object register rule.** This section renders object-to-code, context class, derivation, access, handoff rights, and proof-documentation links from the Single YAML Master. It is shown because phase release must cite registered derived-only objects rather than relying on informal intent. Context class separates Lean objects from tool, generated, database, documentation, governance and reference-gate objects; Significance tier controls the documentation burden from F0 full dossier to F4 mention-only; object status remains release/proof-state only. Legends are local to the object tables that carry colored object-status cells.

### Object table A: current code modules

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| ID  | Object                                       | Context class                            | Tier | Layer                 | Object status | Code status                                                                | Current code modules / derivation site                                                                                                                                                                             | Neutral target location / canonical API          | Phase |
|-----|----------------------------------------------|------------------------------------------|------|-----------------------|---------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------|
| G00 | Generator provenance                         | Lean;<br>Source;<br>Object               | F0   | Foundation            | G             | present as source/<br>seed;<br>derivation still to be<br>canonicalized     | CNNA/PillarA/Arithmetic/Smoke/GeneratedSource.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedMainPath.lean;<br>CNNA/PillarA/Arithmetic/BoundarySource/<br>GeneratedSubstrateRoute.lean                            | CNNA/PillarA/Foundation/GeneratorProvenance.lean | 1.2.2 |
| H00 | A input handoff B->A                         | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Handoff policy        | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized     | CNNA/PillarA/Handoff/Inputs/B_to_A.lean                                                                                                                                                                            | CNNA/PillarA/Handoff/AinputhandoffBA.lean        | 2.3.1 |
| H01 | A input handoff C->A                         | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Handoff policy        | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized     | CNNA/PillarA/Handoff/Inputs/C_to_A.lean                                                                                                                                                                            | CNNA/PillarA/Handoff/AinputhandoffCA.lean        | 2.3.1 |
| H02 | A input handoff D->A                         | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Handoff policy        | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized     | CNNA/PillarA/Handoff/Inputs/D_to_A.lean                                                                                                                                                                            | CNNA/PillarA/Handoff/AinputhandoffDA.lean        | 2.3.1 |
| H03 | A input handoff E->A                         | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Handoff policy        | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized     | CNNA/PillarA/Handoff/Inputs/E_to_A.lean                                                                                                                                                                            | CNNA/PillarA/Handoff/AinputhandoffEA.lean        | 2.3.1 |
| O00 | ExecComplex                                  | Lean;<br>Source;<br>Object               | F1   | Operational<br>scalar | W             | canonically<br>derived-only;<br>consumable as<br>computable scalar<br>path | CNNA/PillarA/Foundation/ExecComplex.lean;<br>CNNA/PillarA/Foundation/ExecComplexBridge.lean;<br>CNNA/PillarA/Foundation/ExecComplexLemmas.lean                                                                     | CNNA/PillarA/Foundation/ExecComplex.lean         | 1.2.1 |
| G01 | substrate: regular<br>branching              | Lean;<br>Source;<br>Object               | F0   | Foundation            | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized     | CNNA/PillarA/Foundation/ConcreteSubstrate.lean;<br>CNNA/PillarA/Arithmetic/BoundarySource/<br>BranchAddressSubstrate.lean;<br>CNNA/PillarA/Arithmetic/BoundarySource/<br>BoundaryStatesubstrate.lean               | CNNA/PillarA/Foundation/Substrate/Regular.lean   | 1.2.3 |
| G02 | substrate: variable<br>branching             | Lean;<br>Source;<br>Object               | F0   | Foundation            | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized     | CNNA/PillarA/Foundation/LevelVariableSubstrate.lean;<br>CNNA/PillarA/Foundation/WeightPolicy.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedDegreeFormulaAuditSM3db2eR0.lean                                  | CNNA/PillarA/Foundation/Substrate/Variable.lean  | 1.2.4 |
| G03 | Level / Cutoff                               | Lean;<br>Source;<br>Object               | F0   | Finite carrier        | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized     | CNNA/PillarA/Arithmetic/BoundarySource/LevelHistoryIndex.lean;<br>CNNA/PillarA/Arithmetic/BoundarySource/LevelStepWitness.lean;<br>CNNA/PillarA/Handoff/Core/Step1MathData.lean                                    | CNNA/PillarA/Foundation/LevelProfile.lean        | 1.2.5 |
| G04 | Carrier / index type                         | Lean;<br>Source;<br>Object               | F0   | Finite carrier        | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized     | CNNA/PillarA/Arithmetic/BoundarySource/<br>RecursiveLevelHistoryCarrier.lean;<br>CNNA/PillarA/Arithmetic/BoundarySource/LevelHistoryIndex.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelStepWitnessSM6.lean | CNNA/PillarA/Foundation/CarrierProfile.lean      | 1.2.6 |
| G05 | Boundary / interface /<br>interior partition | Lean;<br>Source;<br>Object               | F0   | Schur geometry        | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized     | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedBoundaryInteriorPartition.lean;<br>CNNA/PillarA/Finite/CutSpec.lean;<br>CNNA/PillarA/Finite/BoundaryPorts.lean                                                          | CNNA/PillarA/Schur/PartitionProfile.lean         | 1.3.1 |

| ID                  | Object                                    | Context class        | Tier | Layer              | Object status | Code status                                                                                                                                                       | Current code modules / derivation site                                                                                                                                                                              | Neutral target location / canonical API               | Phase                 |
|---------------------|-------------------------------------------|----------------------|------|--------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------|
| <a href="#">G06</a> | Regime: noOuterEnvironment                | Lean; Source; Object | F0   | Regime             | Y             | present as source/seed; derivation still to be canonicalized                                                                                                      | CNNA/PillarA/Handoff/Outputs/A_to_B.lean; CNNA/PillarA/Sectors/SectorSplit.lean; CNNA/PillarA/Sectors/ComplementSectorFamily.lean                                                                                   | CNNA/PillarA/Regime/NoOuter.lean                      | <a href="#">1.3.2</a> |
| <a href="#">G07</a> | Regime: hasOuterEnvironment               | Lean; Source; Object | F0   | Regime             | Y             | present as source/seed; derivation still to be canonicalized                                                                                                      | CNNA/PillarA/Handoff/Outputs/A_to_B.lean; CNNA/PillarA/Sectors/ComplementSectorFamily.lean; CNNA/PillarA/Network/SectorSysEnv.lean                                                                                  | CNNA/PillarA/Regime/HasOuter.lean                     | <a href="#">1.3.2</a> |
| <a href="#">G08</a> | Complement families: UV tail              | Lean; Source; Object | F0   | Regime/Complement  | R             | present as source/seed; derivation still to be canonicalized                                                                                                      | CNNA/PillarA/Sectors/UVSpectralSelector.lean; CNNA/PillarA/Sectors/ComplementSectorFamily.lean; CNNA/PillarA/Handoff/Outputs/A_to_B.lean                                                                            | CNNA/PillarA/Regime/Complement/UVTail.lean            | <a href="#">1.3.3</a> |
| <a href="#">G09</a> | Complement families: IR/parent-path tail  | Lean; Source; Object | F0   | Regime/Complement  | R             | present as source/seed; derivation still to be canonicalized                                                                                                      | CNNA/PillarA/Sectors/ComplementSectorFamily.lean; CNNA/PillarA/Handoff/Outputs/A_to_B.lean                                                                                                                          | CNNA/PillarA/Regime/Complement/IRTail.lean            | <a href="#">1.3.3</a> |
| <a href="#">G10</a> | Complement families: sideways environment | Lean; Source; Object | F0   | Regime/Complement  | R             | present as source/seed; derivation still to be canonicalized                                                                                                      | CNNA/PillarA/Sectors/ComplementSectorFamily.lean; CNNA/PillarA/Network/SectorSysEnv.lean; CNNA/PillarA/Network/SectorChannels.lean                                                                                  | CNNA/PillarA/Regime/Complement/SideEnvironment.lean   | <a href="#">1.3.3</a> |
| <a href="#">CD0</a> | Cayley-Dickson source                     | Lean; Source; Object | F1   | Structural algebra | B             | derived-only/proof-carrying present, but BLUE semantic/API correction remains open; not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Structural/CayleyDickson/Source.lean; CNNA/PillarA/Handoff/CayleyDicksonAdapter.lean                                                                                                                   | CNNA/PillarA/Structural/CayleyDickson/Source.lean     | <a href="#">1.4.1</a> |
| <a href="#">CD1</a> | Scalar part                               | Lean; Source; Object | F1   | Cayley-Dickson     | B             | derived-only/proof-carrying present, but BLUE semantic/API correction remains open; not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalScalarPart.lean                                                                                                                                                    | CNNA/PillarA/Structural/CayleyDickson/ScalarPart.lean | <a href="#">1.4.2</a> |
| <a href="#">CD2</a> | Cayley-Dickson star/conjugation           | Lean; Source; Object | F1   | Cayley-Dickson     | B             | derived-only/proof-carrying present, but BLUE semantic/API correction remains open; not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalStar.lean                                                                                                                                                          | CNNA/PillarA/Structural/CayleyDickson/Star.lean       | <a href="#">1.4.3</a> |
| <a href="#">CD3</a> | CD normSq                                 | Lean; Source; Object | F1   | Cayley-Dickson     | B             | derived-only/proof-carrying present, but BLUE semantic/API correction remains open; not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqDefined.lean; CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqReduction.lean; CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqBlockSystem.lean | CNNA/PillarA/Structural/CayleyDickson/NormSq.lean     | <a href="#">1.4.4</a> |

| ID                  | Object                                   | Context class              | Tier | Layer          | Object status | Code status                                                                                                                                                                                 | Current code modules / derivation site                                                                                                                                                                                                                                                                                                                                                                                                | Neutral target location / canonical API                             | Phase                 |
|---------------------|------------------------------------------|----------------------------|------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------|
| <a href="#">CD4</a> | CD multiplication/product                | Lean;<br>Source;<br>Object | F1   | Cayley-Dickson | B             | derived-only/<br>proof-carrying present,<br>but BLUE semantic/<br>API correction<br>remains open;<br>not public final API<br>until LEG7 and the<br>object natural<br>correction phase close | CNNA/PillarA/Structural/CayleyDickson/<br>S6CanonicalProductSurfaceCandidate.lean;<br>CNNA/PillarA/Structural/CayleyDickson/<br>S6CanonicalProductMultiplicationCandidate.lean                                                                                                                                                                                                                                                        | CNNA/PillarA/Structural/CayleyDickson/Product.lean                  | <a href="#">1.4.5</a> |
| <a href="#">CD5</a> | Alternative law                          | Lean;<br>Source;<br>Object | F1   | Cayley-Dickson | B             | derived-only/<br>proof-carrying present,<br>but BLUE semantic/<br>API correction<br>remains open;<br>not public final API<br>until LEG7 and the<br>object natural<br>correction phase close | CNNA/PillarA/Structural/CayleyDickson/<br>S6AlternativeLawScaffold.lean;<br>CNNA/PillarA/Structural/CayleyDickson/<br>S6FullAlternativeLawBoundary.lean                                                                                                                                                                                                                                                                               | CNNA/PillarA/Structural/CayleyDickson/<br>AlternativeLaw.lean       | <a href="#">1.4.6</a> |
| <a href="#">CD6</a> | Norm multiplicativity                    | Lean;<br>Source;<br>Object | F1   | Cayley-Dickson | B             | derived-only/<br>proof-carrying present,<br>but BLUE semantic/<br>API correction<br>remains open;<br>not public final API<br>until LEG7 and the<br>object natural<br>correction phase close | CNNA/PillarA/Structural/CayleyDickson/<br>S6NormSqMultiplicativity.lean;<br>CNNA/PillarA/Structural/CayleyDickson/<br>S6NormMultiplicativityScaffold.lean                                                                                                                                                                                                                                                                             | CNNA/PillarA/Structural/CayleyDickson/<br>NormMultiplicativity.lean | <a href="#">1.4.7</a> |
| <a href="#">CD7</a> | Hurwitz stop / R,C,H,O<br>classification | Lean;<br>Source;<br>Object | F1   | Cayley-Dickson | B             | derived-only/<br>proof-carrying present,<br>but BLUE semantic/<br>API correction<br>remains open;<br>not public final API<br>until LEG7 and the<br>object natural<br>correction phase close | CNNA/PillarA/Structural/CayleyDickson/S6HurwitzStopSeed.lean;<br>CNNA/PillarA/Structural/CayleyDickson/<br>S6CayleyDicksonIdentification.lean                                                                                                                                                                                                                                                                                         | CNNA/PillarA/Structural/CayleyDickson/HurwitzStop.lean              | <a href="#">1.4.8</a> |
| <a href="#">S00</a> | Dirichlet-Laplacian                      | Lean;<br>Source;<br>Object | F0   | Schur/DtN      | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized                                                                                                                      | CNNA/PillarA/Finite/DirichletLaplacian.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedDirichletEntry.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedDirichletExecEntrySourceSM3bq.lean                                                                                                                                                                                                                                          | CNNA/PillarA/Schur/DirichletLaplacian.lean                          | <a href="#">3.1</a>   |
| <a href="#">S01</a> | Interior block                           | Lean;<br>Source;<br>Object | F0   | Schur/DtN      | B             | derived-only/<br>proof-carrying present,<br>but BLUE semantic/<br>API correction<br>remains open;<br>not public final API<br>until LEG7 and the<br>object natural<br>correction phase close | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorBlock.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorEliminationCarrier.lean                                                                                                                                                                                                                                                                                              | CNNA/PillarA/Schur/InteriorBlock.lean                               | <a href="#">3.2</a>   |
| <a href="#">S02</a> | Interior inverse candidate               | Lean;<br>Source;<br>Object | F0   | Schur/DtN      | B             | derived-only/<br>proof-carrying present,<br>but BLUE semantic/<br>API correction<br>remains open;<br>not public final API<br>until LEG7 and the<br>object natural<br>correction phase close | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorInverseCandidateStart.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorInverseCandidateEntry.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorAccumulatedTraceEvaluation.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorInverseCandidateMatrix.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorInverseCandidateShapeSeparation.lean | CNNA/PillarA/Schur/InteriorInverseCandidate.lean                    | <a href="#">3.3</a>   |

| ID                  | Object                             | Context class              | Tier | Layer                  | Object status | Code status                                                                                                                                                                  | Current code modules / derivation site                                                                                                                                                                                                                                                                            | Neutral target location / canonical API             | Phase               |
|---------------------|------------------------------------|----------------------------|------|------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|---------------------|
| <a href="#">S03</a> | Two-sided inverse certificate      | Lean;<br>Source;<br>Object | F0   | Schur/DtN              | B             | derived-only/<br>proof-carrying present, but BLUE semantic/<br>API correction remains open;<br>not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorProductIdentityProofR4a.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorTwoSidedInvFromSM3db7R.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorProvenTwoSidedInvR5.lean                                                                | CNNA/PillarA/Schur/TwoSidedInverse.lean             | <a href="#">3.4</a> |
| <a href="#">S04</a> | InteriorEliminationData            | Lean;<br>Source;<br>Object | F0   | Schur/DtN              | B             | derived-only/<br>proof-carrying present, but BLUE semantic/<br>API correction remains open;<br>not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedInteriorEliminationCertificateSM3fR.lean;<br>CNNA/PillarA/DtN/DtN.lean                                                                                                                                                                                                 | CNNA/PillarA/Schur/InteriorElimination.lean         | <a href="#">3.5</a> |
| <a href="#">S05</a> | DtN boundary operator              | Lean;<br>Source;<br>Object | F0   | Schur/DtN              | B             | derived-only/<br>proof-carrying present, but BLUE semantic/<br>API correction remains open;<br>not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedTopBoundaryDtNSM3gR.lean;<br>CNNA/PillarA/DtN/DtN.lean                                                                                                                                                                                                                 | CNNA/PillarA/Schur/BoundaryOperator.lean            | <a href="#">3.6</a> |
| <a href="#">S06</a> | GeneralizedDtN / MultiSchur        | Lean;<br>Source;<br>Object | F0   | Schur/DtN              | B             | derived-only/<br>proof-carrying present, but BLUE semantic/<br>API correction remains open;<br>not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedGeneralizedTopBoundaryDtNSM3hR.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedMultiSchurSM4.lean;<br>CNNA/PillarA/Coupling/GeneralizedDtN.lean;<br>CNNA/PillarA/Coupling/MultiSchur.lean                                                                              | CNNA/PillarA/Schur/GeneralizedBoundaryOperator.lean | <a href="#">3.7</a> |
| <a href="#">O01</a> | Boundary matrix over ExecComplex   | Lean;<br>Source;<br>Object | F1   | Operational operator   | B             | derived-only/<br>proof-carrying present, but BLUE semantic/<br>API correction remains open;<br>not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedMultiSchurExecBoundaryOperatorSM4q.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedMultiSchurExecBoundaryOperatorFromSM3fqPlusSM4q.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedLevelHistoryExecMatrixFromSM6bRecheckSM8.lean                               | CNNA/PillarA/Operational/BoundaryMatrix.lean        | <a href="#">4.1</a> |
| <a href="#">O02</a> | Determinant coefficient recurrence | Lean;<br>Source;<br>Object | F1   | Operational polynomial | B             | derived-only/<br>proof-carrying present, but BLUE semantic/<br>API correction remains open;<br>not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedOperationalDeterminantCoefficientRecurrenceSM9e.lean                                                                                                                                                                                                                   | CNNA/PillarA/Operational/DeterminantRecurrence.lean | <a href="#">4.2</a> |
| <a href="#">O03</a> | Operational polynomial profile     | Lean;<br>Source;<br>Object | F1   | Operational polynomial | B             | derived-only/<br>proof-carrying present, but BLUE semantic/<br>API correction remains open;<br>not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedLevelHistoryOperationalCharpolyPolynomialProfileFromSM9b.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedLevelHistoryOperationalCoefficientSourceFromSM9c.lean;<br>CNNA/PillarA/Arithmetic/Smoke/<br>OperationalBoundedPolynomialEvaluationNormalFormSM9ePlus.lean | CNNA/PillarA/Operational/PolynomialProfile.lean     | <a href="#">4.3</a> |



| ID  | Object                                         | Context class           | Tier | Layer                                      | Object status | Code status                                                                                                                                                                  | Current code modules / derivation site                                                                                                                                  | Neutral target location / canonical API                                                         | Phase |
|-----|------------------------------------------------|-------------------------|------|--------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------|
| O04 | Quadratic coefficient window                   | Lean; Source; Object    | F1   | Operational polynomial                     | B             | derived-only/<br>proof-carrying present, but BLUE semantic/<br>API correction remains open;<br>not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedOperationalQuadraticCoefficientWindowFromSM9dSM9e.lean                                                                       | CNNA/PillarA/Operational/QuadraticWindow.lean                                                   | 4.4   |
| O05 | Operational discriminant D_op                  | Lean; Source; Object    | F1   | Arithmetic                                 | B             | derived-only/<br>proof-carrying present, but BLUE semantic/<br>API correction remains open;<br>not public final API until LEG7 and the object natural correction phase close | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedOperationalQuadraticDiscriminantSourceFromSM9g1Window.lean                                                                   | CNNA/PillarA/Operational/QuadraticDiscriminant.lean                                             | 4.5   |
| O06 | Comparison target policy                       | Reference; Gate; Object | F2   | Arithmetic                                 | G             | present, but incomplete/<br>non-canonical                                                                                                                                    | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedOperationalDiscriminantFactorGateFromSM9fFull.lean;<br>current code has target-exclusion/gate only, no general target policy | CNNA/PillarA/Operational/ComparisonTarget.lean                                                  | 4.6   |
| O07 | General finite computation test matrix profile | Lean; Target; Object    | F2   | Operational / test / computation readiness | Y             | planned;<br>no Lean implementation yet                                                                                                                                       | Future module selected by Phase 1.1.1.1.1;<br>must reuse existing Smoke/Generator modules.                                                                              | CNNA/PillarA/Arithmetic/Smoke or Generator test-matrix API, exact path pending module manifest. | 4.0.1 |
| N00 | Fundamental discriminant D_fund                | Lean; Source; Object    | F1   | Number theory                              | G             | present, but incomplete/<br>non-canonical                                                                                                                                    | CNNA/PillarA/Arithmetic/Orders/DiscriminantConvention.lean (special Gaussian/Eisenstein bridge;<br>not general derived D_fund)                                          | CNNA/PillarA/NumberTheory/<br>DiscriminantNormalization.lean                                    | 5.1   |
| N01 | Conductor f                                    | Lean; Source; Object    | F1   | Number theory                              | G             | present as source/<br>seed;<br>derivation still to be canonicalized                                                                                                          | CNNA/PillarA/Arithmetic/Orders/DiscriminantConvention.lean<br>(implicit/bridge status;<br>conductor not canonical)                                                      | CNNA/PillarA/NumberTheory/Conductor.lean                                                        | 5.1   |
| N02 | Quadratic field $K=\mathbb{Q}(\sqrt{D})$       | Lean; Source; Object    | F1   | Number theory                              | Y             | present, but incomplete/<br>non-canonical                                                                                                                                    | CNNA/PillarA/Arithmetic/Orders/QuadraticOrder.lean (special shapes)                                                                                                     | CNNA/PillarA/NumberTheory/QuadraticField.lean                                                   | 5.2   |
| N03 | Quadratic order O_D                            | Lean; Source; Object    | F1   | Number theory                              | Y             | present, but incomplete/<br>non-canonical                                                                                                                                    | CNNA/PillarA/Arithmetic/Orders/QuadraticOrder.lean (special Gaussian/Eisenstein examples)                                                                               | CNNA/PillarA/NumberTheory/QuadraticOrder.lean                                                   | 5.3   |
| N04 | Integral basis / omega_D                       | Lean; Source; Object    | F1   | Number theory                              | Y             | present, but incomplete/<br>non-canonical                                                                                                                                    | CNNA/PillarA/Arithmetic/Orders/QuadraticOrder.lean (halfIntegral shape;<br>basis not general API)                                                                       | CNNA/PillarA/NumberTheory/IntegralBasis.lean                                                    | 5.4   |
| N05 | Trace form                                     | Lean; Target; Object    | F1   | Number theory                              | Y             | missing in code                                                                                                                                                              | no current general module;<br>partial arithmetic in CNNA/PillarA/Arithmetic/Orders/NormForms.lean                                                                       | CNNA/PillarA/NumberTheory/TraceForm.lean                                                        | 5.5   |
| N06 | Norm form                                      | Lean; Source; Object    | F1   | Number theory                              | Y             | present, but incomplete/<br>non-canonical                                                                                                                                    | CNNA/PillarA/Arithmetic/Orders/NormForms.lean (special BinaryQuadraticNormForm)                                                                                         | CNNA/PillarA/NumberTheory/NormForm.lean                                                         | 5.5   |
| N07 | Conjugation                                    | Lean; Target; Object    | F1   | Number theory                              | Y             | missing in code                                                                                                                                                              | no current general module;<br>should live with QuadraticOrder/NormForms                                                                                                 | CNNA/PillarA/NumberTheory/Conjugation.lean                                                      | 5.5   |
| N08 | Unit group $O_D^\times$                        | Lean; Target; Object    | F2   | Number theory/<br>Symmetry                 | Y             | missing in code                                                                                                                                                              | no current derived module;<br>only UnitConventionStatus in NormForms.lean                                                                                               | CNNA/PillarA/NumberTheory/UnitgroupOD.lean                                                      | 6.1   |
| N09 | Class group $Cl(O_D)$                          | Lean; Target; Object    | F2   | Number theory                              | Y             | missing in code                                                                                                                                                              | no current module                                                                                                                                                       | CNNA/PillarA/NumberTheory/ClassgroupClOD.lean                                                   | 6.2   |
| N10 | Class number $h(D)$                            | Lean; Target; Object    | F2   | Number theory                              | Y             | missing in code                                                                                                                                                              | no current module                                                                                                                                                       | CNNA/PillarA/NumberTheory/ClassnumberhD.lean                                                    | 6.2   |



| ID                  | Object                                                 | Context class              | Tier | Layer                 | Object status | Code status                                                                 | Current code modules / derivation site                                                                                                                                                                         | Neutral target location / canonical API                        | Phase                  |
|---------------------|--------------------------------------------------------|----------------------------|------|-----------------------|---------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------|
| <a href="#">N11</a> | Fractional/integral ideas                              | Lean;<br>Target;<br>Object | F2   | Number theory         | Y             | missing in code                                                             | no current module                                                                                                                                                                                              | CNNA/PillarA/NumberTheory/Fractionalintegralideas.lean         | <a href="#">6.2.1</a>  |
| <a href="#">N12</a> | Principal ideal predicate                              | Lean;<br>Target;<br>Object | F2   | Number theory         | Y             | missing in code                                                             | no current module                                                                                                                                                                                              | CNNA/PillarA/NumberTheory/Principalidealpredicate.lean         | <a href="#">6.2.2</a>  |
| <a href="#">N13</a> | Heegner/PID/UFD indicator                              | Lean;<br>Target;<br>Object | F2   | Number theory         | Y             | missing in code                                                             | no current module;<br>old HeegnerDiscriminantBridge in DiscriminantConvention.lean is not full h(D)                                                                                                            | CNNA/PillarA/NumberTheory/HeegnerPIDUFDindicator.lean          | <a href="#">6.3</a>    |
| <a href="#">N14</a> | Prime splitting behavior                               | Lean;<br>Target;<br>Object | F2   | Number theory         | Y             | missing in code                                                             | no current module                                                                                                                                                                                              | CNNA/PillarA/NumberTheory/Primesplittingbehavior.lean          | <a href="#">6.4</a>    |
| <a href="#">N15</a> | Kronecker/Dirichlet character chi_D                    | Lean;<br>Target;<br>Object | F2   | Number theory         | Y             | missing in code                                                             | no current module                                                                                                                                                                                              | CNNA/PillarA/NumberTheory/KroneckerDirichletcharacterchiD.lean | <a href="#">6.4</a>    |
| <a href="#">CD8</a> | Arithmetic-CD compatibility as Schur-duality corollary | Lean;<br>Source;<br>Object | F1   | Bridge                | K             | present as source/seed only;<br>final theorem deferred to BR08/Phase 10.4.2 | CNNA/PillarA/Arithmetic/CayleyDickson/ArithmeticCDBridge.lean;<br>CNNA/PillarA/Arithmetic/Convergence/CDArithmeticCompatibility.lean;<br>CNNA/PillarA/Arithmetic/Convergence/QuaternionRamanujanInterface.lean | CNNA/PillarA/Bridge/ArithmeticCayleyDicksonCompatibility.lean  | <a href="#">10.4.2</a> |
| <a href="#">L00</a> | Lattice embedding<br>Lambda -> C                       | Lean;<br>Target;<br>Object | F2   | Geometry/CM           | Y             | missing in code;<br>EXEC-2<br>mirror-boundary<br>decision recorded          | no current CM-lattice module;<br>Geometry/* is spacetime/foiation, not CM embedding                                                                                                                            | CNNA/PillarA/Geometry/LatticeembeddingLambdaC.lean             | <a href="#">7.1</a>    |
| <a href="#">L01</a> | Lattice basis                                          | Lean;<br>Target;<br>Object | F2   | Geometry/CM           | Y             | missing in code                                                             | no current CM-lattice module                                                                                                                                                                                   | CNNA/PillarA/Geometry/Latticebasis.lean                        | <a href="#">7.1</a>    |
| <a href="#">L02</a> | Gram matrix                                            | Lean;<br>Target;<br>Object | F2   | Geometry              | Y             | missing in code;<br>EXEC-2<br>mirror-boundary<br>decision recorded          | no current CM-lattice Gram module                                                                                                                                                                              | CNNA/PillarA/Geometry/Grammatrix.lean                          | <a href="#">7.2</a>    |
| <a href="#">L03</a> | Lattice determinant / covolume                         | Lean;<br>Target;<br>Object | F2   | Geometry              | Y             | missing in code                                                             | no current CM-lattice determinant/covolume module                                                                                                                                                              | CNNA/PillarA/Geometry/Latticedeterminantcovolume.lean          | <a href="#">7.2</a>    |
| <a href="#">L04</a> | Period ratio tau                                       | Lean;<br>Target;<br>Object | F2   | Modular geometry      | Y             | missing in code;<br>EXEC-2<br>mirror-boundary<br>decision recorded          | no current tau module                                                                                                                                                                                          | CNNA/PillarA/Geometry/Periodratiotau.lean                      | <a href="#">7.3</a>    |
| <a href="#">L05</a> | Upper half-plane condition                             | Lean;<br>Target;<br>Object | F2   | Modular geometry      | Y             | missing in code                                                             | no current upper-half-plane module                                                                                                                                                                             | CNNA/PillarA/Geometry/Upperhalfplanecondition.lean             | <a href="#">7.3</a>    |
| <a href="#">L06</a> | Angle / cosine profile                                 | Lean;<br>Target;<br>Object | F2   | Geometry              | Y             | missing in code                                                             | no current derived angle module;<br>drawings/visual intuition not code source                                                                                                                                  | CNNA/PillarA/Geometry/Anglecosineprofile.lean                  | <a href="#">7.4</a>    |
| <a href="#">L07</a> | Lattice automorphism group                             | Lean;<br>Target;<br>Object | F2   | Geometry/<br>Symmetry | Y             | missing in code                                                             | no current lattice automorphism module                                                                                                                                                                         | CNNA/PillarA/Geometry/Latticeautomorphismgroup.lean            | <a href="#">7.5</a>    |
| <a href="#">M02</a> | Eisenstein series G4,G6                                | Lean;<br>Target;<br>Object | F2   | Modular forms         | Y             | missing in code                                                             | no current Eisenstein-series module                                                                                                                                                                            | CNNA/PillarA/Modular/EisensteinseriesG4G6.lean                 | <a href="#">8.1</a>    |
| <a href="#">M03</a> | Weierstrass invariants g2,g3                           | Lean;<br>Target;<br>Object | F2   | Elliptic functions    | Y             | missing in code                                                             | no current Weierstrass-invariant module                                                                                                                                                                        | CNNA/PillarA/Modular/Weierstrassinvariantsg2g3.lean            | <a href="#">8.2</a>    |
| <a href="#">M04</a> | Modular discriminant Delta_mod                         | Lean;<br>Target;<br>Object | F2   | Modular forms         | Y             | missing in code                                                             | no current modular-discriminant module                                                                                                                                                                         | CNNA/PillarA/Modular/ModulardiscriminantDeltamod.lean          | <a href="#">8.3</a>    |
| <a href="#">M05</a> | j-invariant                                            | Lean;<br>Target;<br>Object | F2   | Modular forms/CM      | Y             | missing in code;<br>EXEC-2<br>mirror-boundary<br>decision recorded          | no current j-invariant module                                                                                                                                                                                  | CNNA/PillarA/Modular/jinvariant.lean                           | <a href="#">8.4</a>    |



| ID                  | Object                               | Context class                 | Tier | Layer                         | Object status | Code status                                                                                                               | Current code modules / derivation site                                                                                                                     | Neutral target location / canonical API                       | Phase                |
|---------------------|--------------------------------------|-------------------------------|------|-------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------|
| <a href="#">SP0</a> | Eigenvalue profile                   | Lean;<br>Source;<br>Object    | F1   | Spectral                      | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized;<br>EXEC-2<br>mirror-boundary<br>decision recorded | CNNA/PillarA/Finite/ExecSpectral/CharpolyCore.lean;<br>CNNA/PillarA/Finite/ExecSpectral/EigenvectorKernel.lean;<br>CNNA/PillarA/Finite/SpectralBridge.lean | CNNA/PillarA/Schur/Eigenvalueprofile.lean                     | <a href="#">10.1</a> |
| <a href="#">SP1</a> | Spectral gap                         | Lean;<br>Source;<br>Object    | F1   | Spectral                      | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized                                                    | CNNA/PillarA/Arithmetic/Graph/SpectralGapSeed.lean;<br>CNNA/PillarA/Finite/ExecSpectral/Certificates.lean                                                  | CNNA/PillarA/Schur/Spectralgap.lean                           | <a href="#">10.1</a> |
| <a href="#">P00</a> | Operator algebra                     | Lean;<br>Source;<br>Object    | F1   | AQFT/finite                   | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized                                                    | CNNA/PillarA/Foundation/MatrixAlgebra.lean;<br>CNNA/PillarA/Finite/StateSpace.lean                                                                         | CNNA/PillarB/Derived/OperatorAlgebraFromA.lean                | <a href="#">10.2</a> |
| <a href="#">P01</a> | Star operation                       | Lean;<br>Source;<br>Object    | F1   | AQFT/finite                   | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized                                                    | CNNA/PillarA/Foundation/ExecComplex.lean;<br>CNNA/PillarA/Foundation/HermitianStructure.lean;<br>CNNA/PillarA/Foundation/MatrixAlgebra.lean                | CNNA/PillarA/Physics/Staroperation.lean                       | <a href="#">10.2</a> |
| <a href="#">P02</a> | State functional                     | Lean;<br>Source;<br>Object    | F1   | AQFT/finite                   | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized                                                    | CNNA/PillarA/Finite/StateSpace.lean                                                                                                                        | CNNA/PillarA/Physics/Statefunctional.lean                     | <a href="#">10.2</a> |
| <a href="#">P03</a> | Gibbs state                          | Lean;<br>Source;<br>Object    | F1   | AQFT/finite                   | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized                                                    | CNNA/PillarA/Finite/GibbsStateSeed.lean                                                                                                                    | CNNA/PillarA/Physics/Gibbsstate.lean                          | <a href="#">10.2</a> |
| <a href="#">P04</a> | Time evolution                       | Lean;<br>Source;<br>Object    | F1   | AQFT/<br>dynamics             | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized                                                    | CNNA/PillarA/Finite/MatrixExponential.lean;<br>CNNA/PillarA/Finite/UnitaryEvolution.lean;<br>CNNA/PillarA/Finite/DynamicsAdapter.lean                      | CNNA/PillarA/Physics/Timeevolution.lean                       | <a href="#">10.2</a> |
| <a href="#">P05</a> | Unitarity certificate                | Lean;<br>Source;<br>Object    | F1   | AQFT/<br>dynamics             | Y             | present, but<br>incomplete/<br>non-canonical                                                                              | CNNA/PillarA/Finite/UnitaryEvolution.lean (finite seed;<br>regime split still pending)                                                                     | CNNA/PillarA/Physics/Unitaritycertificate.lean                | <a href="#">10.2</a> |
| <a href="#">P06</a> | Open channel / effective<br>openness | Lean;<br>Source;<br>Object    | F1   | OQS/dynamics                  | R             | present as source/<br>seed;<br>derivation still to be<br>canonicalized                                                    | CNNA/PillarA/Finite/ChannelSeed.lean;<br>CNNA/PillarA/Network/SectorChannels.lean                                                                          | CNNA/PillarA/Physics/Openchanneleffectiveopenness.lean        | <a href="#">10.2</a> |
| <a href="#">N16</a> | Dirichlet L-function<br>L(s,chi_D)   | Reference;<br>Gate;<br>Object | F2   | Analytic<br>number theory     | K             | missing in code                                                                                                           | no current module                                                                                                                                          | CNNA/PillarA/NumberTheory/<br>DirichletLfunctionLschiD.lean   | <a href="#">9.2</a>  |
| <a href="#">N17</a> | Class number formula                 | Reference;<br>Gate;<br>Object | F2   | Analytic<br>number theory     | K             | missing in code                                                                                                           | no current module                                                                                                                                          | CNNA/PillarA/NumberTheory/Classnumberformula.lean             | <a href="#">9.3</a>  |
| <a href="#">L08</a> | Theta series<br>Theta_Lambda         | Reference;<br>Gate;<br>Object | F2   | Modular<br>forms/<br>geometry | K             | missing in code                                                                                                           | no current theta-series module                                                                                                                             | CNNA/PillarA/Geometry/ThetaseriesThetaLambda.lean             | <a href="#">9.1</a>  |
| <a href="#">M00</a> | Modular group action<br>SL2Z         | Lean;<br>Target;<br>Object    | F2   | Modular forms                 | K             | missing in code                                                                                                           | no current modular-action module                                                                                                                           | CNNA/PillarA/Modular/ModulargroupactionSL2Z.lean              | <a href="#">8.0</a>  |
| <a href="#">M01</a> | Fundamental domain<br>representative | Lean;<br>Target;<br>Object    | F2   | Modular forms                 | K             | missing in code                                                                                                           | no current fundamental-domain module                                                                                                                       | CNNA/PillarA/Modular/<br>Fundamentaldomainrepresentative.lean | <a href="#">8.0</a>  |
| <a href="#">M06</a> | q-expansion of j                     | Lean;<br>Target;<br>Object    | F2   | Modular forms                 | K             | missing in code                                                                                                           | no current q-expansion module                                                                                                                              | CNNA/PillarA/Modular/qexpansionofj.lean                       | <a href="#">8.8</a>  |
| <a href="#">M07</a> | Hilbert class polynomial<br>H_D      | Lean;<br>Target;<br>Object    | F2   | CM                            | K             | missing in code                                                                                                           | no current Hilbert-class-polynomial module                                                                                                                 | CNNA/PillarA/Modular/HilbertclasspolynomialHD.lean            | <a href="#">8.8</a>  |

| ID                  | Object                                    | Context class                            | Tier | Layer                 | Object status | Code status                                                         | Current code modules / derivation site                                                                                                                                | Neutral target location / canonical API                      | Phase                |
|---------------------|-------------------------------------------|------------------------------------------|------|-----------------------|---------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------|
| <a href="#">TM0</a> | Transfermatrix T_k                        | Lean;<br>Source;<br>Object               | F1   | Schur algebra         | K             | present, but incomplete/<br>non-canonical                           | CNNA/PillarA/Arithmetic/Schur/MobiusTransfer.lean (anchor/status only);<br>CNNA/PillarA/Arithmetic/Schur/Recurrence.lean;<br>no full $M_2(\mathbb{Z}[\lambda])$ proof | CNNA/PillarA/Transfer/TransfermatrixTk.lean                  | <a href="#">10.4</a> |
| <a href="#">TM1</a> | Transfermatrix product                    | Lean;<br>Source;<br>Object               | F1   | Schur algebra         | K             | present as source/<br>seed;<br>derivation still to be canonicalized | CNNA/PillarA/Arithmetic/Schur/MobiusTransfer.lean (no product formalization)                                                                                          | CNNA/PillarA/Transfer/Transfermatrixproduct.lean             | <a href="#">10.4</a> |
| <a href="#">TM2</a> | Moebius/continued fraction relation       | Lean;<br>Source;<br>Object               | F1   | Schur algebra         | K             | present as source/<br>seed;<br>derivation still to be canonicalized | CNNA/PillarA/Arithmetic/Schur/MobiusTransfer.lean (route/status; no theoreemically complete Moebius-chain)                                                            | CNNA/PillarA/Transfer/Moebiuscontinuedfractionrelation.lean  | <a href="#">10.4</a> |
| <a href="#">SP2</a> | Heat kernel K(t)                          | Lean;<br>Target;<br>Object               | F2   | Spectral geometry     | K             | missing in code                                                     | no current heat-kernel module;<br>MatrixExponential is related but not HeatKernelProfile                                                                              | CNNA/PillarA/Schur/HeatkernelKt.lean                         | <a href="#">10.3</a> |
| <a href="#">SP3</a> | Spectral zeta zeta_H                      | Lean;<br>Target;<br>Object               | F2   | Spectral geometry     | K             | missing in code                                                     | no current spectral-zeta module                                                                                                                                       | CNNA/PillarA/Schur/SpectralzetazetaH.lean                    | <a href="#">10.3</a> |
| <a href="#">SP4</a> | Spectral dimension                        | Lean;<br>Target;<br>Object               | F2   | Spectral geometry     | K             | missing in code                                                     | no current spectral-dimension module                                                                                                                                  | CNNA/PillarA/Schur/Spectraldimension.lean                    | <a href="#">10.3</a> |
| <a href="#">SP5</a> | Zeta-regularized determinant              | Lean;<br>Target;<br>Object               | F2   | Spectral geometry     | K             | missing in code                                                     | no current zeta-regularized determinant module                                                                                                                        | CNNA/PillarA/Schur/Zetaregularizeddeterminant.lean           | <a href="#">10.3</a> |
| <a href="#">P07</a> | KMS condition                             | Lean;<br>Target;<br>Object               | F2   | Pillar B AQFT         | K             | missing in code                                                     | no current KMS proof module;<br>GibbsStateSeed is prerequisite only                                                                                                   | CNNA/PillarA/Physics/KMScondition.lean                       | <a href="#">13.4</a> |
| <a href="#">P08</a> | GNS Hilbert space                         | Lean;<br>Target;<br>Object               | F2   | Pillar B AQFT         | K             | missing in code                                                     | no current GNS module                                                                                                                                                 | CNNA/PillarA/Physics/GNSHilbertspace.lean                    | <a href="#">13.4</a> |
| <a href="#">P09</a> | Tomita operator S                         | Lean;<br>Target;<br>Object               | F2   | Pillar B AQFT modular | K             | missing in code                                                     | no current Tomita module                                                                                                                                              | CNNA/PillarA/Physics/TomitaoperatorS.lean                    | <a href="#">13.5</a> |
| <a href="#">P10</a> | Modular operator Delta                    | Lean;<br>Target;<br>Object               | F2   | Pillar B AQFT modular | K             | missing in code                                                     | no current modular-operator module                                                                                                                                    | CNNA/PillarA/Physics/ModularoperatorDelta.lean               | <a href="#">13.5</a> |
| <a href="#">P11</a> | Modular conjugation J                     | Lean;<br>Target;<br>Object               | F2   | Pillar B AQFT modular | K             | missing in code                                                     | no current modular-conjugation module                                                                                                                                 | CNNA/PillarA/Physics/ModularconjugationJ.lean                | <a href="#">13.5</a> |
| <a href="#">P12</a> | Bisognano-Wichmann/<br>CPT bridge         | Lean;<br>Target;<br>Object               | F2   | Pillar B AQFT         | K             | missing in code                                                     | no current BW/CPT module                                                                                                                                              | CNNA/PillarA/Physics/BisognanoWichmannCPTbridge.lean         | <a href="#">13.6</a> |
| <a href="#">P13</a> | Spin-statistics                           | Lean;<br>Target;<br>Object               | F2   | Pillar B AQFT         | K             | missing in code                                                     | no current spin-statistics module                                                                                                                                     | CNNA/PillarA/Physics/Spinstatistics.lean                     | <a href="#">13.6</a> |
| <a href="#">P14</a> | Kramers degeneracy                        | Lean;<br>Target;<br>Object               | F2   | Pillar B Physics      | K             | missing in code                                                     | no current Kramers module                                                                                                                                             | CNNA/PillarA/Physics/Kramersdegeneracy.lean                  | <a href="#">13.6</a> |
| <a href="#">P15</a> | Split property                            | Lean;<br>Source;<br>Object               | F1   | Pillar B AQFT         | K             | present, but incomplete/<br>non-canonical                           | CNNA/PillarA/Sectors/SectorSplit.lean;<br>Handoff has split fields;<br>AQFT split property not derived                                                                | CNNA/PillarA/Physics/SplitProperty.lean                      | <a href="#">13.6</a> |
| <a href="#">P16</a> | Local net / Haag-Kastler algebraic axioms | Lean;<br>Target;<br>Object               | F2   | Pillar B AQFT         | K             | missing in code                                                     | no current local-net/HK module;<br>legacy target only                                                                                                                 | CNNA/PillarA/Physics/LocalnetHaagKastleralgebraicaxioms.lean | <a href="#">13.3</a> |
| <a href="#">H10</a> | A->B bright handoff                       | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Handoff policy        | Y             | present as source/<br>seed;<br>derivation still to be canonicalized | CNNA/PillarA/Handoff/Outputs/A_to_B.lean;<br>CNNA/PillarA/Handoff/Core/Step1MathData.lean;<br>CNNA/PillarA/Handoff/Core/SectorExport.lean                             | CNNA/PillarA/Handoff/Outputs/A_to_B.lean                     | <a href="#">11.1</a> |

| ID                    | Object                                        | Context class                            | Tier | Layer                             | Object status | Code status                                                            | Current code modules / derivation site                                                 | Neutral target location / canonical API                            | Phase                |
|-----------------------|-----------------------------------------------|------------------------------------------|------|-----------------------------------|---------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------|
| <a href="#">H11</a>   | A->B prime/dark future handoff                | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Handoff policy                    | R             | missing in code                                                        | no current dark/B-prime handoff module                                                 | CNNA/PillarA/Handoff/ABprimedarkfuturehandoff.lean                 | <a href="#">11.2</a> |
| <a href="#">H12</a>   | A->C handoff                                  | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Handoff policy                    | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized | CNNA/PillarA/Handoff/Outputs/A_to_C.lean                                               | CNNA/PillarA/Handoff/AChandoff.lean                                | <a href="#">11.3</a> |
| <a href="#">H13</a>   | A->D handoff                                  | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Handoff policy                    | Y             | present as source/<br>seed;<br>derivation still to be<br>canonicalized | CNNA/PillarA/Handoff/Outputs/A_to_D.lean                                               | CNNA/PillarA/Handoff/ADhandoff.lean                                | <a href="#">11.4</a> |
| <a href="#">H14</a>   | A->E handoff                                  | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Handoff policy                    | K             | present as source/<br>seed;<br>derivation still to be<br>canonicalized | CNNA/PillarA/Handoff/Outputs/A_to_E.lean                                               | CNNA/PillarA/Handoff/AEhandoff.lean                                | <a href="#">11.5</a> |
| <a href="#">MOON0</a> | j coefficient stream                          | Reference;<br>Gate;<br>Object            | F2   | Moonshine/<br>established<br>math | K             | missing;<br>not in SM9h code                                           | no current Lean module                                                                 | CNNA/PillarA/Arithmetic/Moonshine/<br>JCoefficientStream.lean      | <a href="#">12.1</a> |
| <a href="#">MOON1</a> | Monster dimension<br>comparison profile       | Reference;<br>Gate;<br>Object            | F2   | Moonshine/<br>established<br>math | K             | missing;<br>not in SM9h code                                           | no current Lean module                                                                 | CNNA/PillarA/Arithmetic/Moonshine/<br>MonsterComparison.lean       | <a href="#">12.2</a> |
| <a href="#">MOON2</a> | McKay-Thompson<br>comparison profile          | Reference;<br>Gate;<br>Object            | F2   | Moonshine/<br>established<br>math | K             | missing;<br>not in SM9h code                                           | no current Lean module                                                                 | CNNA/PillarA/Arithmetic/Moonshine/<br>McKayThompsonComparison.lean | <a href="#">12.3</a> |
| <a href="#">B00</a>   | Pillar B input contract<br>from A             | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Pillar B QFT                      | K             | PillarB namespace<br>empty in SM9h;<br>handoff source exists<br>in A   | CNNA/PillarB/BuildAll.lean;<br>CNNA/PillarA/Handoff/Outputs/A_to_B.lean                | CNNA/PillarB/Handoff/BInputFromA.lean                              | <a href="#">13.1</a> |
| <a href="#">B01</a>   | Boundary star-algebra /<br>observable algebra | Lean;<br>Source;<br>Object               | F1   | Pillar B QFT                      | K             | not implemented in<br>PillarB;<br>related finite seeds in<br>A         | CNNA/PillarA/Finite/StateSpace.lean;<br>CNNA/PillarB/BuildAll.lean                     | CNNA/PillarB/QFT/BoundaryStarAlgebra.lean                          | <a href="#">13.2</a> |
| <a href="#">B02</a>   | Local net / QFT algebraic<br>profile          | Lean;<br>Source;<br>Object               | F1   | Pillar B QFT                      | K             | not implemented in<br>PillarB                                          | CNNA/PillarB/BuildAll.lean                                                             | CNNA/PillarB/AQFT/LocalNetFromBoundary.lean                        | <a href="#">13.3</a> |
| <a href="#">B03</a>   | B state/GNS/KMS profile                       | Lean;<br>Source;<br>Object               | F1   | Pillar B QFT                      | K             | not implemented in<br>PillarB;<br>A seeds exist                        | CNNA/PillarA/Finite/GibbsStateSeed.lean;<br>CNNA/PillarA/Finite/MatrixExponential.lean | CNNA/PillarB/QFT/StateGNSKMS.lean                                  | <a href="#">13.4</a> |
| <a href="#">B04</a>   | Tomita-Takesaki modular<br>data               | Lean;<br>Source;<br>Object               | F1   | Pillar B AQFT                     | K             | not implemented in<br>PillarB                                          | CNNA/PillarB/BuildAll.lean                                                             | CNNA/PillarB/AQFT/ModularTheory.lean                               | <a href="#">13.5</a> |
| <a href="#">B05</a>   | CPT/BW/spin-statistics/<br>split gates        | Lean;<br>Source;<br>Object               | F1   | Pillar B AQFT                     | K             | not implemented in<br>PillarB                                          | CNNA/PillarB/BuildAll.lean                                                             | CNNA/PillarB/AQFT/StructuralTheorems.lean                          | <a href="#">13.6</a> |
| <a href="#">B06</a>   | B->A allowed output<br>profile                | Lean;<br>Source;<br>Object               | F1   | Pillar B<br>handoff               | K             | input stub exists on A<br>side only                                    | CNNA/PillarA/Handoff/Inputs/B_to_A.lean                                                | CNNA/PillarB/Handoff/BToAOutput.lean                               | <a href="#">13.7</a> |
| <a href="#">C00</a>   | Pillar C OQS input<br>contract from A         | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Pillar C OQS                      | K             | PillarC namespace<br>empty in SM9h                                     | CNNA/PillarC/BuildAll.lean;<br>CNNA/PillarA/Handoff/Outputs/A_to_C if created later    | CNNA/PillarC/Handoff/CInputFromA.lean                              | <a href="#">14.1</a> |
| <a href="#">C01</a>   | Finite CP/channel profile                     | Lean;<br>Source;<br>Object               | F1   | Pillar C OQS                      | K             | A channel seed exists;<br>PillarC empty                                | CNNA/PillarA/Finite/ChannelSeed.lean;<br>CNNA/PillarC/BuildAll.lean                    | CNNA/PillarC/OQS/FiniteChannelProfile.lean                         | <a href="#">14.2</a> |
| <a href="#">C02</a>   | Effective OQS generator /<br>Lindblad gate    | Lean;<br>Source;<br>Object               | F1   | Pillar C OQS                      | K             | not implemented in<br>PillarC                                          | CNNA/PillarC/BuildAll.lean                                                             | CNNA/PillarC/OQS/EffectiveGenerator.lean                           | <a href="#">14.3</a> |

| ID                    | Object                                           | Context class                            | Tier | Layer                   | Object status | Code status                         | Current code modules / derivation site  | Neutral target location / canonical API        | Phase                |
|-----------------------|--------------------------------------------------|------------------------------------------|------|-------------------------|---------------|-------------------------------------|-----------------------------------------|------------------------------------------------|----------------------|
| <a href="#">C03</a>   | Decoherence/noise profile                        | Lean;<br>Source;<br>Object               | F1   | Pillar C OQS            | K             | not implemented in PillarC          | CNNA/PillarC/BuildAll.lean              | CNNA/PillarC/OQS/DecoherenceProfile.lean       | <a href="#">14.4</a> |
| <a href="#">C04</a>   | C->A allowed output profile                      | Lean;<br>Source;<br>Object               | F1   | Pillar C handoff        | K             | input stub exists on A side only    | CNNA/PillarA/Handoff/Inputs/C_to_A.lean | CNNA/PillarC/Handoff/CToAOutput.lean           | <a href="#">14.5</a> |
| <a href="#">D00</a>   | Pillar D spacetime input contract                | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Pillar D spacetime      | K             | PillarD namespace empty in SM9h     | CNNA/PillarD/BuildAll.lean              | CNNA/PillarD/Handoff/DInputFromA.lean          | <a href="#">15.1</a> |
| <a href="#">D01</a>   | Region/localization/causal-order candidate       | Lean;<br>Source;<br>Object               | F1   | Pillar D spacetime      | K             | not implemented in PillarD          | CNNA/PillarD/BuildAll.lean              | CNNA/PillarD/Spacetime/RegionCausalOrder.lean  | <a href="#">15.2</a> |
| <a href="#">D02</a>   | Metric/coarse-geometry/covariance gate           | Lean;<br>Source;<br>Object               | F1   | Pillar D spacetime      | K             | not implemented in PillarD          | CNNA/PillarD/BuildAll.lean              | CNNA/PillarD/Spacetime/MetricCovariance.lean   | <a href="#">15.3</a> |
| <a href="#">D03</a>   | D->A allowed output profile                      | Lean;<br>Source;<br>Object               | F1   | Pillar D handoff        | K             | input stub exists on A side only    | CNNA/PillarA/Handoff/Inputs/D_to_A.lean | CNNA/PillarD/Handoff/DToAOutput.lean           | <a href="#">15.4</a> |
| <a href="#">E00</a>   | Pillar E matter input contract                   | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Pillar E matter         | K             | PillarE namespace empty in SM9h     | CNNA/PillarE/BuildAll.lean              | CNNA/PillarE/Handoff/EInputProfile.lean        | <a href="#">16.1</a> |
| <a href="#">E01</a>   | Matter sector and internal symmetry candidate    | Lean;<br>Source;<br>Object               | F1   | Pillar E matter         | K             | not implemented in PillarE          | CNNA/PillarE/BuildAll.lean              | CNNA/PillarE/Matter/SectorSymmetry.lean        | <a href="#">16.2</a> |
| <a href="#">E02</a>   | Observable coupling and phenomenology comparison | Reference;<br>Gate;<br>Object            | F3   | Pillar E matter         | K             | not implemented in PillarE          | CNNA/PillarE/BuildAll.lean              | CNNA/PillarE/Matter/ObservableComparison.lean  | <a href="#">16.3</a> |
| <a href="#">E03</a>   | E->A allowed output profile                      | Lean;<br>Source;<br>Object               | F1   | Pillar E handoff        | K             | input stub exists on A side only    | CNNA/PillarA/Handoff/Inputs/E_to_A.lean | CNNA/PillarE/Handoff/EToAOutput.lean           | <a href="#">16.4</a> |
| <a href="#">BR00</a>  | Backreaction input DAG                           | Lean;<br>Target;<br>Object               | F2   | Backreaction governance | K             | partial A input handoff stubs exist | CNNA/PillarA/Handoff/Inputs/*.lean      | CNNA/PillarA/Backreaction/InputDAG.lean        | <a href="#">17</a>   |
| <a href="#">BR01</a>  | B->A backreaction input gate                     | Lean;<br>Source;<br>Object               | F1   | Backreaction governance | K             | input stub exists                   | CNNA/PillarA/Handoff/Inputs/B_to_A.lean | CNNA/PillarA/Backreaction/BInputGate.lean      | <a href="#">17.1</a> |
| <a href="#">BR02</a>  | C->A OQS backreaction input gate                 | Lean;<br>Source;<br>Object               | F1   | Backreaction governance | K             | input stub exists                   | CNNA/PillarA/Handoff/Inputs/C_to_A.lean | CNNA/PillarA/Backreaction/CInputGate.lean      | <a href="#">17.2</a> |
| <a href="#">BR03</a>  | D->A spacetime backreaction input gate           | Lean;<br>Source;<br>Object               | F1   | Backreaction governance | K             | input stub exists                   | CNNA/PillarA/Handoff/Inputs/D_to_A.lean | CNNA/PillarA/Backreaction/DInputGate.lean      | <a href="#">17.3</a> |
| <a href="#">BR04</a>  | E->A matter backreaction input gate              | Lean;<br>Source;<br>Object               | F1   | Backreaction governance | K             | input stub exists                   | CNNA/PillarA/Handoff/Inputs/E_to_A.lean | CNNA/PillarA/Backreaction/EInputGate.lean      | <a href="#">17.4</a> |
| <a href="#">BR05</a>  | Pre/post-emergence switch for A statements       | Lean;<br>Source;<br>Object               | F0   | Backreaction governance | K             | not implemented neutrally           | no current neutral module               | CNNA/PillarA/Backreaction/EmergenceSwitch.lean | <a href="#">17.5</a> |
| <a href="#">SIDE0</a> | Full ToE comparison profile                      | Reference;<br>Gate;<br>Object            | F4   | Side comparison         | K             | not implemented                     | no current Lean module                  | CNNA/Comparisons/FullToE.lean                  | <a href="#">18.1</a> |
| <a href="#">SIDE1</a> | String-theory comparison profile                 | Reference;<br>Gate;<br>Object            | F4   | Side comparison         | K             | not implemented                     | no current Lean module                  | CNNA/Comparisons/StringTheory.lean             | <a href="#">18.2</a> |
| <a href="#">SIDE2</a> | Loop quantum gravity comparison profile          | Reference;<br>Gate;<br>Object            | F4   | Side comparison         | K             | not implemented                     | no current Lean module                  | CNNA/Comparisons/LoopQuantumGravity.lean       | <a href="#">18.3</a> |

| ID                    | Object                                                    | Context class              | Tier | Layer                      | Object status | Code status                                                             | Current code modules / derivation site                                                                                                                                                                                                                                                                                                                                  | Neutral target location / canonical API                                                                                                                | Phase                     |
|-----------------------|-----------------------------------------------------------|----------------------------|------|----------------------------|---------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <a href="#">SIDE3</a> | Noncommutative geometry comparison profile                | Reference; Gate; Object    | F4   | Side comparison            | K             | not implemented                                                         | no current Lean module                                                                                                                                                                                                                                                                                                                                                  | CNNA/Comparisons/NoncommutativeGeometry.lean                                                                                                           | <a href="#">18.4</a>      |
| <a href="#">MOON3</a> | Finite simple group comparison profile                    | Reference; Gate; Object    | F2   | Moonshine/established math | K             | missing; not in SM9h code                                               | no current Lean module                                                                                                                                                                                                                                                                                                                                                  | CNNA/PillarA/Arithmetic/Moonshine/FiniteSimpleGroupComparison.lean                                                                                     | <a href="#">12.4</a>      |
| <a href="#">MOON4</a> | Sporadic/Monster comparison boundary                      | Reference; Gate; Object    | F2   | Moonshine/established math | K             | missing; not in SM9h code                                               | no current Lean module                                                                                                                                                                                                                                                                                                                                                  | CNNA/PillarA/Arithmetic/Moonshine/SporadicMonsterBoundary.lean                                                                                         | <a href="#">12.4</a>      |
| <a href="#">MOON5</a> | Leech lattice comparison profile                          | Reference; Gate; Object    | F2   | Moonshine/Leech lattice    | K             | missing; not in SM9h code                                               | no current Lean module                                                                                                                                                                                                                                                                                                                                                  | CNNA/PillarA/Arithmetic/Moonshine/LeechLatticeComparison.lean                                                                                          | <a href="#">12.5</a>      |
| <a href="#">MOON6</a> | Leech theta/shell coefficient comparison                  | Reference; Gate; Object    | F2   | Moonshine/Leech lattice    | K             | missing; not in SM9h code                                               | no current Lean module                                                                                                                                                                                                                                                                                                                                                  | CNNA/PillarA/Arithmetic/Moonshine/LeechThetaComparison.lean                                                                                            | <a href="#">12.6</a>      |
| <a href="#">MOON7</a> | VOA/moonshine-module comparison profile                   | Reference; Gate; Object    | F2   | Moonshine/VOA              | K             | missing; not in SM9h code                                               | no current Lean module                                                                                                                                                                                                                                                                                                                                                  | CNNA/PillarA/Arithmetic/Moonshine/VOAComparison.lean                                                                                                   | <a href="#">12.7</a>      |
| <a href="#">MOON8</a> | Common-indicator / Hauptmodul bridge-test profile         | Reference; Gate; Object    | F2   | Moonshine/established math | K             | missing; not in SM9h code                                               | no current Lean module                                                                                                                                                                                                                                                                                                                                                  | CNNA/PillarA/Arithmetic/Moonshine/CommonIndicatorBridgeTest.lean                                                                                       | <a href="#">12.8</a>      |
| <a href="#">LEG0</a>  | Consumption and split inventory umbrella                  | Governance; Policy; Object | F3   | Refactor governance        | Y             | planning/audit object; split across 1.1.1.1-1.1.1.5                     | Arithmetic/Notation.lean; repo_inventory/raw_export                                                                                                                                                                                                                                                                                                                     | planning master only; no Lean API until child gates close                                                                                              | <a href="#">1.1.1</a>     |
| <a href="#">LEG1</a>  | Legacy single-constructor status gates                    | Lean; Refactor; Object     | F3   | Legacy quarantine          | Y             | present in legacy monolith; target quarantine missing                   | CNNA/PillarA/Arithmetic/Notation.lean                                                                                                                                                                                                                                                                                                                                   | CNNA/PillarA/Arithmetic/Legacy/StatusGates.lean                                                                                                        | <a href="#">1.1.2</a>     |
| <a href="#">LEG2</a>  | CoreSymbols true notation surface                         | Lean; Refactor; Object     | F3   | Notation                   | Y             | target module missing                                                   | CNNA/PillarA/Arithmetic/Notation.lean; Foundation/Notation.lean; Finite/Notation.lean; Handoff/Notation.lean                                                                                                                                                                                                                                                            | CNNA/PillarA/Arithmetic/Notation/CoreSymbols.lean                                                                                                      | <a href="#">1.1.3</a>     |
| <a href="#">LEG3</a>  | Legacy phase aliases compatibility surface                | Lean; Refactor; Object     | F3   | Legacy quarantine          | Y             | target module missing                                                   | CNNA/PillarA/Arithmetic/Notation.lean; historical SM*/S6* modules                                                                                                                                                                                                                                                                                                       | CNNA/PillarA/Arithmetic/Legacy/PhaseAliases.lean; CNNA/PillarA/Arithmetic/NeutralWrappers.lean                                                         | <a href="#">1.1.4</a>     |
| <a href="#">LEG4</a>  | Neutral import firewall for post-notation phases          | Governance; Policy; Object | F3   | Import governance          | Y             | missing; requires post-split audit                                      | BuildAll.lean; repo_inventory/raw_export/module_imports.tsv                                                                                                                                                                                                                                                                                                             | generated_exports/module inventory views; masters/cnna_roadmap_master.yaml phase gates                                                                 | <a href="#">1.1.5</a>     |
| <a href="#">LEG5</a>  | Declaration consumption classification map                | Governance; Policy; Object | F3   | Planning/API hygiene       | G             | closed generated classification artifact; no Lean source move           | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated_exports/consumption_classification_map.generated.tsv; generated_exports/module_consumption_records.generated.tsv; generated_exports/declaration_consumption_records.generated.tsv; generated_exports/object_consumption_classification.generated.tsv; tables/consumption_classification_map.tex | planning master, generated consumption-classification section, generated module/declaration/object TSV ledgers; optional future Lean audit mirror only | <a href="#">1.1.1.1.4</a> |
| <a href="#">LEG7</a>  | Blue-object semantic correction ledger                    | Governance; Policy; Object | F3   | Planning/API hygiene       | Y             | planned; prevents treating proof-carrying blue code as public final API | object register; Structural/CayleyDickson/*; Arithmetic/Smoke/GeneratedInterior*; Arithmetic/Smoke/*Operational*                                                                                                                                                                                                                                                        | planning master; target phase records 1.4.*, 3.*, 4.*                                                                                                  | <a href="#">1.1.1.2</a>   |
| <a href="#">LEG6</a>  | Public API contract and consumer whitelist                | Governance; Policy; Object | F3   | Planning/API hygiene       | Y             | planned contract; no Lean source move yet                               | masters/cnna_roadmap_master.yaml; planned CoreSymbols/Legacy/NeutralWrappers modules                                                                                                                                                                                                                                                                                    | planning master and later CNNA/PillarA/Arithmetic/Notation/CoreSymbols.lean / neutral wrappers                                                         | <a href="#">1.1.1.3</a>   |
| <a href="#">LEG8</a>  | Pruning and dead-candidate ledger                         | Governance; Policy; Object | F3   | Planning/API hygiene       | Y             | planned; no deletion authorized by this object alone                    | planning master; generated reports; repo_inventory/raw_export                                                                                                                                                                                                                                                                                                           | planning master; later pruning implementation phase only if build evidence exists                                                                      | <a href="#">1.1.1.4</a>   |
| <a href="#">LEG9</a>  | Generated-view authority and implementation-lock contract | Governance; Policy; Object | F3   | Planning/API hygiene       | Y             | planned generated-artifact lock; no Lean build claim                    | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated exports; PDF                                                                                                                                                                                                                                                                                    | README/report and generated planning PDF                                                                                                               | <a href="#">1.1.1.5</a>   |



| ID                    | Object                                                         | Context class              | Tier | Layer                                  | Object status | Code status                                                                                    | Current code modules / derivation site                                                                                                                                                                                                                                                                                | Neutral target location / canonical API                                                                                                                                             | Phase                       |
|-----------------------|----------------------------------------------------------------|----------------------------|------|----------------------------------------|---------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <a href="#">LEG10</a> | Implementation module manifest and semantic placement contract | Governance; Policy; Object | F1   | Planning/API hygiene                   | G             | closed generated-artifact gate; no Lean source move                                            | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated_exports/module_implementation_manifest.generated.tsv; tables/module_implementation_manifest.tex                                                                                                                                               | planning master, generated module-manifest table, and implementation scratchpad                                                                                                     | <a href="#">1.1.1.1.1</a>   |
| <a href="#">LEG11</a> | Reuse/duplication ledger and canonical generator policy        | Governance; Policy; Object | F1   | Planning/API hygiene                   | G             | closed generated-policy gate; no Lean source move                                              | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated_exports/reuse_duplication_ledger.generated.tsv; tables/reuse_duplication_ledger.tex; Arithmetic/Smoke/GeneratedSource.lean; Arithmetic/Smoke/GeneratedMainPath.lean; Actions/GeneratorActionInventory.lean; Actions/ActionFormatDecision.lean | planning master and generated reuse/duplication ledger now; later neutral generator API only if a code phase proves the current Smoke namespace is too narrow while semantics match | <a href="#">1.1.1.1.2</a>   |
| <a href="#">LEG12</a> | Object and size proof-documentation infrastructure             | Governance; Policy; Object | F1   | Legacy/Planning governance             | G             | closed generated proof-documentation infrastructure; no Lean code semantics                    | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated_exports/object_proof_documentation.generated.tsv; tables/object_proof_documentation.tex; tables/object_proof_link_register.tex                                                                                                                | generated object proof documentation section and generator-validated object-register proof:<ID> links                                                                               | <a href="#">1.1.1.1.3</a>   |
| <a href="#">LEG13</a> | Global no-go guardrail contract                                | Governance; Policy; Object | F0   | Planning/API hygiene                   | G             | planning infrastructure implemented in v26 master/generator; no Lean code semantics claimed    | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; tables/global_no_go_guardrails.tex; generated_exports/global_no_go_guardrails.generated.tsv                                                                                                                                                             | top-level generated global no-go guardrail section and inherited scratchpad field                                                                                                   | <a href="#">1.1.1.1.0</a>   |
| <a href="#">LEG14</a> | Concrete computable instance traversal contract                | Governance; Policy; Object | F3   | Workflow / operational executability   | G             | planning/workflow infrastructure present in YAML and generated views; no Lean declaration move | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; implementation_scratchpad                                                                                                                                                                                                                               | Planning workflow contract; optional later Lean audit mirror outside operational semantics                                                                                          | <a href="#">1.1.1.1.0.1</a> |
| <a href="#">LEG15</a> | Mathlib complication and noncomputable-boundary log            | Governance; Policy; Object | F3   | Workflow / mathlib audit               | G             | planning/workflow infrastructure present in YAML and generated views; no Lean declaration move | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; implementation_scratchpad                                                                                                                                                                                                                               | Planning workflow contract; optional later static audit script output                                                                                                               | <a href="#">1.1.1.1.0.1</a> |
| <a href="#">LEG16</a> | Semi-automated existing-code documentation coverage            | Reference; Gate; Object    | F3   | Secondary long-term meta-documentation | K             | future secondary documentation target; no operational code                                     | repo_inventory/raw_export/*.json; generated_exports/*.tsv; tables/object_proof_documentation.tex                                                                                                                                                                                                                      | generated documentation coverage view; optional future non-operational MetaAnalysis/DocumentationCoverage.*                                                                         | <a href="#">19.1</a>        |
| <a href="#">LEG17</a> | Derivation-graph geometry diagnostics                          | Reference; Gate; Object    | F3   | Secondary long-term meta-analysis      | K             | future generated meta-analysis target; no operational Lean code                                | repo_inventory/raw_export/import_edges.json; figures/*.dot/pdf                                                                                                                                                                                                                                                        | generated derivation-geometry view; optional future MetaAnalysis/DerivationGraphGeometry.*                                                                                          | <a href="#">19.2</a>        |
| <a href="#">LEG18</a> | Open-system emergence-path forensics                           | Reference; Gate; Object    | F3   | Secondary long-term meta-analysis      | K             | future secondary target; blocked until A/C derivations are present                             | future A/C modules plus repo_inventory/raw_export path analysis                                                                                                                                                                                                                                                       | generated emergence-path forensic report; optional future MetaAnalysis/OpenSystemPathForensics.*                                                                                    | <a href="#">19.3</a>        |
| <a href="#">LEG19</a> | Reflective CNNA/formalization graph comparison boundary        | Reference; Gate; Object    | F3   | Secondary long-term meta-analysis      | K             | future secondary comparison target; no operational Lean code                                   | generated graph views and proofbook links                                                                                                                                                                                                                                                                             | optional future MetaAnalysis/ReflectiveComparisonBoundary.*                                                                                                                         | <a href="#">19.4</a>        |
| <a href="#">LEG20</a> | EXEC-2 rational ExecComplex boundary policy                    | Lean; Governance; Object   | F1   | Architecture governance                | G             | planning decision; no Lean code yet                                                            | masters/cnna_roadmap_master.yaml; format_decision EXEC-2; workflow_policy EXEC-2                                                                                                                                                                                                                                      | CNNA/PillarA/Operational/ExecComplexBoundaryPolicy.lean (future, if code-level record becomes useful)                                                                               | <a href="#">1.1.1.1.0.2</a> |
| <a href="#">LEG21</a> | Refactor regression and incremental migration protocol         | Lean; Governance; Object   | F1   | Architecture governance                | G             | planning decision; no Lean code yet                                                            | masters/cnna_roadmap_master.yaml; workflow_policy refactor regression                                                                                                                                                                                                                                                 | CNNA/PillarA/Arithmetic/Legacy/RefactorRegressionPolicy.lean (future optional audit record)                                                                                         | <a href="#">1.1.1.1.0.3</a> |
| <a href="#">LEG22</a> | Lean/mathlib version migration policy                          | Lean; Governance; Object   | F1   | Architecture governance                | G             | planning decision; no Lean code yet                                                            | masters/cnna_roadmap_master.yaml; workflow_policy version management                                                                                                                                                                                                                                                  | CNNA/PillarA/Architecture/VersionPolicy.lean (future optional audit record)                                                                                                         | <a href="#">1.1.1.1.0.4</a> |

| ID                    | Object                                                             | Context class              | Tier | Layer                                       | Object status | Code status                                                        | Current code modules / derivation site                                                                                                                                                                | Neutral target location / canonical API                                                               | Phase                        |
|-----------------------|--------------------------------------------------------------------|----------------------------|------|---------------------------------------------|---------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------|
| <a href="#">LEG23</a> | Phase-type-specific scratchpad schema and governance retrospective | Documentation; Object      | F3   | Documentation/governance                    | G             | planning decision; generator updated                               | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py scratchpad renderer                                                                                                                      | generated implementation scratchpad view; future schema validator                                     | <a href="#">1.1.1.1.0.5</a>  |
| <a href="#">LEG24</a> | Phase 3 to Phase 4 consumer-contract direction                     | Lean; Governance; Object   | F1   | Architecture governance                     | G             | planning decision; no Lean code yet                                | masters/cnna_roadmap_master.yaml phase goals 3.6 and 4.1                                                                                                                                              | CNNA/PillarA/Schur/BoundaryOperatorProfile.lean plus CNNA/PillarA/Operational/ExecBoundaryMatrix.lean | <a href="#">1.1.1.1.0.6</a>  |
| <a href="#">LEG25</a> | IdealThread/CutSpec.full extremal-route contract                   | Governance; Policy; Object | F1   | Workflow / route governance                 | W             | planning decision closed; no Lean source changed                   | CNNA/PillarA/Foundation/SubstrateClass.lean; CNNA/PillarA/ToC/Contract.lean; CNNA/PillarA/Finite/CutSpec.lean; CNNA/PillarA/Arithmetic/BoundarySource/ExistingIdealThreadNoGo.lean                    | YAML master plus future route-contract modules if implementation justifies them                       | <a href="#">1.1.1.1.0.7</a>  |
| <a href="#">S07</a>   | IdealThread singleton directed-family profile                      | Lean; Source; Object       | F1   | ToC / substrate profile                     | G             | source theorems present; neutral public API not yet extracted      | CNNA/PillarA/Foundation/SubstrateClass.lean; CNNA/PillarA/ToC/Contract.lean; CNNA/PillarA/Arithmetic/BoundarySource/ExistingIdealThreadNoGo.lean                                                      | CNNA/PillarA/ToC/IdealThreadProfile.lean                                                              | <a href="#">1.2.7</a>        |
| <a href="#">S08</a>   | CutSpec.full maximal-carrier profile                               | Lean; Source; Object       | F1   | Finite approximant / cut profile            | G             | source theorem present; neutral public API not yet extracted       | CNNA/PillarA/Finite/CutSpec.lean; CNNA/PillarA/Arithmetic/Core/FullBoundaryAudit.lean                                                                                                                 | CNNA/PillarA/Finite/FullCutProfile.lean                                                               | <a href="#">1.2.8</a>        |
| <a href="#">BR06</a>  | Boundary-route extrema contract                                    | Lean; Source; Object       | F1   | Schur/DtN route governance                  | G             | planned neutral route contract; source evidence exists in audits   | CNNA/PillarA/Arithmetic/BoundarySource/ExistingIdealThreadNoGo.lean; CNNA/PillarA/Arithmetic/Core/FullBoundaryAudit.lean                                                                              | CNNA/PillarA/Schur/BoundaryRouteContract.lean                                                         | <a href="#">3.0</a>          |
| <a href="#">BR07</a>  | Thread-recursion arithmetic-shadow audit profile                   | Lean; Source; Object       | F1   | Operational arithmetic / route audit        | Y             | recurrence code exists but route provenance still needs audit      | CNNA/PillarA/Arithmetic/Schur/RecursiveSchurSource.lean; CNNA/PillarA/Arithmetic/Schur/MobiusTransfer.lean; CNNA/PillarA/Arithmetic/Schur/CharpolyExec.lean; CNNA/PillarA/Arithmetic/Smoke/*SM9*.lean | CNNA/PillarA/Operational/ThreadRecursionAudit.lean                                                    | <a href="#">4.0</a>          |
| <a href="#">BR09</a>  | Subsystem-induced invariant bifurcation profile                    | Lean; Target; Object       | F1   | Bridge                                      | K             | registered future explanatory bridge target; no canonical code yet | No canonical module yet.                                                                                                                                                                              | CNNA/PillarA/Bridge/SubsystemSchurInvariantBifurcation.lean                                           | <a href="#">10.4.1</a>       |
| <a href="#">BR08</a>  | Horizontal/vertical Schur-duality bridge                           | Lean; Target; Object       | F1   | Bridge                                      | K             | registered future bridge target; no canonical code yet             | No canonical module yet; source audit must inspect Schur/DtN, Arithmetic/Smoke, Structural/CayleyDickson, and Transfer modules.                                                                       | CNNA/PillarA/Bridge/SchurHorizontalVerticalDuality.lean                                               | <a href="#">10.4.1</a>       |
| <a href="#">LEG26</a> | Reflective thread/full-boundary analogy diagnostic boundary        | Governance; Policy; Object | F4   | Secondary long-term diagnostics             | K             | not implemented; future diagnostic/documentation target            | repo_inventory/raw_export/*; future generated documentation views                                                                                                                                     | Generated documentation/diagnostic reports; no core Lean module unless a later theorem requires one.  | <a href="#">19.5</a>         |
| <a href="#">LEG27</a> | Horizontal/vertical Schur-duality dependency-order lock            | Governance; Policy; Object | F1   | Planning / bridge governance                | G             | planning lock closed; no Lean code changed                         | masters/cnna_roadmap_master.yaml                                                                                                                                                                      | Planning only; no Lean API target.                                                                    | <a href="#">1.1.1.1.0.8</a>  |
| <a href="#">LEG28</a> | Heuristic Eisenstein/Gauss lattice test-matrix boundary contract   | Governance; Policy; Object | F1   | Workflow / test governance                  | G             | planning decision lock only; no Lean code yet                      | masters/cnna_roadmap_master.yaml; generated workflow/phase/proofbook views                                                                                                                            | No API. Future tests must be placed by module-manifest gate.                                          | <a href="#">1.1.1.1.0.9</a>  |
| <a href="#">LEG29</a> | General early finite computation test-matrix contract              | Governance; Policy; Object | F1   | Planning / workflow / computation readiness | G             | planning contract only; no Lean code                               | masters/cnna_roadmap_master.yaml                                                                                                                                                                      | Workflow policy plus Phase 4.0.1/O07 test-matrix profile.                                             | <a href="#">1.1.1.1.0.10</a> |
| <a href="#">L09</a>   | Static branching Eisenstein/Gauss lattice test matrix              | Reference; Gate; Object    | F2   | Lattice / test profile                      | Y             | not implemented; planned test/profile object                       | Not yet; placement must be decided by Phase 1.1.1.1.1 and future 8.5 manifest.                                                                                                                        | A compact neutral test/profile module only after semantic placement audit.                            | <a href="#">8.5</a>          |
| <a href="#">L10</a>   | Dynamic branching and L_max lattice test sweep                     | Reference; Gate; Object    | F2   | Lattice / test profile                      | Y             | not implemented; planned test/profile object                       | Not yet; future placement must reuse canonical generator.                                                                                                                                             | Same test/profile surface as L09 unless module-manifest gate decides otherwise.                       | <a href="#">8.6</a>          |
| <a href="#">M08</a>   | Eisenstein/Gauss CM and j special-value test targets               | Reference; Gate; Object    | F2   | Modular / comparison target                 | Y             | not implemented; planned comparison/test target object             | Not yet; future mirror/comparison module after M05.                                                                                                                                                   | Mirror/comparison test target profile, not A-operational API.                                         | <a href="#">8.5</a>          |
| <a href="#">BR10</a>  | Hierarchical versus spiral embedding collision and closure audit   | Reference; Gate; Object    | F2   | Bridge / test audit                         | Y             | not implemented; planned test audit object                         | Not yet; future test/profile module.                                                                                                                                                                  | Consumer-only audit profile.                                                                          | <a href="#">8.6</a>          |

| ID                    | Object                                                              | Context class              | Tier | Layer                                                     | Object status | Code status                                                                 | Current code modules / derivation site                                                                                         | Neutral target location / canonical API                                                                                                 | Phase                        |
|-----------------------|---------------------------------------------------------------------|----------------------------|------|-----------------------------------------------------------|---------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <a href="#">LEG30</a> | Regime-typed early test-matrix contract                             | Governance; Policy; Object | F1   | Planning / workflow / regime governance                   | G             | planning contract only; no Lean code                                        | masters/cnna_roadmap_master.yaml; generated views                                                                              | Workflow policy and future O08 extension of the O07 test-matrix profile.                                                                | <a href="#">1.1.1.1.0.11</a> |
| <a href="#">O08</a>   | Regime-typed finite computation test matrix profile                 | Lean; Target; Object       | F2   | Operational / test / regime readiness                     | Y             | planned; no Lean implementation yet                                         | future module selected by Phase 1.1.1.1.1; must extend/reuse O07 surface                                                       | CNNA/PillarA/Arithmetic/Smoke or Generator test-matrix API; exact path pending module manifest.                                         | <a href="#">4.0.2</a>        |
| <a href="#">N18</a>   | Heegner regime-projection classification profile                    | Lean; Target; Object       | F1   | Number theory / regime comparison                         | Y             | planned; no implementation yet                                              | no current module; may consume old HeegnerDiscriminantBridge only after neutralization                                         | CNNA/PillarA/NumberTheory/HeegnerRegimeProjection.lean or integrated profile after module manifest.                                     | <a href="#">7.2.1</a>        |
| <a href="#">BR11</a>  | IDEAL_infty projection and complement-family Heegner boundary       | Lean; Target; Object       | F1   | Bridge / regime / comparison boundary                     | Y             | planned bridge boundary only                                                | no current module                                                                                                              | Bridge/profile documentation or later small module if required by consumer API.                                                         | <a href="#">7.2.1</a>        |
| <a href="#">BR12</a>  | noOuter REAL-root Eisenstein/Gauss sentinel hypothesis              | Lean; Target; Object       | F1   | Bridge / test sentinel / regime                           | Y             | planned sentinel profile; no implementation yet                             | no current module; future placement with O08/L09/M08 after manifest                                                            | consumer-only sentinel row/profile under the test matrix surface                                                                        | <a href="#">4.0.2</a>        |
| <a href="#">LEG31</a> | Computable scalar architecture contract                             | Governance; Policy; Object | F0   | governance                                                | G             | planning lock                                                               | masters/cnna_roadmap_master.yaml; repo_snapshot audit paths                                                                    | global scalar architecture policy                                                                                                       | <a href="#">1.1.1.1.0.12</a> |
| <a href="#">LEG32</a> | Derived-only exact quadratic provenance contract                    | Governance; Policy; Object | F0   | governance/ scalar                                        | G             | planning lock                                                               | masters/cnna_roadmap_master.yaml                                                                                               | global scalar provenance policy; later ExactQuadraticScalar schema and discriminant-instantiation bridge                                | <a href="#">1.1.1.1.0.13</a> |
| <a href="#">LEG33</a> | Phase 5-10 bifurcation and dependency repair contract               | Governance; Policy; Object | F1   | Planning/ governance                                      | G             | planning contract generated in YAML/ PDF; no Lean object                    | masters/cnna_roadmap_master.yaml                                                                                               | planning/generated views only                                                                                                           | <a href="#">1.1.1.1.0.14</a> |
| <a href="#">O09</a>   | Required exact quadratic operational scalar schema                  | Lean; Target; Object       | F1   | operational scalar                                        | Y             | planned                                                                     | none yet; candidate informed by Mathlib.Algebra.QuadraticAlgebra and future manifest                                           | CNNA.PillarA.Foundation.ExactQuadraticScalar or a natural existing module if manifest proves semantic fit                               | <a href="#">1.2.1.2</a>      |
| <a href="#">O10</a>   | Proof-carrying discriminant-indexed quadratic scalar instance       | Lean; Target; Object       | F1   | operational scalar/ arithmetic bridge                     | Y             | planned                                                                     | none yet; future module names require Phase 1.1.1.1.1 manifest                                                                 | candidate discriminant-to-quadratic scalar instantiation profile/bridge adjacent to operational discriminant and exact quadratic schema | <a href="#">4.5.1</a>        |
| <a href="#">BR13</a>  | ExecComplex to exact-quadratic projection bridge                    | Lean; Target; Object       | F1   | bridge                                                    | Y             | planned                                                                     | none yet                                                                                                                       | candidate bridge API after scalar-layer pilot                                                                                           | <a href="#">4.0.3</a>        |
| <a href="#">BR14</a>  | Discriminant-source to exact-quadratic algebra instantiation bridge | Lean; Target; Object       | F1   | bridge                                                    | Y             | planned                                                                     | none yet                                                                                                                       | future scalar/discriminant bridge after module manifest                                                                                 | <a href="#">4.5.1</a>        |
| <a href="#">CD9</a>   | Cayley-Dickson C-layer noncircularity audit                         | Reference; Gate; Object    | F1   | Cayley-Dickson/bridge                                     | K             | planned                                                                     | CNNA/PillarA/Arithmetic/CayleyDickson/ArithmeticCDBridge.lean                                                                  | future BR/CD bridge only after BR08/CD8                                                                                                 | <a href="#">10.4.3</a>       |
| <a href="#">LEG34</a> | Phase-object-link policy contract                                   | Governance; Policy; Object | F1   | LEG / planning infrastructure                             | G             | planned/ documentation-generator infrastructure closed in v38; no Lean code | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated phase table                                            | phase object_link_policy fields; generated phase-object-link-policy table; validator rule in generate_tables.py                         | <a href="#">1.1.1.1.0.15</a> |
| <a href="#">LEG35</a> | Ontic subsystem-minimality contract                                 | Governance; Policy; Object | F0   | LEG / global ontology and planning infrastructure         | G             | planning/ global-postulate infrastructure closed in v39; no Lean code       | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; tables/global_postulates.tex; global no-go table; proof dossiers | global_postulates.POST1; LEG35 object dossier; GNG22; ontic subsystem-minimality workflow                                               | <a href="#">1.1.1.1.0.16</a> |
| <a href="#">DBR0</a>  | Database-ready CNNA review and documentation structure              | Documentation; Object      | F3   | Main side objective / documentation-review infrastructure | G             | planning/tool infrastructure added in V1.07; no Lean code                   | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated exports and PDF views                                  | future database-ready schema/export layer; optional Repository/scripts/db_export_*; generated review/documentation views                | <a href="#">20</a>           |



| ID                    | Object                                                                      | Context class                | Tier | Layer                                                               | Object status | Code status                                                                                                               | Current code modules / derivation site                                                                                                                                                                                                                 | Neutral target location / canonical API                                                                                                                                                                                            | Phase                     |
|-----------------------|-----------------------------------------------------------------------------|------------------------------|------|---------------------------------------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <a href="#">DBR1</a>  | Expert-review and no-infallibility epistemic rule                           | Documentation; Object        | F1   | Global documentation/review governance                              | G             | planning/tool governance added in V1.07; no Lean code                                                                     | masters/cnna_roadmap_master.yaml; generated_exports/global_no_go_guardrails.generated.tsv; generated PDF                                                                                                                                               | global documentation maxim and review/audit ledger semantics                                                                                                                                                                       | <a href="#">20.1</a>      |
| <a href="#">DBR2</a>  | Database-ready YAML schema and round-trip target                            | Tool; Infrastructure; Object | F1   | Future tool/database infrastructure                                 | K             | future secondary tooling target; no implementation yet                                                                    | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv                                                                                                                                                                                              | future database schema/export scripts and round-trip validation reports                                                                                                                                                            | <a href="#">20.2</a>      |
| <a href="#">DBR3</a>  | Discipline-separated expert-review matrix                                   | Documentation; Object        | F3   | Future review infrastructure                                        | K             | future secondary review target; no implementation yet                                                                     | masters/cnna_roadmap_master.yaml; object_proof_dossiers section                                                                                                                                                                                        | future review_matrix.generated.tsv/json/tex                                                                                                                                                                                        | <a href="#">20.3</a>      |
| <a href="#">DBR4</a>  | Blue/reference-theory audit ledger                                          | Documentation; Object        | F3   | Future reference-theory audit infrastructure                        | K             | future audit target; no implementation yet                                                                                | masters/cnna_roadmap_master.yaml; generated_exports/status_legend.generated.tsv                                                                                                                                                                        | future blue_reference_audit.generated.tsv/json/tex                                                                                                                                                                                 | <a href="#">20.4</a>      |
| <a href="#">DBR5</a>  | Interactive/online database-backed documentation views                      | Tool; Infrastructure; Object | F1   | Future interactive documentation infrastructure                     | K             | future interactive documentation target; no implementation yet                                                            | generated_exports/*.tsv; figures/*.dot/pdf; future DB/export scripts                                                                                                                                                                                   | future online/interactable review/documentation layer                                                                                                                                                                              | <a href="#">20.5</a>      |
| <a href="#">DBR6</a>  | Context-filtered documentation slice generator                              | Generated; Artifact; Object  | F3   | Generated-secondary documentation/review tooling                    | G             | implemented as V1.08 documentation generator; no Lean code                                                                | scripts/generate_context_doc.py; masters/cnna_roadmap_master.yaml; context_docs/dtn_matrix/* sample output                                                                                                                                             | python3 Repository/scripts/generate_context_doc.py "DtN Matrix" --profile dtn-matrix --slug dtn_matrix                                                                                                                             | <a href="#">20.6</a>      |
| <a href="#">DBR7</a>  | Stats front section and tool-interface documentation layer                  | Generated; Artifact; Object  | F3   | Generated-secondary documentation/review tooling                    | G             | implemented as V1.09 documentation generator/display update; no Lean code                                                 | scripts/generate_tables.py; masters/cnna_roadmap_master.yaml; tables/stats.tex; generated_exports/project_stats.generated.tsv                                                                                                                          | python3 Repository/scripts/generate_tables.py; front PDF section Stats                                                                                                                                                             | <a href="#">20.7</a>      |
| <a href="#">LEG36</a> | Semantic module consolidation ledger                                        | Governance; Policy; Object   | F1   | Planning/API hygiene and refactor governance                        | Y             | active audit target; no Lean source move yet                                                                              | masters/cnna_roadmap_master.yaml; repo_snapshot/CNNA/**/*.*.lean; repo_inventory/raw_export/*.json; generated_exports/*.tsv; SEMANTIC_MODULE_CONSOLIDATION_AUDIT_REPORT.md                                                                             | planning master and future generated consolidation ledger; possible later neutral Lean facades only after local build-proved migration                                                                                             | <a href="#">1.1.1.1.5</a> |
| <a href="#">DBR8</a>  | Root artifact retention and cleanup policy                                  | Generated; Artifact; Object  | F3   | Generated-secondary documentation/review tooling                    | G             | implemented as V1.13 root cleanup tooling; no Lean code                                                                   | scripts/generate_full_document.py; scripts/clean_generated_root_artifacts.py; README.md; archive/reports/; masters/cnna_roadmap_master.yaml                                                                                                            | python3 Repository/scripts/generate_full_document.py --compile; python3 Repository/scripts/clean_generated_root_artifacts.py --apply                                                                                               | <a href="#">20.8</a>      |
| <a href="#">DBR9</a>  | Current README and workflow-orientation manual                              | Generated; Artifact; Object  | F3   | Generated-secondary documentation/review tooling                    | G             | implemented as V1.14 README/documentation update; no Lean code                                                            | README.md; masters/cnna_roadmap_master.yaml; archive/reports/V1_14_README_WORKFLOW_ORIENTATION_REPORT.md                                                                                                                                               | README.md as root entry point; command examples for scripts/generate_tables.py, scripts/generate_full_document.py, scripts/generate_context_doc.py, scripts/build_export_script_v1.8.py, scripts/clean_generated_root_artifacts.py | <a href="#">20.9</a>      |
| <a href="#">DBR10</a> | Root Lean/Lake project-control files                                        | Generated; Artifact; Object  | F3   | Build reproducibility and generated-secondary documentation/tooling | G             | implemented as V1.15 root build-entry/toolchain update; no Lean source content changed inside repo_snapshot/CNNA          | CNNA.lean; lakefile.lean; lake-manifest.json; lean-toolchain; Repository/scripts/clean_generated_root_artifacts.py; README.md; Repository/masters/cnna_roadmap_master.yaml                                                                             | Run local Lean build from the package/project root with lake build; Lake uses srcDir := "Repository/repo_snapshot".                                                                                                                | <a href="#">20.10</a>     |
| <a href="#">DBR11</a> | Root license, single-root README, credits and assistant-requirement surface | Generated; Artifact; Object  | F3   | Build reproducibility and generated-secondary documentation/tooling | G             | implemented as V1.17 documentation/license/build-surface update; no Lean source content changed inside repo_snapshot/CNNA | README.md; LICENSE; Repository/scripts/generate_full_document.py; Repository/scripts/generate_context_doc.py; Repository/scripts/generate_tables.py; Repository/scripts/clean_generated_root_artifacts.py; Repository/masters/cnna_roadmap_master.yaml | README.md and LICENSE in package root; python3 Repository/scripts/generate_full_document.py --compile for PDF title/author credits; no maintained README.md under Repository/.                                                     | <a href="#">20.11</a>     |

| ID                    | Object                                                              | Context class               | Tier | Layer                                                                  | Object status | Code status                                                                                                                        | Current code modules / derivation site                                                                                                                                                                                                                                                    | Neutral target location / canonical API                                                                                                                                                                                    | Phase                       |
|-----------------------|---------------------------------------------------------------------|-----------------------------|------|------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <a href="#">DBR12</a> | Generated PDF section-pagebreak layout invariant                    | Generated; Artifact; Object | F3   | Generated-secondary documentation/review tooling                       | G             | implemented as V1.18 PDF generator/layout update; no Lean code                                                                     | Repository/scripts/generate_full_document.py; Repository/scripts/generate_context_doc.py; Repository/masters/cnna_roadmap_master.yaml; README.md; Repository/archive/reports/V1_18_SECTION_PAGEBREAKS_REPORT.md                                                                           | Full docs: python3 scripts/generate_full_document.py --compile. Context docs: python3 scripts/generate_context_doc.py "<topic>" --profile <profile> --slug <slug>. Both generators apply section pagebreaks automatically. | <a href="#">20.12</a>       |
| <a href="#">DBR13</a> | Phase status release policy and red-lock ledger                     | Generated; Artifact; Object | F3   | Planning/governance and generated-secondary documentation/tooling      | G             | implemented as V1.19 planning/governance and generator-validation update; no Lean source content changed inside repo_snapshot/CNNA | Repository/masters/cnna_roadmap_master.yaml; Repository/scripts/generate_tables.py; README.md; Repository/archive/reports/V1_19_PHASE_STATUS_RELEASE_POLICY_REPORT.md                                                                                                                     | python3 Repository/scripts/generate_tables.py validates that no derivation phase remains white and that exactly one yellow phase is active.                                                                                | <a href="#">20.13</a>       |
| <a href="#">DBR14</a> | Static documentation explanations and module-change appendix policy | Generated; Artifact; Object | F3   | Generated-secondary documentation/review tooling                       | G             | implemented as V1.20 documentation generator/static prose update; no Lean code                                                     | Repository/scripts/generate_full_document.py; Repository/scripts/generate_context_doc.py; Repository/scripts/generate_tables.py; Repository/masters/cnna_roadmap_master.yaml; README.md; Repository/archive/reports/V1_20_DOC_SECTION_EXPLANATIONS_AND_MODULE_MANIFEST_APPENDIX_REPORT.md | Full docs and context docs generated by existing scripts; module-change manifest appears as technical appendix after the primary phase/scratchpad sections.                                                                | <a href="#">20.14</a>       |
| <a href="#">LEG37</a> | Current source intake completeness ledger                           | Governance; Policy; Object  | F1   | Planning/API hygiene and pre-code source audit                         | Y             | active audit target; no Lean source move yet                                                                                       | repo_inventory/raw_export/*.json; generated_exports/*consumption*.tsv; masters/cnna_roadmap_master.yaml                                                                                                                                                                                   | masters/cnna_roadmap_master.yaml plus future generated source-intake ledger/export                                                                                                                                         | <a href="#">1.1.1.1.5.1</a> |
| <a href="#">LEG38</a> | Neutral module-name and facade migration map                        | Governance; Policy; Object  | F1   | Planning/API hygiene and compatibility-preserving refactor design      | Y             | planned after LEG37; no Lean move yet                                                                                              | repo_snapshot/CNNA/**/*lean; masters/cnna_roadmap_master.yaml                                                                                                                                                                                                                             | future neutral module/facade map generated from YAML; old modules retained until build-proven migration                                                                                                                    | <a href="#">1.1.1.1.5.2</a> |
| <a href="#">LEG39</a> | Neutral refactor execution certificate                              | Governance; Policy; Object  | F1   | Build reproducibility and compatibility-preserving refactor governance | Y             | planned after LEG38; requires local build evidence when executed                                                                   | future modified Lean modules authorized by LEG38                                                                                                                                                                                                                                          | post-refactor root/import surface with compatibility facades                                                                                                                                                               | <a href="#">1.1.1.1.5.3</a> |
| <a href="#">LEG40</a> | Post-refactor inventory closure certificate                         | Governance; Policy; Object  | F1   | Generated inventory and object/proof-dossier reclassification          | Y             | planned after LEG39; no current Lean move                                                                                          | repo_inventory/raw_export/*.json after refactor; generated_exports/*.tsv                                                                                                                                                                                                                  | fresh post-refactor source truth map                                                                                                                                                                                       | <a href="#">1.1.1.1.5.4</a> |
| <a href="#">LEG41</a> | Pre-substantive-code release certificate                            | Governance; Policy; Object  | F1   | Release governance before new Lean content                             | Y             | planned final gate before 1.1.1.2 may become active                                                                                | masters/cnna_roadmap_master.yaml; generated_exports/pre_implementation_phase_check_results.generated.tsv                                                                                                                                                                                  | release proof for 1.1.1.2 or upstream red blockers                                                                                                                                                                         | <a href="#">1.1.1.1.5.5</a> |
| <a href="#">DBR15</a> | Supabase/Postgres mirror feasibility and schema design              | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling                   | W             | planned DB/tooling object; no implementation script or live DB connection yet                                                      | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                  | Future Supabase/Postgres schemas internal.*, api.*, review.*, admin/import.* plus YAML/export importer and read-only web frontend contract.                                                                                | <a href="#">20.15</a>       |
| <a href="#">DBR16</a> | YAML/exports to Postgres import and roundtrip contract              | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling                   | W             | planned DB/tooling object; no implementation script or live DB connection yet                                                      | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                  | Future Supabase/Postgres schemas internal.*, api.*, review.*, admin/import.* plus YAML/export importer and read-only web frontend contract.                                                                                | <a href="#">20.16</a>       |
| <a href="#">DBR17</a> | Append-only snapshot and checksum model                             | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling                   | W             | planned DB/tooling object; no implementation script or live DB connection yet                                                      | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                  | Future Supabase/Postgres schemas internal.*, api.*, review.*, admin/import.* plus YAML/export importer and read-only web frontend contract.                                                                                | <a href="#">20.17</a>       |

| ID                    | Object                                                                         | Context class               | Tier | Layer                                                | Object status | Code status                                                                   | Current code modules / derivation site                                                                                                                                                                                                                                                             | Neutral target location / canonical API                                                                                                                                                                                 | Phase                   |
|-----------------------|--------------------------------------------------------------------------------|-----------------------------|------|------------------------------------------------------|---------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <a href="#">DBR18</a> | Public read-only API views and RLS policies                                    | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling | W             | planned DB/tooling object; no implementation script or live DB connection yet | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                           | Future Supabase/Postgres schemas internal.*, api.*, review.*, admin/import.* plus YAML/export importer and read-only web frontend contract.                                                                             | <a href="#">20.18</a>   |
| <a href="#">DBR19</a> | cnna-project.net web-frontend data contract                                    | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling | W             | planned DB/tooling object; no implementation script or live DB connection yet | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                           | Future Supabase/Postgres schemas internal.*, api.*, review.*, admin/import.* plus YAML/export importer and read-only web frontend contract.                                                                             | <a href="#">20.19</a>   |
| <a href="#">DBR20</a> | Reviewer annotation and future admin-write model                               | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling | W             | planned DB/tooling object; no implementation script or live DB connection yet | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                           | Future Supabase/Postgres schemas internal.*, api.*, review.*, admin/import.* plus YAML/export importer and read-only web frontend contract.                                                                             | <a href="#">20.20</a>   |
| <a href="#">DBR21</a> | Enlarged generated-PDF typography profile                                      | Generated; Artifact; Object | F3   | Generated-secondary documentation/review tooling     | G             | implemented as V1.23 PDF typography generator update; no Lean code            | Repository/scripts/generate_full_document.py; Repository/scripts/generate_context_doc.py; Repository/scripts/generate_tables.py; Repository/scripts/build_export_script_v1.8.py; README.md; Repository/masters/cnna_roadmap_master.yaml; Repository/archive/reports/V1_23_PDF_FONT_SCALE_REPORT.md | Full docs: python3 scripts/generate_full_document.py -compile. Context docs: python3 scripts/generate_context_doc.py "<topic>" -profile <profile> -slug <slug>. Both use the enlarged typography profile automatically. | <a href="#">20.21</a>   |
| <a href="#">DBR22</a> | Supabase live project discovery and empty-schema baseline audit                | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling | G             | closed read-only discovery baseline                                           | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future Repository/database/supabase/* artifacts                                                                                                                                                                                         | Supabase project aiekahdewadrejrrdajn; schemas internal/api/review/admin; public read-only api views; local/admin importer path.                                                                                        | <a href="#">20.22</a>   |
| <a href="#">DBR23</a> | Supabase initial schema migration package and RLS/read-only policy design      | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling | W             | planned DB/tooling object; no live DB write without explicit authorization    | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future Repository/database/supabase/* artifacts                                                                                                                                                                                         | Supabase project aiekahdewadrejrrdajn; schemas internal/api/review/admin; public read-only api views; local/admin importer path.                                                                                        | <a href="#">20.23</a>   |
| <a href="#">DBR24</a> | Supabase snapshot import package from YAML/generated exports                   | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling | W             | planned DB/tooling object; no live DB write without explicit authorization    | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future Repository/database/supabase/* artifacts                                                                                                                                                                                         | Supabase project aiekahdewadrejrrdajn; schemas internal/api/review/admin; public read-only api views; local/admin importer path.                                                                                        | <a href="#">20.24</a>   |
| <a href="#">DBR25</a> | Explicitly authorized live Supabase schema migration and first snapshot import | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling | W             | planned DB/tooling object; no live DB write without explicit authorization    | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future Repository/database/supabase/* artifacts                                                                                                                                                                                         | Supabase project aiekahdewadrejrrdajn; schemas internal/api/review/admin; public read-only api views; local/admin importer path.                                                                                        | <a href="#">20.25</a>   |
| <a href="#">DBR26</a> | Post-import read-only/RLS/advisor verification and web-frontend handoff        | Database; Mirror; Object    | F1   | Generated-secondary database/web publication tooling | W             | planned DB/tooling object; no live DB write without explicit authorization    | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future Repository/database/supabase/* artifacts                                                                                                                                                                                         | Supabase project aiekahdewadrejrrdajn; schemas internal/api/review/admin; public read-only api views; local/admin importer path.                                                                                        | <a href="#">20.26</a>   |
| <a href="#">GA0</a>   | Horn quaternion/geometric-algebra reference-gate packet                        | Reference; Gate; Object     | F2   | External reference/comparison gate                   | K             | planned comparison/reference target; no Lean module yet                       | No current Lean module; masters/cnna_roadmap_master.yaml scratchpad/source-crosscheck only.                                                                                                                                                                                                        | future CNNA/PillarA/Structural/CayleyDickson/ReferenceGates/QuaternionGAGate.lean if released                                                                                                                           | <a href="#">1.4.9</a>   |
| <a href="#">GA1</a>   | Hamilton unit and Pauli/Cartan sign-convention comparison profile              | Reference; Gate; Object     | F2   | Cayley-Dickson/quaternion reference                  | K             | planned comparison/reference target; no Lean module yet                       | Future comparison may inspect CNNA/PillarA/Structural/CayleyDickson/*.lean after neutral reintegration.                                                                                                                                                                                            | future QuaternionUnitComparisonProfile API under Structural/CayleyDickson/ReferenceGates                                                                                                                                | <a href="#">1.4.9.1</a> |
| <a href="#">GA2</a>   | Pauli bivector/pseudoscalar duality comparison profile                         | Reference; Gate; Object     | F2   | Cayley-Dickson/Clifford reference                    | K             | planned comparison/reference target; no Lean module yet                       | No current Lean module; future reference-gate only.                                                                                                                                                                                                                                                | future CliffordDualityComparisonProfile API under Structural/CayleyDickson/ReferenceGates                                                                                                                               | <a href="#">1.4.9.2</a> |

| ID    | Object                                                        | Context class                | Tier | Layer                                                                     | Object status | Code status                                                                    | Current code modules / derivation site                                                                                                                                                          | Neutral target location / canonical API                                                                                                                                         | Phase   |
|-------|---------------------------------------------------------------|------------------------------|------|---------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| GA3   | Complex-quaternion eight-basis multivector comparison profile | Reference; Gate; Object      | F2   | Biquaternion/geometric-algebra reference                                  | K             | planned comparison/reference target; no Lean module yet                        | Could later relate to 10.4.3 CD9 only after noncircular C-layer audit.                                                                                                                          | future BiquaternionMultivectorComparisonProfile API under ReferenceGates or bridge audit docs                                                                                   | 1.4.9.2 |
| GA4   | Boundary Pauli/Clifford comparison gate                       | Reference; Gate; Object      | F2   | Pillar B boundary algebra reference                                       | K             | planned comparison/reference target; no Lean module yet                        | PillarB empty; future CNNA/PillarB/QFT/BoundaryStarAlgebra.lean consumer output.                                                                                                                | future CNNA/PillarB/QFT/CliffordComparisonGate.lean                                                                                                                             | 13.2.1  |
| GA5   | Dirac/spacetime-algebra comparison target gate                | Reference; Gate; Object      | F2   | Pillar B AQFT/STA reference                                               | K             | planned comparison/reference target; no Lean module yet                        | PillarB empty; no current STA/Dirac module.                                                                                                                                                     | future CNNA/PillarB/AQFT/SpacetimeAlgebraComparisonGate.lean                                                                                                                    | 13.6.1  |
| GA6   | Geometric and conformal algebra side-comparison register      | Reference; Gate; Object      | F2   | Side comparison/reference theory                                          | K             | planned comparison/reference target; no Lean module yet                        | No current Lean module.                                                                                                                                                                         | future CNNA/Comparisons/GeometricAlgebra.lean or generated comparison ledger                                                                                                    | 18.5    |
| DBR27 | Phase/Object context-class taxonomy                           | Tool; Infrastructure; Object | F3   | Generated-secondary planning schema and OOP-style classification boundary | G             | implemented in V1.26 generated planning/tooling schema; no Lean source changed | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; scripts/generate_full_document.py; generated_exports/phase_roadmap.generated.tsv; generated_exports/object_register.generated.tsv | Context class field on every phase/object row plus context_class_taxonomy generated table; future Supabase schema should mirror it as phase_context_class/object_context_class. | 20.27   |

### Object table B: derivation, Exec path, and status

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| ID  | Object                        | Context class                   | Tier | Object status | Canonical CNNA definition                                                                                                                     | Natural provenance                                     | Computed from                                       | ExecComplex/computable path                                                                                                                                    | Priority | Phase |
|-----|-------------------------------|---------------------------------|------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|
| G00 | Generator provenance          | Lean; Source; Object            | F0   | G             | Canonical A-internal generator/provenance record; no pillar consumer generates its own sources.                                               | Generated substrate + proof-carrying Schur/Exec chain. | substrate, branching, level, partition, Schur data. | ExecComplex arises downstream; provenance itself is structural.                                                                                                | high     | 1.2.2 |
| H00 | A input handoff B->A          | Lean; Handoff; Contract; Object | F0   | Y             | BToAInputProfile.                                                                                                                             | Declared handoff input only.                           | B-derived theorem/data, if explicitly allowed.      | May carry non-Exec analytic data; must be typed.                                                                                                               | medium   | 2.3.1 |
| H01 | A input handoff C->A          | Lean; Handoff; Contract; Object | F0   | Y             | CToAInputProfile.                                                                                                                             | Declared handoff input only.                           | C-derived data.                                     | Typed; Exec mirror if numeric.                                                                                                                                 | medium   | 2.3.1 |
| H02 | A input handoff D->A          | Lean; Handoff; Contract; Object | F0   | Y             | DToAInputProfile.                                                                                                                             | Declared handoff input only.                           | D-derived data.                                     | Typed; may include time/dynamics.                                                                                                                              | medium   | 2.3.1 |
| H03 | A input handoff E->A          | Lean; Handoff; Contract; Object | F0   | Y             | EToAInputProfile.                                                                                                                             | Declared handoff input only.                           | E-derived data.                                     | Typed; not default.                                                                                                                                            | low      | 2.3.1 |
| O00 | ExecComplex                   | Lean; Source; Object            | F1   | W             | ExecComplex as rational-pair operational scalar floor QxQ; not a complete Complex layer. It owns executable rational complex arithmetic only. | Already in code.                                       | Rational pair / operational complex implementation. | Direct QxQ executable path. Algebraic irrational coordinates require O09 or mirror-only treatment; analytic Complex is reached only through ExecComplexBridge. | critical | 1.2.1 |
| G01 | substrate: regular branching  | Lean; Source; Object            | F0   | Y             | RegularSubstrateProfile.                                                                                                                      | GeneratedBranchFamily / RegularCell-path.              | code-b, arity, level carrier.                       | No scalar; generates Exec-matrix indices.                                                                                                                      | high     | 1.2.3 |
| G02 | substrate: variable branching | Lean; Source; Object            | F0   | Y             | VariableSubstrateProfile.                                                                                                                     | LevelVariableCell/Policy-path.                         | Branching policy beta, level history.               | No scalar; Exec-Matrix downstream.                                                                                                                             | medium   | 1.2.4 |

| ID  | Object                                       | Context class              | Tier | Object status | Canonical CNNA definition                                                 | Natural provenance                                           | Computed from                                    | ExecComplex/computable path                                              | Priority | Phase                 |
|-----|----------------------------------------------|----------------------------|------|---------------|---------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------|----------|-----------------------|
| G03 | Level / Cutoff                               | Lean;<br>Source;<br>Object | F0   | Y             | LevelProfile with selectedLevel/cutoff/levelSpec.                         | Generator request or existing Step1/<br>Handoff core.        | Request + admissibility proof.                   | Nat;<br>optional ExecComplex-mirror for numeric reports.                 | high     | <a href="#">1.2.5</a> |
| G04 | Carrier / index type                         | Lean;<br>Source;<br>Object | F0   | Y             | CarrierProfile: carrier, Fintype, DecidableEq, cardinality, equiv-to-Fin. | LevelProfile + generated level-history.                      | GeneratedLevelHistoryIndex or neutral Carrier.   | Not Skalar;<br>Matrix via Carrier in ExecComplex.                        | high     | <a href="#">1.2.6</a> |
| G05 | Boundary / interface / interior partition    | Lean;<br>Source;<br>Object | F0   | Y             | PartitionProfile.                                                         | GeneratedBoundaryInteriorPartition / sources near SM3a/SM3b. | Carrier + cut/BoundarySpec.                      | indicator values operational as Bool/Finset; matrix entries Exec.        | high     | <a href="#">1.3.1</a> |
| G06 | Regime:<br>noOuterEnvironment                | Lean;<br>Source;<br>Object | F0   | Y             | RegimeProfile.noOuter.                                                    | Split-/environment proofs from A, not from B.                | Partition + complement/<br>environment proof.    | No Exec-Skalar;<br>affects operator/dynamics Profile.                    | high     | <a href="#">1.3.2</a> |
| G07 | Regime:<br>hasOuterEnvironment               | Lean;<br>Source;<br>Object | F0   | Y             | RegimeProfile.hasOuter.                                                   | Complementfamilien/outer environment proof.                  | Approximant + complement proof.                  | No Exec-Skalar;<br>open matrix/channel Profile downstream.               | high     | <a href="#">1.3.2</a> |
| G08 | Complement families:<br>UV tail              | Lean;<br>Source;<br>Object | F0   | R             | UVComplementProfile.                                                      | Generated complement/provenance;<br>not consumer-invented.   | Approximant + infinite carrier.                  | Structural;<br>effect can in Exec-Operator mixed sein.                   | high     | <a href="#">1.3.3</a> |
| G09 | Complement families:<br>IR/parent-path tail  | Lean;<br>Source;<br>Object | F0   | R             | IRComplementProfile.                                                      | Generated complement/provenance.                             | Approximant + parent/level structure.            | Structural;<br>Exec downstream via operator effect.                      | medium   | <a href="#">1.3.3</a> |
| G10 | Complement families:<br>sideways environment | Lean;<br>Source;<br>Object | F0   | R             | SideEnvironmentProfile.                                                   | Generated complement/provenance.                             | Branching + nonselected sectors.                 | Structural;<br>Exec downstream via operator effect.                      | medium   | <a href="#">1.3.3</a> |
| CD0 | Cayley-Dickson source                        | Lean;<br>Source;<br>Object | F1   | B             | CayleyDicksonSourceProfile.                                               | Structural/CayleyDickson + Handoff adapter.                  | carrier/regularization/schur from closed A data. | Not inherently Exec;<br>arithmetic compatibility path must state mirror. | high     | <a href="#">1.4.1</a> |
| CD1 | Scalar part                                  | Lean;<br>Source;<br>Object | F1   | B             | CDScalarPartProfile.                                                      | CD source.                                                   | CD algebra data.                                 | Exec bridge only if coordinates available.                               | medium   | <a href="#">1.4.2</a> |
| CD2 | Cayley-Dickson star/<br>conjugation          | Lean;<br>Source;<br>Object | F1   | B             | CDStarProfile.                                                            | CD source/scalar part.                                       | CD algebra data.                                 | Coordinate-level mirror optional.                                        | medium   | <a href="#">1.4.3</a> |
| CD3 | CD normSq                                    | Lean;<br>Source;<br>Object | F1   | B             | CDNormSqProfile.                                                          | star + product.                                              | CD algebra.                                      | Exec bridge only after arithmetic compatibility.                         | medium   | <a href="#">1.4.4</a> |
| CD4 | CD multiplication/<br>product                | Lean;<br>Source;<br>Object | F1   | B             | CDProductProfile.                                                         | CD source.                                                   | CD coordinates.                                  | Coordinate calculations possible;<br>not automatic ExecComplex.          | medium   | <a href="#">1.4.5</a> |
| CD5 | Alternative law                              | Lean;<br>Source;<br>Object | F1   | B             | CDAlternativeLawCertificate.                                              | CD product.                                                  | Product associator.                              | Proof-level;<br>computations by coordinates if available.                | medium   | <a href="#">1.4.6</a> |
| CD6 | Norm multiplicativity                        | Lean;<br>Source;<br>Object | F1   | B             | CDNormMultiplicativityCertificate.                                        | CD norm/product.                                             | normSq, product.                                 | Proof-level;<br>arithmetic bridge later.                                 | medium   | <a href="#">1.4.7</a> |
| CD7 | Hurwitz stop / R,C,H,O classification        | Lean;<br>Source;<br>Object | F1   | B             | HurwitzStopProfile.                                                       | CD duties closed.                                            | alternative + norm + division.                   | Not Exec;<br>structural theorem.                                         | medium   | <a href="#">1.4.8</a> |
| S00 | Dirichlet-Laplacian                          | Lean;<br>Source;<br>Object | F0   | Y             | DirichletLaplacianProfile.                                                | Generated Dirichlet data.                                    | Partition + weights + adjacency.                 | Matrix carrier carrier ExecComplex or bridge from Real to ExecComplex.   | high     | <a href="#">3.1</a>   |
| S01 | Interior block                               | Lean;<br>Source;<br>Object | F0   | B             | InteriorBlockProfile.                                                     | Dirichlet-Laplacian + partition.                             | Interior indices + matrix.                       | ExecComplex-matrix or proven bridge.                                     | high     | <a href="#">3.2</a>   |
| S02 | Interior inverse candidate                   | Lean;<br>Source;<br>Object | F0   | B             | InteriorInverseCandidateProfile.                                          | SM3db trace/fold candidate chain.                            | Trace + accumulated entry fold.                  | Matrix ExecComplex/Real with bridge.                                     | high     | <a href="#">3.3</a>   |



| ID  | Object                                         | Context class                 | Tier | Object status | Canonical CNNA definition                                                                                                                                                                                                                                                     | Natural provenance                                                                                                                                    | Computed from                                                                                                                                              | ExecComplex/computable path                                                                                                                                                 | Priority                                    | Phase |
|-----|------------------------------------------------|-------------------------------|------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------|
| S03 | Two-sided inverse certificate                  | Lean;<br>Source;<br>Object    | F0   | B             | Two-sided inverse certificate.                                                                                                                                                                                                                                                | Candidate + product identities.                                                                                                                       | Interior block + candidate.                                                                                                                                | Matrixprodukt-Statements operational pruefbar.                                                                                                                              | high                                        | 3.4   |
| S04 | InteriorEliminationData                        | Lean;<br>Source;<br>Object    | F0   | B             | InteriorEliminationProfile.                                                                                                                                                                                                                                                   | TwoSidedInv + partition + blocks.                                                                                                                     | Interior block, inverse, couplings.                                                                                                                        | ExecComplex Schur-formula downstream.                                                                                                                                       | high                                        | 3.5   |
| S05 | DtN boundary operator                          | Lean;<br>Source;<br>Object    | F0   | B             | BoundaryOperatorProfile.                                                                                                                                                                                                                                                      | Interior elimination / Schur complement.                                                                                                              | Laplacian blocks + inverse certificate. Route must be checked against BR06 before public-final promotion.                                                  | Matrix boundary boundary ExecComplex; Pflicht for Part 2.                                                                                                                   | high                                        | 3.6   |
| S06 | GeneralizedDtN / MultiSchur                    | Lean;<br>Source;<br>Object    | F0   | B             | GeneralizedBoundaryOperatorProfile.                                                                                                                                                                                                                                           | Generated generalized DtN/MultiSchur chain.                                                                                                           | Boundary operator + additional ports.                                                                                                                      | ExecComplex matrix profile.                                                                                                                                                 | high                                        | 3.7   |
| O01 | Boundary matrix over ExecComplex               | Lean;<br>Source;<br>Object    | F1   | B             | ExecMatrixProfile.                                                                                                                                                                                                                                                            | DtN/MultiSchur -> Exec mirror.                                                                                                                        | Boundary operator + Exec bridge.                                                                                                                           | Matrix carrier carrier ExecComplex.                                                                                                                                         | high                                        | 4.1   |
| O02 | Determinant coefficient recurrence             | Lean;<br>Source;<br>Object    | F1   | B             | CoefficientRecurrenceProfile.                                                                                                                                                                                                                                                 | SM9e recurrence.                                                                                                                                      | Exec matrix + recurrence.                                                                                                                                  | Nat -> ExecComplex.                                                                                                                                                         | high                                        | 4.2   |
| O03 | Operational polynomial profile                 | Lean;<br>Source;<br>Object    | F1   | B             | OperationalPolynomialProfile.                                                                                                                                                                                                                                                 | SM9c/SM9d/SM9e chain.                                                                                                                                 | Coefficient recurrence.                                                                                                                                    | Coefficients ExecComplex.                                                                                                                                                   | high                                        | 4.3   |
| O04 | Quadratic coefficient window                   | Lean;<br>Source;<br>Object    | F1   | B             | QuadraticWindowCertificate.                                                                                                                                                                                                                                                   | SM9g1 from recurrence/window.                                                                                                                         | CoefficientAt source.                                                                                                                                      | a,b,c : ExecComplex.                                                                                                                                                        | high                                        | 4.4   |
| O05 | Operational discriminant D_op                  | Lean;<br>Source;<br>Object    | F1   | B             | OperationalDiscriminantCertificate.                                                                                                                                                                                                                                           | SM9h.                                                                                                                                                 | QuadraticWindow.                                                                                                                                           | discriminantValue : ExecComplex.                                                                                                                                            | critical                                    | 4.5   |
| O06 | Comparison target policy                       | Reference;<br>Gate;<br>Object | F2   | G             | ComparisonTargetPolicy. Also governs heuristic lattice test targets: comparison rows may not feed back into operational generation.                                                                                                                                           | theory register + target-source proof.                                                                                                                | D_op + target definition. BR07 must audit thread-recursion provenance before comparison interpretation.                                                    | Operational side may use only rational/integer comparison invariants computed from D_op. General algebraic target data lives in mirror/non-operational layers under EXEC-2. | high                                        | 4.6   |
| O07 | General finite computation test matrix profile | Lean;<br>Target;<br>Object    | F2   | Y             | Reusable proof-carrying finite test matrix over code-defined generated instances, branch modes, L_max rows, operational output witnesses, and optional comparison sentinel columns. v34 adds O08 as the regime-typed extension; O07 remains the generic/non-specialized base. | canonical smoke-test generator; concrete computable instance traversal; operational arithmetic path; generalization of L09/L10 special lattice tests. | code-defined generator/branching API, finite L_max, static/dynamic branch-mode tags, finite carriers/matrices/profiles, executable output witness records. | Core fields are operational finite/rational/integer; mirror comparison columns are explicitly non-operational.                                                              | early before specialized computation claims | 4.0.1 |
| N00 | Fundamental discriminant D_fund                | Lean;<br>Source;<br>Object    | F1   | G             | FundamentalDiscriminantProfile.                                                                                                                                                                                                                                               | Normalization of D_op.                                                                                                                                | D_op, square factor/congreence proof.                                                                                                                      | Integer/rat mirror; ExecComplex embedding for reports.                                                                                                                      | high                                        | 5.1   |
| N01 | Conductor f                                    | Lean;<br>Source;<br>Object    | F1   | G             | ConductorProfile.                                                                                                                                                                                                                                                             | Discriminant normalization.                                                                                                                           | D_op, D_fund.                                                                                                                                              | Nat/Int; ExecComplex mirror optional.                                                                                                                                       | high                                        | 5.1   |
| N02 | Quadratic field K=Q(sqrt(D))                   | Lean;<br>Source;<br>Object    | F1   | Y             | QuadraticFieldProfile.                                                                                                                                                                                                                                                        | D_fund.                                                                                                                                               | D_fund + sign/non-square proof.; exact quadratic coordinates only through O09/O10/BR14 when required                                                       | Not purely ExecComplex; operational embedding optional. Derived-quadratic update: no free sqrt(D); exact coordinates require proof-carrying D-source.                       | medium                                      | 5.2   |
| N03 | Quadratic order O_D                            | Lean;<br>Source;<br>Object    | F1   | Y             | QuadraticOrderProfile.                                                                                                                                                                                                                                                        | D_fund + conductor.                                                                                                                                   | K, f.                                                                                                                                                      | Coordinates can be operational; ring laws proof layer.                                                                                                                      | high                                        | 5.3   |
| N04 | Integral basis / omega_D                       | Lean;<br>Source;<br>Object    | F1   | Y             | IntegralBasisProfile.                                                                                                                                                                                                                                                         | OrderProfile.                                                                                                                                         | D congruence, conductor.; exact quadratic coordinates only through O09/O10/BR14 when required                                                              | Coordinates in rationas/ExecComplex embedding. Derived-quadratic update: no free sqrt(D); exact coordinates require proof-carrying D-source.                                | high                                        | 5.4   |
| N05 | Trace form                                     | Lean;<br>Target;<br>Object    | F1   | Y             | TraceFormProfile.                                                                                                                                                                                                                                                             | Order/field.                                                                                                                                          | Conjugation + basis.                                                                                                                                       | Rational/Exec mirror for coefficients.                                                                                                                                      | medium                                      | 5.5   |

| ID  | Object                                                    | Context class              | Tier | Object status | Canonical CNNA definition                                                                                                                                    | Natural provenance                                                      | Computed from                                                                                                  | ExecComplex/computable path                                                                                                                                                                                                                                                                               | Priority | Phase                  |
|-----|-----------------------------------------------------------|----------------------------|------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------|
| N06 | Norm form                                                 | Lean;<br>Source;<br>Object | F1   | Y             | AlgebraicNormProfile.                                                                                                                                        | Order + conjugation.                                                    | Integral basis.;<br>exact quadratic coordinates only<br>through O09/O10/BR14 when<br>required                  | Integer/ExecComplex values.<br>Derived-quadratic update: no free sqrt(D);<br>exact coordinates require proof-carrying<br>D-source.                                                                                                                                                                        | high     | <a href="#">5.5</a>    |
| N07 | Conjugation                                               | Lean;<br>Target;<br>Object | F1   | Y             | ConjugationProfile.                                                                                                                                          | QuadraticFieldProfile.                                                  | sqrt(D) -> -sqrt(D).                                                                                           | Coordinate-level operational reflection.                                                                                                                                                                                                                                                                  | medium   | <a href="#">5.5</a>    |
| N08 | Unit group $O\_D^*$                                       | Lean;<br>Target;<br>Object | F2   | Y             | UnitGroupProfile.                                                                                                                                            | OrderProfile + norm one classification.                                 | $O\_D$ + norm form.                                                                                            | Units as coordinates;<br>group order operational.                                                                                                                                                                                                                                                         | high     | <a href="#">6.1</a>    |
| N09 | Class group $Cl(O\_D)$                                    | Lean;<br>Target;<br>Object | F2   | Y             | ClassGroupProfile.                                                                                                                                           | OrderProfile + ideas.                                                   | Fractional ideas modulo principal.                                                                             | Finite group data;<br>order mirrored as Nat/Exec.                                                                                                                                                                                                                                                         | medium   | <a href="#">6.2</a>    |
| N10 | Class number $h(D)$                                       | Lean;<br>Target;<br>Object | F2   | Y             | ClassNumberProfile.                                                                                                                                          | ClassGroupProfile.                                                      | $Cl(O\_D)$ .                                                                                                   | Nat/Exec mirror.                                                                                                                                                                                                                                                                                          | medium   | <a href="#">6.2</a>    |
| N11 | Fractional/integral ideas                                 | Lean;<br>Target;<br>Object | F2   | Y             | IdealProfile.                                                                                                                                                | OrderProfile.                                                           | $O\_D$ modules/submodules.                                                                                     | Not Exec;<br>invariants operational.                                                                                                                                                                                                                                                                      | medium   | <a href="#">6.2.1</a>  |
| N12 | Principal ideal predicate                                 | Lean;<br>Target;<br>Object | F2   | Y             | PrincipalIdealProfile.                                                                                                                                       | IdealProfile + generator existence.                                     | Ideas and norm.                                                                                                | Witness coordinates operational if present.                                                                                                                                                                                                                                                               | medium   | <a href="#">6.2.2</a>  |
| N13 | Heegner/PID/UFD indicator                                 | Lean;<br>Target;<br>Object | F2   | Y             | HeegnerIndicatorProfile. It classifies number-theoretic<br>class-number-one/PID/UFD data only;<br>it does not classify CNNA closed/noOuter regime<br>status. | Class number + order data.                                              | $h(D)$ , $D\_fund$ .                                                                                           | Bool/Nat;<br>target list only as comparison.                                                                                                                                                                                                                                                              | medium   | <a href="#">6.3</a>    |
| N14 | Prime splitting behavior                                  | Lean;<br>Target;<br>Object | F2   | Y             | PrimeSplittingProfile.                                                                                                                                       | Kronecker character.                                                    | $D\_fund$ , primes.                                                                                            | Values -1/0/1 as Exec mirror.                                                                                                                                                                                                                                                                             | medium   | <a href="#">6.4</a>    |
| N15 | Kronecker/Dirichlet character $\chi\_D$                   | Lean;<br>Target;<br>Object | F2   | Y             | CharacterProfile.                                                                                                                                            | $D\_fund$ .                                                             | Kronecker symbol.                                                                                              | Function Nat -> {-1,0,1};<br>operational mirror.                                                                                                                                                                                                                                                          | medium   | <a href="#">6.4</a>    |
| CD8 | Arithmetic-CD compatibility as<br>Schur-duality corollary | Lean;<br>Source;<br>Object | F1   | K             | CDArithmeticCompatibilityProfile as a corollary of<br>BR08, not as an independently injected norm analogy.                                                   | Order/norm + CD Profile + horizontal/<br>vertical Schur-duality bridge. | BR08, BR09, CD3/CD6/CD7,<br>O05, and later number-theory<br>norm objects when available.                       | Requires explicit coordinate bridge and<br>concrete finite witness;<br>no compatibility is accepted before the<br>Schur bridge proof.                                                                                                                                                                     | medium   | <a href="#">10.4.2</a> |
| L00 | Lattice embedding<br>$\Lambda \rightarrow C$              | Lean;<br>Target;<br>Object | F2   | Y             | ComplexEmbeddingProfile.                                                                                                                                     | Order + selected embedding.                                             | Integral basis + sqrt(D) branch.;<br>exact quadratic coordinates only<br>through O09/O10/BR14 when<br>required | Mirror-only for general algebraic<br>embeddings involving sqrt(D). Operational<br>return is restricted to rational/integer<br>invariants such as discriminant/norm data<br>when separately proved. Derived-quadratic<br>update: no free sqrt(D);<br>exact coordinates require proof-carrying<br>D-source. | high     | <a href="#">7.1</a>    |
| L01 | Lattice basis                                             | Lean;<br>Target;<br>Object | F2   | Y             | LatticeBasisProfile.                                                                                                                                         | Embedding + integral basis.                                             | OrderProfile.                                                                                                  | Coordinates operational if encoded.                                                                                                                                                                                                                                                                       | high     | <a href="#">7.1</a>    |
| L02 | Gram matrix                                               | Lean;<br>Target;<br>Object | F2   | Y             | GramProfile.                                                                                                                                                 | Lattice basis + scalar product.                                         | Embedding.;<br>exact quadratic coordinates only<br>through O09/O10/BR14 when<br>required                       | Operational only for rational Gram entries<br>or integer/rational invariants such as<br>determinant/discriminant;<br>general algebraic/real Gram data is<br>mirror-only under EXEC-2.<br>Derived-quadratic update: no free sqrt(D);<br>exact coordinates require proof-carrying<br>D-source.              | high     | <a href="#">7.2</a>    |
| L03 | Lattice determinant /<br>covolume                         | Lean;<br>Target;<br>Object | F2   | Y             | CovolumeProfile.                                                                                                                                             | GramProfile.                                                            | Gram.                                                                                                          | Numeric/Exec mirror if exact rational.                                                                                                                                                                                                                                                                    | medium   | <a href="#">7.2</a>    |



| ID  | Object                            | Context class              | Tier | Object status | Canonical CNNA definition                                                                                                | Natural provenance                                                                                                                                   | Computed from                                                                                   | ExecComplex/computable path                                                                                                                                                                                                                                                                                                                                                                | Priority | Phase |
|-----|-----------------------------------|----------------------------|------|---------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|
| L04 | Period ratio tau                  | Lean;<br>Target;<br>Object | F2   | Y             | PeriodRatioProfile.                                                                                                      | Lattice basis + orientation.                                                                                                                         | Embedding, basis.;<br>exact quadratic coordinates only<br>through O09/O10/BR14 when<br>required | General tau is mirror-only. Operational<br>path may record rational special cases or<br>integer/rational invariants only when<br>derived from prior objects and separately<br>proved. Under EXEC-2 tau is mirror/<br>comparison unless a rational/integer<br>invariant is exported. Derived-quadratic<br>update: no free sqrt(D);<br>exact coordinates require proof-carrying<br>D-source. | high     | 7.3   |
| L05 | Upper half-plane<br>condition     | Lean;<br>Target;<br>Object | F2   | Y             | UpperHalfPlaneCertificate.                                                                                               | Period ratio + orientation.                                                                                                                          | tau.                                                                                            | Inequality may need real/mirror layer.                                                                                                                                                                                                                                                                                                                                                     | medium   | 7.3   |
| L06 | Angle / cosine profile            | Lean;<br>Target;<br>Object | F2   | Y             | AngleProfile. Includes future test-matrix<br>consumption by L09/L10;<br>angle values are profiles, not generator inputs. | GramProfile.                                                                                                                                         | Gram entries;<br>cos = <u,v>/sqrt(...).                                                         | Cosine ratio operational;<br>arccos in analytic mirror.                                                                                                                                                                                                                                                                                                                                    | medium   | 7.4   |
| L07 | Lattice automorphism<br>group     | Lean;<br>Target;<br>Object | F2   | Y             | LatticeAutomorphismProfile.                                                                                              | UnitGroup + embedding action.                                                                                                                        | Units + lattice basis.                                                                          | Permutation/action finite operational.                                                                                                                                                                                                                                                                                                                                                     | medium   | 7.5   |
| M02 | Eisenstein series G4,G6           | Lean;<br>Target;<br>Object | F2   | Y             | EisensteinSeriesProfile.                                                                                                 | Lattice/period ratio. v37: retained in<br>modular/CM Path A.                                                                                         | Lattice sums or q-expansion.                                                                    | Truncated sums/q-series operational;<br>exact analytic mirror.                                                                                                                                                                                                                                                                                                                             | medium   | 8.1   |
| M03 | Weierstrass invariants<br>g2,g3   | Lean;<br>Target;<br>Object | F2   | Y             | WeierstrassInvariantProfile.                                                                                             | G4,G6. v37: retained in modular/CM Path<br>A.                                                                                                        | Eisenstein series.                                                                              | Operational if finite/truncated or h=1<br>closed source.                                                                                                                                                                                                                                                                                                                                   | medium   | 8.2   |
| M04 | Modular discriminant<br>Delta_mod | Lean;<br>Target;<br>Object | F2   | Y             | ModularDiscriminantProfile.                                                                                              | g2,g3. v37: retained in modular/CM Path<br>A.                                                                                                        | Weierstrass invariants.                                                                         | ExecComplex if g2,g3 operational.                                                                                                                                                                                                                                                                                                                                                          | medium   | 8.3   |
| M05 | j-invariant                       | Lean;<br>Target;<br>Object | F2   | Y             | JInvariantProfile.                                                                                                       | Delta_mod + g2, or q-expansion. v37:<br>retained in modular/CM Path A.                                                                               | g2, Delta_mod, tau.                                                                             | General j-invariant computation is mirror/<br>non-operational. Operational return is<br>allowed only for proved integer/rational<br>special invariants, e.g. h=1 anchors, and<br>must not source the derivation. Future M08<br>may record j=0 and j=1728 as integer<br>comparison targets;<br>analytic j evaluation remains<br>non-operational unless separately justified.                | medium   | 8.4   |
| SP0 | Eigenvalue profile                | Lean;<br>Source;<br>Object | F1   | Y             | FiniteSpectrumProfile.                                                                                                   | Exec matrix / charpoly. v37: placed on<br>independent spectral/physical Path B after<br>Phase 4.1, not behind modular/j path.                        | Matrix, polynomial.                                                                             | General algebraic eigenvalues are<br>mirror-only. Operational spectral path<br>carries charpoly/coefficients/discriminants/<br>gaps or other rational/integer certificates<br>unless algebraic values are confined to<br>mirror.                                                                                                                                                           | medium   | 10.1  |
| SP1 | Spectral gap                      | Lean;<br>Source;<br>Object | F1   | Y             | SpectralGapProfile.                                                                                                      | FiniteSpectrumProfile. v37: placed on<br>independent spectral/physical Path B after<br>Phase 4.1, not behind modular/j path.                         | Eigenvalues/order.                                                                              | Needs comparison/order;<br>operational if finite exact/approx bridge.                                                                                                                                                                                                                                                                                                                      | medium   | 10.1  |
| P00 | Operator algebra                  | Lean;<br>Source;<br>Object | F1   | Y             | OperatorAlgebraProfile.                                                                                                  | Carrier + matrix algebra + star. v37:<br>placed on independent spectral/physical<br>Path B after Phase 4.1, not behind<br>modular/j path.            | Boundary matrix.                                                                                | Matrix algebra over ExecComplex;<br>C-mirror later.                                                                                                                                                                                                                                                                                                                                        | medium   | 10.2  |
| P01 | Star operation                    | Lean;<br>Source;<br>Object | F1   | Y             | StarOperationProfile.                                                                                                    | ExecComplex conjugation + matrix<br>transpose. v37: placed on independent<br>spectral/physical Path B after Phase 4.1,<br>not behind modular/j path. | Matrix algebra.                                                                                 | Computable over ExecComplex if<br>conjugation defined.                                                                                                                                                                                                                                                                                                                                     | medium   | 10.2  |
| P02 | State functional                  | Lean;<br>Source;<br>Object | F1   | Y             | StateFunctionalProfile.                                                                                                  | Trace/Gibbs from operator profile. v37:<br>placed on independent spectral/physical<br>Path B after Phase 4.1, not behind<br>modular/j path.          | Matrix, weights.                                                                                | Finite trace operational;<br>positivity maybe mirror/proof.                                                                                                                                                                                                                                                                                                                                | medium   | 10.2  |
| P03 | Gibbs state                       | Lean;<br>Source;<br>Object | F1   | Y             | GibbsStateProfile.                                                                                                       | Matrix exponential seed. v37: placed on<br>independent spectral/physical Path B after<br>Phase 4.1, not behind modular/j path.                       | H, beta.                                                                                        | Exp not purely ExecComplex unless<br>truncated/seed;<br>mirror boundary.                                                                                                                                                                                                                                                                                                                   | medium   | 10.2  |

| ID  | Object                              | Context class                 | Tier | Object status | Canonical CNNA definition      | Natural provenance                                                                                                                    | Computed from                            | ExecComplex/computable path                                               | Priority | Phase |
|-----|-------------------------------------|-------------------------------|------|---------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------|----------|-------|
| P04 | Time evolution                      | Lean;<br>Source;<br>Object    | F1   | Y             | TimeEvolutionProfile.          | Operator + regime. v37: placed on independent spectral/physical Path B after Phase 4.1, not behind modular/j path.                    | H, t.                                    | Finite approximations; exp mirror.                                        | medium   | 10.2  |
| P05 | Unitarity certificate               | Lean;<br>Source;<br>Object    | F1   | Y             | UnitarityCertificate.          | Time evolution + noOuter. v37: placed on independent spectral/physical Path B after Phase 4.1, not behind modular/j path.             | H, star, state.                          | Matrix identities operational if finite exact.                            | medium   | 10.2  |
| P06 | Open channel / effective openness   | Lean;<br>Source;<br>Object    | F1   | R             | OpenChannelProfile.            | Regime hasOuter + complement effects. v37: placed on independent spectral/physical Path B after Phase 4.1, not behind modular/j path. | Boundary operator + environment profile. | Krfrom/Lindblad only if derived; Exec matrices if finite.                 | medium   | 10.2  |
| N16 | Dirichlet L-function L(s,chi_D)     | Reference;<br>Gate;<br>Object | F2   | K             | LFunctionProfile.              | CharacterProfile + analytic layer. v37: analytic far target in Path A, not a source for finite spectral/physical Path B.              | chi_D.                                   | Finite truncations operational; exact analytic object mirror.             | low      | 9.2   |
| N17 | Class number formula                | Reference;<br>Gate;<br>Object | F2   | K             | ClassNumberFormulaProfile.     | LFunction + unit group + D. v37: analytic far target in Path A, not a source for finite spectral/physical Path B.                     | chi_D, units, D_fund.                    | Approximation only operational; theorem analytic.                         | low      | 9.3   |
| L08 | Theta series Theta_Lambda           | Reference;<br>Gate;<br>Object | F2   | K             | ThetaSeriesProfile.            | Lattice + norm form.                                                                                                                  | Norm form over lattice.                  | Truncated coefficients operational; full series formal/mirror.            | low      | 9.1   |
| M00 | Modular group action SL2Z           | Lean;<br>Target;<br>Object    | F2   | K             | ModularActionProfile.          | Upper half-plane + integer matrices. v37: retained in modular/CM Path A.                                                              | tau.                                     | Action formulas operational for finite transformations.                   | low      | 8.0   |
| M01 | Fundamental domain representative   | Lean;<br>Target;<br>Object    | F2   | K             | FundamentalDomainProfile.      | ModularActionProfile. v37: retained in modular/CM Path A.                                                                             | tau + reduction algorithm/proof.         | Algorithmic reduction possible.                                           | low      | 8.0   |
| M06 | q-expansion of j                    | Lean;<br>Target;<br>Object    | F2   | K             | JQExpansionProfile.            | tau/q. v37: retained in modular/CM Path A.                                                                                            | q=exp(2pi i tau).                        | Truncated coefficients operational if q path exists; exact series formal. | low      | 8.8   |
| M07 | Hilbert class polynomial H_D        | Lean;<br>Target;<br>Object    | F2   | K             | HilbertClassPolynomialProfile. | Class group + j-values. v37: retained in modular/CM Path A.                                                                           | Cl(O_D), j.                              | Polynomial coefficients operational if constructed.                       | low      | 8.8   |
| TM0 | Transfermatrix T_k                  | Lean;<br>Source;<br>Object    | F1   | K             | TransferMatrixProfile.         | Schur recurrence. v37: moved to Schur/transfer Path B and late BR08 bridge, not modular-form path.                                    | Local recursion step.                    | Entries operational in polynomial/ring representation.                    | low      | 10.4  |
| TM1 | Transfermatrix product              | Lean;<br>Source;<br>Object    | F1   | K             | TransferProductProfile.        | TransferMatrixProfile. v37: moved to Schur/transfer Path B and late BR08 bridge, not modular-form path.                               | Trace/fold/product.                      | Operational finite product.                                               | low      | 10.4  |
| TM2 | Moebius/continued fraction relation | Lean;<br>Source;<br>Object    | F1   | K             | MobiusSchurProfile.            | Transfermatrix. v37: moved to Schur/transfer Path B and late BR08 bridge, not modular-form path.                                      | T_k action.                              | Operational finite map.                                                   | low      | 10.4  |
| SP2 | Heat kernel K(t)                    | Lean;<br>Target;<br>Object    | F2   | K             | HeatKernelProfile.             | Spectrum/operator. v37: placed on independent spectral/physical Path B after Phase 4.1, not behind modular/j path.                    | Eigenvalues or matrix exponential seed.  | Finite approximations operational; real exp mirror.                       | low      | 10.3  |
| SP3 | Spectral zeta zeta_H                | Lean;<br>Target;<br>Object    | F2   | K             | SpectralZetaProfile.           | Spectrum. v37: placed on independent spectral/physical Path B after Phase 4.1, not behind modular/j path.                             | Eigenvalues.                             | Analytic object; finite/truncated operational.                            | low      | 10.3  |
| SP4 | Spectral dimension                  | Lean;<br>Target;<br>Object    | F2   | K             | SpectralDimensionProfile.      | HeatKernelProfile + limit. v37: placed on independent spectral/physical Path B after Phase 4.1, not behind modular/j path.            | K(t).                                    | Limit/mirror; finite estimator operational.                               | low      | 10.3  |
| SP5 | Zeta-regularized determinant        | Lean;<br>Target;<br>Object    | F2   | K             | ZetaDeterminantProfile.        | SpectralZetaProfile. v37: placed on independent spectral/physical Path B after Phase 4.1, not behind modular/j path.                  | zeta_H.                                  | Analytic mirror; finite determinant operational.                          | low      | 10.3  |
| P07 | KMS condition                       | Lean;<br>Target;<br>Object    | F2   | K             | KMSProfile.                    | State + time evolution.                                                                                                               | State, dynamics.                         | Analytic strip not Exec; finite checks/truncations operational.           | low      | 13.4  |
| P08 | GNS Hilbert space                   | Lean;<br>Target;<br>Object    | F2   | K             | GNSProfile.                    | State functional.                                                                                                                     | Algebra quotient/completion.             | Finite quotient operational; completion mirror.                           | low      | 13.4  |

| ID    | Object                                       | Context class                            | Tier | Object status | Canonical CNNA definition                                                                               | Natural provenance                                                                        | Computed from                                                                                                                                         | ExecComplex/computable path                                                                                   | Priority | Phase                |
|-------|----------------------------------------------|------------------------------------------|------|---------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------|----------------------|
| P09   | Tomita operator S                            | Lean;<br>Target;<br>Object               | F2   | K             | TomitaOperatorProfile.                                                                                  | GNS + star.                                                                               | GNS state algebra.                                                                                                                                    | Finite matrix profile possible;<br>closure analytic.                                                          | low      | <a href="#">13.5</a> |
| P10   | Modular operator Delta                       | Lean;<br>Target;<br>Object               | F2   | K             | ModularOperatorProfile.                                                                                 | Tomita S.                                                                                 | S.                                                                                                                                                    | Finite matrix possible;<br>analytic layer.                                                                    | low      | <a href="#">13.5</a> |
| P11   | Modular conjugation J                        | Lean;<br>Target;<br>Object               | F2   | K             | ModularConjugationProfile.                                                                              | Tomita-Takesaki.                                                                          | S, Delta.                                                                                                                                             | Not direct Exec;<br>finite mirror only if polar decomposition<br>derived.                                     | low      | <a href="#">13.5</a> |
| P12   | Bisognano-Wichmann/<br>CPT bridge            | Lean;<br>Target;<br>Object               | F2   | K             | CPTBridgeProfile.                                                                                       | Modular J + spacetime/wedge data.                                                         | J, local net, geometry.                                                                                                                               | Analytic/geometric, not pure Exec.                                                                            | low      | <a href="#">13.6</a> |
| P13   | Spin-statistics                              | Lean;<br>Target;<br>Object               | F2   | K             | SpinStatisticsProfile.                                                                                  | Modular + redations.                                                                      | J, covariance, redations.                                                                                                                             | Not Exec except finite symmetry shadows.                                                                      | low      | <a href="#">13.6</a> |
| P14   | Kramers degeneracy                           | Lean;<br>Target;<br>Object               | F2   | K             | KramersProfile.                                                                                         | Time reversal + state/symmetry.                                                           | Antiunitary T, spectrum.                                                                                                                              | Finite spectral shadows possible.                                                                             | low      | <a href="#">13.6</a> |
| P15   | Split property                               | Lean;<br>Source;<br>Object               | F1   | K             | SplitPropertyProfile.                                                                                   | Operator algebra + collar/regime.                                                         | Local net, inclusion.                                                                                                                                 | Finite type-I checks operational;<br>full split analytic.                                                     | low      | <a href="#">13.6</a> |
| P16   | Local net / Haag-Kastler<br>algebraic axioms | Lean;<br>Target;<br>Object               | F2   | K             | LocalNetProfile.                                                                                        | RegionNet + operator algebra.                                                             | Regions, inclusions, operators.                                                                                                                       | Finite net operational;<br>spacetime axioms need C/D.                                                         | low      | <a href="#">13.3</a> |
| H10   | A->B bright handoff                          | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Y             | AToBBrightOutputProfile.                                                                                | A operational boundary profile.                                                           | Omega, L, regime, operator;<br>no hidden complement.                                                                                                  | ExecComplex operator may be included;<br>B may mirror to C.                                                   | high     | <a href="#">11.1</a> |
| H11   | A->B prime/dark future<br>handoff            | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | R             | AToBDarkOutputProfile.                                                                                  | A complement profile;<br>separate from bright.                                            | Dark/complement data.                                                                                                                                 | Likely not finite Exec-only;<br>must specify.                                                                 | low      | <a href="#">11.2</a> |
| H12   | A->C handoff                                 | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Y             | AToCOutputProfile.                                                                                      | A arithmetic/geometric Profile.                                                           | Carrier, lattice, Gram, tau, regime<br>as allowed.                                                                                                    | ExecComplex mirrors for coordinates where<br>available.                                                       | medium   | <a href="#">11.3</a> |
| H13   | A->D handoff                                 | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Y             | AToDOutputProfile.                                                                                      | Operator/regime/spectral Profile.                                                         | Boundary operator, state seeds,<br>regime.                                                                                                            | Exec matrices;<br>time evolution maybe mirror/analytic.                                                       | medium   | <a href="#">11.4</a> |
| H14   | A->E handoff                                 | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | K             | AToEOutputProfile.                                                                                      | Arithmetic/theory register Profile.                                                       | D, CM, class data, j, etc.                                                                                                                            | Exec mirrors where present.                                                                                   | low      | <a href="#">11.5</a> |
| MOON0 | j coefficient stream                         | Reference;<br>Gate;<br>Object            | F2   | K             | CNNA-derived q-expansion coefficient stream of j<br>after j is derived.                                 | q-expansion of j;<br>modular forms chain.                                                 | M06 q-expansion object.                                                                                                                               | Computable only after q-expansion<br>representation;<br>may require mirror arithmetic for infinite<br>series. | low      | <a href="#">12.1</a> |
| MOON1 | Monster dimension<br>comparison profile      | Reference;<br>Gate;<br>Object            | F2   | K             | Comparison profile between derived j coefficients and<br>established Monster representation dimensions. | j coefficient stream plus external established<br>Monster data as comparison target only. | MOON0 plus external established<br>Monster representation dimension<br>data as comparison target only;<br>no reverse dependency into j<br>derivation. | Comparison layer;<br>no Exec path into foundation.                                                            | low      | <a href="#">12.2</a> |
| MOON2 | McKay-Thompson<br>comparison profile         | Reference;<br>Gate;<br>Object            | F2   | K             | Comparison-only profile for McKay-Thompson series.                                                      | Moonshine comparison layer after j/q data.                                                | MOON0 and optional external<br>named McKay-Thompson/<br>Hauptmodul series as targets.                                                                 | Comparison-only;<br>no foundation Exec source.                                                                | low      | <a href="#">12.3</a> |
| B00   | Pillar B input contract<br>from A            | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | K             | B consumes exactly the A->B bright handoff and<br>nothing else.                                         | A->B handoff H10.                                                                         | H10;<br>exported boundary/operator/<br>regime profiles.                                                                                               | Exec data only if exported in A->B profile.                                                                   | medium   | <a href="#">13.1</a> |

| ID  | Object                                           | Context class                   | Tier | Object status | Canonical CNNA definition                                                                                   | Natural provenance                                                 | Computed from                                    | ExecComplex/computable path                                                                 | Priority | Phase                |
|-----|--------------------------------------------------|---------------------------------|------|---------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------|----------|----------------------|
| B01 | Boundary star-algebra / observable algebra       | Lean; Source; Object            | F1   | K             | Finite *-algebra of boundary observables derived from exported A boundary operator data.                    | A->B bright boundary carrier/operator.                             | B00; ExecBoundaryMatrix; BoundaryOperator.       | Finite matrix algebra can have ExecComplex operational mirror; analytic mirror later.       | medium   | <a href="#">13.2</a> |
| B02 | Local net / QFT algebraic profile                | Lean; Source; Object            | F1   | K             | Net of algebras derived from boundary/region data; algebraic QFT consumer.                                  | B boundary star-algebra plus exported region/localization profile. | B01; A->B region/interface data.                 | Finite operational net first; continuum/AQFT mirror later.                                  | medium   | <a href="#">13.3</a> |
| B03 | B state/GNS/KMS profile                          | Lean; Source; Object            | F1   | K             | State functional, GNS construction, and KMS gate over B observable algebra.                                 | B algebra; exported dynamics/state seeds from A if available.      | B01; P01/P02/P03; Gibbs/time-evolution seeds.    | Finite state/GNS may be operational; KMS analyticity belongs to mirror/far target.          | medium   | <a href="#">13.4</a> |
| B04 | Tomita-Takesaki modular data                     | Lean; Source; Object            | F1   | K             | Tomita operator S, modular operator Delta, and modular conjugation J derived from B GNS/KMS data.           | B03 state/GNS/KMS profile.                                         | B03; local net.                                  | Mostly analytic mirror; finite pre-versions may be operational only if matrices are finite. | medium   | <a href="#">13.5</a> |
| B05 | CPT/BW/ spin-statistics/split gates              | Lean; Source; Object            | F1   | K             | Structural theorem gates derived from B local net and modular data.                                         | B02 local net plus B04 modular data.                               | B02; B04.                                        | Analytic theorem layer; no Exec source except finite prerequisites.                         | low      | <a href="#">13.6</a> |
| B06 | B->A allowed output profile                      | Lean; Source; Object            | F1   | K             | Typed B output that A may consume only through B->A input handoff.                                          | B-derived theorem/data after B phases.                             | B01-B05 if completed.                            | Typed; Exec mirror if numeric.                                                              | medium   | <a href="#">13.7</a> |
| C00 | Pillar C OQS input contract from A               | Lean; Handoff; Contract; Object | F0   | K             | C consumes explicit A->C open/regime/complement outputs.                                                    | A->C handoff; HasOuter regime; complement profile.                 | H12; RegimeProfile; ComplementProfile.           | Exec only through exported finite channel/ operator data.                                   | medium   | <a href="#">14.1</a> |
| C01 | Finite CP/channel profile                        | Lean; Source; Object            | F1   | K             | Completely-positive/channel profile derived from finite exported data.                                      | A finite channel seed and open regime.                             | C00; P06; ChannelSeed.                           | Finite ExecComplex channel path if matrices and Kraus/Choi data are derived.                | medium   | <a href="#">14.2</a> |
| C02 | Effective OQS generator / Lindblad gate          | Lean; Source; Object            | F1   | K             | Effective generator and Lindblad/GKSL gate if derived from finite channel/semigroup data.                   | C01 plus time-evolution/semigroup data.                            | C01; finite dynamics seeds.                      | Operational finite generator first; analytic continuous-time mirror later.                  | medium   | <a href="#">14.3</a> |
| C03 | Decoherence/noise profile                        | Lean; Source; Object            | F1   | K             | Effective decoherence/noise profile derived from complement/environment data.                               | HasOuter regime and C channel/effective dynamics.                  | C00-C02; ComplementProfile.                      | Finite operational noise kernel if exported matrices exist.                                 | medium   | <a href="#">14.4</a> |
| C04 | C->A allowed output profile                      | Lean; Source; Object            | F1   | K             | Typed C output that A may consume only through C->A input.                                                  | C-derived OQS theorem/data.                                        | C01-C03.                                         | Typed; Exec mirror if numeric.                                                              | medium   | <a href="#">14.5</a> |
| D00 | Pillar D spacetime input contract                | Lean; Handoff; Contract; Object | F0   | K             | D consumes explicit A/C geometry, regime, and localization outputs.                                         | A->D handoff; C outputs when available.                            | H13; lattice/region/spectral geometry objects.   | Finite/computable geometry where derived; continuum mirror later.                           | low      | <a href="#">15.1</a> |
| D01 | Region/localization/ causal-order candidate      | Lean; Source; Object            | F1   | K             | Candidate region poset, localization profile, and causal order derived from CNNA relational/geometric data. | D00; region/localization/geometry profiles.                        | A geometry; C open-system profiles if available. | Finite relational/computable representation first.                                          | low      | <a href="#">15.2</a> |
| D02 | Metric/coarse-geometry/ covariance gate          | Lean; Source; Object            | F1   | K             | Metric candidate, coarse geometry, and covariance gate derived from D localization/spectral data.           | D01 plus spectral geometry.                                        | D01; SP0-SP5 if available.                       | Finite/spectral computable approximants; continuum analytic mirror later.                   | low      | <a href="#">15.3</a> |
| D03 | D->A allowed output profile                      | Lean; Source; Object            | F1   | K             | Typed D output that A may consume only through D->A input.                                                  | D-derived geometry/covariance theorem/ data.                       | D01-D02.                                         | Typed; Exec mirror if numeric.                                                              | low      | <a href="#">15.4</a> |
| E00 | Pillar E matter input contract                   | Lean; Handoff; Contract; Object | F0   | K             | E consumes permitted A/B/C/D outputs for matter/ observable comparison.                                     | A->E handoff; B/C/D outputs when available.                        | H14; B/C/D outputs.                              | Exec only for exported finite observable data.                                              | low      | <a href="#">16.1</a> |
| E01 | Matter sector and internal symmetry candidate    | Lean; Source; Object            | F1   | K             | Candidate matter sectors and internal symmetries derived from CNNA symmetry/arithmetic data.                | E00; unit groups, CD path, B/D symmetries when available.          | Units/CD/symmetry objects; B/D outputs.          | Computable only through finite exported symmetry/observable data.                           | low      | <a href="#">16.2</a> |
| E02 | Observable coupling and phenomenology comparison | Reference; Gate; Object         | F3   | K             | Observable coupling and empirical comparison layer, not foundational input.                                 | E01 plus B/C/D observable structures.                              | E01; B/C/D outputs.                              | Finite observed comparison values are external targets, not CNNA sources.                   | low      | <a href="#">16.3</a> |
| E03 | E->A allowed output profile                      | Lean; Source; Object            | F1   | K             | Typed E output that A may consume only through E->A input.                                                  | E-derived matter/phenomenology theorem/ data.                      | E01-E02.                                         | Typed; Exec mirror if numeric.                                                              | low      | <a href="#">16.4</a> |

| ID    | Object                                            | Context class                    | Tier | Object status | Canonical CNNA definition                                                                                                 | Natural provenance                                                                                                 | Computed from                                                                                               | ExecComplex/computable path                                                                                 | Priority | Phase                 |
|-------|---------------------------------------------------|----------------------------------|------|---------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------|-----------------------|
| BR00  | Backreaction input DAG                            | Lean;<br>Target;<br>Object       | F2   | K             | DAG of allowed B/C/D/E -> A input handoffs.                                                                               | Explicit input handoff profiles only.                                                                              | B06;<br>C04;<br>D03;<br>E03.                                                                                | Typed;<br>not necessarily Exec.                                                                             | medium   | <a href="#">17</a>    |
| BR01  | B->A backreaction input gate                      | Lean;<br>Source;<br>Object       | F1   | K             | Gate for B-derived theorem/data entering A.                                                                               | B06.                                                                                                               | B06.                                                                                                        | Typed;<br>Exec if numeric.                                                                                  | medium   | <a href="#">17.1</a>  |
| BR02  | C->A OQS backreaction input gate                  | Lean;<br>Source;<br>Object       | F1   | K             | Gate for C-derived OQS data entering A.                                                                                   | C04.                                                                                                               | C04.                                                                                                        | Typed;<br>Exec if numeric.                                                                                  | medium   | <a href="#">17.2</a>  |
| BR03  | D->A spacetime backreaction input gate            | Lean;<br>Source;<br>Object       | F1   | K             | Gate for D-derived geometry/covariance data entering A.                                                                   | D03.                                                                                                               | D03.                                                                                                        | Typed;<br>Exec if numeric.                                                                                  | low      | <a href="#">17.3</a>  |
| BR04  | E->A matter backreaction input gate               | Lean;<br>Source;<br>Object       | F1   | K             | Gate for E-derived matter/phenomenology data entering A.                                                                  | E03.                                                                                                               | E03.                                                                                                        | Typed;<br>Exec if numeric.                                                                                  | low      | <a href="#">17.4</a>  |
| BR05  | Pre/post-emergence switch for A statements        | Lean;<br>Source;<br>Object       | F0   | K             | Profile distinguishing pre-emergence and post-emergence interpretation of A statements.                                   | Allowed input gates plus emergence criteria.                                                                       | BR00-BR04;<br>regime profile.                                                                               | Mostly structural;<br>Exec if numeric thresholds are derived.                                               | medium   | <a href="#">17.5</a>  |
| SIDE0 | Full ToE comparison profile                       | Reference;<br>Gate;<br>Object    | F4   | K             | Comparison-only placeholder for any future full ToE interpretation that falls out of CNNA.                                | Completed B/C/D/E/backreaction outputs if ever available.                                                          | CNNA mature outputs;<br>no ToE assumptions.                                                                 | No Exec source;<br>comparison target only.                                                                  | low      | <a href="#">18.1</a>  |
| SIDE1 | String-theory comparison profile                  | Reference;<br>Gate;<br>Object    | F4   | K             | Comparison-only placeholder for string-theory-like structures if derivable from CNNA outputs.                             | D/E outputs if ever available.                                                                                     | CNNA outputs;<br>no string assumptions.                                                                     | No Exec source;<br>comparison target only.                                                                  | low      | <a href="#">18.2</a>  |
| SIDE2 | Loop quantum gravity comparison profile           | Reference;<br>Gate;<br>Object    | F4   | K             | Comparison-only placeholder for LQG-like structures if derivable from CNNA outputs.                                       | D spacetime outputs if ever available.                                                                             | CNNA outputs;<br>no LQG assumptions.                                                                        | No Exec source;<br>comparison target only.                                                                  | low      | <a href="#">18.3</a>  |
| SIDE3 | Noncommutative geometry comparison profile        | Reference;<br>Gate;<br>Object    | F4   | K             | Comparison-only placeholder for NCG-like structures if derivable from CNNA outputs.                                       | B algebra and D geometry outputs if ever available.                                                                | CNNA outputs;<br>no NCG axioms as foundation.                                                               | No Exec source;<br>comparison target only.                                                                  | low      | <a href="#">18.4</a>  |
| MOON3 | Finite simple group comparison profile            | Reference;<br>Gate;<br>Object    | F2   | K             | Comparison-only profile for CFSG taxonomy: cyclic prime-order, alternating, Lie-type, sporadic.                           | External established mathematics used only after CNNA has derived a candidate graded/symmetry output.              | CNNA-derived q/graded/<br>symmetry profile plus external comparison taxonomy.                               | Comparison tags can be finite data;<br>no operational foundation source.                                    | low      | <a href="#">12.4</a>  |
| MOON4 | Sporadic/Monster comparison boundary              | Reference;<br>Gate;<br>Object    | F2   | K             | Boundary profile separating sporadic-group/Monster comparison data from CNNA-generative objects.                          | Finite simple group comparison context plus post-j coefficient stream.                                             | MOON0/MOON3 and external Monster representation dimension table as target.                                  | Finite comparison table only;<br>no foundation Exec source.                                                 | low      | <a href="#">12.4</a>  |
| MOON5 | Leech lattice comparison profile                  | Reference;<br>Gate;<br>Object    | F2   | K             | Comparison-only profile for a rootless even unimodular 24-dimensional lattice target.                                     | External Leech-lattice mathematics; CNNA may only compare after deriving an internal lattice/theta-like invariant. | Derived CNNA lattice/theta candidate plus external Leech target data.                                       | Finite Gram/basis/shell data can be represented operationally as comparison tables;<br>not a source object. | low      | <a href="#">12.5</a>  |
| MOON6 | Leech theta/shell coefficient comparison          | Reference;<br>Gate;<br>Object    | F2   | K             | Comparison profile for Leech theta series and shell-count/kissing-number coefficients.                                    | Leech lattice comparison target plus derived CNNA q/theta coefficient stream.                                      | MOON5 and CNNA-derived q/theta stream.                                                                      | Coefficient comparison is computable once both streams are represented;<br>no oracle in derivation.         | low      | <a href="#">12.6</a>  |
| MOON7 | VOA/moonshine-module comparison profile           | Reference;<br>Gate;<br>Object    | F2   | K             | Comparison-only profile for vertex-operator-algebra/moonshine-module structures.                                          | Post-j/Moonshine comparison horizon; possible later graded algebra from B/A outputs.                               | CNNA-derived graded algebra/<br>q-character output, if any, plus external VOA/moonshine-module target data. | Not an operational source;<br>finite graded truncations may be computable comparison artifacts.             | low      | <a href="#">12.7</a>  |
| MOON8 | Common-indicator / Hauptmodul bridge-test profile | Reference;<br>Gate;<br>Object    | F2   | K             | Comparison profile testing whether CNNA-derived q-characters match known Hauptmoduln or McKay-Thompson-type series.       | Thread-motivated bridge-test after j/<br>q-expansion and graded data;<br>not a prior assumption.                   | CNNA-derived q-character/<br>coefficient stream and external normalized Hauptmodul target list.             | Finite coefficient-prefix comparison is computable;<br>equality theorem belongs to mirror/proof layer.      | low      | <a href="#">12.8</a>  |
| LEG0  | Consumption and split inventory umbrella          | Governance;<br>Policy;<br>Object | F3   | Y             | SplitInventoryProfile with declaration-consumption map, not a code object.                                                | repo inventory + Arithmetic/Notation.lean audit + object register.                                                 | static module/declaration/<br>reference summaries.                                                          | n/a: planning/refactor artifact, not mathematical computation.                                              | active   | <a href="#">1.1.1</a> |
| LEG1  | Legacy single-constructor status gates            | Lean;<br>Refactor;<br>Object     | F3   | Y             | Historical single-constructor status enums retained for compatibility but quarantined outside the neutral public surface. | Existing historical declarations in Arithmetic/Notation.lean.                                                      | frozen SM9h legacy status declarations;<br>no new semantics.                                                | n/a: legacy gate/status declarations.                                                                       | near     | <a href="#">1.1.2</a> |



| ID    | Object                                                         | Context class              | Tier | Object status | Canonical CNNA definition                                                                                                                                                                                                                                                                                                                                                                                                                | Natural provenance                                                                                                                                | Computed from                                                                                                                                                         | ExecComplex/computable path                                                                                                                    | Priority | Phase       |
|-------|----------------------------------------------------------------|----------------------------|------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|
| LEG2  | CoreSymbols true notation surface                              | Lean; Refactor; Object     | F3   | Y             | Minimal notation/core-symbol module containing only actual notation/syntax/domain-role symbols needed by later neutral APIs.                                                                                                                                                                                                                                                                                                             | Declarations classified as true notation/core symbols by LEG0.                                                                                    | Notation split inventory.                                                                                                                                             | n/a: syntax/API surface.                                                                                                                       | near     | 1.1.3       |
| LEG3  | Legacy phase aliases compatibility surface                     | Lean; Refactor; Object     | F3   | Y             | Historical phase-semantic aliases and old names are retained behind a legacy module, not exposed as new public API.                                                                                                                                                                                                                                                                                                                      | Existing SM*/S6*/R*/Audit naming surface.                                                                                                         | Notation split inventory and frozen historical modules.                                                                                                               | n/a: compatibility surface.                                                                                                                    | near     | 1.1.4       |
| LEG4  | Neutral import firewall for post-notation phases               | Governance; Policy; Object | F3   | Y             | Audit rule and generated module inventory check that Phase 3+ imports neutral/CoreSymbols surfaces and not Legacy.StatusGates.                                                                                                                                                                                                                                                                                                           | Generated module inventory after the split.                                                                                                       | Lean import graph and phase-object DAG.                                                                                                                               | n/a: import governance artifact.                                                                                                               | near     | 1.1.5       |
| LEG5  | Declaration consumption classification map                     | Governance; Policy; Object | F3   | G             | ConsumptionClassificationMap with classes core-computational, proof-bridge, public-wrapper, legacy-status, legacy-policy, phase-alias, audit-only, blue-semantic-correction, dead-candidate, final-output.                                                                                                                                                                                                                               | Static module graph + symbol/declaration reference summary + direct code audit.                                                                   | repo_inventory/raw_export, repo_snapshot Lean source, object register, LEG10 module manifest, LEG11 reuse/duplication ledger.                                         | not applicable; planning/audit layer only                                                                                                      | critical | 1.1.1.1.4   |
| LEG7  | Blue-object semantic correction ledger                         | Governance; Policy; Object | F3   | Y             | BlueCorrectionLedger mapping each Object status B record to semantic defect, allowed temporary consumers, forbidden public consumers, neutral target API, and exit-to-white condition.                                                                                                                                                                                                                                                   | Object register B rows + consumption classification + TomS/source guardrails where applicable.                                                    | CD/S/O blue object rows and their current code derivation sites.                                                                                                      | not applicable; planning/audit layer only                                                                                                      | critical | 1.1.1.2     |
| LEG6  | Public API contract and consumer whitelist                     | Governance; Policy; Object | F3   | Y             | PublicAPIContract listing allowed neutral modules/declarations and forbidden imports for Phase 3+ and later handoffs.                                                                                                                                                                                                                                                                                                                    | Consumption map + blue correction ledger + access policy.                                                                                         | LEG5, LEG7, object access register, phase dependency table.                                                                                                           | not applicable; planning/audit layer only                                                                                                      | critical | 1.1.1.3     |
| LEG8  | Pruning and dead-candidate ledger                              | Governance; Policy; Object | F3   | Y             | PruningLedger classifying declarations/modules as keep, quarantine, wrap, reexport, archive, or deletion-candidate.                                                                                                                                                                                                                                                                                                                      | Consumption map + public API contract + reference summaries.                                                                                      | LEG5, LEG6, symbol reference summaries, import graph.                                                                                                                 | not applicable; planning/audit layer only                                                                                                      | critical | 1.1.1.4     |
| LEG9  | Generated-view authority and implementation-lock contract      | Governance; Policy; Object | F3   | Y             | GeneratedViewAuthorityLock tying YAML master, generated tables/PDF, and local-build evidence rule to the next implementation prompt.                                                                                                                                                                                                                                                                                                     | Single-YAML rule + generated table validation + PDF render check.                                                                                 | 1.1.1.1-1.1.1.4 records and generate_tables.py output.                                                                                                                | not applicable; planning/audit layer only                                                                                                      | critical | 1.1.1.5     |
| LEG10 | Implementation module manifest and semantic placement contract | Governance; Policy; Object | F1   | G             | ModuleImplementationManifest: for each implementation phase, record exact proposed added modules, modified modules, module budget, semantic placement rationale, existing-module fit check, public/legacy role, import direction, expected consumers, phase type, regression protocol, version impact, concrete-instance traversal, generic test placement, mathlib/noncomputable risk, and rejection reason for alternative placements. | Generated module inventory + phase/object DAG + active scratchpad + frozen Lean source snapshot.                                                  | repo_inventory/raw_export/modules.json, import graph, object register, current phase target objects, implementation scratchpad fields, and existing module semantics. | not applicable; planning/audit layer only                                                                                                      | critical | 1.1.1.1.1   |
| LEG11 | Reuse/duplication ledger and canonical generator policy        | Governance; Policy; Object | F1   | G             | ReuseDuplicationLedger plus CanonicalGeneratorPolicy: no second implementation of an already available semantic operation; generator-like request/output, smoke-test, extraction, and ledger code must reuse or extend the canonical theory/smoke-test generator seam.                                                                                                                                                                   | Existing Smoke/Test generator chain, static symbol/reference summaries, and current phase target semantics.                                       | LEG10, repo_inventory/raw_export/symbol_reference_summary.json, existing Smoke/Test generator modules, and planned target objects.                                    | not applicable; planning/audit layer only                                                                                                      | critical | 1.1.1.1.2   |
| LEG12 | Object and size proof-documentation infrastructure             | Governance; Policy; Object | F1   | G             | Generated proofbook infrastructure that links each registered object/size to a dossier containing source objects, definitions, proof target, proof sketch, and verification/audit status.                                                                                                                                                                                                                                                | Object register + code derivation sites + blue-object correction requirements + generated single-master views.                                    | registered object IDs, phase IDs, derivation-site fields, proof-documentation status fields.                                                                          | No ExecComplex computation; documentation governance only. ExecComplex objects must document operational/mirror boundary in their own dossier. | high     | 1.1.1.1.3   |
| LEG13 | Global no-go guardrail contract                                | Governance; Policy; Object | F0   | G             | GlobalNoGoGuardrailContract: top-level invariant list for operational path, mirror isolation, inverse derivations, formula provenance, backjump discipline, positive-path ordering, public API naming, structural proof strategy, legacy quarantine, Phase 3+ import firewall, and YAML-only roadmap authority.                                                                                                                          | Prior Lean implementation failure modes, user-specified global constraints, legacy quarantine requirement, and single-YAML planning architecture. | global_no_go_guardrails YAML section, workflow_policy inheritance rule, and per-phase global_no_go_inheritance scratchpad field.                                      | No ExecComplex computation; governance-only invariant. Operational ExecComplex phases must still inherit the listed constraints.               | critical | 1.1.1.1.0   |
| LEG14 | Concrete computable instance traversal contract                | Governance; Policy; Object | F3   | G             | Global scratchpad contract requiring every operational computable phase to traverse the path with a concrete finite instance or to declare itself infrastructure-only.                                                                                                                                                                                                                                                                   | Observed risk that computable code can become a proof-shaped skeleton without executable instance pressure.                                       | active phase target declarations; existing generator route; concrete finite substrate/window; expected executable output witness.                                     | Directly governs the operational computable path; not a mathematical input.                                                                    | critical | 1.1.1.1.0.1 |

| ID    | Object                                                             | Context class              | Tier | Object status | Canonical CNNA definition                                                                                                                                                                   | Natural provenance                                                                                                                                              | Computed from                                                                                                                            | ExecComplex/computable path                                                                                                                                                              | Priority            | Phase       |
|-------|--------------------------------------------------------------------|----------------------------|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------|
| LEG15 | Mathlib complication and noncomputable-boundary log                | Governance; Policy; Object | F3   | G             | Per-phase scratchpad log for mathlib import availability, noncomputable leaks, executable-code barriers, mirror-layer isolation, and required natural backjump/refactor layer.              | Prior Lean phases encountered mathlib APIs whose algebraic interfaces compile only noncomputably; operational CNNA must record and isolate these complications. | expected_mathlib_imports; observed compiler/noncomputable errors; operational vs mirror-layer classification; mitigation decision.       | Protects the operational path from hidden noncomputable dependencies; mirror-only usage must be isolated and documented.                                                                 | critical            | 1.1.1.1.0.1 |
| LEG16 | Semi-automated existing-code documentation coverage                | Reference; Gate; Object    | F3   | K             | A generated coverage/documentation ledger tying Lean modules/declarations to object proof dossiers, consumers, and provenance classifications.                                              | repo_inventory/raw_export plus object register, proof dossiers, and declaration consumption classification.                                                     | modules.json, import_edges.json, symbol_reference_summary.json, object_proof_dossiers, implementation_scratchpad.                        | Non-operational documentation metadata; may reference executable witnesses but does not execute them.                                                                                    | secondary-long-term | 19.1        |
| LEG17 | Derivation-graph geometry diagnostics                              | Reference; Gate; Object    | F3   | K             | A non-operational diagnostic profile for the repository derivation graph: import edges, dominators, layers, source-sink paths, bottlenecks, cuts, and discipline clusters.                  | Generated Lean module inventory and graph exports.                                                                                                              | import_edges.json, dominator/layer/path summaries, module_group_dependency_graph, core_robust_semantic_mainline.                         | Graph-metadata computation only; not part of CNNA operational mathematics.                                                                                                               | secondary-long-term | 19.2        |
| LEG18 | Open-system emergence-path forensics                               | Reference; Gate; Object    | F3   | K             | A diagnostic map locating formal proof/module paths from regime/complement/boundary data toward effective open-system dynamics consumers.                                                   | Checked A regime/complement, DtN/Schur, A->C, and C/OQS derivation paths plus graph analysis.                                                                   | future checked object dossiers and source-sink/dominator path analysis over relevant modules.                                            | Non-operational path forensics. It may point to executable witnesses but does not generate OQS behavior.                                                                                 | secondary-long-term | 19.3        |
| LEG19 | Reflective CNNA/formalization graph comparison boundary            | Reference; Gate; Object    | F3   | K             | A boundary object marking any comparison between CNNA object-structure and the formalization graph as meta-level, observational, and non-generative.                                        | Stable CNNA object proof dossiers plus generated graph metrics.                                                                                                 | core object dossiers, discipline subgraphs, graph cuts, layers, signal paths, and cluster analysis after stabilization.                  | Non-operational meta-comparison only.                                                                                                                                                    | secondary-long-term | 19.4        |
| LEG20 | EXEC-2 rational ExecComplex boundary policy                        | Lean; Governance; Object   | F1   | G             | ExecComplexRationalBoundaryPolicy Superseded/refined by LEG31/EXEC-3 for exact quadratic candidate policy.                                                                                  | Review of ExecComplex = Q x Q and later algebraic CM/lattice requirements                                                                                       | O00/O01/O06 plus L00/L02/L04/M05/SP0 boundary risks                                                                                      | Governance object; forces rational/integer operational return only, mirror-only algebraic irrational data                                                                                | high                | 1.1.1.1.0.2 |
| LEG21 | Refactor regression and incremental migration protocol             | Lean; Governance; Object   | F1   | G             | RefactorRegressionProtocol                                                                                                                                                                  | Notation split and legacy-quarantine risk audit                                                                                                                 | module manifest, consumer list, import graph delta, local build evidence                                                                 | Infrastructure-only; no operational witness, but requires build-regression evidence for refactor code packages                                                                           | high                | 1.1.1.1.0.3 |
| LEG22 | Lean/mathlib version migration policy                              | Lean; Governance; Object   | F1   | G             | LeanMathlibVersionPolicy                                                                                                                                                                    | Frozen baseline plus future mathlib/API pressure                                                                                                                | Lean 4.28.0/mathlib snapshot and explicit upgrade gate decisions                                                                         | Infrastructure-only; blocks silent noncomputable/classical/import drift into operational code                                                                                            | high                | 1.1.1.1.0.4 |
| LEG23 | Phase-type-specific scratchpad schema and governance retrospective | Documentation; Object      | F3   | G             | PhaseTypeScratchpadPolicy                                                                                                                                                                   | Review that refactor, derivation, infrastructure and comparison phases need different evidence                                                                  | phase_type, type-specific contract, governance_overhead_review                                                                           | Infrastructure-only; prevents proof-rehearsal ballast in pure refactor phases and missing migration metrics                                                                              | high                | 1.1.1.1.0.5 |
| LEG24 | Phase 3 to Phase 4 consumer-contract direction                     | Lean; Governance; Object   | F1   | G             | Phase3ToPhase4ConsumerContract                                                                                                                                                              | GEN-1 generator policy and Schur/DtN -> Exec arithmetic dependency direction                                                                                    | Phase 3.6 BoundaryOperatorProfile and Phase 4.1 ExecBoundaryMatrix                                                                       | Infrastructure-only; records import direction and avoids duplicate Schur/generator logic                                                                                                 | high                | 1.1.1.1.0.6 |
| LEG25 | IdealThread/CutSpec.full extremal-route contract                   | Governance; Policy; Object | F1   | W             | Planning contract separating the singleton IdealThread route from the maximal CutSpec.full route and marking string-like interpretation as comparison-only.                                 | Code-checked IdealThread and CutSpec.full definitions plus ExistingIdealThreadNoGo and FullBoundaryAudit.                                                       | SubstrateClass.InfiniteThread; ToC.IdealThread.carrier/family/coarsen; Finite.CutSpec.full; boundary-source audits.                      | No operational scalar introduced. Constrains later operational uses of scalar/thread versus full-boundary paths.                                                                         | high                | 1.1.1.1.0.7 |
| S07   | IdealThread singleton directed-family profile                      | Lean; Source; Object       | F1   | G             | Neutral profile of an IdealThread U with carrier(L) = {U.cell L}, its DirectedFamily embedding, rootedness, forward property, and deterministic coarsen fixed-point behavior.               | InfiniteThread + IdealThread.carrier + IdealThread.family in ToC.Contract.                                                                                      | SubstrateClass, InfiniteThread, IdealThread definitions; DeterministicSubstrateClass for coarsen fixed-point theorem.                    | Pure structural finite-set profile; no ExecComplex arithmetic. Concrete instance traversal must use reference/variation thread carriers and show singleton cardinality at finite levels. | high                | 1.2.7       |
| S08   | CutSpec.full maximal-carrier profile                               | Lean; Source; Object       | F1   | G             | Neutral profile of CutSpec.full T where carrier(L) = T.carrier L; represents the maximal finite cut/full-region route.                                                                      | Finite.CutSpec.full and full_carrier theorem.                                                                                                                   | TruncatedFamily carrier and CutSpec carrier_subset proof.                                                                                | Pure finite-set/cut profile; later matrix/operator computations consume the resulting full boundary/region data, not IdealThread singleton data.                                         | high                | 1.2.8       |
| BR06  | Boundary-route extrema contract                                    | Lean; Source; Object       | F1   | G             | Contract separating full-boundary matrix geometry from IdealThread scalar/singleton collapse before Schur/DtN neutral wrappers.                                                             | S07 + S08 + ExistingIdealThreadNoGo + FullBoundaryAudit.                                                                                                        | IdealThread singleton profile, CutSpec.full profile, boundary ports subset/carrier facts, and planned BoundaryOperatorProfile consumers. | Constrains later ExecComplex matrices: nontrivial boundary matrices must come from full/generated boundary route; thread route may only supply scalar audits unless lifted by theorem.   | high                | 3.0         |
| BR07  | Thread-recursion arithmetic-shadow audit profile                   | Lean; Source; Object       | F1   | Y             | Audit profile determining whether scalar Schur/Mobius recurrence data is sourced from IdealThread singleton carriers, full-boundary diagonal entries, or another generated boundary source. | RecursiveSchurSource, MobiusTransfer, CharpolyExec, operational discriminant chain, plus S07/BR06 route separation.                                             | Code references to recurrence source fields, boundaryMatrixAt diagonal entries, transfer parameters, and discriminant outputs.           | Operational only after concrete finite instance traversal proves the recurrence path and executable witness; otherwise mirror/comparison-only.                                           | high                | 4.0         |



| ID    | Object                                                           | Context class              | Tier | Object status | Canonical CNNA definition                                                                                                                                                                                                                                                                                                 | Natural provenance                                                                                                                                                           | Computed from                                                                                                                                                                                             | ExecComplex/computable path                                                                                                                                                                                                  | Priority                                        | Phase        |
|-------|------------------------------------------------------------------|----------------------------|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------|
| BR09  | Subsystem-induced invariant bifurcation profile                  | Lean; Target; Object       | F1   | K             | SubsystemSchurInvariantBifurcationProfile: cutting the infinite substrate creates boundary/interior data whose Schur complements expose horizontal algebraic invariants and vertical arithmetic invariants as two audited output families, without treating the slogan as proof.                                          | Finite CutSpec/subsystem boundary plus multiscale Schur decomposition.                                                                                                       | BR08 plus route-specific invariant records CD7/O05/TM0/S05.                                                                                                                                               | Diagnostic/profile layer; operational witnesses must be concrete finite cuts and scalar recursion examples.                                                                                                                  | low                                             | 10.4.1       |
| BR08  | Horizontal/vertical Schur-duality bridge                         | Lean; Target; Object       | F1   | K             | SchurHorizontalVerticalDualityProfile: horizontal sector Schur elimination and vertical level-recursion Schur elimination are compatible factorizations/aspects of one multiscale Schur decomposition of the derived Dirichlet-Laplacian, with explicit commutation/naturality conditions.                                | Derived DtN boundary operator, sector decomposition, IdealThread/CutSpec route split, transfermatrix recursion, and CD/Hurwitz profile.                                      | S05, BR06, BR07, S07, S08, TM0, TM1, TM2, CD7, O05.                                                                                                                                                       | Bridge theorem may use finite operational witnesses; no algebraic irrationality is admitted except through EXEC-2 allowed mirror/invariant boundaries.                                                                       | medium                                          | 10.4.1       |
| LEG26 | Reflective thread/full-boundary analogy diagnostic boundary      | Governance; Policy; Object | F4   | K             | Boundary object ensuring thread/full-boundary and string-like analogies remain documentation/diagnostic comparisons until formal CNNA objects and proofs are closed.                                                                                                                                                      | LEG25, LT5, repository derivation graph metadata, later S07/S08/BR06/BR07 dossiers.                                                                                          | Closed proof dossiers and generated derivation-graph metadata only after not-before gates.                                                                                                                | None. Diagnostic/meta-documentation only; never an operational input.                                                                                                                                                        | low                                             | 19.5         |
| LEG27 | Horizontal/vertical Schur-duality dependency-order lock          | Governance; Policy; Object | F1   | G             | SchurDualityOrderLock: horizontal-sector Schur and vertical-level Schur may be compared only after their source routes are derived and exported.                                                                                                                                                                          | Review insight plus existing route plan S07/S08/BR06/BR07/CD7/O05/TM0.                                                                                                       | Planning master and dependency order.                                                                                                                                                                     | No operational computation; constrains future bridge phases.                                                                                                                                                                 | high                                            | 1.1.1.1.0.8  |
| LEG28 | Heuristic Eisenstein/Gauss lattice test-matrix boundary contract | Governance; Policy; Object | F1   | G             | A global planning contract saying that post-97 b-ary-tree spiral constructions are test and comparison targets, not generator definitions.                                                                                                                                                                                | TomS-thread post #97 audit plus EXEC-2 rational boundary and canonical generator policy.                                                                                     | post-97 source notes, code-defined branching policy, L_max sweep requirement, CM/j target boundary.                                                                                                       | Operational path records rational/integer finite invariants only; irrational coordinates, tau, sqrt(D), and analytic j evaluation remain mirror/comparison.                                                                  | high                                            | 1.1.1.1.0.9  |
| LEG29 | General early finite computation test-matrix contract            | Governance; Policy; Object | F1   | G             | Global contract requiring early, general finite test matrices for operational computable paths; special Eisenstein/Gauss rows are sentinel rows only.                                                                                                                                                                     | Concrete computable instance traversal contract; v32 heuristic lattice test matrix; user requirement for early non-specialized theory testing.                               | phase manifest, canonical generator policy, finite code-defined inputs, expected executable witness schema, sentinel-row separation.                                                                      | Operational rows may contain only finite/rational/integer fields; mirror-only columns hold tau, sqrt(D), analytic j, or geometric coordinates.                                                                               | immediate governance before code implementation | 1.1.1.1.0.10 |
| L09   | Static branching Eisenstein/Gauss lattice test matrix            | Reference; Gate; Object    | F2   | Y             | Static-branching sentinel extension of the general finite computation test matrix, including Eisenstein/Gauss rows as special comparisons rather than the full test harness. v34 further requires that its Eisenstein/Gauss rows be tagged as noOuter-root sentinel candidates, not as universal Heegner representatives. | Consumes canonical generator/branching profile, L06 angle/cosine profile, L04 tau target, O05 discriminant target, and M05 j target after those are available.               | O07 generic test matrix profile, code-defined b, L_max, finite addresses, selected angle profile, rational cos(theta), closure-depth witness, target D and j comparison row., derived regime tag from O08 | Computable part: b, finite address counts/relations, rational cos(theta), rational/integer Gram/determinant surrogates, D target. Mirror-only: complex coordinates, tau, analytic j.                                         | high                                            | 8.5          |
| L10   | Dynamic branching and L_max lattice test sweep                   | Reference; Gate; Object    | F2   | Y             | Dynamic-branching extension of the general finite computation test matrix over code-defined branch families and L_max values.                                                                                                                                                                                             | Consumes generated branch-family route and the static L09 matrix.                                                                                                            | O07 generic test matrix profile, canonical generator outputs, branch policy at each level, finite address sets, selected angle/sentinel profiles, closure/collision finite relations.                     | Operational outputs are finite relations and rational/integer invariants; mirror outputs remain separated.                                                                                                                   | high                                            | 8.6          |
| M08   | Eisenstein/Gauss CM and j special-value test targets             | Reference; Gate; Object    | F2   | Y             | Comparison target row linking Eisenstein rho to D=-3 and j=0, and Gauss i to D=-4 and j=1728, used only as sentinel columns after generic coverage exists. In v34 these are noOuter-root sentinel targets only if the row carries derived noOuter provenance.                                                             | Consumes L04 tau, O05 D, M05 j-invariant; supported as established special-value targets but not operationally evaluated in ExecComplex. v37: retained in modular/CM Path A. | CM-target records and mirror/comparison layer; not from geometric drawing input.; exact quadratic coordinates only through O09/O10/BR14 when required                                                     | j target values 0 and 1728 are integer comparison targets; analytic j evaluation and tau live outside operational ExecComplex. Derived-quadratic update: no free sqrt(D); exact coordinates require proof-carrying D-source. | high                                            | 8.5          |
| BR10  | Hierarchical versus spiral embedding collision and closure audit | Reference; Gate; Object    | F2   | Y             | Audit profile separating angle-invariant hierarchical address overlays from angle-dependent spiral embedding closure/collision behavior.                                                                                                                                                                                  | Consumes finite address structure, code-defined branching, L09/L10 test rows, and post-97 heuristic source notes.                                                            | O07 generic rows plus finite address words, level L_max, embedding-kind tag, angle profile, finite collision/closure relation witnesses.                                                                  | Finite address/collision relations are operationally computable; geometric coordinates remain mirror-only unless encoded by rational surrogates.                                                                             | medium                                          | 8.6          |
| LEG30 | Regime-typed early test-matrix contract                          | Governance; Policy; Object | F1   | G             | Global contract requiring early finite test rows to carry a derived regime tag: generic, noOuter-root/closed candidate, or HasOuter/complement-projection/comparison. h(D)=1 or Heegner-list membership is never itself noOuter evidence.                                                                                 | v33 O07 general test matrix; noOuter/HasOuter regime discipline; IDEAL_infty projection/complement-family interpretation; post #97 Eisenstein/Gauss sentinel hypothesis.     | planning decision locks, regime split objects, O07 row schema, EXEC-2 boundary, external Heegner/class-number-one information as comparison-only source material.                                         | Operational rows remain finite/rational/integer and carry regime tags; mirror/comparison columns may mention tau, sqrt(D), j, CM or Heegner labels.                                                                          | immediate governance before code implementation | 1.1.1.1.0.11 |

| ID    | Object                                                              | Context class              | Tier | Object status | Canonical CNNA definition                                                                                                                                                                                                                                                          | Natural provenance                                                                                                                                                              | Computed from                                                                                                                                                                    | ExecComplex/computable path                                                                                                                                                                            | Priority                                            | Phase        |
|-------|---------------------------------------------------------------------|----------------------------|------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------|
| O08   | Regime-typed finite computation test matrix profile                 | Lean; Target; Object       | F2   | Y             | Extension of O07 whose rows include branch mode, L_max, executable witness, and a derived regime tag separating noOuter-root closed candidates from HasOuter/complement-projection/comparison cases.                                                                               | O07 generic finite computation test matrix plus noOuter/HasOuter regime split and concrete computable instance traversal.                                                       | canonical generator outputs, finite L_max rows, static/dynamic branch-mode tags, regime split witness, rational/integer operational outputs, optional mirror comparison targets. | Core row fields are operational finite/rational/integer; tau, sqrt(D), analytic j, and general algebraic coordinates remain mirror-only.                                                               | early before specialized computation claims         | 4.0.2        |
| N18   | Heegner regime-projection classification profile                    | Lean; Target; Object       | F1   | Y             | Consumer-only profile classifying class-number-one/Heegner and order-discriminant targets by derived CNNA regime: closed noOuter-root sentinel candidate versus HasOuter/complement-family projection/comparison.                                                                  | ClassNumberProfile, HeegnerIndicatorProfile, O08 regime-typed test matrix, IDEAL_infty projection discipline.                                                                   | h(D), discriminant/order data, units/class group evidence, derived regime tag, complement-family/projection metadata. External Heegner lists are comparison checks, not sources. | Integer/rational invariants such as D and h(D) may be operational after derivation; tau/j/CM analytic interpretation remains mirror/comparison.                                                        | medium/high before Heegner interpretations are used | 7.2.1        |
| BR11  | IDEAL_infty projection and complement-family Heegner boundary       | Lean; Target; Object       | F1   | Y             | Boundary statement that additional Heegner/class-number-one phenomena beyond noOuter-root sentinels, if they appear, are interpreted first as projections/approximations of IDEAL_infty through complement-family/regime choices, not as multiple closed roots.                    | IDEAL_infty projection discipline, complement-family architecture, noOuter/HasOuter split, N18 classification.                                                                  | formal regime data, complement-family metadata, finite approximant rows, class-number-one comparison rows.                                                                       | No direct computation; consumes operational finite rows and rational/integer invariants; mirror-only CM labels remain separated.                                                                       | medium                                              | 7.2.1        |
| BR12  | noOuter REAL-root Eisenstein/Gauss sentinel hypothesis              | Lean; Target; Object       | F1   | Y             | Hypothesis/test profile that Eisenstein D=-3/j=0 and Gauss D=-4/j=1728 are the likely sentinel cases for the closed noOuter/empty-set REAL approximant, not a blanket statement about all Heegner targets.                                                                         | post #97 angle/closure heuristic; noOuter REAL-root regime discipline; O08 regime-typed test matrix; EXEC-2.                                                                    | derived noOuter-root finite rows, closure-depth witness, rational angle/cosine sentinel data, later O05/M08 comparison targets.                                                  | Operational row uses finite/rational/integer witnesses and target integers; tau and analytic j remain mirror-only.                                                                                     | high as early sentinel hypothesis                   | 4.0.2        |
| LEG31 | Computable scalar architecture contract                             | Governance; Policy; Object | F0   | G             | ScalarArchitectureContract: ExecComplex is the rational operational floor; exact quadratic algebraic scalar data require a separate proof-carrying operational layer; analytic Complex/Real remains mirror-only; CD-derived C-layer use is blocked until noncircular bridge proof. | ExecComplex code audit, ExecComplexBridge, CD bridge no-identification status, mathlib Complex/QuadraticAlgebra docs                                                            | O00, LEG20, LEG22, current code import boundaries, mathlib docs                                                                                                                  | Governance object protecting executable path; no direct computation.                                                                                                                                   | critical                                            | 1.1.1.1.0.12 |
| LEG32 | Derived-only exact quadratic provenance contract                    | Governance; Policy; Object | F0   | G             | Contract that the exact quadratic layer is a canonical algebra generated from a previously derived discriminant parameter D, not a free sqrt(D), axiom, analytic Complex object, or Cayley-Dickson early scalar source.                                                            | User-derived-only objection to introducing sqrt(D); scalar architecture review; mathlib QuadraticAlgebra/AdjoinRoot reference models for adjoining generators by relations.     | LEG31, O00, O05/O06 discriminant-source plan, mathlib reference docs, CD9 noncircularity constraint                                                                              | Protects operational computability: exact quadratic coordinates are generated only from proof-carrying D and project/rationalize invariants back to QxQ/Int where allowed.                             | critical                                            | 1.1.1.1.0.13 |
| LEG33 | Phase 5-10 bifurcation and dependency repair contract               | Governance; Policy; Object | F1   | G             | PhasePathBifurcationContract.                                                                                                                                                                                                                                                      | v36 phase/object audit found artificial dependency of SP/P objects on modular-form Phase 8/9; v37 records corrected path split.                                                 | Phase/object register; dependency audit; Phase 4.1 consumer requirements.                                                                                                        | planning-only; no operational scalar content.                                                                                                                                                          | high                                                | 1.1.1.1.0.14 |
| O09   | Required exact quadratic operational scalar schema                  | Lean; Target; Object       | F1   | Y             | Required minimal discriminant-indexed schema for elements $a + b \cdot \theta$ with $\theta^2 = D$ over Q, where D is only a parameter at schema level and must later come from a proof-carrying operational discriminant ledger before instantiation.                             | CM/lattice coordinates require algebraic quadratic data, but derived-only discipline forbids free sqrt(D); the schema is the canonical algebraic wrapper over a later D-source. | LEG31, LEG32, O00; concrete instantiations require O05/O06 via O10/BR14                                                                                                          | Executable coordinate arithmetic over Q if implemented directly/wrapped safely; not a replacement for general Complex and not instantiated without D-source.                                           | critical                                            | 1.2.1.2      |
| O10   | Proof-carrying discriminant-indexed quadratic scalar instance       | Lean; Target; Object       | F1   | Y             | An instantiation of the exact quadratic scalar schema from a ledger-proved $D : \text{Int}$ , producing $\theta$ with $\theta^2 = D$ and coordinate elements $a + b \theta$ only after D has been derived operationally.                                                           | O09 schema cannot be instantiated freely; it must consume O05/O06 or equivalent operational discriminant certificate.                                                           | O05/O06 proof-carrying discriminant ledger, O09 schema, LEG32                                                                                                                    | Coordinates are exact rational pairs tagged by D; projected trace/norm/discriminant/integer invariants may flow to ExecComplex-compatible reports. No analytic sqrt or Complex coordinate is required. | critical                                            | 4.5.1        |
| BR13  | ExecComplex to exact-quadratic projection bridge                    | Lean; Target; Object       | F1   | Y             | Bridge that relates rational ExecComplex outputs, exact quadratic scalar rows, D-source instantiations, and integer/rational invariants without allowing quadratic coordinates to replace generator semantics or appear without provenance.                                        | Operational finite tests need exact algebraic sentinels while preserving QxQ boundary.                                                                                          | O00, O09, O10, BR14, O07, O08, LEG31/LEG32                                                                                                                                       | Operational bridge only for rational/integer invariants and exact quadratic arithmetic if O09 exists; analytic Complex remains mirror-only.                                                            | high                                                | 4.0.3        |
| BR14  | Discriminant-source to exact-quadratic algebra instantiation bridge | Lean; Target; Object       | F1   | Y             | Bridge proving that an exact quadratic scalar instance is generated from a proof-carrying operational discriminant D and that all exported invariants are projections from this instance, not independent formulas.                                                                | Derived-only scalar provenance requirement: $\theta$ is generated by the relation $\theta^2 = D$ after D is computed.                                                           | O05/O06, O09, O10, LEG32                                                                                                                                                         | Exports only rational/integer invariants to QxQ path; keeps quadratic coordinates in the exact quadratic layer.                                                                                        | critical                                            | 4.5.1        |

| ID    | Object                                                     | Context class                | Tier | Object status | Canonical CNNA definition                                                                                                                                                                                                                                                 | Natural provenance                                                                                                                                                                                                                               | Computed from                                                                                                                                                                              | ExecComplex/computable path                                                                                                                               | Priority              | Phase        |
|-------|------------------------------------------------------------|------------------------------|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------|
| CD9   | Cayley-Dickson C-layer noncircularity audit                | Reference; Gate; Object      | F1   | K             | Late audit asking whether a CD-generated complex scalar surface can be compared to or bridge O09/O10 without consuming ExecComplex, analytic Complex, free sqrt(D), or later arithmetic data circularly.                                                                  | User question about deriving C through Cayley-Dickson and avoiding ExecComplex circularity.                                                                                                                                                      | CD7, CD8, BR08, BR09, O09/O10, current ArithmeticCDBridge no-identification ledger                                                                                                         | Not operational until a theorem proves noncircular scalar-surface provenance. Current code keeps CD and Exec/mirror scalar surfaces separate.             | medium                | 10.4.3       |
| LEG34 | Phase-object-link policy contract                          | Governance; Policy; Object   | F1   | G             | Global planning contract classifying every phase-object relation as direct ownership, scope-only umbrella, no-direct-object audit/baseline/closeout, or moved placeholder.                                                                                                | v37 linked-object ambiguity: generator showed blank cells whenever no object had Phase equal to the phase id, even if child phases owned objects or the phase was a valid audit/umbrella.                                                        | phase hierarchy; object register Phase field; generator direct-object map; child phase scopes; moved/audit/closeout markers                                                                | not mathematical computation; generated planning validation only                                                                                          | active-infrastructure | 1.1.1.1.0.15 |
| LEG35 | Ontic subsystem-minimality contract                        | Governance; Policy; Object   | F0   | G             | Global postulate: the only primitive ontic emergence trigger is subsystem-being, i.e. selecting a finite/local subsystem, cut or approximant inside the formal IDEAL_infty provenance whole. All further structures are derived consequences or comparison-only readings. | User-requested ontic-minimality sharpening after regime-typed test-matrix, Schur-duality, and open-system emergence discussions: avoid importing dynamics, physics, environment, stochasticity, arithmetic or geometry as primitive assumptions. | IDEAL_infty formal provenance whole; subsystem/cut/approximant selection; boundary; inside/outside split; complement families; noOuter/HasOuter regime; Schur/DtN effective operator chain | not a numerical object; constrains all computable paths by requiring concrete finite subsystem traversal before derived spectra/arithmetic/physics claims | active-infrastructure | 1.1.1.1.0.16 |
| DBR0  | Database-ready CNNA review and documentation structure     | Documentation; Object        | F3   | G             | A non-operational review/documentation structure that keeps the Single-YAML-Master database-ready and prepares queryable phase/object/module/handoff/theory/review representations.                                                                                       | User instruction plus existing Single-YAML-Master, generated views, module inventory, object register, proof dossiers, status legends, and workflow policies.                                                                                    | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; repo_inventory/raw_export; tables/*.tex; figures/*.dot/pdf.                                                                     | Non-operational documentation/review metadata only; no ExecComplex or mathematical computation is introduced.                                             | main-side-objective   | 20           |
| DBR1  | Expert-review and no-infallibility epistemic rule          | Documentation; Object        | F1   | G             | A global rule that Lean code is the primary truth system but not semantically infallible; generated artifacts are secondary; every relevant claim remains subject to discipline-specific expert review.                                                                   | User instruction; CNNA derived-only methodology; distinction between Lean formal consistency and semantic/domain correctness; GNG23-GNG24.                                                                                                       | global_no_go_guardrails GNG23-GNG24; workflow_policy DB-REVIEW; status_legend review/audit rows; latest_change_overview V1.07.                                                             | Non-operational governance rule only; no computable mathematical path.                                                                                    | immediate-governance  | 20.1         |
| DBR2  | Database-ready YAML schema and round-trip target           | Tool; Infrastructure; Object | F1   | K             | A future schema/export target mapping YAML phases, objects, proof dossiers, modules, handoffs, statuses, theory markings, and review states into relational or graph-friendly tables.                                                                                     | Single-YAML-Master schema and generated TSV/JSON/DOT exports.                                                                                                                                                                                    | cnna_roadmap_master.yaml; generated_exports/*.tsv; repo_inventory/raw_export/*.json; future schema audit.                                                                                  | Non-operational export metadata; not mathematical computation.                                                                                            | secondary-tooling     | 20.2         |
| DBR3  | Discipline-separated expert-review matrix                  | Documentation; Object        | F3   | K             | A future review matrix linking claims, objects, phases, modules, and established-theory references to required specialist disciplines and review evidence/status.                                                                                                         | Object proof dossiers, local theory context, end-theory comparison ledger, blue/reference audit, human expert review inputs.                                                                                                                     | object_proof_dossiers; workflow_policy; status_legend review/audit rows; future discipline review rows.                                                                                    | Review metadata only; no operational computation.                                                                                                         | secondary-review      | 20.3         |
| DBR4  | Blue/reference-theory audit ledger                         | Documentation; Object        | F3   | K             | A future audit ledger for established-theory/reference material that may need explicit blue marking or comparison-only classification.                                                                                                                                    | GNG24, local theory context versus end-theory comparison workflow, external source cross-checks, object register, proof dossiers.                                                                                                                | status_legend; external_source_crosscheck; consumption_classification_map; object_register; future blue-audit rows.                                                                        | Audit metadata only; no operational computation.                                                                                                          | secondary-audit       | 20.4         |
| DBR5  | Interactive/online database-backed documentation views     | Tool; Infrastructure; Object | F1   | K             | A future generated-secondary interface for querying and visualizing phases, modules, objects, DAGs, handoffs, theory markings, and review status online or interactively.                                                                                                 | DBR2 schema/export target, DBR3 review matrix, DBR4 blue/reference audit ledger, generated exports and graph figures.                                                                                                                            | future database export; generated_exports; figures; tables; review ledgers.                                                                                                                | Display/query metadata only; not a CNNA mathematical computation.                                                                                         | secondary-tooling     | 20.5         |
| DBR6  | Context-filtered documentation slice generator             | Generated; Artifact; Object  | F3   | G             | A generator that creates query/profile-specific documentation packets preserving every major documentation section while emitting only rows relevant to the requested context. It consumes the Single-YAML-Master and generated TSV exports only.                         | DB-REVIEW side objective; user requirement for targeted documentation such as DtN matrix documentation and phase-closeout change documentation; generated-secondary/Lean-primary rule.                                                           | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; context_documentation_profiles; scripts/generate_context_doc.py.                                                                | Python tooling path only; not an operational CNNA mathematical computation and not a Lean theorem.                                                        | active-tooling        | 20.6         |
| DBR7  | Stats front section and tool-interface documentation layer | Generated; Artifact; Object  | F3   | G             | A generated front documentation surface that summarizes project facts, embeds the repository module inventory and derivation graphs, records the tool-interface guiding metaphor, and states project licensing notices.                                                   | DB-REVIEW side objective; user requirement for a concise Leitmetapher, a Stats opening section, explicit documentation/code licensing, and integration of the former repository module inventory section.                                        | masters/cnna_roadmap_master.yaml; repo_inventory/raw_export/stats.json; repo_inventory/raw_export/modules.json; generated_exports/*.tsv; scripts/generate_tables.py.                       | Python documentation-generation path only; not an operational CNNA mathematical computation and not a Lean theorem.                                       | active-tooling        | 20.7         |



| ID    | Object                                                                      | Context class               | Tier | Object status | Canonical CNNA definition                                                                                                                                                                                                                                                                                     | Natural provenance                                                                                                                                                                                                                                                   | Computed from                                                                                                                                                                    | ExecComplex/computable path                                                                                                                                                     | Priority                             | Phase       |
|-------|-----------------------------------------------------------------------------|-----------------------------|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------|
| LEG36 | Semantic module consolidation ledger                                        | Governance; Policy; Object  | F1   | Y             | A pre-substantive audit ledger that classifies semantically related Lean modules into safe-merge candidates, facade-preserve candidates, split/keep candidates, and no-merge proof-stage chains. It records natural-layer placement and regression obligations before any code consolidation.                 | Generated module inventory, import graph, declaration consumption map, reuse/duplication ledger, and direct spot-checks of candidate code families.                                                                                                                  | LEG10, LEG11, LEG5, repo_inventory/raw_export/modules.json, import_edges.json, symbol_reference_summary.json, and manual code inspection of high-similarity module clusters.     | not applicable; planning/refactor governance layer only. If later Lean refactor changes executable modules, the regression package must preserve concrete computable witnesses. | critical-before-new-substantive-code | 1.1.1.1.5   |
| DBR8  | Root artifact retention and cleanup policy                                  | Generated; Artifact; Object | F3   | G             | A repository hygiene rule and toolchain pair that keeps the project root source-oriented by archiving historical Markdown reports and treating root PDF/TeX/LaTeX outputs as disposable generated artifacts.                                                                                                  | DB-REVIEW side objective; user requirement that PDFs and TeX need not be retained because they can be generated at any time; root-file audit after V1.12.                                                                                                            | masters/cnna_roadmap_master.yaml; root file inventory; scripts/generate_tables.py; scripts/generate_full_document.py; scripts/clean_generated_root_artifacts.py.                 | Python tooling path only; not an operational CNNA mathematical computation and not a Lean theorem.                                                                              | active-tooling                       | 20.8        |
| DBR9  | Current README and workflow-orientation manual                              | Generated; Artifact; Object | F3   | G             | A source-root README that replaces chronological release notes with a current operational manual for the CNNA planning/documentation tool.                                                                                                                                                                    | DB-REVIEW side objective; user request for a completely new README adapted to the current version; correction that the developed/tested assistant workflow is with ChatGPT, not Claude.                                                                              | masters/cnna_roadmap_master.yaml; generated_exports/project_stats.generated.tsv; existing scripts; user workflow requirements; V1.13 root cleanup policy.                        | Not applicable; README/documentation-tooling layer only. No mathematical CNNA object and no Lean theorem.                                                                       | active-tooling                       | 20.9        |
| DBR10 | Root Lean/Lake project-control files                                        | Generated; Artifact; Object | F3   | G             | The minimal package-root build-entry surface consisting of CNNA.lean, lakefile.lean, lake-manifest.json and lean-toolchain, with README documentation in Repository/ and optionally mirrored at package root. These files are source/control configuration, not disposable generated documentation artifacts. | User correction after V1.14 that the Lean toolchain must be included in the project root; uploaded lakefile.lean and CNNA.lean explicitly reference Repository/repo_snapshot as the source location.                                                                 | Uploaded CNNA.lean, lakefile.lean, lake-manifest.json, lean-toolchain; repo_snapshot/CNNA; scripts/clean_generated_root_artifacts.py; README.md.                                 | Not an ExecComplex mathematical computation. This is build reproducibility/toolchain configuration for local Lean/Lake builds.                                                  | active-tooling                       | 20.10       |
| DBR11 | Root license, single-root README, credits and assistant-requirement surface | Generated; Artifact; Object | F3   | G             | A package-root documentation/license surface that keeps the maintained README only in the root, adds a root LICENSE file, exposes editor/assistant credits, and records the minimum tested ChatGPT setup for assistant-instantiated workflows.                                                                | User requirements after V1.16: add license notices to README and root LICENSE; keep README only in root; add Editor Jan Seeck (antaris) and Assistant ChatGPT (OpenAI) also to generated PDFs; state the minimum requirement as a ChatGPT Plus account with GPT-5.5. | User-provided licensing URLs; README; license_notice table; metadata editor/assistant/minimum assistant requirement; full/context documentation generators; root file inventory. | Not applicable; documentation, licensing and tooling metadata only. No mathematical CNNA object and no Lean theorem.                                                            | active-tooling                       | 20.11       |
| DBR12 | Generated PDF section-pagebreak layout invariant                            | Generated; Artifact; Object | F3   | G             | A documentation-layout invariant requiring every top-level section in generated full and context PDFs to begin on a new page, without changing the underlying Lean/YAML truth boundary.                                                                                                                       | User requirement after V1.17 that each section should automatically start on a new page in generated documentation.                                                                                                                                                  | generate_full_document.py; generate_context_doc.py; metadata/tool version; generated table/context views.                                                                        | Not applicable; documentation layout invariant only. No mathematical CNNA object and no Lean theorem.                                                                           | active-tooling                       | 20.12       |
| DBR13 | Phase status release policy and red-lock ledger                             | Generated; Artifact; Object | F3   | G             | A phase-status governance object enforcing that direct Lean-derivation phases are locked red unless already green, uniquely active yellow, or orange by an explicit predecessor-release check; white is reserved for non-Lean/tooling/refactor/infrastructure/documentation/comparison phases.                | User requirement after V1.18 that all not-yet-derivable Lean-code phases default to red until the plan proves their release from derived-only Lean objects; white is redefined for non-direct-Lean implementation work.                                              | status_legend; phases; implementation_scratchpad.phase_type; pre_implementation_phase_check_id; generate_tables.py validation; existing derived-only/Lean-primary rules.         | Not applicable; planning/status-governance layer only. No mathematical CNNA object and no Lean theorem.                                                                         | active-tooling                       | 20.13       |
| DBR14 | Static documentation explanations and module-change appendix policy         | Generated; Artifact; Object | F3   | G             | A generated-documentation contract: static prose explains each section/table source and purpose, dynamic rows remain generated from YAML/exports, and the module-change manifest is a technical appendix used only for real module/API surface changes.                                                       | User request after V1.19 that the module manifest was confusing and that each section/table should be explained more clearly without making the prose dynamic.                                                                                                       | generate_full_document.py; generate_context_doc.py; generate_tables.py; Master YAML; generated exports; section order and module-manifest scratchpad fields.                     | Not applicable; documentation layout/semantics only. No mathematical CNNA object and no Lean theorem.                                                                           | active-tooling                       | 20.14       |
| LEG37 | Current source intake completeness ledger                                   | Governance; Policy; Object  | F1   | Y             | A complete classification ledger for the current Lean source snapshot: every module/declaration is linked to an object/proof dossier, legacy/audit-only bucket, tool/refactor bucket, or a newly created natural-origin ingestion phase.                                                                      | 1.1.1.1.5.1; repo_inventory/raw_export; generated_exports declaration/module consumption records; current object register.                                                                                                                                           | LEG10, LEG5, repo_inventory/raw_export/modules.json, symbol_reference_summary.json, module/declaration consumption TSVs.                                                         | not applicable; source inventory and classification audit only.                                                                                                                 | critical-before-new-substantive-code | 1.1.1.1.5.1 |
| LEG38 | Neutral module-name and facade migration map                                | Governance; Policy; Object  | F1   | Y             | A migration map assigning phase-semantic or legacy module names to neutral target names, facade modules, no-merge exceptions, and proof-stage preservation constraints.                                                                                                                                       | 1.1.1.1.5.2; closed LEG37 intake ledger; semantic consolidation findings SCF44/SCF45/SCF52.                                                                                                                                                                          | LEG37 plus direct source scan of SM*, S6*, StrengtheningPhase*, BoundarySource, Handoff and Notation/Legacy module families.                                                     | not applicable until execution; planned refactor surface only.                                                                                                                  | critical-before-new-substantive-code | 1.1.1.1.5.2 |

| ID                    | Object                                                 | Context class               | Tier | Object status | Canonical CNNA definition                                                                                                                                                                                                                                                                                                                                                    | Natural provenance                                                                                                           | Computed from                                                                                                                                                    | ExecComplex/computable path                                                                              | Priority                             | Phase                       |
|-----------------------|--------------------------------------------------------|-----------------------------|------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------|
| <a href="#">LEG39</a> | Neutral refactor execution certificate                 | Governance; Policy; Object  | F1   | Y             | A certificate for a bounded neutral-module refactor package: old-module facades, import/declaration delta, consumer breakage audit, and local full-build evidence.                                                                                                                                                                                                           | 1.1.1.1.5.3; LEG38 migration map; refactor regression protocol.                                                              | LEG38, GNG03, GNG26, GNG28-GNG32, local Lean build evidence after refactor.                                                                                      | not an ExecComplex computation; must preserve all executable witnesses.                                  | critical-before-new-substantive-code | <a href="#">1.1.1.1.5.3</a> |
| <a href="#">LEG40</a> | Post-refactor inventory closure certificate            | Governance; Policy; Object  | F1   | Y             | A regenerated inventory closure record after any neutral refactor: module graph, declaration graph, object-register delta, proof-dossier delta, public/API consumer delta, unresolved-gap ledger.                                                                                                                                                                            | 1.1.1.1.5.4; LEG39 closed refactor execution certificate; regenerated inventory/export data.                                 | build_export_script_v1.8.py outputs after refactor, generated_tables.py outputs, object/proof/phase registers.                                                   | not applicable; inventory and release metadata only.                                                     | critical-before-new-substantive-code | <a href="#">1.1.1.1.5.4</a> |
| <a href="#">LEG41</a> | Pre-substantive-code release certificate               | Governance; Policy; Object  | F1   | Y             | A release certificate proving that a new substantive Lean/content phase may become active only after source intake, neutral naming/refactor, post-refactor inventory, proof-dossier updates and red-lock checks close.                                                                                                                                                       | 1.1.1.1.5.5; LEG37-LEG40 closed; DBR13/GNG31 red-lock policy.                                                                | Closed LEG37, LEG38, LEG39, LEG40, object proof dossiers, phase release checks, local build evidence if refactor executed.                                       | not applicable; release certificate only.                                                                | critical-before-new-substantive-code | <a href="#">1.1.1.1.5.5</a> |
| <a href="#">DBR15</a> | Supabase/Postgres mirror feasibility and schema design | Database; Mirror; Object    | F1   | W             | Define the first DB mirror boundary: Lean/YAML remain authoritative; Supabase/Postgres is a public query/visualization mirror only. internal/api/review/admin schema split; minimal tables for db_snapshots, phases, objects, proof_dossiers, guardrails, workflows, modules, declarations, import_edges, phase_object_links, context_profiles, static_findings and reports. | SLTG-06 database-ready CNNA review/documentation structure and the V1.22 Supabase/public-web-frontend architecture decision. | Master YAML, generated TSV exports, repo_inventory/raw_export, status/guardrail/proof-dossier records and future import reports/checksums.                       | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem. | secondary-tooling-sidechain          | <a href="#">20.15</a>       |
| <a href="#">DBR16</a> | YAML/exports to Postgres import and roundtrip contract | Database; Mirror; Object    | F1   | W             | Define a reproducible importer/exporter contract from Master YAML and generated exports into Supabase/Postgres and back to YAML/reportable snapshots. Importer must be local/admin-run with .env secrets, produce import report/checksums, and never require a service_role key in chat or browser.                                                                          | SLTG-06 database-ready CNNA review/documentation structure and the V1.22 Supabase/public-web-frontend architecture decision. | Master YAML, generated TSV exports, repo_inventory/raw_export, status/guardrail/proof-dossier records and future import reports/checksums.                       | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem. | secondary-tooling-sidechain          | <a href="#">20.16</a>       |
| <a href="#">DBR17</a> | Append-only snapshot and checksum model                | Database; Mirror; Object    | F1   | W             | Define snapshot provenance so every public DB row is traceable to a Lean/YAML/tool package state. db_snapshots with snapshot_id, tool_version, lean_data_timestamp, yaml_hash, repo_inventory_hash, created_at, created_by, source_zip_name and optional import_report_hash.                                                                                                 | SLTG-06 database-ready CNNA review/documentation structure and the V1.22 Supabase/public-web-frontend architecture decision. | Master YAML, generated TSV exports, repo_inventory/raw_export, status/guardrail/proof-dossier records and future import reports/checksums.                       | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem. | secondary-tooling-sidechain          | <a href="#">20.17</a>       |
| <a href="#">DBR18</a> | Public read-only API views and RLS policies            | Database; Mirror; Object    | F1   | W             | Define public read-only exposure through api.* views while internal canonical tables remain non-public. anon gets SELECT on api.* views only; no anon INSERT/UPDATE/DELETE; internal.* and admin/import.* remain closed; service_role never appears in frontend/browser.                                                                                                     | SLTG-06 database-ready CNNA review/documentation structure and the V1.22 Supabase/public-web-frontend architecture decision. | Master YAML, generated TSV exports, repo_inventory/raw_export, status/guardrail/proof-dossier records and future import reports/checksums.                       | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem. | secondary-tooling-sidechain          | <a href="#">20.18</a>       |
| <a href="#">DBR19</a> | cnna-project.net web-frontend data contract            | Database; Mirror; Object    | F1   | W             | Define the read-only web frontend contract for public visualization/navigation over Supabase api.* views. Frontend can query public views with anon/publishable key only and render phase map, object register, red/orange release reasons, import/module graph, proof dossiers, blue-debt dashboard and HTML context docs.                                                  | SLTG-06 database-ready CNNA review/documentation structure and the V1.22 Supabase/public-web-frontend architecture decision. | Master YAML, generated TSV exports, repo_inventory/raw_export, status/guardrail/proof-dossier records and future import reports/checksums.                       | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem. | secondary-tooling-sidechain          | <a href="#">20.19</a>       |
| <a href="#">DBR20</a> | Reviewer annotation and future admin-write model       | Database; Mirror; Object    | F1   | W             | Define later authenticated reviewer annotations and controlled admin/import writes without changing the initial public read-only guarantee. review.* for later authenticated comments/annotations; admin/import.* for local importer or future signed Edge Function; no public writes and no manual table-click update as standard workflow.                                 | SLTG-06 database-ready CNNA review/documentation structure and the V1.22 Supabase/public-web-frontend architecture decision. | Master YAML, generated TSV exports, repo_inventory/raw_export, status/guardrail/proof-dossier records and future import reports/checksums.                       | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem. | secondary-tooling-sidechain          | <a href="#">20.20</a>       |
| <a href="#">DBR21</a> | Enlarged generated-PDF typography profile              | Generated; Artifact; Object | F3   | G             | A generated-document layout profile that raises full/context PDF body and table text one typographic step, including explicit TeX size commands and graph label sizes, without changing Lean/YAML data semantics.                                                                                                                                                            | User request after V1.22 to make all generated-document text one size larger.                                                | generate_full_document.py, generate_context_doc.py, generate_tables.py, build_export_script_v1.8.py, generated TeX/PDF outputs and representative render checks. | Not applicable; documentation typography only. No mathematical CNNA object and no Lean theorem.          | active-tooling                       | <a href="#">20.21</a>       |

| ID                    | Object                                                                         | Context class            | Tier | Object status | Canonical CNNA definition                                                                                                                                                                                                                                                                                      | Natural provenance                                                                                                                    | Computed from                                                                                                                             | ExecComplex/computable path                                                                                     | Priority                    | Phase                   |
|-----------------------|--------------------------------------------------------------------------------|--------------------------|------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------|
| <a href="#">DBR22</a> | Supabase live project discovery and empty-schema baseline audit                | Database; Mirror; Object | F1   | G             | Record the reachable Supabase organization/project baseline before any write: CNNA Project; project ref aiekahdewadrejrddajn; ACTIVE_HEALTHY; Postgres 17; no tables in public/internal/api/review/admin. It is a database/tooling object only and cannot release CNNA phases or replace Lean/YAML provenance. | Concrete activation of existing DBR15-DBR20 Supabase/Postgres architecture after live project discovery in organization CNNA Project. | Supabase project discovery, Master YAML, generated TSV exports, repo_inventory/raw_export, future migration/import reports and checksums. | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem.        | secondary-tooling-sidechain | <a href="#">20.22</a>   |
| <a href="#">DBR23</a> | Supabase initial schema migration package and RLS/read-only policy design      | Database; Mirror; Object | F1   | W             | Produce the first migration package for internal/api/review/admin schemas, internal canonical tables, api read-only views, grants/RLS posture, and comments documenting non-truth-source status. It is a database/tooling object only and cannot release CNNA phases or replace Lean/YAML provenance.          | Concrete activation of existing DBR15-DBR20 Supabase/Postgres architecture after live project discovery in organization CNNA Project. | Supabase project discovery, Master YAML, generated TSV exports, repo_inventory/raw_export, future migration/import reports and checksums. | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem.        | secondary-tooling-sidechain | <a href="#">20.23</a>   |
| <a href="#">DBR24</a> | Supabase snapshot import package from YAML/generated exports                   | Database; Mirror; Object | F1   | W             | Define importer contract from Master YAML, generated_exports and repo_inventory into internal tables with snapshot_id, hashes, import report and no Lean mutation. It is a database/tooling object only and cannot release CNNA phases or replace Lean/YAML provenance.                                        | Concrete activation of existing DBR15-DBR20 Supabase/Postgres architecture after live project discovery in organization CNNA Project. | Supabase project discovery, Master YAML, generated TSV exports, repo_inventory/raw_export, future migration/import reports and checksums. | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem.        | secondary-tooling-sidechain | <a href="#">20.24</a>   |
| <a href="#">DBR25</a> | Explicitly authorized live Supabase schema migration and first snapshot import | Database; Mirror; Object | F1   | W             | Execute, only after explicit user authorization, the schema migration and first data import into Supabase project aiekahdewadrejrddajn using controlled admin/tool write path. It is a database/tooling object only and cannot release CNNA phases or replace Lean/YAML provenance.                            | Concrete activation of existing DBR15-DBR20 Supabase/Postgres architecture after live project discovery in organization CNNA Project. | Supabase project discovery, Master YAML, generated TSV exports, repo_inventory/raw_export, future migration/import reports and checksums. | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem.        | secondary-tooling-sidechain | <a href="#">20.25</a>   |
| <a href="#">DBR26</a> | Post-import read-only/RLS/advisor verification and web-frontend handoff        | Database; Mirror; Object | F1   | W             | Verify public read-only access, absence of public writes, Supabase advisor output, table/view inventory, publishable frontend data contract and cnna-project.net handoff. It is a database/tooling object only and cannot release CNNA phases or replace Lean/YAML provenance.                                 | Concrete activation of existing DBR15-DBR20 Supabase/Postgres architecture after live project discovery in organization CNNA Project. | Supabase project discovery, Master YAML, generated TSV exports, repo_inventory/raw_export, future migration/import reports and checksums. | Not applicable; database/publication tooling only. No mathematical CNNA computation and no Lean theorem.        | secondary-tooling-sidechain | <a href="#">20.26</a>   |
| <a href="#">GA0</a>   | Horn quaternion/geometric-algebra reference-gate packet                        | Reference; Gate; Object  | F2   | K             | A documentation-only packet of target patterns extracted from Horn 0709.2238v1: Hamilton quaternion relations, Pauli/Cartan relation, geometric-product basis grades, pseudoscalar duality, pure-quaternion-as-bivector interpretation and complex-quaternion eight-basis bookkeeping.                         | User-requested reading of Horn 0709.2238v1 as a reference/target gate, explicitly not as a CNNA input.                                | Horn 0709.2238v1; external primary/source checks for mathlib Quaternion/CliffordAlgebra and GA formalization; no Lean CNNA computation.   | None. Reference metadata only; no operational computation and no scalar-layer authority.                        | reference-gate              | <a href="#">1.4.9</a>   |
| <a href="#">GA1</a>   | Hamilton unit and Pauli/Cartan sign-convention comparison profile              | Reference; Gate; Object  | F2   | K             | A target profile recording unit-square, anti-commutation and sign-convention checks for Hamilton units and their Pauli/Cartan representation, with explicit separation of complex i and quaternion i.                                                                                                          | Horn sections on quaternions and Pauli matrices; mathlib Quaternion API as later formal target seam.                                  | GA0 and external reference equations only; future comparison waits for CD0-CD7 closure.                                                   | None currently; future checks may be exact algebraic equalities, not ExecComplex calculations.                  | reference-gate              | <a href="#">1.4.9.1</a> |
| <a href="#">GA2</a>   | Pauli bivector/pseudoscalar duality comparison profile                         | Reference; Gate; Object  | F2   | K             | A target profile for geometric-algebra duality: scalar/trivector and vector/bivector pairing, pseudoscalar I = sigma_x sigma_y sigma_z, I^2 = -1, and pure quaternions as oriented-plane/bivector targets.                                                                                                     | Horn duality section and geometric algebra basis-grade discussion.                                                                    | GA0, GA1 and external GA references; no CNNA-derived orientation or dimension is assumed.                                                 | None currently. Future implementation must require CNNA-derived orientation/volume candidate before comparison. | reference-gate              | <a href="#">1.4.9.2</a> |
| <a href="#">GA3</a>   | Complex-quaternion eight-basis multivector comparison profile                  | Reference; Gate; Object  | F2   | K             | A target bookkeeping profile for the eight real components of complex quaternions mapped to geometric-algebra grades: scalar, vector, bivector and pseudoscalar components.                                                                                                                                    | Horn complex-quaternion section and its mapping between biquaternions and geometric-algebra multivectors.                             | GA0-GA2; external reference only, no CNNA scalar surface or C-layer assumption.                                                           | None currently. Explicitly not an ExecComplex replacement or complex scalar import.                             | reference-gate              | <a href="#">1.4.9.2</a> |
| <a href="#">GA4</a>   | Boundary Pauli/Clifford comparison gate                                        | Reference; Gate; Object  | F2   | K             | A comparison gate asking whether generators produced by the CNNA-derived boundary *-algebra satisfy Pauli/Clifford-like square, star and anti-commutation targets.                                                                                                                                             | Horn Pauli/geometric-algebra interpretation; mathlib CliffordAlgebra universal-property API; Pillar-B boundary star-algebra roadmap.  | Future 13.2 boundary *-algebra plus GA0-GA3 target profiles; no direct Pillar-A internals.                                                | Potential finite matrix check only after 13.2 derives the algebra; currently none.                              | reference-gate              | <a href="#">13.2.1</a>  |
| <a href="#">GA5</a>   | Dirac/spacetime-algebra comparison target gate                                 | Reference; Gate; Object  | F2   | K             | A comparison target for later CNNA-derived local-net/modular data against Dirac/spacetime-algebra Clifford signatures, Lorentz-covariance formulations and spin/CPT compatibility patterns.                                                                                                                    | Horn Dirac/quaternion discussion and GA/STA reference literature; placed behind 13.6 theorem gates.                                   | Future 13.5/13.6 outputs plus GA4; no imported Dirac equation, Lorentz group or spacetime metric.                                         | None currently; later finite checks may exist only as mirror diagnostics after B data is derived.               | reference-gate              | <a href="#">13.6.1</a>  |

| ID    | Object                                                   | Context class                | Tier | Object status | Canonical CNNA definition                                                                                                                                                                                                                                   | Natural provenance                                                                                                                                                              | Computed from                                                                                                                                                                                                                         | ExecComplex/computable path                                                                              | Priority                    | Phase |
|-------|----------------------------------------------------------|------------------------------|------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------|-------|
| GA6   | Geometric and conformal algebra side-comparison register | Reference; Gate; Object      | F2   | K             | A side-comparison register for geometric algebra, spacetime algebra and conformal geometric algebra as external target surfaces for already-derived CNNA outputs.                                                                                           | Horn outlook and related GA/CGA reference literature; CNNA side-comparison Phase 18.                                                                                            | GA0-GA5 and future derived CNNA outputs; never from external GA as a foundation.                                                                                                                                                      | None. Documentation/review comparison metadata only.                                                     | secondary-reference         | 18.5  |
| DBR27 | Phase/Object context-class taxonomy                      | Tool; Infrastructure; Object | F3   | G             | Generated-secondary context-class taxonomy: every phase/object row carries an explicit Context class separating Lean mathematical objects, Lean-facing contracts/refactors/governance, tool/database/generated/documentation artifacts and reference gates. | User-requested correction that objects and phases must be classified by overarching context; OOP principle: class/type identity must be visible and separate from state/status. | Schema extension from V1.25 plus second-pass semantic class audit: phase/object rows receive Context class; Lean-facing refactor/contract/governance rows are separated from tool-only rows and mathematical Lean source/target rows. | Not applicable; no mathematical computation. This is schema/classification metadata for review surfaces. | critical-tooling-governance | 20.27 |

### Object table C: access, handoff, and consumer rights

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| ID  | Object                                    | Context class                   | Tier | Object status | Pillar-A access                                                   | Handoff rights                                                                           | May be consumed by                                    | Current code modules / derivation site                                                                                                                                                                   | Phase |
|-----|-------------------------------------------|---------------------------------|------|---------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| G00 | Generator provenance                      | Lean; Source; Object            | F0   | G             | A-internal readable only along import order; no self-handoff use. | Do not export directly; only derived Profile in A->*.                                    | All A-internal Profile; B/C/D/E only through outputs. | CNNA/PillarA/Arithmetic/Smoke/GeneratedSource.lean; CNNA/PillarA/Arithmetic/Smoke/GeneratedMainPath.lean; CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean                            | 1.2.2 |
| H00 | A input handoff B->A                      | Lean; Handoff; Contract; Object | F0   | Y             | A may consume only this input, never A->B.                        | B->A input.                                                                              | A post-emergence rules.                               | CNNA/PillarA/Handoff/Inputs/B_to_A.lean                                                                                                                                                                  | 2.3.1 |
| H01 | A input handoff C->A                      | Lean; Handoff; Contract; Object | F0   | Y             | A may consume only this input, never A->C.                        | C->A input.                                                                              | A post-emergence geometry effects.                    | CNNA/PillarA/Handoff/Inputs/C_to_A.lean                                                                                                                                                                  | 2.3.1 |
| H02 | A input handoff D->A                      | Lean; Handoff; Contract; Object | F0   | Y             | A may consume only this input, never A->D.                        | D->A input.                                                                              | A post-emergence dynamics effects.                    | CNNA/PillarA/Handoff/Inputs/D_to_A.lean                                                                                                                                                                  | 2.3.1 |
| H03 | A input handoff E->A                      | Lean; Handoff; Contract; Object | F0   | Y             | A may consume only this input, never A->E.                        | E->A input.                                                                              | A post-emergence high-level effects.                  | CNNA/PillarA/Handoff/Inputs/E_to_A.lean                                                                                                                                                                  | 2.3.1 |
| O00 | ExecComplex                               | Lean; Source; Object            | F1   | W             | Allowed A-internally and in output Profile.                       | May appear in A->B/C/D/E as operational mirror when semantically released.               | All operational Profile.                              | CNNA/PillarA/Foundation/ExecComplex.lean; CNNA/PillarA/Foundation/ExecComplexBridge.lean; CNNA/PillarA/Foundation/ExecComplexLemmas.lean                                                                 | 1.2.1 |
| G01 | substrate: regular branching              | Lean; Source; Object            | F0   | Y             | A-internal; may be parameterized request-side.                    | A->B only indirectly as carrier/operator; A->C later as geometry source.                 | arithmetic, geometry, handoff bridges.                | CNNA/PillarA/Foundation/ConcreteSubstrate.lean; CNNA/PillarA/Arithmetic/BoundarySource/BranchAddressSubstrate.lean; CNNA/PillarA/Arithmetic/BoundarySource/BoundaryStatesubstrate.lean                   | 1.2.3 |
| G02 | substrate: variable branching             | Lean; Source; Object            | F0   | Y             | A-internal; separated from regular but with common interface.     | A->B indirectly if operator profile is derived.                                          | arithmetic-Scan, regime tests.                        | CNNA/PillarA/Foundation/LevelVariableSubstrate.lean; CNNA/PillarA/Foundation/WeightPolicy.lean; CNNA/PillarA/Arithmetic/Smoke/GeneratedDegreeFormulaAuditSM3db2eR0.lean                                  | 1.2.4 |
| G03 | Level / Cutoff                            | Lean; Source; Object            | F0   | Y             | A-internal; must be fixed before carrier and matrix.              | A->B as selected level only if the consumer needs it; A must not consume its own output. | Carrier, Matrix, Handoff.                             | CNNA/PillarA/Arithmetic/BoundarySource/LevelHistoryIndex.lean; CNNA/PillarA/Arithmetic/BoundarySource/LevelStepWitness.lean; CNNA/PillarA/Handoff/Core/Step1MathData.lean                                | 1.2.5 |
| G04 | Carrier / index type                      | Lean; Source; Object            | F0   | Y             | A-internal; no free reindexing.                                   | A->B as Omega/boundary carrier; A->C as finite-geometry carrier.                         | Matrix, operator algebra, Handoff.                    | CNNA/PillarA/Arithmetic/BoundarySource/RecursiveLevelHistoryCarrier.lean; CNNA/PillarA/Arithmetic/BoundarySource/LevelHistoryIndex.lean; CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelStepWitnessSM6.lean | 1.2.6 |
| G05 | Boundary / interface / interior partition | Lean; Source; Object            | F0   | Y             | A-internal; basis for Schur/DtN.                                  | A->B only boundary/interface abstracted; dark complement not directly to B.              | Schur, DtN, AQFT bright consumer.                     | CNNA/PillarA/Arithmetic/Smoke/GeneratedBoundaryInteriorPartition.lean; CNNA/PillarA/Finite/CutSpec.lean; CNNA/PillarA/Finite/BoundaryPorts.lean                                                          | 1.3.1 |



| ID                  | Object                                    | Context class        | Tier | Object status | Pillar-A access                                                                                                                                                                                                     | Handoff rights                                                                                                         | May be consumed by                                                                                                                                                           | Current code modules / derivation site                                                                                                                                                                              | Phase                 |
|---------------------|-------------------------------------------|----------------------|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <a href="#">G06</a> | Regime: noOuterEnvironment                | Lean; Source; Object | F0   | Y             | derive A-internally; must not be misused as plan status.                                                                                                                                                            | A->B as regime flag allowed; A must not consume A->B.                                                                  | B closed/unitary branch, D dynamics.                                                                                                                                         | CNNA/PillarA/Handoff/Outputs/A_to_B.lean; CNNA/PillarA/Sectors/SectorSplit.lean; CNNA/PillarA/Sectors/ComplementSectorFamily.lean                                                                                   | <a href="#">1.3.2</a> |
| <a href="#">G07</a> | Regime: ha-sOuterEnvironment              | Lean; Source; Object | F0   | Y             | derive A-internally; no stochasticity as input.                                                                                                                                                                     | A->B as regime flag + effective operator; complement not directly.                                                     | B open-system branch, D channels.                                                                                                                                            | CNNA/PillarA/Handoff/Outputs/A_to_B.lean; CNNA/PillarA/Sectors/ComplementSectorFamily.lean; CNNA/PillarA/Network/SectorSysEnv.lean                                                                                  | <a href="#">1.3.2</a> |
| <a href="#">G08</a> | Complement families: UV tail              | Lean; Source; Object | F0   | R             | A-internal; may only come naturally from complement layer.                                                                                                                                                          | Not directly to B-bright; possibly A->B-prime via effective data; B-prime dark later.                                  | Regime, spectral gap, dark sector.                                                                                                                                           | CNNA/PillarA/Sectors/UVSpectralSelector.lean; CNNA/PillarA/Sectors/ComplementSectorFamily.lean; CNNA/PillarA/Handoff/Outputs/A_to_B.lean                                                                            | <a href="#">1.3.3</a> |
| <a href="#">G09</a> | Complement families: IR/parent-path tail  | Lean; Source; Object | F0   | R             | A-internal.                                                                                                                                                                                                         | Not directly to B-bright.                                                                                              | Regime, scale, spectral tests.                                                                                                                                               | CNNA/PillarA/Sectors/ComplementSectorFamily.lean; CNNA/PillarA/Handoff/Outputs/A_to_B.lean                                                                                                                          | <a href="#">1.3.3</a> |
| <a href="#">G10</a> | Complement families: sideways environment | Lean; Source; Object | F0   | R             | A-internal.                                                                                                                                                                                                         | Not directly to B-bright.                                                                                              | open branch, channels.                                                                                                                                                       | CNNA/PillarA/Sectors/ComplementSectorFamily.lean; CNNA/PillarA/Network/SectorSysEnv.lean; CNNA/PillarA/Network/SectorChannels.lean                                                                                  | <a href="#">1.3.3</a> |
| <a href="#">CD0</a> | Cayley-Dickson source                     | Lean; Source; Object | F1   | B             | A-internal; reintegrate as structural consumer of A data, not separate generator. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase. | A->CayleyDickson output only; A must not consume own output. Uncorrected blue form is not handoff-exportable.          | CD structural proofs.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                    | CNNA/PillarA/Structural/CayleyDickson/Source.lean; CNNA/PillarA/Handoff/CayleyDicksonAdapter.lean                                                                                                                   | <a href="#">1.4.1</a> |
| <a href="#">CD1</a> | Scalar part                               | Lean; Source; Object | F1   | B             | A-internal structural; not arithmetic source. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                                     | A->CD only. Uncorrected blue form is not handoff-exportable.                                                           | star/norm.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                               | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalScalarPart.lean                                                                                                                                                    | <a href="#">1.4.2</a> |
| <a href="#">CD2</a> | Cayley-Dickson star/conjugation           | Lean; Source; Object | F1   | B             | A-internal structural. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                                                            | A->CD only; possible analogy to number conjugation only after bridge. Uncorrected blue form is not handoff-exportable. | normSq.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                                  | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalStar.lean                                                                                                                                                          | <a href="#">1.4.3</a> |
| <a href="#">CD3</a> | CD normSq                                 | Lean; Source; Object | F1   | B             | A-internal structural. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                                                            | A->CD only. Uncorrected blue form is not handoff-exportable.                                                           | Hurwitz/division.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                        | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqDefined.lean; CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqReduction.lean; CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqBlockSystem.lean | <a href="#">1.4.4</a> |
| <a href="#">CD4</a> | CD multiplication/product                 | Lean; Source; Object | F1   | B             | A-internal structural. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                                                            | A->CD only. Uncorrected blue form is not handoff-exportable.                                                           | alternative law, norm multiplicativity.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                  | CNNA/PillarA/Structural/CayleyDickson/S6CanonicalProductSurfaceCandidate.lean; CNNA/PillarA/Structural/CayleyDickson/S6CanonicalProductMultiplicationCandidate.lean                                                 | <a href="#">1.4.5</a> |
| <a href="#">CD5</a> | Alternative law                           | Lean; Source; Object | F1   | B             | A-internal structural. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                                                            | A->CD only. Uncorrected blue form is not handoff-exportable.                                                           | Hurwitz stop.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                            | CNNA/PillarA/Structural/CayleyDickson/S6AlternativeLawScaffold.lean; CNNA/PillarA/Structural/CayleyDickson/S6FullAlternativeLawBoundary.lean                                                                        | <a href="#">1.4.6</a> |
| <a href="#">CD6</a> | Norm multiplicativity                     | Lean; Source; Object | F1   | B             | A-internal structural. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                                                            | A->CD only. Uncorrected blue form is not handoff-exportable.                                                           | division.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                                | CNNA/PillarA/Structural/CayleyDickson/S6NormSqMultiplicativity.lean; CNNA/PillarA/Structural/CayleyDickson/S6NormMultiplicativityScaffold.lean                                                                      | <a href="#">1.4.7</a> |
| <a href="#">CD7</a> | Hurwitz stop / R,C,H,O classification     | Lean; Source; Object | F1   | B             | A-internal structural; must be fully reintegrated as closed subproject. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.           | A->CD, later A->E as structural invariant. Uncorrected blue form is not handoff-exportable.                            | Architecture consistency.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption; BR08 horizontal/vertical Schur-duality bridge | CNNA/PillarA/Structural/CayleyDickson/S6HurwitzStopSeed.lean; CNNA/PillarA/Structural/CayleyDickson/S6CayleyDicksonIdentification.lean                                                                              | <a href="#">1.4.8</a> |

| ID  | Object                             | Context class              | Tier | Object status | Pillar-A access                                                                                                                                                                      | Handoff rights                                                                                                                     | May be consumed by                                                                                                                                                            | Current code modules / derivation site                                                                                                                                                                                                                                                                                                                                                                            | Phase |
|-----|------------------------------------|----------------------------|------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| S00 | Dirichlet-Laplacian                | Lean;<br>Source;<br>Object | F0   | Y             | A-internal;<br>consumers may use only Profile.                                                                                                                                       | A->B only after boundary reduction/effective operator.                                                                             | Schur, DtN, spectrum.                                                                                                                                                         | CNNA/PillarA/Finite/DirichletLaplacian.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedDirichletEntry.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedDirichletExecEntrySourceSM3bq.lean                                                                                                                                                                                                                          | 3.1   |
| S01 | Interior block                     | Lean;<br>Source;<br>Object | F0   | B             | Present in SM3 chain, but plan-semantically named. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase. | Do not export directly; results only. Uncorrected blue form is not handoff-exportable.                                             | Interior inverse, Schur.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                  | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorBlock.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorEliminationCarrier.lean                                                                                                                                                                                                                                                                              | 3.2   |
| S02 | Interior inverse candidate         | Lean;<br>Source;<br>Object | F0   | B             | A-internal, legacy plan-semantic. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                  | Do not export directly. Uncorrected blue form is not handoff-exportable.                                                           | TwoSidedInv proof.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                        | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorInverseCandidateStart.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorInverseCandidateEntry.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorAccumulatedTraceEvaluation.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorInverseCandidateMatrix.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorInverseCandidateShapeSeparation.lean | 3.3   |
| S03 | Two-sided inverse certificate      | Lean;<br>Source;<br>Object | F0   | B             | A-internal; consumable as certificate. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.             | Do not export directly; DtN output may be exported. Uncorrected blue form is not handoff-exportable.                               | InteriorEliminationData, DtN.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                             | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorProductIdentityProofR4a.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorTwoSidedInvFromSM3db7R.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorProvenTwoSidedInvR5.lean                                                                                                                                                                            | 3.4   |
| S04 | InteriorEliminationData            | Lean;<br>Source;<br>Object | F0   | B             | A-internal; may be consumed by DtN. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                | Not raw an B; only boundary operator. Uncorrected blue form is not handoff-exportable.                                             | DtN, generalized DtN.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                     | CNNA/PillarA/Arithmetic/Smoke/GeneratedInteriorEliminationCertificateSM3fR.lean;<br>CNNA/PillarA/DtN/DtN.lean                                                                                                                                                                                                                                                                                                     | 3.5   |
| S05 | DtN boundary operator              | Lean;<br>Source;<br>Object | F0   | B             | A-internal; central output. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                        | A->B as L/Operator erlaubt; A->C geometry optional; A may Output not konsumieren. Uncorrected blue form is not handoff-exportable. | arithmetic, B bright AQFT.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption; BR08 horizontal/vertical Schur-duality bridge | CNNA/PillarA/Arithmetic/Smoke/GeneratedTopBoundaryDtNSM3gR.lean;<br>CNNA/PillarA/DtN/DtN.lean                                                                                                                                                                                                                                                                                                                     | 3.6   |
| S06 | GeneralizedDtN / MultiSchur        | Lean;<br>Source;<br>Object | F0   | B             | A-internal; legacy plan-semantic. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                  | A->B only if bright handoff specified; A->D dynamics optional. Uncorrected blue form is not handoff-exportable.                    | arithmetic matrices, Handoff.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                             | CNNA/PillarA/Arithmetic/Smoke/GeneratedGeneralizedTopBoundaryDtNSM3hR.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedMultiSchurSM4.lean;<br>CNNA/PillarA/Coupling/GeneralizedDtN.lean;<br>CNNA/PillarA/Coupling/MultiSchur.lean                                                                                                                                                                                  | 3.7   |
| O01 | Boundary matrix over ExecComplex   | Lean;<br>Source;<br>Object | F1   | B             | A-internal; neutral wrapper missing. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.               | A->B as bright operator; A->C as spectral/geometric source. Uncorrected blue form is not handoff-exportable.                       | Charpoly, algebra, B.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                     | CNNA/PillarA/Arithmetic/Smoke/GeneratedMultiSchurExecBoundaryOperatorSM4q.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedMultiSchurExecBoundaryOperatorFromSM3fqPlusSM4q.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryExecMatrixFromSM6bRecheckSM8.lean                                                                                                                                           | 4.1   |
| O02 | Determinant coefficient recurrence | Lean;<br>Source;<br>Object | F1   | B             | A-internal; historical name. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                       | Not directly an B; arithmeticconsumer. Uncorrected blue form is not handoff-exportable.                                            | Quadratic window, discriminant.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                           | CNNA/PillarA/Arithmetic/Smoke/GeneratedOperationalDeterminantCoefficientRecurrenceSM9e.lean                                                                                                                                                                                                                                                                                                                       | 4.2   |
| O03 | Operational polynomial profile     | Lean;<br>Source;<br>Object | F1   | B             | A-internal; neutralize. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                            | A->C/E only if a polynomial invariant is needed. Uncorrected blue form is not handoff-exportable.                                  | Window, spectrum.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                         | CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryOperationalCharpolyPolynomialProfileFromSM9b.lean;<br>CNNA/PillarA/Arithmetic/Smoke/GeneratedLevelHistoryOperationalCoefficientSourceFromSM9c.lean;<br>CNNA/PillarA/Arithmetic/Smoke/OperationalBoundedPolynomialEvaluationNormalFormSM9ePlus.lean                                                                                                             | 4.3   |
| O04 | Quadratic coefficient window       | Lean;<br>Source;<br>Object | F1   | B             | A-internal; Teil-2-Wrapper first. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase.                  | Not directly an B; A->C optional as invariant. Uncorrected blue form is not handoff-exportable.                                    | Discriminant.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption                                                             | CNNA/PillarA/Arithmetic/Smoke/GeneratedOperationalQuadraticCoefficientWindowFromSM9dSM9e.lean                                                                                                                                                                                                                                                                                                                     | 4.4   |

| ID  | Object                                         | Context class           | Tier | Object status | Pillar-A access                                                                                                                                                                | Handoff rights                                                                                           | May be consumed by                                                                                                                                                      | Current code modules / derivation site                                                                                                                           | Phase |
|-----|------------------------------------------------|-------------------------|------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| O05 | Operational discriminant D_op                  | Lean; Source; Object    | F1   | B             | A-internal; erster frozen Teil-2-Resetpunkt. BLUE quarantine: derived-only evidence may be audited, but semantic/API use must go through the planned neutral correction phase. | A->C/E as arithmetic invariant; not as B-Dynamikquelle. Uncorrected blue form is not handoff-exportable. | Comparison, CM seed.; blue-correction chain only until neutral API closes; no direct Phase 3+/handoff public consumption; BR08 horizontal/vertical Schur-duality bridge | CNNA/PillarA/Arithmetic/Smoke/GeneratedOperationalQuadraticDiscriminantSourceFromSM9g1Window.lean                                                                | 4.5   |
| O06 | Comparison target policy                       | Reference; Gate; Object | F2   | G             | A-internal; only after computed D_op.                                                                                                                                          | A->E meta/report optional; not an B.                                                                     | Tests.                                                                                                                                                                  | CNNA/PillarA/Arithmetic/Smoke/GeneratedOperationalDiscriminantFactorGateFromSM9fFull.lean; current code has target-exclusion/gate only, no general target policy | 4.6   |
| O07 | General finite computation test matrix profile | Lean; Target; Object    | F2   | Y             | A-internal reusable testing object; not a generative axiom.                                                                                                                    | none until proof-carrying rows are derived; comparison columns cannot be handed off as operational data. | Phase 4 wrappers, Phase 8 lattice sentinel tests, refactor regression checks, documentation/proof dossiers., O08 regime-typed extension                                 | Future module selected by Phase 1.1.1.1.1; must reuse existing Smoke/Generator modules.                                                                          | 4.0.1 |
| N00 | Fundamental discriminant D_fund                | Lean; Source; Object    | F1   | G             | A-internal; before CM.                                                                                                                                                         | A->C as number-theoretic invariant.                                                                      | CM order, conductor.                                                                                                                                                    | CNNA/PillarA/Arithmetic/Orders/DiscriminantConvention.lean (special Gaussian/Eisenstein bridge; not general derived D_fund)                                      | 5.1   |
| N01 | Conductor f                                    | Lean; Source; Object    | F1   | G             | A-internal.                                                                                                                                                                    | A->C as arithmetic invariant.                                                                            | Order profile.                                                                                                                                                          | CNNA/PillarA/Arithmetic/Orders/DiscriminantConvention.lean (implicit/bridge status; conductor not canonical)                                                     | 5.1   |
| N02 | Quadratic field K=Q(sqrt(D))                   | Lean; Source; Object    | F1   | Y             | A-internal; not before D_fund.                                                                                                                                                 | A->C, A->E maybe; B not directly.                                                                        | Order, embeddings.                                                                                                                                                      | CNNA/PillarA/Arithmetic/Orders/QuadraticOrder.lean (special shapes)                                                                                              | 5.2   |
| N03 | Quadratic order O_D                            | Lean; Source; Object    | F1   | Y             | A-internal; canonical once.                                                                                                                                                    | A->C as CM seed; A->E as invariant.                                                                      | Norm, units, class group.                                                                                                                                               | CNNA/PillarA/Arithmetic/Orders/QuadraticOrder.lean (special Gaussian/Eisenstein examples)                                                                        | 5.3   |
| N04 | Integral basis / omega_D                       | Lean; Source; Object    | F1   | Y             | A-internal.                                                                                                                                                                    | A->C geometry.                                                                                           | Norm, trace, lattice.                                                                                                                                                   | CNNA/PillarA/Arithmetic/Orders/QuadraticOrder.lean (halfIntegral shape; basis not general API)                                                                   | 5.4   |
| N05 | Trace form                                     | Lean; Target; Object    | F1   | Y             | A-internal.                                                                                                                                                                    | A->C, A->E.                                                                                              | Gram, norm.                                                                                                                                                             | no current general module; partial arithmetic in CNNA/PillarA/Arithmetic/Orders/NormForms.lean                                                                   | 5.5   |
| N06 | Norm form                                      | Lean; Source; Object    | F1   | Y             | A-internal; critical for prime claims.                                                                                                                                         | A->C/E as invariant; not B.                                                                              | Theta, prime splitting.                                                                                                                                                 | CNNA/PillarA/Arithmetic/Orders/NormForms.lean (special BinaryQuadraticNormForm)                                                                                  | 5.5   |
| N07 | Conjugation                                    | Lean; Target; Object    | F1   | Y             | A-internal.                                                                                                                                                                    | A->C if embeddings exported.                                                                             | Norm/trace/star analogies.                                                                                                                                              | no current general module; should live with QuadraticOrder/NormForms                                                                                             | 5.5   |
| N08 | Unit group O_D^*                               | Lean; Target; Object    | F2   | Y             | A-internal; must not be inferred from drawing.                                                                                                                                 | A->C symmetry; A->B only if operator action derived.                                                     | Lattice automorphisms, angles.                                                                                                                                          | no current derived module; only UnitConventionStatus in NormForms.lean                                                                                           | 6.1   |
| N09 | Class group Cl(O_D)                            | Lean; Target; Object    | F2   | Y             | A-internal but complex; far from immediate.                                                                                                                                    | A->C/E as invariant.                                                                                     | Heegner/PID/UFD, Hilbert class polynomial.                                                                                                                              | no current module                                                                                                                                                | 6.2   |
| N10 | Class number h(D)                              | Lean; Target; Object    | F2   | Y             | A-internal; not target table input.                                                                                                                                            | A->C/E.                                                                                                  | Heegner, j algebraicity degree.                                                                                                                                         | no current module                                                                                                                                                | 6.2   |
| N11 | Fractional/integral ideas                      | Lean; Target; Object    | F2   | Y             | A-internal.                                                                                                                                                                    | A->C/E only as abstract invariant.                                                                       | Class group, Hilbert class polynomial.                                                                                                                                  | no current module                                                                                                                                                | 6.2.1 |
| N12 | Principal ideal predicate                      | Lean; Target; Object    | F2   | Y             | A-internal.                                                                                                                                                                    | A->C/E.                                                                                                  | Heegner tests.                                                                                                                                                          | no current module                                                                                                                                                | 6.2.2 |
| N13 | Heegner/PID/UFD indicator                      | Lean; Target; Object    | F2   | Y             | A-internal; after h(D), not before.                                                                                                                                            | A->E report; A->C.                                                                                       | Prime generation claims., N18 regime-projection classification                                                                                                          | no current module; old HeegnerDiscriminantBridge in DiscriminantConvention.lean is not full h(D)                                                                 | 6.3   |
| N14 | Prime splitting behavior                       | Lean; Target; Object    | F2   | Y             | A-internal.                                                                                                                                                                    | A->C/E; B not direct.                                                                                    | Prime/norm interpretation.                                                                                                                                              | no current module                                                                                                                                                | 6.4   |

| ID  | Object                                                 | Context class              | Tier | Object status | Pillar-A access                                                                                  | Handoff rights                                            | May be consumed by                                                                  | Current code modules / derivation site                                                                                                                                                                         | Phase  |
|-----|--------------------------------------------------------|----------------------------|------|---------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| N15 | Kronecker/Dirichlet character chi_D                    | Lean;<br>Target;<br>Object | F2   | Y             | A-internal.                                                                                      | A->C/E.                                                   | L-function, splitting.                                                              | no current module                                                                                                                                                                                              | 6.4    |
| CD8 | Arithmetic-CD compatibility as Schur-duality corollary | Lean;<br>Source;<br>Object | F1   | K             | A-internal bridge after both algebraic and arithmetic Schur routes, not a Phase-6 public output. | A->E only after theorem.                                  | Unification only after BR08 theorem;<br>not by earlier CD/arithmetic source phases. | CNNA/PillarA/Arithmetic/CayleyDickson/ArithmeticCDBridge.lean;<br>CNNA/PillarA/Arithmetic/Convergence/CDArithmeticCompatibility.lean;<br>CNNA/PillarA/Arithmetic/Convergence/QuaternionRamanujanInterface.lean | 10.4.2 |
| L00 | Lattice embedding Lambda -> C                          | Lean;<br>Target;<br>Object | F2   | Y             | A-internal;<br>no drawing input.                                                                 | A->C geometry;<br>maybe B geometric if later.             | Gram, tau, angles.                                                                  | no current CM-lattice module;<br>Geometry/* is spacetime/foliation, not CM embedding                                                                                                                           | 7.1    |
| L01 | Lattice basis                                          | Lean;<br>Target;<br>Object | F2   | Y             | A-internal.                                                                                      | A->C.                                                     | Gram, tau.                                                                          | no current CM-lattice module                                                                                                                                                                                   | 7.1    |
| L02 | Gram matrix                                            | Lean;<br>Target;<br>Object | F2   | Y             | A-internal.                                                                                      | A->C;<br>A->B not direct.                                 | Angles, determinant, automorphisms.                                                 | no current CM-lattice Gram module                                                                                                                                                                              | 7.2    |
| L03 | Lattice determinant / covolume                         | Lean;<br>Target;<br>Object | F2   | Y             | A-internal.                                                                                      | A->C/E.                                                   | Discriminant consistency.                                                           | no current CM-lattice determinant/covolume module                                                                                                                                                              | 7.2    |
| L04 | Period ratio tau                                       | Lean;<br>Target;<br>Object | F2   | Y             | A-internal;<br>prerequisite for j.                                                               | A->C.                                                     | Modular forms, j.;<br>M08 CM/j special-value test targets                           | no current tau module                                                                                                                                                                                          | 7.3    |
| L05 | Upper half-plane condition                             | Lean;<br>Target;<br>Object | F2   | Y             | A-internal.                                                                                      | A->C.                                                     | Modular group, j.                                                                   | no current upper-half-plane module                                                                                                                                                                             | 7.3    |
| L06 | Angle / cosine profile                                 | Lean;<br>Target;<br>Object | F2   | Y             | A-internal after Gram.                                                                           | A->C;<br>maybe report to E.                               | Symmetry tests.;<br>L09/L10 heuristic lattice tests                                 | no current derived angle module;<br>drawings/visual intuition not code source                                                                                                                                  | 7.4    |
| L07 | Lattice automorphism group                             | Lean;<br>Target;<br>Object | F2   | Y             | A-internal.                                                                                      | A->C;<br>A->B only if operator action derived.            | Angles, CD analogies.                                                               | no current lattice automorphism module                                                                                                                                                                         | 7.5    |
| M02 | Eisenstein series G4,G6                                | Lean;<br>Target;<br>Object | F2   | Y             | A-internal after tau/lattice.                                                                    | A->C/E.                                                   | g2,g3,j.                                                                            | no current Eisenstein-series module                                                                                                                                                                            | 8.1    |
| M03 | Weierstrass invariants g2,g3                           | Lean;<br>Target;<br>Object | F2   | Y             | A-internal.                                                                                      | A->C/E.                                                   | Delta_mod, j.                                                                       | no current Weierstrass-invariant module                                                                                                                                                                        | 8.2    |
| M04 | Modular discriminant Delta_mod                         | Lean;<br>Target;<br>Object | F2   | Y             | A-internal;<br>name disambiguation mandatory.                                                    | A->C/E.                                                   | j.                                                                                  | no current modular-discriminant module                                                                                                                                                                         | 8.3    |
| M05 | j-invariant                                            | Lean;<br>Target;<br>Object | F2   | Y             | A-internal after modular prerequisites.                                                          | A->C/E.                                                   | CM comparison, moonshine later.;<br>M08;<br>L09 test matrix                         | no current j-invariant module                                                                                                                                                                                  | 8.4    |
| SP0 | Eigenvalue profile                                     | Lean;<br>Source;<br>Object | F1   | Y             | A-internal after matrix.                                                                         | A->B maybe as spectral data only if specified;<br>A->C/E. | Heat kernel, gaps.;<br>v37 Path B consumers                                         | CNNA/PillarA/Finite/ExecSpectral/CharpolyCore.lean;<br>CNNA/PillarA/Finite/ExecSpectral/EigenvectorKernel.lean;<br>CNNA/PillarA/Finite/SpectralBridge.lean                                                     | 10.1   |
| SP1 | Spectral gap                                           | Lean;<br>Source;<br>Object | F1   | Y             | A-internal.                                                                                      | A->B/D only as allowed profile.                           | Dynamics, UV cut.;<br>v37 Path B consumers                                          | CNNA/PillarA/Arithmetic/Graph/SpectralGapSeed.lean;<br>CNNA/PillarA/Finite/ExecSpectral/Certificates.lean                                                                                                      | 10.1   |
| P00 | Operator algebra                                       | Lean;<br>Source;<br>Object | F1   | Y             | A-internal as profile;<br>B consumes output.                                                     | A->B allowed after bridge;<br>A cannot reimport.          | B bright, finite seeds.;<br>v37 Path B consumers                                    | CNNA/PillarA/Foundation/MatrixAlgebra.lean;<br>CNNA/PillarA/Finite/StateSpace.lean                                                                                                                             | 10.2   |
| P01 | Star operation                                         | Lean;<br>Source;<br>Object | F1   | Y             | A-internal.                                                                                      | A->B part of operator algebra.                            | State positivity, KMS.;<br>v37 Path B consumers                                     | CNNA/PillarA/Foundation/ExecComplex.lean;<br>CNNA/PillarA/Foundation/HermitianStructure.lean;<br>CNNA/PillarA/Foundation/MatrixAlgebra.lean                                                                    | 10.2   |
| P02 | State functional                                       | Lean;<br>Source;<br>Object | F1   | Y             | A-internal/far.                                                                                  | A->B after derivation.                                    | GNS/KMS.;<br>v37 Path B consumers                                                   | CNNA/PillarA/Finite/StateSpace.lean                                                                                                                                                                            | 10.2   |



| ID  | Object                                  | Context class                 | Tier | Object status | Pillar-A access                                                                           | Handoff rights                                                 | May be consumed by                                                             | Current code modules / derivation site                                                                                                                   | Phase |
|-----|-----------------------------------------|-------------------------------|------|---------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| P03 | Gibbs state                             | Lean;<br>Source;<br>Object    | F1   | Y             | A-internal/far.                                                                           | A->B/D.                                                        | KMS, dynamics.;<br>v37 Path B consumers                                        | CNNA/PillarA/Finite/GibbsStateSeed.lean                                                                                                                  | 10.2  |
| P04 | Time evolution                          | Lean;<br>Source;<br>Object    | F1   | Y             | A-internal/far.                                                                           | A->D/B.                                                        | unitarity/open dynamics.;<br>v37 Path B consumers                              | CNNA/PillarA/Finite/MatrixExponential.lean;<br>CNNA/PillarA/Finite/UnitaryEvolution.lean;<br>CNNA/PillarA/Finite/DynamicsAdapter.lean                    | 10.2  |
| P05 | Unitarity certificate                   | Lean;<br>Source;<br>Object    | F1   | Y             | A-internal after noOuter.                                                                 | A->B/D.                                                        | closed branch theorem.;<br>v37 Path B consumers                                | CNNA/PillarA/Finite/UnitaryEvolution.lean (finite seed;<br>regime split still pending)                                                                   | 10.2  |
| P06 | Open channel /<br>effective openness    | Lean;<br>Source;<br>Object    | F1   | R             | A-internal after hasOuter;<br>no stochastic input.                                        | A->D/B as effective data.                                      | OQS consumer.;<br>v37 Path B consumers                                         | CNNA/PillarA/Finite/ChannelSeed.lean;<br>CNNA/PillarA/Network/SectorChannels.lean                                                                        | 10.2  |
| N16 | Dirichlet L-function<br>L(s,chi_D)      | Reference;<br>Gate;<br>Object | F2   | K             | A-internal far target;<br>not immediate.                                                  | A->E only after analytic bridge.                               | Class number formula.                                                          | no current module                                                                                                                                        | 9.2   |
| N17 | Class number<br>formula                 | Reference;<br>Gate;<br>Object | F2   | K             | Far target;<br>not a source for early h(D).                                               | A->E optional.                                                 | Consistency.                                                                   | no current module                                                                                                                                        | 9.3   |
| L08 | Theta series<br>Theta_Lambda            | Reference;<br>Gate;<br>Object | F2   | K             | Far target but registered.                                                                | A->C/E only after profile.                                     | Prime/norm statistics.                                                         | no current theta-series module                                                                                                                           | 9.1   |
| M00 | Modular group<br>action SL2Z            | Lean;<br>Target;<br>Object    | F2   | K             | Far target.                                                                               | A->C.                                                          | j invariance, fundamental domain.                                              | no current modular-action module                                                                                                                         | 8.0   |
| M01 | Fundamental<br>domain<br>representative | Lean;<br>Target;<br>Object    | F2   | K             | Far target.                                                                               | A->C/E.                                                        | j comparison.                                                                  | no current fundamental-domain module                                                                                                                     | 8.0   |
| M06 | q-expansion of j                        | Lean;<br>Target;<br>Object    | F2   | K             | Far target.                                                                               | A->E/report.                                                   | j computation consistency.                                                     | no current q-expansion module                                                                                                                            | 8.8   |
| M07 | Hilbert class<br>polynomial H_D         | Lean;<br>Target;<br>Object    | F2   | K             | Far target.                                                                               | A->C/E.                                                        | class field consistency.                                                       | no current Hilbert-class-polynomial module                                                                                                               | 8.8   |
| TM0 | Transfermatrix T_k                      | Lean;<br>Source;<br>Object    | F1   | K             | A-internal far target;<br>not needed for immediate wrapper.                               | A->E only after proof.                                         | Schur theory explanation.;<br>BR08 horizontal/vertical<br>Schur-duality bridge | CNNA/PillarA/Arithmetic/Schur/MobiusTransfer.lean (anchor/status only);<br>CNNA/PillarA/Arithmetic/Schur/Recurrence.lean;<br>no full M2(Z[lambda]) proof | 10.4  |
| TM1 | Transfermatrix<br>product               | Lean;<br>Source;<br>Object    | F1   | K             | Far target.                                                                               | A->E.                                                          | charpoly connection.;<br>BR08 horizontal/vertical<br>Schur-duality bridge      | CNNA/PillarA/Arithmetic/Schur/MobiusTransfer.lean (no product formalization)                                                                             | 10.4  |
| TM2 | Moebius/continued<br>fraction relation  | Lean;<br>Source;<br>Object    | F1   | K             | Far target.                                                                               | A->E.                                                          | Theoretical bridge.;<br>BR08 horizontal/vertical<br>Schur-duality bridge       | CNNA/PillarA/Arithmetic/Schur/MobiusTransfer.lean (route/status;<br>no theoremicallly complete Moebius-chain)                                            | 10.4  |
| SP2 | Heat kernel K(t)                        | Lean;<br>Target;<br>Object    | F2   | K             | Far target.                                                                               | A->D/E.                                                        | Spectral dimension.;<br>v37 Path B consumers                                   | no current heat-kernel module;<br>MatrixExponential is related but not HeatKernelProfile                                                                 | 10.3  |
| SP3 | Spectral zeta<br>zeta_H                 | Lean;<br>Target;<br>Object    | F2   | K             | Far target.                                                                               | A->E.                                                          | regularized determinant.;<br>v37 Path B consumers                              | no current spectral-zeta module                                                                                                                          | 10.3  |
| SP4 | Spectral dimension                      | Lean;<br>Target;<br>Object    | F2   | K             | Far target.                                                                               | A->D/E.                                                        | physical interpretation.;<br>v37 Path B consumers                              | no current spectral-dimension module                                                                                                                     | 10.3  |
| SP5 | Zeta-regularized<br>determinant         | Lean;<br>Target;<br>Object    | F2   | K             | Far target.                                                                               | A->E.                                                          | consistency.;<br>v37 Path B consumers                                          | no current zeta-regularized determinant module                                                                                                           | 10.3  |
| P07 | KMS condition                           | Lean;<br>Target;<br>Object    | F2   | K             | A may consume only if returned through<br>explicit B->A input;<br>never from A->B output. | B internal / possible B->A output<br>only after explicit gate. | B.                                                                             | no current KMS proof module;<br>GibbsStateSeed is prerequisite only                                                                                      | 13.4  |

| ID    | Object                                    | Context class                            | Tier | Object status | Pillar-A access                                                                     | Handoff rights                                              | May be consumed by                   | Current code modules / derivation site                                                                                                    | Phase                |
|-------|-------------------------------------------|------------------------------------------|------|---------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| P08   | GNS Hilbert space                         | Lean;<br>Target;<br>Object               | F2   | K             | A may consume only if returned through explicit B->A input; never from A->B output. | B internal / possible B->A output only after explicit gate. | Tomita.                              | no current GNS module                                                                                                                     | <a href="#">13.4</a> |
| P09   | Tomita operator S                         | Lean;<br>Target;<br>Object               | F2   | K             | A may consume only if returned through explicit B->A input; never from A->B output. | B internal / possible B->A output only after explicit gate. | Delta,J.                             | no current Tomita module                                                                                                                  | <a href="#">13.5</a> |
| P10   | Modular operator Delta                    | Lean;<br>Target;<br>Object               | F2   | K             | A may consume only if returned through explicit B->A input; never from A->B output. | B internal / possible B->A output only after explicit gate. | KMS/CPT.                             | no current modular-operator module                                                                                                        | <a href="#">13.5</a> |
| P11   | Modular conjugation J                     | Lean;<br>Target;<br>Object               | F2   | K             | A may consume only if returned through explicit B->A input; never from A->B output. | B internal / possible B->A output only after explicit gate. | CPT, spin-statistics.                | no current modular-conjugation module                                                                                                     | <a href="#">13.5</a> |
| P12   | Bisognano-Wichmann/CPT bridge             | Lean;<br>Target;<br>Object               | F2   | K             | A may consume only if returned through explicit B->A input; never from A->B output. | B internal / possible B->A output only after explicit gate. | CPT.                                 | no current BW/CPT module                                                                                                                  | <a href="#">13.6</a> |
| P13   | Spin-statistics                           | Lean;<br>Target;<br>Object               | F2   | K             | A may consume only if returned through explicit B->A input; never from A->B output. | B internal / possible B->A output only after explicit gate. | physics consistency.                 | no current spin-statistics module                                                                                                         | <a href="#">13.6</a> |
| P14   | Kramers degeneracy                        | Lean;<br>Target;<br>Object               | F2   | K             | A may consume only if returned through explicit B->A input; never from A->B output. | B internal / possible B->A output only after explicit gate. | physics tests.                       | no current Kramers module                                                                                                                 | <a href="#">13.6</a> |
| P15   | Split property                            | Lean;<br>Source;<br>Object               | F1   | K             | A may consume only if returned through explicit B->A input; never from A->B output. | B internal / possible B->A output only after explicit gate. | AQFT consistency.                    | CNNA/PillarA/Sectors/SectorSplit.lean;<br>Handoff has split fields;<br>AQFT split property not derived                                    | <a href="#">13.6</a> |
| P16   | Local net / Haag-Kastler algebraic axioms | Lean;<br>Target;<br>Object               | F2   | K             | A may consume only if returned through explicit B->A input; never from A->B output. | B internal / possible B->A output only after explicit gate. | AQFT branch.                         | no current local-net/HK module; legacy target only                                                                                        | <a href="#">13.3</a> |
| H10   | A->B bright handoff                       | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Y             | A forbidden to consume own A->B output.                                             | A->B output only.                                           | Pillar B bright consumer.            | CNNA/PillarA/Handoff/Outputs/A_to_B.lean;<br>CNNA/PillarA/Handoff/Core/Step1MathData.lean;<br>CNNA/PillarA/Handoff/Core/SectorExport.lean | <a href="#">11.1</a> |
| H11   | A->B prime/dark future handoff            | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | R             | A forbidden to consume; not current B.                                              | A->B prime output only, future.                             | Pillar B prime.                      | no current dark/B-prime handoff module                                                                                                    | <a href="#">11.2</a> |
| H12   | A->C handoff                              | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Y             | A forbidden to consume own A->C output.                                             | A->C output.                                                | Pillar C.                            | CNNA/PillarA/Handoff/Outputs/A_to_C.lean                                                                                                  | <a href="#">11.3</a> |
| H13   | A->D handoff                              | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Y             | A forbidden to consume own A->D output.                                             | A->D output.                                                | Pillar D.                            | CNNA/PillarA/Handoff/Outputs/A_to_D.lean                                                                                                  | <a href="#">11.4</a> |
| H14   | A->E handoff                              | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | K             | A forbidden to consume own A->E output.                                             | A->E output.                                                | Pillar E.                            | CNNA/PillarA/Handoff/Outputs/A_to_E.lean                                                                                                  | <a href="#">11.5</a> |
| MOON0 | j coefficient stream                      | Reference;<br>Gate;<br>Object            | F2   | K             | A-internal after modular-form prerequisites; not a source for earlier A.            | May be exported only as comparison output after derivation. | Moonshine comparison side consumers. | no current Lean module                                                                                                                    | <a href="#">12.1</a> |
| MOON1 | Monster dimension comparison profile      | Reference;<br>Gate;<br>Object            | F2   | K             | A may compare after derivation; may not consume as source.                          | External comparison only.                                   | Moonshine comparison; documentation. | no current Lean module                                                                                                                    | <a href="#">12.2</a> |
| MOON2 | McKay-Thompson comparison profile         | Reference;<br>Gate;<br>Object            | F2   | K             | A may not import it into core derivation.                                           | External comparison only.                                   | Moonshine comparison side target.    | no current Lean module                                                                                                                    | <a href="#">12.3</a> |
| B00   | Pillar B input contract from A            | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | K             | A cannot consume this; B consumes A->B.                                             | A->B output consumed by B; B->A only via explicit input.    | Pillar B QFT/AQFT modules.           | CNNA/PillarB/BuildAll.lean;<br>CNNA/PillarA/Handoff/Outputs/A_to_B.lean                                                                   | <a href="#">13.1</a> |



| ID  | Object                                           | Context class                   | Tier | Object status | Pillar-A access                                        | Handoff rights                                               | May be consumed by                                | Current code modules / derivation site                                              | Phase                |
|-----|--------------------------------------------------|---------------------------------|------|---------------|--------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------|----------------------|
| B01 | Boundary star-algebra / observable algebra       | Lean; Source; Object            | F1   | K             | A does not consume B algebra except typed B->A result. | B internal; possible B->A theorem output only if registered. | B local net/state/KMS consumers.                  | CNNA/PillarA/Finite/StateSpace.lean; CNNA/PillarB/BuildAll.lean                     | <a href="#">13.2</a> |
| B02 | Local net / QFT algebraic profile                | Lean; Source; Object            | F1   | K             | Not A-consumable except via B->A.                      | B internal; output only as typed B theorem.                  | B KMS/modular/split phases.                       | CNNA/PillarB/BuildAll.lean                                                          | <a href="#">13.3</a> |
| B03 | B state/GNS/KMS profile                          | Lean; Source; Object            | F1   | K             | A may not use B state unless B->A input.               | B internal; possible B->A theorem output.                    | B modular theory.                                 | CNNA/PillarA/Finite/GibbsStateSeed.lean; CNNA/PillarA/Finite/MatrixExponential.lean | <a href="#">13.4</a> |
| B04 | Tomita-Takesaki modular data                     | Lean; Source; Object            | F1   | K             | A cannot consume except explicit B->A.                 | B theorem output only if registered.                         | CPT/BW/spin-statistics gates.                     | CNNA/PillarB/BuildAll.lean                                                          | <a href="#">13.5</a> |
| B05 | CPT/BW/spin-statistics/split gates               | Lean; Source; Object            | F1   | K             | A cannot consume except explicit B->A.                 | B theorem output only if registered.                         | B->A, documentation, later physics consumers.     | CNNA/PillarB/BuildAll.lean                                                          | <a href="#">13.6</a> |
| B06 | B->A allowed output profile                      | Lean; Source; Object            | F1   | K             | A may consume only after it arrives as B->A.           | B->A input only; never A->B.                                 | A post-emergence/backreaction phases.             | CNNA/PillarA/Handoff/Inputs/B_to_A.lean                                             | <a href="#">13.7</a> |
| C00 | Pillar C OQS input contract from A               | Lean; Handoff; Contract; Object | F0   | K             | A cannot consume C internals except C->A.              | A->C consumed by C.                                          | C OQS modules.                                    | CNNA/PillarC/BuildAll.lean; CNNA/PillarA/Handoff/Outputs/A_to_C if created later    | <a href="#">14.1</a> |
| C01 | Finite CP/channel profile                        | Lean; Source; Object            | F1   | K             | A may not consume except C->A.                         | C internal; C->A only for typed outputs.                     | Effective dynamics/noise phases.                  | CNNA/PillarA/Finite/ChannelSeed.lean; CNNA/PillarC/BuildAll.lean                    | <a href="#">14.2</a> |
| C02 | Effective OQS generator / Lindblad gate          | Lean; Source; Object            | F1   | K             | No A consumption except C->A.                          | C theorem output only if registered.                         | Noise/decoherence and C->A.                       | CNNA/PillarC/BuildAll.lean                                                          | <a href="#">14.3</a> |
| C03 | Decoherence/noise profile                        | Lean; Source; Object            | F1   | K             | No A consumption except C->A.                          | C->A typed OQS input.                                        | Backreaction; phenomenology.                      | CNNA/PillarC/BuildAll.lean                                                          | <a href="#">14.4</a> |
| C04 | C->A allowed output profile                      | Lean; Source; Object            | F1   | K             | A may consume only through C->A.                       | C->A input only; never A->C.                                 | A post-emergence/backreaction phases.             | CNNA/PillarA/Handoff/Inputs/C_to_A.lean                                             | <a href="#">14.5</a> |
| D00 | Pillar D spacetime input contract                | Lean; Handoff; Contract; Object | F0   | K             | A cannot consume D internals except D->A.              | A->D consumed by D.                                          | D spacetime modules.                              | CNNA/PillarD/BuildAll.lean                                                          | <a href="#">15.1</a> |
| D01 | Region/ localization/ causal-order candidate     | Lean; Source; Object            | F1   | K             | No A consumption except D->A.                          | D theorem output only if registered.                         | Metric/covariance and backreaction.               | CNNA/PillarD/BuildAll.lean                                                          | <a href="#">15.2</a> |
| D02 | Metric/ coarse-geometry/ covariance gate         | Lean; Source; Object            | F1   | K             | No A consumption except D->A.                          | D theorem output only if registered.                         | D->A; E matter coupling; comparison side targets. | CNNA/PillarD/BuildAll.lean                                                          | <a href="#">15.3</a> |
| D03 | D->A allowed output profile                      | Lean; Source; Object            | F1   | K             | A may consume only through D->A.                       | D->A input only; never A->D.                                 | A post-emergence/backreaction phases.             | CNNA/PillarA/Handoff/Inputs/D_to_A.lean                                             | <a href="#">15.4</a> |
| E00 | Pillar E matter input contract                   | Lean; Handoff; Contract; Object | F0   | K             | A cannot consume E internals except E->A.              | A->E/B/C/D->E if explicitly allowed.                         | E matter modules.                                 | CNNA/PillarE/BuildAll.lean                                                          | <a href="#">16.1</a> |
| E01 | Matter sector and internal symmetry candidate    | Lean; Source; Object            | F1   | K             | No A consumption except E->A.                          | E theorem output only if registered.                         | Phenomenology comparison.                         | CNNA/PillarE/BuildAll.lean                                                          | <a href="#">16.2</a> |
| E02 | Observable coupling and phenomenology comparison | Reference; Gate; Object         | F3   | K             | No A consumption except E->A.                          | External comparison only unless typed E->A input.            | E->A, documentation.                              | CNNA/PillarE/BuildAll.lean                                                          | <a href="#">16.3</a> |

| ID    | Object                                            | Context class              | Tier | Object status | Pillar-A access                                                                     | Handoff rights                                          | May be consumed by                                               | Current code modules / derivation site              | Phase |
|-------|---------------------------------------------------|----------------------------|------|---------------|-------------------------------------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------|-------|
| E03   | E->A allowed output profile                       | Lean; Source; Object       | F1   | K             | A may consume only through E->A.                                                    | E->A input only; never A->E.                            | A post-emergence/backreaction phases.                            | CNNA/PillarA/Handoff/Inputs/E_to_A.lean             | 16.4  |
| BR00  | Backreaction input DAG                            | Lean; Target; Object       | F2   | K             | A may consume only listed inputs.                                                   | B/C/D/E -> A only.                                      | A post-emergence interpretation.                                 | CNNA/PillarA/Handoff/Inputs/*.lean                  | 17    |
| BR01  | B->A backreaction input gate                      | Lean; Source; Object       | F1   | K             | A may consume through this gate only.                                               | B->A.                                                   | A post-emergence.                                                | CNNA/PillarA/Handoff/Inputs/B_to_A.lean             | 17.1  |
| BR02  | C->A OQS backreaction input gate                  | Lean; Source; Object       | F1   | K             | A may consume through this gate only.                                               | C->A.                                                   | A post-emergence.                                                | CNNA/PillarA/Handoff/Inputs/C_to_A.lean             | 17.2  |
| BR03  | D->A spacetime backreaction input gate            | Lean; Source; Object       | F1   | K             | A may consume through this gate only.                                               | D->A.                                                   | A post-emergence.                                                | CNNA/PillarA/Handoff/Inputs/D_to_A.lean             | 17.3  |
| BR04  | E->A matter backreaction input gate               | Lean; Source; Object       | F1   | K             | A may consume through this gate only.                                               | E->A.                                                   | A post-emergence.                                                | CNNA/PillarA/Handoff/Inputs/E_to_A.lean             | 17.4  |
| BR05  | Pre/post-emergence switch for A statements        | Lean; Source; Object       | F0   | K             | A-internal after input gate validation.                                             | No output; governs A interpretation.                    | A theorem statements after higher-pillar emergence.              | no current neutral module                           | 17.5  |
| SIDE0 | Full ToE comparison profile                       | Reference; Gate; Object    | F4   | K             | Forbidden as source for A or any core derivation.                                   | external comparison only.                               | Documentation/comparison.                                        | no current Lean module                              | 18.1  |
| SIDE1 | String-theory comparison profile                  | Reference; Gate; Object    | F4   | K             | Forbidden as source for A or any core derivation.                                   | external comparison only.                               | Documentation/comparison.                                        | no current Lean module                              | 18.2  |
| SIDE2 | Loop quantum gravity comparison profile           | Reference; Gate; Object    | F4   | K             | Forbidden as source for A or any core derivation.                                   | external comparison only.                               | Documentation/comparison.                                        | no current Lean module                              | 18.3  |
| SIDE3 | Noncommutative geometry comparison profile        | Reference; Gate; Object    | F4   | K             | Forbidden as source for A or any core derivation.                                   | external comparison only.                               | Documentation/comparison.                                        | no current Lean module                              | 18.4  |
| MOON3 | Finite simple group comparison profile            | Reference; Gate; Object    | F2   | K             | A may register as comparison only; not consumable by earlier A derivations.         | External comparison only.                               | Moonshine comparison; later E symmetry comparison if generated.  | no current Lean module                              | 12.4  |
| MOON4 | Sporadic/Monster comparison boundary              | Reference; Gate; Object    | F2   | K             | Comparison only after derivation; forbidden as source.                              | External comparison only.                               | Monster dimension comparison; common-indicator bridge tests.     | no current Lean module                              | 12.4  |
| MOON5 | Leech lattice comparison profile                  | Reference; Gate; Object    | F2   | K             | External comparison only; no Leech data upstream of derivation.                     | External comparison only.                               | Leech theta comparison; VOA comparison.                          | no current Lean module                              | 12.5  |
| MOON6 | Leech theta/shell coefficient comparison          | Reference; Gate; Object    | F2   | K             | Comparison only after CNNA coefficient stream exists.                               | External comparison only.                               | Common-indicator bridge tests; documentation.                    | no current Lean module                              | 12.6  |
| MOON7 | VOA/moonshine-module comparison profile           | Reference; Gate; Object    | F2   | K             | Forbidden as A source; possible comparison consumer after generated algebra exists. | External comparison only; possible B/E consumer target. | Moonshine comparison; later AQFT/string-theory side comparisons. | no current Lean module                              | 12.7  |
| MOON8 | Common-indicator / Hauptmodul bridge-test profile | Reference; Gate; Object    | F2   | K             | Comparison only; not a generator.                                                   | External comparison only.                               | Moonshine/McKay-Thompson comparison; documentation.              | no current Lean module                              | 12.8  |
| LEG0  | Consumption and split inventory umbrella          | Governance; Policy; Object | F3   | Y             | A-internal refactor governance only.                                                | none                                                    | 1.1.1 child gates only                                           | Arithmetic/Notation.lean; repo_inventory/raw_export | 1.1.1 |
| LEG1  | Legacy single-constructor status gates            | Lean; Refactor; Object     | F3   | Y             | Legacy-only; forbidden to Phase 3+ neutral imports.                                 | no handoff                                              | Legacy compatibility modules only                                | CNNA/PillarA/Arithmetic/Notation.lean               | 1.1.2 |

| ID    | Object                                                         | Context class              | Tier | Object status | Pillar-A access                                                                                                  | Handoff rights                                              | May be consumed by                                                                                                                                           | Current code modules / derivation site                                                                                                                                                                                                                                                                                                                                  | Phase      |
|-------|----------------------------------------------------------------|----------------------------|------|---------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| LEG2  | CoreSymbols true notation surface                              | Lean; Refactor; Object     | F3   | Y             | A-internal public neutral import surface.                                                                        | no handoff                                                  | 1.2+ neutral modules; Phase 3+                                                                                                                               | CNNA/PillarA/Arithmetic/Notation.lean; Foundation/Notation.lean; Finite/Notation.lean; Handoff/Notation.lean                                                                                                                                                                                                                                                            | 1.1.3      |
| LEG3  | Legacy phase aliases compatibility surface                     | Lean; Refactor; Object     | F3   | Y             | Legacy-only except explicitly neutral wrappers.                                                                  | no handoff                                                  | old modules; neutral wrapper seam only                                                                                                                       | CNNA/PillarA/Arithmetic/Notation.lean; historical SM*/S6* modules                                                                                                                                                                                                                                                                                                       | 1.1.4      |
| LEG4  | Neutral import firewall for post-notation phases               | Governance; Policy; Object | F3   | Y             | A-internal governance only.                                                                                      | no handoff                                                  | 1.2+ and 3+ only as an import rule                                                                                                                           | BuildAll.lean; repo_inventory/raw_export/module_imports.tsv                                                                                                                                                                                                                                                                                                             | 1.1.5      |
| LEG5  | Declaration consumption classification map                     | Governance; Policy; Object | F3   | G             | A-internal planning/control surface only; not a mathematical generator                                           | none                                                        | 1.1.1.2, 1.1.1.3, 1.1.1.4, 1.1.2, all public-API/correction prompts                                                                                          | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated_exports/consumption_classification_map.generated.tsv; generated_exports/module_consumption_records.generated.tsv; generated_exports/declaration_consumption_records.generated.tsv; generated_exports/object_consumption_classification.generated.tsv; tables/consumption_classification_map.tex | 1.1.1.1.4  |
| LEG7  | Blue-object semantic correction ledger                         | Governance; Policy; Object | F3   | Y             | A-internal planning/control surface only; not a mathematical generator                                           | none                                                        | 1.1.1.3, 1.1.4, 1.1.5, and each natural B-object correction phase                                                                                            | object register; Structural/CayleyDickson/*; Arithmetic/Smoke/GeneratedInterior*; Arithmetic/Smoke/*Operational*                                                                                                                                                                                                                                                        | 1.1.1.2    |
| LEG6  | Public API contract and consumer whitelist                     | Governance; Policy; Object | F3   | Y             | A-internal planning/control surface only; not a mathematical generator                                           | none                                                        | 1.1.2, 1.1.3, 1.1.4, 1.1.5, Phase 3+ import firewall checks                                                                                                  | masters/cnna_roadmap_master.yaml; planned CoreSymbols/Legacy/NeutralWrappers modules                                                                                                                                                                                                                                                                                    | 1.1.1.3    |
| LEG8  | Pruning and dead-candidate ledger                              | Governance; Policy; Object | F3   | Y             | A-internal planning/control surface only; not a mathematical generator                                           | none                                                        | 1.1.5 and future optional pruning phase after quarantine/wrappers close                                                                                      | planning master; generated reports; repo_inventory/raw_export                                                                                                                                                                                                                                                                                                           | 1.1.1.4    |
| LEG9  | Generated-view authority and implementation-lock contract      | Governance; Policy; Object | F3   | Y             | A-internal planning/control surface only; not a mathematical generator                                           | none                                                        | 1.1.2 implementation prompt and post-green closeout workflow                                                                                                 | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated exports; PDF                                                                                                                                                                                                                                                                                    | 1.1.1.5    |
| LEG10 | Implementation module manifest and semantic placement contract | Governance; Policy; Object | F1   | G             | A-internal planning/control surface only; not a mathematical generator                                           | none                                                        | all future implementation prompts; 1.1.1.1.2; 1.1.1.1.3; 1.1.1.3; post-green closeout                                                                        | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated_exports/module_implementation_manifest.generated.tsv; tables/module_implementation_manifest.tex                                                                                                                                                                                                 | 1.1.1.1.1  |
| LEG11 | Reuse/duplication ledger and canonical generator policy        | Governance; Policy; Object | F1   | G             | A-internal governance only; public only as an implementation rule and comparison/audit surface                   | none                                                        | 1.1.1.1.3, 1.1.1.3, Phase 5 experiment request/output, O07 finite computation-test matrix, heuristic lattice test matrices, all future generator-like phases | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated_exports/reuse_duplication_ledger.generated.tsv; tables/reuse_duplication_ledger.tex; Arithmetic/Smoke/GeneratedSource.lean; Arithmetic/Smoke/GeneratedMainPath.lean; Actions/GeneratorActionInventory.lean; Actions/ActionFormatDecision.lean                                                   | 1.1.1.1.2  |
| LEG12 | Object and size proof-documentation infrastructure             | Governance; Policy; Object | F1   | G             | A-internal planning/read-only governance; no mathematical input to generator chain.                              | No handoff export; only documentation evidence may cite it. | 1.1.1.1.4 declaration classification, Phase preparation, blue-object correction, public API contract, review workflow.                                       | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated_exports/object_proof_documentation.generated.tsv; tables/object_proof_documentation.tex; tables/object_proof_link_register.tex                                                                                                                                                                  | 1.1.1.1.3  |
| LEG13 | Global no-go guardrail contract                                | Governance; Policy; Object | F0   | G             | A-internal global governance; constrains all Pillar-A implementation prompts and later consumer/firewall phases. | No mathematical handoff; policy evidence only.              | all phases, all implementation prompts, object proof documentation workflow, closeout workflow, Phase 3+ import firewall checks                              | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; tables/global_no_go_guardrails.tex; generated_exports/global_no_go_guardrails.generated.tsv                                                                                                                                                                                                               | 1.1.1.1.0  |
| LEG14 | Concrete computable instance traversal contract                | Governance; Policy; Object | F3   | G             | A-internal workflow invariant.                                                                                   | No handoff payload.                                         | All future implementation prompts and scratchpad closeout checks.                                                                                            | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; implementation_scratchpad                                                                                                                                                                                                                                                                                 | 1.1.1.1.0. |

| ID    | Object                                                             | Context class              | Tier | Object status | Pillar-A access                                                                         | Handoff rights                                                                                | May be consumed by                                                                                                                                                                                                           | Current code modules / derivation site                                                                                                                                                                | Phase      |
|-------|--------------------------------------------------------------------|----------------------------|------|---------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| LEG15 | Mathlib complication and noncomputable-boundary log                | Governance; Policy; Object | F3   | G             | A-internal workflow invariant.                                                          | No handoff payload.                                                                           | All future implementation prompts, especially ExecComplex, matrix, polynomial, inverse, determinant, and mirror phases.                                                                                                      | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; implementation_scratchpad                                                                                                               | 1.1.1.1.0. |
| LEG16 | Semi-automated existing-code documentation coverage                | Reference; Gate; Object    | F3   | K             | A-internal diagnostic/documentation only; may not alter derived objects.                | No pillar handoff rights.                                                                     | Planning workflow, proofbook workflow, human review only.                                                                                                                                                                    | repo_inventory/raw_export/*.json; generated_exports/*.tsv; tables/object_proof_documentation.tex                                                                                                      | 19.1       |
| LEG17 | Derivation-graph geometry diagnostics                              | Reference; Gate; Object    | F3   | K             | Diagnostic only; cannot introduce A-internal axioms or objects.                         | No pillar handoff rights.                                                                     | Secondary reports, architecture review, pruning/API review.                                                                                                                                                                  | repo_inventory/raw_export/import_edges.json; figures/*.dot/pdf                                                                                                                                        | 19.2       |
| LEG18 | Open-system emergence-path forensics                               | Reference; Gate; Object    | F3   | K             | Diagnostic only; emergence claims must be proved from CNNA objects, not graph metadata. | No handoff rights.                                                                            | Later A/C review and secondary architecture reports.                                                                                                                                                                         | future A/C modules plus repo_inventory/raw_export path analysis                                                                                                                                       | 19.3       |
| LEG19 | Reflective CNNA/formalization graph comparison boundary            | Reference; Gate; Object    | F3   | K             | May constrain interpretation/prose; cannot supply CNNA data.                            | No handoff rights.                                                                            | Secondary research notes and architecture reviews only.                                                                                                                                                                      | generated graph views and proofbook links                                                                                                                                                             | 19.4       |
| LEG20 | EXEC-2 rational ExecComplex boundary policy                        | Lean; Governance; Object   | F1   | G             | A-internal governance; applies to operational/mirror separation                         | No direct handoff; constrains future A->* numeric exports                                     | All ExecComplex, geometry, modular, spectral and arithmetic phases                                                                                                                                                           | masters/cnna_roadmap_master.yaml; format_decision EXEC-2; workflow_policy EXEC-2                                                                                                                      | 1.1.1.1.0. |
| LEG21 | Refactor regression and incremental migration protocol             | Lean; Governance; Object   | F1   | G             | A-internal governance for refactor phases                                               | No direct handoff                                                                             | Phase 1.1.2 and all module-split/quarantine phases                                                                                                                                                                           | masters/cnna_roadmap_master.yaml; workflow_policy refactor regression                                                                                                                                 | 1.1.1.1.0. |
| LEG22 | Lean/mathlib version migration policy                              | Lean; Governance; Object   | F1   | G             | A-internal governance; applies to all imports                                           | No direct handoff                                                                             | All phases touching mathlib imports                                                                                                                                                                                          | masters/cnna_roadmap_master.yaml; workflow_policy version management                                                                                                                                  | 1.1.1.1.0. |
| LEG23 | Phase-type-specific scratchpad schema and governance retrospective | Documentation; Object      | F3   | G             | A-internal planning/documentation                                                       | No direct handoff                                                                             | All phase preparation and closeout workflows                                                                                                                                                                                 | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py scratchpad renderer                                                                                                                      | 1.1.1.1.0. |
| LEG24 | Phase 3 to Phase 4 consumer-contract direction                     | Lean; Governance; Object   | F1   | G             | A-internal; constrains wrapper design                                                   | A->B only later through filtered profiles, not through implementation modules                 | Phase 3.6 and Phase 4.1                                                                                                                                                                                                      | masters/cnna_roadmap_master.yaml phase goals 3.6 and 4.1                                                                                                                                              | 1.1.1.1.0. |
| LEG25 | IdealThread/CutSpec.full extremal-route contract                   | Governance; Policy; Object | F1   | W             | Global workflow/access policy only; no mathematical object export.                      | None.                                                                                         | Phase 1 module-manifest gate, Phase 1.2.7/1.2.8, Phase 3.0, Phase 4.0.                                                                                                                                                       | CNNA/PillarA/Foundation/SubstrateClass.lean; CNNA/PillarA/ToC/Contract.lean; CNNA/PillarA/Finite/CutSpec.lean; CNNA/PillarA/Arithmetic/BoundarySource/ExistingIdealThreadNoGo.lean                    | 1.1.1.1.0. |
| S07   | IdealThread singleton directed-family profile                      | Lean; Source; Object       | F1   | G             | A-internal structural profile.                                                          | None directly; not an A->B boundary matrix.                                                   | Boundary-route audit, scalar/thread-recursion audit, secondary diagnostics.; BR08 horizontal/vertical Schur-duality bridge                                                                                                   | CNNA/PillarA/Foundation/SubstrateClass.lean; CNNA/PillarA/ToC/Contract.lean; CNNA/PillarA/Arithmetic/BoundarySource/ExistingIdealThreadNoGo.lean                                                      | 1.2.7      |
| S08   | CutSpec.full maximal-carrier profile                               | Lean; Source; Object       | F1   | G             | A-internal structural profile.                                                          | No direct handoff; downstream boundary operator may be exported only after Schur/DtN closure. | Full-boundary Schur/DtN route, boundary-route audit, sector/complement geometry.; BR08 horizontal/vertical Schur-duality bridge                                                                                              | CNNA/PillarA/Finite/CutSpec.lean; CNNA/PillarA/Arithmetic/Core/FullBoundaryAudit.lean                                                                                                                 | 1.2.8      |
| BR06  | Boundary-route extrema contract                                    | Lean; Source; Object       | F1   | G             | A-internal route governance.                                                            | None; protects A->B handoff from scalar/thread collapse.                                      | Phase 3 Schur/DtN wrappers and Phase 4 arithmetic wrappers.; BR08 horizontal/vertical Schur-duality bridge                                                                                                                   | CNNA/PillarA/Arithmetic/BoundarySource/ExistingIdealThreadNoGo.lean; CNNA/PillarA/Arithmetic/Core/FullBoundaryAudit.lean                                                                              | 3.0        |
| BR07  | Thread-recursion arithmetic-shadow audit profile                   | Lean; Source; Object       | F1   | Y             | A-internal audit profile.                                                               | None directly; discriminant outputs follow only via operational arithmetic phases.            | Phase 4 arithmetic wrappers, Phase 5 number-theory phases, secondary diagnostics.; BR08 horizontal/vertical Schur-duality bridge; L09/L10 test matrix if thread-recursion arithmetic shadows are later connected to geometry | CNNA/PillarA/Arithmetic/Schur/RecursiveSchurSource.lean; CNNA/PillarA/Arithmetic/Schur/MobiusTransfer.lean; CNNA/PillarA/Arithmetic/Schur/CharpolyExec.lean; CNNA/PillarA/Arithmetic/Smoke/*SM9*.lean | 4.0        |

| ID    | Object                                                           | Context class              | Tier | Object status | Pillar-A access                                                      | Handoff rights                                                                                         | May be consumed by                                                                                                         | Current code modules / derivation site                                                                                          | Phase       |
|-------|------------------------------------------------------------------|----------------------------|------|---------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------|
| BR09  | Subsystem-induced invariant bifurcation profile                  | Lean; Target; Object       | F1   | K             | A-internal explanatory bridge after BR08.                            | No handoff until mathematically proved and downgraded to a precise theorem statement.                  | Long-term derivation-geometry diagnostics; proofbook explanations.                                                         | No canonical module yet.                                                                                                        | 10.4.1      |
| BR08  | Horizontal/vertical Schur-duality bridge                         | Lean; Target; Object       | F1   | K             | A-internal far bridge; cannot be consumed by earlier source phases.  | No A->B/C/D/E handoff until theorem is proved and access policy is reviewed.                           | CD8; secondary derivation-geometry diagnostics; possibly later Pillar-A unification reports.                               | No canonical module yet; source audit must inspect Schur/DtN, Arithmetic/Smoke, Structural/CayleyDickson, and Transfer modules. | 10.4.1      |
| LEG26 | Reflective thread/full-boundary analogy diagnostic boundary      | Governance; Policy; Object | F4   | K             | Comparison-only; forbidden as A-internal generative input.           | None.                                                                                                  | Secondary long-term meta-analysis only.                                                                                    | repo_inventory/raw_export/*; future generated documentation views                                                               | 19.5        |
| LEG27 | Horizontal/vertical Schur-duality dependency-order lock          | Governance; Policy; Object | F1   | G             | Global planning invariant.                                           | No handoff output; constrains A-internal phase order.                                                  | BR08; BR09; CD8; Phase 10.4.1/10.4.2.                                                                                      | masters/cnna_roadmap_master.yaml                                                                                                | 1.1.1.1.0.3 |
| LEG28 | Heuristic Eisenstein/Gauss lattice test-matrix boundary contract | Governance; Policy; Object | F1   | G             | A-internal planning/test governance only.                            | No handoff export.                                                                                     | 8.5, 8.6, 8.7 and later documentation/test closeout phases.                                                                | masters/cnna_roadmap_master.yaml; generated workflow/phase/proofbook views                                                      | 1.1.1.1.0.4 |
| LEG29 | General early finite computation test-matrix contract            | Governance; Policy; Object | F1   | G             | A-internal governance only; not mathematical generator input.        | none                                                                                                   | all future computation/refactor/test phases; especially 1.1.1.1.1, 4.0.1, 8.5, 8.6                                         | masters/cnna_roadmap_master.yaml                                                                                                | 1.1.1.1.0.5 |
| L09   | Static branching Eisenstein/Gauss lattice test matrix            | Reference; Gate; Object    | F2   | Y             | A-internal test consumer; not a source for generator definitions.    | No handoff; comparison/test only.                                                                      | Phase 8.7; documentation; comparison diagnostics. It may not replace O07 generic coverage.                                 | Not yet; placement must be decided by Phase 1.1.1.1.1 and future 8.5 manifest.                                                  | 8.5         |
| L10   | Dynamic branching and L_max lattice test sweep                   | Reference; Gate; Object    | F2   | Y             | A-internal test consumer only.                                       | No handoff.                                                                                            | 8.7 closeout; later derivation-geometry diagnostics.                                                                       | Not yet; future placement must reuse canonical generator.                                                                       | 8.6         |
| M08   | Eisenstein/Gauss CM and j special-value test targets             | Reference; Gate; Object    | F2   | Y             | A-internal comparison consumer only.                                 | No handoff until separately derived and approved; likely documentation/test only.                      | 8.5, 8.7, later modular comparison phases.                                                                                 | Not yet; future mirror/comparison module after M05.                                                                             | 8.5         |
| BR10  | Hierarchical versus spiral embedding collision and closure audit | Reference; Gate; Object    | F2   | Y             | A-internal test/audit consumer; cannot drive generator semantics.    | No handoff.                                                                                            | 8.6, 8.7, secondary derivation-geometry diagnostics.                                                                       | Not yet; future test/profile module.                                                                                            | 8.6         |
| LEG30 | Regime-typed early test-matrix contract                          | Governance; Policy; Object | F1   | G             | A-internal governance only; not a mathematical generator input.      | none                                                                                                   | 1.1.1.1.1, 4.0.2, 7.2.1, 8.5-8.7, proof dossiers and comparison diagnostics                                                | masters/cnna_roadmap_master.yaml; generated views                                                                               | 1.1.1.1.0.6 |
| O08   | Regime-typed finite computation test matrix profile              | Lean; Target; Object       | F2   | Y             | A-internal reusable test object; not a generative axiom.             | none until proof-carrying rows are derived; comparison columns cannot be exported as operational data. | Phase 4 wrappers, regime diagnostics, Phase 8 lattice sentinel tests, Heegner classification, documentation/proof dossiers | future module selected by Phase 1.1.1.1.1; must extend/reuse O07 surface                                                        | 4.0.2       |
| N18   | Heegner regime-projection classification profile                 | Lean; Target; Object       | F1   | Y             | A-internal consumer profile only.                                    | no handoff; comparison/regime documentation only until later approval                                  | 7.2.1, 8.5-8.7, modular comparison diagnostics, secondary long-term derivation-geometry studies                            | no current module; may consume old HeegnerDiscriminantBridge only after neutralization                                          | 7.2.1       |
| BR11  | IDEAL_infty projection and complement-family Heegner boundary    | Lean; Target; Object       | F1   | Y             | A-internal bridge constraint; not an input to generator definitions. | none                                                                                                   | 7.2.1, 8.7, 19.* derivation-geometry diagnostics                                                                           | no current module                                                                                                               | 7.2.1       |
| BR12  | noOuter REAL-root Eisenstein/Gauss sentinel hypothesis           | Lean; Target; Object       | F1   | Y             | A-internal test hypothesis only; not a generator input.              | none                                                                                                   | 4.0.2, 8.5, 8.7, proof dossiers                                                                                            | no current module; future placement with O08/L09/M08 after manifest                                                             | 4.0.2       |
| LEG31 | Computable scalar architecture contract                          | Governance; Policy; Object | F0   | G             | A-internal                                                           | none                                                                                                   | all scalar/CM/lattice/arithmetic implementation phases                                                                     | masters/cnna_roadmap_master.yaml; repo_snapshot audit paths                                                                     | 1.1.1.1.0.7 |



| ID    | Object                                                              | Context class                | Tier | Object status | Pillar-A access                                                                                    | Handoff rights                                                                                 | May be consumed by                                                                                                                            | Current code modules / derivation site                                                                                         | Phase      |
|-------|---------------------------------------------------------------------|------------------------------|------|---------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------|
| LEG32 | Derived-only exact quadratic provenance contract                    | Governance; Policy; Object   | F0   | G             | A-internal                                                                                         | none                                                                                           | O09, O10, BR13, BR14, N02/N04/N06, Lattice/CM tests, CD9                                                                                      | masters/cnna_roadmap_master.yaml                                                                                               | 1.1.1.1.0. |
| LEG33 | Phase 5-10 bifurcation and dependency repair contract               | Governance; Policy; Object   | F1   | G             | global planning invariant                                                                          | none; governance only                                                                          | Phase 5-10 planning; module manifest gate; dependency-depth audit                                                                             | masters/cnna_roadmap_master.yaml                                                                                               | 1.1.1.1.0. |
| O09   | Required exact quadratic operational scalar schema                  | Lean; Target; Object         | F1   | Y             | A-internal                                                                                         | none until proven stable                                                                       | O10/BR14 after D-source; Phase 4.0.3 tests; 5.x/7.x/8.x exact quadratic rows; CD9 only for late comparison                                    | none yet; candidate informed by Mathlib.Algebra.QuadraticAlgebra and future manifest                                           | 1.2.1.2    |
| O10   | Proof-carrying discriminant-indexed quadratic scalar instance       | Lean; Target; Object         | F1   | Y             | A-internal; not before D-source correction                                                         | none until stable; only invariant projections may become handoff candidates later              | N02, N04, N06, L00/L02/L04/L09/L10, M08, BR13                                                                                                 | none yet; future module names require Phase 1.1.1.1.1 manifest                                                                 | 4.5.1      |
| BR13  | ExecComplex to exact-quadratic projection bridge                    | Lean; Target; Object         | F1   | Y             | A-internal                                                                                         | none                                                                                           | O07/O08/O09 tests, N04/N06, M08 comparisons                                                                                                   | none yet                                                                                                                       | 4.0.3      |
| BR14  | Discriminant-source to exact-quadratic algebra instantiation bridge | Lean; Target; Object         | F1   | Y             | A-internal                                                                                         | none                                                                                           | BR13, N02/N04/N06, lattice/CM/test-matrix rows                                                                                                | none yet                                                                                                                       | 4.5.1      |
| CD9   | Cayley-Dickson C-layer noncircularity audit                         | Reference; Gate; Object      | F1   | K             | A-internal; future bridge                                                                          | none                                                                                           | CD8, BR08/BR09, optional future scalar layer comparison                                                                                       | CNNA/PillarA/Arithmetic/CayleyDickson/ArithmeticCDBridge.lean                                                                  | 10.4.3     |
| LEG34 | Phase-object-link policy contract                                   | Governance; Policy; Object   | F1   | G             | planning-only; no Pillar-A mathematical input                                                      | none; documentation/governance only                                                            | all planning phases, phase table generator, implementation manifest gate, proofbook/documentation workflow                                    | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated phase table                                            | 1.1.1.1.0. |
| LEG35 | Ontic subsystem-minimality contract                                 | Governance; Policy; Object   | F0   | G             | global Pillar-A ontic constraint; no mathematical input beyond subsystem-being                     | may constrain all handoffs; exports no physics, arithmetic, geometry or dynamics as primitives | all phases and object dossiers; regime split; complement-family phases; Schur/DtN phases; spectral/physical consumers; documentation workflow | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; tables/global_postulates.tex; global no-go table; proof dossiers | 1.1.1.1.0. |
| DBR0  | Database-ready CNNA review and documentation structure              | Documentation; Object        | F3   | G             | A-internal documentation/review only; may not alter derived objects.                               | No pillar handoff rights.                                                                      | Planning workflow, documentation generator, expert-review process, database/export tooling.                                                   | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; generated exports and PDF views                                  | 20         |
| DBR1  | Expert-review and no-infallibility epistemic rule                   | Documentation; Object        | F1   | G             | Global review constraint; cannot generate A-internal mathematics.                                  | No pillar handoff rights.                                                                      | All phase scratchpads, proof dossiers, documentation outputs, expert-review workflows.                                                        | masters/cnna_roadmap_master.yaml; generated_exports/global_no_go_guardrails.generated.tsv; generated PDF                       | 20.1       |
| DBR2  | Database-ready YAML schema and round-trip target                    | Tool; Infrastructure; Object | F1   | K             | Documentation/query only; no CNNA generator role.                                                  | No pillar handoff rights.                                                                      | Tooling, documentation, expert-review views.                                                                                                  | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv                                                                      | 20.2       |
| DBR3  | Discipline-separated expert-review matrix                           | Documentation; Object        | F3   | K             | Review only; cannot supply A-internal objects.                                                     | No pillar handoff rights.                                                                      | Expert reviewers, documentation workflow, phase closeout audits.                                                                              | masters/cnna_roadmap_master.yaml; object_proof_dossiers section                                                                | 20.3       |
| DBR4  | Blue/reference-theory audit ledger                                  | Documentation; Object        | F3   | K             | Audit/comparison marking only; may not import established theory as A-source.                      | No pillar handoff rights.                                                                      | Documentation workflow, expert-review matrix, comparison-theory ledgers.                                                                      | masters/cnna_roadmap_master.yaml; generated_exports/status_legend.generated.tsv                                                | 20.4       |
| DBR5  | Interactive/online database-backed documentation views              | Tool; Infrastructure; Object | F1   | K             | Documentation and review only; no mathematical source role.                                        | No pillar handoff rights.                                                                      | Human review, online docs, visualization, sharing and exploration workflows.                                                                  | generated_exports/*.tsv; figures/*.dot/pdf; future DB/export scripts                                                           | 20.5       |
| DBR6  | Context-filtered documentation slice generator                      | Generated; Artifact; Object  | F3   | G             | Documentation/review only; may reference Pillar-A rows but cannot generate A-internal mathematics. | No pillar handoff rights.                                                                      | Human review, phase closeout documentation, targeted documentation requests, online/database-ready generated views.                           | scripts/generate_context_doc.py; masters/cnna_roadmap_master.yaml; context_docs/dtn_matrix/* sample output                     | 20.6       |



| ID    | Object                                                                      | Context class               | Tier | Object status | Pillar-A access                                                                                                      | Handoff rights            | May be consumed by                                                                                                                                                | Current code modules / derivation site                                                                                                                                                                                                                                               | Phase      |
|-------|-----------------------------------------------------------------------------|-----------------------------|------|---------------|----------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| DBR7  | Stats front section and tool-interface documentation layer                  | Generated; Artifact; Object | F3   | G             | Documentation/review only; may reference Pillar-A rows and module counts but cannot generate A-internal mathematics. | No pillar handoff rights. | Human review, phase closeout summaries, context documentation slices, database-ready views, online documentation, project onboarding.                             | scripts/generate_tables.py; masters/cnna_roadmap_master.yaml; tables/stats.tex; generated_exports/project_stats.generated.tsv                                                                                                                                                        | 20.7       |
| LEG36 | Semantic module consolidation ledger                                        | Governance; Policy; Object  | F1   | Y             | A-internal refactor/documentation governance only; not a mathematical generator                                      | none                      | 1.1.1.2, 1.1.1.3, public API/firewall phases, pruning/refactor phases, future Lean implementation prompts                                                         | masters/cnna_roadmap_master.yaml; repo_snapshot/CNNA/**/*lean; repo_inventory/raw_export/*.json; generated_exports/*.tsv; SEMANTIC_MODULE_CONSOLIDATION_AUDIT_REPORT.md                                                                                                              | 1.1.1.1.5  |
| DBR8  | Root artifact retention and cleanup policy                                  | Generated; Artifact; Object | F3   | G             | Documentation/review only; no Pillar-A mathematical object access.                                                   | No pillar handoff rights. | Release packaging, documentation generation, root hygiene checks, database-ready review/export workflows, human review.                                           | scripts/generate_full_document.py; scripts/clean_generated_root_artifacts.py; README.md; archive/reports/; masters/cnna_roadmap_master.yaml                                                                                                                                          | 20.8       |
| DBR9  | Current README and workflow-orientation manual                              | Generated; Artifact; Object | F3   | G             | Documentation/review only; no Pillar-A mathematical object access.                                                   | No pillar handoff rights. | Human users, ChatGPT-assisted tool updates, documentation generation, workflow extension, release packaging, expert review onboarding.                            | README.md; masters/cnna_roadmap_master.yaml; archive/reports/V1_14_README_WORKFLOW_ORIENTATION_REPORT.md                                                                                                                                                                             | 20.9       |
| DBR10 | Root Lean/Lake project-control files                                        | Generated; Artifact; Object | F3   | G             | No mathematical access rights; build/toolchain surface only.                                                         | No pillar handoff rights. | Local build workflow, release packaging, root cleanup workflow, README/manual use, future CI/database packaging.                                                  | CNNA.lean; lakefile.lean; lake-manifest.json; lean-toolchain; Repository/scripts/clean_generated_root_artifacts.py; README.md; Repository/masters/cnna_roadmap_master.yaml                                                                                                           | 20.10      |
| DBR11 | Root license, single-root README, credits and assistant-requirement surface | Generated; Artifact; Object | F3   | G             | Documentation/review only; no Pillar-A mathematical object access.                                                   | No pillar handoff rights. | Human users, release packaging, documentation generation, ChatGPT-assisted workflow onboarding, licensing review, future CI/database packaging.                   | README.md; LICENSE; Repository/scripts/generate_full_document.py; Repository/scripts/generate_context_doc.py; Repository/scripts/generate_tables.py; Repository/scripts/clean_generated_root_artifacts.py; Repository/masters/cnna_roadmap_master.yaml                               | 20.11      |
| DBR12 | Generated PDF section-pagebreak layout invariant                            | Generated; Artifact; Object | F3   | G             | Documentation/review only; no Pillar-A mathematical object access.                                                   | No pillar handoff rights. | Human reviewers, ChatGPT-assisted context documentation, generated PDF readers, future online/documentation layout layers.                                        | Repository/scripts/generate_full_document.py; Repository/scripts/generate_context_doc.py; Repository/masters/cnna_roadmap_master.yaml; README.md; Repository/archive/reports/V1_18_SECTION_PAGEBREAKS_REPORT.md                                                                      | 20.12      |
| DBR13 | Phase status release policy and red-lock ledger                             | Generated; Artifact; Object | F3   | G             | Documentation/planning only; no Pillar-A mathematical object access.                                                 | No pillar handoff rights. | Phase preparation, pre-implementation checks, active cursor selection, ChatGPT-assisted planning workflows, README/manual use, future database/status dashboards. | Repository/masters/cnna_roadmap_master.yaml; Repository/scripts/generate_tables.py; README.md; Repository/archive/reports/V1_19_PHASE_STATUS_RELEASE_POLICY_REPORT.md                                                                                                                | 20.13      |
| DBR14 | Static documentation explanations and module-change appendix policy         | Generated; Artifact; Object | F3   | G             | Documentation/review only; no Pillar-A mathematical object access.                                                   | No pillar handoff rights. | Human reviewers, ChatGPT-assisted documentation, phase status review, module refactor audits, future database/online documentation layer.                         | Repository/scripts/generate_full_document.py; Repository/scripts/generate_context_doc.py; Repository/scripts/generate_tables.py; Repository/masters/cnna_roadmap_master.yaml; README.md; Repository/archive/reports/V1_20_DOC_SECTION_EXPLANATIONS_AND_MODULE_MANIFEST_APPENDIX_REPO | 20.14      |
| LEG37 | Current source intake completeness ledger                                   | Governance; Policy; Object  | F1   | Y             | A-internal planning/refactor governance only; no pillar handoff use.                                                 | none                      | 1.1.1.1.5 child phases, 1.1.1.2, public API/firewall phases, future Lean implementation prompts                                                                   | repo_inventory/raw_export/*.json; generated_exports/*consumption*.tsv; masters/cnna_roadmap_master.yaml                                                                                                                                                                              | 1.1.1.1.5. |
| LEG38 | Neutral module-name and facade migration map                                | Governance; Policy; Object  | F1   | Y             | A-internal planning/refactor governance only; no pillar handoff use.                                                 | none                      | 1.1.1.1.5 child phases, 1.1.1.2, public API/firewall phases, future Lean implementation prompts                                                                   | repo_snapshot/CNNA/**/*lean; masters/cnna_roadmap_master.yaml                                                                                                                                                                                                                        | 1.1.1.1.5. |

| ID    | Object                                                                    | Context class               | Tier | Object status | Pillar-A access                                                                     | Handoff rights                                        | May be consumed by                                                                                                                      | Current code modules / derivation site                                                                                                                                                                                                                                                             | Phase       |
|-------|---------------------------------------------------------------------------|-----------------------------|------|---------------|-------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| LEG39 | Neutral refactor execution certificate                                    | Governance; Policy; Object  | F1   | Y             | A-internal planning/refactor governance only; no pillar handoff use.                | none                                                  | 1.1.1.1.5 child phases, 1.1.1.2, public API/firewall phases, future Lean implementation prompts                                         | future modified Lean modules authorized by LEG38                                                                                                                                                                                                                                                   | 1.1.1.1.5.4 |
| LEG40 | Post-refactor inventory closure certificate                               | Governance; Policy; Object  | F1   | Y             | A-internal planning/refactor governance only; no pillar handoff use.                | none                                                  | 1.1.1.1.5 child phases, 1.1.1.2, public API/firewall phases, future Lean implementation prompts                                         | repo_inventory/raw_export/*.json after refactor; generated_exports/*.tsv                                                                                                                                                                                                                           | 1.1.1.1.5.4 |
| LEG41 | Pre-substantive-code release certificate                                  | Governance; Policy; Object  | F1   | Y             | A-internal planning/refactor governance only; no pillar handoff use.                | none                                                  | 1.1.1.1.5 child phases, 1.1.1.2, public API/firewall phases, future Lean implementation prompts                                         | masters/cnna_roadmap_master.yaml; generated_exports/pre_implementation_phase_check_results.generated.tsv                                                                                                                                                                                           | 1.1.1.1.5.4 |
| DBR15 | Supabase/Postgres mirror feasibility and schema design                    | Database; Mirror; Object    | F1   | W             | Documentation/query/publication layer only; no Pillar-A mathematical object access. | No pillar handoff rights and no phase-release rights. | DB mirror planning phases 20.15-20.20, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling.  | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                           | 20.15       |
| DBR16 | YAML/exports to Postgres import and roundtrip contract                    | Database; Mirror; Object    | F1   | W             | Documentation/query/publication layer only; no Pillar-A mathematical object access. | No pillar handoff rights and no phase-release rights. | DB mirror planning phases 20.15-20.20, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling.  | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                           | 20.16       |
| DBR17 | Append-only snapshot and checksum model                                   | Database; Mirror; Object    | F1   | W             | Documentation/query/publication layer only; no Pillar-A mathematical object access. | No pillar handoff rights and no phase-release rights. | DB mirror planning phases 20.15-20.20, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling.  | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                           | 20.17       |
| DBR18 | Public read-only API views and RLS policies                               | Database; Mirror; Object    | F1   | W             | Documentation/query/publication layer only; no Pillar-A mathematical object access. | No pillar handoff rights and no phase-release rights. | DB mirror planning phases 20.15-20.20, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling.  | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                           | 20.18       |
| DBR19 | cnna-project.net web-frontend data contract                               | Database; Mirror; Object    | F1   | W             | Documentation/query/publication layer only; no Pillar-A mathematical object access. | No pillar handoff rights and no phase-release rights. | DB mirror planning phases 20.15-20.20, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling.  | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                           | 20.19       |
| DBR20 | Reviewer annotation and future admin-write model                          | Database; Mirror; Object    | F1   | W             | Documentation/query/publication layer only; no Pillar-A mathematical object access. | No pillar handoff rights and no phase-release rights. | DB mirror planning phases 20.15-20.20, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling.  | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future DB migration/import/frontend artifacts                                                                                                                                                                                           | 20.20       |
| DBR21 | Enlarged generated-PDF typography profile                                 | Generated; Artifact; Object | F3   | G             | Documentation/review only; no Pillar-A mathematical object access.                  | No pillar handoff rights.                             | Human reviewers, generated PDF readers, context documentation, future web/export layout profiles.                                       | Repository/scripts/generate_full_document.py; Repository/scripts/generate_context_doc.py; Repository/scripts/generate_tables.py; Repository/scripts/build_export_script_v1.8.py; README.md; Repository/masters/cnna_roadmap_master.yaml; Repository/archive/reports/V1_23_PDF_FONT_SCALE_REPORT.md | 20.21       |
| DBR22 | Supabase live project discovery and empty-schema baseline audit           | Database; Mirror; Object    | F1   | G             | Documentation/query/publication layer only; no Pillar-A mathematical object access. | No pillar handoff rights and no phase-release rights. | Supabase live import activation phases, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling. | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future Repository/database/supabase/* artifacts                                                                                                                                                                                         | 20.22       |
| DBR23 | Supabase initial schema migration package and RLS/read-only policy design | Database; Mirror; Object    | F1   | W             | Documentation/query/publication layer only; no Pillar-A mathematical object access. | No pillar handoff rights and no phase-release rights. | Supabase live import activation phases, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling. | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future Repository/database/supabase/* artifacts                                                                                                                                                                                         | 20.23       |
| DBR24 | Supabase snapshot import package from YAML/generated exports              | Database; Mirror; Object    | F1   | W             | Documentation/query/publication layer only; no Pillar-A mathematical object access. | No pillar handoff rights and no phase-release rights. | Supabase live import activation phases, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling. | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future Repository/database/supabase/* artifacts                                                                                                                                                                                         | 20.24       |

| ID    | Object                                                                         | Context class                | Tier | Object status | Pillar-A access                                                                               | Handoff rights                                        | May be consumed by                                                                                                                                   | Current code modules / derivation site                                                                                                                                                          | Phase   |
|-------|--------------------------------------------------------------------------------|------------------------------|------|---------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| DBR25 | Explicitly authorized live Supabase schema migration and first snapshot import | Database; Mirror; Object     | F1   | W             | Documentation/query/publication layer only; no Pillar-A mathematical object access.           | No pillar handoff rights and no phase-release rights. | Supabase live import activation phases, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling.              | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future Repository/database/supabase/* artifacts                                                                                      | 20.25   |
| DBR26 | Post-import read-only/RLS/advisor verification and web-frontend handoff        | Database; Mirror; Object     | F1   | W             | Documentation/query/publication layer only; no Pillar-A mathematical object access.           | No pillar handoff rights and no phase-release rights. | Supabase live import activation phases, future cnna-project.net frontend, public read-only API views, expert-review navigation tooling.              | masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; future Repository/database/supabase/* artifacts                                                                                      | 20.26   |
| GA0   | Horn quaternion/geometric-algebra reference-gate packet                        | Reference; Gate; Object      | F2   | K             | Reference/comparison only; prohibited as a source or generator.                               | none                                                  | 1.4.9.1, 1.4.9.2, 13.2.1, 13.6.1, 18.5 as comparison targets only                                                                                    | No current Lean module; masters/cnna_roadmap_master.yaml scratchpad/source-crosscheck only.                                                                                                     | 1.4.9   |
| GA1   | Hamilton unit and Pauli/Cartan sign-convention comparison profile              | Reference; Gate; Object      | F2   | K             | Reference/comparison only; no generator rights.                                               | none                                                  | 1.4.9.2; 13.2.1 comparison gates after CNNA-derived data exists                                                                                      | Future comparison may inspect CNNA/PillarA/Structural/CayleyDickson/*.lean after neutral reintegration.                                                                                         | 1.4.9.1 |
| GA2   | Pauli bivector/pseudoscalar duality comparison profile                         | Reference; Gate; Object      | F2   | K             | Reference/comparison only; no basis/orientation/dimension generator rights.                   | none                                                  | 13.2.1, 13.6.1 and 18.5 as comparison targets only                                                                                                   | No current Lean module; future reference-gate only.                                                                                                                                             | 1.4.9.2 |
| GA3   | Complex-quaternion eight-basis multivector comparison profile                  | Reference; Gate; Object      | F2   | K             | Reference/comparison only; no scalar-layer generator rights.                                  | none                                                  | 10.4.3 CD9 audit, 13.6.1 Dirac/STA comparison, 18.5 side-comparison                                                                                  | Could later relate to 10.4.3 CD9 only after noncircular C-layer audit.                                                                                                                          | 1.4.9.2 |
| GA4   | Boundary Pauli/Clifford comparison gate                                        | Reference; Gate; Object      | F2   | K             | No direct A access; B consumer may only compare against A->B handoff-derived algebra outputs. | none                                                  | 13.6.1 and 18.5 as later comparison/reference gates                                                                                                  | PillarB empty; future CNNA/PillarB/QFT/BoundaryStarAlgebra.lean consumer output.                                                                                                                | 13.2.1  |
| GA5   | Dirac/spacetime-algebra comparison target gate                                 | Reference; Gate; Object      | F2   | K             | No direct A access; comparison only after B consumer chain.                                   | none                                                  | 18.5 side-comparison and later expert-review documentation                                                                                           | PillarB empty; no current STA/Dirac module.                                                                                                                                                     | 13.6.1  |
| GA6   | Geometric and conformal algebra side-comparison register                       | Reference; Gate; Object      | F2   | K             | No A-source rights; side comparison only.                                                     | none                                                  | Documentation workflow, expert review and later side-comparison packets                                                                              | No current Lean module.                                                                                                                                                                         | 18.5    |
| DBR27 | Phase/Object context-class taxonomy                                            | Tool; Infrastructure; Object | F3   | G             | No mathematical Pillar-A access. Planning/control surface only.                               | No pillar handoff rights and no release-proof rights. | phase table; object register; proof dossiers; implementation scratchpad; module-change manifest; future DB/public API schema; implementation prompts | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; scripts/generate_full_document.py; generated_exports/phase_roadmap.generated.tsv; generated_exports/object_register.generated.tsv | 20.27   |

### Object table D: proof-documentation status and proofbook links

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| ID  | Object               | Context class                   | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                         | Proof dossier | Change version     | Change timestamp                    | Phase |
|-----|----------------------|---------------------------------|------|---------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|-------|
| G00 | Generator provenance | Lean; Source; Object            | F0   | G             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.2.2 |
| H00 | A input handoff B->A | Lean; Handoff; Contract; Object | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | H00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 2.3.1 |

| ID  | Object                                    | Context class                            | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                         | Proof dossier | Change version     | Change timestamp                    | Phase |
|-----|-------------------------------------------|------------------------------------------|------|---------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|-------|
| H01 | A input handoff C->A                      | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | H01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 2.3.1 |
| H02 | A input handoff D->A                      | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | H02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 2.3.1 |
| H03 | A input handoff E->A                      | Lean;<br>Handoff;<br>Contract;<br>Object | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | H03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 2.3.1 |
| O00 | ExecComplex                               | Lean;<br>Source;<br>Object               | F1   | W             | partial                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | O00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.2.1 |
| G01 | substrate: regular branching              | Lean;<br>Source;<br>Object               | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.2.3 |
| G02 | substrate: variable branching             | Lean;<br>Source;<br>Object               | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.2.4 |
| G03 | Level / Cutoff                            | Lean;<br>Source;<br>Object               | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.2.5 |
| G04 | Carrier / index type                      | Lean;<br>Source;<br>Object               | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G04           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.2.6 |
| G05 | Boundary / interface / interior partition | Lean;<br>Source;<br>Object               | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G05           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.3.1 |
| G06 | Regime: noOuterEnvironment                | Lean;<br>Source;<br>Object               | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G06           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.3.2 |
| G07 | Regime: hasOuterEnvironment               | Lean;<br>Source;<br>Object               | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G07           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.3.2 |
| G08 | Complement families: UV tail              | Lean;<br>Source;<br>Object               | F0   | R             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G08           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.3.3 |
| G09 | Complement families: IR/parent-path tail  | Lean;<br>Source;<br>Object               | F0   | R             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G09           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.3.3 |
| G10 | Complement families: sideways environment | Lean;<br>Source;<br>Object               | F0   | R             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | G10           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.3.3 |
| CD0 | Cayley-Dickson source                     | Lean;<br>Source;<br>Object               | F1   | B             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | CD0           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.4.1 |
| CD1 | Scalar part                               | Lean;<br>Source;<br>Object               | F1   | B             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | CD1           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.4.2 |
| CD2 | Cayley-Dickson star/conjugation           | Lean;<br>Source;<br>Object               | F1   | B             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | CD2           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.4.3 |
| CD3 | CD normSq                                 | Lean;<br>Source;<br>Object               | F1   | B             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | CD3           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.4.4 |



| ID  | Object                                         | Context class           | Tier | Object status | Proof documentation status                            | Documentation depth requirement                                                                                                                                                                         | Proof dossier | Change version     | Change timestamp                    | Phase |
|-----|------------------------------------------------|-------------------------|------|---------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|-------|
| CD4 | CD multiplication/product                      | Lean; Source; Object    | F1   | B             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | CD4           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.4.5 |
| CD5 | Alternative law                                | Lean; Source; Object    | F1   | B             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | CD5           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.4.6 |
| CD6 | Norm multiplicativity                          | Lean; Source; Object    | F1   | B             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | CD6           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.4.7 |
| CD7 | Hurwitz stop / R,C,H,O classification          | Lean; Source; Object    | F1   | B             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | CD7           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.4.8 |
| S00 | Dirichlet-Laplacian                            | Lean; Source; Object    | F0   | Y             | no                                                    | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | S00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 3.1   |
| S01 | Interior block                                 | Lean; Source; Object    | F0   | B             | no                                                    | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | S01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 3.2   |
| S02 | Interior inverse candidate                     | Lean; Source; Object    | F0   | B             | no                                                    | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | S02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 3.3   |
| S03 | Two-sided inverse certificate                  | Lean; Source; Object    | F0   | B             | no                                                    | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | S03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 3.4   |
| S04 | InteriorEliminationData                        | Lean; Source; Object    | F0   | B             | no                                                    | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | S04           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 3.5   |
| S05 | DtN boundary operator                          | Lean; Source; Object    | F0   | B             | no                                                    | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | S05           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 3.6   |
| S06 | GeneralizedDtN / MultiSchur                    | Lean; Source; Object    | F0   | B             | no                                                    | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | S06           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 3.7   |
| O01 | Boundary matrix over ExecComplex               | Lean; Source; Object    | F1   | B             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | O01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.1   |
| O02 | Determinant coefficient recurrence             | Lean; Source; Object    | F1   | B             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | O02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.2   |
| O03 | Operational polynomial profile                 | Lean; Source; Object    | F1   | B             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | O03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.3   |
| O04 | Quadratic coefficient window                   | Lean; Source; Object    | F1   | B             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | O04           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.4   |
| O05 | Operational discriminant D_op                  | Lean; Source; Object    | F1   | B             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | O05           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.5   |
| O06 | Comparison target policy                       | Reference; Gate; Object | F2   | G             | no                                                    | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | O06           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.6   |
| O07 | General finite computation test matrix profile | Lean; Target; Object    | F2   | Y             | planned dossier; must be filled before implementation | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | O07           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.0.1 |
| N00 | Fundamental discriminant D_fund                | Lean; Source; Object    | F1   | G             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | N00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 5.1   |

| ID  | Object                                                 | Context class              | Tier | Object status | Proof documentation status             | Documentation depth requirement                                                                                                                                                                | Proof dossier | Change version     | Change timestamp                    | Phase  |
|-----|--------------------------------------------------------|----------------------------|------|---------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|--------|
| N01 | Conductor f                                            | Lean;<br>Source;<br>Object | F1   | G             | no                                     | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | N01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 5.1    |
| N02 | Quadratic field $K=\mathbb{Q}(\sqrt{D})$               | Lean;<br>Source;<br>Object | F1   | Y             | no                                     | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | N02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 5.2    |
| N03 | Quadratic order $O_D$                                  | Lean;<br>Source;<br>Object | F1   | Y             | no                                     | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | N03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 5.3    |
| N04 | Integral basis / $\omega_D$                            | Lean;<br>Source;<br>Object | F1   | Y             | no                                     | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | N04           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 5.4    |
| N05 | Trace form                                             | Lean;<br>Target;<br>Object | F1   | Y             | no                                     | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | N05           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 5.5    |
| N06 | Norm form                                              | Lean;<br>Source;<br>Object | F1   | Y             | no                                     | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | N06           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 5.5    |
| N07 | Conjugation                                            | Lean;<br>Target;<br>Object | F1   | Y             | no                                     | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | N07           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 5.5    |
| N08 | Unit group $O_D^\times$                                | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | N08           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 6.1    |
| N09 | Class group $Cl(O_D)$                                  | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | N09           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 6.2    |
| N10 | Class number $h(D)$                                    | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | N10           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 6.2    |
| N11 | Fractional/integral ideas                              | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | N11           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 6.2.1  |
| N12 | Principal ideal predicate                              | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | N12           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 6.2.2  |
| N13 | Heegner/PID/UFD indicator                              | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | N13           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 6.3    |
| N14 | Prime splitting behavior                               | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | N14           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 6.4    |
| N15 | Kronecker/Dirichlet character $\chi_D$                 | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | N15           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 6.4    |
| CD8 | Arithmetic-CD compatibility as Schur-duality corollary | Lean;<br>Source;<br>Object | F1   | K             | no - dependency-order corrected in v31 | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | CD8           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.4.2 |
| L00 | Lattice embedding $\Lambda \rightarrow \mathbb{C}$     | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 7.1    |
| L01 | Lattice basis                                          | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 7.1    |
| L02 | Gram matrix                                            | Lean;<br>Target;<br>Object | F2   | Y             | no                                     | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 7.2    |



| ID  | Object                            | Context class           | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                | Proof dossier | Change version     | Change timestamp                    | Phase |
|-----|-----------------------------------|-------------------------|------|---------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|-------|
| L03 | Lattice determinant / covolume    | Lean; Target; Object    | F2   | Y             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 7.2   |
| L04 | Period ratio tau                  | Lean; Target; Object    | F2   | Y             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L04           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 7.3   |
| L05 | Upper half-plane condition        | Lean; Target; Object    | F2   | Y             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L05           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 7.3   |
| L06 | Angle / cosine profile            | Lean; Target; Object    | F2   | Y             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L06           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 7.4   |
| L07 | Lattice automorphism group        | Lean; Target; Object    | F2   | Y             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L07           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 7.5   |
| M02 | Eisenstein series G4,G6           | Lean; Target; Object    | F2   | Y             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | M02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.1   |
| M03 | Weierstrass invariants g2,g3      | Lean; Target; Object    | F2   | Y             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | M03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.2   |
| M04 | Modular discriminant Delta_mod    | Lean; Target; Object    | F2   | Y             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | M04           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.3   |
| M05 | j-invariant                       | Lean; Target; Object    | F2   | Y             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | M05           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.4   |
| SP0 | Eigenvalue profile                | Lean; Source; Object    | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | SP0           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.1  |
| SP1 | Spectral gap                      | Lean; Source; Object    | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | SP1           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.1  |
| P00 | Operator algebra                  | Lean; Source; Object    | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | P00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.2  |
| P01 | Star operation                    | Lean; Source; Object    | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | P01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.2  |
| P02 | State functional                  | Lean; Source; Object    | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | P02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.2  |
| P03 | Gibbs state                       | Lean; Source; Object    | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | P03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.2  |
| P04 | Time evolution                    | Lean; Source; Object    | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | P04           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.2  |
| P05 | Unitarity certificate             | Lean; Source; Object    | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | P05           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.2  |
| P06 | Open channel / effective openness | Lean; Source; Object    | F1   | R             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | P06           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.2  |
| N16 | Dirichlet L-function L(s,chi_D)   | Reference; Gate; Object | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | N16           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 9.2   |

| ID  | Object                              | Context class           | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                | Proof dossier | Change version     | Change timestamp                    | Phase |
|-----|-------------------------------------|-------------------------|------|---------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|-------|
| N17 | Class number formula                | Reference; Gate; Object | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | N17           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 9.3   |
| L08 | Theta series Theta_Lambda           | Reference; Gate; Object | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L08           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 9.1   |
| M00 | Modular group action SL2Z           | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | M00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.0   |
| M01 | Fundamental domain representative   | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | M01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.0   |
| M06 | q-expansion of j                    | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | M06           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.8   |
| M07 | Hilbert class polynomial H_D        | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | M07           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.8   |
| TM0 | Transfermatrix T_k                  | Lean; Source; Object    | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | TM0           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.4  |
| TM1 | Transfermatrix product              | Lean; Source; Object    | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | TM1           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.4  |
| TM2 | Moebius/continued fraction relation | Lean; Source; Object    | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | TM2           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.4  |
| SP2 | Heat kernel K(t)                    | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | SP2           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.3  |
| SP3 | Spectral zeta zeta_H                | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | SP3           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.3  |
| SP4 | Spectral dimension                  | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | SP4           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.3  |
| SP5 | Zeta-regularized determinant        | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | SP5           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.3  |
| P07 | KMS condition                       | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | P07           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.4  |
| P08 | GNS Hilbert space                   | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | P08           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.4  |
| P09 | Tomita operator S                   | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | P09           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.5  |
| P10 | Modular operator Delta              | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | P10           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.5  |
| P11 | Modular conjugation J               | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | P11           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.5  |
| P12 | Bisognano-Wichmann/CPT bridge       | Lean; Target; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | P12           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.6  |

| ID    | Object                                     | Context class                   | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                         | Proof dossier | Change version     | Change timestamp                    | Phase |
|-------|--------------------------------------------|---------------------------------|------|---------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|-------|
| P13   | Spin-statistics                            | Lean; Target; Object            | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | P13           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.6  |
| P14   | Kramers degeneracy                         | Lean; Target; Object            | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | P14           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.6  |
| P15   | Split property                             | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | P15           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.6  |
| P16   | Local net / Haag-Kastler algebraic axioms  | Lean; Target; Object            | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | P16           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.3  |
| H10   | A->B bright handoff                        | Lean; Handoff; Contract; Object | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | H10           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 11.1  |
| H11   | A->B prime/dark future handoff             | Lean; Handoff; Contract; Object | F0   | R             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | H11           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 11.2  |
| H12   | A->C handoff                               | Lean; Handoff; Contract; Object | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | H12           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 11.3  |
| H13   | A->D handoff                               | Lean; Handoff; Contract; Object | F0   | Y             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | H13           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 11.4  |
| H14   | A->E handoff                               | Lean; Handoff; Contract; Object | F0   | K             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | H14           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 11.5  |
| MOON0 | j coefficient stream                       | Reference; Gate; Object         | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | MOON0         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 12.1  |
| MOON1 | Monster dimension comparison profile       | Reference; Gate; Object         | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | MOON1         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 12.2  |
| MOON2 | McKay-Thompson comparison profile          | Reference; Gate; Object         | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | MOON2         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 12.3  |
| B00   | Pillar B input contract from A             | Lean; Handoff; Contract; Object | F0   | K             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | B00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.1  |
| B01   | Boundary star-algebra / observable algebra | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | B01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.2  |
| B02   | Local net / QFT algebraic profile          | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | B02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.3  |
| B03   | B state/GNS/KMS profile                    | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | B03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.4  |
| B04   | Tomita-Takesaki modular data               | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | B04           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.5  |

| ID   | Object                                           | Context class                   | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                         | Proof dossier | Change version     | Change timestamp                    | Phase |
|------|--------------------------------------------------|---------------------------------|------|---------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|-------|
| B05  | CPT/BW/spin-statistics/split gates               | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | B05           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.6  |
| B06  | B->A allowed output profile                      | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | B06           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 13.7  |
| C00  | Pillar C OQS input contract from A               | Lean; Handoff; Contract; Object | F0   | K             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | C00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 14.1  |
| C01  | Finite CP/channel profile                        | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | C01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 14.2  |
| C02  | Effective OQS generator / Lindblad gate          | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | C02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 14.3  |
| C03  | Decoherence/noise profile                        | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | C03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 14.4  |
| C04  | C->A allowed output profile                      | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | C04           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 14.5  |
| D00  | Pillar D spacetime input contract                | Lean; Handoff; Contract; Object | F0   | K             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | D00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 15.1  |
| D01  | Region/localization/causal-order candidate       | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | D01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 15.2  |
| D02  | Metric/coarse-geometry/covariance gate           | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | D02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 15.3  |
| D03  | D->A allowed output profile                      | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | D03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 15.4  |
| E00  | Pillar E matter input contract                   | Lean; Handoff; Contract; Object | F0   | K             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | E00           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 16.1  |
| E01  | Matter sector and internal symmetry candidate    | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | E01           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 16.2  |
| E02  | Observable coupling and phenomenology comparison | Reference; Gate; Object         | F3   | K             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | E02           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 16.3  |
| E03  | E->A allowed output profile                      | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | E03           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 16.4  |
| BR00 | Backreaction input DAG                           | Lean; Target; Object            | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | BR00          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 17    |
| BR01 | B->A backreaction input gate                     | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | BR01          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 17.1  |
| BR02 | C->A OQS backreaction input gate                 | Lean; Source; Object            | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | BR02          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 17.2  |



| ID    | Object                                            | Context class              | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                         | Proof dossier | Change version     | Change timestamp                    | Phase     |
|-------|---------------------------------------------------|----------------------------|------|---------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|-----------|
| BR03  | D->A spacetime backreaction input gate            | Lean; Source; Object       | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | BR03          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 17.3      |
| BR04  | E->A matter backreaction input gate               | Lean; Source; Object       | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | BR04          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 17.4      |
| BR05  | Pre/post-emergence switch for A statements        | Lean; Source; Object       | F0   | K             | no                         | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | BR05          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 17.5      |
| SIDE0 | Full ToE comparison profile                       | Reference; Gate; Object    | F4   | K             | no                         | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                           | SIDE0         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 18.1      |
| SIDE1 | String-theory comparison profile                  | Reference; Gate; Object    | F4   | K             | no                         | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                           | SIDE1         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 18.2      |
| SIDE2 | Loop quantum gravity comparison profile           | Reference; Gate; Object    | F4   | K             | no                         | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                           | SIDE2         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 18.3      |
| SIDE3 | Noncommutative geometry comparison profile        | Reference; Gate; Object    | F4   | K             | no                         | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                           | SIDE3         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 18.4      |
| MOON3 | Finite simple group comparison profile            | Reference; Gate; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | MOON3         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 12.4      |
| MOON4 | Sporadic/Monster comparison boundary              | Reference; Gate; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | MOON4         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 12.4      |
| MOON5 | Leech lattice comparison profile                  | Reference; Gate; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | MOON5         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 12.5      |
| MOON6 | Leech theta/shell coefficient comparison          | Reference; Gate; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | MOON6         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 12.6      |
| MOON7 | VOA/moonshine-module comparison profile           | Reference; Gate; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | MOON7         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 12.7      |
| MOON8 | Common-indicator / Hauptmodul bridge-test profile | Reference; Gate; Object    | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                           | MOON8         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 12.8      |
| LEG0  | Consumption and split inventory umbrella          | Governance; Policy; Object | F3   | Y             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG0          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1     |
| LEG1  | Legacy single-constructor status gates            | Lean; Refactor; Object     | F3   | Y             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG1          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.2     |
| LEG2  | CoreSymbols true notation surface                 | Lean; Refactor; Object     | F3   | Y             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG2          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.3     |
| LEG3  | Legacy phase aliases compatibility surface        | Lean; Refactor; Object     | F3   | Y             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG3          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.4     |
| LEG4  | Neutral import firewall for post-notation phases  | Governance; Policy; Object | F3   | Y             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG4          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.5     |
| LEG5  | Declaration consumption classification map        | Governance; Policy; Object | F3   | G             | documented-in-v43          | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG5          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.4 |

| ID    | Object                                                             | Context class              | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                         | Proof dossier | Change version     | Change timestamp                    | Phase       |
|-------|--------------------------------------------------------------------|----------------------------|------|---------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|-------------|
| LEG7  | Blue-object semantic correction ledger                             | Governance; Policy; Object | F3   | Y             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG7          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.2     |
| LEG6  | Public API contract and consumer whitelist                         | Governance; Policy; Object | F3   | Y             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG6          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.3     |
| LEG8  | Pruning and dead-candidate ledger                                  | Governance; Policy; Object | F3   | Y             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG8          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.4     |
| LEG9  | Generated-view authority and implementation-lock contract          | Governance; Policy; Object | F3   | Y             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG9          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.5     |
| LEG10 | Implementation module manifest and semantic placement contract     | Governance; Policy; Object | F1   | G             | documented-in-v40          | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | LEG10         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.1   |
| LEG11 | Reuse/duplication ledger and canonical generator policy            | Governance; Policy; Object | F1   | G             | documented-in-v41          | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | LEG11         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.2   |
| LEG12 | Object and size proof-documentation infrastructure                 | Governance; Policy; Object | F1   | G             | documented-in-v42          | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | LEG12         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.3   |
| LEG13 | Global no-go guardrail contract                                    | Governance; Policy; Object | F0   | G             | yes                        | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | LEG13         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0   |
| LEG14 | Concrete computable instance traversal contract                    | Governance; Policy; Object | F3   | G             | yes-infrastructure         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG14         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.1 |
| LEG15 | Mathlib complication and noncomputable-boundary log                | Governance; Policy; Object | F3   | G             | yes-infrastructure         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG15         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.1 |
| LEG16 | Semi-automated existing-code documentation coverage                | Reference; Gate; Object    | F3   | K             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG16         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 19.1        |
| LEG17 | Derivation-graph geometry diagnostics                              | Reference; Gate; Object    | F3   | K             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG17         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 19.2        |
| LEG18 | Open-system emergence-path forensics                               | Reference; Gate; Object    | F3   | K             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG18         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 19.3        |
| LEG19 | Reflective CNNA/formalization graph comparison boundary            | Reference; Gate; Object    | F3   | K             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG19         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 19.4        |
| LEG20 | EXEC-2 rational ExecComplex boundary policy                        | Lean; Governance; Object   | F1   | G             | yes                        | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | LEG20         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.2 |
| LEG21 | Refactor regression and incremental migration protocol             | Lean; Governance; Object   | F1   | G             | yes                        | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | LEG21         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.3 |
| LEG22 | Lean/mathlib version migration policy                              | Lean; Governance; Object   | F1   | G             | yes                        | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | LEG22         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.4 |
| LEG23 | Phase-type-specific scratchpad schema and governance retrospective | Documentation; Object      | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | LEG23         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.5 |
| LEG24 | Phase 3 to Phase 4 consumer-contract direction                     | Lean; Governance; Object   | F1   | G             | yes                        | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | LEG24         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.6 |



| ID    | Object                                                           | Context class              | Tier | Object status | Proof documentation status                            | Documentation depth requirement                                                                                                                                                                | Proof dossier | Change version     | Change timestamp                    | Phase        |
|-------|------------------------------------------------------------------|----------------------------|------|---------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------------------------------|--------------|
| LEG25 | IdealThread/CutSpec.full extremal-route contract                 | Governance; Policy; Object | F1   | W             | yes                                                   | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | LEG25         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.7  |
| S07   | IdealThread singleton directed-family profile                    | Lean; Source; Object       | F1   | G             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | S07           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.2.7        |
| S08   | CutSpec.full maximal-carrier profile                             | Lean; Source; Object       | F1   | G             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | S08           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.2.8        |
| BR06  | Boundary-route extrema contract                                  | Lean; Source; Object       | F1   | G             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | BR06          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 3.0          |
| BR07  | Thread-recursion arithmetic-shadow audit profile                 | Lean; Source; Object       | F1   | Y             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | BR07          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.0          |
| BR09  | Subsystem-induced invariant bifurcation profile                  | Lean; Target; Object       | F1   | K             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | BR09          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.4.1       |
| BR08  | Horizontal/vertical Schur-duality bridge                         | Lean; Target; Object       | F1   | K             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | BR08          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.4.1       |
| LEG26 | Reflective thread/full-boundary analogy diagnostic boundary      | Governance; Policy; Object | F4   | K             | no                                                    | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                  | LEG26         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 19.5         |
| LEG27 | Horizontal/vertical Schur-duality dependency-order lock          | Governance; Policy; Object | F1   | G             | yes                                                   | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | LEG27         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.8  |
| LEG28 | Heuristic Eisenstein/Gauss lattice test-matrix boundary contract | Governance; Policy; Object | F1   | G             | yes                                                   | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | LEG28         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.9  |
| LEG29 | General early finite computation test-matrix contract            | Governance; Policy; Object | F1   | G             | documented as planning invariant                      | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | LEG29         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.10 |
| L09   | Static branching Eisenstein/Gauss lattice test matrix            | Reference; Gate; Object    | F2   | Y             | no                                                    | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L09           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.5          |
| L10   | Dynamic branching and L_max lattice test sweep                   | Reference; Gate; Object    | F2   | Y             | no                                                    | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | L10           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.6          |
| M08   | Eisenstein/Gauss CM and j special-value test targets             | Reference; Gate; Object    | F2   | Y             | no                                                    | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | M08           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.5          |
| BR10  | Hierarchical versus spiral embedding collision and closure audit | Reference; Gate; Object    | F2   | Y             | no                                                    | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | BR10          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 8.6          |
| LEG30 | Regime-typed early test-matrix contract                          | Governance; Policy; Object | F1   | G             | documented as planning invariant                      | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | LEG30         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.11 |
| O08   | Regime-typed finite computation test matrix profile              | Lean; Target; Object       | F2   | Y             | planned dossier; must be filled before implementation | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | O08           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.0.2        |
| N18   | Heegner regime-projection classification profile                 | Lean; Target; Object       | F1   | Y             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | N18           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 7.2.1        |
| BR11  | IDEAL_infty projection and complement-family Heegner boundary    | Lean; Target; Object       | F1   | Y             | no                                                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | BR11          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 7.2.1        |

| ID    | Object                                                              | Context class                | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                         | Proof dossier | Change version     | Change timestamp                    | Phase        |
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| BR12  | noOuter REAL-root Eisenstein/Gauss sentinel hypothesis              | Lean; Target; Object         | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | BR12          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.0.2        |
| LEG31 | Computable scalar architecture contract                             | Governance; Policy; Object   | F0   | G             | yes                        | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | LEG31         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.12 |
| LEG32 | Derived-only exact quadratic provenance contract                    | Governance; Policy; Object   | F0   | G             | yes                        | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | LEG32         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.13 |
| LEG33 | Phase 5-10 bifurcation and dependency repair contract               | Governance; Policy; Object   | F1   | G             | yes                        | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | LEG33         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.14 |
| O09   | Required exact quadratic operational scalar schema                  | Lean; Target; Object         | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | O09           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.2.1.2      |
| O10   | Proof-carrying discriminant-indexed quadratic scalar instance       | Lean; Target; Object         | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | O10           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.5.1        |
| BR13  | ExecComplex to exact-quadratic projection bridge                    | Lean; Target; Object         | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | BR13          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.0.3        |
| BR14  | Discriminant-source to exact-quadratic algebra instantiation bridge | Lean; Target; Object         | F1   | Y             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | BR14          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 4.5.1        |
| CD9   | Cayley-Dickson C-layer noncircularity audit                         | Reference; Gate; Object      | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | CD9           | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 10.4.3       |
| LEG34 | Phase-object-link policy contract                                   | Governance; Policy; Object   | F1   | G             | documented-in-v38          | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | LEG34         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.15 |
| LEG35 | Ontic subsystem-minimality contract                                 | Governance; Policy; Object   | F0   | G             | documented-in-v39          | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. | LEG35         | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 1.1.1.1.0.16 |
| DBR0  | Database-ready CNNA review and documentation structure              | Documentation; Object        | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | DBR0          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 20           |
| DBR1  | Expert-review and no-infallibility epistemic rule                   | Documentation; Object        | F1   | G             | yes                        | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | DBR1          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 20.1         |
| DBR2  | Database-ready YAML schema and round-trip target                    | Tool; Infrastructure; Object | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | DBR2          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 20.2         |
| DBR3  | Discipline-separated expert-review matrix                           | Documentation; Object        | F3   | K             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | DBR3          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 20.3         |
| DBR4  | Blue/reference-theory audit ledger                                  | Documentation; Object        | F3   | K             | no                         | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | DBR4          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 20.4         |
| DBR5  | Interactive/online database-backed documentation views              | Tool; Infrastructure; Object | F1   | K             | no                         | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.          | DBR5          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 20.5         |
| DBR6  | Context-filtered documentation slice generator                      | Generated; Artifact; Object  | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | DBR6          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 20.6         |
| DBR7  | Stats front section and tool-interface documentation layer          | Generated; Artifact; Object  | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                               | DBR7          | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | 20.7         |

| ID                    | Object                                                                      | Context class               | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                | Proof dossier         | Change version     | Change timestamp                    | Phase                       |
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| <a href="#">LEG36</a> | Semantic module consolidation ledger                                        | Governance; Policy; Object  | F1   | Y             | active-prepared-in-V1.12   | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">LEG36</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">1.1.1.1.5</a>   |
| <a href="#">DBR8</a>  | Root artifact retention and cleanup policy                                  | Generated; Artifact; Object | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                      | <a href="#">DBR8</a>  | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.8</a>        |
| <a href="#">DBR9</a>  | Current README and workflow-orientation manual                              | Generated; Artifact; Object | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                      | <a href="#">DBR9</a>  | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.9</a>        |
| <a href="#">DBR10</a> | Root Lean/Lake project-control files                                        | Generated; Artifact; Object | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                      | <a href="#">DBR10</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.10</a>       |
| <a href="#">DBR11</a> | Root license, single-root README, credits and assistant-requirement surface | Generated; Artifact; Object | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                      | <a href="#">DBR11</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.11</a>       |
| <a href="#">DBR12</a> | Generated PDF section-pagebreak layout invariant                            | Generated; Artifact; Object | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                      | <a href="#">DBR12</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.12</a>       |
| <a href="#">DBR13</a> | Phase status release policy and red-lock ledger                             | Generated; Artifact; Object | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                      | <a href="#">DBR13</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.13</a>       |
| <a href="#">DBR14</a> | Static documentation explanations and module-change appendix policy         | Generated; Artifact; Object | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                      | <a href="#">DBR14</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.14</a>       |
| <a href="#">LEG37</a> | Current source intake completeness ledger                                   | Governance; Policy; Object  | F1   | Y             | active-prepared            | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">LEG37</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">1.1.1.1.5.1</a> |
| <a href="#">LEG38</a> | Neutral module-name and facade migration map                                | Governance; Policy; Object  | F1   | Y             | active-prepared            | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">LEG38</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">1.1.1.1.5.2</a> |
| <a href="#">LEG39</a> | Neutral refactor execution certificate                                      | Governance; Policy; Object  | F1   | Y             | active-prepared            | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">LEG39</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">1.1.1.1.5.3</a> |
| <a href="#">LEG40</a> | Post-refactor inventory closure certificate                                 | Governance; Policy; Object  | F1   | Y             | active-prepared            | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">LEG40</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">1.1.1.1.5.4</a> |
| <a href="#">LEG41</a> | Pre-substantive-code release certificate                                    | Governance; Policy; Object  | F1   | Y             | active-prepared            | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">LEG41</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">1.1.1.1.5.5</a> |
| <a href="#">DBR15</a> | Supabase/Postgres mirror feasibility and schema design                      | Database; Mirror; Object    | F1   | W             | planned                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR15</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.15</a>       |
| <a href="#">DBR16</a> | YAML/exports to Postgres import and roundtrip contract                      | Database; Mirror; Object    | F1   | W             | planned                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR16</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.16</a>       |
| <a href="#">DBR17</a> | Append-only snapshot and checksum model                                     | Database; Mirror; Object    | F1   | W             | planned                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR17</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.17</a>       |
| <a href="#">DBR18</a> | Public read-only API views and RLS policies                                 | Database; Mirror; Object    | F1   | W             | planned                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR18</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.18</a>       |
| <a href="#">DBR19</a> | cnna-project.net web-frontend data contract                                 | Database; Mirror; Object    | F1   | W             | planned                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR19</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.19</a>       |
| <a href="#">DBR20</a> | Reviewer annotation and future admin-write model                            | Database; Mirror; Object    | F1   | W             | planned                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR20</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.20</a>       |

| ID                    | Object                                                                         | Context class                | Tier | Object status | Proof documentation status | Documentation depth requirement                                                                                                                                                                | Proof dossier         | Change version     | Change timestamp                    | Phase                   |
|-----------------------|--------------------------------------------------------------------------------|------------------------------|------|---------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------|-------------------------------------|-------------------------|
| <a href="#">DBR21</a> | Enlarged generated-PDF typography profile                                      | Generated; Artifact; Object  | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                      | <a href="#">DBR21</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.21</a>   |
| <a href="#">DBR22</a> | Supabase live project discovery and empty-schema baseline audit                | Database; Mirror; Object     | F1   | G             | yes                        | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR22</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.22</a>   |
| <a href="#">DBR23</a> | Supabase initial schema migration package and RLS/read-only policy design      | Database; Mirror; Object     | F1   | W             | planned                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR23</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.23</a>   |
| <a href="#">DBR24</a> | Supabase snapshot import package from YAML/generated exports                   | Database; Mirror; Object     | F1   | W             | planned                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR24</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.24</a>   |
| <a href="#">DBR25</a> | Explicitly authorized live Supabase schema migration and first snapshot import | Database; Mirror; Object     | F1   | W             | planned                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR25</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.25</a>   |
| <a href="#">DBR26</a> | Post-import read-only/RLS/advisor verification and web-frontend handoff        | Database; Mirror; Object     | F1   | W             | planned                    | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. | <a href="#">DBR26</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.26</a>   |
| <a href="#">GA0</a>   | Horn quaternion/geometric-algebra reference-gate packet                        | Reference; Gate; Object      | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | <a href="#">GA0</a>   | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">1.4.9</a>   |
| <a href="#">GA1</a>   | Hamilton unit and Pauli/Cartan sign-convention comparison profile              | Reference; Gate; Object      | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | <a href="#">GA1</a>   | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">1.4.9.1</a> |
| <a href="#">GA2</a>   | Pauli bivector/pseudoscalar duality comparison profile                         | Reference; Gate; Object      | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | <a href="#">GA2</a>   | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">1.4.9.2</a> |
| <a href="#">GA3</a>   | Complex-quaternion eight-basis multivector comparison profile                  | Reference; Gate; Object      | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | <a href="#">GA3</a>   | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">1.4.9.2</a> |
| <a href="#">GA4</a>   | Boundary Pauli/Clifford comparison gate                                        | Reference; Gate; Object      | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | <a href="#">GA4</a>   | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">13.2.1</a>  |
| <a href="#">GA5</a>   | Dirac/spacetime-algebra comparison target gate                                 | Reference; Gate; Object      | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | <a href="#">GA5</a>   | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">13.6.1</a>  |
| <a href="#">GA6</a>   | Geometric and conformal algebra side-comparison register                       | Reference; Gate; Object      | F2   | K             | no                         | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  | <a href="#">GA6</a>   | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">18.5</a>    |
| <a href="#">DBR27</a> | Phase/Object context-class taxonomy                                            | Tool; Infrastructure; Object | F3   | G             | yes                        | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                      | <a href="#">DBR27</a> | unchanged-by-V1.01 | Lean data 2026-05-10T05:51:42+02:00 | <a href="#">20.27</a>   |



## 17    Object and quantities proof documentation

**What this section shows.**    This generated proofbook section gives one dossier row per registered object or quantity. A dossier is not automatically a Lean proof: it states source objects, definitions, proof targets, proof sketches and verification status. It is shown so expert review can inspect each object without reading the entire repository.

**Object and quantities proof-documentation workflow.**    This section is generated from the object proof-dossier records in the Master YAML. It is the proofbook infrastructure. Every registered object or quantity has a scratchpad-style dossier. A dossier is not automatically a Lean proof; it records the source objects, definition, proof target, proof sketch, and verification status that must be checked against code. Blue objects may be proof-carrying yet semantically/API-unfinished; their dossier must state that gap before any public-final promotion.

### Object or quantity G00 – Generator provenance

**Table legend.**    W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                              |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | G00                                                                                                                                                                                                                                                                                                                 |
| Object status                   | G                                                                                                                                                                                                                                                                                                                   |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                    |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                                  |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                             |
| Documented                      | no                                                                                                                                                                                                                                                                                                                  |
| Phase                           | 1.2.2                                                                                                                                                                                                                                                                                                               |
| Source objects                  | substrate, branching, level, partition, Schur data.                                                                                                                                                                                                                                                                 |
| Definitions                     | Canonical A-internal generator/provenance record;<br>no pillar consumer generates its own sources.                                                                                                                                                                                                                  |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                              |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                             |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object's natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity H00 – A input handoff B->A

**Table legend.**    W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | H00                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                       |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                               |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Documented                      | no                                                                                                                                                                                                      |
| Phase                           | 2.3.1                                                                                                                                                                                                   |
| Source objects                  | B-derived theorem/data, if explicitly allowed.                                                                                                                                                          |
| Definitions                     | BToAInputProfile.                                                                                                                                                                                       |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                            |

|                           |                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                             |
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity H01 – A input handoff C->A

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                              |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">H01</a>                                                                                                                                                                                                                                                                                                 |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                   |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                                                                                                                                           |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                                  |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                             |
| Documented                      | no                                                                                                                                                                                                                                                                                                                  |
| Phase                           | <a href="#">2.3.1</a>                                                                                                                                                                                                                                                                                               |
| Source objects                  | C-derived data.                                                                                                                                                                                                                                                                                                     |
| Definitions                     | CToAInputProfile.                                                                                                                                                                                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                              |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                             |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity H02 – A input handoff D->A

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                              |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">H02</a>                                                                                                                                                                                                                                                                                                 |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                   |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                                                                                                                                           |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                                  |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                             |
| Documented                      | no                                                                                                                                                                                                                                                                                                                  |
| Phase                           | <a href="#">2.3.1</a>                                                                                                                                                                                                                                                                                               |
| Source objects                  | D-derived data.                                                                                                                                                                                                                                                                                                     |
| Definitions                     | DToAInputProfile.                                                                                                                                                                                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                              |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                             |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |



Object or quantity H03 – A input handoff E->A

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | H03                                                                                                                                                                                                                                                                                                              |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | 2.3.1                                                                                                                                                                                                                                                                                                            |
| Source objects                  | E-derived data.                                                                                                                                                                                                                                                                                                  |
| Definitions                     | EToAInputProfile.                                                                                                                                                                                                                                                                                                |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity O00 – ExecComplex

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | O00                                                                                                                                                                                                                                                                                                     |
| Object status                   | W                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | partial                                                                                                                                                                                                                                                                                                 |
| Phase                           | 1.2.1                                                                                                                                                                                                                                                                                                   |
| Source objects                  | Rational pair / operational complex implementation.                                                                                                                                                                                                                                                     |
| Definitions                     | ExecComplex as the single operational scalar basis in Part 2.                                                                                                                                                                                                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity G01 – substrate: regular branching

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | G01                                                                                                                                                                                                                                                                                                              |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | 1.2.3                                                                                                                                                                                                                                                                                                            |
| Source objects                  | code-b, arity, level carrier.                                                                                                                                                                                                                                                                                    |
| Definitions                     | RegularSubstrateProfile.                                                                                                                                                                                                                                                                                         |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity G02 – substrate: variable branching

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | G02                                                                                                                                                                                                                                                                                                              |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | 1.2.4                                                                                                                                                                                                                                                                                                            |
| Source objects                  | Branching policy beta, level history.                                                                                                                                                                                                                                                                            |
| Definitions                     | VariableSubstrateProfile.                                                                                                                                                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity G03 – Level / Cutoff

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field         | Record |
|---------------|--------|
| Object ID     | G03    |
| Object status | Y      |

|                                 |                                                                                                                                                                                                                                                                                                                  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | <a href="#">1.2.5</a>                                                                                                                                                                                                                                                                                            |
| Source objects                  | Request + admissibility proof.                                                                                                                                                                                                                                                                                   |
| Definitions                     | LevelProfile with selectedLevel/cutoff/levelSpec.                                                                                                                                                                                                                                                                |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity G04 – Carrier / index type

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">G04</a>                                                                                                                                                                                                                                                                                              |
| Object status                   | <span style="background-color: #FFFF00;">Y</span>                                                                                                                                                                                                                                                                |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | <a href="#">1.2.6</a>                                                                                                                                                                                                                                                                                            |
| Source objects                  | GeneratedLevelHistoryIndex or neutral Carrier.                                                                                                                                                                                                                                                                   |
| Definitions                     | CarrierProfile: carrier, Fintype, DecidableEq, cardinality, equiv-to-Fin.                                                                                                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity G05 – Boundary / interface / interior partition

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">G05</a>                                                                                                                                                                                     |
| Object status                   | <span style="background-color: #FFFF00;">Y</span>                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |

|                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documented                | no                                                                                                                                                                                                                                                                                                               |
| Phase                     | 1.3.1                                                                                                                                                                                                                                                                                                            |
| Source objects            | Carrier + cut/BoundarySpec.                                                                                                                                                                                                                                                                                      |
| Definitions               | PartitionProfile.                                                                                                                                                                                                                                                                                                |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity G06 – Regime: noOuterEnvironment

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | G06                                                                                                                                                                                                                                                                                                              |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | 1.3.2                                                                                                                                                                                                                                                                                                            |
| Source objects                  | Partition + complement/environment proof.                                                                                                                                                                                                                                                                        |
| Definitions                     | RegimeProfile.noOuter.                                                                                                                                                                                                                                                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity G07 – Regime: hasOuterEnvironment

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | G07                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Documented                      | no                                                                                                                                                                                                      |
| Phase                           | 1.3.2                                                                                                                                                                                                   |
| Source objects                  | Approximant + complement proof.                                                                                                                                                                         |

|                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definitions               | RegimeProfile.hasOuter.                                                                                                                                                                                                                                                                                          |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity G08 – Complement families: UV tail

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <span style="color: blue;">G08</span>                                                                                                                                                                                                                                                                            |
| Object status                   | <span style="background-color: #ffb6c1;">R</span>                                                                                                                                                                                                                                                                |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | <span style="color: blue;">1.3.3</span>                                                                                                                                                                                                                                                                          |
| Source objects                  | Approximant + infinite carrier.                                                                                                                                                                                                                                                                                  |
| Definitions                     | UVComplementProfile.                                                                                                                                                                                                                                                                                             |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity G09 – Complement families: IR/parent-path tail

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <span style="color: blue;">G09</span>                                                                                                                                                                   |
| Object status                   | <span style="background-color: #ffb6c1;">R</span>                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Documented                      | no                                                                                                                                                                                                      |
| Phase                           | <span style="color: blue;">1.3.3</span>                                                                                                                                                                 |
| Source objects                  | Approximant + parent/level structure.                                                                                                                                                                   |
| Definitions                     | IRComplementProfile.                                                                                                                                                                                    |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                  |

|                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity G10 – Complement families: sideways environment

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | G10                                                                                                                                                                                                                                                                                                              |
| Object status                   | R                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | 1.3.3                                                                                                                                                                                                                                                                                                            |
| Source objects                  | Branching + nonselected sectors.                                                                                                                                                                                                                                                                                 |
| Definitions                     | SideEnvironmentProfile.                                                                                                                                                                                                                                                                                          |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity CD0 – Cayley-Dickson source

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | CD0                                                                                                                                                                                            |
| Object status                   | B                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | 1.4.1                                                                                                                                                                                          |
| Source objects                  | carrier/regularization/schur from closed A data.                                                                                                                                               |
| Definitions                     | CayleyDicksonSourceProfile.                                                                                                                                                                    |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                         |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                      |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                        |



|                           |                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Object or quantity CD1 – Scalar part

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | CD1                                                                                                                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 1.4.2                                                                                                                                                                                                                                                                                                   |
| Source objects                  | CD algebra data.                                                                                                                                                                                                                                                                                        |
| Definitions                     | CDScalarPartProfile.                                                                                                                                                                                                                                                                                    |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity CD2 – Cayley-Dickson star/conjugation

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | CD2                                                                                                                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 1.4.3                                                                                                                                                                                                                                                                                                   |
| Source objects                  | CD algebra data.                                                                                                                                                                                                                                                                                        |
| Definitions                     | CDStarProfile.                                                                                                                                                                                                                                                                                          |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity CD3 – CD normSq

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | CD3                                                                                                                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 1.4.4                                                                                                                                                                                                                                                                                                   |
| Source objects                  | CD algebra.                                                                                                                                                                                                                                                                                             |
| Definitions                     | CDNormSqProfile.                                                                                                                                                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity CD4 – CD multiplication/product

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | CD4                                                                                                                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 1.4.5                                                                                                                                                                                                                                                                                                   |
| Source objects                  | CD coordinates.                                                                                                                                                                                                                                                                                         |
| Definitions                     | CDProductProfile.                                                                                                                                                                                                                                                                                       |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity CD5 – Alternative law

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | CD5                                                                                                                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 1.4.6                                                                                                                                                                                                                                                                                                   |
| Source objects                  | Product associator.                                                                                                                                                                                                                                                                                     |
| Definitions                     | CDAAlternativeLawCertificate.                                                                                                                                                                                                                                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity CD6 – Norm multiplicativity

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | CD6                                                                                                                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 1.4.7                                                                                                                                                                                                                                                                                                   |
| Source objects                  | normSq, product.                                                                                                                                                                                                                                                                                        |
| Definitions                     | CDNormMultiplicativityCertificate.                                                                                                                                                                                                                                                                      |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity CD7 – Hurwitz stop / R,C,H,O classification

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field         | Record |
|---------------|--------|
| Object ID     | CD7    |
| Object status | B      |

|                                 |                                                                                                                                                                                                                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 1.4.8                                                                                                                                                                                                                                                                                                   |
| Source objects                  | alternative + norm + division.                                                                                                                                                                                                                                                                          |
| Definitions                     | HurwitzStopProfile.                                                                                                                                                                                                                                                                                     |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity S00 – Dirichlet-Laplacian

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | S00                                                                                                                                                                                                                                                                                                              |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | 3.1                                                                                                                                                                                                                                                                                                              |
| Source objects                  | Partition + weights + adjacency.                                                                                                                                                                                                                                                                                 |
| Definitions                     | DirichletLaplacianProfile.                                                                                                                                                                                                                                                                                       |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity S01 – Interior block

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | S01                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |

|                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documented                | no                                                                                                                                                                                                                                                                                                               |
| Phase                     | 3.2                                                                                                                                                                                                                                                                                                              |
| Source objects            | Interior indices + matrix.                                                                                                                                                                                                                                                                                       |
| Definitions               | InteriorBlockProfile.                                                                                                                                                                                                                                                                                            |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity S02 – Interior inverse candidate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | S02                                                                                                                                                                                                                                                                                                              |
| Object status                   | B                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | 3.3                                                                                                                                                                                                                                                                                                              |
| Source objects                  | Trace + accumulated entry fold.                                                                                                                                                                                                                                                                                  |
| Definitions                     | InteriorInverseCandidateProfile.                                                                                                                                                                                                                                                                                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity S03 – Two-sided inverse certificate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | S03                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Documented                      | no                                                                                                                                                                                                      |
| Phase                           | 3.4                                                                                                                                                                                                     |
| Source objects                  | Interior block + candidate.                                                                                                                                                                             |

|                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definitions               | Two-sided inverse certificate.                                                                                                                                                                                                                                                                                   |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity S04 – InteriorEliminationData

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">S04</a>                                                                                                                                                                                                                                                                                              |
| Object status                   | <span style="background-color: #ADD8E6;"> </span> B                                                                                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | <a href="#">3.5</a>                                                                                                                                                                                                                                                                                              |
| Source objects                  | Interior block, inverse, couplings.                                                                                                                                                                                                                                                                              |
| Definitions                     | InteriorEliminationProfile.                                                                                                                                                                                                                                                                                      |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity S05 – DtN boundary operator

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">S05</a>                                                                                                                                                                                     |
| Object status                   | <span style="background-color: #ADD8E6;"> </span> B                                                                                                                                                     |
| Context class                   | LeanSourceObject                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Documented                      | no                                                                                                                                                                                                      |
| Phase                           | <a href="#">3.6</a>                                                                                                                                                                                     |
| Source objects                  | Laplacian blocks + inverse certificate.                                                                                                                                                                 |
| Definitions                     | BoundaryOperatorProfile.                                                                                                                                                                                |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                               |



|                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity S06 – GeneralizedDtN / MultiSchur

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">S06</a>                                                                                                                                                                                                                                                                                              |
| Object status                   | <span style="background-color: #ADD8E6;"> </span> B                                                                                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | <a href="#">3.7</a>                                                                                                                                                                                                                                                                                              |
| Source objects                  | Boundary operator + additional ports.                                                                                                                                                                                                                                                                            |
| Definitions                     | GeneralizedBoundaryOperatorProfile.                                                                                                                                                                                                                                                                              |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity O01 – Boundary matrix over ExecComplex

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">O01</a>                                                                                                                                                                                                                                                                                     |
| Object status                   | <span style="background-color: #ADD8E6;"> </span> B                                                                                                                                                                                                                                                     |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | <a href="#">4.1</a>                                                                                                                                                                                                                                                                                     |
| Source objects                  | Boundary operator + Exec bridge.                                                                                                                                                                                                                                                                        |
| Definitions                     | ExecMatrixProfile.                                                                                                                                                                                                                                                                                      |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity O02 – Determinant coefficient recurrence

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | O02                                                                                                                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 4.2                                                                                                                                                                                                                                                                                                     |
| Source objects                  | Exec matrix + recurrence.                                                                                                                                                                                                                                                                               |
| Definitions                     | CoefficientRecurrenceProfile.                                                                                                                                                                                                                                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity O03 – Operational polynomial profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | O03                                                                                                                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 4.3                                                                                                                                                                                                                                                                                                     |
| Source objects                  | Coefficient recurrence.                                                                                                                                                                                                                                                                                 |
| Definitions                     | OperationalPolynomialProfile.                                                                                                                                                                                                                                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity O04 – Quadratic coefficient window

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | O04                                                                                                                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 4.4                                                                                                                                                                                                                                                                                                     |
| Source objects                  | CoefficientAt source.                                                                                                                                                                                                                                                                                   |
| Definitions                     | QuadraticWindowCertificate.                                                                                                                                                                                                                                                                             |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity O05 – Operational discriminant D\_op

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | O05                                                                                                                                                                                                                                                                                                     |
| Object status                   | B                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 4.5                                                                                                                                                                                                                                                                                                     |
| Source objects                  | QuadraticWindow.                                                                                                                                                                                                                                                                                        |
| Definitions                     | OperationalDiscriminantCertificate.                                                                                                                                                                                                                                                                     |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity O06 – Comparison target policy

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field         | Record |
|---------------|--------|
| Object ID     | O06    |
| Object status | G      |

|                                 |                                                                                                                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                   | ReferenceGateObject                                                                                                                                                                                                   |
| Significance tier               | F2                                                                                                                                                                                                                    |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                         |
| Documented                      | no                                                                                                                                                                                                                    |
| Phase                           | 4.6                                                                                                                                                                                                                   |
| Source objects                  | D_op + target definition.                                                                                                                                                                                             |
| Definitions                     | ComparisonTargetPolicy.                                                                                                                                                                                               |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                             |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. EXEC-2 audit pending/applied: object must not leak algebraic irrational data into the operational Q x Q path. |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                                              |

### Object or quantity O07 – General finite computation test matrix profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | O07                                                                                                                                                                                      |
| Object status                   | Y                                                                                                                                                                                        |
| Context class                   | LeanTargetObject                                                                                                                                                                         |
| Significance tier               | F2                                                                                                                                                                                       |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.            |
| Documented                      | no - planned proof/test dossier                                                                                                                                                          |
| Phase                           | 4.0.1                                                                                                                                                                                    |
| Source objects                  | LEG29, canonical generator, finite branch/L_max inputs, BR07 arithmetic-shadow audit, Phase-3 boundary profiles where consumed.                                                          |
| Definitions                     | A row consists of input provenance, generated finite instance, branch-mode tag, L_max, operational output witness, verification status, and optional mirror/sentinel comparison columns. |
| Proof target                    | Construct a reusable finite test matrix showing that the operational computation path is traversed by generic code-defined instances before special sentinel interpretation.             |
| Proof sketch                    | Use existing generator outputs to populate finite rows; prove or record executable witness fields directly; add Eisenstein/Gauss as sentinel rows only after generic rows exist.         |
| Verification / audit            | Lean build and row outputs must be checked during implementation; no Matrix.inv/noncomputable operational field; no duplicate generator.                                                 |
| Next documentation action       | Before implementing Phase 4.0.1, fill module manifest, exact row schema, generic coverage set, and mathlib complication log.                                                             |

### Object or quantity N00 – Fundamental discriminant D\_fund

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N00                                                                                                                                                                                            |
| Object status                   | G                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |

|                           |                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documented                | no                                                                                                                                                                                                                                                                                                      |
| Phase                     | 5.1                                                                                                                                                                                                                                                                                                     |
| Source objects            | D_op, square factor/congreence proof.                                                                                                                                                                                                                                                                   |
| Definitions               | FundamentalDiscriminantProfile.                                                                                                                                                                                                                                                                         |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity N01 – Conductor f

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N01                                                                                                                                                                                                                                                                                                     |
| Object status                   | G                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 5.1                                                                                                                                                                                                                                                                                                     |
| Source objects                  | D_op, D_fund.                                                                                                                                                                                                                                                                                           |
| Definitions                     | ConductorProfile.                                                                                                                                                                                                                                                                                       |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity N02 – Quadratic field K=Q(sqrt(D))

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N02                                                                                                                                                                                            |
| Object status                   | Y                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | 5.2                                                                                                                                                                                            |
| Source objects                  | D_fund + sign/non-square proof.                                                                                                                                                                |

|                           |                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definitions               | QuadraticFieldProfile.                                                                                                                                                                                                                                                                                  |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity N03 – Quadratic order O\_D

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">N03</a>                                                                                                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | <a href="#">5.3</a>                                                                                                                                                                                                                                                                                     |
| Source objects                  | K, f.                                                                                                                                                                                                                                                                                                   |
| Definitions                     | QuadraticOrderProfile.                                                                                                                                                                                                                                                                                  |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity N04 – Integral basis / omega\_D

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">N04</a>                                                                                                                                                                            |
| Object status                   | Y                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | <a href="#">5.4</a>                                                                                                                                                                            |
| Source objects                  | D congruence, conductor.                                                                                                                                                                       |
| Definitions                     | IntegralBasisProfile.                                                                                                                                                                          |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                         |



|                           |                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity N05 – Trace form

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N05                                                                                                                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanTargetObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 5.5                                                                                                                                                                                                                                                                                                     |
| Source objects                  | Conjugation + basis.                                                                                                                                                                                                                                                                                    |
| Definitions                     | TraceFormProfile.                                                                                                                                                                                                                                                                                       |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity N06 – Norm form

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N06                                                                                                                                                                                            |
| Object status                   | Y                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | 5.5                                                                                                                                                                                            |
| Source objects                  | Integral basis.                                                                                                                                                                                |
| Definitions                     | AlgebraicNormProfile.                                                                                                                                                                          |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                         |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                      |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                        |

|                           |                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Object or quantity N07 – Conjugation

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N07                                                                                                                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanTargetObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 5.5                                                                                                                                                                                                                                                                                                     |
| Source objects                  | sqrt(D) -> -sqrt(D).                                                                                                                                                                                                                                                                                    |
| Definitions                     | ConjugationProfile.                                                                                                                                                                                                                                                                                     |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity N08 – Unit group O\_D^\*

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N08                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 6.1                                                                                                                                                                           |
| Source objects                  | O_D + norm form.                                                                                                                                                              |
| Definitions                     | UnitGroupProfile.                                                                                                                                                             |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity N09 – Class group Cl(O\_D)

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N09                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 6.2                                                                                                                                                                           |
| Source objects                  | Fractional ideas modulo principal.                                                                                                                                            |
| Definitions                     | ClassGroupProfile.                                                                                                                                                            |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity N10 – Class number h(D)

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N10                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 6.2                                                                                                                                                                           |
| Source objects                  | Cl(O_D).                                                                                                                                                                      |
| Definitions                     | ClassNumberProfile.                                                                                                                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity N11 – Fractional/integral ideas

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field     | Record |
|-----------|--------|
| Object ID | N11    |

|                                 |                                                                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 6.2.1                                                                                                                                                                         |
| Source objects                  | O_D modules/submodules.                                                                                                                                                       |
| Definitions                     | IdealProfile.                                                                                                                                                                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity N12 – Principal ideal predicate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N12                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 6.2.2                                                                                                                                                                         |
| Source objects                  | Ideas and norm.                                                                                                                                                               |
| Definitions                     | PrincipalIdealProfile.                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity N13 – Heegner/PID/UFD indicator

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N13                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |

|                           |                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Phase                     | 6.3                                                                                                                       |
| Source objects            | h(D), D_fund.                                                                                                             |
| Definitions               | HeegnerIndicatorProfile.                                                                                                  |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                    |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions. |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                   |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                  |

### Object or quantity N14 – Prime splitting behavior

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N14                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 6.4                                                                                                                                                                           |
| Source objects                  | D_fund, primes.                                                                                                                                                               |
| Definitions                     | PrimeSplittingProfile.                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity N15 – Kronecker/Dirichlet character chi\_D

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N15                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 6.4                                                                                                                                                                           |
| Source objects                  | Kronecker symbol.                                                                                                                                                             |
| Definitions                     | CharacterProfile.                                                                                                                                                             |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |

|                           |                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions. |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                   |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                  |

### Object or quantity CD8 – Arithmetic-CD compatibility as Schur-duality corollary

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                         |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | CD8                                                                                                                                                                                                                                                                                                                            |
| Object status                   | K                                                                                                                                                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                 |
| Documented                      | no - dependency-order corrected in v31                                                                                                                                                                                                                                                                                         |
| Phase                           | 10.4.2                                                                                                                                                                                                                                                                                                                         |
| Source objects                  | BR08/BR09; CD normSq and norm-multiplicativity data CD3/CD6/CD7; arithmetic discriminant/norm route O05 and later N-objects; transfermatrix route TM0-TM2.                                                                                                                                                                     |
| Definitions                     | CDArithmeticCompatibilityProfile as a corollary of horizontal/vertical Schur-duality, not as a free norm analogy.                                                                                                                                                                                                              |
| Proof target                    | Prove that the CD-side norm structure and arithmetic-side norm/discriminant structure are compatible because both are derived from the same multiscale Schur decomposition, under the exact hypotheses proved in BR08.                                                                                                         |
| Proof sketch                    | Defer proof until BR08 is closed. Do not use source-file similarity, physical analogy, or Hurwitz/CM comparison prose as proof. First export precise norm objects from both routes, then build a coordinate/entry-level bridge.                                                                                                |
| Verification / audit            | Current code modules are source/seed only. The v31 audit corrects the derivation order: Phase 6.5 may collect preconditions but not close CD8.                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When Phase 10.4.2 activates, split if necessary into norm-source alignment, coordinate bridge, and final compatibility theorem. |

### Object or quantity L00 – Lattice embedding Lambda -> C

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | L00                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 7.1                                                                                                                                                                           |
| Source objects                  | Integral basis + sqrt(D) branch.                                                                                                                                              |
| Definitions                     | ComplexEmbeddingProfile.                                                                                                                                                      |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |



|                           |                                                                                                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. EXEC-2 audit pending/applied: object must not leak algebraic irrational data into the operational $\mathbb{Q} \times \mathbb{Q}$ path. |
| Next documentation action | When this phase is prepared, write the mirror-layer definition and the exact rational/integer invariant, if any, allowed to return operationally.                                                                                              |

### Object or quantity L01 – Lattice basis

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | L01                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 7.1                                                                                                                                                                           |
| Source objects                  | OrderProfile.                                                                                                                                                                 |
| Definitions                     | LatticeBasisProfile.                                                                                                                                                          |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity L02 – Gram matrix

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | L02                                                                                                                                                                                                                                            |
| Object status                   | Y                                                                                                                                                                                                                                              |
| Context class                   | LeanTargetObject                                                                                                                                                                                                                               |
| Significance tier               | F2                                                                                                                                                                                                                                             |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                  |
| Documented                      | no                                                                                                                                                                                                                                             |
| Phase                           | 7.2                                                                                                                                                                                                                                            |
| Source objects                  | Embedding.                                                                                                                                                                                                                                     |
| Definitions                     | GramProfile.                                                                                                                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                         |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                      |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. EXEC-2 audit pending/applied: object must not leak algebraic irrational data into the operational $\mathbb{Q} \times \mathbb{Q}$ path. |
| Next documentation action       | When this phase is prepared, write the mirror-layer definition and the exact rational/integer invariant, if any, allowed to return operationally.                                                                                              |

Object or quantity L03 – Lattice determinant / covolume

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | L03                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 7.2                                                                                                                                                                           |
| Source objects                  | Gram.                                                                                                                                                                         |
| Definitions                     | CovolumeProfile.                                                                                                                                                              |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity L04 – Period ratio tau

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | L04                                                                                                                                                                                                                                            |
| Object status                   | Y                                                                                                                                                                                                                                              |
| Context class                   | LeanTargetObject                                                                                                                                                                                                                               |
| Significance tier               | F2                                                                                                                                                                                                                                             |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                  |
| Documented                      | no                                                                                                                                                                                                                                             |
| Phase                           | 7.3                                                                                                                                                                                                                                            |
| Source objects                  | Embedding, basis.                                                                                                                                                                                                                              |
| Definitions                     | PeriodRatioProfile.                                                                                                                                                                                                                            |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                         |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                      |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. EXEC-2 audit pending/applied: object must not leak algebraic irrational data into the operational $\mathbb{Q} \times \mathbb{Q}$ path. |
| Next documentation action       | When this phase is prepared, write the mirror-layer definition and the exact rational/integer invariant, if any, allowed to return operationally.                                                                                              |

Object or quantity L05 – Upper half-plane condition

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field     | Record |
|-----------|--------|
| Object ID | L05    |

|                                 |                                                                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 7.3                                                                                                                                                                           |
| Source objects                  | tau.                                                                                                                                                                          |
| Definitions                     | UpperHalfPlaneCertificate.                                                                                                                                                    |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity L06 – Angle / cosine profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | L06                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 7.4                                                                                                                                                                           |
| Source objects                  | Gram entries; $\cos = \langle u,v \rangle / \sqrt{\dots}$ .                                                                                                                   |
| Definitions                     | AngleProfile.                                                                                                                                                                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity L07 – Lattice automorphism group

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | L07                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |

|                           |                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Documented                | no                                                                                                                        |
| Phase                     | 7.5                                                                                                                       |
| Source objects            | Units + lattice basis.                                                                                                    |
| Definitions               | LatticeAutomorphismProfile.                                                                                               |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                    |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions. |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                   |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                  |

Object or quantity M02 – Eisenstein series G4,G6

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | M02                                                                                                                                                                                      |
| Object status                   | Y                                                                                                                                                                                        |
| Context class                   | LeanTargetObject                                                                                                                                                                         |
| Significance tier               | F2                                                                                                                                                                                       |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.            |
| Documented                      | no                                                                                                                                                                                       |
| Phase                           | 8.1                                                                                                                                                                                      |
| Source objects                  | Lattice sums or q-expansion.                                                                                                                                                             |
| Definitions                     | EisensteinSeriesProfile.                                                                                                                                                                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                   |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: modular/CM path only; cannot gate finite spectral/physical consumers. |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                 |

Object or quantity M03 – Weierstrass invariants g2,g3

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | M03                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 8.2                                                                                                                                                                           |
| Source objects                  | Eisenstein series.                                                                                                                                                            |
| Definitions                     | WeierstrassInvariantProfile.                                                                                                                                                  |

|                           |                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                   |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: modular/CM path only; cannot gate finite spectral/physical consumers. |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                 |

### Object or quantity M04 – Modular discriminant Delta\_mod

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | M04                                                                                                                                                                                      |
| Object status                   | Y                                                                                                                                                                                        |
| Context class                   | LeanTargetObject                                                                                                                                                                         |
| Significance tier               | F2                                                                                                                                                                                       |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.            |
| Documented                      | no                                                                                                                                                                                       |
| Phase                           | 8.3                                                                                                                                                                                      |
| Source objects                  | Weierstrass invariants.                                                                                                                                                                  |
| Definitions                     | ModularDiscriminantProfile.                                                                                                                                                              |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                   |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: modular/CM path only; cannot gate finite spectral/physical consumers. |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                 |

### Object or quantity M05 – j-invariant

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | M05                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 8.4                                                                                                                                                                           |
| Source objects                  | g2, Delta_mod, tau.                                                                                                                                                           |
| Definitions                     | JInvariantProfile.                                                                                                                                                            |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |

|                           |                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. EXEC-2 audit pending/applied: object must not leak algebraic irrational data into the operational $\mathbb{Q} \times \mathbb{Q}$ path. v37 audit: modular/CM path only; cannot gate finite spectral/physical consumers. |
| Next documentation action | When this phase is prepared, write the mirror-layer definition and the exact rational/integer invariant, if any, allowed to return operationally.                                                                                                                                                                               |

### Object or quantity SP0 – Eigenvalue profile

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">SP0</a>                                                                                                                                                                                                                                                                                                                              |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                                                 |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                   |
| Documented                      | no                                                                                                                                                                                                                                                                                                                                               |
| Phase                           | <a href="#">10.1</a>                                                                                                                                                                                                                                                                                                                             |
| Source objects                  | Matrix, polynomial.                                                                                                                                                                                                                                                                                                                              |
| Definitions                     | FiniteSpectrumProfile.                                                                                                                                                                                                                                                                                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. EXEC-2 audit pending/applied: object must not leak algebraic irrational data into the operational $\mathbb{Q} \times \mathbb{Q}$ path. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged.            |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When this phase is prepared, write the mirror-layer definition and the exact rational/integer invariant, if any, allowed to return operationally. |

### Object or quantity SP1 – Spectral gap

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">SP1</a>                                                                                                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | <a href="#">10.1</a>                                                                                                                                                                                                                                                                                    |
| Source objects                  | Eigenvalues/order.                                                                                                                                                                                                                                                                                      |
| Definitions                     | SpectralGapProfile.                                                                                                                                                                                                                                                                                     |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged.                                                                                                          |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |



Object or quantity P00 – Operator algebra

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P00                                                                                                                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 10.2                                                                                                                                                                                                                                                                                                    |
| Source objects                  | Boundary matrix.                                                                                                                                                                                                                                                                                        |
| Definitions                     | OperatorAlgebraProfile.                                                                                                                                                                                                                                                                                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged.                                                                                                          |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity P01 – Star operation

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P01                                                                                                                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 10.2                                                                                                                                                                                                                                                                                                    |
| Source objects                  | Matrix algebra.                                                                                                                                                                                                                                                                                         |
| Definitions                     | StarOperationProfile.                                                                                                                                                                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged.                                                                                                          |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity P02 – State functional

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P02                                                                                                                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 10.2                                                                                                                                                                                                                                                                                                    |
| Source objects                  | Matrix, weights.                                                                                                                                                                                                                                                                                        |
| Definitions                     | StateFunctionalProfile.                                                                                                                                                                                                                                                                                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged.                                                                                                          |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity P03 – Gibbs state

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P03                                                                                                                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 10.2                                                                                                                                                                                                                                                                                                    |
| Source objects                  | H, beta.                                                                                                                                                                                                                                                                                                |
| Definitions                     | GibbsStateProfile.                                                                                                                                                                                                                                                                                      |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged.                                                                                                          |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity P04 – Time evolution

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field         | Record |
|---------------|--------|
| Object ID     | P04    |
| Object status | Y      |

|                                 |                                                                                                                                                                                                                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 10.2                                                                                                                                                                                                                                                                                                    |
| Source objects                  | H, t.                                                                                                                                                                                                                                                                                                   |
| Definitions                     | TimeEvolutionProfile.                                                                                                                                                                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged.                                                                                                          |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity P05 – Unitarity certificate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P05                                                                                                                                                                                                                                                                                                     |
| Object status                   | Y                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 10.2                                                                                                                                                                                                                                                                                                    |
| Source objects                  | H, star, state.                                                                                                                                                                                                                                                                                         |
| Definitions                     | UnitarityCertificate.                                                                                                                                                                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged.                                                                                                          |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity P06 – Open channel / effective openness

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P06                                                                                                                                                                                            |
| Object status                   | R                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |

|                           |                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documented                | no                                                                                                                                                                                                                                                                                                      |
| Phase                     | <a href="#">10.2</a>                                                                                                                                                                                                                                                                                    |
| Source objects            | Boundary operator + environment profile.                                                                                                                                                                                                                                                                |
| Definitions               | OpenChannelProfile.                                                                                                                                                                                                                                                                                     |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged.                                                                                                          |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity N16 – Dirichlet L-function L(s,chi\_D)

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                    |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">N16</a>                                                                                                                                                                                       |
| Object status                   | <span style="background-color: #808080;">K</span>                                                                                                                                                         |
| Context class                   | ReferenceGateObject                                                                                                                                                                                       |
| Significance tier               | F2                                                                                                                                                                                                        |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                             |
| Documented                      | no                                                                                                                                                                                                        |
| Phase                           | <a href="#">9.2</a>                                                                                                                                                                                       |
| Source objects                  | chi_D.                                                                                                                                                                                                    |
| Definitions                     | LFunctionProfile.                                                                                                                                                                                         |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                    |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                 |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: analytic far target only; cannot source early h(D) or finite spectral/physical phases. |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                                  |

### Object or quantity N17 – Class number formula

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">N17</a>                                                                                                                                                           |
| Object status                   | <span style="background-color: #808080;">K</span>                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | <a href="#">9.3</a>                                                                                                                                                           |
| Source objects                  | chi_D, units, D_fund.                                                                                                                                                         |

|                           |                                                                                                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definitions               | ClassNumberFormulaProfile.                                                                                                                                                                                |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                    |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                 |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: analytic far target only; cannot source early h(D) or finite spectral/physical phases. |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                                  |

### Object or quantity L08 – Theta series Theta\_Lambda

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">L08</a>                                                                                                                                                           |
| Object status                   | <span style="background-color: #a9a9a9;">K</span>                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | <a href="#">9.1</a>                                                                                                                                                           |
| Source objects                  | Norm form over lattice.                                                                                                                                                       |
| Definitions                     | ThetaSeriesProfile.                                                                                                                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity M00 – Modular group action SL2Z

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">M00</a>                                                                                                                                                           |
| Object status                   | <span style="background-color: #a9a9a9;">K</span>                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | <a href="#">8.0</a>                                                                                                                                                           |
| Source objects                  | tau.                                                                                                                                                                          |
| Definitions                     | ModularActionProfile.                                                                                                                                                         |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |

|                           |                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: modular/CM path only; cannot gate finite spectral/physical consumers. |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                 |

### Object or quantity M01 – Fundamental domain representative

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | M01                                                                                                                                                                                      |
| Object status                   | K                                                                                                                                                                                        |
| Context class                   | LeanTargetObject                                                                                                                                                                         |
| Significance tier               | F2                                                                                                                                                                                       |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.            |
| Documented                      | no                                                                                                                                                                                       |
| Phase                           | 8.0                                                                                                                                                                                      |
| Source objects                  | tau + reduction algorithm/proof.                                                                                                                                                         |
| Definitions                     | FundamentalDomainProfile.                                                                                                                                                                |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                   |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: modular/CM path only; cannot gate finite spectral/physical consumers. |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                 |

### Object or quantity M06 – q-expansion of j

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | M06                                                                                                                                                                                      |
| Object status                   | K                                                                                                                                                                                        |
| Context class                   | LeanTargetObject                                                                                                                                                                         |
| Significance tier               | F2                                                                                                                                                                                       |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.            |
| Documented                      | no                                                                                                                                                                                       |
| Phase                           | 8.8                                                                                                                                                                                      |
| Source objects                  | q=exp(2pi i tau).                                                                                                                                                                        |
| Definitions                     | JQExpansionProfile.                                                                                                                                                                      |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                   |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: modular/CM path only; cannot gate finite spectral/physical consumers. |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                 |



Object or quantity M07 – Hilbert class polynomial H\_D

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | M07                                                                                                                                                                                      |
| Object status                   | K                                                                                                                                                                                        |
| Context class                   | LeanTargetObject                                                                                                                                                                         |
| Significance tier               | F2                                                                                                                                                                                       |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.            |
| Documented                      | no                                                                                                                                                                                       |
| Phase                           | 8.8                                                                                                                                                                                      |
| Source objects                  | Cl(O_D), j.                                                                                                                                                                              |
| Definitions                     | HilbertClassPolynomialProfile.                                                                                                                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                   |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: modular/CM path only; cannot gate finite spectral/physical consumers. |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                 |

Object or quantity TM0 – Transfermatrix T\_k

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | TM0                                                                                                                                                                                                                                                                                                     |
| Object status                   | K                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 10.4                                                                                                                                                                                                                                                                                                    |
| Source objects                  | Local recursion step.                                                                                                                                                                                                                                                                                   |
| Definitions                     | TransferMatrixProfile.                                                                                                                                                                                                                                                                                  |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity TM1 – Transfermatrix product

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | TM1                                                                                                                                                                                                                                                                                                     |
| Object status                   | K                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 10.4                                                                                                                                                                                                                                                                                                    |
| Source objects                  | Trace/fold/product.                                                                                                                                                                                                                                                                                     |
| Definitions                     | TransferProductProfile.                                                                                                                                                                                                                                                                                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object's natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity TM2 – Moebius/continued fraction relation

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | TM2                                                                                                                                                                                                                                                                                                     |
| Object status                   | K                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 10.4                                                                                                                                                                                                                                                                                                    |
| Source objects                  | T_k action.                                                                                                                                                                                                                                                                                             |
| Definitions                     | MobiusSchurProfile.                                                                                                                                                                                                                                                                                     |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object's natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity SP2 – Heat kernel K(t)

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field         | Record |
|---------------|--------|
| Object ID     | SP2    |
| Object status | K      |

|                                 |                                                                                                                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                   | LeanTargetObject                                                                                                                                                                               |
| Significance tier               | F2                                                                                                                                                                                             |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | 10.3                                                                                                                                                                                           |
| Source objects                  | Eigenvalues or matrix exponential seed.                                                                                                                                                        |
| Definitions                     | HeatKernelProfile.                                                                                                                                                                             |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                         |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                      |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged. |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                       |

### Object or quantity SP3 – Spectral zeta zeta\_H

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | SP3                                                                                                                                                                                            |
| Object status                   | K                                                                                                                                                                                              |
| Context class                   | LeanTargetObject                                                                                                                                                                               |
| Significance tier               | F2                                                                                                                                                                                             |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | 10.3                                                                                                                                                                                           |
| Source objects                  | Eigenvalues.                                                                                                                                                                                   |
| Definitions                     | SpectralZetaProfile.                                                                                                                                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                         |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                      |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged. |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                       |

### Object or quantity SP4 – Spectral dimension

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | SP4                                                                                                                                                                           |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 10.3                                                                                                                                                                          |

|                           |                                                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source objects            | K(t).                                                                                                                                                                                          |
| Definitions               | SpectralDimensionProfile.                                                                                                                                                                      |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                         |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                      |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged. |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                       |

### Object or quantity SP5 – Zeta-regularized determinant

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | SP5                                                                                                                                                                                            |
| Object status                   | K                                                                                                                                                                                              |
| Context class                   | LeanTargetObject                                                                                                                                                                               |
| Significance tier               | F2                                                                                                                                                                                             |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                  |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | 10.3                                                                                                                                                                                           |
| Source objects                  | zeta_H.                                                                                                                                                                                        |
| Definitions                     | ZetaDeterminantProfile.                                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                         |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                      |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists. v37 audit: must verify no Phase 8/9 modular or j dependency unless explicitly bridged. |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                                       |

### Object or quantity P07 – KMS condition

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P07                                                                                                                                                                           |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 13.4                                                                                                                                                                          |
| Source objects                  | State, dynamics.                                                                                                                                                              |
| Definitions                     | KMSProfile.                                                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |

|                           |                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------|
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.     |
| Next documentation action | Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity P08 – GNS Hilbert space

**Table legend.**   W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">P08</a>                                                                                                                                                           |
| Object status                   | <span style="background-color: #a9a9a9;">K</span>                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | <a href="#">13.4</a>                                                                                                                                                          |
| Source objects                  | Algebra quotient/completion.                                                                                                                                                  |
| Definitions                     | GNSProfile.                                                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                  |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose.                                                                   |

### Object or quantity P09 – Tomita operator S

**Table legend.**   W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">P09</a>                                                                                                                                                           |
| Object status                   | <span style="background-color: #a9a9a9;">K</span>                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | <a href="#">13.5</a>                                                                                                                                                          |
| Source objects                  | GNS state algebra.                                                                                                                                                            |
| Definitions                     | TomitaOperatorProfile.                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                  |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose.                                                                   |

Object or quantity P10 – Modular operator Delta

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P10                                                                                                                                                                           |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 13.5                                                                                                                                                                          |
| Source objects                  | S.                                                                                                                                                                            |
| Definitions                     | ModularOperatorProfile.                                                                                                                                                       |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity P11 – Modular conjugation J

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P11                                                                                                                                                                           |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 13.5                                                                                                                                                                          |
| Source objects                  | S, Delta.                                                                                                                                                                     |
| Definitions                     | ModularConjugationProfile.                                                                                                                                                    |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity P12 – Bisognano-Wichmann/CPT bridge

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field     | Record |
|-----------|--------|
| Object ID | P12    |



|                                 |                                                                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object status                   | K                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 13.6                                                                                                                                                                          |
| Source objects                  | J, local net, geometry.                                                                                                                                                       |
| Definitions                     | CPTBridgeProfile.                                                                                                                                                             |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity P13 – Spin-statistics

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P13                                                                                                                                                                           |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 13.6                                                                                                                                                                          |
| Source objects                  | J, covariance, redations.                                                                                                                                                     |
| Definitions                     | SpinStatisticsProfile.                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity P14 – Kramers degeneracy

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P14                                                                                                                                                                           |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |

|                           |                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Phase                     | 13.6                                                                                                                      |
| Source objects            | Antiunitary T, spectrum.                                                                                                  |
| Definitions               | KramersProfile.                                                                                                           |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                    |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions. |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                   |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                  |

### Object or quantity P15 – Split property

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P15                                                                                                                                                                                                                                                                                                     |
| Object status                   | K                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 13.6                                                                                                                                                                                                                                                                                                    |
| Source objects                  | Local net, inclusion.                                                                                                                                                                                                                                                                                   |
| Definitions                     | SplitPropertyProfile.                                                                                                                                                                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity P16 – Local net / Haag-Kastler algebraic axioms

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | P16                                                                                                                                                                           |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 13.3                                                                                                                                                                          |
| Source objects                  | Regions, inclusions, operators.                                                                                                                                               |
| Definitions                     | LocalNetProfile.                                                                                                                                                              |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |

|                           |                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions. |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                   |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                  |

### Object or quantity H10 – A->B bright handoff

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">H10</a>                                                                                                                                                                                                                                                                                              |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | <a href="#">11.1</a>                                                                                                                                                                                                                                                                                             |
| Source objects                  | Omega, L, regime, operator; no hidden complement.                                                                                                                                                                                                                                                                |
| Definitions                     | AToBBBrightOutputProfile.                                                                                                                                                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity H11 – A->B prime/dark future handoff

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">H11</a>                                                                                                                                                                                     |
| Object status                   | R                                                                                                                                                                                                       |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                               |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Documented                      | no                                                                                                                                                                                                      |
| Phase                           | <a href="#">11.2</a>                                                                                                                                                                                    |
| Source objects                  | Dark/complement data.                                                                                                                                                                                   |
| Definitions                     | AToBDarkOutputProfile.                                                                                                                                                                                  |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                 |

|                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Object or quantity H12 – A->C handoff

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">H12</a>                                                                                                                                                                                                                                                                                              |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | <a href="#">11.3</a>                                                                                                                                                                                                                                                                                             |
| Source objects                  | Carrier, lattice, Gram, tau, regime as allowed.                                                                                                                                                                                                                                                                  |
| Definitions                     | AToCOutputProfile.                                                                                                                                                                                                                                                                                               |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity H13 – A->D handoff

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">H13</a>                                                                                                                                                                                                                                                                                              |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | <a href="#">11.4</a>                                                                                                                                                                                                                                                                                             |
| Source objects                  | Boundary operator, state seeds, regime.                                                                                                                                                                                                                                                                          |
| Definitions                     | AToDOutputProfile.                                                                                                                                                                                                                                                                                               |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity H14 – A->E handoff

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | H14                                                                                                                                                                                                                                                                                                              |
| Object status                   | K                                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | 11.5                                                                                                                                                                                                                                                                                                             |
| Source objects                  | D, CM, class data, j, etc.                                                                                                                                                                                                                                                                                       |
| Definitions                     | AToEOutputProfile.                                                                                                                                                                                                                                                                                               |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity MOON0 – j coefficient stream

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | MOON0                                                                                                                                                                         |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 12.1                                                                                                                                                                          |
| Source objects                  | M06 q-expansion object.                                                                                                                                                       |
| Definitions                     | CNNA-derived q-expansion coefficient stream of j after j is derived.                                                                                                          |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity MOON1 – Monster dimension comparison profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field | Record |
|-------|--------|
|-------|--------|

|                                 |                                                                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | MOON1                                                                                                                                                                         |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 12.2                                                                                                                                                                          |
| Source objects                  | MOON0 plus external established Monster representation dimension data as comparison target only;<br>no reverse dependency into j derivation.                                  |
| Definitions                     | Comparison profile between derived j coefficients and established Monster representation dimensions.                                                                          |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                  |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose.                                                                   |

Object or quantity MOON2 – McKay-Thompson comparison profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

|                                 |                                                                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Field                           | Record                                                                                                                                                                        |
| Object ID                       | MOON2                                                                                                                                                                         |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 12.3                                                                                                                                                                          |
| Source objects                  | MOON0 and optional external named McKay-Thompson/Hauptmodul series as targets.                                                                                                |
| Definitions                     | Comparison-only profile for McKay-Thompson series.                                                                                                                            |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                  |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose.                                                                   |

Object or quantity B00 – Pillar B input contract from A

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

|                   |                           |
|-------------------|---------------------------|
| Field             | Record                    |
| Object ID         | B00                       |
| Object status     | K                         |
| Context class     | LeanHandoffContractObject |
| Significance tier | F0                        |



|                                 |                                                                                                                                                                                                                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                             |
| Documented                      | no                                                                                                                                                                                                                                                                                                                  |
| Phase                           | 13.1                                                                                                                                                                                                                                                                                                                |
| Source objects                  | H10;<br>exported boundary/operator/regime profiles.                                                                                                                                                                                                                                                                 |
| Definitions                     | B consumes exactly the A->B bright handoff and nothing else.                                                                                                                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                              |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                             |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity B01 – Boundary star-algebra / observable algebra

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                     |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | B01                                                                                                                                                                                                                                                                                                        |
| Object status                   | K                                                                                                                                                                                                                                                                                                          |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                           |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                         |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                             |
| Documented                      | no                                                                                                                                                                                                                                                                                                         |
| Phase                           | 13.2                                                                                                                                                                                                                                                                                                       |
| Source objects                  | B00;<br>ExecBoundaryMatrix;<br>BoundaryOperator.                                                                                                                                                                                                                                                           |
| Definitions                     | Finite *-algebra of boundary observables derived from exported A boundary operator data.                                                                                                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                     |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                    |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity B02 – Local net / QFT algebraic profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | B02                                                                                                                                                                                            |
| Object status                   | K                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |

|                           |                                                                                                                                                                                                                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documented                | no                                                                                                                                                                                                                                                                                                         |
| Phase                     | <a href="#">13.3</a>                                                                                                                                                                                                                                                                                       |
| Source objects            | B01;<br>A->B region/interface data.                                                                                                                                                                                                                                                                        |
| Definitions               | Net of algebras derived from boundary/region data;<br>algebraic QFT consumer.                                                                                                                                                                                                                              |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                     |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                    |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity B03 – B state/GNS/KMS profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                     |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">B03</a>                                                                                                                                                                                                                                                                                        |
| Object status                   | <span>K</span>                                                                                                                                                                                                                                                                                             |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                           |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                         |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                             |
| Documented                      | no                                                                                                                                                                                                                                                                                                         |
| Phase                           | <a href="#">13.4</a>                                                                                                                                                                                                                                                                                       |
| Source objects                  | B01;<br>P01/P02/P03;<br>Gibbs/time-evolution seeds.                                                                                                                                                                                                                                                        |
| Definitions                     | State functional, GNS construction, and KMS gate over B observable algebra.                                                                                                                                                                                                                                |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                     |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                    |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity B04 – Tomita-Takesaki modular data

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">B04</a>                                                                                                                                                                            |
| Object status                   | <span>K</span>                                                                                                                                                                                 |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |

|                           |                                                                                                                                                                                                                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documented                | no                                                                                                                                                                                                                                                                                                         |
| Phase                     | 13.5                                                                                                                                                                                                                                                                                                       |
| Source objects            | B03;<br>local net.                                                                                                                                                                                                                                                                                         |
| Definitions               | Tomita operator S, modular operator Delta, and modular conjugation J derived from B GNS/KMS data.                                                                                                                                                                                                          |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                     |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                    |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity B05 – CPT/BW/spin-statistics/split gates

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                     |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | B05                                                                                                                                                                                                                                                                                                        |
| Object status                   | K                                                                                                                                                                                                                                                                                                          |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                           |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                         |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                             |
| Documented                      | no                                                                                                                                                                                                                                                                                                         |
| Phase                           | 13.6                                                                                                                                                                                                                                                                                                       |
| Source objects                  | B02;<br>B04.                                                                                                                                                                                                                                                                                               |
| Definitions                     | Structural theorem gates derived from B local net and modular data.                                                                                                                                                                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                     |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                    |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity B06 – B->A allowed output profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | B06                                                                                                                                                                                            |
| Object status                   | K                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | 13.7                                                                                                                                                                                           |

|                           |                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source objects            | B01-B05 if completed.                                                                                                                                                                                                                                                                                   |
| Definitions               | Typed B output that A may consume only through B->A input handoff.                                                                                                                                                                                                                                      |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object's natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity C00 – Pillar C OQS input contract from A

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <span style="color: blue;">C00</span>                                                                                                                                                                                                                                                                            |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                                                                                                                                                                                                                |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                               |
| Phase                           | <span style="color: blue;">14.1</span>                                                                                                                                                                                                                                                                           |
| Source objects                  | H12;<br>RegimeProfile;<br>ComplementProfile.                                                                                                                                                                                                                                                                     |
| Definitions                     | C consumes explicit A->C open/regime/complement outputs.                                                                                                                                                                                                                                                         |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object's natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity C01 – Finite CP/channel profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <span style="color: blue;">C01</span>                                                                                                                                                          |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | <span style="color: blue;">14.2</span>                                                                                                                                                         |

|                           |                                                                                                                                                                                                                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source objects            | C00;<br>P06;<br>ChannelSeed.                                                                                                                                                                                                                                                                               |
| Definitions               | Completely-positive/channel profile derived from finite exported data.                                                                                                                                                                                                                                     |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                     |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                    |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity C02 – Effective OQS generator / Lindblad gate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                     |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <span style="color: blue;">C02</span>                                                                                                                                                                                                                                                                      |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                                                                                                                                                                                                          |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                           |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                         |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                             |
| Documented                      | no                                                                                                                                                                                                                                                                                                         |
| Phase                           | <span style="color: blue;">14.3</span>                                                                                                                                                                                                                                                                     |
| Source objects                  | C01;<br>finite dynamics seeds.                                                                                                                                                                                                                                                                             |
| Definitions                     | Effective generator and Lindblad/GKSL gate if derived from finite channel/semigroup data.                                                                                                                                                                                                                  |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                     |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                    |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity C03 – Decoherence/noise profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <span style="color: blue;">C03</span>                                                                                                                                                          |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | <span style="color: blue;">14.4</span>                                                                                                                                                         |

|                           |                                                                                                                                                                                                                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source objects            | C00-C02;<br>ComplementProfile.                                                                                                                                                                                                                                                                             |
| Definitions               | Effective decoherence/noise profile derived from complement/environment data.                                                                                                                                                                                                                              |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                     |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                    |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity C04 – C->A allowed output profile

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                     |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | C04                                                                                                                                                                                                                                                                                                        |
| Object status                   | K                                                                                                                                                                                                                                                                                                          |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                           |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                         |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                             |
| Documented                      | no                                                                                                                                                                                                                                                                                                         |
| Phase                           | 14.5                                                                                                                                                                                                                                                                                                       |
| Source objects                  | C01-C03.                                                                                                                                                                                                                                                                                                   |
| Definitions                     | Typed C output that A may consume only through C->A input.                                                                                                                                                                                                                                                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                     |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                    |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity D00 – Pillar D spacetime input contract

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | D00                                                                                                                                                                                                     |
| Object status                   | K                                                                                                                                                                                                       |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                               |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Documented                      | no                                                                                                                                                                                                      |
| Phase                           | 15.1                                                                                                                                                                                                    |
| Source objects                  | H13;<br>lattice/region/spectral geometry objects.                                                                                                                                                       |
| Definitions                     | D consumes explicit A/C geometry, regime, and localization outputs.                                                                                                                                     |



|                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                           |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                          |
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity D01 – Region/localization/causal-order candidate

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | D01                                                                                                                                                                                                                                                                                                     |
| Object status                   | K                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 15.2                                                                                                                                                                                                                                                                                                    |
| Source objects                  | A geometry;<br>C open-system profiles if available.                                                                                                                                                                                                                                                     |
| Definitions                     | Candidate region poset, localization profile, and causal order derived from CNNA relational/geometric data.                                                                                                                                                                                             |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity D02 – Metric/coarse-geometry/covariance gate

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | D02                                                                                                                                                                                            |
| Object status                   | K                                                                                                                                                                                              |
| Context class                   | LeanSourceObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | 15.3                                                                                                                                                                                           |
| Source objects                  | D01;<br>SP0-SP5 if available.                                                                                                                                                                  |
| Definitions                     | Metric candidate, coarse geometry, and covariance gate derived from D localization/spectral data.                                                                                              |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                         |

|                           |                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity D03 – D->A allowed output profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | D03                                                                                                                                                                                                                                                                                                     |
| Object status                   | K                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 15.4                                                                                                                                                                                                                                                                                                    |
| Source objects                  | D01-D02.                                                                                                                                                                                                                                                                                                |
| Definitions                     | Typed D output that A may consume only through D->A input.                                                                                                                                                                                                                                              |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity E00 – Pillar E matter input contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | E00                                                                                                                                                                                                     |
| Object status                   | K                                                                                                                                                                                                       |
| Context class                   | LeanHandoffContractObject                                                                                                                                                                               |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Documented                      | no                                                                                                                                                                                                      |
| Phase                           | 16.1                                                                                                                                                                                                    |
| Source objects                  | H14; B/C/D outputs.                                                                                                                                                                                     |
| Definitions                     | E consumes permitted A/B/C/D outputs for matter/observable comparison.                                                                                                                                  |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                 |

|                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Object or quantity E01 – Matter sector and internal symmetry candidate

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | E01                                                                                                                                                                                                                                                                                                     |
| Object status                   | K                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 16.2                                                                                                                                                                                                                                                                                                    |
| Source objects                  | Units/CD/symmetry objects; B/D outputs.                                                                                                                                                                                                                                                                 |
| Definitions                     | Candidate matter sectors and internal symmetries derived from CNNA symmetry/arithmetic data.                                                                                                                                                                                                            |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity E02 – Observable coupling and phenomenology comparison

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | E02                                                                                                                                       |
| Object status                   | K                                                                                                                                         |
| Context class                   | ReferenceGateObject                                                                                                                       |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | no                                                                                                                                        |
| Phase                           | 16.3                                                                                                                                      |
| Source objects                  | E01; B/C/D outputs.                                                                                                                       |
| Definitions                     | Observable coupling and empirical comparison layer, not foundational input.                                                               |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                    |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                 |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                   |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                  |

Object or quantity E03 – E->A allowed output profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | E03                                                                                                                                                                                                                                                                                                     |
| Object status                   | K                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 16.4                                                                                                                                                                                                                                                                                                    |
| Source objects                  | E01-E02.                                                                                                                                                                                                                                                                                                |
| Definitions                     | Typed E output that A may consume only through E->A input.                                                                                                                                                                                                                                              |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity BR00 – Backreaction input DAG

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | BR00                                                                                                                                                                          |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | LeanTargetObject                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 17                                                                                                                                                                            |
| Source objects                  | B06;<br>C04;<br>D03;<br>E03.                                                                                                                                                  |
| Definitions                     | DAG of allowed B/C/D/E -> A input handoffs.                                                                                                                                   |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity BR01 – B->A backreaction input gate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | BR01                                                                                                                                                                                                                                                                                                    |
| Object status                   | K                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 17.1                                                                                                                                                                                                                                                                                                    |
| Source objects                  | B06.                                                                                                                                                                                                                                                                                                    |
| Definitions                     | Gate for B-derived theorem/data entering A.                                                                                                                                                                                                                                                             |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity BR02 – C->A OQS backreaction input gate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | BR02                                                                                                                                                                                                                                                                                                    |
| Object status                   | K                                                                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | 17.2                                                                                                                                                                                                                                                                                                    |
| Source objects                  | C04.                                                                                                                                                                                                                                                                                                    |
| Definitions                     | Gate for C-derived OQS data entering A.                                                                                                                                                                                                                                                                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

Object or quantity BR03 – D->A spacetime backreaction input gate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field         | Record |
|---------------|--------|
| Object ID     | BR03   |
| Object status | K      |

|                                 |                                                                                                                                                                                                                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | <a href="#">17.3</a>                                                                                                                                                                                                                                                                                    |
| Source objects                  | D03.                                                                                                                                                                                                                                                                                                    |
| Definitions                     | Gate for D-derived geometry/covariance data entering A.                                                                                                                                                                                                                                                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity BR04 – E->A matter backreaction input gate

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">BR04</a>                                                                                                                                                                                                                                                                                    |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                                                                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                        |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                                      |
| Phase                           | <a href="#">17.4</a>                                                                                                                                                                                                                                                                                    |
| Source objects                  | E03.                                                                                                                                                                                                                                                                                                    |
| Definitions                     | Gate for E-derived matter/phenomenology data entering A.                                                                                                                                                                                                                                                |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                  |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                                                                                                                                               |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose. |

### Object or quantity BR05 – Pre/post-emergence switch for A statements

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">BR05</a>                                                                                                                                                                                    |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                                                                                                       |
| Context class                   | LeanSourceObject                                                                                                                                                                                        |
| Significance tier               | F0                                                                                                                                                                                                      |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |



|                           |                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documented                | no                                                                                                                                                                                                                                                                                                                  |
| Phase                     | 17.5                                                                                                                                                                                                                                                                                                                |
| Source objects            | BR00-BR04;<br>regime profile.                                                                                                                                                                                                                                                                                       |
| Definitions               | Profile distinguishing pre-emergence and post-emergence interpretation of A statements.                                                                                                                                                                                                                             |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                                                                                                                                              |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.                                                                                                                                                                                        |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                                                                                                                                                             |
| Next documentation action | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose. |

### Object or quantity SIDE0 – Full ToE comparison profile

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | SIDE0                                                                                                                         |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                             |
| Context class                   | ReferenceGateObject                                                                                                           |
| Significance tier               | F4                                                                                                                            |
| Documentation depth requirement | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted. |
| Documented                      | no                                                                                                                            |
| Phase                           | 18.1                                                                                                                          |
| Source objects                  | CNNA mature outputs;<br>no ToE assumptions.                                                                                   |
| Definitions                     | Comparison-only placeholder for any future full ToE interpretation that falls out of CNNA.                                    |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.  |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose.                   |

### Object or quantity SIDE1 – String-theory comparison profile

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | SIDE1                                                                                                                         |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                             |
| Context class                   | ReferenceGateObject                                                                                                           |
| Significance tier               | F4                                                                                                                            |
| Documentation depth requirement | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted. |
| Documented                      | no                                                                                                                            |
| Phase                           | 18.2                                                                                                                          |

|                           |                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Source objects            | CNNA outputs;<br>no string assumptions.                                                                                      |
| Definitions               | Comparison-only placeholder for string-theory-like structures if derivable from CNNA outputs.                                |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                       |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions. |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                      |
| Next documentation action | Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose.                  |

### Object or quantity SIDE2 – Loop quantum gravity comparison profile

**Table legend.** W = white/finished public object;  B = blue/proof-carrying but semantic/API correction still open;  G = green/locally closable;  Y = yellow/known route with prerequisites;  R = red/derivation layer incomplete or unclear;  K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">SIDE2</a>                                                                                                         |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                             |
| Context class                   | ReferenceGateObject                                                                                                           |
| Significance tier               | F4                                                                                                                            |
| Documentation depth requirement | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted. |
| Documented                      | no                                                                                                                            |
| Phase                           | <a href="#">18.3</a>                                                                                                          |
| Source objects                  | CNNA outputs;<br>no LQG assumptions.                                                                                          |
| Definitions                     | Comparison-only placeholder for LQG-like structures if derivable from CNNA outputs.                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence;<br>prose is secondary and must not introduce new assumptions.  |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase;<br>do not duplicate in phase-local prose.                   |

### Object or quantity SIDE3 – Noncommutative geometry comparison profile

**Table legend.** W = white/finished public object;  B = blue/proof-carrying but semantic/API correction still open;  G = green/locally closable;  Y = yellow/known route with prerequisites;  R = red/derivation layer incomplete or unclear;  K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">SIDE3</a>                                                                                                         |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                             |
| Context class                   | ReferenceGateObject                                                                                                           |
| Significance tier               | F4                                                                                                                            |
| Documentation depth requirement | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted. |
| Documented                      | no                                                                                                                            |
| Phase                           | <a href="#">18.4</a>                                                                                                          |
| Source objects                  | CNNA outputs;<br>no NCG axioms as foundation.                                                                                 |
| Definitions                     | Comparison-only placeholder for NCG-like structures if derivable from CNNA outputs.                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                        |

|                           |                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions. |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                   |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                  |

### Object or quantity MOON3 – Finite simple group comparison profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | MOON3                                                                                                                                                                         |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 12.4                                                                                                                                                                          |
| Source objects                  | CNNA-derived q/graded/symmetry profile plus external comparison taxonomy.                                                                                                     |
| Definitions                     | Comparison-only profile for CFSG taxonomy: cyclic prime-order, alternating, Lie-type, sporadic.                                                                               |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity MOON4 – Sporadic/Monster comparison boundary

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | MOON4                                                                                                                                                                         |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 12.4                                                                                                                                                                          |
| Source objects                  | MOON0/MOON3 and external Monster representation dimension table as target.                                                                                                    |
| Definitions                     | Boundary profile separating sporadic-group/Monster comparison data from CNNA-generative objects.                                                                              |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity MOON5 – Leech lattice comparison profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | MOON5                                                                                                                                                                         |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 12.5                                                                                                                                                                          |
| Source objects                  | Derived CNNA lattice/theta candidate plus external Leech target data.                                                                                                         |
| Definitions                     | Comparison-only profile for a rootless even unimodular 24-dimensional lattice target.                                                                                         |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity MOON6 – Leech theta/shell coefficient comparison

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | MOON6                                                                                                                                                                         |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 12.6                                                                                                                                                                          |
| Source objects                  | MOON5 and CNNA-derived q/theta stream.                                                                                                                                        |
| Definitions                     | Comparison profile for Leech theta series and shell-count/kissing-number coefficients.                                                                                        |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

Object or quantity MOON7 – VOA/moonshine-module comparison profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field     | Record |
|-----------|--------|
| Object ID | MOON7  |

|                                 |                                                                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 12.7                                                                                                                                                                          |
| Source objects                  | CNNA-derived graded algebra/q-character output, if any, plus external VOA/moonshine-module target data.                                                                       |
| Definitions                     | Comparison-only profile for vertex-operator-algebra/moonshine-module structures.                                                                                              |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity MOON8 – Common-indicator / Hauptmodul bridge-test profile

**Table legend.**   W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | MOON8                                                                                                                                                                         |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 12.8                                                                                                                                                                          |
| Source objects                  | CNNA-derived q-character/coefficient stream and external normalized Hauptmodul target list.                                                                                   |
| Definitions                     | Comparison profile testing whether CNNA-derived q-characters match known Hauptmoduln or McKay-Thompson-type series.                                                           |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                        |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                                                     |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                                                       |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                                                      |

### Object or quantity LEG0 – Consumption and split inventory umbrella

**Table legend.**   W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG0                                                                                                                                      |
| Object status                   | Y                                                                                                                                         |
| Context class                   | GovernancePolicyObject                                                                                                                    |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | no                                                                                                                                        |

|                           |                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Phase                     | 1.1.1                                                                                                                     |
| Source objects            | static module/declaration/reference summaries.                                                                            |
| Definitions               | SplitInventoryProfile with declaration-consumption map, not a code object.                                                |
| Proof target              | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                    |
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions. |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                   |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                  |

### Object or quantity LEG1 – Legacy single-constructor status gates

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG1                                                                                                                                      |
| Object status                   | Y                                                                                                                                         |
| Context class                   | LeanRefactorObject                                                                                                                        |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | no                                                                                                                                        |
| Phase                           | 1.1.2                                                                                                                                     |
| Source objects                  | frozen SM9h legacy status declarations; no new semantics.                                                                                 |
| Definitions                     | Historical single-constructor status enums retained for compatibility but quarantined outside the neutral public surface.                 |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                    |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                 |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                   |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                  |

### Object or quantity LEG2 – CoreSymbols true notation surface

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG2                                                                                                                                      |
| Object status                   | Y                                                                                                                                         |
| Context class                   | LeanRefactorObject                                                                                                                        |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | no                                                                                                                                        |
| Phase                           | 1.1.3                                                                                                                                     |
| Source objects                  | Notation split inventory.                                                                                                                 |
| Definitions                     | Minimal notation/core-symbol module containing only actual notation/syntax/domain-role symbols needed by later neutral APIs.              |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                    |



|                           |                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions. |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                   |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                  |

### Object or quantity LEG3 – Legacy phase aliases compatibility surface

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">LEG3</a>                                                                                                                      |
| Object status                   | <span style="background-color: #FFFF00;">Y</span>                                                                                         |
| Context class                   | LeanRefactorObject                                                                                                                        |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | no                                                                                                                                        |
| Phase                           | <a href="#">1.1.4</a>                                                                                                                     |
| Source objects                  | Notation split inventory and frozen historical modules.                                                                                   |
| Definitions                     | Historical phase-semantic aliases and old names are retained behind a legacy module, not exposed as new public API.                       |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                    |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                 |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                   |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                  |

### Object or quantity LEG4 – Neutral import firewall for post-notation phases

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">LEG4</a>                                                                                                                      |
| Object status                   | <span style="background-color: #FFFF00;">Y</span>                                                                                         |
| Context class                   | GovernancePolicyObject                                                                                                                    |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | no                                                                                                                                        |
| Phase                           | <a href="#">1.1.5</a>                                                                                                                     |
| Source objects                  | Lean import graph and phase-object DAG.                                                                                                   |
| Definitions                     | Audit rule and generated module inventory check that Phase 3+ imports neutral/CoreSymbols surfaces and not Legacy.StatusGates.            |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                    |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                 |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                   |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                  |

Object or quantity LEG5 – Declaration consumption classification map

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Significance tier               | F3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                                                                                                                                                                                                                                                      |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Phase                           | 1.1.1.1.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Source objects                  | repo_inventory/raw_export/modules.json;<br>symbol_references/declarations.json;<br>symbol_references/references.json;<br>architectural_core/sink_taxonomy.json;<br>object register;<br>object_proof_dossiers;<br>LEG10 module manifest;<br>LEG11 reuse/duplication ledger;<br>LEG12 proof-documentation infrastructure.                                                                                                                                                                        |
| Definitions                     | ConsumptionClassificationMap with classes core-computational, proof-bridge, public-wrapper, legacy-status, legacy-policy, phase-alias, audit-only, blue-semantic-correction, dead-candidate, final-output;<br>generated module/declaration/object consumption records.                                                                                                                                                                                                                         |
| Proof target                    | Classify declarations/modules/registered objects conservatively before blue correction and public API work: BuildAll/import reachability is not semantic consumption;<br>blue objects are never public-final;<br>full records are generated as TSV evidence.                                                                                                                                                                                                                                   |
| Proof sketch                    | Extend scripts/generate_tables.py so the single YAML master defines the permitted classes and the generator validates that exact class set. Read the frozen repo inventory and symbol-reference JSON, classify modules and declarations by deterministic evidence rules, classify object-register rows with B objects forced to blue-semantic-correction, export full TSV ledgers, and render a compact PDF policy/count section. No Lean source move or mathematical generator is introduced. |
| Verification / audit            | v43 generator run validates the consumption_classification_map section and emits consumption_classification_map.generated.tsv, module_consumption_records.generated.tsv, declaration_consumption_records.generated.tsv, and object_consumption_classification.generated.tsv. The blue-object gate is checked in the generator;<br>no Lean source was changed and no new Lean build is claimed from this environment.                                                                           |
| Next documentation action       | Phase 1.1.1.2 consumes the object_consumption_classification ledger to build the blue-object semantic correction ledger;<br>1.1.1.3 later consumes it for public API/consumer whitelist rules.                                                                                                                                                                                                                                                                                                 |

Object or quantity LEG7 – Blue-object semantic correction ledger

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                 |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG7                                                                                                                                                                                   |
| Object status                   | Y                                                                                                                                                                                      |
| Context class                   | GovernancePolicyObject                                                                                                                                                                 |
| Significance tier               | F3                                                                                                                                                                                     |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                              |
| Documented                      | no                                                                                                                                                                                     |
| Phase                           | 1.1.1.2                                                                                                                                                                                |
| Source objects                  | CD/S/O blue object rows and their current code derivation sites.                                                                                                                       |
| Definitions                     | BlueCorrectionLedger mapping each Object status B record to semantic defect, allowed temporary consumers, forbidden public consumers, neutral target API, and exit-to-white condition. |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                                                                 |

|                           |                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions. |
| Verification / audit      | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                   |
| Next documentation action | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                  |

### Object or quantity LEG6 – Public API contract and consumer whitelist

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">LEG6</a>                                                                                                                      |
| Object status                   | <span style="background-color: #FFFF00;">Y</span>                                                                                         |
| Context class                   | GovernancePolicyObject                                                                                                                    |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | no                                                                                                                                        |
| Phase                           | <a href="#">1.1.1.3</a>                                                                                                                   |
| Source objects                  | LEG5, LEG7, object access register, phase dependency table.                                                                               |
| Definitions                     | PublicAPIContract listing allowed neutral modules/declarations and forbidden imports for Phase 3+ and later handoffs.                     |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                    |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                 |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                   |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                  |

### Object or quantity LEG8 – Pruning and dead-candidate ledger

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">LEG8</a>                                                                                                                      |
| Object status                   | <span style="background-color: #FFFF00;">Y</span>                                                                                         |
| Context class                   | GovernancePolicyObject                                                                                                                    |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | no                                                                                                                                        |
| Phase                           | <a href="#">1.1.1.4</a>                                                                                                                   |
| Source objects                  | LEG5, LEG6, symbol reference summaries, import graph.                                                                                     |
| Definitions                     | PruningLedger classifying declarations/modules as keep, quarantine, wrap, reexport, archive, or deletion-candidate.                       |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                    |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                 |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                   |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                  |

Object or quantity LEG9 – Generated-view authority and implementation-lock contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG9                                                                                                                                      |
| Object status                   | Y                                                                                                                                         |
| Context class                   | GovernancePolicyObject                                                                                                                    |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | no                                                                                                                                        |
| Phase                           | 1.1.1.5                                                                                                                                   |
| Source objects                  | 1.1.1.1-1.1.1.4 records and generate_tables.py output.                                                                                    |
| Definitions                     | GeneratedViewAuthorityLock tying YAML master, generated tables/PDF, and local-build evidence rule to the next implementation prompt.      |
| Proof target                    | Not yet documented in v25 proofbook. Must be filled before public-final promotion or handoff exposure.                                    |
| Proof sketch                    | Pending. Use checked code derivation site as primary evidence; prose is secondary and must not introduce new assumptions.                 |
| Verification / audit            | Pending. For blue objects, semantic/API correction remains required even if proof-carrying code exists.                                   |
| Next documentation action       | Fill during the object’s natural correction/implementation phase; do not duplicate in phase-local prose.                                  |

Object or quantity LEG10 – Implementation module manifest and semantic placement contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG10                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                                                                                                           |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Phase                           | 1.1.1.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Source objects                  | repo_inventory/raw_export/modules.json;<br>import_edges.json;<br>symbol_reference_summary.json;<br>phase/object DAG;<br>implementation scratchpad;<br>frozen Lean source snapshot                                                                                                                                                                                                                                                        |
| Definitions                     | ModuleImplementationManifest is the generated scratchpad contract requiring exact added/modified modules, reusable existing modules, semantic placement, rejection of alternative placements, public/legacy/API role, import direction, expected consumers, phase type, regression protocol, version impact, concrete finite traversal, generic test placement, mathlib risk, and noncomputable boundary audit before Lean code changes. |
| Proof target                    | Future implementation phases cannot start with unnamed modules, hidden consumer direction, duplicated generator/test infrastructure, or unclassified mathlib/noncomputable pressure.                                                                                                                                                                                                                                                     |
| Proof sketch                    | Extend the single-master generator so every scratchpad record must carry the module-manifest fields. Render a module-manifest table and TSV from those fields. Mark 1.1.1.1.1 as infrastructure-only closure: no Lean source move, no operational object, no build claim. Set the next active cursor to 1.1.1.1.2 so generator reuse is audited after named module placement exists.                                                     |
| Verification / audit            | scripts/generate_tables.py validates mandatory manifest fields and generated module-manifest artifacts were regenerated in v40. No Lean source was changed and no Lean build is claimed.                                                                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. During 1.1.1.1.2, consume LEG10 to classify generator-like future code as consume, extend, wrap, or genuine semantic split; later Lean phases must fill phase-specific manifest content before edits.                                     |

Object or quantity LEG11 – Reuse/duplication ledger and canonical generator policy

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Phase                           | 1.1.1.1.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Source objects                  | LEG10, repo_inventory/raw_export/symbol_reference_summary.json, existing Smoke/Test generator modules, and planned target objects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Definitions                     | ReuseDuplicationLedger plus CanonicalGeneratorPolicy: no second implementation of an already available semantic operation; generator-like request/output, smoke-test, extraction, and ledger code must reuse or extend the canonical theory/smoke-test generator seam.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof target                    | Close LEG11 as a generated governance ledger: every future generator-like operation must be classified as consume existing, extend existing, wrap existing, or genuine semantic split before Lean implementation; duplicate generator/request/output/extraction/ledger paths are rejected by default.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Proof sketch                    | Use LEG10 module-manifest data and the frozen repo inventory to locate the existing generator seams. The operational generated source/main path is already represented by CNNA.PillarA.Arithmetic.Smoke.GeneratedSource and GeneratedMainPath, with downstream generated SM modules consuming that route. Action-like generator semantics are separately recorded by GeneratorActionInventory and ActionFormatDecision. Documentation and repo-inventory scripts are artifact generators, not CNNA mathematical generators. Therefore LEG11 records one canonical extension rule: same semantics must consume, extend, or wrap the natural existing seam; only a ledger-recorded semantic split may justify a separate generator. |
| Verification / audit            | v41 adds the reuse_duplication_ledger top-level YAML section, generator validation/rendering in scripts/generate_tables.py, tables/reuse_duplication_ledger.tex, and generated_exports/reuse_duplication_ledger.generated.tsv. No Lean source was changed. The user reported the prior Lean build green before this infrastructure-only update; no new Lean build is claimed from this environment.                                                                                                                                                                                                                                                                                                                               |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Phase 1.1.1.1.3 now consumes documented LEG11 while closing object/proof-dossier infrastructure; later implementation prompts must consult the reuse ledger before adding any generator-like code.                                                                                                                                                                                                                                                                                                                                 |

Object or quantity LEG12 – Object and size proof-documentation infrastructure

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                   |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG12                                                                                                                                                                                                                                                    |
| Object status                   | G                                                                                                                                                                                                                                                        |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                   |
| Significance tier               | F1                                                                                                                                                                                                                                                       |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                           |
| Documented                      | yes                                                                                                                                                                                                                                                      |
| Phase                           | 1.1.1.1.3                                                                                                                                                                                                                                                |
| Source objects                  | object register;<br>object_proof_dossiers;<br>LEG10 module manifest gate;<br>LEG11 reuse/duplication policy;<br>blue-object correction requirements;<br>generated single-master views.                                                                   |
| Definitions                     | ObjectProofDossier infrastructure: one generated dossier row per registered object/quantity, with canonical proof:<ID> link from the object register and generated anchor in the proof-documentation section.                                            |
| Proof target                    | Every registered object/quantity has a non-empty proof-documentation status, an exact proof:<ID> proof-dossier link, and exactly one matching object_proof_dossiers row with required source, definition, target, sketch, audit, and next-action fields. |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | scripts/generate_tables.py loads the single YAML master, validates the object register, validates object_proof_dossiers against object IDs, requires exact proof:<ID> object links, checks non-empty required dossier fields, and checks object/phase agreement before generating TSV/PDF tables. No mathematical proof content is inferred from documentation alone.                                         |
| Verification / audit      | v42 generator run succeeded with 199 objects and 199 proof-dossier records. The PDF proof-documentation section and object proof-link table were regenerated. Lean source was not changed and no Lean build is claimed for this documentation-only gate.                                                                                                                                                      |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Consumed by 1.1.1.1.4 for declaration classification. Later object phases must fill no/partial dossiers at their natural source layer and cannot promote blue/public-final objects on name-only documentation. |

### Object or quantity LEG13 – Global no-go guardrail contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG13                                                                                                                                                                                                                                                                                                                                      |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                          |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                     |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                                                         |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                                                    |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                        |
| Phase                           | 1.1.1.1.0                                                                                                                                                                                                                                                                                                                                  |
| Source objects                  | User-specified no-go list; prior implementation constraints; LEG4 import firewall; LEG10 module manifest; LEG11 reuse/duplication policy; LEG12 proofbook workflow.                                                                                                                                                                        |
| Definitions                     | GlobalNoGoGuardrailContract is the top-level invariant set in global_no_go_guardrails. Per-phase no_go_guardrails are local addenda only and cannot weaken the global contract.                                                                                                                                                            |
| Proof target                    | Every phase and workflow record exposes inheritance of the global guardrails before implementation or closeout.                                                                                                                                                                                                                            |
| Proof sketch                    | The YAML master contains a top-level global_no_go_guardrails section, a workflow_policy record for inheritance, a green infrastructure phase 1.1.1.1.0, object LEG13, and a global_no_go_inheritance field in every scratchpad record. The generator validates the section and renders it as a first-class table.                          |
| Verification / audit            | v26 generator validation checks presence of the global section and scratchpad inheritance field. This is planning/documentation infrastructure only; no Lean build is claimed.                                                                                                                                                             |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. During each code phase, cite the global guardrail table and record only phase-specific stricter addenda in local no_go_guardrails. |

### Object or quantity LEG14 – Concrete computable instance traversal contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG14                                                                                                                                     |
| Object status                   | G                                                                                                                                         |
| Context class                   | GovernancePolicyObject                                                                                                                    |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | yes-infrastructure                                                                                                                        |
| Phase                           | 1.1.1.1.0.1                                                                                                                               |
| Source objects                  | global no-go guardrails; implementation scratchpad; canonical generator path; active phase target objects                                 |



|                           |                                                                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definitions               | ConcreteComputableInstanceTraversal records a concrete finite instance, expected executable witness, path endpoint, and failure/localization point if traversal stops.                                   |
| Proof target              | Every operational phase has executable-instance pressure and cannot close as a skeleton-only construction.                                                                                               |
| Proof sketch              | The generator requires the fields in every scratchpad record and renders them in the implementation scratch pad. Active implementation records must fill them nontrivially or state infrastructure-only. |
| Verification / audit      | Checked by generate_tables.py required_scratch_fields and rendered scratchpad output; no Lean build claim.                                                                                               |
| Next documentation action | When the first Lean implementation begins, fill the concrete instance path with the actual generated substrate/window and executable endpoint.                                                           |

### Object or quantity LEG15 – Mathlib complication and noncomputable-boundary log

**Table legend.**   W = white/finished public object;   

B

 = blue/proof-carrying but semantic/API correction still open;   

G

 = green/locally closable;   

Y

 = yellow/known route with prerequisites;   

R

 = red/derivation layer incomplete or unclear;   

K

 = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                             |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">LEG15</a>                                                                                                                                                                              |
| Object status                   | <div>G</div>                                                                                                                                                                                       |
| Context class                   | GovernancePolicyObject                                                                                                                                                                             |
| Significance tier               | F3                                                                                                                                                                                                 |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects.   Avoid proofbook bloat.                                                        |
| Documented                      | yes-infrastructure                                                                                                                                                                                 |
| Phase                           | <a href="#">1.1.1.1.0.1</a>                                                                                                                                                                        |
| Source objects                  | global noncomputable/classical no-go rules;<br>expected mathlib imports;<br>observed compiler barriers;<br>mirror/non-operational isolation policy                                                 |
| Definitions                     | MathlibComplicationLog records import candidates, computability risk, noncomputable/classical leak status, mitigation, and natural refactor layer.                                                 |
| Proof target                    | Mathlib complications are not hidden in operational code and cannot silently weaken the computable-path invariant.                                                                                 |
| Proof sketch                    | Each scratchpad record contains mathlib_complication_log, noncomputable_boundary_audit, and mathlib_mitigation_or_refactor_layer;<br>active phases must record the risk state before code changes. |
| Verification / audit            | Generated scratchpad renders the fields;<br>future implementation closeout must compare them to compiler/build evidence.                                                                           |
| Next documentation action       | Replace generic placeholders by concrete mathlib module names and observed error/warning evidence during each implementation phase.                                                                |

### Object or quantity LEG16 – Semi-automated existing-code documentation coverage

**Table legend.**   W = white/finished public object;   

B

 = blue/proof-carrying but semantic/API correction still open;   

G

 = green/locally closable;   

Y

 = yellow/known route with prerequisites;   

R

 = red/derivation layer incomplete or unclear;   

K

 = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                         |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">LEG16</a>                                                                                                                                                                                                                                          |
| Object status                   | <div>K</div>                                                                                                                                                                                                                                                   |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                            |
| Significance tier               | F3                                                                                                                                                                                                                                                             |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects.   Avoid proofbook bloat.                                                                                                                    |
| Documented                      | no                                                                                                                                                                                                                                                             |
| Phase                           | <a href="#">19.1</a>                                                                                                                                                                                                                                           |
| Source objects                  | modules.json, import_edges.json, symbol_reference_summary.json, object_proof_dossiers, implementation_scratchpad.                                                                                                                                              |
| Definitions                     | A generated coverage/documentation ledger tying Lean modules/declarations to object proof dossiers, consumers, and provenance classifications.                                                                                                                 |
| Proof target                    | Secondary long-term target only.   Before activation, prove/verify that this dossier consumes generated code/proof metadata only and does not introduce CNNA-generative input.                                                                                 |
| Proof sketch                    | Pending.   The eventual proof/documentation route should be: generated repository graph or checked object dossiers -> explicit diagnostic record -> review-only report.   No operational Lean theorem or CNNA object may depend on the diagnostic as an axiom. |

|                           |                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / audit      | Pending. Must audit derived-only boundary and confirm that use is documentation/diagnostic/comparison-only.                                        |
| Next documentation action | Keep black/secondary until the relevant Phase 1.x cleanup and later core derivations are stable;<br>then fill source-specific diagnostic evidence. |

### Object or quantity LEG17 – Derivation-graph geometry diagnostics

**Table legend.** W = white/finished public object; 

B

 = blue/proof-carrying but semantic/API correction still open; 

G

 = green/locally closable; 

Y

 = yellow/known route with prerequisites; 

R

 = red/derivation layer incomplete or unclear; 

K

 = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                     |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">LEG17</a>                                                                                                                                                                                                                                      |
| Object status                   | <div>K</div>                                                                                                                                                                                                                                               |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                        |
| Significance tier               | F3                                                                                                                                                                                                                                                         |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                  |
| Documented                      | no                                                                                                                                                                                                                                                         |
| Phase                           | <a href="#">19.2</a>                                                                                                                                                                                                                                       |
| Source objects                  | import_edges.json, dominator/layer/path summaries, module_group_dependency_graph, core_robust_semantic_mainline.                                                                                                                                           |
| Definitions                     | A non-operational diagnostic profile for the repository derivation graph: import edges, dominators, layers, source-sink paths, bottlenecks, cuts, and discipline clusters.                                                                                 |
| Proof target                    | Secondary long-term target only. Before activation, prove/verify that this dossier consumes generated code/proof metadata only and does not introduce CNNA-generative input.                                                                               |
| Proof sketch                    | Pending. The eventual proof/documentation route should be: generated repository graph or checked object dossiers -> explicit diagnostic record -> review-only report. No operational Lean theorem or CNNA object may depend on the diagnostic as an axiom. |
| Verification / audit            | Pending. Must audit derived-only boundary and confirm that use is documentation/diagnostic/comparison-only.                                                                                                                                                |
| Next documentation action       | Keep black/secondary until the relevant Phase 1.x cleanup and later core derivations are stable;<br>then fill source-specific diagnostic evidence.                                                                                                         |

### Object or quantity LEG18 – Open-system emergence-path forensics

**Table legend.** W = white/finished public object; 

B

 = blue/proof-carrying but semantic/API correction still open; 

G

 = green/locally closable; 

Y

 = yellow/known route with prerequisites; 

R

 = red/derivation layer incomplete or unclear; 

K

 = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                     |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">LEG18</a>                                                                                                                                                                                                                                      |
| Object status                   | <div>K</div>                                                                                                                                                                                                                                               |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                        |
| Significance tier               | F3                                                                                                                                                                                                                                                         |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                  |
| Documented                      | no                                                                                                                                                                                                                                                         |
| Phase                           | <a href="#">19.3</a>                                                                                                                                                                                                                                       |
| Source objects                  | future checked object dossiers and source-sink/dominator path analysis over relevant modules.                                                                                                                                                              |
| Definitions                     | A diagnostic map locating formal proof/module paths from regime/complement/boundary data toward effective open-system dynamics consumers.                                                                                                                  |
| Proof target                    | Secondary long-term target only. Before activation, prove/verify that this dossier consumes generated code/proof metadata only and does not introduce CNNA-generative input.                                                                               |
| Proof sketch                    | Pending. The eventual proof/documentation route should be: generated repository graph or checked object dossiers -> explicit diagnostic record -> review-only report. No operational Lean theorem or CNNA object may depend on the diagnostic as an axiom. |
| Verification / audit            | Pending. Must audit derived-only boundary and confirm that use is documentation/diagnostic/comparison-only.                                                                                                                                                |
| Next documentation action       | Keep black/secondary until the relevant Phase 1.x cleanup and later core derivations are stable;<br>then fill source-specific diagnostic evidence.                                                                                                         |

Object or quantity LEG19 – Reflective CNNA/formalization graph comparison boundary

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                     |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG19                                                                                                                                                                                                                                                      |
| Object status                   | K                                                                                                                                                                                                                                                          |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                        |
| Significance tier               | F3                                                                                                                                                                                                                                                         |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                  |
| Documented                      | no                                                                                                                                                                                                                                                         |
| Phase                           | 19.4                                                                                                                                                                                                                                                       |
| Source objects                  | core object dossiers, discipline subgraphs, graph cuts, layers, signal paths, and cluster analysis after stabilization.                                                                                                                                    |
| Definitions                     | A boundary object marking any comparison between CNNA object-structure and the formalization graph as meta-level, observational, and non-generative.                                                                                                       |
| Proof target                    | Secondary long-term target only. Before activation, prove/verify that this dossier consumes generated code/proof metadata only and does not introduce CNNA-generative input.                                                                               |
| Proof sketch                    | Pending. The eventual proof/documentation route should be: generated repository graph or checked object dossiers -> explicit diagnostic record -> review-only report. No operational Lean theorem or CNNA object may depend on the diagnostic as an axiom. |
| Verification / audit            | Pending. Must audit derived-only boundary and confirm that use is documentation/diagnostic/comparison-only.                                                                                                                                                |
| Next documentation action       | Keep black/secondary until the relevant Phase 1.x cleanup and later core derivations are stable; then fill source-specific diagnostic evidence.                                                                                                            |

Object or quantity LEG20 – EXEC-2 rational ExecComplex boundary policy

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                       |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG20                                                                                                                                                                                                                                                                                        |
| Object status                   | G                                                                                                                                                                                                                                                                                            |
| Context class                   | LeanGovernanceObject                                                                                                                                                                                                                                                                         |
| Significance tier               | F1                                                                                                                                                                                                                                                                                           |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                               |
| Documented                      | yes                                                                                                                                                                                                                                                                                          |
| Phase                           | 1.1.1.1.0.2                                                                                                                                                                                                                                                                                  |
| Source objects                  | O00/O01/O06;<br>L00/L02/L04/M05/SP0;<br>ExecComplex = Q x Q audit                                                                                                                                                                                                                            |
| Definitions                     | Operational Q x Q path;<br>mirror-only algebraic irrational data;<br>rational/integer return invariant                                                                                                                                                                                       |
| Proof target                    | Show every ExecComplex-touching phase has an explicit operational/mirror classification before implementation.                                                                                                                                                                               |
| Proof sketch                    | Planning proof: values not representable in Q x Q are classified as mirror unless reduced to a proved rational/integer invariant.                                                                                                                                                            |
| Verification / audit            | Generator renders format decision, workflow policy, object updates, and green phase 1.1.1.1.0.2;<br>no Lean proof claimed.                                                                                                                                                                   |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. During Phase 7+ preparation, instantiate the policy for each lattice/tau/j/eigenvalue object. |

Object or quantity LEG21 – Refactor regression and incremental migration protocol

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                           |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG21                                                                                                                                                                                                                                                                                            |
| Object status                   | G                                                                                                                                                                                                                                                                                                |
| Context class                   | LeanGovernanceObject                                                                                                                                                                                                                                                                             |
| Significance tier               | F1                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                   |
| Documented                      | yes                                                                                                                                                                                                                                                                                              |
| Phase                           | 1.1.1.1.0.3                                                                                                                                                                                                                                                                                      |
| Source objects                  | Notation.lean audit;<br>wrapper-first policy;<br>global build rule                                                                                                                                                                                                                               |
| Definitions                     | Bounded migration package;<br>declaration migrated count;<br>import graph delta;<br>consumer breakage audit;<br>local full-build evidence                                                                                                                                                        |
| Proof target                    | Show refactor phases cannot close without regression evidence and cannot delete legacy/status material prematurely.                                                                                                                                                                              |
| Proof sketch                    | Planning proof: scratchpad fields and workflow policy require migration metrics and build evidence for refactor closeout.                                                                                                                                                                        |
| Verification / audit            | Generator renders workflow and scratchpad fields;<br>Lean build not run here.                                                                                                                                                                                                                    |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. For Phase 1.1.2, fill concrete declaration families and migration batch size before editing code. |

### Object or quantity LEG22 – Lean/mathlib version migration policy

**Table legend.**   W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                 |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG22                                                                                                                                                                                                                                                                                                  |
| Object status                   | G                                                                                                                                                                                                                                                                                                      |
| Context class                   | LeanGovernanceObject                                                                                                                                                                                                                                                                                   |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                     |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                         |
| Documented                      | yes                                                                                                                                                                                                                                                                                                    |
| Phase                           | 1.1.1.1.0.4                                                                                                                                                                                                                                                                                            |
| Source objects                  | Lean 4.28.0/mathlib baseline;<br>future number-theory API pressure                                                                                                                                                                                                                                     |
| Definitions                     | Baseline freeze;<br>explicit upgrade gate;<br>mirror/non-operational isolation of noncomputable/classical pressure                                                                                                                                                                                     |
| Proof target                    | Prevent silent mathlib version drift or operational noncomputable leakage.                                                                                                                                                                                                                             |
| Proof sketch                    | Planning proof: every phase has lean_mathlib_version_policy and mathlib_complication_log;<br>workflow mandates phase-boundary upgrade gate.                                                                                                                                                            |
| Verification / audit            | Rendered policy exists;<br>no external version claims or Lean build made.                                                                                                                                                                                                                              |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When a future phase needs a mathlib import, cite/verify the exact baseline API or open an upgrade gate. |

Object or quantity LEG23 – Phase-type-specific scratchpad schema and governance retrospective

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG23                                                                                                                                     |
| Object status                   | G                                                                                                                                         |
| Context class                   | DocumentationObject                                                                                                                       |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | yes                                                                                                                                       |
| Phase                           | 1.1.1.1.0.5                                                                                                                               |
| Source objects                  | Scratchpad schema;<br>governance-to-math ratio concern;<br>refactor vs derivation evidence mismatch                                       |
| Definitions                     | phase_type plus type-specific contract and governance_overhead_review                                                                     |
| Proof target                    | Make scratchpad evidence fit the phase type and force a retrospective after first code refactor.                                          |
| Proof sketch                    | Planning proof: generator requires/renders phase_type fields and workflows define type-specific obligations.                              |
| Verification / audit            | Rendered scratchpad includes the new fields;<br>enforcement is schema-level, not Lean-level.                                              |
| Next documentation action       | After Phase 1.1.2, record which governance gates prevented concrete errors and whether to flatten Phase 3+.                               |

Object or quantity LEG24 – Phase 3 to Phase 4 consumer-contract direction

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG24                                                                                                                                                                                                                                                                                     |
| Object status                   | G                                                                                                                                                                                                                                                                                         |
| Context class                   | LeanGovernanceObject                                                                                                                                                                                                                                                                      |
| Significance tier               | F1                                                                                                                                                                                                                                                                                        |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                            |
| Documented                      | yes                                                                                                                                                                                                                                                                                       |
| Phase                           | 1.1.1.1.0.6                                                                                                                                                                                                                                                                               |
| Source objects                  | Phase 3.6 BoundaryOperatorProfile;<br>Phase 4.1 ExecBoundaryMatrix;<br>GEN-1 generator policy                                                                                                                                                                                             |
| Definitions                     | Phase 3 exports consumer-ready profile;<br>Phase 4 consumes it;<br>no reverse import and no duplicate generator/Schur logic                                                                                                                                                               |
| Proof target                    | Show wrapper order respects derivation while allowing downstream consumer shape to constrain public API.                                                                                                                                                                                  |
| Proof sketch                    | Planning proof: phase goals and workflow require contract direction fields before either phase is prepared.                                                                                                                                                                               |
| Verification / audit            | Rendered phase goals updated for 3.6 and 4.1;<br>no Lean code changed.                                                                                                                                                                                                                    |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When Phase 3.6 is prepared, write the exact carrier/operator fields expected by Phase 4.1. |

Object or quantity LEG25 – IdealThread/CutSpec.full extremal-route contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                              |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG25                                                                                                                                                                                                                                                                                                               |
| Object status                   | W                                                                                                                                                                                                                                                                                                                   |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                              |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                  |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                      |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                 |
| Phase                           | 1.1.1.1.0.7                                                                                                                                                                                                                                                                                                         |
| Source objects                  | SubstrateClass.InfiniteThread;<br>ToC.IdealThread.carrier/family/coarsen lemmas;<br>Finite.CutSpec.full;<br>ExistingIdealThreadNoGo;<br>FullBoundaryAudit.                                                                                                                                                          |
| Definitions                     | IdealThread route: carrier L = singleton {cell L};<br>CutSpec.full route: carrier L = T.carrier L;<br>string analogy: comparison-only interpretation.                                                                                                                                                               |
| Proof target                    | Planning-level proof that these routes must not be conflated before implementation.                                                                                                                                                                                                                                 |
| Proof sketch                    | The code facts show singleton collapse for IdealThread and full carrier for CutSpec.full. Therefore any nontrivial full-boundary matrix must use full/generated boundary data, not the existing IdealThread carrier, unless a later theorem constructs a lift.                                                      |
| Verification / audit            | Code references checked in repo_snapshot. No Lean build run;<br>this is a planning contract based on existing source.                                                                                                                                                                                               |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When S07/S08/BR06 activate, replace this summary by code-line proof dossiers and concrete finite instance witnesses. |

Object or quantity S07 – IdealThread singleton directed-family profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                             |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | S07                                                                                                                                                                                                                                                                                                |
| Object status                   | G                                                                                                                                                                                                                                                                                                  |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                   |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                 |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                     |
| Documented                      | no                                                                                                                                                                                                                                                                                                 |
| Phase                           | 1.2.7                                                                                                                                                                                                                                                                                              |
| Source objects                  | InfiniteThread;<br>IdealThread.carrier;<br>IdealThread.family;<br>DeterministicSubstrateClass.                                                                                                                                                                                                     |
| Definitions                     | Neutral structure containing singleton carrier, family embedding, rooted/forward facts, and deterministic coarsen fixed-point.                                                                                                                                                                     |
| Proof target                    | Prove/export carrier singleton, family carrier equality, rootedness/forward property, and coarsen_carrier_eq_prev under deterministic substrate.                                                                                                                                                   |
| Proof sketch                    | Mostly projection/rfl proofs from ToC.Contract. The deterministic coarsen proof uses parent uniqueness. Avoid phase-semantic aliases and avoid large eliminator proofs.                                                                                                                            |
| Verification / audit            | Source theorems exist;<br>neutral API and concrete finite instance traversal not yet extracted.                                                                                                                                                                                                    |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. During Phase 1.2.7, record exact module manifest and concrete reference/variation thread instances. |



Object or quantity S08 – CutSpec.full maximal-carrier profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | S08                                                                                                                                                                                                                                                                                                                    |
| Object status                   | G                                                                                                                                                                                                                                                                                                                      |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                                                       |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                     |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                         |
| Documented                      | no                                                                                                                                                                                                                                                                                                                     |
| Phase                           | 1.2.8                                                                                                                                                                                                                                                                                                                  |
| Source objects                  | TruncatedFamily;<br>CutSpec.full;<br>RegionCore;<br>BoundaryPorts;<br>FullBoundaryAudit.                                                                                                                                                                                                                               |
| Definitions                     | Neutral structure exposing (CutSpec.full T).carrier L = T.carrier L and full-region route facts.                                                                                                                                                                                                                       |
| Proof target                    | Export full_carrier and sufficient route facts for Phase 3.0 without duplicating CutSpec implementation.                                                                                                                                                                                                               |
| Proof sketch                    | The core theorem is rfl from CutSpec.full. Additional facts should be projections/constructor lemmas from RegionCore/BoundaryPorts if needed.                                                                                                                                                                          |
| Verification / audit            | Source theorem exists;<br>no neutral API yet.                                                                                                                                                                                                                                                                          |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. During Phase 1.2.8, decide whether this belongs in existing Finite/CutSpec wrappers or a compact FullCutProfile module. |

Object or quantity BR06 – Boundary-route extrema contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                   |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | BR06                                                                                                                                                                                                                                                                     |
| Object status                   | G                                                                                                                                                                                                                                                                        |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                         |
| Significance tier               | F1                                                                                                                                                                                                                                                                       |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                           |
| Documented                      | no                                                                                                                                                                                                                                                                       |
| Phase                           | 3.0                                                                                                                                                                                                                                                                      |
| Source objects                  | S07;<br>S08;<br>ExistingIdealThreadNoGo;<br>FullBoundaryAudit;<br>BoundaryOperatorProfile consumer needs.                                                                                                                                                                |
| Definitions                     | Route contract classifying a proof or computation as thread-scalar, full-boundary, or generated-intermediate.                                                                                                                                                            |
| Proof target                    | Before Schur/DtN neutral wrappers, prove/record that nontrivial matrix geometry is sourced from full/generated boundary data and not from IdealThread singleton route.                                                                                                   |
| Proof sketch                    | Use S07 singleton facts plus ExistingIdealThreadNoGo to block non-singleton claims from IdealThread;<br>use S08 full-carrier facts to authorize full-region matrix route.                                                                                                |
| Verification / audit            | Pending Phase 3.0 preparation;<br>must include concrete finite instance traversal and import firewall.                                                                                                                                                                   |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Prepare exact module manifest for BoundaryRouteContract before Phase 3.1. |

Object or quantity BR07 – Thread-recursion arithmetic-shadow audit profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | BR07                                                                                                                                                                                                                                                                                  |
| Object status                   | Y                                                                                                                                                                                                                                                                                     |
| Context class                   | LeanSourceObject                                                                                                                                                                                                                                                                      |
| Significance tier               | F1                                                                                                                                                                                                                                                                                    |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                        |
| Documented                      | no                                                                                                                                                                                                                                                                                    |
| Phase                           | 4.0                                                                                                                                                                                                                                                                                   |
| Source objects                  | RecursiveSchurSource;<br>MobiusTransfer;<br>CharpolyExec;<br>S07;<br>BR06;<br>operational discriminant chain.                                                                                                                                                                         |
| Definitions                     | Audit profile locating the scalar recurrence source: IdealThread singleton, full-boundary diagonal, or generated intermediate route.                                                                                                                                                  |
| Proof target                    | Show the precise code provenance of sigma/recurrence/transfer entries before interpreting arithmetic outputs as thread-generated.                                                                                                                                                     |
| Proof sketch                    | Trace recurrenceLedger.recursiveSource, boundaryMatrixAt diagonal use, MobiusTransfer parameters, and discriminant output fields. Do not accept D=-3/-4/string claims until the route is code-derived.                                                                                |
| Verification / audit            | Pending Phase 4.0;<br>no claim of completed route provenance yet.                                                                                                                                                                                                                     |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When Phase 4.0 activates, fill source-to-CNNA mapping and concrete executable witness. |

Object or quantity LEG26 – Reflective thread/full-boundary analogy diagnostic boundary

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                        |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG26                                                                                                                                         |
| Object status                   | K                                                                                                                                             |
| Context class                   | GovernancePolicyObject                                                                                                                        |
| Significance tier               | F4                                                                                                                                            |
| Documentation depth requirement | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                 |
| Documented                      | no                                                                                                                                            |
| Phase                           | 19.5                                                                                                                                          |
| Source objects                  | LEG25;<br>LT5;<br>S07;<br>S08;<br>BR06;<br>BR07;<br>derivation-graph metadata.                                                                |
| Definitions                     | Secondary diagnostic boundary for string-like/thread-full analogies.                                                                          |
| Proof target                    | Ensure analogy-based reports remain downstream documentation and never become CNNA-generative inputs.                                         |
| Proof sketch                    | Require closed proof dossiers for underlying objects before comparison;<br>generated graph data may diagnose but not create CNNA mathematics. |
| Verification / audit            | Future-only;<br>not-before gates not closed.                                                                                                  |

|                           |                                                                                    |
|---------------------------|------------------------------------------------------------------------------------|
| Next documentation action | Keep black/side target until after Schur/DtN and arithmetic dossiers are complete. |
|---------------------------|------------------------------------------------------------------------------------|

Object or quantity LEG27 – Horizontal/vertical Schur-duality dependency-order lock

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                       |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG27                                                                                                                                                                                                                                                                                                        |
| Object status                   | G                                                                                                                                                                                                                                                                                                            |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                       |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                           |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                               |
| Documented                      | yes                                                                                                                                                                                                                                                                                                          |
| Phase                           | 1.1.1.1.0.8                                                                                                                                                                                                                                                                                                  |
| Source objects                  | Review insight;<br>route objects S05/O05/TM0/CD7/BR06/BR07;<br>existing v30 route locks.                                                                                                                                                                                                                     |
| Definitions                     | SchurDualityOrderLock.                                                                                                                                                                                                                                                                                       |
| Proof target                    | Show at the planning level that a horizontal/vertical Schur bridge is a future theorem target with strict source dependencies and no backward input rights.                                                                                                                                                  |
| Proof sketch                    | The lock is closed by adding GNG14, BR08/BR09/CD8 dependency corrections, and phases 10.4.1/10.4.2 after TM0. No mathematical theorem is asserted here.                                                                                                                                                      |
| Verification / audit            | Validate that no white phase precedes the active cursor and that later bridge phases depend on source routes rather than vice versa.                                                                                                                                                                         |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When BR08 activates, replace the planning-level proof sketch with checked code and concrete finite witnesses. |

Object or quantity BR08 – Horizontal/vertical Schur-duality bridge

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                            |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | BR08                                                                                                                                                                                                                                              |
| Object status                   | K                                                                                                                                                                                                                                                 |
| Context class                   | LeanTargetObject                                                                                                                                                                                                                                  |
| Significance tier               | F1                                                                                                                                                                                                                                                |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                    |
| Documented                      | no                                                                                                                                                                                                                                                |
| Phase                           | 10.4.1                                                                                                                                                                                                                                            |
| Source objects                  | S05 DtN boundary operator;<br>BR06 boundary route extrema;<br>BR07 thread-recursion arithmetic shadow;<br>S07/S08 route profiles;<br>TM0-TM2 transfermatrix chain;<br>CD7 Hurwitz stop;<br>O05 operational discriminant.                          |
| Definitions                     | SchurHorizontalVerticalDualityProfile with explicit horizontal block/sector elimination map, vertical level-recursion map, and a comparison/commutation or factorization statement over one derived Dirichlet-Laplacian.                          |
| Proof target                    | Prove that horizontal sector Schur-complementing and vertical level-recursive Schur-complementing are compatible factorizations/aspects of the same multiscale Schur decomposition, or split/obstruct the statement at the first precise failure. |

|                           |                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Positive route: first instantiate one finite generated tree/cut;<br>build the full Dirichlet-Laplacian with both sector and level block indices;<br>express both elimination orders;<br>prove entrywise equality/naturality where defined. Avoid importing physical analogy or CD/arithmetic conclusions as assumptions.                                    |
| Verification / audit      | Requires local Lean theorem evidence and concrete computable instance traversal. Must audit whether the two decompositions actually commute, require side conditions, or only share a weaker universal source.                                                                                                                                              |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Prepare only after S05/O05/TM0/CD7/BR06/BR07 are neutral and documented;<br>then split into smaller lemmas if the block-index product structure is too deep. |

### Object or quantity BR09 – Subsystem-induced invariant bifurcation profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                    |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | BR09                                                                                                                                                                                                                                                      |
| Object status                   | K                                                                                                                                                                                                                                                         |
| Context class                   | LeanTargetObject                                                                                                                                                                                                                                          |
| Significance tier               | F1                                                                                                                                                                                                                                                        |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                            |
| Documented                      | no                                                                                                                                                                                                                                                        |
| Phase                           | 10.4.1                                                                                                                                                                                                                                                    |
| Source objects                  | BR08 plus finite cut/subsystem boundary data and route-specific invariant records.                                                                                                                                                                        |
| Definitions                     | SubsystemSchurInvariantBifurcationProfile.                                                                                                                                                                                                                |
| Proof target                    | Document, after BR08, which invariant families are outputs of horizontal Schur route and which are outputs of vertical Schur route, without conflating explanation with proof.                                                                            |
| Proof sketch                    | Use proofbook diagrams/tables and checked code references to show source paths. The profile is explanatory until backed by BR08 theorem objects.                                                                                                          |
| Verification / audit            | Audit that this profile remains downstream and cannot create formulas or public API names.                                                                                                                                                                |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill after BR08, not during Phase 3/4 source construction. |

### Object or quantity LEG28 – Heuristic Eisenstein/Gauss lattice test-matrix boundary contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG28                                                                                                                                                                                                                                           |
| Object status                   | G                                                                                                                                                                                                                                               |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                          |
| Significance tier               | F1                                                                                                                                                                                                                                              |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                  |
| Documented                      | yes                                                                                                                                                                                                                                             |
| Phase                           | 1.1.1.1.0.9                                                                                                                                                                                                                                     |
| Source objects                  | EXEC-2, canonical generator policy, post-97 source audit, CM/j target boundary.                                                                                                                                                                 |
| Definitions                     | The post-97 spiral b-ary tree construction is accepted only as a future test-matrix and comparison profile, never as generator input.                                                                                                           |
| Proof target                    | Show as a planning invariant that future lattice tests must consume code-defined branching and derived CM/j objects rather than defining them.                                                                                                  |
| Proof sketch                    | The decision follows from the existing derived-only and EXEC-2 constraints: angle profiles and complex coordinates are not operational generator data; they can be compared after the finite generator, lattice, and CM/j target objects exist. |
| Verification / audit            | Checked against post #97 content: b-ary addresses, b>=2 heuristic, 120/90 degree profiles, L=3/L=4 closure claims, hierarchical vs spiral distinction. No Lean code changed.                                                                    |

|                           |                                                                                                                                                                                                                                                                                                                            |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When 8.5 starts, fill concrete rows and module manifest; keep all analytic/irrational material in mirror/comparison fields. |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Object or quantity LEG29 – General early finite computation test-matrix contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                            |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG29                                                                                                                                                                                                                                                                                             |
| Object status                   | G                                                                                                                                                                                                                                                                                                 |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                            |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                    |
| Documented                      | yes - planning invariant                                                                                                                                                                                                                                                                          |
| Phase                           | 1.1.1.1.0.10                                                                                                                                                                                                                                                                                      |
| Source objects                  | LEG14, LEG15, LEG28, canonical generator policy, v32 L09/L10/M08/BR10 sentinel-test objects.                                                                                                                                                                                                      |
| Definitions                     | A generic finite test matrix is a family of code-defined finite input rows with operational output witnesses; sentinel rows are optional comparison rows and cannot replace generic coverage.                                                                                                     |
| Proof target                    | Show as a planning invariant that early computation phases cannot close on special examples alone.                                                                                                                                                                                                |
| Proof sketch                    | Insert the invariant before active implementation; make it global through workflow and guardrail sections; route specialized Eisenstein/Gauss material through O07/L09/L10 rather than through generator definitions.                                                                             |
| Verification / audit            | Check YAML has one active cursor, no white predecessor before active phase, O07/L09 separation, and GNG16.                                                                                                                                                                                        |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When Phase 1.1.1.1.1 prepares module manifests, record exact test module placement and row schema. |

Object or quantity L09 – Static branching Eisenstein/Gauss lattice test matrix

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                               |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | L09                                                                                                                                                                                                                                                                  |
| Object status                   | Y                                                                                                                                                                                                                                                                    |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                                  |
| Significance tier               | F2                                                                                                                                                                                                                                                                   |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                        |
| Documented                      | no                                                                                                                                                                                                                                                                   |
| Phase                           | 8.5                                                                                                                                                                                                                                                                  |
| Source objects                  | code-defined static b, L_max, L06, L04, O05, M05; O07 generic finite computation test matrix profile                                                                                                                                                                 |
| Definitions                     | Finite rows for Eisenstein and Gauss test profiles using rational/integer operational invariants.                                                                                                                                                                    |
| Proof target                    | Add static Eisenstein/Gauss sentinel rows to the general O07 test matrix; do not treat them as a complete test suite.                                                                                                                                                |
| Proof sketch                    | Pending. Must use checked finite code/test rows as primary evidence; forum geometry is a heuristic source, not proof.                                                                                                                                                |
| Verification / audit            | Pending. Must check code-defined branching, no duplicate generator logic, EXEC-2 boundary, and no noncomputable leak into operational path. Audit must confirm that special Eisenstein/Gauss rows are sentinels inside generic coverage, not the only passing tests. |

|                           |                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| Next documentation action | After O07 exists, specify generic row set first, then static sentinel rows and expected mirror/comparison targets. |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|

### Object or quantity L10 – Dynamic branching and L\_max lattice test sweep

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                               |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | L10                                                                                                                                                                                                                                                                  |
| Object status                   | Y                                                                                                                                                                                                                                                                    |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                                  |
| Significance tier               | F2                                                                                                                                                                                                                                                                   |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                        |
| Documented                      | no                                                                                                                                                                                                                                                                   |
| Phase                           | 8.6                                                                                                                                                                                                                                                                  |
| Source objects                  | canonical generated branch-family route, L09;<br>O07 generic finite computation test matrix profile                                                                                                                                                                  |
| Definitions                     | Dynamic-branching extension of L09 over legal code-defined branch policies.                                                                                                                                                                                          |
| Proof target                    | Extend O07 to dynamic branching and L_max sweeps without duplicating generator logic or specializing to CM targets.                                                                                                                                                  |
| Proof sketch                    | Pending. Must use checked finite code/test rows as primary evidence;<br>forum geometry is a heuristic source, not proof.                                                                                                                                             |
| Verification / audit            | Pending. Must check code-defined branching, no duplicate generator logic, EXEC-2 boundary, and no noncomputable leak into operational path. Audit must confirm that special Eisenstein/Gauss rows are sentinels inside generic coverage, not the only passing tests. |
| Next documentation action       | Fill during the phase natural implementation;<br>include concrete test matrix rows and pass/fail criteria.                                                                                                                                                           |

### Object or quantity M08 – Eisenstein/Gauss CM and j special-value test targets

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | M08                                                                                                                                                                                                                                                                                                                                                      |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                                                        |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                                                                                                                      |
| Significance tier               | F2                                                                                                                                                                                                                                                                                                                                                       |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                            |
| Documented                      | no                                                                                                                                                                                                                                                                                                                                                       |
| Phase                           | 8.5                                                                                                                                                                                                                                                                                                                                                      |
| Source objects                  | L04, O05, M05;<br>O07 generic finite computation test matrix profile                                                                                                                                                                                                                                                                                     |
| Definitions                     | Sentinel comparison values for CM/j special cases, available only after the generic finite computation matrix and relevant CM/j objects exist.                                                                                                                                                                                                           |
| Proof target                    | Record special-value targets only after CM and j objects are available;<br>do not operationally evaluate analytic j in ExecComplex.                                                                                                                                                                                                                      |
| Proof sketch                    | Pending. Must use checked finite code/test rows as primary evidence;<br>forum geometry is a heuristic source, not proof.                                                                                                                                                                                                                                 |
| Verification / audit            | Pending. Must check code-defined branching, no duplicate generator logic, EXEC-2 boundary, and no noncomputable leak into operational path. Audit must confirm that special Eisenstein/Gauss rows are sentinels inside generic coverage, not the only passing tests. v37 audit: modular/CM path only;<br>cannot gate finite spectral/physical consumers. |



|                           |                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------|
| Next documentation action | Fill during the phase natural implementation;<br>include concrete test matrix rows and pass/fail criteria. |
|---------------------------|------------------------------------------------------------------------------------------------------------|

### Object or quantity BR10 – Hierarchical versus spiral embedding collision and closure audit

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                               |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">BR10</a>                                                                                                                                                                                                                                                 |
| Object status                   | Y                                                                                                                                                                                                                                                                    |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                                  |
| Significance tier               | F2                                                                                                                                                                                                                                                                   |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                        |
| Documented                      | no                                                                                                                                                                                                                                                                   |
| Phase                           | <a href="#">8.6</a>                                                                                                                                                                                                                                                  |
| Source objects                  | finite address structure, L09/L10, post #97 notes;<br>O07 generic finite computation test matrix profile                                                                                                                                                             |
| Definitions                     | Audit separating angle-invariant hierarchical overlays from angle-dependent spiral closure/collision behavior.                                                                                                                                                       |
| Proof target                    | Audit hierarchical-vs-spiral closure/collision behavior as a diagnostic overlay on generic finite rows.                                                                                                                                                              |
| Proof sketch                    | Pending. Must use checked finite code/test rows as primary evidence;<br>forum geometry is a heuristic source, not proof.                                                                                                                                             |
| Verification / audit            | Pending. Must check code-defined branching, no duplicate generator logic, EXEC-2 boundary, and no noncomputable leak into operational path. Audit must confirm that special Eisenstein/Gauss rows are sentinels inside generic coverage, not the only passing tests. |
| Next documentation action       | Fill during the phase natural implementation;<br>include concrete test matrix rows and pass/fail criteria.                                                                                                                                                           |

### Object or quantity LEG30 – Regime-typed early test-matrix contract

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                  |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">LEG30</a>                                                                                                                                                                                                                                                   |
| Object status                   | G                                                                                                                                                                                                                                                                       |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                  |
| Significance tier               | F1                                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                          |
| Documented                      | yes                                                                                                                                                                                                                                                                     |
| Phase                           | <a href="#">1.1.1.1.0.11</a>                                                                                                                                                                                                                                            |
| Source objects                  | O07, noOuter/HasOuter discipline, EXEC-2, IDEAL_infty projection discipline, post #97                                                                                                                                                                                   |
| Definitions                     | Global workflow contract for derived regime tags on test rows.                                                                                                                                                                                                          |
| Proof target                    | Prevent false promotion of Heegner/class-number-one or Eisenstein/Gauss comparison data into closed/noOuter generator evidence.                                                                                                                                         |
| Proof sketch                    | Planning proof: insert a green order lock before active implementation, add global no-go GNG17, add O08/N18/BR11/BR12, and make later phases consume the schema rather than define regimes from it.                                                                     |
| Verification / audit            | Audit in generated phase/object/workflow tables;<br>no Lean code.                                                                                                                                                                                                       |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Use during Phase 1.1.1.1.1 module manifest and later O08 implementation. |

Object or quantity O08 – Regime-typed finite computation test matrix profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | O08                                                                                                                                                                                             |
| Object status                   | Y                                                                                                                                                                                               |
| Context class                   | LeanTargetObject                                                                                                                                                                                |
| Significance tier               | F2                                                                                                                                                                                              |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                   |
| Documented                      | no                                                                                                                                                                                              |
| Phase                           | 4.0.2                                                                                                                                                                                           |
| Source objects                  | O07;<br>noOuter/HasOuter split;<br>canonical generator outputs                                                                                                                                  |
| Definitions                     | Row schema with branch mode, L_max, executable witness, derived regime tag, and optional comparison columns.                                                                                    |
| Proof target                    | Show early finite computations are both general and regime-aware.                                                                                                                               |
| Proof sketch                    | Pending Lean implementation. Expected proof path: build row constructors from canonical generator output and regime witness; prove sentinel/comparison columns cannot alter operational regime. |
| Verification / audit            | Pending local Lean build and row witness audit.                                                                                                                                                 |
| Next documentation action       | Fill before Phase 4.0.2 implementation.                                                                                                                                                         |

Object or quantity N18 – Heegner regime-projection classification profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | N18                                                                                                                                                                                                                                                  |
| Object status                   | Y                                                                                                                                                                                                                                                    |
| Context class                   | LeanTargetObject                                                                                                                                                                                                                                     |
| Significance tier               | F1                                                                                                                                                                                                                                                   |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                       |
| Documented                      | no                                                                                                                                                                                                                                                   |
| Phase                           | 7.2.1                                                                                                                                                                                                                                                |
| Source objects                  | ClassNumberProfile, HeegnerIndicatorProfile, O08, BR11                                                                                                                                                                                               |
| Definitions                     | Consumer-only classification of h(D)=1/Heegner targets by derived CNNA regime.                                                                                                                                                                       |
| Proof target                    | Separate number-theoretic class-number-one from closed noOuter physics/regime status.                                                                                                                                                                |
| Proof sketch                    | Pending. h(D)=1 can support PID/UFD classification in number theory after derivation, but not noOuter closure. Other Heegner targets must be projection/comparison rows unless a noOuter derivation exists.                                          |
| Verification / audit            | Check against class-number data and external list only after CNNA data are derived.                                                                                                                                                                  |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill after 7.2 and before any Heegner interpretation. |

Object or quantity BR11 – IDEAL\_infty projection and complement-family Heegner boundary

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field     | Record |
|-----------|--------|
| Object ID | BR11   |

|                                 |                                                                                                                                                                                                                        |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object status                   | Y                                                                                                                                                                                                                      |
| Context class                   | LeanTargetObject                                                                                                                                                                                                       |
| Significance tier               | F1                                                                                                                                                                                                                     |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                         |
| Documented                      | no                                                                                                                                                                                                                     |
| Phase                           | 7.2.1                                                                                                                                                                                                                  |
| Source objects                  | IDEAL_infty, complement families, noOuter/HasOuter split, N18                                                                                                                                                          |
| Definitions                     | Boundary that additional Heegner phenomena are first treated as projection/complement-family cases.                                                                                                                    |
| Proof target                    | Prevent multiple closed-root interpretations without derived noOuter proof.                                                                                                                                            |
| Proof sketch                    | Pending. Must show any non-root Heegner row carries a projection/complement-family provenance or remains comparison-only.                                                                                              |
| Verification / audit            | Audit against O08/N18 rows and regime tags.                                                                                                                                                                            |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill together with N18. |

### Object or quantity BR12 – noOuter REAL-root Eisenstein/Gauss sentinel hypothesis

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                               |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | BR12                                                                                                                                                                                                                                 |
| Object status                   | Y                                                                                                                                                                                                                                    |
| Context class                   | LeanTargetObject                                                                                                                                                                                                                     |
| Significance tier               | F1                                                                                                                                                                                                                                   |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                       |
| Documented                      | no                                                                                                                                                                                                                                   |
| Phase                           | 4.0.2                                                                                                                                                                                                                                |
| Source objects                  | post #97;<br>O08;<br>M08;<br>O05;<br>noOuter root regime                                                                                                                                                                             |
| Definitions                     | Sentinel hypothesis for D=-3/D=-4 as closed noOuter REAL-root candidates.                                                                                                                                                            |
| Proof target                    | Test whether the noOuter empty-set REAL approximant uniquely matches Eisenstein/Gauss sentinels, without assuming the result.                                                                                                        |
| Proof sketch                    | Pending. Use finite noOuter-root rows and compare D/j targets only after O05/M08 exist.                                                                                                                                              |
| Verification / audit            | Audit must reject use of the heuristic as generator input and keep tau/j mirror-only.                                                                                                                                                |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during 4.0.2/8.5 implementation. |

### Object or quantity LEG31 – Computable scalar architecture contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field             | Record                 |
|-------------------|------------------------|
| Object ID         | LEG31                  |
| Object status     | G                      |
| Context class     | GovernancePolicyObject |
| Significance tier | F0                     |

|                                 |                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                                                            |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                                |
| Phase                           | 1.1.1.1.0.12                                                                                                                                                                                                                                                                                                                                       |
| Source objects                  | O00, LEG20, LEG22, current code audit, mathlib Complex and QuadraticAlgebra docs                                                                                                                                                                                                                                                                   |
| Definitions                     | Top-level scalar policy separating rational ExecComplex, required exact quadratic schema/provenance, analytic Complex mirror, and late CD noncircularity audit.                                                                                                                                                                                    |
| Proof target                    | Prevent false assumption that ExecComplex is a complete C layer or that CD can replace it before proof.                                                                                                                                                                                                                                            |
| Proof sketch                    | Code audit shows ExecComplex is QxQ and bridge maps to Complex;<br>CD bridge currently records no identification. Mathlib audit shows QuadraticAlgebra is coordinate-based and AdjoinRoot adjoins a root through a quotient pattern. Therefore the CNNA exact quadratic layer must be discriminant-indexed and proof-carrying, not a free sqrt(D). |
| Verification / audit            | Generated table;<br>no Lean build. Future verification in 1.2.1.1/1.2.1.2/4.0.3/CD9.                                                                                                                                                                                                                                                               |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. Use LEG32 in Phase 1.1.1.1.1 module manifest before proposing scalar modules.                                                              |

### Object or quantity LEG32 – Derived-only exact quadratic provenance contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG32                                                                                                                                                                                                                                                                                                                            |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                           |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                                               |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                                          |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                              |
| Phase                           | 1.1.1.1.0.13                                                                                                                                                                                                                                                                                                                     |
| Source objects                  | LEG31, O00, O05/O06 plan, mathlib QuadraticAlgebra and AdjoinRoot reference docs, CD9 noncircularity rule                                                                                                                                                                                                                        |
| Definitions                     | No-free-sqrt scalar provenance contract. The exact quadratic layer is a relation-generated algebra over a derived D, not an axiom or analytic square root.                                                                                                                                                                       |
| Proof target                    | Prevent O09 from becoming a postulated algebraic-number source. Ensure every exact quadratic instance cites a proof-carrying D-source.                                                                                                                                                                                           |
| Proof sketch                    | Use dependency order: first operational Schur/DtN/charpoly/determinant produces D;<br>then O09 constructs a two-coordinate algebra with theta^2 = D;<br>then BR14 projects trace/norm/invariants. Missing D forces a backjump to O05/O06.                                                                                        |
| Verification / audit            | Planning lock only. Web/mathlib references support the construction pattern;<br>Lean implementation deferred to O09/O10/BR14 phases.                                                                                                                                                                                             |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. During Phase 1.1.1.1.1, scalar-related module manifests must classify schema versus D-instantiation versus mirror proof. |

### Object or quantity O09 – Required exact quadratic operational scalar schema

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | O09                                                                                                                                                                                            |
| Object status                   | Y                                                                                                                                                                                              |
| Context class                   | LeanTargetObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |

|                           |                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase                     | 1.2.1.2                                                                                                                                                                                                                                                                                                 |
| Source objects            | O00, O05, N02, N04, N06, LEG31                                                                                                                                                                                                                                                                          |
| Definitions               | Required parameterized exact quadratic schema for a+b theta with theta^2=D. No concrete D is introduced here; instantiation belongs to O10/BR14.                                                                                                                                                        |
| Proof target              | Support exact imaginary-quadratic computations while preserving derived-only provenance and not promoting analytic Complex to operational path.                                                                                                                                                         |
| Proof sketch              | Pending. Expected route: define/wrap two-coordinate algebra; prove coordinate addition/multiplication/conjugation/trace/norm; keep D as parameter; require O10 for concrete D-ledger rows.                                                                                                              |
| Verification / audit      | Pending local Lean build, mathlib import check, and executable witness tests.                                                                                                                                                                                                                           |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill before any schema module named ExactQuadraticScalar/ExactQuadraticComplex or equivalent is created. |

### Object or quantity O10 – Proof-carrying discriminant-indexed quadratic scalar instance

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                         |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | O10                                                                                                                                                                                                                                                                            |
| Object status                   | Y                                                                                                                                                                                                                                                                              |
| Context class                   | LeanTargetObject                                                                                                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                 |
| Documented                      | no                                                                                                                                                                                                                                                                             |
| Phase                           | 4.5.1                                                                                                                                                                                                                                                                          |
| Source objects                  | O05/O06 discriminant ledger, O09 schema, LEG32                                                                                                                                                                                                                                 |
| Definitions                     | Instance of ExactQuadraticScalar at ledger.D with theta^2 = D; coordinate elements a+b theta; proof-carrying link to D-source.                                                                                                                                                 |
| Proof target                    | Show exact quadratic coordinates are generated from the operational discriminant and not independently assumed.                                                                                                                                                                |
| Proof sketch                    | Pending. Expected route: consume D certificate; define instance; prove theta-square relation; derive conjugation/trace/norm formulas by coordinate equations; record executable witnesses.                                                                                     |
| Verification / audit            | Pending local Lean build and finite computation rows after O05 correction and O09 schema.                                                                                                                                                                                      |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Do not implement before O09 schema and O05/O06 public correction are available. |

### Object or quantity BR13 – ExecComplex to exact-quadratic projection bridge

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | BR13                                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                                              |
| Context class                   | LeanTargetObject                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |

|                           |                                                                                                                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documented                | no                                                                                                                                                                                                                |
| Phase                     | 4.0.3                                                                                                                                                                                                             |
| Source objects            | O00, O09, O10, BR14, O07, O08, BR12                                                                                                                                                                               |
| Definitions               | Bridge from exact quadratic computations and D-source instantiations to rational/integer operational invariants and ExecComplex-compatible reports.                                                               |
| Proof target              | Ensure exact quadratic layer enriches tests without redefining generator or rational floor.                                                                                                                       |
| Proof sketch              | Pending. Use row constructors and projection lemmas: exact quadratic coordinate -> trace/norm/discriminant/integer invariant -> allowed operational column.                                                       |
| Verification / audit      | Pending finite computation matrix rows and local build.                                                                                                                                                           |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during 4.0.3. |

### Object or quantity BR14 – Discriminant-source to exact-quadratic algebra instantiation bridge

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                              |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | BR14                                                                                                                                                                                                                                |
| Object status                   | Y                                                                                                                                                                                                                                   |
| Context class                   | LeanTargetObject                                                                                                                                                                                                                    |
| Significance tier               | F1                                                                                                                                                                                                                                  |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                      |
| Documented                      | no                                                                                                                                                                                                                                  |
| Phase                           | 4.5.1                                                                                                                                                                                                                               |
| Source objects                  | O05/O06, O09, O10, LEG32                                                                                                                                                                                                            |
| Definitions                     | Bridge from proof-carrying D-source to exact quadratic algebra instance and from that instance to allowed invariant projections.                                                                                                    |
| Proof target                    | Every exported quadratic invariant has a provenance path through the D-ledger and O09/O10, not through a free formula.                                                                                                              |
| Proof sketch                    | Pending. Expected route: define bridge record with D-source, exact quadratic instance, and projection lemmas; prove trace/norm/discriminant projection equations structurally.                                                      |
| Verification / audit            | Pending after 4.5 and 1.2.1.2 are available.                                                                                                                                                                                        |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill during Phase 4.5.1 preparation. |

### Object or quantity CD9 – Cayley-Dickson C-layer noncircularity audit

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | CD9                                                                                                                                                                                            |
| Object status                   | K                                                                                                                                                                                              |
| Context class                   | ReferenceGateObject                                                                                                                                                                            |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |
| Phase                           | 10.4.3                                                                                                                                                                                         |
| Source objects                  | CD7, CD8, BR08, BR09, O09, ArithmeticCDBridge                                                                                                                                                  |
| Definitions                     | Late audit determining whether CD supplies a genuine scalar C surface comparable to O09/O10 without serving as early scalar source.                                                            |
| Proof target                    | Prove noncircularity before any CD-derived C layer is consumed operationally; it cannot supply sqrt(D) for O09/O10.                                                                            |



|                           |                                                                                                                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Pending. Must show CD surface is generated from closed CD/Schur data and not from ExecComplex/Complex mirror or later arithmetic consumers; otherwise mark blocked/comparison-only.                                                                                |
| Verification / audit      | Pending after BR08/CD8 closure.                                                                                                                                                                                                                                    |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Do not use for early scalar implementation; revisit only in 10.4.3. |

### Object or quantity LEG33 – Phase 5-10 bifurcation and dependency repair contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG33                                                                                                                                                                                                                                                                                                                                                                    |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                                                        |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                                                   |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                                                       |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                                           |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                                                      |
| Phase                           | 1.1.1.1.0.14                                                                                                                                                                                                                                                                                                                                                             |
| Source objects                  | v36 phase table; object register; Phase 4.1 ExecBoundaryMatrix; M/SP/P/TM/N/L object groups                                                                                                                                                                                                                                                                              |
| Definitions                     | Planning contract separating Path A (number-theory/geometric/modular/analytic) from Path B (spectral/physical/transfer/late bridges).                                                                                                                                                                                                                                    |
| Proof target                    | Show that SP0/SP1/P00-P06 and TM0-TM2 do not require Phase 8/9 modular/j data as prerequisites, while M00-M07 belong to the modular path and L08/N16/N17 to analytic far targets.                                                                                                                                                                                        |
| Proof sketch                    | Compare each object’s Computed from/Natural provenance fields. Spectral/physical objects consume ExecBoundaryMatrix, finite matrix algebra, star/state/regime data; modular objects consume tau/lattice/Eisenstein/Delta/j; analytic far targets consume theta/chi_D/L-function data. Therefore the old linear chain imposed an artificial bottleneck and must be split. |
| Verification / audit            | Generated v37 phase/object tables must show M objects in Phase 8, analytic L/theta objects in Phase 9, and SP/P/TM objects in Phase 10 with no Phase 8/9 dependency unless a bridge explicitly records it.                                                                                                                                                               |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. During Phase 1.1.1.1.1 module manifest, verify proposed modules preserve this path split and do not import modular APIs into finite spectral/physical code.               |

### Object or quantity LEG34 – Phase-object-link policy contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG34                                                                                                                                                                                                                                                   |
| Object status                   | G                                                                                                                                                                                                                                                       |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                  |
| Significance tier               | F1                                                                                                                                                                                                                                                      |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                          |
| Documented                      | yes                                                                                                                                                                                                                                                     |
| Phase                           | 1.1.1.1.0.15                                                                                                                                                                                                                                            |
| Source objects                  | v37 phase table; object register; direct object map; phase hierarchy                                                                                                                                                                                    |
| Definitions                     | A phase has direct linked objects iff object.Phase equals phase.Phase. If the direct set is empty, the phase must declare scope_only, no_direct_object, or moved. Scope-only child objects may be displayed as a summary but do not transfer ownership. |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof target              | Every phase row in the generated table has an explicit object-link interpretation;<br>no blank Linked objects cell can be mistaken for missing plan data.                                                                                                                                                                                                                                                        |
| Proof sketch              | Compute direct objects from the object register, compute descendant object scope from the phase hierarchy, classify moved placeholders from routing text, classify direct phases by direct ownership, classify remaining descendant-bearing phases as scope-only, and classify the rest as no-direct-object audit/baseline/closeout. Validate that empty-direct phases are never rendered as unqualified blanks. |
| Verification / audit      | Generator validation fails if a phase declares direct ownership without direct objects, declares scope_only without child scope, or has an invalid policy kind. The phase table renders explanatory text instead of empty cells.                                                                                                                                                                                 |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. During every future phase/object move, update the object Phase field and phase object-link policy together;<br>never leave a newly empty phase without explicit policy.                                           |

Object or quantity LEG35 – Ontic subsystem-minimality contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG35                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Significance tier               | F0                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                                                                                                                                                                                                         |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Phase                           | 1.1.1.1.0.16                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Source objects                  | IDEAL_infty formal provenance whole;<br>subsystem/cut/approximant;<br>complement-family/regime split;<br>Schur/DtN effective operator path                                                                                                                                                                                                                                                                                                      |
| Definitions                     | Subsystem-being is the only primitive ontic emergence trigger: a finite/local subsystem, cut or approximant is selected inside IDEAL_infty. IDEAL_infty remains the formal relational/provenance whole. Boundary, inside/outside, complement families, regimes, operators, spectra, algebra, arithmetic, geometry, dynamics and physics are downstream derived consequences or comparison-only interpretations.                                 |
| Proof target                    | Prevent any phase from importing dynamics, physics, stochasticity, geometry, arithmetic, operator algebra, environment or unitarity as primitive data beyond subsystem-being.                                                                                                                                                                                                                                                                   |
| Proof sketch                    | Declare POST1 as global postulate. Route every later non-subsystem object through source-object chains beginning with cut/approximant and passing through boundary/regime/complement and Schur/DtN or a certified later bridge. Add GNG22 so missing structure forces backjump to the natural generating layer rather than consumer-side patching.                                                                                              |
| Verification / audit            | Generated global-postulate table, GNG22, workflow policy and LEG35 dossier must be present before active implementation. Future phase audits must state whether physical/arithmetic/geometric claims are derived or comparison-only.                                                                                                                                                                                                            |
| Next documentation action       | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. When implementation starts, require the module manifest and proof dossiers to record the subsystem/cut provenance chain for every consumer that mentions dynamics, geometry, arithmetic, stochasticity, openness, unitarity or physics. |

Object or quantity DBR0 – Database-ready CNNA review and documentation structure

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR0                                                                                                                                      |
| Object status                   | G                                                                                                                                         |
| Context class                   | DocumentationObject                                                                                                                       |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | yes                                                                                                                                       |
| Phase                           | 20                                                                                                                                        |

|                           |                                                                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source objects            | masters/cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export;<br>tables/*.tex;<br>figures/*.dot/pdf.                                                                                                               |
| Definitions               | A non-operational review/documentation structure that keeps the Single-YAML-Master database-ready and prepares queryable phase/object/module/handoff/theory/review representations.                                                              |
| Proof target              | Documentation/governance object: verify that it is review/query infrastructure only, not a CNNA-generative source. For DBR1 also verify immediate expert-review/no-infallibility enforcement.                                                    |
| Proof sketch              | Closed as documentation/governance route: YAML master + generated exports + workflow/GNG rows -> DB-REVIEW side objective -> future database-ready views. This proves no database layer is a CNNA generator;<br>it is a review/query layer only. |
| Verification / audit      | V1.07 YAML contains Phase 20, DBR0-DBR5 objects, scratchpads, GNG23-GNG24, workflow row, and generated tables. No Lean source changed.                                                                                                           |
| Next documentation action | Use future Phase 20.2+ to implement schema/export only after review links and IDs are stable.                                                                                                                                                    |

### Object or quantity DBR1 – Expert-review and no-infallibility epistemic rule

**Table legend.**   W = white/finished public object;     B = blue/proof-carrying but semantic/API correction still open;     G = green/locally closable;     Y = yellow/known route with prerequisites;     R = red/derivation layer incomplete or unclear;     K = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                        |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <span style="color: blue;">DBR1</span>                                                                                                                                                                                                                                                        |
| Object status                   | <span style="background-color: #90EE90;">G</span>                                                                                                                                                                                                                                             |
| Context class                   | DocumentationObject                                                                                                                                                                                                                                                                           |
| Significance tier               | F1                                                                                                                                                                                                                                                                                            |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                |
| Documented                      | yes                                                                                                                                                                                                                                                                                           |
| Phase                           | <span style="color: blue;">20.1</span>                                                                                                                                                                                                                                                        |
| Source objects                  | global_no_go_guardrails GNG23-GNG24;<br>workflow_policy DB-REVIEW;<br>status_legend review/audit rows;<br>latest_change_overview V1.07.                                                                                                                                                       |
| Definitions                     | A global rule that Lean code is the primary truth system but not semantically infallible;<br>generated artifacts are secondary;<br>every relevant claim remains subject to discipline-specific expert review.                                                                                 |
| Proof target                    | Documentation/governance object: verify that it is review/query infrastructure only, not a CNNA-generative source. For DBR1 also verify immediate expert-review/no-infallibility enforcement.                                                                                                 |
| Proof sketch                    | Closed as global governance rule: GNG23-GNG24 and DB-REVIEW workflow state Lean-primary/generated-secondary semantics and expert-review obligation. This is not a mathematical theorem and not a Lean axiom.                                                                                  |
| Verification / audit            | V1.07 generated guardrail/workflow/status tables include no-infallibility and blue-audit obligations.                                                                                                                                                                                         |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Apply immediately in future phase closeouts, proof dossiers, and blue/reference-theory audits. |

### Object or quantity DBR2 – Database-ready YAML schema and round-trip target

**Table legend.**   W = white/finished public object;     B = blue/proof-carrying but semantic/API correction still open;     G = green/locally closable;     Y = yellow/known route with prerequisites;     R = red/derivation layer incomplete or unclear;     K = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <span style="color: blue;">DBR2</span>                                                                                                                                                         |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                                                                                              |
| Context class                   | ToolInfrastructureObject                                                                                                                                                                       |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | no                                                                                                                                                                                             |

|                           |                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase                     | 20.2                                                                                                                                                                                                                                                                                     |
| Source objects            | cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future schema audit.                                                                                                                                                                       |
| Definitions               | A future schema/export target mapping YAML phases, objects, proof dossiers, modules, handoffs, statuses, theory markings, and review states into relational or graph-friendly tables.                                                                                                    |
| Proof target              | Documentation/governance object: verify that it is review/query infrastructure only, not a CNNA-generative source. For DBR1 also verify immediate expert-review/no-infallibility enforcement.                                                                                            |
| Proof sketch              | Pending. Future tooling/review phase must show this object consumes YAML/Lean-generated metadata only and remains generated-secondary documentation/review infrastructure.                                                                                                               |
| Verification / audit      | Pending. Must verify round-trip/reference integrity and no CNNA-generative role before activation.                                                                                                                                                                                       |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Keep black/secondary until the corresponding Phase 20.x tooling/audit phase is activated. |

### Object or quantity DBR3 – Discipline-separated expert-review matrix

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                        |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR3                                                                                                                                                                                          |
| Object status                   | K                                                                                                                                                                                             |
| Context class                   | DocumentationObject                                                                                                                                                                           |
| Significance tier               | F3                                                                                                                                                                                            |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                     |
| Documented                      | no                                                                                                                                                                                            |
| Phase                           | 20.3                                                                                                                                                                                          |
| Source objects                  | object_proof_dossiers;<br>workflow_policy;<br>status_legend review/audit rows;<br>future discipline review rows.                                                                              |
| Definitions                     | A future review matrix linking claims, objects, phases, modules, and established-theory references to required specialist disciplines and review evidence/status.                             |
| Proof target                    | Documentation/governance object: verify that it is review/query infrastructure only, not a CNNA-generative source. For DBR1 also verify immediate expert-review/no-infallibility enforcement. |
| Proof sketch                    | Pending. Future tooling/review phase must show this object consumes YAML/Lean-generated metadata only and remains generated-secondary documentation/review infrastructure.                    |
| Verification / audit            | Pending. Must verify round-trip/reference integrity and no CNNA-generative role before activation.                                                                                            |
| Next documentation action       | Keep black/secondary until the corresponding Phase 20.x tooling/audit phase is activated.                                                                                                     |

### Object or quantity DBR4 – Blue/reference-theory audit ledger

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR4                                                                                                                                      |
| Object status                   | K                                                                                                                                         |
| Context class                   | DocumentationObject                                                                                                                       |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | no                                                                                                                                        |
| Phase                           | 20.4                                                                                                                                      |

|                           |                                                                                                                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source objects            | status_legend;<br>external_source_crosscheck;<br>consumption_classification_map;<br>object_register;<br>future blue-audit rows.                                                               |
| Definitions               | A future audit ledger for established-theory/reference material that may need explicit blue marking or comparison-only classification.                                                        |
| Proof target              | Documentation/governance object: verify that it is review/query infrastructure only, not a CNNA-generative source. For DBR1 also verify immediate expert-review/no-infallibility enforcement. |
| Proof sketch              | Pending. Future tooling/review phase must show this object consumes YAML/Lean-generated metadata only and remains generated-secondary documentation/review infrastructure.                    |
| Verification / audit      | Pending. Must verify round-trip/reference integrity and no CNNA-generative role before activation.                                                                                            |
| Next documentation action | Keep black/secondary until the corresponding Phase 20.x tooling/audit phase is activated.                                                                                                     |

### Object or quantity DBR5 – Interactive/online database-backed documentation views

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">DBR5</a>                                                                                                                                                                                                                                                                     |
| Object status                   | <span style="background-color: #a9a9a9;">K</span>                                                                                                                                                                                                                                        |
| Context class                   | ToolInfrastructureObject                                                                                                                                                                                                                                                                 |
| Significance tier               | F1                                                                                                                                                                                                                                                                                       |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                           |
| Documented                      | no                                                                                                                                                                                                                                                                                       |
| Phase                           | <a href="#">20.5</a>                                                                                                                                                                                                                                                                     |
| Source objects                  | future database export;<br>generated_exports;<br>figures;<br>tables;<br>review ledgers.                                                                                                                                                                                                  |
| Definitions                     | A future generated-secondary interface for querying and visualizing phases, modules, objects, DAGs, handoffs, theory markings, and review status online or interactively.                                                                                                                |
| Proof target                    | Documentation/governance object: verify that it is review/query infrastructure only, not a CNNA-generative source. For DBR1 also verify immediate expert-review/no-infallibility enforcement.                                                                                            |
| Proof sketch                    | Pending. Future tooling/review phase must show this object consumes YAML/Lean-generated metadata only and remains generated-secondary documentation/review infrastructure.                                                                                                               |
| Verification / audit            | Pending. Must verify round-trip/reference integrity and no CNNA-generative role before activation.                                                                                                                                                                                       |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Keep black/secondary until the corresponding Phase 20.x tooling/audit phase is activated. |

### Object or quantity DBR6 – Context-filtered documentation slice generator

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">DBR6</a>                                                                                                                      |
| Object status                   | <span style="background-color: #90ee90;">G</span>                                                                                         |
| Context class                   | GeneratedArtifactObject                                                                                                                   |
| Significance tier               | F3                                                                                                                                        |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Documented                      | yes                                                                                                                                       |
| Phase                           | <a href="#">20.6</a>                                                                                                                      |

|                           |                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source objects            | Single YAML master;<br>generated_exports/*.tsv;<br>DBR0-DBR5;<br>context_documentation_profiles;<br>generate_context_doc.py.                                                                                                                                                                                                                        |
| Definitions               | A generated-secondary documentation packet that retains the official section topology but filters rows in each section by a query/profile relevance predicate. Each output includes a filter manifest and section hit summary.                                                                                                                      |
| Proof target              | Show that the slice generator is a reproducible view over existing YAML/export data only;<br>it does not edit Lean, does not introduce CNNA-generative objects, and visibly preserves expert-review/no-infallibility warnings.                                                                                                                      |
| Proof sketch              | The script loads cnna_roadmap_master.yaml, optional curated profiles, and generated TSV exports, computes relevance scores for rows, emits a manifest plus TeX/PDF view, and labels the output as generated-secondary. Because every row is copied from existing records, the slice is an index/filter view rather than a new semantic certificate. |
| Verification / audit      | V1.08 generator validates by running generate_tables.py first, then generate_context_doc.py for the dtn-matrix profile. The sample context_docs/dtn_matrix packet records matched/total rows per section and states the Lean-primary/expert-review warning.                                                                                         |
| Next documentation action | Use this function for targeted documentation requests and phase-closeout deltas. Add more curated profiles only when their term sets are reviewable and do not hide established-theory imports.                                                                                                                                                     |

### Object or quantity DBR7 – Stats front section and tool-interface documentation layer

**Table legend.**   W = white/finished public object;     B = blue/proof-carrying but semantic/API correction still open;     G = green/locally closable;     Y = yellow/known route with prerequisites;     R = red/derivation layer incomplete or unclear;     K = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">DBR7</a>                                                                                                                                                                                                                                                                                                                                                |
| Object status                   | <span style="background-color: #90EE90;"> </span> G                                                                                                                                                                                                                                                                                                                 |
| Context class                   | GeneratedArtifactObject                                                                                                                                                                                                                                                                                                                                             |
| Significance tier               | F3                                                                                                                                                                                                                                                                                                                                                                  |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects.   Avoid proofbook bloat.                                                                                                                                                                                                                         |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                                                 |
| Phase                           | <a href="#">20.7</a>                                                                                                                                                                                                                                                                                                                                                |
| Source objects                  | Single YAML master;<br>repo inventory JSON;<br>generated exports;<br>DBR0-DBR6;<br>GNG23-GNG25.                                                                                                                                                                                                                                                                     |
| Definitions                     | A generated-secondary documentation layer that opens the full document with project facts, license notices, and an operational interface metaphor before embedding the repository module inventory and derivation graphs.                                                                                                                                           |
| Proof target                    | Show that the front Stats layer is a reproducible view over existing YAML/repo-inventory data and that the metaphor is explicitly bounded as non-ontological and non-theorematic.                                                                                                                                                                                   |
| Proof sketch                    | The generator computes counts from the YAML master and repo_inventory/raw_export JSON plus Lean file statistics, exports project_stats.generated.tsv, renders a compact Stats table, then includes the existing module inventory/graph view. The metaphor and license notices are stored as explicit YAML records and rendered as documentation, not as Lean input. |
| Verification / audit            | V1.09 validation runs generate_tables.py, checks the generated project_stats table and the full PDF, and confirms that the old standalone module-inventory section is removed from the main TeX while its content is embedded under Stats.                                                                                                                          |
| Next documentation action       | When a future database representation exists, map project_stats, guiding_metaphors, and license_notice into database tables/views without changing their generated-secondary status.                                                                                                                                                                                |

### Object or quantity LEG36 – Semantic module consolidation ledger

**Table legend.**   W = white/finished public object;     B = blue/proof-carrying but semantic/API correction still open;     G = green/locally closable;     Y = yellow/known route with prerequisites;     R = red/derivation layer incomplete or unclear;     K = black/registered side or future target.   Context class is orthogonal to status.

| Field             | Record                                              |
|-------------------|-----------------------------------------------------|
| Object ID         | <a href="#">LEG36</a>                               |
| Object status     | <span style="background-color: #FFFF00;"> </span> Y |
| Context class     | GovernancePolicyObject                              |
| Significance tier | F1                                                  |



|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                                                                                                                                                                                                                                             |
| Documented                      | active-prepared                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Phase                           | 1.1.1.1.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Source objects                  | LEG10 module manifest;<br>LEG11 reuse/duplication ledger;<br>LEG5 consumption classification map;<br>repo_inventory/raw_export modules/import_edges/symbol references;<br>direct code spot-checks of candidate clusters.                                                                                                                                                                                                                                                                                                                                                   |
| Definitions                     | Compatibility-preserving merge/no-merge ledger for semantically related Lean modules. It distinguishes true duplicate/schema families from proof-stage chains where module granularity carries derivation provenance.                                                                                                                                                                                                                                                                                                                                                      |
| Proof target                    | Before any new substantive Lean implementation, close a bounded audit stating which module families should be consolidated, which require only facade aliases, which must remain separated, and which later refactor package may safely execute the changes.                                                                                                                                                                                                                                                                                                               |
| Proof sketch                    | Use the generated import/declaration inventory to identify module clusters, then spot-check candidate code. High-confidence candidates include clone-like BoundarySource substrate ledger modules and Handoff input/output schema aliases. Smoke/SM and Cayley-Dickson S6 chains are treated as no-merge-until-proven because their file boundaries encode proof/phase progression. Any accepted consolidation must preserve old imports through facades first, then record import graph delta, declaration delta, consumer breakage audit, and local full-build evidence. |
| Verification / audit            | V1.12 records the audit as an active gate, adds SCF43-SCF45, GNG26, RD08, a workflow row, and a report. No Lean source was moved. Local Lean build evidence is required for any later actual consolidation.                                                                                                                                                                                                                                                                                                                                                                |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Run the active Phase 1.1.1.1.5 audit to fill the ledger with exact candidate rows and decide whether to open a bounded Lean refactor execution phase before returning to 1.1.1.2.                                                                                                                                                                                           |

### Object or quantity DBR8 – Root artifact retention and cleanup policy

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR8                                                                                                                                                                                                                                                                                                                                                                                     |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                                                                        |
| Context class                   | GeneratedArtifactObject                                                                                                                                                                                                                                                                                                                                                                  |
| Significance tier               | F3                                                                                                                                                                                                                                                                                                                                                                                       |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                                                                                                                                                |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                                                                      |
| Phase                           | 20.8                                                                                                                                                                                                                                                                                                                                                                                     |
| Source objects                  | Single YAML master;<br>DBR0-DBR7;<br>GNG23-GNG27;<br>root artifact audit;<br>user instruction that PDFs/TeX need not be retained because they are regenerable.                                                                                                                                                                                                                           |
| Definitions                     | A generated-secondary repository hygiene layer: README.md stays in the project root, historical Markdown reports move to archive/reports/, and root PDF/TeX/LaTeX build products are deleted or regenerated on demand under a disposable output directory.                                                                                                                               |
| Proof target                    | Show that no root PDF/TeX/build products or historical reports are required as active sources and that the regeneration path remains explicit.                                                                                                                                                                                                                                           |
| Proof sketch                    | The YAML master and generated tables/exports are the maintained documentation data. Full PDFs and TeX wrappers can be reconstructed from scripts/generate_full_document.py after scripts/generate_tables.py. Historical Markdown reports are not needed for generation and are moved to archive/reports/ for traceability. The cleanup script enforces this classification mechanically. |
| Verification / audit            | V1.13 audits root file extensions, adds GNG27 and SCF46, adds generator/cleanup scripts, regenerates tables, verifies optional full-document compilation under generated_docs/full/, and removes root PDF/TeX/auxiliary artifacts from the packaged root.                                                                                                                                |
| Next documentation action       | For future releases, keep generated full PDFs/TeX out of the root package unless explicitly requested as a deliverable; use context docs or full doc generation on demand.                                                                                                                                                                                                               |

### Object or quantity DBR9 – Current README and workflow-orientation manual

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field | Record |
|-------|--------|
|-------|--------|

|                                 |                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR9                                                                                                                                                                                                                                                                                                                               |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                  |
| Context class                   | GeneratedArtifactObject                                                                                                                                                                                                                                                                                                            |
| Significance tier               | F3                                                                                                                                                                                                                                                                                                                                 |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                                                                                          |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                |
| Phase                           | 20.9                                                                                                                                                                                                                                                                                                                               |
| Source objects                  | Single YAML master;<br>DBR0-DBR8;<br>workflow_policy;<br>context_documentation_profiles;<br>root artifact policy;<br>user correction that tested/developed assistant workflow is ChatGPT-only.                                                                                                                                     |
| Definitions                     | A concise README that explains the current repository layout, Lean-primary truth discipline, manual documentation commands, context-slice generation, ChatGPT-assisted workflow model, workflow extensibility, scalability vision, and license split.                                                                              |
| Proof target                    | Show that the README is an entry-point manual rather than a chronological changelog and that it does not misrepresent Claude as a tested/developed workflow component.                                                                                                                                                             |
| Proof sketch                    | Replace the old release-note-style README with a current manual. Keep generated artifacts secondary, keep Lean-code primary, state that the integrated assistant workflow was developed/tested with ChatGPT, and present Claude or other tools only as external review/discussion channels if used outside this validated package. |
| Verification / audit            | V1.14 rewrites README.md, updates metadata/latest-change records, regenerates YAML-derived exports with generate_tables.py, and checks that no Lean source was modified.                                                                                                                                                           |
| Next documentation action       | Keep README as compact current manual. Move detailed historical notes to archive/reports or generated documentation rather than re-growing a chronological README.                                                                                                                                                                 |

### Object or quantity DBR10 – Root Lean/Lake project-control files

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR10                                                                                                                                                                                                                                                                                                                                                                |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                                                    |
| Context class                   | GeneratedArtifactObject                                                                                                                                                                                                                                                                                                                                              |
| Significance tier               | F3                                                                                                                                                                                                                                                                                                                                                                   |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                                                                                                                            |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                                                  |
| Phase                           | 20.10                                                                                                                                                                                                                                                                                                                                                                |
| Source objects                  | DBR8 root artifact retention policy;<br>uploaded root Lean/Lake files;<br>repo_snapshot/CNNA source tree;<br>README/manual build commands.                                                                                                                                                                                                                           |
| Definitions                     | A package-root build-control set: CNNA.lean top-level compatibility import file, lakefile.lean package definition pointing the CNNA library at Repository/repo_snapshot, lake-manifest.json dependency lock, and lean-toolchain Lean version pin.                                                                                                                    |
| Proof target                    | Show that root cleanup must distinguish generated documentation artifacts from Lean/Lake source-control files, preserving the latter for reproducible local builds.                                                                                                                                                                                                  |
| Proof sketch                    | The V1.13/V1.14 cleanup policy correctly removed generated PDF/TeX/report clutter, but over-stated the root invariant as README-only. The corrected invariant retains README.md plus the four Lean/Lake project-control files. The cleanup script keep-list is amended so future root hygiene does not delete build configuration.                                   |
| Verification / audit            | V1.15 copies the uploaded root files into the package root above Repository/, updates README and YAML records, amends clean_generated_root_artifacts.py defensively, regenerates tables/exports, runs cleanup inside Repository/, and checks that the package root contains the Lean/Lake build-control files while Repository root has no PDF/TeX/LaTeX byproducts. |
| Next documentation action       | If further root source-control files become necessary, classify them explicitly in DBR10/GNG28 rather than broadening the root policy to arbitrary artifacts.                                                                                                                                                                                                        |

### Object or quantity DBR11 – Root license, single-root README, credits and assistant-requirement surface

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR11                                                                                                                                                                                                                                                                                                                                                  |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                                      |
| Context class                   | GeneratedArtifactObject                                                                                                                                                                                                                                                                                                                                |
| Significance tier               | F3                                                                                                                                                                                                                                                                                                                                                     |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                                                                                                              |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                                    |
| Phase                           | 20.11                                                                                                                                                                                                                                                                                                                                                  |
| Source objects                  | DBR8 root artifact policy;<br>DBR9 README surface;<br>DBR10 Lean/Lake build-entry files;<br>user licensing URLs and credit/minimum-requirement instructions.                                                                                                                                                                                           |
| Definitions                     | The maintained README is a root-only manual;<br>LICENSE is a root-only legal notice file with dual project-authored license scopes;<br>generated PDFs carry explicit Editor and Assistant credits;<br>the tested assistant workflow states its minimum ChatGPT Plus/GPT-5.5 requirement.                                                               |
| Proof target                    | Show that the package root exposes the legal/documentation entry surface without duplicating README files and that generated PDFs visibly include the requested credits.                                                                                                                                                                               |
| Proof sketch                    | Separate legal/source-control files from generated artifacts: README.md and LICENSE are source/control documentation in root;<br>generated PDF/TeX files are disposable under generated_docs/full/ or context_docs/. Update generator author fields and stats metadata so PDF output carries the same credit semantics without becoming primary truth. |
| Verification / audit            | V1.17 removes Repository/README.md and repo_inventory/raw_export/README.md, adds root LICENSE, updates root README, regenerates tables, compiles a full PDF with the new title/author credits, regenerates context documentation, and audits that README.md appears only in package root.                                                              |
| Next documentation action       | If multilingual README or alternative assistant workflows are later validated, add them as explicit output/workflow layers rather than weakening the current English-first, ChatGPT-tested statement.                                                                                                                                                  |

### Object or quantity DBR12 – Generated PDF section-pagebreak layout invariant

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                            |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR12                                                                                                                                                                                                                                             |
| Object status                   | G                                                                                                                                                                                                                                                 |
| Context class                   | GeneratedArtifactObject                                                                                                                                                                                                                           |
| Significance tier               | F3                                                                                                                                                                                                                                                |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                         |
| Documented                      | yes                                                                                                                                                                                                                                               |
| Phase                           | 20.12                                                                                                                                                                                                                                             |
| Source objects                  | DBR6 context documentation generator;<br>DBR9 README/manual;<br>DBR11 PDF credit surface;<br>user section-pagebreak requirement.                                                                                                                  |
| Definitions                     | Each generated documentation top-level section, including starred context-preamble sections and normal content sections, starts on a new page via a generator-level LaTeX section hook.                                                           |
| Proof target                    | Show that both full and context PDF generators enforce the section-pagebreak invariant while preserving the same source tables, context matching, credits and generated-secondary warnings.                                                       |
| Proof sketch                    | Patch the TeX preambles in generate_full_document.py and generate_context_doc.py to load etoolbox and prepend clearpage to section commands;<br>remove one-off manual post-TOC pagebreaks so the invariant is generator-level rather than ad hoc. |
| Verification / audit            | V1.18 regenerates tables, compiles the full PDF and DtN context PDF, checks generated TeX for the section hook, renders early pages, and audits that no Lean source under repo_snapshot/CNNA changed.                                             |
| Next documentation action       | If subsection-level page isolation is later desired, add it as a separate layout profile;<br>V1.18 intentionally applies only to top-level documentation sections.                                                                                |

Object or quantity DBR13 – Phase status release policy and red-lock ledger

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR13                                                                                                                                                                                                                                                                                                 |
| Object status                   | G                                                                                                                                                                                                                                                                                                     |
| Context class                   | GeneratedArtifactObject                                                                                                                                                                                                                                                                               |
| Significance tier               | F3                                                                                                                                                                                                                                                                                                    |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                                                             |
| Documented                      | yes                                                                                                                                                                                                                                                                                                   |
| Phase                           | 20.13                                                                                                                                                                                                                                                                                                 |
| Source objects                  | Status legend;<br>pre-implementation phase check workflow;<br>implementation scratchpad phase_type;<br>user red/yellow/orange/white policy requirement;<br>Lean-primary and derived-only guardrails.                                                                                                  |
| Definitions                     | G = closed;<br>Y = unique active implementable phase;<br>O = implementable after named predecessor closure;<br>R = locked Lean-derivation phase lacking release proof;<br>W = non-direct Lean derivation work such as refactor, tool, infrastructure, documentation, or comparison.                   |
| Proof target                    | Show that the planning table now encodes default red locking for unreleased Lean-derivation phases and that white no longer means unchecked Lean derivation.                                                                                                                                          |
| Proof sketch                    | Use implementation_scratchpad.phase_type as the classification seam. Convert all W/derivation phases to R. Keep W for refactor/infrastructure/documentation/comparison phases. Patch generator validation so a W derivation phase fails generation and exactly one Y phase remains the active cursor. |
| Verification / audit            | V1.19 converts 81 derivation phases from W to R, keeps the active phase 1.1.1.1.5 yellow and 1.1.1.2 orange, regenerates tables, compiles representative PDFs, and checks status counts.                                                                                                              |
| Next documentation action       | For any red phase, add a pre-implementation check that references the prerequisite derived-only Lean objects before changing it to yellow or orange.                                                                                                                                                  |

Object or quantity DBR14 – Static documentation explanations and module-change appendix policy

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR14                                                                                                                                                                                                                                                                                                                                 |
| Object status                   | G                                                                                                                                                                                                                                                                                                                                     |
| Context class                   | GeneratedArtifactObject                                                                                                                                                                                                                                                                                                               |
| Significance tier               | F3                                                                                                                                                                                                                                                                                                                                    |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                                                                                             |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                                   |
| Phase                           | 20.14                                                                                                                                                                                                                                                                                                                                 |
| Source objects                  | DBR6 context documentation generator;<br>DBR12 section-pagebreak layout;<br>DBR13 phase-status release policy;<br>user clarification about module-manifest confusion and section/table explanations.                                                                                                                                  |
| Definitions                     | A documentation generator policy in which static section prose explains what data is shown, where it comes from and why it is displayed;<br>only table rows remain dynamic. The old implementation module manifest is renamed and moved to a technical appendix for actual module/API changes only.                                   |
| Proof target                    | Show that full and context documentation now explain major sections without creating a second phase table or duplicating release logic.                                                                                                                                                                                               |
| Proof sketch                    | Patch full generator section text, context generator section descriptions/order, and module-manifest static paragraph. Move the module-change manifest after phase table and scratchpad in the full document. Keep all table contents generated from YAML/exports. Validate with table generation and representative PDF compilation. |
| Verification / audit            | V1.20 regenerates tables, compiles full PDF and DtN context PDF, checks generated TeX for new section titles/explanations, renders representative pages, and verifies no Lean source under repo_snapshot/CNNA changed.                                                                                                                |

|                           |                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Next documentation action | If later sections become too verbose, trim static prose in generators rather than moving explanations into dynamic YAML rows. |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|

Object or quantity LEG37 – Current source intake completeness ledger

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG37                                                                                                                                                                                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                                               |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                          |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                              |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                  |
| Documented                      | active-prepared                                                                                                                                                                                                                                                                                                                                 |
| Phase                           | 1.1.1.1.5.1                                                                                                                                                                                                                                                                                                                                     |
| Source objects                  | LEG10, LEG5, repo_inventory/raw_export/modules.json, symbol_reference_summary.json, module/declaration consumption TSVs.                                                                                                                                                                                                                        |
| Definitions                     | A complete classification ledger for the current Lean source snapshot: every module/declaration is linked to an object/proof dossier, legacy/audit-only bucket, tool/refactor bucket, or a newly created natural-origin ingestion phase.                                                                                                        |
| Proof target                    | Close the corresponding pre-substantive-code readiness condition before new Lean content implementation is released.                                                                                                                                                                                                                            |
| Proof sketch                    | Use generated inventory, direct source scan, object/proof-dossier register and local build evidence where required. If the condition cannot be closed, create or refine an upstream red blocker at the natural source layer instead of patching a consumer phase.                                                                               |
| Verification / audit            | V1.21 creates the readiness sequence and records the gap explicitly. Closure requires future local Lean/build/inventory evidence; no Lean source is changed in V1.21.                                                                                                                                                                           |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill with exact candidate rows and evidence during the corresponding child phase; do not mark 1.1.1.2 active until this dossier chain is closed. |

Object or quantity LEG38 – Neutral module-name and facade migration map

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG38                                                                                                                                                                                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                                               |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                          |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                              |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                  |
| Documented                      | active-prepared                                                                                                                                                                                                                                                                                                                                 |
| Phase                           | 1.1.1.1.5.2                                                                                                                                                                                                                                                                                                                                     |
| Source objects                  | LEG37 plus direct source scan of SM*, S6*, StrengtheningPhase*, BoundarySource, Handoff and Notation/Legacy module families.                                                                                                                                                                                                                    |
| Definitions                     | A migration map assigning phase-semantic or legacy module names to neutral target names, facade modules, no-merge exceptions, and proof-stage preservation constraints.                                                                                                                                                                         |
| Proof target                    | Close the corresponding pre-substantive-code readiness condition before new Lean content implementation is released.                                                                                                                                                                                                                            |
| Proof sketch                    | Use generated inventory, direct source scan, object/proof-dossier register and local build evidence where required. If the condition cannot be closed, create or refine an upstream red blocker at the natural source layer instead of patching a consumer phase.                                                                               |
| Verification / audit            | V1.21 creates the readiness sequence and records the gap explicitly. Closure requires future local Lean/build/inventory evidence; no Lean source is changed in V1.21.                                                                                                                                                                           |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill with exact candidate rows and evidence during the corresponding child phase; do not mark 1.1.1.2 active until this dossier chain is closed. |

Object or quantity LEG39 – Neutral refactor execution certificate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG39                                                                                                                                                                                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                                               |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                          |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                              |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                  |
| Documented                      | active-prepared                                                                                                                                                                                                                                                                                                                                 |
| Phase                           | 1.1.1.1.5.3                                                                                                                                                                                                                                                                                                                                     |
| Source objects                  | LEG38, GNG03, GNG26, GNG28-GNG32, local Lean build evidence after refactor.                                                                                                                                                                                                                                                                     |
| Definitions                     | A certificate for a bounded neutral-module refactor package: old-module facades, import/declaration delta, consumer breakage audit, and local full-build evidence.                                                                                                                                                                              |
| Proof target                    | Close the corresponding pre-substantive-code readiness condition before new Lean content implementation is released.                                                                                                                                                                                                                            |
| Proof sketch                    | Use generated inventory, direct source scan, object/proof-dossier register and local build evidence where required. If the condition cannot be closed, create or refine an upstream red blocker at the natural source layer instead of patching a consumer phase.                                                                               |
| Verification / audit            | V1.21 creates the readiness sequence and records the gap explicitly. Closure requires future local Lean/build/inventory evidence; no Lean source is changed in V1.21.                                                                                                                                                                           |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill with exact candidate rows and evidence during the corresponding child phase; do not mark 1.1.1.2 active until this dossier chain is closed. |

Object or quantity LEG40 – Post-refactor inventory closure certificate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG40                                                                                                                                                                                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                                               |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                          |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                              |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                  |
| Documented                      | active-prepared                                                                                                                                                                                                                                                                                                                                 |
| Phase                           | 1.1.1.1.5.4                                                                                                                                                                                                                                                                                                                                     |
| Source objects                  | build_export_script_v1.8.py outputs after refactor, generated_tables.py outputs, object/proof/phase registers.                                                                                                                                                                                                                                  |
| Definitions                     | A regenerated inventory closure record after any neutral refactor: module graph, declaration graph, object-register delta, proof-dossier delta, public/API consumer delta, unresolved-gap ledger.                                                                                                                                               |
| Proof target                    | Close the corresponding pre-substantive-code readiness condition before new Lean content implementation is released.                                                                                                                                                                                                                            |
| Proof sketch                    | Use generated inventory, direct source scan, object/proof-dossier register and local build evidence where required. If the condition cannot be closed, create or refine an upstream red blocker at the natural source layer instead of patching a consumer phase.                                                                               |
| Verification / audit            | V1.21 creates the readiness sequence and records the gap explicitly. Closure requires future local Lean/build/inventory evidence; no Lean source is changed in V1.21.                                                                                                                                                                           |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill with exact candidate rows and evidence during the corresponding child phase; do not mark 1.1.1.2 active until this dossier chain is closed. |

Object or quantity LEG41 – Pre-substantive-code release certificate

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.



| Field                           | Record                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | LEG41                                                                                                                                                                                                                                                                                                                                           |
| Object status                   | Y                                                                                                                                                                                                                                                                                                                                               |
| Context class                   | GovernancePolicyObject                                                                                                                                                                                                                                                                                                                          |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                              |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                  |
| Documented                      | active-prepared                                                                                                                                                                                                                                                                                                                                 |
| Phase                           | 1.1.1.1.5.5                                                                                                                                                                                                                                                                                                                                     |
| Source objects                  | Closed LEG37, LEG38, LEG39, LEG40, object proof dossiers, phase release checks, local build evidence if refactor executed.                                                                                                                                                                                                                      |
| Definitions                     | A release certificate proving that a new substantive Lean/content phase may become active only after source intake, neutral naming/refactor, post-refactor inventory, proof-dossier updates and red-lock checks close.                                                                                                                          |
| Proof target                    | Close the corresponding pre-substantive-code readiness condition before new Lean content implementation is released.                                                                                                                                                                                                                            |
| Proof sketch                    | Use generated inventory, direct source scan, object/proof-dossier register and local build evidence where required. If the condition cannot be closed, create or refine an upstream red blocker at the natural source layer instead of patching a consumer phase.                                                                               |
| Verification / audit            | V1.21 creates the readiness sequence and records the gap explicitly. Closure requires future local Lean/build/inventory evidence; no Lean source is changed in V1.21.                                                                                                                                                                           |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. Fill with exact candidate rows and evidence during the corresponding child phase; do not mark 1.1.1.2 active until this dossier chain is closed. |

### Object or quantity DBR15 – Supabase/Postgres mirror feasibility and schema design

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR15                                                                                                                                                                                                                                                                                                                                                              |
| Object status                   | W                                                                                                                                                                                                                                                                                                                                                                  |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                                                 |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                                     |
| Documented                      | planned                                                                                                                                                                                                                                                                                                                                                            |
| Phase                           | 20.15                                                                                                                                                                                                                                                                                                                                                              |
| Source objects                  | SLTG-06; DBR0-DBR14; Master YAML; generated exports; Supabase official docs on API security/RLS/service keys as external implementation reference only.                                                                                                                                                                                                            |
| Definitions                     | internal/api/review/admin schema split; minimal tables for db_snapshots, phases, objects, proof_dossiers, guardrails, workflows, modules, declarations, import_edges, phase_object_links, context_profiles, static_findings and reports.                                                                                                                           |
| Proof target                    | Specify the DB/web-publication object without giving it CNNA truth-source status or public write authority.                                                                                                                                                                                                                                                        |
| Proof sketch                    | Keep Lean/YAML as source; export snapshots into DB; expose only read-only public views; preserve snapshot provenance and checksums; forbid browser service_role keys and public writes; keep admin/import writes local or later signed/admin-only.                                                                                                                 |
| Verification / audit            | Planned. Closure requires schema/import/RLS/frontend contract review, no secrets in repo/chat, and roundtrip/export evidence. V1.22 records planning objects only.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When the pre-code gate stabilizes or DB work becomes active, create migration/importer artifacts and a Supabase context documentation slice from these DBR records. |

Object or quantity DBR16 – YAML/exports to Postgres import and roundtrip contract

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR16                                                                                                                                                                                                                                                                                                                                                              |
| Object status                   | W                                                                                                                                                                                                                                                                                                                                                                  |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                                                 |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                                     |
| Documented                      | planned                                                                                                                                                                                                                                                                                                                                                            |
| Phase                           | 20.16                                                                                                                                                                                                                                                                                                                                                              |
| Source objects                  | SLTG-06;<br>DBR0-DBR14;<br>Master YAML;<br>generated exports;<br>Supabase official docs on API security/RLS/service keys as external implementation reference only.                                                                                                                                                                                                |
| Definitions                     | Importer must be local/admin-run with .env secrets, produce import report/checksums, and never require a service_role key in chat or browser.                                                                                                                                                                                                                      |
| Proof target                    | Specify the DB/web-publication object without giving it CNNA truth-source status or public write authority.                                                                                                                                                                                                                                                        |
| Proof sketch                    | Keep Lean/YAML as source;<br>export snapshots into DB;<br>expose only read-only public views;<br>preserve snapshot provenance and checksums;<br>forbid browser service_role keys and public writes;<br>keep admin/import writes local or later signed/admin-only.                                                                                                  |
| Verification / audit            | Planned. Closure requires schema/import/RLS/frontend contract review, no secrets in repo/chat, and roundtrip/export evidence. V1.22 records planning objects only.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When the pre-code gate stabilizes or DB work becomes active, create migration/importer artifacts and a Supabase context documentation slice from these DBR records. |

Object or quantity DBR17 – Append-only snapshot and checksum model

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR17                                                                                                                                                                                          |
| Object status                   | W                                                                                                                                                                                              |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                           |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | planned                                                                                                                                                                                        |
| Phase                           | 20.17                                                                                                                                                                                          |
| Source objects                  | SLTG-06;<br>DBR0-DBR14;<br>Master YAML;<br>generated exports;<br>Supabase official docs on API security/RLS/service keys as external implementation reference only.                            |
| Definitions                     | db_snapshots with snapshot_id, tool_version, lean_data_timestamp, yaml_hash, repo_inventory_hash, created_at, created_by, source_zip_name and optional import_report_hash.                     |
| Proof target                    | Specify the DB/web-publication object without giving it CNNA truth-source status or public write authority.                                                                                    |

|                           |                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Keep Lean/YAML as source;<br>export snapshots into DB;<br>expose only read-only public views;<br>preserve snapshot provenance and checksums;<br>forbid browser service_role keys and public writes;<br>keep admin/import writes local or later signed/admin-only.                                                                                                  |
| Verification / audit      | Planned. Closure requires schema/import/RLS/frontend contract review, no secrets in repo/chat, and roundtrip/export evidence. V1.22 records planning objects only.                                                                                                                                                                                                 |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When the pre-code gate stabilizes or DB work becomes active, create migration/importer artifacts and a Supabase context documentation slice from these DBR records. |

### Object or quantity DBR18 – Public read-only API views and RLS policies

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">DBR18</a>                                                                                                                                                                                                                                                                                                                                              |
| Object status                   | W                                                                                                                                                                                                                                                                                                                                                                  |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                                                 |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                                     |
| Documented                      | planned                                                                                                                                                                                                                                                                                                                                                            |
| Phase                           | <a href="#">20.18</a>                                                                                                                                                                                                                                                                                                                                              |
| Source objects                  | SLTG-06;<br>DBR0-DBR14;<br>Master YAML;<br>generated exports;<br>Supabase official docs on API security/RLS/service keys as external implementation reference only.                                                                                                                                                                                                |
| Definitions                     | anon gets SELECT on api.* views only;<br>no anon INSERT/UPDATE/DELETE;<br>internal.* and admin/import.* remain closed;<br>service_role never appears in frontend/browser.                                                                                                                                                                                          |
| Proof target                    | Specify the DB/web-publication object without giving it CNNA truth-source status or public write authority.                                                                                                                                                                                                                                                        |
| Proof sketch                    | Keep Lean/YAML as source;<br>export snapshots into DB;<br>expose only read-only public views;<br>preserve snapshot provenance and checksums;<br>forbid browser service_role keys and public writes;<br>keep admin/import writes local or later signed/admin-only.                                                                                                  |
| Verification / audit            | Planned. Closure requires schema/import/RLS/frontend contract review, no secrets in repo/chat, and roundtrip/export evidence. V1.22 records planning objects only.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When the pre-code gate stabilizes or DB work becomes active, create migration/importer artifacts and a Supabase context documentation slice from these DBR records. |

### Object or quantity DBR19 – cnna-project.net web-frontend data contract

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">DBR19</a>                                                                                                                                                                          |
| Object status                   | W                                                                                                                                                                                              |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                           |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |

|                           |                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documented                | planned                                                                                                                                                                                                                                                                                                                                                            |
| Phase                     | 20.19                                                                                                                                                                                                                                                                                                                                                              |
| Source objects            | SLTG-06;<br>DBR0-DBR14;<br>Master YAML;<br>generated exports;<br>Supabase official docs on API security/RLS/service keys as external implementation reference only.                                                                                                                                                                                                |
| Definitions               | Frontend can query public views with anon/publishable key only and render phase map, object register, red/orange release reasons, import/module graph, proof dossiers, blue-debt dashboard and HTML context docs.                                                                                                                                                  |
| Proof target              | Specify the DB/web-publication object without giving it CNNA truth-source status or public write authority.                                                                                                                                                                                                                                                        |
| Proof sketch              | Keep Lean/YAML as source;<br>export snapshots into DB;<br>expose only read-only public views;<br>preserve snapshot provenance and checksums;<br>forbid browser service_role keys and public writes;<br>keep admin/import writes local or later signed/admin-only.                                                                                                  |
| Verification / audit      | Planned. Closure requires schema/import/RLS/frontend contract review, no secrets in repo/chat, and roundtrip/export evidence. V1.22 records planning objects only.                                                                                                                                                                                                 |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When the pre-code gate stabilizes or DB work becomes active, create migration/importer artifacts and a Supabase context documentation slice from these DBR records. |

Object or quantity DBR20 – Reviewer annotation and future admin-write model

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR20                                                                                                                                                                                                                                                                                                                                                              |
| Object status                   | W                                                                                                                                                                                                                                                                                                                                                                  |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                                                                                                                                                                                               |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                                                 |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                                     |
| Documented                      | planned                                                                                                                                                                                                                                                                                                                                                            |
| Phase                           | 20.20                                                                                                                                                                                                                                                                                                                                                              |
| Source objects                  | SLTG-06;<br>DBR0-DBR14;<br>Master YAML;<br>generated exports;<br>Supabase official docs on API security/RLS/service keys as external implementation reference only.                                                                                                                                                                                                |
| Definitions                     | review.* for later authenticated comments/annotations;<br>admin/import.* for local importer or future signed Edge Function;<br>no public writes and no manual table-click update as standard workflow.                                                                                                                                                             |
| Proof target                    | Specify the DB/web-publication object without giving it CNNA truth-source status or public write authority.                                                                                                                                                                                                                                                        |
| Proof sketch                    | Keep Lean/YAML as source;<br>export snapshots into DB;<br>expose only read-only public views;<br>preserve snapshot provenance and checksums;<br>forbid browser service_role keys and public writes;<br>keep admin/import writes local or later signed/admin-only.                                                                                                  |
| Verification / audit            | Planned. Closure requires schema/import/RLS/frontend contract review, no secrets in repo/chat, and roundtrip/export evidence. V1.22 records planning objects only.                                                                                                                                                                                                 |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. When the pre-code gate stabilizes or DB work becomes active, create migration/importer artifacts and a Supabase context documentation slice from these DBR records. |

Object or quantity DBR21 – Enlarged generated-PDF typography profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                             |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR21                                                                                                                                                                                                                                                                                                              |
| Object status                   | G                                                                                                                                                                                                                                                                                                                  |
| Context class                   | GeneratedArtifactObject                                                                                                                                                                                                                                                                                            |
| Significance tier               | F3                                                                                                                                                                                                                                                                                                                 |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                                                                          |
| Documented                      | yes                                                                                                                                                                                                                                                                                                                |
| Phase                           | 20.21                                                                                                                                                                                                                                                                                                              |
| Source objects                  | DBR12 section-pagebreak layout invariant;<br>DBR14 documentation section explanations;<br>DBR21 typography request;<br>existing full/context/table generators.                                                                                                                                                     |
| Definitions                     | Full and context PDF generators use 11pt A3 landscape document classes;<br>generated table fragments bump explicit TeX sizes one step: scriptsize to footnotesize, footnotesize to small, small to normalsize, tiny to scriptsize;<br>generated graph labels are bumped by one point where controlled by the tool. |
| Proof target                    | Show that PDF typography enlargement is implemented at generator level and does not edit disposable PDF outputs or change Lean/YAML row semantics.                                                                                                                                                                 |
| Proof sketch                    | Patch generator sources, regenerate tables/exports, compile representative full/context PDFs, inspect generated TeX for bumped size commands and render early pages to ensure no obvious clipping/black-glyph failure.                                                                                             |
| Verification / audit            | V1.23 regenerates tables, compiles the full PDF and a Supabase context PDF, checks generated TeX for documentclass[11pt] and larger table size commands, renders representative pages, and audits no Lean source under repo_snapshot/CNNA changed.                                                                 |
| Next documentation action       | If readability still needs adjustment, add an explicit PDF typography profile table instead of ad-hoc per-table changes.                                                                                                                                                                                           |

Object or quantity DBR22 – Supabase live project discovery and empty-schema baseline audit

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR22                                                                                                                                                                                                                 |
| Object status                   | G                                                                                                                                                                                                                     |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                                                  |
| Significance tier               | F1                                                                                                                                                                                                                    |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                        |
| Documented                      | yes                                                                                                                                                                                                                   |
| Phase                           | 20.22                                                                                                                                                                                                                 |
| Source objects                  | DBR15-DBR20;<br>GNG34;<br>live Supabase discovery;<br>Master YAML;<br>generated_exports;<br>repo_inventory/raw_export.                                                                                                |
| Definitions                     | Record the reachable Supabase organization/project baseline before any write: CNNA Project;<br>project ref aiekahdewadrejrrdajn;<br>ACTIVE_HEALTHY;<br>Postgres 17;<br>no tables in public/internal/api/review/admin. |
| Proof target                    | Closed by read-only discovery. No schema/data write. This is the baseline evidence for later migration/import phases.                                                                                                 |
| Proof sketch                    | Keep DB activation as generated-secondary tooling: every imported row carries snapshot/provenance, public access is read-only, writes require explicit authorization, and no Lean source file is changed.             |

|                           |                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / audit      | For 20.22: read-only Supabase organization/project/table discovery only. For later phases: require migration/import report, checksum report, advisor output and public-read-only tests.                                                                                                                                                  |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. After explicit authorization, record actual migration/import result and compare live DB inventory against planned schema/import contract. |

### Object or quantity DBR23 – Supabase initial schema migration package and RLS/read-only policy design

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">DBR23</a>                                                                                                                                                                                                                                                                                                                    |
| Object status                   | W                                                                                                                                                                                                                                                                                                                                        |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                                                                                                                                                                     |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                       |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                           |
| Documented                      | planned                                                                                                                                                                                                                                                                                                                                  |
| Phase                           | <a href="#">20.23</a>                                                                                                                                                                                                                                                                                                                    |
| Source objects                  | DBR15-DBR20;<br>GNG34;<br>live Supabase discovery;<br>Master YAML;<br>generated_exports;<br>repo_inventory/raw_export.                                                                                                                                                                                                                   |
| Definitions                     | Produce the first migration package for internal/api/review/admin schemas, internal canonical tables, api read-only views, grants/RLS posture, and comments documenting non-truth-source status.                                                                                                                                         |
| Proof target                    | Planned tooling artifact. No live migration is applied until explicit user authorization.                                                                                                                                                                                                                                                |
| Proof sketch                    | Keep DB activation as generated-secondary tooling: every imported row carries snapshot/provenance, public access is read-only, writes require explicit authorization, and no Lean source file is changed.                                                                                                                                |
| Verification / audit            | For 20.22: read-only Supabase organization/project/table discovery only. For later phases: require migration/import report, checksum report, advisor output and public-read-only tests.                                                                                                                                                  |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. After explicit authorization, record actual migration/import result and compare live DB inventory against planned schema/import contract. |

### Object or quantity DBR24 – Supabase snapshot import package from YAML/generated exports

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">DBR24</a>                                                                                                                                                                          |
| Object status                   | W                                                                                                                                                                                              |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                           |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | planned                                                                                                                                                                                        |
| Phase                           | <a href="#">20.24</a>                                                                                                                                                                          |
| Source objects                  | DBR15-DBR20;<br>GNG34;<br>live Supabase discovery;<br>Master YAML;<br>generated_exports;<br>repo_inventory/raw_export.                                                                         |
| Definitions                     | Define importer contract from Master YAML, generated_exports and repo_inventory into internal tables with snapshot_id, hashes, import report and no Lean mutation.                             |



|                           |                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof target              | Planned tooling artifact. It prepares data import only; no DB data write until explicit authorization.                                                                                                                                                                                                                                   |
| Proof sketch              | Keep DB activation as generated-secondary tooling: every imported row carries snapshot/provenance, public access is read-only, writes require explicit authorization, and no Lean source file is changed.                                                                                                                                |
| Verification / audit      | For 20.22: read-only Supabase organization/project/table discovery only. For later phases: require migration/import report, checksum report, advisor output and public-read-only tests.                                                                                                                                                  |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. After explicit authorization, record actual migration/import result and compare live DB inventory against planned schema/import contract. |

### Object or quantity DBR25 – Explicitly authorized live Supabase schema migration and first snapshot import

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">DBR25</a>                                                                                                                                                                                                                                                                                                                    |
| Object status                   | W                                                                                                                                                                                                                                                                                                                                        |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                                                                                                                                                                     |
| Significance tier               | F1                                                                                                                                                                                                                                                                                                                                       |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                           |
| Documented                      | planned                                                                                                                                                                                                                                                                                                                                  |
| Phase                           | <a href="#">20.25</a>                                                                                                                                                                                                                                                                                                                    |
| Source objects                  | DBR15-DBR20;<br>GNG34;<br>live Supabase discovery;<br>Master YAML;<br>generated_exports;<br>repo_inventory/raw_export.                                                                                                                                                                                                                   |
| Definitions                     | Execute, only after explicit user authorization, the schema migration and first data import into Supabase project aiekahdewadrejrrdajn using controlled admin/tool write path.                                                                                                                                                           |
| Proof target                    | Future live DB-write phase. It writes only Supabase schema/data, never Lean files, and must produce migration/import reports.                                                                                                                                                                                                            |
| Proof sketch                    | Keep DB activation as generated-secondary tooling: every imported row carries snapshot/provenance, public access is read-only, writes require explicit authorization, and no Lean source file is changed.                                                                                                                                |
| Verification / audit            | For 20.22: read-only Supabase organization/project/table discovery only. For later phases: require migration/import report, checksum report, advisor output and public-read-only tests.                                                                                                                                                  |
| Next documentation action       | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. After explicit authorization, record actual migration/import result and compare live DB inventory against planned schema/import contract. |

### Object or quantity DBR26 – Post-import read-only/RLS/advisor verification and web-frontend handoff

**Table legend.** W = white/finished public object;   B = blue/proof-carrying but semantic/API correction still open;   G = green/locally closable;   Y = yellow/known route with prerequisites;   R = red/derivation layer incomplete or unclear;   K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">DBR26</a>                                                                                                                                                                          |
| Object status                   | W                                                                                                                                                                                              |
| Context class                   | DatabaseMirrorObject                                                                                                                                                                           |
| Significance tier               | F1                                                                                                                                                                                             |
| Documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Documented                      | planned                                                                                                                                                                                        |
| Phase                           | <a href="#">20.26</a>                                                                                                                                                                          |

|                           |                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source objects            | DBR15-DBR20;<br>GNG34;<br>live Supabase discovery;<br>Master YAML;<br>generated_exports;<br>repo_inventory/raw_export.                                                                                                                                                                                                                   |
| Definitions               | Verify public read-only access, absence of public writes, Supabase advisor output, table/view inventory, publishable frontend data contract and cnna-project.net handoff.                                                                                                                                                                |
| Proof target              | Future verification/handoff phase after live import;<br>no Lean files and no phase-release authority.                                                                                                                                                                                                                                    |
| Proof sketch              | Keep DB activation as generated-secondary tooling: every imported row carries snapshot/provenance, public access is read-only, writes require explicit authorization, and no Lean source file is changed.                                                                                                                                |
| Verification / audit      | For 20.22: read-only Supabase organization/project/table discovery only. For later phases: require migration/import report, checksum report, advisor output and public-read-only tests.                                                                                                                                                  |
| Next documentation action | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. After explicit authorization, record actual migration/import result and compare live DB inventory against planned schema/import contract. |

### Object or quantity GA0 – Horn quaternion/geometric-algebra reference-gate packet

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                                                                 |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | GA0                                                                                                                                                                                                                                                                                    |
| Object status                   | K                                                                                                                                                                                                                                                                                      |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                                                    |
| Significance tier               | F2                                                                                                                                                                                                                                                                                     |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                          |
| Documented                      | no                                                                                                                                                                                                                                                                                     |
| Phase                           | 1.4.9                                                                                                                                                                                                                                                                                  |
| Source objects                  | Horn 0709.2238v1;<br>external primary/source checks for mathlib Quaternion/CliffordAlgebra and GA formalization;<br>no Lean CNNA computation.                                                                                                                                          |
| Definitions                     | A documentation-only packet of target patterns extracted from Horn 0709.2238v1: Hamilton quaternion relations, Pauli/Cartan relation, geometric-product basis grades, pseudoscalar duality, pure-quaternion-as-bivector interpretation and complex-quaternion eight-basis bookkeeping. |
| Proof target                    | Future proof/documentation target: show that the referenced CNNA-derived objects satisfy or fail the external target profile, with the external source classified as comparison-only and with no handoff rights.                                                                       |
| Proof sketch                    | Current state is a reference-gate scratchpad, not a Lean proof. Implementation must first expose CNNA-derived predecessor data, then encode the comparison profile structurally and prove/falsify it without importing external theory as generator.                                   |
| Verification / audit            | No Lean code introduced in V1.25;<br>generator validation and table/PDF generation are the only checks. Expert review required before implementation.                                                                                                                                  |
| Next documentation action       | When phase 1.4.9 becomes implementable, replace this placeholder with exact Lean module/declaration links and sign/provenance audit.                                                                                                                                                   |

### Object or quantity GA1 – Hamilton unit and Pauli/Cartan sign-convention comparison profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | GA1                                                                                                                                                                           |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |

|                           |                                                                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase                     | 1.4.9.1                                                                                                                                                                                                                                              |
| Source objects            | GA0 and external reference equations only;<br>future comparison waits for CD0-CD7 closure.                                                                                                                                                           |
| Definitions               | A target profile recording unit-square, anti-commutation and sign-convention checks for Hamilton units and their Pauli/Cartan representation, with explicit separation of complex i and quaternion i.                                                |
| Proof target              | Future proof/documentation target: show that the referenced CNNA-derived objects satisfy or fail the external target profile, with the external source classified as comparison-only and with no handoff rights.                                     |
| Proof sketch              | Current state is a reference-gate scratchpad, not a Lean proof. Implementation must first expose CNNA-derived predecessor data, then encode the comparison profile structurally and prove/falsify it without importing external theory as generator. |
| Verification / audit      | No Lean code introduced in V1.25;<br>generator validation and table/PDF generation are the only checks. Expert review required before implementation.                                                                                                |
| Next documentation action | When phase 1.4.9.1 becomes implementable, replace this placeholder with exact Lean module/declaration links and sign/provenance audit.                                                                                                               |

### Object or quantity GA2 – Pauli bivector/pseudoscalar duality comparison profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | GA2                                                                                                                                                                                                                                                  |
| Object status                   | K                                                                                                                                                                                                                                                    |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                  |
| Significance tier               | F2                                                                                                                                                                                                                                                   |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                        |
| Documented                      | no                                                                                                                                                                                                                                                   |
| Phase                           | 1.4.9.2                                                                                                                                                                                                                                              |
| Source objects                  | GA0, GA1 and external GA references;<br>no CNNA-derived orientation or dimension is assumed.                                                                                                                                                         |
| Definitions                     | A target profile for geometric-algebra duality: scalar/trivector and vector/bivector pairing, pseudoscalar I = sigma_x sigma_y sigma_z, I^2 = -1, and pure quaternions as oriented-plane/bivector targets.                                           |
| Proof target                    | Future proof/documentation target: show that the referenced CNNA-derived objects satisfy or fail the external target profile, with the external source classified as comparison-only and with no handoff rights.                                     |
| Proof sketch                    | Current state is a reference-gate scratchpad, not a Lean proof. Implementation must first expose CNNA-derived predecessor data, then encode the comparison profile structurally and prove/falsify it without importing external theory as generator. |
| Verification / audit            | No Lean code introduced in V1.25;<br>generator validation and table/PDF generation are the only checks. Expert review required before implementation.                                                                                                |
| Next documentation action       | When phase 1.4.9.2 becomes implementable, replace this placeholder with exact Lean module/declaration links and sign/provenance audit.                                                                                                               |

### Object or quantity GA3 – Complex-quaternion eight-basis multivector comparison profile

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | GA3                                                                                                                                                                           |
| Object status                   | K                                                                                                                                                                             |
| Context class                   | ReferenceGateObject                                                                                                                                                           |
| Significance tier               | F2                                                                                                                                                                            |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Documented                      | no                                                                                                                                                                            |
| Phase                           | 1.4.9.2                                                                                                                                                                       |
| Source objects                  | GA0-GA2;<br>external reference only, no CNNA scalar surface or C-layer assumption.                                                                                            |

|                           |                                                                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definitions               | A target bookkeeping profile for the eight real components of complex quaternions mapped to geometric-algebra grades: scalar, vector, bivector and pseudoscalar components.                                                                          |
| Proof target              | Future proof/documentation target: show that the referenced CNNA-derived objects satisfy or fail the external target profile, with the external source classified as comparison-only and with no handoff rights.                                     |
| Proof sketch              | Current state is a reference-gate scratchpad, not a Lean proof. Implementation must first expose CNNA-derived predecessor data, then encode the comparison profile structurally and prove/falsify it without importing external theory as generator. |
| Verification / audit      | No Lean code introduced in V1.25;<br>generator validation and table/PDF generation are the only checks. Expert review required before implementation.                                                                                                |
| Next documentation action | When phase 1.4.9.2 becomes implementable, replace this placeholder with exact Lean module/declaration links and sign/provenance audit.                                                                                                               |

### Object or quantity GA4 – Boundary Pauli/Clifford comparison gate

**Table legend.**   W = white/finished public object;     B = blue/proof-carrying but semantic/API correction still open;     G = green/locally closable;     Y = yellow/known route with prerequisites;     R = red/derivation layer incomplete or unclear;     K = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">GA4</a>                                                                                                                                                                                                                                  |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                                                                                                                                                    |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                  |
| Significance tier               | F2                                                                                                                                                                                                                                                   |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                        |
| Documented                      | no                                                                                                                                                                                                                                                   |
| Phase                           | <a href="#">13.2.1</a>                                                                                                                                                                                                                               |
| Source objects                  | Future 13.2 boundary *-algebra plus GA0-GA3 target profiles;<br>no direct Pillar-A internals.                                                                                                                                                        |
| Definitions                     | A comparison gate asking whether generators produced by the CNNA-derived boundary *-algebra satisfy Pauli/Clifford-like square, star and anti-commutation targets.                                                                                   |
| Proof target                    | Future proof/documentation target: show that the referenced CNNA-derived objects satisfy or fail the external target profile, with the external source classified as comparison-only and with no handoff rights.                                     |
| Proof sketch                    | Current state is a reference-gate scratchpad, not a Lean proof. Implementation must first expose CNNA-derived predecessor data, then encode the comparison profile structurally and prove/falsify it without importing external theory as generator. |
| Verification / audit            | No Lean code introduced in V1.25;<br>generator validation and table/PDF generation are the only checks. Expert review required before implementation.                                                                                                |
| Next documentation action       | When phase 13.2.1 becomes implementable, replace this placeholder with exact Lean module/declaration links and sign/provenance audit.                                                                                                                |

### Object or quantity GA5 – Dirac/spacetime-algebra comparison target gate

**Table legend.**   W = white/finished public object;     B = blue/proof-carrying but semantic/API correction still open;     G = green/locally closable;     Y = yellow/known route with prerequisites;     R = red/derivation layer incomplete or unclear;     K = black/registered side or future target.   Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | <a href="#">GA5</a>                                                                                                                                                                                              |
| Object status                   | <span style="background-color: #A9A9A9;">K</span>                                                                                                                                                                |
| Context class                   | ReferenceGateObject                                                                                                                                                                                              |
| Significance tier               | F2                                                                                                                                                                                                               |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                    |
| Documented                      | no                                                                                                                                                                                                               |
| Phase                           | <a href="#">13.6.1</a>                                                                                                                                                                                           |
| Source objects                  | Future 13.5/13.6 outputs plus GA4;<br>no imported Dirac equation, Lorentz group or spacetime metric.                                                                                                             |
| Definitions                     | A comparison target for later CNNA-derived local-net/modular data against Dirac/spacetime-algebra Clifford signatures, Lorentz-covariance formulations and spin/CPT compatibility patterns.                      |
| Proof target                    | Future proof/documentation target: show that the referenced CNNA-derived objects satisfy or fail the external target profile, with the external source classified as comparison-only and with no handoff rights. |

|                           |                                                                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof sketch              | Current state is a reference-gate scratchpad, not a Lean proof. Implementation must first expose CNNA-derived predecessor data, then encode the comparison profile structurally and prove/falsify it without importing external theory as generator. |
| Verification / audit      | No Lean code introduced in V1.25; generator validation and table/PDF generation are the only checks. Expert review required before implementation.                                                                                                   |
| Next documentation action | When phase 13.6.1 becomes implementable, replace this placeholder with exact Lean module/declaration links and sign/provenance audit.                                                                                                                |

### Object or quantity GA6 – Geometric and conformal algebra side-comparison register

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | GA6                                                                                                                                                                                                                                                  |
| Object status                   | K                                                                                                                                                                                                                                                    |
| Context class                   | ReferenceGateObject                                                                                                                                                                                                                                  |
| Significance tier               | F2                                                                                                                                                                                                                                                   |
| Documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                        |
| Documented                      | no                                                                                                                                                                                                                                                   |
| Phase                           | 18.5                                                                                                                                                                                                                                                 |
| Source objects                  | GA0-GA5 and future derived CNNA outputs; never from external GA as a foundation.                                                                                                                                                                     |
| Definitions                     | A side-comparison register for geometric algebra, spacetime algebra and conformal geometric algebra as external target surfaces for already-derived CNNA outputs.                                                                                    |
| Proof target                    | Future proof/documentation target: show that the referenced CNNA-derived objects satisfy or fail the external target profile, with the external source classified as comparison-only and with no handoff rights.                                     |
| Proof sketch                    | Current state is a reference-gate scratchpad, not a Lean proof. Implementation must first expose CNNA-derived predecessor data, then encode the comparison profile structurally and prove/falsify it without importing external theory as generator. |
| Verification / audit            | No Lean code introduced in V1.25; generator validation and table/PDF generation are the only checks. Expert review required before implementation.                                                                                                   |
| Next documentation action       | When phase 18.5 becomes implementable, replace this placeholder with exact Lean module/declaration links and sign/provenance audit.                                                                                                                  |

### Object or quantity DBR27 – Phase/Object context-class taxonomy

**Table legend.** W = white/finished public object; B = blue/proof-carrying but semantic/API correction still open; G = green/locally closable; Y = yellow/known route with prerequisites; R = red/derivation layer incomplete or unclear; K = black/registered side or future target. Context class is orthogonal to status.

| Field                           | Record                                                                                                                                                                                                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object ID                       | DBR27                                                                                                                                                                                                                                                |
| Object status                   | G                                                                                                                                                                                                                                                    |
| Context class                   | ToolInfrastructureObject                                                                                                                                                                                                                             |
| Significance tier               | F3                                                                                                                                                                                                                                                   |
| Documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                            |
| Documented                      | yes                                                                                                                                                                                                                                                  |
| Phase                           | 20.27                                                                                                                                                                                                                                                |
| Source objects                  | LEG34;<br>DBR13;<br>status_legend;<br>phase_object_link_policy;<br>object register;<br>phase table;<br>user-requested OOP-style class separation                                                                                                     |
| Definitions                     | PhaseContextClass and ObjectContextClass are orthogonal metadata fields. They identify Lean/tool/reference/database/generated/governance/documentation class boundaries while Object status and Phase status continue to encode release/proof state. |

|                           |                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof target              | Complete register coverage and visible rendering: every phase/object has exactly one context class from the taxonomy, and generated phase/object/proof/scratchpad tables show the class beside status-sensitive fields.                                                                                                                      |
| Proof sketch              | Generated schema enforces non-empty Context class on phases and objects. The taxonomy is rendered before phase/object registers. Second-pass semantic audit separates LeanSourceObject/LeanTargetObject from LeanRefactorObject, LeanHandoffContractObject, LeanGovernanceObject and non-Lean tool/reference/database/documentation classes. |
| Verification / audit      | python3 scripts/generate_tables.py validates presence of context_class_taxonomy and Context class fields;<br>PDF generation renders the taxonomy and all phase/object tables with Context class columns. No Lean source is changed by this tooling/schema phase.                                                                             |
| Next documentation action | Mirror context classes into future Supabase schema and require future phase/object additions to declare a class at creation time.                                                                                                                                                                                                            |



18 Phase table in natural derivation order

**What this section shows.** This is the primary roadmap table. It is the authoritative generated view for phase status, dependencies, linked objects, current code sites, target modules/objects, proof goals and implementation scratchpad links. Lean-derivation phases are red by default until released by derived-only object evidence; yellow is the unique active phase; orange is implementable after stated predecessor closure; white is reserved for non-direct-Lean tooling, infrastructure, documentation, comparison or refactor work.

**Phase table rule.** This is the primary roadmap and release table. Green means closed/derived-only or governance-closed. Yellow is the unique active implementable phase. Orange means implementable only after the stated predecessor closure. Red is the default for unreleased Lean-derivation phases until they cite sufficient derived-only object/proof evidence. White is reserved for non-direct-Lean tooling, infrastructure, documentation, comparison, packaging, or refactor work. Context class is the orthogonal type/class field; Phase documentation tier is the orthogonal documentation-depth field: it distinguishes LeanDerivationPhase/LeanRefactorPhase from ToolInfrastructurePhase, DatabaseMirrorPhase, ReferenceGatePhase and generated/documentation classes. The module-change appendix must never replace this release logic.

Single phase table: derivation order, dependencies, code sites, proof goals, and implementation links

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

|     | Phase                                             | Status   | Context class                     | Doc tier | Linked objects                                                                                                                                                                                                                                      | Main goal                                                                                                                       | Source objects/<br>dependencies                                         | Definitions/objects                                                                                                                                                                                                                            | Current code modules                                                                                                                        | Target modules/<br>objects                                                                                                                                                                                            | Proof goals                                                                                                                                                                                                                                                                                                                  | How imple-<br>mented                   |
|-----|---------------------------------------------------|----------|-----------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 0   | <span style="background-color: #90EE90;"> </span> | <b>G</b> | BaselinePhase                     | F3       | <i>no direct object:</i><br>baseline/audit/<br>closeout/<br>infrastructure<br>phase with no<br>registered object<br>ownership                                                                                                                       | Freeze the SM9h<br>baseline unchanged                                                                                           | green SM9h zip                                                          | frozen legacy baseline                                                                                                                                                                                                                         | CNNA/PillarA/Arithmetic/<br>BuildAll.lean;<br>Smoke/* through GeneratedOpera-<br>tionalQuadraticDiscriminantSource-<br>FromSM9g1Window.lean | no new definitions                                                                                                                                                                                                    | reference unchanged;<br>no refactor in the original                                                                                                                                                                                                                                                                          | <a href="#">implemented<br/>record</a> |
| 1   | <span style="background-color: #90EE90;"> </span> | <b>G</b> | Lean;<br>Refactor;<br>Phase       | F0       | <i>scope-only;</i> child<br>objects: LEG0,<br>LEG13, LEG14,<br>LEG15, LEG20,<br>LEG21, LEG22,<br>LEG23, LEG24,<br>LEG25, LEG27,<br>LEG28, LEG29,<br>LEG30, LEG31,<br>LEG32, LEG33,<br>LEG34, LEG10,<br>LEG11, LEG12,<br>LEG5, LEG7,<br>LEG6;<br>... | Neutral code surface<br>umbrella;<br>decomposition and<br>active branch fixed<br>before object intake<br>and new<br>mathematics | 0;<br>static code audit;<br>frozen SM9h baseline                        | Legacy/Neutral wrapper<br>policy;<br>notation split;<br>phase-semantic-name<br>quarantine;<br>foundation/regime/CD<br>wrapper gates                                                                                                            | Arithmetic/Notation.lean;<br>all SM*/S6* modules                                                                                            | neutral API according to<br>object register;<br>no new mathematics<br>before import surface is<br>clean                                                                                                               | Decomposition-closed umbrella.<br>This is not a claim that all<br>descendant Lean refactors are<br>implemented;<br>it records that the child<br>ordering and active cursor are<br>fixed so no white predecessor<br>stands before the next active<br>phase. Implementation closure<br>is delegated to descendants<br>1.1-1.4. | <a href="#">implemented<br/>record</a> |
| 1.1 | <span style="background-color: #90EE90;"> </span> | <b>G</b> | Tool;<br>Infrastructure;<br>Phase | F0       | <i>scope-only;</i> child<br>objects: LEG0,<br>LEG13, LEG14,<br>LEG15, LEG20,<br>LEG21, LEG22,<br>LEG23, LEG24,<br>LEG25, LEG27,<br>LEG28, LEG29,<br>LEG30, LEG31,<br>LEG32, LEG33,<br>LEG34, LEG10,<br>LEG11, LEG12,<br>LEG5, LEG7,<br>LEG6;<br>... | Notation split and<br>legacy/API cleanup<br>umbrella;<br>child gates fixed in<br>derivation order                               | 1;<br>Arithmetic/<br>Notation.lean audit;<br>YAML single-master<br>rule | Declaration consumption<br>ledger;<br>blue-object semantic<br>correction ledger;<br>public API contract;<br>pruning/dead-candidate<br>ledger;<br>Legacy.StatusGates<br>quarantine;<br>CoreSymbols;<br>Legacy phase aliases;<br>import firewall | Arithmetic/Notation.lean;<br>Foundation/Notation.lean;<br>Finite/Notation.lean;<br>Handoff/Notation.lean;<br>generated TeX/PDF views        | CNNA/PillarA/<br>Arithmetic/Notation/<br>CoreSymbols.lean;<br>CNNA/PillarA/<br>Arithmetic/Legacy/<br>StatusGates.lean;<br>CNNA/PillarA/<br>Arithmetic/Legacy/<br>PhaseAliases.lean;<br>generated v22 PDF/TSV<br>views | Decomposition-closed umbrella.<br>Do not implement here;<br>execute the child gates in order.<br>This row is green only as a<br>routing/order lock, not as<br>completed Lean code<br>movement.                                                                                                                               | <a href="#">implemented<br/>record</a> |

| Phase   | Status | Context                    | class | Doc tier | Linked objects                                                                                                                                                                             | Main goal                                                                                                                                                             | Source objects/<br>dependencies                                                                                                                          | Definitions/objects                                                                                                                                                                                                                                                                                                                                                                                                                      | Current code modules                                                                                    | Target modules/<br>objects                                                                                                                                        | Proof goals                                                                                                                                                                                                                                                                                                                                                         | How imple-<br>mented               |
|---------|--------|----------------------------|-------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1.1.1   | G      | Documentation; F0<br>Phase |       |          | LEG0                                                                                                                                                                                       | Pre-code governance umbrella: module manifest, reuse, consumption classification, proofbook infrastructure, blue correction, API and pruning gates                    | 1.1;<br>static module graph; declaration/reference inventory;<br>blue object register; Arithmetic/Notation.lean audit; YAML single-master rule           | module implementation manifest;<br>semantic placement contract;<br>reuse/duplication ledger; canonical generator policy;<br>declaration consumption classes;<br>object proof documentation infrastructure;<br>blue-object semantic correction ledger;<br>public API contract; pruning/dead-candidate ledger;<br>generated-view authority rule;<br>semantic module consolidation audit and compatibility-preserving merge/no-merge ledger | Arithmetic/Notation.lean; README.md;<br>masters/<br>cnna_roadmap_master.yaml; generated tables/PDF      | masters/<br>cnna_roadmap_master.yaml; generated exports;<br>CONSUMPTION_AND_REUSE_OBJECT_PRECORRECTION_REPORT.md; no Lean declaration move yet                    | Decomposition-closed umbrella. It fixes that module/reuse governance and proofbook declaration classification, blue-object public correction, API exposure, and pruning. Remaining implementation closure is delegated to child gates. V1.12 inserts Phase 1.1.1.1.5 as a pre-substantive refactor/consolidation gate before the blue-object content phase resumes. | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Documentation; F0<br>Phase |       |          | <i>scope-only</i> ; child objects: LEG13, LEG14, LEG15, LEG20, LEG21, LEG22, LEG23, LEG24, LEG25, LEG27, LEG28, LEG29, LEG30, LEG31, LEG32, LEG33, LEG34, LEG10, LEG11, LEG12, LEG5, LEG36 | Module/declaration consumption classification umbrella with architecture-decision locks, pre-implementation module manifest, reuse governance, and proofbook ordering | 1.1.1;<br>repo_inventory/raw_export; Arithmetic/Notation.lean; symbol reference summary;<br>prior static code audit; existing Smoke/Test generator chain | Global no-go contract; concrete instance traversal;<br>EXEC-2 rational-boundary policy;<br>refactor regression protocol;<br>Lean/mathlib version policy;<br>phase-type scratchpads; Phase 3->4 consumer contract;<br>declaration consumption ledger;<br>blue-object correction ledger;<br>public API contract; pruning ledger.;<br>semantic module consolidation audit and compatibility-preserving merge/no-merge ledger                | repo_snapshot/**/*.*.lean; repo_inventory/raw_export/*.*.json; generated_exports/*.*.tsv                | masters/<br>cnna_roadmap_master.yaml; generated module-manifest/consumption records; MODULE_MANIFEST_AND_REUSE_GOVERNANCE_REPORT.md; no Lean declaration move yet | Decomposition-closed umbrella. Architecture decision locks 1.1.1.1.0-1.1.1.1.0.14 must be green before the active module-manifest gate. This is closure. V1.12 inserts Phase 1.1.1.1.5 as a pre-substantive refactor/consolidation gate before the blue-object content phase resumes.                                                                               | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Lean; Governance; Phase    | F0    |          | LEG13                                                                                                                                                                                      | Global no-go guardrail contract and inheritance lock                                                                                                                  | 1.1.1.1;<br>operational-path constraints from prior Lean implementation history;<br>legacy quarantine requirements; YAML single-master rule              | GlobalNoGoGuardrailContract; global_no_go_guardrails top-level YAML section; scratchpad inheritance field;<br>workflow rule that local no-go lists are addenda only                                                                                                                                                                                                                                                                      | masters/<br>cnna_roadmap_master.yaml; scripts/generate_tables.py; all implementation_scratchpad records | LEG13 Global no-go guardrail contract; generated global_no_go_guardrails table; scratchpad global_no_go_inheritance rows                                          | Planning-infrastructure gate. Promote the operational no-go list from phase-local scratchpad content to a global invariant that applies to every phase and every future code-edit prompt. Close this before module-manifest implementation starts so no child phase can silently weaken the rules.                                                                  | <a href="#">implemented record</a> |

| Phase   | Status | Context                       | class | Doc tier | Linked objects                                | Main goal                                                                               | Source objects/<br>dependencies                                                                                                                                              | Definitions/objects                                                                                                                                                                                                                                                                   | Current code modules                                                                                          | Target modules/<br>objects                                                                                                                                           | Proof goals                                                                                                                                                                                                                                                                                                                                                                                  | How imple-<br>mented               |
|---------|--------|-------------------------------|-------|----------|-----------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase | F3    |          | <a href="#">LEG14</a> , <a href="#">LEG15</a> | Concrete computable instance traversal and mathlib-complication scratchpad contract     | 1.1.1.1.0; global operational-path no-go rules; prior SM/AR implementation history; mathlib computability complications observed around polynomial/matrix/noncomputable APIs | ConcreteComputableInstanceRehearsal; ComputablePathNonSkeletonContract; MathlibComplicationLog; NoncomputableBoundaryAnnouncement; per-phase scratchpad fields for concrete instance path, expected executability witness, mathlib complication status, and mitigation/refactor layer | mathlib/rsa; toml/contract; scripts/generate_tables.py; implementation_scratchpad records; no Lean code moved | LEG14 Concrete computable instance traversal contract; LEG15 Mathlib complication and noncomputable-boundary log; generated scratchpad fields; workflow-policy entry | Planning-infrastructure gate. Close before any implementation starts: the operational computable path may not remain a formal skeleton. Each implementation phase must name at least one concrete instance or explain why the phase is infrastructure-only, record how that instance traverses the path, and log mathlib/noncomputable complications with their isolation or refactor layer. | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase | F1    |          | <a href="#">LEG20</a>                         | ExecComplex rational-boundary and mirror-only algebraic irrationality contract (EXEC-2) | 1.1.1.1.0.1; O00/O01/O06; L00/L02/L04; M05; SP0                                                                                                                              | EXEC-2 policy: operational ExecComplex remains rational $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrationalities live in mirror/non-operational layers; only proved rational/integer invariants may return operationally                                                            | Planning master; future ExecComplex arithmetic and geometry/modular modules                                   | LEG20; format decision EXEC-2; updated ExecComplex/computable-path notes for O06/L00/L02/L04/M05/SP0                                                                 | Close the architecture decision before any Phase 7+ algebraic-geometry object is prepared. This does not prove CM geometry; it prevents implicit leakage of $\sqrt{D}$ , $\tau$ , algebraic eigenvalues, or $j$ -values into the operational $\mathbb{Q} \times \mathbb{Q}$ path.                                                                                                            | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase | F1    |          | <a href="#">LEG21</a>                         | Refactor regression protocol and incremental migration policy                           | 1.1.1.1.0.2; Notation.lean audit; global build rule                                                                                                                          | wrapper-first migration; no-red intermediate policy; declarations_migrated_count/import_graph_delta/consumer_breakage_audit metrics                                                                                                                                                   | Arithmetic/Notation.lean; future CoreSymbols/Legacy modules; BuildAll and import consumers                    | LEG21; scratchpad refactor fields; regression protocol workflow                                                                                                      | Before the Notation split starts, choose a safe incremental mode: add neutral wrappers next to legacy, migrate consumers in bounded batches, and require local full-build evidence after each submitted code package.                                                                                                                                                                        | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase | F1    |          | <a href="#">LEG22</a>                         | Lean/mathlib version baseline and upgrade-gate policy                                   | 1.1.1.1.0.3; Lean 4.28.0/mathlib4 baseline; future number-theory imports                                                                                                     | version-freeze baseline; phase-boundary upgrade gates; mirror/non-operational isolation for noncomputable mathlib pressure                                                                                                                                                            | No code change; future imports in number theory, ring theory, spectral and modular phases                     | LEG22; scratchpad lean_mathlib_version_policy field; workflow policy for upgrades                                                                                    | Record that implementation uses the frozen baseline unless an explicit phase-boundary upgrade gate is opened. Any upgrade is a separate review/build event, not an implicit import adjustment inside a math phase.                                                                                                                                                                           | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase | F3    |          | <a href="#">LEG23</a>                         | Phase-type-specific scratchpad schema and governance-depth review gate                  | 1.1.1.1.0.4; current scratchpad schema; governance-to-math ratio review                                                                                                      | phase_type = baseline/refactor/infrastructure/derivation/comparison/documentation; type-specific required evidence; governance-depth retrospective after first real code change                                                                                                       | Planning master and generated scratchpad renderer                                                             | LEG23; phase_type scratchpad fields; governance_overhead_review field                                                                                                | Avoid dead scratchpad ballast: refactor phases must record migration metrics; derivation phases must record proof rehearsal and tactics; comparison phases must record derived-only boundary. After Phase 1.1.2, review whether governance depth should be flattened for Phase 3+.                                                                                                           | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase | F1    |          | <a href="#">LEG24</a>                         | Phase 3 to Phase 4 consumer-contract direction lock                                     | 1.1.1.1.0.5; GEN-1 canonical generator policy; Phase 3.6 BoundaryOperatorProfile; Phase 4.1 ExecBoundaryMatrix                                                               | BoundaryOperatorProfile exports a consumer-ready contract for the operational generator path; Phase 4 consumes Phase 3 output; Phase 3 does not import Phase 4 implementation                                                                                                         | Future Schur/DtN neutral wrappers; future Operational/ExecComplex wrappers                                    | LEG24; updated Phase 3.6/4.1 phase goals and scratchpad consumer_contract_direction fields                                                                           | Keep the dependency direction mathematically correct while making the downstream consumer contract visible early: Phase 3.6 must publish the boundary/operator profile in the shape Phase 4 needs, without importing Phase 4.                                                                                                                                                                | <a href="#">implemented record</a> |

| Phase Status |   | Context class                 | Doc tier | Linked objects        | Main goal                                                                                                                                               | Source objects/<br>dependencies                                                                                                                                                                                         | Definitions/objects                                                                                                                                                                      | Current code modules                                                                                                                                                                                                                                                | Target modules/<br>objects                                                                                                                                                                                                    | Proof goals                                                                                                                                                                                                                                                                                                                                                                                   | How imple-<br>mented                   |
|--------------|---|-------------------------------|----------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 1.1.1.1      | G | Lean;<br>Governance;<br>Phase | F1       | <a href="#">LEG25</a> | IdealThread/<br>CutSpec.full<br>extremal-route<br>contract and<br>string-analogy<br>boundary lock                                                       | Code-checked<br>definitions:<br>InfiniteThread/<br>IdealThread,<br>IdealThread.carrier/<br>family/coarsen,<br>CutSpec.full, Ex-<br>istingIdealThreadNoGo,<br>FullBoundaryAudit.                                         | LEG25;<br>IdealThread singleton<br>route;<br>CutSpec.full maximal<br>route;<br>scalar-collapse warning;<br>comparison-only<br>string-analogy boundary.                                   | CNNA/PillarA/Foundation/<br>SubstrateClass.lean;<br>CNNA/PillarA/ToC/Contract.lean;<br>CNNA/PillarA/Finite/CutSpec.lean;<br>CNNA/PillarA/Arithmetic/<br>BoundarySource/<br>ExistingIdealThreadNoGo.lean;<br>CNNA/PillarA/Arithmetic/Core/<br>FullBoundaryAudit.lean | No Lean module in this<br>decision gate. Later<br>neutral modules: CNNA/<br>PillarA/ToC/<br>IdealThreadProfile.lean;<br>CNNA/PillarA/Finite/<br>FullCutProfile.lean;<br>CNNA/PillarA/Schur/<br>BoundaryRouteContract.lean     | Record that IdealThread is<br>singleton per level and<br>coarsen-fixed under<br>deterministic substrates, while<br>CutSpec.full carries T.carrier.<br>Do not use IdealThread as a<br>full-boundary source;<br>keep string-like analogy as<br>interpretation/comparison only.                                                                                                                  | <a href="#">implemented<br/>record</a> |
| 1.1.1.1      | G | Lean;<br>Governance;<br>Phase | F1       | <a href="#">LEG27</a> | Horizontal/vertical<br>Schur-complement<br>duality<br>dependency-order<br>lock                                                                          | Review insight;<br>existing Schur/CD/<br>arithmetic plan;<br>no Lean code<br>movement.                                                                                                                                  | LEG27;<br>GNG14;<br>BR08/BR09/CD8<br>dependency-order<br>correction;<br>one-generator bridge<br>boundary                                                                                 | Planning-only decision gate;<br>existing code not modified.                                                                                                                                                                                                         | masters/<br>cnna_roadmap_master.yaml<br>generated format<br>decisions, workflow policy,<br>object proof dossiers                                                                                                              | record that horizontal sector<br>Schur and vertical level Schur<br>may be investigated only after<br>their separate routes are<br>neutralized/proved;<br>forbid using the duality as an<br>input to earlier phases                                                                                                                                                                            | <a href="#">implemented<br/>record</a> |
| 1.1.1.1      | G | Lean;<br>Governance;<br>Phase | F1       | <a href="#">LEG28</a> | Heuristic Eisenstein/<br>Gauss lattice<br>test-matrix boundary<br>and ordering lock                                                                     | 1.1.1.1.0.2 EXEC-2;<br>1.1.1.1.0.6 Phase-3 to<br>Phase-4 consumer<br>direction;<br>1.1.1.1.0.7 IdealThread/<br>CutSpec route split;<br>TomS thread post #97                                                             | LEG28 heuristic lattice<br>test-matrix contract;<br>angle/cosine, closure<br>depth, CM-point, D,<br>Gram, j special-value<br>test targets;<br>operational-vs-mirror<br>split             | No new Lean module yet;<br>future tests must inspect existing<br>generator/branching/level APIs and<br>use the code-defined b, not the forum<br>drawing convention.                                                                                                 | Planning target only now.<br>Future code target must<br>be decided by module<br>manifest, likely a compact<br>test/profile module under<br>the neutral lattice/<br>arithmetic test surface<br>rather than a new<br>generator. | Lock that post-97 Eisenstein/<br>Gauss geometry is not a<br>generator definition. It<br>becomes a future test matrix:<br>static and dynamic<br>code-defined branching over<br>several L_max values,<br>computing rational/integer<br>operational invariants and<br>comparing mirror-only CM/j<br>targets after their source<br>objects exist.                                                 | <a href="#">implemented<br/>record</a> |
| 1.1.1.1      | G | Lean;<br>Governance;<br>Phase | F1       | <a href="#">LEG29</a> | General early finite<br>computation<br>test-matrix contract<br>and<br>non-specialization<br>lock                                                        | 1.1.1.1.0.9;<br>concrete computable<br>instance traversal<br>contract;<br>canonical generator<br>policy;<br>EXEC-2;<br>heuristic lattice sentinel<br>rows from post #97                                                 | LEG29;<br>O07;<br>generic finite test matrix<br>schema;<br>sentinel-row policy;<br>static/dynamic<br>branch-mode coverage;<br>L_max sweep coverage;<br>executable-witness<br>requirement | Planning/YAML only;<br>no Lean code change.                                                                                                                                                                                                                         | Future test modules must<br>consume the canonical<br>generator/branching API<br>and existing computation<br>modules;<br>exact module names fixed<br>by Phase 1.1.1.1.1<br>manifests before<br>implementation.                 | Lock that Eisenstein/Gauss are<br>sentinel rows inside a general<br>test matrix, not the whole test<br>purpose;<br>every computable operational<br>path must have finite<br>non-skeletal instances before<br>promotion.                                                                                                                                                                       | <a href="#">implemented<br/>record</a> |
| 1.1.1.1      | G | Lean;<br>Governance;<br>Phase | F1       | <a href="#">LEG30</a> | Regime-typed early<br>test matrix and<br>Heegner-projection<br>lock                                                                                     | 1.1.1.1.0.10 general test<br>matrix;<br>noOuter/HasOuter<br>regime discipline;<br>EXEC-2;<br>IDEAL_infty/<br>projection discipline;<br>post #97 heuristic;<br>external<br>class-number-one list as<br>comparison only   | LEG30, O08, N18,<br>BR11, BR12;<br>regime-tagged test rows;<br>noOuter-root sentinels;<br>HasOuter/<br>complement-projection<br>rows;<br>Heegner comparison<br>targets                   | Planning/YAML only;<br>no Lean code change.                                                                                                                                                                                                                         | Future implementation<br>extends O07 test-matrix<br>profile or its natural<br>owning module;<br>exact module names fixed<br>by Phase 1.1.1.1.1<br>manifest.                                                                   | Lock that Eisenstein/Gauss<br>may be early noOuter/<br>REAL-root sentinel candidates,<br>while additional Heegner/<br>class-number-one targets are<br>not closed-system outputs by<br>default but<br>complement-projection/<br>comparison rows unless their<br>regime is derived. Prevent<br>h(D)=1 from being used as<br>noOuter evidence.                                                   | <a href="#">implemented<br/>record</a> |
| 1.1.1.1      | G | Lean;<br>Governance;<br>Phase | F0       | <a href="#">LEG31</a> | Computable scalar<br>architecture decision:<br>ExecComplex<br>boundary, required<br>exact quadratic layer,<br>and Cayley-Dickson<br>noncircularity lock | O00, LEG20, LEG22,<br>LEG30;<br>Foundation/<br>ExecComplex.lean;<br>Foundation/<br>ExecComplexBridge.lean;<br>Arithmetic/<br>CayleyDickson/<br>ArithmeticCDBridge.lean;<br>mathlib Complex and<br>QuadraticAlgebra docs | LEG31, O09, BR13,<br>CD9                                                                                                                                                                 | Planning/master only;<br>code facts audited from<br>repo_snapshot paths and repo<br>inventory;<br>no Lean code change.                                                                                                                                              | Global scalar architecture<br>contract before<br>implementation module<br>manifest;<br>later candidate modules<br>must be proposed by<br>Phase 1.1.1.1.1 before code<br>exists.                                               | Lock the decision that<br>ExecComplex is not a complete<br>C layer;<br>general Complex/Real analytic<br>data remain mirror-only;<br>an exact quadratic algebraic<br>layer is required for CM/lattice<br>coordinates but cannot be<br>instantiated without a<br>proof-carrying D-source;<br>CD-derived C cannot be<br>consumed as a replacement<br>unless noncircular provenance<br>is proved. | <a href="#">implemented<br/>record</a> |

| Phase   | Status | Context                       | class | Doc tier | Linked objects        | Main goal                                                                                                     | Source objects/<br>dependencies                                                                                                                                                                               | Definitions/objects                                                                                                                                                                                   | Current code modules                                                                                                                 | Target modules/<br>objects                                                                                                                                                                        | Proof goals                                                                                                                                                                                                                                                                                                                                                                      | How imple-<br>mented               |
|---------|--------|-------------------------------|-------|----------|-----------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase |       | F0       | <a href="#">LEG32</a> | Derived-only exact quadratic algebra provenance lock: no free sqrt(D)                                         | 1.1.1.1.0.12 scalar architecture;<br>O00 ExecComplex floor;<br>O05/O06 operational discriminant path;<br>mathlib<br>QuadraticAlgebra and AdjoinRoot reference docs;<br>Cayley-Dickson noncircularity lock     | LEG32;<br>O09 reinterpreted as a required discriminant-indexed exact quadratic algebra schema;<br>O10/BR14 future proof-carrying instantiation gate;<br>GNG19 no-free-sqrt rule                       | Planning/master only;<br>no Lean code change.                                                                                        | masters/<br>cnna_roadmap_master.yaml<br>generated scalar architecture decisions, guardrails, proof dossiers, and future phase contracts                                                           | Lock that sqrt(D) is never introduced as an axiom or external number. The operational route is: Schur/DtN/charpoly/determinant -> proof-carrying D : Int -> canonical algebra with $\theta^2 = D$ over Q -> exact coordinate computations -> rational/integer invariant projection. If D is missing, backjump to the discriminant source rather than patching a scalar consumer. | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase |       | F1       | <a href="#">LEG33</a> | Phase 5-10 dependency bifurcation lock: number-theory/modular path separated from spectral/physical consumers | v36 phase/object register;<br>object phase audit for N/L/M/SP/P/TM/BR/CD groups;<br>Phase 4.1 ExecBoundaryMatrix dependency;<br>Phase 8/9 modular-form chain                                                  | LEG33 phase-path bifurcation and dependency repair contract;<br>GNG20 no artificial modular-form bottleneck for physical/spectral finite consumers                                                    | masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/<br>phase_roadmap.generated.tsv;<br>object register;<br>no Lean code      | Corrected Phase 5-10 roadmap: Path A number-theory/geometric/modular/analytic targets; Path B spectral/physical/transfer/bridge targets                                                           | Planning proof: SP0/SP1/P00-P06 consume Phase 4.1/finite boundary matrices and regime data, not Phase 8 modular/j objects. M00-M07 belong to the modular path. TM0-TM2 belong to Schur/transfer path and feed BR08 rather than modular-form phases.                                                                                                                              | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase |       | F1       | <a href="#">LEG34</a> | Global phase-object-link policy: no ambiguous empty Linked-objects cells                                      | v37 phase table;<br>object register;<br>generator rule object.Phase == phase.Phase;<br>phase hierarchy;<br>umbrella/routing/closeout/moved placeholders                                                       | LEG34 phase-object-link-policy contract;<br>object_link_policy kind direct/scope_only/<br>no_direct_object/moved;<br>validator rule that empty direct object lists require explicit non-direct policy | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>generated phase table;<br>object register;<br>no Lean source | Phase table Linked objects cells render direct object links, child-scope summaries, no-direct-object explanations, or moved placeholders;<br>generated policy table;<br>hard generator validation | Planning-infrastructure proof: a blank Linked objects cell is forbidden unless the phase explicitly declares why it has no direct object ownership. Direct ownership remains object.Phase == phase.Phase; umbrella phases may show child scope but do not steal object ownership from child phases.                                                                              | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase |       | F0       | <a href="#">LEG35</a> | Global ontic subsystem-minimality postulate and emergence-order lock                                          | IDEAL_infty formal provenance whole;<br>subsystem/cut/approximant discipline;<br>noOuter/HasOuter regime split;<br>complement-family contract;<br>Schur/DtN emergence chain;<br>prior global no-go guardrails | LEG35 ontic subsystem-minimality contract;<br>POST1 global postulate;<br>GNG22 no non-subsystem ontic primitives;<br>allowed primitive = subsystem-being only                                         | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>generated global postulate table;<br>no Lean source          | Global postulate rendered before workflows and phase table;<br>object/proof-dossier record LEG35;<br>scratchpad inheritance and no-go enforcement for later phases                                | Planning/architecture proof: dynamics, spectra, algebra, arithmetic, geometry, stochasticity, openness, unitarity and physical interpretation are not primitive inputs. They must be derived from subsystem/cut, complement/regime and Schur/DtN consequences inside IDEAL_infty formal provenance.                                                                              | <a href="#">implemented record</a> |



| Phase   | Status | Context                       | class | Doc tier | Linked objects        | Main goal                                                                     | Source objects/<br>dependencies                                                                                                                                                                                                                                                        | Definitions/objects                                                                                                                                                                                      | Current code modules                                                                                                                                                                                                                                             | Target modules/<br>objects                                                                                                                                                                                            | Proof goals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | How imple-<br>mented               |
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| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase | F1    |          | <a href="#">LEG10</a> | Implementation module manifest and semantic placement gate                    | 1.1.1.1.0-1.1.1.1.0.8; generated module inventory; static code audit; current active cursor; 1.1.1.1.0.10 general test-matrix contract; 1.1.1.1.0.11 regime-typed test-matrix lock; 1.1.1.1.0.13 derived quadratic algebra provenance lock; 1.1.1.1.0.15 phase-object-link policy lock | implementation module manifest; semantic placement decisions; module budget; phase_type obligations; refactor/derivation/ comparison-specific evidence fields                                            | masters/<br>cnna_roadmap_master.yaml;<br>repo_inventory/raw_export/<br>modules.json;<br>repo_snapshot/**/*lean                                                                                                                                                   | LEG10 ModuleImplementa-<br>tionManifest;<br>generated scratchpad<br>contract;<br>no Lean code move yet                                                                                                                | Produce a compact pre-code manifest: exact new/modified modules, semantic placement, reuse decision, consumer direction, phase_type obligations, regression protocol, version impact, and no duplicate generator logic. No Lean source is changed in this gate. Include generic finite test placement for upcoming computation modules; special sentinel tests are insufficient. The manifest must also decide where regime-tagged generic rows, noOuter-root sentinel rows, and HasOuter/ complement-projection comparison rows live without duplicating the generator. Scalar modules must distinguish parameterized schema from proof-carrying D-instantiation. The manifest must also state whether the target implementation phase owns direct objects, consumes child-scope objects, or is objectless infrastructure under the global phase-object-link policy. | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Lean;<br>Governance;<br>Phase | F1    |          | <a href="#">LEG11</a> | Reuse and duplication audit with canonical theory/smoke-test generator policy | 1.1.1.1.1; existing Smoke/Test generator modules; repo symbol/reference summaries; planned phase outputs                                                                                                                                                                               | ReuseDuplicationLedger; CanonicalGeneratorPolicy; Arithmetic/Smoke/*Generated*; single-source generator extension rule; duplicate implementation ban                                                     | Arithmetic/Smoke/*Generator*;<br>Arithmetic/Smoke/*Generated*;<br>repo_inventory/raw_export/<br>symbol_reference_summary.json                                                                                                                                    | LEG11<br>ReuseDuplicationLedger<br>and<br>CanonicalGeneratorPolicy;<br>no second generator module unless the manifest proves a genuine semantic split                                                                 | Closed as a planning/API governance gate: generator-like code must consume, extend, or wrap the canonical seam recorded in LEG11; parallel request/output, extraction, ledger, or smoke-test generators are blocked unless a genuine semantic split is explicitly recorded in the reuse ledger.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <a href="#">implemented record</a> |
| 1.1.1.1 | G      | Documentation; F1<br>Phase    |       |          | <a href="#">LEG12</a> | Object and quantities proof-documentation infrastructure gate                 | 1.1.1.1.2; object register; code derivation sites; blue-object correction ledger requirements                                                                                                                                                                                          | ObjectProofDossier; ObjectProofDocumentation; proof-dossier anchor/link; per-object fields for documented yes/no, source objects, definitions, proof target, proof sketch, and verification/audit result | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>generated_exports/<br>object_proof_documentation.generated_table;<br>tables/<br>object_proof_documentation.tex;<br>tables/<br>object_proof_link_register.tex;<br>generated PDF/TSV views | object_proof_dossiers<br>top-level YAML section;<br>object proof<br>documentation generated table;<br>object-register<br>proof-documentation<br>columns;<br>workflow policy entry for documentation as third workflow | Closed as an infrastructure-only gate. The generator validates one object-proof-dossier row per registered object/quantity, exact object-register proof links of the form proof:<ID>, matching object names/phases, and non-empty proof-documentation fields. Documentation may still be no/partial until the natural object phase fills the mathematical proof content; blue objects remain blocked from public-final promotion until semantic correction and dossier verification are both recorded.                                                                                                                                                                                                                                                                                                                                                                | <a href="#">implemented record</a> |



| Phase Status |   | Context                       | class | Doc tier | Linked objects | Main goal                                                                                                                                                     | Source objects/<br>dependencies                                                                                                                                                                                                                                                        | Definitions/objects                                                                                                                                                                                                                                                        | Current code modules                                                                                                                                                                                                                                                                                                                                                                                                                              | Target modules/<br>objects                                                                                                                            | Proof goals                                                                                                                                                                                                                                                                                          | How imple-<br>mented   |
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| 1.1.1.1      | G | Lean;<br>Governance;<br>Phase |       | F3       | LEG5           | Declaration<br>consumption<br>classification map<br>finalization                                                                                              | 1.1.1.1.3;<br>LEG10 module<br>manifest;<br>LEG11 reuse/generator<br>audit;<br>proof-documentation<br>infrastructure;<br>declaration/reference<br>inventory                                                                                                                             | consumption_classification<br>remap_snapshot/**/*.<br>lean;<br>repo_inventory/raw_export/*.<br>json;<br>generated_exports/*.<br>tsv                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                   | LEG5 Declaration<br>consumption classification<br>map;<br>generated module/<br>declaration/object<br>consumption records;<br>no Lean declaration move | LEG5 closed as a generated<br>classification policy plus<br>module/declaration/object<br>TSV exports. Semantic<br>consumption is separated from<br>BuildAll/import reachability;<br>blue objects remain blocked<br>from public-final classification.                                                 | implemented<br>record  |
| 1.1.1.1      | W | Lean;<br>Governance;<br>Phase |       | F1       | LEG36          | Pre-substantive-code<br>readiness umbrella:<br>semantic<br>consolidation, full<br>IST intake, neutral<br>naming, refactor,<br>post-scan, and<br>release proof | 1.1.1.1.1 module<br>manifest;<br>1.1.1.1.2 reuse/<br>duplication ledger;<br>1.1.1.1.4 declaration<br>intake, neutral module<br>consumption map;<br>repo_inventory/<br>raw_export modules/<br>import_edges/symbol<br>references;<br>direct code spot-checks<br>of clone/schema families | LEG36 umbrella plus<br>LEG37-LEG40 readiness<br>ledgers: semantic<br>module consolidation,<br>full IST source/object<br>intake, neutral module<br>naming/facades,<br>post-refactor inventory<br>re-scan, and pre-code<br>release certificate.                              | repo_snapshot/CNNA/PillarA/<br>Arithmetic/BoundarySource/<br>{BoundaryStateSubstrate,BranchAddressSubstrate,HistoryExpandedSubstrate,InterfaceExposureSubstrate}.lean;<br>repo_snapshot/CNNA/PillarA/<br>Handoff/Outputs/<br>A_to_{C,D,E}.lean;<br>repo_snapshot/CNNA/PillarA/<br>Handoff/Inputs/<br>{B,C,D,E}_to_A.lean;<br>Smoke/GeneratedInterior*/<br>Operational* chains;<br>Structural/CayleyDickson/S6* chain;<br>Arithmetic/Notation.lean | LEG36, LEG37, LEG38,<br>LEG39, LEG40;<br>mathematics until child<br>sequence closes.                                                                  | Umbrella only. The active<br>cursor is delegated to child<br>proofs. SubstrateExposureSubstrate}.lean;<br>not a release proof for new<br>Lean content;<br>it records the pre-code<br>readiness sequence.                                                                                             | planned<br>scratch pad |
| 1.1.1.1      | Y | Lean;<br>Governance;<br>Phase |       | F1       | LEG37          | Full IST module/<br>declaration/object<br>intake completeness<br>audit before any new<br>substantive code                                                     | 1.1.1.1.5 umbrella;<br>LEG10 module<br>manifest;<br>LEG5 declaration<br>consumption map;<br>repo_inventory/<br>raw_export modules/<br>import_edges/symbol<br>references;<br>current object register;<br>direct repo_snapshot/<br>CNNA scan.                                            | LEG37<br>CurrentSourceIntake-<br>CompletenessLedger:<br>every existing Lean<br>module/declaration is<br>either linked to a<br>registered object/proof<br>dossier, classified as<br>tool/legacy/audit-only,<br>or explicitly scheduled<br>for object/register<br>ingestion. | repo_snapshot/CNNA/**/*.<br>lean;<br>repo_inventory/raw_export/*.<br>json;<br>generated_exports/<br>module_consumption_records.generated.<br>tsv;<br>generated_exports/<br>declaration_consumption_records.generated.<br>tsv;<br>generated_exports/<br>object_register.generated.<br>tsv                                                                                                                                                          | LEG37;<br>exact missing-object/<br>overloaded-module/<br>neutralization<br>backlog;<br>noted as source move yet.                                      | Prove/audit that the IST<br>codebase has no unclassified<br>source surface before<br>neutralization/refactor starts.<br>If a module/declaration cannot<br>be mapped to a registered<br>object, legacy/audit bucket, or<br>new ingestion phase, create the<br>missing phase at its natural<br>origin. | active scratch<br>pad  |
| 1.1.1.1      | O | Lean;<br>Governance;<br>Phase |       | F1       | LEG38          | Neutral<br>module-name and<br>namespace/facade<br>migration plan                                                                                              | 1.1.1.1.5.1;<br>LEG37 closed intake<br>ledger;<br>SCF44/SCF45/SCF52<br>findings;<br>phase-semantic and<br>proof-stage<br>module-name audit.                                                                                                                                            | LEG38 NeutralModule-<br>NameAndFacadeMigra-<br>tionMap: canonical<br>neutral target names,<br>old-module facade plan,<br>no-merge/proof-stage<br>exceptions, and import/<br>API migration order.                                                                           | Current phase-semantic module<br>families: Smoke/SM*, Structural/<br>CayleyDickson/S6*,<br>StrengtheningPhase* audits,<br>BoundarySource clones, Handoff<br>input/output wrappers, Notation/<br>Legacy surfaces.                                                                                                                                                                                                                                  | LEG38;<br>planned neutral module/<br>API surface;<br>facade-first migration<br>package;<br>no code move until<br>1.1.1.1.5.3.                         | Prove/audit that every planned<br>module rename/merge/split<br>has natural-layer placement,<br>compatibility facades, public/<br>API impact statement, and no<br>loss of proof/provenance<br>granularity.                                                                                            | planned<br>scratch pad |
| 1.1.1.1      | O | Lean;<br>Governance;<br>Phase |       | F1       | LEG39          | Compatibility-<br>preserving refactor<br>execution gate for<br>neutral module<br>surface                                                                      | 1.1.1.1.5.2;<br>LEG38 neutral/facade<br>migration map;<br>refactor regression<br>protocol GNG03/<br>GNG26/<br>GNG28-GNG32.                                                                                                                                                             | LEG39 NeutralRefac-<br>torExecutionCertificate:<br>bounded Lean refactor<br>package, old-module<br>facades, import/<br>declaration delta,<br>consumer breakage<br>audit, and local<br>full-build evidence.                                                                 | Only modules authorized by LEG38;<br>no mathematical strengthening<br>beyond surface/facade/refactor<br>changes.                                                                                                                                                                                                                                                                                                                                  | LEG39;<br>local-build-backed refactor<br>closeout;<br>regenerated inventory<br>input for 1.1.1.1.5.4.                                                 | Execute only the approved<br>neutralization/refactor<br>package. Close only after local<br>full build, declaration/<br>consumer diff, old-import<br>compatibility plan, and proof<br>that the refactor preserved or<br>improved auditability.                                                        | planned<br>scratch pad |
| 1.1.1.1      | O | Lean;<br>Governance;<br>Phase |       | F1       | LEG40          | Post-refactor<br>inventory re-scan and<br>IST-object/<br>proof-dossier<br>reclassification                                                                    | 1.1.1.1.5.3;<br>LEG39 refactor<br>execution certificate;<br>regenerated<br>build_export_script_v1.8.<br>tsv                                                                                                                                                                            | LEG40<br>PostRefactorInvento-<br>ryClosureCertificate:<br>fresh module/import/<br>declaration inventory,<br>object-register delta,<br>proof-dossier delta,<br>status/color audit, and<br>unresolved-gap ledger.                                                            | repo_inventory/raw_export<br>regenerated after refactor;<br>generated_exports regenerated;<br>object/proof/phase registers.                                                                                                                                                                                                                                                                                                                       | LEG40;<br>post-refactor source truth<br>map;<br>unresolved gaps turned<br>into red locked phases or<br>explicitly white tool/<br>refactor tasks.      | Prove/audit that the refactored<br>IST codebase is fully<br>reclassified and that no stale<br>module name, stale object<br>mapping, or hidden public<br>consumer remains before any<br>content phase is released.                                                                                    | planned<br>scratch pad |

| Phase Status |   | Context class                 | Doc tier | Linked objects                                                                                                                                                                                                          | Main goal                                                                                   | Source objects/<br>dependencies                                                                                                                                                                                                                                                  | Definitions/objects                                                                                                                                                                                                    | Current code modules                                                                                                                                                              | Target modules/<br>objects                                                                                                | Proof goals                                                                                                                                                                                                                                                                                                                                         | How imple-<br>mented                                   |
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| 1.1.1.1      | O | Lean;<br>Governance;<br>Phase | F1       | <a href="#">LEG41</a>                                                                                                                                                                                                   | Pre-substantive-code<br>release certificate for<br>1.1.1.2 and later<br>Lean content phases | 1.1.1.1.5.4;<br>LEG37-LEG40 closed;<br>DBR13/GNG31<br>red-lock policy;<br>all required object proof<br>dossiers updated.                                                                                                                                                         | LEG41 PreSubstantive-<br>CodeReleaseCertificate:<br>explicit release proof<br>that 1.1.1.2 may become<br>active only after intake,<br>neutralization, refactor<br>and post-scan closure.                               | No Lean source target except<br>generated release ledger;<br>consumes registers and build<br>evidence.                                                                            | LEG41;<br>release proof for 1.1.1.2 or<br>new red blockers at<br>natural origin.                                          | Before turning 1.1.1.2 yellow,<br>prove by references to closed<br>LEG37-LEG40 and registered<br>derived-only Lean objects/<br>proof dossiers that the phase<br>can consume a stable neutral<br>API surface. Otherwise keep<br>1.1.1.2 locked/orange and<br>spawn upstream red blockers.                                                            | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.1.2      | O | Lean;<br>Governance;<br>Phase | F3       | <a href="#">LEG7</a>                                                                                                                                                                                                    | Blue-object semantic<br>correction ledger                                                   | 1.1.1.1.5.5<br>pre-substantive-code<br>release certificate;<br>LEG37-LEG41 closed;<br>stable neutral module/<br>API surface;<br>updated proof dossiers;<br>object register rows<br>with Object status B;<br>TomS guardrails where<br>arithmetic/number<br>theory touches appear. | blue_correction_ledger:<br>for each B object record<br>correction phase, allowed<br>temporary consumers,<br>forbidden direct<br>consumers, semantic<br>mismatch, neutral API<br>target, and exit-to-white<br>condition | Structural/CayleyDickson/*;<br>Arithmetic/Smoke/<br>GeneratedInterior*;<br>Arithmetic/Smoke/*Operational*;<br>object register                                                     | LEG7 blue-object<br>semantic correction ledger;<br>updated B-object access/<br>consumer clauses;<br>no Lean code move yet | Only after 1.1.1.1.5.5 closes:<br>implement blue-object semantic<br>correction ledger against the<br>stable neutral API/source<br>surface. No blue object may be<br>released unless its referenced<br>Lean object/proof dossier is<br>derived-only or explicitly<br>remains blocked.                                                                | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.1.3      | W | Lean;<br>Governance;<br>Phase | F3       | <a href="#">LEG6</a>                                                                                                                                                                                                    | Public API contract<br>and consumer<br>whitelist                                            | 1.1.1.1.1;<br>1.1.1.1.2;<br>1.1.1.1.3;<br>1.1.1.1.4;<br>1.1.1.2;<br>phase/object register;<br>access policy                                                                                                                                                                      | public_api_contract;<br>consumer_whitelist;<br>forbidden_import_set;<br>post-1.1 visible module<br>list;<br>per-public-module<br>semantic placement<br>provenance                                                      | Arithmetic/Notation.lean;<br>planned CoreSymbols/Legacy/<br>NeutralWrappers modules;<br>BuildAll/Umbrella modules                                                                 | LEG6 public API contract;<br>LEG9 consumer whitelist/<br>firewall inputs                                                  | Declare exactly which names/<br>modules Phase 3+ may see and<br>why each belongs in its chosen<br>module. All legacy status/<br>policy gates, phase aliases,<br>audit-only declarations, dead<br>candidates, duplicate generator<br>seams, uncorrected blue<br>objects, and undocumented<br>public objects are excluded<br>from the public surface. | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.1.4      | W | Lean;<br>Governance;<br>Phase | F3       | <a href="#">LEG8</a>                                                                                                                                                                                                    | Pruning and<br>dead-candidate<br>ledger before deletion                                     | public API contract<br>from 1.1.1.3;<br>declaration classification<br>from 1.1.1.1.4;<br>object proof dossiers<br>from 1.1.1.1.3;<br>repo semantic/reference<br>summaries                                                                                                        | pruning_ledger with<br>keep, quarantine, wrap,<br>reexport, archive, and<br>deletion-candidate<br>classes                                                                                                              | all legacy/audit/SM modules<br>classified as non-public;<br>BuildAll/Umbrella-only leaves;<br>phase-status ledgers                                                                | LEG8 pruning/<br>dead-candidate ledger;<br>no deletions during this<br>phase                                              | No declaration is deleted solely<br>because it is not on the<br>mainline. Deletion candidates<br>require no public contract, no<br>proof-bridge role, no legacy/<br>audit role, and local green<br>build evidence after the actual<br>removal phase.                                                                                                | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.1.5      | W | Lean;<br>Governance;<br>Phase | F3       | <a href="#">LEG9</a>                                                                                                                                                                                                    | Generated-view<br>authority and<br>inventory lock for<br>implementation<br>prompts          | 1.1.1.1.1-1.1.1.1.4;<br>1.1.1.2-1.1.1.4;<br>YAML single-master<br>rule;<br>generated table/PDF<br>pipeline                                                                                                                                                                       | implementation prompt<br>contract for 1.1.2-1.1.5;<br>module manifest rule;<br>generator-reuse rule;<br>generated view<br>authority;<br>local-build evidence rule                                                      | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>README.md;<br>generated PDF/TSV                                                                           | updated README/<br>report/PDF;<br>no Lean declaration move<br>yet                                                         | Close the audit/planning gate<br>with reproducible generated<br>artifacts and one active next<br>cursor. Future implementation<br>prompts are invalid until they<br>include the module-name/<br>budget manifest and reuse/<br>generator audit. This phase<br>does not claim Lean build<br>success.                                                  | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.1.5      | W | Lean;<br>Governance;<br>Phase | F2       | <i>no direct object:</i><br>tooling/<br>governance<br>subphase;<br>it prepares<br>generated views/<br>ledgers and owns<br>no CNNA<br>mathematical<br>object until a<br>future object row<br>is explicitly<br>registered | Tool-mode split and<br>lean-focus contract                                                  | 1.1.1.5;<br>SCF37;<br>workflow policy;<br>repo inventory;<br>object register;<br>scratchpad                                                                                                                                                                                      | Define audit-full,<br>phase-check, and<br>lean-focus modes so<br>routine Lean work<br>consumes compact<br>implementation packets<br>instead of full PDF/<br>table scans.                                               | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>repo_inventory/raw_export;<br>generated_exports;<br>tables;<br>figures;<br>tool_state if introduced later | workflow-policy row;<br>scaling-plan TSV/TeX;<br>no Lean code                                                             | Tooling/governance proof goal:<br>implement the Tool-mode split<br>and lean-focus contract without<br>modifying Lean sources unless<br>a later explicit Lean phase<br>accepts that change. Ensure<br>theory semantics: No theory<br>classification yet;<br>this phase only fixes the mode<br>boundary.                                              | <a href="#">planned</a><br><a href="#">scratch pad</a> |

| Phase   | Status | Context                       | class | Doc | tier | Linked objects                                                                                                                                                                                                          | Main goal                                            | Source objects/<br>dependencies                                                             | Definitions/objects                                                                                                                                                                                                                                  | Current code modules                                                                                                                                                              | Target modules/<br>objects                                              | Proof goals                                                                                                                                                                                                                                                                                                                                                                                                        | How imple-<br>mented                                   |
|---------|--------|-------------------------------|-------|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 1.1.1.5 | W      | Lean;<br>Governance;<br>Phase | F2    |     |      | <i>no direct object:</i><br>tooling/<br>governance<br>subphase;<br>it prepares<br>generated views/<br>ledgers and owns<br>no CNNA<br>mathematical<br>object until a<br>future object row<br>is explicitly<br>registered | Implementation<br>Packet Generator                   | 1.1.1.5;<br>SCF37;<br>workflow policy;<br>repo inventory;<br>object register;<br>scratchpad | Generate the phase-local<br>packet with goal, source<br>objects, producer/<br>consumer modules, red<br>blockers, yellow risks,<br>forbidden shortcuts,<br>expected artifacts, and<br>comparison-only<br>warnings.                                    | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>repo_inventory/raw_export;<br>generated_exports;<br>tables;<br>figures;<br>tool_state if introduced later | active-phase<br>implementation packet<br>TSV/JSON/TeX                   | Tooling/governance proof goal:<br>implement the Implementation<br>Packet Generator without<br>modifying Lean sources unless<br>a later explicit Lean phase<br>accepts that change. Ensure<br>theory semantics: Consumes<br>module-theory/end-theory<br>ledgers once available;<br>before then marks theory<br>context as unknown-not-input.                                                                        | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.1.5 | W      | Lean;<br>Governance;<br>Phase | F2    |     |      | <i>no direct object:</i><br>tooling/<br>governance<br>subphase;<br>it prepares<br>generated views/<br>ledgers and owns<br>no CNNA<br>mathematical<br>object until a<br>future object row<br>is explicitly<br>registered | Incremental Lean<br>inventory and<br>file-hash cache | 1.1.1.5;<br>SCF37;<br>workflow policy;<br>repo inventory;<br>object register;<br>scratchpad | Avoid rescanning/<br>rendering everything by<br>caching file hashes,<br>module declarations,<br>import edges, and<br>symbol-reference<br>candidates between<br>Lean-data updates.                                                                    | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>repo_inventory/raw_export;<br>generated_exports;<br>tables;<br>figures;<br>tool_state if introduced later | file-hash cache;<br>declaration cache;<br>inventory-delta TSV           | Tooling/governance proof goal:<br>implement the Incremental<br>Lean inventory and file-hash<br>cache without modifying Lean<br>sources unless a later explicit<br>Lean phase accepts that<br>change. Ensure theory<br>semantics: May infer rough<br>local background categories<br>from imports, but must mark<br>them heuristic until<br>semantically confirmed.                                                  | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.1.5 | W      | Lean;<br>Governance;<br>Phase | F2    |     |      | <i>no direct object:</i><br>tooling/<br>governance<br>subphase;<br>it prepares<br>generated views/<br>ledgers and owns<br>no CNNA<br>mathematical<br>object until a<br>future object row<br>is explicitly<br>registered | Module theory<br>context ledger                      | 1.1.1.5;<br>SCF37;<br>workflow policy;<br>repo inventory;<br>object register;<br>scratchpad | Track local established<br>mathematical<br>background per module<br>as formal language,<br>proof method, library/<br>API context,<br>implementation<br>substrate, or comparison<br>lemma context.                                                    | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>repo_inventory/raw_export;<br>generated_exports;<br>tables;<br>figures;<br>tool_state if introduced later | module-theory-context<br>ledger TSV;<br>active-module slice             | Tooling/governance proof goal:<br>implement the Module theory<br>context ledger without<br>modifying Lean sources unless<br>a later explicit Lean phase<br>accepts that change. Ensure<br>theory semantics: Local<br>established theory is allowed<br>formalization context, never<br>ontic CNNA generator.<br>Examples: Matrix, Finset,<br>Polynomial, graph/linear<br>algebra, Schur-style algebraic<br>methods. | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.1.5 | W      | Lean;<br>Governance;<br>Phase | F2    |     |      | <i>no direct object:</i><br>tooling/<br>governance<br>subphase;<br>it prepares<br>generated views/<br>ledgers and owns<br>no CNNA<br>mathematical<br>object until a<br>future object row<br>is explicitly<br>registered | End-theory<br>comparison ledger                      | 1.1.1.5;<br>SCF37;<br>workflow policy;<br>repo inventory;<br>object register;<br>scratchpad | Track large established<br>target theories as<br>comparison-only<br>horizons: AQFT/QFT,<br>OQS, modular forms,<br>CM, elliptic functions/<br>curves, L-functions,<br>Moonshine,<br>renormalization,<br>operator algebras,<br>spectral geometry, etc. | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>repo_inventory/raw_export;<br>generated_exports;<br>tables;<br>figures;<br>tool_state if introduced later | end-theory-comparison<br>ledger TSV;<br>comparison-boundary<br>table    | Tooling/governance proof goal:<br>implement the End-theory<br>comparison ledger without<br>modifying Lean sources unless<br>a later explicit Lean phase<br>accepts that change. Ensure<br>theory semantics: End theories<br>are target structures,<br>consistency checks, terminology<br>alignment, or obstruction/<br>comparison mirrors;<br>forbidden as CNNA producers.                                         | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.1.5 | W      | Lean;<br>Governance;<br>Phase | F2    |     |      | <i>no direct object:</i><br>tooling/<br>governance<br>subphase;<br>it prepares<br>generated views/<br>ledgers and owns<br>no CNNA<br>mathematical<br>object until a<br>future object row<br>is explicitly<br>registered | Active object graph<br>slices                        | 1.1.1.5;<br>SCF37;<br>workflow policy;<br>repo inventory;<br>object register;<br>scratchpad | Render small<br>producer-consumer/<br>provenance<br>neighborhoods around<br>the active phase and<br>active CNNA objects<br>rather than dumping the<br>full object/declaration<br>graph into the PDF.                                                 | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>repo_inventory/raw_export;<br>generated_exports;<br>tables;<br>figures;<br>tool_state if introduced later | active-object-<br>neighborhood DOT/SVG/<br>PDF;<br>graph-slice TSV/JSON | Tooling/governance proof goal:<br>implement the Active object<br>graph slices without modifying<br>Lean sources unless a later<br>explicit Lean phase accepts<br>that change. Ensure theory<br>semantics: Graph edges<br>distinguish internal provenance,<br>local formalization context, and<br>external comparison-only<br>references.                                                                           | <a href="#">planned</a><br><a href="#">scratch pad</a> |

| Phase   | Status | Context                           | class | Doc | tier | Linked objects                                                                                                                                                                                                          | Main goal                                                                                           | Source objects/<br>dependencies                                                                                    | Definitions/objects                                                                                                                                                                                                    | Current code modules                                                                                                                                                              | Target modules/<br>objects                                                                                                                                                                                                                                                       | Proof goals                                                                                                                                                                                                                                                                                                                                                                             | How imple-<br>mented                                   |
|---------|--------|-----------------------------------|-------|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 1.1.1.5 | W      | Lean;<br>Governance;<br>Phase     | F2    |     |      | <i>no direct object:</i><br>tooling/<br>governance<br>subphase;<br>it prepares<br>generated views/<br>ledgers and owns<br>no CNNA<br>mathematical<br>object until a<br>future object row<br>is explicitly<br>registered | Red-blocker<br>addressing-phase<br>generator                                                        | 1.1.1.5;<br>SCF37;<br>workflow policy;<br>repo inventory;<br>object register;<br>scratchpad                        | When the<br>pre-implementation<br>thought experiment<br>marks a future phase<br>red, propose an<br>upstream addressing<br>phase at the earliest<br>natural source layer<br>before the blocked phase<br>becomes active. | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>repo_inventory/raw_export;<br>generated_exports;<br>tables;<br>figures;<br>tool_state if introduced later | red-blocker phase-proposal<br>TSV;<br>accepted phase rows only<br>after approval                                                                                                                                                                                                 | Tooling/governance proof goal:<br>implement the Red-blocker<br>addressing-phase generator<br>without modifying Lean sources<br>unless a later explicit Lean<br>phase accepts that change.<br>Ensure theory semantics: If the<br>blocker is an imported<br>end-theory assumption, route<br>to internal derivation/<br>comparison-boundary phase<br>rather than patching the<br>consumer. | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.2   | W      | Lean;<br>Refactor;<br>Phase       | F3    |     |      | <a href="#">LEG1</a>                                                                                                                                                                                                    | Quarantine<br>single-constructor<br>legacy status gates                                             | 1.1.1.1.1-1.1.1.1.4;<br>refactor regression<br>protocol 1.1.1.1.0.3;<br>module manifest and<br>public API contract | Legacy.StatusGates<br>module;<br>optional<br>Legacy.StatusEnums<br>alias if the declaration<br>family is too large                                                                                                     | Arithmetic/Notation.lean;<br>historical SM*/R*/Audit status<br>declarations                                                                                                       | CNNA/PillarA/<br>Arithmetic/Legacy/<br>StatusGates.lean                                                                                                                                                                                                                          | Quarantine classified legacy<br>status gates via wrapper-first/<br>incremental migration.<br>Preserve build compatibility;<br>after every delivered code<br>package the local full build<br>must be green before the phase<br>can close.                                                                                                                                                | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.3   | W      | Lean;<br>Refactor;<br>Phase       | F3    |     |      | <a href="#">LEG2</a>                                                                                                                                                                                                    | Create true<br>CoreSymbols/<br>Notation<br>micro-surface                                            | 1.1.1.3;<br>public API contract;<br>true notation bucket                                                           | small notation/<br>core-symbol surface;<br>no plan-semantic<br>declarations                                                                                                                                            | Arithmetic/Notation.lean;<br>Foundation/Notation.lean;<br>Finite/Notation.lean;<br>Handoff/Notation.lean                                                                          | CNNA/PillarA/<br>Arithmetic/Notation/<br>CoreSymbols.lean                                                                                                                                                                                                                        | Keep the module small and<br>syntactic/domain-role based.<br>No proof-heavy objects and no<br>historical SM phase aliases.                                                                                                                                                                                                                                                              | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.4   | W      | Lean;<br>Refactor;<br>Phase       | F3    |     |      | <a href="#">LEG3</a>                                                                                                                                                                                                    | Separate legacy<br>phase aliases from<br>neutral wrappers and<br>blue-object<br>correction wrappers | 1.1.1.2;<br>1.1.1.3;<br>blue-object semantic<br>correction ledger;<br>Legacy.PhaseAliases<br>bucket                | Legacy.PhaseAliases;<br>neutral public wrappers;<br>blue-object correction<br>wrapper contracts                                                                                                                        | Arithmetic/Notation.lean;<br>historical SM*/S6* alias surfaces                                                                                                                    | CNNA/PillarA/<br>Arithmetic/Legacy/<br>PhaseAliases.lean;<br>CNNA/PillarA/<br>Arithmetic/<br>NeutralWrappers.lean                                                                                                                                                                | Old SM/AR names may remain<br>as legacy aliases. Public<br>wrappers must expose role/<br>object names only. Blue objects<br>get neutral wrapper contracts<br>but not final white status until<br>their natural correction phase<br>closes.                                                                                                                                              | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.1.5   | W      | Tool;<br>Infrastructure;<br>Phase | F3    |     |      | <a href="#">LEG4</a>                                                                                                                                                                                                    | Post-split import<br>firewall and Phase-3<br>visibility gate                                        | 1.1.1.3;<br>1.1.2;<br>1.1.3;<br>1.1.4;<br>pruning ledger from<br>1.1.1.4                                           | import firewall;<br>forbidden<br>Legacy.StatusGates<br>consumers;<br>generated audit update                                                                                                                            | BuildAll.lean;<br>all imports touching Arithmetic/<br>Notation.lean;<br>repo_inventory/raw_export                                                                                 | updated generated module<br>inventory;<br>phase 3 source<br>dependencies may import<br>only neutral/CoreSymbols<br>surfaces                                                                                                                                                      | Prove/check the import<br>firewall: Phase 3+ must<br>consume only public neutral<br>API modules, never<br>Legacy.StatusGates,<br>Legacy.PhaseAliases,<br>dead-candidate ledgers, or blue<br>objects before semantic<br>correction.                                                                                                                                                      | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.2     | W      | Lean;<br>Refactor;<br>Phase       | F0    |     |      | <i>scope-only</i> ; child<br>objects: O00,<br>O09, G00, G01,<br>G02, G03, G04,<br>S07, S08                                                                                                                              | Foundation and<br>generator Profile<br>neutralization                                               | 1.1.5;<br>current foundation/<br>generator modules;<br>preliminary object IDs<br>O00/G00-G04                       | ExecComplexPath;<br>GeneratorProvenance;<br>substrateProfile;<br>LevelProfile;<br>CarrierProfile                                                                                                                       | GeneratedSource.lean;<br>GeneratedMainPath.lean;<br>BoundarySource/*.lean;<br>Foundation/ExecComplex*.lean                                                                        | CNNA/PillarA/<br>Foundation/<br>ExecComplex.lean;<br>CNNA/PillarA/<br>Foundation/<br>GeneratorProvenance.lean;<br>CNNA/PillarA/<br>Foundation/Substrate/<br>*.lean;<br>CNNA/PillarA/<br>Foundation/<br>LevelProfile.lean;<br>CNNA/PillarA/<br>Foundation/<br>CarrierProfile.lean | No reindex/level/carrier<br>without proof profile.<br>ExecComplex remains the<br>operational scalar path;<br>no noncomputable Complex<br>replacement.                                                                                                                                                                                                                                   | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.2.1   | W      | Lean;<br>Refactor;<br>Phase       | F1    |     |      | <a href="#">O00</a>                                                                                                                                                                                                     | Neutralize<br>ExecComplex<br>operational scalar<br>boundary                                         | 1.2;<br>current ExecComplex<br>modules                                                                             | ExecComplexPath and<br>computability boundary                                                                                                                                                                          | CNNA/PillarA/Foundation/<br>ExecComplex.lean;<br>ExecComplexBridge.lean;<br>ExecComplexLemmas.lean                                                                                | CNNA/PillarA/<br>Foundation/<br>ExecComplex.lean                                                                                                                                                                                                                                 | Expose only the computable<br>operational scalar boundary<br>and any mirror bridge<br>explicitly;<br>no analytic shortcut is allowed.                                                                                                                                                                                                                                                   | <a href="#">planned</a><br><a href="#">scratch pad</a> |



| Phase   | Status | Context                       | class | Doc tier | Linked objects                                                                                                                | Main goal                                                                                                 | Source objects/<br>dependencies                                                                                                                       | Definitions/objects                                                                                                                                               | Current code modules                                                                                                                                               | Target modules/<br>objects                                                                                                                                      | Proof goals                                                                                                                                                                                                                                                                                                                                                                                                                                                       | How imple-<br>mented                                   |
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| 1.2.1.1 | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <i>no direct object:</i><br>baseline/audit/<br>closeout/<br>infrastructure<br>phase with no<br>registered object<br>ownership | ExecComplex<br>capability and<br>limitation audit                                                         | O00, LEG20, LEG31;<br>current ExecComplex<br>API and bridge modules                                                                                   | O00/O09 boundary<br>checklist                                                                                                                                     | CNNA/PillarA/Foundation/<br>ExecComplex.lean;<br>CNNA/PillarA/Foundation/<br>ExecComplexBridge.lean                                                                | No new module by default;<br>possible future CNNA/<br>PillarA/Foundation/<br>ExactQuadraticComplex.lean<br>only after manifest<br>approval                      | Classify which operations are<br>fully executable over QxQ,<br>which are exact quadratic but<br>not QxQ, and which are<br>analytic mirror-only. V1.19<br>status policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                                                                                                                                                 | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 1.2.1.2 | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">O09</a>                                                                                                           | Required exact<br>quadratic scalar<br>schema:<br>discriminant-indexed<br>minimal operational<br>interface | O09, LEG31, LEG32;<br>mathlib<br>QuadraticAlgebra over<br>Q as reference;<br>no concrete<br>D-instantiation before<br>O05/O06 or equivalent<br>ledger | ExactQuadraticScalar<br>schema parameterized<br>by D;<br>theta symbol with<br>relation $\theta^2 = D$ ;<br>coordinate operations<br>over Q;<br>no free $\sqrt{D}$ | None yet                                                                                                                                                           | Candidate schema module<br>only if Phase 1.1.1.1.1/<br>1.1.1.1.2 prove no existing<br>module owns the<br>semantics;<br>no concrete instance<br>without O10/BR14 | Construct or wrap the minimal<br>two-coordinate exact quadratic<br>schema and prove its<br>coordinate laws. This phase<br>must not assert that any<br>particular D has been<br>generated;<br>it only prepares the schema<br>consumed later by the<br>discriminant-source<br>instantiation gate. V1.19 status<br>policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 1.2.2   | W      | Lean;<br>Refactor;<br>Phase   | F0    |          | <a href="#">G00</a>                                                                                                           | Neutralize generator<br>provenance source<br>profile                                                      | 1.2.1;<br>generated source/<br>main-path modules                                                                                                      | GeneratorProvenance                                                                                                                                               | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedSource.lean;<br>GeneratedMainPath.lean;<br>BoundarySource/<br>GeneratedSubstrateRoute.lean                              | CNNA/PillarA/<br>Foundation/<br>GeneratorProvenance.lean                                                                                                        | Forward existing provenance<br>data through a neutral source<br>profile;<br>do not make substrate/level<br>choices in this module.                                                                                                                                                                                                                                                                                                                                | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.2.3   | W      | Lean;<br>Refactor;<br>Phase   | F0    |          | <a href="#">G01</a>                                                                                                           | Neutralize regular<br>substrate profile                                                                   | 1.2.2;<br>regular branching<br>substrate modules                                                                                                      | substrate: regular<br>branching                                                                                                                                   | CNNA/PillarA/Foundation/<br>ConcreteSubstrate.lean;<br>Arithmetic/BoundarySource/<br>BranchAddressSubstrate.lean;<br>BoundaryStateSubstrate.lean                   | CNNA/PillarA/<br>Foundation/Substrate/<br>Regular.lean                                                                                                          | Separate regular branching<br>data from variable branching<br>data;<br>no geometric angle/lattice<br>assumption enters here.                                                                                                                                                                                                                                                                                                                                      | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.2.4   | W      | Lean;<br>Refactor;<br>Phase   | F0    |          | <a href="#">G02</a>                                                                                                           | Neutralize variable<br>substrate profile                                                                  | 1.2.2;<br>variable branching and<br>weight-policy modules                                                                                             | substrate: variable<br>branching                                                                                                                                  | CNNA/PillarA/Foundation/<br>LevelVariableSubstrate.lean;<br>Foundation/WeightPolicy.lean;<br>GeneratedDegreeFormulaAuditSM3db2eR0.lean                             | CNNA/PillarA/<br>Foundation/Substrate/<br>Variable.lean                                                                                                         | Keep variable branching<br>proof-carrying and explicit;<br>no class/instance oracle hides<br>the generated data.                                                                                                                                                                                                                                                                                                                                                  | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.2.5   | W      | Lean;<br>Refactor;<br>Phase   | F0    |          | <a href="#">G03</a>                                                                                                           | Neutralize level and<br>cutoff profile                                                                    | 1.2.2-1.2.4;<br>level-history modules                                                                                                                 | Level / Cutoff                                                                                                                                                    | CNNA/PillarA/Arithmetic/<br>BoundarySource/<br>LevelHistoryIndex.lean;<br>LevelStepWitness.lean;<br>Handoff/Core/Step1MathData.lean                                | CNNA/PillarA/<br>Foundation/<br>LevelProfile.lean                                                                                                               | Level selection must be<br>provenance-carrying and<br>cannot be conflated with later<br>experiment requests.                                                                                                                                                                                                                                                                                                                                                      | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.2.6   | W      | Lean;<br>Refactor;<br>Phase   | F0    |          | <a href="#">G04</a>                                                                                                           | Neutralize carrier/<br>index profile                                                                      | 1.2.5;<br>recursive level-history<br>carrier modules                                                                                                  | Carrier / index type                                                                                                                                              | CNNA/PillarA/Arithmetic/<br>BoundarySource/<br>RecursiveLevelHistoryCarrier.lean;<br>LevelHistoryIndex.lean;<br>GeneratedLevelStepWitnessSM6.lean                  | CNNA/PillarA/<br>Foundation/<br>CarrierProfile.lean                                                                                                             | Expose carrier, Fintype/<br>DecidableEq/cardinality/<br>equivalence data explicitly;<br>avoid high-arity index<br>eliminators.                                                                                                                                                                                                                                                                                                                                    | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 1.2.7   | W      | Lean;<br>Refactor;<br>Phase   | F1    |          | <a href="#">S07</a>                                                                                                           | Neutralize<br>IdealThread<br>singleton<br>directed-family<br>profile                                      | SubstrateClass;<br>InfiniteThread;<br>IdealThread.carrier;<br>IdealThread.family;<br>deterministic coarsening<br>lemmas.                              | S07 IdealThread<br>singleton directed-family<br>profile.                                                                                                          | CNNA/PillarA/Foundation/<br>SubstrateClass.lean;<br>CNNA/PillarA/ToC/Contract.lean;<br>CNNA/PillarA/Arithmetic/<br>BoundarySource/<br>ExistingIdealThreadNoGo.lean | CNNA/PillarA/ToC/<br>IdealThreadProfile.lean                                                                                                                    | Expose neutral theorems:<br>carrier $L = \{\text{cell } L\}$ ;<br>family carrier equals singleton<br>carrier;<br>family is rooted/forward;<br>deterministic<br>$\text{coarsenSet}(\text{carrier}(L+1)) =$<br>carrier L. No public<br>phase-semantic names.                                                                                                                                                                                                        | <a href="#">planned</a><br><a href="#">scratch pad</a> |

| Phase | Status | Context                     | class | Doc tier | Linked objects                                                                                               | Main goal                                                                     | Source objects/<br>dependencies                                                                         | Definitions/objects                                                                 | Current code modules                                                                                                                                                                                        | Target modules/<br>objects                                                                                                                                        | Proof goals                                                                                                                                                                                    | How imple-<br>mented                    |
|-------|--------|-----------------------------|-------|----------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 1.2.8 | W      | Lean;<br>Refactor;<br>Phase | F1    |          | <a href="#">S08</a>                                                                                          | Neutralize<br>CutSpec.full<br>maximal-carrier/<br>full-approximant<br>profile | TruncatedFamily;<br>CutSpec.full;<br>RegionCore.ofCutSpec;<br>BoundaryPorts.full;<br>FullBoundaryAudit. | S08 CutSpec.full<br>maximal cut profile.                                            | CNNA/PillarA/Finite/CutSpec.lean;<br>CNNA/PillarA/Finite/<br>RegionCore.lean;<br>CNNA/PillarA/Finite/<br>BoundaryPorts.lean;<br>CNNA/PillarA/Arithmetic/Core/<br>FullBoundaryAudit.lean                     | CNNA/PillarA/Finite/<br>FullCutProfile.lean                                                                                                                       | Expose neutral theorem that<br>(CutSpec.full T).carrier L =<br>T.carrier L. Record that full<br>cut is the full-region route and<br>must not be conflated with<br>IdealThread singleton route. | <a href="#">planned<br/>scratch pad</a> |
| 1.3   | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <i>scope-only</i> ; child<br>objects: G05,<br>G06, G07, G08,<br>G09, G10                                     | Regime, partition,<br>and complement<br>Profile neutralization                | 1.2.6;<br>current partition/<br>regime/complement<br>modules;<br>preliminary object IDs<br>G05-G10      | PartitionProfile;<br>RegimeProfile;<br>ComplementProfile                            | GeneratedBoundaryInteriorPartition.lean;<br>CNNA/PillarA/Schur/<br>Sectors/*.lean;<br>Network/*.lean;<br>A_to_B.lean                                                                                        | CNNA/PillarA/Schur/<br>PartitionProfile.lean;<br>CNNA/PillarA/Regime/<br>*.lean;<br>CNNA/PillarA/Regime/<br>Complement/*.lean                                     | noOuter/hasOuter are<br>A-derived regimes.<br>Complement data is A-internal<br>and must not leak directly to<br>bright Pillar B.                                                               | <a href="#">planned<br/>scratch pad</a> |
| 1.3.1 | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <a href="#">G05</a>                                                                                          | Neutralize boundary/<br>interface/interior<br>partition profile               | 1.3;<br>generated boundary/<br>interior partition<br>modules                                            | Boundary / interface /<br>interior partition                                        | CNNA/PillarA/Arithmetic/Smoke/<br>GeneratedBoundaryInteriorPartition.lean;<br>CNNA/PillarA/Finite/CutSpec.lean;<br>CNNA/PillarA/Finite/<br>BoundaryPorts.lean                                               | CNNA/PillarA/Schur/<br>PartitionProfile.lean                                                                                                                      | Partition is a source layer for<br>Schur/DtN, not a<br>consumer-side convention.                                                                                                               | <a href="#">planned<br/>scratch pad</a> |
| 1.3.2 | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <a href="#">G06</a> , <a href="#">G07</a>                                                                    | Neutralize noOuter/<br>hasOuter regime split                                  | 1.3.1;<br>sector/complement/<br>environment modules                                                     | Regime:<br>noOuterEnvironment;<br>Regime:<br>hasOuterEnvironment                    | CNNA/PillarA/Handoff/Outputs/<br>A_to_B.lean;<br>CNNA/PillarA/Sectors/<br>SectorSplit.lean;<br>CNNA/PillarA/Sectors/<br>ComplementSectorFamily.lean;<br>CNNA/PillarA/Network/<br>SectorSysEnv.lean          | CNNA/PillarA/Regime/<br>NoOuter.lean;<br>CNNA/PillarA/Regime/<br>HasOuter.lean                                                                                    | Keep noOuter closed/root<br>branch separate from<br>hasOuter/open branch. No<br>stochasticity or nonunitarity is<br>genericized.                                                               | <a href="#">planned<br/>scratch pad</a> |
| 1.3.3 | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <a href="#">G08</a> , <a href="#">G09</a> , <a href="#">G10</a>                                              | Neutralize<br>complement-family<br>profiles                                   | 1.3.2;<br>complement-family<br>source modules                                                           | Complement families:<br>UV tail;<br>IR/parent-path tail;<br>sideways environment    | CNNA/PillarA/Sectors/<br>UVSpectralSelector.lean;<br>CNNA/PillarA/Sectors/<br>ComplementSectorFamily.lean;<br>CNNA/PillarA/Network/<br>SectorChannels.lean;<br>CNNA/PillarA/Handoff/Outputs/<br>A_to_B.lean | CNNA/PillarA/Regime/<br>Complement/UVTail.lean;<br>CNNA/PillarA/Regime/<br>Complement/IRTail.lean;<br>CNNA/PillarA/Regime/<br>Complement/<br>SideEnvironment.lean | Complement families are<br>derived A-internal profiles;<br>they are not exported as direct<br>bright-sector data.                                                                              | <a href="#">planned<br/>scratch pad</a> |
| 1.4   | W      | Lean;<br>Refactor;<br>Phase | F1    |          | <i>scope-only</i> ; child<br>objects: CD0,<br>CD1, CD2, CD3,<br>CD4, CD5, CD6,<br>CD7, GA0, GA1,<br>GA2, GA3 | Cayley-Dickson<br>structural path<br>neutral reintegration                    | 1.1.5;<br>current Cayley-Dickson<br>structural modules;<br>preliminary object IDs<br>CD0-CD7            | CD source;<br>star;<br>normSq;<br>product;<br>alternative/norm laws;<br>HurwitzStop | Structural/CayleyDickson/*.lean;<br>Arithmetic/CayleyDickson/*.lean;<br>Handoff/CayleyDicksonAdapter.lean                                                                                                   | CNNA/PillarA/<br>Structural/<br>CayleyDickson/*.lean                                                                                                              | CD remains an A-internal<br>structural consumer/bridge;<br>no unproved identification of<br>R/C/H/O with arithmetic<br>sectors.                                                                | <a href="#">planned<br/>scratch pad</a> |
| 1.4.1 | W      | Lean;<br>Refactor;<br>Phase | F1    |          | <a href="#">CD0</a>                                                                                          | Neutralize<br>Cayley-Dickson<br>source profile                                | 1.4                                                                                                     | Cayley-Dickson source                                                               | CNNA/PillarA/Structural/<br>CayleyDickson/Source.lean;<br>CNNA/PillarA/Handoff/<br>CayleyDicksonAdapter.lean                                                                                                | CNNA/PillarA/<br>Structural/<br>CayleyDickson/<br>Source.lean                                                                                                     | Forward the existing CD source<br>through a neutral profile<br>without turning it into an<br>arithmetic generator.                                                                             | <a href="#">planned<br/>scratch pad</a> |
| 1.4.2 | W      | Lean;<br>Refactor;<br>Phase | F1    |          | <a href="#">CD1</a>                                                                                          | Neutralize scalar<br>part profile                                             | 1.4.1                                                                                                   | Scalar part                                                                         | CNNA/PillarA/Structural/<br>CayleyDickson/<br>S6CanonicalScalarPart.lean                                                                                                                                    | CNNA/PillarA/<br>Structural/<br>CayleyDickson/<br>ScalarPart.lean                                                                                                 | Scalar part must be<br>downstream of CD source and<br>must not be used as sector<br>ontology.                                                                                                  | <a href="#">planned<br/>scratch pad</a> |
| 1.4.3 | W      | Lean;<br>Refactor;<br>Phase | F1    |          | <a href="#">CD2</a>                                                                                          | Neutralize<br>Cayley-Dickson star/<br>conjugation profile                     | 1.4.1-1.4.2                                                                                             | Cayley-Dickson star/<br>conjugation                                                 | CNNA/PillarA/Structural/<br>CayleyDickson/S6CanonicalStar.lean                                                                                                                                              | CNNA/PillarA/<br>Structural/<br>CayleyDickson/Star.lean                                                                                                           | Star/conjugation proof is<br>forwarded structurally;<br>no arithmetic identification is<br>introduced.                                                                                         | <a href="#">planned<br/>scratch pad</a> |
| 1.4.4 | W      | Lean;<br>Refactor;<br>Phase | F1    |          | <a href="#">CD3</a>                                                                                          | Neutralize CD<br>normSq profile                                               | 1.4.3                                                                                                   | CD normSq                                                                           | CNNA/PillarA/Structural/<br>CayleyDickson/<br>S6CanonicalNormSqDefined.lean;<br>S6CanonicalNormSqReduction.lean;<br>S6CanonicalNormSqBlockSystem.lean                                                       | CNNA/PillarA/<br>Structural/<br>CayleyDickson/<br>NormSq.lean                                                                                                     | NormSq remains a CD<br>structural certificate until a<br>later arithmetic bridge is<br>proved.                                                                                                 | <a href="#">planned<br/>scratch pad</a> |
| 1.4.5 | W      | Lean;<br>Refactor;<br>Phase | F1    |          | <a href="#">CD4</a>                                                                                          | Neutralize CD<br>product profile                                              | 1.4.1;<br>1.4.3                                                                                         | CD multiplication/<br>product                                                       | CNNA/PillarA/Structural/<br>CayleyDickson/<br>S6CanonicalProductSurfaceCandidate.lean;<br>S6CanonicalProductMultiplicationCandidate.lean                                                                    | CNNA/PillarA/<br>Structural/<br>CayleyDickson/<br>Product.lean                                                                                                    | Product data is re-exposed<br>without changing algebraic<br>semantics.                                                                                                                         | <a href="#">planned<br/>scratch pad</a> |



| Phase   | Status | Context                      | class | Doc                                                                                                                            | tier | Linked objects | Main goal                                                                                                      | Source objects/<br>dependencies                                                                                                                                                                                       | Definitions/objects                                                                                                                                                | Current code modules                                                                                                      | Target modules/<br>objects                                                                                                                     | Proof goals                                                                                                                                                                                                                                                                                                                                                        | How imple-<br>mented                    |
|---------|--------|------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------|------|----------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 1.4.6   | W      | Lean;<br>Refactor;<br>Phase  | F1    | <a href="#">CD5</a>                                                                                                            |      |                | Neutralize<br>alternative-law<br>certificate                                                                   | 1.4.5                                                                                                                                                                                                                 | Alternative law                                                                                                                                                    | CNNA/PillarA/Structural/<br>CayleyDickson/<br>S6AlternativeLawScaffold.lean;<br>S6FullAlternativeLawBoundary.lean         | CNNA/PillarA/<br>Structural/<br>CayleyDickson/<br>AlternativeLaw.lean                                                                          | Forward the certificate only if<br>the exact product/star/norm<br>provenance is visible.                                                                                                                                                                                                                                                                           | <a href="#">planned<br/>scratch pad</a> |
| 1.4.7   | W      | Lean;<br>Refactor;<br>Phase  | F1    | <a href="#">CD6</a>                                                                                                            |      |                | Neutralize<br>norm-multiplicativity<br>certificate                                                             | 1.4.4;<br>1.4.5                                                                                                                                                                                                       | Norm multiplicativity                                                                                                                                              | CNNA/PillarA/Structural/<br>CayleyDickson/<br>S6NormSqMultiplicativity.lean;<br>S6NormMultiplicativityScaffold.lean       | CNNA/PillarA/<br>Structural/<br>CayleyDickson/<br>NormMultiplicativity.lean                                                                    | Do not use multiplicativity to<br>infer arithmetic sector<br>semantics before CD8.                                                                                                                                                                                                                                                                                 | <a href="#">planned<br/>scratch pad</a> |
| 1.4.8   | W      | Lean;<br>Refactor;<br>Phase  | F1    | <a href="#">CD7</a>                                                                                                            |      |                | Neutralize<br>Hurwitz-stop/<br>classification<br>boundary                                                      | 1.4.6;<br>1.4.7                                                                                                                                                                                                       | Hurwitz stop / R,C,H,O<br>classification                                                                                                                           | CNNA/PillarA/Structural/<br>CayleyDickson/<br>S6HurwitzStopSeed.lean;<br>S6CayleyDicksonIdentification.lean               | CNNA/PillarA/<br>Structural/<br>CayleyDickson/<br>HurwitzStop.lean                                                                             | Classification is a structural<br>boundary result, not a license<br>to identify arithmetic sectors<br>without a later bridge.                                                                                                                                                                                                                                      | <a href="#">planned<br/>scratch pad</a> |
| 1.4.9   | W      | Reference;<br>Gate;<br>Phase | F2    | <a href="#">GA0</a>                                                                                                            |      |                | Quaternion/<br>geometric-algebra<br>external reference<br>gateway for the<br>Cayley-Dickson<br>structural path | 1.4.8;<br>CD0-CD7;<br>Horn 0709.2238v1;<br>Hamilton quaternion<br>relations;<br>Pauli/Cartan sign<br>conventions;<br>mathlib Quaternion and<br>CliffordAlgebra<br>documentation;<br>comparison-only, no<br>CNNA input | GA0;<br>umbrella reference<br>packet for Hamilton<br>units, Pauli/Cartan<br>representation, bivector<br>duality and biquaternion<br>multivector target<br>profiles | No current Lean module;<br>external PDF and<br>documentation-only source notes                                            | GA0;<br>optional future CNNA/<br>PillarA/Structural/<br>CayleyDickson/<br>ReferenceGates/<br>QuaternionGAGate.lean<br>only after release check | Record what must later be<br>compared if the CNNA CD<br>path produces quaternion-like<br>or Clifford-like structure: unit<br>squares, anti-commutation,<br>orientation/sign convention,<br>pseudoscalar square and<br>eight-basis profile. The gate<br>must never identify CD levels<br>with R/C/H/O or Pauli/<br>Clifford objects before a<br>CNNA-derived proof. | <a href="#">planned<br/>scratch pad</a> |
| 1.4.9.1 | W      | Reference;<br>Gate;<br>Phase | F2    | <a href="#">GA1</a>                                                                                                            |      |                | Hamilton unit and<br>Pauli/Cartan<br>sign-convention<br>comparison profile                                     | 1.4.9;<br>CD0-CD7;<br>Horn quaternion/Pauli<br>sections;<br>mathlib Quaternion<br>API;<br>no operational import<br>yet                                                                                                | GA1                                                                                                                                                                | No current Lean module;<br>compare later against neutral CD<br>source/product/star modules                                | GA1;<br>optional future QuaternionUnitComparisonProfile<br>API                                                                                 | Later, after CD structure is<br>closed, compare only the target<br>identities $i^2=j^2=k^2=ijk=-1$<br>and anti-commutation patterns<br>plus sign-convention choices.<br>The comparison must<br>distinguish complex i from<br>quaternion i and must not<br>create arithmetic-sector<br>identifications.                                                             | <a href="#">planned<br/>scratch pad</a> |
| 1.4.9.2 | W      | Reference;<br>Gate;<br>Phase | F2    | <a href="#">GA2, GA3</a>                                                                                                       |      |                | Bivector/<br>pseudoscalar duality<br>and<br>complex-quaternion<br>multivector<br>comparison profile            | 1.4.9;<br>1.4.9.1;<br>Horn duality and<br>complex-quaternion<br>sections;<br>CliffordAlgebra<br>reference APIs;<br>no CNNA input                                                                                      | GA2, GA3                                                                                                                                                           | No current Lean module;<br>future comparison only after a<br>derived orientation/volume/<br>pseudoscalar candidate exists | GA2, GA3;<br>optional future CliffordDualityComparisonProfile<br>and BiquaternionMultivectorProfile APIs                                       | Record target duality facts $I = \sigma_x \sigma_y \sigma_z$ , $I^2 = -1$ , vector/bivector pairing<br>and eight-component<br>multivector bookkeeping. Do<br>not let the external GA basis<br>choose CNNA basis,<br>orientation, dimension,<br>complex scalar layer or physics<br>semantics.                                                                       | <a href="#">planned<br/>scratch pad</a> |
| 2       | W      | Lean;<br>Refactor;<br>Phase  | F0    | <i>scope-only</i> ; child<br>objects: H00,<br>H01, H02, H03                                                                    |      |                | Canonical object<br>register and<br>code-module map<br>after refactor                                          | 1 closed, i.e. 1.1.5 +<br>1.2.6 + 1.3.3 + 1.4.8;<br>neutral wrapper surface;<br>theory register;<br>code audit                                                                                                        | all relevant canonical<br>objects G/S/O/N/L/M/<br>TM/SP/CD/H/P                                                                                                     | masters/object_register.tsv;<br>masters/phase_roadmap.tsv                                                                 | neutral API according to<br>object register for: all<br>relevant objects G/S/O/<br>N/L/M/TM/SP/CD/H/P                                          | define each object exactly once;<br>current module, target API,<br>handoff rights, status, and Exec<br>path per row                                                                                                                                                                                                                                                | <a href="#">planned<br/>scratch pad</a> |
| 2.1     | W      | Lean;<br>Refactor;<br>Phase  | F2    | <i>no direct object</i> :<br>baseline/audit/<br>closeout/<br>infrastructure<br>phase with no<br>registered object<br>ownership |      |                | Object DAG and<br>single-definition rule                                                                       | 2;<br>neutral wrapper surface                                                                                                                                                                                         | G00 and all register IDs                                                                                                                                           | masters/object_register.tsv                                                                                               | neutral API according to<br>object register for: G00<br>and all register IDs                                                                   | every object has natural<br>provenance;<br>no consumer defines its own<br>source                                                                                                                                                                                                                                                                                   | <a href="#">planned<br/>scratch pad</a> |
| 2.2     | W      | Lean;<br>Refactor;<br>Phase  | F2    | <i>no direct object</i> :<br>baseline/audit/<br>closeout/<br>infrastructure<br>phase with no<br>registered object<br>ownership |      |                | Record current<br>derivation sites from<br>the code                                                            | 2.1;<br>neutral wrapper surface;<br>SM9h code tree                                                                                                                                                                    | code-module field                                                                                                                                                  | all listed CNNA/PillarA/*.lean<br>Module                                                                                  | neutral API according to<br>object register for:<br>code-module field                                                                          | each register row states where<br>the derivation lives or that it is<br>missing/incomplete                                                                                                                                                                                                                                                                         | <a href="#">planned<br/>scratch pad</a> |

| Phase | Status | Context                     | class | Doc tier | Linked objects                                                                                                                 | Main goal                                                                                                                              | Source objects/<br>dependencies                                                                               | Definitions/objects                                                                             | Current code modules                                                                                                                                                                     | Target modules/<br>objects                                                                                                                                | Proof goals                                                                                                                                                                                                                                                                                                        | How imple-<br>mented                                   |
|-------|--------|-----------------------------|-------|----------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 2.3   | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <i>scope-only</i> ; child<br>objects: H00,<br>H01, H02, H03                                                                    | Access and handoff<br>governance                                                                                                       | 2.1;<br>H00-H14;<br>neutral handoff surface                                                                   | A-internal, A-in:*,<br>A->*, forbidden-to-A                                                     | Handoff/Inputs/*.lean;<br>Handoff/Outputs/*.lean                                                                                                                                         | neutral API according to<br>object register for:<br>A-internal, A-in:*, A->*,<br>forbidden-to-A                                                           | A consumes no own outputs;<br>inputs B/C/D/E -> A are the<br>only backreaction sources                                                                                                                                                                                                                             | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 2.3.1 | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <a href="#">H00</a> , <a href="#">H01</a> , <a href="#">H02</a> ,<br><a href="#">H03</a>                                       | Derive/register<br>object group: A<br>input handoff B->A,<br>A input handoff<br>C->A, A input<br>handoff D->A, A<br>input handoff E->A | 2.3                                                                                                           | A input handoff B->A;<br>A input handoff C->A;<br>A input handoff D->A;<br>A input handoff E->A | CNNA/PillarA/Handoff/Inputs/<br>B_to_A.lean;<br>CNNA/PillarA/Handoff/Inputs/<br>C_to_A.lean;<br>CNNA/PillarA/Handoff/Inputs/<br>D_to_A.lean;<br>...                                      | CNNA/PillarA/Handoff/<br>AinputhandoffBA.lean;<br>CNNA/PillarA/Handoff/<br>AinputhandoffCA.lean;<br>CNNA/PillarA/Handoff/<br>AinputhandoffDA.lean;<br>... | establish the listed object(s) in<br>natural dependency order;<br>split further if proof<br>dependencies are too deep                                                                                                                                                                                              | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 2.4   | W      | Lean;<br>Refactor;<br>Phase | F2    |          | <i>no direct object</i> :<br>baseline/audit/<br>closeout/<br>infrastructure<br>phase with no<br>registered object<br>ownership | Split rule for phases<br>and subphases                                                                                                 | 2.1-2.3                                                                                                       | DAG-depth gate                                                                                  | masters/phase_roadmap.tsv                                                                                                                                                                | neutral API according to<br>object register for:<br>DAG-depth gate                                                                                        | split a phase when<br>prerequisites/proof kinds/<br>generation layers are too deep<br>or mixed                                                                                                                                                                                                                     | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 3     | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <i>scope-only</i> ; child<br>objects: BR06,<br>S00, S01, S02,<br>S03, S04, S05,<br>S06                                         | Schur/DtN core in<br>neutral derivation<br>order                                                                                       | 1.2.6-1.3.3;<br>S00-S06;<br>notation/import firewall<br>1.1.5 closed;<br>BR06 route separation<br>before 3.1. | DirichletLaplacian -><br>InteriorBlock -> Inverse<br>-> Elimination -> DtN<br>-> generalized    | Smoke/<br>GeneratedDirichletEntry.lean;<br>GeneratedInterior*.lean;<br>GeneratedTopBoundaryDtNSM3gR.lean;<br>GeneratedGeneralizedTopBoundaryDtNSM3gR.lean                                | neutral API according to<br>object register for:<br>DirichletLaplacian -><br>InteriorBlock -> Inverse<br>Elimination -> DtN -><br>generalized             | encapsulate the existing<br>derived-only chain by domain<br>role;<br>do not reinvent product/inverse<br>formulas;<br>route classification must be<br>complete before wrapping<br>boundary operators.                                                                                                               | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 3.0   | W      | Lean;<br>Refactor;<br>Phase | F1    |          | <a href="#">BR06</a>                                                                                                           | Boundary-source<br>route separation:<br>full-cut matrix<br>geometry versus<br>IdealThread scalar<br>collapse                           | S07;<br>S08;<br>ExistingIdealThreadNoGo;<br>FullBoundaryAudit;<br>Phase 3 Schur/DtN<br>profiles.              | BR06 Boundary-route<br>extrema contract.                                                        | CNNA/PillarA/Arithmetic/<br>BoundarySource/<br>ExistingIdealThreadNoGo.lean;<br>CNNA/PillarA/Arithmetic/Core/<br>FullBoundaryAudit.lean;<br>CNNA/PillarA/Finite/*;<br>CNNA/PillarA/DtN/* | CNNA/PillarA/Schur/<br>BoundaryRouteContract.lean                                                                                                         | Before wrapping Schur/DtN,<br>state which proofs consume the<br>full CutSpec route and which<br>are only scalar/thread-route<br>audits. A nontrivial boundary<br>matrix must be sourced from<br>full/generated boundary data,<br>not from an IdealThread<br>singleton unless a separate<br>lifting theorem exists. | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 3.1   | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <a href="#">S00</a>                                                                                                            | Dirichlet Laplacian<br>and partition                                                                                                   | G04-G05/S00                                                                                                   | DirichletLaplacianProfile                                                                       | Finite/DirichletLaplacian.lean;<br>Smoke/GeneratedDirichletEntry.lean                                                                                                                    | neutral API according to<br>object register for:<br>DirichletLaplacianProfile                                                                             | operator source from partition,<br>weights, adjacency;<br>clarify Exec mirror                                                                                                                                                                                                                                      | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 3.2   | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <a href="#">S01</a>                                                                                                            | InteriorBlock                                                                                                                          | 3.1/S01                                                                                                       | InteriorBlockProfile                                                                            | Smoke/GeneratedInteriorBlock.lean                                                                                                                                                        | neutral API according to<br>object register for:<br>InteriorBlockProfile                                                                                  | block entries proof-carrying<br>from Dirichlet profile;<br>no free matrix                                                                                                                                                                                                                                          | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 3.3   | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <a href="#">S02</a>                                                                                                            | Interior inverse<br>candidate and trace/<br>fold                                                                                       | 3.2/S02                                                                                                       | InteriorInverseCandidateProfile                                                                 | Smoke/<br>GeneratedInteriorInverseCandidate*.lean;<br>GeneratedInteriorAccumulatedTraceEvaluationCandidateProfile                                                                        | neutral API according to<br>object register for: Inter-<br>GeneratedInteriorCandidateProfile                                                              | candidate from trace/fold, not<br>Matrix.inv                                                                                                                                                                                                                                                                       | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 3.4   | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <a href="#">S03</a>                                                                                                            | Two-sided inverse<br>certificate                                                                                                       | 3.3/S03                                                                                                       | Two-sided inverse<br>certificate                                                                | GeneratedInteriorTwoSidedInvFromSM3gR.lean;<br>GeneratedInteriorProvenTwoSidedInvFromSM3gR.lean                                                                                          | neutral API according to<br>object register for:<br>Two-sided inverse<br>certificate                                                                      | left and right products from<br>existing provenance chain                                                                                                                                                                                                                                                          | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 3.5   | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <a href="#">S04</a>                                                                                                            | InteriorEliminationData                                                                                                                | 3.4/S04                                                                                                       | InteriorEliminationProfile                                                                      | GeneratedInteriorEliminationCertificateSM3gR.lean                                                                                                                                        | neutral API according to<br>object register for:<br>InteriorEliminationProfile                                                                            | Schur elimination data from<br>TwoSidedInv                                                                                                                                                                                                                                                                         | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 3.6   | W      | Lean;<br>Refactor;<br>Phase | F0    |          | <a href="#">S05</a>                                                                                                            | DtN<br>BoundaryOperator                                                                                                                | 3.5/S05                                                                                                       | BoundaryOperatorProfile                                                                         | GeneratedTopBoundaryDtNSM3gR.lean;<br>DtN/DtN.lean                                                                                                                                       | CNNA/PillarA/Schur/<br>BoundaryOperatorProfile.lean;<br>consumer-contract notes<br>for Operational/<br>ExecComplex phase 4.1                              | Boundary operator as a profile<br>derivable inside A and later<br>exportable. The wrapper must<br>also publish the<br>consumer-ready shape needed<br>by Phase 4.1, without<br>importing Phase 4 or<br>depending on Exec arithmetic<br>implementation.                                                              | <a href="#">planned</a><br><a href="#">scratch pad</a> |

| Phase | Status | Context                       | class | Doc tier | Linked objects                                                                                                        | Main goal                                                                                                         | Source objects/<br>dependencies                                                                                                         | Definitions/objects                                                                                                                                                                             | Current code modules                                                                                                                                                                                                        | Target modules/<br>objects                                                                                                                                                                                                                                                        | Proof goals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | How imple-<br>mented                    |
|-------|--------|-------------------------------|-------|----------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 3.7   | W      | Lean;<br>Refactor;<br>Phase   | F0    |          | <a href="#">S06</a>                                                                                                   | GeneralizedDtN/<br>MultiSchur                                                                                     | 3.6/S06                                                                                                                                 | GeneralizedBoundaryOperatorProfile                                                                                                                                                              | GeneralizedTopBoundaryDtNMultiSchurSM4.lean;<br>GeneratedMultiSchurCoupling/*.lean                                                                                                                                          | NSM3hRAPI according to<br>object register for:<br>GeneralizedBoundaryOperatorProfile                                                                                                                                                                                              | MultiSchur only as an<br>extension of the<br>BoundaryOperatorProfile                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <a href="#">planned<br/>scratch pad</a> |
| 4     | W      | Lean;<br>Refactor;<br>Phase   | F1    |          | <i>scope-only</i> ; child<br>objects: BR07,<br>O07, O08, BR12,<br>BR13, O01, O02,<br>O03, O04, O05,<br>O10, BR14, O06 | Operational<br>ExecComplex<br>arithmetic path from<br>BoundaryOperator                                            | 3.6-3.7;<br>O01-O06;<br>BR07 thread-recursion<br>audit before 4.1.                                                                      | BoundaryMatrix -><br>recurrence -><br>polynomial -> window<br>-> D_op -> target<br>policy                                                                                                       | GeneratedMultiSchurExecBoundaryOperator*.lean;<br>GeneratedOperationalDeterminantCoefficientRegister*.lean<br>GeneratedOperationalQuadratic*.lean                                                                           | Operational API according to<br>Object Register for<br>BoundaryMatrix -><br>recurrence -> polynomial<br>-> window -> D_op -><br>target policy                                                                                                                                     | neutralize SM9h core: all<br>values via ExecComplex, no<br>target values as sources;<br>recurrence provenance must be<br>audited before arithmetic<br>interpretation.                                                                                                                                                                                                                                                                                                                                                         | <a href="#">planned<br/>scratch pad</a> |
| 4.0   | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">BR07</a>                                                                                                  | Thread-recursion<br>arithmetic-shadow<br>audit before<br>operational<br>arithmetic wrappers                       | S07;<br>BR06;<br>RecursiveSchurSource;<br>MobiusTransfer;<br>CharpolyExec;<br>operational discriminant<br>chain.                        | BR07 Thread-recursion<br>arithmetic-shadow audit<br>profile.                                                                                                                                    | CNNA/PillarA/Arithmetic/Schur/<br>RecursiveSchurSource.lean;<br>CNNA/PillarA/Arithmetic/Schur/<br>MobiusTransfer.lean;<br>CNNA/PillarA/Arithmetic/Schur/<br>CharpolyExec.lean;<br>CNNA/PillarA/Arithmetic/<br>Notation.lean | CNNA/PillarA/<br>Operational/<br>ThreadRecursionAudit.lean                                                                                                                                                                                                                        | Audit whether the scalar<br>Schur/Mobius recurrence is<br>actually tied to IdealThread<br>singleton data, full-boundary<br>diagonal data, or an<br>intermediate generated source.<br>Keep D=-3/-4 claims as derived<br>arithmetic outputs only after<br>the chain is code-proven and<br>not as string-analogy input.<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                                        | <a href="#">blocked<br/>scratch pad</a> |
| 4.0.1 | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">O07</a>                                                                                                   | General finite<br>computation test<br>matrix before<br>operational<br>arithmetic wrappers                         | 4.0;<br>LEG14;<br>LEG15;<br>LEG29;<br>BR07;<br>canonical generator<br>policy;<br>Phase-3 consumer<br>boundary profile when<br>available | O07 general finite<br>computation test matrix<br>profile;<br>generic rows over<br>code-defined finite<br>instances;<br>sentinel-row slots for<br>Eisenstein/Gauss and<br>later CM/j comparisons | Future implementation must reuse<br>existing Smoke/Generator/<br>Arithmetic modules;<br>exact module placement decided by<br>Phase 1.1.1.1.1 manifest.                                                                      | One compact reusable<br>test-matrix module or<br>integration into an<br>existing smoke-test<br>generator module;<br>no duplicate generator;<br>candidate API: finite input<br>row, branch-mode tag,<br>L_max, expected<br>executable witness,<br>optional comparison<br>sentinel. | Show that the operational<br>computation path is not just a<br>skeleton by running finite<br>instances through the canonical<br>generator and recording<br>proof-carrying output rows<br>before specialized CM/lattice<br>tests are interpreted. V1.19<br>status policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                                                                                                            | <a href="#">blocked<br/>scratch pad</a> |
| 4.0.2 | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">O08</a> , <a href="#">BR12</a>                                                                            | Regime-typed finite<br>computation test<br>matrix: noOuter-root<br>sentinel versus<br>HasOuter projection<br>rows | 4.0.1/O07;<br>1.3.2 noOuter/HasOuter<br>regime split;<br>LEG30/O08;<br>canonical generator<br>policy;<br>EXEC-2                         | O08 regime-typed test<br>matrix profile;<br>BR12 noOuter-root<br>Eisenstein/Gauss<br>sentinel hypothesis;<br>generic HasOuter/<br>complement-projection<br>row schema                           | Future implementation must reuse<br>the O07/canonical smoke-test<br>generator surface;<br>exact module placement decided by<br>Phase 1.1.1.1.1 manifest.                                                                    | Prefer extension of the<br>compact reusable O07 test<br>matrix module/profile;<br>no separate Heegner or<br>lattice generator.                                                                                                                                                    | Run finite generic rows through<br>the computable path and<br>record a derived regime tag.<br>Only the empty-set/noOuter<br>REAL-root branch may be<br>tested as closed/unitary;<br>Eisenstein/Gauss are sentinel<br>candidates there. Other CM/<br>Heegner rows remain<br>HasOuter/projection/<br>comparison rows until a<br>noOuter derivation is proved.<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange. | <a href="#">blocked<br/>scratch pad</a> |

| Phase | Status | Context                       | class | Doc tier | Linked objects | Main goal                                                                             | Source objects/<br>dependencies                                                                                         | Definitions/objects                                                                                                                | Current code modules                                                                                                                                              | Target modules/<br>objects                                                                                                                                                           | Proof goals                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | How imple-<br>mented                |
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| 4.0.3 | R      | Lean;<br>Derivation;<br>Phase | F1    |          | BR13           | Scalar-layer computation test rows: ExecComplex vs exact quadratic extension boundary | O07, O08, O09, LEG31, BR12; O09 schema and O10/BR14 D-instantiation if exact quadratic coordinates are used             | Test rows for rational ExecComplex outputs and exact quadratic candidate outputs                                                   | Existing arithmetic/smoke-test generator modules after neutralization                                                                                             | Use canonical early finite computation matrix; no duplicate generator                                                                                                                | Run generic finite rows first; then test whether Eisenstein/Gauss sentinel rows require exact quadratic coordinates or can close via rational/integer invariants only. Exact quadratic rows must cite a proof-carrying D-source; no free sqrt(D). V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                                                        | <a href="#">blocked scratch pad</a> |
| 4.1   | W      | Lean;<br>Refactor;<br>Phase   | F1    |          | O01            | ExecBoundaryMatrix                                                                    | 3.6/O01 plus consumer-contract direction lock 1.1.1.1.0.6                                                               | ExecMatrixProfile                                                                                                                  | GeneratedMultiSchurExecBoundaryOperatorFromABSM36qPngSM4q<br>GeneratedLevelHistoryExecMatrixFromSM46bRegisterSM48lean<br>ExecMatrixProfile                        | Matrix carrier over ExecComplex from BoundaryOperator provenance; consume the Phase-3.6 boundary/operator profile contract rather than reconstructing Schur/DtN data.                | <a href="#">planned scratch pad</a>                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     |
| 4.2   | W      | Lean;<br>Refactor;<br>Phase   | F1    |          | O02            | Coefficient recurrence                                                                | 4.1/O02                                                                                                                 | DeterminantRecurrenceProfile                                                                                                       | GeneratedOperationalDeterminantCoefficientRegisterForDeterminantRecurrenceProfile                                                                                 | Operational coefficient path without noncomputable Polynomial                                                                                                                        | <a href="#">planned scratch pad</a>                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     |
| 4.3   | W      | Lean;<br>Refactor;<br>Phase   | F1    |          | O03            | Polynomial profile                                                                    | 4.2/O03                                                                                                                 | OperationalPolynomialProfile                                                                                                       | GeneratedLevelHistoryOperationalCharacteristicPolynomialProfileFromSM9blean<br>GeneratedLevelHistoryOperationalCoefficientRegisterForOperationalPolynomialProfile | Polynomial profile from SM9blean coefficients; no Mathlib-charpoly oracle as operational path                                                                                        | <a href="#">planned scratch pad</a>                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     |
| 4.4   | W      | Lean;<br>Refactor;<br>Phase   | F1    |          | O04            | Quadratic window                                                                      | 4.3/O04                                                                                                                 | QuadraticWindowCertificate                                                                                                         | GeneratedOperationalQuadraticCoefficientWindowFromSM9blean<br>object register for: QuadraticWindowCertificate                                                     | Quadratic window from coefficient source; no target comparison                                                                                                                       | <a href="#">planned scratch pad</a>                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     |
| 4.5   | W      | Lean;<br>Refactor;<br>Phase   | F1    |          | O05            | Operational discriminant D_op                                                         | 4.4/O05                                                                                                                 | OperationalDiscriminantCertificate                                                                                                 | GeneratedOperationalQuadraticDiscriminantSourceFromSM9blean<br>object register for: OperationalDiscriminantCertificate                                            | Operational discriminant D_op from SM9blean; no target comparison                                                                                                                    | <a href="#">planned scratch pad</a>                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     |
| 4.5.1 | R      | Lean;<br>Derivation;<br>Phase | F1    |          | O10, BR14      | Proof-carrying discriminant-to-exact-quadratic instantiation gate                     | 4.5 Operational discriminant D_op; O05/O06 or equivalent discriminant ledger; O09 schema; LEG32; no-free-sqrt guardrail | O10 proof-carrying discriminant source; BR14 discriminant-to-quadratic algebra instantiation bridge; allowed invariant projections | Future neutral operational discriminant module and exact quadratic schema; no module fixed before manifest.                                                       | Natural target likely a bridge/profile module adjacent to the operational discriminant and exact quadratic schema, but exact names must be fixed by module manifest and reuse audit. | Given a proof-carrying operational D, construct the corresponding exact quadratic algebra instance with $\theta^2 = D$ and prove that trace/norm/conjugation/projected invariants are derived from that D. If D is absent or semantically blue/uncorrected, the phase must remain blocked or backjump to O05 correction. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. | <a href="#">blocked scratch pad</a> |
| 4.6   | W      | Reference;<br>Gate;<br>Phase  | F2    |          | O06            | Comparison target policy                                                              | 4.5/O06                                                                                                                 | ComparisonTargetProfile                                                                                                            | GeneratedOperationalDiscriminantFactorGateFromSM48blean<br>currently only blocks targets                                                                          | target values only as consumers; no Eisenstein/Gauss-Namen im Request                                                                                                                | <a href="#">planned scratch pad</a>                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     |

| Phase | Status | Context                       | class | Doc tier | Linked objects                                                                        | Main goal                                                                                                                                    | Source objects/<br>dependencies                                                                                                 | Definitions/objects                                                                                                                      | Current code modules                                                                                                                                                                                                                        | Target modules/<br>objects                                                                                                                                  | Proof goals                                                                                                                                                                                                                                                                                                                                                                                                       | How imple-<br>mented                                   |
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| 5     | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <i>scope-only</i> ; child<br>objects: N00,<br>N01, N02, N03,<br>N04, N05, N06,<br>N07 | Number-theory base<br>from operational<br>discriminant:<br>D_fund, conductor,<br>quadratic field/order,<br>basis, trace/norm/<br>conjugation | 4.5 Operational<br>discriminant D_op;<br>4.5.1 exact-quadratic<br>instantiation gate when<br>coordinates are needed;<br>N00-N07 | N00-N07: fundamental<br>discriminant, conductor,<br>quadratic field,<br>quadratic order, integral<br>basis, trace, norm,<br>conjugation. | Arithmetic/Smoke operational<br>discriminant/window sources;<br>future NumberTheory neutral<br>modules                                                                                                                                      | CNNA/PillarA/<br>NumberTheory/* base<br>profiles;<br>O09/O10 only as<br>D-indexed coordinate layer<br>when required                                         | Base number-theory phase, not<br>experiment infrastructure. It<br>consumes the operational<br>discriminant path and produces<br>the first algebraic-number<br>profiles without depending on<br>modular forms, lattices or<br>physics consumers. V1.19<br>status policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 5.1   | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">N00</a> , <a href="#">N01</a>                                             | Fundamental<br>discriminant and<br>conductor<br>normalization from<br>D_op                                                                   | 4.5/O05;<br>O06 target policy;<br>5 umbrella                                                                                    | N00 Fundamental<br>discriminant D_fund;<br>N01 conductor f                                                                               | CNNA/PillarA/Arithmetic/Orders/<br>DiscriminantConvention.lean (special<br>Gaussian/Eisenstein bridge;<br>not general derived D_fund);<br>CNNA/PillarA/Arithmetic/Orders/<br>DiscriminantConvention.lean<br>(implicit/bridge status;<br>... | CNNA/PillarA/<br>NumberTheory/<br>DiscriminantNormalization.lean<br>CNNA/PillarA/<br>NumberTheory/<br>Conductor.lean                                        | Normalize operational D_op to<br>D_fund and conductor with<br>explicit divisibility/<br>square-factor conditions;<br>no class group or modular<br>input. V1.19 status policy:<br>locked red until a release proof<br>references the prerequisite<br>derived-only Lean objects and<br>turns this phase yellow or<br>orange.                                                                                        | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 5.2   | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">N02</a>                                                                   | Quadratic field<br>profile from<br>proof-carrying<br>discriminant                                                                            | 5.1;<br>O09 schema and O10/<br>BR14 D-instantiation if<br>exact quadratic<br>coordinates are used                               | N02 Quadratic field<br>K=Q(sqrt(D)) as<br>derived algebraic profile;<br>no free sqrt(D)                                                  | CNNA/PillarA/Arithmetic/Orders/<br>QuadraticOrder.lean (special shapes)                                                                                                                                                                     | CNNA/PillarA/<br>NumberTheory/<br>QuadraticField.lean                                                                                                       | Represent K via derived D and<br>exact quadratic schema/mirror<br>boundary;<br>do not use analytic Complex as<br>generator. V1.19 status policy:<br>locked red until a release proof<br>references the prerequisite<br>derived-only Lean objects and<br>turns this phase yellow or<br>orange.                                                                                                                     | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 5.3   | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">N03</a>                                                                   | Quadratic order<br>profile from D_fund<br>and conductor                                                                                      | 5.1;<br>5.2;<br>O10/BR14 if coordinate<br>witnesses are needed                                                                  | N03 Quadratic order<br>O_D                                                                                                               | CNNA/PillarA/Arithmetic/Orders/<br>QuadraticOrder.lean (special<br>Gaussian/Eisenstein examples)                                                                                                                                            | CNNA/PillarA/<br>NumberTheory/<br>QuadraticOrder.lean                                                                                                       | Define order from conductor/<br>maximal-order profile;<br>operationally store<br>proof-carrying data, not<br>axiomatic field elements. V1.19<br>status policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                                                                                                          | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 5.4   | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">N04</a>                                                                   | Integral basis /<br>omega_D profile                                                                                                          | 5.1;<br>5.3                                                                                                                     | N04 Integral basis /<br>omega_D                                                                                                          | CNNA/PillarA/Arithmetic/Orders/<br>QuadraticOrder.lean (halfIntegral<br>shape;<br>basis not general API)                                                                                                                                    | CNNA/PillarA/<br>NumberTheory/<br>IntegralBasis.lean                                                                                                        | Derive basis case split from D<br>mod 4 and conductor data;<br>exact quadratic theta is used<br>only through D-indexed<br>schema. V1.19 status policy:<br>locked red until a release proof<br>references the prerequisite<br>derived-only Lean objects and<br>turns this phase yellow or<br>orange.                                                                                                               | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 5.5   | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">N05</a> , <a href="#">N06</a> , <a href="#">N07</a>                       | Trace, norm and<br>conjugation forms                                                                                                         | 5.2-5.4                                                                                                                         | N05 Trace form;<br>N06 Norm form;<br>N07 Conjugation                                                                                     | no current general module;<br>partial arithmetic in CNNA/PillarA/<br>Arithmetic/Orders/NormForms.lean;<br>CNNA/PillarA/Arithmetic/Orders/<br>NormForms.lean (special<br>BinaryQuadraticNormForm);<br>...                                    | CNNA/PillarA/<br>NumberTheory/<br>TraceForm.lean;<br>CNNA/PillarA/<br>NumberTheory/<br>NormForm.lean;<br>CNNA/PillarA/<br>NumberTheory/<br>Conjugation.lean | Prove coordinate formulas from<br>the quadratic schema/order<br>data;<br>rational/integer projections<br>may feed operational tests.<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                                                                                                            | <a href="#">blocked</a><br><a href="#">scratch pad</a> |



| Phase | Status | Context                       | class | Doc tier | Linked objects                                                                        | Main goal                                                                                                    | Source objects/<br>dependencies                                                                  | Definitions/objects                                                                                                                                                    | Current code modules                                                                            | Target modules/<br>objects                                                                                                   | Proof goals                                                                                                                                                                                                                                                                                                              | How imple-<br>mented                                   |
|-------|--------|-------------------------------|-------|----------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 6     | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <i>scope-only</i> ; child<br>objects: N08,<br>N09, N10, N11,<br>N12, N13, N14,<br>N15 | Units, class group,<br>ideals, class number,<br>prime splitting and<br>character from the<br>quadratic order | 5;<br>N00-N07;<br>noOuter/HasOuter<br>regime split only where<br>explicitly used by N18<br>later | N08-N15: units, class<br>group, class number,<br>ideals, principal<br>predicate, Heegner/PID/<br>UFD indicator, prime<br>splitting, Kronecker/<br>Dirichlet character. | future NumberTheory neutral<br>modules;<br>existing arithmetic seeds only as<br>audited sources | CNNA/PillarA/<br>NumberTheory/* units/<br>classes/splitting profiles                                                         | Second number-theory phase.<br>It consumes the quadratic<br>order/basis/trace/norm layer.<br>It does not produce lattice<br>geometry or modular forms.<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 6.1   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">N08</a>                                                                   | Unit group of the<br>quadratic order                                                                         | 5.3/N03;<br>5.5/N07;<br>N08                                                                      | N08 UnitGroupProfile                                                                                                                                                   | Orders/DiscriminantConvention.lean<br>special only                                              | neutral API according to<br>object register for:<br>DiscriminantNormaliza-<br>tionProfile,<br>ConductorProfile               | Derive units from order/<br>conjugation/norm;<br>keep CM special-unit facts<br>proof-carrying and<br>phase-bound. V1.19 status<br>policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                                      | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 6.2   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">N09</a> , <a href="#">N10</a>                                             | Class group, class<br>number and ideal<br>profiles                                                           | 5.3/N03;<br>5.5/N05-N07;<br>N09-N12                                                              | N09 ClassGroupProfile;<br>N10<br>ClassNumberProfile;<br>N11 ideals;<br>N12 principal ideal<br>predicate                                                                | Orders/QuadraticOrder.lean                                                                      | neutral API according to<br>object register for:<br>QuadraticFieldProfile,<br>QuadraticOrderProfile,<br>IntegralBasisProfile | Register ideal/class-group<br>machinery after the order<br>exists;<br>do not use analytic<br>class-number formula as source.<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                           | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 6.2.1 | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">N11</a>                                                                   | Fractional and<br>integral ideal profile                                                                     | 6.2;<br>N11                                                                                      | N11 Fractional/integral<br>ideal profile                                                                                                                               | no current module                                                                               | CNNA/PillarA/<br>NumberTheory/<br>FractionalIntegralIdeas.lean                                                               | Define ideal data consumed by<br>class group and principal<br>predicate. V1.19 status policy:<br>locked red until a release proof<br>references the prerequisite<br>derived-only Lean objects and<br>turns this phase yellow or<br>orange.                                                                               | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 6.2.2 | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">N12</a>                                                                   | Principal ideal<br>predicate                                                                                 | 6.2.1;<br>N12                                                                                    | N12 Principal ideal<br>predicate                                                                                                                                       | no current module                                                                               | CNNA/PillarA/<br>NumberTheory/<br>PrincipalIdealPredicate.lean                                                               | Predicate over derived ideals/<br>order;<br>do not collapse principal to<br>arbitrary class-number-one<br>claims. V1.19 status policy:<br>locked red until a release proof<br>references the prerequisite<br>derived-only Lean objects and<br>turns this phase yellow or<br>orange.                                      | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 6.3   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">N13</a>                                                                   | Heegner/PID/UFD<br>indicator after<br>class-number data                                                      | 6.2;<br>N10-N13                                                                                  | N13 Heegner/PID/UFD<br>indicator                                                                                                                                       | Orders/NormForms.lean partial/<br>special                                                       | neutral API according to<br>object register for:<br>TraceFormProfile,<br>AlgebraicNormProfile,<br>ConjugationProfile         | Record class-number-one/PID/<br>UFD indicator as<br>number-theory data only;<br>no closed noOuter inference<br>without regime proof. V1.19<br>status policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                   | <a href="#">blocked</a><br><a href="#">scratch pad</a> |



| Phase | Status | Context                       | class | Doc tier | Linked objects                                                                                             | Main goal                                                                                                       | Source objects/<br>dependencies                                                                                                       | Definitions/objects                                                                                                                                      | Current code modules                                                                                                                                                                                                                   | Target modules/<br>objects                                                                                                           | Proof goals                                                                                                                                                                                                                                                                                                                  | How imple-<br>mented                                   |
|-------|--------|-------------------------------|-------|----------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 6.4   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">N14</a> , <a href="#">N15</a>                                                                  | Prime splitting and Kronecker/Dirichlet character                                                               | 5.5;<br>6.2;<br>N14-N15                                                                                                               | N14<br>PrimeSplittingProfile;<br>N15 KroneckerCharac-<br>terProfile                                                                                      | no current module                                                                                                                                                                                                                      | CNNA/PillarA/<br>NumberTheory/<br>Primesplittingbehavior.lean;<br>CNNA/PillarA/<br>NumberTheory/<br>KroneckerDirichletcharacter.lean | Derive splitting/character profiles from discriminant/order/ideal data; irreducible-vs-prime guardrail applies. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                                           | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 6.5   | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <i>no direct object</i> : baseline/audit/closeout/infrastructure phase with no registered object ownership | Arithmetic-CD compatibility source-site audit and bridge-precondition register, not final compatibility theorem | 6;<br>CD3/CD6/CD7;<br>O05 when available;<br>final theorem deferred to 10.4.2 after BR08                                              | precondition audit for CD8;<br>no public-final Arithmetic-CD theorem in this phase                                                                       | CNNA/PillarA/Arithmetic/<br>CayleyDickson/<br>ArithmeticCDBridge.lean;<br>CNNA/PillarA/Arithmetic/<br>Convergence/<br>CDArithmeticCompatibility.lean;<br>CNNA/PillarA/Arithmetic/<br>Convergence/<br>QuaternionRamanujanInterface.lean | CNNA/PillarA/Bridge/<br>ArithmeticCayleyDicksonCompatibility.lean                                                                    | Only source-site audit: CD8 is incompatible and must wait for Schur-duality bridge BR08/BR09. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                                                             | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 7     | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <i>scope-only</i> ; child objects: L00, L01, L02, L03, N18, BR11, L04, L05, L06, L07                       | CM lattice geometry from the quadratic order and exact quadratic coordinates                                    | 5.2-5.5;<br>6.1 only for automorphism/unit input;<br>L00-L07;<br>O09/O10 when irrational coordinates are required                     | L00-L07: embedding, lattice basis, Gram matrix, determinant/covolume, tau, upper half-plane condition, angle/cosine profile, lattice automorphism group. | no current canonical lattice modules; heuristic drawings remain comparison only                                                                                                                                                        | CNNA/PillarA/<br>Geometry/* lattice profiles                                                                                         | Geometric path after the order/basis/norm layer. It does not depend on modular forms or physical consumers; general irrational coordinates use O09/O10, not free sqrt(D). V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 7.1   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">L00</a> , <a href="#">L01</a>                                                                  | CM embedding and lattice basis                                                                                  | 5.2-5.4;<br>L00-L01;<br>O09/O10 if coordinate witnesses are needed                                                                    | L00<br>CMEmbeddingProfile;<br>L01 LatticeBasisProfile                                                                                                    | no derived module; UnitConventionStatus only                                                                                                                                                                                           | neutral API according to object register for: UnitGroupProfile                                                                       | Construct embedding/basis from order and exact quadratic schema/mirror boundary; no heuristic angle as input. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                                             | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 7.2   | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">L02</a> , <a href="#">L03</a>                                                                  | Gram matrix, determinant and covolume                                                                           | 7.1;<br>L02-L03;<br>N06 norm form                                                                                                     | L02 GramProfile;<br>L03 lattice determinant/covolume                                                                                                     | no current module                                                                                                                                                                                                                      | neutral API according to object register for: ClassGroupProfile, ClassNumberProfile, IdealProfile                                    | Compute Gram/determinant projections from derived basis/norm data; rationals/integers may be operational. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                                                 | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 7.2.1 | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">N18</a> , <a href="#">BR11</a>                                                                 | Heegner/class-number-one regime classification after class-number data                                          | 6.3<br>ClassNumberProfile/<br>Heegner indicator;<br>N13;<br>O08 regime-typed test matrix;<br>EXEC-2;<br>noOuter/HasOuter regime split | N18 Heegner regime-projection classification profile; BR11 IDEAL_infty projection boundary                                                               | No current implementation; future placement decided by manifest and natural NumberTheory/test-profile module ownership.                                                                                                                | Consumer-only classification/profile module if needed; otherwise integrated into N13/O08 documentation/test rows.                    | Bridge number-theory class-number-one data to regime classification only after both class-number and regime data exist; no h(D)=1 -> closed noOuter shortcut. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.             | <a href="#">blocked</a><br><a href="#">scratch pad</a> |

| Phase | Status | Context                       | class | Doc tier | Linked objects                                                                                 | Main goal                                                                           | Source objects/<br>dependencies                                                | Definitions/objects                                                                                                                                                                                               | Current code modules                                                                               | Target modules/<br>objects                                                                                                                        | Proof goals                                                                                                                                                                                                                                                                                                       | How imple-<br>mented                                   |
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| 7.3   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">L04</a> , <a href="#">L05</a>                                                      | Period ratio tau and upper-half-plane condition                                     | 7.1-7.2; L04-L05; O09/O10 if exact quadratic coordinates are used              | L04 PeriodRatioProfile; L05 UpperHalf-PlaneCondition                                                                                                                                                              | no current module                                                                                  | neutral API according to object register for: PrimeSplittingProfile, KroneckerCharacterProfile                                                    | Derive tau profile from lattice basis; analytic upper-half-plane semantics remain proof/mirror-boundary unless operationally projected. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                        | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 7.4   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">L06</a>                                                                            | Angle / cosine profile                                                              | 7.2-7.3; L06; O09/O10 for non-rational coordinates                             | L06 Angle/cosine profile                                                                                                                                                                                          | no current derived angle module; drawings/visual intuition not code source                         | CNNA/PillarA/Geometry/Anglecosineprofile.lean                                                                                                     | Use rational special cosines where available; general angle is comparison/mirror unless an exact algebraic projection is proved. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                               | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 7.5   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">L07</a>                                                                            | Lattice automorphism group                                                          | 7.1-7.4; 6.1 units if applicable; L07                                          | L07 LatticeAutomor-<br>phismProfile                                                                                                                                                                               | no current lattice automorphism module                                                             | CNNA/PillarA/Geometry/<br>Latticeautomorphismgroup.lean                                                                                           | Aut profile consumes lattice and unit data; Eisenstein/Gauss special symmetries are sentinel rows, not generator definitions. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                  | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 8     | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <i>scope-only</i> ; child objects: M00, M01, M02, M03, M04, M05, L09, M08, L10, BR10, M06, M07 | Modular forms, CM j-invariant targets, and heuristic lattice test matrix            | 7; 6.2/6.3 when class-group/Heegner data are used; M00-M08; L09-L10/BR10 tests | M00-M08 plus L09/L10/BR10: modular action/fundamental domain, Eisenstein series, Weierstrass invariants, modular discriminant, j, q-expansion, Hilbert class polynomial, and Eisenstein/Gauss sentinel test rows. | no current canonical modular-form modules; TomS/post-97 heuristic remains comparison/test material | CNNA/PillarA/Modular/<br>* and Geometry<br>test-matrix profiles                                                                                   | Modular path consumes lattice/number-theory data. It does not gate spectral/physical finite consumers unless an explicit object dependency says so. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.            | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 8.0   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">M00</a> , <a href="#">M01</a>                                                      | Modular coordinate prerequisites: SL2Z action and fundamental-domain representative | 7.3 tau/ upper-half-plane profile; M00-M01; modular path only                  | M00 Modular group action SL2Z; M01 Fundamental domain representative                                                                                                                                              | no current modular-action/fundamental-domain module                                                | CNNA/PillarA/Modular/<br>ModularAction.lean;<br>CNNA/PillarA/Modular/<br>FundamentalDomain.lean<br>or existing modular module if manifest says so | Define modular action/domain as modular-path consumers of tau and integer matrices. They are not prerequisites for finite spectral/operator-algebra consumers. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 8.1   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">M02</a>                                                                            | Eisenstein series profile                                                           | 7.1-7.3; 8.0 when modular-action data are used; M02                            | M02 EisensteinSeriesProfile                                                                                                                                                                                       | no current module                                                                                  | neutral API according to object register for: CMEmbeddingProfile, LatticeBasisProfile                                                             | Formal/modular profile after lattice/tau; analytic/noncomputable series stay mirror/comparison unless truncated operational witnesses are supplied. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.            | <a href="#">blocked</a><br><a href="#">scratch pad</a> |

| Phase | Status | Context                       | class | Doc tier | Linked objects                                                                                                                | Main goal                                                                                                                              | Source objects/<br>dependencies                                                                                                                                      | Definitions/objects                                                                                                                                                                          | Current code modules                                                                                                                                                              | Target modules/<br>objects                                                                                                             | Proof goals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | How imple-<br>mented                    |
|-------|--------|-------------------------------|-------|----------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 8.2   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">M03</a>                                                                                                           | Weierstrass<br>invariants g2/g3                                                                                                        | 8.1;<br>M03;<br>O09/O10 if exact<br>quadratic coordinates<br>are consumed                                                                                            | M03 WeierstrassInvari-<br>antProfile                                                                                                                                                         | no current module                                                                                                                                                                 | neutral API according to<br>object register for:<br>GramProfile, TauProfile,<br>AngleProfile                                           | Derive g2/g3 profile from<br>Eisenstein/lattice data;<br>do not source earlier lattice/<br>number-theory definitions.<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                                                                                                                                                                                                                                                                                                              | <a href="#">blocked<br/>scratch pad</a> |
| 8.3   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">M04</a>                                                                                                           | Modular discriminant<br>Delta_mod                                                                                                      | 8.2;<br>M04                                                                                                                                                          | M04 ModularDiscrimi-<br>nantProfile                                                                                                                                                          | no current module                                                                                                                                                                 | neutral API according to<br>object register for: Lat-<br>ticeAutomorphismProfile                                                       | Disambiguate modular Delta<br>from operational discriminant<br>D_op;<br>this is modular-path data only.<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                                                                                                                                                                                                                                                                                                                            | <a href="#">blocked<br/>scratch pad</a> |
| 8.4   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">M05</a>                                                                                                           | j-invariant profile<br>and special-value<br>comparison targets                                                                         | 8.2-8.3;<br>M05;<br>M08 for sentinel target<br>rows                                                                                                                  | M05 JInvariantProfile;<br>special j targets remain<br>comparison outputs                                                                                                                     | no current j-invariant module                                                                                                                                                     | CNNA/PillarA/Modular/<br>jinvariant.lean                                                                                               | General analytic j is mirror/<br>comparison. Operational<br>outputs allowed only as proved<br>rational/integer special<br>invariants and never as input to<br>generator/discriminant<br>definitions. V1.19 status policy:<br>locked red until a release proof<br>references the prerequisite<br>derived-only Lean objects and<br>turns this phase yellow or<br>orange.                                                                                                                                                                                                                               | <a href="#">blocked<br/>scratch pad</a> |
| 8.5   | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <a href="#">L09</a> , <a href="#">M08</a>                                                                                     | Eisenstein/Gauss<br>sentinel rows inside<br>the general lattice/<br>computation test<br>matrix: static<br>branching and<br>L_max sweep | 4.0.1/O07 generic test<br>harness;<br>4.0.2/O08 regime rows;<br>7.4/L06;<br>8.4/M05/M08;<br>O05;<br>EXEC-2;<br>O09/O10 if exact<br>quadratic coordinates<br>are used | L09 static-branching<br>sentinel extension;<br>M08 CM/j special-value<br>target rows;<br>branch b from code;<br>L_max rows include<br>generic coverage plus<br>Eisenstein/Gauss<br>sentinels | To be determined by Phase 1.1.1.1.1<br>module manifest. Candidate must<br>reuse existing generated substrate/<br>smoke-test generator and existing<br>lattice/CM target wrappers. | A compact test/profile<br>module only if placement<br>audit proves no existing<br>module naturally owns it;<br>no duplicate generator. | Sentinel rows test noOuter/<br>HasOuter hypotheses and<br>heuristic closures;<br>they do not define branching,<br>carrier, angle or generator<br>semantics.                                                                                                                                                                                                                                                                                                                                                                                                                                          | <a href="#">planned<br/>scratch pad</a> |
| 8.6   | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <a href="#">L10</a> , <a href="#">BR10</a>                                                                                    | Dynamic branching<br>and code-defined b<br>lattice test-matrix<br>extension                                                            | 8.5;<br>4.0.1/O07;<br>generated branch-family<br>route;<br>static test matrix green;<br>canonical generator<br>policy                                                | L10 dynamic branching/<br>L_max test sweep;<br>BR10<br>hierarchical-vs-spiral<br>collision and closure<br>audit profile                                                                      | To be determined by module<br>manifest;<br>must consume the one canonical<br>generator and branch policy, never<br>introduce a parallel b-ary tree<br>implementation.             | Same test/profile surface<br>as 8.5 unless semantic<br>placement audit proves a<br>separate dynamic profile<br>module is necessary.    | Extend the generic test matrix<br>to dynamic branching without<br>duplicating generator logic;<br>keep Eisenstein/Gauss sentinel<br>rows separate from generic<br>dynamic coverage.                                                                                                                                                                                                                                                                                                                                                                                                                  | <a href="#">planned<br/>scratch pad</a> |
| 8.7   | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <i>no direct object:</i><br>baseline/audit/<br>closeout/<br>infrastructure<br>phase with no<br>registered object<br>ownership | Heuristic lattice<br>test-matrix closeout<br>and public-boundary<br>audit                                                              | 8.5;<br>8.6;<br>object proof dossiers<br>L09/L10/M08/BR10                                                                                                            | public/non-public test<br>matrix boundary;<br>comparison-only CM/j<br>claims;<br>failure ledger for<br>non-derived or<br>noncomputable tests                                                 | Implemented test/profile modules<br>from 8.5-8.6                                                                                                                                  | No new mathematics;<br>closeout ledger and<br>generated proof-dossier<br>updates.                                                      | Prove or audit that the test<br>matrix is a consumer-only<br>artifact: it consumes<br>code-defined generator, lattice,<br>CM, and j target objects, and<br>cannot feed back into generator<br>semantics or public Pillar-A<br>handoff data. Closeout must<br>explicitly report whether<br>Eisenstein/Gauss passed as<br>noOuter-root sentinels and<br>whether any other Heegner/<br>CM rows remain projection/<br>comparison-only. V1.19 status<br>policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange. | <a href="#">blocked<br/>scratch pad</a> |

| Phase | Status | Context                       | class | Doc tier | Linked objects                                                                                                                          | Main goal                                                                                                 | Source objects/<br>dependencies                                                                                                                          | Definitions/objects                                                                                          | Current code modules                                                                                                                                  | Target modules/<br>objects                                                                                                                | Proof goals                                                                                                                                                                                                                                                                                                                                                  | How imple-<br>mented                                   |
|-------|--------|-------------------------------|-------|----------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 8.8   | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">M06</a> , <a href="#">M07</a>                                                                                               | q-expansion and Hilbert class polynomial modular far targets                                              | 8.4 j profile;<br>6.2 class group/class number;<br>M06-M07;<br>analytic/mirror boundary                                                                  | M06 q-expansion of j;<br>M07 Hilbert class polynomial H_D                                                    | no current q-expansion or Hilbert-class-polynomial module                                                                                             | CNNA/PillarA/Modular/JQExpansion.lean;<br>CNNA/PillarA/Modular/HilbertClassPolynomial.lean or equivalent after manifest                   | Keep q-expansion/H_D in the modular/CM path. They are far targets and must not gate finite spectral/physical consumers. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                                                                   | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 9     | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <i>scope-only</i> ; child objects: L08, N16, N17                                                                                        | Theta, Dirichlet L-functions, and analytic class-number formula far targets                               | 6.4 character;<br>7 lattice/norm data;<br>8 modular/j data only where explicitly used;<br>L08/N16/N17                                                    | L08 theta series;<br>N16 Dirichlet L-function;<br>N17 class-number formula;<br>analytic comparison boundary. | no canonical analytic-number-theory modules                                                                                                           | CNNA/PillarA/Geometry/ThetaSeries.lean;<br>CNNA/PillarA/NumberTheory/LFunction.lean;<br>CNNA/PillarA/NumberTheory/ClassNumberFormula.lean | Analytic far path. It may compare with earlier class-number data but cannot source early h(D), splitting, spectral, or physical definitions.                                                                                                                                                                                                                 | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 9.1   | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <a href="#">L08</a>                                                                                                                     | Theta series of the derived lattice                                                                       | 7.2/7.5;<br>L08;<br>norm/lattice data                                                                                                                    | L08 ThetaSeriesProfile                                                                                       | no current module                                                                                                                                     | neutral API according to object register for: EisensteinSeriesProfile, WeierstrassInvariantProfile                                        | Define theta profile as a lattice/norm consumer; full series may be formal/mirror, finite coefficients can be operational test outputs.                                                                                                                                                                                                                      | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 9.2   | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <a href="#">N16</a>                                                                                                                     | Dirichlet L-function profile for chi_D                                                                    | 6.4/N15;<br>N16;<br>analytic/mirror boundary                                                                                                             | N16 Dirichlet L-function L(s,chi_D)                                                                          | no current module                                                                                                                                     | neutral API according to object register for: ModularDiscriminantProfile, JInvariantProfile                                               | Register analytic L-function only after character; finite truncations may be comparison rows, not source for early arithmetic.                                                                                                                                                                                                                               | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 9.3   | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <a href="#">N17</a>                                                                                                                     | Class-number formula comparison profile                                                                   | 9.2/N16;<br>6.1 units;<br>6.2 class number;<br>5.1 D_fund;<br>N17                                                                                        | N17 ClassNumberFormulaProfile                                                                                | CNNA/PillarA/Finite/StateSpace.lean;<br>CNNA/PillarA/Finite/GibbsStateSeed.lean                                                                       | CNNA/PillarA/Physics/Statefunctional.lean;<br>CNNA/PillarA/Physics/Gibbsstate.lean                                                        | Comparison theorem/profile only. It cannot be used to define h(D) upstream.                                                                                                                                                                                                                                                                                  | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 9.4   | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <i>no direct object</i> : baseline/audit/closeout/infrastructure phase with no registered object ownership                              | Analytic mirror/noncomputable boundary audit for theta and L-functions                                    | 9.1-9.3;<br>LEG15;<br>lean/mathlib version policy                                                                                                        | mirror-only analytic objects;<br>finite truncation allowances;<br>mathlib complication ledger                | CNNA/PillarA/Finite/MatrixExponential.lean;<br>CNNA/PillarA/Finite/UnitaryEvolution.lean;<br>CNNA/PillarA/Finite/DynamicsAdapter.lean;<br>...         | CNNA/PillarA/Physics/Timeevolution.lean;<br>CNNA/PillarA/Physics/Unitaritycertificate.lean                                                | Classify which analytic constructions are operational finite rows, formal mirror profiles, or forbidden operational imports.                                                                                                                                                                                                                                 | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 9.5   | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <i>no direct object</i> : baseline/audit/closeout/infrastructure phase with no registered object ownership                              | Theta/L-function/class-number formula closeout and failure ledger                                         | 9.1-9.4                                                                                                                                                  | analytic far-target closeout;<br>failure/backjump records                                                    | CNNA/PillarA/Finite/ChannelSeed.lean;<br>CNNA/PillarA/Network/SectorChannels.lean                                                                     | CNNA/PillarA/Physics/Openchanneleffectiveopenness.lean                                                                                    | Close or defer analytic targets without blocking Phase 10 finite spectral/physical path.                                                                                                                                                                                                                                                                     | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 10    | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <i>scope-only</i> ; child objects: SP0, SP1, P00, P01, P02, P03, P04, P05, P06, SP2, SP3, SP4, SP5, TM0, TM1, TM2, BR09, BR08, CD8, CD9 | Spectral, finite physical, transfermatrix, and Schur-bridge consumers after operational boundary matrices | 4.1 ExecBoundaryMatrix;<br>3.6/3.7 BoundaryOperator/GeneralizedDtN where needed;<br>regime split for P05/P06;<br>not Phase 8/9 unless explicitly bridged | SP0-SP5, P00-P06, TM0-TM2, BR08/BR09/CD8/CD9. Modular forms and q/H_D are not in this phase.                 | Finite/ExecSpectral/*;<br>Foundation/MatrixAlgebra.lean;<br>Finite/StateSpace.lean;<br>Arithmetic/Schur/MobiusTransfer.lean;<br>finite dynamics seeds | CNNA/PillarA/Spectral/*;<br>CNNA/PillarA/Physics/*;<br>CNNA/PillarA/Transfer/*;<br>late Bridge modules                                    | Independent Path B after Phase 4. Spectral/operator-algebra objects consume finite matrices and regimes, not modular-form/j prerequisites. Transfermatrix consumes Schur recurrence and feeds BR08 later. V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |





| Phase  | Status | Context                       | class | Doc tier | Linked objects                                                                                                                | Main goal                                                                                        | Source objects/<br>dependencies                                                               | Definitions/objects                                                                                      | Current code modules                                                                                                                                                                                                                                                                         | Target modules/<br>objects                                                                                                                                            | Proof goals                                                                                                                                                                                                                                                                                                                                                                              | How imple-<br>mented                    |
|--------|--------|-------------------------------|-------|----------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 10.4.2 | R      | Lean;<br>Derivation;<br>Phase | F1    |          | CD8                                                                                                                           | Arithmetic-CD<br>compatibility as<br>Schur-duality<br>corollary                                  | BR08;<br>BR09;<br>CD3;<br>CD6;<br>CD7;<br>O05;<br>N03/N04/N05 as<br>available;<br>TM0/TM1/TM2 | CD8                                                                                                      | CNNA/PillarA/Arithmetic/<br>CayleyDickson/<br>ArithmeticCDBridge.lean;<br>CNNA/PillarA/Arithmetic/<br>Convergence/<br>CDArithmeticCompatibility.lean;<br>CNNA/PillarA/Arithmetic/<br>Convergence/<br>QuaternionRamanujanInterface.lean<br>(source/seed only, not final canonical<br>theorem) | CNNA/PillarA/Bridge/<br>ArithmeticCayleyDicksonCompatibility.lean                                                                                                     | derive the algebraic norm/CD<br>compatibility only as a<br>consequence of the closed<br>horizontal/vertical Schur<br>bridge;<br>do not assert compatibility<br>from analogy or from parallel<br>source-site similarity V1.19<br>status policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange. | <a href="#">blocked<br/>scratch pad</a> |
| 10.4.3 | W      | Reference;<br>Gate;<br>Phase  | F1    |          | CD9                                                                                                                           | Cayley-Dickson<br>scalar-surface<br>noncircularity and<br>possible C-layer<br>bridge             | CD7, CD8, BR08,<br>BR09, O09, BR13                                                            | CD9                                                                                                      | CayleyDickson structural and<br>arithmetic bridge modules                                                                                                                                                                                                                                    | Later bridge module only<br>after BR08/CD8<br>prerequisites close                                                                                                     | Prove or reject whether a<br>CD-generated complex scalar<br>surface can be used as a<br>C-layer without consuming<br>ExecComplex/analytic mirror<br>or later arithmetic data<br>circularly.                                                                                                                                                                                              | <a href="#">planned<br/>scratch pad</a> |
| 10.5   | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <i>no direct object:</i><br>baseline/audit/<br>closeout/<br>infrastructure<br>phase with no<br>registered object<br>ownership | Spectral/physical<br>and transfer path<br>closeout before late<br>bridges and Pillar-B<br>export | 10.1-10.4;<br>P00-P06;<br>SP0-SP5;<br>TM0-TM2                                                 | path closeout;<br>A->B readiness audit;<br>no modular-form<br>dependencies unless<br>explicitly recorded | no current heat-kernel module;<br>MatrixExponential is related but not<br>HeatKernelProfile;<br>no current spectral-zeta module;<br>...                                                                                                                                                      | CNNA/PillarA/Schur/<br>HeatkernelKt.lean;<br>CNNA/PillarA/Schur/<br>SpectralzetazetaH.lean;<br>CNNA/PillarA/Schur/<br>Spectraldimension.lean;<br>...                  | Check all Path B objects have<br>minimal dependencies and do<br>not import modular/j/<br>L-function modules by<br>convenience. V1.19 status<br>policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                                                                                         | <a href="#">blocked<br/>scratch pad</a> |
| 10.6   | W      | Documentation;<br>Phase       | F3    |          | <i>moved:</i> historical<br>placeholder;<br>object ownership<br>moved to the<br>cited downstream<br>phase                     | Moved to Phase 13:<br>AQFT KMS/GNS/<br>Tomita-Takesaki<br>consumer targets                       | 13.4-13.5;<br>retained only as<br>historical routing note                                     | No active objects here;<br>P07-P11 are owned by<br>Phase 13.                                             | no current KMS proof module;<br>GibbsStateSeed is prerequisite only;<br>no current GNS module;<br>...                                                                                                                                                                                        | CNNA/PillarA/Physics/<br>KMScondition.lean;<br>CNNA/PillarA/Physics/<br>GNSHilbertspace.lean;<br>CNNA/PillarA/Physics/<br>TomitaoperatorS.lean;<br>...                | Prevent duplicate Phase 10<br>ownership of Pillar-B AQFT<br>objects;<br>Phase 10 only prepares finite<br>A-side seeds.                                                                                                                                                                                                                                                                   | <a href="#">planned<br/>scratch pad</a> |
| 10.7   | W      | Documentation;<br>Phase       | F3    |          | <i>moved:</i> historical<br>placeholder;<br>object ownership<br>moved to the<br>cited downstream<br>phase                     | Moved to Phase 13:<br>CPT/spin-statistics/<br>Kramers/split/<br>local-net consumer<br>targets    | 13.3-13.6;<br>retained only as<br>historical routing note                                     | No active objects here;<br>P12-P16 are owned by<br>Phase 13.                                             | no current BW/CPT module;<br>no current spin-statistics module;<br>no current Kramers module;<br>...                                                                                                                                                                                         | CNNA/PillarA/Physics/<br>BisognanoWichmannCPTbridge.lean;<br>CNNA/PillarA/Physics/<br>Spinstatistics.lean;<br>CNNA/PillarA/Physics/<br>Kramersdegeneracy.lean;<br>... | Prevent duplicate Phase 10<br>ownership of Pillar-B AQFT<br>objects;<br>Phase 13 owns B-consumer<br>roadmap.                                                                                                                                                                                                                                                                             | <a href="#">planned<br/>scratch pad</a> |
| 11     | R      | Lean;<br>Derivation;<br>Phase | F0    |          | <i>scope-only;</i> child<br>objects: H10,<br>H11, H12, H13,<br>H14                                                            | handoff bridges from<br>canonical A-objects                                                      | 2.3;<br>relevant outputs                                                                      | A_to_B/C/D/E,<br>A_to_CD;<br>input constraints                                                           | Handoff/Inputs/*.lean;<br>Handoff/Outputs/*.lean;<br>Handoff/CayleyDicksonAdapter.lean                                                                                                                                                                                                       | neutral API according to<br>object register for:<br>A_to_B/C/D/E,<br>A_to_CD;<br>input constraints                                                                    | Outputs only for consumers;<br>A itself forbidden-to-A V1.19<br>status policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                                                                                                                                                                 | <a href="#">blocked<br/>scratch pad</a> |
| 11.1   | R      | Lean;<br>Derivation;<br>Phase | F0    |          | H10                                                                                                                           | Derive/register<br>object group: A->B<br>bright handoff                                          | 11                                                                                            | A->B bright handoff                                                                                      | CNNA/PillarA/Handoff/Outputs/<br>A_to_B.lean;<br>CNNA/PillarA/Handoff/Core/<br>Step1MathData.lean;<br>CNNA/PillarA/Handoff/Core/<br>SectorExport.lean                                                                                                                                        | CNNA/PillarA/Handoff/<br>Outputs/A_to_B.lean                                                                                                                          | establish the listed object(s) in<br>natural dependency order;<br>split further if proof<br>dependencies are too deep<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                                                                                                  | <a href="#">blocked<br/>scratch pad</a> |



| Phase | Status | Context class                 | Doc tier | Linked objects                                                                                                                 | Main goal                                                                                                                             | Source objects/<br>dependencies                                                         | Definitions/objects                                                                                                                                                                                                                           | Current code modules                         | Target modules/<br>objects                                                                    | Proof goals                                                                                                                                                                                                                                                                             | How imple-<br>mented                                   |
|-------|--------|-------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 11.2  | R      | Lean;<br>Derivation;<br>Phase | F0       | <a href="#">H11</a>                                                                                                            | Derive/register<br>object group: A->B<br>prime/dark future<br>handoff                                                                 | 11                                                                                      | A->B prime/dark future<br>handoff                                                                                                                                                                                                             | no current dark/B-prime handoff<br>module    | CNNA/PillarA/Handoff/<br>ABprimedarkfuturehandoff.lean                                        | establish the listed object(s) in<br>natural dependency order;<br>split further if proof<br>dependencies are too deep<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 11.3  | R      | Lean;<br>Derivation;<br>Phase | F0       | <a href="#">H12</a>                                                                                                            | Derive/register<br>object group: A->C<br>handoff                                                                                      | 11                                                                                      | A->C handoff                                                                                                                                                                                                                                  | CNNA/PillarA/Handoff/Outputs/<br>A_to_C.lean | CNNA/PillarA/Handoff/<br>AChandoff.lean                                                       | establish the listed object(s) in<br>natural dependency order;<br>split further if proof<br>dependencies are too deep<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 11.4  | R      | Lean;<br>Derivation;<br>Phase | F0       | <a href="#">H13</a>                                                                                                            | Derive/register<br>object group: A->D<br>handoff                                                                                      | 11                                                                                      | A->D handoff                                                                                                                                                                                                                                  | CNNA/PillarA/Handoff/Outputs/<br>A_to_D.lean | CNNA/PillarA/Handoff/<br>ADhandoff.lean                                                       | establish the listed object(s) in<br>natural dependency order;<br>split further if proof<br>dependencies are too deep<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 11.5  | R      | Lean;<br>Derivation;<br>Phase | F0       | <a href="#">H14</a>                                                                                                            | Derive/register<br>object group: A->E<br>handoff                                                                                      | 11                                                                                      | A->E handoff                                                                                                                                                                                                                                  | CNNA/PillarA/Handoff/Outputs/<br>A_to_E.lean | CNNA/PillarA/Handoff/<br>AEhandoff.lean                                                       | establish the listed object(s) in<br>natural dependency order;<br>split further if proof<br>dependencies are too deep<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 12    | W      | Reference;<br>Gate;<br>Phase  | F2       | <i>scope-only</i> ; child<br>objects:<br>MOON0,<br>MOON1,<br>MOON2,<br>MOON3,<br>MOON4,<br>MOON5,<br>MOON6,<br>MOON7,<br>MOON8 | Established<br>mathematics beyond<br>j: finite simple<br>groups, Leech/VOA,<br>Monster/Moonshine<br>and post-j arithmetic<br>register | 9;<br>modular forms and<br>j-invariant objects;<br>no Monster input into<br>foundations | Moonshine comparison<br>objects;<br>finite-simple-group<br>comparison context;<br>Leech lattice/theta<br>comparison;<br>VOA/moonshine module<br>comparison;<br>McKay-Thompson<br>placeholders;<br>Monster representation<br>comparison policy | no current Lean module in SM9h<br>codebase   | CNNA/PillarA/<br>Arithmetic/Moonshine/<br>*.lean;<br>comparison-only, after j/<br>q-expansion | comparison-only unless CNNA<br>has already derived j/<br>q-expansion and the required<br>graded structure;<br>no Monster, Leech, VOA,<br>L-function, or string-theory<br>input into Pillar A foundation                                                                                 | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 12.1  | R      | Lean;<br>Derivation;<br>Phase | F2       | <a href="#">MOON0</a>                                                                                                          | q-expansion<br>coefficient stream for<br>j as established<br>arithmetic output                                                        | 9.2;<br>10.3;<br>q-expansion object M06                                                 | j(q) coefficient stream;<br>coefficient provenance;<br>Exec/mirror boundary                                                                                                                                                                   | no current Lean module in SM9h<br>codebase   | CNNA/PillarA/<br>Arithmetic/Moonshine/<br>JCoefficientStream.lean                             | define coefficient stream only<br>after j/q-expansion is derived;<br>no coefficient table as source<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                   | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 12.2  | W      | Reference;<br>Gate;<br>Phase  | F2       | <a href="#">MOON1</a>                                                                                                          | Moonshine<br>comparison profile                                                                                                       | 12.1;<br>established j coefficient<br>stream                                            | MoonshineComparisonProfile;<br>MonsterDimensionComparison                                                                                                                                                                                     | no current Lean module in SM9h<br>codebase   | CNNA/PillarA/<br>Arithmetic/Moonshine/<br>MoonshineComparisonProfile.lean                     | compare CNNA-derived j<br>coefficients to established<br>Monster dimensions without<br>importing Monster data into<br>earlier phases                                                                                                                                                    | <a href="#">planned</a><br><a href="#">scratch pad</a> |

| Phase | Status | Context                       | class | Doc tier | Linked objects                                                                                                                                        | Main goal                                                                | Source objects/<br>dependencies                                           | Definitions/objects                                                                      | Current code modules                                                                                      | Target modules/<br>objects                                                   | Proof goals                                                                                                                                                                                                                                                                                                                   | How imple-<br>mented                    |
|-------|--------|-------------------------------|-------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 12.3  | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">MOON2</a>                                                                                                                                 | McKay-Thompson<br>side profile                                           | 12.2;<br>modular function side<br>targets                                 | McKayThompsonComparisonProfile                                                           | no current Lean module in<br>codebase                                                                     | CNNA/PillarA/<br>Arithmetic/Moonshine/<br>McKayThompsonComparisonProfile     | register comparison-only<br>targets;<br>no effect on generator/<br>provenance chain V1.19 status<br>policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                                                                         | <a href="#">blocked<br/>scratch pad</a> |
| 12.4  | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <a href="#">MOON3</a> ,<br><a href="#">MOON4</a>                                                                                                      | Finite simple group<br>comparison context<br>after modular/j layer       | 12.1-12.3;<br>external CFSG<br>references only as<br>comparison targets   | FiniteSimpleGroupComparisonProfile                                                       | no current Lean module                                                                                    | CNNA/PillarA/<br>Arithmetic/Moonshine/<br>FiniteSimpleGroupComparisonProfile | register CFSG only as<br>taxonomy for comparison;<br>no classification theorem<br>imported as A source                                                                                                                                                                                                                        | <a href="#">planned<br/>scratch pad</a> |
| 12.5  | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <a href="#">MOON5</a>                                                                                                                                 | Leech lattice<br>comparison profile                                      | 12.4;<br>CNNA-derived graded/<br>lattice-like output if<br>ever available | LeechLatticeComparisonProfile                                                            | no current Lean module                                                                                    | CNNA/PillarA/<br>Arithmetic/Moonshine/<br>LeechLatticeComparisonProfile      | compare only after CNNA<br>generates a lattice/theta-like<br>invariant;<br>do not inject 24-dimensional<br>Leech data upstream                                                                                                                                                                                                | <a href="#">planned<br/>scratch pad</a> |
| 12.6  | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <a href="#">MOON6</a>                                                                                                                                 | Leech theta/shell<br>and kissing-number<br>comparison stream             | 12.5;<br>derived theta/<br>q-character stream if<br>available             | LeechThetaComparisonProfile;<br>ShellCoefficientComparisonProfile                        | no current Lean module                                                                                    | CNNA/PillarA/<br>Arithmetic/Moonshine/<br>LeechThetaComparisonProfile        | compare derived q/theta<br>coefficients to established Leech<br>shell data;<br>comparison-only, no coefficient<br>oracle                                                                                                                                                                                                      | <a href="#">planned<br/>scratch pad</a> |
| 12.7  | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <a href="#">MOON7</a>                                                                                                                                 | Vertex-operator-<br>algebra /<br>moonshine-module<br>comparison boundary | 12.1-12.6;<br>B/AQFT-like graded<br>algebra if later derived              | VOAComparisonProfile;<br>MoonshineModuleComparisonProfile;<br>ChiralCFTComparisonProfile | no current Lean module                                                                                    | CNNA/PillarA/<br>Arithmetic/Moonshine/<br>VOAComparisonProfile               | VOA is a comparison/consumer<br>target unless CNNA first<br>derives the required graded<br>algebraic operations;<br>no string-theory or VOA axiom<br>import                                                                                                                                                                   | <a href="#">planned<br/>scratch pad</a> |
| 12.8  | R      | Lean;<br>Derivation;<br>Phase | F2    |          | <a href="#">MOON8</a>                                                                                                                                 | Common-indicator /<br>Hauptmodul<br>bridge-test register                 | 12.1-12.7;<br>CNNA-derived<br>q-character stream                          | CommonIndicatorComparisonProfile;<br>HauptmodulBridgeTestProfile                         | no current Lean module                                                                                    | CNNA/PillarA/<br>Arithmetic/Moonshine/<br>CommonIndicatorBridgeTestProfile   | test whether derived<br>q-characters match known<br>Hauptmodul/<br>McKay-Thompson-type series;<br>failure is diagnostic, not<br>obstruction to earlier CNNA<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange. | <a href="#">blocked<br/>scratch pad</a> |
| 13    | R      | Lean;<br>Derivation;<br>Phase | F0    |          | <i>scope-only</i> ; child<br>objects: B00,<br>B01, P16, B02,<br>P07, P08, B03,<br>P09, P10, P11,<br>B04, P12, P13,<br>P14, P15, B05,<br>B06, GA4, GA5 | Pillar B QFT/AQFT<br>consumer roadmap<br>from A outputs                  | 11.1;<br>A->B bright handoff;<br>no direct access to A<br>internals       | B input contract;<br>QFT/AQFT algebraic<br>consumer objects                              | CNNA/PillarB/BuildAll.lean<br>currently empty namespace                                                   | CNNA/PillarB/QFT/<br>*.lean;<br>CNNA/PillarB/AQFT/<br>*.lean                 | derive QFT/AQFT only from<br>explicit A->B handoff objects;<br>no independent B generator<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                                                                   | <a href="#">blocked<br/>scratch pad</a> |
| 13.1  | R      | Lean;<br>Derivation;<br>Phase | F0    |          | <a href="#">B00</a>                                                                                                                                   | B input contract<br>from A bright<br>boundary handoff                    | 11.1;<br>H10;<br>A->B handoff rights                                      | BInputFromA;<br>BrightBoundaryInputProfile                                               | CNNA/PillarB/BuildAll.lean<br>currently empty;<br>CNNA/PillarA/Handoff/Outputs/<br>A_to_B.lean            | CNNA/PillarB/Handoff/<br>BInputFromA.lean                                    | prove B consumes exactly<br>exported A->B data and not A<br>internals V1.19 status policy:<br>locked red until a release proof<br>references the prerequisite<br>derived-only Lean objects and<br>turns this phase yellow or<br>orange.                                                                                       | <a href="#">blocked<br/>scratch pad</a> |
| 13.2  | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">B01</a>                                                                                                                                   | Boundary<br>star-algebra /<br>observable algebra                         | 13.1;<br>ExecBoundaryMatrix;<br>BoundaryOperator                          | finite boundary<br>*-algebra;<br>observable algebra<br>profile                           | CNNA/PillarA/Finite/<br>StateSpace.lean;<br>CNNA/PillarA/Finite/<br>ExecSpectral/*.lean;<br>PillarB empty | CNNA/PillarB/QFT/<br>BoundaryStarAlgebra.lean                                | construct algebra from A<br>exported finite boundary data;<br>prove closure/star/unit<br>properties V1.19 status policy:<br>locked red until a release proof<br>references the prerequisite<br>derived-only Lean objects and<br>turns this phase yellow or<br>orange.                                                         | <a href="#">blocked<br/>scratch pad</a> |

| Phase  | Status | Context                           | class | Doc tier | Linked objects                                                                                                 | Main goal                                                                                             | Source objects/<br>dependencies                                                                                                                                                                     | Definitions/objects                                                          | Current code modules                                                                     | Target modules/<br>objects                                                               | Proof goals                                                                                                                                                                                                                                                                                                                  | How imple-<br>mented                    |
|--------|--------|-----------------------------------|-------|----------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 13.2.1 | W      | Reference;<br>Gate;<br>Phase      | F2    |          | <a href="#">GA4</a>                                                                                            | Pauli/Clifford<br>boundary-star-<br>algebra comparison<br>gate                                        | 13.2;<br>A->B bright boundary<br>handoff;<br>finite boundary<br>*-algebra;<br>GA0-GA3 as reference<br>targets only;<br>mathlib CliffordAlgebra<br>documentation;<br>no direct Pillar-A<br>internals | GA4                                                                          | PillarB empty;<br>no current comparison module                                           | GA4;<br>optional future CNNA/<br>PillarB/QFT/<br>CliffordComparisonGate.lean             | After B derives the boundary<br>observable *-algebra, check<br>whether CNNA-derived<br>generators satisfy Pauli/<br>Clifford-like square and<br>anti-commutation target<br>patterns. This is a consistency/<br>comparison gate only, not a<br>source of B generators or a<br>shortcut to spin/QM semantics.                  | <a href="#">planned<br/>scratch pad</a> |
| 13.3   | R      | Lean;<br>Derivation;<br>Phase     | F1    |          | <a href="#">P16</a> , <a href="#">B02</a>                                                                      | Local net and<br>Haag-Kastler<br>algebraic profile                                                    | 13.2;<br>region/localization data<br>from A->B                                                                                                                                                      | LocalNet;<br>inclusion/isotony;<br>locality where finite data<br>supports it | PillarB empty;<br>legacy REALOQS not imported                                            | CNNA/PillarB/AQFT/<br>LocalNetFromBoundary.lean                                          | derive local net as consumer<br>theorem;<br>separate algebraic HK from<br>geometric/spacetime axioms<br>reserved for C/D V1.19 status<br>policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                                   | <a href="#">blocked<br/>scratch pad</a> |
| 13.4   | R      | Lean;<br>Derivation;<br>Phase     | F1    |          | <a href="#">P07</a> , <a href="#">P08</a> , <a href="#">B03</a>                                                | State, GNS, and<br>KMS consumer layer                                                                 | 13.2;<br>Gibbs/state seeds;<br>time evolution when<br>exported                                                                                                                                      | state functional;<br>GNS pre-Hilbert/Hilbert<br>profile;<br>KMS condition    | CNNA/PillarA/Finite/<br>GibbsStateSeed.lean;<br>MatrixExponential.lean;<br>PillarB empty | CNNA/PillarB/QFT/<br>StateGNSKMS.lean                                                    | build finite state/GNS/KMS<br>mirror from exported operator<br>data;<br>mark analytic completions as<br>mirror/far-target V1.19 status<br>policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                                  | <a href="#">blocked<br/>scratch pad</a> |
| 13.5   | R      | Lean;<br>Derivation;<br>Phase     | F1    |          | <a href="#">P09</a> , <a href="#">P10</a> , <a href="#">P11</a> ,<br><a href="#">B04</a>                       | Tomita-Takesaki and<br>modular data                                                                   | 13.4;<br>KMS/GNS profile                                                                                                                                                                            | Tomita S;<br>modular operator Delta;<br>modular conjugation J                | PillarB empty                                                                            | CNNA/PillarB/AQFT/<br>ModularTheory.lean                                                 | derive modular objects only<br>after state/GNS prerequisites;<br>no free modular conjugation<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                                                               | <a href="#">blocked<br/>scratch pad</a> |
| 13.6   | W      | Tool;<br>Infrastructure;<br>Phase | F1    |          | <a href="#">P12</a> , <a href="#">P13</a> , <a href="#">P14</a> ,<br><a href="#">P15</a> , <a href="#">B05</a> | CPT, spin-statistics,<br>Kramers and<br>split-property gates                                          | 13.5;<br>local net;<br>modular data                                                                                                                                                                 | CPT/BW bridge;<br>spin-statistics;<br>Kramers;<br>split property             | PillarB empty                                                                            | CNNA/PillarB/AQFT/<br>StructuralTheorems.lean                                            | register theorem gates;<br>prove only when hypotheses are<br>derived from B consumer chain<br>and A handoff                                                                                                                                                                                                                  | <a href="#">planned<br/>scratch pad</a> |
| 13.6.1 | W      | Reference;<br>Gate;<br>Phase      | F2    |          | <a href="#">GA5</a>                                                                                            | Dirac/<br>spacetime-algebra<br>comparison target<br>behind CPT,<br>spin-statistics and<br>split gates | 13.6;<br>GA4;<br>local net/modular data<br>if later derived;<br>Horn Dirac/quaternion<br>notes;<br>Hestenes<br>spacetime-algebra<br>references;<br>comparison-only                                  | GA5                                                                          | PillarB empty;<br>no current STA/Dirac comparison<br>module                              | GA5;<br>optional future CNNA/<br>PillarB/AQFT/<br>SpacetimeAlgebraComparisonGate.lean    | Once Pillar B has its own local<br>net, modular data and<br>structural-theorem hypotheses,<br>compare against Dirac/STA<br>targets such as gamma/Clifford<br>signatures, Lorentz-covariance<br>formulations and<br>spin-statistics/CPT<br>compatibility. No Dirac<br>equation or Lorentz structure is<br>imported into CNNA. | <a href="#">planned<br/>scratch pad</a> |
| 13.7   | R      | Lean;<br>Derivation;<br>Phase     | F1    |          | <a href="#">B06</a>                                                                                            | B->A input handoff<br>preparation                                                                     | 13.1-13.6;<br>explicit backreaction<br>policy                                                                                                                                                       | BToAAllowedInputProfile                                                      | CNNA/PillarA/Handoff/Inputs/<br>B_to_A.lean                                              | CNNA/PillarB/Handoff/<br>BToAOutput.lean;<br>CNNA/PillarA/Handoff/<br>Inputs/B_to_A.lean | B may export only explicitly<br>typed results back to A;<br>A must not consume A->B<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                                                                        | <a href="#">blocked<br/>scratch pad</a> |

| Phase | Status | Context                       | class | Doc tier | Linked objects                                                      | Main goal                                                                            | Source objects/<br>dependencies                                                       | Definitions/objects                                              | Current code modules                                                                | Target modules/<br>objects                                                               | Proof goals                                                                                                                                                                                                                                                                                             | How imple-<br>mented                                   |
|-------|--------|-------------------------------|-------|----------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 14    | R      | Lean;<br>Derivation;<br>Phase | F0    |          | <i>scope-only</i> ; child<br>objects: C00,<br>C01, C02, C03,<br>C04 | Pillar C OQS<br>consumer roadmap<br>from A regimes and<br>B-compatible<br>interfaces | 11.3;<br>HasOuter regime;<br>C input handoff;<br>optional B state/<br>channel outputs | C input contract;<br>open-system channels;<br>effective openness | CNNA/PillarC/BuildAll.lean<br>currently empty;<br>A finite channel seeds exist      | CNNA/PillarC/OQS/<br>*.lean                                                              | derive OQS from explicit A->C<br>handoff and exported finite<br>channel data;<br>no stochastic ontology injected<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                      | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 14.1  | R      | Lean;<br>Derivation;<br>Phase | F0    |          | <a href="#">C00</a>                                                 | C input contract<br>from A regime and<br>complement profile                          | 11.3;<br>RegimeProfile;<br>ComplementProfile                                          | CInputFromA;<br>OpenSubsystemRequest                             | CNNA/PillarC/BuildAll.lean<br>currently empty;<br>CNNA/PillarA/Sectors/*.lean       | CNNA/PillarC/Handoff/<br>CInputFromA.lean                                                | prove C sees only exported<br>open-system profiles;<br>dark complement remains A<br>provenance unless explicitly<br>exported V1.19 status policy:<br>locked red until a release proof<br>references the prerequisite<br>derived-only Lean objects and<br>turns this phase yellow or<br>orange.          | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 14.2  | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">C01</a>                                                 | Finite channel /<br>CP-map profile                                                   | 14.1;<br>open channel seed;<br>ExecComplex matrix<br>path                             | CPMapProfile;<br>Kraus/Stinespring finite<br>mirror if derived   | CNNA/PillarA/Finite/<br>ChannelSeed.lean;<br>DynamicsAdapter.lean;<br>PillarC empty | CNNA/PillarC/OQS/<br>FiniteChannelProfile.lean                                           | derive complete positivity/trace<br>behavior from finite exported<br>data, not from assumed noise<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                                     | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 14.3  | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">C02</a>                                                 | Effective dynamics<br>and Lindblad/GKSL<br>gate                                      | 14.2;<br>time evolution/channel<br>semigroup data                                     | EffectiveGenerator;<br>Lindblad/GKSL gate                        | PillarC empty                                                                       | CNNA/PillarC/OQS/<br>EffectiveGenerator.lean                                             | separate finite operational<br>generator from analytic<br>semigroup mirror;<br>no continuous-time assumption<br>without derived limit V1.19<br>status policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 14.4  | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">C03</a>                                                 | Decoherence/noise<br>observables as<br>derived effective<br>quantities               | 14.2-14.3;<br>HasOuter regime                                                         | DecoherenceProfile;<br>NoiseKernelProfile                        | PillarC empty                                                                       | CNNA/PillarC/OQS/<br>DecoherenceProfile.lean                                             | derive effective noise from<br>environment/complement<br>profile;<br>no external stochastic source<br>V1.19 status policy: locked red<br>until a release proof references<br>the prerequisite derived-only<br>Lean objects and turns this<br>phase yellow or orange.                                    | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 14.5  | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">C04</a>                                                 | C->A input handoff<br>preparation                                                    | 14.1-14.4;<br>explicit backreaction<br>policy                                         | CToAAllowedInputProfile                                          | CNNA/PillarA/Handoff/Inputs/<br>C_to_A.lean                                         | CNNA/PillarC/Handoff/<br>CToAOutput.lean;<br>CNNA/PillarA/Handoff/<br>Inputs/C_to_A.lean | C may export only explicit<br>OQS-derived profiles back to A<br>through C->A V1.19 status<br>policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                                                          | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 15    | R      | Lean;<br>Derivation;<br>Phase | F0    |          | <i>scope-only</i> ; child<br>objects: D00,<br>D01, D02, D03         | Pillar D spacetime/<br>covariance roadmap                                            | 11.4;<br>A geometry/CM/lattice<br>outputs;<br>C OQS when available                    | spacetime/localization/<br>covariance consumer<br>objects        | CNNA/PillarD/BuildAll.lean<br>currently empty                                       | CNNA/PillarD/<br>Spacetime/*.lean                                                        | derive spacetime structures<br>only from A/C outputs;<br>no GR/spacetime continuum<br>postulated as input V1.19<br>status policy: locked red until a<br>release proof references the<br>prerequisite derived-only Lean<br>objects and turns this phase<br>yellow or orange.                             | <a href="#">blocked</a><br><a href="#">scratch pad</a> |

| Phase | Status | Context                       | class | Doc tier                                              | Linked objects                                         | Main goal                                                       | Source objects/<br>dependencies                        | Definitions/objects                     | Current code modules           | Target modules/<br>objects                                                    | Proof goals                                                                                                                                                                                                                                                     | How imple-<br>mented                |
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| 15.1  | R      | Lean;<br>Derivation;<br>Phase | F0    | <a href="#">D00</a>                                   | D input contract from A/C geometry exports             | 11.4; CM/lattice/region profiles; optional C channel geometry   | DInputFromA; GeometryExportProfile                     | CNNA/PillarD/BuildAll.lean              | currently empty                | CNNA/PillarD/Handoff/DInputFromA.lean                                         | prove D consumes only exported geometry/regime profiles V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                                      | <a href="#">blocked scratch pad</a> |
| 15.2  | R      | Lean;<br>Derivation;<br>Phase | F1    | <a href="#">D01</a>                                   | Region, localization, and causal-order candidates      | 15.1; region/localization data                                  | RegionPoset; LocalizationProfile; CausalOrderCandidate | PillarD empty                           |                                | CNNA/PillarD/Spacetime/RegionCausalOrder.lean                                 | derive causal/localization order from CNNA relational/geometric data; no continuum assumption V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                | <a href="#">blocked scratch pad</a> |
| 15.3  | R      | Lean;<br>Derivation;<br>Phase | F1    | <a href="#">D02</a>                                   | Metric/<br>coarse-geometry and covariance gates        | 15.2; spectral/geometry profiles                                | MetricCandidate; CoarseGeometryProfile; CovarianceGate | PillarD empty                           |                                | CNNA/PillarD/Spacetime/MetricCovariance.lean                                  | prove metric/covariance only when induced by derived geometry/spectral data V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                  | <a href="#">blocked scratch pad</a> |
| 15.4  | R      | Lean;<br>Derivation;<br>Phase | F1    | <a href="#">D03</a>                                   | D->A input handoff preparation                         | 15.1-15.3; explicit backreaction policy                         | DToAAllowedInputProfile                                | CNNA/PillarA/Handoff/Inputs/D_to_A.lean |                                | CNNA/PillarD/Handoff/DToAOutput.lean; CNNA/PillarA/Handoff/Inputs/D_to_A.lean | D may export only explicit geometry/covariance results back to A through D->A V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                | <a href="#">blocked scratch pad</a> |
| 16    | R      | Lean;<br>Derivation;<br>Phase | F0    | <i>scope-only</i> ; child objects: E00, E01, E02, E03 | Pillar E matter and phenomenology roadmap              | 11.5; B/C/D outputs as available; A arithmetic/symmetry outputs | matter/sector/observable consumer objects              | CNNA/PillarE/BuildAll.lean              | currently empty                | CNNA/PillarE/Matter/*.lean                                                    | derive matter-sector candidates from established CNNA outputs; no Standard Model data injected as foundation V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. | <a href="#">blocked scratch pad</a> |
| 16.1  | R      | Lean;<br>Derivation;<br>Phase | F0    | <a href="#">E00</a>                                   | E input contract from A/B/C/D outputs                  | 11.5; B/C/D outputs when available                              | EInputProfile; MatterRequestProfile                    | CNNA/PillarE/BuildAll.lean              | currently empty                | CNNA/PillarE/Handoff/EInputProfile.lean                                       | prove E consumes only permitted handoff outputs V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                                              | <a href="#">blocked scratch pad</a> |
| 16.2  | R      | Lean;<br>Derivation;<br>Phase | F1    | <a href="#">E01</a>                                   | Matter sector and internal symmetry candidates         | 16.1; arithmetic/CD/unit/symmetry outputs                       | MatterSectorCandidate; InternalSymmetryCandidate       | PillarE empty                           |                                | CNNA/PillarE/Matter/SectorSymmetry.lean                                       | derive candidates from CNNA symmetry/arithmetic data; compare to established physics later V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                   | <a href="#">blocked scratch pad</a> |
| 16.3  | W      | Reference;<br>Gate;<br>Phase  | F3    | <a href="#">E02</a>                                   | Observable coupling and phenomenology comparison layer | 16.2; B/C/D observable structures                               | ObservableCouplingProfile                              | PillarE empty                           | PhenomenologyComparisonProfile | CNNA/PillarE/Matter/ObservableComparison.lean                                 | separate derived observables from empirical comparison targets                                                                                                                                                                                                  | <a href="#">planned scratch pad</a> |



| Phase | Status | Context                       | class | Doc tier | Linked objects                                                     | Main goal                                                 | Source objects/<br>dependencies                                           | Definitions/objects                                                       | Current code modules                                                  | Target modules/<br>objects                                                               | Proof goals                                                                                                                                                                                                                                      | How imple-<br>mented                                   |
|-------|--------|-------------------------------|-------|----------|--------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 16.4  | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">E03</a>                                                | E->A input handoff preparation                            | 16.1-16.3;<br>explicit backreaction policy                                | EToAAllowedInputProfile                                                   | CNNA/PillarA/Handoff/Inputs/<br>E_to_A.lean                           | CNNA/PillarE/Handoff/<br>EToAOutput.lean;<br>CNNA/PillarA/Handoff/<br>Inputs/E_to_A.lean | E may export only explicit matter/phenomenology-derived data back to A through E->A V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.           | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 17    | R      | Lean;<br>Derivation;<br>Phase | F0    |          | <a href="#">BR00</a>                                               | Backreaction roadmap through explicit input handoffs only | 13.7;<br>14.5;<br>15.4;<br>16.4;<br>A post-emergence rules                | BackreactionInputDAG;<br>emergence switch;<br>no self-handoff consumption | CNNA/PillarA/Handoff/Inputs/<br>*.lean;<br>ProofObligationIII modules | CNNA/PillarA/<br>Backreaction/*.lean                                                     | derive backreaction only as typed inputs B->A/C->A/<br>D->A/E->A;<br>never from A own outputs V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 17.1  | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">BR01</a>                                               | B->A backreaction gate                                    | 13.7;<br>BToAAllowedInputProfile                                          | BToABackreactionInputGate                                                 | CNNA/PillarA/Handoff/Inputs/<br>B_to_A.lean                           | CNNA/PillarA/<br>Backreaction/<br>BInputGate.lean                                        | verify A consumes only B->A data and only after B emergence V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                   | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 17.2  | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">BR02</a>                                               | C->A OQS backreaction gate                                | 14.5;<br>CToAAllowedInputProfile                                          | CToABackreactionInputGate                                                 | CNNA/PillarA/Handoff/Inputs/<br>C_to_A.lean                           | CNNA/PillarA/<br>Backreaction/<br>CInputGate.lean                                        | verify open-system effects return only through typed C->A data V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 17.3  | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">BR03</a>                                               | D->A spacetime backreaction gate                          | 15.4;<br>DToAAllowedInputProfile                                          | DToABackreactionInputGate                                                 | CNNA/PillarA/Handoff/Inputs/<br>D_to_A.lean                           | CNNA/PillarA/<br>Backreaction/<br>DInputGate.lean                                        | verify geometry/covariance effects return only through typed D->A data V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                        | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 17.4  | R      | Lean;<br>Derivation;<br>Phase | F1    |          | <a href="#">BR04</a>                                               | E->A matter backreaction gate                             | 16.4;<br>EToAAllowedInputProfile                                          | EToABackreactionInputGate                                                 | CNNA/PillarA/Handoff/Inputs/<br>E_to_A.lean                           | CNNA/PillarA/<br>Backreaction/<br>EInputGate.lean                                        | verify matter/phenomenology effects return only through typed E->A data V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                       | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 17.5  | R      | Lean;<br>Derivation;<br>Phase | F0    |          | <a href="#">BR05</a>                                               | Pre-/post-emergence regime switch for A statements        | 17.1-17.4;<br>emergence criteria                                          | EmergenceSwitchProfile;<br>PostEmergenceAInterpretation                   | no current neutral module;<br>unilateral governance in handoff plans  | CNNA/PillarA/<br>Backreaction/<br>EmergenceSwitch.lean                                   | make irreversible regime changes explicit and non-circular V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange.                                    | <a href="#">blocked</a><br><a href="#">scratch pad</a> |
| 18    | W      | Reference;<br>Gate;<br>Phase  | F2    |          | <i>scope-only</i> ; child objects: SIDE0, SIDE1, SIDE2, SIDE3, GA6 | Side-comparison register beyond the CNNA focus            | 12;<br>13-17 outputs if available;<br>external frameworks not foundations | comparison-only side targets: ToE, strings, LQG, noncommutative geometry  | no current Lean module                                                | CNNA/Comparisons/<br>*.lean                                                              | register only as possible consumers of CNNA outputs; never as assumptions                                                                                                                                                                        | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 18.1  | W      | Reference;<br>Gate;<br>Phase  | F4    |          | <a href="#">SIDE0</a>                                              | Full ToE comparison placeholder                           | 17 outputs if ever sufficiently complete                                  | FullToEComparisonProfile                                                  | no current Lean module                                                | CNNA/Comparisons/<br>FullToE.lean                                                        | comparison-only;<br>no effect on core derivation                                                                                                                                                                                                 | <a href="#">planned</a><br><a href="#">scratch pad</a> |



| Phase |   | Status                       | Context class | Doc tier                                                                      | Linked objects                                                                                | Main goal                                                                                     | Source objects/<br>dependencies                                                                                                                        | Definitions/objects                                                                                                                                                   | Current code modules                                                                                                                                | Target modules/<br>objects                                                                                                                                                        | Proof goals                                                                                                                                                                                                                                                                                          | How imple-<br>mented                    |
|-------|---|------------------------------|---------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 18.2  | W | Reference;<br>Gate;<br>Phase | F4            | <a href="#">SIDE1</a>                                                         | String-theory<br>comparison<br>placeholder                                                    | String-theory<br>comparison<br>placeholder                                                    | D/E outputs if ever<br>available                                                                                                                       | StringTheoryComparisonProfile                                                                                                                                         | No current Lean module                                                                                                                              | CNNA/Comparisons/<br>StringTheory.lean                                                                                                                                            | comparison-only;<br>no string data in CNNA<br>foundation                                                                                                                                                                                                                                             | <a href="#">planned<br/>scratch pad</a> |
| 18.3  | W | Reference;<br>Gate;<br>Phase | F4            | <a href="#">SIDE2</a>                                                         | Loop-quantum-<br>gravity comparison<br>placeholder                                            | Loop-quantum-<br>gravity comparison<br>placeholder                                            | D spacetime outputs if<br>ever available                                                                                                               | LoopQuantumGravityComparisonProfile                                                                                                                                   | No current Lean module                                                                                                                              | CNNA/Comparisons/<br>LoopQuantumGravity.lean                                                                                                                                      | comparison-only;<br>no LQG input to CNNA<br>foundation                                                                                                                                                                                                                                               | <a href="#">planned<br/>scratch pad</a> |
| 18.4  | W | Reference;<br>Gate;<br>Phase | F4            | <a href="#">SIDE3</a>                                                         | Noncommutative-<br>geometry comparison<br>placeholder                                         | Noncommutative-<br>geometry comparison<br>placeholder                                         | B/D algebra/geometry<br>outputs if ever available                                                                                                      | NoncommutativeGeometryComparisonProfile                                                                                                                               | No current Lean module                                                                                                                              | CNNA/Comparisons/<br>NoncommutativeGeometry.lean                                                                                                                                  | comparison-only;<br>no NCG axiom import into<br>CNNA foundation                                                                                                                                                                                                                                      | <a href="#">planned<br/>scratch pad</a> |
| 18.5  | W | Reference;<br>Gate;<br>Phase | F2            | <a href="#">GA6</a>                                                           | Geometric and<br>conformal algebra<br>side-comparison<br>register                             | Geometric and<br>conformal algebra<br>side-comparison<br>register                             | 18;<br>GA0-GA5 if available;<br>Horn outlook;<br>Hestenes/Sobczyk and<br>GA formalization<br>references;<br>external frameworks are<br>not foundations | GA6                                                                                                                                                                   | No current Lean module                                                                                                                              | GA6;<br>optional future CNNA/<br>Comparisons/<br>GeometricAlgebra.lean                                                                                                            | Register geometric algebra,<br>spacetime algebra and<br>conformal geometric algebra as<br>possible later comparison<br>surfaces for CNNA outputs.<br>They may organize questions<br>and target gates but cannot set<br>dimension, metric, dynamics,<br>matter, spacetime or algebraic<br>primitives. | <a href="#">planned<br/>scratch pad</a> |
| 19    | W | Reference;<br>Gate;<br>Phase | F3            | <i>scope-only</i> ; child<br>objects: LEG16,<br>LEG17, LEG18,<br>LEG19, LEG26 | Secondary long-term<br>meta-documentation<br>and<br>derivation-geometry<br>roadmap            | Secondary long-term<br>meta-documentation<br>and<br>derivation-geometry<br>roadmap            | Generated repo<br>inventory;<br>proof dossiers;<br>object register;<br>Phase 1.x API cleanup;<br>later A/B/C/D/E<br>checked outputs                    | Semi-automated<br>existing-code<br>documentation;<br>derivation-graph<br>geometry;<br>emergence-path<br>forensics;<br>reflective CNNA/<br>formalization<br>comparison | repo_inventory/raw_export/*.json;<br>generated figures;<br>module inventory;<br>proofbook dossiers                                                  | No Lean operational<br>modules initially;<br>generated documentation/<br>analysis views only;<br>possible future<br>MetaAnalysis/*<br>non-operational files after<br>core closure | Register secondary long-term<br>goals without changing active<br>implementation order. These<br>goals are diagnostic/<br>comparison/documentation<br>targets and may not feed<br>CNNA-generative assumptions.                                                                                        | <a href="#">planned<br/>scratch pad</a> |
| 19.1  | W | Reference;<br>Gate;<br>Phase | F3            | <a href="#">LEG16</a>                                                         | Semi-automated<br>existing-code<br>documentation inside<br>the YAML/<br>proofbook tool        | Semi-automated<br>existing-code<br>documentation inside<br>the YAML/<br>proofbook tool        | Phase 1.1.1.1.4<br>declaration<br>consumption<br>classification;<br>object proof dossiers;<br>module inventory;<br>symbol reference<br>summary         | Existing-code<br>documentation coverage<br>ledger;<br>module-to-object<br>proof-dossier links;<br>undocumented<br>declaration-family queue                            | repo_inventory/raw_export/<br>modules.json;<br>symbol_reference_summary.json;<br>object_proof_dossiers;<br>implementation_scratchpad                | LEG16;<br>generated existing-code<br>documentation coverage<br>view;<br>no new Lean module<br>unless later justified by<br>manifest gate                                          | Ensure Bestandscode<br>documentation can be planned<br>and tracked without relying on<br>post-hoc interpretation. Every<br>relevant code artifact should<br>eventually have a consumer/<br>provenance/dossier<br>classification.                                                                     | <a href="#">planned<br/>scratch pad</a> |
| 19.2  | W | Reference;<br>Gate;<br>Phase | F3            | <a href="#">LEG17</a>                                                         | Derivation-graph<br>geometry and<br>bottleneck analysis of<br>the formalized theory           | Derivation-graph<br>geometry and<br>bottleneck analysis of<br>the formalized theory           | import_edges.json;<br>dominator analysis;<br>layer decomposition;<br>source-sink paths;<br>module-group graph;<br>mainline forensics                   | Graph-geometry<br>diagnostics: dominators,<br>weak mainline edges,<br>discipline clusters,<br>bottleneck/cut<br>candidates, optional<br>Cheeger-style indicators      | repo_inventory/raw_export/<br>import_edges.json;<br>figures/<br>module_group_dependency_graph.pdf;<br>figures/<br>core_robust_semantic_mainline.pdf | LEG17;<br>generated meta-analysis<br>views only at first                                                                                                                          | Study the geometry of the<br>derivation network as a<br>secondary diagnostic layer after<br>core derivations stabilize. No<br>graph metric may become an<br>ontic CNNA input.                                                                                                                        | <a href="#">planned<br/>scratch pad</a> |
| 19.3  | W | Reference;<br>Gate;<br>Phase | F3            | <a href="#">LEG18</a>                                                         | Formal<br>emergence-path<br>forensics for<br>open-system<br>dynamics                          | Formal<br>emergence-path<br>forensics for<br>open-system<br>dynamics                          | A regime/complement<br>objects;<br>DtN/Schur handoff<br>chain;<br>A->C interface;<br>C/OQS consumer path;<br>graph source-sink/<br>dominator data      | Open-system<br>emergence-path<br>localization;<br>boundary/complement-<br>to-effective-dynamics<br>path diagnostics;<br>proof-bottleneck map                          | future checked A/C modules plus<br>repo_inventory/raw_export path<br>analysis                                                                       | LEG18;<br>future diagnostic report<br>only;<br>no operational CNNA<br>object generated from<br>graph metadata                                                                     | Later localize where open<br>effective dynamics emerges in<br>the formal derivation chain,<br>while requiring actual CNNA<br>proofs for any physical/<br>mathematical emergence<br>statement.                                                                                                        | <a href="#">planned<br/>scratch pad</a> |
| 19.4  | W | Reference;<br>Gate;<br>Phase | F3            | <a href="#">LEG19</a>                                                         | Reflective<br>comparison of CNNA<br>object-structure with<br>formalization graph<br>structure | Reflective<br>comparison of CNNA<br>object-structure with<br>formalization graph<br>structure | Core CNNA objects and<br>stable public API;<br>discipline subgraphs;<br>object proof dossiers;<br>graph metrics                                        | Meta-level comparison<br>notes between theory<br>layering/boundaries/<br>operators/signal flow<br>and repository graph<br>layering/cuts/paths/<br>clusters            | generated graph views and proofbook<br>links;<br>no operational Lean code                                                                           | LEG19;<br>optional future<br>non-operational<br>MetaAnalysis report/view                                                                                                          | Record self-similarity or<br>structural resonance only as a<br>far-target observation. It is<br>never a premise for the CNNA<br>derivation chain.                                                                                                                                                    | <a href="#">planned<br/>scratch pad</a> |

| Phase | Status | Context                     | class | Doc tier | Linked objects        | Main goal                                                                                                                                             | Source objects/<br>dependencies                                                                                                                           | Definitions/objects                                                                                                                                                                                                                                                      | Current code modules                                                                                                                         | Target modules/<br>objects                                                                                                                                                                                                       | Proof goals                                                                                                                                                                                                                           | How imple-<br>mented                    |
|-------|--------|-----------------------------|-------|----------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 19.5  | W      | Documentation; F4 Phase     |       |          | <a href="#">LEG26</a> | Thread/<br>full-boundary<br>derivation-geometry<br>and string-analogy<br>diagnostic side-road                                                         | S07;<br>S08;<br>BR06;<br>BR07;<br>S05;<br>S06;<br>LT5;<br>proof dossiers and<br>derivation graph<br>metadata.                                             | LEG26 Reflective<br>thread/full-boundary<br>analogy diagnostic<br>boundary.                                                                                                                                                                                              | repo_inventory/raw_export/*;<br>future neutral ToC/FullCut/Schur<br>route profiles.                                                          | Documentation/diagnostic<br>generated views only<br>unless a later derived-only<br>theorem needs a code<br>module.                                                                                                               | Document analogy boundaries<br>and graph diagnostics after<br>object closure. No generative<br>input, no premature physical<br>theorem, no bypass of<br>derived-only route.                                                           | <a href="#">planned<br/>scratch pad</a> |
| 20    | G      | Documentation; F1 Phase     |       |          | <a href="#">DBR0</a>  | DB-REVIEW main<br>side objective:<br>database-ready<br>CNNA review and<br>documentation<br>structure with<br>immediate<br>expert-review<br>principles | Single YAML master;<br>generated repo<br>inventory;<br>status legends;<br>object/proof dossiers;<br>GNG23-GNG24;<br>workflow_policy<br>DB-REVIEW row      | DBR0 database-ready<br>review/documentation<br>structure;<br>child phases 20.1-20.5;<br>immediate epistemic rule<br>that Lean is primary but<br>not semantically<br>infallible and generated<br>views are secondary;<br>DBR6 context<br>documentation slice<br>generator | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>generated_exports/*.tsv;<br>tables/*.tex;<br>generated PDF           | DBR0;<br>DBR1-DBR5;<br>future database schema/<br>export scripts;<br>discipline-review matrix;<br>blue/reference-theory<br>audit ledger;<br>interactive documentation<br>views;<br>DBR6 context-filtered<br>documentation slices | Governance-closed side<br>objective. Technical database<br>implementation is not claimed;<br>the immediate principle is<br>closed: no artifact is infallible,<br>Lean is primary truth system,<br>expert review remains<br>mandatory. | <a href="#">implemented<br/>record</a>  |
| 20.1  | G      | Documentation; F1 Phase     |       |          | <a href="#">DBR1</a>  | Immediate epistemic<br>expert-review rule<br>and no-infallibility<br>maxim                                                                            | 20;<br>user instruction;<br>Lean-primary rule;<br>generated-secondary<br>rule;<br>GNG23;<br>GNG24                                                         | DBR1 expert-review and<br>no-infallibility rule:<br>green Lean builds and<br>generated artifacts<br>remain semantically<br>reviewable;<br>all relevant claims<br>require specialized<br>expert review                                                                    | masters/<br>cnna_roadmap_master.yaml;<br>global_no_go_guardrails;<br>workflow_policy;<br>latest_change_overview;<br>generated tables/PDF     | DBR1;<br>GNG23;<br>GNG24;<br>DB-REVIEW workflow<br>row;<br>review/audit status legend<br>rows                                                                                                                                    | Closed as governance/tool rule.<br>No Lean source changed. The<br>rule constrains all future<br>documentation and phase/<br>object interpretation<br>immediately.                                                                     | <a href="#">implemented<br/>record</a>  |
| 20.2  | W      | Tool; Infrastructure; Phase | F1    |          | <a href="#">DBR2</a>  | Master-YAML<br>database schema<br>preparation and<br>round-trip-safe<br>identifiers                                                                   | 20;<br>DBR0;<br>YAML phase/object/<br>proof-dossier schemas;<br>generated TSV exports;<br>module inventory                                                | DBR2 database-ready<br>schema target: stable<br>IDs, foreign-key-like<br>phase/object/proof<br>links, review status<br>fields, theory/reference<br>markers, generated-view<br>provenance                                                                                 | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>generated_exports/*.tsv                                              | future SQL/SQLite/graph<br>schema export;<br>schema consistency checks;<br>round-trip report                                                                                                                                     | Future tooling phase. Prove/<br>verify that exported schema is<br>secondary documentation/<br>review metadata and<br>round-trips without changing<br>Lean/YAML truth semantics.                                                       | <a href="#">planned<br/>scratch pad</a> |
| 20.3  | W      | Documentation; F3 Phase     |       |          | <a href="#">DBR3</a>  | Discipline-separated<br>expert-review matrix                                                                                                          | 20.1;<br>DBR1;<br>object proof dossiers;<br>local theory context<br>ledger;<br>end-theory comparison<br>ledger;<br>blue/reference markings                | DBR3 review matrix<br>target: claims/objects/<br>phases linked to required<br>expert disciplines and<br>review evidence/status                                                                                                                                           | masters/<br>cnna_roadmap_master.yaml;<br>object_proof_dossiers;<br>workflow_policy                                                           | discipline review matrix<br>TSV/TeX/JSON;<br>review-required ledger;<br>expert-review evidence<br>slots                                                                                                                          | Future documentation/audit<br>phase. It must make review<br>obligations visible without<br>declaring final semantic truth<br>or importing external theory as<br>generator input.                                                      | <a href="#">planned<br/>scratch pad</a> |
| 20.4  | W      | Documentation; F3 Phase     |       |          | <a href="#">DBR4</a>  | Blue/<br>reference-theory<br>marking audit<br>backlog                                                                                                 | 20.1;<br>GNG24;<br>local theory context vs<br>end-theory comparison<br>workflow;<br>existing blue object<br>register;<br>TomS/source<br>cross-check table | DBR4 blue/reference<br>audit ledger: identify<br>established-theory/<br>reference material that is<br>missing explicit blue/<br>comparison marking                                                                                                                       | masters/<br>cnna_roadmap_master.yaml;<br>status_legend;<br>external_source_crosscheck;<br>consumption_classification_map;<br>object_register | blue-audit-needed ledger;<br>reference-theory marking<br>checklist;<br>discipline-specific review<br>queue                                                                                                                       | Future audit phase. Missing<br>blue marks remain audit debt;<br>blue means comparison/<br>reference/review horizon only,<br>not CNNA-generative input.                                                                                | <a href="#">planned<br/>scratch pad</a> |
| 20.5  | W      | Tool; Infrastructure; Phase | F1    |          | <a href="#">DBR5</a>  | Interactive/online<br>documentation and<br>database-backed<br>exploration views                                                                       | 20.2;<br>20.3;<br>20.4;<br>DBR2-DBR4;<br>generated TSV/JSON/<br>DOT/PDF exports                                                                           | DBR5 interactive<br>documentation target:<br>queryable phase/object/<br>module/DAG/review<br>views for online display<br>and collaborative expert<br>review                                                                                                              | generated_exports/*.tsv;<br>figures/*.dot/pdf;<br>tables/*.tex;<br>future database/export scripts                                            | interactive documentation<br>JSON;<br>graph/database views;<br>online review dashboards;<br>discipline-filtered query<br>surfaces                                                                                                | Future tooling phase.<br>Interactive views must remain<br>generated-secondary and may<br>not bypass Lean-primary<br>semantics or expert review<br>obligations.                                                                        | <a href="#">planned<br/>scratch pad</a> |

| Phase | Status | Context class                    | Doc tier | Linked objects        | Main goal                                                                                      | Source objects/<br>dependencies                                                                                                                                                                                                    | Definitions/objects                                                                                                                                                                                                                        | Current code modules                                                                                                                                                                                                                                                         | Target modules/<br>objects                                                                                                                                     | Proof goals                                                                                                                                                                                                                                                                                    | How imple-<br>mented               |
|-------|--------|----------------------------------|----------|-----------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 20.6  | G      | Generated;<br>Artifact;<br>Phase | F3       | <a href="#">DBR6</a>  | Context-filtered documentation slices: all sections retained, only query-relevant rows emitted | 20;<br>DBR0-DBR5;<br>Single YAML master; generated_exports/*.tsv;<br>context_documentation_fullfile; scripts/generate_context_doc.py                                                                                               | DBR6 context documentation slice generator: generated-secondary packet that preserves the full documentation section topology while filtering each section to context-relevant rows by query/profile.                                      | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_context_doc.py;<br>generated_exports/*.tsv;<br>context_docs/<slug>/*                                                                                                                                               | DBR6;<br>context_filter_manifest.json;<br>context_section_summary.tsv;<br><slug>_context_documentation.tex;<br>curated profile dtn-matrix                      | Closed as tooling/documentation function. It must not create a second truth source, every slice is a filtered view over YAML/export data and remains subject to Lean-primary and expert-review rules.                                                                                          | <a href="#">implemented record</a> |
| 20.7  | G      | Generated;<br>Artifact;<br>Phase | F3       | <a href="#">DBR7</a>  | Stats front section, tool-interface guiding metaphor, and project licensing notices            | 20;<br>DBR0-DBR6;<br>Single YAML master; repo_inventory/raw_export/*.json; generated_exports/*.tsv;<br>module inventory and derivation graphs                                                                                      | DBR7 documentation-front surface: generated project facts, embedded module inventory/ graphs, concise tool-interface metaphor, and documentation/code license notice.                                                                      | masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>tables/stats.tex;<br>tables/project_stats_table.tex;<br>tables/module_inventory.tex;<br>README.md                                                                                                    | DBR7;<br>front Stats section; guiding_metaphors; license_notice;<br>project_stats.generated.tsv                                                                | Closed as documentation/tooling update. The Stats section must remain generated-secondary and must visibly connect repository facts to the module inventory without becoming a second truth source. The metaphor is operational only, and licensing notices do not certify third-party rights. | <a href="#">implemented record</a> |
| 20.8  | G      | Generated;<br>Artifact;<br>Phase | F3       | <a href="#">DBR8</a>  | Root artifact retention and generated-output cleanup policy                                    | 20;<br>DBR0-DBR7;<br>GNG23-GNG27;<br>Single YAML master; generated tables/exports;<br>root file audit of V1.12 package                                                                                                             | DBR8 root artifact retention policy: source-oriented root, archived historical reports, disposable generated PDF/TeX/LaTeX outputs, and explicit regeneration commands.                                                                    | Repository root; README.md;<br>scripts/generate_full_document.py;<br>scripts/clean_generated_root_artifacts.py;<br>archive/reports/;<br>generated_docs/full/ (disposable output path)                                                                                        | DBR8;<br>GNG27;<br>SCF46;<br>cleanup script;<br>full-document generator script;<br>archived root reports;<br>root without retained PDF/TeX/build artifacts     | Closed as tooling/governance cleanup. Generated documentation remains reproducible from YAML/tables/scripts but is not retained in the root as active source. Markdown reports are archival, not authoritative.                                                                                | <a href="#">implemented record</a> |
| 20.9  | G      | Generated;<br>Artifact;<br>Phase | F3       | <a href="#">DBR9</a>  | Current README and workflow-orientation manual                                                 | 20;<br>DBR0-DBR8;<br>Single YAML master; generated tables/exports;<br>context-documentation generator;<br>root cleanup policy;<br>user requirement that tested/developed workflow references must name ChatGPT rather than Claude. | DBR9 current README surface: compact non-chronological project manual explaining truth status, manual documentation generation, context slices, ChatGPT-assisted workflows, scalable workflow extension, repository layout, and licensing. | README.md;<br>masters/<br>cnna_roadmap_master.yaml;<br>scripts/generate_tables.py;<br>scripts/generate_full_document.py;<br>scripts/generate_context_doc.py;<br>scripts/clean_generated_root_artifacts.py                                                                    | DBR9;<br>README.md;<br>V1_14_README_WORKFLOW_ORIENTATION_REPORT.md;<br>latest_change_overview update                                                           | Closed as documentation/tooling update. The README WORKFLOW_ORIENTATION_REPORT.md; current tool surface without presenting generated documentation as truth and without claiming that the workflow was tested/developed with Claude.                                                           | <a href="#">implemented record</a> |
| 20.10 | G      | Generated;<br>Artifact;<br>Phase | F3       | <a href="#">DBR10</a> | Root Lean/Lake project-control files and build-entry retention policy                          | 20;<br>DBR0-DBR9;<br>DBR8 root artifact policy;<br>package-root CNNA.lean, lakefile.lean, lake-manifest.json, lean-toolchain;<br>repo_snapshot/CNNA source tree.                                                                   | DBR10 package-root Lean toolchain entry surface: Lake/Lean project-control files retained above Repository/ while generated documentation artifacts remain disposable.                                                                     | CNNA.lean;<br>lakefile.lean;<br>lake-manifest.json;<br>lean-toolchain;<br>Repository/scripts/clean_generated_root_artifacts.py;<br>README.md;<br>Repository/masters/<br>cnna_roadmap_master.yaml                                                                             | DBR10;<br>GNG28;<br>SCF47;<br>root Lake build entry files; amended cleanup policy;<br>V1_15_ROOT_LEAN_TOOLCHAIN_PDF_TSV_REPORT                                 | Closed as tooling/build-surface correction. Lean/Lake project-control files are source/control inputs and must be retained in the package root; generated PDF/TSV REPORT products and historical reports remain non-root secondary artifacts.                                                  | <a href="#">implemented record</a> |
| 20.11 | G      | Generated;<br>Artifact;<br>Phase | F3       | <a href="#">DBR11</a> | Root license, single-root README, credits, and assistant-requirement documentation             | 20;<br>DBR0-DBR10;<br>root README;<br>license_notice table;<br>full/context documentation generators;<br>user requirement that README exists only in root and credits appear also in generated PDFs.                               | DBR11 license and credit surface: dual-license root file, README license notices, single-root README invariant, PDF title/author credits, and minimum tested ChatGPT setup statement.                                                      | README.md;<br>LICENSE;<br>Repository/scripts/generate_full_document.py;<br>Repository/scripts/generate_context_doc.py;<br>Repository/scripts/generate_tables.py;<br>Repository/scripts/clean_generated_root_artifacts.py;<br>Repository/masters/<br>cnna_roadmap_master.yaml | DBR11;<br>GNG29;<br>SCF48;<br>root LICENSE;<br>root-only README;<br>PDF author credits; updated cleanup policy;<br>V1_17_LICENSE_ROOT_README_CREDITS_REPORT.md | Closed as documentation/tooling update. Licensing and credits must be visible at the package root and in generated PDFs while avoiding duplicate active README files inside subdirectories.                                                                                                    | <a href="#">implemented record</a> |

| Phase Status |   | Context class                    | Doc tier | Linked objects        | Main goal                                                                                  | Source objects/<br>dependencies                                                                                                                                                                                                                                                                       | Definitions/objects                                                                                                                                                                                                                                                                             | Current code modules                                                                                                                                                                                                | Target modules/<br>objects                                                                                                                                                                   | Proof goals                                                                                                                                                                                                                                                                                                                                                                                 | How imple-<br>mented                |
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| 20.12        | G | Generated;<br>Artifact;<br>Phase | F3       | <a href="#">DBR12</a> | Section-pagebreak layout for generated full and context PDFs                               | 20;<br>DBR0-DBR11;<br>full/context documentation generators;<br>user requirement that each documentation section automatically starts on a new page.                                                                                                                                                  | DBR12<br>section-pagebreak PDF layout invariant:<br>generated full and context PDF top-level sections start on fresh pages while preserving generated-secondary truth status.                                                                                                                   | Repository/scripts/<br>generate_full_document.py;<br>Repository/scripts/<br>generate_context_doc.py;<br>README.md;<br>Repository/masters/<br>cnna_roadmap_master.yaml                                               | DBR12;<br>GNG30;<br>SCF49;<br>V1_18_SECTION_PAGEBREAKS_DOCUMENT_GENERATORS<br>updated generated PDFs/<br>context docs                                                                        | Closed as documentation/<br>layout tooling update. The full-document and BREAKS_DOCUMENT_GENERATORS must automatically insert a page boundary before each top-level section so review packets are easier to navigate and section scopes are visually separated.                                                                                                                             | <a href="#">implemented record</a>  |
| 20.13        | G | Generated;<br>Artifact;<br>Phase | F3       | <a href="#">DBR13</a> | Phase status release semantics:<br>red-by-default for unreleased<br>Lean-derivation phases | 20;<br>DBR0-DBR12;<br>status_legend;<br>phase table;<br>implementation scratchpad;<br>pre-implementation check workflow;<br>user requirement that unreleased Lean-code derivation phases are red and white is reserved for non-Lean/<br>tooling/refactor/<br>documentation/<br>infrastructure phases. | DBR13 phase status release policy: a<br>Lean-derivation phase is green if closed, yellow if uniquely active and implementable now, orange if implementable after listed predecessor closure, red if not yet released, and white only if not a direct Lean-code derivation implementation phase. | Repository/masters/<br>cnna_roadmap_master.yaml;<br>Repository/scripts/<br>generate_tables.py;<br>README.md;<br>generated phase/status tables                                                                       | DBR13;<br>GNG31;<br>SCF50;<br>converted W-derivation phases to R;<br>updated validation so derivation phases may not remain white;<br>V1_19_PHASE_STATUS_DERIVATION_POLICY_REPORT.md         | Closed as planning/governance update. The phase table must no longer allow unreleased direct Lean-derivation phases to remain white. Release requires explicit prerequisite/object evidence and a pre-implementation check before new policy status.                                                                                                                                        | <a href="#">implemented record</a>  |
| 20.14        | G | Generated;<br>Artifact;<br>Phase | F3       | <a href="#">DBR14</a> | Static section explanations and module-change manifest appendix semantics                  | 20;<br>DBR0-DBR13;<br>generated full/context documentation;<br>user request to explain sections/tables and reduce module-manifest confusion.                                                                                                                                                          | DBR14 documentation explanation layer and module-change appendix policy: static source/<br>purpose prose around dynamic tables, and module manifest demoted to technical appendix for actual Lean module/API changes only.                                                                      | README.md;<br>Repository/scripts/<br>generate_full_document.py;<br>Repository/scripts/<br>generate_context_doc.py;<br>Repository/scripts/<br>generate_tables.py;<br>Repository/masters/<br>cnna_roadmap_master.yaml | DBR14;<br>GNG32;<br>SCF51;<br>V1_20_DOC_SECTION_EXPLANATIONS_RANDS_MODULE_MANIFEST_APPENDIX_REPORT.md<br>updated full/context docs                                                           | Closed as documentation/<br>tooling update. The main phase table remains the EXPLANATIONS_RANDS_MODULE_MANIFEST_APPENDIX_REPORT module-change manifest is an appendix and not a second phase system. Static explanatory prose is generated by scripts, while table rows remain dynamic.                                                                                                     | <a href="#">implemented record</a>  |
| 20.15        | W | Database;<br>Mirror;<br>Phase    | F1       | <a href="#">DBR15</a> | Supabase/Postgres mirror feasibility and schema design                                     | 20;<br>SLTG-06<br>database-ready CNNA review/documentation structure;<br>DBR0-DBR14<br>generated-secondary tooling;<br>Supabase/Postgres public read-only mirror concept;<br>current Master YAML and generated exports.                                                                               | DBR15: internal/api/<br>review/admin schema split;<br>minimal tables for db_snapshots, phases, objects, proof_dossiers, guardrails, workflows, modules, declarations, import_edges, phase_object_links, context_profiles, static_findings and reports.                                          | Repository/masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future database/import scripts not yet implemented                                             | DBR15;<br>future Repository/<br>database/supabase migrations/importers/<br>frontend contract artifacts;<br>no Lean modules;<br>consumed/refined by concrete activation phases<br>20.22-20.26 | Planned non-blocking DB/<br>tooling phase. Schema design only;<br>no live DB connection and no data import yet. Supabase/Postgres may mirror and publish CNNA planning metadata, but it may not release phases or supply CNNA truth. V1.24 audit: this remains a broad architecture contract;<br>concrete live-project activation is routed through 20.22-20.26 to avoid duplicating scope. | <a href="#">planned scratch pad</a> |
| 20.16        | W | Database;<br>Mirror;<br>Phase    | F1       | <a href="#">DBR16</a> | YAML/exports to Postgres import and roundtrip contract                                     | 20;<br>SLTG-06<br>database-ready CNNA review/documentation structure;<br>DBR0-DBR14<br>generated-secondary tooling;<br>Supabase/Postgres public read-only mirror concept;<br>current Master YAML and generated exports.                                                                               | DBR16: Importer must be local/admin-run with .env secrets, produce import report/checksums, and never require a service_role key in chat or browser.                                                                                                                                            | Repository/masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future database/import scripts not yet implemented                                             | DBR16;<br>future Repository/<br>database/supabase migrations/importers/<br>frontend contract artifacts;<br>no Lean modules;<br>consumed/refined by concrete activation phases<br>20.22-20.26 | Planned non-blocking DB/<br>tooling phase. Roundtrip contract only;<br>no live DB write yet. Supabase/Postgres may mirror and publish CNNA planning metadata, but it may not release phases or supply CNNA truth. V1.24 audit: this remains a broad architecture contract;<br>concrete live-project activation is routed through 20.22-20.26 to avoid duplicating scope.                    | <a href="#">planned scratch pad</a> |



| Phase | Status | Context                       | class | Doc tier              | Linked objects                                         | Main goal                                                                                                                                                                                                                                 | Source objects/<br>dependencies                                                                                                                                                                                                           | Definitions/objects                                                                                                                                                                                                                                 | Current code modules                                                                                                                                                       | Target modules/<br>objects                                                                                                                                                                         | Proof goals                                                                                                                                                                                                                                                                                                                                                                                                                                            | How imple-<br>mented                                   |
|-------|--------|-------------------------------|-------|-----------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 20.17 | W      | Database;<br>Mirror;<br>Phase | F1    | <a href="#">DBR17</a> | Append-only<br>snapshot and<br>checksum model          | 20;<br>SLTG-06<br>database-ready CNNA<br>review/documentation<br>structure;<br>DBR0-DBR14<br>generated-secondary<br>tooling;<br>Supabase/Postgres<br>public read-only mirror<br>concept;<br>current Master YAML<br>and generated exports. | 20;<br>SLTG-06<br>database-ready CNNA<br>review/documentation<br>structure;<br>DBR0-DBR14<br>generated-secondary<br>tooling;<br>Supabase/Postgres<br>public read-only mirror<br>concept;<br>current Master YAML<br>and generated exports. | DBR17: db_snapshots<br>with snapshot_id,<br>tool_version,<br>lean_data_timestamp,<br>yaml_hash,<br>repo_inventory_hash,<br>created_at, created_by,<br>source_zip_name and<br>optional<br>import_report_hash.                                        | Repository/masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future database/import scripts not<br>yet implemented | DBR17;<br>future Repository/<br>database/supabase<br>migrations/importers/<br>frontend contract artifacts;<br>no Lean modules;<br>consumed/refined by<br>concrete activation phases<br>20.22-20.26 | Planned non-blocking DB/<br>tooling phase. Snapshot model<br>only;<br>no live DB write yet.<br>Supabase/Postgres may mirror<br>and publish CNNA planning<br>metadata, but it may not<br>release phases or supply CNNA<br>truth. V1.24 audit: this<br>remains a broad architecture<br>contract;<br>concrete live-project activation<br>is routed through 20.22-20.26<br>to avoid duplicating scope.                                                     | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 20.18 | W      | Database;<br>Mirror;<br>Phase | F1    | <a href="#">DBR18</a> | Public read-only API<br>views and RLS<br>policies      | 20;<br>SLTG-06<br>database-ready CNNA<br>review/documentation<br>structure;<br>DBR0-DBR14<br>generated-secondary<br>tooling;<br>Supabase/Postgres<br>public read-only mirror<br>concept;<br>current Master YAML<br>and generated exports. | 20;<br>SLTG-06<br>database-ready CNNA<br>review/documentation<br>structure;<br>DBR0-DBR14<br>generated-secondary<br>tooling;<br>Supabase/Postgres<br>public read-only mirror<br>concept;<br>current Master YAML<br>and generated exports. | DBR18: anon gets<br>SELECT on api.* views<br>only;<br>no anon INSERT/<br>UPDATE/DELETE;<br>internal.* and admin/<br>import.* remain closed;<br>service_role never<br>appears in frontend/<br>browser.                                               | Repository/masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future database/import scripts not<br>yet implemented | DBR18;<br>future Repository/<br>database/supabase<br>migrations/importers/<br>frontend contract artifacts;<br>no Lean modules;<br>consumed/refined by<br>concrete activation phases<br>20.22-20.26 | Planned non-blocking DB/<br>tooling phase. RLS/API policy<br>design only;<br>no Supabase project secrets<br>stored in this package.<br>Supabase/Postgres may mirror<br>and publish CNNA planning<br>metadata, but it may not<br>release phases or supply CNNA<br>truth. V1.24 audit: this<br>remains a broad architecture<br>contract;<br>concrete live-project activation<br>is routed through 20.22-20.26<br>to avoid duplicating scope.             | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 20.19 | W      | Database;<br>Mirror;<br>Phase | F1    | <a href="#">DBR19</a> | cnna-project.net<br>web-frontend data<br>contract      | 20;<br>SLTG-06<br>database-ready CNNA<br>review/documentation<br>structure;<br>DBR0-DBR14<br>generated-secondary<br>tooling;<br>Supabase/Postgres<br>public read-only mirror<br>concept;<br>current Master YAML<br>and generated exports. | 20;<br>SLTG-06<br>database-ready CNNA<br>review/documentation<br>structure;<br>DBR0-DBR14<br>generated-secondary<br>tooling;<br>Supabase/Postgres<br>public read-only mirror<br>concept;<br>current Master YAML<br>and generated exports. | DBR19: Frontend can<br>query public views with<br>anon/publishable key<br>only and render phase<br>map, object register,<br>red/orange release<br>reasons, import/module<br>graph, proof dossiers,<br>blue-debt dashboard and<br>HTML context docs. | Repository/masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future database/import scripts not<br>yet implemented | DBR19;<br>future Repository/<br>database/supabase<br>migrations/importers/<br>frontend contract artifacts;<br>no Lean modules;<br>consumed/refined by<br>concrete activation phases<br>20.22-20.26 | Planned non-blocking DB/<br>tooling phase. Frontend<br>contract only;<br>no deployed site code yet.<br>Supabase/Postgres may mirror<br>and publish CNNA planning<br>metadata, but it may not<br>release phases or supply CNNA<br>truth. V1.24 audit: this<br>remains a broad architecture<br>contract;<br>concrete live-project activation<br>is routed through 20.22-20.26<br>to avoid duplicating scope.                                             | <a href="#">planned</a><br><a href="#">scratch pad</a> |
| 20.20 | W      | Database;<br>Mirror;<br>Phase | F1    | <a href="#">DBR20</a> | Reviewer annotation<br>and future<br>admin-write model | 20;<br>SLTG-06<br>database-ready CNNA<br>review/documentation<br>structure;<br>DBR0-DBR14<br>generated-secondary<br>tooling;<br>Supabase/Postgres<br>public read-only mirror<br>concept;<br>current Master YAML<br>and generated exports. | 20;<br>SLTG-06<br>database-ready CNNA<br>review/documentation<br>structure;<br>DBR0-DBR14<br>generated-secondary<br>tooling;<br>Supabase/Postgres<br>public read-only mirror<br>concept;<br>current Master YAML<br>and generated exports. | DBR20: review.* for<br>later authenticated<br>comments/annotations;<br>admin/import.* for local<br>importer or future<br>signed Edge Function;<br>no public writes and no<br>manual table-click<br>update as standard<br>workflow.                  | Repository/masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future database/import scripts not<br>yet implemented | DBR20;<br>future Repository/<br>database/supabase<br>migrations/importers/<br>frontend contract artifacts;<br>no Lean modules;<br>consumed/refined by<br>concrete activation phases<br>20.22-20.26 | Planned non-blocking DB/<br>tooling phase. Future role<br>model only;<br>not active until read-only<br>mirror and snapshot contract<br>are stable. Supabase/Postgres<br>may mirror and publish CNNA<br>planning metadata, but it may<br>not release phases or supply<br>CNNA truth. V1.24 audit: this<br>remains a broad architecture<br>contract;<br>concrete live-project activation<br>is routed through 20.22-20.26<br>to avoid duplicating scope. | <a href="#">planned</a><br><a href="#">scratch pad</a> |

| Phase Status |   | Context class                    | Doc tier | Linked objects        | Main goal                                                                               | Source objects/<br>dependencies                                                                                                                                                                               | Definitions/objects                                                                                                                                                                                                                                 | Current code modules                                                                                                                                                                                                                                                       | Target modules/<br>objects                                                                                                                                      | Proof goals                                                                                                                                                                                                                                                                           | How imple-<br>mented                    |
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| 20.21        | G | Generated;<br>Artifact;<br>Phase | F3       | <a href="#">DBR21</a> | Generated PDF<br>typography scale<br>increase                                           | 20;<br>DBR12<br>section-pagebreak<br>layout invariant;<br>full/context<br>documentation<br>generators;<br>table generator;<br>user request to make all<br>generated<br>documentation text one<br>step larger. | DBR21 enlarged PDF<br>typography profile:<br>generated full/context<br>PDFs use 11pt<br>document classes and<br>one-step-larger explicit<br>TeX text-size commands<br>for tables, legends, stats,<br>explanations and graph<br>labels.              | Repository/scripts/<br>generate_full_document.py;<br>Repository/scripts/<br>generate_context_doc.py;<br>Repository/scripts/<br>generate_tables.py;<br>Repository/scripts/<br>build_export_script_v1.8.py;<br>README.md;<br>Repository/masters/<br>cnna_roadmap_master.yaml | DBR21;<br>GNG35;<br>SCF54;<br>V1_23_PDF_FONT_SCALING_REPORT; must make all<br>regenerated full/context<br>PDFs                                                  | Closed as<br>documentation-layout tooling<br>update. The generator-level<br>font report must make all<br>generated PDF text visibly one<br>step larger while preserving<br>table contents, context filters,<br>section-pagebreak behavior and<br>generated-secondary truth<br>status. | <a href="#">implemented<br/>record</a>  |
| 20.22        | G | Database;<br>Mirror;<br>Phase    | F1       | <a href="#">DBR22</a> | Supabase live project<br>discovery and<br>empty-schema<br>baseline audit                | 20.15-20.20 existing<br>Supabase/Postgres DB/<br>tooling contracts;<br>live read-only Supabase<br>discovery;<br>Master YAML;<br>generated_exports;<br>repo_inventory/<br>raw_export;<br>GNG34;<br>GNG36.      | DBR22: Record the<br>reachable Supabase<br>organization/project<br>baseline before any<br>write: CNNA Project;<br>project ref<br>aiekahdewadrejrrdajn;<br>ACTIVE_HEALTHY;<br>Postgres 17;<br>no tables in public/<br>internal/api/review/<br>admin. | Repository/masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future Repository/database/<br>supabase/* artifacts;<br>no Lean modules                                                                               | DBR22;<br>Repository/database/<br>supabase migration/<br>import/frontend-contract<br>artifacts;<br>Supabase project<br>aiekahdewadrejrrdajn;<br>no Lean modules | Closed by read-only discovery.<br>No schema/data write. This is<br>the baseline evidence for later<br>migration/import phases.                                                                                                                                                        | <a href="#">implemented<br/>record</a>  |
| 20.23        | W | Database;<br>Mirror;<br>Phase    | F1       | <a href="#">DBR23</a> | Supabase initial<br>schema migration<br>package and RLS/<br>read-only policy<br>design  | 20.15-20.20 existing<br>Supabase/Postgres DB/<br>tooling contracts;<br>live read-only Supabase<br>discovery;<br>Master YAML;<br>generated_exports;<br>repo_inventory/<br>raw_export;<br>GNG34;<br>GNG36.      | DBR23: Produce the<br>first migration package<br>for internal/api/review/<br>admin schemas, internal<br>canonical tables, api<br>read-only views, grants/<br>RLS posture, and<br>comments documenting<br>non-truth-source status.                   | Repository/masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future Repository/database/<br>supabase/* artifacts;<br>no Lean modules                                                                               | DBR23;<br>Repository/database/<br>supabase migration/<br>import/frontend-contract<br>artifacts;<br>Supabase project<br>aiekahdewadrejrrdajn;<br>no Lean modules | Planned tooling artifact. No<br>live migration is applied until<br>explicit user authorization.                                                                                                                                                                                       | <a href="#">planned<br/>scratch pad</a> |
| 20.24        | W | Database;<br>Mirror;<br>Phase    | F1       | <a href="#">DBR24</a> | Supabase snapshot<br>import package from<br>YAML/generated<br>exports                   | 20.15-20.20 existing<br>Supabase/Postgres DB/<br>tooling contracts;<br>live read-only Supabase<br>discovery;<br>Master YAML;<br>generated_exports;<br>repo_inventory/<br>raw_export;<br>GNG34;<br>GNG36.      | DBR24: Define importer<br>contract from Master<br>YAML,<br>generated_exports and<br>repo_inventory into<br>internal tables with<br>snapshot_id, hashes,<br>import report and no<br>Lean mutation.                                                   | Repository/masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future Repository/database/<br>supabase/* artifacts;<br>no Lean modules                                                                               | DBR24;<br>Repository/database/<br>supabase migration/<br>import/frontend-contract<br>artifacts;<br>Supabase project<br>aiekahdewadrejrrdajn;<br>no Lean modules | Planned tooling artifact. It<br>prepares data import only;<br>no DB data write until explicit<br>authorization.                                                                                                                                                                       | <a href="#">planned<br/>scratch pad</a> |
| 20.25        | W | Database;<br>Mirror;<br>Phase    | F1       | <a href="#">DBR25</a> | Explicitly authorized<br>live Supabase schema<br>migration and first<br>snapshot import | 20.15-20.20 existing<br>Supabase/Postgres DB/<br>tooling contracts;<br>live read-only Supabase<br>discovery;<br>Master YAML;<br>generated_exports;<br>repo_inventory/<br>raw_export;<br>GNG34;<br>GNG36.      | DBR25: Execute, only<br>after explicit user<br>authorization, the<br>schema migration and<br>first data import into<br>Supabase project<br>aiekahdewadrejrrdajn<br>using controlled admin/<br>tool write path.                                      | Repository/masters/<br>cnna_roadmap_master.yaml;<br>generated_exports/*.tsv;<br>repo_inventory/raw_export/*.json;<br>future Repository/database/<br>supabase/* artifacts;<br>no Lean modules                                                                               | DBR25;<br>Repository/database/<br>supabase migration/<br>import/frontend-contract<br>artifacts;<br>Supabase project<br>aiekahdewadrejrrdajn;<br>no Lean modules | Future live DB-write phase. It<br>writes only Supabase schema/<br>data, never Lean files, and<br>must produce migration/<br>import reports.                                                                                                                                           | <a href="#">planned<br/>scratch pad</a> |



| Phase | Status | Context class                     | Doc tier | Linked objects        | Main goal                                                                                   | Source objects/<br>dependencies                                                                                                                                          | Definitions/objects                                                                                                                                                                             | Current code modules                                                                                                                                                     | Target modules/<br>objects                                                                                                               | Proof goals                                                                                                                                                                                                      | How imple-<br>mented                |
|-------|--------|-----------------------------------|----------|-----------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 20.26 | W      | Database;<br>Mirror;<br>Phase     | F1       | <a href="#">DBR26</a> | Post-import read-only/RLS/advisor verification and web-frontend handoff                     | 20.15-20.20 existing Supabase/Postgres DB/tooling contracts; live read-only Supabase discovery; Master YAML; generated_exports; repo_inventory/raw_export; GNG34; GNG36. | DBR26: Verify public read-only access, absence of public writes, Supabase advisor output, table/view inventory, publishable frontend data contract and cnna-project.net handoff.                | Repository/masters/cnna_roadmap_master.yaml; generated_exports/*.tsv; repo_inventory/raw_export/*.json; future Repository/database/supabase/* artifacts; no Lean modules | DBR26; Repository/database/supabase migration/import/frontend-contract artifacts; Supabase project aiekahdewadrejrrdajn; no Lean modules | Future verification/handoff phase after live import; no Lean files and no phase-release authority.                                                                                                               | <a href="#">planned scratch pad</a> |
| 20.27 | G      | Tool;<br>Infrastructure;<br>Phase | F3       | <a href="#">DBR27</a> | OOP-style phase/object context-class taxonomy and table-wide Lean/tool/reference separation | LEG34; DBR13; object register; phase table; proof dossiers; generated table pipeline; user-requested OOP-style classification principle                                  | PhaseContextClass; ObjectContextClass; LeanSourceObject; LeanTargetObject; ToolInfrastructureObject; ReferenceGateObject; DatabaseMirrorObject; GeneratedArtifactObject; GovernancePolicyObject | masters/cnna_roadmap_master.yaml; scripts/generate_tables.py; scripts/generate_full_document.py                                                                          | context_class_taxonomy table; Context class column in phase table, object tables, proof dossiers, scratchpads and module-change manifest | Every phase and object row carries an explicit context class; status remains release/proof-state logic only; tool/database/reference/generated/governance classes cannot be read as Lean objects or Lean phases. | <a href="#">implemented record</a>  |

19 Implementation scratch pad

**What this section shows.** This section records how each phase was prepared, implemented or closed. It is the detailed audit trail behind the phase table: changed files, proof strategy, expected tactics, build status, no-go checks, dependency audit and post-implementation notes. It is shown after the phase table because it explains the evidence behind phase-state changes.

Phase 0 – Freeze the SM9h baseline unchanged

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                                          |
| Context class                          | BaselinePhase                                                                                                                                                                                                              |
| Phase documentation tier               | F3                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                  |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | completed                                                                                                                                                                                                                  |
| Preparation status                     | completed-baseline-freeze                                                                                                                                                                                                  |
| Phase type                             | baseline                                                                                                                                                                                                                   |
| Phase-type-specific contract           | Freeze/reference phase: record source package and do not introduce new code.                                                                                                                                               |
| Title / state                          | implemented / closed                                                                                                                                                                                                       |
| Preparation workflow                   | No source-preparation workflow was needed. The phase freezes the last locally green SM9h package as immutable baseline evidence.                                                                                           |
| Source-to-CNNA mapping                 | Existing SM9h modules are not renamed in the baseline. They are mapped as legacy derivation sites and later wrapped by neutral target modules.                                                                             |
| Dependency-depth audit                 | No new dependencies are introduced. Baseline freeze depends only on user-confirmed green SM9h code and static package inspection.                                                                                          |
| Phase split decision                   | No split. This is a governance freeze, not a mathematical implementation phase.                                                                                                                                            |
| Positive path plan                     | Preserve the green baseline and build neutral wrappers later instead of mutating historical modules.                                                                                                                       |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                           |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation  | Not a Lean proof phase. The rehearsal is architectural: treat SM9h as immutable input and verify that later phases consume it through neutral interfaces.                                                                  |
| Expected tactics                       | None;<br>no Lean proofs are added in phase 0.                                                                                                                                                                              |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                |
| ExecComplex/computable plan            | ExecComplex baseline is preserved;<br>no analytic or mirror replacement is introduced.                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                        |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                    |
| Expected executable witness            | not specified yet                                                                                                                                                                                                          |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                       |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                  |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bridge gap status                    | No bridge gap is closed here. Known gaps are registered for later phases.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Natural backjump/refactor layer      | If a later neutral wrapper cannot consume SM9h cleanly, backjump to the neutral interface phase, not to this frozen baseline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| How implemented                      | The SM9h code package is treated as a frozen baseline. No new Lean declarations are introduced in this phase. The existing historical modules remain unchanged and are referenced only as legacy derivation sites. Part 2 will add neutral wrapper/import surfaces later instead of renaming or editing the frozen baseline directly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Verification / evidence              | Baseline object is the provided green SM9h package. Current derivation sites include CNNA/PillarA/Arithmetic/BuildAll.lean, Smoke/* through GeneratedOperationalQuadraticDiscriminantSourceFromSM9g1Window.lean, plus the already existing Schur, DtN, Finite, Handoff, and Cayley-Dickson legacy modules. I did not run Lean in this environment; the green status is taken from the supplied/local confirmed baseline and static inspection.                                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check       | Closed as governance baseline: no code mutation, no new formulas, no new consumer access, no Lean build run by assistant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Implementation notes                 | This record is deliberately non-code: it documents freeze semantics. Any later refactor must preserve import compatibility and must not mutate the Part 1 baseline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Source materials / references        | <ul style="list-style-type: none"> <li>user-confirmed green SM9h package</li> <li>static inspection of v17 roadmap data and legacy code-module references</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>ExecComplex operational matrix/discriminant baseline</li> <li>legacy Schur/DtN source chain</li> <li>legacy Handoff/Finite/AQFT/Cayley-Dickson reference surfaces</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 1 – Neutral code surface umbrella; decomposition and active branch fixed before object intake and new mathematics

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | completed                                                                                                                                                                                               |
| Preparation status                    | decomposition-closed                                                                                                                                                                                    |
| Phase type                            | refactor                                                                                                                                                                                                |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                               |
| Title / state                          | Neutral code surface before object intake and new mathematics                                                                                                                                                                                                         |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                           |
| Source-to-CNNA mapping                 | Umbrella phase: does not itself introduce public declarations. Descendant phases map legacy code to neutral surfaces.                                                                                                                                                 |
| Dependency-depth audit                 | Phase 0 is closed. This umbrella is intentionally not active because it mixes notation, foundation, regime, and CD wrapper work.                                                                                                                                      |
| Phase split decision                   | Split required. Phase 1 decomposes into 1.1 notation/legacy quarantine, 1.2 foundation/generator, 1.3 regime/complement, 1.4 CD structural reintegration.                                                                                                             |
| Positive path plan                     | Implement descendants in order. Never start Phase 5/number theory or Phase 3 Schur/DtN before 1.1 import firewall and relevant foundation/regime wrappers are closed.                                                                                                 |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | Umbrella has no Lean proof. Descendants must use definitional forwarding, constructors, field projections, and import checks.                                                                                                                                         |
| Expected tactics                       | n/a for umbrella; descendant records specify tactics.                                                                                                                                                                                                                 |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                               |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                              |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                  |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                                                           |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                              |
| How implemented                        | Planning/decomposition closure only: the child DAG, ordering rule, and active cursor were fixed in the YAML master. No Lean refactor or theorem implementation is claimed for this umbrella row.                                                                      |
| Verification / evidence                | Generated YAML/table/PDF validation for the governance decomposition. Lean build was not run in this environment.                                                                                                                                                     |
| Build result                           | No Lean build run; governance/documentation-only closure.                                                                                                                                                                                                             |
| Post-implementation plan check         | Satisfied as a routing/order lock: no white predecessor remains on the active branch before the unique yellow active phase.                                                                                                                                           |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Green here means decomposition-closed umbrella. Descendant implementation phases remain responsible for code movement and proof closure.                                                                                                                              |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source materials / references        | <ul style="list-style-type: none"> <li>SM9h green zip / legacy code tree</li> <li>static code audit</li> <li>v22 phase 1.1 split refinement</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>umbrella refactor only; closed by descendants 1.1-1.4</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global <code>_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import <code>Legacy.StatusGates</code></li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

**Phase 1.1 – Notation split and legacy/API cleanup umbrella; child gates fixed in derivation order**

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                       |
| Context class                         | ToolInfrastructurePhase                                                                                                                                                                                 |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | completed                                                                                                                                                                                               |
| Preparation status                    | decomposition-closed                                                                                                                                                                                    |
| Phase type                            | infrastructure                                                                                                                                                                                          |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness; close schema/decision prerequisites before code.                                                                                 |
| Title / state                         | Notation split and legacy-status quarantine                                                                                                                                                             |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.             |
| Source-to-CNNA mapping                | LEG0 is produced first. LEG1-LEG4 consume LEG0 in sequence. Mathematical objects are not introduced in this branch.                                                                                     |
| Dependency-depth audit                | Dependencies for preparation are complete, but implementation depth is mixed: inventory, quarantine, notation extraction, alias routing, import firewall.                                               |
| Phase split decision                  | Split into 1.1.1-1.1.5. Keeping all of this in one phase would create an oversized refactor and obscure import failures.                                                                                |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive path plan                     | Close the split map first;<br>then quarantine status gates;<br>then create CoreSymbols;<br>then route aliases/wrappers;<br>finally run the import firewall audit.                                                                                                     |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                           |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | Mostly no mathematical proofs. Expected Lean work is module movement, reexports, definitional aliases, exact forwarding, and import graph verification.                                                                                                               |
| Expected tactics                       | exact, rfl, constructor, field projections, named theorem forwarding;<br>avoid broad simp/simpa except final trivial closure.                                                                                                                                         |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                            |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                           |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                        |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Planning/decomposition closure only: the child DAG, ordering rule, and active cursor were fixed in the YAML master. No Lean refactor or theorem implementation is claimed for this umbrella row.                                                                      |
| Verification / evidence                | Generated YAML/table/PDF validation for the governance decomposition. Lean build was not run in this environment.                                                                                                                                                     |
| Build result                           | No Lean build run;<br>governance/documentation-only closure.                                                                                                                                                                                                          |
| Post-implementation plan check         | Satisfied as a routing/order lock: no white predecessor remains on the active branch before the unique yellow active phase.                                                                                                                                           |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Green here means decomposition-closed umbrella. Descendant implementation phases remain responsible for code movement and proof closure.                                                                                                                              |
| Source materials / references          | <ul style="list-style-type: none"> <li>Arithmetic/Notation.lean static audit: 5095 lines, 903 declarations, 0 notation/syntax/macro declarations by grep</li> <li>code-audit finding: 749 single-constructor status enums</li> <li>YAML single-master rule</li> </ul> |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"> <li>LEG0 Notation split inventory</li> <li>LEG1 Legacy.StatusGates</li> <li>LEG2 CoreSymbols true notation surface</li> <li>LEG3 Legacy phase aliases</li> <li>LEG4 Neutral import firewall</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global__no__go__guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.1.1 – Pre-code governance umbrella: module manifest, reuse, consumption classification, proofbook infrastructure, blue correction, API and pruning gates

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #90EE90; border: 1px solid black; padding: 0 2px;">G</span>                                                                                                              |
| Context class                         | DocumentationPhase                                                                                                                                                                                      |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | completed                                                                                                                                                                                               |
| Preparation status                    | decomposition-closed                                                                                                                                                                                    |
| Phase type                            | documentation                                                                                                                                                                                           |
| Phase-type-specific contract          | Documentation phase: record dossier coverage, verification/audit status, and links to code/object records; no mathematical object is generated by documentation alone.                                  |
| Title / state                         | Consumption classification and blue-object correction umbrella                                                                                                                                          |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.             |
| Source-to-CNNA mapping                | This umbrella binds LEG0/LEG5/LEG6/LEG7/LEG8/LEG9. It prevents code moves before consumption and semantic API roles are classified.                                                                     |
| Dependency-depth audit                | Too broad for one implementation step; split into 1.1.1.1-1.1.1.5.                                                                                                                                      |
| Phase split decision                  | Split required by v23: classification, blue correction ledger, public API contract, pruning ledger, generated-view lock.                                                                                |
| Positive path plan                    | Proceed through child gates only. No deletion, wrapper move, or status quarantine until classification and contracts exist.                                                                             |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.       |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.             |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                              |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No Lean proof. Structural validation only: exactly one active child, complete scratchpad records, generated tables/PDF compile.                                                                                                                                       |
| Expected tactics                       | No Lean tactics expected. If a small Lean audit module is later added, use only constants/strings/records with exact/rfl where applicable.                                                                                                                            |
| Expected mathlib imports               | No new mathlib imports. No Lean imports are required for this planning/audit gate.                                                                                                                                                                                    |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                              |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                                                           |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Planning/decomposition closure only: the child DAG, ordering rule, and active cursor were fixed in the YAML master. No Lean refactor or theorem implementation is claimed for this umbrella row.                                                                      |
| Verification / evidence                | Generated YAML/table/PDF validation for the governance decomposition. Lean build was not run in this environment.                                                                                                                                                     |
| Build result                           | No Lean build run;<br>governance/documentation-only closure.                                                                                                                                                                                                          |
| Post-implementation plan check         | Satisfied as a routing/order lock: no white predecessor remains on the active branch before the unique yellow active phase.                                                                                                                                           |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Green here means decomposition-closed umbrella. Descendant implementation phases remain responsible for code movement and proof closure.                                                                                                                              |
| Source materials / references          | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml v23</li> <li>repo_inventory/raw_export</li> <li>object register blue rows</li> <li>static declaration/reference audit</li> </ul>                                                              |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>consumption classification gate</li> <li>blue-object semantic correction gate</li> <li>public API contract gate</li> <li>pruning ledger gate</li> <li>generated-view authority lock</li> </ul>                                 |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>• no <code>@[simp]</code> attributes in the operational path</li> <li>• no <code>noncomputable</code> in the operational path</li> <li>• no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>• no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>• no free formula without proof-carrying entry structure</li> <li>• no consumer-side patch when the missing structure belongs earlier</li> <li>• no premature obstruction before a serious positive construction attempt</li> <li>• no phase-semantic public API names</li> <li>• no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>• do not delete legacy status enums during quarantine</li> <li>• do not let Phase 3+ import <code>Legacy.StatusGates</code></li> <li>• do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Phase 1.1.1.1 – Module/declaration consumption classification umbrella with architecture-decision locks, pre-implementation module manifest, reuse governance, and proofbook ordering**

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                                                                   |
| Context class                         | DocumentationPhase                                                                                                                                                                                                                                  |
| Phase documentation tier              | F0                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | completed                                                                                                                                                                                                                                           |
| Preparation status                    | decomposition-closed                                                                                                                                                                                                                                |
| Phase type                            | documentation                                                                                                                                                                                                                                       |
| Phase-type-specific contract          | Documentation phase: record dossier coverage, verification/audit status, and links to code/object records; no mathematical object is generated by documentation alone.                                                                              |
| Title / state                         | module/declaration classification umbrella with module-manifest and reuse governance                                                                                                                                                                |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.                                                                            |
| Source-to-CNNA mapping                | This umbrella maps code hygiene to LEG10, LEG11, and LEG5. It no longer directly closes declaration classification.                                                                                                                                 |
| Dependency-depth audit                | Split required. Module placement/reuse must be decided before declaration classification, because whether code belongs in an existing module is part of semantic consumption.                                                                       |
| Phase split decision                  | Split into 1.1.1.1.1 module manifest, 1.1.1.1.2 reuse/generator audit, and 1.1.1.1.3 proof-documentation infrastructure and 1.1.1.1.4 declaration classification finalization.                                                                      |
| Positive path plan                    | Close 1.1.1.1.1 first;<br>then reuse/generator audit;<br>then classify declarations using the two governance ledgers.                                                                                                                               |
| Module implementation manifest        | Umbrella only. Child 1.1.1.1.1 defines the required manifest format.                                                                                                                                                                                |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                          |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                  |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No Lean proof in umbrella. Treat as planning contract.                                                                                                                                                                                                                |
| Expected tactics                       | No Lean tactics expected. If the classification is mirrored in Lean later, use explicit records and exact/rfl;<br>no simp-driven folding.                                                                                                                             |
| Expected mathlib imports               | No new mathlib imports expected;<br>if a later Lean audit file is introduced, verify imports against mathlib 4.28.0 first.                                                                                                                                            |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                                                                |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | A declaration family with unclear usage is a classification gap, not a proof gap. It must block movement/deletion until classified.                                                                                                                                   |
| Natural backjump/refactor layer        | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                              |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Planning/decomposition closure only: the child DAG, ordering rule, and active cursor were fixed in the YAML master. No Lean refactor or theorem implementation is claimed for this umbrella row.                                                                      |
| Verification / evidence                | Generated YAML/table/PDF validation for the governance decomposition. Lean build was not run in this environment.                                                                                                                                                     |
| Proof strategy                         | Use green architecture locks through 1.1.1.1.0.14 before active module implementation;<br>then classify modules/declarations.                                                                                                                                         |
| Build result                           | No Lean build run;<br>governance/documentation-only closure.                                                                                                                                                                                                          |
| Post-implementation plan check         | Satisfied as a routing/order lock: no white predecessor remains on the active branch before the unique yellow active phase.                                                                                                                                           |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | Green here means decomposition-closed umbrella. Descendant implementation phases remain responsible for code movement and proof closure.                                                                                                                              |
| Source materials / references          | <ul style="list-style-type: none"> <li>repo_inventory/raw_export</li> <li>Arithmetic/Notation.lean audit</li> <li>symbol/reference summary</li> <li>existing Smoke-Test Generator modules</li> </ul>                                                                  |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>ModuleImplementationManifest</li> <li>ReuseDuplicationLedger</li> <li>CanonicalGeneratorPolicy</li> <li>ConsumptionClassificationMap</li> </ul>                                                                                |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not infer semantic consumption from <code>BuildAll</code> or umbrella import reachability</li> <li>do not delete anything in the classification phase</li> <li>do not mark blue objects white merely because they are proof-carrying</li> <li>do not route blue-object corrections into <code>Legacy.StatusGates</code></li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 1.1.1.1.0 – Global no-go guardrail contract and inheritance lock

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                       |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | completed                                                                                                                                                                                               |
| Preparation status                    | completed                                                                                                                                                                                               |
| Phase type                            | infrastructure                                                                                                                                                                                          |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness; close schema/decision prerequisites before code.                                                                                 |
| Title / state                         | Global no-go guardrail contract and inheritance lock                                                                                                                                                    |
| Preparation workflow                  | Global guardrail infrastructure insertion before active implementation phase.                                                                                                                           |
| Source-to-CNNA mapping                | User global no-go list -> LEG13;<br>workflow inheritance -> <code>workflow_policy</code> ;<br>phase enforcement -> <code>implementation_scratchpad.global__no_go_inheritance</code> .                   |
| Dependency-depth audit                | Must precede 1.1.1.1.1 because module manifests and all later implementation prompts need the global constraints already active.                                                                        |
| Phase split decision                  | Inserted as 1.1.1.1.0 rather than merging into 1.1.1.1.1 because guardrail inheritance is a cross-cutting invariant, not module-manifest content.                                                       |
| Positive path plan                    | Close global guardrail infrastructure first, then continue with compact module manifest and semantic placement gate.                                                                                    |
| Module implementation manifest        | No code modules introduced by this gate.                                                                                                                                                                |
| Semantic placement decision           | Governance belongs in YAML master and generated views unless a future code-level policy module is justified.                                                                                            |
| Reuse and duplication audit           | No duplicate implementation authorized; decision reduces future duplication risk.                                                                                                                       |
| Canonical generator policy            | Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation.                                                                                                |
| Module compactness rule               | Do not split code solely for this planning decision; use YAML/workflow unless a reusable code-level policy module is later justified.                                                                   |

|                                        |                                                                                                                                                                                                                            |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation  | If every scratchpad record carries global_no_go_inheritance and generator renders top-level guardrails, local no_go fields cannot be interpreted as exhaustive phase-only rules.                                           |
| Expected tactics                       | Planning validation only: schema presence checks, generated table output, active-front order check.                                                                                                                        |
| Expected mathlib imports               | None;<br>no Lean code in this phase.                                                                                                                                                                                       |
| ExecComplex/computable plan            | None;<br>governance only.                                                                                                                                                                                                  |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                        |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                    |
| Expected executable witness            | not specified yet                                                                                                                                                                                                          |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                       |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                  |
| Bridge gap status                      | Closed for planning: global list exists and is inherited. Future code phases still need actual code audits.                                                                                                                |
| Natural backjump/refactor layer        | If a future phase violates a global no-go, backjump to its natural source layer or mirror isolation, not to consumer patching.                                                                                             |
| Implementation prompt contract         | Every future implementation prompt must cite global_no_go_guardrails and then state only additional phase-local no-go constraints.                                                                                         |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                |
| How implemented                        | Added top-level global_no_go_guardrails, workflow inheritance rule, LEG13, phase 1.1.1.1.0, proof dossier, generated table, and global_no_go_inheritance field across scratchpad records.                                  |
| Verification / evidence                | Planning generator validates global_no_go_guardrails and scratchpad inheritance. PDF/TSV generated from YAML. No Lean build was run or claimed.                                                                            |
| Proof strategy                         | Treat the guardrail list as a global invariant and make local scratchpad no-go lists inherit it explicitly.                                                                                                                |
| Build result                           | No Lean build. Planning table generation and PDF compilation only.                                                                                                                                                         |
| Post-implementation plan check         | Active cursor remains 1.1.1.1.1;<br>no white predecessor stands before it because 1.1.1.1.0 is green planning infrastructure.                                                                                              |
| Artifact packaging rule                | Return complete v26 zip, compiled PDF, and concise report. State that no Lean build was executed.                                                                                                                          |
| Implementation notes                   | This is a planning/workflow infrastructure close, not a Lean refactor. It prevents future phase-local no-go fields from replacing the global operational constraints.                                                      |
| Source materials / references          | <ul style="list-style-type: none"> <li>User-specified global no-go list in conversation</li> <li>Existing v25 workflow policy</li> <li>Existing phase-local scratchpad no_go_guardrails fields</li> </ul>                  |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>GlobalNoGoGuardrailContract</li> <li>GlobalNoGoInheritanceField</li> <li>PhaseLocalAddendumRule</li> </ul>                                                                          |
| Implemented modules                    | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml;</li> <li>scripts/generate_tables.py;</li> <li>generated tables/PDF/TSV</li> </ul>                                                                 |
| Added modules                          | <ul style="list-style-type: none"> <li>No Lean modules. Added YAML top-level section and generated planning table.</li> </ul>                                                                                              |



|                            |                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modified modules           | <ul style="list-style-type: none"> <li>masters/cnna__roadmap__master.yaml;</li> <li>scripts/generate_tables.py;</li> <li>main TeX wrapper</li> </ul>                                                                                                                                                                                                                              |
| Forbidden patterns checked | <ul style="list-style-type: none"> <li>No operational Lean code touched</li> <li>No local no-go list allowed to supersede global list</li> <li>No old TeX/PDF source-of-truth regression</li> </ul>                                                                                                                                                                               |
| No-go guardrails           | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>Do not leave global constraints only inside phase-local scratchpad records.</li> <li>Do not permit local no_go_guardrails to weaken global rules.</li> <li>Do not claim Lean build success for this planning update.</li> </ul> |

### Phase 1.1.1.1.0.1 – Concrete computable instance traversal and mathlib-complication scratchpad contract

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                    |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                                                                                         |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F3                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                       |
| Implementation status                 | completed                                                                                                                                                                                                                                                                 |
| Preparation status                    | completed                                                                                                                                                                                                                                                                 |
| Phase type                            | infrastructure                                                                                                                                                                                                                                                            |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness; close schema/decision prerequisites before code.                                                                                                                                                   |
| Title / state                         | Concrete computable instance traversal and mathlib-complication scratchpad contract                                                                                                                                                                                       |
| Preparation workflow                  | Infrastructure gate inserted after global no-go inheritance and before active module-manifest implementation.                                                                                                                                                             |
| Source-to-CNNA mapping                | LEG14 and LEG15 are workflow/control objects. They do not add CNNA mathematics; they force future operational phases to expose executable instance traversal and mathlib computability risks before code edits.                                                           |
| Dependency-depth audit                | Depends only on closed 1.1.1.1.0 global no-go inheritance. It must be green before active phase 1.1.1.1.1 because module manifests must know whether planned code is infrastructure-only or operational/executable.                                                       |
| Phase split decision                  | New small predecessor gate added. Do not split further unless a separate generated static-audit table is later introduced.                                                                                                                                                |
| Positive path plan                    | Keep positive construction first, but require instance pressure: generated substrate/window -> operational declarations -> concrete endpoint witness. If the path cannot be traversed, record the earliest natural missing source layer instead of patching the consumer. |
| Module implementation manifest        | No code modules introduced by this gate.                                                                                                                                                                                                                                  |
| Semantic placement decision           | Governance belongs in YAML master and generated views unless a future code-level policy module is justified.                                                                                                                                                              |
| Reuse and duplication audit           | No duplicate implementation authorized; decision reduces future duplication risk.                                                                                                                                                                                         |
| Canonical generator policy            | Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation.                                                                                                                                                                  |
| Module compactness rule               | Do not split code solely for this planning decision; use YAML/workflow unless a reusable code-level policy module is later justified.                                                                                                                                     |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                         |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                               |

|                                        |                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                              |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                        |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                           |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                     |
| Proof rehearsal before implementation  | For a future implementation: choose concrete finite instance; compute through the intended operational path; compare endpoint shape to proof target; if mathlib blocks executability, isolate mirror use or backjump.                                                                   |
| Expected tactics                       | No Lean tactics now. Future phases should prefer constructor/projection proofs and explicit finite-instance lemmas over fragile rewrite chains.                                                                                                                                         |
| Expected mathlib imports               | None for this planning gate. Future phases must record concrete mathlib import candidates and computability status.                                                                                                                                                                     |
| ExecComplex/computable plan            | No ExecComplex object introduced. Future ExecComplex phases must include a concrete finite instance traversal through the ExecComplex boundary/mirror seam.                                                                                                                             |
| Concrete computable instance traversal | Infrastructure-only gate: no operational Lean object introduced. The generated requirement is itself the traversal contract for future operational phases.                                                                                                                              |
| Computable path non-skeleton status    | closed for planning infrastructure; future operational phases must state concrete instance traversal before code edits                                                                                                                                                                  |
| Expected executable witness            | Required future field; for this gate the witness is generated scratchpad visibility, not a Lean executable term.                                                                                                                                                                        |
| Mathlib complication log               | Planning gate only. Records known risk class: mathlib algebraic interfaces may force noncomputable/classical dependencies; future phases must log concrete module/API names.                                                                                                            |
| Noncomputable boundary audit           | No Lean code changed. Future operational phases must separate operational code from mirror/non-operational noncomputable uses.                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib forces noncomputability in operational route, isolate in mirror layer or derive/implement an operational replacement at the natural source layer.                                                                                                                            |
| Bridge gap status                      | Closed as planning infrastructure. Mathematical bridges remain phase-specific.                                                                                                                                                                                                          |
| Natural backjump/refactor layer        | If concrete traversal fails because an earlier generated datum is missing, backjump to the earliest natural generator/source layer. If mathlib only provides noncomputable APIs, route them to mirror/non-operational layer or build an operational replacement.                        |
| Implementation prompt contract         | Before any Lean code edit, report whether the phase is infrastructure-only or operational. If operational, provide <code>concrete_instance_traversal</code> and <code>expected_executable_witness</code> , plus <code>mathlib_complication_log/noncomputable_boundary_audit</code> .    |
| Global no-go inheritance               | yes; all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                   |
| How implemented                        | Planning infrastructure added to the YAML master and generator: every scratchpad record now contains concrete-instance traversal and mathlib/noncomputable complication fields. No Lean code moved.                                                                                     |
| Verification / evidence                | <code>generate_tables.py</code> validates the new required fields and renders them in <code>implementation_scratchpad.tex</code> ; artifact generation only, no Lean build.                                                                                                             |
| Proof strategy                         | No Lean proof. Infrastructure proof is that the generator requires and renders the fields for every phase record.                                                                                                                                                                       |
| Build result                           | No Lean build run; generated planning artifacts only.                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After this gate, active phase 1.1.1.1.1 inherits the fields and must keep them visible in the scratchpad.                                                                                                                                                                               |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts. Lean build success may be recorded only from actual/current build or user-reported local build; otherwise local Lean build is required.                                                                 |
| Implementation notes                   | This closes the workflow-contract gap: operational computable code must be exercised by concrete instances and mathlib barriers must be logged before implementation.                                                                                                                   |
| Source materials / references          | <ul style="list-style-type: none"> <li>prior SM/AR implementation history</li> <li><code>global_no_go_guardrails</code></li> <li><code>implementation_scratchpad</code> records</li> <li>mathlib computability complication examples from earlier polynomial/matrix attempts</li> </ul> |

|                                      |                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"> <li>ConcreteComputableInstanceTraversal</li> <li>ComputablePathNonSkeletonContract</li> <li>ExpectedExecutableWitness</li> <li>MathlibComplicationLog</li> <li>NoncomputableBoundaryAudit</li> <li>MathlibMitigationOrRefactorLayer</li> </ul>                                                                                  |
| Modified modules                     | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> <li>README.md</li> </ul>                                                                                                                                                                                                                          |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>global no-go inheritance retained</li> <li>operational noncomputable/classical risk must be logged</li> <li>no Matrix.inv/external inverse oracle weakening</li> <li>no skeleton-only operational closeout</li> </ul>                                                                                                       |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no operational code phase may close as a skeleton-only path</li> <li>no unlogged mathlib noncomputable/classical complication</li> <li>no hidden mirror dependency inside operational executable path</li> </ul> |

### Phase 1.1.1.1.0.2 – ExecComplex rational-boundary and mirror-only algebraic irrationality contract (EXEC-2)

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                     |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3; border: 1px solid #000; padding: 0 2px;">G</span>                                                                                                                                                                                                                  |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                                                                                                        |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                        |
| Implementation status                 | completed                                                                                                                                                                                                                                                                                                  |
| Preparation status                    | completed                                                                                                                                                                                                                                                                                                  |
| Phase type                            | infrastructure                                                                                                                                                                                                                                                                                             |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness; close schema/decision prerequisites before code.                                                                                                                                                                                    |
| Title / state                         | EXEC-2 rational-boundary and mirror-only algebraic irrationality contract                                                                                                                                                                                                                                  |
| Preparation workflow                  | Architecture decision recorded before active implementation gate.                                                                                                                                                                                                                                          |
| Source-to-CNNA mapping                | Decision mapped to LEG object and workflow/format-decision rows; no ontic CNNA input introduced.                                                                                                                                                                                                           |
| Dependency-depth audit                | Depends on previous green architecture gate before active phase 1.1.1.1.1; closes a review-identified decision gap without changing Lean code.                                                                                                                                                             |
| Phase split decision                  | Inserted as green pre-active infrastructure gate under 1.1.1.1.0.x so no white predecessor remains before the active cursor.                                                                                                                                                                               |
| Positive path plan                    | Keep operational arithmetic through Q x Q ExecComplex for concrete rational/integer invariants. Put sqrt(D), general tau, algebraic eigenvalues, general lattice vectors and analytic modular data in mirror/non-operational layers. Permit return only for explicitly proved rational/integer invariants. |
| Module implementation manifest        | No code modules introduced by this gate.                                                                                                                                                                                                                                                                   |

|                                        |                                                                                                                                                                                                                               |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semantic placement decision            | Governance belongs in YAML master and future optional architecture policy modules only if code-level records become useful.                                                                                                   |
| Reuse and duplication audit            | No duplicate implementation authorized;<br>decision reduces future duplication risk.                                                                                                                                          |
| Canonical generator policy             | Generator-like logic must consume the canonical smoke/theory generator path;<br>no parallel implementation.                                                                                                                   |
| Module compactness rule                | Do not split code solely for this planning decision;<br>encode as workflow/format decision unless a reusable code-level policy module is later justified.                                                                     |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                             |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                   |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Closed: operational paths end at $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrationalities are mirror-only;<br>proved rational/integer invariants may return operationally.                                               |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                 |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation  | Check each affected object: is the proposed value representable in $\mathbb{Q} \times \mathbb{Q}$ ? If no, mark mirror-only;<br>if a rational/integer invariant is needed, state its proof target and source objects.         |
| Expected tactics                       | None;<br>no Lean proof in this planning gate.                                                                                                                                                                                 |
| Expected mathlib imports               | None for planning gate.                                                                                                                                                                                                       |
| ExecComplex/computable plan            | EXEC-2 closed at planning level: $\mathbb{Q} \times \mathbb{Q}$ is not extended now;<br>algebraic extensions are not silently introduced.                                                                                     |
| Concrete computable instance traversal | Infrastructure-only gate: no operational Lean object introduced. Later operational phases must provide concrete finite instance traversal.                                                                                    |
| Computable path non-skeleton status    | infrastructure-only;<br>this gate adds obligations that prevent later operational skeletons.                                                                                                                                  |
| Expected executable witness            | None for planning gate.                                                                                                                                                                                                       |
| Mathlib complication log               | No direct mathlib import in this planning gate. Future phases must log complications.                                                                                                                                         |
| Noncomputable boundary audit           | No direct Lean boundary touched;<br>future phases must audit noncomputable/classical pressure.                                                                                                                                |
| Mathlib mitigation/refactor layer      | If future mathlib pressure appears, isolate in mirror/non-operational layer or backjump to earliest natural source layer.                                                                                                     |
| Bridge gap status                      | Closed as planning decision;<br>implementation bridges remain future phase obligations.                                                                                                                                       |
| Natural backjump/refactor layer        | If implementation later violates this decision, backjump to the earliest natural layer recorded by the relevant workflow.                                                                                                     |
| Implementation prompt contract         | Future implementation prompts must cite this gate when relevant.                                                                                                                                                              |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                   |
| How implemented                        | Planning/architecture gate closed in YAML master;<br>no Lean source changed.                                                                                                                                                  |
| Verification / evidence                | Generated tables/PDF render the gate. No Lean build was run or claimed.                                                                                                                                                       |
| Proof strategy                         | Schema-level planning proof, not Lean theorem.                                                                                                                                                                                |
| Build result                           | No Lean build run;<br>planning artifact generated only.                                                                                                                                                                       |
| Post-implementation plan check         | Satisfied at planning level;<br>active cursor remains 1.1.1.1.1.                                                                                                                                                              |
| Artifact packaging rule                | Return complete zip/PDF/report for planning updates. Lean build success may be recorded only from an actual current/user-reported build.                                                                                      |
| Implementation notes                   | This is a pre-implementation decision gate;<br>it constrains later code phases without modifying code itself.                                                                                                                 |

|                                      |                                                                                                                                                                      |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source materials / references        | <ul style="list-style-type: none"> <li>v28 code/theory/review validation</li> <li>current YAML master</li> <li>static repo inventory and previous reports</li> </ul> |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>architecture decision gate</li> <li>workflow policy</li> <li>proof-dossier object</li> </ul>                                  |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>inherits global no-go guardrails; no local weakening allowed</li> </ul>                                                       |

### Phase 1.1.1.1.0.3 – Refactor regression protocol and incremental migration policy

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
 Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                            |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                                                 |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                               |
| Phase documentation tier              | F1                                                                                                                                                                                                                                |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                    |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                               |
| Implementation status                 | completed                                                                                                                                                                                                                         |
| Preparation status                    | completed                                                                                                                                                                                                                         |
| Phase type                            | infrastructure                                                                                                                                                                                                                    |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness; close schema/decision prerequisites before code.                                                                                                           |
| Title / state                         | Refactor regression protocol and incremental migration policy                                                                                                                                                                     |
| Preparation workflow                  | Architecture decision recorded before active implementation gate.                                                                                                                                                                 |
| Source-to-CNNA mapping                | Decision mapped to LEG object and workflow/format-decision rows; no ontic CNNA input introduced.                                                                                                                                  |
| Dependency-depth audit                | Depends on previous green architecture gate before active phase 1.1.1.1.1; closes a review-identified decision gap without changing Lean code.                                                                                    |
| Phase split decision                  | Inserted as green pre-active infrastructure gate under 1.1.1.1.0.x so no white predecessor remains before the active cursor.                                                                                                      |
| Positive path plan                    | Use wrapper-first migration for Notation/Legacy cleanup: introduce neutral wrappers, migrate consumers in bounded batches, keep compatibility shims until consumers are moved, and require local full-build evidence per package. |
| Module implementation manifest        | No code modules introduced by this gate.                                                                                                                                                                                          |
| Semantic placement decision           | Governance belongs in YAML master and future optional architecture policy modules only if code-level records become useful.                                                                                                       |
| Reuse and duplication audit           | No duplicate implementation authorized; decision reduces future duplication risk.                                                                                                                                                 |
| Canonical generator policy            | Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation.                                                                                                                          |
| Module compactness rule               | Do not split code solely for this planning decision; encode as workflow/format decision unless a reusable code-level policy module is later justified.                                                                            |
| Refactor regression protocol          | Closed: no refactor phase may close without declaration-count/import-delta/consumer-breakage audit and local full-build evidence.                                                                                                 |

|                                        |                                                                                                                                                                                                                            |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refactor migration metrics             | Template required: declarations_migrated_count;<br>modules_added;<br>modules_modified;<br>imports_changed;<br>import_graph_delta;<br>consumer_breakage_audit;<br>compatibility_aliases_kept;<br>build_evidence_source.     |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                           |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation  | For each migration batch: list declarations moved, imports changed, consumers touched, compatibility aliases kept, and expected local build command/evidence.                                                              |
| Expected tactics                       | None;<br>no Lean proof in this planning gate.                                                                                                                                                                              |
| Expected mathlib imports               | None for planning gate.                                                                                                                                                                                                    |
| ExecComplex/computable plan            | None for this planning/infrastructure gate unless explicitly stated.                                                                                                                                                       |
| Concrete computable instance traversal | Infrastructure-only gate: no operational Lean object introduced. Later operational phases must provide concrete finite instance traversal.                                                                                 |
| Computable path non-skeleton status    | infrastructure-only;<br>this gate adds obligations that prevent later operational skeletons.                                                                                                                               |
| Expected executable witness            | None for planning gate.                                                                                                                                                                                                    |
| Mathlib complication log               | No direct mathlib import in this planning gate. Future phases must log complications.                                                                                                                                      |
| Noncomputable boundary audit           | No direct Lean boundary touched;<br>future phases must audit noncomputable/classical pressure.                                                                                                                             |
| Mathlib mitigation/refactor layer      | If future mathlib pressure appears, isolate in mirror/non-operational layer or backjump to earliest natural source layer.                                                                                                  |
| Bridge gap status                      | Closed as planning decision;<br>implementation bridges remain future phase obligations.                                                                                                                                    |
| Natural backjump/refactor layer        | If implementation later violates this decision, backjump to the earliest natural layer recorded by the relevant workflow.                                                                                                  |
| Implementation prompt contract         | Future implementation prompts must cite this gate when relevant.                                                                                                                                                           |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                |
| How implemented                        | Planning/architecture gate closed in YAML master;<br>no Lean source changed.                                                                                                                                               |
| Verification / evidence                | Generated tables/PDF render the gate. No Lean build was run or claimed.                                                                                                                                                    |
| Proof strategy                         | Schema-level planning proof, not Lean theorem.                                                                                                                                                                             |
| Build result                           | No Lean build run;<br>planning artifact generated only.                                                                                                                                                                    |
| Post-implementation plan check         | Satisfied at planning level;<br>active cursor remains 1.1.1.1.1.                                                                                                                                                           |
| Artifact packaging rule                | Return complete zip/PDF/report for planning updates. Lean build success may be recorded only from an actual current/user-reported build.                                                                                   |
| Implementation notes                   | This is a pre-implementation decision gate;<br>it constrains later code phases without modifying code itself.                                                                                                              |
| Source materials / references          | <ul style="list-style-type: none"> <li>v28 code/theory/review validation</li> <li>current YAML master</li> <li>static repo inventory and previous reports</li> </ul>                                                       |



|                                      |                                                                                                                                 |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"><li>architecture decision gate</li><li>workflow policy</li><li>proof-dossier object</li></ul> |
| No-go guardrails                     | <ul style="list-style-type: none"><li>inherits global no-go guardrails;<br/>no local weakening allowed</li></ul>                |

Phase 1.1.1.1.0.4 – Lean/mathlib version baseline and upgrade-gate policy

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                            |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                 |
| Context class                         | LeanGovernancePhase                                                                                                                                                                               |
| Phase documentation tier              | F1                                                                                                                                                                                                |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.    |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                               |
| Implementation status                 | completed                                                                                                                                                                                         |
| Preparation status                    | completed                                                                                                                                                                                         |
| Phase type                            | infrastructure                                                                                                                                                                                    |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness;<br>close schema/decision prerequisites before code.                                                                        |
| Title / state                         | Lean/mathlib version baseline and upgrade-gate policy                                                                                                                                             |
| Preparation workflow                  | Architecture decision recorded before active implementation gate.                                                                                                                                 |
| Source-to-CNNA mapping                | Decision mapped to LEG object and workflow/format-decision rows;<br>no ontic CNNA input introduced.                                                                                               |
| Dependency-depth audit                | Depends on previous green architecture gate before active phase 1.1.1.1.1;<br>closes a review-identified decision gap without changing Lean code.                                                 |
| Phase split decision                  | Inserted as green pre-active infrastructure gate under 1.1.1.1.0.x so no white predecessor remains before the active cursor.                                                                      |
| Positive path plan                    | Use the frozen Lean 4.28.0/mathlib baseline for implementation. If a phase needs newer API, open a separate phase-boundary upgrade gate instead of silently changing imports.                     |
| Module implementation manifest        | No code modules introduced by this gate.                                                                                                                                                          |
| Semantic placement decision           | Governance belongs in YAML master and future optional architecture policy modules only if code-level records become useful.                                                                       |
| Reuse and duplication audit           | No duplicate implementation authorized;<br>decision reduces future duplication risk.                                                                                                              |
| Canonical generator policy            | Generator-like logic must consume the canonical smoke/theory generator path;<br>no parallel implementation.                                                                                       |
| Module compactness rule               | Do not split code solely for this planning decision;<br>encode as workflow/format decision unless a reusable code-level policy module is later justified.                                         |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout. |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.       |
| Lean/mathlib version policy           | Closed: baseline freeze by default;<br>upgrades only at explicit phase-boundary gates with separate build/review evidence.                                                                        |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                       |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                     |
| Governance-overhead review            | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                            |

|                                        |                                                                                                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof rehearsal before implementation  | Before adding a mathlib import: verify baseline availability, record computability/noncomputable pressure, and decide operational vs mirror placement.               |
| Expected tactics                       | None;<br>no Lean proof in this planning gate.                                                                                                                        |
| Expected mathlib imports               | None for planning gate.                                                                                                                                              |
| ExecComplex/computable plan            | None for this planning/infrastructure gate unless explicitly stated.                                                                                                 |
| Concrete computable instance traversal | Infrastructure-only gate: no operational Lean object introduced. Later operational phases must provide concrete finite instance traversal.                           |
| Computable path non-skeleton status    | infrastructure-only;<br>this gate adds obligations that prevent later operational skeletons.                                                                         |
| Expected executable witness            | None for planning gate.                                                                                                                                              |
| Mathlib complication log               | No direct mathlib import in this planning gate. Future phases must log complications.                                                                                |
| Noncomputable boundary audit           | No direct Lean boundary touched;<br>future phases must audit noncomputable/classical pressure.                                                                       |
| Mathlib mitigation/refactor layer      | If future mathlib pressure appears, isolate in mirror/non-operational layer or backjump to earliest natural source layer.                                            |
| Bridge gap status                      | Closed as planning decision;<br>implementation bridges remain future phase obligations.                                                                              |
| Natural backjump/refactor layer        | If implementation later violates this decision, backjump to the earliest natural layer recorded by the relevant workflow.                                            |
| Implementation prompt contract         | Future implementation prompts must cite this gate when relevant.                                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                          |
| How implemented                        | Planning/architecture gate closed in YAML master;<br>no Lean source changed.                                                                                         |
| Verification / evidence                | Generated tables/PDF render the gate. No Lean build was run or claimed.                                                                                              |
| Proof strategy                         | Schema-level planning proof, not Lean theorem.                                                                                                                       |
| Build result                           | No Lean build run;<br>planning artifact generated only.                                                                                                              |
| Post-implementation plan check         | Satisfied at planning level;<br>active cursor remains 1.1.1.1.1.                                                                                                     |
| Artifact packaging rule                | Return complete zip/PDF/report for planning updates. Lean build success may be recorded only from an actual current/user-reported build.                             |
| Implementation notes                   | This is a pre-implementation decision gate;<br>it constrains later code phases without modifying code itself.                                                        |
| Source materials / references          | <ul style="list-style-type: none"> <li>v28 code/theory/review validation</li> <li>current YAML master</li> <li>static repo inventory and previous reports</li> </ul> |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>architecture decision gate</li> <li>workflow policy</li> <li>proof-dossier object</li> </ul>                                  |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>inherits global no-go guardrails;<br/>no local weakening allowed</li> </ul>                                                   |

### Phase 1.1.1.1.0.5 – Phase-type-specific scratchpad schema and governance-depth review gate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field         | Record                                            |
|---------------|---------------------------------------------------|
| Phase status  | <span style="background-color: #d9ead3;">G</span> |
| Context class | LeanGovernancePhase                               |

|                                        |                                                                                                                                                                                                                            |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase documentation tier               | F3                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                  |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | completed                                                                                                                                                                                                                  |
| Preparation status                     | completed                                                                                                                                                                                                                  |
| Phase type                             | infrastructure                                                                                                                                                                                                             |
| Phase-type-specific contract           | Infrastructure/planning phase: justify absence of operational witness; close schema/decision prerequisites before code.                                                                                                    |
| Title / state                          | Phase-type-specific scratchpad schema and governance-depth review gate                                                                                                                                                     |
| Preparation workflow                   | Architecture decision recorded before active implementation gate.                                                                                                                                                          |
| Source-to-CNNA mapping                 | Decision mapped to LEG object and workflow/format-decision rows; no ontic CNNA input introduced.                                                                                                                           |
| Dependency-depth audit                 | Depends on previous green architecture gate before active phase 1.1.1.1.1; closes a review-identified decision gap without changing Lean code.                                                                             |
| Phase split decision                   | Inserted as green pre-active infrastructure gate under 1.1.1.1.0.x so no white predecessor remains before the active cursor.                                                                                               |
| Positive path plan                     | Record phase_type for every phase and render type-specific fields. Use different required evidence for refactor, derivation, infrastructure, documentation and comparison phases.                                          |
| Module implementation manifest         | No code modules introduced by this gate.                                                                                                                                                                                   |
| Semantic placement decision            | Governance belongs in YAML master and future optional architecture policy modules only if code-level records become useful.                                                                                                |
| Reuse and duplication audit            | No duplicate implementation authorized; decision reduces future duplication risk.                                                                                                                                          |
| Canonical generator policy             | Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation.                                                                                                                   |
| Module compactness rule                | Do not split code solely for this planning decision; encode as workflow/format decision unless a reusable code-level policy module is later justified.                                                                     |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                           |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review             | Closed as future-retrospective requirement: after Phase 1.1.2, record which governance gates prevented actual errors and decide whether Phase 3+ can flatten structure.                                                    |
| Proof rehearsal before implementation  | Inspect a refactor phase and a derivation phase: the refactor must not fake proof rehearsal; the derivation must not omit proof/tactic/mathlib evidence.                                                                   |
| Expected tactics                       | None; no Lean proof in this planning gate.                                                                                                                                                                                 |
| Expected mathlib imports               | None for planning gate.                                                                                                                                                                                                    |
| ExecComplex/computable plan            | None for this planning/infrastructure gate unless explicitly stated.                                                                                                                                                       |
| Concrete computable instance traversal | Infrastructure-only gate: no operational Lean object introduced. Later operational phases must provide concrete finite instance traversal.                                                                                 |
| Computable path non-skeleton status    | infrastructure-only; this gate adds obligations that prevent later operational skeletons.                                                                                                                                  |
| Expected executable witness            | None for planning gate.                                                                                                                                                                                                    |
| Mathlib complication log               | No direct mathlib import in this planning gate. Future phases must log complications.                                                                                                                                      |
| Noncomputable boundary audit           | No direct Lean boundary touched; future phases must audit noncomputable/classical pressure.                                                                                                                                |
| Mathlib mitigation/refactor layer      | If future mathlib pressure appears, isolate in mirror/non-operational layer or backjump to earliest natural source layer.                                                                                                  |

|                                      |                                                                                                                                                                      |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bridge gap status                    | Closed as planning decision;<br>implementation bridges remain future phase obligations.                                                                              |
| Natural backjump/refactor layer      | If implementation later violates this decision, backjump to the earliest natural layer recorded by the relevant workflow.                                            |
| Implementation prompt contract       | Future implementation prompts must cite this gate when relevant.                                                                                                     |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                          |
| How implemented                      | Planning/architecture gate closed in YAML master;<br>no Lean source changed.                                                                                         |
| Verification / evidence              | Generated tables/PDF render the gate. No Lean build was run or claimed.                                                                                              |
| Proof strategy                       | Schema-level planning proof, not Lean theorem.                                                                                                                       |
| Build result                         | No Lean build run;<br>planning artifact generated only.                                                                                                              |
| Post-implementation plan check       | Satisfied at planning level;<br>active cursor remains 1.1.1.1.1.                                                                                                     |
| Artifact packaging rule              | Return complete zip/PDF/report for planning updates. Lean build success may be recorded only from an actual current/user-reported build.                             |
| Implementation notes                 | This is a pre-implementation decision gate;<br>it constrains later code phases without modifying code itself.                                                        |
| Source materials / references        | <ul style="list-style-type: none"> <li>v28 code/theory/review validation</li> <li>current YAML master</li> <li>static repo inventory and previous reports</li> </ul> |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>architecture decision gate</li> <li>workflow policy</li> <li>proof-dossier object</li> </ul>                                  |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>inherits global no-go guardrails;<br/>no local weakening allowed</li> </ul>                                                   |

### Phase 1.1.1.1.0.6 – Phase 3 to Phase 4 consumer-contract direction lock

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                              |
| Context class                         | LeanGovernancePhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | completed                                                                                                                                                                                      |
| Preparation status                    | completed                                                                                                                                                                                      |
| Phase type                            | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness;<br>close schema/decision prerequisites before code.                                                                     |
| Title / state                         | Phase 3 to Phase 4 consumer-contract direction lock                                                                                                                                            |
| Preparation workflow                  | Architecture decision recorded before active implementation gate.                                                                                                                              |
| Source-to-CNNA mapping                | Decision mapped to LEG object and workflow/format-decision rows;<br>no ontic CNNA input introduced.                                                                                            |

|                                        |                                                                                                                                                                                                                               |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependency-depth audit                 | Depends on previous green architecture gate before active phase 1.1.1.1.1;<br>closes a review-identified decision gap without changing Lean code.                                                                             |
| Phase split decision                   | Inserted as green pre-active infrastructure gate under 1.1.1.1.0.x so no white predecessor remains before the active cursor.                                                                                                  |
| Positive path plan                     | Make Phase 3.6 publish a BoundaryOperatorProfile shaped for Phase 4.1. Keep import direction one-way: Phase 4 consumes Phase 3;<br>Phase 3 never imports Phase 4.                                                             |
| Module implementation manifest         | No code modules introduced by this gate.                                                                                                                                                                                      |
| Semantic placement decision            | Governance belongs in YAML master and future optional architecture policy modules only if code-level records become useful.                                                                                                   |
| Reuse and duplication audit            | No duplicate implementation authorized;<br>decision reduces future duplication risk.                                                                                                                                          |
| Canonical generator policy             | Generator-like logic must consume the canonical smoke/theory generator path;<br>no parallel implementation.                                                                                                                   |
| Module compactness rule                | Do not split code solely for this planning decision;<br>encode as workflow/format decision unless a reusable code-level policy module is later justified.                                                                     |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                             |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                   |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                           |
| Consumer-contract direction            | Closed: Phase 3.6 exports consumer-ready profile;<br>Phase 4.1 consumes;<br>no reverse import and no duplicate generator/Schur implementation.                                                                                |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation  | At Phase 3.6 preparation, write the exact fields Phase 4.1 needs. At Phase 4.1 preparation, show it consumes those fields and does not reconstruct Schur/DtN internals.                                                       |
| Expected tactics                       | None;<br>no Lean proof in this planning gate.                                                                                                                                                                                 |
| Expected mathlib imports               | None for planning gate.                                                                                                                                                                                                       |
| ExecComplex/computable plan            | None for this planning/infrastructure gate unless explicitly stated.                                                                                                                                                          |
| Concrete computable instance traversal | Infrastructure-only gate: no operational Lean object introduced. Later operational phases must provide concrete finite instance traversal.                                                                                    |
| Computable path non-skeleton status    | infrastructure-only;<br>this gate adds obligations that prevent later operational skeletons.                                                                                                                                  |
| Expected executable witness            | None for planning gate.                                                                                                                                                                                                       |
| Mathlib complication log               | No direct mathlib import in this planning gate. Future phases must log complications.                                                                                                                                         |
| Noncomputable boundary audit           | No direct Lean boundary touched;<br>future phases must audit noncomputable/classical pressure.                                                                                                                                |
| Mathlib mitigation/refactor layer      | If future mathlib pressure appears, isolate in mirror/non-operational layer or backjump to earliest natural source layer.                                                                                                     |
| Bridge gap status                      | Closed as planning decision;<br>implementation bridges remain future phase obligations.                                                                                                                                       |
| Natural backjump/refactor layer        | If implementation later violates this decision, backjump to the earliest natural layer recorded by the relevant workflow.                                                                                                     |
| Implementation prompt contract         | Future implementation prompts must cite this gate when relevant.                                                                                                                                                              |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                   |
| How implemented                        | Planning/architecture gate closed in YAML master;<br>no Lean source changed.                                                                                                                                                  |
| Verification / evidence                | Generated tables/PDF render the gate. No Lean build was run or claimed.                                                                                                                                                       |
| Proof strategy                         | Schema-level planning proof, not Lean theorem.                                                                                                                                                                                |
| Build result                           | No Lean build run;<br>planning artifact generated only.                                                                                                                                                                       |

|                                      |                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Post-implementation plan check       | Satisfied at planning level;<br>active cursor remains 1.1.1.1.1.                                                                                                 |
| Artifact packaging rule              | Return complete zip/PDF/report for planning updates. Lean build success may be recorded only from an actual current/user-reported build.                         |
| Implementation notes                 | This is a pre-implementation decision gate;<br>it constrains later code phases without modifying code itself.                                                    |
| Source materials / references        | <ul style="list-style-type: none"><li>v28 code/theory/review validation</li><li>current YAML master</li><li>static repo inventory and previous reports</li></ul> |
| Extracted established theory objects | <ul style="list-style-type: none"><li>architecture decision gate</li><li>workflow policy</li><li>proof-dossier object</li></ul>                                  |
| No-go guardrails                     | <ul style="list-style-type: none"><li>inherits global no-go guardrails;<br/>no local weakening allowed</li></ul>                                                 |

Phase 1.1.1.1.0.7 – IdealThread/CutSpec.full extremal-route contract and string-analogy boundary lock

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3; border: 1px solid #000; padding: 0 2px;">G</span>                                                                                                           |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                 |
| Phase documentation tier              | F1                                                                                                                                                                                                  |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.      |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                 |
| Implementation status                 | completed                                                                                                                                                                                           |
| Preparation status                    | completed                                                                                                                                                                                           |
| Phase type                            | infrastructure                                                                                                                                                                                      |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness;<br>close schema/decision prerequisites before code.                                                                          |
| Title / state                         | IdealThread/CutSpec.full extremal-route contract and string-analogy boundary lock                                                                                                                   |
| Preparation workflow                  | Architecture decision recorded before active implementation gate.                                                                                                                                   |
| Source-to-CNNA mapping                | Mapped verified code facts to a workflow/format-decision row, LEG proof dossier, and future neutral route phases;<br>no analogy supplies CNNA-generative data.                                      |
| Dependency-depth audit                | Depends on previous green architecture gates and closes an additional review-identified decision gap before active phase 1.1.1.1.1.                                                                 |
| Phase split decision                  | Inserted as green pre-active infrastructure gate under 1.1.1.1.0.x so no white predecessor remains before the active cursor.                                                                        |
| Positive path plan                    | Close the route decision before any module manifest: IdealThread is the singleton/thread route;<br>CutSpec.full is the full-carrier route;<br>the string-like reading is a downstream analogy only. |
| Module implementation manifest        | No code modules introduced by this gate;<br>future module candidates are S07/S08/BR06/BR07 phases.                                                                                                  |
| Semantic placement decision           | Place code-backed route facts in later neutral structural modules;<br>place string-like interpretation only in secondary/documentation side roads.                                                  |
| Reuse and duplication audit           | Reuse existing ToC.Contract, CutSpec.full, ExistingIdealThreadNoGo, FullBoundaryAudit, RecursiveSchurSource and MobiusTransfer;<br>no duplicate route implementation authorized.                    |
| Canonical generator policy            | Generator-like logic remains on the canonical smoke/theory generator path. IdealThread/string analogy may not instantiate a parallel generator.                                                     |



|                                        |                                                                                                                                                                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module compactness rule                | Do not split code solely for this planning decision;<br>encode as workflow/format decision unless a reusable code-level policy module is later justified.                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                    |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                          |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.        |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                |
| Consumer-contract direction            | Later phases consume route profiles one-way: S07/S08 feed BR06;<br>BR06 feeds Phase 3/4;<br>secondary analogy consumes dossiers only and never feeds A-generation.                                                                   |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                               |
| Proof rehearsal before implementation  | Code facts: IdealThread.carrier is a singleton;<br>IdealThread.family embeds it as DirectedFamily;<br>deterministic coarsen gives previous singleton;<br>CutSpec.full carrier is T.carrier. Therefore route conflation is forbidden. |
| Expected tactics                       | None;<br>no Lean proof in this planning gate.                                                                                                                                                                                        |
| Expected mathlib imports               | None for planning gate.                                                                                                                                                                                                              |
| ExecComplex/computable plan            | None for this planning/infrastructure gate unless explicitly stated.                                                                                                                                                                 |
| Concrete computable instance traversal | Infrastructure-only gate;<br>it requires later S07/S08/BR06/BR07 phases to show concrete finite reference/variation/full-cut instances before operational use.                                                                       |
| Computable path non-skeleton status    | infrastructure-only;<br>explicitly prevents later operational skeletons by requiring route-specific finite witnesses.                                                                                                                |
| Expected executable witness            | None for this planning gate;<br>later route phases must name the finite instance.                                                                                                                                                    |
| Mathlib complication log               | No new mathlib import in this planning gate.                                                                                                                                                                                         |
| Noncomputable boundary audit           | No Lean boundary touched;<br>later route phases must audit noncomputable/classical pressure if they import additional mathlib APIs.                                                                                                  |
| Mathlib mitigation/refactor layer      | If future mathlib pressure appears, isolate in mirror/non-operational layer or backjump to earliest natural source layer.                                                                                                            |
| Bridge gap status                      | Closed as planning decision;<br>implementation bridges remain future phase obligations.                                                                                                                                              |
| Natural backjump/refactor layer        | If implementation later violates this decision, backjump to the earliest natural layer recorded by the relevant workflow.                                                                                                            |
| Implementation prompt contract         | Future implementation prompts must cite this gate when relevant.                                                                                                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                          |
| How implemented                        | Planning/architecture gate closed in YAML master;<br>no Lean source changed.                                                                                                                                                         |
| Verification / evidence                | Existing Lean source was inspected for route facts;<br>generated tables/PDF render the gate. No Lean build was run or claimed.                                                                                                       |
| Proof strategy                         | Schema-level planning proof, not Lean theorem.                                                                                                                                                                                       |
| Build result                           | No Lean build run;<br>planning artifact generated only.                                                                                                                                                                              |
| Post-implementation plan check         | Satisfied at planning level;<br>active cursor remains 1.1.1.1.1.                                                                                                                                                                     |
| Artifact packaging rule                | Return complete zip/PDF/report for planning updates. Lean build success may be recorded only from an actual current/user-reported build.                                                                                             |
| Implementation notes                   | This is a pre-implementation route decision gate. It constrains later code phases without moving Lean declarations.                                                                                                                  |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source materials / references        | <ul style="list-style-type: none"> <li>• <code>CNNA/PillarA/Foundation/SubstrateClass.lean</code></li> <li>• <code>CNNA/PillarA/ToC/Contract.lean</code></li> <li>• <code>CNNA/PillarA/Finite/CutSpec.lean</code></li> <li>• <code>CNNA/PillarA/Arithmetic/BoundarySource/ExistingIdealThreadNoGo.lean</code></li> <li>• <code>CNNA/PillarA/Arithmetic/Core/FullBoundaryAudit.lean</code></li> </ul> |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>• <code>IdealThread</code> singleton route</li> <li>• <code>CutSpec.full</code> maximal route</li> <li>• Boundary-route extrema contract</li> <li>• String-analogy comparison boundary</li> </ul>                                                                                                                                                             |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase adds no weakening.</li> <li>• no string-like analogy as CNNA-generative input</li> <li>• do not use <code>IdealThread</code> singleton route as full-boundary matrix source without a lifting theorem</li> </ul>                                                         |

### Phase 1.1.1.1.0.8 – Horizontal/vertical Schur-complement duality dependency-order lock

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3; border: 1px solid #000; padding: 0 2px;">G</span>                                                                                                                                                                        |
| Context class                         | <code>LeanGovernancePhase</code>                                                                                                                                                                                                                                 |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                               |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                   |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                               |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                              |
| Implementation status                 | completed                                                                                                                                                                                                                                                        |
| Preparation status                    | completed                                                                                                                                                                                                                                                        |
| Phase type                            | infrastructure                                                                                                                                                                                                                                                   |
| Phase-type-specific contract          | Infrastructure/planning phase: close bridge-order and no-input constraints before code; no operational witness required.                                                                                                                                         |
| Title / state                         | Horizontal/vertical Schur-complement duality dependency-order lock                                                                                                                                                                                               |
| Preparation workflow                  | Architecture decision recorded before active implementation gate.                                                                                                                                                                                                |
| Source-to-CNNA mapping                | Mapped verified code facts to a workflow/format-decision row, LEG proof dossier, and future neutral route phases; no analogy supplies CNNA-generative data.                                                                                                      |
| Dependency-depth audit                | The bridge requires S05, O05, TM0-TM2, CD7, BR06, BR07, S07 and S08. Since these are later or unclosed, only the order lock is green; the bridge itself is white/future.                                                                                         |
| Phase split decision                  | Inserted as green pre-active infrastructure gate 1.1.1.1.0.8; actual mathematics placed later as 10.4.1 and 10.4.2 so no white predecessor appears before the active cursor.                                                                                     |
| Positive path plan                    | Preserve the positive insight by registering BR08/BR09 and by moving CD8 final proof after BR08. Earlier phases export data; they do not consume the duality conclusion.                                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                   |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy_coexistence</code> status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                       |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                               |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consumer-contract direction            | One-way only: earlier route phases export S05/O05/TM0/CD7/BR06/BR07 to BR08; BR08/CD8 may not backfeed into their definitions.                                                                                                                                                                                                                                                           |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                                                      |
| Proof rehearsal before implementation  | Planning proof: if a theorem depends on vertical transfermatrix data TM0 and horizontal boundary operator S05, it cannot be scheduled before those routes. Therefore the bridge is locked as a later consumer of both routes.                                                                                                                                                            |
| Expected tactics                       | None; no Lean proof in this planning gate.                                                                                                                                                                                                                                                                                                                                               |
| Expected mathlib imports               | None for planning gate.                                                                                                                                                                                                                                                                                                                                                                  |
| ExecComplex/computable plan            | None for this planning/infrastructure gate unless explicitly stated.                                                                                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only gate; it requires later S07/S08/BR06/BR07 phases to show concrete finite reference/variation/full-cut instances before operational use.                                                                                                                                                                                                                              |
| Computable path non-skeleton status    | infrastructure-only; explicitly prevents later operational skeletons by requiring route-specific finite witnesses.                                                                                                                                                                                                                                                                       |
| Expected executable witness            | None for this planning gate; later route phases must name the finite instance.                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No new mathlib import in this planning gate.                                                                                                                                                                                                                                                                                                                                             |
| Noncomputable boundary audit           | No Lean boundary touched; later route phases must audit noncomputable/classical pressure if they import additional mathlib APIs.                                                                                                                                                                                                                                                         |
| Mathlib mitigation/refactor layer      | If future mathlib pressure appears, isolate in mirror/non-operational layer or backjump to earliest natural source layer.                                                                                                                                                                                                                                                                |
| Bridge gap status                      | Closed as an order/governance decision; mathematical bridge remains BR08 future target.                                                                                                                                                                                                                                                                                                  |
| Natural backjump/refactor layer        | If implementation later violates this decision, backjump to the earliest natural layer recorded by the relevant workflow.                                                                                                                                                                                                                                                                |
| Implementation prompt contract         | Future implementation prompts must cite this gate when relevant.                                                                                                                                                                                                                                                                                                                         |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                 |
| How implemented                        | Planning/architecture gate closed in YAML master; no Lean source changed.                                                                                                                                                                                                                                                                                                                |
| Verification / evidence                | Planning master updated and generated views validate the dependency-order lock. No Lean build was run or claimed.                                                                                                                                                                                                                                                                        |
| Build result                           | No Lean build required for planning gate; PDF/table generation only.                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Satisfied at planning level; active cursor remains 1.1.1.1.1.                                                                                                                                                                                                                                                                                                                            |
| Artifact packaging rule                | Return complete zip/PDF/report for planning updates. Lean build success may be recorded only from an actual current/user-reported build.                                                                                                                                                                                                                                                 |
| Implementation notes                   | This planning gate accepts the Schur-unification insight only as a dependency-ordered future theorem target. It adds no Lean definitions.                                                                                                                                                                                                                                                |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase adds no weakening.</li> <li>do not use horizontal/vertical Schur-duality as an input to Phase 1.4, 3, 4, 5, or 6</li> <li>do not close CD8 before BR08</li> <li>do not convert subsystem/string analogy into a theorem without checked Schur decomposition proof</li> </ul> |

### Phase 1.1.1.1.0.9 – Heuristic Eisenstein/Gauss lattice test-matrix boundary and ordering lock

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                    | Record                                                                                    |
|--------------------------|-------------------------------------------------------------------------------------------|
| Phase status             | <span style="background-color: #d9ead3; border: 1px solid #000; padding: 0 2px;">G</span> |
| Context class            | LeanGovernancePhase                                                                       |
| Phase documentation tier | F1                                                                                        |

|                                        |                                                                                                                                                                                                                                                                        |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                         |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                     |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                    |
| Implementation status                  | completed                                                                                                                                                                                                                                                              |
| Preparation status                     | completed                                                                                                                                                                                                                                                              |
| Phase type                             | infrastructure                                                                                                                                                                                                                                                         |
| Phase-type-specific contract           | infrastructure decision                                                                                                                                                                                                                                                |
| Title / state                          | Heuristic Eisenstein/Gauss lattice test-matrix boundary and ordering lock                                                                                                                                                                                              |
| Preparation workflow                   | Infrastructure / planning gate.                                                                                                                                                                                                                                        |
| Source-to-CNNA mapping                 | post #97 maps to a future consumer-only test matrix: L09/L10/M08/BR10. It does not define b, the generator, tau, or j.                                                                                                                                                 |
| Dependency-depth audit                 | The test matrix must come after generator/branching, EXEC-2, lattice angle/cosine, tau/CM and j target phases. Therefore it is registered now as a decision lock and implemented later as 8.5-8.7.                                                                     |
| Phase split decision                   | Split into green planning lock 1.1.1.1.0.9 plus future phases 8.5 static test matrix, 8.6 dynamic branch sweep, and 8.7 closeout/public-boundary audit.                                                                                                                |
| Positive path plan                     | Use the positive path: construct explicit test records and only then compare against special cases;<br>do not start by obstruction.                                                                                                                                    |
| Refactor regression protocol           | No refactor performed in this phase;<br>future code tests must keep the full local build green after each wrapper/test addition.                                                                                                                                       |
| Refactor migration metrics             | Not applicable until implementation.                                                                                                                                                                                                                                   |
| Lean/mathlib version policy            | Lean/mathlib baseline remains 4.28.0 unless an explicit upgrade gate is opened.                                                                                                                                                                                        |
| ExecComplex boundary policy            | EXEC-2 applies: QxQ operational boundary;<br>irrational geometry and j analytic values are mirror/comparison targets only.                                                                                                                                             |
| Consumer-contract direction            | Tests consume generator/geometry/CM outputs;<br>they do not define those outputs.                                                                                                                                                                                      |
| Governance-overhead review             | This gate prevents a likely false implementation path: injecting angle heuristics into the generator rather than testing them as comparison targets.                                                                                                                   |
| Proof rehearsal before implementation  | For future finite tests: build rows from code-defined b and L_max;<br>compute address counts/relations and rational cos/Gram/D surrogates;<br>compare mirror-only tau/j targets after their source objects exist.                                                      |
| Expected tactics                       | Finite structural constructors, explicit Fin/Nat/address enumeration, relation witness construction, projection lemmas;<br>no analytic complex simplification in operational path.                                                                                     |
| Expected mathlib imports               | Use existing project imports first;<br>future CM/j mirror tests may require mathlib modular/complex APIs only in non-operational layers after import existence is checked.                                                                                             |
| ExecComplex/computable plan            | Operational path may record only rational/integer outputs: branching code value, finite addresses, collision relations, cos(theta), integer/rational Gram invariants, D target. Complex coordinates, tau, sqrt(D), and analytic j evaluation remain mirror/comparison. |
| Concrete computable instance traversal | Required for future implementation: at least b=2 with L_max=3 and 4, plus one code-defined dynamic-branching case if available;<br>b=0/1 as degenerate controls only if the code exposes them legally.                                                                 |
| Computable path non-skeleton status    | Future test implementation must actually traverse finite addresses and compute expected finite invariants, not merely instantiate empty profile records.                                                                                                               |
| Expected executable witness            | Future witness: finite test matrix rows for Eisenstein and Gauss cases over selected L_max values, using the canonical generator/branching API.                                                                                                                        |
| Mathlib complication log               | Likely complication: analytic j/tau/sqrt(D) APIs may be noncomputable;<br>keep those in mirror/comparison layer and record any import/noncomputable boundary explicitly.                                                                                               |
| Noncomputable boundary audit           | Operational tests must avoid noncomputable analytic complex evaluation. Mirror tests may state comparison targets but must not feed operational definitions.                                                                                                           |
| Mathlib mitigation/refactor layer      | Use rational/integer surrogates in operational layer;<br>isolate analytic or algebraic-irrational material in mirror/profile modules.                                                                                                                                  |
| Bridge gap status                      | Open until L06/L04/M05 and the generator route are separately closed.                                                                                                                                                                                                  |
| Natural backjump/refactor layer        | If tests require new generator semantics, backjump to the canonical generator layer;<br>do not patch the test module.                                                                                                                                                  |
| Implementation prompt contract         | Before code: provide module manifest, reuse audit, concrete finite instances, and test matrix rows;<br>do not create duplicate generators.                                                                                                                             |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                            |

|                                      |                                                                                                                                                                                                                                                                                                               |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How implemented                      | Decision lock added to the YAML master and generated views;<br>no Lean code was changed.                                                                                                                                                                                                                      |
| Verification / evidence              | YAML/master validation and generated-view compilation;<br>no Lean build run.                                                                                                                                                                                                                                  |
| Build result                         | No Lean build executed.                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check       | After implementation, verify the test matrix uses code-defined branching and did not duplicate generator logic.                                                                                                                                                                                               |
| Artifact packaging rule              | Return complete zip;<br>no Lean-build success may be claimed unless actually run.                                                                                                                                                                                                                             |
| Source materials / references        | <ul style="list-style-type: none"> <li>TomS thread post #97 / antaris b-ary tree Eisenstein-Gauss heuristic construction</li> <li>Existing EXEC-2 rational-boundary policy</li> <li>Canonical generator/reuse policy</li> </ul>                                                                               |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>b-ary address words over code-defined branching</li> <li>angle profiles 120 degrees / 90 degrees</li> <li>closure-depth targets L=3 / L=4</li> <li>CM targets rho and i</li> <li>j targets 0 and 1728</li> <li>hierarchical-vs-spiral embedding distinction</li> </ul> |
| Modified modules                     | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>generated tables/PDF views</li> </ul>                                                                                                                                                                                        |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no generator semantics imported from angle drawings</li> <li>no noncomputable analytic j in operational path</li> <li>no duplicate b-ary generator implementation</li> </ul>                                                                                           |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>Inherit global guardrails.</li> <li>Do not use geometric heuristic as generator input.</li> <li>Do not put irrational tau/coordinates in the operational ExecComplex path.</li> </ul>                                                                                  |

### Phase 1.1.1.1.0.10 – General early finite computation test-matrix contract and non-specialization lock

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                              |
| Context class                         | LeanGovernancePhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | completed                                                                                                                                                                                      |
| Preparation status                    | closed-infrastructure-contract                                                                                                                                                                 |
| Phase type                            | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract          | Global test-governance contract;<br>it adds required fields to later derivation/refactor manifests but does not itself implement Lean code.                                                    |
| Title / state                         | General early finite computation test-matrix contract and non-specialization lock                                                                                                              |
| Preparation workflow                  | General finite computation-test workflow;<br>extends the concrete computable instance traversal workflow.                                                                                      |

|                                        |                                                                                                                                                                                                                                                                                    |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source-to-CNNA mapping                 | The post-97 Eisenstein/Gauss rows map to sentinel rows;<br>the governing CNNA object is now the general finite computation test matrix O07. L09/L10/M08/BR10 remain specialized later consumers.                                                                                   |
| Dependency-depth audit                 | Must precede Phase 1.1.1.1.1 because module manifests must decide where generic tests live before code is written. It does not depend on later CM/j phases except for sentinel comparison columns.                                                                                 |
| Phase split decision                   | Added as its own green infrastructure gate because the requirement changes test semantics globally, not only Phase 8.5.                                                                                                                                                            |
| Positive path plan                     | Use generic tests first: construct finite generated instances, run the operational/computable path, export rational/integer/finite witnesses, then attach sentinel rows such as Eisenstein/Gauss as comparison rows where dependencies permit.                                     |
| Refactor regression protocol           | No code movement here. Later refactors must keep generic test rows buildable as migration sentinels.                                                                                                                                                                               |
| Refactor migration metrics             | Future refactor phases must report declarations migrated plus generic/sentinel test rows still passing.                                                                                                                                                                            |
| Lean/mathlib version policy            | Version impacts are logged per test field. A mathlib upgrade may be required only at phase boundaries, never inside a half-migrated test harness.                                                                                                                                  |
| ExecComplex boundary policy            | EXEC-2 binding: generic operational fields are finite/rational/integer;<br>non-rational geometry remains mirror/comparison.                                                                                                                                                        |
| Consumer-contract direction            | Tests consume canonical generator and operational profiles. They do not define generator semantics.                                                                                                                                                                                |
| Governance-overhead review             | This gate is justified only if it prevents special-case tunnel vision during the first computation phases;<br>review after first real code-test package.                                                                                                                           |
| Proof rehearsal before implementation  | A future test closes only if at least one generic row family is generated from code-defined data, not hand-coded to the expected CM result. Sentinel rows may fail without invalidating the generic test harness;<br>generic rows failing blocks promotion of the computable path. |
| Expected tactics                       | Not a Lean proof phase. Future Lean tests should prefer explicit constructors, finite enumerations, and proof-carrying row records over rewriting chains.                                                                                                                          |
| Expected mathlib imports               | No new mathlib import now. Future test modules must log whether finite matrix, polynomial, or number-theory APIs force noncomputable pressure.                                                                                                                                     |
| ExecComplex/computable plan            | Operational test fields remain finite/rational/integer. Algebraic coordinates, tau, sqrt(D), analytic j, and non-rational angles are mirror/comparison fields only.                                                                                                                |
| Concrete computable instance traversal | Defines the global obligation: every operational computable phase must include or point to finite generated rows that actually traverse the path, not only typecheck a skeleton.                                                                                                   |
| Computable path non-skeleton status    | strengthened: a path is non-skeletal only when a concrete finite input produces a proof-carrying executable witness or an explicitly documented failure row.                                                                                                                       |
| Expected executable witness            | Generic row witness: finite input record -> output record with rational/integer/finite fields and proof/status flags;<br>sentinel comparison fields are separate.                                                                                                                  |
| Mathlib complication log               | Future rows must log mathlib/noncomputable barriers per field, especially matrix inverses, polynomial APIs, discriminant APIs, and analytic j/elliptic APIs.                                                                                                                       |
| Noncomputable boundary audit           | Any noncomputable pressure must be isolated to mirror/comparison layer or trigger a natural refactor;<br>it cannot leak into operational row generation.                                                                                                                           |
| Mathlib mitigation/refactor layer      | Prefer executable finite constructors and proof-carrying row records. Use mirror layer for analytic/algebraic-irrational targets.                                                                                                                                                  |
| Bridge gap status                      | This closes a planning gap: v32 had specialized lattice-test phases, but lacked a general early test-matrix obligation for all computable paths.                                                                                                                                   |
| Natural backjump/refactor layer        | If a future test requires a duplicate generator or hand-coded special branch logic, backjump to the canonical generator layer or module manifest instead of patching in the test module.                                                                                           |
| Implementation prompt contract         | Before implementing a computation phase, specify the generic test matrix rows, code-defined input generator, finite instance coverage, expected executable witness, and sentinel rows separately.                                                                                  |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                        |
| How implemented                        | Added as a planning/infrastructure contract. It generalizes the v32 Eisenstein/Gauss test idea: those rows become sentinel cases in a broader finite test matrix over the code-defined generator/branching API, static/dynamic branch modes, L_max rows, and executable witnesses. |
| Verification / evidence                | YAML validation target: phase inserted before active cursor;<br>no white predecessors before active cursor;<br>LEG29/O07 registered with proof dossiers;<br>workflow/global guardrail/format decisions updated.                                                                    |
| Proof strategy                         | Planning proof: check that generic finite test-matrix obligations are required before special sentinel rows can close;<br>check that the phase/order does not feed heuristic data into generator semantics.                                                                        |
| Build result                           | No Lean build. PDF generation only after table regeneration.                                                                                                                                                                                                                       |
| Post-implementation plan check         | After the first code-change phase, review whether generic finite tests caught an architectural error and whether the harness should be flattened for later phases.                                                                                                                 |
| Artifact packaging rule                | Return full zip;<br>do not claim Lean build success unless executed locally/currently.                                                                                                                                                                                             |
| Implementation notes                   | This is not a mathematical theorem and not a Lean implementation. It prevents the first computation phases from becoming special-case demonstrations instead of general computable-path tests.                                                                                     |
| Source materials / references          | <ul style="list-style-type: none"> <li>v32 heuristic Eisenstein/Gauss test-matrix lock</li> <li>user clarification: early theory testing now, general rather than overspecialized</li> <li>existing canonical generator and smoke-test generator policy</li> </ul>                 |



|                                      |                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"><li>generic finite instance row</li><li>static branching mode</li><li>dynamic branching mode</li><li>L_max sweep row</li><li>expected executable witness</li><li>sentinel row for Eisenstein/Gauss only after generic coverage exists</li></ul>                                           |
| Modified modules                     | <ul style="list-style-type: none"><li>masters/cnna_roadmap_master.yaml</li><li>scripts/generate_tables.py</li><li>generated TeX/PDF views</li></ul>                                                                                                                                                                         |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>no special-case-only test matrix</li><li>no duplicate generator for testing</li><li>no CM/j target as operational input</li><li>no angle/lattice heuristic as branching definition</li></ul>                                                                                          |
| No-go guardrails                     | <ul style="list-style-type: none"><li>Do not use Eisenstein/Gauss as the whole test harness.</li><li>Do not duplicate generator logic for tests.</li><li>Do not turn mirror/CM/j columns into operational inputs.</li><li>Do not promote a computable path whose only passing examples are hand-picked sentinels.</li></ul> |

Phase 1.1.1.1.0.11 – Regime-typed early test matrix and Heegner-projection lock

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                  |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                                                       |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                                     |
| Phase documentation tier              | F1                                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                     |
| Implementation status                 | completed                                                                                                                                                                                                                               |
| Preparation status                    | closed                                                                                                                                                                                                                                  |
| Phase type                            | infrastructure                                                                                                                                                                                                                          |
| Phase-type-specific contract          | infrastructure lock for regime-typed tests                                                                                                                                                                                              |
| Title / state                         | Regime-typed early test matrix and Heegner-projection lock                                                                                                                                                                              |
| Preparation workflow                  | regime-typed general early finite computation test workflow                                                                                                                                                                             |
| Source-to-CNNA mapping                | Eisenstein/Gauss map to closed-root sentinel candidates only after noOuter/empty-set REAL-root route exists. Other Heegner or CM targets map to complement-projection/comparison rows unless a noOuter derivation is separately proven. |
| Dependency-depth audit                | No early mathematics consumes the classification as an input. It is a test/reporting schema before implementation and a proof target later.                                                                                             |
| Phase split decision                  | split from v33 O07 because generic-vs-sentinel is not enough; regime typing is a separate semantic axis.                                                                                                                                |
| Positive path plan                    | First run generic finite rows; then annotate rows by derived regime; then interpret Eisenstein/Gauss as noOuter-root sentinels and all other Heegner/CM rows as projections/comparison until proven otherwise.                          |
| Module implementation manifest        | pending Phase 1.1.1.1.1                                                                                                                                                                                                                 |
| Semantic placement decision           | test/schema layer, not generator layer                                                                                                                                                                                                  |

|                                        |                                                                                                                                                                                                                                                   |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reuse and duplication audit            | must reuse O07/canonical smoke-test generator;<br>no parallel Heegner test generator                                                                                                                                                              |
| Canonical generator policy             | consume the one canonical generator only                                                                                                                                                                                                          |
| Module compactness rule                | do not create a new module unless existing test/profile module cannot own the semantics                                                                                                                                                           |
| Refactor regression protocol           | no code movement in this gate;<br>future test implementation must preserve full local build green status                                                                                                                                          |
| Refactor migration metrics             | not applicable at planning gate                                                                                                                                                                                                                   |
| Lean/mathlib version policy            | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                                                                      |
| ExecComplex boundary policy            | EXEC-2 enforced                                                                                                                                                                                                                                   |
| Consumer-contract direction            | test matrix consumes regime/generator outputs;<br>regime/generator definitions do not consume tests                                                                                                                                               |
| Governance-overhead review             | review after Phase 1.1.2 whether this lock prevented false closed-system classification.                                                                                                                                                          |
| Proof rehearsal before implementation  | Prove/record row provenance: generator output -> finite witness -> regime tag -> optional comparison target. A row may not change the regime tag based on h(D), j, or external Heegner membership.                                                |
| Expected tactics                       | structural constructors/projections;<br>explicit row records;<br>no fragile rewrites;<br>no inverse oracle.                                                                                                                                       |
| Expected mathlib imports               | none at planning gate;<br>future number-theory phases may require mathlib version review.                                                                                                                                                         |
| ExecComplex/computable plan            | Operational fields remain finite/rational/integer. tau, sqrt(D), analytic j and general CM fields stay mirror-only under EXEC-2.                                                                                                                  |
| Concrete computable instance traversal | required later: every row must contain actual finite generator traversal evidence or remain unclosed                                                                                                                                              |
| Computable path non-skeleton status    | schema lock only now;<br>future rows must provide executable witnesses                                                                                                                                                                            |
| Expected executable witness            | finite row witness with branch mode, L_max, regime tag, rational/integer outputs, optional comparison target                                                                                                                                      |
| Mathlib complication log               | record any class number/CM/j mathlib dependency or noncomputable boundary explicitly                                                                                                                                                              |
| Noncomputable boundary audit           | Heegner/class-number and analytic j data are comparison/mirror until operational invariant is derived                                                                                                                                             |
| Mathlib mitigation/refactor layer      | use a mirror layer or later version-policy gate if mathlib forces noncomputable definitions                                                                                                                                                       |
| Bridge gap status                      | open until noOuter/HasOuter regime split and O07 implementation exist                                                                                                                                                                             |
| Natural backjump/refactor layer        | if a row needs regime data not yet produced, backjump to regime/source layer rather than patching the test consumer.                                                                                                                              |
| Implementation prompt contract         | Before Lean code, module manifest must state exact module placement and prove no duplicate generator/test harness.                                                                                                                                |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                       |
| How implemented                        | Completed as planning/infrastructure lock in YAML;<br>no Lean code change.                                                                                                                                                                        |
| Verification / evidence                | Generated-table validation pending/run after patch;<br>no Lean build.                                                                                                                                                                             |
| Proof strategy                         | planning-level dependency/order lock;<br>implementation proof obligations deferred to target phase                                                                                                                                                |
| Build result                           | No Lean build executed or claimed.                                                                                                                                                                                                                |
| Post-implementation plan check         | Check whether row classification forced an actual implementation decision;<br>update governance overhead review.                                                                                                                                  |
| Artifact packaging rule                | Return complete zip;<br>Lean build local unless actually executed here.                                                                                                                                                                           |
| Implementation notes                   | Added planning lock separating noOuter-root Eisenstein/Gauss sentinel candidates from HasOuter/complement-projection Heegner comparison rows.                                                                                                     |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA noOuter/HasOuter regime split;<br/>EXEC-2;<br/>general early finite test-matrix contract;<br/>post #97 heuristic;<br/>class-number-one/Heegner comparison list as external comparison only</li> </ul> |

|                                      |                                                                                                                                                                                                                                               |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Regime tag;<br/>noOuter root sentinel;<br/>HasOuter/complement-projection row;<br/>Heegner class-number-one comparison target;<br/>IDEAL_infty projection label</li> </ul>                             |
| Implemented modules                  | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                                                        |
| Added modules                        | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                                    |
| Modified modules                     | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml only</li> </ul>                                                                                                                                                       |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                                                                                          |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>Local additions: no Heegner-list membership as noOuter evidence;<br/>no closed/unitary interpretation for complement-projection rows;<br/>no Eisenstein/Gauss-only evidence for generality.</li> </ul> |
| Proposed added modules               | <ul style="list-style-type: none"> <li>none at decision gate;<br/>future target likely existing O07 test/profile surface</li> </ul>                                                                                                           |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml;<br/>generated views</li> </ul>                                                                                                                                       |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>canonical smoke-test generator;<br/>regime split modules after Phase 1.3.2;<br/>O07 test matrix</li> </ul>                                                                                             |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>Heegner row schema;<br/>CM sentinel profile;<br/>closed-system sentinel tags</li> </ul>                                                                                                                |

**Phase 1.1.1.1.0.12 – Computable scalar architecture decision: ExecComplex boundary, required exact quadratic layer, and Cayley-Dickson noncircularity lock**

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                       |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | completed                                                                                                                                                                                               |
| Preparation status                    | closed                                                                                                                                                                                                  |
| Phase type                            | infrastructure                                                                                                                                                                                          |
| Phase-type-specific contract          | infrastructure scalar architecture obligations                                                                                                                                                          |
| Title / state                         | Computable scalar architecture decision: ExecComplex boundary, required exact quadratic layer, and Cayley-Dickson noncircularity lock                                                                   |
| Preparation workflow                  | computable scalar architecture workflow                                                                                                                                                                 |
| Source-to-CNNA mapping                | ExecComplex remains O00. Exact quadratic layer is O09 if needed. CD-derived C layer is CD9/10.4.3 only, not a replacement for O00.                                                                      |
| Dependency-depth audit                | Scalar architecture decisions must precede code implementation. Exact quadratic or CD scalar routes may not feed earlier generator definitions.                                                         |
| Phase split decision                  | split out because ExecComplex limits affect Phase 1.2, 4, 5, 7, 8 and CD bridges differently.                                                                                                           |

|                                        |                                                                                                                                                                                                          |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive path plan                     | Keep current QxQ path first;<br>plan exact quadratic support as required for CM/lattice coordinates, but split schema from D-instantiation and keep analytic Complex mirror-only.                        |
| Module implementation manifest         | pending Phase 1.1.1.1.1                                                                                                                                                                                  |
| Semantic placement decision            | scalar-foundation / bridge governance, not generator semantics                                                                                                                                           |
| Reuse and duplication audit            | must check existing ExecComplex, ExecComplexBridge, Schur.CharpolyC, ArithmeticCDBridge, and mathlib QuadraticAlgebra before proposing new module                                                        |
| Canonical generator policy             | scalar layer must consume canonical generator outputs only;<br>cannot define generator semantics                                                                                                         |
| Module compactness rule                | do not create ExactQuadraticComplex unless existing foundation/bridge modules cannot carry the semantics cleanly                                                                                         |
| Refactor regression protocol           | no code movement in this gate                                                                                                                                                                            |
| Refactor migration metrics             | not applicable at planning gate                                                                                                                                                                          |
| Lean/mathlib version policy            | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                             |
| ExecComplex boundary policy            | EXEC-3/4 enforced: QxQ floor;<br>exact quadratic required downstream;<br>no free sqrt(D);<br>analytic Complex mirror-only.                                                                               |
| Consumer-contract direction            | scalar layer consumes generator/arithmetic source data;<br>generator definitions do not consume scalar comparison/mirror data                                                                            |
| Governance-overhead review             | review after first scalar/module manifest whether this lock forced a concrete decision.                                                                                                                  |
| Proof rehearsal before implementation  | Proof by dependency order: before any scalar code, classify each scalar requirement as QxQ, exact quadratic, invariant projection, or mirror-only.                                                       |
| Expected tactics                       | explicit structures, constructor/projection lemmas, ring-normal-form only for coordinate arithmetic, no Matrix.inv and no consumer-side patch.                                                           |
| Expected mathlib imports               | Mathlib.Algebra.QuadraticAlgebra only if pilot approved;<br>Mathlib.Data.Complex.Basic remains bridge/mirror-only;<br>NumberField modules require version-policy gate.                                   |
| ExecComplex/computable plan            | QxQ computations remain primary. Exact quadratic computations are not optional for later CM/lattice coordinates, but O09 is only a parameterized schema until O10/BR14 supply a proof-carrying D-source. |
| Concrete computable instance traversal | required later: QxQ baseline row, Q(i) row, Q(sqrt(-3)) row, and at least one generic D row if O09 is implemented                                                                                        |
| Computable path non-skeleton status    | future implementation must provide executable witnesses;<br>planning gate only now                                                                                                                       |
| Expected executable witness            | finite row with rational ExecComplex output plus optional exact quadratic coordinate and rational/integer invariant projection                                                                           |
| Mathlib complication log               | Record Complex/Real noncomputable boundary, QuadraticAlgebra instance diamond warning, NumberField formal/non-executable dependency, and any required version-gate.                                      |
| Noncomputable boundary audit           | General C/R and algebraic closure remain mirror/formal. Exact quadratic candidate must avoid noncomputable definitions on operational path.                                                              |
| Mathlib mitigation/refactor layer      | Use project-owned explicit Q-coordinate structure or narrowly wrapped QuadraticAlgebra;<br>isolate formal mathlib NumberField in mirror/proof layer.                                                     |
| Bridge gap status                      | Open until O09/BR13/CD9 are implemented or rejected.                                                                                                                                                     |
| Natural backjump/refactor layer        | If exact quadratic data are needed, backjump to scalar-foundation layer rather than patching arithmetic/lattice consumers.                                                                               |
| Implementation prompt contract         | Before Lean code, module manifest must state exact module placement, imported mathlib modules, and why no existing module owns the semantics.                                                            |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                              |
| How implemented                        | Completed as planning/infrastructure lock in YAML;<br>no Lean code change.                                                                                                                               |
| Verification / evidence                | Generated-table validation pending/run after patch;<br>no Lean build.                                                                                                                                    |
| Proof strategy                         | planning-level dependency/order lock;<br>implementation proof obligations deferred to target phase                                                                                                       |
| Build result                           | No Lean build executed or claimed.                                                                                                                                                                       |
| Post-implementation plan check         | Check whether the scalar architecture lock prevented analytic/noncomputable leakage or duplicate scalar implementation.                                                                                  |
| Artifact packaging rule                | Return complete zip;<br>Lean build local unless actually executed here.                                                                                                                                  |
| Implementation notes                   | Added EXEC-3 scalar architecture lock and future O09/BR13/CD9 phases.                                                                                                                                    |

|                                      |                                                                                                                                                                                                                               |
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| Source materials / references        | <ul style="list-style-type: none"> <li>Code audit of ExecComplex/ExecComplexBridge/ArithmeticCDBridge plus mathlib docs for Complex and QuadraticAlgebra.</li> </ul>                                                          |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Rational ExecComplex; analytic Complex mirror; exact quadratic candidate; CD noncircularity audit.</li> </ul>                                                                          |
| Implemented modules                  | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                                        |
| Added modules                        | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                    |
| Modified modules                     | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml only</li> </ul>                                                                                                                                       |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                                                                          |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>Local additions: no full Complex operational replacement; no CD-derived C-layer before noncircularity proof; no exact quadratic module without reuse/manifest audit.</li> </ul>        |
| Proposed added modules               | <ul style="list-style-type: none"> <li>none at decision gate; possible CNNA/PillarA/Foundation/ExactQuadraticComplex.lean only after manifest</li> </ul>                                                                      |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml; generated views</li> </ul>                                                                                                                           |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Foundation/ExecComplex.lean; Foundation/ExecComplexBridge.lean; Arithmetic/Schur/CharpolyC.lean; Arithmetic/CayleyDickson/ArithmeticCDBridge.lean; mathlib QuadraticAlgebra</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>alternate complex scalar, algebraic complex, CD complex, mirror Complex, exact quadratic norm/trace/conjugation</li> </ul>                                                             |

### Phase 1.1.1.1.0.13 – Derived-only exact quadratic algebra provenance lock: no free sqrt(D)

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                                       |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | completed                                                                                                                                                                                                               |
| Preparation status                    | closed                                                                                                                                                                                                                  |
| Phase type                            | infrastructure                                                                                                                                                                                                          |
| Phase-type-specific contract          | infrastructure phase: obligations recorded in proof and scratchpad fields.                                                                                                                                              |
| Title / state                         | Derived-only exact quadratic algebra provenance lock: no free sqrt(D)                                                                                                                                                   |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Source-to-CNNA mapping                | QuadraticAlgebra/AdjoinRoot supply reference patterns for adjoining a generator by a relation; CNNA uses them only after D has been derived by the operational Schur/DtN/arithmetic path.                               |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependency-depth audit                 | This gate must be green before the active module manifest: exact quadratic support is required later, but only downstream of O05/O06 or equivalent proof-carrying D-source. CD and analytic Complex cannot be upstream scalar sources.                                |
| Phase split decision                   | Closed decision gate.                                                                                                                                                                                                                                                 |
| Positive path plan                     | Do not postpone the need for exact quadratic data; plan it as required. But implement only a parameterized schema first, then instantiate it only after the operational discriminant ledger produces D.                                                               |
| Refactor regression protocol           | Apply global refactor regression protocol if code changes.                                                                                                                                                                                                            |
| Refactor migration metrics             | Pending or not applicable.                                                                                                                                                                                                                                            |
| Lean/mathlib version policy            | Use frozen Lean/mathlib baseline unless a phase-boundary upgrade gate is opened.                                                                                                                                                                                      |
| ExecComplex boundary policy            | EXEC-3/4 enforced: QxQ floor; exact quadratic required but discriminant-indexed; no free sqrt(D).                                                                                                                                                                     |
| Consumer-contract direction            | No consumer-side patch; missing D-source belongs earlier.                                                                                                                                                                                                             |
| Governance-overhead review             | Pending or not applicable.                                                                                                                                                                                                                                            |
| Proof rehearsal before implementation  | Given D from a ledger, define theta formally by a coordinate algebra with $\theta^2 = D$ . Prove coordinate addition, multiplication, conjugation, trace, norm, and allowed projection lemmas. No theorem may mention a free sqrt(D) object.                          |
| Expected tactics                       | No Lean tactics; planning lock only.                                                                                                                                                                                                                                  |
| Expected mathlib imports               | No import in this planning gate. Later implementation may audit Mathlib.Algebra.QuadraticAlgebra; AdjoinRoot is proof/mirror reference unless an executable wrapper is explicitly justified.                                                                          |
| ExecComplex/computable plan            | ExecComplex keeps rational/integer shadows. O09 carries exact quadratic coordinates only with a D-source; BR13/BR14 project trace/norm/discriminant/integer invariants back to allowed operational rows.                                                              |
| Concrete computable instance traversal | Later required rows: D=-1 for Gauss-like Q(i) if code source produces it, D=-3 for Eisenstein, D=-4 depending on discriminant convention, and at least one generic nonsquare D; every row must cite its D-ledger.                                                     |
| Computable path non-skeleton status    | planning lock only; operational witnesses required later                                                                                                                                                                                                              |
| Expected executable witness            | Future witnesses: D=-3, D=-4 and generic D rows only after proof-carrying D-source.                                                                                                                                                                                   |
| Mathlib complication log               | QuadraticAlgebra has coordinate operations suitable as reference; AdjoinRoot quotient construction may introduce noncomputable instances and is therefore not automatically operational.                                                                              |
| Noncomputable boundary audit           | No analytic Complex, Real, algebraic closure, or uninspected AdjoinRoot field structure may become operational. Any noncomputable mathlib dependency belongs to mirror/proof layer unless isolated.                                                                   |
| Mathlib mitigation/refactor layer      | Prefer project-owned executable coordinate layer or tightly audited mathlib wrapper.                                                                                                                                                                                  |
| Bridge gap status                      | Open theorem work is registered in O09/O10/BR14. The planning lock itself is closed.                                                                                                                                                                                  |
| Natural backjump/refactor layer        | If a D-source is missing, backjump to discriminant source, not to scalar consumer.                                                                                                                                                                                    |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                              |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | Completed for planning lock only.                                                                                                                                                                                                                                     |
| Artifact packaging rule                | Return a full zip; do not claim Lean build unless executed or explicitly user-reported green.                                                                                                                                                                         |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| Source materials / references          | <ul style="list-style-type: none"> <li>Mathlib.Algebra.QuadraticAlgebra.Defs</li> <li>Mathlib.RingTheory.AdjoinRoot</li> <li>existing ExecComplex and operational discriminant plan</li> <li>Cayley-Dickson noncircularity lock</li> </ul>                            |



|                                      |                                                                                                                                                                                                                                                                            |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"> <li>proof-carrying discriminant source D</li> <li>canonical theta with <math>\text{theta}^2 = D</math></li> <li>ExactQuadraticScalar D schema</li> <li>no-free-sqrt guardrail</li> <li>rational/integer invariant projection</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>no free sqrt(D)</li> <li>no O09 instance without proof-carrying D-source</li> <li>no analytic Complex operational replacement</li> <li>no Cayley-Dickson C-source before CD9 noncircularity proof</li> </ul>                        |

**Phase 1.1.1.1.0.14 – Phase 5-10 dependency bifurcation lock: number-theory/modular path separated from spectral/physical consumers**

**Table legend.**   G = green/closed and derived-only;   W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   O = orange/implementable after stated predecessor closure;   Y = yellow/active and implementable now;   R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                               |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;"> </span> G                                                                                                                                                                                  |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                                  |
| Phase documentation tier              | F1                                                                                                                                                                                                                                   |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                       |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                   |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                  |
| Implementation status                 | completed                                                                                                                                                                                                                            |
| Preparation status                    | completed                                                                                                                                                                                                                            |
| Phase type                            | infrastructure                                                                                                                                                                                                                       |
| Phase-type-specific contract          | infrastructure contract: dependency and object placement must be explicit and minimal.                                                                                                                                               |
| Title / state                         | Phase 5-10 dependency bifurcation lock: number-theory/modular path separated from spectral/physical consumers                                                                                                                        |
| Preparation workflow                  | phase dependency bifurcation and semantic placement workflow                                                                                                                                                                         |
| Source-to-CNNA mapping                | Object phase placement must match actual source objects, not legacy narrative phase titles.                                                                                                                                          |
| Dependency-depth audit                | No spectral/physical object may depend on modular forms or j unless its proof target explicitly consumes them.                                                                                                                       |
| Phase split decision                  | Split because phase 9 mixed modular and physical consumers and phase 10 mixed modular, spectral, transfer, analytic and bridge targets.                                                                                              |
| Positive path plan                    | Keep parallel paths after Phase 4;<br>close each path on its own proof-carrying prerequisites.                                                                                                                                       |
| Module implementation manifest        | No Lean implementation manifest was required when this already-closed/planned infrastructure row was created;<br>under closed LEG10, any future code change touching this phase must fill exact added/modified modules before edits. |
| Semantic placement decision           | Existing placement remains historical/planning-only. Future module placement must be revalidated by LEG10 before code changes.                                                                                                       |
| Reuse and duplication audit           | Not a generator implementation. If later code resembles an existing generator/API, consume LEG11 before edits.                                                                                                                       |
| Canonical generator policy            | No generator-like code in this record. Future generator-like extensions must route through LEG11.                                                                                                                                    |
| Module compactness rule               | No one-phase-one-file inference;<br>future modules require semantic/API/import-boundary justification.                                                                                                                               |
| Refactor regression protocol          | incremental wrapper mode;<br>full local build green after each migration step                                                                                                                                                        |
| Refactor migration metrics            | track object reassignments, phase dependency changes, and consumer-path imports                                                                                                                                                      |
| Lean/mathlib version policy           | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                                                         |
| ExecComplex boundary policy           | Phase 4.1 finite matrix is enough for spectral/physical finite consumers;<br>modular/CM analytics remain separate.                                                                                                                   |

|                                        |                                                                                                                                                                                 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consumer-contract direction            | Phase 10 consumes Phase 4.1 directly;<br>Phase 8/9 do not gate Phase 10 unless a specific object bridge says so.                                                                |
| Governance-overhead review             | Review after first Phase 10 finite consumer whether the bifurcation prevented a false wait on Phase 8/9.                                                                        |
| Proof rehearsal before implementation  | For each object, list minimal existing sources and reject any extra phase dependency not used by the proof target.                                                              |
| Expected tactics                       | planning/generator validation now;<br>later Lean proofs use constructor/projection lemmas over source objects only.                                                             |
| Expected mathlib imports               | none for planning lock                                                                                                                                                          |
| ExecComplex/computable plan            | Spectral/physical finite profiles consume Phase 4.1 ExecBoundaryMatrix;<br>modular forms/CM consume lattice/number-theory path separately.                                      |
| Concrete computable instance traversal | Future finite test rows must run separately for arithmetic/modular and spectral/physical consumers.                                                                             |
| Computable path non-skeleton status    | Must be demonstrated by finite rows over Phase 4.1 matrix for spectral/physical path and by D/CM rows for modular path.                                                         |
| Expected executable witness            | At least one finite ExecBoundaryMatrix spectral profile row and at least one D-to-modular comparison row when those phases activate.                                            |
| Mathlib complication log               | Record if modular-form/mathlib APIs are attempted in spectral/physical path;<br>that is a placement smell unless explicitly justified.                                          |
| Noncomputable boundary audit           | Analytic modular/j/L-function objects remain mirror/comparison and cannot block finite spectral/operator algebra profiles.                                                      |
| Mathlib mitigation/refactor layer      | Use separate path modules;<br>do not import modular APIs into finite physical consumers for convenience.                                                                        |
| Bridge gap status                      | Open bridge objects remain late and cannot force early path sequentialization.                                                                                                  |
| Natural backjump/refactor layer        | If a consumer needs a source from another path, add an explicit bridge dependency rather than moving the whole phase behind that path.                                          |
| Implementation prompt contract         | Before implementing any phase, module manifest must state exact source phase IDs and must not inherit unrelated predecessor chains.                                             |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                     |
| How implemented                        | Planning-only architecture/order update;<br>no Lean code moved.                                                                                                                 |
| Verification / evidence                | Generated-table validation after v37 patch;<br>no Lean build.                                                                                                                   |
| Proof strategy                         | Treat phase dependency as provenance obligation;<br>split unrelated consumers rather than sequentializing them through non-required objects.                                    |
| Build result                           | No Lean build executed or claimed.                                                                                                                                              |
| Post-implementation plan check         | After first Phase 4/10 implementation, verify that no M-object import or j target appears in the physical path unless explicitly justified.                                     |
| Artifact packaging rule                | Return complete zip;<br>Lean build local unless actually executed here.                                                                                                         |
| Implementation notes                   | Records dependency and topic-boundary decision before code implementation.                                                                                                      |
| Source materials / references          | <ul style="list-style-type: none"> <li>v36 phase/object register</li> <li>code audit module inventory</li> <li>review identifying artificial modular-form bottleneck</li> </ul> |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>independent number-theory/modular path</li> <li>independent spectral/physical path</li> <li>explicit consumer dependencies</li> </ul>    |
| Implemented modules                    | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                          |
| Added modules                          | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                          |
| Modified modules                       | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml only</li> </ul>                                                                                         |
| Forbidden patterns checked             | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                            |

|                  |                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"><li>No artificial dependency bottleneck;<br/>no phase-title/object mismatch;<br/>no modular-form gate for spectral/physical finite objects.</li></ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 1.1.1.1.0.15 – Global phase-object-link policy: no ambiguous empty Linked-objects cells

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #90EE90; border: 1px solid black; padding: 0 2px;">G</span>                                                                                                     |
| Context class                         | LeanGovernancePhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | completed                                                                                                                                                                                      |
| Preparation status                    | completed                                                                                                                                                                                      |
| Phase type                            | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract          | Global generator/view consistency phase: must create validation/display logic rather than mathematical declarations.                                                                           |
| Title / state                         | Global phase-object-link policy and no-empty-linked-object-cell contract                                                                                                                       |
| Preparation workflow                  | Phase-object-link policy infrastructure was inserted before the active module-manifest phase so implementation cannot start with ambiguous phase/object table semantics.                       |
| Source-to-CNNA mapping                | Generator display problem -> global planning invariant LEG34;<br>direct object ownership remains object.Phase == phase.Phase;<br>child objects are only scope summaries.                       |
| Dependency-depth audit                | Depends only on existing planning data and generator output;<br>no mathematical phase is re-ordered.                                                                                           |
| Phase split decision                  | Added as 1.1.1.1.0.15 because it is a global view/validation lock that must precede implementation manifest work.                                                                              |
| Positive path plan                    | Use generated validation, not manual interpretation, to keep phase/object ownership visible and auditable.                                                                                     |
| Module implementation manifest        | No Lean module manifest;<br>generator/documentation-only phase.                                                                                                                                |
| Semantic placement decision           | Belongs to global planning infrastructure, not to mathematical object phases.                                                                                                                  |
| Reuse and duplication audit           | Reuses existing phase hierarchy and object register. No duplicate object registry created.                                                                                                     |
| Canonical generator policy            | The table generator is the canonical renderer of phase-object link semantics.                                                                                                                  |
| Module compactness rule               | Keep policy in the generator and YAML fields;<br>avoid a separate manual table source.                                                                                                         |
| Refactor regression protocol          | View regression: no unqualified blank Linked objects cell may remain after table generation.                                                                                                   |
| Refactor migration metrics            | Policies assigned to every phase;<br>direct/scope_only/no_direct_object/moved counts recorded in report.                                                                                       |
| Lean/mathlib version policy           | not applicable                                                                                                                                                                                 |
| ExecComplex boundary policy           | not applicable                                                                                                                                                                                 |
| Consumer-contract direction           | Planning views consume object register and phase hierarchy;<br>object register does not consume table-rendered child scope.                                                                    |
| Governance-overhead review            | Justified because the prior table display could falsely suggest missing objects or broken phase coverage.                                                                                      |
| Proof rehearsal before implementation | Show that every phase has one of four cases: direct, scope_only, no_direct_object, moved. Each case has a different rendering and validator obligation.                                        |
| Expected tactics                      | Python generator validation;<br>YAML structural audit;<br>no Lean tactics.                                                                                                                     |

|                                        |                                                                                                                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected mathlib imports               | none                                                                                                                                                                                                                                             |
| ExecComplex/computable plan            | not applicable;<br>planning infrastructure only                                                                                                                                                                                                  |
| Concrete computable instance traversal | Generator run over all phases/objects is the concrete planning instance.                                                                                                                                                                         |
| Computable path non-skeleton status    | not applicable to mathematical computation;<br>generator validation is concrete over the current master.                                                                                                                                         |
| Expected executable witness            | python scripts/generate_tables.py produces policy-aware phase table and TSV exports.                                                                                                                                                             |
| Mathlib complication log               | none                                                                                                                                                                                                                                             |
| Noncomputable boundary audit           | none;<br>no Lean code                                                                                                                                                                                                                            |
| Mathlib mitigation/refactor layer      | none                                                                                                                                                                                                                                             |
| Bridge gap status                      | closes the display/ownership gap that made scope-only phases look empty                                                                                                                                                                          |
| Natural backjump/refactor layer        | If a phase unexpectedly loses all direct objects, update the object Phase assignment or declare the correct non-direct policy before generating views.                                                                                           |
| Implementation prompt contract         | Before code work, verify that the phase has a clear object-link policy and that the module manifest references direct or child-scope objects correctly.                                                                                          |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                      |
| How implemented                        | Planning generator and YAML schema updated: phases carry Object link policy/explanation/child scope;<br>phase table renders direct objects, scope-only child summaries, no-direct-object explanations, or moved placeholders.                    |
| Verification / evidence                | YAML has explicit policy fields for every phase;<br>generator validates allowed policy kinds and renders the phase table without raw blank Linked objects cells.                                                                                 |
| Proof strategy                         | Treat phase-object display as a generated invariant: empty direct-object sets require explicit phase policy, not silence.                                                                                                                        |
| Build result                           | PDF view generated;<br>no Lean build executed.                                                                                                                                                                                                   |
| Post-implementation plan check         | Future generator runs must fail or visibly explain if a phase has no direct linked object.                                                                                                                                                       |
| Artifact packaging rule                | Return complete zip and PDF view;<br>do not claim Lean build.                                                                                                                                                                                    |
| Implementation notes                   | This is planning/documentation infrastructure only. It does not change Lean code and does not alter object ownership. Direct ownership remains controlled exclusively by the object register Phase field.                                        |
| Source materials / references          | <ul style="list-style-type: none"><li>v37 phase table linked-object ambiguity</li><li>object register Phase field semantics</li><li>phase hierarchy/child phase ownership</li></ul>                                                              |
| Extracted established theory objects   | <ul style="list-style-type: none"><li>LEG34 Phase-object-link policy contract</li></ul>                                                                                                                                                          |
| Implemented modules                    | <ul style="list-style-type: none"><li>masters/cnna_roadmap_master.yaml</li><li>scripts/generate_tables.py</li><li>tables/phase_table.tex</li><li>tables/phase_object_link_policy.tex</li></ul>                                                   |
| Modified modules                       | <ul style="list-style-type: none"><li>masters/cnna_roadmap_master.yaml</li><li>scripts/generate_tables.py</li><li>README.md</li><li>CNNA_Planning_and_Documentation_Tool_v38_Phase_Object_Link_Policy_A3_English.tex</li></ul>                   |
| Forbidden patterns checked             | <ul style="list-style-type: none"><li>no object ownership aggregation into umbrella phases</li><li>no blank Linked objects cell without policy</li><li>no child-object scope treated as direct ownership</li><li>no Lean source change</li></ul> |

|                                      |                                                                                                                                                                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails                     | <ul style="list-style-type: none"> <li>No blank Linked objects cell without explicit policy</li> <li>No umbrella phase may claim child objects as direct ownership</li> <li>No object may be moved by display aggregation alone</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                     |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>phase_object_map</li> <li>phase hierarchy validator</li> <li>object register Phase field</li> </ul>                                                                                                 |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>Linked objects display aggregation</li> <li>manual child-object lists drifting from object register</li> </ul>                                                                                      |

### Phase 1.1.1.1.0.16 – Global ontic subsystem-minimality postulate and emergence-order lock

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                           |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                                                                |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                                              |
| Phase documentation tier              | F0                                                                                                                                                                                                                                               |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                               |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                              |
| Implementation status                 | completed                                                                                                                                                                                                                                        |
| Preparation status                    | completed                                                                                                                                                                                                                                        |
| Phase type                            | infrastructure                                                                                                                                                                                                                                   |
| Phase-type-specific contract          | Global ontology/provenance lock: creates planning constraints, not mathematical Lean declarations.                                                                                                                                               |
| Title / state                         | Global ontic subsystem-minimality postulate and emergence-order lock                                                                                                                                                                             |
| Preparation workflow                  | Inserted as a green decision lock before module-manifest implementation, so code planning cannot start with ambiguous ontic premises.                                                                                                            |
| Source-to-CNNA mapping                | User formulation -> POST1 global postulate;<br>allowed primitive = subsystem/cut/approximant selection within IDEAL_infty;<br>derived consequences = boundary, complement, Schur/DtN, spectra, algebra, arithmetic, geometry, dynamics, physics. |
| Dependency-depth audit                | Depends only on already established planning/regime/complement discipline. It does not reorder mathematics;<br>it constrains all future derivation and consumer phases.                                                                          |
| Phase split decision                  | Added as 1.1.1.1.0.16 because this is a global architecture/postulate lock that must precede active implementation.                                                                                                                              |
| Positive path plan                    | Use subsystem-being as the minimal emergence trigger. Let Schur/DtN/complement/regime machinery generate all later structures rather than importing them.                                                                                        |
| Module implementation manifest        | No Lean module manifest;<br>planning/postulate infrastructure only.                                                                                                                                                                              |
| Semantic placement decision           | Belongs to global postulates and no-go guardrails before active implementation.                                                                                                                                                                  |
| Reuse and duplication audit           | Reuses global no-go, workflow, proof-dossier and generated-table infrastructure;<br>no duplicate roadmap source.                                                                                                                                 |
| Canonical generator policy            | The table generator is the canonical renderer of POST1 and its supporting enforcement objects.                                                                                                                                                   |
| Module compactness rule               | Keep the postulate in one top-level section plus LEG35/GNG22 references;<br>avoid scattering prose copies.                                                                                                                                       |
| Refactor regression protocol          | Generated views must keep POST1 visible and GNG22 inherited;<br>no code refactor may remove the global postulate from generated documentation.                                                                                                   |

|                                        |                                                                                                                                                                                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refactor migration metrics             | One postulate, one object, one phase lock, one no-go, one workflow added.                                                                                                                                                                                                 |
| Lean/mathlib version policy            | not applicable                                                                                                                                                                                                                                                            |
| ExecComplex boundary policy            | ExecComplex is downstream of subsystem-derived computation;<br>no scalar layer is ontic primitive.                                                                                                                                                                        |
| Consumer-contract direction            | Consumer phases consume derived operator/spectral/algebraic/arithmetic data only after subsystem/cut provenance exists.                                                                                                                                                   |
| Governance-overhead review             | Justified because the rule prevents the project from importing the very structures it aims to derive.                                                                                                                                                                     |
| Proof rehearsal before implementation  | Check any object that mentions dynamics, spectra, algebra, arithmetic, geometry, stochasticity, openness, unitarity or physics. If its source chain does not begin at subsystem/cut and pass through the required generating layers, mark it comparison-only or backjump. |
| Expected tactics                       | YAML/generator validation;<br>proof-dossier audit;<br>no Lean tactics in this planning phase.                                                                                                                                                                             |
| Expected mathlib imports               | none                                                                                                                                                                                                                                                                      |
| ExecComplex/computable plan            | ExecComplex remains downstream computational carrier;<br>it does not supply ontic primitives. Concrete computable paths must start from finite subsystem instances.                                                                                                       |
| Concrete computable instance traversal | Planning-level concrete instance: trace a finite subsystem/cut -> boundary -> complement/regime -> Schur/DtN -> derived consumer chain for later active phases.                                                                                                           |
| Computable path non-skeleton status    | The postulate explicitly requires concrete subsystem instances before computational consumer claims;<br>it rejects skeleton-only computational paths.                                                                                                                     |
| Expected executable witness            | python scripts/generate_tables.py renders POST1 and exports global_postulates.generated.tsv.                                                                                                                                                                              |
| Mathlib complication log               | none for the postulate. Later mathlib complications must not be solved by importing non-derived primitives.                                                                                                                                                               |
| Noncomputable boundary audit           | No Lean code. Later noncomputable/mirror boundaries must remain downstream of derived subsystem provenance.                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | none here;<br>later phases must backjump to natural generating layer if mathlib forces inappropriate primitives.                                                                                                                                                          |
| Bridge gap status                      | closes the ontic-primitivity gap: prevents later phases from treating physics/dynamics/arithmetic/geometry/environment as inputs.                                                                                                                                         |
| Natural backjump/refactor layer        | If a consumer needs a non-subsystem structure without provenance, backjump to the cut/regime/complement/Schur source layer rather than patching the consumer.                                                                                                             |
| Implementation prompt contract         | Before implementing any phase, state which structures are primitive, derived, or comparison-only. Only subsystem-being may be primitive.                                                                                                                                  |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                               |
| How implemented                        | Added POST1 global postulate, LEG35 object/dossier, GNG22 global no-go, ontic subsystem-minimality workflow and generated global-postulate rendering before the active implementation phase.                                                                              |
| Verification / evidence                | YAML contains global_postulates.POST1, LEG35, GNG22, and a green phase 1.1.1.1.0.16 before the active cursor. Generator validates and renders the global postulate table.                                                                                                 |
| Proof strategy                         | Treat ontology as a global provenance invariant: non-subsystem structures must be classified as derived consequences or comparison-only readings, never as primitive inputs.                                                                                              |
| Build result                           | PDF view generated;<br>no Lean build executed.                                                                                                                                                                                                                            |
| Post-implementation plan check         | Future phase closures must record derived provenance for all non-subsystem structures or mark them comparison-only.                                                                                                                                                       |
| Artifact packaging rule                | Return complete zip and PDF view;<br>do not claim Lean build.                                                                                                                                                                                                             |
| Implementation notes                   | Planning/documentation infrastructure only. It does not introduce Lean axioms. The postulate constrains the plan: subsystem-being is the only primitive emergence trigger;<br>all other structures require derived provenance.                                            |
| Source materials / references          | <ul style="list-style-type: none"> <li>User ontic-minimality instruction</li> <li>IDEAL_infty formal provenance background</li> <li>noOuter/HasOuter regime split</li> <li>subsystem/cut/complement-family discipline</li> <li>Schur/DtN emergence chain</li> </ul>       |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>POST1 CNNA ontic subsystem-minimality</li> <li>LEG35 Ontic subsystem-minimality contract</li> <li>GNG22 no non-subsystem ontic primitive</li> </ul>                                                                                |



|                                      |                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implemented modules                  | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> <li>tables/global_postulates.tex</li> <li>tables/global_no_go_guardrails.tex</li> </ul>                                                                                                                                        |
| Modified modules                     | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> <li>README.md</li> <li>CNNA_Planning_and_Documentation_Tool_v39_Ontic_Subsystem_Minimality_A3_English.tex</li> </ul>                                                                                                           |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no dynamics as primitive</li> <li>no physics/AQFT as primitive</li> <li>no stochasticity/noise as primitive</li> <li>no geometry/arithmetic/operator algebra as primitive</li> <li>no environment as primitive beyond complement derived from subsystem cut</li> <li>no Lean axiom introduced</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>No dynamics, physics, stochasticity, geometry, arithmetic, operator algebra, environment or unitarity as primitive beyond subsystem-being</li> <li>No comparison analogy may supply an ontic input</li> <li>No consumer-side patch for missing emergence provenance</li> </ul>                           |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                          |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>global_no_go_guardrails renderer</li> <li>workflow_policy renderer</li> <li>object_proof_dossiers</li> <li>phase table</li> </ul>                                                                                                                                                                        |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>ontic primitive</li> <li>subsystem-being</li> <li>emergence trigger</li> </ul>                                                                                                                                                                                                                           |

### Phase 1.1.1.1.1 – Implementation module manifest and semantic placement gate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
 Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                              |
| Context class                         | LeanGovernancePhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | completed                                                                                                                                                                                      |
| Preparation status                    | completed                                                                                                                                                                                      |
| Phase type                            | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness; close schema/decision prerequisites before code.                                                                        |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title / state                          | Implementation module manifest and semantic placement gate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Preparation workflow                   | Active module-manifest gate after global guardrail inheritance lock.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Source-to-CNNA mapping                 | LEG10 governs all later implementation phases. It is not mathematics; it is a provenance/API hygiene contract preventing code fragmentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | Depends on green parents 1.1.1.1, 1.1.1.1.0, and decision locks 1.1.1.1.0.1-1.1.1.1.0.13. No implementation may start until module names/budget, phase_type, regression mode, version impact, consumer direction, concrete-instance status, and mathlib complication status are explicit.                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Phase split decision                   | New child gate inserted before 1.1.1.1.2 and 1.1.1.1.3. No further split unless a separate generated manifest table becomes necessary.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Positive path plan                     | Define the manifest: for each future implementation phase, list proposed added modules, modified modules, exact names, semantic placement, phase_type, type-specific obligations, reuse/extension/rejection decision, public/legacy/API role, import direction, regression protocol, version impact, and expected downstream consumers. Include a rule that any future bridge theorem must list whether it is an upstream generator input, downstream consumer, or comparison-only diagnostic. Additionally, every proposed computation module must include a generic finite test-matrix placement decision: whether it reuses O07, extends the canonical smoke-test generator, or documents why no executable test applies yet. |
| Module implementation manifest         | Closed contract: every scratchpad record must expose proposed_added_modules, proposed_modified_modules, reusable_existing_modules, semantic placement, rejection/fit decision, public/legacy/API role, import direction, expected downstream consumers, phase_type obligations, refactor regression protocol, version impact, concrete-instance traversal, generic test placement, mathlib complication log, and noncomputable boundary audit before Lean edits.                                                                                                                                                                                                                                                                 |
| Semantic placement decision            | The manifest is planning infrastructure and belongs to the single YAML master plus generated views. It is not a Lean module and must not be implemented as phase-local code. Future code modules are created only when an existing semantic module cannot be safely extended or wrapped.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Reuse and duplication audit            | Phase 1.1.1.1.1 names reusable candidates but defers semantic generator classification to 1.1.1.1.2. Duplicate implementation remains blocked already at manifest level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Canonical generator policy             | Any generator-like phase must consume this manifest, then pass through 1.1.1.1.2 before Lean code. The manifest table must state whether O07/canonical smoke-test generator is reused, extended, or rejected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Module compactness rule                | Closed rule: phase granularity is not file granularity. A later phase must justify every new module by semantic ownership, API/firewall direction, or unavoidable import split.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Refactor regression protocol           | This gate does not refactor code, but it must require every later refactor manifest to declare wrapper-first/incremental vs atomic mode. Default for 1.1.2 is wrapper-first incremental with local full-build evidence per package.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Refactor migration metrics             | The active gate must define the metric template, not fill counts yet: declarations_migrated_count; modules_added/modified; imports_changed; import_graph_delta; consumer_breakage_audit; compatibility_aliases; build evidence source.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Lean/mathlib version policy            | Every module manifest must state whether it stays on Lean 4.28.0/mathlib baseline, uses no mathlib change, or requires a separate upgrade gate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| ExecComplex boundary policy            | EXEC-2/3/4 enforced: rational QxQ floor; exact quadratic layer is required later for CM/lattice coordinates but may only be instantiated from a proof-carrying discriminant ledger; analytic Complex and CD C-surfaces are not early scalar sources.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Consumer-contract direction            | Every proposed public wrapper must state its expected consumers and import direction. BoundaryOperatorProfile must prepare for Phase 4.1 without importing Phase 4. For BR08/CD8 specifically, earlier route modules export data downstream and must not import the bridge.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Governance-overhead review             | This gate is retained because it blocks concrete observed failure modes: one-phase-one-file proliferation, duplicate generator surfaces, hidden import direction, and unclassified mathlib/noncomputable pressure. Review flattening only after the first real code/refactor package closes.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Proof rehearsal before implementation  | For a future phase, rehearse: target object -> existing module semantic scan -> either extend existing module or justify new module -> record consumer role -> only then implement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Expected tactics                       | No Lean tactics expected. If mirrored as Lean audit constants later, use records with exact/rfl fields only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | None. This planning gate uses generated repo inventory, not new mathlib imports.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| ExecComplex/computable plan            | No ExecComplex object introduced. The rule applies to future ExecComplex-touching phases: reuse existing boundary modules unless semantic placement proves otherwise.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Concrete computable instance traversal | This active gate is infrastructure-only: it will not introduce operational Lean code. For every later operational phase, the manifest must include a concrete finite instance path from generated input to intended endpoint. This phase must plan generic finite test traversal, not only special sentinels, for every upcoming computable path.                                                                                                                                                                                                                                                                                                                                                                                |
| Computable path non-skeleton status    | active infrastructure gate; it prevents later skeleton-only operational phases by requiring concrete instance traversal fields in every scratchpad.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Expected executable witness            | For this gate: generated manifest fields. For later operational phases: a named concrete finite generated instance and executable endpoint object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Mathlib complication log               | No new mathlib import in this gate. Later manifests must list expected mathlib imports and whether each is executable, noncomputable-only, classical-only, or mirror-layer-only. Scalar architecture check added: any proposed new module must classify Complex/Real/QuadraticAlgebra dependencies under LEG31. Derived-quadratic update: any proposed QuadraticAlgebra/AdjoinRoot/ExactQuadratic module must state whether it is a schema, D-instantiation, proof/mirror reference, or executable coordinate layer.                                                                                                                                                                                                             |
| Noncomputable boundary audit           | No Lean operational code changed. Later phases must document noncomputable leakage risk before implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Mathlib mitigation/refactor layer      | If a proposed implementation needs noncomputable mathlib APIs, the manifest must route them to an allowed mirror/non-operational module or backjump to an operational replacement layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Bridge gap status                      | No mathematical bridge. Missing semantic placement data blocks implementation, not mathematics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                      |                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Natural backjump/refactor layer      | If implementation wants a new module but semantics fit an existing module, backjump to the existing module or neutral wrapper seam. If import direction prevents reuse, backjump to the earliest natural API/firewall layer.                                                                                                                               |
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation.                          |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                |
| How implemented                      | Closed as planning/generator infrastructure: the YAML scratchpad now has mandatory module-manifest fields, generate_tables.py validates them, and generated table/TSV views expose the compact module-placement contract before Lean edits.                                                                                                                |
| Verification / evidence              | scripts/generate_tables.py executed successfully for v40;<br>module_implementation_manifest table/TSV regenerated;<br>no Lean source changed;<br>no Lean build claimed.                                                                                                                                                                                    |
| Proof strategy                       | No Lean proof. Planning proof is generator-enforced semantic placement against the module inventory and scratchpad schema.                                                                                                                                                                                                                                 |
| Build result                         | Generated planning artifacts and PDF were regenerated;<br>no Lean build executed or claimed.                                                                                                                                                                                                                                                               |
| Post-implementation plan check       | Closed in v40. Active cursor moved to 1.1.1.1.2 reuse/generator audit;<br>later implementation phases must fill the manifest before code changes.                                                                                                                                                                                                          |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                             |
| Implementation notes                 | Infrastructure-only closure. The gate creates no mathematical object, moves no Lean module, and exists to prevent code fragmentation before later implementation phases.                                                                                                                                                                                   |
| Source materials / references        | <ul style="list-style-type: none"> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/import_edges.json</li> <li>repo_inventory/raw_export/symbol_reference_summary.json</li> <li>repo_snapshot/**/*.lean</li> <li>object register and phase DAG</li> <li>v30 IdealThread/CutSpec.full route contract</li> </ul>                |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>ModuleImplementationManifest</li> <li>SemanticPlacementDecision</li> <li>ModuleBudget</li> <li>ExistingModuleFitCheck</li> <li>Public/Legacy/API role classification</li> </ul>                                                                                                                                     |
| Implemented modules                  | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> <li>tables/module_implementation_manifest.tex</li> <li>generated_exports/module_implementation_manifest.generated.tsv</li> <li>README.md</li> <li>CNNA_Planning_and_Documentation_Tool_v40_Module_Manifest_Gate_A3_English.tex</li> </ul> |
| Modified modules                     | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> <li>README.md</li> <li>CNNA_Planning_and_Documentation_Tool_v40_Module_Manifest_Gate_A3_English.tex</li> </ul>                                                                                                                            |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source change</li> <li>no @[simp] attribute introduced</li> <li>no noncomputable operational code introduced</li> <li>no classical helper introduced</li> <li>no duplicate generator introduced</li> <li>no Lean build success claim without evidence</li> </ul>                                            |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global__no_go_guardrails apply; this phase list is local addenda only.</li> <li>no implementation phase may start without exact module count and names</li> <li>no one-phase-one-file proliferation by default</li> <li>no new module when an existing semantic module can be extended safely</li> <li>no public wrapper without declared consumer role</li> <li>no Lean build success claim without actual/current evidence</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> <li>README.md</li> </ul>                                                                                                                                                                                                                                                                                                                                                                  |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/Notation.lean as source to split, not as target monolith</li> <li>existing Smoke/Test generator modules as candidates for generator-like work</li> </ul>                                                                                                                                                                                                                                                                                    |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>new generator/request/output helper names without canonical generator route</li> <li>IdealThread.carrier/family singleton facts</li> <li>CutSpec.full carrier facts</li> <li>RecursiveSchurSource/MobiusTransfer recurrence logic</li> <li>string-like analogy used as generator</li> </ul>                                                                                                                                                                         |

### Phase 1.1.1.1.2 – Reuse and duplication audit with canonical theory/smoke-test generator policy

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked

Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                     |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                                                                                                                          |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                                                                                                        |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                        |
| Implementation status                 | completed                                                                                                                                                                                                                                                                                                  |
| Preparation status                    | completed                                                                                                                                                                                                                                                                                                  |
| Phase type                            | infrastructure                                                                                                                                                                                                                                                                                             |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness; close schema/decision prerequisites before code.                                                                                                                                                                                    |
| Title / state                         | Reuse and duplication audit with canonical theory/smoke-test generator policy                                                                                                                                                                                                                              |
| Preparation workflow                  | Use LEG10 module manifest, symbol/reference summaries, existing Smoke/Test generator modules, and planned target semantics to decide consume/extend/wrap/genuine-split for every generator-like future phase.                                                                                              |
| Source-to-CNNA mapping                | LEG11 is now a generated governance object. It does not generate mathematics; it constrains future CNNA-derived-only implementations so that generator-like code extends the natural existing seam instead of forking provenance.                                                                          |
| Dependency-depth audit                | Closed after LEG10 because the module manifest now names reusable modules and duplicate-risk symbols. The audit classifies the existing smoke/generated source seam, action inventory seam, documentation generator, and future experiment/request-output work before any new generator code is permitted. |
| Phase split decision                  | No split needed for this gate. A later code phase may create a neutral generator API only if a concrete implementation phase proves the current Smoke namespace is too narrow while semantics are the same.                                                                                                |
| Positive path plan                    | Closed positive path: identify existing seams -> classify allowed use as consume/extend/wrap/genuine split -> record blocked duplication -> render ledger -> move active cursor to proof-documentation infrastructure.                                                                                     |
| Module implementation manifest        | Closed contract consumed from LEG10. v41 adds no Lean module; it extends the planning generator and YAML master to render the reuse/duplication ledger.                                                                                                                                                    |

|                                        |                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semantic placement decision            | The policy belongs in the single YAML master and generated planning views. It must not be implemented as a phase-local Lean status module; later neutral generator code is allowed only after a concrete module-manifest proof.                                                                                                   |
| Reuse and duplication audit            | Closed: generator-like operations are classified by ledger rows as consume existing, extend existing, wrap existing, artifact-only split, or genuine semantic split. Duplicate generator implementations are blocked by default.                                                                                                  |
| Canonical generator policy             | Closed rule: the Smoke/GeneratedSource-MainPath seam is canonical for generated ToC/smoke semantics; action-like semantics reuse GeneratorActionInventory/ActionFormatDecision; documentation/inventory scripts are artifact generators and comparison/audit only.                                                                |
| Module compactness rule                | Closed rule: a generator policy row is not a new Lean file. Later code should prefer one coherent reusable generator/API seam plus wrappers over phase-local generator files.                                                                                                                                                     |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                 |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                       |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                        |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                          |
| Consumer-contract direction            | Exports a planning/API policy to 1.1.1.1.3, 1.1.1.3, Phase 5, O07, heuristic lattice tests, and all generator-like phases. It does not export mathematical data to downstream pillars.                                                                                                                                            |
| Governance-overhead review             | Retained: this gate blocks a concrete failure mode, namely duplicate generator/provenance surfaces. Review flattening only after the first real generator-consuming code phase closes under the ledger.                                                                                                                           |
| Proof rehearsal before implementation  | For future generator-like targets: first cite the ledger row; if semantics match the Smoke/GeneratedSource-MainPath seam, consume or extend it; if artifact-only, keep it in scripts/YAML; if action-like, consume GeneratorActionInventory/ActionFormatDecision; only then permit a recorded genuine semantic split.             |
| Expected tactics                       | No Lean tactics expected.                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | None.                                                                                                                                                                                                                                                                                                                             |
| ExecComplex/computable plan            | No ExecComplex object introduced. Future generated ExecComplex data must route through the canonical generator path where semantics match and remain under EXEC-2 operational Q x Q boundary rules.                                                                                                                               |
| Concrete computable instance traversal | Infrastructure-only; no operational finite instance is introduced. Future operational generator phases must provide concrete finite traversal through the canonical seam.                                                                                                                                                         |
| Computable path non-skeleton status    | not applicable for this planning gate; later operational generator code must provide executable witness rows before closure.                                                                                                                                                                                                      |
| Expected executable witness            | none for this infrastructure gate; future O07/Phase-5-like generator consumers must name one.                                                                                                                                                                                                                                     |
| Mathlib complication log               | No Lean/mathlib import is introduced by this phase.                                                                                                                                                                                                                                                                               |
| Noncomputable boundary audit           | No noncomputable boundary introduced; no Lean operational path changed.                                                                                                                                                                                                                                                           |
| Mathlib mitigation/refactor layer      | Not needed for this gate. If later generator code introduces mathlib/noncomputable pressure, isolate it at the earliest natural source layer or mirror/comparison layer.                                                                                                                                                          |
| Bridge gap status                      | Closed for governance. A possible future neutral generator API remains an implementation gap only if later code needs a namespace-neutral wrapper; it is not a license to create a parallel generator.                                                                                                                            |
| Natural backjump/refactor layer        | If a future consumer wants a new generator, backjump to the natural existing seam: Smoke/GeneratedSource-MainPath for generated ToC/smoke semantics, Actions/GeneratorActionInventory for action candidates, scripts/YAML for documentation artifacts, or a ledger-recorded genuine semantic split.                               |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation. |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                          |
| How implemented                        | Closed in v41 as a generated reuse/duplication ledger plus canonical generator policy. Added a YAML reuse_duplication_ledger section, generator-script validation/rendering, a PDF section, and TSV export. No Lean source was edited.                                                                                            |
| Verification / evidence                | python3 scripts/generate_tables.py completed successfully after adding reuse_duplication_ledger validation/rendering. pdflatex was run for the v41 PDF. Prior Lean build was user-reported green; this phase changed no Lean source and no Lean build is claimed from this environment.                                           |
| Proof strategy                         | Infrastructure proof by generated ledger coverage and validation: the YAML master owns the policy rows; generate_tables.py validates required columns and emits PDF/TSV views. No Lean proof is introduced.                                                                                                                       |
| Build result                           | User reported prior Lean build green. This phase changed no Lean source; no Lean build executed or claimed here.                                                                                                                                                                                                                  |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Post-implementation plan check       | LEG11 is documented and green;<br>generated ledger artifacts exist;<br>active cursor moves to 1.1.1.1.3 proof-documentation infrastructure. Later Phase 5/O07/heuristic lattice/generator-like work must pass through this policy before code.                                                                                                                                                                                                                                                                         |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                         |
| Implementation notes                 | LEG11 now blocks parallel generator/request/output/extraction/ledger infrastructure by default. Future generator-like phases must classify their relation to the canonical seam before implementation.                                                                                                                                                                                                                                                                                                                 |
| Source materials / references        | <ul style="list-style-type: none"><li>• LEG10 module manifest</li><li>• repo_inventory/raw_export/modules.json</li><li>• repo_inventory/raw_export/symbol_reference_summary.json</li><li>• repo_inventory/raw_export/symbol_references/declarations.json</li><li>• CNNA.PillarA.Arithmetic.Smoke.GeneratedSource</li><li>• CNNA.PillarA.Arithmetic.Smoke.GeneratedMainPath</li><li>• CNNA.PillarA.Arithmetic.Actions.GeneratorActionInventory</li><li>• CNNA.PillarA.Arithmetic.Actions.ActionFormatDecision</li></ul> |
| Extracted established theory objects | <ul style="list-style-type: none"><li>• ReuseDuplicationLedger</li><li>• CanonicalGeneratorPolicy</li><li>• single-generator extension rule</li></ul>                                                                                                                                                                                                                                                                                                                                                                  |
| Implemented modules                  | <ul style="list-style-type: none"><li>• masters/cnna_roadmap_master.yaml</li><li>• scripts/generate_tables.py</li><li>• tables/reuse_duplication_ledger.tex</li><li>• tables/reuse_duplication_ledger_table.tex</li><li>• generated_exports/reuse_duplication_ledger.generated.tsv</li><li>• CNNA_Planning_and_Documentation_Tool_v41_Reuse_Duplication_Generator_Policy_A3_English.tex</li></ul>                                                                                                                      |
| Modified modules                     | <ul style="list-style-type: none"><li>• masters/cnna_roadmap_master.yaml</li><li>• scripts/generate_tables.py</li><li>• README.md</li></ul>                                                                                                                                                                                                                                                                                                                                                                            |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>• no Lean source edits</li><li>• no new Lean generator module</li><li>• no parallel request/output infrastructure</li><li>• no duplicate extraction or ledger implementation</li><li>• single YAML source for reuse/duplication ledger</li></ul>                                                                                                                                                                                                                                 |
| No-go guardrails                     | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li><li>• no duplicate generator</li><li>• no parallel request/output infrastructure</li><li>• no reimplementation of existing extraction/ledger logic</li><li>• no hidden generator in a downstream consumer module</li></ul>                                                                                                                                               |
| Proposed modified modules            | <ul style="list-style-type: none"><li>• masters/cnna_roadmap_master.yaml</li><li>• scripts/generate_tables.py</li><li>• README.md</li></ul>                                                                                                                                                                                                                                                                                                                                                                            |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reusable existing modules            | <ul style="list-style-type: none"> <li>• <code>CNNA.PillarA.Arithmetic.Smoke.GeneratedSource</code></li> <li>• <code>CNNA.PillarA.Arithmetic.Smoke.GeneratedMainPath</code></li> <li>• <code>CNNA.PillarA.Arithmetic.Actions.GeneratorActionInventory</code></li> <li>• <code>CNNA.PillarA.Arithmetic.Actions.ActionFormatDecision</code></li> <li>• <code>scripts/generate_tables.py</code></li> <li>• <code>scripts/build_export_script_v1.8.py</code></li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>• experiment request/output generator</li> <li>• phase-local extraction generator</li> <li>• parallel ledger generator</li> <li>• second smoke-test generator</li> <li>• independent documentation/proofbook source outside YAML</li> </ul>                                                                                                                                                                                  |

### Phase 1.1.1.1.3 – Object and quantities proof-documentation infrastructure gate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                                                                                                                                                                                                                                 |
| Context class                         | DocumentationPhase                                                                                                                                                                                                                                                                                                                                                                                                |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                                                                                                                                |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                                                                                    |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                                                                                |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                                                                               |
| Implementation status                 | completed                                                                                                                                                                                                                                                                                                                                                                                                         |
| Preparation status                    | completed                                                                                                                                                                                                                                                                                                                                                                                                         |
| Phase type                            | documentation                                                                                                                                                                                                                                                                                                                                                                                                     |
| Phase-type-specific contract          | Documentation phase: record dossier coverage, verification/audit status, and links to code/object records; no mathematical object is generated by documentation alone.                                                                                                                                                                                                                                            |
| Title / state                         | Object and quantities proof-documentation infrastructure gate                                                                                                                                                                                                                                                                                                                                                     |
| Preparation workflow                  | Documentation is now a third workflow next to phase preparation and implementation. For every object or quantity, keep a proof dossier with source objects, definitions, proof target, proof sketch, and verification result.                                                                                                                                                                                     |
| Source-to-CNNA mapping                | LEG12 governs proof documentation. It does not introduce mathematical content; it makes object-level proof obligations trackable and linkable from the object register.                                                                                                                                                                                                                                           |
| Dependency-depth audit                | Now depends on closed 1.1.1.1.2/LEG11. Must precede final public declaration classification and blue-object promotion; otherwise duplicate generator decisions and undocumented proof-carrying blue objects could leak into the public surface.                                                                                                                                                                   |
| Phase split decision                  | Inserted between reuse/generator audit and final declaration classification. No Lean module is added unless the documentation infrastructure is later mirrored in code.                                                                                                                                                                                                                                           |
| Positive path plan                    | Implemented. The proofbook infrastructure now consumes the object register and validates exact proof:<ID> dossier links for every registered object/quantity. The generated object register table exposes proof-documentation status and dossier anchors; the generated proof-documentation section carries source objects, definitions, proof targets, proof sketch, verification/audit, and next action fields. |
| Module implementation manifest        | No Lean modules added. Planning/generator artifacts only: YAML master, table generator, generated object proof documentation table/export, PDF view.                                                                                                                                                                                                                                                              |
| Semantic placement decision           | Documentation infrastructure belongs in the YAML/generator layer, not scattered into phase prose or Lean modules.                                                                                                                                                                                                                                                                                                 |
| Reuse and duplication audit           | Consumes closed LEG11. The proofbook is generated from the single YAML master and existing object register; no independent proofbook file or duplicate object list is allowed.                                                                                                                                                                                                                                    |
| Canonical generator policy            | No CNNA mathematical generator introduced. The documentation generator is an artifact generator only and cannot provide mathematical input to Pillar A.                                                                                                                                                                                                                                                           |
| Module compactness rule               | A new documentation section does not imply new Lean modules. It is a generated view over object register data.                                                                                                                                                                                                                                                                                                    |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                                                                                 |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                                                                                       |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                                                                                                                                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                                                                                                                                                           |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                                                                                                                                                                                                                                      |
| Governance-overhead review             | LEG12 caught and removed stale raw Proof dossier references that lacked the canonical proof:<ID> prefix, making future public-API classification checkable instead of prose-dependent.                                                                                                                                                                                                                                                                                                             |
| Proof rehearsal before implementation  | For each object: identify source objects -> state CNNA definition -> state proof goal -> cite proof-carrying code or mark missing -> verify semantic/API status -> decide documented yes/no/partial.                                                                                                                                                                                                                                                                                               |
| Expected tactics                       | No Lean tactics expected. If proof-dossier records later get mirrored in Lean, use explicit structures and exact/rfl fields; no simp-driven folding.                                                                                                                                                                                                                                                                                                                                               |
| Expected mathlib imports               | None for planning infrastructure. Object-level mathematical proofs must verify imports in their own phase.                                                                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object introduced. For ExecComplex-touching objects, the dossier must distinguish operational computable path from mirror/non-operational interpretation.                                                                                                                                                                                                                                                                                                                           |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                          |
| Bridge gap status                      | Infrastructure gap closed. Remaining no/partial proof dossiers are object-local documentation/proof gaps and must be filled at their natural correction/implementation phases; they do not block the infrastructure gate itself.                                                                                                                                                                                                                                                                   |
| Natural backjump/refactor layer        | If an object cannot be documented from its supposed sources, backjump to the earliest natural source object or blue-correction phase, not to the consumer/export layer.                                                                                                                                                                                                                                                                                                                            |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow and additionally update proof-dossier status for every object introduced, corrected, exported, or consumed as a public proof-relevant size.                                                                                                                                                                                                                                                                                             |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| How implemented                        | Closed in v42 by turning the existing proofbook view into a stricter generated infrastructure gate. scripts/generate_tables.py now validates one proof-dossier row per object ID, non-empty required dossier fields, object-register proof-documentation status, exact Proof dossier links of the form proof:<ID>, and matching object names/phases between the object register and object_proof_dossiers. The YAML master was normalized to satisfy those invariants. No Lean source was changed. |
| Verification / evidence                | python3 scripts/generate_tables.py succeeded with 189 phases, 199 objects, 189 scratch-pad records, and 199 proof-dossier records. v42 PDF compilation succeeded with pdflatex. Rendered PDF pages around the object/proof-documentation region were inspected. No new Lean build was run or claimed; the user-reported green build is the Lean baseline.                                                                                                                                          |
| Proof strategy                         | No Lean proof. Structural validation path: object register IDs -> expected proof:<ID> link -> matching object_proof_dossiers row -> generated anchor/table/export -> PDF view. Mathematical proof content remains per-object and code-checked in later natural phases.                                                                                                                                                                                                                             |
| Build result                           | Lean not run in this environment. User reported the pre-phase Lean build green; v42 generator and PDF build succeeded.                                                                                                                                                                                                                                                                                                                                                                             |
| Post-implementation plan check         | Closed. Generated object tables contain proof-documentation status and dossier links for every object or quantity; object_proof_dossiers coverage is generator-validated. Active cursor proceeds to 1.1.1.1.4 declaration classification finalization.                                                                                                                                                                                                                                             |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                     |
| Implementation notes                   | This closes only the documentation infrastructure. It does not mark all mathematical objects as proved. Dossiers with Documented=no/partial remain explicit proof/documentation obligations, and blue objects still require semantic/API correction before public-final promotion.                                                                                                                                                                                                                 |
| Source materials / references          | <ul style="list-style-type: none"> <li>object register</li> <li>current code derivation-site fields</li> <li>blue-object correction requirements</li> <li>YAML single-master rule</li> </ul>                                                                                                                                                                                                                                                                                                       |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"><li>ObjectProofDossier</li><li>ObjectProofDocumentationStatus</li><li>proof-dossier anchor</li><li>documented yes/no field</li><li>source object / definition / target / proof / verification slots</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Implemented modules                  | <ul style="list-style-type: none"><li>masters/cnna__roadmap__master.yaml</li><li>scripts/generate_tables.py</li><li>tables/object_register.tex</li><li>tables/object_proof_documentation.tex</li><li>tables/object_proof_link_register.tex</li><li>generated_exports/object_proof_documentation.generated.tsv</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Modified modules                     | <ul style="list-style-type: none"><li>masters/cnna__roadmap__master.yaml</li><li>scripts/generate_tables.py</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>no Lean source changed</li><li>no new @[simp] attribute introduced</li><li>no operational noncomputable/classical helper introduced</li><li>no independent proofbook source file created</li><li>no duplicate object list outside the YAML master</li><li>blue objects not promoted by documentation alone</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| No-go guardrails                     | <ul style="list-style-type: none"><li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li><li>no @[simp] attributes in the operational path</li><li>no noncomputable in the operational path</li><li>no classical helper unless isolated in allowed mirror/non-operational layer</li><li>no Matrix.inv or external inverse oracle in operational derivations</li><li>no free formula without proof-carrying entry structure</li><li>no consumer-side patch when the missing structure belongs earlier</li><li>no premature obstruction before a serious positive construction attempt</li><li>no phase-semantic public API names</li><li>no fragile rewrite chain when a structural constructor/projection proof is available</li><li>do not mark an object documented merely because it has a name or phase row</li><li>do not promote blue objects to white/public-final without semantic correction and proof-dossier verification</li><li>do not duplicate proof explanations in free prose outside the generated object proof section</li></ul> |
| Proposed modified modules            | <ul style="list-style-type: none"><li>masters/cnna__roadmap__master.yaml</li><li>scripts/generate_tables.py</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Reusable existing modules            | <ul style="list-style-type: none"><li>tables/object_register.tex</li><li>generated_exports/object_register.generated.tsv</li><li>scripts/generate_tables.py</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"><li>independent proofbook file not generated from YAML</li><li>phase-local object proof prose duplicated in scratchpad</li><li>object list copied outside object register</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Phase 1.1.1.1.4 – Declaration consumption classification map finalization

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                           |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span style="background-color: #d9ead3;">G</span>                                                                                                                                                                                                                                |
| Context class                          | LeanGovernancePhase                                                                                                                                                                                                                                                              |
| Phase documentation tier               | F3                                                                                                                                                                                                                                                                               |
| Phase documentation depth requirement  | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                                        |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                               |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                              |
| Implementation status                  | completed                                                                                                                                                                                                                                                                        |
| Preparation status                     | closed                                                                                                                                                                                                                                                                           |
| Phase type                             | refactor                                                                                                                                                                                                                                                                         |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                                          |
| Title / state                          | Declaration consumption classification map finalization                                                                                                                                                                                                                          |
| Preparation workflow                   | Use LEG10/LEG11 plus declaration/reference inventory to classify declarations and modules.                                                                                                                                                                                       |
| Source-to-CNNA mapping                 | LEG5 now closes after module-placement, reuse decisions, and proof-dossier anchors, so public API classification is not blind to code modularity or object-proof documentation.                                                                                                  |
| Dependency-depth audit                 | Depends on 1.1.1.1.1, 1.1.1.1.2, and 1.1.1.1.3. Without the proof-dossier infrastructure, classification may falsely make proof-carrying but semantically unfinished blue objects public-final.                                                                                  |
| Phase split decision                   | Renumbered from 1.1.1.1.3 to 1.1.1.1.4 because proof-documentation infrastructure must exist before final declaration/public-consumption classification.                                                                                                                         |
| Positive path plan                     | Classify declarations as core-computational, proof-bridge, public-wrapper, legacy-status, legacy-policy, phase-alias, audit-only, blue-semantic-correction, dead-candidate, or final-output, and require object-proof documentation links for public object or quantity outputs. |
| Module implementation manifest         | Consumes LEG10: the classification artifact is generated in planning tables/exports only; no new Lean module and no declaration movement.                                                                                                                                        |
| Semantic placement decision            | LEG5 remains a planning/API-hygiene artifact. Full records are generated_exports TSVs; PDF receives only the compact policy/count view.                                                                                                                                          |
| Reuse and duplication audit            | Consumes LEG11: generator-like classification is implemented as an extension of scripts/generate_tables.py, not a second inventory/export generator.                                                                                                                             |
| Canonical generator policy             | Use existing single-master table generator as the canonical seam for planning ledgers; future Lean audit mirror is optional and must not duplicate semantics.                                                                                                                    |
| Module compactness rule                | Classification operates at module/declaration/object record granularity but stores no huge declaration table in the PDF.                                                                                                                                                         |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                |
| Refactor migration metrics             | No Lean declarations migrated; import_graph_delta=none for Lean source; wrapper coexistence unchanged; generated planning artifacts only.                                                                                                                                        |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                       |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                               |
| Consumer-contract direction            | Exports generated TSVs for 1.1.1.2 blue-object correction and 1.1.1.3 public API/consumer whitelist; does not grant handoff rights.                                                                                                                                              |
| Governance-overhead review             | Useful overhead: LEG10/LEG11/LEG12 prevented a name-only classifier by forcing module placement, duplicate-policy, and proof-dossier gates before classification.                                                                                                                |
| Proof rehearsal before implementation  | Module reachability is not semantic consumption. For each class, cite direct consumer evidence, public API role, or legacy/audit reason.                                                                                                                                         |
| Expected tactics                       | No Lean tactics expected.                                                                                                                                                                                                                                                        |
| Expected mathlib imports               | None.                                                                                                                                                                                                                                                                            |
| ExecComplex/computable plan            | No ExecComplex object introduced.                                                                                                                                                                                                                                                |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                              |

|                                      |                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computable path non-skeleton status  | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                           |
| Expected executable witness          | not specified yet                                                                                                                                                                                                                                                                                                                 |
| Mathlib complication log             | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                              |
| Noncomputable boundary audit         | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                      |
| Mathlib mitigation/refactor layer    | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                         |
| Bridge gap status                    | If an item cannot be classified, keep it out of public API and route to the natural source/correction phase.                                                                                                                                                                                                                      |
| Natural backjump/refactor layer      | Backjump to module manifest if classification depends on placement;<br>backjump to generator policy if duplicate generator semantics are detected.                                                                                                                                                                                |
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation. |
| Global no-go inheritance             | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                          |
| How implemented                      | Closed as v43 infrastructure/generator gate. Added a top-level <code>consumption_classification_map</code> policy, extended <code>scripts/generate_tables.py</code> to validate the exact class set, and generated module/declaration/object consumption TSV records from the frozen repo inventory and object register.          |
| Verification / evidence              | <code>scripts/generate_tables.py</code> completed on the v43 master and generated full module, declaration, and object consumption ledgers. The PDF renders the policy and aggregate counts. The generator forces B-status registered objects into blue-semantic-correction and blocks public/final promotion in this gate.       |
| Proof strategy                       | No Lean proof. Deterministic generator validation and classification evidence from repo inventory, symbol-reference summaries, sink taxonomy, object register, and proof-dossier status.                                                                                                                                          |
| Build result                         | Planning generator and PDF compilation performed in this environment;<br>prior Lean build was user-reported green. No new Lean build is claimed because no Lean source was changed.                                                                                                                                               |
| Post-implementation plan check       | After closure, advance to blue-object semantic correction ledger.                                                                                                                                                                                                                                                                 |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                    |
| Implementation notes                 | No Lean source was moved or edited. Full declaration records live in <code>generated_exports/declaration_consumption_records.generated.tsv</code> ; the PDF intentionally renders only compact policy/count tables.                                                                                                               |
| Source materials / references        | <ul style="list-style-type: none"> <li>LEG10 module manifest</li> <li>LEG11 reuse/generator audit</li> <li>object proof documentation infrastructure</li> <li>declaration/reference inventory</li> </ul>                                                                                                                          |
| Extracted established theory objects | <ul style="list-style-type: none"> <li><code>ConsumptionClassificationMap</code></li> <li>declaration classes</li> <li>semantic consumption vs syntactic reachability distinction</li> </ul>                                                                                                                                      |
| Implemented modules                  | <ul style="list-style-type: none"> <li><code>scripts/generate_tables.py</code></li> <li><code>masters/cnna_roadmap_master.yaml</code></li> <li><code>tables/consumption_classification_map.tex</code></li> <li><code>generated_exports/*consumption*.tsv</code></li> </ul>                                                        |
| Modified modules                     | <ul style="list-style-type: none"> <li><code>scripts/generate_tables.py</code></li> <li><code>masters/cnna_roadmap_master.yaml</code></li> <li><code>README.md</code></li> <li><code>CNNA_Planning_and_Documentation_Tool_v43_Declaration_Consumption_Classification_Map_A3_English.tex</code></li> </ul>                         |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source modified</li> <li>no BuildAll/import reachability used as semantic consumption evidence</li> <li>blue objects forced away from public-final classes</li> <li>full declaration dump exported as TSV, not hand-curated prose</li> </ul>                                       |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no__go__guardrails</code> apply; this phase list is local addenda only.</li> <li>no BuildAll reachability as semantic consumption evidence</li> <li>no constructor-count-only quarantine</li> <li>no deletion before pruning ledger</li> <li>no public export for uncorrected blue objects</li> </ul>                                         |
| Reusable existing modules            | <ul style="list-style-type: none"> <li><code>scripts/generate_tables.py</code></li> <li><code>repo__inventory/raw__export/modules.json</code></li> <li><code>repo__inventory/raw__export/symbol__references/declarations.json</code></li> <li><code>repo__inventory/raw__export/symbol__references/references.json</code></li> <li><code>repo__inventory/raw__export/architectural__core/sink__taxonomy.json</code></li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>BuildAll reachability as false semantic evidence</li> <li>SM phase aliases promoted as public API</li> <li>blue object treated as final-output by documentation alone</li> </ul>                                                                                                                                                                                                        |

**Phase 1.1.1.1.5 – Pre-substantive-code readiness umbrella: semantic consolidation, full IST intake, neutral naming, refactor, post-scan, and release proof**

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                                                                                                                                         |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                            |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | umbrella-delegated-to-child-sequence                                                                                                                                                                                                                                                                                                      |
| Preparation status                    | active-prepared                                                                                                                                                                                                                                                                                                                           |
| Phase type                            | refactor                                                                                                                                                                                                                                                                                                                                  |
| Phase-type-specific contract          | Refactor phase: wrapper-first/facade-first, bounded migration package, import/declaration diff, consumer-breakage audit, and local full-build evidence before closeout.                                                                                                                                                                   |
| Title / state                         | Pre-substantive-code readiness umbrella                                                                                                                                                                                                                                                                                                   |
| Preparation workflow                  | Semantic module consolidation and compatibility-preserving refactor workflow.                                                                                                                                                                                                                                                             |
| Source-to-CNNA mapping                | This is not a mathematical CNNA object. It maps repository structure to refactor governance: only semantically identical representation layers may be consolidated; proof-producing layers remain split unless their proof/provenance information is preserved by a better canonical module.                                              |
| Dependency-depth audit                | Prerequisites closed: module manifest LEG10, reuse/generator policy LEG11, proofbook infrastructure LEG12, declaration consumption map LEG5, phase-object policy LEG21, and refactor regression protocol GNG03. Blocking item: a per-cluster ledger is not yet closed, so 1.1.1.2 is now orange rather than active.                       |
| Phase split decision                  | Create one active audit gate 1.1.1.1.5 now. Only if the ledger finds safe changes should it open bounded execution subphases, e.g. Handoff feedback wrappers, BoundarySource candidate ledgers, or notation-facade normalization. Do not split by every file family yet.                                                                  |
| Positive path plan                    | Fill LEG36 with candidate clusters: high-confidence consolidate/facade families, medium-risk schema abstractions, explicit no-merge proof-stage families, and deferred notation-facade normalization. Then either close with no Lean changes or open one small refactor execution phase with old-module facades and local build evidence. |
| Module implementation manifest        | No Lean module added now. Potential future modules, if authorized: <code>CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean</code> ; <code>CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean</code> ; <code>CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean</code> . Old modules remain as facades initially.                |
| Semantic placement decision           | The audit belongs after declaration consumption classification and before blue-object content correction. This is the natural layer because it decides public/import surfaces before new semantic content is added.                                                                                                                       |
| Reuse and duplication audit           | RD08 records the consolidation decision rule. Same-semantics wrappers may be unified; proof-stage modules must remain separated until a stronger canonical proof module preserves their role.                                                                                                                                             |
| Canonical generator policy            | No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.                                                                                                                                               |



|                                        |                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module compactness rule                | Slim modules only when statement/provenance auditability is preserved or improved. Small clone wrappers are candidates; long proof chains are not automatically candidates; a monolithic notation facade may require splitting rather than merging.                                                                                                                   |
| Refactor regression protocol           | Mandatory before any Lean merge: retain old modules as facades in pass 1; record declarations _migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.                                                                                                                                |
| Refactor migration metrics             | Not yet executed. Initial audit candidates: BoundarySource four substrate ledger clones; Handoff three A_to_{C,D,E} output aliases; Handoff four {B,C,D,E}_to_A feedback-input clones; no-merge candidates: SM generated proof chains, Cayley-Dickson S6 chain; watch item: Arithmetic/Notation monolith.                                                             |
| Lean/mathlib version policy            | Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed; refactor-only changes must compile locally before closeout.                                                                                                                                                                                                                              |
| ExecComplex boundary policy            | EXEC-2 remains unchanged. Consolidation may not move rational executable data behind noncomputable or analytic mirrors.                                                                                                                                                                                                                                               |
| Consumer-contract direction            | Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.                                                                                                                                                                   |
| Governance-overhead review             | This gate is justified because concrete clone families exist, but it must remain compact: one ledger, no broad rewrite, no full code normalization before the active content path.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | For each candidate cluster prove/check: same semantic operation, same or explicitly mapped public declarations, no consumer lost, import direction preserved or improved, natural generating layer chosen, old imports kept through facades, full local build green, and statement/provenance granularity not lost.                                                   |
| Expected tactics                       | If later Lean facades are created, expect explicit abbrev/def wrappers, namespace-preserving aliases, extensionality reuse where existing, and rfl proofs. Avoid tactic-heavy rewrites; no @[simp] promotion in operational path.                                                                                                                                     |
| Expected mathlib imports               | No new mathlib imports expected for audit. Any later facade/refactor should reuse existing imports; new imports require Lean 4.28.0 verification.                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | No ExecComplex changes in the audit. If Arithmetic/Notation or operational modules are touched later, preserve all concrete executable witnesses and rational-boundary policy.                                                                                                                                                                                        |
| Concrete computable instance traversal | Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.                                                                                                                                                     |
| Computable path non-skeleton status    | Audit-only. A later refactor may not skeletonize operational paths; it must preserve concrete generated source, DtN/Schur, determinant, and discriminant witnesses.                                                                                                                                                                                                   |
| Expected executable witness            | For audit: generated module inventory and candidate report. For later code refactor: unchanged downstream Lean build plus regenerated module/declaration inventory showing no lost public declarations unless explicitly facaded.                                                                                                                                     |
| Mathlib complication log               | No mathlib complication in audit. Later code movement may expose import-minimization issues; resolve by preserving existing import surfaces first.                                                                                                                                                                                                                    |
| Noncomputable boundary audit           | No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths.                                                                                                                                                                                                                                      |
| Mathlib mitigation/refactor layer      | If new imports are needed, prefer local facade modules or existing imported modules rather than widening mathlib dependency surface.                                                                                                                                                                                                                                  |
| Bridge gap status                      | Current bridge gap is governance-level: exact consolidation candidates are not yet fully ledgered. No mathematical bridge is missing. Do not patch consumers; place any new facade at the natural source layer of the duplicate/schema family.                                                                                                                        |
| Natural backjump/refactor layer        | BoundarySource candidate ledgers belong under Arithmetic/BoundarySource; Handoff input/output wrapper consolidation belongs under Handoff/Inputs or Outputs; notation normalization belongs at notation facade boundaries; proof-stage chains stay in their proof-stage folders unless a canonical proof module is naturally introduced.                              |
| Implementation prompt contract         | Before writing any new substantive Lean code, complete the LEG36 audit. If refactor is authorized, produce a bounded migration packet: old-module facades, changed imports, declaration delta, consumer-breakage audit, generated inventory diff, and user/local Lean build result.                                                                                   |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG26)                                                                                                                                                                                                                                                                                                              |
| How implemented                        | V1.21 delegates the active cursor to 1.1.1.1.5.1 and expands the old semantic consolidation gate into a five-step readiness chain before 1.1.1.2.                                                                                                                                                                                                                     |
| Verification / evidence                | Static repository facts: 287 Lean modules and zero missing internal import edges from repo_inventory; direct spot-checks found clone/schema families in BoundarySource substrate ledgers and Handoff input/output wrappers; long SM and Cayley-Dickson chains remain proof-stage candidates for keep/no-merge until a dedicated provenance-preservation proof exists. |
| Proof strategy                         | Refactor-governance proof by finite inventory plus targeted code inspection: identify clusters, classify safe/unsafe consolidation, require compatibility facades and regression metrics before any Lean file is moved or merged.                                                                                                                                     |
| Build result                           | No Lean build run; no Lean source changed. Python table generation and PDF compilation only.                                                                                                                                                                                                                                                                          |
| Post-implementation plan check         | After audit closure, update phase/object statuses. If no code merge is executed, return cursor to 1.1.1.2. If a refactor execution phase opens, require local full-build evidence before any new substantive code.                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with updated YAML, generated tables/exports, report, and PDF. Do not claim Lean build success unless actually run or user-reported.                                                                                                                                                                                                             |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation notes                 | This phase is deliberately before 1.1.1.2. Its output is not a code merge, but a precise ledger deciding whether a later bounded Lean refactor phase is warranted. The default is no merge unless semantic equivalence, natural-layer placement, facade compatibility, and regression evidence are explicit.                                                                                                                                                                      |
| Source materials / references        | <ul style="list-style-type: none"> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/import_edges.json</li> <li>repo_inventory/raw_export/symbol_reference_summary.json</li> <li>BoundarySource substrate candidate modules</li> <li>Handoff input/output wrapper modules</li> <li>Smoke/SM generated proof-chain modules</li> <li>Structural/CayleyDickson/S6 proof-chain modules</li> <li>Arithmetic/Notation.lean size/role observation</li> </ul> |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>safe-merge candidate cluster</li> <li>facade-preserve candidate cluster</li> <li>proof-stage no-merge cluster</li> <li>module compactness regression metric</li> <li>natural-layer merge placement rule</li> </ul>                                                                                                                                                                                                                         |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> <li>Repository/SEMANTIC_MODULE_CONSOLIDATION_AUDIT_REPORT.md</li> <li>Repository/CNNA_Planning_and_Documentation_Tool_V1_12_Data_2026-05-10T05-51-42_CEST_A3_English.tex</li> </ul>                                                                                                                                                                                    |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source move in the audit phase</li> <li>no merge by filename similarity alone</li> <li>no deletion of old import surfaces without facade migration</li> <li>no merging proof-stage chains that encode derivation/provenance order</li> <li>no new semantic truth layer from the audit ledger</li> </ul>                                                                                                                            |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG03, GNG21, GNG23-GNG26.</li> <li>no semantic merge without natural-layer placement</li> <li>no deletion of compatibility imports in the first refactor pass</li> <li>no proof-stage chain merge by naming similarity</li> <li>no code slimming that reduces statement/provenance auditability</li> </ul>                                                                |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li> <li>CNNA/PillarA/Handoff/Core/SectorExport.lean</li> <li>CNNA/PillarA/Handoff/Outputs/Closed.lean</li> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/consumption_classification_map.generated.tsv</li> </ul>                                                                                                                                    |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>BoundarySource.*Substrate.ledger and ledger_* theorems</li> <li>AToCHandoffStrong/AToDHandoffStrong/AToEHandoffStrong aliases</li> <li>BToAInput/CToAInput/DToAInput/EToAInput feedback structures and ext/handoff projections</li> </ul>                                                                                                                                                                                                  |

### Phase 1.1.1.1.5.1 – Full IST module/declaration/object intake completeness audit before any new substantive code

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field        | Record                                                                                    |
|--------------|-------------------------------------------------------------------------------------------|
| Phase status | <span style="background-color: #fff2cc; border: 1px solid #000; padding: 0 2px;">Y</span> |

|                                        |                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                          | LeanGovernancePhase                                                                                                                                                                                                                                                                                                 |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                      |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                  |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                 |
| Implementation status                  | active-prepared                                                                                                                                                                                                                                                                                                     |
| Preparation status                     | active-prepared                                                                                                                                                                                                                                                                                                     |
| Phase type                             | refactor                                                                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Pre-substantive-code readiness/refactor gate; not a new mathematical derivation phase.                                                                                                                                                                                                                              |
| Title / state                          | Full IST module/declaration/object intake completeness audit                                                                                                                                                                                                                                                        |
| Preparation workflow                   | Semantic module consolidation and compatibility-preserving refactor workflow.                                                                                                                                                                                                                                       |
| Source-to-CNNA mapping                 | This is not a mathematical CNNA object. It maps repository structure to refactor governance: only semantically identical representation layers may be consolidated; proof-producing layers remain split unless their proof/provenance information is preserved by a better canonical module.                        |
| Dependency-depth audit                 | V1.21 identifies this phase as part of the hard pre-code readiness chain before 1.1.1.2.                                                                                                                                                                                                                            |
| Phase split decision                   | Explicit child split under 1.1.1.1.5; avoids treating a broad semantic-consolidation umbrella as sufficient proof for new content code.                                                                                                                                                                             |
| Positive path plan                     | Execute and close this gate constructively; only create obstruction/red blockers if the positive closure attempt identifies missing source/proof data.                                                                                                                                                              |
| Module implementation manifest         | No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean; CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean; CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.                                    |
| Semantic placement decision            | Natural placement is before blue-object semantic correction and any new Lean content; do not patch later consumer phases.                                                                                                                                                                                           |
| Reuse and duplication audit            | RD08 records the consolidation decision rule. Same-semantics wrappers may be unified; proof-stage modules must remain separated until a stronger canonical proof module preserves their role.                                                                                                                       |
| Canonical generator policy             | No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.                                                                                                                         |
| Module compactness rule                | Preserve or improve auditability; neutralize names and consolidate only with facades/regression evidence.                                                                                                                                                                                                           |
| Refactor regression protocol           | Mandatory before any Lean merge: retain old modules as facades in pass 1; record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.                                                                               |
| Refactor migration metrics             | Not yet executed. Initial audit candidates: BoundarySource four substrate ledger clones; Handoff three A_to_{C,D,E} output aliases; Handoff four {B,C,D,E}_to_A feedback-input clones; no-merge candidates: SM generated proof chains, Cayley-Dickson S6 chain; watch item: Arithmetic/Notation monolith.           |
| Lean/mathlib version policy            | Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed; refactor-only changes must compile locally before closeout.                                                                                                                                                                            |
| ExecComplex boundary policy            | EXEC-2 remains unchanged. Consolidation may not move rational executable data behind noncomputable or analytic mirrors.                                                                                                                                                                                             |
| Consumer-contract direction            | Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.                                                                                                                 |
| Governance-overhead review             | This gate is justified because concrete clone families exist, but it must remain compact: one ledger, no broad rewrite, no full code normalization before the active content path.                                                                                                                                  |
| Proof rehearsal before implementation  | For each candidate cluster prove/check: same semantic operation, same or explicitly mapped public declarations, no consumer lost, import direction preserved or improved, natural generating layer chosen, old imports kept through facades, full local build green, and statement/provenance granularity not lost. |
| Expected tactics                       | If later Lean facades are created, expect explicit abbrev/def wrappers, namespace-preserving aliases, extensionality reuse where existing, and rfl proofs. Avoid tactic-heavy rewrites; no @[simp] promotion in operational path.                                                                                   |
| Expected mathlib imports               | No new mathlib imports expected for audit. Any later facade/refactor should reuse existing imports; new imports require Lean 4.28.0 verification.                                                                                                                                                                   |
| ExecComplex/computable plan            | No ExecComplex changes in the audit. If Arithmetic/Notation or operational modules are touched later, preserve all concrete executable witnesses and rational-boundary policy.                                                                                                                                      |
| Concrete computable instance traversal | Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.                                                                                                   |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computable path non-skeleton status  | Audit-only. A later refactor may not skeletonize operational paths; it must preserve concrete generated source, DtN/Schur, determinant, and discriminant witnesses.                                                                                                                                                                                                                                                |
| Expected executable witness          | For audit: generated module inventory and candidate report. For later code refactor: unchanged downstream Lean build plus regenerated module/declaration inventory showing no lost public declarations unless explicitly facaded.                                                                                                                                                                                  |
| Mathlib complication log             | No mathlib complication in audit. Later code movement may expose import-minimization issues; resolve by preserving existing import surfaces first.                                                                                                                                                                                                                                                                 |
| Noncomputable boundary audit         | No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths.                                                                                                                                                                                                                                                                                   |
| Mathlib mitigation/refactor layer    | If new imports are needed, prefer local facade modules or existing imported modules rather than widening mathlib dependency surface.                                                                                                                                                                                                                                                                               |
| Bridge gap status                    | Current bridge gap is governance-level: exact consolidation candidates are not yet fully ledgered. No mathematical bridge is missing. Do not patch consumers; place any new facade at the natural source layer of the duplicate/schema family.                                                                                                                                                                     |
| Natural backjump/refactor layer      | BoundarySource candidate ledgers belong under Arithmetic/BoundarySource; Handoff input/output wrapper consolidation belongs under Handoff/Inputs or Outputs; notation normalization belongs at notation facade boundaries; proof-stage chains stay in their proof-stage folders unless a canonical proof module is naturally introduced.                                                                           |
| Implementation prompt contract       | Before writing any new substantive Lean code, complete the LEG36 audit. If refactor is authorized, produce a bounded migration packet: old-module facades, changed imports, declaration delta, consumer-breakage audit, generated inventory diff, and user/local Lean build result.                                                                                                                                |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG26)                                                                                                                                                                                                                                                                                                                                                        |
| How implemented                      | V1.21 planning update only: Audit every current source module/declaration/object mapping before planning code movement.                                                                                                                                                                                                                                                                                            |
| Verification / evidence              | No Lean source changed in V1.21. Closure requires exact inventory/refactor/build evidence recorded in the corresponding proof dossier and scratchpad.                                                                                                                                                                                                                                                              |
| Proof strategy                       | Close the phase by explicit references to derived-only Lean objects/proof dossiers, source inventory exports, and local build evidence where Lean refactor is executed. If proof fails, spawn upstream red blocker at natural source layer.                                                                                                                                                                        |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table/PDF generation only.                                                                                                                                                                                                                                                                                                                                    |
| Post-implementation plan check       | Do not advance to 1.1.1.2 until the complete 1.1.1.1.5.1-1.1.1.1.5.5 chain closes.                                                                                                                                                                                                                                                                                                                                 |
| Artifact packaging rule              | Return a full zip with updated YAML, exports, report and generated docs; do not claim Lean build.                                                                                                                                                                                                                                                                                                                  |
| Implementation notes                 | Audit every current source module/declaration/object mapping before planning code movement.                                                                                                                                                                                                                                                                                                                        |
| Source materials / references        | <ul style="list-style-type: none"> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/*.tsv</li> <li>repo_snapshot/CNNA/**/*.lean</li> <li>masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>LEG37</li> </ul>                                                                                                                                                                                                                                                                                                                                                            |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> <li>Repository/archive/reports/V1_21_PRE_CODE_READINESS_DERIVATION_ORDER_AUDIT_REPORT.md</li> </ul>                                                                                                                                                                                                     |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source move in the audit phase</li> <li>no merge by filename similarity alone</li> <li>no deletion of old import surfaces without facade migration</li> <li>no merging proof-stage chains that encode derivation/provenance order</li> <li>no new semantic truth layer from the audit ledger</li> </ul>                                                             |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG03, GNG21, GNG23-GNG26.</li> <li>no semantic merge without natural-layer placement</li> <li>no deletion of compatibility imports in the first refactor pass</li> <li>no proof-stage chain merge by naming similarity</li> <li>no code slimming that reduces statement/provenance auditability</li> </ul> |

|                                      |                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reusable existing modules            | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li> <li>CNNA/PillarA/Handoff/Core/SectorExport.lean</li> <li>CNNA/PillarA/Handoff/Outputs/Closed.lean</li> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/consumption_classification_map.generated.tsv</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>phase-semantic module names;</li> <li>unclassified declarations;</li> <li>duplicate wrapper modules;</li> <li>stale object/proof mappings.</li> </ul>                                                                                                                                                   |

### Phase 1.1.1.1.5.2 – Neutral module-name and namespace/facade migration plan

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #fff2cc;">O</span>                                                                                                                                                                                                                                            |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                                                                                          |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                           |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                               |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                           |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                          |
| Implementation status                 | implementable-after-listed-predecessor                                                                                                                                                                                                                                                       |
| Preparation status                    | prepared-after-predecessor                                                                                                                                                                                                                                                                   |
| Phase type                            | refactor                                                                                                                                                                                                                                                                                     |
| Phase-type-specific contract          | Pre-substantive-code readiness/refactor gate; not a new mathematical derivation phase.                                                                                                                                                                                                       |
| Title / state                         | Neutral module-name and facade migration plan                                                                                                                                                                                                                                                |
| Preparation workflow                  | Semantic module consolidation and compatibility-preserving refactor workflow.                                                                                                                                                                                                                |
| Source-to-CNNA mapping                | This is not a mathematical CNNA object. It maps repository structure to refactor governance: only semantically identical representation layers may be consolidated; proof-producing layers remain split unless their proof/provenance information is preserved by a better canonical module. |
| Dependency-depth audit                | V1.21 identifies this phase as part of the hard pre-code readiness chain before 1.1.1.2.                                                                                                                                                                                                     |
| Phase split decision                  | Explicit child split under 1.1.1.1.5; avoids treating a broad semantic-consolidation umbrella as sufficient proof for new content code.                                                                                                                                                      |
| Positive path plan                    | Execute and close this gate constructively; only create obstruction/red blockers if the positive closure attempt identifies missing source/proof data.                                                                                                                                       |
| Module implementation manifest        | No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean; CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean; CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.             |
| Semantic placement decision           | Natural placement is before blue-object semantic correction and any new Lean content; do not patch later consumer phases.                                                                                                                                                                    |
| Reuse and duplication audit           | RD08 records the consolidation decision rule. Same-semantics wrappers may be unified; proof-stage modules must remain separated until a stronger canonical proof module preserves their role.                                                                                                |
| Canonical generator policy            | No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.                                                                                                  |
| Module compactness rule               | Preserve or improve auditability; neutralize names and consolidate only with facades/regression evidence.                                                                                                                                                                                    |
| Refactor regression protocol          | Mandatory before any Lean merge: retain old modules as facades in pass 1; record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.                                                        |

|                                        |                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refactor migration metrics             | Not yet executed. Initial audit candidates: BoundarySource four substrate ledger clones; Handoff three A_to_{C,D,E} output aliases; Handoff four {B,C,D,E}_to_A feedback-input clones; no-merge candidates: SM generated proof chains, Cayley-Dickson S6 chain; watch item: Arithmetic/Notation monolith.                                |
| Lean/mathlib version policy            | Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed; refactor-only changes must compile locally before closeout.                                                                                                                                                                                                 |
| ExecComplex boundary policy            | EXEC-2 remains unchanged. Consolidation may not move rational executable data behind noncomputable or analytic mirrors.                                                                                                                                                                                                                  |
| Consumer-contract direction            | Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.                                                                                                                                      |
| Governance-overhead review             | This gate is justified because concrete clone families exist, but it must remain compact: one ledger, no broad rewrite, no full code normalization before the active content path.                                                                                                                                                       |
| Proof rehearsal before implementation  | For each candidate cluster prove/check: same semantic operation, same or explicitly mapped public declarations, no consumer lost, import direction preserved or improved, natural generating layer chosen, old imports kept through facades, full local build green, and statement/provenance granularity not lost.                      |
| Expected tactics                       | If later Lean facades are created, expect explicit abbrev/def wrappers, namespace-preserving aliases, extensionality reuse where existing, and rfl proofs. Avoid tactic-heavy rewrites; no @[simp] promotion in operational path.                                                                                                        |
| Expected mathlib imports               | No new mathlib imports expected for audit. Any later facade/refactor should reuse existing imports; new imports require Lean 4.28.0 verification.                                                                                                                                                                                        |
| ExecComplex/computable plan            | No ExecComplex changes in the audit. If Arithmetic/Notation or operational modules are touched later, preserve all concrete executable witnesses and rational-boundary policy.                                                                                                                                                           |
| Concrete computable instance traversal | Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.                                                                                                                        |
| Computable path non-skeleton status    | Audit-only. A later refactor may not skeletonize operational paths; it must preserve concrete generated source, DtN/Schur, determinant, and discriminant witnesses.                                                                                                                                                                      |
| Expected executable witness            | For audit: generated module inventory and candidate report. For later code refactor: unchanged downstream Lean build plus regenerated module/declaration inventory showing no lost public declarations unless explicitly facaded.                                                                                                        |
| Mathlib complication log               | No mathlib complication in audit. Later code movement may expose import-minimization issues; resolve by preserving existing import surfaces first.                                                                                                                                                                                       |
| Noncomputable boundary audit           | No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths.                                                                                                                                                                                                         |
| Mathlib mitigation/refactor layer      | If new imports are needed, prefer local facade modules or existing imported modules rather than widening mathlib dependency surface.                                                                                                                                                                                                     |
| Bridge gap status                      | Current bridge gap is governance-level: exact consolidation candidates are not yet fully ledgered. No mathematical bridge is missing. Do not patch consumers; place any new facade at the natural source layer of the duplicate/schema family.                                                                                           |
| Natural backjump/refactor layer        | BoundarySource candidate ledgers belong under Arithmetic/BoundarySource; Handoff input/output wrapper consolidation belongs under Handoff/Inputs or Outputs; notation normalization belongs at notation facade boundaries; proof-stage chains stay in their proof-stage folders unless a canonical proof module is naturally introduced. |
| Implementation prompt contract         | Before writing any new substantive Lean code, complete the LEG36 audit. If refactor is authorized, produce a bounded migration packet: old-module facades, changed imports, declaration delta, consumer-breakage audit, generated inventory diff, and user/local Lean build result.                                                      |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG26)                                                                                                                                                                                                                                                                              |
| How implemented                        | V1.21 planning update only: Plan neutral names/facades/no-merge exceptions after LEG37 closes.                                                                                                                                                                                                                                           |
| Verification / evidence                | No Lean source changed in V1.21. Closure requires exact inventory/refactor/build evidence recorded in the corresponding proof dossier and scratchpad.                                                                                                                                                                                    |
| Proof strategy                         | Close the phase by explicit references to derived-only Lean objects/proof dossiers, source inventory exports, and local build evidence where Lean refactor is executed. If proof fails, spawn upstream red blocker at natural source layer.                                                                                              |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table/PDF generation only.                                                                                                                                                                                                                                                          |
| Post-implementation plan check         | Do not advance to 1.1.1.2 until the complete 1.1.1.5.1-1.1.1.5.5 chain closes.                                                                                                                                                                                                                                                           |
| Artifact packaging rule                | Return a full zip with updated YAML, exports, report and generated docs; do not claim Lean build.                                                                                                                                                                                                                                        |
| Implementation notes                   | Plan neutral names/facades/no-merge exceptions after LEG37 closes.                                                                                                                                                                                                                                                                       |
| Source materials / references          | <ul style="list-style-type: none"> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/*.tsv</li> <li>repo_snapshot/CNNA/**/*.lean</li> <li>masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                      |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"> <li>LEG38</li> </ul>                                                                                                                                                                                                                                                                                                                                                            |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> <li>Repository/archive/reports/V1_21_PRE_CODE_READINESS_DERIVATION_ORDER_AUDIT_REPORT.md</li> </ul>                                                                                                                                                                                                     |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source move in the audit phase</li> <li>no merge by filename similarity alone</li> <li>no deletion of old import surfaces without facade migration</li> <li>no merging proof-stage chains that encode derivation/provenance order</li> <li>no new semantic truth layer from the audit ledger</li> </ul>                                                             |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG03, GNG21, GNG23-GNG26.</li> <li>no semantic merge without natural-layer placement</li> <li>no deletion of compatibility imports in the first refactor pass</li> <li>no proof-stage chain merge by naming similarity</li> <li>no code slimming that reduces statement/provenance auditability</li> </ul> |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li> <li>CNNA/PillarA/Handoff/Core/SectorExport.lean</li> <li>CNNA/PillarA/Handoff/Outputs/Closed.lean</li> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/consumption_classification_map.generated.tsv</li> </ul>                                                                     |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>phase-semantic module names; unclassified declarations; duplicate wrapper modules; stale object/proof mappings.</li> </ul>                                                                                                                                                                                                                                                  |

### Phase 1.1.1.1.5.3 – Compatibility-preserving refactor execution gate for neutral module surface

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #fce4d6;">O</span>                                                                                                                                                                                                                                            |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                                                                                          |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                           |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                               |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                           |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                          |
| Implementation status                 | implementable-after-listed-predecessor                                                                                                                                                                                                                                                       |
| Preparation status                    | prepared-after-predecessor                                                                                                                                                                                                                                                                   |
| Phase type                            | refactor                                                                                                                                                                                                                                                                                     |
| Phase-type-specific contract          | Pre-substantive-code readiness/refactor gate; not a new mathematical derivation phase.                                                                                                                                                                                                       |
| Title / state                         | Compatibility-preserving refactor execution gate                                                                                                                                                                                                                                             |
| Preparation workflow                  | Semantic module consolidation and compatibility-preserving refactor workflow.                                                                                                                                                                                                                |
| Source-to-CNNA mapping                | This is not a mathematical CNNA object. It maps repository structure to refactor governance: only semantically identical representation layers may be consolidated; proof-producing layers remain split unless their proof/provenance information is preserved by a better canonical module. |

|                                        |                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependency-depth audit                 | V1.21 identifies this phase as part of the hard pre-code readiness chain before 1.1.1.2.                                                                                                                                                                                                                                                          |
| Phase split decision                   | Explicit child split under 1.1.1.1.5;<br>avoids treating a broad semantic-consolidation umbrella as sufficient proof for new content code.                                                                                                                                                                                                        |
| Positive path plan                     | Execute and close this gate constructively;<br>only create obstruction/red blockers if the positive closure attempt identifies missing source/proof data.                                                                                                                                                                                         |
| Module implementation manifest         | No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean;<br>CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean;<br>CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.                                                            |
| Semantic placement decision            | Natural placement is before blue-object semantic correction and any new Lean content;<br>do not patch later consumer phases.                                                                                                                                                                                                                      |
| Reuse and duplication audit            | RD08 records the consolidation decision rule. Same-semantics wrappers may be unified;<br>proof-stage modules must remain separated until a stronger canonical proof module preserves their role.                                                                                                                                                  |
| Canonical generator policy             | No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.                                                                                                                                                       |
| Module compactness rule                | Preserve or improve auditability;<br>neutralize names and consolidate only with facades/regression evidence.                                                                                                                                                                                                                                      |
| Refactor regression protocol           | Mandatory before any Lean merge: retain old modules as facades in pass 1;<br>record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.                                                                                                          |
| Refactor migration metrics             | Not yet executed. Initial audit candidates: BoundarySource four substrate ledger clones;<br>Handoff three A_to_{C,D,E} output aliases;<br>Handoff four {B,C,D,E}_to_A feedback-input clones;<br>no-merge candidates: SM generated proof chains, Cayley-Dickson S6 chain;<br>watch item: Arithmetic/Notation monolith.                             |
| Lean/mathlib version policy            | Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed;<br>refactor-only changes must compile locally before closeout.                                                                                                                                                                                                       |
| ExecComplex boundary policy            | EXEC-2 remains unchanged. Consolidation may not move rational executable data behind noncomputable or analytic mirrors.                                                                                                                                                                                                                           |
| Consumer-contract direction            | Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.                                                                                                                                               |
| Governance-overhead review             | This gate is justified because concrete clone families exist, but it must remain compact: one ledger, no broad rewrite, no full code normalization before the active content path.                                                                                                                                                                |
| Proof rehearsal before implementation  | For each candidate cluster prove/check: same semantic operation, same or explicitly mapped public declarations, no consumer lost, import direction preserved or improved, natural generating layer chosen, old imports kept through facades, full local build green, and statement/provenance granularity not lost.                               |
| Expected tactics                       | If later Lean facades are created, expect explicit abbrev/def wrappers, namespace-preserving aliases, extensionality reuse where existing, and rfl proofs. Avoid tactic-heavy rewrites;<br>no @[simp] promotion in operational path.                                                                                                              |
| Expected mathlib imports               | No new mathlib imports expected for audit. Any later facade/refactor should reuse existing imports;<br>new imports require Lean 4.28.0 verification.                                                                                                                                                                                              |
| ExecComplex/computable plan            | No ExecComplex changes in the audit. If Arithmetic/Notation or operational modules are touched later, preserve all concrete executable witnesses and rational-boundary policy.                                                                                                                                                                    |
| Concrete computable instance traversal | Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.                                                                                                                                 |
| Computable path non-skeleton status    | Audit-only. A later refactor may not skeletonize operational paths;<br>it must preserve concrete generated source, DtN/Schur, determinant, and discriminant witnesses.                                                                                                                                                                            |
| Expected executable witness            | For audit: generated module inventory and candidate report. For later code refactor: unchanged downstream Lean build plus regenerated module/declaration inventory showing no lost public declarations unless explicitly facaded.                                                                                                                 |
| Mathlib complication log               | No mathlib complication in audit. Later code movement may expose import-minimization issues;<br>resolve by preserving existing import surfaces first.                                                                                                                                                                                             |
| Noncomputable boundary audit           | No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths.                                                                                                                                                                                                                  |
| Mathlib mitigation/refactor layer      | If new imports are needed, prefer local facade modules or existing imported modules rather than widening mathlib dependency surface.                                                                                                                                                                                                              |
| Bridge gap status                      | Current bridge gap is governance-level: exact consolidation candidates are not yet fully ledgered. No mathematical bridge is missing. Do not patch consumers;<br>place any new facade at the natural source layer of the duplicate/schema family.                                                                                                 |
| Natural backjump/refactor layer        | BoundarySource candidate ledgers belong under Arithmetic/BoundarySource;<br>Handoff input/output wrapper consolidation belongs under Handoff/Inputs or Outputs;<br>notation normalization belongs at notation facade boundaries;<br>proof-stage chains stay in their proof-stage folders unless a canonical proof module is naturally introduced. |
| Implementation prompt contract         | Before writing any new substantive Lean code, complete the LEG36 audit. If refactor is authorized, produce a bounded migration packet: old-module facades, changed imports, declaration delta, consumer-breakage audit, generated inventory diff, and user/local Lean build result.                                                               |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG26)                                                                                                                                                                                                                                                                                                                                                        |
| How implemented                      | V1.21 planning update only: Execute only approved neutral refactor with facades and local build evidence.                                                                                                                                                                                                                                                                                                          |
| Verification / evidence              | No Lean source changed in V1.21. Closure requires exact inventory/refactor/build evidence recorded in the corresponding proof dossier and scratchpad.                                                                                                                                                                                                                                                              |
| Proof strategy                       | Close the phase by explicit references to derived-only Lean objects/proof dossiers, source inventory exports, and local build evidence where Lean refactor is executed. If proof fails, spawn upstream red blocker at natural source layer.                                                                                                                                                                        |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table/PDF generation only.                                                                                                                                                                                                                                                                                                                                    |
| Post-implementation plan check       | Do not advance to 1.1.1.2 until the complete 1.1.1.5.1-1.1.1.5.5 chain closes.                                                                                                                                                                                                                                                                                                                                     |
| Artifact packaging rule              | Return a full zip with updated YAML, exports, report and generated docs;<br>do not claim Lean build.                                                                                                                                                                                                                                                                                                               |
| Implementation notes                 | Execute only approved neutral refactor with facades and local build evidence.                                                                                                                                                                                                                                                                                                                                      |
| Source materials / references        | <ul style="list-style-type: none"> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/*.tsv</li> <li>repo_snapshot/CNNA/**/*.lean</li> <li>masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>LEG39</li> </ul>                                                                                                                                                                                                                                                                                                                                                            |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> <li>Repository/archive/reports/V1_21_PRE_CODE_READINESS_DERIVATION_ORDER_AUDIT_REPORT.md</li> </ul>                                                                                                                                                                                                     |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source move in the audit phase</li> <li>no merge by filename similarity alone</li> <li>no deletion of old import surfaces without facade migration</li> <li>no merging proof-stage chains that encode derivation/provenance order</li> <li>no new semantic truth layer from the audit ledger</li> </ul>                                                             |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG03, GNG21, GNG23-GNG26.</li> <li>no semantic merge without natural-layer placement</li> <li>no deletion of compatibility imports in the first refactor pass</li> <li>no proof-stage chain merge by naming similarity</li> <li>no code slimming that reduces statement/provenance auditability</li> </ul> |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li> <li>CNNA/PillarA/Handoff/Core/SectorExport.lean</li> <li>CNNA/PillarA/Handoff/Outputs/Closed.lean</li> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/consumption_classification_map.generated.tsv</li> </ul>                                                                     |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>phase-semantic module names;</li> <li>unclassified declarations;</li> <li>duplicate wrapper modules;</li> <li>stale object/proof mappings.</li> </ul>                                                                                                                                                                                                                       |

### Phase 1.1.1.1.5.4 – Post-refactor inventory re-scan and IST-object/proof-dossier reclassification

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field | Record |
|-------|--------|
|-------|--------|

|                                        |                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | O                                                                                                                                                                                                                                                                                                                   |
| Context class                          | LeanGovernancePhase                                                                                                                                                                                                                                                                                                 |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                      |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                  |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                 |
| Implementation status                  | implementable-after-listed-predecessor                                                                                                                                                                                                                                                                              |
| Preparation status                     | prepared-after-predecessor                                                                                                                                                                                                                                                                                          |
| Phase type                             | refactor                                                                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Pre-substantive-code readiness/refactor gate; not a new mathematical derivation phase.                                                                                                                                                                                                                              |
| Title / state                          | Post-refactor inventory re-scan and reclassification                                                                                                                                                                                                                                                                |
| Preparation workflow                   | Semantic module consolidation and compatibility-preserving refactor workflow.                                                                                                                                                                                                                                       |
| Source-to-CNNA mapping                 | This is not a mathematical CNNA object. It maps repository structure to refactor governance: only semantically identical representation layers may be consolidated; proof-producing layers remain split unless their proof/provenance information is preserved by a better canonical module.                        |
| Dependency-depth audit                 | V1.21 identifies this phase as part of the hard pre-code readiness chain before 1.1.1.2.                                                                                                                                                                                                                            |
| Phase split decision                   | Explicit child split under 1.1.1.1.5; avoids treating a broad semantic-consolidation umbrella as sufficient proof for new content code.                                                                                                                                                                             |
| Positive path plan                     | Execute and close this gate constructively; only create obstruction/red blockers if the positive closure attempt identifies missing source/proof data.                                                                                                                                                              |
| Module implementation manifest         | No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean; CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean; CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.                                    |
| Semantic placement decision            | Natural placement is before blue-object semantic correction and any new Lean content; do not patch later consumer phases.                                                                                                                                                                                           |
| Reuse and duplication audit            | RD08 records the consolidation decision rule. Same-semantics wrappers may be unified; proof-stage modules must remain separated until a stronger canonical proof module preserves their role.                                                                                                                       |
| Canonical generator policy             | No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.                                                                                                                         |
| Module compactness rule                | Preserve or improve auditability; neutralize names and consolidate only with facades/regression evidence.                                                                                                                                                                                                           |
| Refactor regression protocol           | Mandatory before any Lean merge: retain old modules as facades in pass 1; record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.                                                                               |
| Refactor migration metrics             | Not yet executed. Initial audit candidates: BoundarySource four substrate ledger clones; Handoff three A_to_{C,D,E} output aliases; Handoff four {B,C,D,E}_to_A feedback-input clones; no-merge candidates: SM generated proof chains, Cayley-Dickson S6 chain; watch item: Arithmetic/Notation monolith.           |
| Lean/mathlib version policy            | Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed; refactor-only changes must compile locally before closeout.                                                                                                                                                                            |
| ExecComplex boundary policy            | EXEC-2 remains unchanged. Consolidation may not move rational executable data behind noncomputable or analytic mirrors.                                                                                                                                                                                             |
| Consumer-contract direction            | Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.                                                                                                                 |
| Governance-overhead review             | This gate is justified because concrete clone families exist, but it must remain compact: one ledger, no broad rewrite, no full code normalization before the active content path.                                                                                                                                  |
| Proof rehearsal before implementation  | For each candidate cluster prove/check: same semantic operation, same or explicitly mapped public declarations, no consumer lost, import direction preserved or improved, natural generating layer chosen, old imports kept through facades, full local build green, and statement/provenance granularity not lost. |
| Expected tactics                       | If later Lean facades are created, expect explicit abbrev/def wrappers, namespace-preserving aliases, extensionality reuse where existing, and rfl proofs. Avoid tactic-heavy rewrites; no @[simp] promotion in operational path.                                                                                   |
| Expected mathlib imports               | No new mathlib imports expected for audit. Any later facade/refactor should reuse existing imports; new imports require Lean 4.28.0 verification.                                                                                                                                                                   |
| ExecComplex/computable plan            | No ExecComplex changes in the audit. If Arithmetic/Notation or operational modules are touched later, preserve all concrete executable witnesses and rational-boundary policy.                                                                                                                                      |
| Concrete computable instance traversal | Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.                                                                                                   |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computable path non-skeleton status  | Audit-only. A later refactor may not skeletonize operational paths; it must preserve concrete generated source, DtN/Schur, determinant, and discriminant witnesses.                                                                                                                                                                                                                                                |
| Expected executable witness          | For audit: generated module inventory and candidate report. For later code refactor: unchanged downstream Lean build plus regenerated module/declaration inventory showing no lost public declarations unless explicitly facaded.                                                                                                                                                                                  |
| Mathlib complication log             | No mathlib complication in audit. Later code movement may expose import-minimization issues; resolve by preserving existing import surfaces first.                                                                                                                                                                                                                                                                 |
| Noncomputable boundary audit         | No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths.                                                                                                                                                                                                                                                                                   |
| Mathlib mitigation/refactor layer    | If new imports are needed, prefer local facade modules or existing imported modules rather than widening mathlib dependency surface.                                                                                                                                                                                                                                                                               |
| Bridge gap status                    | Current bridge gap is governance-level: exact consolidation candidates are not yet fully ledgered. No mathematical bridge is missing. Do not patch consumers; place any new facade at the natural source layer of the duplicate/schema family.                                                                                                                                                                     |
| Natural backjump/refactor layer      | BoundarySource candidate ledgers belong under Arithmetic/BoundarySource; Handoff input/output wrapper consolidation belongs under Handoff/Inputs or Outputs; notation normalization belongs at notation facade boundaries; proof-stage chains stay in their proof-stage folders unless a canonical proof module is naturally introduced.                                                                           |
| Implementation prompt contract       | Before writing any new substantive Lean code, complete the LEG36 audit. If refactor is authorized, produce a bounded migration packet: old-module facades, changed imports, declaration delta, consumer-breakage audit, generated inventory diff, and user/local Lean build result.                                                                                                                                |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG26)                                                                                                                                                                                                                                                                                                                                                        |
| How implemented                      | V1.21 planning update only: Regenerate and reconcile module/declaration/object/proof inventory after refactor.                                                                                                                                                                                                                                                                                                     |
| Verification / evidence              | No Lean source changed in V1.21. Closure requires exact inventory/refactor/build evidence recorded in the corresponding proof dossier and scratchpad.                                                                                                                                                                                                                                                              |
| Proof strategy                       | Close the phase by explicit references to derived-only Lean objects/proof dossiers, source inventory exports, and local build evidence where Lean refactor is executed. If proof fails, spawn upstream red blocker at natural source layer.                                                                                                                                                                        |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table/PDF generation only.                                                                                                                                                                                                                                                                                                                                    |
| Post-implementation plan check       | Do not advance to 1.1.1.2 until the complete 1.1.1.1.5.1-1.1.1.1.5.5 chain closes.                                                                                                                                                                                                                                                                                                                                 |
| Artifact packaging rule              | Return a full zip with updated YAML, exports, report and generated docs; do not claim Lean build.                                                                                                                                                                                                                                                                                                                  |
| Implementation notes                 | Regenerate and reconcile module/declaration/object/proof inventory after refactor.                                                                                                                                                                                                                                                                                                                                 |
| Source materials / references        | <ul style="list-style-type: none"> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/*.tsv</li> <li>repo_snapshot/CNNA/**/*.lean</li> <li>masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>LEG40</li> </ul>                                                                                                                                                                                                                                                                                                                                                            |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> <li>Repository/archive/reports/V1_21_PRE_CODE_READINESS_DERIVATION_ORDER_AUDIT_REPORT.md</li> </ul>                                                                                                                                                                                                     |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source move in the audit phase</li> <li>no merge by filename similarity alone</li> <li>no deletion of old import surfaces without facade migration</li> <li>no merging proof-stage chains that encode derivation/provenance order</li> <li>no new semantic truth layer from the audit ledger</li> </ul>                                                             |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG03, GNG21, GNG23-GNG26.</li> <li>no semantic merge without natural-layer placement</li> <li>no deletion of compatibility imports in the first refactor pass</li> <li>no proof-stage chain merge by naming similarity</li> <li>no code slimming that reduces statement/provenance auditability</li> </ul> |

|                                      |                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reusable existing modules            | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li> <li>CNNA/PillarA/Handoff/Core/SectorExport.lean</li> <li>CNNA/PillarA/Handoff/Outputs/Closed.lean</li> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/consumption_classification_map.generated.tsv</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>phase-semantic module names;</li> <li>unclassified declarations;</li> <li>duplicate wrapper modules;</li> <li>stale object/proof mappings.</li> </ul>                                                                                                                                                   |

### Phase 1.1.1.1.5.5 – Pre-substantive-code release certificate for 1.1.1.2 and later Lean content phases

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #fff2cc;">O</span>                                                                                                                                                                                                                                            |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                                                                                          |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                           |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                               |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                           |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                          |
| Implementation status                 | implementable-after-listed-predecessor                                                                                                                                                                                                                                                       |
| Preparation status                    | prepared-after-predecessor                                                                                                                                                                                                                                                                   |
| Phase type                            | refactor                                                                                                                                                                                                                                                                                     |
| Phase-type-specific contract          | Pre-substantive-code readiness/refactor gate; not a new mathematical derivation phase.                                                                                                                                                                                                       |
| Title / state                         | Pre-substantive-code release certificate                                                                                                                                                                                                                                                     |
| Preparation workflow                  | Semantic module consolidation and compatibility-preserving refactor workflow.                                                                                                                                                                                                                |
| Source-to-CNNA mapping                | This is not a mathematical CNNA object. It maps repository structure to refactor governance: only semantically identical representation layers may be consolidated; proof-producing layers remain split unless their proof/provenance information is preserved by a better canonical module. |
| Dependency-depth audit                | V1.21 identifies this phase as part of the hard pre-code readiness chain before 1.1.1.2.                                                                                                                                                                                                     |
| Phase split decision                  | Explicit child split under 1.1.1.1.5; avoids treating a broad semantic-consolidation umbrella as sufficient proof for new content code.                                                                                                                                                      |
| Positive path plan                    | Execute and close this gate constructively; only create obstruction/red blockers if the positive closure attempt identifies missing source/proof data.                                                                                                                                       |
| Module implementation manifest        | No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean; CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean; CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.             |
| Semantic placement decision           | Natural placement is before blue-object semantic correction and any new Lean content; do not patch later consumer phases.                                                                                                                                                                    |
| Reuse and duplication audit           | RD08 records the consolidation decision rule. Same-semantics wrappers may be unified; proof-stage modules must remain separated until a stronger canonical proof module preserves their role.                                                                                                |
| Canonical generator policy            | No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.                                                                                                  |
| Module compactness rule               | Preserve or improve auditability; neutralize names and consolidate only with facades/regression evidence.                                                                                                                                                                                    |
| Refactor regression protocol          | Mandatory before any Lean merge: retain old modules as facades in pass 1; record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.                                                        |



|                                        |                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refactor migration metrics             | Not yet executed. Initial audit candidates: BoundarySource four substrate ledger clones; Handoff three A_to_{C,D,E} output aliases; Handoff four {B,C,D,E}_to_A feedback-input clones; no-merge candidates: SM generated proof chains, Cayley-Dickson S6 chain; watch item: Arithmetic/Notation monolith.                                |
| Lean/mathlib version policy            | Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed; refactor-only changes must compile locally before closeout.                                                                                                                                                                                                 |
| ExecComplex boundary policy            | EXEC-2 remains unchanged. Consolidation may not move rational executable data behind noncomputable or analytic mirrors.                                                                                                                                                                                                                  |
| Consumer-contract direction            | Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.                                                                                                                                      |
| Governance-overhead review             | This gate is justified because concrete clone families exist, but it must remain compact: one ledger, no broad rewrite, no full code normalization before the active content path.                                                                                                                                                       |
| Proof rehearsal before implementation  | For each candidate cluster prove/check: same semantic operation, same or explicitly mapped public declarations, no consumer lost, import direction preserved or improved, natural generating layer chosen, old imports kept through facades, full local build green, and statement/provenance granularity not lost.                      |
| Expected tactics                       | If later Lean facades are created, expect explicit abbrev/def wrappers, namespace-preserving aliases, extensionality reuse where existing, and rfl proofs. Avoid tactic-heavy rewrites; no @[simp] promotion in operational path.                                                                                                        |
| Expected mathlib imports               | No new mathlib imports expected for audit. Any later facade/refactor should reuse existing imports; new imports require Lean 4.28.0 verification.                                                                                                                                                                                        |
| ExecComplex/computable plan            | No ExecComplex changes in the audit. If Arithmetic/Notation or operational modules are touched later, preserve all concrete executable witnesses and rational-boundary policy.                                                                                                                                                           |
| Concrete computable instance traversal | Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.                                                                                                                        |
| Computable path non-skeleton status    | Audit-only. A later refactor may not skeletonize operational paths; it must preserve concrete generated source, DtN/Schur, determinant, and discriminant witnesses.                                                                                                                                                                      |
| Expected executable witness            | For audit: generated module inventory and candidate report. For later code refactor: unchanged downstream Lean build plus regenerated module/declaration inventory showing no lost public declarations unless explicitly facaded.                                                                                                        |
| Mathlib complication log               | No mathlib complication in audit. Later code movement may expose import-minimization issues; resolve by preserving existing import surfaces first.                                                                                                                                                                                       |
| Noncomputable boundary audit           | No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths.                                                                                                                                                                                                         |
| Mathlib mitigation/refactor layer      | If new imports are needed, prefer local facade modules or existing imported modules rather than widening mathlib dependency surface.                                                                                                                                                                                                     |
| Bridge gap status                      | Current bridge gap is governance-level: exact consolidation candidates are not yet fully ledgered. No mathematical bridge is missing. Do not patch consumers; place any new facade at the natural source layer of the duplicate/schema family.                                                                                           |
| Natural backjump/refactor layer        | BoundarySource candidate ledgers belong under Arithmetic/BoundarySource; Handoff input/output wrapper consolidation belongs under Handoff/Inputs or Outputs; notation normalization belongs at notation facade boundaries; proof-stage chains stay in their proof-stage folders unless a canonical proof module is naturally introduced. |
| Implementation prompt contract         | Before writing any new substantive Lean code, complete the LEG36 audit. If refactor is authorized, produce a bounded migration packet: old-module facades, changed imports, declaration delta, consumer-breakage audit, generated inventory diff, and user/local Lean build result.                                                      |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG26)                                                                                                                                                                                                                                                                              |
| How implemented                        | V1.21 planning update only: Release 1.1.1.2 only after LEG37-LEG40 and proof-dossier updates close.                                                                                                                                                                                                                                      |
| Verification / evidence                | No Lean source changed in V1.21. Closure requires exact inventory/refactor/build evidence recorded in the corresponding proof dossier and scratchpad.                                                                                                                                                                                    |
| Proof strategy                         | Close the phase by explicit references to derived-only Lean objects/proof dossiers, source inventory exports, and local build evidence where Lean refactor is executed. If proof fails, spawn upstream red blocker at natural source layer.                                                                                              |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table/PDF generation only.                                                                                                                                                                                                                                                          |
| Post-implementation plan check         | Do not advance to 1.1.1.2 until the complete 1.1.1.5.1-1.1.1.5.5 chain closes.                                                                                                                                                                                                                                                           |
| Artifact packaging rule                | Return a full zip with updated YAML, exports, report and generated docs; do not claim Lean build.                                                                                                                                                                                                                                        |
| Implementation notes                   | Release 1.1.1.2 only after LEG37-LEG40 and proof-dossier updates close.                                                                                                                                                                                                                                                                  |
| Source materials / references          | <ul style="list-style-type: none"> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/*.tsv</li> <li>repo_snapshot/CNNA/**/*.lean</li> <li>masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                      |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"> <li>LEG41</li> </ul>                                                                                                                                                                                                                                                                                                                                                            |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> <li>Repository/archive/reports/V1_21_PRE_CODE_READINESS_DERIVATION_ORDER_AUDIT_REPORT.md</li> </ul>                                                                                                                                                                                                     |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source move in the audit phase</li> <li>no merge by filename similarity alone</li> <li>no deletion of old import surfaces without facade migration</li> <li>no merging proof-stage chains that encode derivation/provenance order</li> <li>no new semantic truth layer from the audit ledger</li> </ul>                                                             |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG03, GNG21, GNG23-GNG26.</li> <li>no semantic merge without natural-layer placement</li> <li>no deletion of compatibility imports in the first refactor pass</li> <li>no proof-stage chain merge by naming similarity</li> <li>no code slimming that reduces statement/provenance auditability</li> </ul> |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li> <li>CNNA/PillarA/Handoff/Core/SectorExport.lean</li> <li>CNNA/PillarA/Handoff/Outputs/Closed.lean</li> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/consumption_classification_map.generated.tsv</li> </ul>                                                                     |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>phase-semantic module names; unclassified declarations; duplicate wrapper modules; stale object/proof mappings.</li> </ul>                                                                                                                                                                                                                                                  |

### Phase 1.1.1.2 – Blue-object semantic correction ledger

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                   |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #fce4d6;">O</span>                                                                                                                        |
| Context class                         | LeanGovernancePhase                                                                                                                                                      |
| Phase documentation tier              | F3                                                                                                                                                                       |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                       |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                      |
| Implementation status                 | prepared-after-pre-code-readiness-chain                                                                                                                                  |
| Preparation status                    | active-prepared                                                                                                                                                          |
| Phase type                            | refactor                                                                                                                                                                 |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.  |
| Title / state                         | Blue-object semantic correction ledger after semantic module consolidation gate                                                                                          |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries. |
| Source-to-CNNA mapping                | LEG7 maps every B object to its correction phase and public-neutral API. The B object remains proof-carrying but semantically quarantined until corrected.               |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependency-depth audit                 | Now depends on 1.1.1.1.5.5 and closed LEG37-LEG41 readiness chain before activation. Now additionally depends on closure of 1.1.1.1.5 semantic module consolidation audit before activation. Feasible as infrastructure-only after v43/v44: LEG5 declaration/object consumption classification, LEG12 proof-dossier anchors, LEG10 module manifest, and LEG11 reuse policy are closed. Blocking dependency for code changes: if a blue object lacks a dossier/code-site/source object, backjump to object-proof documentation or consumption classification rather than patching LEG7. |
| Phase split decision                   | Do not split before the first LEG7 ledger generation. Split only if the generated B-object list shows distinct correction families requiring separate natural phases, e.g. Cayley-Dickson, Schur/DtN, and operational arithmetic.                                                                                                                                                                                                                                                                                                                                                      |
| Positive path plan                     | Positive path: generate/validate the LEG7 ledger for all B objects from existing object register, proof dossiers, and object_consumption_classification export; record semantic/API defect, allowed temporary consumers, forbidden direct/public consumers, correction phase, and exit-to-white condition for each row.                                                                                                                                                                                                                                                                |
| Module implementation manifest         | Infrastructure/planning gate expected. Use existing YAML/generator tables unless a code phase proves a Lean audit mirror is necessary.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Semantic placement decision            | The ledger belongs in the planning master/generated tables first. It must not patch consumers or move Lean declarations before the correction semantics are recorded.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Reuse and duplication audit            | Consume LEG11 and LEG5;<br>do not create a parallel object-status or classifier generator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Canonical generator policy             | Extend scripts/generate_tables.py if generated views are needed;<br>otherwise keep the blue correction ledger in YAML as a governance source.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Module compactness rule                | Classify by object family and correction phase. Do not generate one prose subsection per Lean declaration unless a correction really depends on it.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                                                                                                                                                                                                                                                      |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                                                                                                                                                                                                                                                            |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                                                                                                                                                                                                                                             |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                                                                                                                                                                                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Proof rehearsal before implementation  | Infrastructure proof rehearsal: each blue object must satisfy B-status -> blue-semantic-correction class -> has proof dossier -> has current derivation site -> has forbidden public consumers -> has natural correction phase -> has exit-to-white condition. Failure localizes to dossier/classification source, not to public API/pruning consumers.                                                                                                                                                                                                                                |
| Expected tactics                       | No Lean proof tactics expected unless a small audit module is introduced. For such modules, prefer explicit records and exact/rfl proofs only.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | No new mathlib imports expected;<br>if a later Lean audit file is introduced, verify imports against mathlib 4.28.0 first.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Expected executable witness            | Not required for this infrastructure-only ledger phase;<br>the witness is generated TSV/TeX consistency over existing object/proof/classification records. If Lean code is proposed, a separate finite executable witness must be named before implementation.                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No mathlib complication expected for ledger-only implementation;<br>no new Lean/mathlib import should be introduced. Any proposed Lean audit file must verify imports against Lean 4.28.0 before code starts.                                                                                                                                                                                                                                                                                                                                                                          |
| Noncomputable boundary audit           | No operational or mirror mathematical code is introduced in the planned ledger-only route;<br>noncomputable/classical pressure would indicate that the phase has drifted into a later mathematical correction phase.                                                                                                                                                                                                                                                                                                                                                                   |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Bridge gap status                      | No mathematical bridge should be invented here. Current known open item SCF26 is a ledger-completeness requirement for 1.1.1.2, not a request for new mathematics.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                               |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation.                                                                                                                                                                                                                                                      |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| How implemented                        | Not completed and no longer the active cursor in V1.12. It becomes implementable in the derivation chain after Phase 1.1.1.1.5 closes.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Feasibility remains valid from v44, but V1.12 inserted a pre-substantive semantic module consolidation gate because code inspection found clone/schema module families that should be audited before new content code.                                                                                                                                                                                                                                                                                                                                                                 |
| Post-implementation plan check         | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Implementation notes                 | Resume only after LEG36 is closed or after a bounded refactor execution phase has completed with local build evidence. The blue-object ledger must then consume the stabilized module/API surface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Source materials / references        | <ul style="list-style-type: none"><li>object register blue rows</li><li>declaration classification from 1.1.1.1.4</li><li>proof-dossier infrastructure from 1.1.1.1.3</li><li>TomS guardrails where arithmetic/number theory touches appear</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Extracted established theory objects | <ul style="list-style-type: none"><li>blue_correction_ledger</li><li>temporary allowed consumer set</li><li>forbidden direct consumer set</li><li>exit-to-white condition per B object</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| No-go guardrails                     | <ul style="list-style-type: none"><li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li><li>no @[simp] attributes in the operational path</li><li>no noncomputable in the operational path</li><li>no classical helper unless isolated in allowed mirror/non-operational layer</li><li>no Matrix.inv or external inverse oracle in operational derivations</li><li>no free formula without proof-carrying entry structure</li><li>no consumer-side patch when the missing structure belongs earlier</li><li>no premature obstruction before a serious positive construction attempt</li><li>no phase-semantic public API names</li><li>no fragile rewrite chain when a structural constructor/projection proof is available</li><li>no white status for a blue object without a closed correction phase</li><li>no direct handoff/export of blue objects as final API</li></ul> |
| Reusable existing modules            | <ul style="list-style-type: none"><li>masters/cnna_roadmap_master.yaml</li><li>scripts/generate_tables.py</li><li>generated_exports/object_consumption_classification.generated.tsv</li><li>tables/object_proof_documentation.tex</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"><li>parallel blue-status table disconnected from object register</li><li>white status without correction proof</li><li>public export of B object through wrapper reachability</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Phase 1.1.1.3 – Public API contract and consumer whitelist

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #fff2cc;">W</span>                                                                                         |
| Context class                         | LeanGovernancePhase                                                                                                                       |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                        |
| Preparation status                    | planned-not-active                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase type                             | refactor                                                                                                                                                                                                                                                                                                                          |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                                                                                           |
| Title / state                          | Public API contract and consumer whitelist                                                                                                                                                                                                                                                                                        |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.                                                                                                                                                          |
| Source-to-CNNA mapping                 | LEG6/LEG9 define which modules and declarations may be seen by Phase 3+ and later handoffs.                                                                                                                                                                                                                                       |
| Dependency-depth audit                 | Depends on module manifest, reuse audit, proof-dossier infrastructure, declaration classification, and blue correction. Public API exposure is invalid if any public object lacks a proof-dossier link or is still blue/semantically unfinished.                                                                                  |
| Phase split decision                   | Atomic contract phase; movement happens only in 1.1.2-1.1.5.                                                                                                                                                                                                                                                                      |
| Positive path plan                     | Declare the visible post-cleanup API and consumer whitelist. Exclude Legacy.StatusGates, phase aliases, audit-only material, duplicate generator seams, uncorrected blue objects, and any public object without proof-documentation status.                                                                                       |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                 |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                       |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                        |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                               |
| Proof rehearsal before implementation  | Verify every downstream phase consumes either a white/yellow planned object or a corrected neutral wrapper, never status/audit/phase aliases.                                                                                                                                                                                     |
| Expected tactics                       | No Lean proof tactics expected unless a small audit module is introduced. For such modules, prefer explicit records and exact/rfl proofs only.                                                                                                                                                                                    |
| Expected mathlib imports               | No new mathlib imports expected; if a later Lean audit file is introduced, verify imports against mathlib 4.28.0 first.                                                                                                                                                                                                           |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                                                                                                                            |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                               |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                                                                              |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                 |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                              |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                      |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                         |
| Bridge gap status                      | No mathematical bridge should be invented here. If a blue object needs mathematics, route it to its natural correction phase.                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                                                                                          |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation. |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                          |
| How implemented                        | No completed implementation record yet. This phase is planned by v23 and must remain inactive until its predecessor gate closes.                                                                                                                                                                                                  |
| Verification / evidence                | Not applicable yet; this is a planning/preparation gate until it becomes the unique active phase.                                                                                                                                                                                                                                 |
| Post-implementation plan check         | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                               |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                    |
| Implementation notes                   | Do not implement by broad deletion. First produce the specified ledger/contract and update generated views.                                                                                                                                                                                                                       |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source materials / references        | <ul style="list-style-type: none"> <li>1.1.1.1 consumption map</li> <li>1.1.1.2 blue correction ledger</li> <li>access_policy table</li> <li>LEG10 module manifest</li> <li>LEG11 reuse/duplication ledger</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>public_api_contract</li> <li>consumer_whitelist</li> <li>forbidden_import_set</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not let BuildAll/Umbrella imports count as public API</li> <li>do not expose Legacy.* to Phase 3+</li> </ul> |

### Phase 1.1.1.4 – Pruning and dead-candidate ledger before deletion

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                               |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                    |
| Context class                         | LeanGovernancePhase                                                                                                                                                                  |
| Phase documentation tier              | F3                                                                                                                                                                                   |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                            |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                   |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                  |
| Implementation status                 | planned-not-active                                                                                                                                                                   |
| Preparation status                    | planned-not-active                                                                                                                                                                   |
| Phase type                            | refactor                                                                                                                                                                             |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.              |
| Title / state                         | Pruning and dead-candidate ledger before deletion                                                                                                                                    |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.             |
| Source-to-CNNA mapping                | LEG8 records pruning classes. Deletion is explicitly deferred to a later implementation phase after build evidence.                                                                  |
| Dependency-depth audit                | Depends on public API contract, declaration classification, and proof-dossier infrastructure; pruning may not delete objects needed for documentation, audit, or proof-bridge roles. |
| Phase split decision                  | Atomic ledger phase; actual deletion remains prohibited here.                                                                                                                        |
| Positive path plan                    | Classify conservatively; keep proof bridges and legacy audit records unless a later green-build removal confirms deletion safety.                                                    |



|                                        |                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                 |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                       |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                        |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                          |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                               |
| Proof rehearsal before implementation  | A declaration can be deletion-candidate only if it has no public contract, proof bridge, legacy audit role, or blue-correction role.                                                                                                                                                                                              |
| Expected tactics                       | No Lean proof tactics expected unless a small audit module is introduced. For such modules, prefer explicit records and exact/rfl proofs only.                                                                                                                                                                                    |
| Expected mathlib imports               | No new mathlib imports expected; if a later Lean audit file is introduced, verify imports against mathlib 4.28.0 first.                                                                                                                                                                                                           |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                                                                                                                            |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                               |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                                                                              |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                 |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                              |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                      |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                         |
| Bridge gap status                      | No mathematical bridge should be invented here. If a blue object needs mathematics, route it to its natural correction phase.                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                                                                                          |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation. |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                          |
| How implemented                        | No completed implementation record yet. This phase is planned by v23 and must remain inactive until its predecessor gate closes.                                                                                                                                                                                                  |
| Verification / evidence                | Not applicable yet; this is a planning/preparation gate until it becomes the unique active phase.                                                                                                                                                                                                                                 |
| Post-implementation plan check         | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                               |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                    |
| Implementation notes                   | Do not implement by broad deletion. First produce the specified ledger/contract and update generated views.                                                                                                                                                                                                                       |
| Source materials / references          | <ul style="list-style-type: none"> <li>1.1.1.1 consumption map</li> <li>1.1.1.3 public API contract</li> <li>semantic/reference summaries</li> <li>LEG10 module manifest</li> <li>LEG11 reuse/duplication ledger</li> </ul>                                                                                                       |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>pruning_ledger classes: keep, quarantine, wrap, reexport, archive, deletion-candidate</li> </ul>                                                                                                                                                                                           |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global__no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>no deletion without a later local green build</li> <li>no deletion of provenance/audit evidence merely because it is not public API</li> </ul> |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 1.1.1.5 – Generated-view authority and inventory lock for implementation prompts

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
 Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                      |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                           |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                         |
| Phase documentation tier              | F3                                                                                                                                                                                                                          |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                   |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                          |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                         |
| Implementation status                 | planned-not-active                                                                                                                                                                                                          |
| Preparation status                    | planned-not-active                                                                                                                                                                                                          |
| Phase type                            | refactor                                                                                                                                                                                                                    |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                     |
| Title / state                         | Generated-view authority and inventory lock                                                                                                                                                                                 |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.                                                    |
| Source-to-CNNA mapping                | Locks the planning artifact so 1.1.2 can be implemented from the classification contracts.                                                                                                                                  |
| Dependency-depth audit                | Depends on all previous child gates and generated table/PDF validation.                                                                                                                                                     |
| Phase split decision                  | No split needed unless generated report and PDF authority diverge.                                                                                                                                                          |
| Positive path plan                    | Regenerate tables/PDF/report and set next active cursor to 1.1.2 only after module manifest, reuse/generator audit, declaration classification, blue correction ledger, public API contract, and pruning ledger are closed. |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                           |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                 |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.  |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                               |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                         |
| Proof rehearsal before implementation | Validate single YAML source, no stale masters, exactly one active phase, all scratchpad records complete, PDF renders.                                                                                                      |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected tactics                       | No Lean proof tactics expected unless a small audit module is introduced. For such modules, prefer explicit records and exact/rfl proofs only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expected mathlib imports               | No new mathlib imports expected;<br>if a later Lean audit file is introduced, verify imports against mathlib 4.28.0 first.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Bridge gap status                      | No mathematical bridge should be invented here. If a blue object needs mathematics, route it to its natural correction phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Natural backjump/refactor layer        | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| How implemented                        | No completed implementation record yet. This phase is planned by v23 and must remain inactive until its predecessor gate closes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Verification / evidence                | Not applicable yet;<br>this is a planning/preparation gate until it becomes the unique active phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Post-implementation plan check         | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation notes                   | Do not implement by broad deletion. First produce the specified ledger/contract and update generated views.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Source materials / references          | <ul style="list-style-type: none"> <li>1.1.1.1.1-1.1.1.1.4</li> <li>1.1.1.2-1.1.1.4</li> <li>YAML single-master rule</li> <li>generated table/PDF pipeline</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>implementation_prompt_contract for 1.1.2-1.1.5</li> <li>generated-view authority record</li> <li>local-build evidence reminder</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not claim Lean build success for planning-only artifacts</li> </ul> |

### Phase 1.1.1.5.1 – Tool-mode split and lean-focus contract

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                          |
| Context class                          | LeanGovernancePhase                                                                                                                                                                                                        |
| Phase documentation tier               | F2                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                              |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | planned-not-active                                                                                                                                                                                                         |
| Preparation status                     | planned-not-active                                                                                                                                                                                                         |
| Phase type                             | infrastructure                                                                                                                                                                                                             |
| Phase-type-specific contract           | Infrastructure phase: update YAML/generator/generated exports, validate single-source constraints, and render PDF without claiming Lean build success.                                                                     |
| Title / state                          | Tool-mode split and lean-focus contract                                                                                                                                                                                    |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.                                                   |
| Source-to-CNNA mapping                 | No theory classification yet; this phase only fixes the mode boundary.                                                                                                                                                     |
| Dependency-depth audit                 | Depends on 1.1.1.5 and earlier governance gates. It must not block the current active 1.1.1.2 blue-correction ledger; it is a future scaling/tooling refinement.                                                           |
| Phase split decision                   | Atomic planned subphase under 1.1.1.5. Split only if implementation would otherwise mix Lean-code changes with tool/PDF/YAML changes.                                                                                      |
| Positive path plan                     | Implement by extending Python/YAML/generated exports first. Do not add Lean code. If a Lean-side audit mirror becomes necessary, create a separate later Lean phase with an explicit finite witness and import policy.     |
| Module implementation manifest         | Tooling subphase. Proposed generated artifacts: workflow-policy row; scaling-plan TSV/TeX; no Lean code                                                                                                                    |
| Semantic placement decision            | Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.                                                                                                      |
| Reuse and duplication audit            | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.                                                                                  |
| Canonical generator policy             | Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.                                                                                         |
| Module compactness rule                | Keep PDF slices compact; full data remains in generated TSV/JSON exports.                                                                                                                                                  |
| Refactor regression protocol           | Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.                                                                                                  |
| Refactor migration metrics             | No Lean declaration migration. Record generator/export changes only when the phase becomes active.                                                                                                                         |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                         |
| Consumer-contract direction            | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.                                                                                             |
| Governance-overhead review             | This phase exists to reduce governance overhead during Lean work by pushing routine collection/rendering into Python.                                                                                                      |
| Proof rehearsal before implementation  | Validate generator output, single YAML source, exactly one active phase, unchanged Lean-data timestamp, and section-specific rendering. This is a tooling proof rehearsal, not a Lean theorem.                             |
| Expected tactics                       | No Lean tactics expected. If a later Lean mirror is explicitly introduced, use explicit records/constructors and avoid @[simp], noncomputable operational paths, classical helpers, and broad fragile rewrites.            |
| Expected mathlib imports               | No new mathlib imports expected for the planned Python/YAML tooling route.                                                                                                                                                 |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                     |
| Concrete computable instance traversal | Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.                                                                                    |
| Computable path non-skeleton status    | not-applicable-tooling-only; no operational mathematical computation is introduced                                                                                                                                         |
| Expected executable witness            | Python generator run plus generated TSV/TeX/PDF artifacts; no Lean executable witness required.                                                                                                                            |
| Mathlib complication log               | No mathlib complication expected because this is a Python/YAML/PDF tooling phase.                                                                                                                                          |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noncomputable boundary audit         | No Lean operational path is modified;<br>any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Mathlib mitigation/refactor layer    | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Bridge gap status                    | No mathematical bridge is introduced here. Local background theory and large end-theory comparison roles are tracked as metadata/ledgers, not as CNNA generators.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Natural backjump/refactor layer      | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| How implemented                      | No completed implementation record yet. This future tooling/governance phase was introduced by Tool V1.04 and must remain inactive until its predecessor gates and active cursor allow it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Verification / evidence              | Not applicable yet;<br>phase is a planned Tool V1.04 implementation-scaling subphase. Lean-data timestamp remains unchanged.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Post-implementation plan check       | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Implementation notes                 | Define audit-full, phase-check, and lean-focus modes so routine Lean work consumes compact implementation packets instead of full PDF/table scans. Full PDF regeneration is not part of ordinary Lean coding; use compact packets unless a governance or audit phase explicitly requests audit-full.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Source materials / references        | <ul style="list-style-type: none"> <li>• Tool V1.04 implementation-scaling plan</li> <li>• SCF37</li> <li>• workflow policy</li> <li>• repo_inventory/raw_export</li> <li>• object register</li> <li>• implementation scratchpad</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>• Tool-mode split and lean-focus contract</li> <li>• No theory classification yet;<br/>this phase only fixes the mode boundary.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: top-level global_no_go_guardrails apply;<br/>this phase list is local addenda only.</li> <li>• no @[simp] attributes in the operational path</li> <li>• no noncomputable in the operational path</li> <li>• no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>• no Matrix.inv or external inverse oracle in operational derivations</li> <li>• no free formula without proof-carrying entry structure</li> <li>• no consumer-side patch when the missing structure belongs earlier</li> <li>• no premature obstruction before a serious positive construction attempt</li> <li>• no phase-semantic public API names</li> <li>• no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>• do not claim Lean build success for planning-only artifacts</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>• masters/cnna_roadmap_master.yaml</li> <li>• scripts/generate_tables.py</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>• scripts/generate_tables.py</li> <li>• repo_inventory/raw_export/modules.json</li> <li>• repo_inventory/raw_export/symbol_references/*.json</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                      |                                                                                                                                                                                                  |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>parallel planning master</li> <li>manual duplicate theory ledger</li> <li>full graph dumped into PDF</li> <li>end theory used as CNNA producer</li> </ul> |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 1.1.1.5.2 – Implementation Packet Generator

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                          |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                        |
| Phase documentation tier              | F2                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                              |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                 | planned-not-active                                                                                                                                                                                                         |
| Preparation status                    | planned-not-active                                                                                                                                                                                                         |
| Phase type                            | infrastructure                                                                                                                                                                                                             |
| Phase-type-specific contract          | Infrastructure phase: update YAML/generator/generated exports, validate single-source constraints, and render PDF without claiming Lean build success.                                                                     |
| Title / state                         | Implementation Packet Generator                                                                                                                                                                                            |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.                                                   |
| Source-to-CNNA mapping                | Consumes module-theory/end-theory ledgers once available; before then marks theory context as unknown-not-input.                                                                                                           |
| Dependency-depth audit                | Depends on 1.1.1.5 and earlier governance gates. It must not block the current active 1.1.1.2 blue-correction ledger; it is a future scaling/tooling refinement.                                                           |
| Phase split decision                  | Atomic planned subphase under 1.1.1.5. Split only if implementation would otherwise mix Lean-code changes with tool/PDF/YAML changes.                                                                                      |
| Positive path plan                    | Implement by extending Python/YAML/generated exports first. Do not add Lean code. If a Lean-side audit mirror becomes necessary, create a separate later Lean phase with an explicit finite witness and import policy.     |
| Module implementation manifest        | Tooling subphase. Proposed generated artifacts: active-phase implementation packet TSV/JSON/TeX                                                                                                                            |
| Semantic placement decision           | Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.                                                                                                      |
| Reuse and duplication audit           | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.                                                                                  |
| Canonical generator policy            | Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.                                                                                         |
| Module compactness rule               | Keep PDF slices compact; full data remains in generated TSV/JSON exports.                                                                                                                                                  |
| Refactor regression protocol          | Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.                                                                                                  |
| Refactor migration metrics            | No Lean declaration migration. Record generator/export changes only when the phase becomes active.                                                                                                                         |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction           | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.                                                                                             |
| Governance-overhead review            | This phase exists to reduce governance overhead during Lean work by pushing routine collection/rendering into Python.                                                                                                      |
| Proof rehearsal before implementation | Validate generator output, single YAML source, exactly one active phase, unchanged Lean-data timestamp, and section-specific rendering. This is a tooling proof rehearsal, not a Lean theorem.                             |
| Expected tactics                      | No Lean tactics expected. If a later Lean mirror is explicitly introduced, use explicit records/constructors and avoid @[simp], noncomputable operational paths, classical helpers, and broad fragile rewrites.            |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected mathlib imports               | No new mathlib imports expected for the planned Python/YAML tooling route.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Concrete computable instance traversal | Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Computable path non-skeleton status    | not-applicable-tooling-only;<br>no operational mathematical computation is introduced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | Python generator run plus generated TSV/TeX/PDF artifacts;<br>no Lean executable witness required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Mathlib complication log               | No mathlib complication expected because this is a Python/YAML/PDF tooling phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Noncomputable boundary audit           | No Lean operational path is modified;<br>any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Bridge gap status                      | No mathematical bridge is introduced here. Local background theory and large end-theory comparison roles are tracked as metadata/ledgers, not as CNNA generators.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Natural backjump/refactor layer        | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| How implemented                        | No completed implementation record yet. This future tooling/governance phase was introduced by Tool V1.04 and must remain inactive until its predecessor gates and active cursor allow it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Verification / evidence                | Not applicable yet;<br>phase is a planned Tool V1.04 implementation-scaling subphase. Lean-data timestamp remains unchanged.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check         | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation notes                   | Generate the phase-local packet with goal, source objects, producer/consumer modules, red blockers, yellow risks, forbidden shortcuts, expected artifacts, and comparison-only warnings. Packet should fit on a small number of pages/rows and link to full TSV/JSON exports rather than embedding full inventories.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Source materials / references          | <ul style="list-style-type: none"> <li>Tool V1.04 implementation-scaling plan</li> <li>SCF37</li> <li>workflow policy</li> <li>repo_inventory/raw_export</li> <li>object register</li> <li>implementation scratchpad</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Implementation Packet Generator</li> <li>Consumes module-theory/end-theory ledgers once available; before then marks theory context as unknown-not-input.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not claim Lean build success for planning-only artifacts</li> </ul> |

|                                      |                                                                                                                                                                                                  |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposed modified modules            | <ul style="list-style-type: none"> <li>masters/cnna__roadmap__master.yaml</li> <li>scripts/generate_tables.py</li> </ul>                                                                         |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul>         |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>parallel planning master</li> <li>manual duplicate theory ledger</li> <li>full graph dumped into PDF</li> <li>end theory used as CNNA producer</li> </ul> |

### Phase 1.1.1.5.3 – Incremental Lean inventory and file-hash cache

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                 |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                      |
| Context class                         | LeanGovernancePhase                                                                                                                                                                                                    |
| Phase documentation tier              | F2                                                                                                                                                                                                                     |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                     |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                    |
| Implementation status                 | planned-not-active                                                                                                                                                                                                     |
| Preparation status                    | planned-not-active                                                                                                                                                                                                     |
| Phase type                            | infrastructure                                                                                                                                                                                                         |
| Phase-type-specific contract          | Infrastructure phase: update YAML/generator/generated exports, validate single-source constraints, and render PDF without claiming Lean build success.                                                                 |
| Title / state                         | Incremental Lean inventory and file-hash cache                                                                                                                                                                         |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.                                               |
| Source-to-CNNA mapping                | May infer rough local background categories from imports, but must mark them heuristic until semantically confirmed.                                                                                                   |
| Dependency-depth audit                | Depends on 1.1.1.5 and earlier governance gates. It must not block the current active 1.1.1.2 blue-correction ledger; it is a future scaling/tooling refinement.                                                       |
| Phase split decision                  | Atomic planned subphase under 1.1.1.5. Split only if implementation would otherwise mix Lean-code changes with tool/PDF/YAML changes.                                                                                  |
| Positive path plan                    | Implement by extending Python/YAML/generated exports first. Do not add Lean code. If a Lean-side audit mirror becomes necessary, create a separate later Lean phase with an explicit finite witness and import policy. |
| Module implementation manifest        | Tooling subphase. Proposed generated artifacts: file-hash cache; declaration cache; inventory-delta TSV                                                                                                                |
| Semantic placement decision           | Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.                                                                                                  |
| Reuse and duplication audit           | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.                                                                              |
| Canonical generator policy            | Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.                                                                                     |
| Module compactness rule               | Keep PDF slices compact; full data remains in generated TSV/JSON exports.                                                                                                                                              |
| Refactor regression protocol          | Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.                                                                                              |
| Refactor migration metrics            | No Lean declaration migration. Record generator/export changes only when the phase becomes active.                                                                                                                     |

|                                        |                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                        |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                  |
| Consumer-contract direction            | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.                                                                                                                                                                                                    |
| Governance-overhead review             | This phase exists to reduce governance overhead during Lean work by pushing routine collection/rendering into Python.                                                                                                                                                                                                             |
| Proof rehearsal before implementation  | Validate generator output, single YAML source, exactly one active phase, unchanged Lean-data timestamp, and section-specific rendering. This is a tooling proof rehearsal, not a Lean theorem.                                                                                                                                    |
| Expected tactics                       | No Lean tactics expected. If a later Lean mirror is explicitly introduced, use explicit records/constructors and avoid @[simp], noncomputable operational paths, classical helpers, and broad fragile rewrites.                                                                                                                   |
| Expected mathlib imports               | No new mathlib imports expected for the planned Python/YAML tooling route.                                                                                                                                                                                                                                                        |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                                                                                                                            |
| Concrete computable instance traversal | Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.                                                                                                                                                                                           |
| Computable path non-skeleton status    | not-applicable-tooling-only;<br>no operational mathematical computation is introduced                                                                                                                                                                                                                                             |
| Expected executable witness            | Python generator run plus generated TSV/TeX/PDF artifacts;<br>no Lean executable witness required.                                                                                                                                                                                                                                |
| Mathlib complication log               | No mathlib complication expected because this is a Python/YAML/PDF tooling phase.                                                                                                                                                                                                                                                 |
| Noncomputable boundary audit           | No Lean operational path is modified;<br>any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase.                                                                                                                                                                             |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                         |
| Bridge gap status                      | No mathematical bridge is introduced here. Local background theory and large end-theory comparison roles are tracked as metadata/ledgers, not as CNNA generators.                                                                                                                                                                 |
| Natural backjump/refactor layer        | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                                                                                          |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                       |
| How implemented                        | No completed implementation record yet. This future tooling/governance phase was introduced by Tool V1.04 and must remain inactive until its predecessor gates and active cursor allow it.                                                                                                                                        |
| Verification / evidence                | Not applicable yet;<br>phase is a planned Tool V1.04 implementation-scaling subphase. Lean-data timestamp remains unchanged.                                                                                                                                                                                                      |
| Post-implementation plan check         | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                               |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                    |
| Implementation notes                   | Avoid rescanning/rendering everything by caching file hashes, module declarations, import edges, and symbol-reference candidates between Lean-data updates. Only changed Lean files and dependent generated summaries are refreshed;<br>full inventory rebuild remains available as audit-full.                                   |
| Source materials / references          | <ul style="list-style-type: none"> <li>Tool V1.04 implementation-scaling plan</li> <li>SCF37</li> <li>workflow policy</li> <li>repo_inventory/raw_export</li> <li>object register</li> <li>implementation scratchpad</li> </ul>                                                                                                   |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Incremental Lean inventory and file-hash cache</li> <li>May infer rough local background categories from imports, but must mark them heuristic until semantically confirmed.</li> </ul>                                                                                                    |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not claim Lean build success for planning-only artifacts</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li><code>masters/cnna_roadmap_master.yaml</code></li> <li><code>scripts/generate_tables.py</code></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Reusable existing modules            | <ul style="list-style-type: none"> <li><code>scripts/generate_tables.py</code></li> <li><code>repo_inventory/raw_export/modules.json</code></li> <li><code>repo_inventory/raw_export/symbol_references/*.json</code></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>parallel planning master</li> <li>manual duplicate theory ledger</li> <li>full graph dumped into PDF</li> <li>end theory used as CNNA producer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### Phase 1.1.1.5.4 – Module theory context ledger

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                            |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f2f2f2;">W</span>                                                                                                                                 |
| Context class                         | LeanGovernancePhase                                                                                                                                                               |
| Phase documentation tier              | F2                                                                                                                                                                                |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.     |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                               |
| Implementation status                 | planned-not-active                                                                                                                                                                |
| Preparation status                    | planned-not-active                                                                                                                                                                |
| Phase type                            | infrastructure                                                                                                                                                                    |
| Phase-type-specific contract          | Infrastructure phase: update YAML/generator/generated exports, validate single-source constraints, and render PDF without claiming Lean build success.                            |
| Title / state                         | Module theory context ledger                                                                                                                                                      |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.          |
| Source-to-CNNA mapping                | Local established theory is allowed formalization context, never ontic CNNA generator. Examples: Matrix, Finset, Polynomial, graph/linear algebra, Schur-style algebraic methods. |
| Dependency-depth audit                | Depends on 1.1.1.5 and earlier governance gates. It must not block the current active 1.1.1.2 blue-correction ledger; it is a future scaling/tooling refinement.                  |
| Phase split decision                  | Atomic planned subphase under 1.1.1.5. Split only if implementation would otherwise mix Lean-code changes with tool/PDF/YAML changes.                                             |

|                                        |                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive path plan                     | Implement by extending Python/YAML/generated exports first. Do not add Lean code. If a Lean-side audit mirror becomes necessary, create a separate later Lean phase with an explicit finite witness and import policy.                                                                                                                                 |
| Module implementation manifest         | Tooling subphase. Proposed generated artifacts: module-theory-context ledger TSV; active-module slice                                                                                                                                                                                                                                                  |
| Semantic placement decision            | Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.                                                                                                                                                                                                                                  |
| Reuse and duplication audit            | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.                                                                                                                                                                                                              |
| Canonical generator policy             | Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.                                                                                                                                                                                                                     |
| Module compactness rule                | Keep PDF slices compact; full data remains in generated TSV/JSON exports.                                                                                                                                                                                                                                                                              |
| Refactor regression protocol           | Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.                                                                                                                                                                                                                              |
| Refactor migration metrics             | No Lean declaration migration. Record generator/export changes only when the phase becomes active.                                                                                                                                                                                                                                                     |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                     |
| Consumer-contract direction            | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.                                                                                                                                                                                                                         |
| Governance-overhead review             | This phase exists to reduce governance overhead during Lean work by pushing routine collection/rendering into Python.                                                                                                                                                                                                                                  |
| Proof rehearsal before implementation  | Validate generator output, single YAML source, exactly one active phase, unchanged Lean-data timestamp, and section-specific rendering. This is a tooling proof rehearsal, not a Lean theorem.                                                                                                                                                         |
| Expected tactics                       | No Lean tactics expected. If a later Lean mirror is explicitly introduced, use explicit records/constructors and avoid @[simp], noncomputable operational paths, classical helpers, and broad fragile rewrites.                                                                                                                                        |
| Expected mathlib imports               | No new mathlib imports expected for the planned Python/YAML tooling route.                                                                                                                                                                                                                                                                             |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                                                                                                                                                 |
| Concrete computable instance traversal | Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.                                                                                                                                                                                                                |
| Computable path non-skeleton status    | not-applicable-tooling-only;<br>no operational mathematical computation is introduced                                                                                                                                                                                                                                                                  |
| Expected executable witness            | Python generator run plus generated TSV/TeX/PDF artifacts;<br>no Lean executable witness required.                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No mathlib complication expected because this is a Python/YAML/PDF tooling phase.                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No Lean operational path is modified;<br>any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase.                                                                                                                                                                                                  |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                              |
| Bridge gap status                      | No mathematical bridge is introduced here. Local background theory and large end-theory comparison roles are tracked as metadata/ledgers, not as CNNA generators.                                                                                                                                                                                      |
| Natural backjump/refactor layer        | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                                                                                                               |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation.                      |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                            |
| How implemented                        | No completed implementation record yet. This future tooling/governance phase was introduced by Tool V1.04 and must remain inactive until its predecessor gates and active cursor allow it.                                                                                                                                                             |
| Verification / evidence                | Not applicable yet;<br>phase is a planned Tool V1.04 implementation-scaling subphase. Lean-data timestamp remains unchanged.                                                                                                                                                                                                                           |
| Post-implementation plan check         | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                         |
| Implementation notes                   | Track local established mathematical background per module as formal language, proof method, library/API context, implementation substrate, or comparison lemma context. Start with active/relevant modules only;<br>do not manually annotate all modules at once. Python proposes rows from imports;<br>semantic closure is explicit and phase-bound. |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source materials / references        | <ul style="list-style-type: none"> <li>Tool V1.04 implementation-scaling plan</li> <li>SCF37</li> <li>workflow policy</li> <li>repo_inventory/raw_export</li> <li>object register</li> <li>implementation scratchpad</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Module theory context ledger</li> <li>Local established theory is allowed formalization context, never ontic CNNA generator. Examples: Matrix, Finset, Polynomial, graph/linear algebra, Schur-style algebraic methods.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not claim Lean build success for planning-only artifacts</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>parallel planning master</li> <li>manual duplicate theory ledger</li> <li>full graph dumped into PDF</li> <li>end theory used as CNNA producer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### Phase 1.1.1.5.5 – End-theory comparison ledger

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                        |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                             |
| Context class                         | LeanGovernancePhase                                                                                                                                                           |
| Phase documentation tier              | F2                                                                                                                                                                            |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                            |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                           |
| Implementation status                 | planned-not-active                                                                                                                                                            |
| Preparation status                    | planned-not-active                                                                                                                                                            |
| Phase type                            | infrastructure                                                                                                                                                                |



|                                        |                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase-type-specific contract           | Infrastructure phase: update YAML/generator/generated exports, validate single-source constraints, and render PDF without claiming Lean build success.                                                                                                                                                                            |
| Title / state                          | End-theory comparison ledger                                                                                                                                                                                                                                                                                                      |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.                                                                                                                                                          |
| Source-to-CNNA mapping                 | End theories are target structures, consistency checks, terminology alignment, or obstruction/comparison mirrors; forbidden as CNNA producers.                                                                                                                                                                                    |
| Dependency-depth audit                 | Depends on 1.1.1.5 and earlier governance gates. It must not block the current active 1.1.1.2 blue-correction ledger; it is a future scaling/tooling refinement.                                                                                                                                                                  |
| Phase split decision                   | Atomic planned subphase under 1.1.1.5. Split only if implementation would otherwise mix Lean-code changes with tool/PDF/YAML changes.                                                                                                                                                                                             |
| Positive path plan                     | Implement by extending Python/YAML/generated exports first. Do not add Lean code. If a Lean-side audit mirror becomes necessary, create a separate later Lean phase with an explicit finite witness and import policy.                                                                                                            |
| Module implementation manifest         | Tooling subphase. Proposed generated artifacts: end-theory-comparison ledger TSV; comparison-boundary table                                                                                                                                                                                                                       |
| Semantic placement decision            | Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.                                                                                                                                                                                                             |
| Reuse and duplication audit            | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.                                                                                                                                                                                         |
| Canonical generator policy             | Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.                                                                                                                                                                                                |
| Module compactness rule                | Keep PDF slices compact; full data remains in generated TSV/JSON exports.                                                                                                                                                                                                                                                         |
| Refactor regression protocol           | Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.                                                                                                                                                                                                         |
| Refactor migration metrics             | No Lean declaration migration. Record generator/export changes only when the phase becomes active.                                                                                                                                                                                                                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                        |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                |
| Consumer-contract direction            | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.                                                                                                                                                                                                    |
| Governance-overhead review             | This phase exists to reduce governance overhead during Lean work by pushing routine collection/rendering into Python.                                                                                                                                                                                                             |
| Proof rehearsal before implementation  | Validate generator output, single YAML source, exactly one active phase, unchanged Lean-data timestamp, and section-specific rendering. This is a tooling proof rehearsal, not a Lean theorem.                                                                                                                                    |
| Expected tactics                       | No Lean tactics expected. If a later Lean mirror is explicitly introduced, use explicit records/constructors and avoid @[simp], noncomputable operational paths, classical helpers, and broad fragile rewrites.                                                                                                                   |
| Expected mathlib imports               | No new mathlib imports expected for the planned Python/YAML tooling route.                                                                                                                                                                                                                                                        |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                                                                                                                            |
| Concrete computable instance traversal | Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.                                                                                                                                                                                           |
| Computable path non-skeleton status    | not-applicable-tooling-only;<br>no operational mathematical computation is introduced                                                                                                                                                                                                                                             |
| Expected executable witness            | Python generator run plus generated TSV/TeX/PDF artifacts;<br>no Lean executable witness required.                                                                                                                                                                                                                                |
| Mathlib complication log               | No mathlib complication expected because this is a Python/YAML/PDF tooling phase.                                                                                                                                                                                                                                                 |
| Noncomputable boundary audit           | No Lean operational path is modified;<br>any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase.                                                                                                                                                                             |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                         |
| Bridge gap status                      | No mathematical bridge is introduced here. Local background theory and large end-theory comparison roles are tracked as metadata/ledgers, not as CNNA generators.                                                                                                                                                                 |
| Natural backjump/refactor layer        | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                                                                                          |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                       |
| How implemented                        | No completed implementation record yet. This future tooling/governance phase was introduced by Tool V1.04 and must remain inactive until its predecessor gates and active cursor allow it.                                                                                                                                        |
| Verification / evidence                | Not applicable yet;<br>phase is a planned Tool V1.04 implementation-scaling subphase. Lean-data timestamp remains unchanged.                                                                                                                                                                                                      |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Post-implementation plan check       | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation notes                 | Track large established target theories as comparison-only horizons: AQFT/QFT, OQS, modular forms, CM, elliptic functions/curves, L-functions, Moonshine, renormalization, operator algebras, spectral geometry, etc. Only record end theories that constrain current or near-future phases; do not build encyclopedia entries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Source materials / references        | <ul style="list-style-type: none"> <li>Tool V1.04 implementation-scaling plan</li> <li>SCF37</li> <li>workflow policy</li> <li>repo_inventory/raw_export</li> <li>object register</li> <li>implementation scratchpad</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>End-theory comparison ledger</li> <li>End theories are target structures, consistency checks, terminology alignment, or obstruction/comparison mirrors; forbidden as CNNA producers.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not claim Lean build success for planning-only artifacts</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>parallel planning master</li> <li>manual duplicate theory ledger</li> <li>full graph dumped into PDF</li> <li>end theory used as CNNA producer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### Phase 1.1.1.5.6 – Active object graph slices

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                    | Record              |
|--------------------------|---------------------|
| Phase status             | W                   |
| Context class            | LeanGovernancePhase |
| Phase documentation tier | F2                  |

|                                        |                                                                                                                                                                                                                            |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                              |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | planned-not-active                                                                                                                                                                                                         |
| Preparation status                     | planned-not-active                                                                                                                                                                                                         |
| Phase type                             | infrastructure                                                                                                                                                                                                             |
| Phase-type-specific contract           | Infrastructure phase: update YAML/generator/generated exports, validate single-source constraints, and render PDF without claiming Lean build success.                                                                     |
| Title / state                          | Active object graph slices                                                                                                                                                                                                 |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.                                                   |
| Source-to-CNNA mapping                 | Graph edges distinguish internal provenance, local formalization context, and external comparison-only references.                                                                                                         |
| Dependency-depth audit                 | Depends on 1.1.1.5 and earlier governance gates. It must not block the current active 1.1.1.2 blue-correction ledger; it is a future scaling/tooling refinement.                                                           |
| Phase split decision                   | Atomic planned subphase under 1.1.1.5. Split only if implementation would otherwise mix Lean-code changes with tool/PDF/YAML changes.                                                                                      |
| Positive path plan                     | Implement by extending Python/YAML/generated exports first. Do not add Lean code. If a Lean-side audit mirror becomes necessary, create a separate later Lean phase with an explicit finite witness and import policy.     |
| Module implementation manifest         | Tooling subphase. Proposed generated artifacts: active-object-neighborhood DOT/SVG/PDF; graph-slice TSV/JSON                                                                                                               |
| Semantic placement decision            | Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.                                                                                                      |
| Reuse and duplication audit            | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.                                                                                  |
| Canonical generator policy             | Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.                                                                                         |
| Module compactness rule                | Keep PDF slices compact; full data remains in generated TSV/JSON exports.                                                                                                                                                  |
| Refactor regression protocol           | Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.                                                                                                  |
| Refactor migration metrics             | No Lean declaration migration. Record generator/export changes only when the phase becomes active.                                                                                                                         |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                         |
| Consumer-contract direction            | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.                                                                                             |
| Governance-overhead review             | This phase exists to reduce governance overhead during Lean work by pushing routine collection/rendering into Python.                                                                                                      |
| Proof rehearsal before implementation  | Validate generator output, single YAML source, exactly one active phase, unchanged Lean-data timestamp, and section-specific rendering. This is a tooling proof rehearsal, not a Lean theorem.                             |
| Expected tactics                       | No Lean tactics expected. If a later Lean mirror is explicitly introduced, use explicit records/constructors and avoid @[simp], noncomputable operational paths, classical helpers, and broad fragile rewrites.            |
| Expected mathlib imports               | No new mathlib imports expected for the planned Python/YAML tooling route.                                                                                                                                                 |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                     |
| Concrete computable instance traversal | Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.                                                                                    |
| Computable path non-skeleton status    | not-applicable-tooling-only;<br>no operational mathematical computation is introduced                                                                                                                                      |
| Expected executable witness            | Python generator run plus generated TSV/TeX/PDF artifacts;<br>no Lean executable witness required.                                                                                                                         |
| Mathlib complication log               | No mathlib complication expected because this is a Python/YAML/PDF tooling phase.                                                                                                                                          |
| Noncomputable boundary audit           | No Lean operational path is modified;<br>any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase.                                                                      |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                  |
| Bridge gap status                      | No mathematical bridge is introduced here. Local background theory and large end-theory comparison roles are tracked as metadata/ledgers, not as CNNA generators.                                                          |
| Natural backjump/refactor layer        | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                   |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| How implemented                      | No completed implementation record yet. This future tooling/governance phase was introduced by Tool V1.04 and must remain inactive until its predecessor gates and active cursor allow it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Verification / evidence              | Not applicable yet;<br>phase is a planned Tool V1.04 implementation-scaling subphase. Lean-data timestamp remains unchanged.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check       | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation notes                 | Render small producer-consumer/provenance neighborhoods around the active phase and active CNNA objects rather than dumping the full object/declaration graph into the PDF. Depth 1-2 graph slices only in PDF;<br>full graph exported as DOT/JSON for optional audit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Source materials / references        | <ul style="list-style-type: none"> <li>Tool V1.04 implementation-scaling plan</li> <li>SCF37</li> <li>workflow policy</li> <li>repo_inventory/raw_export</li> <li>object register</li> <li>implementation scratchpad</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Active object graph slices</li> <li>Graph edges distinguish internal provenance, local formalization context, and external comparison-only references.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not claim Lean build success for planning-only artifacts</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>parallel planning master</li> <li>manual duplicate theory ledger</li> <li>full graph dumped into PDF</li> <li>end theory used as CNNA producer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### Phase 1.1.1.5.7 – Red-blocker addressing-phase generator

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                          |
| Context class                          | LeanGovernancePhase                                                                                                                                                                                                        |
| Phase documentation tier               | F2                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                              |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | planned-not-active                                                                                                                                                                                                         |
| Preparation status                     | planned-not-active                                                                                                                                                                                                         |
| Phase type                             | infrastructure                                                                                                                                                                                                             |
| Phase-type-specific contract           | Infrastructure phase: update YAML/generator/generated exports, validate single-source constraints, and render PDF without claiming Lean build success.                                                                     |
| Title / state                          | Red-blocker addressing-phase generator                                                                                                                                                                                     |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Only write records belonging to this phase; register later-phase theory as missing-source or future-target entries.                                                   |
| Source-to-CNNA mapping                 | If the blocker is an imported end-theory assumption, route to internal derivation/comparison-boundary phase rather than patching the consumer.                                                                             |
| Dependency-depth audit                 | Depends on 1.1.1.5 and earlier governance gates. It must not block the current active 1.1.1.2 blue-correction ledger; it is a future scaling/tooling refinement.                                                           |
| Phase split decision                   | Atomic planned subphase under 1.1.1.5. Split only if implementation would otherwise mix Lean-code changes with tool/PDF/YAML changes.                                                                                      |
| Positive path plan                     | Implement by extending Python/YAML/generated exports first. Do not add Lean code. If a Lean-side audit mirror becomes necessary, create a separate later Lean phase with an explicit finite witness and import policy.     |
| Module implementation manifest         | Tooling subphase. Proposed generated artifacts: red-blocker phase-proposal TSV; accepted phase rows only after approval                                                                                                    |
| Semantic placement decision            | Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.                                                                                                      |
| Reuse and duplication audit            | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.                                                                                  |
| Canonical generator policy             | Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.                                                                                         |
| Module compactness rule                | Keep PDF slices compact; full data remains in generated TSV/JSON exports.                                                                                                                                                  |
| Refactor regression protocol           | Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.                                                                                                  |
| Refactor migration metrics             | No Lean declaration migration. Record generator/export changes only when the phase becomes active.                                                                                                                         |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                           |
| Consumer-contract direction            | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.                                                                                             |
| Governance-overhead review             | This phase exists to reduce governance overhead during Lean work by pushing routine collection/rendering into Python.                                                                                                      |
| Proof rehearsal before implementation  | Validate generator output, single YAML source, exactly one active phase, unchanged Lean-data timestamp, and section-specific rendering. This is a tooling proof rehearsal, not a Lean theorem.                             |
| Expected tactics                       | No Lean tactics expected. If a later Lean mirror is explicitly introduced, use explicit records/constructors and avoid @[simp], noncomputable operational paths, classical helpers, and broad fragile rewrites.            |
| Expected mathlib imports               | No new mathlib imports expected for the planned Python/YAML tooling route.                                                                                                                                                 |
| ExecComplex/computable plan            | No ExecComplex object is introduced. This gate only classifies consumption/API status.                                                                                                                                     |
| Concrete computable instance traversal | Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.                                                                                    |
| Computable path non-skeleton status    | not-applicable-tooling-only; no operational mathematical computation is introduced                                                                                                                                         |
| Expected executable witness            | Python generator run plus generated TSV/TeX/PDF artifacts; no Lean executable witness required.                                                                                                                            |
| Mathlib complication log               | No mathlib complication expected because this is a Python/YAML/PDF tooling phase.                                                                                                                                          |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noncomputable boundary audit         | No Lean operational path is modified;<br>any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Mathlib mitigation/refactor layer    | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Bridge gap status                    | No mathematical bridge is introduced here. Local background theory and large end-theory comparison roles are tracked as metadata/ledgers, not as CNNA generators.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Natural backjump/refactor layer      | Backjump to the earliest natural layer: consumption classifier, blue-object correction ledger, public API contract, pruning ledger, Legacy.StatusGates, CoreSymbols, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, verify module manifest/reuse audit before adding Lean code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| How implemented                      | No completed implementation record yet. This future tooling/governance phase was introduced by Tool V1.04 and must remain inactive until its predecessor gates and active cursor allow it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Verification / evidence              | Not applicable yet;<br>phase is a planned Tool V1.04 implementation-scaling subphase. Lean-data timestamp remains unchanged.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check       | After closure, update linked objects, generated exports/PDF, and advance exactly one active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation notes                 | When the pre-implementation thought experiment marks a future phase red, propose an upstream addressing phase at the earliest natural source layer before the blocked phase becomes active. Generate concise phase proposals only for red blockers; yellow risks remain notes in the implementation packet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Source materials / references        | <ul style="list-style-type: none"> <li>Tool V1.04 implementation-scaling plan</li> <li>SCF37</li> <li>workflow policy</li> <li>repo_inventory/raw_export</li> <li>object register</li> <li>implementation scratchpad</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Red-blocker addressing-phase generator</li> <li>If the blocker is an imported end-theory assumption, route to internal derivation/comparison-boundary phase rather than patching the consumer.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not claim Lean build success for planning-only artifacts</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



|                                      |                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"><li>• parallel planning master</li><li>• manual duplicate theory ledger</li><li>• full graph dumped into PDF</li><li>• end theory used as CNNA producer</li></ul> |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 1.1.2 – Quarantine single-constructor legacy status gates

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F3                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                               |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | [‘LEG0 split inventory from 1.1.1’, ‘Arithmetic/Notation.lean legacy status declarations’]                                                                                                                              |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Title / state                         | Quarantine single-constructor legacy status gates                                                                                                                                                                       |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                             |
| Source-to-CNNA mapping                | Consumes classification bucket from 1.1.1.1.4 and public API contract from 1.1.1.3; moves only legacy-status/legacy-policy single-constructor gates into quarantine.                                                    |
| Dependency-depth audit                | Do not begin until the consumption map and public API contract are closed. A single-constructor inductive can still be a proof bridge; classification, not constructor count alone, decides.                            |
| Phase split decision                  | Create Legacy/StatusGates.lean; move or reexport status declarations; update legacy imports only; do not expose this module to Phase 3+.                                                                                |
| Positive path plan                    | Move/reexport only classified legacy gates; preserve compatibility; do not delete; keep Phase 3+ firewall intact.                                                                                                       |
| Refactor regression protocol          | Required before implementation: wrapper-first incremental migration; bounded declaration batches; preserve compatibility aliases; no legacy status deletion; require local full-build evidence after delivered package. |
| Refactor migration metrics            | To fill before code: declarations_migrated_count; single-constructor status gates moved; aliases retained; imports changed; consumer modules updated; BuildAll impact; local build evidence source.                     |
| Lean/mathlib version policy           | No new mathlib imports expected. If any appear, stop and justify against Lean 4.28.0 baseline or open a separate gate.                                                                                                  |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                      |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consumer-contract direction            | Phase 3+ must not import Legacy.StatusGates;<br>quarantine module is legacy-only with no public-final consumer path.                                                                                                                                                  |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | rfl/exact for definitional aliases;<br>no broad simplification.                                                                                                                                                                                                       |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                            |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                           |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                        |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                          |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                          |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                            |
| Source materials / references          | <ul style="list-style-type: none"> <li>1.1.1.1 consumption map</li> <li>1.1.1.3 public API contract</li> <li>LEG1 Legacy single-constructor status/policy gates</li> </ul>                                                                                            |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Legacy.StatusGates quarantine bucket</li> <li>legacy-policy single-constructor inductive bucket where applicable</li> </ul>                                                                                                    |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import <code>Legacy.StatusGates</code></li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> <li>do not classify by name pattern alone</li> <li>do not quarantine proof-bridge or blue-correction declarations</li> </ul> |
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### Phase 1.1.3 – Create true CoreSymbols/Notation micro-surface

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFFFFF; border: 1px solid black; padding: 0 2px;">W</span>                                                                                                                                                                       |
| Context class                         | <code>LeanRefactorPhase</code>                                                                                                                                                                                                                                   |
| Phase documentation tier              | F3                                                                                                                                                                                                                                                               |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                                        |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                               |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                              |
| Implementation status                 | <span style="background-color: #FFFFFF; border: 1px solid black; padding: 0 2px;">planned-not-active</span>                                                                                                                                                      |
| Preparation status                    | <span style="background-color: #FFFFFF; border: 1px solid black; padding: 0 2px;">planned-not-active</span>                                                                                                                                                      |
| Phase type                            | refactor                                                                                                                                                                                                                                                         |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                          |
| Title / state                         | Create true CoreSymbols/Notation micro-surface                                                                                                                                                                                                                   |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                      |
| Source-to-CNNA mapping                | Requires status-gate quarantine so CoreSymbols is not contaminated by single-constructor status enums.                                                                                                                                                           |
| Dependency-depth audit                | No further split unless CoreSymbols still has unrelated foundation/finite/handoff responsibilities after classification.                                                                                                                                         |
| Phase split decision                  | Create the minimal CoreSymbols surface and redirect neutral imports to it.                                                                                                                                                                                       |
| Positive path plan                    | Mostly module/import closure. Any public constant should be definitional and domain-role named.                                                                                                                                                                  |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                   |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.                         |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                               |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                    |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Proof rehearsal before implementation  | rfl/exact/constructor if any thin wrapper is needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Source materials / references          | <ul style="list-style-type: none"> <li>LEG0 true notation/core-symbol bucket</li> <li>post-1.1.2 monolith state</li> <li>LEG2 CoreSymbols true notation surface</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Declarations classified as genuine notation/core symbols move to CNNA/PillarA/Arithmetic/Notation/CoreSymbols.lean. Anything plan-semantic remains legacy.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply;<br/>this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

Phase 1.1.4 – Separate legacy phase aliases from neutral wrappers and blue-object correction wrappers

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                          |
| Context class                          | LeanRefactorPhase                                                                                                                                                                                                          |
| Phase documentation tier               | F3                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                  |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | planned-not-active                                                                                                                                                                                                         |
| Preparation status                     | planned-not-active                                                                                                                                                                                                         |
| Phase type                             | refactor                                                                                                                                                                                                                   |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                    |
| Title / state                          | Separate legacy phase aliases from neutral wrappers                                                                                                                                                                        |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                |
| Source-to-CNNA mapping                 | Requires CoreSymbols so neutral wrappers import a clean notation surface. Requires status gates quarantined.                                                                                                               |
| Dependency-depth audit                 | Split only if neutral wrappers require nontrivial theorem transport; then create predecessor source wrapper rather than proving in the alias layer.                                                                        |
| Phase split decision                   | Keep old code compiling through legacy aliases while new code imports neutral wrappers only.                                                                                                                               |
| Positive path plan                     | Forward old theorem/data by exact names or field projections; no semantic reinterpretation in alias layer.                                                                                                                 |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation  | exact existing_lemma;<br>rfl for abbreviation;<br>constructor/field projections for wrapper structures.                                                                                                                    |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                    |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                   |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                        |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                    |
| Expected executable witness            | not specified yet                                                                                                                                                                                                          |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                       |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                  |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Natural backjump/refactor layer      | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                      | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Verification / evidence              | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Post-implementation plan check       | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Implementation notes                 | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Source materials / references        | <ul style="list-style-type: none"> <li>LEG0 phase-alias and neutral-wrapper buckets</li> <li>post-1.1.3 CoreSymbols module</li> <li>LEG3 Legacy phase aliases compatibility surface</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Historical SM*/S6*/R*/Audit names are isolated in Legacy/PhaseAliases. NeutralWrappers may forward selected old derivations under domain-role names only.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.1.5 – Post-split import firewall and Phase-3 visibility gate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                         |
| Context class                         | ToolInfrastructurePhase                                                                                                                   |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                        |
| Preparation status                    | planned-not-active                                                                                                                        |
| Phase type                            | infrastructure                                                                                                                            |



|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase-type-specific contract           | Infrastructure/planning phase: justify absence of operational witness; close schema/decision prerequisites before code.                                                                                                                                               |
| Title / state                          | Post-split import firewall and Phase-3 visibility gate                                                                                                                                                                                                                |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                           |
| Source-to-CNNA mapping                 | Depends on all previous 1.1 subphases and regenerated module inventory. This is the gate before 1.2 and 3 can be trusted.                                                                                                                                             |
| Dependency-depth audit                 | No further split unless multiple forbidden import families remain; then backjump to the bucket causing the leak.                                                                                                                                                      |
| Phase split decision                   | Regenerate module inventory; search import graph for forbidden Legacy.StatusGates consumers; update phase 1.2/1.3/3 dependencies only after the firewall passes.                                                                                                      |
| Positive path plan                     | Structural import graph proof-by-audit, plus Lean build evidence supplied locally by user or actual environment if available.                                                                                                                                         |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No proof tactics expected beyond exact/rfl for any final import wrappers.                                                                                                                                                                                             |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                               |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                              |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                                                           |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                          |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit; no Lean build evidence has been recorded for this phase.                                                                                                                                             |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                            |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source materials / references        | <ul style="list-style-type: none"> <li>post-1.1.2-1.1.4 module tree</li> <li>generated module inventory script</li> <li>phase-object DAG</li> <li>LEG4 Neutral import firewall</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>LEG4 is the generated/audited rule that Phase 3+ sees CoreSymbols/NeutralWrappers but not Legacy.StatusGates.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.2 – Foundation and generator Profile neutralization

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                      |
| Preparation status                    | planned-prepared                                                                                                                                                                                        |
| Phase type                            | refactor                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                 |
| Title / state                         | Foundation and generator Profile neutralization                                                                                                                                                         |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.             |
| Source-to-CNNA mapping                | Phase 1.2 establishes or neutralizes: ExecComplexPath; GeneratorProvenance; substrateProfile; LevelProfile; CarrierProfile. Object links are generated from the object register.                        |
| Dependency-depth audit                | Depends on 1.1.5; current foundation/generator modules; preliminary object IDs O00/G00-G04. This record exists so the phase can be prepared without reopening Phase 1.1.                                |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                           |

|                                        |                                                                                                                                                                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Foundation/ExecComplex.lean;<br>CNNA/PillarA/Foundation/GeneratorProvenance.lean;<br>CNNA/PillarA/Foundation/Substrate/*.lean;<br>CNNA/PillarA/Foundation/LevelProfile.lean;<br>CNNA/PillarA/Foundation/CarrierProfile.lean. |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                 |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                     |
| Proof rehearsal before implementation  | No reindex/level/carrier without proof profile. ExecComplex remains the operational scalar path;<br>no noncomputable Complex replacement.                                                                                                                                                  |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                                                    |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                                                 |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                                                |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                        |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                    |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                          |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                       |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                  |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                                             |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                        |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                      |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                               |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                               |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.                          |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                 |
| Source materials / references          | <ul style="list-style-type: none"> <li>GeneratedSource.lean;<br/>GeneratedMainPath.lean;<br/>BoundarySource/*.lean;<br/>Foundation/ExecComplex*.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                        |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"><li>ExecComplexPath;<br/>GeneratorProvenance;<br/>substrateProfile;<br/>LevelProfile;<br/>CarrierProfile</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                     | <ul style="list-style-type: none"><li>GLOBAL-INHERITED: top-level global <code>_no_go_guardrails</code> apply; this phase list is local addenda only.</li><li>no <code>@[simp]</code> attributes in the operational path</li><li>no <code>noncomputable</code> in the operational path</li><li>no classical helper unless isolated in allowed mirror/non-operational layer</li><li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li><li>no free formula without proof-carrying entry structure</li><li>no consumer-side patch when the missing structure belongs earlier</li><li>no premature obstruction before a serious positive construction attempt</li><li>no phase-semantic public API names</li><li>no fragile rewrite chain when a structural constructor/projection proof is available</li><li>do not delete legacy status enums during quarantine</li><li>do not let Phase 3+ import <code>Legacy.StatusGates</code></li><li>do not treat old TeX/PDF files as maintained roadmap sources</li></ul> |

Phase 1.2.1 – Neutralize ExecComplex operational scalar boundary

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                                                                |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                                                                |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                               |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                   |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                               |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                              |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                               |
| Preparation status                    | planned-prepared                                                                                                                                                                                                                                                 |
| Phase type                            | refactor                                                                                                                                                                                                                                                         |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                          |
| Title / state                         | Neutralize ExecComplex operational scalar boundary                                                                                                                                                                                                               |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                      |
| Source-to-CNNA mapping                | Phase 1.2.1 establishes or neutralizes: ExecComplexPath and computability boundary. Object links are generated from the object register.                                                                                                                         |
| Dependency-depth audit                | Depends on 1.2; current ExecComplex modules. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                                        |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                                    |
| Positive path plan                    | Implement only after dependencies close. Target: CNNA/PillarA/Foundation/ExecComplex.lean.                                                                                                                                                                       |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                   |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.                         |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | Expose only the computable operational scalar boundary and any mirror bridge explicitly; no analytic shortcut is allowed.                                                                                                                                             |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                               |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                               |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                              |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                                                           |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                          |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit; no Lean build evidence has been recorded for this phase.                                                                                                                                             |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                            |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Foundation/ExecComplex.lean;<br/>ExecComplexBridge.lean;<br/>ExecComplexLemmas.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>ExecComplexPath and computability boundary</li> </ul>                                                                                                                                                                          |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: top-level global__no_go_guardrails apply; this phase list is local addenda only.</li><li>• no @[simp] attributes in the operational path</li><li>• no noncomputable in the operational path</li><li>• no classical helper unless isolated in allowed mirror/non-operational layer</li><li>• no Matrix.inv or external inverse oracle in operational derivations</li><li>• no free formula without proof-carrying entry structure</li><li>• no consumer-side patch when the missing structure belongs earlier</li><li>• no premature obstruction before a serious positive construction attempt</li><li>• no phase-semantic public API names</li><li>• no fragile rewrite chain when a structural constructor/projection proof is available</li><li>• do not delete legacy status enums during quarantine</li><li>• do not let Phase 3+ import Legacy.StatusGates</li><li>• do not treat old TeX/PDF files as maintained roadmap sources</li></ul> |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 1.2.1.1 – ExecComplex capability and limitation audit

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                  |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                                 |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                     |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                 |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                |
| Implementation status                 | not_started                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                       |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                         |
| Phase-type-specific contract          | derivation scalar architecture obligations                                                                                                                                                                                                                                                                         |
| Title / state                         | ExecComplex capability and limitation audit                                                                                                                                                                                                                                                                        |
| Preparation workflow                  | computable scalar architecture workflow                                                                                                                                                                                                                                                                            |
| Source-to-CNNA mapping                | O00 dossier gets exact capability matrix and limitations.                                                                                                                                                                                                                                                          |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Scalar architecture decisions must precede code implementation. Exact quadratic or CD scalar routes may not feed earlier generator definitions.                 |
| Phase split decision                  | split out because ExecComplex limits affect Phase 1.2, 4, 5, 7, 8 and CD bridges differently.                                                                                                                                                                                                                      |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Keep current QxQ path first; add exact quadratic pilot only if needed by finite tests and no duplicate module exists; keep analytic Complex mirror-only. |
| Module implementation manifest        | pending Phase 1.1.1.1.1                                                                                                                                                                                                                                                                                            |
| Semantic placement decision           | scalar-foundation / bridge governance, not generator semantics                                                                                                                                                                                                                                                     |
| Reuse and duplication audit           | must check existing ExecComplex, ExecComplexBridge, Schur.CharpolyC, ArithmeticCDBridge, and mathlib QuadraticAlgebra before proposing new module                                                                                                                                                                  |
| Canonical generator policy            | scalar layer must consume canonical generator outputs only; cannot define generator semantics                                                                                                                                                                                                                      |
| Module compactness rule               | do not create ExactQuadraticComplex unless existing foundation/bridge modules cannot carry the semantics cleanly                                                                                                                                                                                                   |
| Refactor regression protocol          | incremental wrapper mode; full local build green after each migration step                                                                                                                                                                                                                                         |



|                                        |                                                                                                                                                                                                                                                                 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refactor migration metrics             | track declarations migrated, imports changed, executable witnesses added                                                                                                                                                                                        |
| Lean/mathlib version policy            | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                                                                                    |
| ExecComplex boundary policy            | EXEC-3 enforced: QxQ floor, exact quadratic optional, analytic Complex mirror-only                                                                                                                                                                              |
| Consumer-contract direction            | scalar layer consumes generator/arithmetic source data;<br>generator definitions do not consume scalar comparison/mirror data                                                                                                                                   |
| Governance-overhead review             | review after first scalar/module manifest whether this lock forced a concrete decision.                                                                                                                                                                         |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Show coordinate equations for all O00 operations and prove unsupported requirements are redirected to O09 or mirror. |
| Expected tactics                       | explicit structures, constructor/projection lemmas, ring-normal-form only for coordinate arithmetic, no Matrix.inv and no consumer-side patch.                                                                                                                  |
| Expected mathlib imports               | Mathlib.Algebra.QuadraticAlgebra only if pilot approved;<br>Mathlib.Data.Complex.Basic remains bridge/mirror-only;<br>NumberField modules require version-policy gate.                                                                                          |
| ExecComplex/computable plan            | Direct QxQ path only.                                                                                                                                                                                                                                           |
| Concrete computable instance traversal | required later: QxQ baseline row, Q(i) row, Q(sqrt(-3)) row, and at least one generic D row if O09 is implemented                                                                                                                                               |
| Computable path non-skeleton status    | must be shown by concrete row witnesses                                                                                                                                                                                                                         |
| Expected executable witness            | finite row with rational ExecComplex output plus optional exact quadratic coordinate and rational/integer invariant projection                                                                                                                                  |
| Mathlib complication log               | Record Complex/Real noncomputable boundary, QuadraticAlgebra instance diamond warning, NumberField formal/non-executable dependency, and any required version-gate.                                                                                             |
| Noncomputable boundary audit           | General C/R and algebraic closure remain mirror/formal. Exact quadratic candidate must avoid noncomputable definitions on operational path.                                                                                                                     |
| Mathlib mitigation/refactor layer      | Use project-owned explicit Q-coordinate structure or narrowly wrapped QuadraticAlgebra;<br>isolate formal mathlib NumberField in mirror/proof layer.                                                                                                            |
| Bridge gap status                      | O09 candidate remains open.                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | If exact quadratic data are needed, backjump to scalar-foundation layer rather than patching arithmetic/lattice consumers.                                                                                                                                      |
| Implementation prompt contract         | Before Lean code, module manifest must state exact module placement, imported mathlib modules, and why no existing module owns the semantics.                                                                                                                   |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                     |
| How implemented                        | Not implemented.                                                                                                                                                                                                                                                |
| Verification / evidence                | Generated-table validation pending/run after patch;<br>no Lean build.                                                                                                                                                                                           |
| Proof strategy                         | Show coordinate equations for all O00 operations and prove unsupported requirements are redirected to O09 or mirror.                                                                                                                                            |
| Build result                           | No Lean build executed or claimed.                                                                                                                                                                                                                              |
| Post-implementation plan check         | Check whether the scalar architecture lock prevented analytic/noncomputable leakage or duplicate scalar implementation.                                                                                                                                         |
| Artifact packaging rule                | Return complete zip;<br>Lean build local unless actually executed here.                                                                                                                                                                                         |
| Implementation notes                   | Future audit of O00 before scalar expansion.                                                                                                                                                                                                                    |
| Source materials / references          | <ul style="list-style-type: none"> <li>Current ExecComplex and bridge modules; EXEC-2/EXEC-3.</li> </ul>                                                                                                                                                        |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>QxQ executable operations;<br/>bridge-only Complex operations;<br/>unsupported algebraic irrational coordinates.</li> </ul>                                                                                              |
| Implemented modules                    | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                                                                          |
| Added modules                          | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                                                      |
| Modified modules                       | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                                                      |
| Forbidden patterns checked             | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                                                                                                            |

|                                      |                                                                                                                                                                                                                                               |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails                     | <ul style="list-style-type: none"> <li>No silent widening of ExecComplex semantics.</li> </ul>                                                                                                                                                |
| Proposed added modules               | <ul style="list-style-type: none"> <li>none at decision gate;<br/>possible CNNA/PillarA/Foundation/ExactQuadraticComplex.lean only after manifest</li> </ul>                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>pending manifest</li> </ul>                                                                                                                                                                            |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Foundation/ExecComplex.lean;<br/>Foundation/ExecComplexBridge.lean;<br/>Arithmetic/Schur/CharpolyC.lean;<br/>Arithmetic/CayleyDickson/ArithmeticCDBridge.lean;<br/>mathlib QuadraticAlgebra</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>alternate complex scalar, algebraic complex, CD complex, mirror Complex, exact quadratic norm/trace/conjugation</li> </ul>                                                                             |

### Phase 1.2.1.2 – Required exact quadratic scalar schema: discriminant-indexed minimal operational interface

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid #000; padding: 0 2px;">R</span>                                                                                                                                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                    |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                                                                     |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                         |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                     |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                    |
| Implementation status                 | not_started                                                                                                                                                                                                                                                                                                                                            |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                           |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                             |
| Phase-type-specific contract          | derivation scalar architecture obligations                                                                                                                                                                                                                                                                                                             |
| Title / state                         | Required exact quadratic scalar schema: discriminant-indexed minimal operational interface                                                                                                                                                                                                                                                             |
| Preparation workflow                  | computable scalar architecture workflow                                                                                                                                                                                                                                                                                                                |
| Source-to-CNNA mapping                | If implemented, O09 owns exact quadratic operations; ExecComplex remains rational floor.                                                                                                                                                                                                                                                               |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Scalar architecture decisions must precede code implementation. Exact quadratic or CD scalar routes may not feed earlier generator definitions.                                                     |
| Phase split decision                  | split out because ExecComplex limits affect Phase 1.2, 4, 5, 7, 8 and CD bridges differently.                                                                                                                                                                                                                                                          |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Implement or wrap only a parameterized schema. Do not instantiate with D in this phase. The concrete D-instantiation belongs to Phase 4.5.1 after an operational discriminant ledger exists. |
| Module implementation manifest        | pending Phase 1.1.1.1.1                                                                                                                                                                                                                                                                                                                                |
| Semantic placement decision           | scalar-foundation / bridge governance, not generator semantics                                                                                                                                                                                                                                                                                         |
| Reuse and duplication audit           | must check existing ExecComplex, ExecComplexBridge, Schur.CharpolyC, ArithmeticCDBridge, and mathlib QuadraticAlgebra before proposing new module                                                                                                                                                                                                      |
| Canonical generator policy            | scalar layer must consume canonical generator outputs only; cannot define generator semantics                                                                                                                                                                                                                                                          |
| Module compactness rule               | do not create ExactQuadraticComplex unless existing foundation/bridge modules cannot carry the semantics cleanly                                                                                                                                                                                                                                       |
| Refactor regression protocol          | incremental wrapper mode; full local build green after each migration step                                                                                                                                                                                                                                                                             |
| Refactor migration metrics            | track declarations migrated, imports changed, executable witnesses added                                                                                                                                                                                                                                                                               |

|                                        |                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lean/mathlib version policy            | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                                                                                                                                         |
| ExecComplex boundary policy            | Exact quadratic schema sits above QxQ as a discriminant-indexed operational extension; no free sqrt(D).                                                                                                                                                                                                              |
| Consumer-contract direction            | scalar layer consumes generator/arithmetic source data; generator definitions do not consume scalar comparison/mirror data                                                                                                                                                                                           |
| Governance-overhead review             | review after first scalar/module manifest whether this lock forced a concrete decision.                                                                                                                                                                                                                              |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. For arbitrary D, prove coordinate operations and formulas. Avoid analytic square roots; theta is the generator of the algebraic relation, not an external Complex number. |
| Expected tactics                       | explicit structures, constructor/projection lemmas, ring-normal-form only for coordinate arithmetic, no Matrix.inv and no consumer-side patch.                                                                                                                                                                       |
| Expected mathlib imports               | Prefer project-owned coordinate structure if mathlib dependencies cause noncomputable/classical leakage; otherwise audit Mathlib.Algebra.QuadraticAlgebra. AdjoinRoot remains proof/reference unless executable status is verified.                                                                                  |
| ExecComplex/computable plan            | O09 is required for exact quadratic coordinates, but it does not replace ExecComplex. Only projected rational/integer invariants flow back to QxQ reports.                                                                                                                                                           |
| Concrete computable instance traversal | Schema-level witnesses: symbolic arbitrary D, plus later planned special rows D=-1/-3/-4 only after D-ledger. This phase may include hand-written closed coordinate examples but cannot claim CNNA provenance for D.                                                                                                 |
| Computable path non-skeleton status    | must be shown by concrete row witnesses                                                                                                                                                                                                                                                                              |
| Expected executable witness            | finite row with rational ExecComplex output plus optional exact quadratic coordinate and rational/integer invariant projection                                                                                                                                                                                       |
| Mathlib complication log               | Audit QuadraticAlgebra coordinate operations, basis/noncomputable declarations, and version stability. AdjoinRoot quotient construction includes noncomputable instances in docs and must not silently enter operational path.                                                                                       |
| Noncomputable boundary audit           | No analytic sqrt, Real, Complex exponential, elliptic functions, or uninspected quotient field structure in operational code.                                                                                                                                                                                        |
| Mathlib mitigation/refactor layer      | Use project-owned explicit Q-coordinate structure or narrowly wrapped QuadraticAlgebra; isolate formal mathlib NumberField in mirror/proof layer.                                                                                                                                                                    |
| Bridge gap status                      | Schema can close only as parameterized. Concrete use remains open until 4.5.1 O10/BR14.                                                                                                                                                                                                                              |
| Natural backjump/refactor layer        | If exact quadratic data are needed, backjump to scalar-foundation layer rather than patching arithmetic/lattice consumers.                                                                                                                                                                                           |
| Implementation prompt contract         | Before Lean code, module manifest must state exact module placement, imported mathlib modules, and why no existing module owns the semantics.                                                                                                                                                                        |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                             |
| How implemented                        | Not implemented.                                                                                                                                                                                                                                                                                                     |
| Verification / evidence                | Generated-table validation pending/run after patch; no Lean build.                                                                                                                                                                                                                                                   |
| Proof strategy                         | Pilot Q(i), Q(sqrt(-3)), and generic nonsquare D; prove norm/trace/conjugation formulas and executable witnesses.                                                                                                                                                                                                    |
| Build result                           | No Lean build executed or claimed.                                                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | Check whether the scalar architecture lock prevented analytic/noncomputable leakage or duplicate scalar implementation.                                                                                                                                                                                              |
| Artifact packaging rule                | Return complete zip; Lean build local unless actually executed here.                                                                                                                                                                                                                                                 |
| Implementation notes                   | Future decision whether to add a compact exact quadratic operational layer.                                                                                                                                                                                                                                          |
| Source materials / references          | <ul style="list-style-type: none"> <li>Mathlib.Algebra.QuadraticAlgebra</li> <li>Mathlib.RingTheory.AdjoinRoot as quotient/reference</li> <li>O05/O06 operational discriminant source plan</li> <li>LEG32 no-free-sqrt provenance lock</li> </ul>                                                                    |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>discriminant-indexed schema</li> <li>theta with <math>\text{theta}^2 = D</math></li> <li>coordinate addition/multiplication</li> <li>conjugation</li> <li>trace</li> <li>norm</li> <li>integer/rational projection lemmas</li> </ul>                                          |

|                                      |                                                                                                                                                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implemented modules                  | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                                        |
| Added modules                        | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                    |
| Modified modules                     | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                    |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                                                                          |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>No duplicate algebraic complex implementation; no noncomputable operational dependency.</li> </ul>                                                                                     |
| Proposed added modules               | <ul style="list-style-type: none"> <li>none at decision gate; possible CNNA/PillarA/Foundation/ExactQuadraticComplex.lean only after manifest</li> </ul>                                                                      |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>pending manifest</li> </ul>                                                                                                                                                            |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Foundation/ExecComplex.lean; Foundation/ExecComplexBridge.lean; Arithmetic/Schur/CharpolyC.lean; Arithmetic/CayleyDickson/ArithmeticCDBridge.lean; mathlib QuadraticAlgebra</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>alternate complex scalar, algebraic complex, CD complex, mirror Complex, exact quadratic norm/trace/conjugation</li> </ul>                                                             |

### Phase 1.2.2 – Neutralize generator provenance source profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                      |
| Preparation status                    | planned-prepared                                                                                                                                                                                        |
| Phase type                            | refactor                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                 |
| Title / state                         | Neutralize generator provenance source profile                                                                                                                                                          |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.             |
| Source-to-CNNA mapping                | Phase 1.2.2 establishes or neutralizes: GeneratorProvenance. Object links are generated from the object register.                                                                                       |
| Dependency-depth audit                | Depends on 1.2.1; generated source/main-path modules. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                      |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                           |
| Positive path plan                    | Implement only after dependencies close. Target: CNNA/PillarA/Foundation/GeneratorProvenance.lean.                                                                                                      |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.       |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.             |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | Forward existing provenance data through a neutral source profile; do not make substrate/level choices in this module.                                                                                                                                                |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                               |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                               |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                              |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                  |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                                                           |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                              |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                          |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit; no Lean build evidence has been recorded for this phase.                                                                                                                                             |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                            |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/Smoke/GeneratedSource.lean; GeneratedMainPath.lean; BoundarySource/GeneratedSubstrateRoute.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                         |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>GeneratorProvenance</li> </ul>                                                                                                                                                                                                 |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import <code>Legacy.StatusGates</code></li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |
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### Phase 1.2.3 – Neutralize regular substrate profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                                                                |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                                                                |
| Phase documentation tier              | F0                                                                                                                                                                                                                                                               |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                               |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                              |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                               |
| Preparation status                    | planned-prepared                                                                                                                                                                                                                                                 |
| Phase type                            | refactor                                                                                                                                                                                                                                                         |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                          |
| Title / state                         | Neutralize regular substrate profile                                                                                                                                                                                                                             |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                      |
| Source-to-CNNA mapping                | Phase 1.2.3 establishes or neutralizes: substrate: regular branching. Object links are generated from the object register.                                                                                                                                       |
| Dependency-depth audit                | Depends on 1.2.2; regular branching substrate modules. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                              |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                                    |
| Positive path plan                    | Implement only after dependencies close. Target: CNNA/PillarA/Foundation/Substrate/Regular.lean.                                                                                                                                                                 |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                   |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.                         |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                         |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                    |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                              |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof rehearsal before implementation  | Separate regular branching data from variable branching data;<br>no geometric angle/lattice assumption enters here.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Foundation/ConcreteSubstrate.lean;<br/>Arithmetic/BoundarySource/BranchAddressSubstrate.lean;<br/>BoundaryStateSubstrate.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>substrate: regular branching</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply;<br/>this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |



Phase 1.2.4 – Neutralize variable substrate profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                          |
| Context class                          | LeanRefactorPhase                                                                                                                                                                                                          |
| Phase documentation tier               | F0                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                    |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | planned-not-active                                                                                                                                                                                                         |
| Preparation status                     | planned-prepared                                                                                                                                                                                                           |
| Phase type                             | refactor                                                                                                                                                                                                                   |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                    |
| Title / state                          | Neutralize variable substrate profile                                                                                                                                                                                      |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                |
| Source-to-CNNA mapping                 | Phase 1.2.4 establishes or neutralizes: substrate: variable branching. Object links are generated from the object register.                                                                                                |
| Dependency-depth audit                 | Depends on 1.2.2; variable branching and weight-policy modules. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                               |
| Phase split decision                   | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                              |
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Foundation/Substrate/Variable.lean.                                                                                                                          |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation  | Keep variable branching proof-carrying and explicit; no class/instance oracle hides the generated data.                                                                                                                    |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                    |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                    |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                   |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                        |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                       |
| Expected executable witness            | not specified yet                                                                                                                                                                                                          |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                       |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                  |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Natural backjump/refactor layer      | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                      | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Verification / evidence              | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Post-implementation plan check       | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Implementation notes                 | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Source materials / references        | <ul style="list-style-type: none"> <li>CNNA/PillarA/Foundation/LevelVariableSubstrate.lean;<br/>Foundation/WeightPolicy.lean;<br/>GeneratedDegreeFormulaAuditSM3db2eR0.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>substrate: variable branching</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.2.5 – Neutralize level and cutoff profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                      |
| Preparation status                    | planned-prepared                                                                                                                                                                                        |
| Phase type                            | refactor                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                 |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title / state                          | Neutralize level and cutoff profile                                                                                                                                                                                                                                   |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                           |
| Source-to-CNNA mapping                 | Phase 1.2.5 establishes or neutralizes: Level / Cutoff. Object links are generated from the object register.                                                                                                                                                          |
| Dependency-depth audit                 | Depends on 1.2.2-1.2.4; level-history modules. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                                           |
| Phase split decision                   | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                                         |
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Foundation/LevelProfile.lean.                                                                                                                                                                           |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                              |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | Level selection must be provenance-carrying and cannot be conflated with later experiment requests.                                                                                                                                                                   |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                               |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                               |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                              |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                  |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                                                           |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                              |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                          |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit; no Lean build evidence has been recorded for this phase.                                                                                                                                             |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                            |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/LevelHistoryIndex.lean; LevelStepWitness.lean; Handoff/Core/Step1MathData.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                           |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Level / Cutoff</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import <code>Legacy.StatusGates</code></li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.2.6 – Neutralize carrier/index profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                                                   |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                                                   |
| Phase documentation tier              | F0                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                  |
| Preparation status                    | planned-prepared                                                                                                                                                                                                                                    |
| Phase type                            | refactor                                                                                                                                                                                                                                            |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                             |
| Title / state                         | Neutralize carrier/index profile                                                                                                                                                                                                                    |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                         |
| Source-to-CNNA mapping                | Phase 1.2.6 establishes or neutralizes: Carrier / index type. Object links are generated from the object register.                                                                                                                                  |
| Dependency-depth audit                | Depends on 1.2.5; recursive level-history carrier modules. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                             |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                       |
| Positive path plan                    | Implement only after dependencies close. Target: CNNA/PillarA/Foundation/CarrierProfile.lean.                                                                                                                                                       |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.            |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                  |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | Expose carrier, Fintype/DecidableEq/cardinality/equivalence data explicitly;<br>avoid high-arity index eliminators.                                                                                                                                                   |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                               |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                            |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                           |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                        |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                          |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                          |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                            |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/RecursiveLevelHistoryCarrier.lean;<br/>LevelHistoryIndex.lean;<br/>GeneratedLevelStepWitnessSM6.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                     |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Carrier / index type</li> </ul>                                                                                                                                                                                                |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li><li>• no <code>@[simp]</code> attributes in the operational path</li><li>• no <code>noncomputable</code> in the operational path</li><li>• no classical helper unless isolated in allowed mirror/non-operational layer</li><li>• no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li><li>• no free formula without proof-carrying entry structure</li><li>• no consumer-side patch when the missing structure belongs earlier</li><li>• no premature obstruction before a serious positive construction attempt</li><li>• no phase-semantic public API names</li><li>• no fragile rewrite chain when a structural constructor/projection proof is available</li><li>• do not delete legacy status enums during quarantine</li><li>• do not let Phase 3+ import <code>Legacy.StatusGates</code></li><li>• do not treat old TeX/PDF files as maintained roadmap sources</li></ul> |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 1.2.7 – Neutralize IdealThread singleton directed-family profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #fff2cc;">W</span>                                                                                                                                                                       |
| Context class                         | <code>LeanRefactorPhase</code>                                                                                                                                                                                          |
| Phase documentation tier              | F1                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor/derivation-preparation phase: record module/declaration migration metrics, import delta, consumer audit, and concrete route witness before implementation.                                                     |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must include route-dependency audit and module placement check before activation.                                                                                                                              |
| Proof rehearsal before implementation | Prove/export singleton carrier and family facts by projections/rfl where possible; deterministic coarsen by existing theorem. Concrete instance: reference/variation thread at finite L.                                |
| Bridge gap status                     | Pending. Any missing lifting theorem or route bridge must create or activate the earlier natural source phase.                                                                                                          |

Phase 1.2.8 – Neutralize CutSpec.full maximal-carrier/full-approximant profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #fff2cc;">W</span>                                                                                                                                              |
| Context class                         | <code>LeanRefactorPhase</code>                                                                                                                                                                 |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |



|                                       |                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor/derivation-preparation phase: record module/declaration migration metrics, import delta, consumer audit, and concrete route witness before implementation.                                                     |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must include route-dependency audit and module placement check before activation.                                                                                                                              |
| Proof rehearsal before implementation | Use full_carrier/rfl for core theorem;<br>add only necessary projection lemmas for later Phase 3.0.                                                                                                                     |
| Bridge gap status                     | Pending. Any missing lifting theorem or route bridge must create or activate the earlier natural source phase.                                                                                                          |

### Phase 1.3 – Regime, partition, and complement Profile neutralization

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                          |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                          |
| Phase documentation tier              | F0                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                    |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                 | planned-not-active                                                                                                                                                                                                         |
| Preparation status                    | planned-prepared                                                                                                                                                                                                           |
| Phase type                            | refactor                                                                                                                                                                                                                   |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                    |
| Title / state                         | Regime, partition, and complement Profile neutralization                                                                                                                                                                   |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                |
| Source-to-CNNA mapping                | Phase 1.3 establishes or neutralizes: PartitionProfile;<br>RegimeProfile;<br>ComplementProfile. Object links are generated from the object register.                                                                       |
| Dependency-depth audit                | Depends on 1.2.6;<br>current partition/regime/complement modules;<br>preliminary object IDs G05-G10. This record exists so the phase can be prepared without reopening Phase 1.1.                                          |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                              |
| Positive path plan                    | Implement only after dependencies close. Target: CNNA/PillarA/Schur/PartitionProfile.lean;<br>CNNA/PillarA/Regime/*.lean;<br>CNNA/PillarA/Regime/Complement/*.lean.                                                        |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review            | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                     |
| Proof rehearsal before implementation | noOuter/hasOuter are A-derived regimes. Complement data is A-internal and must not leak directly to bright Pillar B.                                                                                                       |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Source materials / references          | <ul style="list-style-type: none"> <li>GeneratedBoundaryInteriorPartition.lean;<br/>Sectors/*.lean;<br/>Network/*.lean;<br/>A_to_B.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>PartitionProfile;<br/>RegimeProfile;<br/>ComplementProfile</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply;<br/>this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

Phase 1.3.1 – Neutralize boundary/interface/interior partition profile

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                          |
| Context class                          | LeanRefactorPhase                                                                                                                                                                                                          |
| Phase documentation tier               | F0                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                    |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | planned-not-active                                                                                                                                                                                                         |
| Preparation status                     | planned-prepared                                                                                                                                                                                                           |
| Phase type                             | refactor                                                                                                                                                                                                                   |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                    |
| Title / state                          | Neutralize boundary/interface/interior partition profile                                                                                                                                                                   |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                |
| Source-to-CNNA mapping                 | Phase 1.3.1 establishes or neutralizes: Boundary / interface / interior partition. Object links are generated from the object register.                                                                                    |
| Dependency-depth audit                 | Depends on 1.3;<br>generated boundary/interior partition modules. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                             |
| Phase split decision                   | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                              |
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Schur/PartitionProfile.lean.                                                                                                                                 |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                     |
| Proof rehearsal before implementation  | Partition is a source layer for Schur/DtN, not a consumer-side convention.                                                                                                                                                 |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                    |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                 |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                        |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                    |
| Expected executable witness            | not specified yet                                                                                                                                                                                                          |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                       |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                  |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                             |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                        |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                      | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Verification / evidence              | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Post-implementation plan check       | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Implementation notes                 | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Source materials / references        | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/Smoke/GeneratedBoundaryInteriorPartition.lean;</li> <li>CNNA/PillarA/Finite/CutSpec.lean;</li> <li>CNNA/PillarA/Finite/BoundaryPorts.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Boundary / interface / interior partition</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.3.2 – Neutralize noOuter/hasOuter regime split

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                      |
| Preparation status                    | planned-prepared                                                                                                                                                                                        |
| Phase type                            | refactor                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                 |
| Title / state                         | Neutralize noOuter/hasOuter regime split                                                                                                                                                                |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                           |
| Source-to-CNNA mapping                 | Phase 1.3.2 establishes or neutralizes: Regime: noOuterEnvironment;<br>Regime: hasOuterEnvironment. Object links are generated from the object register.                                                                                                              |
| Dependency-depth audit                 | Depends on 1.3.1;<br>sector/complement/environment modules. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                              |
| Phase split decision                   | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                                         |
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Regime/NoOuter.lean;<br>CNNA/PillarA/Regime/HasOuter.lean.                                                                                                                                              |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                           |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | Keep noOuter closed/root branch separate from hasOuter/open branch. No stochasticity or nonunitarity is genericized.                                                                                                                                                  |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                               |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                            |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                           |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                        |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                          |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                          |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                            |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source materials / references        | <ul style="list-style-type: none"> <li>CNNA/PillarA/Handoff/Outputs/A_to_B.lean;<br/>CNNA/PillarA/Sectors/SectorSplit.lean;<br/>CNNA/PillarA/Sectors/ComplementSectorFamily.lean;<br/>CNNA/PillarA/Network/SectorSysEnv.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Regime: noOuterEnvironment;<br/>Regime: hasOuterEnvironment</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.3.3 – Neutralize complement-family profiles

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f2f2f2;">W</span>                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | <span style="background-color: #fff2cc;">planned-not-active</span>                                                                                                                                      |
| Preparation status                    | <span style="background-color: #fff2cc;">planned-prepared</span>                                                                                                                                        |
| Phase type                            | refactor                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                 |
| Title / state                         | Neutralize complement-family profiles                                                                                                                                                                   |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.             |
| Source-to-CNNA mapping                | Phase 1.3.3 establishes or neutralizes: Complement families: UV tail; IR/parent-path tail; sideways environment. Object links are generated from the object register.                                   |
| Dependency-depth audit                | Depends on 1.3.2; complement-family source modules. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                        |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                           |

|                                        |                                                                                                                                                                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Regime/Complement/UVTail.lean;<br>CNNA/PillarA/Regime/Complement/IRTail.lean;<br>CNNA/PillarA/Regime/Complement/SideEnvironment.lean.                                                                                                  |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                    |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                          |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                           |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                          |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                                        |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                               |
| Proof rehearsal before implementation  | Complement families are derived A-internal profiles;<br>they are not exported as direct bright-sector data.                                                                                                                                                                                          |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                                                              |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                                                           |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                                                          |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                  |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                              |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                    |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                 |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                         |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                            |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                  |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                          |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                         |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                         |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                  |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                       |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                           |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Sectors/UVSpectralSelector.lean;<br/>CNNA/PillarA/Sectors/ComplementSectorFamily.lean;<br/>CNNA/PillarA/Network/SectorChannels.lean;<br/>CNNA/PillarA/Handoff/Outputs/A_to_B.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul> |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Complement families: UV tail;<br/>IR/parent-path tail;<br/>sideways environment</li> </ul>                                                                                                                                                                    |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li><li>• no <code>@[simp]</code> attributes in the operational path</li><li>• no <code>noncomputable</code> in the operational path</li><li>• no classical helper unless isolated in allowed mirror/non-operational layer</li><li>• no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li><li>• no free formula without proof-carrying entry structure</li><li>• no consumer-side patch when the missing structure belongs earlier</li><li>• no premature obstruction before a serious positive construction attempt</li><li>• no phase-semantic public API names</li><li>• no fragile rewrite chain when a structural constructor/projection proof is available</li><li>• do not delete legacy status enums during quarantine</li><li>• do not let Phase 3+ import <code>Legacy.StatusGates</code></li><li>• do not treat old TeX/PDF files as maintained roadmap sources</li></ul> |
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Phase 1.4 – Cayley-Dickson structural path neutral reintegration

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #fff2cc;">W</span>                                                                                                                                                                                                                          |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                                                                          |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                        |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                         |
| Preparation status                    | planned-prepared                                                                                                                                                                                                                                                           |
| Phase type                            | refactor                                                                                                                                                                                                                                                                   |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                                    |
| Title / state                         | Cayley-Dickson structural path neutral reintegration                                                                                                                                                                                                                       |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                                |
| Source-to-CNNA mapping                | Phase 1.4 establishes or neutralizes: CD source; star; normSq; product; alternative/norm laws; HurwitzStop. Object links are generated from the object register. Horn/GA profiles are routed through GA0-GA6 comparison objects only, with no generator or handoff rights. |
| Dependency-depth audit                | Depends on 1.1.5; current Cayley-Dickson structural modules; preliminary object IDs CD0-CD7. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                  |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                                              |
| Positive path plan                    | Implement only after dependencies close. Target: CNNA/PillarA/Structural/CayleyDickson/*.lean.                                                                                                                                                                             |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                             |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation.           |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.                                   |



|                                        |                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                      |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                                                           |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                     |
| Proof rehearsal before implementation  | CD remains an A-internal structural consumer/bridge; no unproved identification of R/C/H/O with arithmetic sectors. V1.25 adds Horn/GA material only as downstream target-gate/scratchpad context; it cannot be used as a source object or release proof.                                                               |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                                                                                 |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                                                                                 |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                                                                                |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                     |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                 |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                       |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                    |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                            |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                               |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                                                                                                             |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                   |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                             |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                            |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit; no Lean build evidence has been recorded for this phase.                                                                                                                                                                                               |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                     |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                          |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                              |
| Source materials / references          | <ul style="list-style-type: none"> <li>Structural/CayleyDickson/*.lean;<br/>Arithmetic/CayleyDickson/*.lean;<br/>Handoff/CayleyDicksonAdapter.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> <li>Horn 0709.2238v1 quaternion/geometric-algebra reference-gate intake (V1.25; comparison-only)</li> </ul> |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>CD source;<br/>star;<br/>normSq;<br/>product;<br/>alternative/norm laws;<br/>HurwitzStop</li> </ul>                                                                                                                                                                              |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>• no <code>@[simp]</code> attributes in the operational path</li> <li>• no <code>noncomputable</code> in the operational path</li> <li>• no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>• no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>• no free formula without proof-carrying entry structure</li> <li>• no consumer-side patch when the missing structure belongs earlier</li> <li>• no premature obstruction before a serious positive construction attempt</li> <li>• no phase-semantic public API names</li> <li>• no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>• do not delete legacy status enums during quarantine</li> <li>• do not let Phase 3+ import <code>Legacy.StatusGates</code></li> <li>• do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 1.4.1 – Neutralize Cayley-Dickson source profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                                                                |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                                                                |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                               |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                   |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                               |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                              |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                               |
| Preparation status                    | planned-prepared                                                                                                                                                                                                                                                 |
| Phase type                            | refactor                                                                                                                                                                                                                                                         |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                          |
| Title / state                         | Neutralize Cayley-Dickson source profile                                                                                                                                                                                                                         |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                      |
| Source-to-CNNA mapping                | Phase 1.4.1 establishes or neutralizes: Cayley-Dickson source. Object links are generated from the object register.                                                                                                                                              |
| Dependency-depth audit                | Depends on 1.4. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                                                                     |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                                    |
| Positive path plan                    | Implement only after dependencies close. Target: CNNA/PillarA/Structural/CayleyDickson/Source.lean.                                                                                                                                                              |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                   |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.                         |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                         |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                    |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                              |
| Proof rehearsal before implementation | Forward the existing CD source through a neutral profile without turning it into an arithmetic generator.                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Structural/CayleyDickson/Source.lean;<br/>CNNA/PillarA/Handoff/CayleyDicksonAdapter.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Cayley-Dickson source</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.4.2 – Neutralize scalar part profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                                                                     |
| Context class                          | LeanRefactorPhase                                                                                                                                                                                                                                                     |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                    |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                        |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                    |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                   |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                    |
| Preparation status                     | planned-prepared                                                                                                                                                                                                                                                      |
| Phase type                             | refactor                                                                                                                                                                                                                                                              |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                               |
| Title / state                          | Neutralize scalar part profile                                                                                                                                                                                                                                        |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                           |
| Source-to-CNNA mapping                 | Phase 1.4.2 establishes or neutralizes: Scalar part. Object links are generated from the object register.                                                                                                                                                             |
| Dependency-depth audit                 | Depends on 1.4.1. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                                                                        |
| Phase split decision                   | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                                         |
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Structural/CayleyDickson/ScalarPart.lean.                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | Scalar part must be downstream of CD source and must not be used as sector ontology.                                                                                                                                                                                  |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                               |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                               |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                              |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                  |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                                                           |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                              |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How implemented                      | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Verification / evidence              | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Post-implementation plan check       | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Implementation notes                 | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Source materials / references        | <ul style="list-style-type: none"> <li>CNNA/PillarA/Structural/CayleyDickson/S6CanonicalScalarPart.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Scalar part</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.4.3 – Neutralize Cayley-Dickson star/conjugation profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                              |
| Context class                         | LeanRefactorPhase                                                                                                                                                                              |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | planned-prepared                                                                                                                                                                               |
| Phase type                            | refactor                                                                                                                                                                                       |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                        |
| Title / state                         | Neutralize Cayley-Dickson star/conjugation profile                                                                                                                                             |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.    |
| Source-to-CNNA mapping                | Phase 1.4.3 establishes or neutralizes: Cayley-Dickson star/conjugation. Object links are generated from the object register.                                                                  |
| Dependency-depth audit                | Depends on 1.4.1-1.4.2. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                           |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                  |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Structural/CayleyDickson/Star.lean.                                                                                                                                                                     |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | Star/conjugation proof is forwarded structurally; no arithmetic identification is introduced.                                                                                                                                                                         |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                               |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                               |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                              |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                                                           |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                          |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit; no Lean build evidence has been recorded for this phase.                                                                                                                                             |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                            |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Structural/CayleyDickson/S6CanonicalStar.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                      |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Cayley-Dickson star/conjugation</li> </ul>                                                                                                                                                                                     |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: top-level <code>global__no__go__guardrails</code> apply; this phase list is local addenda only.</li> <li>• no <code>@[simp]</code> attributes in the operational path</li> <li>• no <code>noncomputable</code> in the operational path</li> <li>• no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>• no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>• no free formula without proof-carrying entry structure</li> <li>• no consumer-side patch when the missing structure belongs earlier</li> <li>• no premature obstruction before a serious positive construction attempt</li> <li>• no phase-semantic public API names</li> <li>• no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>• do not delete legacy status enums during quarantine</li> <li>• do not let Phase 3+ import <code>Legacy.StatusGates</code></li> <li>• do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 1.4.4 – Neutralize CD normSq profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                                                                |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                                                                |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                               |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                   |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                               |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                              |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                               |
| Preparation status                    | planned-prepared                                                                                                                                                                                                                                                 |
| Phase type                            | refactor                                                                                                                                                                                                                                                         |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                          |
| Title / state                         | Neutralize CD normSq profile                                                                                                                                                                                                                                     |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                      |
| Source-to-CNNA mapping                | Phase 1.4.4 establishes or neutralizes: CD normSq. Object links are generated from the object register.                                                                                                                                                          |
| Dependency-depth audit                | Depends on 1.4.3. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                                                                   |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                                    |
| Positive path plan                    | Implement only after dependencies close. Target: CNNA/PillarA/Structural/CayleyDickson/NormSq.lean.                                                                                                                                                              |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                   |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.                         |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                         |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                    |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                              |
| Proof rehearsal before implementation | NormSq remains a CD structural certificate until a later arithmetic bridge is proved.                                                                                                                                                                            |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Structural/CayleyDickson/S6CanonicalNormSqDefined.lean;<br/>S6CanonicalNormSqReduction.lean;<br/>S6CanonicalNormSqBlockSystem.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>CD normSq</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

Phase 1.4.5 – Neutralize CD product profile

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                          |
| Context class                          | LeanRefactorPhase                                                                                                                                                                                                          |
| Phase documentation tier               | F1                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | planned-not-active                                                                                                                                                                                                         |
| Preparation status                     | planned-prepared                                                                                                                                                                                                           |
| Phase type                             | refactor                                                                                                                                                                                                                   |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                    |
| Title / state                          | Neutralize CD product profile                                                                                                                                                                                              |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                |
| Source-to-CNNA mapping                 | Phase 1.4.5 establishes or neutralizes: CD multiplication/product. Object links are generated from the object register.                                                                                                    |
| Dependency-depth audit                 | Depends on 1.4.1;<br>1.4.3. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                   |
| Phase split decision                   | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                              |
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Structural/CayleyDickson/Product.lean.                                                                                                                       |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation  | Product data is re-exposed without changing algebraic semantics.                                                                                                                                                           |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                    |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                    |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                   |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                        |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                    |
| Expected executable witness            | not specified yet                                                                                                                                                                                                          |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                       |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                  |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                        |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                      | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Verification / evidence              | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Post-implementation plan check       | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Implementation notes                 | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Source materials / references        | <ul style="list-style-type: none"> <li>CNNA/PillarA/Structural/CayleyDickson/S6CanonicalProductSurfaceCandidate.lean;<br/>S6CanonicalProductMultiplicationCandidate.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>CD multiplication/product</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.4.6 – Neutralize alternative-law certificate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked

Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                              |
| Context class                         | LeanRefactorPhase                                                                                                                                                                              |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | planned-prepared                                                                                                                                                                               |
| Phase type                            | refactor                                                                                                                                                                                       |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                        |
| Title / state                         | Neutralize alternative-law certificate                                                                                                                                                         |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                           |
| Source-to-CNNA mapping                 | Phase 1.4.6 establishes or neutralizes: Alternative law. Object links are generated from the object register.                                                                                                                                                         |
| Dependency-depth audit                 | Depends on 1.4.5. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                                                                        |
| Phase split decision                   | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                                         |
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Structural/CayleyDickson/AlternativeLaw.lean.                                                                                                                                                           |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | Forward the certificate only if the exact product/star/norm provenance is visible.                                                                                                                                                                                    |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                               |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                               |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                              |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No mathematical bridge expected; missing names/imports must be localized to the natural source/split layer.                                                                                                                                                           |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                          |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit; no Lean build evidence has been recorded for this phase.                                                                                                                                             |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                            |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Structural/CayleyDickson/S6AlternativeLawScaffold.lean; S6FullAlternativeLawBoundary.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                          |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Alternative law</li> </ul>                                                                                                                                                                                                     |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>• no <code>@[simp]</code> attributes in the operational path</li> <li>• no <code>noncomputable</code> in the operational path</li> <li>• no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>• no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>• no free formula without proof-carrying entry structure</li> <li>• no consumer-side patch when the missing structure belongs earlier</li> <li>• no premature obstruction before a serious positive construction attempt</li> <li>• no phase-semantic public API names</li> <li>• no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>• do not delete legacy status enums during quarantine</li> <li>• do not let Phase 3+ import <code>Legacy.StatusGates</code></li> <li>• do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 1.4.7 – Neutralize norm-multiplicativity certificate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                                                                |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                                                                |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                               |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                   |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                               |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                              |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                               |
| Preparation status                    | planned-prepared                                                                                                                                                                                                                                                 |
| Phase type                            | refactor                                                                                                                                                                                                                                                         |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                          |
| Title / state                         | Neutralize norm-multiplicativity certificate                                                                                                                                                                                                                     |
| Preparation workflow                  | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                      |
| Source-to-CNNA mapping                | Phase 1.4.7 establishes or neutralizes: Norm multiplicativity. Object links are generated from the object register.                                                                                                                                              |
| Dependency-depth audit                | Depends on 1.4.4; 1.4.5. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                                                            |
| Phase split decision                  | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                                    |
| Positive path plan                    | Implement only after dependencies close. Target: CNNA/PillarA/Structural/CayleyDickson/NormMultiplicativity.lean.                                                                                                                                                |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                   |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.                         |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                 |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                    |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                              |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof rehearsal before implementation  | Do not use multiplicativity to infer arithmetic sector semantics before CD8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expected mathlib imports               | No new mathlib imports expected;<br>prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00;<br>preserve existing computability boundaries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Bridge gap status                      | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| How implemented                        | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Verification / evidence                | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Implementation notes                   | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Source materials / references          | <ul style="list-style-type: none"> <li>CNNA/PillarA/Structural/CayleyDickson/S6NormSqMultiplicativity.lean;<br/>S6NormMultiplicativityScaffold.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Norm multiplicativity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply;<br/>this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |



Phase 1.4.8 – Neutralize Hurwitz-stop/classification boundary

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                     |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                                                          |
| Context class                          | LeanRefactorPhase                                                                                                                                                                                                                                          |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                        |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                         |
| Preparation status                     | planned-prepared                                                                                                                                                                                                                                           |
| Phase type                             | refactor                                                                                                                                                                                                                                                   |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                    |
| Title / state                          | Neutralize Hurwitz-stop/classification boundary                                                                                                                                                                                                            |
| Preparation workflow                   | Phase-bound preparation from the single YAML master. Collect only source material needed by this phase; register later-phase material as future/missing-source rather than placing it here.                                                                |
| Source-to-CNNA mapping                 | Phase 1.4.8 establishes or neutralizes: Hurwitz stop / R,C,H,O classification. Object links are generated from the object register. Horn/GA profiles are routed through GA0-GA6 comparison objects only, with no generator or handoff rights.              |
| Dependency-depth audit                 | Depends on 1.4.6;<br>1.4.7. This record exists so the phase can be prepared without reopening Phase 1.1.                                                                                                                                                   |
| Phase split decision                   | Prepared as its own subphase because it has a distinct source/proof kind or generation layer.                                                                                                                                                              |
| Positive path plan                     | Implement only after dependencies close. Target: CNNA/PillarA/Structural/CayleyDickson/HurwitzStop.lean.                                                                                                                                                   |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                 |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                   |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                        |
| Proof rehearsal before implementation  | Classification is a structural boundary result, not a license to identify arithmetic sectors without a later bridge. V1.25 adds Horn/GA material only as downstream target-gate/scratchpad context; it cannot be used as a source object or release proof. |
| Expected tactics                       | Prefer exact, rfl, constructor, field projection, small calc blocks, and explicit theorem forwarding. Avoid high-arity index eliminators and broad simp/simpa surfaces.                                                                                    |
| Expected mathlib imports               | No new mathlib imports expected; prefer existing CNNA imports. Verify any new import against mathlib 4.28.0 before use.                                                                                                                                    |
| ExecComplex/computable plan            | No ExecComplex object is introduced unless the phase explicitly touches O00; preserve existing computability boundaries.                                                                                                                                   |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                        |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                    |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                          |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                       |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                  |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bridge gap status                    | No mathematical bridge expected;<br>missing names/imports must be localized to the natural source/split layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Natural backjump/refactor layer      | Backjump to the earliest natural split bucket: inventory, Legacy.StatusGates, CoreSymbols, Legacy.PhaseAliases, or import firewall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                      | No completed implementation record yet. This phase has a prepared scratch-pad entry but must not be marked green until implementation/build evidence exists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Verification / evidence              | Preparation-only evidence from the YAML master and static audit;<br>no Lean build evidence has been recorded for this phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Post-implementation plan check       | After implementation/build evidence: update modified/added modules, regenerated module inventory, linked objects, phase status, and exactly one next active cursor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Implementation notes                 | Use this record to constrain the next implementation pass. Do not broaden the phase during implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Source materials / references        | <ul style="list-style-type: none"> <li>CNNA/PillarA/Structural/CayleyDickson/S6HurwitzStopSeed.lean;<br/>S6CayleyDicksonIdentification.lean</li> <li>v22 corrected Phase-1 dependency hierarchy</li> <li>Horn 0709.2238v1 quaternion/geometric-algebra reference-gate intake (V1.25; comparison-only)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Hurwitz stop / R,C,H,O classification</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>do not delete legacy status enums during quarantine</li> <li>do not let Phase 3+ import Legacy.StatusGates</li> <li>do not treat old TeX/PDF files as maintained roadmap sources</li> </ul> |

### Phase 1.4.9 – Quaternion/geometric-algebra external reference gateway for the Cayley-Dickson structural path

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                        |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                             |
| Context class                         | ReferenceGatePhase                                                                                                                                                            |
| Phase documentation tier              | F2                                                                                                                                                                            |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                            |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                           |
| Implementation status                 | planned-not-active                                                                                                                                                            |

|                                        |                                                                                                                                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                     | planned-prepared                                                                                                                                                                                                          |
| Phase type                             | comparison                                                                                                                                                                                                                |
| Phase-type-specific contract           | Comparison/reference phase: records external target criteria and scratchpad information; it cannot supply CNNA premises or release mathematical phases.                                                                   |
| Title / state                          | Quaternion/geometric-algebra external reference gateway for the Cayley-Dickson structural path                                                                                                                            |
| Preparation workflow                   | External-source intake as comparison/reference gate with derived-only boundary.                                                                                                                                           |
| Source-to-CNNA mapping                 | Maps external quaternion/GA facts to target-gate rows GA0-GA3. The mapping is explicitly comparison-only and cannot choose CNNA algebra, orientation or scalar layer.                                                     |
| Dependency-depth audit                 | Depends on CD neutral reintegration 1.4.1-1.4.8 and later CD-derived structure; does not depend on or unlock active pre-code phase 1.1.1.1.5.1.                                                                           |
| Phase split decision                   | Split into narrow comparison gates so CD, Pillar-B boundary algebra, Dirac/STA and side-comparison targets do not collapse into one ambiguous external-theory phase.                                                      |
| Positive path plan                     | Do not implement proof code yet. Keep target invariants in scratchpad; when predecessor data is closed, add a small neutral comparison API and prove/falsify explicitly.                                                  |
| Module implementation manifest         | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.                                                                                     |
| Semantic placement decision            | Natural placement is comparison/reference-gate layer, not a foundational CNNA derivation layer.                                                                                                                           |
| Reuse and duplication audit            | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.                                                                                  |
| Canonical generator policy             | The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.                                                                                          |
| Module compactness rule                | Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists.                                                                         |
| Refactor regression protocol           | No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.                                                                                |
| Refactor migration metrics             | No Lean migration. Future comparison module metrics: declarations added, imports added, consumer count, and target/equality proofs closed.                                                                                |
| Lean/mathlib version policy            | Baseline remains pinned Lean/mathlib snapshot; live documentation is only a source-discovery hint until verified locally against the pin.                                                                                 |
| ExecComplex boundary policy            | Do not replace ExecComplex or CNNA scalar provenance with external complex quaternions or analytic Complex.                                                                                                               |
| Consumer-contract direction            | Comparison consumers may read CNNA-derived outputs; CNNA derivation phases may not consume the comparison profile as a generator.                                                                                         |
| Governance-overhead review             | Added overhead is justified because sign, unit and semantic ambiguities are high-risk; gates are narrow to avoid broad external-theory creep.                                                                             |
| Proof rehearsal before implementation  | When CD0-CD7 are closed, instantiate a small comparison profile and check target identities against CD-derived operations. Failure becomes a documented non-match or natural-origin backjump, not a patch.                |
| Expected tactics                       | Future Lean work should use explicit structures, field/projection equalities, constructor-based proofs, local case splits and finite matrix entry proofs; avoid broad simp/simpa collapse and opaque native_decide goals. |
| Expected mathlib imports               | Reference candidates only: Mathlib.Algebra.Quaternion, Mathlib.Analysis.Quaternion, Mathlib.LinearAlgebra.CliffordAlgebra.Basic. Verify against the pinned Lean/mathlib snapshot before any implementation.               |
| ExecComplex/computable plan            | No ExecComplex object introduced. If finite numeric checks are later needed, keep them mirror/diagnostic unless an exact CNNA-derived computable path is proved.                                                          |
| Concrete computable instance traversal | Concrete traversal currently consists of source equation/profile extraction and target-object registration; no CNNA runtime traversal yet.                                                                                |
| Computable path non-skeleton status    | Reference skeleton only. It must not be presented as computable CNNA evidence until future exact instances are constructed.                                                                                               |
| Expected executable witness            | For V1.25: generated TSV/TEX/PDF artifacts. Future Lean witness: explicit comparison structure instantiated from CNNA-derived objects.                                                                                    |
| Mathlib complication log               | Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0; implementation must recheck exact names and theorem availability.                                    |
| Noncomputable boundary audit           | No noncomputable Lean code introduced. Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed.                                                                   |
| Mathlib mitigation/refactor layer      | If mathlib Clifford/Quaternion APIs pull noncomputable/classical pressure, keep them in mirror/comparison modules or define a minimal finite target profile instead.                                                      |
| Bridge gap status                      | Bridge intentionally open. The external reference describes target shapes; CNNA must derive any matching structure internally before comparison.                                                                          |
| Natural backjump/refactor layer        | If the comparison needs missing structure, patch only the earliest natural CNNA layer where that structure is generated, not the comparison/consumer phase.                                                               |
| Implementation prompt contract         | Future implementation prompt must state comparison-only status, predecessor objects, no new generator axioms, no external-theory imports unless pinned and isolated, and exact target equalities/failures.                |
| Global no-go inheritance               | All global no-go guardrails apply, especially derived-only, positive-path-first, patch-at-natural-origin and reference-theory-as-target-only rules.                                                                       |
| How implemented                        | Planned reference/target-gate integration only. V1.25 records source reading, object links, proof-dossier placeholders and scratchpad material; no Lean code is implemented.                                              |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / evidence              | YAML generator validation and generated table/PDF outputs after V1.25 update. No Lean build evidence because no Lean source changed.                                                                                                                                                                                                                                                                           |
| Proof strategy                       | Positive path: wait for the natural CNNA predecessor objects, then encode the smallest comparison profile and prove target equality/failure from CNNA data only.                                                                                                                                                                                                                                               |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check       | After any future comparison implementation, record whether the target matched, failed, or required a natural-origin backjump;<br>do not smooth failure into a weakened claim.                                                                                                                                                                                                                                  |
| Artifact packaging rule              | Package YAML, generated tables, generated PDF and report. Do not package external PDFs as generated project truth unless they are user-uploaded source material.                                                                                                                                                                                                                                               |
| Implementation notes                 | External theory is recorded only as comparison material. It may suggest target invariants, sign audits and future checks, but cannot release phases or provide CNNA generators.                                                                                                                                                                                                                                |
| Source materials / references        | <ul style="list-style-type: none"> <li>Horn 0709.2238v1 uploaded PDF</li> <li>mathlib Quaternion documentation</li> <li>mathlib CliffordAlgebra.Basic documentation</li> <li>Wieser/Song Lean GA formalization article</li> <li>Hestenes unified-language reference</li> </ul>                                                                                                                                 |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Hamilton unit identities</li> <li>Pauli/Cartan representation sign choices</li> <li>geometric algebra grade basis</li> <li>pseudoscalar duality</li> <li>pure quaternion as bivector</li> <li>biquaternion eight-basis profile</li> </ul>                                                                                                                               |
| Added modules                        | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no external GA/quaternion theorem used as CNNA source</li> <li>no phase release from uploaded/reference paper</li> <li>no Pauli/Clifford/Dirac object imported into Pillar A or B as generator</li> <li>no free complex scalar layer introduced from biquaternions</li> <li>no Lean import added before natural predecessor closure and version-pinned check</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply.</li> <li>GNG37: quaternion/GA references are comparison-only and have no generator or handoff rights.</li> </ul>                                                                                                                                                                                                   |
| Proposed added modules               | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Structural/CayleyDickson/*.lean</li> <li>Arithmetic/CayleyDickson/*.lean</li> <li>PillarB placeholder modules</li> <li>scripts/generate_tables.py</li> <li>scripts/generate_full_document.py</li> </ul>                                                                                                                                                                 |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>none identified</li> </ul>                                                                                                                                                                                                                                                                                                                                              |

### Phase 1.4.9.1 – Hamilton unit and Pauli/Cartan sign-convention comparison profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                    |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                         |
| Context class                          | ReferenceGatePhase                                                                                                                                                                                                        |
| Phase documentation tier               | F2                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                        |
| Preparation status                     | planned-prepared                                                                                                                                                                                                          |
| Phase type                             | comparison                                                                                                                                                                                                                |
| Phase-type-specific contract           | Comparison/reference phase: records external target criteria and scratchpad information; it cannot supply CNNA premises or release mathematical phases.                                                                   |
| Title / state                          | Hamilton unit and Pauli/Cartan sign-convention comparison profile                                                                                                                                                         |
| Preparation workflow                   | External-source intake as comparison/reference gate with derived-only boundary.                                                                                                                                           |
| Source-to-CNNA mapping                 | Records exactly which identities and signs a CNNA-derived quaternion-like candidate must later satisfy; it supplies no candidate itself.                                                                                  |
| Dependency-depth audit                 | Blocked until CD source/product/star/norm profiles have neutral canonical APIs. It is not allowed to precede those derivations.                                                                                           |
| Phase split decision                   | Split into narrow comparison gates so CD, Pillar-B boundary algebra, Dirac/STA and side-comparison targets do not collapse into one ambiguous external-theory phase.                                                      |
| Positive path plan                     | Do not implement proof code yet. Keep target invariants in scratchpad; when predecessor data is closed, add a small neutral comparison API and prove/falsify explicitly.                                                  |
| Module implementation manifest         | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.                                                                                     |
| Semantic placement decision            | Natural placement is comparison/reference-gate layer, not a foundational CNNA derivation layer.                                                                                                                           |
| Reuse and duplication audit            | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.                                                                                  |
| Canonical generator policy             | The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.                                                                                          |
| Module compactness rule                | Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists.                                                                         |
| Refactor regression protocol           | No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.                                                                                |
| Refactor migration metrics             | No Lean migration. Future comparison module metrics: declarations added, imports added, consumer count, and target/equality proofs closed.                                                                                |
| Lean/mathlib version policy            | Baseline remains pinned Lean/mathlib snapshot; live documentation is only a source-discovery hint until verified locally against the pin.                                                                                 |
| ExecComplex boundary policy            | Do not replace ExecComplex or CNNA scalar provenance with external complex quaternions or analytic Complex.                                                                                                               |
| Consumer-contract direction            | Comparison consumers may read CNNA-derived outputs; CNNA derivation phases may not consume the comparison profile as a generator.                                                                                         |
| Governance-overhead review             | Added overhead is justified because sign, unit and semantic ambiguities are high-risk; gates are narrow to avoid broad external-theory creep.                                                                             |
| Proof rehearsal before implementation  | Future proof should define a CNNA candidate triple from CD data, then prove/falsify square/product/anti-commutation equalities and record sign convention explicitly.                                                     |
| Expected tactics                       | Future Lean work should use explicit structures, field/projection equalities, constructor-based proofs, local case splits and finite matrix entry proofs; avoid broad simp/simpa collapse and opaque native_decide goals. |
| Expected mathlib imports               | Reference candidates only: Mathlib.Algebra.Quaternion, Mathlib.Analysis.Quaternion, Mathlib.LinearAlgebra.CliffordAlgebra.Basic. Verify against the pinned Lean/mathlib snapshot before any implementation.               |
| ExecComplex/computable plan            | No ExecComplex object introduced. If finite numeric checks are later needed, keep them mirror/diagnostic unless an exact CNNA-derived computable path is proved.                                                          |
| Concrete computable instance traversal | Concrete traversal currently consists of source equation/profile extraction and target-object registration; no CNNA runtime traversal yet.                                                                                |
| Computable path non-skeleton status    | Reference skeleton only. It must not be presented as computable CNNA evidence until future exact instances are constructed.                                                                                               |
| Expected executable witness            | For V1.25: generated TSV/TEX/PDF artifacts. Future Lean witness: explicit comparison structure instantiated from CNNA-derived objects.                                                                                    |
| Mathlib complication log               | Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0; implementation must recheck exact names and theorem availability.                                    |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noncomputable boundary audit         | No noncomputable Lean code introduced. Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed.                                                                                                                                                                                                                                                        |
| Mathlib mitigation/refactor layer    | If mathlib Clifford/Quaternion APIs pull noncomputable/classical pressure, keep them in mirror/comparison modules or define a minimal finite target profile instead.                                                                                                                                                                                                                                           |
| Bridge gap status                    | Bridge intentionally open. The external reference describes target shapes; CNNA must derive any matching structure internally before comparison.                                                                                                                                                                                                                                                               |
| Natural backjump/refactor layer      | If the comparison needs missing structure, patch only the earliest natural CNNA layer where that structure is generated, not the comparison/consumer phase.                                                                                                                                                                                                                                                    |
| Implementation prompt contract       | Future implementation prompt must state comparison-only status, predecessor objects, no new generator axioms, no external-theory imports unless pinned and isolated, and exact target equalities/failures.                                                                                                                                                                                                     |
| Global no-go inheritance             | All global no-go guardrails apply, especially derived-only, positive-path-first, patch-at-natural-origin and reference-theory-as-target-only rules.                                                                                                                                                                                                                                                            |
| How implemented                      | Planned reference/target-gate integration only. V1.25 records source reading, object links, proof-dossier placeholders and scratchpad material; no Lean code is implemented.                                                                                                                                                                                                                                   |
| Verification / evidence              | YAML generator validation and generated table/PDF outputs after V1.25 update. No Lean build evidence because no Lean source changed.                                                                                                                                                                                                                                                                           |
| Proof strategy                       | Positive path: wait for the natural CNNA predecessor objects, then encode the smallest comparison profile and prove target equality/failure from CNNA data only.                                                                                                                                                                                                                                               |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check       | After any future comparison implementation, record whether the target matched, failed, or required a natural-origin backjump; do not smooth failure into a weakened claim.                                                                                                                                                                                                                                     |
| Artifact packaging rule              | Package YAML, generated tables, generated PDF and report. Do not package external PDFs as generated project truth unless they are user-uploaded source material.                                                                                                                                                                                                                                               |
| Implementation notes                 | External theory is recorded only as comparison material. It may suggest target invariants, sign audits and future checks, but cannot release phases or provide CNNA generators.                                                                                                                                                                                                                                |
| Source materials / references        | <ul style="list-style-type: none"> <li>Horn quaternion and Pauli-matrix sections</li> <li>mathlib Quaternion API documentation</li> </ul>                                                                                                                                                                                                                                                                      |
| Extracted established theory objects | <ul style="list-style-type: none"> <li><math>i^2=j^2=k^2=ijk=-1</math> target</li> <li>anti-commutation target</li> <li>complex-i vs quaternion-i separation</li> <li>left/right-handed sign convention audit</li> </ul>                                                                                                                                                                                       |
| Added modules                        | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no external GA/quaternion theorem used as CNNA source</li> <li>no phase release from uploaded/reference paper</li> <li>no Pauli/Clifford/Dirac object imported into Pillar A or B as generator</li> <li>no free complex scalar layer introduced from biquaternions</li> <li>no Lean import added before natural predecessor closure and version-pinned check</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply.</li> <li>GNG37: quaternion/GA references are comparison-only and have no generator or handoff rights.</li> </ul>                                                                                                                                                                                                   |
| Proposed added modules               | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Structural/CayleyDickson/*.lean</li> <li>Arithmetic/CayleyDickson/*.lean</li> <li>PillarB placeholder modules</li> <li>scripts/generate_tables.py</li> <li>scripts/generate_full_document.py</li> </ul>                                                                                                                                                                 |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>none identified</li> </ul>                                                                                                                                                                                                                                                                                                                                              |

Phase 1.4.9.2 – Bivector/pseudoscalar duality and complex-quaternion multivector comparison profile

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                    |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                         |
| Context class                          | ReferenceGatePhase                                                                                                                                                                                                        |
| Phase documentation tier               | F2                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                        |
| Preparation status                     | planned-prepared                                                                                                                                                                                                          |
| Phase type                             | comparison                                                                                                                                                                                                                |
| Phase-type-specific contract           | Comparison/reference phase: records external target criteria and scratchpad information; it cannot supply CNNA premises or release mathematical phases.                                                                   |
| Title / state                          | Bivector/pseudoscalar duality and complex-quaternion multivector comparison profile                                                                                                                                       |
| Preparation workflow                   | External-source intake as comparison/reference gate with derived-only boundary.                                                                                                                                           |
| Source-to-CNNA mapping                 | Maps GA duality to target profiles GA2-GA3. CNNA must derive any orientation, volume element, dimension or scalar-layer candidate first.                                                                                  |
| Dependency-depth audit                 | Depends on 1.4.9.1 and future CNNA-derived orientation/volume candidates. It must not create a complex layer for CD9.                                                                                                     |
| Phase split decision                   | Split into narrow comparison gates so CD, Pillar-B boundary algebra, Dirac/STA and side-comparison targets do not collapse into one ambiguous external-theory phase.                                                      |
| Positive path plan                     | Do not implement proof code yet. Keep target invariants in scratchpad; when predecessor data is closed, add a small neutral comparison API and prove/falsify explicitly.                                                  |
| Module implementation manifest         | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.                                                                                     |
| Semantic placement decision            | Natural placement is comparison/reference-gate layer, not a foundational CNNA derivation layer.                                                                                                                           |
| Reuse and duplication audit            | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.                                                                                  |
| Canonical generator policy             | The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.                                                                                          |
| Module compactness rule                | Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists.                                                                         |
| Refactor regression protocol           | No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.                                                                                |
| Refactor migration metrics             | No Lean migration. Future comparison module metrics: declarations added, imports added, consumer count, and target/equality proofs closed.                                                                                |
| Lean/mathlib version policy            | Baseline remains pinned Lean/mathlib snapshot; live documentation is only a source-discovery hint until verified locally against the pin.                                                                                 |
| ExecComplex boundary policy            | Do not replace ExecComplex or CNNA scalar provenance with external complex quaternions or analytic Complex.                                                                                                               |
| Consumer-contract direction            | Comparison consumers may read CNNA-derived outputs; CNNA derivation phases may not consume the comparison profile as a generator.                                                                                         |
| Governance-overhead review             | Added overhead is justified because sign, unit and semantic ambiguities are high-risk; gates are narrow to avoid broad external-theory creep.                                                                             |
| Proof rehearsal before implementation  | Future proof should start from a CNNA-derived pseudoscalar/volume candidate, then check square, commutation and dual-pairing identities. For biquaternions, compare only after CD9 noncircularity is settled.             |
| Expected tactics                       | Future Lean work should use explicit structures, field/projection equalities, constructor-based proofs, local case splits and finite matrix entry proofs; avoid broad simp/simpa collapse and opaque native_decide goals. |
| Expected mathlib imports               | Reference candidates only: Mathlib.Algebra.Quaternion, Mathlib.Analysis.Quaternion, Mathlib.LinearAlgebra.CliffordAlgebra.Basic. Verify against the pinned Lean/mathlib snapshot before any implementation.               |
| ExecComplex/computable plan            | No ExecComplex object introduced. If finite numeric checks are later needed, keep them mirror/diagnostic unless an exact CNNA-derived computable path is proved.                                                          |
| Concrete computable instance traversal | Concrete traversal currently consists of source equation/profile extraction and target-object registration; no CNNA runtime traversal yet.                                                                                |
| Computable path non-skeleton status    | Reference skeleton only. It must not be presented as computable CNNA evidence until future exact instances are constructed.                                                                                               |
| Expected executable witness            | For V1.25: generated TSV/TEX/PDF artifacts. Future Lean witness: explicit comparison structure instantiated from CNNA-derived objects.                                                                                    |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathlib complication log             | Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0; implementation must recheck exact names and theorem availability.                                                                                                                                                                                                                         |
| Noncomputable boundary audit         | No noncomputable Lean code introduced. Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed.                                                                                                                                                                                                                                                        |
| Mathlib mitigation/refactor layer    | If mathlib Clifford/Quaternion APIs pull noncomputable/classical pressure, keep them in mirror/comparison modules or define a minimal finite target profile instead.                                                                                                                                                                                                                                           |
| Bridge gap status                    | Bridge intentionally open. The external reference describes target shapes; CNNA must derive any matching structure internally before comparison.                                                                                                                                                                                                                                                               |
| Natural backjump/refactor layer      | If the comparison needs missing structure, patch only the earliest natural CNNA layer where that structure is generated, not the comparison/consumer phase.                                                                                                                                                                                                                                                    |
| Implementation prompt contract       | Future implementation prompt must state comparison-only status, predecessor objects, no new generator axioms, no external-theory imports unless pinned and isolated, and exact target equalities/failures.                                                                                                                                                                                                     |
| Global no-go inheritance             | All global no-go guardrails apply, especially derived-only, positive-path-first, patch-at-natural-origin and reference-theory-as-target-only rules.                                                                                                                                                                                                                                                            |
| How implemented                      | Planned reference/target-gate integration only. V1.25 records source reading, object links, proof-dossier placeholders and scratchpad material; no Lean code is implemented.                                                                                                                                                                                                                                   |
| Verification / evidence              | YAML generator validation and generated table/PDF outputs after V1.25 update. No Lean build evidence because no Lean source changed.                                                                                                                                                                                                                                                                           |
| Proof strategy                       | Positive path: wait for the natural CNNA predecessor objects, then encode the smallest comparison profile and prove target equality/failure from CNNA data only.                                                                                                                                                                                                                                               |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check       | After any future comparison implementation, record whether the target matched, failed, or required a natural-origin backjump; do not smooth failure into a weakened claim.                                                                                                                                                                                                                                     |
| Artifact packaging rule              | Package YAML, generated tables, generated PDF and report. Do not package external PDFs as generated project truth unless they are user-uploaded source material.                                                                                                                                                                                                                                               |
| Implementation notes                 | External theory is recorded only as comparison material. It may suggest target invariants, sign audits and future checks, but cannot release phases or provide CNNA generators.                                                                                                                                                                                                                                |
| Source materials / references        | <ul style="list-style-type: none"> <li>Horn duality and complex-quaternion sections</li> <li>mathlib CliffordAlgebra.Basic documentation</li> <li>Hestenes/Sobczyk GA references</li> <li>Wieser/Song Lean GA formalization</li> </ul>                                                                                                                                                                         |
| Extracted established theory objects | <ul style="list-style-type: none"> <li><math>I = \sigma_x \sigma_y \sigma_z</math> target</li> <li><math>I^2 = -1</math> target</li> <li>vector-bivector duality target</li> <li>scalar/vector/bivector/pseudoscalar eight-basis bookkeeping</li> </ul>                                                                                                                                                        |
| Added modules                        | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no external GA/quaternion theorem used as CNNA source</li> <li>no phase release from uploaded/reference paper</li> <li>no Pauli/Clifford/Dirac object imported into Pillar A or B as generator</li> <li>no free complex scalar layer introduced from biquaternions</li> <li>no Lean import added before natural predecessor closure and version-pinned check</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply.</li> <li>GNG37: quaternion/GA references are comparison-only and have no generator or handoff rights.</li> </ul>                                                                                                                                                                                                   |
| Proposed added modules               | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |



|                                      |                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Structural/CayleyDickson/*.lean</li> <li>Arithmetic/CayleyDickson/*.lean</li> <li>PillarB placeholder modules</li> <li>scripts/generate_tables.py</li> <li>scripts/generate_full_document.py</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>none identified</li> </ul>                                                                                                                                                                              |

## Phase 2 – Canonical object register and code-module map after refactor

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

## Phase 2.1 – Object DAG and single-definition rule

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |

|                                       |                                                                                       |
|---------------------------------------|---------------------------------------------------------------------------------------|
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase. |

## Phase 2.2 – Record current derivation sites from the code

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

## Phase 2.3 – Access and handoff governance

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 2.3.1 – Derive/register object group: A input handoff B->A, A input handoff C->A, A input handoff D->A, A input handoff E->A

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 2.4 – Split rule for phases and subphases

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 3 – Schur/DtN core in neutral derivation order

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |

|                                       |                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 3.0 – Boundary-source route separation: full-cut matrix geometry versus IdealThread scalar collapse

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f2f2f2;">W</span>                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F1                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor/derivation-preparation phase: record module/declaration migration metrics, import delta, consumer audit, and concrete route witness before implementation.                                                     |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must include route-dependency audit and module placement check before activation.                                                                                                                              |
| Proof rehearsal before implementation | Use S07 singleton and no-go proofs to reject non-singleton claims from IdealThread; use S08 full-carrier facts to authorize full boundary route. Name concrete finite full-cut and thread examples.                     |
| Bridge gap status                     | Pending. Any missing lifting theorem or route bridge must create or activate the earlier natural source phase.                                                                                                          |

### Phase 3.1 – Dirichlet Laplacian and partition

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f2f2f2;">W</span>                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                            |

|                                       |                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 3.2 – InteriorBlock

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 3.3 – Interior inverse candidate and trace/fold

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |

|                   |                                                                                       |
|-------------------|---------------------------------------------------------------------------------------|
| Bridge gap status | Pending. Any missing bridge must create or activate the earlier natural source phase. |
|-------------------|---------------------------------------------------------------------------------------|

Phase 3.4 – Two-sided inverse certificate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

Phase 3.5 – InteriorEliminationData

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

Phase 3.6 – DtN BoundaryOperator

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.



| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 3.7 – GeneralizedDtN/MultiSchur

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F0                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 4 – Operational ExecComplex arithmetic path from BoundaryOperator

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                              |
| Context class                         | LeanRefactorPhase                                                                                                                                                                              |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |



|                                       |                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 4.0 – Thread-recursion arithmetic-shadow audit before operational arithmetic wrappers

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                                                                                                                                                      |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                             |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                                              |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                  |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                              |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                             |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                              |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                    |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                      |
| Phase-type-specific contract          | Refactor/derivation-preparation phase: record module/declaration migration metrics, import delta, consumer audit, and concrete route witness before implementation.                                                                                                                                                             |
| Title / state                         | Thread-recursion arithmetic-shadow audit before operational arithmetic wrappers                                                                                                                                                                                                                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                         |
| Source-to-CNNA mapping                | Not prepared yet;<br>when activated, map only code-backed route facts and mark analogies comparison-only.                                                                                                                                                                                                                       |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must include route-dependency audit and module placement check before activation.                                                                                   |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                               |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Trace scalar recurrence provenance before interpreting arithmetic outputs. Keep D=-3/-4/CM claims on operational discriminant/number-theory route, not on analogy. |
| Module implementation manifest        | Candidate: CNNA/PillarA/Operational/ThreadRecursionAudit.lean after checking if existing Schur modules can host the audit compactly.                                                                                                                                                                                            |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                               |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                     |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                      |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                     |
| Consumer-contract direction           | Route direction must be explicit before implementation: thread/full profiles feed Schur/arithmetic audits;<br>analogy/diagnostics never feed generator code.                                                                                                                                                                    |
| Governance-overhead review            | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                          |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Follow recurrenceLedger.recursiveSource fields to boundaryMatrixAt or generated source;<br>prove route classification before arithmetic promotion. Concrete witness required for a finite level/history.                                               |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                 |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                 |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                 |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                               |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                           |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                 |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                              |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                      |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                         |
| Bridge gap status                      | Pending. Any missing lifting theorem or route bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                    |
| Natural backjump/refactor layer        | If a needed route structure is missing, backjump to S07/S08/BR06 rather than patching consumer modules.                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                             |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                       |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                    |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                                                                    |
| Source materials / references          | <ul style="list-style-type: none"> <li>RecursiveSchurSource</li> <li>MobiusTransfer</li> <li>CharpolyExec</li> <li>Operational discriminant SM9 chain</li> </ul>                                                                                                                                                                                                                                  |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>sigma recurrence source</li> <li>Mobius transfer parameters</li> <li>discriminant output route</li> </ul>                                                                                                                                                                                                                                                  |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no string-like analogy as CNNA-generative input</li> <li>no full-boundary matrix claim from IdealThread singleton route without a lifting theorem</li> <li>no duplicate implementation of existing ToC/CutSpec/Schur recurrence code</li> </ul> |

### Phase 4.0.1 – General finite computation test matrix before operational arithmetic wrappers

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                    | Record                                            |
|--------------------------|---------------------------------------------------|
| Phase status             | <span style="background-color: #ffcdd2;">R</span> |
| Context class            | LeanDerivationPhase                               |
| Phase documentation tier | F2                                                |

|                                        |                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                                           |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                                      |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                                     |
| Implementation status                  | planned                                                                                                                                                                                                                                                                                                                                                                 |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                                            |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                              |
| Phase-type-specific contract           | Derivation/test phase: proof-carrying computation rows required before object promotion.                                                                                                                                                                                                                                                                                |
| Title / state                          | General finite computation test matrix before operational arithmetic wrappers                                                                                                                                                                                                                                                                                           |
| Preparation workflow                   | General finite computation-test workflow after Phase 4.0 audit.                                                                                                                                                                                                                                                                                                         |
| Source-to-CNNA mapping                 | Generic rows map to O07. Eisenstein/Gauss rows map later to L09/M08/BR10 only as sentinels and comparison checks.                                                                                                                                                                                                                                                       |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Not active until 4.0 closes. It may consume Phase-3 boundary profiles and generator outputs but must not reconstruct earlier Schur/DtN logic.                                                                        |
| Phase split decision                   | Inserted between arithmetic-shadow audit and operational wrappers so the first calculations are tested generally before special comparison interpretation.                                                                                                                                                                                                              |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Start with small finite code-defined cases across L_max and branch-mode, then enlarge coverage. Keep sentinel rows separate from generic coverage.                                                         |
| Module implementation manifest         | Candidate: CNNA/PillarA/Operational/ThreadRecursionAudit.lean after checking if existing Schur modules can host the audit compactly.                                                                                                                                                                                                                                    |
| Refactor regression protocol           | The test harness later serves as regression sentinel for wrapper migrations.                                                                                                                                                                                                                                                                                            |
| Refactor migration metrics             | not applicable until implementation.                                                                                                                                                                                                                                                                                                                                    |
| Lean/mathlib version policy            | Use frozen baseline unless a phase-boundary upgrade gate is opened.                                                                                                                                                                                                                                                                                                     |
| ExecComplex boundary policy            | EXEC-2 binding.                                                                                                                                                                                                                                                                                                                                                         |
| Consumer-contract direction            | Consumes generator/Phase-3 profiles;<br>does not define them.                                                                                                                                                                                                                                                                                                           |
| Governance-overhead review             | Review after first implementation whether generic test obligation caught issues.                                                                                                                                                                                                                                                                                        |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Each row should state input provenance, generated finite carrier/matrix/profile, executable output, and verification status. A generic-row failure blocks;<br>a sentinel mismatch becomes a comparison/failure ledger entry. |
| Expected tactics                       | Explicit constructors, finite records, direct projections, small arithmetic certificates;<br>avoid fragile rewrite-only proofs.                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Likely existing finite/matrix/arithmetic imports;<br>any polynomial/matrix API noncomputability must be logged.                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Only rational/integer/finite fields operational. Non-rational geometry fields are mirror-only and optional.                                                                                                                                                                                                                                                             |
| Concrete computable instance traversal | Required: at least one generic finite row family traverses the actual computation path;<br>special sentinels cannot be the only traversal.                                                                                                                                                                                                                              |
| Computable path non-skeleton status    | planned: non-skeleton status requires generated finite input plus executable output record.                                                                                                                                                                                                                                                                             |
| Expected executable witness            | O07 row output record with finite/rational/integer fields and proof/status evidence.                                                                                                                                                                                                                                                                                    |
| Mathlib complication log               | to be filled before implementation;<br>include noncomputable risks for matrix/polynomial APIs.                                                                                                                                                                                                                                                                          |
| Noncomputable boundary audit           | No operational row may require noncomputable data. Mirror comparison columns are isolated.                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | Use explicit finite constructors;<br>if mathlib API is noncomputable, introduce an operational substitute or mirror-only column.                                                                                                                                                                                                                                        |
| Bridge gap status                      | Provides the general testing bridge missing from v32 before the specialized lattice test matrix.                                                                                                                                                                                                                                                                        |
| Natural backjump/refactor layer        | If row generation needs duplicate tree/generator code, refactor the canonical generator rather than adding local test logic.                                                                                                                                                                                                                                            |
| Implementation prompt contract         | Specify exact module count/name, input row schema, generic coverage, sentinel rows, and expected executable witness before code is written.                                                                                                                                                                                                                             |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                             |
| How implemented                        | Planned only. The module manifest must decide whether this is a new compact reusable test module or an extension of the canonical smoke-test generator.                                                                                                                                                                                                                 |
| Verification / evidence                | pending until Phase 4.0/3.6 dependencies and module manifest close.                                                                                                                                                                                                                                                                                                     |

|                                      |                                                                                                                                                                                                                                          |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof strategy                       | Use proof-carrying row records. First generic finite rows generated by code-defined parameters; then optional sentinel rows. Avoid Matrix.inv and noncomputable APIs in operational row generation.                                      |
| Build result                         | not run                                                                                                                                                                                                                                  |
| Post-implementation plan check       | Update O07 dossier and decide whether L09/L10 can consume the generic harness.                                                                                                                                                           |
| Artifact packaging rule              | Return full zip;<br>Lean build must be run/verified separately unless executed in current environment.                                                                                                                                   |
| Implementation notes                 | This phase prevents early computations from being validated only on Eisenstein/Gauss or other hand-picked rows.                                                                                                                          |
| Source materials / references        | <ul style="list-style-type: none"> <li>LEG29 general early finite test-matrix contract</li> <li>canonical smoke-test generator policy</li> <li>BR07 thread-recursion arithmetic-shadow audit</li> </ul>                                  |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>finite generated input row</li> <li>branch-mode tag static/dynamic</li> <li>L_max row</li> <li>operational output witness</li> <li>sentinel comparison row</li> </ul>                             |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>No special-case-only row set.</li> <li>No duplicate generator implementation.</li> <li>No noncomputable operational witness.</li> <li>No sentinel comparison as proof of general path.</li> </ul> |

### Phase 4.0.2 – Regime-typed finite computation test matrix: noOuter-root sentinel versus HasOuter projection rows

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                 |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                            |
| Implementation status                 | planned                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                   |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                     |
| Phase-type-specific contract          | test phase must emit row records and pass/fail criteria                                                                                                                                                                                                                                        |
| Title / state                         | Regime-typed finite computation test matrix                                                                                                                                                                                                                                                    |
| Preparation workflow                  | regime-typed general early finite computation test workflow                                                                                                                                                                                                                                    |
| Source-to-CNNA mapping                | Eisenstein/Gauss map to closed-root sentinel candidates only after noOuter/empty-set REAL-root route exists. Other Heegner or CM targets map to complement-projection/comparison rows unless a noOuter derivation is separately proven.                                                        |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. No early mathematics consumes the classification as an input. It is a test/reporting schema before implementation and a proof target later. |
| Phase split decision                  | split from v33 O07 because generic-vs-sentinel is not enough;<br>regime typing is a separate semantic axis.                                                                                                                                                                                    |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. First run generic finite rows;<br>then annotate rows by derived regime;<br>then interpret Eisenstein/Gauss as noOuter-root sentinels and all other Heegner/CM rows as projections/comparison until proven otherwise. |
| Module implementation manifest         | pending Phase 1.1.1.1.1                                                                                                                                                                                                                                                                                                                                                           |
| Semantic placement decision            | test/schema layer, not generator layer                                                                                                                                                                                                                                                                                                                                            |
| Reuse and duplication audit            | must reuse O07/canonical smoke-test generator;<br>no parallel Heegner test generator                                                                                                                                                                                                                                                                                              |
| Canonical generator policy             | consume the one canonical generator only                                                                                                                                                                                                                                                                                                                                          |
| Module compactness rule                | do not create a new module unless existing test/profile module cannot own the semantics                                                                                                                                                                                                                                                                                           |
| Refactor regression protocol           | no code movement in this gate;<br>future test implementation must preserve full local build green status                                                                                                                                                                                                                                                                          |
| Refactor migration metrics             | not applicable at planning gate                                                                                                                                                                                                                                                                                                                                                   |
| Lean/mathlib version policy            | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                                                                                                                                                                                                      |
| ExecComplex boundary policy            | EXEC-2 enforced                                                                                                                                                                                                                                                                                                                                                                   |
| Consumer-contract direction            | test matrix consumes regime/generator outputs;<br>regime/generator definitions do not consume tests                                                                                                                                                                                                                                                                               |
| Governance-overhead review             | review after Phase 1.1.2 whether this lock prevented false closed-system classification.                                                                                                                                                                                                                                                                                          |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Prove/record row provenance: generator output -> finite witness -> regime tag -> optional comparison target. A row may not change the regime tag based on h(D), j, or external Heegner membership.                                     |
| Expected tactics                       | structural constructors/projections;<br>explicit row records;<br>no fragile rewrites;<br>no inverse oracle.                                                                                                                                                                                                                                                                       |
| Expected mathlib imports               | none at planning gate;<br>future number-theory phases may require mathlib version review.                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Operational fields remain finite/rational/integer. tau, sqrt(D), analytic j and general CM fields stay mirror-only under EXEC-2.                                                                                                                                                                                                                                                  |
| Concrete computable instance traversal | required later: every row must contain actual finite generator traversal evidence or remain unclosed                                                                                                                                                                                                                                                                              |
| Computable path non-skeleton status    | schema lock only now;<br>future rows must provide executable witnesses                                                                                                                                                                                                                                                                                                            |
| Expected executable witness            | finite row witness with branch mode, L_max, regime tag, rational/integer outputs, optional comparison target                                                                                                                                                                                                                                                                      |
| Mathlib complication log               | record any class number/CM/j mathlib dependency or noncomputable boundary explicitly                                                                                                                                                                                                                                                                                              |
| Noncomputable boundary audit           | Heegner/class-number and analytic j data are comparison/mirror until operational invariant is derived                                                                                                                                                                                                                                                                             |
| Mathlib mitigation/refactor layer      | use a mirror layer or later version-policy gate if mathlib forces noncomputable definitions                                                                                                                                                                                                                                                                                       |
| Bridge gap status                      | open until noOuter/HasOuter regime split and O07 implementation exist                                                                                                                                                                                                                                                                                                             |
| Natural backjump/refactor layer        | if a row needs regime data not yet produced, backjump to regime/source layer rather than patching the test consumer.                                                                                                                                                                                                                                                              |
| Implementation prompt contract         | Before Lean code, module manifest must state exact module placement and prove no duplicate generator/test harness.                                                                                                                                                                                                                                                                |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                       |
| How implemented                        | Future test phase extending O07 with derived regime tags and closed-root/projection separation.                                                                                                                                                                                                                                                                                   |
| Verification / evidence                | YAML/planning-level validation and generated views only;<br>no Lean build.                                                                                                                                                                                                                                                                                                        |
| Proof strategy                         | planning-level dependency/order lock;<br>implementation proof obligations deferred to target phase                                                                                                                                                                                                                                                                                |
| Build result                           | No Lean build executed or claimed.                                                                                                                                                                                                                                                                                                                                                |
| Post-implementation plan check         | Check whether row classification forced an actual implementation decision;<br>update governance overhead review.                                                                                                                                                                                                                                                                  |
| Artifact packaging rule                | Return complete zip;<br>Lean build local unless actually executed here.                                                                                                                                                                                                                                                                                                           |

|                                      |                                                                                                                                                                                                                                       |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation notes                 | Future test phase extending O07 with derived regime tags and closed-root/projection separation.                                                                                                                                       |
| Source materials / references        | <ul style="list-style-type: none"> <li>CNNA noOuter/HasOuter regime split; EXEC-2; general early finite test-matrix contract; post #97 heuristic; class-number-one/Heegner comparison list as external comparison only</li> </ul>     |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Regime tag; noOuter root sentinel; HasOuter/complement-projection row; Heegner class-number-one comparison target; IDEAL_infty projection label</li> </ul>                                     |
| Implemented modules                  | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                                                |
| Added modules                        | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                            |
| Modified modules                     | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml only</li> </ul>                                                                                                                                               |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                                                                                  |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>Local additions: no Heegner-list membership as noOuter evidence; no closed/unitary interpretation for complement-projection rows; no Eisenstein/Gauss-only evidence for generality.</li> </ul> |
| Proposed added modules               | <ul style="list-style-type: none"> <li>none at decision gate; future target likely existing O07 test/profile surface</li> </ul>                                                                                                       |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml; generated views</li> </ul>                                                                                                                                   |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>canonical smoke-test generator; regime split modules after Phase 1.3.2; O07 test matrix</li> </ul>                                                                                             |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>Heegner row schema; CM sentinel profile; closed-system sentinel tags</li> </ul>                                                                                                                |

### Phase 4.0.3 – Scalar-layer computation test rows: ExecComplex vs exact quadratic extension boundary

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
 Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | not_started                                                                                                                                                                                    |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                   |
| Phase type                            | derivation                                                                                                                                                                                     |



|                                        |                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase-type-specific contract           | derivation scalar architecture obligations                                                                                                                                                                                                                                                                         |
| Title / state                          | Scalar-layer computation test rows: ExecComplex vs exact quadratic extension boundary                                                                                                                                                                                                                              |
| Preparation workflow                   | computable scalar architecture workflow                                                                                                                                                                                                                                                                            |
| Source-to-CNNA mapping                 | Finite row schema maps outputs into rational/integer invariants for operational closure.                                                                                                                                                                                                                           |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Scalar architecture decisions must precede code implementation. Exact quadratic or CD scalar routes may not feed earlier generator definitions.                 |
| Phase split decision                   | split out because ExecComplex limits affect Phase 1.2, 4, 5, 7, 8 and CD bridges differently.                                                                                                                                                                                                                      |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Keep current QxQ path first; add exact quadratic pilot only if needed by finite tests and no duplicate module exists; keep analytic Complex mirror-only. |
| Module implementation manifest         | pending Phase 1.1.1.1.1                                                                                                                                                                                                                                                                                            |
| Semantic placement decision            | scalar-foundation / bridge governance, not generator semantics                                                                                                                                                                                                                                                     |
| Reuse and duplication audit            | must check existing ExecComplex, ExecComplexBridge, Schur.CharpolyC, ArithmeticCDBridge, and mathlib QuadraticAlgebra before proposing new module                                                                                                                                                                  |
| Canonical generator policy             | scalar layer must consume canonical generator outputs only; cannot define generator semantics                                                                                                                                                                                                                      |
| Module compactness rule                | do not create ExactQuadraticComplex unless existing foundation/bridge modules cannot carry the semantics cleanly                                                                                                                                                                                                   |
| Refactor regression protocol           | incremental wrapper mode; full local build green after each migration step                                                                                                                                                                                                                                         |
| Refactor migration metrics             | track declarations migrated, imports changed, executable witnesses added                                                                                                                                                                                                                                           |
| Lean/mathlib version policy            | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                                                                                                                                       |
| ExecComplex boundary policy            | QxQ rational floor; exact quadratic coordinates require O09 plus O10/BR14 proof-carrying D-source; no free sqrt(D).                                                                                                                                                                                                |
| Consumer-contract direction            | scalar layer consumes generator/arithmetic source data; generator definitions do not consume scalar comparison/mirror data                                                                                                                                                                                         |
| Governance-overhead review             | review after first scalar/module manifest whether this lock forced a concrete decision.                                                                                                                                                                                                                            |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Run generic rows first; sentinel rows second; compare QxQ-only closure with O09 exact quadratic closure.                                                                |
| Expected tactics                       | explicit structures, constructor/projection lemmas, ring-normal-form only for coordinate arithmetic, no Matrix.inv and no consumer-side patch.                                                                                                                                                                     |
| Expected mathlib imports               | Mathlib.Algebra.QuadraticAlgebra only if pilot approved; Mathlib.Data.Complex.Basic remains bridge/mirror-only; NumberField modules require version-policy gate.                                                                                                                                                   |
| ExecComplex/computable plan            | Operational outputs must remain executable and project back to allowed invariants.                                                                                                                                                                                                                                 |
| Concrete computable instance traversal | required later: QxQ baseline row, Q(i) row, Q(sqrt(-3)) row, and at least one generic D row if O09 is implemented                                                                                                                                                                                                  |
| Computable path non-skeleton status    | must be shown by concrete row witnesses                                                                                                                                                                                                                                                                            |
| Expected executable witness            | finite row with rational ExecComplex output plus optional exact quadratic coordinate and rational/integer invariant projection                                                                                                                                                                                     |
| Mathlib complication log               | Record Complex/Real noncomputable boundary, QuadraticAlgebra instance diamond warning, NumberField formal/non-executable dependency, and any required version-gate.                                                                                                                                                |
| Noncomputable boundary audit           | General C/R and algebraic closure remain mirror/formal. Exact quadratic candidate must avoid noncomputable definitions on operational path. Exact quadratic use must avoid analytic sqrt/Complex and cite D-source ledger.                                                                                         |
| Mathlib mitigation/refactor layer      | Use project-owned explicit Q-coordinate structure or narrowly wrapped QuadraticAlgebra; isolate formal mathlib NumberField in mirror/proof layer.                                                                                                                                                                  |
| Bridge gap status                      | O09/BR13 open.                                                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | If exact quadratic data are needed, backjump to scalar-foundation layer rather than patching arithmetic/lattice consumers.                                                                                                                                                                                         |
| Implementation prompt contract         | Before Lean code, module manifest must state exact module placement, imported mathlib modules, and why no existing module owns the semantics.                                                                                                                                                                      |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                           |
| How implemented                        | Not implemented.                                                                                                                                                                                                                                                                                                   |



|                                      |                                                                                                                                                                                                                                               |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / evidence              | Generated-table validation pending/run after patch;<br>no Lean build.                                                                                                                                                                         |
| Proof strategy                       | Run generic rows first;<br>sentinel rows second;<br>compare QxQ-only closure with O09 exact quadratic closure.                                                                                                                                |
| Build result                         | No Lean build executed or claimed.                                                                                                                                                                                                            |
| Post-implementation plan check       | Check whether the scalar architecture lock prevented analytic/noncomputable leakage or duplicate scalar implementation.                                                                                                                       |
| Artifact packaging rule              | Return complete zip;<br>Lean build local unless actually executed here.                                                                                                                                                                       |
| Implementation notes                 | Future finite tests deciding whether quadratic coordinates are operationally required or only comparison columns.                                                                                                                             |
| Source materials / references        | <ul style="list-style-type: none"> <li>O07/O08, O09, BR12.</li> </ul>                                                                                                                                                                         |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Test rows over QxQ and exact quadratic candidates.</li> </ul>                                                                                                                                          |
| Implemented modules                  | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                                                        |
| Added modules                        | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                                    |
| Modified modules                     | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                                    |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                                                                                          |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>No sentinel-only closure;<br/>no mirror-only tau/j used as operational witness.</li> </ul>                                                                                                             |
| Proposed added modules               | <ul style="list-style-type: none"> <li>none at decision gate;<br/>possible CNNA/PillarA/Foundation/ExactQuadraticComplex.lean only after manifest</li> </ul>                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>pending manifest</li> </ul>                                                                                                                                                                            |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Foundation/ExecComplex.lean;<br/>Foundation/ExecComplexBridge.lean;<br/>Arithmetic/Schur/CharpolyC.lean;<br/>Arithmetic/CayleyDickson/ArithmeticCDBridge.lean;<br/>mathlib QuadraticAlgebra</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>alternate complex scalar, algebraic complex, CD complex, mirror Complex, exact quadratic norm/trace/conjugation</li> </ul>                                                                             |

### Phase 4.1 – ExecBoundaryMatrix

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                              |
| Context class                         | LeanRefactorPhase                                                                                                                                                                              |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | not-prepared                                                                                                                                                                                   |

|                                       |                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 4.2 – Coefficient recurrence

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F1                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 4.3 – Polynomial profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F1                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |

|                   |                                                                                       |
|-------------------|---------------------------------------------------------------------------------------|
| Bridge gap status | Pending. Any missing bridge must create or activate the earlier natural source phase. |
|-------------------|---------------------------------------------------------------------------------------|

Phase 4.4 – Quadratic window

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F1                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

Phase 4.5 – Operational discriminant D\_op

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | LeanRefactorPhase                                                                                                                                                                                                       |
| Phase documentation tier              | F1                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | refactor                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

Phase 4.5.1 – Proof-carrying discriminant-to-exact-quadratic instantiation gate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | R                                                                                                                                                                                                                                                                                                                                    |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                  |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                                                                                   |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                       |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                   |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                  |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                   |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                         |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                           |
| Phase-type-specific contract           | derivation phase: obligations recorded in proof and scratchpad fields.                                                                                                                                                                                                                                                               |
| Title / state                          | Proof-carrying discriminant-to-exact-quadratic instantiation gate                                                                                                                                                                                                                                                                    |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                              |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                                                                                    |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Not before O05/O06 or equivalent operational discriminant ledger is semantically corrected/public enough for consumption. Must consume O09 schema but not define the schema here. |
| Phase split decision                   | No split decided yet.                                                                                                                                                                                                                                                                                                                |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Use the D ledger as sole input, instantiate exact quadratic algebra, prove projection lemmas. If D-source is missing, jump back to discriminant phase.                     |
| Refactor regression protocol           | Apply global refactor regression protocol if code changes.                                                                                                                                                                                                                                                                           |
| Refactor migration metrics             | Pending or not applicable.                                                                                                                                                                                                                                                                                                           |
| Lean/mathlib version policy            | Use frozen Lean/mathlib baseline unless a phase-boundary upgrade gate is opened.                                                                                                                                                                                                                                                     |
| ExecComplex boundary policy            | ExecComplex remains Q x Q; exact quadratic algebraic data require a proof-carrying discriminant source.                                                                                                                                                                                                                              |
| Consumer-contract direction            | No consumer-side patch; missing D-source belongs earlier.                                                                                                                                                                                                                                                                            |
| Governance-overhead review             | Pending or not applicable.                                                                                                                                                                                                                                                                                                           |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Construct instance parameterized by ledger.D. Prove theta-square relation by definition/projection, then norm/trace/conjugation formulas by coordinate calculation.                       |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                    |
| Expected mathlib imports               | Same as O09 if used; avoid new NumberField/Complex imports unless mirror-only and gated by version policy.                                                                                                                                                                                                                           |
| ExecComplex/computable plan            | Input D and output invariant projections are operational/rational/integer; exact quadratic coordinates remain in O09/O10 layer and do not rewrite ExecComplex.                                                                                                                                                                       |
| Concrete computable instance traversal | At minimum: one generic D-ledger row and the Eisenstein/Gauss sentinel D rows when/if the operational path derives their D. Every row must cite its ledger source.                                                                                                                                                                   |
| Computable path non-skeleton status    | not prepared                                                                                                                                                                                                                                                                                                                         |
| Expected executable witness            | Not prepared yet.                                                                                                                                                                                                                                                                                                                    |
| Mathlib complication log               | Record any need for QuadraticAlgebra lemmas versus project-owned coordinate lemmas; no noncomputable quotient/field instances on operational path.                                                                                                                                                                                   |
| Noncomputable boundary audit           | If instantiation requires noncomputable mathlib structure, split mirror proof from executable coordinate witness.                                                                                                                                                                                                                    |
| Mathlib mitigation/refactor layer      | Pending.                                                                                                                                                                                                                                                                                                                             |
| Bridge gap status                      | Open until O05 is corrected and O09 schema exists; this phase may reveal earlier D-source corrections.                                                                                                                                                                                                                               |
| Natural backjump/refactor layer        | Pending.                                                                                                                                                                                                                                                                                                                             |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                |

|                                      |                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                    |
| How implemented                      | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                              |
| Verification / evidence              | Not applicable yet.                                                                                                                                                                                            |
| Post-implementation plan check       | Pending.                                                                                                                                                                                                       |
| Artifact packaging rule              | Return a full zip;<br>do not claim Lean build unless executed or explicitly user-reported green.                                                                                                               |
| Implementation notes                 | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                 |
| Source materials / references        | <ul style="list-style-type: none"><li>O05 operational discriminant certificate</li><li>O09 exact quadratic schema</li><li>LEG32 no-free-sqrt provenance contract</li></ul>                                     |
| Extracted established theory objects | <ul style="list-style-type: none"><li>proof-carrying D-source</li><li><math>\theta^2 = D</math> instance</li><li>trace/norm/conjugation projection</li><li>allowed rational/integer invariant export</li></ul> |

Phase 4.6 – Comparison target policy

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

Phase 5 – Number-theory base from operational discriminant: D\_fund, conductor, quadratic field/order, basis, trace/norm/conjugation

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | R                                                                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |

|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                                                                                                                   |
| Title / state                          | Number-theory base from operational discriminant: D_fund, conductor, quadratic field/order, basis, trace/norm/conjugation                                                                                                                                                                                                                                 |
| Preparation workflow                   | number-theory base derivation workflow                                                                                                                                                                                                                                                                                                                    |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                     |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.                                                                                         |



|                      |                                                                                                                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation notes | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                          |
| No-go guardrails     | <ul style="list-style-type: none"><li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li></ul> |

Phase 5.1 – Fundamental discriminant and conductor normalization from D\_op

**Table legend.**   G = green/closed and derived-only;   W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   O = orange/implementable after stated predecessor closure;   Y = yellow/active and implementable now;   R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | R                                                                                                                                                                                                                                                                                                                                                         |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                            |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Fundamental discriminant and conductor normalization from D_op                                                                                                                                                                                                                                                                                            |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                  |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                                                                                                      |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |



|                                   |                                                                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathlib complication log          | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit      | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                 | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract    | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance          | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                   | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence           | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check    | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule           | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes              | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                  | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 5.2 – Quadratic field profile from proof-carrying discriminant

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                         |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                            |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                         | Quadratic field profile from proof-carrying discriminant                                                                                                                                                                                                                                                                                                  |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ExecComplex boundary policy            | QxQ rational floor;<br>exact quadratic coordinates require O09 plus O10/BR14 proof-carrying D-source;<br>no free sqrt(D).                                                                                                                                             |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                           |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet. Exact quadratic use must avoid analytic sqrt/Complex and cite D-source ledger.                                                                                                                           |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                         |

### Phase 5.3 – Quadratic order profile from D\_fund and conductor

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                   |

|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Quadratic order profile from D_fund and conductor                                                                                                                                                                                                                                                                                                         |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                               |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.                                                                                         |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                            |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                                                                                                             |

Phase 5.4 – Integral basis / omega\_D profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span style="background-color: #f4cccc; border: 1px solid #000; padding: 0 2px;">R</span>                                                                                                                                                                                                                                                                 |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                            |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Integral basis / omega_D profile                                                                                                                                                                                                                                                                                                                          |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                  |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                                                                                                      |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |

|                                |                                                                                                                                                                                                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation prompt contract | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance       | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence        | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule        | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes           | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails               | <ul style="list-style-type: none"><li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li></ul>                                                           |

Phase 5.5 – Trace, norm and conjugation forms

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                         |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                            |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                         | Trace, norm and conjugation forms                                                                                                                                                                                                                                                                                                                         |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                  |
| Consumer-contract direction           | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 6 – Units, class group, ideals, class number, prime splitting and character from the quadratic order

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                       |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                     |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                            |
| Phase type                            | derivation                                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Title / state                         | Units, class group, ideals, class number, prime splitting and character from the quadratic order                                                                                                                        |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                        |



|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                  |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                            |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                            |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                                                                                                                 |

### Phase 6.1 – Unit group of the quadratic order

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field        | Record                                            |
|--------------|---------------------------------------------------|
| Phase status | <span style="background-color: #f4cccc;">R</span> |



|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Unit group of the quadratic order                                                                                                                                                                                                                                                                                                                         |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                     |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |

|                                |                                                                                                                                                                                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / evidence        | Not applicable yet.                                                                                                                                                                                                                                            |
| Post-implementation plan check | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                             |
| Artifact packaging rule        | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required. |
| Implementation notes           | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                 |
| No-go guardrails               | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                      |

### Phase 6.2 – Class group, class number and ideal profiles

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                         |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                         | Class group, class number and ideal profiles                                                                                                                                                                                                                                                                                                              |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                  |
| Consumer-contract direction           | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                              |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 6.2.1 – Fractional and integral ideal profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid #000; padding: 0 2px;">R</span>                                                                                                                                                                                                                                                                 |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                         | Fractional and integral ideal profile                                                                                                                                                                                                                                                                                                                     |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                              |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                 |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 6.2.2 – Principal ideal predicate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                        |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; padding: 0 2px;">R</span>                                                                                                             |
| Context class                         | LeanDerivationPhase                                                                                                                                                           |
| Phase documentation tier              | F2                                                                                                                                                                            |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                            |

|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Principal ideal predicate                                                                                                                                                                                                                                                                                                                                 |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                     |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.                                                                                         |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                            |

|                  |                                                                                                                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"><li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li></ul> |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 6.3 – Heegner/PID/UFD indicator after class-number data

**Table legend.**   G = green/closed and derived-only;   W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   O = orange/implementable after stated predecessor closure;   Y = yellow/active and implementable now;   R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | R                                                                                                                                                                                                                                                                                                                                                         |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Heegner/PID/UFD indicator after class-number data                                                                                                                                                                                                                                                                                                         |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                  |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                                                                                                      |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |



|                                   |                                                                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noncomputable boundary audit      | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                 | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract    | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance          | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                              |
| How implemented                   | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence           | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check    | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule           | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes              | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                  | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 6.4 – Prime splitting and Kronecker/Dirichlet character

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                         |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                                                                                                                   |
| Title / state                         | Prime splitting and Kronecker/Dirichlet character                                                                                                                                                                                                                                                                                                         |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                            |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation.                                                                                          |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                        |



|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                           |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 6.5 – Arithmetic-CD compatibility source-site audit and bridge-precondition register, not final compatibility theorem

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                     |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                   |
| Phase type                            | derivation                                                                                                                                                                                     |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.      |

|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title / state                          | Arithmetic-CD compatibility source-site audit and bridge-precondition register, not final compatibility theorem                                                                                                                                                                                                                                           |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                               |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Audit source sites for CD norm and arithmetic norm data. Do not prove compatibility here;<br>record missing dependencies and route them to BR08/10.4.2.                                                        |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Final compatibility bridge moved to BR08/CD8 in phases 10.4.1/10.4.2.                                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.                                                                                         |
| Implementation notes                   | This phase is no longer the final Arithmetic-CD compatibility theorem. It only registers source sites and preconditions for CD8, whose final proof is deferred to 10.4.2 after BR08.                                                                                                                                                                      |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                                                                                                             |

Phase 7 – CM lattice geometry from the quadratic order and exact quadratic coordinates

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | R                                                                                                                                                                                                                                                                                                                                                         |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                            |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                                                                                                                   |
| Title / state                          | CM lattice geometry from the quadratic order and exact quadratic coordinates                                                                                                                                                                                                                                                                              |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | EXEC-2 likely relevant: algebraic irrational geometry/spectral/modular values are mirror-only unless a rational/integer invariant is separately proved for operational return.                                                                                                                                                                            |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |

|                                |                                                                                                                                                                                                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation prompt contract | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance       | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence        | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule        | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes           | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails               | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 7.1 – CM embedding and lattice basis

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                         |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                         | CM embedding and lattice basis                                                                                                                                                                                                                                                                                                                            |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy           | EXEC-2 likely relevant: algebraic irrational geometry/spectral/modular values are mirror-only unless a rational/integer invariant is separately proved for operational return.                                                                                                                                                                            |
| Consumer-contract direction           | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 7.2 – Gram matrix, determinant and covolume

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;"> </span> <b>R</b>                                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                     |
| Phase documentation tier              | F1                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                            |
| Phase type                            | derivation                                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Title / state                         | Gram matrix, determinant and covolume                                                                                                                                                                                   |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                        |

|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | EXEC-2 likely relevant: algebraic irrational geometry/spectral/modular values are mirror-only unless a rational/integer invariant is separately proved for operational return.                                                                                                                                                                            |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                            |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                            |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                                                                                                                 |

### Phase 7.2.1 – Heegner/class-number-one regime classification after class-number data

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field        | Record                                                            |
|--------------|-------------------------------------------------------------------|
| Phase status | <span style="background-color: #f4cccc; padding: 0 2px;">R</span> |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                                               |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                                                                                                                                |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                                                    |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                                                |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                                               |
| Implementation status                  | planned                                                                                                                                                                                                                                                                                                                                                                           |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                                                      |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                                        |
| Phase-type-specific contract           | test phase must emit row records and pass/fail criteria                                                                                                                                                                                                                                                                                                                           |
| Title / state                          | Heegner/class-number-one regime classification after class-number data                                                                                                                                                                                                                                                                                                            |
| Preparation workflow                   | regime-typed general early finite computation test workflow                                                                                                                                                                                                                                                                                                                       |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                                                  |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise.                         |
| Phase split decision                   | split from v33 O07 because generic-vs-sentinel is not enough;<br>regime typing is a separate semantic axis.                                                                                                                                                                                                                                                                       |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. First run generic finite rows;<br>then annotate rows by derived regime;<br>then interpret Eisenstein/Gauss as noOuter-root sentinels and all other Heegner/CM rows as projections/comparison until proven otherwise. |
| Module implementation manifest         | pending Phase 1.1.1.1.1                                                                                                                                                                                                                                                                                                                                                           |
| Semantic placement decision            | test/schema layer, not generator layer                                                                                                                                                                                                                                                                                                                                            |
| Reuse and duplication audit            | must reuse O07/canonical smoke-test generator;<br>no parallel Heegner test generator                                                                                                                                                                                                                                                                                              |
| Canonical generator policy             | consume the one canonical generator only                                                                                                                                                                                                                                                                                                                                          |
| Module compactness rule                | do not create a new module unless existing test/profile module cannot own the semantics                                                                                                                                                                                                                                                                                           |
| Refactor regression protocol           | no code movement in this gate;<br>future test implementation must preserve full local build green status                                                                                                                                                                                                                                                                          |
| Refactor migration metrics             | not applicable at planning gate                                                                                                                                                                                                                                                                                                                                                   |
| Lean/mathlib version policy            | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                                                                                                                                                                                                      |
| ExecComplex boundary policy            | EXEC-2 enforced                                                                                                                                                                                                                                                                                                                                                                   |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                                                       |
| Governance-overhead review             | review after Phase 1.1.2 whether this lock prevented false closed-system classification.                                                                                                                                                                                                                                                                                          |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Prove/record row provenance: generator output -> finite witness -> regime tag -> optional comparison target. A row may not change the regime tag based on h(D), j, or external Heegner membership.                                     |
| Expected tactics                       | structural constructors/projections;<br>explicit row records;<br>no fragile rewrites;<br>no inverse oracle.                                                                                                                                                                                                                                                                       |
| Expected mathlib imports               | none at planning gate;<br>future number-theory phases may require mathlib version review.                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Operational fields remain finite/rational/integer. tau, sqrt(D), analytic j and general CM fields stay mirror-only under EXEC-2.                                                                                                                                                                                                                                                  |
| Concrete computable instance traversal | required later: every row must contain actual finite generator traversal evidence or remain unclosed                                                                                                                                                                                                                                                                              |
| Computable path non-skeleton status    | schema lock only now;<br>future rows must provide executable witnesses                                                                                                                                                                                                                                                                                                            |
| Expected executable witness            | finite row witness with branch mode, L_max, regime tag, rational/integer outputs, optional comparison target                                                                                                                                                                                                                                                                      |



|                                      |                                                                                                                                                                                                                                                   |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathlib complication log             | record any class number/CM/j mathlib dependency or noncomputable boundary explicitly                                                                                                                                                              |
| Noncomputable boundary audit         | Heegner/class-number and analytic j data are comparison/mirror until operational invariant is derived                                                                                                                                             |
| Mathlib mitigation/refactor layer    | use a mirror layer or later version-policy gate if mathlib forces noncomputable definitions                                                                                                                                                       |
| Bridge gap status                    | open until noOuter/HasOuter regime split and O07 implementation exist                                                                                                                                                                             |
| Natural backjump/refactor layer      | if a row needs regime data not yet produced, backjump to regime/source layer rather than patching the test consumer.                                                                                                                              |
| Implementation prompt contract       | Before Lean code, module manifest must state exact module placement and prove no duplicate generator/test harness.                                                                                                                                |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                       |
| How implemented                      | Future number-theory/regime consumer phase: h(D)=1 is not closed/noOuter evidence;<br>classify after class-number and regime data exist.                                                                                                          |
| Verification / evidence              | YAML/planning-level validation and generated views only;<br>no Lean build.                                                                                                                                                                        |
| Proof strategy                       | planning-level dependency/order lock;<br>implementation proof obligations deferred to target phase                                                                                                                                                |
| Build result                         | No Lean build executed or claimed.                                                                                                                                                                                                                |
| Post-implementation plan check       | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                |
| Artifact packaging rule              | Return complete zip;<br>Lean build local unless actually executed here.                                                                                                                                                                           |
| Implementation notes                 | Future number-theory/regime consumer phase: h(D)=1 is not closed/noOuter evidence;<br>classify after class-number and regime data exist.                                                                                                          |
| Source materials / references        | <ul style="list-style-type: none"> <li>CNNA noOuter/HasOuter regime split;<br/>EXEC-2;<br/>general early finite test-matrix contract;<br/>post #97 heuristic;<br/>class-number-one/Heegner comparison list as external comparison only</li> </ul> |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Regime tag;<br/>noOuter root sentinel;<br/>HasOuter/complement-projection row;<br/>Heegner class-number-one comparison target;<br/>IDEAL_infty projection label</li> </ul>                                 |
| Implemented modules                  | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                                                            |
| Added modules                        | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                                        |
| Modified modules                     | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml only</li> </ul>                                                                                                                                                           |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                                                                                              |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                     |
| Proposed added modules               | <ul style="list-style-type: none"> <li>none at decision gate;<br/>future target likely existing O07 test/profile surface</li> </ul>                                                                                                               |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml;<br/>generated views</li> </ul>                                                                                                                                           |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>canonical smoke-test generator;<br/>regime split modules after Phase 1.3.2;<br/>O07 test matrix</li> </ul>                                                                                                 |

|                                      |                                                                                                                                |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>Heegner row schema;<br/>CM sentinel profile;<br/>closed-system sentinel tags</li> </ul> |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|

### Phase 7.3 – Period ratio tau and upper-half-plane condition

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                         |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                                                                                                                                                   |
| Title / state                          | Period ratio tau and upper-half-plane condition                                                                                                                                                                                                                                                                                                           |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | EXEC-2 likely relevant: algebraic irrational geometry/spectral/modular values are mirror-only unless a rational/integer invariant is separately proved for operational return.                                                                                                                                                                            |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                                                                                                      |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |

|                                   |                                                                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noncomputable boundary audit      | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                 | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract    | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance          | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                              |
| How implemented                   | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence           | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check    | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule           | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes              | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                  | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 7.4 – Angle / cosine profile

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                         |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                         | Angle / cosine profile                                                                                                                                                                                                                                                                                                                                    |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                            |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy_coexistence</code> status before implementation.                                                                                          |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy           | EXEC-2 likely relevant: algebraic irrational geometry/spectral/modular values are mirror-only unless a rational/integer invariant is separately proved for operational return.                                                                                                                                                                            |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                           |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                         |

### Phase 7.5 – Lattice automorphism group

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                |
| Context class                         | LeanDerivationPhase                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                        |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational. |

|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title / state                          | Lattice automorphism group                                                                                                                                                                                                                                                                                                                                |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | EXEC-2 likely relevant: algebraic irrational geometry/spectral/modular values are mirror-only unless a rational/integer invariant is separately proved for operational return.                                                                                                                                                                            |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                                                                                                      |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                  |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                            |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                            |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                                                                                                                 |

### Phase 8 – Modular forms, CM j-invariant targets, and heuristic lattice test matrix

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | R                                                                                                                                                                                                                                                                                                                                                         |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Modular forms, CM j-invariant targets, and heuristic lattice test matrix                                                                                                                                                                                                                                                                                  |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | EXEC-2 likely relevant: algebraic irrational geometry/spectral/modular values are mirror-only unless a rational/integer invariant is separately proved for operational return.                                                                                                                                                                            |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |



|                                |                                                                                                                                                                                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How implemented                | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                              |
| Verification / evidence        | Not applicable yet.                                                                                                                                                                                                                                            |
| Post-implementation plan check | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                             |
| Artifact packaging rule        | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required. |
| Implementation notes           | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                 |
| No-go guardrails               | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                      |

### Phase 8.0 – Modular coordinate prerequisites: SL2Z action and fundamental-domain representative

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                         |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | not_started                                                                                                                                                                                                                                                                                                                                               |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | derivation contract: dependency and object placement must be explicit and minimal.                                                                                                                                                                                                                                                                        |
| Title / state                         | Modular coordinate prerequisites: SL2Z action and fundamental-domain representative                                                                                                                                                                                                                                                                       |
| Preparation workflow                  | phase dependency bifurcation and semantic placement workflow                                                                                                                                                                                                                                                                                              |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Split because phase 9 mixed modular and physical consumers and phase 10 mixed modular, spectral, transfer, analytic and bridge targets.                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Keep parallel paths after Phase 4; close each path on its own proof-carrying prerequisites.                                                                                                     |
| Module implementation manifest        | No Lean implementation manifest was required when this already-closed/planned infrastructure row was created; under closed LEG10, any future code change touching this phase must fill exact added/modified modules before edits.                                                                                                                         |
| Semantic placement decision           | Existing placement remains historical/planning-only. Future module placement must be revalidated by LEG10 before code changes.                                                                                                                                                                                                                            |
| Reuse and duplication audit           | Not a generator implementation. If later code resembles an existing generator/API, consume LEG11 before edits.                                                                                                                                                                                                                                            |
| Canonical generator policy            | No generator-like code in this record. Future generator-like extensions must route through LEG11.                                                                                                                                                                                                                                                         |
| Module compactness rule               | No one-phase-one-file inference; future modules require semantic/API/import-boundary justification.                                                                                                                                                                                                                                                       |
| Refactor regression protocol          | incremental wrapper mode; full local build green after each migration step                                                                                                                                                                                                                                                                                |
| Refactor migration metrics            | track object reassignments, phase dependency changes, and consumer-path imports                                                                                                                                                                                                                                                                           |
| Lean/mathlib version policy           | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                                                                                                                                                                              |
| ExecComplex boundary policy           | Phase 4.1 finite matrix is enough for spectral/physical finite consumers; modular/CM analytics remain separate.                                                                                                                                                                                                                                           |



|                                        |                                                                                                                                                                                                                                                               |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                   |
| Governance-overhead review             | Review after first Phase 10 finite consumer whether the bifurcation prevented a false wait on Phase 8/9.                                                                                                                                                      |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. For each object, list minimal existing sources and reject any extra phase dependency not used by the proof target. |
| Expected tactics                       | planning/generator validation now;<br>later Lean proofs use constructor/projection lemmas over source objects only.                                                                                                                                           |
| Expected mathlib imports               | none for planning lock                                                                                                                                                                                                                                        |
| ExecComplex/computable plan            | Spectral/physical finite profiles consume Phase 4.1 ExecBoundaryMatrix;<br>modular forms/CM consume lattice/number-theory path separately.                                                                                                                    |
| Concrete computable instance traversal | Future finite test rows must run separately for arithmetic/modular and spectral/physical consumers.                                                                                                                                                           |
| Computable path non-skeleton status    | Must be demonstrated by finite rows over Phase 4.1 matrix for spectral/physical path and by D/CM rows for modular path.                                                                                                                                       |
| Expected executable witness            | At least one finite ExecBoundaryMatrix spectral profile row and at least one D-to-modular comparison row when those phases activate.                                                                                                                          |
| Mathlib complication log               | Record if modular-form/mathlib APIs are attempted in spectral/physical path;<br>that is a placement smell unless explicitly justified.                                                                                                                        |
| Noncomputable boundary audit           | Analytic modular/j/L-function objects remain mirror/comparison and cannot block finite spectral/operator algebra profiles.                                                                                                                                    |
| Mathlib mitigation/refactor layer      | Use separate path modules;<br>do not import modular APIs into finite physical consumers for convenience.                                                                                                                                                      |
| Bridge gap status                      | Open bridge objects remain late and cannot force early path sequentialization.                                                                                                                                                                                |
| Natural backjump/refactor layer        | If a consumer needs a source from another path, add an explicit bridge dependency rather than moving the whole phase behind that path.                                                                                                                        |
| Implementation prompt contract         | Before implementing any phase, module manifest must state exact source phase IDs and must not inherit unrelated predecessor chains.                                                                                                                           |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                   |
| How implemented                        | Not implemented.                                                                                                                                                                                                                                              |
| Verification / evidence                | Generated-table validation after v37 patch;<br>no Lean build.                                                                                                                                                                                                 |
| Proof strategy                         | Treat phase dependency as provenance obligation;<br>split unrelated consumers rather than sequentializing them through non-required objects.                                                                                                                  |
| Build result                           | No Lean build executed or claimed.                                                                                                                                                                                                                            |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                            |
| Artifact packaging rule                | Return complete zip;<br>Lean build local unless actually executed here.                                                                                                                                                                                       |
| Implementation notes                   | Records dependency and topic-boundary decision before code implementation.                                                                                                                                                                                    |
| Source materials / references          | <ul style="list-style-type: none"> <li>v36 phase/object register</li> <li>code audit module inventory</li> <li>review identifying artificial modular-form bottleneck</li> </ul>                                                                               |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>independent number-theory/modular path</li> <li>independent spectral/physical path</li> <li>explicit consumer dependencies</li> </ul>                                                                                  |
| Implemented modules                    | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                                                                        |
| Added modules                          | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                                                                        |
| Modified modules                       | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                                                    |
| Forbidden patterns checked             | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                                                                                                          |

|                  |                                                                                                                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"><li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li></ul> |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 8.1 – Eisenstein series profile

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | R                                                                                                                                                                                                                                                                                                                                                         |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Eisenstein series profile                                                                                                                                                                                                                                                                                                                                 |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | EXEC-2 likely relevant: algebraic irrational geometry/spectral/modular values are mirror-only unless a rational/integer invariant is separately proved for operational return.                                                                                                                                                                            |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                                                                                                      |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |

|                                   |                                                                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathlib mitigation/refactor layer | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                 | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract    | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance          | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                   | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence           | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check    | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule           | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes              | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                  | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 8.2 – Weierstrass invariants g2/g3

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #ffcdd2; border: 1px solid #000; padding: 0 2px;">R</span>                                                                                                                                                                                                                                                                 |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                         | Weierstrass invariants g2/g3                                                                                                                                                                                                                                                                                                                              |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy           | QxQ rational floor; exact quadratic coordinates require O09 plus O10/BR14 proof-carrying D-source; no free sqrt(D).                                                                                                                                                                                                                                       |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                           |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet. Exact quadratic use must avoid analytic sqrt/Complex and cite D-source ledger.                                                                                                                           |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 8.3 – Modular discriminant Delta\_mod

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                |
| Context class                         | LeanDerivationPhase                                                                                                                                                                       |
| Phase documentation tier              | F2                                                                                                                                                                                        |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational. |

|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title / state                          | Modular discriminant Delta_mod                                                                                                                                                                                                                                                                                                                            |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | EXEC-2 likely relevant: algebraic irrational geometry/spectral/modular values are mirror-only unless a rational/integer invariant is separately proved for operational return.                                                                                                                                                                            |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.                                                                                         |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                            |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                                                                                                             |

### Phase 8.4 – j-invariant profile and special-value comparison targets

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | R                                                                                                                                                                                                                                                                                                                                                         |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | j-invariant profile and special-value comparison targets                                                                                                                                                                                                                                                                                                  |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | EXEC-2 likely relevant: algebraic irrational geometry/spectral/modular values are mirror-only unless a rational/integer invariant is separately proved for operational return.                                                                                                                                                                            |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |



|                                |                                                                                                                                                                                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How implemented                | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                              |
| Verification / evidence        | Not applicable yet.                                                                                                                                                                                                                                            |
| Post-implementation plan check | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                             |
| Artifact packaging rule        | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required. |
| Implementation notes           | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                 |
| No-go guardrails               | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                      |

### Phase 8.5 – Eisenstein/Gauss sentinel rows inside the general lattice/computation test matrix: static branching and L\_max sweep

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                 |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                      |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                     |
| Phase documentation tier              | F2                                                                                                                                                                                                     |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                     |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                    |
| Implementation status                 | not-started                                                                                                                                                                                            |
| Preparation status                    | not-prepared                                                                                                                                                                                           |
| Phase type                            | comparison                                                                                                                                                                                             |
| Phase-type-specific contract          | derivation/test phase: requires concrete rows, executable finite witnesses, and explicit mirror/operational boundary.                                                                                  |
| Preparation workflow                  | Prepare when predecessor objects are closed.                                                                                                                                                           |
| Dependency-depth audit                | v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Proof rehearsal before implementation | Pending until predecessor objects are available.                                                                                                                                                       |
| Bridge gap status                     | Open: source objects not yet closed.                                                                                                                                                                   |

### Phase 8.6 – Dynamic branching and code-defined b lattice test-matrix extension

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                        |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                             |
| Context class                         | ReferenceGatePhase                                                                                                                                                            |
| Phase documentation tier              | F2                                                                                                                                                                            |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                            |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                           |
| Implementation status                 | not-started                                                                                                                                                                   |
| Preparation status                    | not-prepared                                                                                                                                                                  |
| Phase type                            | comparison                                                                                                                                                                    |
| Phase-type-specific contract          | derivation/test phase: requires concrete rows, executable finite witnesses, and explicit mirror/operational boundary.                                                         |
| Preparation workflow                  | Prepare when predecessor objects are closed.                                                                                                                                  |



|                                       |                                                                                                                                                                                                        |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependency-depth audit                | v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Proof rehearsal before implementation | Pending until predecessor objects are available.                                                                                                                                                       |
| Bridge gap status                     | Open: source objects not yet closed.                                                                                                                                                                   |

Phase 8.7 – Heuristic lattice test-matrix closeout and public-boundary audit

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | R                                                                                                                                                                                                                                                                                                                                                         |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                            |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | not-started                                                                                                                                                                                                                                                                                                                                               |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | documentation/closeout: audit that tests are consumer-only and update proof dossiers.                                                                                                                                                                                                                                                                     |
| Title / state                          | Heuristic lattice test-matrix closeout and public-boundary audit                                                                                                                                                                                                                                                                                          |
| Preparation workflow                   | Prepare when predecessor objects are closed.                                                                                                                                                                                                                                                                                                              |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Already split from the decision lock to keep implementation after source objects exist.                                                                                                                                                                                                                                                                   |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Implement finite rows and witnesses first, then compare against Eisenstein/Gauss targets. Obstructions only after concrete row construction fails.                                              |
| Refactor regression protocol           | No refactor performed in this phase; future code tests must keep the full local build green after each wrapper/test addition.                                                                                                                                                                                                                             |
| Refactor migration metrics             | Not applicable until implementation.                                                                                                                                                                                                                                                                                                                      |
| Lean/mathlib version policy            | Lean/mathlib baseline remains 4.28.0 unless an explicit upgrade gate is opened.                                                                                                                                                                                                                                                                           |
| ExecComplex boundary policy            | Only rational/integer invariants operational; tau/j analytic values are mirror/comparison.                                                                                                                                                                                                                                                                |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | This gate prevents a likely false implementation path: injecting angle heuristics into the generator rather than testing them as comparison targets.                                                                                                                                                                                                      |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending until predecessor objects are available.                                                                                                                                                               |
| Expected tactics                       | Structural constructors/projections, explicit finite-index calculations, and finite relation witnesses; no fragile rewrite chain.                                                                                                                                                                                                                         |
| Expected mathlib imports               | Use existing project imports first; future CM/j mirror tests may require mathlib modular/complex APIs only in non-operational layers after import existence is checked.                                                                                                                                                                                   |
| ExecComplex/computable plan            | Operational path may record only rational/integer outputs: branching code value, finite addresses, collision relations, cos(theta), integer/rational Gram invariants, D target. Complex coordinates, tau, sqrt(D), and analytic j evaluation remain mirror/comparison.                                                                                    |
| Concrete computable instance traversal | Required at preparation: include b=2 static, L_max values spanning 1-4, closure rows for 120/90 degree profiles, and at least one legal dynamic-branching case for 8.6.                                                                                                                                                                                   |
| Computable path non-skeleton status    | Future test implementation must actually traverse finite addresses and compute expected finite invariants, not merely instantiate empty profile records.                                                                                                                                                                                                  |
| Expected executable witness            | Finite test-matrix output rows, not just theorem stubs.                                                                                                                                                                                                                                                                                                   |

|                                      |                                                                                                                                                                                                               |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathlib complication log             | Record any noncomputable/j/tau/mathlib obstacle and isolate it outside operational path.                                                                                                                      |
| Noncomputable boundary audit         | Mandatory before implementation closes.                                                                                                                                                                       |
| Mathlib mitigation/refactor layer    | Use rational/integer surrogates in operational layer;<br>isolate analytic or algebraic-irrational material in mirror/profile modules.                                                                         |
| Bridge gap status                    | Open: source objects not yet closed.                                                                                                                                                                          |
| Natural backjump/refactor layer      | If tests require new generator semantics, backjump to the canonical generator layer;<br>do not patch the test module.                                                                                         |
| Implementation prompt contract       | Before code: provide module manifest, reuse audit, concrete finite instances, and test matrix rows;<br>do not create duplicate generators.                                                                    |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                   |
| How implemented                      | Planning record only. No Lean code has been changed in this phase.                                                                                                                                            |
| Verification / evidence              | YAML/master validation and generated-view compilation;<br>no Lean build run.                                                                                                                                  |
| Build result                         | No Lean build executed.                                                                                                                                                                                       |
| Post-implementation plan check       | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                            |
| Artifact packaging rule              | Return complete zip;<br>no Lean-build success may be claimed unless actually run.                                                                                                                             |
| Source materials / references        | <ul style="list-style-type: none"> <li>post #97 source note</li> <li>future closed generator/lattice/CM/j objects</li> </ul>                                                                                  |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>test matrix rows</li> <li>finite branch/address witnesses</li> <li>angle/cosine profiles</li> <li>CM/j targets</li> </ul>                                              |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul> |

### Phase 8.8 – q-expansion and Hilbert class polynomial modular far targets

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                        |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                             |
| Context class                         | LeanDerivationPhase                                                                                                                                                           |
| Phase documentation tier              | F2                                                                                                                                                                            |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                            |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                           |
| Implementation status                 | not_started                                                                                                                                                                   |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                  |
| Phase type                            | derivation                                                                                                                                                                    |
| Phase-type-specific contract          | comparison contract: dependency and object placement must be explicit and minimal.                                                                                            |
| Title / state                         | q-expansion and Hilbert class polynomial modular far targets                                                                                                                  |
| Preparation workflow                  | phase dependency bifurcation and semantic placement workflow                                                                                                                  |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                              |

|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Split because phase 9 mixed modular and physical consumers and phase 10 mixed modular, spectral, transfer, analytic and bridge targets.                                                                                                                                                                                                                   |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Keep parallel paths after Phase 4; close each path on its own proof-carrying prerequisites.                                                                                                     |
| Module implementation manifest         | No Lean implementation manifest was required when this already-closed/planned infrastructure row was created; under closed LEG10, any future code change touching this phase must fill exact added/modified modules before edits.                                                                                                                         |
| Semantic placement decision            | Existing placement remains historical/planning-only. Future module placement must be revalidated by LEG10 before code changes.                                                                                                                                                                                                                            |
| Reuse and duplication audit            | Not a generator implementation. If later code resembles an existing generator/API, consume LEG11 before edits.                                                                                                                                                                                                                                            |
| Canonical generator policy             | No generator-like code in this record. Future generator-like extensions must route through LEG11.                                                                                                                                                                                                                                                         |
| Module compactness rule                | No one-phase-one-file inference; future modules require semantic/API/import-boundary justification.                                                                                                                                                                                                                                                       |
| Refactor regression protocol           | incremental wrapper mode; full local build green after each migration step                                                                                                                                                                                                                                                                                |
| Refactor migration metrics             | track object reassignments, phase dependency changes, and consumer-path imports                                                                                                                                                                                                                                                                           |
| Lean/mathlib version policy            | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                                                                                                                                                                              |
| ExecComplex boundary policy            | Phase 4.1 finite matrix is enough for spectral/physical finite consumers; modular/CM analytics remain separate.                                                                                                                                                                                                                                           |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Review after first Phase 10 finite consumer whether the bifurcation prevented a false wait on Phase 8/9.                                                                                                                                                                                                                                                  |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. For each object, list minimal existing sources and reject any extra phase dependency not used by the proof target.                                                                                             |
| Expected tactics                       | planning/generator validation now; later Lean proofs use constructor/projection lemmas over source objects only.                                                                                                                                                                                                                                          |
| Expected mathlib imports               | none for planning lock                                                                                                                                                                                                                                                                                                                                    |
| ExecComplex/computable plan            | Spectral/physical finite profiles consume Phase 4.1 ExecBoundaryMatrix; modular forms/CM consume lattice/number-theory path separately.                                                                                                                                                                                                                   |
| Concrete computable instance traversal | Future finite test rows must run separately for arithmetic/modular and spectral/physical consumers.                                                                                                                                                                                                                                                       |
| Computable path non-skeleton status    | Must be demonstrated by finite rows over Phase 4.1 matrix for spectral/physical path and by D/CM rows for modular path.                                                                                                                                                                                                                                   |
| Expected executable witness            | At least one finite ExecBoundaryMatrix spectral profile row and at least one D-to-modular comparison row when those phases activate.                                                                                                                                                                                                                      |
| Mathlib complication log               | Record if modular-form/mathlib APIs are attempted in spectral/physical path; that is a placement smell unless explicitly justified.                                                                                                                                                                                                                       |
| Noncomputable boundary audit           | Analytic modular/j/L-function objects remain mirror/comparison and cannot block finite spectral/operator algebra profiles.                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | Use separate path modules; do not import modular APIs into finite physical consumers for convenience.                                                                                                                                                                                                                                                     |
| Bridge gap status                      | Open bridge objects remain late and cannot force early path sequentialization.                                                                                                                                                                                                                                                                            |
| Natural backjump/refactor layer        | If a consumer needs a source from another path, add an explicit bridge dependency rather than moving the whole phase behind that path.                                                                                                                                                                                                                    |
| Implementation prompt contract         | Before implementing any phase, module manifest must state exact source phase IDs and must not inherit unrelated predecessor chains.                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                  |
| How implemented                        | Not implemented.                                                                                                                                                                                                                                                                                                                                          |
| Verification / evidence                | Generated-table validation after v37 patch; no Lean build.                                                                                                                                                                                                                                                                                                |
| Proof strategy                         | Treat phase dependency as provenance obligation; split unrelated consumers rather than sequentializing them through non-required objects.                                                                                                                                                                                                                 |
| Build result                           | No Lean build executed or claimed.                                                                                                                                                                                                                                                                                                                        |

|                                      |                                                                                                                                                                                                               |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Post-implementation plan check       | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                            |
| Artifact packaging rule              | Return complete zip;<br>Lean build local unless actually executed here.                                                                                                                                       |
| Implementation notes                 | Records dependency and topic-boundary decision before code implementation.                                                                                                                                    |
| Source materials / references        | <ul style="list-style-type: none"> <li>v36 phase/object register</li> <li>code audit module inventory</li> <li>review identifying artificial modular-form bottleneck</li> </ul>                               |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>independent number-theory/modular path</li> <li>independent spectral/physical path</li> <li>explicit consumer dependencies</li> </ul>                                  |
| Implemented modules                  | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                        |
| Added modules                        | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                        |
| Modified modules                     | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                    |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                                                          |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul> |

## Phase 9 – Theta, Dirichlet L-functions, and analytic class-number formula far targets

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise.                  |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 9.1 – Theta series of the derived lattice

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field        | Record |
|--------------|--------|
| Phase status | W      |

|                                       |                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise.                  |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 9.2 – Dirichlet L-function profile for chi\_D

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise.                  |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 9.3 – Class-number formula comparison profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                        |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                             |
| Context class                         | ReferenceGatePhase                                                                                                                                                            |
| Phase documentation tier              | F2                                                                                                                                                                            |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                            |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                           |

|                                       |                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise.                  |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 9.4 – Analytic mirror/noncomputable boundary audit for theta and L-functions

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise.                  |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 9.5 – Theta/L-function/class-number formula closeout and failure ledger

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |



|                                       |                                                                                                                                                                                                        |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependency-depth audit                | v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                 |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                  |

### Phase 10 – Spectral, finite physical, transfermatrix, and Schur-bridge consumers after operational boundary matrices

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span style="background-color: #FFB6C1;">R</span>                                                                                                                                                                                                                                                                                                         |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                            |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Spectral, finite physical, transfermatrix, and Schur-bridge consumers after operational boundary matrices                                                                                                                                                                                                                                                 |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                  |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                                                                                                      |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |



|                                   |                                                                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathlib complication log          | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit      | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                 | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract    | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance          | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                   | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence           | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check    | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule           | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes              | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                  | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 10.1 – Finite spectral profile from ExecBoundaryMatrix

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                         |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                            |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                         | Finite spectral profile from ExecBoundaryMatrix                                                                                                                                                                                                                                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                 |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 10.2 – Finite operator algebra, star, state, dynamics, unitarity and open-channel profiles

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                   |
| Phase type                            | derivation                                                                                                                                                                                     |

|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Finite operator algebra, star, state, dynamics, unitarity and open-channel profiles                                                                                                                                                                                                                                                                       |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                     |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.                                                                                         |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                            |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                                                                                                             |

Phase 10.3 – Heat kernel and spectral geometry far targets

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span style="background-color: #f4cccc; border: 1px solid #000; padding: 0 2px;">R</span>                                                                                                                                                                                                                                                                 |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier               | F2                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                          | Heat kernel and spectral geometry far targets                                                                                                                                                                                                                                                                                                             |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                 | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                             |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                        |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                                                                                                      |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |

|                                |                                                                                                                                                                                                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation prompt contract | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance       | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence        | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule        | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes           | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |
| No-go guardrails               | <ul style="list-style-type: none"><li>Global no-go inherited;<br/>additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li></ul>                                                           |

Phase 10.4 – Transfermatrix formalization from Schur recurrence

**Table legend.**   G = green/closed and derived-only;   W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   O = orange/implementable after stated predecessor closure;   Y = yellow/active and implementable now;   R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | R                                                                                                                                                                                                                                                                                                                                                         |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                       |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                            |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                       |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                                                                                                        |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                              |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                 |
| Title / state                         | Transfermatrix formalization from Schur recurrence                                                                                                                                                                                                                                                                                                        |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                   |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split;<br>phase titles and object groups must match the actual mathematical source dependencies.                                                                                                                                                                                                          |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                            |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                                                                                                             |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                               |
| Consumer-contract direction           | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.                                                                                                                                                                                               |
| Governance-overhead review            | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                    |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers. Also prepare downstream export shape for BR08 without consuming BR08.                                                                  |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                             |

### Phase 10.4.1 – Multiscale Schur-duality bridge: horizontal sector elimination versus vertical level recursion

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                          |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                               |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                             |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                                              |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                  |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                              |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                             |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                                              |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                    |
| Phase type                            | derivation                                                                                                                                                                                                                                                                      |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                       |
| Title / state                         | Multiscale Schur-duality bridge: horizontal sector elimination versus vertical level recursion                                                                                                                                                                                  |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                         |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                                               |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Not prepared. Must verify that S05, O05, TM0-TM2, CD7, BR06, BR07, S07, and S08 are closed and documented before activation. |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase split decision                   | Expected to split: (a) common block-index product structure, (b) horizontal elimination theorem, (c) vertical recursion theorem, (d) comparison/commutation theorem, (e) obstruction or weakened naturality statement if strict commutation fails.                                                                                                                                                                                                                                                                                      |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Try the positive construction first on a concrete finite generated tree/cut with both sector and level block indices. Express both elimination orders from the same Dirichlet-Laplacian and attempt an entrywise/factorization proof.                                                                                                                                         |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                                                                                                                                                                                                       |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                                                                                                                                                                                                             |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                                                                                                                                                                                              |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                                                                                                                                                                                                |
| Consumer-contract direction            | Consumes S05/O05/TM0/CD7/BR06/BR07/S07/S08 only. Produces BR08/BR09. Must not be imported by earlier route definitions.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                                                                                                                                                                                                     |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Start from the full derived Dirichlet-Laplacian. Introduce two orthogonal decompositions: sector blocks (bright/interface/dark) and level blocks (thread/recursion depth). Define horizontal Schur elimination and vertical Schur recursion as operations on this same matrix. State exact equality, commutation, or functorial factorization only after the domains/codomain indices match. |
| Expected tactics                       | Structural record construction;<br>explicit block/index equivalences;<br>matrix-entry expansion from constructors;<br>local finite-instance witnesses;<br>avoid fragile rewrite chains and generic Matrix.inv.                                                                                                                                                                                                                                                                                                                          |
| Expected mathlib imports               | Matrix block/index support already used by Schur/DtN modules; verify against Lean 4.28.0 when activated. Any mathlib noncomputable pressure must be logged and isolated.                                                                                                                                                                                                                                                                                                                                                                |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Concrete computable instance traversal | Required when active: name one finite generated b-ary approximant/cut, compute both eliminations through the same matrix, and compare outputs.                                                                                                                                                                                                                                                                                                                                                                                          |
| Computable path non-skeleton status    | must provide concrete finite witness before any operational claim                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Expected executable witness            | A finite generated tree/cut witness for horizontal and vertical Schur elimination comparison.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                               |
| Bridge gap status                      | Planned bridge gap. If a needed common block-index structure is missing, backjump to the earliest natural Schur/DtN source phase rather than patching BR08.                                                                                                                                                                                                                                                                                                                                                                             |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                   |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                                                                                                                                                                                                       |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                          |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                                                                                                                                                                                                          |
| Source materials / references          | <ul style="list-style-type: none"> <li>Checked source routes when available: S05/BR06 horizontal boundary route; O05/BR07 vertical arithmetic route; TM0-TM2 transfermatrix route; CD7 Hurwitz boundary. External Schur-complement facts may be used only as proof guidance, not CNNA input.</li> </ul>                                                                                                                                                                                                                                 |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>BR08 Horizontal/vertical Schur-duality bridge</li> <li>BR09 Subsystem-induced invariant bifurcation profile</li> </ul>                                                                                                                                                                                                                                                                                                                                                                           |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no__go__guardrails</code> apply; this phase list is local addenda only.</li> <li>no Schur-duality theorem before source routes are closed</li> <li>no analogy as proof</li> <li>no consumer-side patch if common block-index structure belongs earlier</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> </ul> |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 10.4.2 – Arithmetic-CD compatibility as Schur-duality corollary

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked

Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                          |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                        |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                                                                                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                        |
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                                                                                                         |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                               |
| Phase type                             | derivation                                                                                                                                                                                                                                                                                                                                                 |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                  |
| Title / state                          | Arithmetic-CD compatibility as Schur-duality corollary                                                                                                                                                                                                                                                                                                     |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                                                                    |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Not prepared. Must wait for BR08 and route-specific norm/discriminant objects.                                                                                                                          |
| Phase split decision                   | Likely split into CD norm-source alignment, arithmetic norm-source alignment, coordinate/entry bridge, and final compatibility theorem.                                                                                                                                                                                                                    |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Derive compatibility as a corollary of BR08. Do not assert it from CD/arithmetic resemblance.                                                                                                    |
| Refactor regression protocol           | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                             |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation.                                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                 |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                         |
| Consumer-contract direction            | Consumes BR08/BR09 and route norm objects; produces CD8 only for downstream unification, not for earlier route construction.                                                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                        |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Use BR08 to identify common Schur source. Then show the CD normSq and arithmetic norm/discriminant profiles are projections/invariants of that source under their respective route maps, with exact hypotheses. |
| Expected tactics                       | Record projections, explicit coordinate maps, theorem fields from BR08, finite witness checks, structural proofs over constructor data.                                                                                                                                                                                                                    |
| Expected mathlib imports               | Depends on actual norm/order formalization chosen in earlier number-theory phases; verify against Lean 4.28.0 at activation.                                                                                                                                                                                                                               |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                          |
| Concrete computable instance traversal | Required when active: at least one finite Schur-derived arithmetic example with matching CD-side norm data.                                                                                                                                                                                                                                                |

|                                      |                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computable path non-skeleton status  | must provide concrete finite witness before compatibility is public-final                                                                                                                                                                                                                                                               |
| Expected executable witness          | Finite route witness linking a Schur-derived discriminant/norm and CD normSq data under the BR08 hypotheses.                                                                                                                                                                                                                            |
| Mathlib complication log             | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                    |
| Noncomputable boundary audit         | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                            |
| Mathlib mitigation/refactor layer    | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                               |
| Bridge gap status                    | Deferred until BR08 exists;<br>if BR08 only gives a weakened statement, CD8 must inherit the weakened scope.                                                                                                                                                                                                                            |
| Natural backjump/refactor layer      | Not prepared yet.                                                                                                                                                                                                                                                                                                                       |
| Implementation prompt contract       | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                   |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                             |
| How implemented                      | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                       |
| Verification / evidence              | Not applicable yet.                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check       | Not available yet.                                                                                                                                                                                                                                                                                                                      |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                          |
| Implementation notes                 | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                          |
| Source materials / references        | <ul style="list-style-type: none"> <li>BR08/BR09 proof output;<br/>CD normSq/norm-multiplicativity route;<br/>arithmetic norm/discriminant route;<br/>transfermatrix route.</li> </ul>                                                                                                                                                  |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>CD8 Arithmetic-CD compatibility as Schur-duality corollary</li> </ul>                                                                                                                                                                                                                            |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>do not close CD8 before BR08</li> <li>do not treat CD norm and arithmetic norm analogy as proof</li> <li>no public handoff exposure before theorem proof and access review</li> </ul> |

### Phase 10.4.3 – Cayley-Dickson scalar-surface noncircularity and possible C-layer bridge

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                              |
| Context class                         | ReferenceGatePhase                                                                                                                                                                             |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | not_started                                                                                                                                                                                    |
| Preparation status                    | planned                                                                                                                                                                                        |
| Phase type                            | comparison                                                                                                                                                                                     |
| Phase-type-specific contract          | comparison scalar architecture obligations                                                                                                                                                     |
| Title / state                         | Cayley-Dickson scalar-surface noncircularity and possible C-layer bridge                                                                                                                       |
| Preparation workflow                  | computable scalar architecture workflow                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source-to-CNNA mapping                 | CD9 records whether CD supplies a genuine scalar layer or only comparison/audit data. Horn/GA profiles are routed through GA0-GA6 comparison objects only, with no generator or handoff rights.                                                                                         |
| Dependency-depth audit                 | Scalar architecture decisions must precede code implementation. Exact quadratic or CD scalar routes may not feed earlier generator definitions.                                                                                                                                         |
| Phase split decision                   | split out because ExecComplex limits affect Phase 1.2, 4, 5, 7, 8 and CD bridges differently.                                                                                                                                                                                           |
| Positive path plan                     | Keep current QxQ path first;<br>add exact quadratic pilot only if needed by finite tests and no duplicate module exists;<br>keep analytic Complex mirror-only.                                                                                                                          |
| Module implementation manifest         | pending Phase 1.1.1.1.1                                                                                                                                                                                                                                                                 |
| Semantic placement decision            | scalar-foundation / bridge governance, not generator semantics                                                                                                                                                                                                                          |
| Reuse and duplication audit            | must check existing ExecComplex, ExecComplexBridge, Schur.CharpolyC, ArithmeticCDBridge, and mathlib QuadraticAlgebra before proposing new module                                                                                                                                       |
| Canonical generator policy             | scalar layer must consume canonical generator outputs only;<br>cannot define generator semantics                                                                                                                                                                                        |
| Module compactness rule                | do not create ExactQuadraticComplex unless existing foundation/bridge modules cannot carry the semantics cleanly                                                                                                                                                                        |
| Refactor regression protocol           | incremental wrapper mode;<br>full local build green after each migration step                                                                                                                                                                                                           |
| Refactor migration metrics             | track declarations migrated, imports changed, executable witnesses added                                                                                                                                                                                                                |
| Lean/mathlib version policy            | 4.28.0 baseline unless version gate explicitly opens upgrade                                                                                                                                                                                                                            |
| ExecComplex boundary policy            | EXEC-3 enforced: QxQ floor, exact quadratic optional, analytic Complex mirror-only                                                                                                                                                                                                      |
| Consumer-contract direction            | scalar layer consumes generator/arithmetic source data;<br>generator definitions do not consume scalar comparison/mirror data                                                                                                                                                           |
| Governance-overhead review             | review after first scalar/module manifest whether this lock forced a concrete decision.                                                                                                                                                                                                 |
| Proof rehearsal before implementation  | Only after Schur-duality/CD compatibility closure, prove CD surface generated independently of ExecComplex and not consuming later arithmetic. V1.25 adds Horn/GA material only as downstream target-gate/scratchpad context;<br>it cannot be used as a source object or release proof. |
| Expected tactics                       | explicit structures, constructor/projection lemmas, ring-normal-form only for coordinate arithmetic, no Matrix.inv and no consumer-side patch.                                                                                                                                          |
| Expected mathlib imports               | Mathlib.Algebra.QuadraticAlgebra only if pilot approved;<br>Mathlib.Data.Complex.Basic remains bridge/mirror-only;<br>NumberField modules require version-policy gate.                                                                                                                  |
| ExecComplex/computable plan            | No operational scalar consumption unless bridge is proved.                                                                                                                                                                                                                              |
| Concrete computable instance traversal | required later: QxQ baseline row, Q(i) row, Q(sqrt(-3)) row, and at least one generic D row if O09 is implemented                                                                                                                                                                       |
| Computable path non-skeleton status    | must be shown by concrete row witnesses                                                                                                                                                                                                                                                 |
| Expected executable witness            | finite row with rational ExecComplex output plus optional exact quadratic coordinate and rational/integer invariant projection                                                                                                                                                          |
| Mathlib complication log               | Record Complex/Real noncomputable boundary, QuadraticAlgebra instance diamond warning, NumberField formal/non-executable dependency, and any required version-gate.                                                                                                                     |
| Noncomputable boundary audit           | General C/R and algebraic closure remain mirror/formal. Exact quadratic candidate must avoid noncomputable definitions on operational path.                                                                                                                                             |
| Mathlib mitigation/refactor layer      | Use project-owned explicit Q-coordinate structure or narrowly wrapped QuadraticAlgebra;<br>isolate formal mathlib NumberField in mirror/proof layer.                                                                                                                                    |
| Bridge gap status                      | Open until late bridge prerequisites close.                                                                                                                                                                                                                                             |
| Natural backjump/refactor layer        | If exact quadratic data are needed, backjump to scalar-foundation layer rather than patching arithmetic/lattice consumers.                                                                                                                                                              |
| Implementation prompt contract         | Before Lean code, module manifest must state exact module placement, imported mathlib modules, and why no existing module owns the semantics.                                                                                                                                           |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                             |
| How implemented                        | Not implemented.                                                                                                                                                                                                                                                                        |
| Verification / evidence                | Generated-table validation pending/run after patch;<br>no Lean build.                                                                                                                                                                                                                   |
| Proof strategy                         | Only after Schur-duality/CD compatibility closure, prove CD surface generated independently of ExecComplex and not consuming later arithmetic.                                                                                                                                          |
| Build result                           | No Lean build executed or claimed.                                                                                                                                                                                                                                                      |
| Post-implementation plan check         | Check whether the scalar architecture lock prevented analytic/noncomputable leakage or duplicate scalar implementation.                                                                                                                                                                 |
| Artifact packaging rule                | Return complete zip;<br>Lean build local unless actually executed here.                                                                                                                                                                                                                 |

|                                      |                                                                                                                                                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation notes                 | Late bridge audit for possible CD-derived C layer.                                                                                                                                                                            |
| Source materials / references        | <ul style="list-style-type: none"> <li>CD7/CD8/BR08/BR09 and ArithmeticCDBridge no-identification status.</li> <li>Horn 0709.2238v1 quaternion/geometric-algebra reference-gate intake (V1.25; comparison-only)</li> </ul>    |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>CD scalar surface candidate; noncircularity proof; bridge to O09/O00 if possible.</li> </ul>                                                                                           |
| Implemented modules                  | <ul style="list-style-type: none"> <li>none</li> </ul>                                                                                                                                                                        |
| Added modules                        | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                    |
| Modified modules                     | <ul style="list-style-type: none"> <li>none yet</li> </ul>                                                                                                                                                                    |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>Global no-go guardrails inherited.</li> </ul>                                                                                                                                          |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>No retroactive use of CD story as source of ExecComplex or O09.</li> </ul>                                                                                                             |
| Proposed added modules               | <ul style="list-style-type: none"> <li>none at decision gate; possible CNNA/PillarA/Foundation/ExactQuadraticComplex.lean only after manifest</li> </ul>                                                                      |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>pending manifest</li> </ul>                                                                                                                                                            |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Foundation/ExecComplex.lean; Foundation/ExecComplexBridge.lean; Arithmetic/Schur/CharpolyC.lean; Arithmetic/CayleyDickson/ArithmeticCDBridge.lean; mathlib QuadraticAlgebra</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>alternate complex scalar, algebraic complex, CD complex, mirror Complex, exact quadratic norm/trace/conjugation</li> </ul>                                                             |

### Phase 10.5 – Spectral/physical and transfer path closeout before late bridges and Pillar-B export

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                       |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                     |
| Phase documentation tier              | F1                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                            |
| Phase type                            | derivation                                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Title / state                         | Spectral/physical and transfer path closeout before late bridges and Pillar-B export                                                                                                                                    |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Source-to-CNNA mapping                | Object placement follows the corrected v37 path split; phase titles and object groups must match the actual mathematical source dependencies.                                                                           |

|                                        |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise. |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                                                                                                               |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                         |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                               |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                  |
| Consumer-contract direction            | Use minimal source objects: Path B consumes Phase 4.1 directly; Path A consumes D/order/lattice/modular prerequisites; bridges are explicit and late.                                                                                                                                                                                                     |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                       |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                         |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                       |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                   |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                 |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                     |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                         |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                       |
| Post-implementation plan check         | After implementation, verify imports and objects stayed within the v37 path split.                                                                                                                                                                                                                                                                        |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                            |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                            |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>Global no-go inherited; additionally no artificial modular-form/j dependency for finite spectral/physical consumers and no phase-title/object mismatch.</li> </ul>                                                                                                                                                 |

### Phase 10.6 – Moved to Phase 13: AQFT KMS/GNS/Tomita-Takesaki consumer targets

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field        | Record |
|--------------|--------|
| Phase status | W      |

|                                       |                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                         | DocumentationPhase                                                                                                                                                                                                      |
| Phase documentation tier              | F3                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                               |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | documentation                                                                                                                                                                                                           |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise.                  |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 10.7 – Moved to Phase 13: CPT/spin-statistics/Kramers/split/local-net consumer targets

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFFFFF; border: 1px solid black; padding: 0 2px;">W</span>                                                                                                                              |
| Context class                         | DocumentationPhase                                                                                                                                                                                                      |
| Phase documentation tier              | F3                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                               |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | documentation                                                                                                                                                                                                           |
| Phase-type-specific contract          | Refactor phase: record module/declaration migration metrics, import delta, consumer breakage audit, wrapper coexistence, and local full-build evidence before closeout.                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | v37 repaired Phase 5-10 dependencies: Path A number-theory/geometric/modular/analytic and Path B spectral/physical/transfer are parallel after Phase 4 unless a specific object bridge says otherwise.                  |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 11 – handoff bridges from canonical A-objects

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFB6C1; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |



|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation status                  | planned-not-active                                                                                                                                                                                                                                                    |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | handoff bridges from canonical A-objects                                                                                                                                                                                                                              |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                        |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                           |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global__no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 11.1 – Derive/register object group: A->B bright handoff

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFB6C1;">R</span>                                                                                                                                                                          |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                        |
| Phase documentation tier              | F0                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                    |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                 | planned-not-active                                                                                                                                                                                                         |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                               |
| Phase type                            | derivation                                                                                                                                                                                                                 |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                  |
| Title / state                         | Derive/register object group: A->B bright handoff                                                                                                                                                                          |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.    |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                          |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.         |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                          |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                          |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                          |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                          |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                          |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 11.2 – Derive/register object group: A->B prime/dark future handoff

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFB6C1; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                      |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                            |
| Phase type                            | derivation                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.               |
| Title / state                         | Derive/register object group: A->B prime/dark future handoff                                                                                                                                            |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                           |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                              |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 11.3 – Derive/register object group: A->C handoff

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                   |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F0                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                  |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                           |
| Title / state                         | Derive/register object group: A->C handoff                                                                                                                                                                                                          |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                             |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                   |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                  |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                          |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                 |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                   |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                   |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                   |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 11.4 – Derive/register object group: A->D handoff

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFB6C1; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                      |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                            |
| Phase type                            | derivation                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.               |
| Title / state                         | Derive/register object group: A->D handoff                                                                                                                                                              |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                           |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                              |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                     |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                   |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                        |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 11.5 – Derive/register object group: A->E handoff

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                   |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F0                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned-not-active                                                                                                                                                                                                                                  |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                           |
| Title / state                         | Derive/register object group: A->E handoff                                                                                                                                                                                                          |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                             |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                   |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                  |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                          |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                 |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                   |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                   |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                   |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

## Phase 12 – Established mathematics beyond j: finite simple groups, Leech/VOA, Monster/Moonshine and post-j arithmetic register

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                        |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                             |
| Context class                         | ReferenceGatePhase                                                                                                                                                            |
| Phase documentation tier              | F2                                                                                                                                                                            |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                            |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                           |
| Implementation status                 | planned-not-active                                                                                                                                                            |
| Preparation status                    | source-audited-not-active                                                                                                                                                     |
| Phase type                            | comparison                                                                                                                                                                    |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                 |
| Title / state                         | Established mathematics beyond j: finite simple groups, Leech/VOA, Monster/Moonshine and post-j arithmetic register                                                           |

|                                        |                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                                                                             |
| Source-to-CNNA mapping                 | Thread data are roadmap/reference material only. CNNA must first derive j/q-expansion or a graded/lattice/algebraic stream from Pillar-A provenance; only then may finite simple groups, Leech, VOA, Monster, Hauptmoduln or McKay-Thompson series be used as comparison targets.                                   |
| Dependency-depth audit                 | Deep and intentionally late. Current plan already has j/Moonshine placeholders, but TomS thread requires finer comparison-only subphases for finite simple groups, Leech/theta, VOA/moonshine module, and common-indicator/Hauptmodul tests.                                                                        |
| Phase split decision                   | Split phase 12 into 12.1-12.8: keep 12.1-12.3 for j/Moonshine/McKay-Thompson; add 12.4-12.8 for finite-simple-group context, Leech lattice, Leech theta, VOA/moonshine module, and common-indicator bridge tests.                                                                                                   |
| Positive path plan                     | Maintain the positive derived-only path: build A-derived j/q data first; compare to external Moonshine structures later; do not use thread claims, Monster data, Leech coefficients, L-functions, or string-theory/VOA facts as upstream axioms.                                                                    |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                   |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                         |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                          |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                                                       |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                 |
| Proof rehearsal before implementation  | No Lean implementation yet. Formalization would start with small comparison records and computable coefficient prefixes; analytic/classification/VOA content remains behind explicit mirror/comparison boundaries.                                                                                                  |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                   |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                   |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                   |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                 |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                             |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                   |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                        |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                           |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                               |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                   |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                               |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                         |
| How implemented                        | No implementation record. The phase is a planning row only and must remain untouched until it becomes the unique active frontier.                                                                                                                                                                                   |
| Verification / evidence                | Not applicable yet.                                                                                                                                                                                                                                                                                                 |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                  |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                      |
| Implementation notes                   | When activated, first check dependency depth and split into subphases if the target consumes too many objects or proof layers.                                                                                                                                                                                      |
| Source materials / references          | <ul style="list-style-type: none"> <li>TomS astronews thread pages 1-5 (#1-#95), April-May 2026</li> <li>Borcherds 1992 Inventiones: Monstrous moonshine and monstrous Lie superalgebras</li> <li>Warwick WMI overview of CFSG taxonomy</li> <li>Borcherds 1985 Proc. R. Soc. Lond. A: The Leech lattice</li> </ul> |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"> <li>j/q-expansion coefficient stream</li> <li>L-functions as indirect/derived via modular forms, not Riemann-zeta-first</li> <li>finite simple group taxonomy</li> <li>sporadic/Monster comparison</li> <li>Leech lattice/rootless lattice comparison</li> <li>Leech theta/shell coefficient comparison</li> <li>VOA/moonshine module comparison</li> <li>Hauptmodul/McKay-Thompson common-indicator bridge test</li> </ul>                                                                                                                                                                                                                                                                                                                                       |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 12.1 – q-expansion coefficient stream for j as established arithmetic output

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFB6C1; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                                                 |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                        |
| Phase documentation tier              | F2                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                              |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                 | planned                                                                                                                                                                                                                    |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                               |
| Phase type                            | derivation                                                                                                                                                                                                                 |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                  |
| Title / state                         | q-expansion coefficient stream for j as established arithmetic output                                                                                                                                                      |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.    |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                          |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.         |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                          |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 12.2 – Moonshine comparison profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F2                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                                 |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 12.3 – McKay-Thompson side profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                          |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                        |
| Phase documentation tier              | F2                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                              |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                 | planned                                                                                                                                                                                                                    |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                               |
| Phase type                            | derivation                                                                                                                                                                                                                 |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                  |
| Title / state                         | McKay-Thompson side profile                                                                                                                                                                                                |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.    |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                          |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.         |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                          |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 12.4 – Finite simple group comparison context after modular/j layer

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field        | Record |
|--------------|--------|
| Phase status | W      |



|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                          | ReferenceGatePhase                                                                                                                                                                                                                                                    |
| Phase documentation tier               | F2                                                                                                                                                                                                                                                                    |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                         |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                    |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                   |
| Implementation status                  | planned                                                                                                                                                                                                                                                               |
| Preparation status                     | source-audited-not-active                                                                                                                                                                                                                                             |
| Phase type                             | comparison                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                                                                         |
| Title / state                          | Finite simple group comparison context after modular/j layer                                                                                                                                                                                                          |
| Preparation workflow                   | Source-preparation workflow completed at roadmap level only: extract thread objects, classify as derived target vs comparison-only, map to phase/object IDs, and keep formal proof rehearsal for activation time.                                                     |
| Source-to-CNNA mapping                 | CFSG is a comparison taxonomy for possible CNNA-derived symmetry/graded objects;<br>it is not an A-source theorem.                                                                                                                                                    |
| Dependency-depth audit                 | Very deep. Requires completed A-derived j/q-expansion or later graded/lattice/algebra output before any comparison is meaningful. Therefore it remains black/white horizon material, not an active proof target.                                                      |
| Phase split decision                   | Keep as a planned comparison-only subphase;<br>split further when activated if formal objects require deep algebraic prerequisites.                                                                                                                                   |
| Positive path plan                     | Do not start by falsifying or importing an established target. First derive CNNA data from A provenance;<br>then perform finite-prefix or theorem-level comparison against the registered external target.                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                 |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | Pending until all predecessor CNNA objects are available. Expected shape: define comparison records, prove coefficient/table equality for finite prefixes if computable, and isolate analytic/global equality as mirror theorem if needed.                            |
| Expected tactics                       | Use explicit records, finite list/vector comparisons, Nat/Int arithmetic certificates, and theorem forwarding. Avoid broad simplification or opaque imported classification assumptions.                                                                              |
| Expected mathlib imports               | To be verified against mathlib 4.28.0 at activation time. Likely areas: algebra/group/basic finite groups for comparison tags, polynomials/series for q-prefixes, no large external theorem import in operational path.                                               |
| ExecComplex/computable plan            | Any comparison prefix must be computable data. Infinite series, analytic equality, VOA structures, and classification theorems remain mirror/comparison layers unless separately derived.                                                                             |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No immediate Lean bridge required. Missing structures belong earlier: j/q-expansion, graded q-character, lattice/theta candidate, or B-derived algebraic consumer, depending on the subphase.                                                                         |
| Natural backjump/refactor layer        | If activation reveals missing data, backjump to the earliest natural generator: modular q-expansion for MOON0, graded-character source before MOON8, lattice/theta source before MOON5-6, or B algebra before MOON7.                                                  |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master after TomS thread cross-check.                                                                                                                                             |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / evidence              | Pending;<br>phase is not active. Source material was classified but no Lean proof is claimed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Proof strategy                       | To be specified when this phase becomes active. Positive path: derive CNNA-side stream/object first, then compare to external target; never import target data as a generator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Build result                         | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Post-implementation plan check       | Not available yet. When implemented, mark object status white only if provenance, access rights, and comparison/derivation boundary are explicit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Artifact packaging rule              | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Implementation notes                 | Roadmap placeholder;<br>does not assert the target is proven or derivable yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Source materials / references        | <ul style="list-style-type: none"> <li>TomS #43-#56 finite simple group overview</li> <li>Warwick WMI CFSG overview</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>CFSG taxonomy tags</li> <li>cyclic prime-order groups</li> <li>alternating groups</li> <li>groups of Lie type</li> <li>sporadic groups</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>no external established structure as CNNA source</li> <li>no Monster/Leech/VOA/string-theory axiom import into Pillar A</li> <li>no L-function shortcut before modular/character provenance exists</li> </ul> |

### Phase 12.5 – Leech lattice comparison profile

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                        |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                             |
| Context class                         | ReferenceGatePhase                                                                                                                                                            |
| Phase documentation tier              | F2                                                                                                                                                                            |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                            |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                           |
| Implementation status                 | planned                                                                                                                                                                       |
| Preparation status                    | source-audited-not-active                                                                                                                                                     |
| Phase type                            | comparison                                                                                                                                                                    |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                 |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title / state                          | Leech lattice comparison profile                                                                                                                                                                                                                                      |
| Preparation workflow                   | Source-preparation workflow completed at roadmap level only: extract thread objects, classify as derived target vs comparison-only, map to phase/object IDs, and keep formal proof rehearsal for activation time.                                                     |
| Source-to-CNNA mapping                 | Leech data are registered only as target fingerprints for later derived lattice/theta-like CNNA outputs.                                                                                                                                                              |
| Dependency-depth audit                 | Very deep. Requires completed A-derived j/q-expansion or later graded/lattice/algebra output before any comparison is meaningful. Therefore it remains black/white horizon material, not an active proof target.                                                      |
| Phase split decision                   | Keep as a planned comparison-only subphase;<br>split further when activated if formal objects require deep algebraic prerequisites.                                                                                                                                   |
| Positive path plan                     | Do not start by falsifying or importing an established target. First derive CNNA data from A provenance;<br>then perform finite-prefix or theorem-level comparison against the registered external target.                                                            |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                           |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | Pending until all predecessor CNNA objects are available. Expected shape: define comparison records, prove coefficient/table equality for finite prefixes if computable, and isolate analytic/global equality as mirror theorem if needed.                            |
| Expected tactics                       | Use explicit records, finite list/vector comparisons, Nat/Int arithmetic certificates, and theorem forwarding. Avoid broad simplification or opaque imported classification assumptions.                                                                              |
| Expected mathlib imports               | To be verified against mathlib 4.28.0 at activation time. Likely areas: algebra/group/basic finite groups for comparison tags, polynomials/series for q-prefixes, no large external theorem import in operational path.                                               |
| ExecComplex/computable plan            | Any comparison prefix must be computable data. Infinite series, analytic equality, VOA structures, and classification theorems remain mirror/comparison layers unless separately derived.                                                                             |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No immediate Lean bridge required. Missing structures belong earlier: j/q-expansion, graded q-character, lattice/theta candidate, or B-derived algebraic consumer, depending on the subphase.                                                                         |
| Natural backjump/refactor layer        | If activation reveals missing data, backjump to the earliest natural generator: modular q-expansion for MOON0, graded-character source before MOON8, lattice/theta source before MOON5-6, or B algebra before MOON7.                                                  |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master after TomS thread cross-check.                                                                                                                                             |
| Verification / evidence                | Pending;<br>phase is not active. Source material was classified but no Lean proof is claimed.                                                                                                                                                                         |
| Proof strategy                         | To be specified when this phase becomes active. Positive path: derive CNNA-side stream/object first, then compare to external target;<br>never import target data as a generator.                                                                                     |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet. When implemented, mark object status white only if provenance, access rights, and comparison/derivation boundary are explicit.                                                                                                                     |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven or derivable yet.                                                                                                                                                                                        |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source materials / references        | <ul style="list-style-type: none"> <li>TomS #84/#93 Leech lattice discussion</li> <li>Borcherds 1985 The Leech lattice</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>rootless Leech lattice target</li> <li>24-dimensional even unimodular lattice comparison tag</li> <li>Niemeier/Leech no-root warning</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>no external established structure as CNNA source</li> <li>no Monster/Leech/VOA/string-theory axiom import into Pillar A</li> <li>no L-function shortcut before modular/character provenance exists</li> </ul> |

### Phase 12.6 – Leech theta/shell and kissing-number comparison stream

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                            |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                 |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                |
| Phase documentation tier              | F2                                                                                                                                                                                                                |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                     |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                               |
| Implementation status                 | planned                                                                                                                                                                                                           |
| Preparation status                    | source-audited-not-active                                                                                                                                                                                         |
| Phase type                            | comparison                                                                                                                                                                                                        |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                     |
| Title / state                         | Leech theta/shell and kissing-number comparison stream                                                                                                                                                            |
| Preparation workflow                  | Source-preparation workflow completed at roadmap level only: extract thread objects, classify as derived target vs comparison-only, map to phase/object IDs, and keep formal proof rehearsal for activation time. |
| Source-to-CNNA mapping                | The theta/shell numbers may be finite comparison targets only after CNNA produces its own coefficient stream.                                                                                                     |
| Dependency-depth audit                | Very deep. Requires completed A-derived j/q-expansion or later graded/lattice/algebra output before any comparison is meaningful. Therefore it remains black/white horizon material, not an active proof target.  |
| Phase split decision                  | Keep as a planned comparison-only subphase; split further when activated if formal objects require deep algebraic prerequisites.                                                                                  |
| Positive path plan                    | Do not start by falsifying or importing an established target. First derive CNNA data from A provenance; then perform finite-prefix or theorem-level comparison against the registered external target.           |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                 |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                       |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | Pending until all predecessor CNNA objects are available. Expected shape: define comparison records, prove coefficient/table equality for finite prefixes if computable, and isolate analytic/global equality as mirror theorem if needed.                            |
| Expected tactics                       | Use explicit records, finite list/vector comparisons, $\mathbb{N}$ / $\mathbb{Z}$ arithmetic certificates, and theorem forwarding. Avoid broad simplification or opaque imported classification assumptions.                                                          |
| Expected mathlib imports               | To be verified against mathlib 4.28.0 at activation time. Likely areas: algebra/group/basic finite groups for comparison tags, polynomials/series for q-prefixes, no large external theorem import in operational path.                                               |
| ExecComplex/computable plan            | Any comparison prefix must be computable data. Infinite series, analytic equality, VOA structures, and classification theorems remain mirror/comparison layers unless separately derived.                                                                             |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No immediate Lean bridge required. Missing structures belong earlier: j/q-expansion, graded q-character, lattice/theta candidate, or B-derived algebraic consumer, depending on the subphase.                                                                         |
| Natural backjump/refactor layer        | If activation reveals missing data, backjump to the earliest natural generator: modular q-expansion for MOON0, graded-character source before MOON8, lattice/theta source before MOON5-6, or B algebra before MOON7.                                                  |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                              |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master after TomS thread cross-check.                                                                                                                                             |
| Verification / evidence                | Pending;<br>phase is not active. Source material was classified but no Lean proof is claimed.                                                                                                                                                                         |
| Proof strategy                         | To be specified when this phase becomes active. Positive path: derive CNNA-side stream/object first, then compare to external target; never import target data as a generator.                                                                                        |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet. When implemented, mark object status white only if provenance, access rights, and comparison/derivation boundary are explicit.                                                                                                                     |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven or derivable yet.                                                                                                                                                                                        |
| Source materials / references          | <ul style="list-style-type: none"> <li>TomS #93 Leech theta/shell discussion</li> <li>Borcherds 1985 The Leech lattice</li> </ul>                                                                                                                                     |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>theta series comparison</li> <li>shell coefficient stream</li> <li>196560 minimal-vector/kissing-number target as comparison only</li> </ul>                                                                                   |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: top-level global__no_go_guardrails apply; this phase list is local addenda only.</li> <li>• no @[simp] attributes in the operational path</li> <li>• no noncomputable in the operational path</li> <li>• no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>• no Matrix.inv or external inverse oracle in operational derivations</li> <li>• no free formula without proof-carrying entry structure</li> <li>• no consumer-side patch when the missing structure belongs earlier</li> <li>• no premature obstruction before a serious positive construction attempt</li> <li>• no phase-semantic public API names</li> <li>• no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>• no external established structure as CNNA source</li> <li>• no Monster/Leech/VOA/string-theory axiom import into Pillar A</li> <li>• no L-function shortcut before modular/character provenance exists</li> </ul> |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 12.7 – Vertex-operator-algebra / moonshine-module comparison boundary

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                          |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                         |
| Phase documentation tier              | F2                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                              |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                 | planned                                                                                                                                                                                                                    |
| Preparation status                    | source-audited-not-active                                                                                                                                                                                                  |
| Phase type                            | comparison                                                                                                                                                                                                                 |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                              |
| Title / state                         | Vertex-operator-algebra / moonshine-module comparison boundary                                                                                                                                                             |
| Preparation workflow                  | Source-preparation workflow completed at roadmap level only: extract thread objects, classify as derived target vs comparison-only, map to phase/object IDs, and keep formal proof rehearsal for activation time.          |
| Source-to-CNNA mapping                | VOA and string-theory context are late comparison/consumer targets, not generative inputs.                                                                                                                                 |
| Dependency-depth audit                | Very deep. Requires completed A-derived j/q-expansion or later graded/lattice/algebra output before any comparison is meaningful. Therefore it remains black/white horizon material, not an active proof target.           |
| Phase split decision                  | Keep as a planned comparison-only subphase; split further when activated if formal objects require deep algebraic prerequisites.                                                                                           |
| Positive path plan                    | Do not start by falsifying or importing an established target. First derive CNNA data from A provenance; then perform finite-prefix or theorem-level comparison against the registered external target.                    |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof rehearsal before implementation  | Pending until all predecessor CNNA objects are available. Expected shape: define comparison records, prove coefficient/table equality for finite prefixes if computable, and isolate analytic/global equality as mirror theorem if needed.                            |
| Expected tactics                       | Use explicit records, finite list/vector comparisons, Nat/Int arithmetic certificates, and theorem forwarding. Avoid broad simplification or opaque imported classification assumptions.                                                                              |
| Expected mathlib imports               | To be verified against mathlib 4.28.0 at activation time. Likely areas: algebra/group/basic finite groups for comparison tags, polynomials/series for q-prefixes, no large external theorem import in operational path.                                               |
| ExecComplex/computable plan            | Any comparison prefix must be computable data. Infinite series, analytic equality, VOA structures, and classification theorems remain mirror/comparison layers unless separately derived.                                                                             |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | No immediate Lean bridge required. Missing structures belong earlier: j/q-expansion, graded q-character, lattice/theta candidate, or B-derived algebraic consumer, depending on the subphase.                                                                         |
| Natural backjump/refactor layer        | If activation reveals missing data, backjump to the earliest natural generator: modular q-expansion for MOON0, graded-character source before MOON8, lattice/theta source before MOON5-6, or B algebra before MOON7.                                                  |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master after TomS thread cross-check.                                                                                                                                             |
| Verification / evidence                | Pending;<br>phase is not active. Source material was classified but no Lean proof is claimed.                                                                                                                                                                         |
| Proof strategy                         | To be specified when this phase becomes active. Positive path: derive CNNA-side stream/object first, then compare to external target; never import target data as a generator.                                                                                        |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet. When implemented, mark object status white only if provenance, access rights, and comparison/derivation boundary are explicit.                                                                                                                     |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven or derivable yet.                                                                                                                                                                                        |
| Source materials / references          | <ul style="list-style-type: none"> <li>TomS #64/#82/#85 VOA and string-theory context</li> <li>Borcherds 1992 Monstrous moonshine and monstrous Lie superalgebras</li> </ul>                                                                                          |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>VOA comparison profile</li> <li>moonshine module comparison</li> <li>chiral CFT/string compactification side context</li> </ul>                                                                                                |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no__go__guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> <li>no external established structure as CNNA source</li> <li>no <code>Monster/Leech/VOA/string-theory</code> axiom import into Pillar A</li> <li>no L-function shortcut before modular/character provenance exists</li> </ul> |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 12.8 – Common-indicator / Hauptmodul bridge-test register

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                                                                                                                   |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                                                                                                                                 |
| Phase documentation tier              | F2                                                                                                                                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                                                                                                                                                                       |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                                                                                                                 |
| Implementation status                 | planned                                                                                                                                                                                                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                                                                                                                           |
| Title / state                         | Common-indicator / Hauptmodul bridge-test register                                                                                                                                                                                                                                                                                                                  |
| Preparation workflow                  | Source-preparation workflow completed at roadmap level only: extract thread objects, classify as derived target vs comparison-only, map to phase/object IDs, and keep formal proof rehearsal for activation time.                                                                                                                                                   |
| Source-to-CNNA mapping                | The common indicator is a post-derived comparison test: derived CNNA q-character first, known Hauptmodul/McKay-Thompson target second.                                                                                                                                                                                                                              |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Very deep. Requires completed A-derived j/q-expansion or later graded/lattice/algebra output before any comparison is meaningful. Therefore it remains black/white horizon material, not an active proof target. |
| Phase split decision                  | Keep as a planned comparison-only subphase; split further when activated if formal objects require deep algebraic prerequisites.                                                                                                                                                                                                                                    |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Do not start by falsifying or importing an established target. First derive CNNA data from A provenance; then perform finite-prefix or theorem-level comparison against the registered external target.   |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation.                                                                                                    |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                          |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                            |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                                                                                                       |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending until all predecessor CNNA objects are available. Expected shape: define comparison records, prove coefficient/table equality for finite prefixes if computable, and isolate analytic/global equality as mirror theorem if needed. |
| Expected tactics                       | Use explicit records, finite list/vector comparisons, Nat/Int arithmetic certificates, and theorem forwarding. Avoid broad simplification or opaque imported classification assumptions.                                                                                                                                                                                              |
| Expected mathlib imports               | To be verified against mathlib 4.28.0 at activation time. Likely areas: algebra/group/basic finite groups for comparison tags, polynomials/series for q-prefixes, no large external theorem import in operational path.                                                                                                                                                               |
| ExecComplex/computable plan            | Any comparison prefix must be computable data. Infinite series, analytic equality, VOA structures, and classification theorems remain mirror/comparison layers unless separately derived.                                                                                                                                                                                             |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                             |
| Bridge gap status                      | No immediate Lean bridge required. Missing structures belong earlier: j/q-expansion, graded q-character, lattice/theta candidate, or B-derived algebraic consumer, depending on the subphase.                                                                                                                                                                                         |
| Natural backjump/refactor layer        | If activation reveals missing data, backjump to the earliest natural generator: modular q-expansion for MOON0, graded-character source before MOON8, lattice/theta source before MOON5-6, or B algebra before MOON7.                                                                                                                                                                  |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                 |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master after TomS thread cross-check.                                                                                                                                                                                                                                                             |
| Verification / evidence                | Pending;<br>phase is not active. Source material was classified but no Lean proof is claimed.                                                                                                                                                                                                                                                                                         |
| Proof strategy                         | To be specified when this phase becomes active. Positive path: derive CNNA-side stream/object first, then compare to external target;<br>never import target data as a generator.                                                                                                                                                                                                     |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet. When implemented, mark object status white only if provenance, access rights, and comparison/derivation boundary are explicit.                                                                                                                                                                                                                                     |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.                                                                                                                     |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven or derivable yet.                                                                                                                                                                                                                                                                                                        |
| Source materials / references          | <ul style="list-style-type: none"> <li>TomS #67/#70/#71 and #64 Moonshine discussion</li> <li>Borcherds 1992;<br/>Conway-Norton/Borcherds moonshine context</li> </ul>                                                                                                                                                                                                                |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Hauptmodul bridge test</li> <li>McKay-Thompson finite-prefix comparison</li> <li>common indicator across modular/finite-group/lattice structures</li> </ul>                                                                                                                                                                                    |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: top-level global__no_go_guardrails apply; this phase list is local addenda only.</li><li>• no @[simp] attributes in the operational path</li><li>• no noncomputable in the operational path</li><li>• no classical helper unless isolated in allowed mirror/non-operational layer</li><li>• no Matrix.inv or external inverse oracle in operational derivations</li><li>• no free formula without proof-carrying entry structure</li><li>• no consumer-side patch when the missing structure belongs earlier</li><li>• no premature obstruction before a serious positive construction attempt</li><li>• no phase-semantic public API names</li><li>• no fragile rewrite chain when a structural constructor/projection proof is available</li><li>• no external established structure as CNNA source</li><li>• no Monster/Leech/VOA/string-theory axiom import into Pillar A</li><li>• no L-function shortcut before modular/character provenance exists</li></ul> |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 13 – Pillar B QFT/AQFT consumer roadmap from A outputs

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid #000; padding: 0 2px;">R</span>                                                                                                                                  |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                        |
| Phase documentation tier              | F0                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                    |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                 | planned                                                                                                                                                                                                                    |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                               |
| Phase type                            | derivation                                                                                                                                                                                                                 |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                  |
| Title / state                         | Pillar B QFT/AQFT consumer roadmap from A outputs                                                                                                                                                                          |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.    |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                          |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.         |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                          |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                          |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 13.1 – B input contract from A bright boundary handoff

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                       |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                    |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                   |
| Implementation status                  | planned                                                                                                                                                                                                                                                               |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | B input contract from A bright boundary handoff                                                                                                                                                                                                                       |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                        |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                 |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Artifact packaging rule | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation notes    | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| No-go guardrails        | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 13.2 – Boundary star-algebra / observable algebra

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                                                                                       |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                              |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                               |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                   |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                               |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                              |
| Implementation status                 | planned                                                                                                                                                                                                                                                          |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                                     |
| Phase type                            | derivation                                                                                                                                                                                                                                                       |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                        |
| Title / state                         | Boundary star-algebra / observable algebra                                                                                                                                                                                                                       |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                          |
| Source-to-CNNA mapping                | Not prepared yet. Horn/GA profiles are routed through GA0-GA6 comparison objects only, with no generator or handoff rights.                                                                                                                                      |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                               |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                                |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                      |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                   |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and <code>legacy/coexistence</code> status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                       |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                               |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                    |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                              |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts. V1.25 adds Horn/GA material only as downstream target-gate/scratchpad context; it cannot be used as a source object or release proof.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Source materials / references          | <ul style="list-style-type: none"> <li>Horn 0709.2238v1 quaternion/geometric-algebra reference-gate intake (V1.25; comparison-only)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 13.2.1 – Pauli/Clifford boundary-star-algebra comparison gate

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.



| Field                                  | Record                                                                                                                                                                                                                    |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                         |
| Context class                          | ReferenceGatePhase                                                                                                                                                                                                        |
| Phase documentation tier               | F2                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                        |
| Preparation status                     | planned-prepared                                                                                                                                                                                                          |
| Phase type                             | comparison                                                                                                                                                                                                                |
| Phase-type-specific contract           | Comparison/reference phase: records external target criteria and scratchpad information; it cannot supply CNNA premises or release mathematical phases.                                                                   |
| Title / state                          | Pauli/Clifford boundary-star-algebra comparison gate                                                                                                                                                                      |
| Preparation workflow                   | External-source intake as comparison/reference gate with derived-only boundary.                                                                                                                                           |
| Source-to-CNNA mapping                 | Maps external Pauli/Clifford patterns to a later B-consumer check after boundary *-algebra is derived from A->B handoff.                                                                                                  |
| Dependency-depth audit                 | Locked behind 13.2 and A->B handoff data. It cannot inspect A internals or supply B generators.                                                                                                                           |
| Phase split decision                   | Split into narrow comparison gates so CD, Pillar-B boundary algebra, Dirac/STA and side-comparison targets do not collapse into one ambiguous external-theory phase.                                                      |
| Positive path plan                     | Do not implement proof code yet. Keep target invariants in scratchpad; when predecessor data is closed, add a small neutral comparison API and prove/falsify explicitly.                                                  |
| Module implementation manifest         | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.                                                                                     |
| Semantic placement decision            | Natural placement is comparison/reference-gate layer, not a foundational CNNA derivation layer.                                                                                                                           |
| Reuse and duplication audit            | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.                                                                                  |
| Canonical generator policy             | The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.                                                                                          |
| Module compactness rule                | Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists.                                                                         |
| Refactor regression protocol           | No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.                                                                                |
| Refactor migration metrics             | No Lean migration. Future comparison module metrics: declarations added, imports added, consumer count, and target/equality proofs closed.                                                                                |
| Lean/mathlib version policy            | Baseline remains pinned Lean/mathlib snapshot; live documentation is only a source-discovery hint until verified locally against the pin.                                                                                 |
| ExecComplex boundary policy            | Do not replace ExecComplex or CNNA scalar provenance with external complex quaternions or analytic Complex.                                                                                                               |
| Consumer-contract direction            | Comparison consumers may read CNNA-derived outputs; CNNA derivation phases may not consume the comparison profile as a generator.                                                                                         |
| Governance-overhead review             | Added overhead is justified because sign, unit and semantic ambiguities are high-risk; gates are narrow to avoid broad external-theory creep.                                                                             |
| Proof rehearsal before implementation  | Future proof should derive B generators from boundary algebra, then compare against Clifford target relations through a minimal profile or mathlib Clifford lift only if version-pinned and noncircular.                  |
| Expected tactics                       | Future Lean work should use explicit structures, field/projection equalities, constructor-based proofs, local case splits and finite matrix entry proofs; avoid broad simp/simpa collapse and opaque native_decide goals. |
| Expected mathlib imports               | Reference candidates only: Mathlib.Algebra.Quaternion, Mathlib.Analysis.Quaternion, Mathlib.LinearAlgebra.CliffordAlgebra.Basic. Verify against the pinned Lean/mathlib snapshot before any implementation.               |
| ExecComplex/computable plan            | No ExecComplex object introduced. If finite numeric checks are later needed, keep them mirror/diagnostic unless an exact CNNA-derived computable path is proved.                                                          |
| Concrete computable instance traversal | Concrete traversal currently consists of source equation/profile extraction and target-object registration; no CNNA runtime traversal yet.                                                                                |
| Computable path non-skeleton status    | Reference skeleton only. It must not be presented as computable CNNA evidence until future exact instances are constructed.                                                                                               |
| Expected executable witness            | For V1.25: generated TSV/TEX/PDF artifacts. Future Lean witness: explicit comparison structure instantiated from CNNA-derived objects.                                                                                    |
| Mathlib complication log               | Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0; implementation must recheck exact names and theorem availability.                                    |
| Noncomputable boundary audit           | No noncomputable Lean code introduced. Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed.                                                                   |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathlib mitigation/refactor layer    | If mathlib Clifford/Quaternion APIs pull noncomputable/classical pressure, keep them in mirror/comparison modules or define a minimal finite target profile instead.                                                                                                                                                                                                                                           |
| Bridge gap status                    | Bridge intentionally open. The external reference describes target shapes; CNNA must derive any matching structure internally before comparison.                                                                                                                                                                                                                                                               |
| Natural backjump/refactor layer      | If the comparison needs missing structure, patch only the earliest natural CNNA layer where that structure is generated, not the comparison/consumer phase.                                                                                                                                                                                                                                                    |
| Implementation prompt contract       | Future implementation prompt must state comparison-only status, predecessor objects, no new generator axioms, no external-theory imports unless pinned and isolated, and exact target equalities/failures.                                                                                                                                                                                                     |
| Global no-go inheritance             | All global no-go guardrails apply, especially derived-only, positive-path-first, patch-at-natural-origin and reference-theory-as-target-only rules.                                                                                                                                                                                                                                                            |
| How implemented                      | Planned reference/target-gate integration only. V1.25 records source reading, object links, proof-dossier placeholders and scratchpad material; no Lean code is implemented.                                                                                                                                                                                                                                   |
| Verification / evidence              | YAML generator validation and generated table/PDF outputs after V1.25 update. No Lean build evidence because no Lean source changed.                                                                                                                                                                                                                                                                           |
| Proof strategy                       | Positive path: wait for the natural CNNA predecessor objects, then encode the smallest comparison profile and prove target equality/failure from CNNA data only.                                                                                                                                                                                                                                               |
| Build result                         | No Lean build run; no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                                                                                                                 |
| Post-implementation plan check       | After any future comparison implementation, record whether the target matched, failed, or required a natural-origin backjump; do not smooth failure into a weakened claim.                                                                                                                                                                                                                                     |
| Artifact packaging rule              | Package YAML, generated tables, generated PDF and report. Do not package external PDFs as generated project truth unless they are user-uploaded source material.                                                                                                                                                                                                                                               |
| Implementation notes                 | External theory is recorded only as comparison material. It may suggest target invariants, sign audits and future checks, but cannot release phases or provide CNNA generators.                                                                                                                                                                                                                                |
| Source materials / references        | <ul style="list-style-type: none"> <li>Horn Pauli/GA sections</li> <li>mathlib CliffordAlgebra.Basic documentation</li> <li>Wieser/Song Lean GA formalization</li> </ul>                                                                                                                                                                                                                                       |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>boundary generator square/anti-commutation target</li> <li>Clifford universal-property seam</li> <li>Pauli coordinate/operator ambiguity warning</li> </ul>                                                                                                                                                                                                             |
| Added modules                        | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no external GA/quaternion theorem used as CNNA source</li> <li>no phase release from uploaded/reference paper</li> <li>no Pauli/Clifford/Dirac object imported into Pillar A or B as generator</li> <li>no free complex scalar layer introduced from biquaternions</li> <li>no Lean import added before natural predecessor closure and version-pinned check</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply.</li> <li>GNG37: quaternion/GA references are comparison-only and have no generator or handoff rights.</li> </ul>                                                                                                                                                                                                   |
| Proposed added modules               | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Structural/CayleyDickson/*.lean</li> <li>Arithmetic/CayleyDickson/*.lean</li> <li>PillarB placeholder modules</li> <li>scripts/generate_tables.py</li> <li>scripts/generate_full_document.py</li> </ul>                                                                                                                                                                 |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>none identified</li> </ul>                                                                                                                                                                                                                                                                                                                                              |

Phase 13.3 – Local net and Haag-Kastler algebraic profile

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | R                                                                                                                                                                                                                                                                     |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                   |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                    |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                        |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                    |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                   |
| Implementation status                  | planned                                                                                                                                                                                                                                                               |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | Local net and Haag-Kastler algebraic profile                                                                                                                                                                                                                          |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                           |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                              |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                                                                  |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global no-go inheritance       | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| How implemented                | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Verification / evidence        | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Proof strategy                 | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Build result                   | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Post-implementation plan check | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Artifact packaging rule        | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation notes           | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| No-go guardrails               | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 13.4 – State, GNS, and KMS consumer layer

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid #ccc; padding: 0 2px;">R</span>                                                                                                                               |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                     |
| Phase documentation tier              | F1                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                                 |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                            |
| Phase type                            | derivation                                                                                                                                                                                                              |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                               |
| Title / state                         | State, GNS, and KMS consumer layer                                                                                                                                                                                      |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                       |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.      |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                       |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                             |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li> <li>no @[simp] attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no Matrix.inv or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

Phase 13.5 – Tomita-Takesaki and modular data

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                                                                |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                                     |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                                                                   |
| Phase documentation tier               | F1                                                                                                                                                                                                                                                                    |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                                        |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                                                    |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                                   |
| Implementation status                  | planned                                                                                                                                                                                                                                                               |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | Tomita-Takesaki and modular data                                                                                                                                                                                                                                      |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                           |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global no-go inheritance       | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| How implemented                | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Verification / evidence        | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Proof strategy                 | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Build result                   | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Post-implementation plan check | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Artifact packaging rule        | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation notes           | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| No-go guardrails               | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 13.6 – CPT, spin-statistics, Kramers and split-property gates

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ToolInfrastructurePhase                                                                                                                                                                                                 |
| Phase documentation tier              | F1                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                          |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                                 |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | infrastructure                                                                                                                                                                                                          |
| Phase-type-specific contract          | Infrastructure/planning phase: justify absence of operational witness; close schema/decision prerequisites before code.                                                                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts. V1.25 adds Horn/GA material only as downstream target-gate/scratchpad context; it cannot be used as a source object or release proof.                            |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |



Phase 13.6.1 – Dirac/spacetime-algebra comparison target behind CPT, spin-statistics and split gates

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                          |
| Context class                          | ReferenceGatePhase                                                                                                                                                                                                         |
| Phase documentation tier               | F2                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                              |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | planned-not-active                                                                                                                                                                                                         |
| Preparation status                     | planned-prepared                                                                                                                                                                                                           |
| Phase type                             | comparison                                                                                                                                                                                                                 |
| Phase-type-specific contract           | Comparison/reference phase: records external target criteria and scratchpad information; it cannot supply CNNA premises or release mathematical phases.                                                                    |
| Title / state                          | Dirac/spacetime-algebra comparison target behind CPT, spin-statistics and split gates                                                                                                                                      |
| Preparation workflow                   | External-source intake as comparison/reference gate with derived-only boundary.                                                                                                                                            |
| Source-to-CNNA mapping                 | Maps Dirac/STA material to a late Pillar-B target gate only after local net, modular data and structural theorem hypotheses are CNNA-derived.                                                                              |
| Dependency-depth audit                 | Depends on 13.5/13.6 and GA4; does not unblock any Pillar-B derivation. No Lorentz/Dirac/spacetime structure is imported.                                                                                                  |
| Phase split decision                   | Split into narrow comparison gates so CD, Pillar-B boundary algebra, Dirac/STA and side-comparison targets do not collapse into one ambiguous external-theory phase.                                                       |
| Positive path plan                     | Do not implement proof code yet. Keep target invariants in scratchpad; when predecessor data is closed, add a small neutral comparison API and prove/falsify explicitly.                                                   |
| Module implementation manifest         | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.                                                                                      |
| Semantic placement decision            | Natural placement is comparison/reference-gate layer, not a foundational CNNA derivation layer.                                                                                                                            |
| Reuse and duplication audit            | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.                                                                                   |
| Canonical generator policy             | The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.                                                                                           |
| Module compactness rule                | Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists.                                                                          |
| Refactor regression protocol           | No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.                                                                                 |
| Refactor migration metrics             | No Lean migration. Future comparison module metrics: declarations added, imports added, consumer count, and target/equality proofs closed.                                                                                 |
| Lean/mathlib version policy            | Baseline remains pinned Lean/mathlib snapshot; live documentation is only a source-discovery hint until verified locally against the pin.                                                                                  |
| ExecComplex boundary policy            | Do not replace ExecComplex or CNNA scalar provenance with external complex quaternions or analytic Complex.                                                                                                                |
| Consumer-contract direction            | Comparison consumers may read CNNA-derived outputs; CNNA derivation phases may not consume the comparison profile as a generator.                                                                                          |
| Governance-overhead review             | Added overhead is justified because sign, unit and semantic ambiguities are high-risk; gates are narrow to avoid broad external-theory creep.                                                                              |
| Proof rehearsal before implementation  | Future proof compares CNNA-derived B structures with gamma/STA target signatures and structural theorem hypotheses; any mismatch becomes an explicit comparison failure.                                                   |
| Expected tactics                       | Future Lean work should use explicit structures, field/projection equalities, constructor-based proofs, local case splits and finite matrix entry proofs; avoid broad simp/simpa collapse and opaque native__decide goals. |
| Expected mathlib imports               | Reference candidates only: Mathlib.Algebra.Quaternion, Mathlib.Analysis.Quaternion, Mathlib.LinearAlgebra.CliffordAlgebra.Basic. Verify against the pinned Lean/mathlib snapshot before any implementation.                |
| ExecComplex/computable plan            | No ExecComplex object introduced. If finite numeric checks are later needed, keep them mirror/diagnostic unless an exact CNNA-derived computable path is proved.                                                           |
| Concrete computable instance traversal | Concrete traversal currently consists of source equation/profile extraction and target-object registration; no CNNA runtime traversal yet.                                                                                 |



|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computable path non-skeleton status  | Reference skeleton only. It must not be presented as computable CNNA evidence until future exact instances are constructed.                                                                                                                                                                                                                                                                                    |
| Expected executable witness          | For V1.25: generated TSV/TEX/PDF artifacts. Future Lean witness: explicit comparison structure instantiated from CNNA-derived objects.                                                                                                                                                                                                                                                                         |
| Mathlib complication log             | Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0; implementation must recheck exact names and theorem availability.                                                                                                                                                                                                                         |
| Noncomputable boundary audit         | No noncomputable Lean code introduced. Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed.                                                                                                                                                                                                                                                        |
| Mathlib mitigation/refactor layer    | If mathlib Clifford/Quaternion APIs pull noncomputable/classical pressure, keep them in mirror/comparison modules or define a minimal finite target profile instead.                                                                                                                                                                                                                                           |
| Bridge gap status                    | Bridge intentionally open. The external reference describes target shapes; CNNA must derive any matching structure internally before comparison.                                                                                                                                                                                                                                                               |
| Natural backjump/refactor layer      | If the comparison needs missing structure, patch only the earliest natural CNNA layer where that structure is generated, not the comparison/consumer phase.                                                                                                                                                                                                                                                    |
| Implementation prompt contract       | Future implementation prompt must state comparison-only status, predecessor objects, no new generator axioms, no external-theory imports unless pinned and isolated, and exact target equalities/failures.                                                                                                                                                                                                     |
| Global no-go inheritance             | All global no-go guardrails apply, especially derived-only, positive-path-first, patch-at-natural-origin and reference-theory-as-target-only rules.                                                                                                                                                                                                                                                            |
| How implemented                      | Planned reference/target-gate integration only. V1.25 records source reading, object links, proof-dossier placeholders and scratchpad material; no Lean code is implemented.                                                                                                                                                                                                                                   |
| Verification / evidence              | YAML generator validation and generated table/PDF outputs after V1.25 update. No Lean build evidence because no Lean source changed.                                                                                                                                                                                                                                                                           |
| Proof strategy                       | Positive path: wait for the natural CNNA predecessor objects, then encode the smallest comparison profile and prove target equality/failure from CNNA data only.                                                                                                                                                                                                                                               |
| Build result                         | No Lean build run; no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                                                                                                                 |
| Post-implementation plan check       | After any future comparison implementation, record whether the target matched, failed, or required a natural-origin backjump; do not smooth failure into a weakened claim.                                                                                                                                                                                                                                     |
| Artifact packaging rule              | Package YAML, generated tables, generated PDF and report. Do not package external PDFs as generated project truth unless they are user-uploaded source material.                                                                                                                                                                                                                                               |
| Implementation notes                 | External theory is recorded only as comparison material. It may suggest target invariants, sign audits and future checks, but cannot release phases or provide CNNA generators.                                                                                                                                                                                                                                |
| Source materials / references        | <ul style="list-style-type: none"> <li>Horn Dirac/quaternion discussion</li> <li>Hestenes spacetime-algebra references</li> <li>mathlib CliffordAlgebra.Basic documentation</li> </ul>                                                                                                                                                                                                                         |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Dirac/Clifford signature target</li> <li>Lorentz-invariance comparison target</li> <li>spin/CPT compatibility target</li> <li>STA comparison surface</li> </ul>                                                                                                                                                                                                         |
| Added modules                        | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no external GA/quaternion theorem used as CNNA source</li> <li>no phase release from uploaded/reference paper</li> <li>no Pauli/Clifford/Dirac object imported into Pillar A or B as generator</li> <li>no free complex scalar layer introduced from biquaternions</li> <li>no Lean import added before natural predecessor closure and version-pinned check</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply.</li> <li>GNG37: quaternion/GA references are comparison-only and have no generator or handoff rights.</li> </ul>                                                                                                                                                                                                   |
| Proposed added modules               | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |

|                                      |                                                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Structural/CayleyDickson/*.lean</li> <li>Arithmetic/CayleyDickson/*.lean</li> <li>PillarB placeholder modules</li> <li>scripts/generate_tables.py</li> <li>scripts/generate_full_document.py</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>none identified</li> </ul>                                                                                                                                                                              |

### Phase 13.7 – B->A input handoff preparation

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | R                                                                                                                                                                                                                          |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                        |
| Phase documentation tier               | F1                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | planned                                                                                                                                                                                                                    |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                               |
| Phase type                             | derivation                                                                                                                                                                                                                 |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                  |
| Title / state                          | B->A input handoff preparation                                                                                                                                                                                             |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.    |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.         |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                          |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                          |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                          |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                          |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                          |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                        |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                       |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected executable witness       | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log          | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit      | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                 | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract    | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance          | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                   | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence           | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                    | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                      | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check    | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule           | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes              | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                  | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

## Phase 14 – Pillar C OQS consumer roadmap from A regimes and B-compatible interfaces

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                       |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                 |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                            |
| Phase type                            | derivation                                                                                                                                                                                              |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | Pillar C OQS consumer roadmap from A regimes and B-compatible interfaces                                                                                                                                                                                              |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                           |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                      |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                         |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 14.1 – C input contract from A regime and complement profile

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFB6C1;">R</span>                                                                                                                                                                                                   |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F0                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned                                                                                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                           |
| Title / state                         | C input contract from A regime and complement profile                                                                                                                                                                                               |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                             |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                   |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                  |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.            |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                 |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                   |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                   |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                   |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| No-go guardrails                       | <ul style="list-style-type: none"><li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li><li>no <code>@[simp]</code> attributes in the operational path</li><li>no noncomputable in the operational path</li><li>no classical helper unless isolated in allowed mirror/non-operational layer</li><li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li><li>no free formula without proof-carrying entry structure</li><li>no consumer-side patch when the missing structure belongs earlier</li><li>no premature obstruction before a serious positive construction attempt</li><li>no phase-semantic public API names</li><li>no fragile rewrite chain when a structural constructor/projection proof is available</li></ul> |

Phase 14.2 – Finite channel / CP-map profile

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned                                                                                                                                                                                        |



|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | Finite channel / CP-map profile                                                                                                                                                                                                                                       |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                           |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                         |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: top-level global__no_go_guardrails apply; this phase list is local addenda only.</li><li>• no @[simp] attributes in the operational path</li><li>• no noncomputable in the operational path</li><li>• no classical helper unless isolated in allowed mirror/non-operational layer</li><li>• no Matrix.inv or external inverse oracle in operational derivations</li><li>• no free formula without proof-carrying entry structure</li><li>• no consumer-side patch when the missing structure belongs earlier</li><li>• no premature obstruction before a serious positive construction attempt</li><li>• no phase-semantic public API names</li><li>• no fragile rewrite chain when a structural constructor/projection proof is available</li></ul> |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 14.3 – Effective dynamics and Lindblad/GKSL gate

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                          |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                        |
| Phase documentation tier              | F1                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                 | planned                                                                                                                                                                                                                    |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                               |
| Phase type                            | derivation                                                                                                                                                                                                                 |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                  |
| Title / state                         | Effective dynamics and Lindblad/GKSL gate                                                                                                                                                                                  |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.    |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                          |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.         |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                          |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                          |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                          |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                          |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                          |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 14.4 – Decoherence/noise observables as derived effective quantities

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned                                                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | Decoherence/noise observables as derived effective quantities                                                                                                                                                                                                         |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                        |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                 |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                         |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 14.5 – C->A input handoff preparation

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFB6C1;"> </span> <b>R</b>                                                                                                                                                                                          |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                      |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned                                                                                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                           |
| Title / state                         | C->A input handoff preparation                                                                                                                                                                                                                      |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                             |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                   |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                  |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.            |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                 |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                   |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                   |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                   |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 15 – Pillar D spacetime/covariance roadmap

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                 |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | Pillar D spacetime/covariance roadmap                                                                                                                                                                                                                                 |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                        |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                 |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                         |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 15.1 – D input contract from A/C geometry exports

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                   |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F0                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned                                                                                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                           |
| Title / state                         | D input contract from A/C geometry exports                                                                                                                                                                                                          |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                             |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                   |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                  |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                          |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                 |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                   |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                   |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                   |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 15.2 – Region, localization, and causal-order candidates

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned                                                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | Region, localization, and causal-order candidates                                                                                                                                                                                                                     |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                           |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                         |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 15.3 – Metric/coarse-geometry and covariance gates

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                   |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                      |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned                                                                                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                           |
| Title / state                         | Metric/coarse-geometry and covariance gates                                                                                                                                                                                                         |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                             |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                   |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                  |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.            |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                 |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                   |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                   |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                   |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 15.4 – D->A input handoff preparation

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned                                                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | D->A input handoff preparation                                                                                                                                                                                                                                        |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                        |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                 |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                         |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 16 – Pillar E matter and phenomenology roadmap

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFB6C1;"> </span> <b>R</b>                                                                                                                                                                                          |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F0                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned                                                                                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                           |
| Title / state                         | Pillar E matter and phenomenology roadmap                                                                                                                                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                             |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                   |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                  |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.            |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                 |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                   |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                   |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                   |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 16.1 – E input contract from A/B/C/D outputs

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                       |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                 |



|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | E input contract from A/B/C/D outputs                                                                                                                                                                                                                                 |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                        |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                 |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                         |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 16.2 – Matter sector and internal symmetry candidates

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFB6C1; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                                                                          |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                      |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned                                                                                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                           |
| Title / state                         | Matter sector and internal symmetry candidates                                                                                                                                                                                                      |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                             |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                   |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                  |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.            |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                 |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                   |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                   |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                   |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 16.3 – Observable coupling and phenomenology comparison layer

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                         |
| Context class                         | ReferenceGatePhase                                                                                                                        |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |
| Implementation status                 | planned                                                                                                                                   |

|                                       |                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 16.4 – E->A input handoff preparation

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                     |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span style="background-color: #f4cccc; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                                                 |
| Context class                          | LeanDerivationPhase                                                                                                                                                                                                        |
| Phase documentation tier               | F1                                                                                                                                                                                                                         |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                  | planned                                                                                                                                                                                                                    |
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                               |
| Phase type                             | derivation                                                                                                                                                                                                                 |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                  |
| Title / state                          | E->A input handoff preparation                                                                                                                                                                                             |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.    |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                          |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.         |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                          |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                          |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                          |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                          |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                          |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                        |
| Computable path non-skeleton status    | not-assessed-yet; must be explicit before operational implementation                                                                                                                                                       |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected executable witness       | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Mathlib complication log          | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Noncomputable boundary audit      | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Bridge gap status                 | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Natural backjump/refactor layer   | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Implementation prompt contract    | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Global no-go inheritance          | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| How implemented                   | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Verification / evidence           | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Proof strategy                    | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Build result                      | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Post-implementation plan check    | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Artifact packaging rule           | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Implementation notes              | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| No-go guardrails                  | <ul style="list-style-type: none"><li>GLOBAL-INHERITED: top-level global_no_go_guardrails apply; this phase list is local addenda only.</li><li>no @[simp] attributes in the operational path</li><li>no noncomputable in the operational path</li><li>no classical helper unless isolated in allowed mirror/non-operational layer</li><li>no Matrix.inv or external inverse oracle in operational derivations</li><li>no free formula without proof-carrying entry structure</li><li>no consumer-side patch when the missing structure belongs earlier</li><li>no premature obstruction before a serious positive construction attempt</li><li>no phase-semantic public API names</li><li>no fragile rewrite chain when a structural constructor/projection proof is available</li></ul> |

Phase 17 – Backreaction roadmap through explicit input handoffs only

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc; border: 1px solid black; padding: 0 2px;">R</span>                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                     |
| Phase documentation tier              | F0                                                                                                                                                                                                      |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                 |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                            |
| Phase type                            | derivation                                                                                                                                                                                              |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | Backreaction roadmap through explicit input handoffs only                                                                                                                                                                                                             |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                                           |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.                                            |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                   |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.        |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                         |



|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 17.1 – B->A backreaction gate

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #FFB6C1;"> </span> <b>R</b>                                                                                                                                                                                          |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F1                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                                                      |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned                                                                                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                           |
| Title / state                         | B->A backreaction gate                                                                                                                                                                                                                              |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                             |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                   |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                  |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.            |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                 |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                   |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                   |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                   |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 17.2 – C->A OQS backreaction gate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned                                                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | C->A OQS backreaction gate                                                                                                                                                                                                                                            |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                        |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                                           |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                         |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: top-level global__no_go_guardrails apply; this phase list is local addenda only.</li><li>• no @[simp] attributes in the operational path</li><li>• no noncomputable in the operational path</li><li>• no classical helper unless isolated in allowed mirror/non-operational layer</li><li>• no Matrix.inv or external inverse oracle in operational derivations</li><li>• no free formula without proof-carrying entry structure</li><li>• no consumer-side patch when the missing structure belongs earlier</li><li>• no premature obstruction before a serious positive construction attempt</li><li>• no phase-semantic public API names</li><li>• no fragile rewrite chain when a structural constructor/projection proof is available</li></ul> |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 17.3 – D->A spacetime backreaction gate

**Table legend.**   G = green/closed and derived-only;   W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   O = orange/implementable after stated predecessor closure;   Y = yellow/active and implementable now;   R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | R                                                                                                                                                                                                                          |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                        |
| Phase documentation tier              | F1                                                                                                                                                                                                                         |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                        |
| Implementation status                 | planned                                                                                                                                                                                                                    |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                               |
| Phase type                            | derivation                                                                                                                                                                                                                 |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                  |
| Title / state                         | D->A spacetime backreaction gate                                                                                                                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.    |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                          |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.         |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                          |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                |
| Refactor regression protocol          | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                          |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated. |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains Q x Q; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                   |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                              |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                        |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                          |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                          |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                          |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                          |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 17.4 – E->A matter backreaction gate

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                              |
| Context class                         | LeanDerivationPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned                                                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                     | red-locked-by-release-policy                                                                                                                                                                                                                                          |
| Phase type                             | derivation                                                                                                                                                                                                                                                            |
| Phase-type-specific contract           | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                                             |
| Title / state                          | E->A matter backreaction gate                                                                                                                                                                                                                                         |
| Preparation workflow                   | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                                               |
| Source-to-CNNA mapping                 | Not prepared yet.                                                                                                                                                                                                                                                     |
| Dependency-depth audit                 | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                                    |
| Phase split decision                   | Not prepared yet.                                                                                                                                                                                                                                                     |
| Positive path plan                     | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check;<br>only then may this phase become yellow or orange. Not prepared yet.                                                                                        |
| Refactor regression protocol           | Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                                                     |
| Refactor migration metrics             | Not recorded yet. Refactor phases must fill declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, and legacy/coexistence status before implementation.                                                                           |
| Lean/mathlib version policy            | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.                                         |
| ExecComplex boundary policy            | Apply EXEC-2 when relevant: operational ExecComplex remains $\mathbb{Q} \times \mathbb{Q}$ ;<br>algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                 |
| Consumer-contract direction            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                                         |
| Governance-overhead review             | Pending until first real refactor/code-change package closes;<br>then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                                |
| Proof rehearsal before implementation  | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                                     |
| Expected tactics                       | Not prepared yet.                                                                                                                                                                                                                                                     |
| Expected mathlib imports               | Not prepared yet.                                                                                                                                                                                                                                                     |
| ExecComplex/computable plan            | Not prepared yet.                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                   |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                               |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                     |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                  |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                             |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                 |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                     |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation. |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                           |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                           |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                      |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                 |
| Build result                           | not run                                                                                                                                                                                                                                                               |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                    |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build;<br>otherwise state that local Lean build is required.     |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                         |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global__no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no <code>noncomputable</code> in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 17.5 – Pre-/post-emergence regime switch for A statements

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f4cccc;">R</span>                                                                                                                                                                                                   |
| Context class                         | LeanDerivationPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F0                                                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Full dossier required before promotion or consumption: definition, natural provenance, exact source objects, proof target, proof sketch, verification/audit, consumer rights, and semantic uncertainty.                                             |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                 |
| Implementation status                 | planned                                                                                                                                                                                                                                             |
| Preparation status                    | red-locked-by-release-policy                                                                                                                                                                                                                        |
| Phase type                            | derivation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Derivation phase: record source objects, definitions, proof target, positive construction path, expected tactics, mathlib imports, concrete computable instance traversal if operational.                                                           |
| Title / state                         | Pre-/post-emergence regime switch for A statements                                                                                                                                                                                                  |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code.                             |
| Source-to-CNNA mapping                | Not prepared yet.                                                                                                                                                                                                                                   |
| Dependency-depth audit                | V1.19 status policy: locked red until a release proof references the prerequisite derived-only Lean objects and turns this phase yellow or orange. Pending. Must be completed before this phase may become active.                                  |
| Phase split decision                  | Not prepared yet.                                                                                                                                                                                                                                   |
| Positive path plan                    | First prove/release the prerequisite derived-only Lean objects and record the pre-implementation check; only then may this phase become yellow or orange. Not prepared yet.                                                                         |
| Refactor regression protocol          | Not applicable unless <code>phase_type</code> is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.                                      |
| Refactor migration metrics            | Not recorded yet. Refactor phases must fill <code>declarations_migrated_count</code> , <code>imports_changed</code> , <code>import_graph_delta</code> , <code>consumer_breakage_audit</code> , and legacy/coexistence status before implementation. |
| Lean/mathlib version policy           | Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; <code>noncomputable</code> /classical pressure must be logged and isolated.            |
| ExecComplex boundary policy           | Apply EXEC-2 when relevant: operational ExecComplex remains $Q \times Q$ ; algebraic irrational data stays mirror-only unless a rational/integer invariant is separately proved.                                                                    |
| Consumer-contract direction           | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.                                                                                       |
| Governance-overhead review            | Pending until first real refactor/code-change package closes; then record which governance gates prevented concrete errors and whether later phases can be flatter.                                                                                 |
| Proof rehearsal before implementation | No implementation rehearsal authorized while red. Required rehearsal starts with prerequisite-object proof references and a release check. Pending. Must be written before implementation starts.                                                   |
| Expected tactics                      | Not prepared yet.                                                                                                                                                                                                                                   |
| Expected mathlib imports              | Not prepared yet.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan           | Not prepared yet.                                                                                                                                                                                                                                   |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Computable path non-skeleton status    | not-assessed-yet;<br>must be explicit before operational implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected executable witness            | not specified yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No phase-specific mathlib complication recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Noncomputable boundary audit           | No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mathlib mitigation/refactor layer      | If mathlib introduces noncomputable/classical pressure, isolate in allowed mirror/non-operational layer or backjump to the earliest natural source layer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Natural backjump/refactor layer        | Not prepared yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Implementation prompt contract         | Use the standardized positive-path implementation workflow: respect all plan sections, import/consumer discipline, derived-only and computable operational path, no forbidden shortcuts, package the full modified code, and verify plan points after implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented yet. Scratch-pad placeholder generated from the single YAML roadmap master.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Verification / evidence                | Pending;<br>phase is not active.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proof strategy                         | To be specified when this phase becomes active;<br>split into subphases if dependencies are too deep.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Build result                           | not run                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post-implementation plan check         | Not available yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artifact packaging rule                | Return a complete zip with all modified files and generated planning artifacts when applicable. Lean build success may be recorded only from an actual current build or from the user-reported local build; otherwise state that local Lean build is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation notes                   | Roadmap placeholder;<br>does not assert the target is proven.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| No-go guardrails                       | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: top-level <code>global_no_go_guardrails</code> apply; this phase list is local addenda only.</li> <li>no <code>@[simp]</code> attributes in the operational path</li> <li>no noncomputable in the operational path</li> <li>no classical helper unless isolated in allowed mirror/non-operational layer</li> <li>no <code>Matrix.inv</code> or external inverse oracle in operational derivations</li> <li>no free formula without proof-carrying entry structure</li> <li>no consumer-side patch when the missing structure belongs earlier</li> <li>no premature obstruction before a serious positive construction attempt</li> <li>no phase-semantic public API names</li> <li>no fragile rewrite chain when a structural constructor/projection proof is available</li> </ul> |

### Phase 18 – Side-comparison register beyond the CNNA focus

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                        |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f2f2f2;">W</span>                                                                                                                             |
| Context class                         | ReferenceGatePhase                                                                                                                                                            |
| Phase documentation tier              | F2                                                                                                                                                                            |
| Phase documentation depth requirement | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                            |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                           |
| Implementation status                 | <span style="background-color: #f2f2f2;">planned</span>                                                                                                                       |



|                                       |                                                                                                                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts. V1.25 adds Horn/GA material only as downstream target-gate/scratchpad context; it cannot be used as a source object or release proof.                            |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 18.1 – Full ToE comparison placeholder

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F4                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                                 |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 18.2 – String-theory comparison placeholder

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F4                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                                 |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |

|                                       |                                                                                       |
|---------------------------------------|---------------------------------------------------------------------------------------|
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                       |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase. |

### Phase 18.3 – Loop-quantum-gravity comparison placeholder

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F4                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                                 |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 18.4 – Noncommutative-geometry comparison placeholder

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                       |
| Context class                         | ReferenceGatePhase                                                                                                                                                                                                      |
| Phase documentation tier              | F4                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned                                                                                                                                                                                                                 |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | comparison                                                                                                                                                                                                              |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.                                                                                           |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must be completed before this phase may become active.                                                                                                                                                         |
| Proof rehearsal before implementation | Pending. Must be written before implementation starts.                                                                                                                                                                  |
| Bridge gap status                     | Pending. Any missing bridge must create or activate the earlier natural source phase.                                                                                                                                   |

### Phase 18.5 – Geometric and conformal algebra side-comparison register

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                                    |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                         |
| Context class                          | ReferenceGatePhase                                                                                                                                                                                                        |
| Phase documentation tier               | F2                                                                                                                                                                                                                        |
| Phase documentation depth requirement  | Standard dossier: short definition, source objects, proof target, implementation location, and next documentation action. No large narrative unless it blocks implementation.                                             |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                        |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                       |
| Implementation status                  | planned-not-active                                                                                                                                                                                                        |
| Preparation status                     | planned-prepared                                                                                                                                                                                                          |
| Phase type                             | comparison                                                                                                                                                                                                                |
| Phase-type-specific contract           | Comparison/reference phase: records external target criteria and scratchpad information; it cannot supply CNNA premises or release mathematical phases.                                                                   |
| Title / state                          | Geometric and conformal algebra side-comparison register                                                                                                                                                                  |
| Preparation workflow                   | External-source intake as comparison/reference gate with derived-only boundary.                                                                                                                                           |
| Source-to-CNNA mapping                 | Maps GA/CGA to Phase 18 side-comparison, not to Pillar-A/B generators. It may later help organize review questions and diagrams.                                                                                          |
| Dependency-depth audit                 | Depends on GA0-GA5 and future CNNA outputs. It is non-blocking and cannot set spacetime dimension, metric, dynamics or matter.                                                                                            |
| Phase split decision                   | Split into narrow comparison gates so CD, Pillar-B boundary algebra, Dirac/STA and side-comparison targets do not collapse into one ambiguous external-theory phase.                                                      |
| Positive path plan                     | Do not implement proof code yet. Keep target invariants in scratchpad; when predecessor data is closed, add a small neutral comparison API and prove/falsify explicitly.                                                  |
| Module implementation manifest         | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.                                                                                     |
| Semantic placement decision            | Natural placement is comparison/reference-gate layer, not a foundational CNNA derivation layer.                                                                                                                           |
| Reuse and duplication audit            | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.                                                                                  |
| Canonical generator policy             | The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.                                                                                          |
| Module compactness rule                | Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists.                                                                         |
| Refactor regression protocol           | No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.                                                                                |
| Refactor migration metrics             | No Lean migration. Future comparison module metrics: declarations added, imports added, consumer count, and target/equality proofs closed.                                                                                |
| Lean/mathlib version policy            | Baseline remains pinned Lean/mathlib snapshot; live documentation is only a source-discovery hint until verified locally against the pin.                                                                                 |
| ExecComplex boundary policy            | Do not replace ExecComplex or CNNA scalar provenance with external complex quaternions or analytic Complex.                                                                                                               |
| Consumer-contract direction            | Comparison consumers may read CNNA-derived outputs; CNNA derivation phases may not consume the comparison profile as a generator.                                                                                         |
| Governance-overhead review             | Added overhead is justified because sign, unit and semantic ambiguities are high-risk; gates are narrow to avoid broad external-theory creep.                                                                             |
| Proof rehearsal before implementation  | Future work prepares generated comparison documents from closed CNNA outputs, then asks whether GA/CGA offers a faithful target representation or only a heuristic analogy.                                               |
| Expected tactics                       | Future Lean work should use explicit structures, field/projection equalities, constructor-based proofs, local case splits and finite matrix entry proofs; avoid broad simp/simpa collapse and opaque native_decide goals. |
| Expected mathlib imports               | Reference candidates only: Mathlib.Algebra.Quaternion, Mathlib.Analysis.Quaternion, Mathlib.LinearAlgebra.CliffordAlgebra.Basic. Verify against the pinned Lean/mathlib snapshot before any implementation.               |
| ExecComplex/computable plan            | No ExecComplex object introduced. If finite numeric checks are later needed, keep them mirror/diagnostic unless an exact CNNA-derived computable path is proved.                                                          |
| Concrete computable instance traversal | Concrete traversal currently consists of source equation/profile extraction and target-object registration; no CNNA runtime traversal yet.                                                                                |
| Computable path non-skeleton status    | Reference skeleton only. It must not be presented as computable CNNA evidence until future exact instances are constructed.                                                                                               |
| Expected executable witness            | For V1.25: generated TSV/TEX/PDF artifacts. Future Lean witness: explicit comparison structure instantiated from CNNA-derived objects.                                                                                    |
| Mathlib complication log               | Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0; implementation must recheck exact names and theorem availability.                                    |
| Noncomputable boundary audit           | No noncomputable Lean code introduced. Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed.                                                                   |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mathlib mitigation/refactor layer    | If mathlib Clifford/Quaternion APIs pull noncomputable/classical pressure, keep them in mirror/comparison modules or define a minimal finite target profile instead.                                                                                                                                                                                                                                           |
| Bridge gap status                    | Bridge intentionally open. The external reference describes target shapes; CNNA must derive any matching structure internally before comparison.                                                                                                                                                                                                                                                               |
| Natural backjump/refactor layer      | If the comparison needs missing structure, patch only the earliest natural CNNA layer where that structure is generated, not the comparison/consumer phase.                                                                                                                                                                                                                                                    |
| Implementation prompt contract       | Future implementation prompt must state comparison-only status, predecessor objects, no new generator axioms, no external-theory imports unless pinned and isolated, and exact target equalities/failures.                                                                                                                                                                                                     |
| Global no-go inheritance             | All global no-go guardrails apply, especially derived-only, positive-path-first, patch-at-natural-origin and reference-theory-as-target-only rules.                                                                                                                                                                                                                                                            |
| How implemented                      | Planned reference/target-gate integration only. V1.25 records source reading, object links, proof-dossier placeholders and scratchpad material; no Lean code is implemented.                                                                                                                                                                                                                                   |
| Verification / evidence              | YAML generator validation and generated table/PDF outputs after V1.25 update. No Lean build evidence because no Lean source changed.                                                                                                                                                                                                                                                                           |
| Proof strategy                       | Positive path: wait for the natural CNNA predecessor objects, then encode the smallest comparison profile and prove target equality/failure from CNNA data only.                                                                                                                                                                                                                                               |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                                                                                                              |
| Post-implementation plan check       | After any future comparison implementation, record whether the target matched, failed, or required a natural-origin backjump; do not smooth failure into a weakened claim.                                                                                                                                                                                                                                     |
| Artifact packaging rule              | Package YAML, generated tables, generated PDF and report. Do not package external PDFs as generated project truth unless they are user-uploaded source material.                                                                                                                                                                                                                                               |
| Implementation notes                 | External theory is recorded only as comparison material. It may suggest target invariants, sign audits and future checks, but cannot release phases or provide CNNA generators.                                                                                                                                                                                                                                |
| Source materials / references        | <ul style="list-style-type: none"> <li>Horn outlook on spacetime and conformal GA</li> <li>Hestenes/Sobczyk Clifford Algebra to Geometric Calculus</li> <li>Wieser/Song Lean GA formalization</li> </ul>                                                                                                                                                                                                       |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>geometric algebra side-comparison surface</li> <li>spacetime algebra side-comparison surface</li> <li>conformal GA side-comparison surface</li> </ul>                                                                                                                                                                                                                   |
| Added modules                        | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no external GA/quaternion theorem used as CNNA source</li> <li>no phase release from uploaded/reference paper</li> <li>no Pauli/Clifford/Dirac object imported into Pillar A or B as generator</li> <li>no free complex scalar layer introduced from biquaternions</li> <li>no Lean import added before natural predecessor closure and version-pinned check</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply.</li> <li>GNG37: quaternion/GA references are comparison-only and have no generator or handoff rights.</li> </ul>                                                                                                                                                                                                   |
| Proposed added modules               | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC_ALGEBRA_REFERENCE_GATE_REPORT.md</li> </ul>                                                                                                                                                                                                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Structural/CayleyDickson/*.lean</li> <li>Arithmetic/CayleyDickson/*.lean</li> <li>PillarB placeholder modules</li> <li>scripts/generate_tables.py</li> <li>scripts/generate_full_document.py</li> </ul>                                                                                                                                                                 |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>none identified</li> </ul>                                                                                                                                                                                                                                                                                                                                              |

Phase 19 – Secondary long-term meta-documentation and derivation-geometry roadmap

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                         |
| Context class                         | ReferenceGatePhase                                                                                                                        |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |
| Implementation status                 | not-started                                                                                                                               |
| Preparation status                    | not-prepared                                                                                                                              |
| Phase type                            | comparison                                                                                                                                |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.             |
| Preparation workflow                  | Secondary long-term workflow. Keep out of active implementation until not-before gates are satisfied.                                     |
| Dependency-depth audit                | White secondary target; prerequisites intentionally not closed yet.                                                                       |
| Proof rehearsal before implementation | Pending until activation. Rehearse documentation/diagnostic path, not a mathematical shortcut.                                            |
| Bridge gap status                     | Blocked by not-before gates, not by a Lean proof gap.                                                                                     |

Phase 19.1 – Semi-automated existing-code documentation inside the YAML/proofbook tool

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                         |
| Context class                         | ReferenceGatePhase                                                                                                                        |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |
| Implementation status                 | not-started                                                                                                                               |
| Preparation status                    | not-prepared                                                                                                                              |
| Phase type                            | comparison                                                                                                                                |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.             |
| Preparation workflow                  | Secondary long-term workflow. Keep out of active implementation until not-before gates are satisfied.                                     |
| Dependency-depth audit                | White secondary target; prerequisites intentionally not closed yet.                                                                       |
| Proof rehearsal before implementation | Pending until activation. Rehearse documentation/diagnostic path, not a mathematical shortcut.                                            |
| Bridge gap status                     | Blocked by not-before gates, not by a Lean proof gap.                                                                                     |

Phase 19.2 – Derivation-graph geometry and bottleneck analysis of the formalized theory

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field | Record |
|-------|--------|
|-------|--------|

|                                       |                                                                                                                                           |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                         |
| Context class                         | ReferenceGatePhase                                                                                                                        |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |
| Implementation status                 | not-started                                                                                                                               |
| Preparation status                    | not-prepared                                                                                                                              |
| Phase type                            | comparison                                                                                                                                |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.             |
| Preparation workflow                  | Secondary long-term workflow. Keep out of active implementation until not-before gates are satisfied.                                     |
| Dependency-depth audit                | White secondary target; prerequisites intentionally not closed yet.                                                                       |
| Proof rehearsal before implementation | Pending until activation. Rehearse documentation/diagnostic path, not a mathematical shortcut.                                            |
| Bridge gap status                     | Blocked by not-before gates, not by a Lean proof gap.                                                                                     |

### Phase 19.3 – Formal emergence-path forensics for open-system dynamics

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                         |
| Context class                         | ReferenceGatePhase                                                                                                                        |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |
| Implementation status                 | not-started                                                                                                                               |
| Preparation status                    | not-prepared                                                                                                                              |
| Phase type                            | comparison                                                                                                                                |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs.             |
| Preparation workflow                  | Secondary long-term workflow. Keep out of active implementation until not-before gates are satisfied.                                     |
| Dependency-depth audit                | White secondary target; prerequisites intentionally not closed yet.                                                                       |
| Proof rehearsal before implementation | Pending until activation. Rehearse documentation/diagnostic path, not a mathematical shortcut.                                            |
| Bridge gap status                     | Blocked by not-before gates, not by a Lean proof gap.                                                                                     |

### Phase 19.4 – Reflective comparison of CNNA object-structure with formalization graph structure

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                         |
| Context class                         | ReferenceGatePhase                                                                                                                        |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |



|                                       |                                                                                                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Record change version                 | unchanged-by-V1.01                                                                                                            |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                           |
| Implementation status                 | not-started                                                                                                                   |
| Preparation status                    | not-prepared                                                                                                                  |
| Phase type                            | comparison                                                                                                                    |
| Phase-type-specific contract          | Comparison/secondary phase: record derived-only boundary and prevent comparison targets from becoming CNNA-generative inputs. |
| Preparation workflow                  | Secondary long-term workflow. Keep out of active implementation until not-before gates are satisfied.                         |
| Dependency-depth audit                | White secondary target;<br>prerequisites intentionally not closed yet.                                                        |
| Proof rehearsal before implementation | Pending until activation. Rehearse documentation/diagnostic path, not a mathematical shortcut.                                |
| Bridge gap status                     | Blocked by not-before gates, not by a Lean proof gap.                                                                         |

### Phase 19.5 – Thread/full-boundary derivation-geometry and string-analogy diagnostic side-road

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #f2f2f2;">W</span>                                                                                                                                                                       |
| Context class                         | DocumentationPhase                                                                                                                                                                                                      |
| Phase documentation tier              | F4                                                                                                                                                                                                                      |
| Phase documentation depth requirement | Mention and pointer only: name, context class, status, and reason for non-consumption. No full proof dossier unless promoted.                                                                                           |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                      |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                     |
| Implementation status                 | planned-not-active                                                                                                                                                                                                      |
| Preparation status                    | not-prepared                                                                                                                                                                                                            |
| Phase type                            | documentation                                                                                                                                                                                                           |
| Phase-type-specific contract          | Documentation/comparison phase: keep diagnostic material downstream and non-generative.                                                                                                                                 |
| Preparation workflow                  | Planned phase. Use source-preparation workflow before implementation: attach reliable sources, extract established objects, audit dependencies, split if necessary, and fill proof rehearsal before changing Lean code. |
| Dependency-depth audit                | Pending. Must include route-dependency audit and module placement check before activation.                                                                                                                              |
| Proof rehearsal before implementation | No Lean proof target initially;<br>proof obligation is negative governance: no diagnostic output may feed back into CNNA generation.                                                                                    |
| Bridge gap status                     | Pending. Any missing lifting theorem or route bridge must create or activate the earlier natural source phase.                                                                                                          |

### Phase 20 – DB-REVIEW main side objective: database-ready CNNA review and documentation structure with immediate expert-review principles

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                              |
| Context class                         | DocumentationPhase                                                                                                                                                                             |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | completed                                                                                                                                                                                      |
| Preparation status                    | completed-governance-rule                                                                                                                                                                      |



|                                        |                                                                                                                                                                                                       |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase type                             | documentation                                                                                                                                                                                         |
| Phase-type-specific contract           | Documentation/tooling governance phase: output is review/query infrastructure, not mathematical derivation.                                                                                           |
| Title / state                          | DB-REVIEW side objective governance row                                                                                                                                                               |
| Preparation workflow                   | DB-REVIEW database-ready documentation and expert-review workflow.                                                                                                                                    |
| Source-to-CNNA mapping                 | Maps documentation/database/review infrastructure to generated-secondary metadata only. It cannot supply CNNA-generative mathematical or physical inputs.                                             |
| Dependency-depth audit                 | No Lean dependency introduced. Depends on YAML master and generated views only.                                                                                                                       |
| Phase split decision                   | Split into immediate principles (20.1) and later technical/database/review phases (20.2-20.5).                                                                                                        |
| Positive path plan                     | Apply expert-review/no-infallibility principles immediately;<br>later add schema/export/review tooling without touching Lean truth semantics.                                                         |
| Module implementation manifest         | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.                                                                                                    |
| Semantic placement decision            | DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.                                                                                            |
| Reuse and duplication audit            | Reuse Single-YAML-Master and existing generator/export seams;<br>do not create a second authoritative planning source.                                                                                |
| Canonical generator policy             | YAML remains the manual planning source;<br>generated DB/TSV/PDF/graph views remain secondary and reproducible.                                                                                       |
| Module compactness rule                | Keep Lean implementation packets compact;<br>DB/audit-full exploration must not overload routine code-writing phases.                                                                                 |
| Refactor regression protocol           | No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.                                                                                            |
| Refactor migration metrics             | No declaration migration;<br>no imports changed;<br>no Lean modules moved.                                                                                                                            |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                    |
| ExecComplex boundary policy            | No ExecComplex change. Operational scalar boundaries remain governed by EXEC-2 when relevant.                                                                                                         |
| Consumer-contract direction            | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.                                                                              |
| Governance-overhead review             | DB-REVIEW is non-blocking for Lean coding;<br>principles are mandatory, but technical DB work must remain compact and phase-bound.                                                                    |
| Proof rehearsal before implementation  | For completed governance rows, validation is generator/YAML consistency and explicit guardrails. Future rows must rehearse schema round-trip and no-generator boundary.                               |
| Expected tactics                       | None;<br>documentation/tooling phase, not Lean proof phase.                                                                                                                                           |
| Expected mathlib imports               | None added.                                                                                                                                                                                           |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                         |
| Concrete computable instance traversal | Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.                                                                    |
| Computable path non-skeleton status    | non-operational metadata path;<br>no computable CNNA object claimed                                                                                                                                   |
| Expected executable witness            | For future DB phases: reproducible export/round-trip script output, not a Lean mathematical witness.                                                                                                  |
| Mathlib complication log               | None;<br>no mathlib imports or Lean code changes.                                                                                                                                                     |
| Noncomputable boundary audit           | No noncomputable Lean boundary involved;<br>semantic review is a documentation governance rule.                                                                                                       |
| Mathlib mitigation/refactor layer      | Not applicable unless future Lean-readable metadata tooling is added;<br>then isolate outside operational path.                                                                                       |
| Bridge gap status                      | No mathematical bridge closed. Review gaps and missing blue marks remain explicit audit debt.                                                                                                         |
| Natural backjump/refactor layer        | If a future view needs semantic truth not present in Lean/YAML metadata, backjump to the relevant Lean/object proof source, not to the database/view consumer.                                        |
| Implementation prompt contract         | For future DB-REVIEW phases: preserve Lean-primary/generated-secondary semantics, keep expert review mandatory, and do not turn established theory or database views into CNNA inputs.                |
| Global no-go inheritance               | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                           |
| How implemented                        | Added as a main side objective in the YAML phase DAG with direct DBR0 object and child phases 20.1-20.5. Technical database work remains later/non-blocking, but review principles apply immediately. |
| Verification / evidence                | YAML contains Phase 20, DBR0 proof dossier, SLTG-06, GNG23-GNG24, DB-REVIEW workflow, and generated exports/tables.                                                                                   |

|                                      |                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof strategy                       | Governance/documentation proof: preserve Lean-primary/generated-secondary boundary and explicit review obligations.                                                                                                                                                                                                                                       |
| Build result                         | No Lean build run;<br>no Lean source changed. Generator/PDF regeneration only.                                                                                                                                                                                                                                                                            |
| Post-implementation plan check       | Completed rows changed YAML/generator views only. Future rows must record schema/export artifacts, review status coverage, and no Lean-source mutation unless explicitly requested.                                                                                                                                                                       |
| Artifact packaging rule              | Return a complete zip with the updated YAML, generated tables/exports, report, and PDF when applicable. Do not claim Lean build success unless actually run.                                                                                                                                                                                              |
| Implementation notes                 | Side objective is closed as a governance/roadmap insertion, not as a database implementation.                                                                                                                                                                                                                                                             |
| Source materials / references        | <ul style="list-style-type: none"> <li>• user instruction for DB-REVIEW main side objective</li> <li>• Single YAML master</li> <li>• generated exports/tables</li> <li>• object proof dossiers</li> <li>• global no-go guardrails</li> </ul>                                                                                                              |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>• Lean-primary truth-system rule</li> <li>• generated-secondary documentation rule</li> <li>• discipline-specific expert-review obligation</li> <li>• blue/reference marking audit debt</li> </ul>                                                                                                                 |
| Modified modules                     | <ul style="list-style-type: none"> <li>• Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                           |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>• no Lean source edit</li> <li>• no database/export object used as CNNA generator</li> <li>• no semantic infallibility claim</li> <li>• no established-theory input hidden behind blue/reference marking</li> </ul>                                                                                                |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23 and GNG24.</li> <li>• no 100-percent truth declaration</li> <li>• no generated artifact as semantic certificate</li> <li>• no ignored missing blue/reference-theory marking</li> <li>• no database/view layer as CNNA generator</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>• Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                           |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>• Repository/scripts/generate_tables.py</li> <li>• Repository/generated_exports/*.tsv</li> <li>• Repository/tables/*.tex</li> </ul>                                                                                                                                                                                |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>• second-source-of-truth risk</li> <li>• generated-view-as-truth risk</li> <li>• unmarked-established-theory risk</li> </ul>                                                                                                                                                                                       |

### Phase 20.1 – Immediate epistemic expert-review rule and no-infallibility maxim

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                              |
| Context class                         | DocumentationPhase                                                                                                                                                                             |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |

|                                        |                                                                                                                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                     |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                    |
| Implementation status                  | completed                                                                                                                                                                              |
| Preparation status                     | completed-governance-rule                                                                                                                                                              |
| Phase type                             | documentation                                                                                                                                                                          |
| Phase-type-specific contract           | Documentation/tooling governance phase: output is review/query infrastructure, not mathematical derivation.                                                                            |
| Title / state                          | Immediate expert-review and no-infallibility rule                                                                                                                                      |
| Preparation workflow                   | DB-REVIEW database-ready documentation and expert-review workflow.                                                                                                                     |
| Source-to-CNNA mapping                 | Maps documentation/database/review infrastructure to generated-secondary metadata only. It cannot supply CNNA-generative mathematical or physical inputs.                              |
| Dependency-depth audit                 | No Lean dependency introduced. Depends on YAML master and generated views only.                                                                                                        |
| Phase split decision                   | Split into immediate principles (20.1) and later technical/database/review phases (20.2-20.5).                                                                                         |
| Positive path plan                     | Apply expert-review/no-infallibility principles immediately;<br>later add schema/export/review tooling without touching Lean truth semantics.                                          |
| Module implementation manifest         | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.                                                                                     |
| Semantic placement decision            | DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.                                                                             |
| Reuse and duplication audit            | Reuse Single-YAML-Master and existing generator/export seams;<br>do not create a second authoritative planning source.                                                                 |
| Canonical generator policy             | YAML remains the manual planning source;<br>generated DB/TSV/PDF/graph views remain secondary and reproducible.                                                                        |
| Module compactness rule                | Keep Lean implementation packets compact;<br>DB/audit-full exploration must not overload routine code-writing phases.                                                                  |
| Refactor regression protocol           | No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.                                                                             |
| Refactor migration metrics             | No declaration migration;<br>no imports changed;<br>no Lean modules moved.                                                                                                             |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                     |
| ExecComplex boundary policy            | No ExecComplex change. Operational scalar boundaries remain governed by EXEC-2 when relevant.                                                                                          |
| Consumer-contract direction            | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.                                                               |
| Governance-overhead review             | DB-REVIEW is non-blocking for Lean coding;<br>principles are mandatory, but technical DB work must remain compact and phase-bound.                                                     |
| Proof rehearsal before implementation  | For completed governance rows, validation is generator/YAML consistency and explicit guardrails. Future rows must rehearse schema round-trip and no-generator boundary.                |
| Expected tactics                       | None;<br>documentation/tooling phase, not Lean proof phase.                                                                                                                            |
| Expected mathlib imports               | None added.                                                                                                                                                                            |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                          |
| Concrete computable instance traversal | Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.                                                     |
| Computable path non-skeleton status    | non-operational metadata path;<br>no computable CNNA object claimed                                                                                                                    |
| Expected executable witness            | For future DB phases: reproducible export/round-trip script output, not a Lean mathematical witness.                                                                                   |
| Mathlib complication log               | None;<br>no mathlib imports or Lean code changes.                                                                                                                                      |
| Noncomputable boundary audit           | No noncomputable Lean boundary involved;<br>semantic review is a documentation governance rule.                                                                                        |
| Mathlib mitigation/refactor layer      | Not applicable unless future Lean-readable metadata tooling is added;<br>then isolate outside operational path.                                                                        |
| Bridge gap status                      | No mathematical bridge closed. Review gaps and missing blue marks remain explicit audit debt.                                                                                          |
| Natural backjump/refactor layer        | If a future view needs semantic truth not present in Lean/YAML metadata, backjump to the relevant Lean/object proof source, not to the database/view consumer.                         |
| Implementation prompt contract         | For future DB-REVIEW phases: preserve Lean-primary/generated-secondary semantics, keep expert review mandatory, and do not turn established theory or database views into CNNA inputs. |

|                                      |                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                     |
| How implemented                      | Added GNG23-GNG24 and DBR1 to enforce Lean-primary/generated-secondary semantics, no 100-percent truth declaration, expert review, and blue/reference audit debt.                                                                                                                                                                               |
| Verification / evidence              | YAML contains GNG23-GNG24, DBR1 proof dossier, review/audit status legend rows, and latest-change overview entry.                                                                                                                                                                                                                               |
| Proof strategy                       | Governance/documentation proof: preserve Lean-primary/generated-secondary boundary and explicit review obligations.                                                                                                                                                                                                                             |
| Build result                         | No Lean build run;<br>no Lean source changed. Generator/PDF regeneration only.                                                                                                                                                                                                                                                                  |
| Post-implementation plan check       | Completed rows changed YAML/generator views only. Future rows must record schema/export artifacts, review status coverage, and no Lean-source mutation unless explicitly requested.                                                                                                                                                             |
| Artifact packaging rule              | Return a complete zip with the updated YAML, generated tables/exports, report, and PDF when applicable. Do not claim Lean build success unless actually run.                                                                                                                                                                                    |
| Implementation notes                 | Principle applies immediately to every future phase, proof dossier, generated view, and theory/reference marking.                                                                                                                                                                                                                               |
| Source materials / references        | <ul style="list-style-type: none"> <li>user instruction for DB-REVIEW main side objective</li> <li>Single YAML master</li> <li>generated exports/tables</li> <li>object proof dossiers</li> <li>global no-go guardrails</li> </ul>                                                                                                              |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Lean-primary truth-system rule</li> <li>generated-secondary documentation rule</li> <li>discipline-specific expert-review obligation</li> <li>blue/reference marking audit debt</li> </ul>                                                                                                               |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                   |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source edit</li> <li>no database/export object used as CNNA generator</li> <li>no semantic infallibility claim</li> <li>no established-theory input hidden behind blue/reference marking</li> </ul>                                                                                              |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23 and GNG24.</li> <li>no 100-percent truth declaration</li> <li>no generated artifact as semantic certificate</li> <li>no ignored missing blue/reference-theory marking</li> <li>no database/view layer as CNNA generator</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                                                   |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Repository/scripts/generate_tables.py</li> <li>Repository/generated_exports/*.tsv</li> <li>Repository/tables/*.tex</li> </ul>                                                                                                                                                                            |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>second-source-of-truth risk</li> <li>generated-view-as-truth risk</li> <li>unmarked-established-theory risk</li> </ul>                                                                                                                                                                                   |

### Phase 20.2 – Master-YAML database schema preparation and round-trip-safe identifiers

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field        | Record |
|--------------|--------|
| Phase status | W      |

|                                        |                                                                                                                                                                                                |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                          | ToolInfrastructurePhase                                                                                                                                                                        |
| Phase documentation tier               | F1                                                                                                                                                                                             |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                  | planned                                                                                                                                                                                        |
| Preparation status                     | planned                                                                                                                                                                                        |
| Phase type                             | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract           | Documentation/tooling governance phase: output is review/query infrastructure, not mathematical derivation.                                                                                    |
| Title / state                          | Database schema and round-trip preparation                                                                                                                                                     |
| Preparation workflow                   | DB-REVIEW database-ready documentation and expert-review workflow.                                                                                                                             |
| Source-to-CNNA mapping                 | Maps documentation/database/review infrastructure to generated-secondary metadata only. It cannot supply CNNA-generative mathematical or physical inputs.                                      |
| Dependency-depth audit                 | Future phase must audit YAML fields, generated exports, IDs, references, and review links before implementation.                                                                               |
| Phase split decision                   | Split into immediate principles (20.1) and later technical/database/review phases (20.2-20.5).                                                                                                 |
| Positive path plan                     | Apply expert-review/no-infallibility principles immediately;<br>later add schema/export/review tooling without touching Lean truth semantics.                                                  |
| Module implementation manifest         | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.                                                                                             |
| Semantic placement decision            | DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.                                                                                     |
| Reuse and duplication audit            | Reuse Single-YAML-Master and existing generator/export seams;<br>do not create a second authoritative planning source.                                                                         |
| Canonical generator policy             | YAML remains the manual planning source;<br>generated DB/TSV/PDF/graph views remain secondary and reproducible.                                                                                |
| Module compactness rule                | Keep Lean implementation packets compact;<br>DB/audit-full exploration must not overload routine code-writing phases.                                                                          |
| Refactor regression protocol           | No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.                                                                                     |
| Refactor migration metrics             | No declaration migration;<br>no imports changed;<br>no Lean modules moved.                                                                                                                     |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                             |
| ExecComplex boundary policy            | No ExecComplex change. Operational scalar boundaries remain governed by EXEC-2 when relevant.                                                                                                  |
| Consumer-contract direction            | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.                                                                       |
| Governance-overhead review             | DB-REVIEW is non-blocking for Lean coding;<br>principles are mandatory, but technical DB work must remain compact and phase-bound.                                                             |
| Proof rehearsal before implementation  | For completed governance rows, validation is generator/YAML consistency and explicit guardrails. Future rows must rehearse schema round-trip and no-generator boundary.                        |
| Expected tactics                       | None;<br>documentation/tooling phase, not Lean proof phase.                                                                                                                                    |
| Expected mathlib imports               | None added.                                                                                                                                                                                    |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                  |
| Concrete computable instance traversal | Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.                                                             |
| Computable path non-skeleton status    | non-operational metadata path;<br>no computable CNNA object claimed                                                                                                                            |
| Expected executable witness            | For future DB phases: reproducible export/round-trip script output, not a Lean mathematical witness.                                                                                           |
| Mathlib complication log               | None;<br>no mathlib imports or Lean code changes.                                                                                                                                              |
| Noncomputable boundary audit           | No noncomputable Lean boundary involved;<br>semantic review is a documentation governance rule.                                                                                                |
| Mathlib mitigation/refactor layer      | Not applicable unless future Lean-readable metadata tooling is added;<br>then isolate outside operational path.                                                                                |

|                                      |                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bridge gap status                    | No mathematical bridge closed. Review gaps and missing blue marks remain explicit audit debt.                                                                                                                                                                                                                                                             |
| Natural backjump/refactor layer      | If a future view needs semantic truth not present in Lean/YAML metadata, backjump to the relevant Lean/object proof source, not to the database/view consumer.                                                                                                                                                                                            |
| Implementation prompt contract       | For future DB-REVIEW phases: preserve Lean-primary/generated-secondary semantics, keep expert review mandatory, and do not turn established theory or database views into CNNA inputs.                                                                                                                                                                    |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                               |
| How implemented                      | Not implemented yet. Planned documentation/tooling phase;<br>no Lean source change is authorized by this row.                                                                                                                                                                                                                                             |
| Verification / evidence              | Pending future phase activation. Current evidence is the V1.07 YAML registration and generated planning views only.                                                                                                                                                                                                                                       |
| Proof strategy                       | Future documentation/tooling proof: verify metadata-only status, round-trip integrity, and no CNNA-generative role.                                                                                                                                                                                                                                       |
| Post-implementation plan check       | Completed rows changed YAML/generator views only. Future rows must record schema/export artifacts, review status coverage, and no Lean-source mutation unless explicitly requested.                                                                                                                                                                       |
| Artifact packaging rule              | Return a complete zip with the updated YAML, generated tables/exports, report, and PDF when applicable. Do not claim Lean build success unless actually run.                                                                                                                                                                                              |
| Implementation notes                 | Future database/schema phase. Must preserve YAML as manual source and Lean as primary truth system.                                                                                                                                                                                                                                                       |
| Source materials / references        | <ul style="list-style-type: none"> <li>• user instruction for DB-REVIEW main side objective</li> <li>• Single YAML master</li> <li>• generated exports/tables</li> <li>• object proof dossiers</li> <li>• global no-go guardrails</li> </ul>                                                                                                              |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>• Lean-primary truth-system rule</li> <li>• generated-secondary documentation rule</li> <li>• discipline-specific expert-review obligation</li> <li>• blue/reference marking audit debt</li> </ul>                                                                                                                 |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>• no Lean source edit</li> <li>• no database/export object used as CNNA generator</li> <li>• no semantic infallibility claim</li> <li>• no established-theory input hidden behind blue/reference marking</li> </ul>                                                                                                |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23 and GNG24.</li> <li>• no 100-percent truth declaration</li> <li>• no generated artifact as semantic certificate</li> <li>• no ignored missing blue/reference-theory marking</li> <li>• no database/view layer as CNNA generator</li> </ul> |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>• Repository/scripts/generate_tables.py</li> <li>• Repository/generated_exports/*.tsv</li> <li>• Repository/tables/*.tex</li> </ul>                                                                                                                                                                                |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>• second-source-of-truth risk</li> <li>• generated-view-as-truth risk</li> <li>• unmarked-established-theory risk</li> </ul>                                                                                                                                                                                       |

### Phase 20.3 – Discipline-separated expert-review matrix

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field         | Record             |
|---------------|--------------------|
| Phase status  | W                  |
| Context class | DocumentationPhase |



|                                        |                                                                                                                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase documentation tier               | F3                                                                                                                                                                      |
| Phase documentation depth requirement  | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                               |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                      |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                     |
| Implementation status                  | planned                                                                                                                                                                 |
| Preparation status                     | planned                                                                                                                                                                 |
| Phase type                             | documentation                                                                                                                                                           |
| Phase-type-specific contract           | Documentation/tooling governance phase: output is review/query infrastructure, not mathematical derivation.                                                             |
| Title / state                          | Discipline-separated expert-review matrix                                                                                                                               |
| Preparation workflow                   | DB-REVIEW database-ready documentation and expert-review workflow.                                                                                                      |
| Source-to-CNNA mapping                 | Maps documentation/database/review infrastructure to generated-secondary metadata only. It cannot supply CNNA-generative mathematical or physical inputs.               |
| Dependency-depth audit                 | Future phase must audit YAML fields, generated exports, IDs, references, and review links before implementation.                                                        |
| Phase split decision                   | Split into immediate principles (20.1) and later technical/database/review phases (20.2-20.5).                                                                          |
| Positive path plan                     | Apply expert-review/no-infallibility principles immediately;<br>later add schema/export/review tooling without touching Lean truth semantics.                           |
| Module implementation manifest         | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.                                                                      |
| Semantic placement decision            | DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.                                                              |
| Reuse and duplication audit            | Reuse Single-YAML-Master and existing generator/export seams;<br>do not create a second authoritative planning source.                                                  |
| Canonical generator policy             | YAML remains the manual planning source;<br>generated DB/TSV/PDF/graph views remain secondary and reproducible.                                                         |
| Module compactness rule                | Keep Lean implementation packets compact;<br>DB/audit-full exploration must not overload routine code-writing phases.                                                   |
| Refactor regression protocol           | No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.                                                              |
| Refactor migration metrics             | No declaration migration;<br>no imports changed;<br>no Lean modules moved.                                                                                              |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                      |
| ExecComplex boundary policy            | No ExecComplex change. Operational scalar boundaries remain governed by EXEC-2 when relevant.                                                                           |
| Consumer-contract direction            | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.                                                |
| Governance-overhead review             | DB-REVIEW is non-blocking for Lean coding;<br>principles are mandatory, but technical DB work must remain compact and phase-bound.                                      |
| Proof rehearsal before implementation  | For completed governance rows, validation is generator/YAML consistency and explicit guardrails. Future rows must rehearse schema round-trip and no-generator boundary. |
| Expected tactics                       | None;<br>documentation/tooling phase, not Lean proof phase.                                                                                                             |
| Expected mathlib imports               | None added.                                                                                                                                                             |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                           |
| Concrete computable instance traversal | Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.                                      |
| Computable path non-skeleton status    | non-operational metadata path;<br>no computable CNNA object claimed                                                                                                     |
| Expected executable witness            | For future DB phases: reproducible export/round-trip script output, not a Lean mathematical witness.                                                                    |
| Mathlib complication log               | None;<br>no mathlib imports or Lean code changes.                                                                                                                       |
| Noncomputable boundary audit           | No noncomputable Lean boundary involved;<br>semantic review is a documentation governance rule.                                                                         |
| Mathlib mitigation/refactor layer      | Not applicable unless future Lean-readable metadata tooling is added;<br>then isolate outside operational path.                                                         |
| Bridge gap status                      | No mathematical bridge closed. Review gaps and missing blue marks remain explicit audit debt.                                                                           |



|                                      |                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Natural backjump/refactor layer      | If a future view needs semantic truth not present in Lean/YAML metadata, backjump to the relevant Lean/object proof source, not to the database/view consumer.                                                                                                                                                                                      |
| Implementation prompt contract       | For future DB-REVIEW phases: preserve Lean-primary/generated-secondary semantics, keep expert review mandatory, and do not turn established theory or database views into CNNA inputs.                                                                                                                                                              |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                         |
| How implemented                      | Not implemented yet. Planned documentation/tooling phase;<br>no Lean source change is authorized by this row.                                                                                                                                                                                                                                       |
| Verification / evidence              | Pending future phase activation. Current evidence is the V1.07 YAML registration and generated planning views only.                                                                                                                                                                                                                                 |
| Proof strategy                       | Future documentation/tooling proof: verify metadata-only status, round-trip integrity, and no CNNA-generative role.                                                                                                                                                                                                                                 |
| Post-implementation plan check       | Completed rows changed YAML/generator views only. Future rows must record schema/export artifacts, review status coverage, and no Lean-source mutation unless explicitly requested.                                                                                                                                                                 |
| Artifact packaging rule              | Return a complete zip with the updated YAML, generated tables/exports, report, and PDF when applicable. Do not claim Lean build success unless actually run.                                                                                                                                                                                        |
| Implementation notes                 | Future review-matrix phase. Must keep review status explicit by discipline and claim/object/phase.                                                                                                                                                                                                                                                  |
| Source materials / references        | <ul style="list-style-type: none"><li>• user instruction for DB-REVIEW main side objective</li><li>• Single YAML master</li><li>• generated exports/tables</li><li>• object proof dossiers</li><li>• global no-go guardrails</li></ul>                                                                                                              |
| Extracted established theory objects | <ul style="list-style-type: none"><li>• Lean-primary truth-system rule</li><li>• generated-secondary documentation rule</li><li>• discipline-specific expert-review obligation</li><li>• blue/reference marking audit debt</li></ul>                                                                                                                |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>• no Lean source edit</li><li>• no database/export object used as CNNA generator</li><li>• no semantic infallibility claim</li><li>• no established-theory input hidden behind blue/reference marking</li></ul>                                                                                               |
| No-go guardrails                     | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23 and GNG24.</li><li>• no 100-percent truth declaration</li><li>• no generated artifact as semantic certificate</li><li>• no ignored missing blue/reference-theory marking</li><li>• no database/view layer as CNNA generator</li></ul> |
| Reusable existing modules            | <ul style="list-style-type: none"><li>• Repository/scripts/generate_tables.py</li><li>• Repository/generated_exports/*.tsv</li><li>• Repository/tables/*.tex</li></ul>                                                                                                                                                                              |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"><li>• second-source-of-truth risk</li><li>• generated-view-as-truth risk</li><li>• unmarked-established-theory risk</li></ul>                                                                                                                                                                                     |

Phase 20.4 – Blue/reference-theory marking audit backlog

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                    | Record             |
|--------------------------|--------------------|
| Phase status             | W                  |
| Context class            | DocumentationPhase |
| Phase documentation tier | F3                 |

|                                        |                                                                                                                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase documentation depth requirement  | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                               |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                      |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                     |
| Implementation status                  | planned                                                                                                                                                                 |
| Preparation status                     | planned                                                                                                                                                                 |
| Phase type                             | documentation                                                                                                                                                           |
| Phase-type-specific contract           | Documentation/tooling governance phase: output is review/query infrastructure, not mathematical derivation.                                                             |
| Title / state                          | Blue/reference-theory marking audit backlog                                                                                                                             |
| Preparation workflow                   | DB-REVIEW database-ready documentation and expert-review workflow.                                                                                                      |
| Source-to-CNNA mapping                 | Maps documentation/database/review infrastructure to generated-secondary metadata only. It cannot supply CNNA-generative mathematical or physical inputs.               |
| Dependency-depth audit                 | Future phase must audit YAML fields, generated exports, IDs, references, and review links before implementation.                                                        |
| Phase split decision                   | Split into immediate principles (20.1) and later technical/database/review phases (20.2-20.5).                                                                          |
| Positive path plan                     | Apply expert-review/no-infallibility principles immediately;<br>later add schema/export/review tooling without touching Lean truth semantics.                           |
| Module implementation manifest         | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.                                                                      |
| Semantic placement decision            | DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.                                                              |
| Reuse and duplication audit            | Reuse Single-YAML-Master and existing generator/export seams;<br>do not create a second authoritative planning source.                                                  |
| Canonical generator policy             | YAML remains the manual planning source;<br>generated DB/TSV/PDF/graph views remain secondary and reproducible.                                                         |
| Module compactness rule                | Keep Lean implementation packets compact;<br>DB/audit-full exploration must not overload routine code-writing phases.                                                   |
| Refactor regression protocol           | No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.                                                              |
| Refactor migration metrics             | No declaration migration;<br>no imports changed;<br>no Lean modules moved.                                                                                              |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                      |
| ExecComplex boundary policy            | No ExecComplex change. Operational scalar boundaries remain governed by EXEC-2 when relevant.                                                                           |
| Consumer-contract direction            | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.                                                |
| Governance-overhead review             | DB-REVIEW is non-blocking for Lean coding;<br>principles are mandatory, but technical DB work must remain compact and phase-bound.                                      |
| Proof rehearsal before implementation  | For completed governance rows, validation is generator/YAML consistency and explicit guardrails. Future rows must rehearse schema round-trip and no-generator boundary. |
| Expected tactics                       | None;<br>documentation/tooling phase, not Lean proof phase.                                                                                                             |
| Expected mathlib imports               | None added.                                                                                                                                                             |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                           |
| Concrete computable instance traversal | Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.                                      |
| Computable path non-skeleton status    | non-operational metadata path;<br>no computable CNNA object claimed                                                                                                     |
| Expected executable witness            | For future DB phases: reproducible export/round-trip script output, not a Lean mathematical witness.                                                                    |
| Mathlib complication log               | None;<br>no mathlib imports or Lean code changes.                                                                                                                       |
| Noncomputable boundary audit           | No noncomputable Lean boundary involved;<br>semantic review is a documentation governance rule.                                                                         |
| Mathlib mitigation/refactor layer      | Not applicable unless future Lean-readable metadata tooling is added;<br>then isolate outside operational path.                                                         |
| Bridge gap status                      | No mathematical bridge closed. Review gaps and missing blue marks remain explicit audit debt.                                                                           |
| Natural backjump/refactor layer        | If a future view needs semantic truth not present in Lean/YAML metadata, backjump to the relevant Lean/object proof source, not to the database/view consumer.          |

|                                      |                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation prompt contract       | For future DB-REVIEW phases: preserve Lean-primary/generated-secondary semantics, keep expert review mandatory, and do not turn established theory or database views into CNNA inputs.                                                                                                                                                          |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                     |
| How implemented                      | Not implemented yet. Planned documentation/tooling phase;<br>no Lean source change is authorized by this row.                                                                                                                                                                                                                                   |
| Verification / evidence              | Pending future phase activation. Current evidence is the V1.07 YAML registration and generated planning views only.                                                                                                                                                                                                                             |
| Proof strategy                       | Future documentation/tooling proof: verify metadata-only status, round-trip integrity, and no CNNA-generative role.                                                                                                                                                                                                                             |
| Post-implementation plan check       | Completed rows changed YAML/generator views only. Future rows must record schema/export artifacts, review status coverage, and no Lean-source mutation unless explicitly requested.                                                                                                                                                             |
| Artifact packaging rule              | Return a complete zip with the updated YAML, generated tables/exports, report, and PDF when applicable. Do not claim Lean build success unless actually run.                                                                                                                                                                                    |
| Implementation notes                 | Future audit phase. Missing blue markings remain visible audit debt;<br>blue marks comparison/reference only.                                                                                                                                                                                                                                   |
| Source materials / references        | <ul style="list-style-type: none"> <li>user instruction for DB-REVIEW main side objective</li> <li>Single YAML master</li> <li>generated exports/tables</li> <li>object proof dossiers</li> <li>global no-go guardrails</li> </ul>                                                                                                              |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Lean-primary truth-system rule</li> <li>generated-secondary documentation rule</li> <li>discipline-specific expert-review obligation</li> <li>blue/reference marking audit debt</li> </ul>                                                                                                               |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source edit</li> <li>no database/export object used as CNNA generator</li> <li>no semantic infallibility claim</li> <li>no established-theory input hidden behind blue/reference marking</li> </ul>                                                                                              |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23 and GNG24.</li> <li>no 100-percent truth declaration</li> <li>no generated artifact as semantic certificate</li> <li>no ignored missing blue/reference-theory marking</li> <li>no database/view layer as CNNA generator</li> </ul> |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Repository/scripts/generate_tables.py</li> <li>Repository/generated_exports/*.tsv</li> <li>Repository/tables/*.tex</li> </ul>                                                                                                                                                                            |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>second-source-of-truth risk</li> <li>generated-view-as-truth risk</li> <li>unmarked-established-theory risk</li> </ul>                                                                                                                                                                                   |

### Phase 20.5 – Interactive/online documentation and database-backed exploration views

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                    | Record                  |
|--------------------------|-------------------------|
| Phase status             | W                       |
| Context class            | ToolInfrastructurePhase |
| Phase documentation tier | F1                      |

|                                        |                                                                                                                                                                                                |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                  | planned                                                                                                                                                                                        |
| Preparation status                     | planned                                                                                                                                                                                        |
| Phase type                             | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract           | Documentation/tooling governance phase: output is review/query infrastructure, not mathematical derivation.                                                                                    |
| Title / state                          | Interactive/online database-backed documentation views                                                                                                                                         |
| Preparation workflow                   | DB-REVIEW database-ready documentation and expert-review workflow.                                                                                                                             |
| Source-to-CNNA mapping                 | Maps documentation/database/review infrastructure to generated-secondary metadata only. It cannot supply CNNA-generative mathematical or physical inputs.                                      |
| Dependency-depth audit                 | Future phase must audit YAML fields, generated exports, IDs, references, and review links before implementation.                                                                               |
| Phase split decision                   | Split into immediate principles (20.1) and later technical/database/review phases (20.2-20.5).                                                                                                 |
| Positive path plan                     | Apply expert-review/no-infallibility principles immediately;<br>later add schema/export/review tooling without touching Lean truth semantics.                                                  |
| Module implementation manifest         | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.                                                                                             |
| Semantic placement decision            | DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.                                                                                     |
| Reuse and duplication audit            | Reuse Single-YAML-Master and existing generator/export seams;<br>do not create a second authoritative planning source.                                                                         |
| Canonical generator policy             | YAML remains the manual planning source;<br>generated DB/TSV/PDF/graph views remain secondary and reproducible.                                                                                |
| Module compactness rule                | Keep Lean implementation packets compact;<br>DB/audit-full exploration must not overload routine code-writing phases.                                                                          |
| Refactor regression protocol           | No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.                                                                                     |
| Refactor migration metrics             | No declaration migration;<br>no imports changed;<br>no Lean modules moved.                                                                                                                     |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                             |
| ExecComplex boundary policy            | No ExecComplex change. Operational scalar boundaries remain governed by EXEC-2 when relevant.                                                                                                  |
| Consumer-contract direction            | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.                                                                       |
| Governance-overhead review             | DB-REVIEW is non-blocking for Lean coding;<br>principles are mandatory, but technical DB work must remain compact and phase-bound.                                                             |
| Proof rehearsal before implementation  | For completed governance rows, validation is generator/YAML consistency and explicit guardrails. Future rows must rehearse schema round-trip and no-generator boundary.                        |
| Expected tactics                       | None;<br>documentation/tooling phase, not Lean proof phase.                                                                                                                                    |
| Expected mathlib imports               | None added.                                                                                                                                                                                    |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                  |
| Concrete computable instance traversal | Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.                                                             |
| Computable path non-skeleton status    | non-operational metadata path;<br>no computable CNNA object claimed                                                                                                                            |
| Expected executable witness            | For future DB phases: reproducible export/round-trip script output, not a Lean mathematical witness.                                                                                           |
| Mathlib complication log               | None;<br>no mathlib imports or Lean code changes.                                                                                                                                              |
| Noncomputable boundary audit           | No noncomputable Lean boundary involved;<br>semantic review is a documentation governance rule.                                                                                                |
| Mathlib mitigation/refactor layer      | Not applicable unless future Lean-readable metadata tooling is added;<br>then isolate outside operational path.                                                                                |
| Bridge gap status                      | No mathematical bridge closed. Review gaps and missing blue marks remain explicit audit debt.                                                                                                  |
| Natural backjump/refactor layer        | If a future view needs semantic truth not present in Lean/YAML metadata, backjump to the relevant Lean/object proof source, not to the database/view consumer.                                 |

|                                      |                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation prompt contract       | For future DB-REVIEW phases: preserve Lean-primary/generated-secondary semantics, keep expert review mandatory, and do not turn established theory or database views into CNNA inputs.                                                                                                                                                          |
| Global no-go inheritance             | yes;<br>all global_no_go_guardrails inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                     |
| How implemented                      | Not implemented yet. Planned documentation/tooling phase;<br>no Lean source change is authorized by this row.                                                                                                                                                                                                                                   |
| Verification / evidence              | Pending future phase activation. Current evidence is the V1.07 YAML registration and generated planning views only.                                                                                                                                                                                                                             |
| Proof strategy                       | Future documentation/tooling proof: verify metadata-only status, round-trip integrity, and no CNNA-generative role.                                                                                                                                                                                                                             |
| Post-implementation plan check       | Completed rows changed YAML/generator views only. Future rows must record schema/export artifacts, review status coverage, and no Lean-source mutation unless explicitly requested.                                                                                                                                                             |
| Artifact packaging rule              | Return a complete zip with the updated YAML, generated tables/exports, report, and PDF when applicable. Do not claim Lean build success unless actually run.                                                                                                                                                                                    |
| Implementation notes                 | Future interactive/online documentation phase. Views remain generated-secondary and may not certify semantic truth.                                                                                                                                                                                                                             |
| Source materials / references        | <ul style="list-style-type: none"> <li>user instruction for DB-REVIEW main side objective</li> <li>Single YAML master</li> <li>generated exports/tables</li> <li>object proof dossiers</li> <li>global no-go guardrails</li> </ul>                                                                                                              |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Lean-primary truth-system rule</li> <li>generated-secondary documentation rule</li> <li>discipline-specific expert-review obligation</li> <li>blue/reference marking audit debt</li> </ul>                                                                                                               |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source edit</li> <li>no database/export object used as CNNA generator</li> <li>no semantic infallibility claim</li> <li>no established-theory input hidden behind blue/reference marking</li> </ul>                                                                                              |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23 and GNG24.</li> <li>no 100-percent truth declaration</li> <li>no generated artifact as semantic certificate</li> <li>no ignored missing blue/reference-theory marking</li> <li>no database/view layer as CNNA generator</li> </ul> |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Repository/scripts/generate_tables.py</li> <li>Repository/generated_exports/*.tsv</li> <li>Repository/tables/*.tex</li> </ul>                                                                                                                                                                            |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>second-source-of-truth risk</li> <li>generated-view-as-truth risk</li> <li>unmarked-established-theory risk</li> </ul>                                                                                                                                                                                   |

### Phase 20.6 – Context-filtered documentation slices: all sections retained, only query-relevant rows emitted

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #c8e6c9;">G</span>                                                                                         |
| Context class                         | GeneratedArtifactPhase                                                                                                                    |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |

|                                        |                                                                                                                                                                                                                                     |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                                  |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                 |
| Implementation status                  | completed                                                                                                                                                                                                                           |
| Preparation status                     | completed-tooling-function                                                                                                                                                                                                          |
| Phase type                             | documentation                                                                                                                                                                                                                       |
| Phase-type-specific contract           | Documentation/tooling governance phase: output is review/query infrastructure, not mathematical derivation.                                                                                                                         |
| Title / state                          | Context-filtered documentation slice generator                                                                                                                                                                                      |
| Preparation workflow                   | DB-REVIEW database-ready documentation and expert-review workflow.                                                                                                                                                                  |
| Source-to-CNNA mapping                 | Maps documentation slicing to review/navigation only. Filtered rows may reveal CNNA objects/modules, but the slice itself contributes no CNNA-generative mathematical content.                                                      |
| Dependency-depth audit                 | No Lean dependency introduced. The generator depends only on PyYAML, pdflatex for optional PDF output, the Single-YAML-Master, and generated TSV exports.                                                                           |
| Phase split decision                   | Implemented as a DB-REVIEW tooling phase 20.6 rather than as a Lean/math phase or as a separate authoritative documentation source.                                                                                                 |
| Positive path plan                     | Use context slices for requests such as DtN Matrix documentation and for phase-closeout delta documentation. Future profiles can be added when the vocabulary is curated and auditable.                                             |
| Module implementation manifest         | Added Repository/scripts/generate_context_doc.py. No Lean module manifest change.                                                                                                                                                   |
| Semantic placement decision            | Belongs to DB-REVIEW generated-secondary documentation tooling. It is not a Pillar-A mathematical derivation layer and not a database truth layer.                                                                                  |
| Reuse and duplication audit            | Reuses the Single-YAML-Master and generated_exports. Does not create a second master; output manifest records the source and filter terms.                                                                                          |
| Canonical generator policy             | Context documents are reproducible generated views. Curated profiles live in YAML so they can later round-trip into the database representation.                                                                                    |
| Module compactness rule                | Keeps targeted documentation compact by filtering rows per context while retaining section topology. Reduces overhead during Lean implementation closeouts.                                                                         |
| Refactor regression protocol           | No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.                                                                                                                          |
| Refactor migration metrics             | No declaration migration;<br>no imports changed;<br>no Lean modules moved.                                                                                                                                                          |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                                  |
| ExecComplex boundary policy            | No ExecComplex change. Operational scalar boundaries remain governed by EXEC-2 when relevant.                                                                                                                                       |
| Consumer-contract direction            | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.                                                                                                            |
| Governance-overhead review             | Positive: reduces review overhead by creating small context packets. Risk: filter misses can hide relevant rows; mitigated by manifest terms, hit counts, and expert-review warning.                                                |
| Proof rehearsal before implementation  | Check that every context packet includes all major sections, row counts, source kind, filter terms, manifest, and generated-secondary warning. Do not treat absence of a matched row as proof of irrelevance without expert review. |
| Expected tactics                       | None;<br>Python documentation/tooling phase, not Lean proof phase.                                                                                                                                                                  |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                       |
| Concrete computable instance traversal | Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.                                                                                                  |
| Computable path non-skeleton status    | non-operational metadata path;<br>no computable CNNA object claimed                                                                                                                                                                 |
| Expected executable witness            | For future DB phases: reproducible export/round-trip script output, not a Lean mathematical witness.                                                                                                                                |
| Mathlib complication log               | None;<br>no mathlib imports or Lean code changes.                                                                                                                                                                                   |
| Noncomputable boundary audit           | No noncomputable Lean boundary involved;<br>semantic review is a documentation governance rule.                                                                                                                                     |
| Mathlib mitigation/refactor layer      | Not applicable unless future Lean-readable metadata tooling is added;<br>then isolate outside operational path.                                                                                                                     |
| Bridge gap status                      | No mathematical bridge closed. Context filtering may expose bridge gaps more efficiently, but every gap remains subject to Lean/object/proof-dossier review.                                                                        |
| Natural backjump/refactor layer        | If a context slice misses relevant content, fix the profile/metadata at the natural documentation schema layer;<br>if the missing content is semantic, backjump to the relevant Lean/object proof source.                           |
| Implementation prompt contract         | For context-doc updates: regenerate tables first, then generate the context slice from YAML/exports;<br>preserve all major sections, include manifest/hit counts, and never claim semantic completeness without expert review.      |



|                                      |                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global no-go inheritance             | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG24)                                                                                                                                                                                                                                                                                         |
| How implemented                      | Added <code>scripts/generate_context_doc.py</code> plus <code>context_documentation_profiles</code> . The generator creates query/profile-specific documentation packets with all major sections retained and only context-relevant rows emitted.                                                                                                                |
| Verification / evidence              | Ran <code>scripts/generate_tables.py</code> and <code>scripts/generate_context_doc.py "DtN Matrix" --profile dtn-matrix --slug dtn_matrix</code> . The sample <code>context_docs/dtn_matrix</code> packet contains a manifest, section summary, TeX source, compile log, and PDF if <code>pdflatex</code> succeeds.                                              |
| Proof strategy                       | Tooling proof by reproducible filtering: consume only YAML/export rows, retain section topology, emit filter manifest and hit counts, and state Lean-primary/expert-review warning in every context document.                                                                                                                                                    |
| Build result                         | No Lean build run;<br>no Lean source changed. Python <code>generator/tables/context-doc</code> generation only.                                                                                                                                                                                                                                                  |
| Post-implementation plan check       | V1.08 adds a reusable context-doc generator and a DtN sample packet. Future phase closeouts should use context slices to document changed context instead of forcing full-document review unless a full PDF is explicitly needed.                                                                                                                                |
| Artifact packaging rule              | Return a complete zip with updated YAML, scripts, generated tables/exports, <code>context_docs</code> sample, report, and full PDF when applicable. Do not claim Lean build success unless actually run.                                                                                                                                                         |
| Implementation notes                 | This is a generated-secondary documentation view, not a truth source. It is intended for targeted requests and phase-closeout deltas without regenerating or reading the entire full documentation every time.                                                                                                                                                   |
| Source materials / references        | <ul style="list-style-type: none"><li>• user instruction for DB-REVIEW main side objective</li><li>• Single YAML master</li><li>• generated exports/tables</li><li>• object proof dossiers</li><li>• global no-go guardrails</li></ul>                                                                                                                           |
| Extracted established theory objects | <ul style="list-style-type: none"><li>• context-filtered documentation slice</li><li>• filter manifest</li><li>• curated dtn-matrix profile</li><li>• section hit summary</li><li>• generated-secondary view warning</li></ul>                                                                                                                                   |
| Implemented modules                  | <ul style="list-style-type: none"><li>• <code>Repository/scripts/generate_context_doc.py</code></li></ul>                                                                                                                                                                                                                                                        |
| Added modules                        | <ul style="list-style-type: none"><li>• <code>Repository/scripts/generate_context_doc.py</code></li></ul>                                                                                                                                                                                                                                                        |
| Modified modules                     | <ul style="list-style-type: none"><li>• <code>Repository/masters/cnna_roadmap_master.yaml</code></li><li>• <code>README.md</code></li></ul>                                                                                                                                                                                                                      |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>• no Lean source edit</li><li>• no database/export object used as CNNA generator</li><li>• no semantic infallibility claim</li><li>• no established-theory input hidden behind blue/reference marking</li></ul>                                                                                                            |
| No-go guardrails                     | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: all <code>global_no_go_guardrails</code> apply, especially GNG23 and GNG24.</li><li>• no 100-percent truth declaration</li><li>• no generated artifact as semantic certificate</li><li>• no ignored missing blue/reference-theory marking</li><li>• no database/view layer as CNNA generator</li></ul> |
| Proposed added modules               | <ul style="list-style-type: none"><li>• <code>Repository/scripts/generate_context_doc.py</code></li></ul>                                                                                                                                                                                                                                                        |
| Proposed modified modules            | <ul style="list-style-type: none"><li>• <code>Repository/masters/cnna_roadmap_master.yaml</code></li><li>• <code>README.md</code></li></ul>                                                                                                                                                                                                                      |
| Reusable existing modules            | <ul style="list-style-type: none"><li>• <code>Repository/scripts/generate_tables.py</code></li><li>• <code>Repository/generated_exports/*.tsv</code></li><li>• <code>Repository/masters/cnna_roadmap_master.yaml</code></li></ul>                                                                                                                                |



|                                      |                                                                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>second-source-of-truth risk</li> <li>profile-overfiltering risk</li> <li>false-negative relevance risk</li> <li>generated-view-as-truth risk</li> </ul> |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 20.7 – Stats front section, tool-interface guiding metaphor, and project licensing notices

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                                                                 |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3; border: 1px solid black; padding: 0 2px;">G</span>                                                                                                                                                             |
| Context class                         | GeneratedArtifactPhase                                                                                                                                                                                                                                 |
| Phase documentation tier              | F3                                                                                                                                                                                                                                                     |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                                                                              |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                                                     |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                                                    |
| Implementation status                 | completed                                                                                                                                                                                                                                              |
| Preparation status                    | completed-tooling-function                                                                                                                                                                                                                             |
| Phase type                            | documentation                                                                                                                                                                                                                                          |
| Phase-type-specific contract          | Documentation/tooling governance phase: output is review/interface infrastructure, not mathematical derivation.                                                                                                                                        |
| Title / state                         | Stats front section, tool-interface guiding metaphor, and licensing notices                                                                                                                                                                            |
| Preparation workflow                  | DB-REVIEW database-ready documentation and expert-review workflow.                                                                                                                                                                                     |
| Source-to-CNNA mapping                | Maps the analogy to documentation/review practice only: active phase as bright object; rest of Lean code and unaudited semantics as complement; YAML/registers/graphs/database views as interface. No CNNA-generative mathematical data is introduced. |
| Dependency-depth audit                | No Lean dependency introduced. The generator depends on YAML, repo inventory JSON, source-file counts, and existing generated module inventory tables.                                                                                                 |
| Phase split decision                  | Implemented as DB-REVIEW documentation phase 20.7; not a Lean/math phase and not a database implementation phase.                                                                                                                                      |
| Positive path plan                    | Use Stats as a compact orientation surface and the tool-interface metaphor as a reminder that every active phase must expose its boundary, hidden complement risks, blue markings, and expert-review duties.                                           |
| Module implementation manifest        | Modified Repository/scripts/generate_tables.py only; no Lean module manifest change.                                                                                                                                                                   |
| Semantic placement decision           | Belongs to DB-REVIEW generated-secondary documentation tooling. It is not a Pillar-A mathematical derivation layer, not a database truth layer, and not a formal CNNA theorem.                                                                         |
| Reuse and duplication audit           | Reuses the existing module inventory table instead of adding a duplicate standalone module-inventory section. Adds only a generated project_stats view and YAML records for metaphor/license.                                                          |
| Canonical generator policy            | Stats rows are reproducible generated views. Guiding metaphors and license notices live in YAML so they can later round-trip into database views.                                                                                                      |
| Module compactness rule               | Keeps the front section compact: short facts table plus the existing inventory/graph view. Avoids duplicating the old module-inventory section.                                                                                                        |
| Refactor regression protocol          | No Lean refactor. Future tooling must preserve generated-view reproducibility, YAML single-source rule, and old-module-inventory-under-Stats placement.                                                                                                |
| Refactor migration metrics            | No declaration migration; no imports changed; no Lean modules moved.                                                                                                                                                                                   |
| Lean/mathlib version policy           | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                                                     |
| ExecComplex boundary policy           | No ExecComplex change. Operational scalar boundaries remain governed by EXEC-2 when relevant.                                                                                                                                                          |
| Consumer-contract direction           | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.                                                                                                                               |
| Governance-overhead review            | Positive: improves orientation and makes expert-review/no-infallibility context visible at the front. Risk: users may overread the metaphor; mitigated by GNG25 and explicit boundary text.                                                            |
| Proof rehearsal before implementation | Check that Stats uses generated counts, includes the former module inventory/graphs, states licensing, and keeps the metaphor explicitly non-ontological and non-theorematic.                                                                          |
| Expected tactics                      | None; Python documentation/tooling phase, not Lean proof phase.                                                                                                                                                                                        |

|                                        |                                                                                                                                                                                                                                                                                                              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected mathlib imports               | None added.                                                                                                                                                                                                                                                                                                  |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                                                                                |
| Concrete computable instance traversal | Not an operational Lean computation. Concrete traversal is metadata path: <code>YAML/repo inventory/source counts</code> -> generated TSV/TeX/PDF -> future database view.                                                                                                                                   |
| Computable path non-skeleton status    | non-operational metadata path;<br>no computable CNNA object claimed                                                                                                                                                                                                                                          |
| Expected executable witness            | <code>project_stats.generated.tsv</code> and the front Stats PDF section, not a Lean mathematical witness.                                                                                                                                                                                                   |
| Mathlib complication log               | None;<br>no mathlib imports or Lean code changes.                                                                                                                                                                                                                                                            |
| Noncomputable boundary audit           | No noncomputable Lean boundary involved;<br>semantic review is a documentation governance rule.                                                                                                                                                                                                              |
| Mathlib mitigation/refactor layer      | Not applicable unless future Lean-readable metadata tooling is added;<br>then isolate outside operational path.                                                                                                                                                                                              |
| Bridge gap status                      | No mathematical bridge closed. The interface metaphor may help reveal bridge gaps, but it cannot close them.                                                                                                                                                                                                 |
| Natural backjump/refactor layer        | If Stats counts are wrong, fix the repository-inventory/generator layer. If the metaphor is misused as a theorem or ontology, enforce GNG25 and correct the documentation layer.                                                                                                                             |
| Implementation prompt contract         | For future front-documentation edits: keep Stats compact;<br>embed generated module inventory under Stats;<br>avoid duplicate standalone sections;<br>keep Lean-primary/expert-review/no-infallibility warnings active.                                                                                      |
| Global no-go inheritance               | yes;<br>all <code>global_no_go_guardrails</code> inherited (GNG01-GNG25)                                                                                                                                                                                                                                     |
| How implemented                        | Added <code>guiding_metaphors</code> and <code>license_notice</code> records to the Single-YAML-Master. Extended <code>scripts/generate_tables.py</code> to compute <code>project_stats.generated.tsv</code> and render a front Stats section that embeds the former module inventory and derivation graphs. |
| Verification / evidence                | Run <code>scripts/generate_tables.py</code> , compile the V1.09 full PDF, and spot-render opening pages to verify that Stats appears at the beginning, the old standalone module-inventory section is removed, and module inventory content is retained under Stats.                                         |
| Proof strategy                         | Tooling proof by reproducible rendering: project facts are computed from <code>YAML/repo-inventory/source</code> files;<br>the old module inventory is included under Stats;<br>the guiding metaphor is explicitly bounded as review architecture only;<br>license notices are concise scope records.        |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table generation and PDF compilation only.                                                                                                                                                                                                              |
| Post-implementation plan check         | V1.09 makes the document start with project facts and an explicit interface/review frame. Future context-doc and phase-closeout outputs should be able to reuse the same Stats/export data.                                                                                                                  |
| Artifact packaging rule                | Return a complete zip with updated YAML, scripts, generated tables/exports, report, full PDF, and existing context-doc samples when applicable. Do not claim Lean build success unless actually run.                                                                                                         |
| Implementation notes                   | The Stats section is a generated-secondary front surface. It is useful for orientation and review, but it does not certify semantic truth. The tool-interface principle is a Leitmetapher with operative review consequences, not a mathematical CNNA theorem.                                               |
| Source materials / references          | <ul style="list-style-type: none"> <li>• user instruction for tool-interface guiding metaphor</li> <li>• user licensing references CC BY 4.0 and MIT</li> <li>• Single YAML master</li> <li>• <code>repo_inventory/raw_export/*.json</code></li> <li>• generated exports/tables</li> </ul>                   |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>• tool-interface principle</li> <li>• project stats front section</li> <li>• documentation license notice</li> <li>• code license notice</li> <li>• generated-secondary project-fact table</li> </ul>                                                                 |
| Implemented modules                    | <ul style="list-style-type: none"> <li>• <code>Repository/scripts/generate_tables.py</code></li> </ul>                                                                                                                                                                                                       |

|                                      |                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna__roadmap__master.yaml</li> <li>Repository/scripts/generate__tables.py</li> <li>README.md</li> <li>Repository/CNNA_Planning_and_Documentation_Tool_V1_09_Data_2026-05-10T05-51-42_CEST_A3_English.tex</li> </ul>                                                              |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source edit</li> <li>no generated stats as semantic certificate</li> <li>no metaphor as CNNA theorem</li> <li>no license notice used to override upstream third-party rights</li> <li>no second truth system</li> </ul>                                                                      |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global__no__go__guardrails apply, especially GNG23, GNG24 and GNG25.</li> <li>no 100-percent truth declaration</li> <li>no generated artifact as semantic certificate</li> <li>no metaphor as formal CNNA theorem</li> <li>no database/view layer as CNNA generator</li> </ul> |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/scripts/generate__tables.py</li> <li>Repository/masters/cnna__roadmap__master.yaml</li> <li>README.md</li> </ul>                                                                                                                                                                          |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>Repository/scripts/generate__tables.py</li> <li>Repository/repo__inventory/raw__export/*.json</li> <li>Repository/tables/module__inventory.tex</li> </ul>                                                                                                                                            |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>second-source-of-truth risk</li> <li>metaphor-as-theorem risk</li> <li>project-stats-as-truth-certificate risk</li> <li>license-overreach risk</li> </ul>                                                                                                                                            |

### Phase 20.8 – Root artifact retention and generated-output cleanup policy

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #90EE90; border: 1px solid black; padding: 0 2px;">G</span>                                                |
| Context class                         | GeneratedArtifactPhase                                                                                                                    |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |
| Implementation status                 | completed                                                                                                                                 |
| Preparation status                    | completed-tooling-function                                                                                                                |
| Phase type                            | documentation                                                                                                                             |
| Phase-type-specific contract          | Documentation/tooling governance phase: output is repository hygiene infrastructure, not mathematical derivation.                         |
| Title / state                         | Root artifact retention and generated-output cleanup policy                                                                               |
| Preparation workflow                  | Root artifact retention and generated-output cleanup workflow.                                                                            |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source-to-CNNA mapping                 | Maps repository hygiene to Lean-primary/generated-secondary discipline: artifacts that can be regenerated are not CNNA truth sources; archived reports are trace records; YAML/code remain maintained sources.                                                                                                                                                                                    |
| Dependency-depth audit                 | No Lean dependency introduced. Tool dependencies are Python stdlib, PyYAML already used by generate_tables.py, and optional pdflatex for full documentation generation.                                                                                                                                                                                                                           |
| Phase split decision                   | Implemented as DB-REVIEW documentation/tooling phase 20.8, not as a Lean refactor and not as a database implementation phase.                                                                                                                                                                                                                                                                     |
| Positive path plan                     | Keep the root clean before returning to the active content path. Future phase closeouts should generate context docs or full docs only on demand and avoid accumulating root artifacts.                                                                                                                                                                                                           |
| Module implementation manifest         | Added two Python tool scripts only; no Lean module manifest change.                                                                                                                                                                                                                                                                                                                               |
| Semantic placement decision            | Belongs to DB-REVIEW generated-secondary documentation/tooling. It is a retention policy, not a Pillar-A mathematical derivation layer.                                                                                                                                                                                                                                                           |
| Reuse and duplication audit            | Reuses existing generation pipeline; avoids duplicating versioned full-doc TeX/PDF in root. Adds a single canonical full-document generator path.                                                                                                                                                                                                                                                 |
| Canonical generator policy             | Full document TeX/PDF must be generated from YAML/tables by script and treated as disposable unless explicitly requested as an external artifact.                                                                                                                                                                                                                                                 |
| Module compactness rule                | Reduces package clutter and clarifies active source boundaries without changing Lean source structure.                                                                                                                                                                                                                                                                                            |
| Refactor regression protocol           | No Lean refactor. Verify source/control files remain, reports are archived, and root generated artifacts are absent after cleanup.                                                                                                                                                                                                                                                                |
| Refactor migration metrics             | No declaration migration; no imports changed; no Lean modules moved. Root report files moved to archive/reports; root generated PDFs/TeX/build products removed.                                                                                                                                                                                                                                  |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                                                                                                                                                                                                |
| ExecComplex boundary policy            | No ExecComplex change.                                                                                                                                                                                                                                                                                                                                                                            |
| Consumer-contract direction            | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.                                                                                                                                                                                                                                                                          |
| Governance-overhead review             | Positive: reduces root clutter and prevents generated artifacts from being mistaken for sources. Risk: losing convenient PDFs; mitigated by explicit regeneration script and context-documentation workflow.                                                                                                                                                                                      |
| Proof rehearsal before implementation  | Run root audit, add generator/cleanup scripts, regenerate tables, optionally compile full PDF outside root, execute cleanup, and verify root artifact invariants.                                                                                                                                                                                                                                 |
| Expected tactics                       | None; Python/documentation-tooling phase, not Lean proof phase.                                                                                                                                                                                                                                                                                                                                   |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                                                                                                                                                                                       |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Not an operational Lean computation. Concrete traversal is filesystem classification: root artifacts -> archive or delete -> regenerate on demand.                                                                                                                                                                                                                                                |
| Computable path non-skeleton status    | non-operational metadata/tooling path; no computable CNNA object claimed                                                                                                                                                                                                                                                                                                                          |
| Expected executable witness            | scripts/generate_full_document.py, scripts/clean_generated_root_artifacts.py, root cleanup report JSON, and absence of root PDF/TeX artifacts after cleanup.                                                                                                                                                                                                                                      |
| Mathlib complication log               | None; no mathlib imports or Lean code changes.                                                                                                                                                                                                                                                                                                                                                    |
| Noncomputable boundary audit           | No noncomputable Lean boundary involved; only generated documentation artifacts are classified.                                                                                                                                                                                                                                                                                                   |
| Mathlib mitigation/refactor layer      | Not applicable.                                                                                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | No mathematical bridge closed. The bridge here is toolchain hygiene between YAML source data and disposable generated outputs.                                                                                                                                                                                                                                                                    |
| Natural backjump/refactor layer        | If root artifacts reappear, fix release packaging/cleanup workflow, not the Lean source tree. If a report becomes source-like, promote its content into YAML rather than keeping it as an active root Markdown file.                                                                                                                                                                              |
| Implementation prompt contract         | For future packaging: do not retain full PDFs/TeX in root; archive reports; keep README in root; regenerate documentation on demand from YAML/tables/scripts.                                                                                                                                                                                                                                     |
| Global no-go inheritance               | yes; all global_no_go_guardrails inherited (GNG01-GNG27)                                                                                                                                                                                                                                                                                                                                          |
| How implemented                        | Audited V1.12 project-root artifacts, added GNG27/SCF46/DBR8, introduced scripts/generate_full_document.py for disposable full-document generation under generated_docs/full/, introduced scripts/clean_generated_root_artifacts.py for root cleanup, and moved historical Markdown reports to archive/reports/ while removing root PDF/TeX/LaTeX build products.                                 |
| Verification / evidence                | Root extension audit before cleanup found 36 Markdown reports, 20 PDFs, 20 TeX files, 19 aux files, 22 logs, 8 out files, 11 toc files and latexmk artifacts. V1.13 validates YAML, regenerates tables/exports, compiles an optional full document under generated_docs/full/, then verifies the packaged root has README.md as the only Markdown file and no root PDF/TeX/LaTeX build artifacts. |

|                                      |                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proof strategy                       | Classify root files by epistemic role: source/control files stay active; generated full-doc PDFs/TeX and LaTeX byproducts are disposable; historical Markdown reports are archived for traceability. Preserve regeneration commands instead of retaining rendered artifacts.                                   |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table generation and optional PDF compilation only.                                                                                                                                                                                                       |
| Post-implementation plan check       | V1.13 makes generated documentation disposable and root hygiene explicit. The active cursor remains 1.1.1.1.5.                                                                                                                                                                                                 |
| Artifact packaging rule              | Return a complete zip with YAML, scripts, tables/exports, repo inventory, repo snapshot, archived reports, and no retained root PDF/TeX/LaTeX byproducts unless the user explicitly asks for a PDF deliverable.                                                                                                |
| Implementation notes                 | This is repository hygiene, not CNNA mathematics. PDFs/TeX are disposable because they can be regenerated from YAML/tables/scripts. Markdown reports are archival evidence and should not crowd the active root or be mistaken for authoritative sources.                                                      |
| Source materials / references        | <ul style="list-style-type: none"><li>• user instruction about root Markdown/PDF/TeX artifacts</li><li>• V1.12 root file inventory</li><li>• Single YAML master</li><li>• generated tables/exports</li><li>• README.md</li></ul>                                                                               |
| Extracted established theory objects | <ul style="list-style-type: none"><li>• root artifact retention policy</li><li>• disposable full-document output path</li><li>• Markdown report archive</li><li>• root cleanup script</li><li>• full-document generator script</li></ul>                                                                       |
| Implemented modules                  | <ul style="list-style-type: none"><li>• Repository/scripts/generate_full_document.py</li><li>• Repository/scripts/clean_generated_root_artifacts.py</li></ul>                                                                                                                                                  |
| Added modules                        | <ul style="list-style-type: none"><li>• Repository/scripts/generate_full_document.py</li><li>• Repository/scripts/clean_generated_root_artifacts.py</li></ul>                                                                                                                                                  |
| Modified modules                     | <ul style="list-style-type: none"><li>• Repository/masters/cnna_roadmap_master.yaml</li><li>• README.md</li></ul>                                                                                                                                                                                              |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>• no Lean source edit</li><li>• no generated PDF/TeX as active root source</li><li>• no historical report as authoritative source</li><li>• no deletion of YAML/scripts/repo_snapshot/repo_inventory/tables/generated_exports</li><li>• no second truth system</li></ul> |
| No-go guardrails                     | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23-GNG24 and GNG27.</li><li>• no root generated PDF/TeX retention</li><li>• no historical report as active source</li><li>• no hidden deletion of source/control data</li></ul>                     |
| Proposed added modules               | <ul style="list-style-type: none"><li>• Repository/scripts/generate_full_document.py</li><li>• Repository/scripts/clean_generated_root_artifacts.py</li></ul>                                                                                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"><li>• Repository/masters/cnna_roadmap_master.yaml</li><li>• README.md</li></ul>                                                                                                                                                                                              |
| Reusable existing modules            | <ul style="list-style-type: none"><li>• Repository/scripts/generate_tables.py</li><li>• Repository/scripts/generate_context_doc.py</li><li>• Repository/tables/*.tex</li><li>• Repository/generated_exports/*.tsv</li></ul>                                                                                    |

|                                      |                                                                                                                                                                                                                 |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>• root PDF/TEX clutter</li> <li>• historical report as false source</li> <li>• generated-secondary artifact treated as authoritative</li> <li>• package bloat</li> </ul> |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 20.9 – Current README and workflow-orientation manual

**Table legend.**   **G** = green/closed and derived-only;   **W** = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   **O** = orange/implementable after stated predecessor closure;   **Y** = yellow/active and implementable now;   **R** = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                         |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;"> </span> <b>G</b>                                                                                                                     |
| Context class                         | GeneratedArtifactPhase                                                                                                                                                         |
| Phase documentation tier              | F3                                                                                                                                                                             |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                      |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                            |
| Implementation status                 | completed                                                                                                                                                                      |
| Preparation status                    | completed-tooling-documentation                                                                                                                                                |
| Phase type                            | documentation                                                                                                                                                                  |
| Phase-type-specific contract          | Documentation phase: clarify current tool use, commands, workflow contract, and epistemic status without changing Lean semantics.                                              |
| Title / state                         | Current README and workflow-orientation manual                                                                                                                                 |
| Preparation workflow                  | README/workflow-orientation update workflow.                                                                                                                                   |
| Source-to-CNNA mapping                | Maps the tool workflow to an interface layer over Lean-primary code and generated-secondary documentation without creating a CNNA mathematical input.                          |
| Dependency-depth audit                | No Lean dependency introduced. Documentation commands depend on Python scripts and optional pdflatex only.                                                                     |
| Phase split decision                  | Implemented as DB-REVIEW documentation/tooling phase 20.9, not as a Lean implementation phase and not as database implementation.                                              |
| Positive path plan                    | Use the new README as root entry point before returning to the active 1.1.1.1.5 module-consolidation gate and then 1.1.1.2.                                                    |
| Module implementation manifest        | No Lean modules added/modified. Documentation files only: README.md and V1_14 report; YAML metadata/governance records updated.                                                |
| Semantic placement decision           | Natural placement is DB-REVIEW side objective because this is a documentation/workflow entry-surface update.                                                                   |
| Reuse and duplication audit           | Reuses existing scripts and generated views; no new generator required.                                                                                                        |
| Canonical generator policy            | README documents existing canonical generators instead of creating alternative documentation sources.                                                                          |
| Module compactness rule               | Keep README compact and current; place detailed history in archive/reports or generated docs.                                                                                  |
| Refactor regression protocol          | No Lean refactor. Regression check is generate_tables.py plus root cleanliness and README content audit.                                                                       |
| Refactor migration metrics            | No declaration/module migration; no Lean imports changed.                                                                                                                      |
| Lean/mathlib version policy           | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                             |
| ExecComplex boundary policy           | No ExecComplex change or consumption.                                                                                                                                          |
| Consumer-contract direction           | Human/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.                                                                                   |
| Governance-overhead review            | Positive: replaces chronologically bloated README with a compact current manual. Risk: README could drift; mitigated by keeping source of truth in YAML and generated exports. |
| Proof rehearsal before implementation | Read current README, replace with compact manual, update master records, regenerate tables, verify root cleanliness and absence of Lean changes.                               |
| Expected tactics                      | None; documentation-tooling phase, not Lean proof phase.                                                                                                                       |
| Expected mathlib imports              | None added.                                                                                                                                                                    |



|                                        |                                                                                                                                                                                                                                                                        |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                                          |
| Concrete computable instance traversal | No mathematical traversal;<br>documentation-tooling surface only.                                                                                                                                                                                                      |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>README documents tool commands only.                                                                                                                                                                                       |
| Expected executable witness            | Successful Python generation of tables/exports;<br>no Lean executable witness required.                                                                                                                                                                                |
| Mathlib complication log               | None;<br>no Lean/mathlib import touched.                                                                                                                                                                                                                               |
| Noncomputable boundary audit           | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                                                        |
| Mathlib mitigation/refactor layer      | Not applicable;<br>documentation tooling only.                                                                                                                                                                                                                         |
| Bridge gap status                      | No mathematical bridge closed. The bridge here is user-facing workflow clarity.                                                                                                                                                                                        |
| Natural backjump/refactor layer        | If README starts accumulating historical release notes again, move history into archive/reports or generated docs;<br>keep README current and compact.                                                                                                                 |
| Implementation prompt contract         | Future tool packages should keep README as a current manual, not a chronological changelog. Assistant workflow claims must match actually tested/developed integrations.                                                                                               |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and root-artifact hygiene.                                                                                                                                                           |
| How implemented                        | Replaced the chronological V1.13 README with a current operational manual covering project facts, truth status, manual documentation generation, context documentation, ChatGPT-assisted workflows, workflow extensibility, repository layout, cleanup, and licensing. |
| Verification / evidence                | README was rewritten directly. YAML metadata, latest-change overview, object register, proof dossier, and scratchpad were updated. generate_tables.py was run successfully. No Lean source was edited.                                                                 |
| Proof strategy                         | Turn README into a stable user manual: current commands first, workflow contract second, history in archive. Preserve the Lean-primary/generated-secondary/expert-review split.                                                                                        |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table generation only.                                                                                                                                                                                            |
| Post-implementation plan check         | V1.14 is a documentation/tooling closeout. Active mathematical/content cursor remains 1.1.1.1.5.                                                                                                                                                                       |
| Artifact packaging rule                | Full updated zip package;<br>no generated full PDF/TeX retained in root.                                                                                                                                                                                               |
| Implementation notes                   | The README explicitly says the integrated assistant workflow was developed and tested with ChatGPT. It does not present Claude as a validated toolchain component.                                                                                                     |
| Source materials / references          | <ul style="list-style-type: none"> <li>• user README request</li> <li>• user correction about ChatGPT-only development/testing</li> <li>• V1.13 README</li> <li>• Single YAML master</li> <li>• generated stats/exports</li> </ul>                                     |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>• README entry surface</li> <li>• ChatGPT-assisted workflow contract</li> <li>• manual generation command surface</li> <li>• workflow extensibility/scalability principle</li> </ul>                                            |
| Implemented modules                    | <ul style="list-style-type: none"> <li>• README.md</li> <li>• Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                   |
| Added modules                          | <ul style="list-style-type: none"> <li>• Repository/archive/reports/V1_14_README_WORKFLOW_ORIENTATION_REPORT.md</li> </ul>                                                                                                                                             |
| Modified modules                       | <ul style="list-style-type: none"> <li>• README.md</li> <li>• Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                   |



|                                      |                                                                                                                                                                                                                                                                                |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source edit</li> <li>no Claude-tested claim</li> <li>no generated artifact as truth source</li> <li>no chronological README bloat</li> <li>no root PDF/TeX retention</li> </ul>                                                 |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23-GNG27.</li> <li>no Claude-tested/developed claim for this workflow</li> <li>no second truth system</li> <li>no generated PDF/TeX as active root source</li> </ul> |
| Proposed added modules               | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_14_README_WORKFLOW_ORIENTATION_REPORT.md</li> </ul>                                                                                                                                                       |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>README.md</li> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                               |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>scripts/generate_full_document.py</li> <li>scripts/generate_context_doc.py</li> <li>scripts/build_export_script_v1.8.py</li> <li>scripts/clean_generated_root_artifacts.py</li> </ul>               |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>Low;<br/>README consolidates command/workflow explanations formerly scattered across release notes.</li> </ul>                                                                                                                          |

### Phase 20.10 – Root Lean/Lake project-control files and build-entry retention policy

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                     |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                          |
| Context class                         | GeneratedArtifactPhase                                                                                                                                                                     |
| Phase documentation tier              | F3                                                                                                                                                                                         |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                  |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                         |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                        |
| Implementation status                 | completed                                                                                                                                                                                  |
| Preparation status                    | completed-tooling-function                                                                                                                                                                 |
| Phase type                            | infrastructure                                                                                                                                                                             |
| Phase-type-specific contract          | Infrastructure/tooling phase: preserve build reproducibility while keeping generated documentation disposable.                                                                             |
| Title / state                         | Root Lean/Lake project-control files and build-entry retention policy                                                                                                                      |
| Preparation workflow                  | Root artifact retention and Lean/Lake build-entry preservation workflow.                                                                                                                   |
| Source-to-CNNA mapping                | Maps build reproducibility into DB-REVIEW tooling: root Lean/Lake files are not mathematical CNNA generators, but are required for local verification of the Lean-primary source snapshot. |
| Dependency-depth audit                | No new Lean import dependency inside repo_snapshot/CNNA. Package-root lakefile requires mathlib v4.28.0 and uses Repository/repo_snapshot as srcDir.                                       |
| Phase split decision                  | Implemented as DB-REVIEW tooling phase 20.10, separate from mathematical/content cursor 1.1.1.1.5.                                                                                         |
| Positive path plan                    | Preserve reproducible local build entry before returning to the semantic module consolidation gate and then blue-object ledger work.                                                       |
| Module implementation manifest        | Added package-root Lean/Lake project-control files and amended cleanup script; no Lean modules under repo_snapshot/CNNA changed.                                                           |

|                                        |                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semantic placement decision            | Natural placement is DB-REVIEW/build-reproducibility tooling at package-root boundary, not a mathematical pillar or CNNA generator.                                                                                                                                                                                                   |
| Reuse and duplication audit            | Root CNNA.lean duplicates the top-level import surface only as compatibility entry;<br>Lake source root remains repo_snapshot via srcDir.                                                                                                                                                                                             |
| Canonical generator policy             | Use existing canonical generators: generate_tables.py for YAML views, cleanup script for root hygiene. No alternate truth source.                                                                                                                                                                                                     |
| Module compactness rule                | Root remains compact: source/control files only, not generated outputs.                                                                                                                                                                                                                                                               |
| Refactor regression protocol           | No Lean refactor. Regression check is root file audit and cleanup keep-list test.                                                                                                                                                                                                                                                     |
| Refactor migration metrics             | No module migration;<br>root entry files added only. Import/declaration graph unchanged.                                                                                                                                                                                                                                              |
| Lean/mathlib version policy            | Root toolchain and manifest retain Lean/mathlib v4.28.0 as supplied.                                                                                                                                                                                                                                                                  |
| ExecComplex boundary policy            | No ExecComplex boundary affected.                                                                                                                                                                                                                                                                                                     |
| Consumer-contract direction            | Human/local-build consumers only;<br>no pillar handoff rights.                                                                                                                                                                                                                                                                        |
| Governance-overhead review             | Positive and low overhead: restores four necessary build-control files while keeping generated artifact clutter out of root.                                                                                                                                                                                                          |
| Proof rehearsal before implementation  | Copy build-control files, amend cleanup keep-list, update docs/YAML, regenerate tables, execute cleanup, verify allowed root set.                                                                                                                                                                                                     |
| Expected tactics                       | None;<br>build/tooling phase, not Lean proof phase.                                                                                                                                                                                                                                                                                   |
| Expected mathlib imports               | No source import added;<br>root lakefile pins mathlib v4.28.0.                                                                                                                                                                                                                                                                        |
| ExecComplex/computable plan            | No ExecComplex path introduced.                                                                                                                                                                                                                                                                                                       |
| Concrete computable instance traversal | Concrete traversal is root file inventory and cleanup classification over the actual package root.                                                                                                                                                                                                                                    |
| Computable path non-skeleton status    | Not an operational CNNA computable path;<br>nevertheless the build-entry files are concrete root files, not placeholders.                                                                                                                                                                                                             |
| Expected executable witness            | Successful Python table generation and root cleanup keep-list test;<br>no Lean executable witness required.                                                                                                                                                                                                                           |
| Mathlib complication log               | No Lean/mathlib source complication introduced;<br>root manifest/toolchain pin supplied build environment.                                                                                                                                                                                                                            |
| Noncomputable boundary audit           | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                                                                                                                       |
| Mathlib mitigation/refactor layer      | If mathlib/toolchain changes later, update root Lean/Lake files and manifest in a separate build-entry phase.                                                                                                                                                                                                                         |
| Bridge gap status                      | No mathematical bridge closed;<br>build reproducibility gap closed.                                                                                                                                                                                                                                                                   |
| Natural backjump/refactor layer        | Root cleanup policy is the natural layer;<br>do not patch packaging ad hoc or hide Lake files under generated folders.                                                                                                                                                                                                                |
| Implementation prompt contract         | Future packages must retain CNNA.lean, lakefile.lean, lake-manifest.json and lean-toolchain in the package root above Repository/. Generated docs remain outside active root unless explicitly requested.                                                                                                                             |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and corrected root artifact policy.                                                                                                                                                                                                                 |
| How implemented                        | Copied the user-provided CNNA.lean, lakefile.lean, lake-manifest.json and lean-toolchain into the package/project root above Repository/;<br>amended the root cleanup script to preserve them;<br>updated README and master records to distinguish Lean/Lake project-control files from disposable generated documentation artifacts. |
| Verification / evidence                | Python table generation succeeds;<br>clean_generated_root_artifacts.py --apply leaves Repository root clean;<br>final package-root audit shows CNNA.lean, lakefile.lean, lake-manifest.json and lean-toolchain present above Repository/ with no package-root PDF/TeX/LaTeX build artifacts.                                          |
| Proof strategy                         | Classify root files by reproducibility role: generated documentation remains disposable;<br>Lean/Lake project-control files are maintained build inputs and must be retained in root.                                                                                                                                                 |
| Build result                           | No Lean build run;<br>no repo_snapshot/CNNA Lean source changed. Python table generation and cleanup-root audit only.                                                                                                                                                                                                                 |
| Post-implementation plan check         | V1.15 restores root build entry. Active mathematical/content cursor remains 1.1.1.1.5.                                                                                                                                                                                                                                                |
| Artifact packaging rule                | Full updated zip package must include package-root Lean/Lake files and no retained package-root PDF/TeX/LaTeX byproducts.                                                                                                                                                                                                             |
| Implementation notes                   | This corrects the V1.14 README-only root invariant. The true package-root invariant is source/control oriented: Lean/Lake build-control files stay;<br>generated docs and historical reports do not. Repository/ remains the tool/documentation workspace.                                                                            |

|                                      |                                                                                                                                                                                                                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source materials / references        | <ul style="list-style-type: none"> <li>user-provided CNNA.lean</li> <li>user-provided lakefile.lean</li> <li>user-provided lake-manifest.json</li> <li>user-provided lean-toolchain</li> <li>V1.14 README/root cleanup policy</li> </ul>                                                   |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>root Lean/Lake build-entry file set</li> <li>Lake srcDir Repository/repo_snapshot build surface</li> <li>toolchain version pin</li> <li>dependency manifest lock</li> </ul>                                                                         |
| Implemented modules                  | <ul style="list-style-type: none"> <li>CNNA.lean</li> <li>lakefile.lean</li> <li>lake-manifest.json</li> <li>lean-toolchain</li> <li>Repository/scripts/clean_generated_root_artifacts.py</li> </ul>                                                                                       |
| Added modules                        | <ul style="list-style-type: none"> <li>CNNA.lean</li> <li>lakefile.lean</li> <li>lake-manifest.json</li> <li>lean-toolchain</li> </ul>                                                                                                                                                     |
| Modified modules                     | <ul style="list-style-type: none"> <li>README.md</li> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>Repository/scripts/clean_generated_root_artifacts.py</li> </ul>                                                                                                             |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no deletion of Lean/Lake root build files</li> <li>no generated PDF/TeX as root source</li> <li>no historical report as active root source</li> <li>no Lean source content edit under repo_snapshot/CNNA</li> <li>no second truth system</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23-GNG28.</li> <li>do not delete root Lean/Lake project-control files</li> <li>do not treat generated documentation as root source</li> </ul>                                    |
| Proposed added modules               | <ul style="list-style-type: none"> <li>CNNA.lean</li> <li>lakefile.lean</li> <li>lake-manifest.json</li> <li>lean-toolchain</li> <li>Repository/archive/reports/V1_15_ROOT_LEAN_TOOLCHAIN_FILES_REPORT.md</li> </ul>                                                                       |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>README.md</li> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>Repository/scripts/clean_generated_root_artifacts.py</li> </ul>                                                                                                             |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>scripts/clean_generated_root_artifacts.py</li> <li>README.md</li> </ul>                                                                                                                                         |

|                                      |                                                                                                                                                                                  |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duplicate-risk symbols or operations | <div><ul style="list-style-type: none"><li>Low;<br/>root CNNA.lean is a compatibility entry and repo_snapshot/CNNA.lean remains Lake source root through srcDir.</li></ul></div> |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 20.11 – Root license, single-root README, credits, and assistant-requirement documentation

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                         |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span style="background-color: #d9ead3;">G</span>                                                                                                              |
| Context class                          | GeneratedArtifactPhase                                                                                                                                         |
| Phase documentation tier               | F3                                                                                                                                                             |
| Phase documentation depth requirement  | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                      |
| Record change version                  | unchanged-by-V1.01                                                                                                                                             |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                            |
| Implementation status                  | completed                                                                                                                                                      |
| Preparation status                     | completed-tooling-documentation                                                                                                                                |
| Phase type                             | documentation                                                                                                                                                  |
| Phase-type-specific contract           | Documentation phase: clarify license, credits, requirements and root entry point without changing Lean semantics.                                              |
| Title / state                          | Root license, single-root README, credits, and assistant-requirement documentation                                                                             |
| Preparation workflow                   | Root license, README and credit-surface workflow.                                                                                                              |
| Source-to-CNNA mapping                 | Maps licensing and assistant-workflow metadata into generated-secondary review tooling;<br>no CNNA mathematical generator is introduced.                       |
| Dependency-depth audit                 | No Lean dependency introduced. PDF generation depends on existing Python scripts and optional pdflatex only.                                                   |
| Phase split decision                   | Implemented as DB-REVIEW tooling phase 20.11 after DBR10 because it modifies package-root release surface and generated documentation metadata.                |
| Positive path plan                     | Return to active 1.1.1.1.5 semantic module consolidation gate after packaging/documentation surface is closed.                                                 |
| Module implementation manifest         | No Lean modules added/modified. Documentation/license/tool scripts and YAML only.                                                                              |
| Semantic placement decision            | Natural placement is DB-REVIEW/root documentation and license surface, not a mathematical pillar or CNNA generator.                                            |
| Reuse and duplication audit            | Closed for README surface: duplicate active README copies removed;<br>license/credits centralized at package root and generated from metadata/table for PDFs.  |
| Canonical generator policy             | PDF credits are generated from metadata;<br>license table remains generated from the Single YAML master.                                                       |
| Module compactness rule                | Keep package root compact and unambiguous: README.md, LICENSE, .gitignore, Lean/Lake files, Repository/.                                                       |
| Refactor regression protocol           | No Lean refactor. Regression checks are generate_tables.py, PDF compile, and README-location audit.                                                            |
| Refactor migration metrics             | No Lean module migration;<br>removed duplicate README files only. Import/declaration graph unchanged.                                                          |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                             |
| ExecComplex boundary policy            | No ExecComplex boundary affected.                                                                                                                              |
| Consumer-contract direction            | Human/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.                                                                   |
| Governance-overhead review             | Positive and low overhead: one root LICENSE file, one root README, compact metadata rows and generator author fields;<br>no new mathematical phase complexity. |
| Proof rehearsal before implementation  | Patch README/LICENSE/YAML/scripts, remove duplicate README files, regenerate tables, compile PDFs, audit root and README locations.                            |
| Expected tactics                       | None;<br>documentation/tooling phase, not Lean proof phase.                                                                                                    |
| Expected mathlib imports               | None added.                                                                                                                                                    |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                  |
| Concrete computable instance traversal | Concrete traversal is root file inventory, README-location audit, and generated PDF front-matter check.                                                        |

|                                      |                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computable path non-skeleton status  | Not an operational/computable CNNA path;<br>concrete output is root LICENSE, root README-only invariant and generated PDF credit surface.                                                                                                                                                                                                                              |
| Expected executable witness          | Successful Python table generation and PDF compilation;<br>no Lean executable witness required.                                                                                                                                                                                                                                                                        |
| Mathlib complication log             | No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.                                                                                                                                                                                                                                                                                  |
| Noncomputable boundary audit         | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                                                                                                                                                        |
| Mathlib mitigation/refactor layer    | If toolchain/license boundaries around dependencies change later, update LICENSE and root build-entry records in a separate phase.                                                                                                                                                                                                                                     |
| Bridge gap status                    | No mathematical bridge closed;<br>release/documentation metadata gap closed.                                                                                                                                                                                                                                                                                           |
| Natural backjump/refactor layer      | Root packaging/documentation surface is the natural layer;<br>do not spread README/license authority into subdirectories.                                                                                                                                                                                                                                              |
| Implementation prompt contract       | Future releases must keep README and LICENSE in package root, update PDF credits from metadata, and avoid duplicate active README surfaces.                                                                                                                                                                                                                            |
| Global no-go inheritance             | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and root surface rules.                                                                                                                                                                                                                                                              |
| How implemented                      | Added root LICENSE, removed duplicate README files from subdirectories, updated root README license/workflow sections, updated PDF generators to show Editor and Assistant credits, and recorded the minimum tested ChatGPT Plus/GPT-5.5 setup.                                                                                                                        |
| Verification / evidence              | generate_tables.py succeeds;<br>full PDF and DtN context PDF compile with updated author credits;<br>grep confirms README.md exists only at package root;<br>no Lean source under repo_snapshot/CNNA changed.                                                                                                                                                          |
| Proof strategy                       | Treat root README/LICENSE and PDF credits as source/control documentation metadata;<br>remove duplicate active README surfaces;<br>keep all generated PDFs secondary and disposable.                                                                                                                                                                                   |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table generation and PDF generation only.                                                                                                                                                                                                                                                                         |
| Post-implementation plan check       | V1.17 is a documentation/tooling closeout. Active mathematical/content cursor remains 1.1.1.1.5.                                                                                                                                                                                                                                                                       |
| Artifact packaging rule              | Full updated zip package;<br>README.md and LICENSE at package root;<br>no duplicate README.md under Repository/ or raw export folders.                                                                                                                                                                                                                                 |
| Implementation notes                 | This is a documentation/tooling governance update only. It changes release/package surfaces and generated PDF front matter, not CNNA mathematical content.                                                                                                                                                                                                             |
| Source materials / references        | <ul style="list-style-type: none"> <li>• user license request</li> <li>• user root-only README request</li> <li>• user editor/assistant credit request</li> <li>• user minimum ChatGPT Plus/GPT-5.5 requirement</li> <li>• V1.16 README and master YAML</li> </ul>                                                                                                     |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>• root license surface</li> <li>• single-root README invariant</li> <li>• PDF author-credit metadata</li> <li>• minimum tested assistant setup statement</li> </ul>                                                                                                                                                             |
| Implemented modules                  | <ul style="list-style-type: none"> <li>• README.md</li> <li>• LICENSE</li> <li>• Repository/scripts/generate_full_document.py</li> <li>• Repository/scripts/generate_context_doc.py</li> <li>• Repository/scripts/generate_tables.py</li> <li>• Repository/scripts/clean_generated_root_artifacts.py</li> <li>• Repository/masters/cnna_roadmap_master.yaml</li> </ul> |

|                                      |                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added modules                        | <ul style="list-style-type: none"><li>LICENSE</li><li>Repository/archive/reports/V1_17_LICENSE_ROOT_README_CREDITS_REPORT.md</li></ul>                                                                                                                                                                                           |
| Modified modules                     | <ul style="list-style-type: none"><li>README.md</li><li>Repository/scripts/generate_full_document.py</li><li>Repository/scripts/generate_context_doc.py</li><li>Repository/scripts/generate_tables.py</li><li>Repository/scripts/clean_generated_root_artifacts.py</li><li>Repository/masters/cnna_roadmap_master.yaml</li></ul> |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>no duplicate active README.md outside package root</li><li>no generated PDF/TeX in package root</li><li>no Lean source edit</li><li>no license ambiguity</li><li>no claim that documentation is primary truth</li></ul>                                                                    |
| No-go guardrails                     | <ul style="list-style-type: none"><li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23-GNG29.</li><li>no duplicate active README</li><li>no hidden license/credit surface</li><li>no generated artifact as primary truth</li></ul>                                                                          |
| Proposed added modules               | <ul style="list-style-type: none"><li>LICENSE</li><li>Repository/archive/reports/V1_17_LICENSE_ROOT_README_CREDITS_REPORT.md</li></ul>                                                                                                                                                                                           |
| Proposed modified modules            | <ul style="list-style-type: none"><li>README.md</li><li>Repository/scripts/generate_full_document.py</li><li>Repository/scripts/generate_context_doc.py</li><li>Repository/scripts/generate_tables.py</li><li>Repository/scripts/clean_generated_root_artifacts.py</li><li>Repository/masters/cnna_roadmap_master.yaml</li></ul> |
| Reusable existing modules            | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>scripts/generate_full_document.py</li><li>scripts/generate_context_doc.py</li><li>scripts/clean_generated_root_artifacts.py</li><li>README.md</li></ul>                                                                                                 |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"><li>Closed for README: maintained README.md exists only in package root. No Lean symbols affected.</li></ul>                                                                                                                                                                                   |

Phase 20.12 – Section-pagebreak layout for generated full and context PDFs

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                         |
| Context class                         | GeneratedArtifactPhase                                                                                                                    |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |



|                                        |                                                                                                                                                                                                                                               |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation status                  | completed                                                                                                                                                                                                                                     |
| Preparation status                     | completed-tooling-documentation                                                                                                                                                                                                               |
| Phase type                             | documentation                                                                                                                                                                                                                                 |
| Phase-type-specific contract           | Documentation phase: improve PDF navigability without changing Lean semantics or context-filter row selection.                                                                                                                                |
| Title / state                          | Section-pagebreak layout for generated full and context PDFs                                                                                                                                                                                  |
| Preparation workflow                   | Generated PDF section-pagebreak workflow.                                                                                                                                                                                                     |
| Source-to-CNNA mapping                 | Maps a review-layout requirement into generated-secondary tooling;<br>no CNNA mathematical generator is introduced.                                                                                                                           |
| Dependency-depth audit                 | No Lean dependency introduced. TeX generation depends on existing Python scripts and pdflatex with etoolbox package.                                                                                                                          |
| Phase split decision                   | Implemented as DB-REVIEW tooling phase 20.12 after DBR11 because it modifies generated documentation layout rather than mathematical content.                                                                                                 |
| Positive path plan                     | Return to active 1.1.1.1.5 semantic module consolidation gate after layout tooling is closed.                                                                                                                                                 |
| Module implementation manifest         | No Lean modules added/modified. Documentation generator scripts and YAML only.                                                                                                                                                                |
| Semantic placement decision            | Natural placement is DB-REVIEW/documentation generator layer, not a mathematical pillar or CNNA generator.                                                                                                                                    |
| Reuse and duplication audit            | Closed for layout: the pagebreak invariant is centralized in generators and not repeated in disposable outputs.                                                                                                                               |
| Canonical generator policy             | Section-pagebreak behavior is generated from the full/context generator preambles.                                                                                                                                                            |
| Module compactness rule                | Keep the layout behavior centralized in generators rather than duplicated across generated artifacts.                                                                                                                                         |
| Refactor regression protocol           | Regression checks are generate_tables.py, full PDF compile, context PDF compile, generated TeX hook audit, and early-page rendering.                                                                                                          |
| Refactor migration metrics             | No Lean module migration. Generated PDF page count may increase due to section isolation.                                                                                                                                                     |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                                            |
| ExecComplex boundary policy            | No ExecComplex boundary affected.                                                                                                                                                                                                             |
| Consumer-contract direction            | Human/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.                                                                                                                                                  |
| Governance-overhead review             | Low overhead: one generator-level macro in each PDF generator and one tracked documentation object/guardrail.                                                                                                                                 |
| Proof rehearsal before implementation  | Patch generator macros, regenerate tables, compile representative PDFs, render early pages, and audit no Lean source changed.                                                                                                                 |
| Expected tactics                       | None;<br>documentation/tooling phase, not Lean proof phase.                                                                                                                                                                                   |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                                   |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                 |
| Concrete computable instance traversal | Concrete traversal is full-document PDF generation and DtN context-document generation.                                                                                                                                                       |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete output is compiled full/context PDF layout with top-level section starts on new pages.                                                                                                   |
| Expected executable witness            | Successful Python table generation and PDF compilation;<br>no Lean executable witness required.                                                                                                                                               |
| Mathlib complication log               | No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.                                                                                                                                                         |
| Noncomputable boundary audit           | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | If LaTeX package availability becomes an issue, replace etoolbox with an equivalent built-in LaTeX hook in a later tooling phase.                                                                                                             |
| Bridge gap status                      | No mathematical bridge closed;<br>generated documentation layout gap closed.                                                                                                                                                                  |
| Natural backjump/refactor layer        | Documentation generators are the natural layer;<br>do not edit disposable TeX/PDF output manually.                                                                                                                                            |
| Implementation prompt contract         | Future PDF layout changes must be implemented in the generators and tested on full plus context outputs.                                                                                                                                      |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and section-pagebreak invariant rules.                                                                                                                      |
| How implemented                        | Updated the full-document and context-document generators to load etoolbox and prepend clearpage to top-level section commands;<br>removed the old one-off post-TOC clearpage so section starts are governed uniformly by the generator hook. |



|                                      |                                                                                                                                                                                                                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / evidence              | <p>generate_tables.py succeeds;</p> <p>generated full/documentation TeX contains pretocmd section clearpage hook;</p> <p>full PDF and DtN context PDF compile;</p> <p>early pages were rendered for visual sanity;</p> <p>no Lean source under repo_snapshot/CNNA changed.</p>         |
| Proof strategy                       | <p>Treat section pagination as a generator-level invariant;</p> <p>verify by compiling and rendering representative generated PDFs rather than editing disposable PDFs directly.</p>                                                                                                   |
| Build result                         | <p>No Lean build run;</p> <p>no Lean source changed. Python table generation and PDF generation only.</p>                                                                                                                                                                              |
| Post-implementation plan check       | V1.18 is a documentation-layout closeout. Active mathematical/content cursor remains 1.1.1.1.5.                                                                                                                                                                                        |
| Artifact packaging rule              | <p>Full updated zip package;</p> <p>no generated full/context PDFs in root;</p> <p>generated representative PDFs may exist under generated_docs/ or context_docs/.</p>                                                                                                                 |
| Implementation notes                 | This is a layout/readability update for generated-secondary documentation. It does not alter context matching, table generation, Lean source, or CNNA mathematical semantics.                                                                                                          |
| Source materials / references        | <ul style="list-style-type: none"> <li>• user request: every section should automatically start on a new page</li> <li>• V1.17 generators</li> <li>• V1.17 README and master YAML</li> </ul>                                                                                           |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>• section-pagebreak layout invariant</li> <li>• generator-level TeX hook</li> <li>• full/context documentation layout profile</li> </ul>                                                                                                        |
| Implemented modules                  | <ul style="list-style-type: none"> <li>• Repository/scripts/generate_full_document.py</li> <li>• Repository/scripts/generate_context_doc.py</li> <li>• README.md</li> <li>• Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                     |
| Added modules                        | <ul style="list-style-type: none"> <li>• Repository/archive/reports/V1_18_SECTION_PAGEBREAKS_REPORT.md</li> </ul>                                                                                                                                                                      |
| Modified modules                     | <ul style="list-style-type: none"> <li>• Repository/scripts/generate_full_document.py</li> <li>• Repository/scripts/generate_context_doc.py</li> <li>• README.md</li> <li>• Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                     |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>• no manual edits to disposable generated PDFs</li> <li>• no generated PDF/TeX in package root</li> <li>• no Lean source edit</li> <li>• no change to context filter semantics</li> <li>• no claim that layout improves truth status</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23-GNG30.</li> <li>• no generated artifact as primary truth</li> <li>• layout changes do not certify semantics</li> </ul>                                                  |
| Proposed added modules               | <ul style="list-style-type: none"> <li>• Repository/archive/reports/V1_18_SECTION_PAGEBREAKS_REPORT.md</li> </ul>                                                                                                                                                                      |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>• Repository/scripts/generate_full_document.py</li> <li>• Repository/scripts/generate_context_doc.py</li> <li>• README.md</li> <li>• Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                     |

|                                      |                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_full_document.py</li> <li>scripts/generate_context_doc.py</li> <li>scripts/generate_tables.py</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>No Lean symbols affected. Avoid duplicate layout rules outside the generators.</li> </ul>                                 |

### Phase 20.13 – Phase status release semantics: red-by-default for unreleased Lean-derivation phases

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                                                            |
| Context class                         | GeneratedArtifactPhase                                                                                                                                                                       |
| Phase documentation tier              | F3                                                                                                                                                                                           |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                                                    |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                           |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                          |
| Implementation status                 | completed                                                                                                                                                                                    |
| Preparation status                    | completed-tooling-governance                                                                                                                                                                 |
| Phase type                            | infrastructure                                                                                                                                                                               |
| Phase-type-specific contract          | Infrastructure/status-governance phase: alter planning color semantics and generator validation without changing Lean semantics.                                                             |
| Title / state                         | Phase status release semantics: red-by-default for unreleased Lean-derivation phases                                                                                                         |
| Preparation workflow                  | Phase status release and red-lock workflow.                                                                                                                                                  |
| Source-to-CNNA mapping                | Maps Lean-derived-only partial-order discipline into phase-status governance; no CNNA mathematical generator is introduced.                                                                  |
| Dependency-depth audit                | No Lean dependency introduced. The policy depends on existing proof dossiers/object register and future pre-implementation checks for releases.                                              |
| Phase split decision                  | Implemented as DB-REVIEW/status-governance phase 20.13 after PDF/layout tooling because it changes phase color semantics globally.                                                           |
| Positive path plan                    | Keep 1.1.1.1.5 yellow; after it closes, release 1.1.1.2 only if its prerequisite objects and ledger evidence are explicit. Red derivation phases stay locked until their own release checks. |
| Module implementation manifest        | No Lean modules added/modified. Master YAML and generator validation only.                                                                                                                   |
| Semantic placement decision           | Natural placement is planning/status-governance layer, not a mathematical pillar or CNNA generator.                                                                                          |
| Reuse and duplication audit           | Status policy is centralized in generator validation and master YAML, avoiding per-phase duplicate rules.                                                                                    |
| Canonical generator policy            | generate_tables.py is the canonical validator for status semantics.                                                                                                                          |
| Module compactness rule               | One policy object and guardrail govern all phase statuses; no separate status prose per red phase beyond automatic lock note.                                                                |
| Refactor regression protocol          | No Lean refactor. Regression checks are generate_tables.py, status-count audit, and full/context documentation generation.                                                                   |
| Refactor migration metrics            | Converted 81 phases from W to R. No Lean imports or declarations changed.                                                                                                                    |
| Lean/mathlib version policy           | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                           |
| ExecComplex boundary policy           | No ExecComplex boundary affected.                                                                                                                                                            |
| Consumer-contract direction           | Planning/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.                                                                                              |
| Governance-overhead review            | Necessary: makes missing derivability explicit. Overhead controlled by using phase_type and generator validation instead of manual per-phase prose.                                          |
| Proof rehearsal before implementation | Audit all W phases by phase_type; convert derivation to R; validate unique Y; compile generated views; inspect status counts.                                                                |

|                                        |                                                                                                                                                                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected tactics                       | None;<br>documentation/tooling phase, not Lean proof phase.                                                                                                                                                                                                       |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                                                       |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                                     |
| Concrete computable instance traversal | Concrete traversal is all phase rows and scratchpad phase_type values.                                                                                                                                                                                            |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete output is status conversion plus generator validation.                                                                                                                                                       |
| Expected executable witness            | Successful Python table generation and status-count audit;<br>no Lean executable witness required.                                                                                                                                                                |
| Mathlib complication log               | No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.                                                                                                                                                                             |
| Noncomputable boundary audit           | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                                                   |
| Mathlib mitigation/refactor layer      | Not applicable;<br>status policy only.                                                                                                                                                                                                                            |
| Bridge gap status                      | No mathematical bridge closed;<br>phase-status release semantics closed.                                                                                                                                                                                          |
| Natural backjump/refactor layer        | Status-governance generator/master YAML is the natural layer;<br>do not patch individual consumer phases ad hoc.                                                                                                                                                  |
| Implementation prompt contract         | Future implementation prompts must not start from red derivation phases;<br>first create/close the release proof referencing derived-only Lean objects.                                                                                                           |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and GNG31.                                                                                                                                                                      |
| How implemented                        | Converted 81 W/derivation phases to R, updated status legend, added DBR13/GNG31/SCF50/workflow, and patched generate_tables.py validation so no future derivation phase can remain white.                                                                         |
| Verification / evidence                | generate_tables.py succeeds after conversion;<br>status counts are generated in project stats;<br>full/context documentation can be regenerated from updated tables;<br>no Lean source under repo_snapshot/CNNA changed.                                          |
| Proof strategy                         | Use phase_type as the release gate: derivation phases require derived-only Lean-object release evidence before yellow/orange;<br>non-direct-Lean work remains white if not active/closed.                                                                         |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table generation and PDF generation only.                                                                                                                                                                    |
| Post-implementation plan check         | V1.19 is a status-governance closeout. Active mathematical/content cursor remains 1.1.1.1.5.                                                                                                                                                                      |
| Artifact packaging rule                | Full updated zip package;<br>generated PDFs may exist under generated_docs/context_docs but not in package root.                                                                                                                                                  |
| Implementation notes                   | This is a planning/status-governance update. It does not implement red phases;<br>it locks them until release evidence references prerequisite derived-only Lean objects.                                                                                         |
| Source materials / references          | <ul style="list-style-type: none"> <li>• user phase-status policy request</li> <li>• status_legend</li> <li>• phases</li> <li>• implementation_scratchpad</li> <li>• pre_implementation_phase_check_results</li> <li>• generate_tables.py validation</li> </ul>   |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>• red-lock status invariant</li> <li>• white non-Lean phase invariant</li> <li>• yellow unique active cursor</li> <li>• orange release-after-predecessor status</li> <li>• derived-only object release evidence</li> </ul> |

|                                      |                                                                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implemented modules                  | <ul style="list-style-type: none"><li>Repository/masters/cnna__roadmap__master.yaml</li><li>Repository/scripts/generate__tables.py</li><li>README.md</li></ul>                                                                                      |
| Added modules                        | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_19_PHASE_STATUS_RELEASE_POLICY_REPORT.md</li></ul>                                                                                                                              |
| Modified modules                     | <ul style="list-style-type: none"><li>Repository/masters/cnna__roadmap__master.yaml</li><li>Repository/scripts/generate__tables.py</li><li>README.md</li></ul>                                                                                      |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>no white derivation phases</li><li>no red phase treated as active</li><li>no release without pre-implementation evidence</li><li>no Lean source edit</li><li>no generated artifact as primary truth</li></ul> |
| No-go guardrails                     | <ul style="list-style-type: none"><li>GLOBAL-INHERITED: all global__no__go__guardrails apply, especially GNG31.</li><li>no white derivation phases</li><li>no red-to-yellow/orange release without prerequisite-object evidence</li></ul>           |
| Proposed added modules               | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_19_PHASE_STATUS_RELEASE_POLICY_REPORT.md</li></ul>                                                                                                                              |
| Proposed modified modules            | <ul style="list-style-type: none"><li>Repository/masters/cnna__roadmap__master.yaml</li><li>Repository/scripts/generate__tables.py</li><li>README.md</li></ul>                                                                                      |
| Reusable existing modules            | <ul style="list-style-type: none"><li>scripts/generate__tables.py</li><li>pre__implementation__phase__check__results</li><li>status__legend</li><li>implementation__scratchpad</li></ul>                                                            |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"><li>Closed by central status validation; no Lean symbols affected.</li></ul>                                                                                                                                      |

Phase 20.14 – Static section explanations and module-change manifest appendix semantics

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3; padding: 0 2px;">G</span>                                                                         |
| Context class                         | GeneratedArtifactPhase                                                                                                                    |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |
| Implementation status                 | completed                                                                                                                                 |
| Preparation status                    | completed-tooling-documentation                                                                                                           |
| Phase type                            | documentation                                                                                                                             |
| Phase-type-specific contract          | Documentation phase: clarify section purpose/data source and module-manifest authority without changing Lean semantics.                   |

|                                        |                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title / state                          | Static section explanations and module-change manifest appendix semantics                                                                                                                                                                                                                                                                              |
| Preparation workflow                   | Documentation section explanation and appendix workflow.                                                                                                                                                                                                                                                                                               |
| Source-to-CNNA mapping                 | Maps documentation-readability requirements into generated-secondary tooling; no CNNA mathematical generator is introduced.                                                                                                                                                                                                                            |
| Dependency-depth audit                 | No Lean dependency introduced. Documentation generation depends on existing Python scripts and pdflatex only.                                                                                                                                                                                                                                          |
| Phase split decision                   | Implemented as DB-REVIEW documentation/tooling phase 20.14 after the phase-status release policy because it clarifies how the generated documentation should be read.                                                                                                                                                                                  |
| Positive path plan                     | Return to active 1.1.1.1.5 semantic module consolidation gate after closing documentation readability/authority boundaries.                                                                                                                                                                                                                            |
| Module implementation manifest         | Module manifest remains generated, but is now explained as technical appendix rather than a primary reading table.                                                                                                                                                                                                                                     |
| Semantic placement decision            | Natural placement is DB-REVIEW/documentation generator layer, not a mathematical pillar or CNNA generator.                                                                                                                                                                                                                                             |
| Reuse and duplication audit            | Closed for documentation semantics: primary phase table and module-change appendix roles are separated.                                                                                                                                                                                                                                                |
| Canonical generator policy             | Static section explanations are generated from full/context generator code; dynamic table rows continue to come from Master YAML or generated exports.                                                                                                                                                                                                 |
| Module compactness rule                | Keep static explanations concise and centralized in generators; keep module-change table narrow and appendix-only.                                                                                                                                                                                                                                     |
| Refactor regression protocol           | Regression checks are generate_tables.py, full PDF compile, context PDF compile, generated TeX title/order audit, and representative rendering.                                                                                                                                                                                                        |
| Refactor migration metrics             | No Lean module migration. Full/context PDF page counts may change due to explanation prose.                                                                                                                                                                                                                                                            |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                                                                                                                                                     |
| ExecComplex boundary policy            | No ExecComplex boundary affected.                                                                                                                                                                                                                                                                                                                      |
| Consumer-contract direction            | Human/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.                                                                                                                                                                                                                                                           |
| Governance-overhead review             | Medium but justified: extra static prose reduces user confusion and prevents the module appendix from being read as a second control surface.                                                                                                                                                                                                          |
| Proof rehearsal before implementation  | Patch static prose in generators, regenerate tables, compile representative PDFs, render representative pages, and audit no Lean source changed.                                                                                                                                                                                                       |
| Expected tactics                       | None; documentation/tooling phase, not Lean proof phase.                                                                                                                                                                                                                                                                                               |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                                                                                                                                            |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                                                                                                                          |
| Concrete computable instance traversal | Concrete traversal is full-document PDF generation and DtN context-document generation.                                                                                                                                                                                                                                                                |
| Computable path non-skeleton status    | Not an operational/computable CNNA path; concrete output is compiled full/context PDFs with static explanations and dynamic tables.                                                                                                                                                                                                                    |
| Expected executable witness            | Successful Python table generation and PDF compilation; no Lean executable witness required.                                                                                                                                                                                                                                                           |
| Mathlib complication log               | No Lean/mathlib source complication introduced; root manifest/toolchain unchanged.                                                                                                                                                                                                                                                                     |
| Noncomputable boundary audit           | No Lean source edited; no noncomputable boundary introduced.                                                                                                                                                                                                                                                                                           |
| Mathlib mitigation/refactor layer      | Not applicable.                                                                                                                                                                                                                                                                                                                                        |
| Bridge gap status                      | No mathematical bridge closed; documentation reading-order and authority-boundary gap closed.                                                                                                                                                                                                                                                          |
| Natural backjump/refactor layer        | Documentation generators and table static paragraphs are the natural layer; do not duplicate static explanation prose as dynamic YAML data rows.                                                                                                                                                                                                       |
| Implementation prompt contract         | Future documentation-clarity changes should modify generator-owned static prose and keep data rows generated from YAML/exports.                                                                                                                                                                                                                        |
| Global no-go inheritance               | All global no-go guardrails apply; especially Lean-primary/generated-secondary, red-by-default phase release and module-change appendix rules.                                                                                                                                                                                                         |
| How implemented                        | Updated full-document generator with static source/purpose prose before each major section; updated context-document generator with static per-section explanations and moved the module manifest to a technical appendix; updated module-manifest table paragraph and phase-table rule in generate_tables.py; updated README and Master YAML records. |

|                                      |                                                                                                                                                                                                                                                                                          |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verification / evidence              | generate_tables.py succeeds;<br>generated full and context TeX include the new appendix title and section explanations;<br>full PDF and DtN context PDF compile;<br>representative pages rendered;<br>no Lean source under repo_snapshot/CNNA changed.                                   |
| Proof strategy                       | Treat section explanations as static generator-owned prose and table rows as dynamic YAML/export data. Verify by regenerating docs and checking the module manifest no longer appears as a main release section.                                                                         |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table generation and PDF generation only.                                                                                                                                                                                           |
| Post-implementation plan check       | V1.20 is a documentation readability/authority-boundary closeout. Active mathematical/content cursor remains 1.1.1.1.5.                                                                                                                                                                  |
| Artifact packaging rule              | Full updated zip package;<br>representative generated docs may exist under generated_docs/ and context_docs/ but not in package root.                                                                                                                                                    |
| Implementation notes                 | This is documentation/tooling semantics only. It clarifies reading order and data provenance without changing any Lean code, status counts, object derivations, or table row generation logic.                                                                                           |
| Source materials / references        | <ul style="list-style-type: none"><li>• user request: module-manifest table is confusing</li><li>• user request: explain sections/tables more clearly</li><li>• V1.19 full/context generators and table paragraphs</li></ul>                                                             |
| Extracted established theory objects | <ul style="list-style-type: none"><li>• static section explanation layer</li><li>• module-change appendix policy</li><li>• primary phase-table release surface</li></ul>                                                                                                                 |
| Implemented modules                  | <ul style="list-style-type: none"><li>• Repository/scripts/generate_full_document.py</li><li>• Repository/scripts/generate_context_doc.py</li><li>• Repository/scripts/generate_tables.py</li><li>• README.md</li><li>• Repository/masters/cnna_roadmap_master.yaml</li></ul>            |
| Added modules                        | <ul style="list-style-type: none"><li>• Repository/archive/reports/V1_20_DOC_SECTION_EXPLANATIONS_AND_MODULE_MANIFEST_APPENDIX_REPORT.md</li></ul>                                                                                                                                       |
| Modified modules                     | <ul style="list-style-type: none"><li>• Repository/scripts/generate_full_document.py</li><li>• Repository/scripts/generate_context_doc.py</li><li>• Repository/scripts/generate_tables.py</li><li>• README.md</li><li>• Repository/masters/cnna_roadmap_master.yaml</li></ul>            |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>• no second phase table from module manifest</li><li>• no dynamic YAML rows for static section explanations</li><li>• no Lean source edit</li><li>• no change to phase status semantics</li><li>• no generated artifact as primary truth</li></ul> |
| No-go guardrails                     | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23-GNG32.</li><li>• no second phase table</li><li>• no generated artifact as primary truth</li><li>• no dynamic data rows for static section-purpose prose</li></ul>          |
| Proposed added modules               | <ul style="list-style-type: none"><li>• Repository/archive/reports/V1_20_DOC_SECTION_EXPLANATIONS_AND_MODULE_MANIFEST_APPENDIX_REPORT.md</li></ul>                                                                                                                                       |



|                                      |                                                                                                                                                                                                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/scripts/generate_full_document.py</li> <li>Repository/scripts/generate_context_doc.py</li> <li>Repository/scripts/generate_tables.py</li> <li>README.md</li> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul> |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_full_document.py</li> <li>scripts/generate_context_doc.py</li> <li>scripts/generate_tables.py</li> </ul>                                                                                                          |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>Closed for module manifest: it is no longer positioned as a main phase/release table. No Lean symbols affected.</li> </ul>                                                                                                         |

### Phase 20.15 – Supabase/Postgres mirror feasibility and schema design

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                              |
| Context class                         | DatabaseMirrorPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | planned-sidechain                                                                                                                                                                              |
| Phase type                            | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract          | Infrastructure/tooling phase: database/web-publication planning only, no Lean derivation and no DB truth authority.                                                                            |
| Title / state                         | Supabase/Postgres mirror feasibility and schema design                                                                                                                                         |
| Preparation workflow                  | Supabase public read-only mirror and web-frontend workflow.                                                                                                                                    |
| Source-to-CNNA mapping                | Maps CNNA planning metadata into a public query/visualization layer. It is not a mathematical CNNA input and not a Lean replacement.                                                           |
| Dependency-depth audit                | Depends on stable YAML identifiers, generated exports, repo inventory and root packaging discipline. Does not block active Lean/content pre-code readiness chain.                              |
| Phase split decision                  | Kept as white DB/tooling sidechain phases 20.15-20.20 so DB planning is visible but does not preempt the active 1.1.1.1.5.1 cursor.                                                            |
| Positive path plan                    | Design DB mirror in staged order: schema -> import/roundtrip -> snapshots/checksums -> read-only api views/RLS -> web frontend contract -> later reviewer/admin model.                         |
| Module implementation manifest        | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.                                                               |
| Semantic placement decision           | Natural placement is DB-REVIEW/generated-secondary publication infrastructure, not a CNNA mathematical pillar.                                                                                 |
| Reuse and duplication audit           | Reuse existing YAML/export/context-doc data; do not create a second canonical planning model in DB without roundtrip contract.                                                                 |
| Canonical generator policy            | Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.                                                                                |
| Module compactness rule               | Keep DB schema normalized and queryable; avoid large JSON blobs/PDFs/ZIPs as primary DB content.                                                                                               |
| Refactor regression protocol          | No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.                                                                                         |
| Refactor migration metrics            | No migration executed. Future metrics: tables imported, rows imported, snapshot hash, RLS test count, anon write-denial tests.                                                                 |
| Lean/mathlib version policy           | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                             |
| ExecComplex boundary policy           | No ExecComplex change.                                                                                                                                                                         |
| Consumer-contract direction           | Consumers are public read-only frontend/API users and internal admin/import tooling only; no pillar handoff or mathematical consumer rights.                                                   |



|                                        |                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Governance-overhead review             | Planned sidechain is justified by long-term public visualization/query value, but it must remain non-blocking and sparse until YAML/Lean surfaces stabilize.                                                                                                                                                                                                   |
| Proof rehearsal before implementation  | Before activation, rehearse importer locally using a disposable database, check hashes, verify anon read-only views, and prove that every public row traces to a snapshot_id.                                                                                                                                                                                  |
| Expected tactics                       | None;<br>database/tooling phase, not Lean proof phase.                                                                                                                                                                                                                                                                                                         |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                                                                                                                                                    |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                                                                                                                                  |
| Concrete computable instance traversal | No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.                                                                                                                                                                                                                                   |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete future output is schema/import/view/frontend contract, not a mathematical witness.                                                                                                                                                                                                                        |
| Expected executable witness            | Future activation requires SQL migrations, import script, snapshot hash report, RLS/grant test report and public-view query examples.                                                                                                                                                                                                                          |
| Mathlib complication log               | None;<br>no Lean/mathlib code changes.                                                                                                                                                                                                                                                                                                                         |
| Noncomputable boundary audit           | No Lean noncomputable boundary involved.                                                                                                                                                                                                                                                                                                                       |
| Mathlib mitigation/refactor layer      | Not applicable.                                                                                                                                                                                                                                                                                                                                                |
| Bridge gap status                      | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                                                                                                                                                                                                          |
| Natural backjump/refactor layer        | If DB requirements conflict with YAML schema, patch the YAML/export schema at the natural documentation/tooling layer;<br>do not alter Lean derivations for DB convenience.                                                                                                                                                                                    |
| Implementation prompt contract         | When activated, generate migrations/importer/frontend contracts from this phase. Do not ask for or store service_role keys in chat or repo;<br>use local .env/admin execution.                                                                                                                                                                                 |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and future Supabase read-only guardrails.                                                                                                                                                                                                                                    |
| How implemented                        | Not implemented. V1.22 records the phase as a non-blocking DB/tooling sidechain for future Supabase/Postgres public read-only mirror and web-frontend work.                                                                                                                                                                                                    |
| Verification / evidence                | Planning record only. No live Supabase connection, no DB write, no frontend deployment and no Lean source change were performed in V1.22.                                                                                                                                                                                                                      |
| Proof strategy                         | Planning/governance proof by explicit scope separation: Lean/YAML source -> snapshot/import -> Supabase read-only mirror -> web frontend. DB rows never become truth-source rows.                                                                                                                                                                              |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                                                              |
| Post-implementation plan check         | V1.22 only records the sidechain. Active mathematical/content cursor remains 1.1.1.1.5.1.                                                                                                                                                                                                                                                                      |
| Artifact packaging rule                | Full updated zip package;<br>no secrets, no .env, no live DB credentials, no generated DB dumps with sensitive keys.                                                                                                                                                                                                                                           |
| Implementation notes                   | internal/api/review/admin schema split;<br>minimal tables for db_snapshots, phases, objects, proof_dossiers, guardrails, workflows, modules, declarations, import_edges, phase_object_links, context_profiles, static_findings and reports. This is<br>generated-secondary/publication tooling and cannot release CNNA phases or replace Lean/YAML provenance. |
| Source materials / references          | <ul style="list-style-type: none"> <li>Master YAML</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> <li>Supabase API/RLS/key docs as external implementation reference</li> <li>user cnna-project.net web-frontend vision</li> </ul>                                                                                            |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>DBR15</li> <li>public read-only mirror</li> <li>append-only snapshot</li> <li>api.* read-only view</li> <li>internal.* canonical tables</li> <li>review.* annotations</li> <li>admin/import.* importer path</li> </ul>                                                                                                  |
| Modified modules                       | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul>                                                                                                                                                                                                                                               |

|                                      |                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Supabase row as CNNA truth source</li> <li>no public write permissions</li> <li>no browser service_role key</li> <li>no DB phase release authority</li> <li>no DB update without snapshot/checksum/import report</li> <li>no large PDFs/ZIPs in DB</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG34 once added.</li> <li>No Supabase row is a CNNA truth source.</li> <li>No public writes.</li> <li>No browser service_role key.</li> </ul>                                                   |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> <li>scripts/generate_tables.py</li> </ul>                                                                                                       |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>second truth system risk</li> <li>manual DB table edits</li> <li>public write policy leak</li> <li>service_role key leak</li> <li>snapshot provenance loss</li> </ul>                                                                                            |

### Phase 20.16 – YAML/exports to Postgres import and roundtrip contract

**Table legend.**   = green/closed and derived-only;   W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   O = orange/implementable after stated predecessor closure;   Y = yellow/active and implementable now;   R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                              |
| Context class                         | DatabaseMirrorPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | planned-sidechain                                                                                                                                                                              |
| Phase type                            | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract          | Infrastructure/tooling phase: database/web-publication planning only, no Lean derivation and no DB truth authority.                                                                            |
| Title / state                         | YAML/exports to Postgres import and roundtrip contract                                                                                                                                         |
| Preparation workflow                  | Supabase public read-only mirror and web-frontend workflow.                                                                                                                                    |
| Source-to-CNNA mapping                | Maps CNNA planning metadata into a public query/visualization layer. It is not a mathematical CNNA input and not a Lean replacement.                                                           |
| Dependency-depth audit                | Depends on stable YAML identifiers, generated exports, repo inventory and root packaging discipline. Does not block active Lean/content pre-code readiness chain.                              |
| Phase split decision                  | Kept as white DB/tooling sidechain phases 20.15-20.20 so DB planning is visible but does not preempt the active 1.1.1.1.5.1 cursor.                                                            |
| Positive path plan                    | Design DB mirror in staged order: schema -> import/roundtrip -> snapshots/checksums -> read-only api views/RLS -> web frontend contract -> later reviewer/admin model.                         |
| Module implementation manifest        | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.                                                               |
| Semantic placement decision           | Natural placement is DB-REVIEW/generated-secondary publication infrastructure, not a CNNA mathematical pillar.                                                                                 |

|                                        |                                                                                                                                                                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reuse and duplication audit            | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.                                                                                                                                   |
| Canonical generator policy             | Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.                                                                                                                                                     |
| Module compactness rule                | Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content.                                                                                                                                                                 |
| Refactor regression protocol           | No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.                                                                                                                                                              |
| Refactor migration metrics             | No migration executed. Future metrics: tables imported, rows imported, snapshot hash, RLS test count, anon write-denial tests.                                                                                                                                      |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                                                                  |
| ExecComplex boundary policy            | No ExecComplex change.                                                                                                                                                                                                                                              |
| Consumer-contract direction            | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.                                                                                                                     |
| Governance-overhead review             | Planned sidechain is justified by long-term public visualization/query value, but it must remain non-blocking and sparse until YAML/Lean surfaces stabilize.                                                                                                        |
| Proof rehearsal before implementation  | Before activation, rehearse importer locally using a disposable database, check hashes, verify anon read-only views, and prove that every public row traces to a snapshot_id.                                                                                       |
| Expected tactics                       | None;<br>database/tooling phase, not Lean proof phase.                                                                                                                                                                                                              |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                                       |
| Concrete computable instance traversal | No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.                                                                                                                                        |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete future output is schema/import/view/frontend contract, not a mathematical witness.                                                                                                                             |
| Expected executable witness            | Future activation requires SQL migrations, import script, snapshot hash report, RLS/grant test report and public-view query examples.                                                                                                                               |
| Mathlib complication log               | None;<br>no Lean/mathlib code changes.                                                                                                                                                                                                                              |
| Noncomputable boundary audit           | No Lean noncomputable boundary involved.                                                                                                                                                                                                                            |
| Mathlib mitigation/refactor layer      | Not applicable.                                                                                                                                                                                                                                                     |
| Bridge gap status                      | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                                                                                                               |
| Natural backjump/refactor layer        | If DB requirements conflict with YAML schema, patch the YAML/export schema at the natural documentation/tooling layer;<br>do not alter Lean derivations for DB convenience.                                                                                         |
| Implementation prompt contract         | When activated, generate migrations/importer/frontend contracts from this phase. Do not ask for or store service_role keys in chat or repo;<br>use local .env/admin execution.                                                                                      |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and future Supabase read-only guardrails.                                                                                                                                         |
| How implemented                        | Not implemented. V1.22 records the phase as a non-blocking DB/tooling sidechain for future Supabase/Postgres public read-only mirror and web-frontend work.                                                                                                         |
| Verification / evidence                | Planning record only. No live Supabase connection, no DB write, no frontend deployment and no Lean source change were performed in V1.22.                                                                                                                           |
| Proof strategy                         | Planning/governance proof by explicit scope separation: Lean/YAML source -> snapshot/import -> Supabase read-only mirror -> web frontend. DB rows never become truth-source rows.                                                                                   |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                   |
| Post-implementation plan check         | V1.22 only records the sidechain. Active mathematical/content cursor remains 1.1.1.1.5.1.                                                                                                                                                                           |
| Artifact packaging rule                | Full updated zip package;<br>no secrets, no .env, no live DB credentials, no generated DB dumps with sensitive keys.                                                                                                                                                |
| Implementation notes                   | Importer must be local/admin-run with .env secrets, produce import report/checksums, and never require a service_role key in chat or browser. This is generated-secondary/publication tooling and cannot release CNNA phases or replace Lean/YAML provenance.       |
| Source materials / references          | <ul style="list-style-type: none"> <li>Master YAML</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> <li>Supabase API/RLS/key docs as external implementation reference</li> <li>user cnna-project.net web-frontend vision</li> </ul> |

|                                      |                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extracted established theory objects | <ul style="list-style-type: none"> <li>• DBR16</li> <li>• public read-only mirror</li> <li>• append-only snapshot</li> <li>• api.* read-only view</li> <li>• internal.* canonical tables</li> <li>• review.* annotations</li> <li>• admin/import.* importer path</li> </ul>                                         |
| Modified modules                     | <ul style="list-style-type: none"> <li>• Repository/masters/cnna_roadmap_master.yaml</li> <li>• README.md</li> </ul>                                                                                                                                                                                                |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>• no Supabase row as CNNA truth source</li> <li>• no public write permissions</li> <li>• no browser service_role key</li> <li>• no DB phase release authority</li> <li>• no DB update without snapshot/checksum/import report</li> <li>• no large PDFs/ZIPs in DB</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>• GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG34 once added.</li> <li>• No Supabase row is a CNNA truth source.</li> <li>• No public writes.</li> <li>• No browser service_role key.</li> </ul>                                                       |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>• masters/cnna_roadmap_master.yaml</li> <li>• generated_exports/*.tsv</li> <li>• repo_inventory/raw_export/*.json</li> <li>• scripts/generate_tables.py</li> </ul>                                                                                                           |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>• second truth system risk</li> <li>• manual DB table edits</li> <li>• public write policy leak</li> <li>• service_role key leak</li> <li>• snapshot provenance loss</li> </ul>                                                                                              |

### Phase 20.17 – Append-only snapshot and checksum model

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                              |
| Context class                         | DatabaseMirrorPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | planned-sidechain                                                                                                                                                                              |
| Phase type                            | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract          | Infrastructure/tooling phase: database/web-publication planning only, no Lean derivation and no DB truth authority.                                                                            |

|                                        |                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title / state                          | Append-only snapshot and checksum model                                                                                                                                           |
| Preparation workflow                   | Supabase public read-only mirror and web-frontend workflow.                                                                                                                       |
| Source-to-CNNA mapping                 | Maps CNNA planning metadata into a public query/visualization layer. It is not a mathematical CNNA input and not a Lean replacement.                                              |
| Dependency-depth audit                 | Depends on stable YAML identifiers, generated exports, repo inventory and root packaging discipline. Does not block active Lean/content pre-code readiness chain.                 |
| Phase split decision                   | Kept as white DB/tooling sidechain phases 20.15-20.20 so DB planning is visible but does not preempt the active 1.1.1.1.5.1 cursor.                                               |
| Positive path plan                     | Design DB mirror in staged order: schema -> import/roundtrip -> snapshots/checksums -> read-only api views/RLS -> web frontend contract -> later reviewer/admin model.            |
| Module implementation manifest         | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.                                                  |
| Semantic placement decision            | Natural placement is DB-REVIEW/generated-secondary publication infrastructure, not a CNNA mathematical pillar.                                                                    |
| Reuse and duplication audit            | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.                                                 |
| Canonical generator policy             | Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.                                                                   |
| Module compactness rule                | Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content.                                                                               |
| Refactor regression protocol           | No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.                                                                            |
| Refactor migration metrics             | No migration executed. Future metrics: tables imported, rows imported, snapshot hash, RLS test count, anon write-denial tests.                                                    |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                |
| ExecComplex boundary policy            | No ExecComplex change.                                                                                                                                                            |
| Consumer-contract direction            | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.                                   |
| Governance-overhead review             | Planned sidechain is justified by long-term public visualization/query value, but it must remain non-blocking and sparse until YAML/Lean surfaces stabilize.                      |
| Proof rehearsal before implementation  | Before activation, rehearse importer locally using a disposable database, check hashes, verify anon read-only views, and prove that every public row traces to a snapshot_id.     |
| Expected tactics                       | None;<br>database/tooling phase, not Lean proof phase.                                                                                                                            |
| Expected mathlib imports               | None added.                                                                                                                                                                       |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                     |
| Concrete computable instance traversal | No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.                                                      |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete future output is schema/import/view/frontend contract, not a mathematical witness.                                           |
| Expected executable witness            | Future activation requires SQL migrations, import script, snapshot hash report, RLS/grant test report and public-view query examples.                                             |
| Mathlib complication log               | None;<br>no Lean/mathlib code changes.                                                                                                                                            |
| Noncomputable boundary audit           | No Lean noncomputable boundary involved.                                                                                                                                          |
| Mathlib mitigation/refactor layer      | Not applicable.                                                                                                                                                                   |
| Bridge gap status                      | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                             |
| Natural backjump/refactor layer        | If DB requirements conflict with YAML schema, patch the YAML/export schema at the natural documentation/tooling layer;<br>do not alter Lean derivations for DB convenience.       |
| Implementation prompt contract         | When activated, generate migrations/importer/frontend contracts from this phase. Do not ask for or store service_role keys in chat or repo;<br>use local .env/admin execution.    |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and future Supabase read-only guardrails.                                                       |
| How implemented                        | Not implemented. V1.22 records the phase as a non-blocking DB/tooling sidechain for future Supabase/Postgres public read-only mirror and web-frontend work.                       |
| Verification / evidence                | Planning record only. No live Supabase connection, no DB write, no frontend deployment and no Lean source change were performed in V1.22.                                         |
| Proof strategy                         | Planning/governance proof by explicit scope separation: Lean/YAML source -> snapshot/import -> Supabase read-only mirror -> web frontend. DB rows never become truth-source rows. |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                 |
| Post-implementation plan check         | V1.22 only records the sidechain. Active mathematical/content cursor remains 1.1.1.1.5.1.                                                                                         |

|                                      |                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Artifact packaging rule              | Full updated zip package;<br>no secrets, no .env, no live DB credentials, no generated DB dumps with sensitive keys.                                                                                                                                                                                    |
| Implementation notes                 | db_snapshots with snapshot_id, tool_version, lean_data_timestamp, yaml_hash, repo_inventory_hash, created_at, created_by, source_zip_name and optional import_report_hash. This is generated-secondary/publication tooling and cannot release CNNA phases or replace Lean/YAML provenance.              |
| Source materials / references        | <ul style="list-style-type: none"> <li>Master YAML</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> <li>Supabase API/RLS/key docs as external implementation reference</li> <li>user cnna-project.net web-frontend vision</li> </ul>                                     |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>DBR17</li> <li>public read-only mirror</li> <li>append-only snapshot</li> <li>api.* read-only view</li> <li>internal.* canonical tables</li> <li>review.* annotations</li> <li>admin/import.* importer path</li> </ul>                                           |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul>                                                                                                                                                                                        |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Supabase row as CNNA truth source</li> <li>no public write permissions</li> <li>no browser service_role key</li> <li>no DB phase release authority</li> <li>no DB update without snapshot/checksum/import report</li> <li>no large PDFs/ZIPs in DB</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG34 once added.</li> <li>No Supabase row is a CNNA truth source.</li> <li>No public writes.</li> <li>No browser service_role key.</li> </ul>                                                   |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> <li>scripts/generate_tables.py</li> </ul>                                                                                                       |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>second truth system risk</li> <li>manual DB table edits</li> <li>public write policy leak</li> <li>service_role key leak</li> <li>snapshot provenance loss</li> </ul>                                                                                            |

### Phase 20.18 – Public read-only API views and RLS policies

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field        | Record |
|--------------|--------|
| Phase status | W      |



|                                        |                                                                                                                                                                                                |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context class                          | DatabaseMirrorPhase                                                                                                                                                                            |
| Phase documentation tier               | F1                                                                                                                                                                                             |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                  | planned-not-active                                                                                                                                                                             |
| Preparation status                     | planned-sidechain                                                                                                                                                                              |
| Phase type                             | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract           | Infrastructure/tooling phase: database/web-publication planning only, no Lean derivation and no DB truth authority.                                                                            |
| Title / state                          | Public read-only API views and RLS policies                                                                                                                                                    |
| Preparation workflow                   | Supabase public read-only mirror and web-frontend workflow.                                                                                                                                    |
| Source-to-CNNA mapping                 | Maps CNNA planning metadata into a public query/visualization layer. It is not a mathematical CNNA input and not a Lean replacement.                                                           |
| Dependency-depth audit                 | Depends on stable YAML identifiers, generated exports, repo inventory and root packaging discipline. Does not block active Lean/content pre-code readiness chain.                              |
| Phase split decision                   | Kept as white DB/tooling sidechain phases 20.15-20.20 so DB planning is visible but does not preempt the active 1.1.1.1.5.1 cursor.                                                            |
| Positive path plan                     | Design DB mirror in staged order: schema -> import/roundtrip -> snapshots/checksums -> read-only api views/RLS -> web frontend contract -> later reviewer/admin model.                         |
| Module implementation manifest         | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.                                                               |
| Semantic placement decision            | Natural placement is DB-REVIEW/generated-secondary publication infrastructure, not a CNNA mathematical pillar.                                                                                 |
| Reuse and duplication audit            | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.                                                              |
| Canonical generator policy             | Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.                                                                                |
| Module compactness rule                | Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content.                                                                                            |
| Refactor regression protocol           | No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.                                                                                         |
| Refactor migration metrics             | No migration executed. Future metrics: tables imported, rows imported, snapshot hash, RLS test count, anon write-denial tests.                                                                 |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                             |
| ExecComplex boundary policy            | No ExecComplex change.                                                                                                                                                                         |
| Consumer-contract direction            | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.                                                |
| Governance-overhead review             | Planned sidechain is justified by long-term public visualization/query value, but it must remain non-blocking and sparse until YAML/Lean surfaces stabilize.                                   |
| Proof rehearsal before implementation  | Before activation, rehearse importer locally using a disposable database, check hashes, verify anon read-only views, and prove that every public row traces to a snapshot_id.                  |
| Expected tactics                       | None;<br>database/tooling phase, not Lean proof phase.                                                                                                                                         |
| Expected mathlib imports               | None added.                                                                                                                                                                                    |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                  |
| Concrete computable instance traversal | No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.                                                                   |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete future output is schema/import/view/frontend contract, not a mathematical witness.                                                        |
| Expected executable witness            | Future activation requires SQL migrations, import script, snapshot hash report, RLS/grant test report and public-view query examples.                                                          |
| Mathlib complication log               | None;<br>no Lean/mathlib code changes.                                                                                                                                                         |
| Noncomputable boundary audit           | No Lean noncomputable boundary involved.                                                                                                                                                       |
| Mathlib mitigation/refactor layer      | Not applicable.                                                                                                                                                                                |
| Bridge gap status                      | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                                          |
| Natural backjump/refactor layer        | If DB requirements conflict with YAML schema, patch the YAML/export schema at the natural documentation/tooling layer;<br>do not alter Lean derivations for DB convenience.                    |



|                                      |                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation prompt contract       | When activated, generate migrations/importer/frontend contracts from this phase. Do not ask for or store service_role keys in chat or repo; use local .env/admin execution.                                                                                                                                  |
| Global no-go inheritance             | All global no-go guardrails apply; especially Lean-primary/generated-secondary and future Supabase read-only guardrails.                                                                                                                                                                                     |
| How implemented                      | Not implemented. V1.22 records the phase as a non-blocking DB/tooling sidechain for future Supabase/Postgres public read-only mirror and web-frontend work.                                                                                                                                                  |
| Verification / evidence              | Planning record only. No live Supabase connection, no DB write, no frontend deployment and no Lean source change were performed in V1.22.                                                                                                                                                                    |
| Proof strategy                       | Planning/governance proof by explicit scope separation: Lean/YAML source -> snapshot/import -> Supabase read-only mirror -> web frontend. DB rows never become truth-source rows.                                                                                                                            |
| Build result                         | No Lean build run; no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                               |
| Post-implementation plan check       | V1.22 only records the sidechain. Active mathematical/content cursor remains 1.1.1.1.5.1.                                                                                                                                                                                                                    |
| Artifact packaging rule              | Full updated zip package; no secrets, no .env, no live DB credentials, no generated DB dumps with sensitive keys.                                                                                                                                                                                            |
| Implementation notes                 | anon gets SELECT on api.* views only; no anon INSERT/UPDATE/DELETE; internal.* and admin/import.* remain closed; service_role never appears in frontend/browser. This is generated-secondary/publication tooling and cannot release CNNA phases or replace Lean/YAML provenance.                             |
| Source materials / references        | <ul style="list-style-type: none"><li>• Master YAML</li><li>• generated_exports/*.tsv</li><li>• repo_inventory/raw_export/*.json</li><li>• Supabase API/RLS/key docs as external implementation reference</li><li>• user cnna-project.net web-frontend vision</li></ul>                                      |
| Extracted established theory objects | <ul style="list-style-type: none"><li>• DBR18</li><li>• public read-only mirror</li><li>• append-only snapshot</li><li>• api.* read-only view</li><li>• internal.* canonical tables</li><li>• review.* annotations</li><li>• admin/import.* importer path</li></ul>                                          |
| Modified modules                     | <ul style="list-style-type: none"><li>• Repository/masters/cnna_roadmap_master.yaml</li><li>• README.md</li></ul>                                                                                                                                                                                            |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>• no Supabase row as CNNA truth source</li><li>• no public write permissions</li><li>• no browser service_role key</li><li>• no DB phase release authority</li><li>• no DB update without snapshot/checksum/import report</li><li>• no large PDFs/ZIPs in DB</li></ul> |
| No-go guardrails                     | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG34 once added.</li><li>• No Supabase row is a CNNA truth source.</li><li>• No public writes.</li><li>• No browser service_role key.</li></ul>                                                     |
| Reusable existing modules            | <ul style="list-style-type: none"><li>• masters/cnna_roadmap_master.yaml</li><li>• generated_exports/*.tsv</li><li>• repo_inventory/raw_export/*.json</li><li>• scripts/generate_tables.py</li></ul>                                                                                                         |

|                                      |                                                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"><li>• second truth system risk</li><li>• manual DB table edits</li><li>• public write policy leak</li><li>• service_role key leak</li><li>• snapshot provenance loss</li></ul> |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Phase 20.19 – cnna-project.net web-frontend data contract

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                              |
| Context class                         | DatabaseMirrorPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | planned-sidechain                                                                                                                                                                              |
| Phase type                            | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract          | Infrastructure/tooling phase: database/web-publication planning only, no Lean derivation and no DB truth authority.                                                                            |
| Title / state                         | cnna-project.net web-frontend data contract                                                                                                                                                    |
| Preparation workflow                  | Supabase public read-only mirror and web-frontend workflow.                                                                                                                                    |
| Source-to-CNNA mapping                | Maps CNNA planning metadata into a public query/visualization layer. It is not a mathematical CNNA input and not a Lean replacement.                                                           |
| Dependency-depth audit                | Depends on stable YAML identifiers, generated exports, repo inventory and root packaging discipline. Does not block active Lean/content pre-code readiness chain.                              |
| Phase split decision                  | Kept as white DB/tooling sidechain phases 20.15-20.20 so DB planning is visible but does not preempt the active 1.1.1.1.5.1 cursor.                                                            |
| Positive path plan                    | Design DB mirror in staged order: schema -> import/roundtrip -> snapshots/checksums -> read-only api views/RLS -> web frontend contract -> later reviewer/admin model.                         |
| Module implementation manifest        | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.                                                               |
| Semantic placement decision           | Natural placement is DB-REVIEW/generated-secondary publication infrastructure, not a CNNA mathematical pillar.                                                                                 |
| Reuse and duplication audit           | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.                                                              |
| Canonical generator policy            | Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.                                                                                |
| Module compactness rule               | Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content.                                                                                            |
| Refactor regression protocol          | No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.                                                                                         |
| Refactor migration metrics            | No migration executed. Future metrics: tables imported, rows imported, snapshot hash, RLS test count, anon write-denial tests.                                                                 |
| Lean/mathlib version policy           | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                             |
| ExecComplex boundary policy           | No ExecComplex change.                                                                                                                                                                         |
| Consumer-contract direction           | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.                                                |
| Governance-overhead review            | Planned sidechain is justified by long-term public visualization/query value, but it must remain non-blocking and sparse until YAML/Lean surfaces stabilize.                                   |
| Proof rehearsal before implementation | Before activation, rehearse importer locally using a disposable database, check hashes, verify anon read-only views, and prove that every public row traces to a snapshot_id.                  |
| Expected tactics                      | None;<br>database/tooling phase, not Lean proof phase.                                                                                                                                         |
| Expected mathlib imports              | None added.                                                                                                                                                                                    |
| ExecComplex/computable plan           | No ExecComplex object introduced or consumed.                                                                                                                                                  |

|                                        |                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concrete computable instance traversal | No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.                                                                                                                                                                                                      |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete future output is schema/import/view/frontend contract, not a mathematical witness.                                                                                                                                                                                           |
| Expected executable witness            | Future activation requires SQL migrations, import script, snapshot hash report, RLS/grant test report and public-view query examples.                                                                                                                                                                                             |
| Mathlib complication log               | None;<br>no Lean/mathlib code changes.                                                                                                                                                                                                                                                                                            |
| Noncomputable boundary audit           | No Lean noncomputable boundary involved.                                                                                                                                                                                                                                                                                          |
| Mathlib mitigation/refactor layer      | Not applicable.                                                                                                                                                                                                                                                                                                                   |
| Bridge gap status                      | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                                                                                                                                                                             |
| Natural backjump/refactor layer        | If DB requirements conflict with YAML schema, patch the YAML/export schema at the natural documentation/tooling layer;<br>do not alter Lean derivations for DB convenience.                                                                                                                                                       |
| Implementation prompt contract         | When activated, generate migrations/importer/frontend contracts from this phase. Do not ask for or store service_role keys in chat or repo;<br>use local .env/admin execution.                                                                                                                                                    |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and future Supabase read-only guardrails.                                                                                                                                                                                                       |
| How implemented                        | Not implemented. V1.22 records the phase as a non-blocking DB/tooling sidechain for future Supabase/Postgres public read-only mirror and web-frontend work.                                                                                                                                                                       |
| Verification / evidence                | Planning record only. No live Supabase connection, no DB write, no frontend deployment and no Lean source change were performed in V1.22.                                                                                                                                                                                         |
| Proof strategy                         | Planning/governance proof by explicit scope separation: Lean/YAML source -> snapshot/import -> Supabase read-only mirror -> web frontend. DB rows never become truth-source rows.                                                                                                                                                 |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                                 |
| Post-implementation plan check         | V1.22 only records the sidechain. Active mathematical/content cursor remains 1.1.1.1.5.1.                                                                                                                                                                                                                                         |
| Artifact packaging rule                | Full updated zip package;<br>no secrets, no .env, no live DB credentials, no generated DB dumps with sensitive keys.                                                                                                                                                                                                              |
| Implementation notes                   | Frontend can query public views with anon/publishable key only and render phase map, object register, red/orange release reasons, import/module graph, proof dossiers, blue-debt dashboard and HTML context docs. This is generated-secondary/publication tooling and cannot release CNNA phases or replace Lean/YAML provenance. |
| Source materials / references          | <ul style="list-style-type: none"> <li>Master YAML</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> <li>Supabase API/RLS/key docs as external implementation reference</li> <li>user cnna-project.net web-frontend vision</li> </ul>                                                               |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>DBR19</li> <li>public read-only mirror</li> <li>append-only snapshot</li> <li>api.* read-only view</li> <li>internal.* canonical tables</li> <li>review.* annotations</li> <li>admin/import.* importer path</li> </ul>                                                                     |
| Modified modules                       | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul>                                                                                                                                                                                                                  |
| Forbidden patterns checked             | <ul style="list-style-type: none"> <li>no Supabase row as CNNA truth source</li> <li>no public write permissions</li> <li>no browser service_role key</li> <li>no DB phase release authority</li> <li>no DB update without snapshot/checksum/import report</li> <li>no large PDFs/ZIPs in DB</li> </ul>                           |

|                                      |                                                                                                                                                                                                                                                            |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No-go guardrails                     | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: all global__no__go_guardrails apply, especially GNG34 once added.</li><li>• No Supabase row is a CNNA truth source.</li><li>• No public writes.</li><li>• No browser service_role key.</li></ul> |
| Reusable existing modules            | <ul style="list-style-type: none"><li>• masters/cnna_roadmap_master.yaml</li><li>• generated_exports/*.tsv</li><li>• repo_inventory/raw_export/*.json</li><li>• scripts/generate_tables.py</li></ul>                                                       |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"><li>• second truth system risk</li><li>• manual DB table edits</li><li>• public write policy leak</li><li>• service_role key leak</li><li>• snapshot provenance loss</li></ul>                                           |

Phase 20.20 – Reviewer annotation and future admin-write model

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                              |
| Context class                         | DatabaseMirrorPhase                                                                                                                                                                            |
| Phase documentation tier              | F1                                                                                                                                                                                             |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history. |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                             |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                            |
| Implementation status                 | planned-not-active                                                                                                                                                                             |
| Preparation status                    | planned-sidechain                                                                                                                                                                              |
| Phase type                            | infrastructure                                                                                                                                                                                 |
| Phase-type-specific contract          | Infrastructure/tooling phase: database/web-publication planning only, no Lean derivation and no DB truth authority.                                                                            |
| Title / state                         | Reviewer annotation and future admin-write model                                                                                                                                               |
| Preparation workflow                  | Supabase public read-only mirror and web-frontend workflow.                                                                                                                                    |
| Source-to-CNNA mapping                | Maps CNNA planning metadata into a public query/visualization layer. It is not a mathematical CNNA input and not a Lean replacement.                                                           |
| Dependency-depth audit                | Depends on stable YAML identifiers, generated exports, repo inventory and root packaging discipline. Does not block active Lean/content pre-code readiness chain.                              |
| Phase split decision                  | Kept as white DB/tooling sidechain phases 20.15-20.20 so DB planning is visible but does not preempt the active 1.1.1.1.5.1 cursor.                                                            |
| Positive path plan                    | Design DB mirror in staged order: schema -> import/roundtrip -> snapshots/checksums -> read-only api views/RLS -> web frontend contract -> later reviewer/admin model.                         |
| Module implementation manifest        | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.                                                               |
| Semantic placement decision           | Natural placement is DB-REVIEW/generated-secondary publication infrastructure, not a CNNA mathematical pillar.                                                                                 |
| Reuse and duplication audit           | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.                                                              |
| Canonical generator policy            | Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.                                                                                |
| Module compactness rule               | Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content.                                                                                            |
| Refactor regression protocol          | No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.                                                                                         |
| Refactor migration metrics            | No migration executed. Future metrics: tables imported, rows imported, snapshot hash, RLS test count, anon write-denial tests.                                                                 |
| Lean/mathlib version policy           | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                             |

|                                        |                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ExecComplex boundary policy            | No ExecComplex change.                                                                                                                                                                                                                                                                                                 |
| Consumer-contract direction            | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.                                                                                                                                                                        |
| Governance-overhead review             | Planned sidechain is justified by long-term public visualization/query value, but it must remain non-blocking and sparse until YAML/Lean surfaces stabilize.                                                                                                                                                           |
| Proof rehearsal before implementation  | Before activation, rehearse importer locally using a disposable database, check hashes, verify anon read-only views, and prove that every public row traces to a snapshot_id.                                                                                                                                          |
| Expected tactics                       | None;<br>database/tooling phase, not Lean proof phase.                                                                                                                                                                                                                                                                 |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                                                                                                            |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                                                                                          |
| Concrete computable instance traversal | No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.                                                                                                                                                                                           |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete future output is schema/import/view/frontend contract, not a mathematical witness.                                                                                                                                                                                |
| Expected executable witness            | Future activation requires SQL migrations, import script, snapshot hash report, RLS/grant test report and public-view query examples.                                                                                                                                                                                  |
| Mathlib complication log               | None;<br>no Lean/mathlib code changes.                                                                                                                                                                                                                                                                                 |
| Noncomputable boundary audit           | No Lean noncomputable boundary involved.                                                                                                                                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | Not applicable.                                                                                                                                                                                                                                                                                                        |
| Bridge gap status                      | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                                                                                                                                                                  |
| Natural backjump/refactor layer        | If DB requirements conflict with YAML schema, patch the YAML/export schema at the natural documentation/tooling layer;<br>do not alter Lean derivations for DB convenience.                                                                                                                                            |
| Implementation prompt contract         | When activated, generate migrations/importer/frontend contracts from this phase. Do not ask for or store service_role keys in chat or repo;<br>use local .env/admin execution.                                                                                                                                         |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and future Supabase read-only guardrails.                                                                                                                                                                                            |
| How implemented                        | Not implemented. V1.22 records the phase as a non-blocking DB/tooling sidechain for future Supabase/Postgres public read-only mirror and web-frontend work.                                                                                                                                                            |
| Verification / evidence                | Planning record only. No live Supabase connection, no DB write, no frontend deployment and no Lean source change were performed in V1.22.                                                                                                                                                                              |
| Proof strategy                         | Planning/governance proof by explicit scope separation: Lean/YAML source -> snapshot/import -> Supabase read-only mirror -> web frontend. DB rows never become truth-source rows.                                                                                                                                      |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                      |
| Post-implementation plan check         | V1.22 only records the sidechain. Active mathematical/content cursor remains 1.1.1.1.5.1.                                                                                                                                                                                                                              |
| Artifact packaging rule                | Full updated zip package;<br>no secrets, no .env, no live DB credentials, no generated DB dumps with sensitive keys.                                                                                                                                                                                                   |
| Implementation notes                   | review.* for later authenticated comments/annotations;<br>admin/import.* for local importer or future signed Edge Function;<br>no public writes and no manual table-click update as standard workflow. This is generated-secondary/publication tooling and cannot release CNNA phases or replace Lean/YAML provenance. |
| Source materials / references          | <ul style="list-style-type: none"> <li>Master YAML</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> <li>Supabase API/RLS/key docs as external implementation reference</li> <li>user cnna-project.net web-frontend vision</li> </ul>                                                    |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>DBR20</li> <li>public read-only mirror</li> <li>append-only snapshot</li> <li>api.* read-only view</li> <li>internal.* canonical tables</li> <li>review.* annotations</li> <li>admin/import.* importer path</li> </ul>                                                          |

|                                      |                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modified modules                     | <ul style="list-style-type: none"><li>Repository/masters/cnna__roadmap__master.yaml</li><li>README.md</li></ul>                                                                                                                                                                                   |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>no Supabase row as CNNA truth source</li><li>no public write permissions</li><li>no browser service__role key</li><li>no DB phase release authority</li><li>no DB update without snapshot/checksum/import report</li><li>no large PDFs/ZIPs in DB</li></ul> |
| No-go guardrails                     | <ul style="list-style-type: none"><li>GLOBAL-INHERITED: all global__no__go__guardrails apply, especially GNG34 once added.</li><li>No Supabase row is a CNNA truth source.</li><li>No public writes.</li><li>No browser service__role key.</li></ul>                                              |
| Reusable existing modules            | <ul style="list-style-type: none"><li>masters/cnna__roadmap__master.yaml</li><li>generated__exports/*.tsv</li><li>repo__inventory/raw__export/*.json</li><li>scripts/generate__tables.py</li></ul>                                                                                                |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"><li>second truth system risk</li><li>manual DB table edits</li><li>public write policy leak</li><li>service__role key leak</li><li>snapshot provenance loss</li></ul>                                                                                           |

Phase 20.21 – Generated PDF typography scale increase

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                         |
| Context class                         | GeneratedArtifactPhase                                                                                                                    |
| Phase documentation tier              | F3                                                                                                                                        |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat. |
| Record change version                 | unchanged-by-V1.01                                                                                                                        |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                       |
| Implementation status                 | completed                                                                                                                                 |
| Preparation status                    | completed-tooling-documentation                                                                                                           |
| Phase type                            | documentation                                                                                                                             |
| Phase-type-specific contract          | Documentation phase: improve PDF readability without changing Lean semantics, YAML row semantics or context-filter row selection.         |
| Title / state                         | Generated PDF typography scale increase                                                                                                   |
| Preparation workflow                  | Generated PDF typography profile workflow.                                                                                                |
| Source-to-CNNA mapping                | Maps a review-layout/readability requirement into generated-secondary tooling; no CNNA mathematical generator is introduced.              |
| Dependency-depth audit                | No Lean dependency introduced. TeX generation depends on existing Python scripts and pdflatex only.                                       |
| Phase split decision                  | Implemented as DB-REVIEW tooling phase 20.21 because it modifies generated documentation layout rather than mathematical content.         |
| Positive path plan                    | Return to active 1.1.1.1.5.1 IST intake gate after typography tooling is closed.                                                          |



|                                        |                                                                                                                                                                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module implementation manifest         | No Lean modules added/modified. Documentation generator scripts and YAML only.                                                                                                                                                                                      |
| Semantic placement decision            | Natural placement is DB-REVIEW/documentation generator layer, not a mathematical pillar or CNNA generator.                                                                                                                                                          |
| Reuse and duplication audit            | Closed for typography: size profile is centralized in generators and not repeated by manual PDF edits.                                                                                                                                                              |
| Canonical generator policy             | Typography behavior is generated from full/context/table generator source, not from edited PDF files.                                                                                                                                                               |
| Module compactness rule                | Do not split table generators solely for typography;<br>use one central size-bump policy unless later profiles require explicit configuration.                                                                                                                      |
| Refactor regression protocol           | Regression checks are generate_tables.py, full PDF compile, context PDF compile, generated TeX size-profile audit and representative rendering.                                                                                                                     |
| Refactor migration metrics             | No Lean module migration. Generated PDF page count may increase due to larger fonts and wrapping.                                                                                                                                                                   |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                                                                  |
| ExecComplex boundary policy            | No ExecComplex boundary affected.                                                                                                                                                                                                                                   |
| Consumer-contract direction            | Human/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.                                                                                                                                                                        |
| Governance-overhead review             | Low overhead: centralized generator/table size profile;<br>possible downside is increased page count, accepted as readability tradeoff.                                                                                                                             |
| Proof rehearsal before implementation  | Patch generator size profile, regenerate tables, compile representative PDFs, render sample pages, and audit no Lean source changed.                                                                                                                                |
| Expected tactics                       | None;<br>documentation/tooling phase, not Lean proof phase.                                                                                                                                                                                                         |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                                       |
| Concrete computable instance traversal | Concrete traversal is full-document PDF generation and Supabase context-document generation.                                                                                                                                                                        |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete output is compiled full/context PDF layout with enlarged text.                                                                                                                                                 |
| Expected executable witness            | Successful Python table generation and PDF compilation;<br>no Lean executable witness required.                                                                                                                                                                     |
| Mathlib complication log               | No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.                                                                                                                                                                               |
| Noncomputable boundary audit           | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                                                     |
| Mathlib mitigation/refactor layer      | Not applicable;<br>documentation tooling only.                                                                                                                                                                                                                      |
| Bridge gap status                      | No mathematical bridge closed;<br>generated PDF readability/layout gap closed.                                                                                                                                                                                      |
| Natural backjump/refactor layer        | Documentation generators and table generator are the natural layer;<br>do not edit disposable PDFs manually.                                                                                                                                                        |
| Implementation prompt contract         | Future font/layout changes must be encoded as generator profiles and tested on at least one full plus one context PDF.                                                                                                                                              |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and typography profile rules.                                                                                                                                                     |
| How implemented                        | Updated full/context PDF generators to use 11pt A3 landscape document classes;<br>bumped explicit table and legend TeX size commands one step in generate_tables.py;<br>bumped generated graph label sizes where the tool controls DOT output.                      |
| Verification / evidence                | generate_tables.py succeeds;<br>generated full and context PDFs compile;<br>generated TeX contains documentclass[11pt] and larger table-size commands;<br>representative pages were rendered for visual sanity;<br>no Lean source under repo_snapshot/CNNA changed. |
| Proof strategy                         | Treat typography as a generator-level profile;<br>prove by regeneration and render inspection rather than manual PDF editing.                                                                                                                                       |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table generation and PDF generation only.                                                                                                                                                                      |
| Post-implementation plan check         | V1.23 is a documentation-layout closeout. Active mathematical/content cursor remains 1.1.1.1.5.1.                                                                                                                                                                   |



|                                      |                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Artifact packaging rule              | Full updated zip package;<br>no generated full/context PDFs in package root;<br>generated representative PDFs may exist under generated_docs/ or context_docs/.                                                                                                                                                                        |
| Implementation notes                 | This is a generated-secondary readability/layout update only. It may increase PDF page counts and table wrapping, but must not change row content, phase status, context filtering, Lean code or proof semantics.                                                                                                                      |
| Source materials / references        | <ul style="list-style-type: none"><li>• user request: make all generated-document text one size larger</li><li>• V1.22 generators</li><li>• V1.22 README and master YAML</li></ul>                                                                                                                                                     |
| Extracted established theory objects | <ul style="list-style-type: none"><li>• enlarged PDF typography profile</li><li>• one-step TeX size-command bump</li><li>• 11pt document-class profile</li><li>• graph-label size bump</li></ul>                                                                                                                                       |
| Implemented modules                  | <ul style="list-style-type: none"><li>• Repository/scripts/generate_full_document.py</li><li>• Repository/scripts/generate_context_doc.py</li><li>• Repository/scripts/generate_tables.py</li><li>• Repository/scripts/build_export_script_v1.8.py</li><li>• README.md</li><li>• Repository/masters/cnna_roadmap_master.yaml</li></ul> |
| Added modules                        | <ul style="list-style-type: none"><li>• Repository/archive/reports/V1_23_PDF_FONT_SCALE_REPORT.md</li></ul>                                                                                                                                                                                                                            |
| Modified modules                     | <ul style="list-style-type: none"><li>• Repository/scripts/generate_full_document.py</li><li>• Repository/scripts/generate_context_doc.py</li><li>• Repository/scripts/generate_tables.py</li><li>• Repository/scripts/build_export_script_v1.8.py</li><li>• README.md</li><li>• Repository/masters/cnna_roadmap_master.yaml</li></ul> |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>• no manual edits to disposable generated PDFs</li><li>• no Lean source edit</li><li>• no context-filter semantic change</li><li>• no phase-status semantic change</li><li>• no generated artifact as primary truth</li></ul>                                                                    |
| No-go guardrails                     | <ul style="list-style-type: none"><li>• GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG23-GNG35.</li><li>• no generated artifact as primary truth</li><li>• font changes do not certify semantics</li></ul>                                                                                                        |
| Proposed added modules               | <ul style="list-style-type: none"><li>• Repository/archive/reports/V1_23_PDF_FONT_SCALE_REPORT.md</li></ul>                                                                                                                                                                                                                            |
| Proposed modified modules            | <ul style="list-style-type: none"><li>• Repository/scripts/generate_full_document.py</li><li>• Repository/scripts/generate_context_doc.py</li><li>• Repository/scripts/generate_tables.py</li><li>• Repository/scripts/build_export_script_v1.8.py</li><li>• README.md</li><li>• Repository/masters/cnna_roadmap_master.yaml</li></ul> |
| Reusable existing modules            | <ul style="list-style-type: none"><li>• scripts/generate_full_document.py</li><li>• scripts/generate_context_doc.py</li><li>• scripts/generate_tables.py</li></ul>                                                                                                                                                                     |

Duplicate-risk symbols or operations

- No Lean symbols affected. Avoid duplicate typography hacks outside the generators.

Phase 20.22 – Supabase live project discovery and empty-schema baseline audit

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                              |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | <span>G</span>                                                                                                                                                                                                      |
| Context class                          | DatabaseMirrorPhase                                                                                                                                                                                                 |
| Phase documentation tier               | F1                                                                                                                                                                                                                  |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                      |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                  |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                 |
| Implementation status                  | completed                                                                                                                                                                                                           |
| Preparation status                     | completed-readonly-discovery                                                                                                                                                                                        |
| Phase type                             | infrastructure                                                                                                                                                                                                      |
| Phase-type-specific contract           | Infrastructure/tooling sidechain: Supabase stores generated-secondary planning metadata for public read-only query/visualization; no Lean semantic authority.                                                       |
| Title / state                          | Supabase live project discovery and empty-schema baseline audit                                                                                                                                                     |
| Preparation workflow                   | Supabase live import activation workflow.                                                                                                                                                                           |
| Source-to-CNNA mapping                 | Maps CNNA generated-secondary metadata to Supabase as public query/visualization data. It is not a mathematical CNNA input and not a Lean replacement.                                                              |
| Dependency-depth audit                 | Depends on stable YAML identifiers, generated exports, repo inventory, root packaging discipline, and explicit user authorization for live writes. Does not block the active Lean/content pre-code readiness chain. |
| Phase split decision                   | Existing 20.15-20.20 remain broad contracts; V1.24 adds concrete live-project activation phases after checking those existing phases to avoid scope duplication.                                                    |
| Positive path plan                     | Proceed from read-only baseline -> migration package -> import package -> explicit live execution -> read-only/RLS/frontend verification.                                                                           |
| Module implementation manifest         | No Lean modules. Future DB artifacts may be SQL migrations/import scripts/frontend contract files under Repository/database/supabase.                                                                               |
| Semantic placement decision            | Natural placement is DB-REVIEW/Supabase infrastructure sidechain, not a Lean derivation phase.                                                                                                                      |
| Reuse and duplication audit            | Existing 20.15-20.20 were checked and kept; new phases are concrete activation subchain, not duplicates.                                                                                                            |
| Canonical generator policy             | Supabase rows must be generated from Master YAML/generated_exports/repo_inventory via canonical importer, not manual table edits.                                                                                   |
| Module compactness rule                | Keep DB artifacts isolated under Repository/database/supabase and avoid leaking DB concerns into Lean module names.                                                                                                 |
| Refactor regression protocol           | No Lean refactor. Regression checks are generated tables/PDF, DB schema/import dry-run, live inventory/advisor output after authorized execution.                                                                   |
| Refactor migration metrics             | No Lean module migration. Future DB migration metrics: schemas/tables/views/policies/row counts/import hashes.                                                                                                      |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                  |
| ExecComplex boundary policy            | No ExecComplex boundary affected.                                                                                                                                                                                   |
| Consumer-contract direction            | Consumers are public web/query/review tooling only. No pillar handoff or mathematical consumer rights are granted.                                                                                                  |
| Governance-overhead review             | Moderate but justified: direct DB access now exists, so broad DB phases need concrete activation contracts before any write.                                                                                        |
| Proof rehearsal before implementation  | Dry-run schema/import artifacts, then when authorized apply migration/import, check tables/views, advisor output, anon read-only access and no public writes.                                                       |
| Expected tactics                       | None; database/tooling phase, not Lean proof phase.                                                                                                                                                                 |
| Expected mathlib imports               | None added.                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                       |
| Concrete computable instance traversal | Concrete traversal is Supabase project discovery, planned schema/import tables, and later live table/view verification.                                                                                             |
| Computable path non-skeleton status    | Not an operational/computable CNNA path; concrete output is DB tooling metadata and later Supabase rows with snapshot provenance.                                                                                   |

|                                      |                                                                                                                                                                                                                                                        |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected executable witness          | For planning phases: generate_tables.py/PDF generation. For future live phases: migration/import report, row counts, advisor output and read-only test results.                                                                                        |
| Mathlib complication log             | No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.                                                                                                                                                                  |
| Noncomputable boundary audit         | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                                        |
| Mathlib mitigation/refactor layer    | Not applicable;<br>database/tooling layer only.                                                                                                                                                                                                        |
| Bridge gap status                    | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                                                                                                  |
| Natural backjump/refactor layer      | If DB requirements conflict with YAML/export structure, patch the YAML/export schema or DB importer layer;<br>do not alter Lean derivations for DB convenience.                                                                                        |
| Implementation prompt contract       | Future live DB work must be explicit: apply migration/import only after user authorization, write only to Supabase, never edit Lean files, and produce report/checksums/advisor output.                                                                |
| Global no-go inheritance             | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and Supabase public-read-only rules.                                                                                                                                 |
| How implemented                      | Performed read-only Supabase discovery through connected tool: listed organization CNNA Project, project aiekahdewadrejrrdajn in eu-north-1 as ACTIVE_HEALTHY, and confirmed no tables in public/internal/api/review/admin before migration/import.    |
| Verification / evidence              | Supabase list_organizations/list_projects/list_tables read-only calls;<br>no migration, no SQL write, no data import and no Lean source change.                                                                                                        |
| Proof strategy                       | Separate broad DB architecture contracts 20.15-20.20 from concrete live activation phases 20.22-20.26;<br>require snapshot_id, hashes, import reports and explicit authorization for writes.                                                           |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                      |
| Post-implementation plan check       | Supabase DB sidechain remains W/G tooling and does not alter active mathematical/content cursor 1.1.1.1.5.1.                                                                                                                                           |
| Artifact packaging rule              | Package migration/import/frontend-contract artifacts in Repository/database/supabase or reports;<br>do not package secrets, service_role keys or generated DB credentials.                                                                             |
| Implementation notes                 | Supabase/Postgres DB/tooling sidechain only. It writes no Lean files and cannot act as CNNA truth system or phase-release authority.                                                                                                                   |
| Source materials / references        | <ul style="list-style-type: none"><li>Existing phases 20.15-20.20</li><li>Supabase project discovery</li><li>Master YAML</li><li>generated_exports/*.tsv</li><li>repo_inventory/raw_export/*.json</li><li>official Supabase RLS/API guidance</li></ul> |
| Extracted established theory objects | <ul style="list-style-type: none"><li>Supabase live project baseline</li><li>schema migration package</li><li>snapshot import package</li><li>live import execution contract</li><li>RLS/read-only verification contract</li></ul>                     |
| Implemented modules                  | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li></ul>                                                                                                                                                            |
| Added modules                        | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_24_SUPABASE_LIVE_IMPORT_ACTIVATION_PLAN_REPORT.md</li></ul>                                                                                                                        |
| Modified modules                     | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li><li>README.md</li></ul>                                                                                                                                          |

|                                      |                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source edit</li> <li>no Supabase row as CNNA truth source</li> <li>no public write permissions</li> <li>no browser service_role key</li> <li>no DB write without explicit authorization</li> <li>no manual production table-click update as standard workflow</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG34 and GNG36.</li> <li>no public writes</li> <li>no DB truth-source status</li> <li>no service_role key in browser/repo/chat</li> </ul>                                                                       |
| Proposed added modules               | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_24_SUPABASE_LIVE_IMPORT_ACTIVATION_PLAN_REPORT.md</li> </ul>                                                                                                                                                                                       |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul>                                                                                                                                                                                                        |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>scripts/generate_context_doc.py</li> <li>scripts/generate_full_document.py</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> </ul>                                                                             |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>Existing broad DBR15-DBR20 contracts vs new concrete activation phases; resolved by explicit broad-vs-activation split.</li> </ul>                                                                                                                                               |

### Phase 20.23 – Supabase initial schema migration package and RLS/read-only policy design

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                              |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                   |
| Context class                         | DatabaseMirrorPhase                                                                                                                                                                                                 |
| Phase documentation tier              | F1                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                      |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                 |
| Implementation status                 | planned-not-active                                                                                                                                                                                                  |
| Preparation status                    | planned-sidechain                                                                                                                                                                                                   |
| Phase type                            | infrastructure                                                                                                                                                                                                      |
| Phase-type-specific contract          | Infrastructure/tooling sidechain: Supabase stores generated-secondary planning metadata for public read-only query/visualization; no Lean semantic authority.                                                       |
| Title / state                         | Supabase initial schema migration package and RLS/read-only policy design                                                                                                                                           |
| Preparation workflow                  | Supabase live import activation workflow.                                                                                                                                                                           |
| Source-to-CNNA mapping                | Maps CNNA generated-secondary metadata to Supabase as public query/visualization data. It is not a mathematical CNNA input and not a Lean replacement.                                                              |
| Dependency-depth audit                | Depends on stable YAML identifiers, generated exports, repo inventory, root packaging discipline, and explicit user authorization for live writes. Does not block the active Lean/content pre-code readiness chain. |
| Phase split decision                  | Existing 20.15-20.20 remain broad contracts; V1.24 adds concrete live-project activation phases after checking those existing phases to avoid scope duplication.                                                    |

|                                        |                                                                                                                                                                                              |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive path plan                     | Proceed from read-only baseline -> migration package -> import package -> explicit live execution -> read-only/RLS/frontend verification.                                                    |
| Module implementation manifest         | No Lean modules. Future DB artifacts may be SQL migrations/import scripts/frontend contract files under Repository/database/supabase.                                                        |
| Semantic placement decision            | Natural placement is DB-REVIEW/Supabase infrastructure sidechain, not a Lean derivation phase.                                                                                               |
| Reuse and duplication audit            | Existing 20.15-20.20 were checked and kept;<br>new phases are concrete activation subchain, not duplicates.                                                                                  |
| Canonical generator policy             | Supabase rows must be generated from Master YAML/generated_exports/repo_inventory via canonical importer, not manual table edits.                                                            |
| Module compactness rule                | Keep DB artifacts isolated under Repository/database/supabase and avoid leaking DB concerns into Lean module names.                                                                          |
| Refactor regression protocol           | No Lean refactor. Regression checks are generated tables/PDF, DB schema/import dry-run, live inventory/advisor output after authorized execution.                                            |
| Refactor migration metrics             | No Lean module migration. Future DB migration metrics: schemas/tables/views/policies/row counts/import hashes.                                                                               |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                           |
| ExecComplex boundary policy            | No ExecComplex boundary affected.                                                                                                                                                            |
| Consumer-contract direction            | Consumers are public web/query/review tooling only. No pillar handoff or mathematical consumer rights are granted.                                                                           |
| Governance-overhead review             | Moderate but justified: direct DB access now exists, so broad DB phases need concrete activation contracts before any write.                                                                 |
| Proof rehearsal before implementation  | Dry-run schema/import artifacts, then when authorized apply migration/import, check tables/views, advisor output, anon read-only access and no public writes.                                |
| Expected tactics                       | None;<br>database/tooling phase, not Lean proof phase.                                                                                                                                       |
| Expected mathlib imports               | None added.                                                                                                                                                                                  |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                |
| Concrete computable instance traversal | Concrete traversal is Supabase project discovery, planned schema/import tables, and later live table/view verification.                                                                      |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete output is DB tooling metadata and later Supabase rows with snapshot provenance.                                                         |
| Expected executable witness            | For planning phases: generate_tables.py/PDF generation. For future live phases: migration/import report, row counts, advisor output and read-only test results.                              |
| Mathlib complication log               | No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.                                                                                                        |
| Noncomputable boundary audit           | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                              |
| Mathlib mitigation/refactor layer      | Not applicable;<br>database/tooling layer only.                                                                                                                                              |
| Bridge gap status                      | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                                        |
| Natural backjump/refactor layer        | If DB requirements conflict with YAML/export structure, patch the YAML/export schema or DB importer layer;<br>do not alter Lean derivations for DB convenience.                              |
| Implementation prompt contract         | Future live DB work must be explicit: apply migration/import only after user authorization, write only to Supabase, never edit Lean files, and produce report/checksums/advisor output.      |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and Supabase public-read-only rules.                                                                       |
| How implemented                        | Planned. Will produce initial SQL migration package only;<br>no live apply_migration until explicit authorization.                                                                           |
| Verification / evidence                | Planned evidence: SQL migration file, schema/table/view inventory expectation, no secret material.                                                                                           |
| Proof strategy                         | Separate broad DB architecture contracts 20.15-20.20 from concrete live activation phases 20.22-20.26;<br>require snapshot_id, hashes, import reports and explicit authorization for writes. |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                            |
| Post-implementation plan check         | Supabase DB sidechain remains W/G tooling and does not alter active mathematical/content cursor 1.1.1.1.5.1.                                                                                 |
| Artifact packaging rule                | Package migration/import/frontend-contract artifacts in Repository/database/supabase or reports;<br>do not package secrets, service_role keys or generated DB credentials.                   |
| Implementation notes                   | Supabase/Postgres DB/tooling sidechain only. It writes no Lean files and cannot act as CNNA truth system or phase-release authority.                                                         |

|                                      |                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source materials / references        | <ul style="list-style-type: none"><li>Existing phases 20.15-20.20</li><li>Supabase project discovery</li><li>Master YAML</li><li>generated_exports/*.tsv</li><li>repo_inventory/raw_export/*.json</li><li>official Supabase RLS/API guidance</li></ul>                                                           |
| Extracted established theory objects | <ul style="list-style-type: none"><li>Supabase live project baseline</li><li>schema migration package</li><li>snapshot import package</li><li>live import execution contract</li><li>RLS/read-only verification contract</li></ul>                                                                               |
| Modified modules                     | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li><li>README.md</li></ul>                                                                                                                                                                                                    |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>no Lean source edit</li><li>no Supabase row as CNNA truth source</li><li>no public write permissions</li><li>no browser service_role key</li><li>no DB write without explicit authorization</li><li>no manual production table-click update as standard workflow</li></ul> |
| No-go guardrails                     | <ul style="list-style-type: none"><li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG34 and GNG36.</li><li>no public writes</li><li>no DB truth-source status</li><li>no service_role key in browser/repo/chat</li></ul>                                                                     |
| Proposed added modules               | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_24_SUPABASE_LIVE_IMPORT_ACTIVATION_PLAN_REPORT.md</li></ul>                                                                                                                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li><li>README.md</li></ul>                                                                                                                                                                                                    |
| Reusable existing modules            | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>scripts/generate_context_doc.py</li><li>scripts/generate_full_document.py</li><li>generated_exports/*.tsv</li><li>repo_inventory/raw_export/*.json</li></ul>                                                                            |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"><li>Existing broad DBR15-DBR20 contracts vs new concrete activation phases; resolved by explicit broad-vs-activation split.</li></ul>                                                                                                                                          |

Phase 20.24 – Supabase snapshot import package from YAML/generated exports

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                    | Record              |
|--------------------------|---------------------|
| Phase status             | W                   |
| Context class            | DatabaseMirrorPhase |
| Phase documentation tier | F1                  |



|                                        |                                                                                                                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                      |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                  |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                 |
| Implementation status                  | planned-not-active                                                                                                                                                                                                  |
| Preparation status                     | planned-sidechain                                                                                                                                                                                                   |
| Phase type                             | infrastructure                                                                                                                                                                                                      |
| Phase-type-specific contract           | Infrastructure/tooling sidechain: Supabase stores generated-secondary planning metadata for public read-only query/visualization; no Lean semantic authority.                                                       |
| Title / state                          | Supabase snapshot import package from YAML/generated exports                                                                                                                                                        |
| Preparation workflow                   | Supabase live import activation workflow.                                                                                                                                                                           |
| Source-to-CNNA mapping                 | Maps CNNA generated-secondary metadata to Supabase as public query/visualization data. It is not a mathematical CNNA input and not a Lean replacement.                                                              |
| Dependency-depth audit                 | Depends on stable YAML identifiers, generated exports, repo inventory, root packaging discipline, and explicit user authorization for live writes. Does not block the active Lean/content pre-code readiness chain. |
| Phase split decision                   | Existing 20.15-20.20 remain broad contracts;<br>V1.24 adds concrete live-project activation phases after checking those existing phases to avoid scope duplication.                                                 |
| Positive path plan                     | Proceed from read-only baseline -> migration package -> import package -> explicit live execution -> read-only/RLS/frontend verification.                                                                           |
| Module implementation manifest         | No Lean modules. Future DB artifacts may be SQL migrations/import scripts/frontend contract files under Repository/database/supabase.                                                                               |
| Semantic placement decision            | Natural placement is DB-REVIEW/Supabase infrastructure sidechain, not a Lean derivation phase.                                                                                                                      |
| Reuse and duplication audit            | Existing 20.15-20.20 were checked and kept;<br>new phases are concrete activation subchain, not duplicates.                                                                                                         |
| Canonical generator policy             | Supabase rows must be generated from Master YAML/generated_exports/repo_inventory via canonical importer, not manual table edits.                                                                                   |
| Module compactness rule                | Keep DB artifacts isolated under Repository/database/supabase and avoid leaking DB concerns into Lean module names.                                                                                                 |
| Refactor regression protocol           | No Lean refactor. Regression checks are generated tables/PDF, DB schema/import dry-run, live inventory/advisor output after authorized execution.                                                                   |
| Refactor migration metrics             | No Lean module migration. Future DB migration metrics: schemas/tables/views/policies/row counts/import hashes.                                                                                                      |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                  |
| ExecComplex boundary policy            | No ExecComplex boundary affected.                                                                                                                                                                                   |
| Consumer-contract direction            | Consumers are public web/query/review tooling only. No pillar handoff or mathematical consumer rights are granted.                                                                                                  |
| Governance-overhead review             | Moderate but justified: direct DB access now exists, so broad DB phases need concrete activation contracts before any write.                                                                                        |
| Proof rehearsal before implementation  | Dry-run schema/import artifacts, then when authorized apply migration/import, check tables/views, advisor output, anon read-only access and no public writes.                                                       |
| Expected tactics                       | None;<br>database/tooling phase, not Lean proof phase.                                                                                                                                                              |
| Expected mathlib imports               | None added.                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                       |
| Concrete computable instance traversal | Concrete traversal is Supabase project discovery, planned schema/import tables, and later live table/view verification.                                                                                             |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete output is DB tooling metadata and later Supabase rows with snapshot provenance.                                                                                |
| Expected executable witness            | For planning phases: generate_tables.py/PDF generation. For future live phases: migration/import report, row counts, advisor output and read-only test results.                                                     |
| Mathlib complication log               | No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.                                                                                                                               |
| Noncomputable boundary audit           | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                     |
| Mathlib mitigation/refactor layer      | Not applicable;<br>database/tooling layer only.                                                                                                                                                                     |
| Bridge gap status                      | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                                                               |
| Natural backjump/refactor layer        | If DB requirements conflict with YAML/export structure, patch the YAML/export schema or DB importer layer;<br>do not alter Lean derivations for DB convenience.                                                     |
| Implementation prompt contract         | Future live DB work must be explicit: apply migration/import only after user authorization, write only to Supabase, never edit Lean files, and produce report/checksums/advisor output.                             |



|                                      |                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global no-go inheritance             | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and Supabase public-read-only rules.                                                                                                                                                                                           |
| How implemented                      | Planned. Will produce importer/seed contract reading YAML/generated_exports/repo_inventory and writing snapshot-tagged internal rows only after authorization.                                                                                                                                                   |
| Verification / evidence              | Planned evidence: importer script, dry-run report, generated SQL or row-count manifest, source hashes.                                                                                                                                                                                                           |
| Proof strategy                       | Separate broad DB architecture contracts 20.15-20.20 from concrete live activation phases 20.22-20.26;<br>require snapshot_id, hashes, import reports and explicit authorization for writes.                                                                                                                     |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                |
| Post-implementation plan check       | Supabase DB sidechain remains W/G tooling and does not alter active mathematical/content cursor 1.1.1.1.5.1.                                                                                                                                                                                                     |
| Artifact packaging rule              | Package migration/import/frontend-contract artifacts in Repository/database/supabase or reports;<br>do not package secrets, service_role keys or generated DB credentials.                                                                                                                                       |
| Implementation notes                 | Supabase/Postgres DB/tooling sidechain only. It writes no Lean files and cannot act as CNNA truth system or phase-release authority.                                                                                                                                                                             |
| Source materials / references        | <ul style="list-style-type: none"><li>Existing phases 20.15-20.20</li><li>Supabase project discovery</li><li>Master YAML</li><li>generated_exports/*.tsv</li><li>repo_inventory/raw_export/*.json</li><li>official Supabase RLS/API guidance</li></ul>                                                           |
| Extracted established theory objects | <ul style="list-style-type: none"><li>Supabase live project baseline</li><li>schema migration package</li><li>snapshot import package</li><li>live import execution contract</li><li>RLS/read-only verification contract</li></ul>                                                                               |
| Modified modules                     | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li><li>README.md</li></ul>                                                                                                                                                                                                    |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>no Lean source edit</li><li>no Supabase row as CNNA truth source</li><li>no public write permissions</li><li>no browser service_role key</li><li>no DB write without explicit authorization</li><li>no manual production table-click update as standard workflow</li></ul> |
| No-go guardrails                     | <ul style="list-style-type: none"><li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG34 and GNG36.</li><li>no public writes</li><li>no DB truth-source status</li><li>no service_role key in browser/repo/chat</li></ul>                                                                     |
| Proposed added modules               | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_24_SUPABASE_LIVE_IMPORT_ACTIVATION_PLAN_REPORT.md</li></ul>                                                                                                                                                                                  |
| Proposed modified modules            | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li><li>README.md</li></ul>                                                                                                                                                                                                    |
| Reusable existing modules            | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>scripts/generate_context_doc.py</li><li>scripts/generate_full_document.py</li><li>generated_exports/*.tsv</li><li>repo_inventory/raw_export/*.json</li></ul>                                                                            |

|                                      |                                                                                                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>Existing broad DBR15-DBR20 contracts vs new concrete activation phases; resolved by explicit broad-vs-activation split.</li> </ul> |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Phase 20.25 – Explicitly authorized live Supabase schema migration and first snapshot import

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                  | Record                                                                                                                                                                                                              |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                           | W                                                                                                                                                                                                                   |
| Context class                          | DatabaseMirrorPhase                                                                                                                                                                                                 |
| Phase documentation tier               | F1                                                                                                                                                                                                                  |
| Phase documentation depth requirement  | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                      |
| Record change version                  | unchanged-by-V1.01                                                                                                                                                                                                  |
| Record change timestamp                | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                 |
| Implementation status                  | planned-not-active                                                                                                                                                                                                  |
| Preparation status                     | planned-sidechain                                                                                                                                                                                                   |
| Phase type                             | infrastructure                                                                                                                                                                                                      |
| Phase-type-specific contract           | Infrastructure/tooling sidechain: Supabase stores generated-secondary planning metadata for public read-only query/visualization; no Lean semantic authority.                                                       |
| Title / state                          | Explicitly authorized live Supabase schema migration and first snapshot import                                                                                                                                      |
| Preparation workflow                   | Supabase live import activation workflow.                                                                                                                                                                           |
| Source-to-CNNA mapping                 | Maps CNNA generated-secondary metadata to Supabase as public query/visualization data. It is not a mathematical CNNA input and not a Lean replacement.                                                              |
| Dependency-depth audit                 | Depends on stable YAML identifiers, generated exports, repo inventory, root packaging discipline, and explicit user authorization for live writes. Does not block the active Lean/content pre-code readiness chain. |
| Phase split decision                   | Existing 20.15-20.20 remain broad contracts; V1.24 adds concrete live-project activation phases after checking those existing phases to avoid scope duplication.                                                    |
| Positive path plan                     | Proceed from read-only baseline -> migration package -> import package -> explicit live execution -> read-only/RLS/frontend verification.                                                                           |
| Module implementation manifest         | No Lean modules. Future DB artifacts may be SQL migrations/import scripts/frontend contract files under Repository/database/supabase.                                                                               |
| Semantic placement decision            | Natural placement is DB-REVIEW/Supabase infrastructure sidechain, not a Lean derivation phase.                                                                                                                      |
| Reuse and duplication audit            | Existing 20.15-20.20 were checked and kept; new phases are concrete activation subchain, not duplicates.                                                                                                            |
| Canonical generator policy             | Supabase rows must be generated from Master YAML/generated_exports/repo_inventory via canonical importer, not manual table edits.                                                                                   |
| Module compactness rule                | Keep DB artifacts isolated under Repository/database/supabase and avoid leaking DB concerns into Lean module names.                                                                                                 |
| Refactor regression protocol           | No Lean refactor. Regression checks are generated tables/PDF, DB schema/import dry-run, live inventory/advisor output after authorized execution.                                                                   |
| Refactor migration metrics             | No Lean module migration. Future DB migration metrics: schemas/tables/views/policies/row counts/import hashes.                                                                                                      |
| Lean/mathlib version policy            | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                  |
| ExecComplex boundary policy            | No ExecComplex boundary affected.                                                                                                                                                                                   |
| Consumer-contract direction            | Consumers are public web/query/review tooling only. No pillar handoff or mathematical consumer rights are granted.                                                                                                  |
| Governance-overhead review             | Moderate but justified: direct DB access now exists, so broad DB phases need concrete activation contracts before any write.                                                                                        |
| Proof rehearsal before implementation  | Dry-run schema/import artifacts, then when authorized apply migration/import, check tables/views, advisor output, anon read-only access and no public writes.                                                       |
| Expected tactics                       | None; database/tooling phase, not Lean proof phase.                                                                                                                                                                 |
| Expected mathlib imports               | None added.                                                                                                                                                                                                         |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                       |
| Concrete computable instance traversal | Concrete traversal is Supabase project discovery, planned schema/import tables, and later live table/view verification.                                                                                             |
| Computable path non-skeleton status    | Not an operational/computable CNNA path; concrete output is DB tooling metadata and later Supabase rows with snapshot provenance.                                                                                   |

|                                      |                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected executable witness          | For planning phases: generate_tables.py/PDF generation. For future live phases: migration/import report, row counts, advisor output and read-only test results.                                                                                                                                                         |
| Mathlib complication log             | No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.                                                                                                                                                                                                                                   |
| Noncomputable boundary audit         | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                                                                                                         |
| Mathlib mitigation/refactor layer    | Not applicable;<br>database/tooling layer only.                                                                                                                                                                                                                                                                         |
| Bridge gap status                    | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                                                                                                                                                                   |
| Natural backjump/refactor layer      | If DB requirements conflict with YAML/export structure, patch the YAML/export schema or DB importer layer;<br>do not alter Lean derivations for DB convenience.                                                                                                                                                         |
| Implementation prompt contract       | Future live DB work must be explicit: apply migration/import only after user authorization, write only to Supabase, never edit Lean files, and produce report/checksums/advisor output.                                                                                                                                 |
| Global no-go inheritance             | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and Supabase public-read-only rules.                                                                                                                                                                                                  |
| How implemented                      | Planned future DB-write phase. Requires explicit user authorization in the conversation before apply_migration/execute_sql writes.                                                                                                                                                                                      |
| Verification / evidence              | Future evidence: migration result, import report, row counts, snapshot_id, checksums, no Lean file diff.                                                                                                                                                                                                                |
| Proof strategy                       | Separate broad DB architecture contracts 20.15-20.20 from concrete live activation phases 20.22-20.26;<br>require snapshot_id, hashes, import reports and explicit authorization for writes.                                                                                                                            |
| Build result                         | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                                                                                       |
| Post-implementation plan check       | Supabase DB sidechain remains W/G tooling and does not alter active mathematical/content cursor 1.1.1.1.5.1.                                                                                                                                                                                                            |
| Artifact packaging rule              | Package migration/import/frontend-contract artifacts in Repository/database/supabase or reports;<br>do not package secrets, service_role keys or generated DB credentials.                                                                                                                                              |
| Implementation notes                 | Supabase/Postgres DB/tooling sidechain only. It writes no Lean files and cannot act as CNNA truth system or phase-release authority.                                                                                                                                                                                    |
| Source materials / references        | <ul style="list-style-type: none"> <li>Existing phases 20.15-20.20</li> <li>Supabase project discovery</li> <li>Master YAML</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> <li>official Supabase RLS/API guidance</li> </ul>                                                           |
| Extracted established theory objects | <ul style="list-style-type: none"> <li>Supabase live project baseline</li> <li>schema migration package</li> <li>snapshot import package</li> <li>live import execution contract</li> <li>RLS/read-only verification contract</li> </ul>                                                                                |
| Modified modules                     | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul>                                                                                                                                                                                                        |
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source edit</li> <li>no Supabase row as CNNA truth source</li> <li>no public write permissions</li> <li>no browser service_role key</li> <li>no DB write without explicit authorization</li> <li>no manual production table-click update as standard workflow</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG34 and GNG36.</li> <li>no public writes</li> <li>no DB truth-source status</li> <li>no service_role key in browser/repo/chat</li> </ul>                                                                       |

|                                      |                                                                                                                                                                                                                                             |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposed added modules               | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_24_SUPABASE_LIVE_IMPORT_ACTIVATION_PLAN_REPORT.md</li> </ul>                                                                                                           |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul>                                                                                                                            |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>scripts/generate_context_doc.py</li> <li>scripts/generate_full_document.py</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> </ul> |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>Existing broad DBR15-DBR20 contracts vs new concrete activation phases; resolved by explicit broad-vs-activation split.</li> </ul>                                                                   |

## Phase 20.26 – Post-import read-only/RLS/advisor verification and web-frontend handoff

**Table legend.**   = green/closed and derived-only;   = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;   = orange/implementable after stated predecessor closure;   = yellow/active and implementable now;   = red/locked

Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                                                                              |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | W                                                                                                                                                                                                                   |
| Context class                         | DatabaseMirrorPhase                                                                                                                                                                                                 |
| Phase documentation tier              | F1                                                                                                                                                                                                                  |
| Phase documentation depth requirement | Extended dossier: definition, source chain, main lemmas/proof obligations, dependency risks, and downstream consumers. Enough detail for later implementation without re-reading chat history.                      |
| Record change version                 | unchanged-by-V1.01                                                                                                                                                                                                  |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                                                                                 |
| Implementation status                 | planned-not-active                                                                                                                                                                                                  |
| Preparation status                    | planned-sidechain                                                                                                                                                                                                   |
| Phase type                            | infrastructure                                                                                                                                                                                                      |
| Phase-type-specific contract          | Infrastructure/tooling sidechain: Supabase stores generated-secondary planning metadata for public read-only query/visualization; no Lean semantic authority.                                                       |
| Title / state                         | Post-import read-only/RLS/advisor verification and web-frontend handoff                                                                                                                                             |
| Preparation workflow                  | Supabase live import activation workflow.                                                                                                                                                                           |
| Source-to-CNNA mapping                | Maps CNNA generated-secondary metadata to Supabase as public query/visualization data. It is not a mathematical CNNA input and not a Lean replacement.                                                              |
| Dependency-depth audit                | Depends on stable YAML identifiers, generated exports, repo inventory, root packaging discipline, and explicit user authorization for live writes. Does not block the active Lean/content pre-code readiness chain. |
| Phase split decision                  | Existing 20.15-20.20 remain broad contracts; V1.24 adds concrete live-project activation phases after checking those existing phases to avoid scope duplication.                                                    |
| Positive path plan                    | Proceed from read-only baseline -> migration package -> import package -> explicit live execution -> read-only/RLS/frontend verification.                                                                           |
| Module implementation manifest        | No Lean modules. Future DB artifacts may be SQL migrations/import scripts/frontend contract files under Repository/database/supabase.                                                                               |
| Semantic placement decision           | Natural placement is DB-REVIEW/Supabase infrastructure sidechain, not a Lean derivation phase.                                                                                                                      |
| Reuse and duplication audit           | Existing 20.15-20.20 were checked and kept; new phases are concrete activation subchain, not duplicates.                                                                                                            |
| Canonical generator policy            | Supabase rows must be generated from Master YAML/generated_exports/repo_inventory via canonical importer, not manual table edits.                                                                                   |
| Module compactness rule               | Keep DB artifacts isolated under Repository/database/supabase and avoid leaking DB concerns into Lean module names.                                                                                                 |
| Refactor regression protocol          | No Lean refactor. Regression checks are generated tables/PDF, DB schema/import dry-run, live inventory/advisor output after authorized execution.                                                                   |
| Refactor migration metrics            | No Lean module migration. Future DB migration metrics: schemas/tables/views/policies/row counts/import hashes.                                                                                                      |
| Lean/mathlib version policy           | No Lean/mathlib version change. Lean data timestamp remains fixed.                                                                                                                                                  |
| ExecComplex boundary policy           | No ExecComplex boundary affected.                                                                                                                                                                                   |

|                                        |                                                                                                                                                                                                                                                               |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consumer-contract direction            | Consumers are public web/query/review tooling only. No pillar handoff or mathematical consumer rights are granted.                                                                                                                                            |
| Governance-overhead review             | Moderate but justified: direct DB access now exists, so broad DB phases need concrete activation contracts before any write.                                                                                                                                  |
| Proof rehearsal before implementation  | Dry-run schema/import artifacts, then when authorized apply migration/import, check tables/views, advisor output, anon read-only access and no public writes.                                                                                                 |
| Expected tactics                       | None;<br>database/tooling phase, not Lean proof phase.                                                                                                                                                                                                        |
| Expected mathlib imports               | None added.                                                                                                                                                                                                                                                   |
| ExecComplex/computable plan            | No ExecComplex object introduced or consumed.                                                                                                                                                                                                                 |
| Concrete computable instance traversal | Concrete traversal is Supabase project discovery, planned schema/import tables, and later live table/view verification.                                                                                                                                       |
| Computable path non-skeleton status    | Not an operational/computable CNNA path;<br>concrete output is DB tooling metadata and later Supabase rows with snapshot provenance.                                                                                                                          |
| Expected executable witness            | For planning phases: generate_tables.py/PDF generation. For future live phases: migration/import report, row counts, advisor output and read-only test results.                                                                                               |
| Mathlib complication log               | No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.                                                                                                                                                                         |
| Noncomputable boundary audit           | No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                                               |
| Mathlib mitigation/refactor layer      | Not applicable;<br>database/tooling layer only.                                                                                                                                                                                                               |
| Bridge gap status                      | No mathematical bridge closed. The bridge is publication/query tooling between YAML exports and a public read-only database/frontend.                                                                                                                         |
| Natural backjump/refactor layer        | If DB requirements conflict with YAML/export structure, patch the YAML/export schema or DB importer layer;<br>do not alter Lean derivations for DB convenience.                                                                                               |
| Implementation prompt contract         | Future live DB work must be explicit: apply migration/import only after user authorization, write only to Supabase, never edit Lean files, and produce report/checksums/advisor output.                                                                       |
| Global no-go inheritance               | All global no-go guardrails apply;<br>especially Lean-primary/generated-secondary and Supabase public-read-only rules.                                                                                                                                        |
| How implemented                        | Planned future verification phase after live DB import. Will verify anon read-only views, no public writes, advisors and frontend data contract.                                                                                                              |
| Verification / evidence                | Future evidence: list_tables, get_advisors, public view checks, failed write attempts if safe, frontend contract artifact.                                                                                                                                    |
| Proof strategy                         | Separate broad DB architecture contracts 20.15-20.20 from concrete live activation phases 20.22-20.26;<br>require snapshot_id, hashes, import reports and explicit authorization for writes.                                                                  |
| Build result                           | No Lean build run;<br>no Lean source changed. Python table/PDF generation only after YAML update.                                                                                                                                                             |
| Post-implementation plan check         | Supabase DB sidechain remains W/G tooling and does not alter active mathematical/content cursor 1.1.1.1.5.1.                                                                                                                                                  |
| Artifact packaging rule                | Package migration/import/frontend-contract artifacts in Repository/database/supabase or reports;<br>do not package secrets, service_role keys or generated DB credentials.                                                                                    |
| Implementation notes                   | Supabase/Postgres DB/tooling sidechain only. It writes no Lean files and cannot act as CNNA truth system or phase-release authority.                                                                                                                          |
| Source materials / references          | <ul style="list-style-type: none"> <li>Existing phases 20.15-20.20</li> <li>Supabase project discovery</li> <li>Master YAML</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> <li>official Supabase RLS/API guidance</li> </ul> |
| Extracted established theory objects   | <ul style="list-style-type: none"> <li>Supabase live project baseline</li> <li>schema migration package</li> <li>snapshot import package</li> <li>live import execution contract</li> <li>RLS/read-only verification contract</li> </ul>                      |
| Modified modules                       | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul>                                                                                                                                              |

|                                      |                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forbidden patterns checked           | <ul style="list-style-type: none"> <li>no Lean source edit</li> <li>no Supabase row as CNNA truth source</li> <li>no public write permissions</li> <li>no browser service_role key</li> <li>no DB write without explicit authorization</li> <li>no manual production table-click update as standard workflow</li> </ul> |
| No-go guardrails                     | <ul style="list-style-type: none"> <li>GLOBAL-INHERITED: all global_no_go_guardrails apply, especially GNG34 and GNG36.</li> <li>no public writes</li> <li>no DB truth-source status</li> <li>no service_role key in browser/repo/chat</li> </ul>                                                                       |
| Proposed added modules               | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_24_SUPABASE_LIVE_IMPORT_ACTIVATION_PLAN_REPORT.md</li> </ul>                                                                                                                                                                                       |
| Proposed modified modules            | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul>                                                                                                                                                                                                        |
| Reusable existing modules            | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>scripts/generate_context_doc.py</li> <li>scripts/generate_full_document.py</li> <li>generated_exports/*.tsv</li> <li>repo_inventory/raw_export/*.json</li> </ul>                                                                             |
| Duplicate-risk symbols or operations | <ul style="list-style-type: none"> <li>Existing broad DBR15-DBR20 contracts vs new concrete activation phases; resolved by explicit broad-vs-activation split.</li> </ul>                                                                                                                                               |

### Phase 20.27 – OOP-style phase/object context-class taxonomy and table-wide Lean/tool/reference separation

**Table legend.** G = green/closed and derived-only; W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison; O = orange/implementable after stated predecessor closure; Y = yellow/active and implementable now; R = red/locked Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Field                                 | Record                                                                                                                                                       |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase status                          | <span style="background-color: #d9ead3;">G</span>                                                                                                            |
| Context class                         | ToolInfrastructurePhase                                                                                                                                      |
| Phase documentation tier              | F3                                                                                                                                                           |
| Phase documentation depth requirement | Compact record: what exists, why it exists, owning phase, source location, and whether it may affect Lean objects. Avoid proofbook bloat.                    |
| Record change version                 | unchanged-by-V1.01                                                                                                                                           |
| Record change timestamp               | Lean data 2026-05-10T05:51:42+02:00                                                                                                                          |
| Implementation status                 | completed                                                                                                                                                    |
| Preparation status                    | closed-tooling-governance                                                                                                                                    |
| Phase type                            | infrastructure                                                                                                                                               |
| Phase-type-specific contract          | ToolInfrastructurePhase: schema/tooling phase; no direct Lean derivation release.                                                                            |
| Title / state                         | OOP-style phase/object context-class taxonomy and table-wide Lean/tool/reference separation                                                                  |
| Preparation workflow                  | Schema/classification patch in the Single YAML Master followed by generated-table regeneration.                                                              |
| Source-to-CNNA mapping                | The OOP analogy is used only at planning-schema level: classes give type boundaries and allowed consumers; they do not create mathematical CNNA entities.    |
| Dependency-depth audit                | Depends on existing status legend, phase-object-link policy, object register, proof dossiers and generated table pipeline; no Lean dependency is introduced. |



|                                        |                                                                                                                                                                                                          |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase split decision                   | Kept as one tooling phase 20.27 because it changes schema/table semantics only;<br>no mathematical derivation phase is spawned.                                                                          |
| Positive path plan                     | Classify all existing rows first, then make context visible in generated phase/object/proof/scratchpad surfaces;<br>later DB schema can mirror these fields.                                             |
| Module implementation manifest         | Records table-renderer and YAML schema changes only;<br>no new Lean modules.                                                                                                                             |
| Semantic placement decision            | Placed under 20.* because this is a planning/documentation tool schema change, not a Pillar-A mathematical object.                                                                                       |
| Reuse and duplication audit            | Reuses existing status legend, phase-object-link and object register instead of creating a parallel object register.                                                                                     |
| Canonical generator policy             | Context classes are type tags for review surfaces;<br>Lean code remains the only primary truth system.                                                                                                   |
| Module compactness rule                | No new script besides small patch helper;<br>permanent changes are in the existing generator scripts.                                                                                                    |
| Refactor regression protocol           | Generated tables before/after are regenerated from the same YAML source;<br>old status columns remain intact.                                                                                            |
| Refactor migration metrics             | Adds one phase, one object, one proof dossier and context-class fields to all existing phase/object rows.                                                                                                |
| Lean/mathlib version policy            | No mathlib dependency change.                                                                                                                                                                            |
| ExecComplex boundary policy            | No ExecComplex change.                                                                                                                                                                                   |
| Consumer-contract direction            | Consumers must read Context class before using status/link fields.                                                                                                                                       |
| Governance-overhead review             | Adds a single orthogonal class field rather than duplicating registers;<br>overhead is justified by preventing category errors.                                                                          |
| Proof rehearsal before implementation  | Completeness condition: count(phases with Context class)=count(phases) and count(objects with Context class)=count(objects);<br>class names must be from taxonomy.                                       |
| Expected tactics                       | not applicable;<br>no Lean proof                                                                                                                                                                         |
| Expected mathlib imports               | none                                                                                                                                                                                                     |
| ExecComplex/computable plan            | not applicable                                                                                                                                                                                           |
| Concrete computable instance traversal | Complete generated-row traversal, not a mathematical instance traversal.                                                                                                                                 |
| Computable path non-skeleton status    | not applicable;<br>classification is metadata, not ExecComplex computation.                                                                                                                              |
| Expected executable witness            | Generated TSV/PDF rows contain non-empty Context class for each phase/object.                                                                                                                            |
| Mathlib complication log               | none                                                                                                                                                                                                     |
| Noncomputable boundary audit           | No Lean noncomputable boundary touched.                                                                                                                                                                  |
| Mathlib mitigation/refactor layer      | none                                                                                                                                                                                                     |
| Bridge gap status                      | No bridge opened;<br>this is a meta-schema boundary that prevents false bridges from tool/reference classes into Lean classes.                                                                           |
| Natural backjump/refactor layer        | Generated planning schema and table renderer, not Lean consumers.                                                                                                                                        |
| Implementation prompt contract         | Future implementation prompts must preserve Context class and may not infer Lean derivability from Object status or Phase status alone.                                                                  |
| Global no-go inheritance               | Inherits all global no-go guardrails;<br>especially no tool/reference source becomes CNNA input.                                                                                                         |
| How implemented                        | Added explicit context-class taxonomy fields to every phase and object row;<br>extended generated phase/object/proof/scratchpad/module-manifest tables so status and context are orthogonal and visible. |
| Verification / evidence                | generate_tables.py validation and full A3 PDF generation after the V1.26 patch;<br>no Lean source files changed.                                                                                         |
| Proof strategy                         | Generated-schema proof by complete coverage: every registered phase and object must carry Context class;<br>validators reject missing classes;<br>generated tables render the field next to status.      |
| Build result                           | No Lean build;<br>documentation/tool generation only.                                                                                                                                                    |
| Post-implementation plan check         | All tables carrying phase/object logic now expose context classes. DB/publication phases remain secondary;<br>Lean derivation phases remain red/yellow/orange/green by release evidence.                 |



|                                      |                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Artifact packaging rule              | Return full zip and generated PDF;<br>no Lean-build claim.                                                                                                                                                                                                                                              |
| Implementation notes                 | This is a planning/tooling classification phase. It applies object-oriented class semantics to the planning layer: each phase/object has a class, inherited boundary, truth role and consumer constraint. It is not a mathematical CNNA generator.                                                      |
| Source materials / references        | <ul style="list-style-type: none"><li>User request: classify objects and phases by overarching context using object-oriented-programming principles.</li></ul>                                                                                                                                          |
| Extracted established theory objects | <ul style="list-style-type: none"><li>PhaseContextClass</li><li>ObjectContextClass</li><li>LeanSourceObject</li><li>LeanTargetObject</li><li>ToolInfrastructureObject</li><li>ReferenceGateObject</li><li>DatabaseMirrorObject</li><li>GeneratedArtifactObject</li><li>GovernancePolicyObject</li></ul> |
| Implemented modules                  | <ul style="list-style-type: none"><li>masters/cnna_roadmap_master.yaml</li><li>scripts/generate_tables.py</li><li>scripts/generate_full_document.py</li></ul>                                                                                                                                           |
| Modified modules                     | <ul style="list-style-type: none"><li>masters/cnna_roadmap_master.yaml</li><li>scripts/generate_tables.py</li><li>scripts/generate_full_document.py</li></ul>                                                                                                                                           |
| Forbidden patterns checked           | <ul style="list-style-type: none"><li>No Lean source change</li><li>No phase-release shortcut</li><li>No tool/reference/database object promoted to Lean object</li><li>No status/context conflation</li></ul>                                                                                          |
| No-go guardrails                     | <ul style="list-style-type: none"><li>GNG38: status/context orthogonality</li><li>GNG34: DB mirror is not a truth source</li><li>GNG37: external GA/quaternion material is comparison-only</li></ul>                                                                                                    |

20    Technical appendix: module-change manifest and semantic placement gate

**What this section shows.**    This appendix is not a second phase table. It is a narrow module-change ledger used only when a phase proposes new Lean modules, module moves, module merges, public API changes or import-surface changes. Ordinary planning, documentation and tool phases should be governed by the main phase table plus scratchpad instead. The appendix is kept for auditability of real code-surface changes, not for normal roadmap reading.

**Module-change manifest rule.**    This appendix is not a second phase table and does not release phases. It is a narrow technical ledger used only when a phase proposes new Lean modules, module moves, module merges, public API changes, import-surface changes, generator seams, or compatibility facades. Its rows are generated from implementation-scratchpad module/refactor fields; phases without module/API changes do not need meaningful rows here. The main phase table remains the primary status and release surface.

**Table legend.**    G = green/closed and derived-only;    W = white/non-direct-Lean tooling/refactor/infrastructure/documentation/comparison;    O = orange/implementable after stated predecessor closure;    Y = yellow/active and implementable now;    R = red/locked  
Lean-derivation phase; release requires derived-only object/proof evidence. Context class is orthogonal to status.

| Phase     | Status                                            | Phase type     | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness                                                                                                                                                                                                                                                                  | Reuse / generator / duplicate risks                                                                                                                                                            | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------|---------------------------------------------------|----------------|-------------------------------|---------------|------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1.1.1   | <span style="background-color: #d9ead3;">G</span> | documentation  | Documentation;<br>Phase       | —             | —                | —                         | Umbrella only. Child 1.1.1.1.1 defines the required manifest format.                                                                                                                                                                                                                                |                                                                                                                                                                                                | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 1.1.1.1.0 | <span style="background-color: #d9ead3;">G</span> | infrastructure | Lean;<br>Governance;<br>Phase | —             | —                | —                         | No code modules introduced by this gate.;<br>Governance belongs in YAML master and generated views unless a future code-level policy module is justified.;<br>Do not split code solely for this planning decision; use YAML/workflow unless a reusable code-level policy module is later justified. | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation. | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase     | Status | Phase type     | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                  | Reuse / generator / duplicate risks                                                                                                                                                            | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|--------|----------------|-------------------------------|---------------|------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —             | —                | —                         | No code modules introduced by this gate.;<br>Governance belongs in YAML master and generated views unless a future code-level policy module is justified.;<br>Do not split code solely for this planning decision; use YAML/workflow unless a reusable code-level policy module is later justified.                                 | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation. | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only gate: no operational Lean object introduced. The generated requirement is itself the traversal contract for future operational phases.;<br>Planning gate only. Records known risk class: mathlib algebraic interfaces may force noncomputable/classical dependencies; future phases must log concrete module/API names.;<br>No Lean code changed. Future operational phases must separate operational code from mirror/non-operational noncomputable uses. |
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —             | —                | —                         | No code modules introduced by this gate.;<br>Governance belongs in YAML master and future optional architecture policy modules only if code-level records become useful.;<br>Do not split code solely for this planning decision; encode as workflow/format decision unless a reusable code-level policy module is later justified. | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation. | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only gate: no operational Lean object introduced. Later operational phases must provide concrete finite instance traversal.;<br>No direct mathlib import in this planning gate. Future phases must log complications.;<br>No direct Lean boundary touched; future phases must audit noncomputable/classical pressure.                                                                                                                                           |

| Phase     | Status | Phase type     | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                  | Reuse / generator / duplicate risks                                                                                                                                                            | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------|--------|----------------|-------------------------------|---------------|------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —             | —                | —                         | No code modules introduced by this gate.;<br>Governance belongs in YAML master and future optional architecture policy modules only if code-level records become useful.;<br>Do not split code solely for this planning decision; encode as workflow/format decision unless a reusable code-level policy module is later justified. | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation. | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Closed: no refactor phase may close without declaration-count/import-delta/consumer-breakage audit and local full-build evidence.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only gate: no operational Lean object introduced. Later operational phases must provide concrete finite instance traversal.;<br>No direct mathlib import in this planning gate. Future phases must log complications.;<br>No direct Lean boundary touched; future phases must audit noncomputable/classical pressure.                                                                 |
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —             | —                | —                         | No code modules introduced by this gate.;<br>Governance belongs in YAML master and future optional architecture policy modules only if code-level records become useful.;<br>Do not split code solely for this planning decision; encode as workflow/format decision unless a reusable code-level policy module is later justified. | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation. | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Closed: baseline freeze by default; upgrades only at explicit phase-boundary gates with separate build/review evidence.;<br>Infrastructure-only gate: no operational Lean object introduced. Later operational phases must provide concrete finite instance traversal.;<br>No direct mathlib import in this planning gate. Future phases must log complications.;<br>No direct Lean boundary touched; future phases must audit noncomputable/classical pressure.                                                                                                       |
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —             | —                | —                         | No code modules introduced by this gate.;<br>Governance belongs in YAML master and future optional architecture policy modules only if code-level records become useful.;<br>Do not split code solely for this planning decision; encode as workflow/format decision unless a reusable code-level policy module is later justified. | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation. | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only gate: no operational Lean object introduced. Later operational phases must provide concrete finite instance traversal.;<br>No direct mathlib import in this planning gate. Future phases must log complications.;<br>No direct Lean boundary touched; future phases must audit noncomputable/classical pressure. |

| Phase     | Status | Phase type     | Context class                 | Added modules                                                                                                                   | Modified modules                                                                               | Reusable existing modules                                                                                                      | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                                | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                  | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|--------|----------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —                                                                                                                               | —                                                                                              | —                                                                                                                              | No code modules introduced by this gate.;<br>Governance belongs in YAML master and future optional architecture policy modules only if code-level records become useful.;<br>Do not split code solely for this planning decision; encode as workflow/format decision unless a reusable code-level policy module is later justified.                                                                               | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation.                                                                                                                                       | Closed: Phase 3.6 exports consumer-ready profile;<br>Phase 4.1 consumes;<br>no reverse import and no duplicate generator/Schur implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only gate: no operational Lean object introduced. Later operational phases must provide concrete finite instance traversal.;<br>No direct mathlib import in this planning gate. Future phases must log complications.;<br>No direct Lean boundary touched; future phases must audit noncomputable/classical pressure.                                        |
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —                                                                                                                               | —                                                                                              | —                                                                                                                              | No code modules introduced by this gate; future module candidates are S07/S08/BR06/BR07 phases.;<br>Place code-backed route facts in later neutral structural modules;<br>place string-like interpretation only in secondary/documentation side roads.;<br>Do not split code solely for this planning decision; encode as workflow/format decision unless a reusable code-level policy module is later justified. | Reuse existing ToC.Contract, CutSpec.full, ExistingIdealThreadNoGo, FullBoundaryAudit, RecursiveSchurSource and MobiusTransfer;<br>no duplicate route implementation authorized.;<br>Generator-like logic remains on the canonical smoke/theory generator path. IdealThread/string analogy may not instantiate a parallel generator. | Later phases consume route profiles one-way: S07/S08 feed BR06;<br>BR06 feeds Phase 3/4;<br>secondary analogy consumes dossiers only and never feeds A-generation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only gate;<br>it requires later S07/S08/BR06/BR07 phases to show concrete finite reference/variation/full-cut instances before operational use.;<br>No new mathlib import in this planning gate.;<br>No Lean boundary touched;<br>later route phases must audit noncomputable/classical pressure if they import additional mathlib APIs. |
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | <ul style="list-style-type: none"> <li>none at decision gate; future target likely existing O07 test/profile surface</li> </ul> | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_masters/generated views</li> </ul> | <ul style="list-style-type: none"> <li>canonical smoke-test regime split modules after Phase 1.3.2; O07 test matrix</li> </ul> | pending Phase 1.1.1.1.1;<br>test/schema layer, not generator layer;<br>do not create a new module unless existing test/profile module cannot own the semantics                                                                                                                                                                                                                                                    | must reuse O07/canonical smoke-test generator;<br>no parallel Heegner test generator;<br>consume the one canonical generator only; Heegner row schema; CM sentinel profile; closed-system sentinel tags                                                                                                                              | test matrix consumes regime/generator outputs; regime/generator definitions do not consume tests;<br>no code movement in this gate; future test implementation must preserve full local build green status;<br>4.28.0 baseline unless version gate explicitly opens upgrade;<br>required later: every row must contain actual finite generator traversal evidence or remain unclosed;<br>record any class number/CM/j mathlib dependency or noncomputable boundary explicitly;<br>Heegner/class-number and analytic j data are comparison/mirror until operational invariant is derived                                                                                                                                                                                                                                                                                                                                                                                |

| Phase     | Status | Phase type     | Context class                 | Added modules                                                                                                                                            | Modified modules                                                                                                       | Reusable existing modules                                                                                                                                                                                                      | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                           | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | <ul style="list-style-type: none"> <li>none at decision gate; possible CNNA/PillarA/Foundation/ExactQuadraticComplex.lean only after manifest</li> </ul> | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.generated views</li> </ul>                          | <ul style="list-style-type: none"> <li>Foundation/ExactComplex.lean; Foundation/ExecComplexBridge.lean; Arithmetic/Schur/CharpolyC.lean; Arithmetic/CayleyDickson/ArithmeticCDBridge.lean; mathlib QuadraticAlgebra</li> </ul> | <p>pending Phase 1.1.1.1.1;</p> <p>scalar-foundation / bridge governance, not generator semantics;</p> <p>do not create ExactQuadraticComplex unless existing foundation/bridge modules cannot carry the semantics cleanly</p>                                                                                                                                                                                                                                                                            | <p>must check existing ExecComplex, ExecComplexBridge, Schur.CharpolyC, ArithmeticCDBridge, and mathlib QuadraticAlgebra before proposing new module;</p> <p>scalar layer must consume canonical generator outputs only;</p> <p>cannot define generator semantics;</p> <p>alternate complex scalar, algebraic complex, CD complex, mirror Complex, exact quadratic norm/trace/conjugation</p> | <p>scalar layer consumes generator/arithmetic source data;</p> <p>generator definitions do not consume scalar comparison/mirror data;</p> <p>no code movement in this gate;</p> <p>4.28.0 baseline unless version gate explicitly opens upgrade;</p> <p>required later: QxQ baseline row, Q(i) row, Q(sqrt(-3)) row, and at least one generic D row if O09 is implemented;</p> <p>Record Complex/Real noncomputable boundary, QuadraticAlgebra instance diamond warning, NumberField formal/non-executable dependency, and any required version-gate.;</p> <p>General C/R and algebraic closure remain mirror/formal. Exact quadratic candidate must avoid noncomputable definitions on operational path.</p>                                |
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —                                                                                                                                                        | —                                                                                                                      | —                                                                                                                                                                                                                              | <p>No Lean implementation manifest was required when this already-closed/planned infrastructure row was created;</p> <p>under closed LEG10, any future code change touching this phase must fill exact added/modified modules before edits.;</p> <p>Existing placement remains historical/planning-only. Future module placement must be revalidated by LEG10 before code changes.;</p> <p>No one-phase-one-file inference;</p> <p>future modules require semantic/API/import-boundary justification.</p> | <p>Not a generator implementation. If later code resembles an existing generator/API, consume LEG11 before edits.;</p> <p>No generator-like code in this record. Future generator-like extensions must route through LEG11.</p>                                                                                                                                                               | <p>Phase 10 consumes Phase 4.1 directly;</p> <p>Phase 8/9 do not gate Phase 10 unless a specific object bridge says so.;</p> <p>incremental wrapper mode;</p> <p>full local build green after each migration step;</p> <p>4.28.0 baseline unless version gate explicitly opens upgrade;</p> <p>Future finite test rows must run separately for arithmetic/modular and spectral/physical consumers.;</p> <p>Record if modular-form/mathlib APIs are attempted in spectral/physical path;</p> <p>that is a placement smell unless explicitly justified.;</p> <p>Analytic modular/j/L-function objects remain mirror/comparison and cannot block finite spectral/operator algebra profiles.</p>                                                 |
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —                                                                                                                                                        | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>masters/cnna_roadmap_master.yaml</li> </ul> | <ul style="list-style-type: none"> <li>phase_object_map</li> <li>phase hierarchy validator</li> <li>object register Phase field</li> </ul>                                                                                     | <p>No Lean module manifest;</p> <p>generator/documentation-only phase.;</p> <p>Belongs to global planning infrastructure, not to mathematical object phases.;</p> <p>Keep policy in the generator and YAML fields;</p> <p>avoid a separate manual table source.</p>                                                                                                                                                                                                                                       | <p>Reuses existing phase hierarchy and object register. No duplicate object registry created.;</p> <p>The table generator is the canonical renderer of phase-object link semantics.;</p> <p>Linked objects display aggregation;</p> <p>manual child-object lists drifting from object register</p>                                                                                            | <p>Planning views consume object register and phase hierarchy;</p> <p>object register does not consume table-rendered child scope.;</p> <p>View regression: no unqualified blank Linked objects cell may remain after table generation.;</p> <p>not applicable;</p> <p>Generator run over all phases/objects is the concrete planning instance.;</p> <p>none;</p> <p>none;</p> <p>no Lean code</p>                                                                                                                                                                                                                                                                                                                                           |
| 1.1.1.1.0 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —                                                                                                                                                        | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>masters/cnna_roadmap_master.yaml</li> </ul> | <ul style="list-style-type: none"> <li>global_no_go_guardrails renderer</li> <li>workflow_policy renderer</li> <li>global_proof_dossiers</li> <li>phase table</li> </ul>                                                       | <p>No Lean module manifest;</p> <p>planning/postulate infrastructure only.;</p> <p>Belongs to global postulates and no-go guardrails before active implementation.;</p> <p>Keep the postulate in one top-level section plus LEG35/GNG22 references;</p> <p>avoid scattering prose copies.</p>                                                                                                                                                                                                             | <p>Reuses global no-go, workflow, proof-dossier and generated-table infrastructure;</p> <p>no duplicate roadmap source.;</p> <p>The table generator is the canonical renderer of POST1 and its supporting enforcement objects.;</p> <p>ontic primitive;</p> <p>subsystem-being;</p> <p>emergence trigger</p>                                                                                  | <p>Consumer phases consume derived operator/spectral/algebraic/arithmetic data only after subsystem/cut provenance exists.;</p> <p>Generated views must keep POST1 visible and GNG22 inherited;</p> <p>no code refactor may remove the global postulate from generated documentation.;</p> <p>not applicable;</p> <p>Planning-level concrete instance: trace a finite subsystem/cut -&gt; boundary -&gt; complement/regime -&gt; Schur/DtN -&gt; derived consumer chain for later active phases.;</p> <p>none for the postulate. Later mathlib complications must not be solved by importing non-derived primitives.;</p> <p>No Lean code. Later noncomputable/mirror boundaries must remain downstream of derived subsystem provenance.</p> |



| Phase     | Status | Phase type     | Context class                 | Added modules | Modified modules                                                                                                                          | Reusable existing modules                                                                                                                                                                                                                                                                                                                                              | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| 1.1.1.1.1 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —             | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> <li>README.md</li> </ul> | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/Notation.lean as source to split, not as target monolith</li> <li>existing Smoke/Test generator modules as candidates for generator-like work</li> </ul>                                                                                                                                                | <p>Closed contract: every scratchpad record must expose proposed_added_modules, proposed_modified_modules, reusable_existing_modules, semantic placement, rejection/fit decision, public/legacy/API role, import direction, expected downstream consumers, phase_type obligations, refactor regression protocol, version impact, concrete-instance traversal, generic test placement, mathlib complication log, and noncomputable boundary audit before Lean edits.; The manifest is planning infrastructure and belongs to the single YAML master plus generated views. It is not a Lean module and must not be implemented as phase-local code. Future code modules are created only when an existing semantic module cannot be safely extended or wrapped.; Closed rule: phase granularity is not file granularity. A later phase must justify every new module by semantic ownership, API/firewall direction, or unavoidable import split.</p> | <p>Phase 1.1.1.1.1 names reusable candidates but defers semantic generator classification to 1.1.1.1.2. Duplicate implementation remains blocked already at manifest level.; Any generator-like phase must consume this manifest, then pass through 1.1.1.1.2 before Lean code. The manifest table must state whether O07/canonical smoke-test generator is reused, extended, or rejected.; new generator/request/output helper names without canonical generator route; IdealThread.carrier/family singleton facts; CutSpec.full carrier facts; RecursiveSchurSource/MobiusTransfer recurrence logic; string-like analogy used as generator</p>                                                    | <p>Every proposed public wrapper must state its expected consumers and import direction. BoundaryOperatorProfile must prepare for Phase 4.1 without importing Phase 4. For BR08/CD8 specifically, earlier route modules export data downstream and must not import the bridge.; This gate does not refactor code, but it must require every later refactor manifest to declare wrapper-first/incremental vs atomic mode. Default for 1.1.2 is wrapper-first incremental with local full-build evidence per package.; Every module manifest must state whether it stays on Lean 4.28.0/mathlib baseline, uses no mathlib change, or requires a separate upgrade gate.; This active gate is infrastructure-only: it will not introduce operational Lean code. For every later operational phase, the manifest must include a concrete finite instance path from generated input to intended endpoint. This phase must plan generic finite test traversal, not only special sentinels, for every upcoming computable path.; No new mathlib import in this gate. Later manifests must list expected mathlib imports and whether each is executable, noncomputable-only, classical-only, or mirror-layer-only. Scalar architecture check added: any proposed new module must classify Complex/Real/QuadraticAlgebra dependencies under LEG31. Derived-quadratic update: any proposed QuadraticAlgebra/AdjoinRoot/ExactQuadratic module must state whether it is a schema, D-instantiation, proof/mirror reference, or executable coordinate layer.; No Lean operational code changed. Later phases must document noncomputable leakage risk before implementation.</p> |
| 1.1.1.1.2 | G      | infrastructure | Lean;<br>Governance;<br>Phase | —             | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> <li>README.md</li> </ul> | <ul style="list-style-type: none"> <li>CNNA.PillarA.Arithmetic.Smoke.GeneratedSource</li> <li>CNNA.PillarA.Arithmetic.Smoke.GeneratedMainPath</li> <li>CNNA.PillarA.Arithmetic.ActionsGeneratorAsAPhaseLocal</li> <li>CNNA.PillarA.Arithmetic.Actions.ActionFormatDecision</li> <li>scripts/generate_tables.py</li> <li>scripts/build_export_script_v1.8.py</li> </ul> | <p>Closed contract consumed from LEG10. v41 adds no Lean module; It extends the planning generator and YAML master to render the reuse/duplication ledger.; The policy belongs in the single YAML master and generated planning views. It must not be implemented as a phase-local Lean status module; later neutral generator code is allowed only after a concrete module-manifest proof.; Closed rule: a generator policy row is not a new Lean file. Later code should prefer one coherent reusable generator/API seam plus wrappers over phase-local generator files.</p>                                                                                                                                                                                                                                                                                                                                                                     | <p>Closed: generator-like operations are classified by ledger rows as consume existing, extend existing, wrap existing, artifact-only split, or genuine semantic split. Duplicate generator implementations are blocked by default.; Closed rule: the Smoke/GeneratedSource-MainPath seam is canonical for generated ToC/smoke semantics; action-like semantics reuse GeneratorActionInventory/ActionFormatDecision; documentation/inventory scripts are artifact generators and comparison/audit only.; experiment request/output generator; phase-local extraction generator; parallel ledger generator; second smoke-test generator; independent documentation/proofbook source outside YAML</p> | <p>Exports a planning/API policy to 1.1.1.1.3, 1.1.1.3, Phase 5, O07, heuristic lattice tests, and all generator-like phases. It does not export mathematical data to downstream pillars.; Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.; Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.; Infrastructure-only; no operational finite instance is introduced. Future operational generator phases must provide concrete finite traversal through the canonical seam.; No Lean/mathlib import is introduced by this phase.; No noncomputable boundary introduced; no Lean operational path changed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



| Phase     | Status | Phase type    | Context class                 | Added modules | Modified modules | Reusable existing modules                                                                                                                                                                                                                                                                                                                   | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                           | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 1.1.1.1.3 | G      | documentation | Documentation;<br>Phase       | —             |                  | <ul style="list-style-type: none"><li>masters/cnna_roadmap_master.yaml</li><li>scripts/generate_tables.py</li></ul> <ul style="list-style-type: none"><li>tables/object_register.tex</li><li>generated_exports/object_register.generated.tsv</li><li>scripts/generate_tables.py</li></ul>                                                   | No Lean modules added. Planning/generator artifacts only: YAML master, table generator, generated object proof documentation table/export, PDF view.; Documentation infrastructure belongs in the YAML/generator layer, not scattered into phase prose or Lean modules.;<br>A new documentation section does not imply new Lean modules. It is a generated view over object register data.                   | Consumes closed LEG11. The proofbook is generated from the single YAML master and existing object register;<br>no independent proofbook file or duplicate object list is allowed.;<br>No CNNA mathematical generator introduced. The documentation generator is an artifact generator only and cannot provide mathematical input to Pillar A.; independent proofbook file not generated from YAML;<br>phase-local object proof prose duplicated in scratchpad;<br>object list copied outside object register | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 1.1.1.1.4 | G      | refactor      | Lean;<br>Governance;<br>Phase | —             | —                | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>repo_inventory/raw_export/modules.json</li><li>repo_inventory/raw_export/symbol_references/declarations.json</li><li>repo_inventory/raw_export/symbol_references/references.json</li><li>repo_inventory/raw_export/architectural_core/sink_taxonomy.json</li></ul> | Consumes LEG10: the classification artifact is generated in planning tables/exports only; no new Lean module and no declaration movement.; LEG5 remains a planning/API-hygiene artifact. Full records are generated_exports TSVs; PDF receives only the compact policy/count view.; Classification operates at module/declaration/object record granularity but stores no huge declaration table in the PDF. | Consumes LEG11: generator-like classification is implemented as an extension of scripts/generate_tables.py, not a second inventory/export generator.; Use existing single-master table generator as the canonical seam for planning ledgers; future Lean audit mirror is optional and must not duplicate semantics.; BuildAll reachability as false semantic evidence;<br>SM phase aliases promoted as public API; blue object treated as final-output by documentation alone                                | Exports generated TSVs for 1.1.1.2 blue-object correction and 1.1.1.3 public API/consumer whitelist;<br>does not grant handoff rights.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet.                        |

| Phase     | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules                                                                                                                                                                                                                                                                                                                    | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| 1.1.1.1.5 | W      | refactor   | Lean;<br>Governance;<br>Phase | —             | —                | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li> <li>CNNA/PillarA/Handoff/Core/SectorExport.lean</li> <li>CNNA/PillarA/Handoff/Outputs/Closed.lean</li> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/consumption_classification_maintainability</li> </ul> | <p>No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean; CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean; CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.;</p> <p>The audit belongs after declaration consumption classification and before blue-object content correction. This is the natural layer because it decides public/import surfaces before new semantic content is added.;</p> <p>Slim modules only when statement/provenance maintainability is preserved or improved. Small clone wrappers are candidates;</p> <p>long proof chains are not automatically candidates; a monolithic notation facade may require splitting rather than merging.</p> | <p>RD08 records the consolidation decision rule. Same-semantics wrappers may be unified;</p> <p>proof-stage modules must remain separated until a stronger canonical proof module preserves their role.;</p> <p>No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.;</p> <p>BoundarySource.*Substrate.ledger and ledger_* theorems;</p> <p>AToCHandoffStrong/AToDHandoffStrong/AToEHandoffStrong aliases;</p> <p>BToAInput/CToAInput/DToAInput/EToAInput feedback structures and ext/handoff projections</p> | <p>Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.;</p> <p>Mandatory before any Lean merge: retain old modules as facades in pass 1;</p> <p>record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.;</p> <p>Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed;</p> <p>refactor-only changes must compile locally before closeout.;</p> <p>Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.;</p> <p>No mathlib complication in audit. Later code movement may expose import-minimization issues;</p> <p>resolve by preserving existing import surfaces first.;</p> <p>No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths.</p>                                                                                    |
| 1.1.1.2   | O      | refactor   | Lean;<br>Governance;<br>Phase | —             | —                | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> <li>generated_exports/object_consumption_classification_generated.tsv</li> <li>tables/object_proof_documentation.tex</li> </ul>                                                                                             | <p>Infrastructure/planning gate expected. Use existing YAML/generator tables unless a code phase proves a Lean audit mirror is necessary.;</p> <p>The ledger belongs in the planning master/generated tables first. It must not patch consumers or move Lean declarations before the correction semantics are recorded.;</p> <p>Classify by object family and correction phase. Do not generate one prose subsection per Lean declaration unless a correction really depends on it.</p>                                                                                                                                                                                                                                                                                                   | <p>Consume LEG11 and LEG5;</p> <p>do not create a parallel object-status or classifier generator.;</p> <p>Extend scripts/generate_tables.py if generated views are needed;</p> <p>otherwise keep the blue correction ledger in YAML as a governance source.;</p> <p>parallel blue-status table disconnected from object register;</p> <p>white status without correction proof;</p> <p>public export of B object through wrapper reachability</p>                                                                                                                                                                                                    | <p>No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;</p> <p>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;</p> <p>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;</p> <p>noncomputable/classical pressure must be logged and isolated.;</p> <p>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;</p> <p>No mathlib complication expected for ledger-only implementation;</p> <p>no new Lean/mathlib import should be introduced. Any proposed Lean audit file must verify imports against Lean 4.28.0 before code starts.;</p> <p>No operational or mirror mathematical code is introduced in the planned ledger-only route;</p> <p>noncomputable/classical pressure would indicate that the phase has drifted into a later mathematical correction phase.</p> |

| Phase     | Status | Phase type     | Context class                 | Added modules | Modified modules                                                                                                       | Reusable existing modules                                                                                                                                                                | Manifest / placement / compactness                                                                                                                                                                                                                                                                        | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                                   | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 1.1.1.5.1 | W      | infrastructure | Lean;<br>Governance;<br>Phase | —             | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> </ul> | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul> | Tooling subphase. Proposed generated artifacts: workflow-policy row; scaling-plan TSV/TeX; no Lean code; Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.; Keep PDF slices compact; full data remains in generated TSV/JSON exports. | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.; Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.; parallel planning master; manual duplicate theory ledger; full graph dumped into PDF; end theory used as CNNA producer | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.; Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.; Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.; Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.; No mathlib complication expected because this is a Python/YAML/PDF tooling phase.; No Lean operational path is modified; any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase. |
| 1.1.1.5.2 | W      | infrastructure | Lean;<br>Governance;<br>Phase | —             | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> </ul> | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul> | Tooling subphase. Proposed generated artifacts: active-phase implementation packet TSV/JSON/TeX; Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.; Keep PDF slices compact; full data remains in generated TSV/JSON exports.         | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.; Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.; parallel planning master; manual duplicate theory ledger; full graph dumped into PDF; end theory used as CNNA producer | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.; Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.; Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.; Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.; No mathlib complication expected because this is a Python/YAML/PDF tooling phase.; No Lean operational path is modified; any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase. |
| 1.1.1.5.3 | W      | infrastructure | Lean;<br>Governance;<br>Phase | —             | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> </ul> | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul> | Tooling subphase. Proposed generated artifacts: file-hash cache; declaration cache; inventory-delta TSV; Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.; Keep PDF slices compact; full data remains in generated TSV/JSON exports. | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.; Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.; parallel planning master; manual duplicate theory ledger; full graph dumped into PDF; end theory used as CNNA producer | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.; Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.; Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.; Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.; No mathlib complication expected because this is a Python/YAML/PDF tooling phase.; No Lean operational path is modified; any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase. |

| Phase     | Status | Phase type     | Context class                 | Added modules | Modified modules                                                                                                       | Reusable existing modules                                                                                                                                                                | Manifest / placement / compactness                                                                                                                                                                                                                                                                             | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                                   | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 1.1.1.5.4 | W      | infrastructure | Lean;<br>Governance;<br>Phase | —             | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> </ul> | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul> | Tooling subphase. Proposed generated artifacts: module-theory-context ledger TSV; active-module slice; Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.; Keep PDF slices compact; full data remains in generated TSV/JSON exports.        | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.; Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.; parallel planning master; manual duplicate theory ledger; full graph dumped into PDF; end theory used as CNNA producer | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.; Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.; Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.; Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.; No mathlib complication expected because this is a Python/YAML/PDF tooling phase.; No Lean operational path is modified; any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase. |
| 1.1.1.5.5 | W      | infrastructure | Lean;<br>Governance;<br>Phase | —             | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> </ul> | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul> | Tooling subphase. Proposed generated artifacts: end-theory-comparison ledger TSV; comparison-boundary table; Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.; Keep PDF slices compact; full data remains in generated TSV/JSON exports.  | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.; Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.; parallel planning master; manual duplicate theory ledger; full graph dumped into PDF; end theory used as CNNA producer | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.; Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.; Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.; Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.; No mathlib complication expected because this is a Python/YAML/PDF tooling phase.; No Lean operational path is modified; any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase. |
| 1.1.1.5.6 | W      | infrastructure | Lean;<br>Governance;<br>Phase | —             | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_master.yaml</li> <li>scripts/generate_tables.py</li> </ul> | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>repo_inventory/raw_export/modules.json</li> <li>repo_inventory/raw_export/symbol_references/*.json</li> </ul> | Tooling subphase. Proposed generated artifacts: active-object-neighborhood DOT/SVG/PDF; graph-slice TSV/JSON; Belongs under 1.1.1.5 generated-view authority/implementation-lock tooling. It must not patch mathematical consumers.; Keep PDF slices compact; full data remains in generated TSV/JSON exports. | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.; Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.; parallel planning master; manual duplicate theory ledger; full graph dumped into PDF; end theory used as CNNA producer | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.; Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.; Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.; Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.; No mathlib complication expected because this is a Python/YAML/PDF tooling phase.; No Lean operational path is modified; any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase. |



| Phase     | Status | Phase type     | Context class                 | Added modules                                                                                                                                             | Modified modules                                                 | Reusable existing modules                                                                                                                                                                                                                                       | Manifest / placement / compactness                                                                                                                                                                        | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                               | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 1.1.1.5.7 | W      | infrastructure | Lean;<br>Governance;<br>Phase | —                                                                                                                                                         |                                                                  | <ul style="list-style-type: none"><li>masters/<br/>cnna_roadmap_master.yaml</li><li>scripts/<br/>generate_tables.py</li></ul>                                                                                                                                   | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>repo_inventory/<br/>raw_export/modules.json</li><li>repo_inventory/<br/>raw_export/<br/>symbol_references/*.json</li></ul>       | Tooling subphase. Proposed generated artifacts: red-blocker phase-proposal TSV; accepted phase rows only after approval; Belongs under 1.1.1.5 generated-view authority/ implementation-lock tooling. It must not patch mathematical consumers.; Keep PDF slices compact; full data remains in generated TSV/JSON exports.                                        | Reuse existing single YAML master and scripts/generate_tables.py. Avoid parallel masters or a second manually maintained planning source.; Extend the existing Python generator in the smallest possible way; exports are generated TSV/JSON/TeX, not manually edited tables.; parallel planning master; manual duplicate theory ledger; full graph dumped into PDF; end theory used as CNNA producer                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Future Lean phases may consume only compact implementation packets and the relevant ledger slices, not broad manual PDF scans.; Not a Lean refactor. If Python generator is changed, run generator validation and render-check the affected PDF sections.; Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.; Not an operational Lean phase. The concrete tooling witness is generated exports and PDF rendering over the current single YAML master.; No mathlib complication expected because this is a Python/YAML/PDF tooling phase.; No Lean operational path is modified; any noncomputable/classical concern indicates accidental drift into Lean code and requires a separate phase. |
| 1.2.1.1   | R      | derivation     | Lean;<br>Derivation;<br>Phase | <ul style="list-style-type: none"><li>none at decision gate; possible CNNA/ PillarA/ Foundation/ ExactQuadraticComplex.lean only after manifest</li></ul> | <ul style="list-style-type: none"><li>pending manifest</li></ul> | <ul style="list-style-type: none"><li>Foundation/<br/>ExecComplex.lean;<br/>Foundation/<br/>ExecComplexBridge.lean;<br/>Arithmetic/Schur/<br/>CharpolyC.lean;<br/>Arithmetic/CayleyDickson/<br/>ArithmeticCDBridge.lean;<br/>mathlib QuadraticAlgebra</li></ul> | pending Phase 1.1.1.1.1; scalar-foundation / bridge governance, not generator semantics; do not create ExactQuadraticComplex unless existing foundation/bridge modules cannot carry the semantics cleanly | must check existing ExecComplex, ExecComplexBridge, Schur.CharpolyC, ArithmeticCDBridge, and mathlib QuadraticAlgebra before proposing new module; scalar layer must consume canonical generator outputs only; cannot define generator semantics; alternate complex scalar, algebraic complex, CD complex, mirror Complex, exact quadratic norm/trace/conjugation | scalar layer consumes generator/arithmetic source data; generator definitions do not consume scalar comparison/mirror data; incremental wrapper mode; full local build green after each migration step; 4.28.0 baseline unless version gate explicitly opens upgrade; required later: QxQ baseline row, Q(i) row, Q(sqrt(-3)) row, and at least one generic D row if O09 is implemented; Record Complex/Real noncomputable boundary, QuadraticAlgebra instance diamond warning, NumberField formal/non-executable dependency, and any required version-gate.; General C/R and algebraic closure remain mirror/formal. Exact quadratic candidate must avoid noncomputable definitions on operational path.                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1.2.1.2   | R      | derivation     | Lean;<br>Derivation;<br>Phase | <ul style="list-style-type: none"><li>none at decision gate; possible CNNA/ PillarA/ Foundation/ ExactQuadraticComplex.lean only after manifest</li></ul> | <ul style="list-style-type: none"><li>pending manifest</li></ul> | <ul style="list-style-type: none"><li>Foundation/<br/>ExecComplex.lean;<br/>Foundation/<br/>ExecComplexBridge.lean;<br/>Arithmetic/Schur/<br/>CharpolyC.lean;<br/>Arithmetic/CayleyDickson/<br/>ArithmeticCDBridge.lean;<br/>mathlib QuadraticAlgebra</li></ul> | pending Phase 1.1.1.1.1; scalar-foundation / bridge governance, not generator semantics; do not create ExactQuadraticComplex unless existing foundation/bridge modules cannot carry the semantics cleanly | must check existing ExecComplex, ExecComplexBridge, Schur.CharpolyC, ArithmeticCDBridge, and mathlib QuadraticAlgebra before proposing new module; scalar layer must consume canonical generator outputs only; cannot define generator semantics; alternate complex scalar, algebraic complex, CD complex, mirror Complex, exact quadratic norm/trace/conjugation | scalar layer consumes generator/arithmetic source data; generator definitions do not consume scalar comparison/mirror data; incremental wrapper mode; full local build green after each migration step; 4.28.0 baseline unless version gate explicitly opens upgrade; Schema-level witnesses: symbolic arbitrary D, plus later planned special rows D=-1/-3/-4 only after D-ledger. This phase may include hand-written closed coordinate examples but cannot claim CNNA provenance for D.; Audit QuadraticAlgebra coordinate operations, basis/noncomputable declarations, and version stability. AdjoinRoot quotient construction includes noncomputable instances in docs and must not silently enter operational path.; No analytic sqrt, Real, Complex exponential, elliptic functions, or uninspected quotient field structure in operational code. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Phase | Status | Phase type | Context class               | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness                                                                                                     | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 1.2.7 | W      | refactor   | Lean;<br>Refactor;<br>Phase | —             | —                | —                         | Candidate: CNNA/PillarA/ToC/<br>IdealThreadProfile.lean only if not better placed as a small section in existing ToC/Contract wrapper. |                                     | Route direction must be explicit before implementation: thread/full profiles feed Schur/ arithmetic audits;<br>analogy/diagnostics never feed generator code.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 1.2.8 | W      | refactor   | Lean;<br>Refactor;<br>Phase | —             | —                | —                         | Candidate: CNNA/PillarA/Finite/<br>FullCutProfile.lean or a compact extension near CutSpec wrapper after module-manifest gate.         |                                     | Route direction must be explicit before implementation: thread/full profiles feed Schur/ arithmetic audits;<br>analogy/diagnostics never feed generator code.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 3.0   | W      | refactor   | Lean;<br>Refactor;<br>Phase | —             | —                | —                         | Candidate: CNNA/PillarA/Schur/<br>BoundaryRouteContract.lean if module-manifest gate justifies a new module.                           |                                     | Route direction must be explicit before implementation: thread/full profiles feed Schur/ arithmetic audits;<br>analogy/diagnostics never feed generator code.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules                                                                                                                        | Modified modules                                                                                 | Reusable existing modules                                                                                                                         | Manifest / placement / compactness                                                                                                                                 | Reuse / generator / duplicate risks                                                                                                                                                                              | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 4.0   | R      | derivation | Lean;<br>Derivation;<br>Phase | —                                                                                                                                    | —                                                                                                | —                                                                                                                                                 | Candidate: CNNA/PillarA/Operational/<br>ThreadRecursionAudit.lean after checking if existing<br>Schur modules can host the audit compactly.                        |                                                                                                                                                                                                                  | Route direction must be explicit before implementation: thread/full profiles feed Schur/ arithmetic audits;<br>analogy/diagnostics never feed generator code.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 4.0.1 | R      | derivation | Lean;<br>Derivation;<br>Phase | —                                                                                                                                    | —                                                                                                | —                                                                                                                                                 | Candidate: CNNA/PillarA/Operational/<br>ThreadRecursionAudit.lean after checking if existing<br>Schur modules can host the audit compactly.                        |                                                                                                                                                                                                                  | Consumes generator/Phase-3 profiles;<br>does not define them.;<br>The test harness later serves as regression sentinel for wrapper migrations.;<br>Use frozen baseline unless a phase-boundary upgrade gate is opened.;<br>Required: at least one generic finite row family traverses the actual computation path;<br>special sentinels cannot be the only traversal.;<br>to be filled before implementation;<br>include noncomputable risks for matrix/<br>polynomial APIs.;<br>No operational row may require noncomputable data. Mirror comparison columns are isolated.                                                                                                                                                                                                                                                                                                             |
| 4.0.2 | R      | derivation | Lean;<br>Derivation;<br>Phase | <ul style="list-style-type: none"><li>• none at decision gate; future target likely existing O07 test/<br/>profile surface</li></ul> | <ul style="list-style-type: none"><li>• masters/<br/>cnna_roadmap_mastegenerated views</li></ul> | <ul style="list-style-type: none"><li>• canonical smoke-test generator;<br/>regime split modules after Phase 1.3.2;<br/>O07 test matrix</li></ul> | pending Phase 1.1.1.1.1;<br>test/schema layer, not generator layer;<br>do not create a new module unless existing test/<br>profile module cannot own the semantics | must reuse O07/canonical smoke-test generator;<br>no parallel Heegner test generator;<br>consume the one canonical generator only;<br>Heegner row schema;<br>CM sentinel profile;<br>closed-system sentinel tags | test matrix consumes regime/generator outputs;<br>regime/generator definitions do not consume tests;<br>no code movement in this gate;<br>future test implementation must preserve full local build green status;<br>4.28.0 baseline unless version gate explicitly opens upgrade;<br>required later: every row must contain actual finite generator traversal evidence or remain unclosed;<br>record any class number/CM/j mathlib dependency or noncomputable boundary explicitly;<br>Heegner/class-number and analytic j data are comparison/mirror until operational invariant is derived                                                                                                                                                                                                                                                                                           |



| Phase | Status | Phase type | Context class                 | Added modules                                                                                                                                            | Modified modules                                                   | Reusable existing modules                                                                                                                                                                                                     | Manifest / placement / compactness                                                                                                                                                                              | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                        | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 4.0.3 | R      | derivation | Lean;<br>Derivation;<br>Phase | <ul style="list-style-type: none"> <li>none at decision gate; possible CNNA/PillarA/Foundation/ExactQuadraticComplex.lean only after manifest</li> </ul> | <ul style="list-style-type: none"> <li>pending manifest</li> </ul> | <ul style="list-style-type: none"> <li>Foundation/ExecComplex.lean; Foundation/ExecComplexBridge.lean; Arithmetic/Schur/CharpolyC.lean; Arithmetic/CayleyDickson/ArithmeticCDBridge.lean; mathlib QuadraticAlgebra</li> </ul> | pending Phase 1.1.1.1.1;<br>scalar-foundation / bridge governance, not generator semantics;<br>do not create ExactQuadraticComplex unless existing foundation/bridge modules cannot carry the semantics cleanly | must check existing ExecComplex, ExecComplexBridge, Schur.CharpolyC, ArithmeticCDBridge, and mathlib QuadraticAlgebra before proposing new module;<br>scalar layer must consume canonical generator outputs only;<br>cannot define generator semantics;<br>alternate complex scalar, algebraic complex, CD complex, mirror Complex, exact quadratic norm/trace/conjugation | scalar layer consumes generator/arithmetic source data;<br>generator definitions do not consume scalar comparison/mirror data;<br>incremental wrapper mode;<br>full local build green after each migration step;<br>4.28.0 baseline unless version gate explicitly opens upgrade;<br>required later: QxQ baseline row, Q(i) row, Q(sqrt(-3)) row, and at least one generic D row if O09 is implemented;<br>Record Complex/Real noncomputable boundary, QuadraticAlgebra instance diamond warning, NumberField formal/non-executable dependency, and any required version-gate.;<br>General C/R and algebraic closure remain mirror/formal. Exact quadratic candidate must avoid noncomputable definitions on operational path. Exact quadratic use must avoid analytic sqrt/Complex and cite D-source ledger.                                                                         |
| 4.5.1 | R      | derivation | Lean;<br>Derivation;<br>Phase | —                                                                                                                                                        | —                                                                  | —                                                                                                                                                                                                                             |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                            | No consumer-side patch;<br>missing D-source belongs earlier.;<br>Apply global refactor regression protocol if code changes.;<br>Use frozen Lean/mathlib baseline unless a phase-boundary upgrade gate is opened.;<br>At minimum: one generic D-ledger row and the Eisenstein/Gauss sentinel D rows when/if the operational path derives their D. Every row must cite its ledger source.;<br>Record any need for QuadraticAlgebra lemmas versus project-owned coordinate lemmas;<br>no noncomputable quotient/field instances on operational path.;<br>If instantiation requires noncomputable mathlib structure, split mirror proof from executable coordinate witness.                                                                                                                                                                                                               |
| 5     | R      | derivation | Lean;<br>Derivation;<br>Phase | —                                                                                                                                                        | —                                                                  | —                                                                                                                                                                                                                             |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| 5.1   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet.                                                                                |
| 5.2   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. Exact quadratic use must avoid analytic sqrt/Complex and cite D-source ledger. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 5.3   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 5.4   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 5.5   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 6     | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 6.1   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 6.2   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 6.2.1 | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 6.2.2 | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |



| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 6.3   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 6.4   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |



| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 6.5   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 7     | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules                                                                                                                   | Modified modules                                                                            | Reusable existing modules                                                                                                                 | Manifest / placement / compactness                                                                                                                             | Reuse / generator / duplicate risks                                                                                                                                                                              | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 7.1   | R      | derivation | Lean;<br>Derivation;<br>Phase | —                                                                                                                               | —                                                                                           | —                                                                                                                                         |                                                                                                                                                                |                                                                                                                                                                                                                  | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 7.2   | R      | derivation | Lean;<br>Derivation;<br>Phase | —                                                                                                                               | —                                                                                           | —                                                                                                                                         |                                                                                                                                                                |                                                                                                                                                                                                                  | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 7.2.1 | R      | derivation | Lean;<br>Derivation;<br>Phase | <ul style="list-style-type: none"> <li>none at decision gate; future target likely existing O07 test/profile surface</li> </ul> | <ul style="list-style-type: none"> <li>masters/cnna_roadmap_mastegenerated views</li> </ul> | <ul style="list-style-type: none"> <li>canonical smoke-test generator; regime split modules after Phase 1.3.2; O07 test matrix</li> </ul> | pending Phase 1.1.1.1.1;<br>test/schema layer, not generator layer;<br>do not create a new module unless existing test/profile module cannot own the semantics | must reuse O07/canonical smoke-test generator;<br>no parallel Heegner test generator;<br>consume the one canonical generator only;<br>Heegner row schema;<br>CM sentinel profile;<br>closed-system sentinel tags | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>no code movement in this gate;<br>future test implementation must preserve full local build green status;<br>4.28.0 baseline unless version gate explicitly opens upgrade;<br>required later: every row must contain actual finite generator traversal evidence or remain unclosed;<br>record any class number/CM/j mathlib dependency or noncomputable boundary explicitly;<br>Heegner/class-number and analytic j data are comparison/mirror until operational invariant is derived                                                                                                                                                                                                                                 |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 7.3   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 7.4   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Reuse / generator / duplicate risks                                                                                                                                                                                     | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 7.5   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                         | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 8     | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                         | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 8.0   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         | No Lean implementation manifest was required when this already-closed/planned infrastructure row was created;<br>under closed LEG10, any future code change touching this phase must fill exact added/modified modules before edits.;<br>Existing placement remains historical/planning-only. Future module placement must be revalidated by LEG10 before code changes.;<br>No one-phase-one-file inference;<br>future modules require semantic/API/import-boundary justification. | Not a generator implementation. If later code resembles an existing generator/API, consume LEG11 before edits.;<br>No generator-like code in this record.<br>Future generator-like extensions must route through LEG11. | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>incremental wrapper mode;<br>full local build green after each migration step;<br>4.28.0 baseline unless version gate explicitly opens upgrade;<br>Future finite test rows must run separately for arithmetic/modular and spectral/physical consumers.;<br>Record if modular-form/mathlib APIs are attempted in spectral/physical path;<br>that is a placement smell unless explicitly justified.;<br>Analytic modular/j/L-function objects remain mirror/comparison and cannot block finite spectral/operator algebra profiles.                                                                                                                                                                                      |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| 8.1   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet.                                                                                |
| 8.2   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. Exact quadratic use must avoid analytic sqrt/Complex and cite D-source ledger. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 8.3   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 8.4   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 8.7   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>No refactor performed in this phase;<br>future code tests must keep the full local build green after each wrapper/test addition.;<br>Lean/mathlib baseline remains 4.28.0 unless an explicit upgrade gate is opened.;<br>Required at preparation: include b=2 static, L_max values spanning 1-4, closure rows for 120/90 degree profiles, and at least one legal dynamic-branching case for 8.6.;<br>Record any noncomputable/j/tau/mathlib obstacle and isolate it outside operational path.;<br>Mandatory before implementation closes.                                                                                                                                                                             |



| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Reuse / generator / duplicate risks                                                                                                                                                                                     | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 8.8   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         | No Lean implementation manifest was required when this already-closed/planned infrastructure row was created;<br>under closed LEG10, any future code change touching this phase must fill exact added/modified modules before edits.;<br>Existing placement remains historical/planning-only. Future module placement must be revalidated by LEG10 before code changes.;<br>No one-phase-one-file inference;<br>future modules require semantic/API/import-boundary justification. | Not a generator implementation. If later code resembles an existing generator/API, consume LEG11 before edits.;<br>No generator-like code in this record.<br>Future generator-like extensions must route through LEG11. | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>incremental wrapper mode;<br>full local build green after each migration step; 4.28.0 baseline unless version gate explicitly opens upgrade;<br>Future finite test rows must run separately for arithmetic/modular and spectral/physical consumers.;<br>Record if modular-form/mathlib APIs are attempted in spectral/physical path; that is a placement smell unless explicitly justified.;<br>Analytic modular/j/L-function objects remain mirror/comparison and cannot block finite spectral/operator algebra profiles.                                                                                                                                                                                            |
| 10    | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                         | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 10.1  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                         | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |



| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 10.2  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 10.3  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase  | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 10.4   | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 10.4.1 | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Consumes S05/O05/TM0/CD7/BR06/BR07/S07/S08 only. Produces BR08/BR09. Must not be imported by earlier route definitions.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Required when active: name one finite generated b-ary approximant/cut, compute both eliminations through the same matrix, and compare outputs.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet.                                          |
| 10.4.2 | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | Consumes BR08/BR09 and route norm objects; produces CD8 only for downstream unification, not for earlier route construction.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Required when active: at least one finite Schur-derived arithmetic example with matching CD-side norm data.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet.                                                                        |

| Phase  | Status | Phase type | Context class                 | Added modules                                                                                                                                            | Modified modules                                                   | Reusable existing modules                                                                                                                                                                                                     | Manifest / placement / compactness                                                                                                                                                                              | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                        | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 10.4.3 | W      | comparison | Reference;<br>Gate;<br>Phase  | <ul style="list-style-type: none"> <li>none at decision gate; possible CNNA/PillarA/Foundation/ExactQuadraticComplex.lean only after manifest</li> </ul> | <ul style="list-style-type: none"> <li>pending manifest</li> </ul> | <ul style="list-style-type: none"> <li>Foundation/ExecComplex.lean; Foundation/ExecComplexBridge.lean; Arithmetic/Schur/CharpolyC.lean; Arithmetic/CayleyDickson/ArithmeticCDBridge.lean; mathlib QuadraticAlgebra</li> </ul> | pending Phase 1.1.1.1.1;<br>scalar-foundation / bridge governance, not generator semantics;<br>do not create ExactQuadraticComplex unless existing foundation/bridge modules cannot carry the semantics cleanly | must check existing ExecComplex, ExecComplexBridge, Schur.CharpolyC, ArithmeticCDBridge, and mathlib QuadraticAlgebra before proposing new module;<br>scalar layer must consume canonical generator outputs only;<br>cannot define generator semantics;<br>alternate complex scalar, algebraic complex, CD complex, mirror Complex, exact quadratic norm/trace/conjugation | scalar layer consumes generator/arithmetic source data;<br>generator definitions do not consume scalar comparison/mirror data;<br>incremental wrapper mode;<br>full local build green after each migration step;<br>4.28.0 baseline unless version gate explicitly opens upgrade;<br>required later: QxQ baseline row, Q(i) row, Q(sqrt(-3)) row, and at least one generic D row if O09 is implemented;<br>Record Complex/Real noncomputable boundary, QuadraticAlgebra instance diamond warning, NumberField formal/non-executable dependency, and any required version-gate.;<br>General C/R and algebraic closure remain mirror/formal. Exact quadratic candidate must avoid noncomputable definitions on operational path.                                                                                                                                                          |
| 10.5   | R      | derivation | Lean;<br>Derivation;<br>Phase | —                                                                                                                                                        | —                                                                  | —                                                                                                                                                                                                                             |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                            | Use minimal source objects: Path B consumes Phase 4.1 directly;<br>Path A consumes D/order/lattice/modular prerequisites;<br>bridges are explicit and late.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet.   |
| 11     | R      | derivation | Lean;<br>Derivation;<br>Phase | —                                                                                                                                                        | —                                                                  | —                                                                                                                                                                                                                             |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                            | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 11.1  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 11.2  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 11.3  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 11.4  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 11.5  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 12.1  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 12.3  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 12.8  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 13    | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |



| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 13.1  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 13.2  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 13.3  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |



| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 13.4  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 13.5  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 13.7  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 14    | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 14.1  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 14.2  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 14.3  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 14.4  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 14.5  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 15    | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 15.1  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 15.2  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 15.3  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 15.4  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 16    | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 16.1  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 16.2  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 16.4  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |



| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 17    | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 17.1  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 17.2  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |



| Phase | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness | Reuse / generator / duplicate risks | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 17.3  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 17.4  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |
| 17.5  | R      | derivation | Lean;<br>Derivation;<br>Phase | —             | —                | —                         |                                    |                                     | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.;<br>No phase-specific mathlib complication recorded yet.;<br>No phase-specific noncomputable-boundary audit recorded yet. |

| Phase | Status | Phase type | Context class                | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness                                                                                                                                                                                                                                                                  | Reuse / generator / duplicate risks                                                                                                                                                            | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 19    | W      | comparison | Reference;<br>Gate;<br>Phase | —             | —                | —                         | No code modules introduced by this gate.;<br>Governance belongs in YAML master and generated views unless a future code-level policy module is justified.;<br>Do not split code solely for this planning decision; use YAML/workflow unless a reusable code-level policy module is later justified. | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation. | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure/diagnostic-only until activation. If future work introduces executable graph metrics, record concrete input JSON and output diagnostic witness.;<br>No mathlib use at planning stage. Future non-operational Lean audit code must record any noncomputable/classical pressure.;<br>No operational Lean code. Graph metadata may be computed by scripts; it cannot enter operational CNNA derivations. |
| 19.1  | W      | comparison | Reference;<br>Gate;<br>Phase | —             | —                | —                         | No code modules introduced by this gate.;<br>Governance belongs in YAML master and generated views unless a future code-level policy module is justified.;<br>Do not split code solely for this planning decision; use YAML/workflow unless a reusable code-level policy module is later justified. | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation. | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure/diagnostic-only until activation. If future work introduces executable graph metrics, record concrete input JSON and output diagnostic witness.;<br>No mathlib use at planning stage. Future non-operational Lean audit code must record any noncomputable/classical pressure.;<br>No operational Lean code. Graph metadata may be computed by scripts; it cannot enter operational CNNA derivations. |

| Phase | Status | Phase type | Context class                | Added modules | Modified modules | Reusable existing modules | Manifest / placement / compactness                                                                                                                                                                                                                                                                  | Reuse / generator / duplicate risks                                                                                                                                                            | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 19.2  | W      | comparison | Reference;<br>Gate;<br>Phase | —             | —                | —                         | No code modules introduced by this gate.;<br>Governance belongs in YAML master and generated views unless a future code-level policy module is justified.;<br>Do not split code solely for this planning decision; use YAML/workflow unless a reusable code-level policy module is later justified. | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation. | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure/diagnostic-only until activation. If future work introduces executable graph metrics, record concrete input JSON and output diagnostic witness.;<br>No mathlib use at planning stage. Future non-operational Lean audit code must record any noncomputable/classical pressure.;<br>No operational Lean code. Graph metadata may be computed by scripts; it cannot enter operational CNNA derivations. |
| 19.3  | W      | comparison | Reference;<br>Gate;<br>Phase | —             | —                | —                         | No code modules introduced by this gate.;<br>Governance belongs in YAML master and generated views unless a future code-level policy module is justified.;<br>Do not split code solely for this planning decision; use YAML/workflow unless a reusable code-level policy module is later justified. | No duplicate implementation authorized; decision reduces future duplication risk.;<br>Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation. | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.;<br>Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.;<br>Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline;<br>noncomputable/classical pressure must be logged and isolated.;<br>Infrastructure/diagnostic-only until activation. If future work introduces executable graph metrics, record concrete input JSON and output diagnostic witness.;<br>No mathlib use at planning stage. Future non-operational Lean audit code must record any noncomputable/classical pressure.;<br>No operational Lean code. Graph metadata may be computed by scripts; it cannot enter operational CNNA derivations. |

| Phase | Status | Phase type    | Context class          | Added modules | Modified modules                                                                                 | Reusable existing modules                                                                                                                                                   | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                 | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                            | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 19.4  | W      | comparison    | Reference; Gate; Phase | —             | —                                                                                                | —                                                                                                                                                                           | No code modules introduced by this gate.; Governance belongs in YAML master and generated views unless a future code-level policy module is justified.; Do not split code solely for this planning decision; use YAML/workflow unless a reusable code-level policy module is later justified.                                      | No duplicate implementation authorized; decision reduces future duplication risk.; Generator-like logic must consume the canonical smoke/theory generator path; no parallel implementation.                                                                                                                                    | No special cross-phase consumer contract recorded yet. If a later phase consumes this output, record export shape and import direction before implementation.; Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.; Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.; Infrastructure/diagnostic-only until activation. If future work introduces executable graph metrics, record concrete input JSON and output diagnostic witness.; No mathlib use at planning stage. Future non-operational Lean audit code must record any noncomputable/classical pressure.; No operational Lean code. Graph metadata may be computed by scripts; it cannot enter operational CNNA derivations. |
| 19.5  | W      | documentation | Documentation; Phase   | —             | —                                                                                                | —                                                                                                                                                                           | No core Lean module planned; generated documentation/diagnostic reports only unless later theorem requires one.                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                | Route direction must be explicit before implementation: thread/full profiles feed Schur/arithmetic audits; analogy/diagnostics never feed generator code.; Not applicable unless phase_type is refactor or the phase changes imports/declarations. If refactor applies: wrapper-first, bounded migration package, local full-build evidence before closeout.; Baseline is Lean 4.28.0/mathlib snapshot unless an explicit phase-boundary upgrade gate is opened. New imports must be verified against the chosen baseline; noncomputable/classical pressure must be logged and isolated.; Infrastructure-only or not prepared yet. Before operational code implementation, name a concrete finite instance and the path from input to output.; No phase-specific mathlib complication recorded yet.; No phase-specific noncomputable-boundary audit recorded yet.                                                                                                                                            |
| 20    | G      | documentation | Documentation; Phase   | —             | <ul style="list-style-type: none"> <li>Repository/ masters/ cнна_roadmap_master.yaml;</li> </ul> | <ul style="list-style-type: none"> <li>Repository/scripts/ generate_tables.py</li> <li>Repository/yaml/ generated_exports/*.tsv</li> <li>Repository/tables/*.tex</li> </ul> | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.; DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.; Keep Lean implementation packets compact; DB/audit-full exploration must not overload routine code-writing phases. | Reuse Single-YAML-Master and existing generator/export seams; do not create a second authoritative planning source.; YAML remains the manual planning source; generated DB/TSV/PDF/graph views remain secondary and reproducible.; second-source-of-truth risk; generated-view-as-truth risk; unmarked-established-theory risk | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.; No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.; No Lean/mathlib version change. Lean data timestamp remains fixed.; Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.; None; no mathlib imports or Lean code changes.; No noncomputable Lean boundary involved; semantic review is a documentation governance rule.                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Phase | Status | Phase type     | Context class                     | Added modules | Modified modules | Reusable existing modules                                                                                                                                                                  | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                 | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                            | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 20.1  | G      | documentation  | Documentation;<br>Phase           | —             |                  | <ul style="list-style-type: none"><li>Repository/<br/>masters/<br/>cnna_roadmap_master.yaml</li><li>Repository/<br/>generated_exports/*.tsv</li><li>Repository/<br/>tables/*.tex</li></ul> | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.; DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.; Keep Lean implementation packets compact; DB/audit-full exploration must not overload routine code-writing phases. | Reuse Single-YAML-Master and existing generator/export seams; do not create a second authoritative planning source.; YAML remains the manual planning source; generated DB/TSV/PDF/graph views remain secondary and reproducible.; second-source-of-truth risk; generated-view-as-truth risk; unmarked-established-theory risk | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.; No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.; No Lean/mathlib version change. Lean data timestamp remains fixed.; Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.; None; no mathlib imports or Lean code changes.; No noncomputable Lean boundary involved; semantic review is a documentation governance rule. |
| 20.2  | W      | infrastructure | Tool;<br>Infrastructure;<br>Phase | —             | —                | <ul style="list-style-type: none"><li>Repository/scripts/<br/>generate_tables.py</li><li>Repository/<br/>generated_exports/*.tsv</li><li>Repository/tables/*.tex</li></ul>                 | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.; DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.; Keep Lean implementation packets compact; DB/audit-full exploration must not overload routine code-writing phases. | Reuse Single-YAML-Master and existing generator/export seams; do not create a second authoritative planning source.; YAML remains the manual planning source; generated DB/TSV/PDF/graph views remain secondary and reproducible.; second-source-of-truth risk; generated-view-as-truth risk; unmarked-established-theory risk | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.; No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.; No Lean/mathlib version change. Lean data timestamp remains fixed.; Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.; None; no mathlib imports or Lean code changes.; No noncomputable Lean boundary involved; semantic review is a documentation governance rule. |
| 20.3  | W      | documentation  | Documentation;<br>Phase           | —             | —                | <ul style="list-style-type: none"><li>Repository/scripts/<br/>generate_tables.py</li><li>Repository/<br/>generated_exports/*.tsv</li><li>Repository/tables/*.tex</li></ul>                 | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.; DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.; Keep Lean implementation packets compact; DB/audit-full exploration must not overload routine code-writing phases. | Reuse Single-YAML-Master and existing generator/export seams; do not create a second authoritative planning source.; YAML remains the manual planning source; generated DB/TSV/PDF/graph views remain secondary and reproducible.; second-source-of-truth risk; generated-view-as-truth risk; unmarked-established-theory risk | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.; No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.; No Lean/mathlib version change. Lean data timestamp remains fixed.; Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.; None; no mathlib imports or Lean code changes.; No noncomputable Lean boundary involved; semantic review is a documentation governance rule. |
| 20.4  | W      | documentation  | Documentation;<br>Phase           | —             | —                | <ul style="list-style-type: none"><li>Repository/scripts/<br/>generate_tables.py</li><li>Repository/<br/>generated_exports/*.tsv</li><li>Repository/tables/*.tex</li></ul>                 | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.; DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.; Keep Lean implementation packets compact; DB/audit-full exploration must not overload routine code-writing phases. | Reuse Single-YAML-Master and existing generator/export seams; do not create a second authoritative planning source.; YAML remains the manual planning source; generated DB/TSV/PDF/graph views remain secondary and reproducible.; second-source-of-truth risk; generated-view-as-truth risk; unmarked-established-theory risk | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.; No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.; No Lean/mathlib version change. Lean data timestamp remains fixed.; Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.; None; no mathlib imports or Lean code changes.; No noncomputable Lean boundary involved; semantic review is a documentation governance rule. |



| Phase | Status | Phase type     | Context class                     | Added modules                                                                                                                                                | Modified modules                                                                                                                                                | Reusable existing modules                                                                                                                                                                                                | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                                    | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                                                                                                      | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 20.5  | W      | infrastructure | Tool;<br>Infrastructure;<br>Phase | —                                                                                                                                                            | —                                                                                                                                                               | <ul style="list-style-type: none"> <li>Repository/scripts/generate_tables.py</li> <li>Repository/generated_exports/*.tsv</li> <li>Repository/tables/*.tex</li> </ul>                                                     | No Lean module manifest change. Future DB/export scripts require separate manifest if implemented.; DB-REVIEW belongs to documentation/review/tooling infrastructure, not to Pillar-A mathematical derivation.; Keep Lean implementation packets compact; DB/audit-full exploration must not overload routine code-writing phases.                                                                                    | Reuse Single-YAML-Master and existing generator/export seams; do not create a second authoritative planning source.; YAML remains the manual planning source; generated DB/TSV/PDF/graph views remain secondary and reproducible.; second-source-of-truth risk; generated-view-as-truth risk; unmarked-established-theory risk                                                                                                                                           | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.; No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.; No Lean/mathlib version change. Lean data timestamp remains fixed.; Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.; None; no mathlib imports or Lean code changes.; No noncomputable Lean boundary involved; semantic review is a documentation governance rule.                                                                         |
| 20.6  | G      | documentation  | Generated;<br>Artifact;<br>Phase  | <ul style="list-style-type: none"> <li>Repository/scripts/generate_context_doc.py</li> </ul>                                                                 | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul>                                                | <ul style="list-style-type: none"> <li>Repository/scripts/generate_tables.py</li> <li>Repository/generated_exports/*.tsv</li> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                 | Added Repository/scripts/generate_context_doc.py. No Lean module manifest change.; Belongs to DB-REVIEW generated-secondary documentation tooling. It is not a Pillar-A mathematical derivation layer and not a database truth layer.; Keeps targeted documentation compact by filtering rows per context while retaining section topology. Reduces overhead during Lean implementation closeouts.                    | Reuses the Single-YAML-Master and generated_exports. Does not create a second master; output manifest records the source and filter terms.; Context documents are reproducible generated views. Curated profiles live in YAML so they can later round-trip into the database representation.; second-source-of-truth risk; profile-overfiltering risk; false-negative relevance risk; generated-view-as-truth risk                                                       | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.; No Lean refactor. Future tooling must preserve generated-view reproducibility and YAML single-source rule.; No Lean/mathlib version change. Lean data timestamp remains fixed.; Not an operational Lean computation. Concrete traversal is metadata path: YAML row -> generated TSV/TeX/PDF -> future DB row/view.; None; no mathlib imports or Lean code changes.; No noncomputable Lean boundary involved; semantic review is a documentation governance rule.                                                                         |
| 20.7  | G      | documentation  | Generated;<br>Artifact;<br>Phase  | —                                                                                                                                                            | <ul style="list-style-type: none"> <li>Repository/scripts/generate_tables.py</li> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul> | <ul style="list-style-type: none"> <li>Repository/scripts/generate_tables.py</li> <li>Repository/repo_inventory/raw_export/*.json</li> <li>Repository/tables/module_inventory.tex</li> </ul>                             | Modified Repository/scripts/generate_tables.py only; no Lean module manifest change.; Belongs to DB-REVIEW generated-secondary documentation tooling. It is not a Pillar-A mathematical derivation layer, not a database truth layer, and not a formal CNNA theorem.; Keeps the front section compact: short facts table plus the existing inventory/graph view. Avoids duplicating the old module-inventory section. | Reuses the existing module inventory table instead of adding a duplicate standalone module-inventory section. Adds only a generated project_stats view and YAML records for metaphor/license.; Stats rows are reproducible generated views. Guiding metaphors and license notices live in YAML so they can later round-trip into database views.; second-source-of-truth risk; metaphor-as-theorem risk; project-stats-as-truth-certificate risk; license-overreach risk | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.; No Lean refactor. Future tooling must preserve generated-view reproducibility, YAML single-source rule, and old-module-inventory-under-Stats placement.; No Lean/mathlib version change. Lean data timestamp remains fixed.; Not an operational Lean computation. Concrete traversal is metadata path: YAML/repo inventory/source counts -> generated TSV/TeX/PDF -> future database view.; None; no mathlib imports or Lean code changes.; No noncomputable Lean boundary involved; semantic review is a documentation governance rule. |
| 20.8  | G      | documentation  | Generated;<br>Artifact;<br>Phase  | <ul style="list-style-type: none"> <li>Repository/scripts/generate_full_document.py</li> <li>Repository/scripts/clean_generated_root_artifacts.py</li> </ul> | <ul style="list-style-type: none"> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>README.md</li> </ul>                                                | <ul style="list-style-type: none"> <li>Repository/scripts/generate_tables.py</li> <li>Repository/scripts/generate_context_doc.py</li> <li>Repository/tables/*.tex</li> <li>Repository/generated_exports/*.tsv</li> </ul> | Added two Python tool scripts only; no Lean module manifest change.; Belongs to DB-REVIEW generated-secondary documentation/tooling. It is a retention policy, not a Pillar-A mathematical derivation layer.; Reduces package clutter and clarifies active source boundaries without changing Lean source structure.                                                                                                  | Reuses existing generation pipeline; avoids duplicating versioned full-doc TeX/PDF in root. Adds a single canonical full-document generator path.; Full document TeX/PDF must be generated from YAML/tables by script and treated as disposable unless explicitly requested as an external artifact.; root PDF/TEX clutter; historical report as false source; generated-secondary artifact treated as authoritative; package bloat                                      | All consumers are documentation/review/tooling consumers. No pillar handoff or mathematical consumer rights are granted.; No Lean refactor. Verify source/control files remain, reports are archived, and root generated artifacts are absent after cleanup.; No Lean/mathlib version change. Lean data timestamp remains fixed.; Not an operational Lean computation. Concrete traversal is filesystem classification: root artifacts -> archive or delete -> regenerate on demand.; None; no mathlib imports or Lean code changes.; No noncomputable Lean boundary involved; only generated documentation artifacts are classified.                              |

| Phase | Status | Phase type     | Context class                    | Added modules                                                                                                                                                                                                        | Modified modules                                                                                                                                                                                                                                                                                                                        | Reusable existing modules                                                                                                                                                                                                                                        | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                         | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                                                             | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| 20.9  | G      | documentation  | Generated;<br>Artifact;<br>Phase | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_14_README</li> </ul>                                                                                                                            | <ul style="list-style-type: none"> <li>README.md</li> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>scripts/generate_full_document.py</li> <li>scripts/generate_context_doc.py</li> <li>scripts/build_export_script_v1.8.py</li> <li>scripts/clean_generated_root_artifacts.py</li> </ul> | <p>No Lean modules added/modified. Documentation files only: README.md and V1_14 report; YAML metadata/governance records updated.; Natural placement is DB-REVIEW side objective because this is a documentation/workflow entry-surface update.;</p> <p>Keep README compact and current; place detailed history in archive/reports or generated docs.</p>                 | <p>Reuses existing scripts and generated views; no new generator required.;</p> <p>README documents existing canonical generators instead of creating alternative documentation sources.;</p> <p>Low;</p> <p>README consolidates command/workflow explanations formerly scattered across release notes.</p>                                                                                                                     | <p>Human/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.;</p> <p>No Lean refactor. Regression check is generate_tables.py plus root cleanliness and README content audit.;</p> <p>No Lean/mathlib version change. Lean data timestamp remains fixed.;</p> <p>No mathematical traversal; documentation-tooling surface only.;</p> <p>None;</p> <p>no Lean/mathlib import touched.;</p> <p>No Lean source edited;</p> <p>no noncomputable boundary introduced.</p>                                                                                                   |
| 20.10 | G      | infrastructure | Generated;<br>Artifact;<br>Phase | <ul style="list-style-type: none"> <li>CNNA.lean</li> <li>lakefile.lean</li> <li>lake-manifest.json</li> <li>lean-toolchain</li> <li>Repository/archive/reports/V1_15_ROOT_LEAN_TOOLCHAIN_FILES_REPORT.md</li> </ul> | <ul style="list-style-type: none"> <li>README.md</li> <li>Repository/masters/cnna_roadmap_master.yaml</li> <li>Repository/scripts/clean_generated_root_artifacts.py</li> </ul>                                                                                                                                                          | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>scripts/clean_generated_root_artifacts.py</li> <li>README.md</li> </ul>                                                                                                               | <p>Added package-root Lean/Lake project-control files and amended cleanup script;</p> <p>no Lean modules under repo_snapshot/CNNA changed.;</p> <p>Natural placement is DB-REVIEW/build-reproducibility tooling at package-root boundary, not a mathematical pillar or CNNA generator.;</p> <p>Root remains compact: source/control files only, not generated outputs.</p> | <p>Root CNNA.lean duplicates the top-level import surface only as compatibility entry;</p> <p>Lake source root remains repo_snapshot via srcDir.;</p> <p>Use existing canonical generators: generate_tables.py for YAML views, cleanup script for root hygiene. No alternate truth source.;</p> <p>Low;</p> <p>root CNNA.lean is a compatibility entry and repo_snapshot/CNNA.lean remains Lake source root through srcDir.</p> | <p>Human/local-build consumers only; no pillar handoff rights.;</p> <p>No Lean refactor. Regression check is root file audit and cleanup keep-list test.;</p> <p>Root toolchain and manifest retain Lean/mathlib v4.28.0 as supplied.;</p> <p>Concrete traversal is root file inventory and cleanup classification over the actual package root.;</p> <p>No Lean/mathlib source complication introduced;</p> <p>root manifest/toolchain pin supplied build environment.;</p> <p>No Lean source edited;</p> <p>no noncomputable boundary introduced.</p>                                                   |
| 20.11 | G      | documentation  | Generated;<br>Artifact;<br>Phase | <ul style="list-style-type: none"> <li>LICENSE</li> <li>Repository/archive/reports/V1_17_LICENSE</li> </ul>                                                                                                          | <ul style="list-style-type: none"> <li>README.md</li> <li>Repository/scripts/generate_full_document.py</li> <li>Repository/scripts/generate_context_doc.py</li> <li>Repository/scripts/generate_tables.py</li> <li>Repository/scripts/clean_generated_root_artifacts.py</li> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul> | <ul style="list-style-type: none"> <li>scripts/generate_tables.py</li> <li>scripts/generate_full_document.py</li> <li>scripts/generate_context_doc.py</li> <li>scripts/generate_tables.py</li> <li>README.md</li> </ul>                                          | <p>No Lean modules added/modified. Documentation/license/tool scripts and YAML only.;</p> <p>Natural placement is DB-REVIEW/root documentation and license surface, not a mathematical pillar or CNNA generator.;</p> <p>Keep package root compact and unambiguous: README.md, LICENSE, .gitignore, Lean/Lake files, Repository/.</p>                                      | <p>Closed for README surface: duplicate active README copies removed;</p> <p>license/credits centralized at package root and generated from metadata/table for PDFs.;</p> <p>PDF credits are generated from metadata;</p> <p>license table remains generated from the Single YAML master.;</p> <p>Closed for README: maintained README.md exists only in package root.</p> <p>No Lean symbols affected.</p>                     | <p>Human/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.;</p> <p>No Lean refactor. Regression checks are generate_tables.py, PDF compile, and README-location audit.;</p> <p>No Lean/mathlib version change. Lean data timestamp remains fixed.;</p> <p>Concrete traversal is root file inventory, README-location audit, and generated PDF front-matter check.;</p> <p>No Lean/mathlib source complication introduced;</p> <p>root manifest/toolchain unchanged.;</p> <p>No Lean source edited;</p> <p>no noncomputable boundary introduced.</p>                  |
| 20.12 | G      | documentation  | Generated;<br>Artifact;<br>Phase | <ul style="list-style-type: none"> <li>Repository/archive/reports/V1_18_SECTION_PAGEBREAKS_REPORT.md</li> </ul>                                                                                                      | <ul style="list-style-type: none"> <li>Repository/scripts/generate_full_document.py</li> <li>Repository/scripts/generate_context_doc.py</li> <li>README.md</li> <li>Repository/masters/cnna_roadmap_master.yaml</li> </ul>                                                                                                              | <ul style="list-style-type: none"> <li>scripts/generate_full_document.py</li> <li>scripts/generate_context_doc.py</li> <li>scripts/generate_tables.py</li> </ul>                                                                                                 | <p>No Lean modules added/modified. Documentation generator scripts and YAML only.;</p> <p>Natural placement is DB-REVIEW/documentation generator layer, not a mathematical pillar or CNNA generator.;</p> <p>Keep the layout behavior centralized in generators rather than duplicated across generated artifacts.</p>                                                     | <p>Closed for layout: the pagebreak invariant is centralized in generators and not repeated in disposable outputs.;</p> <p>Section-pagebreak behavior is generated from the full/context generator preambles.;</p> <p>No Lean symbols affected. Avoid duplicate layout rules outside the generators.</p>                                                                                                                        | <p>Human/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.;</p> <p>Regression checks are generate_tables.py, full PDF compile, context PDF compile, generated TeX hook audit, and early-page rendering.;</p> <p>No Lean/mathlib version change. Lean data timestamp remains fixed.;</p> <p>Concrete traversal is full-document PDF generation and DtN context-document generation.;</p> <p>No Lean/mathlib source complication introduced;</p> <p>root manifest/toolchain unchanged.;</p> <p>No Lean source edited;</p> <p>no noncomputable boundary introduced.</p> |



| Phase     | Status | Phase type     | Context class                    | Added modules                                                                                                                                    | Modified modules                                                                                                                                                                                                                                                    | Reusable existing modules                                                                                                                                                                                                                                                                                                                | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 20.13     | G      | infrastructure | Generated;<br>Artifact;<br>Phase | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_19_PHASE_STATUS_RELEASE_POLICY_REPORT.md</li></ul>                           | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li><li>Repository/scripts/generate_tables.py</li><li>README.md</li></ul>                                                                                                         | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>scripts/generate_full_document.py</li><li>scripts/generate_context_doc.py</li><li>scripts/generate_tables.py</li></ul>                                                                                                                                          | No Lean modules added/modified. Master YAML and generator validation only.;<br>Natural placement is planning/status-governance check results as mathematical pillar or CNNA generator.;<br>One policy object and guardrail govern all phase statuses;<br>no separate status prose per red phase beyond automatic lock note.                                                                                                                                                                                                              | Status policy is centralized in generator validation and master YAML, avoiding per-phase duplicate rules.;<br>generate_tables.py is the canonical validator for status semantics.;<br>Closed by central status validation;<br>no Lean symbols affected.                                                                                                                                                                                                                                                                       | Planning/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.;<br>No Lean refactor. Regression checks are generate_tables.py, status-count audit, and full/context documentation generation.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>Concrete traversal is all phase rows and scratchpad phase_type values.;<br>No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.;<br>No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 20.14     | G      | documentation  | Generated;<br>Artifact;<br>Phase | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_20_DOC_SECTION_EXPLANATIONS_AND_MODULE_MANIFEST_APPENDIX_REPORT.md</li></ul> | <ul style="list-style-type: none"><li>Repository/scripts/generate_full_document.py</li><li>Repository/scripts/generate_context_doc.py</li><li>Repository/scripts/generate_tables.py</li><li>README.md</li><li>Repository/masters/cnna_roadmap_master.yaml</li></ul> | <ul style="list-style-type: none"><li>scripts/generate_full_document.py</li><li>scripts/generate_context_doc.py</li><li>scripts/generate_tables.py</li></ul>                                                                                                                                                                             | Module manifest remains generated, but is now explained as technical appendix rather than a primary reading table.;<br>Natural placement is DB-REVIEW/documentation generator layer, not a mathematical pillar or CNNA generator.;<br>Keep static explanations concise and centralized in generators;<br>keep module-change table narrow and appendix-only.                                                                                                                                                                              | Closed for documentation semantics:<br>primary phase table and module-change appendix roles are separated.;<br>Static section explanations are generated from full/context generator code;<br>dynamic table rows continue to come from Master YAML or generated exports.;<br>Closed for module manifest: it is no longer positioned as a main phase/release table.<br>No Lean symbols affected.                                                                                                                               | Human/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.;<br>Regression checks are generate_tables.py, full PDF compile, context PDF compile, generated TeX title/order audit, and representative rendering.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>Concrete traversal is full-document PDF generation and DtN context-document generation.;<br>No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.;<br>No Lean source edited;<br>no noncomputable boundary introduced.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1.1.1.1.5 | Y      | refactor       | Lean;<br>Governance;<br>Phase    | —                                                                                                                                                | —                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"><li>CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li><li>CNNA/PillarA/Handoff/Core/SectorExport.lean</li><li>CNNA/PillarA/Handoff/Outputs/Closed.lean</li><li>repo_inventory/raw_export/*.json</li><li>generated_exports/consumption_classification_map.generated.tsv</li></ul> | No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean;<br>CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean;<br>CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.;<br>Natural placement is before blue-object semantic correction and any new Lean content;<br>do not patch later consumer phases.;<br>Preserve or improve auditability;<br>neutralize names and consolidate only with facades/regression evidence. | RD08 records the consolidation decision rule. Same-semantics wrappers may be unified;<br>proof-stage modules must remain separated until a stronger canonical proof module preserves their role.;<br>No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.;<br>phase-semantic module names;<br>unclassified declarations;<br>duplicate wrapper modules;<br>stale object/proof mappings. | Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.;<br>Mandatory before any Lean merge: retain old modules as facades in pass 1;<br>record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.;<br>Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed;<br>refactor-only changes must compile locally before closeout.;<br>Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.;<br>No mathlib complication in audit. Later code movement may expose import-minimization issues;<br>resolve by preserving existing import surfaces first.;<br>No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths. |

| Phase     | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules                                                                                                                                                                                                                                                                                                                          | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| 1.1.1.1.5 | O      | refactor   | Lean;<br>Governance;<br>Phase | —             | —                | <ul style="list-style-type: none"><li>• CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li><li>• CNNA/PillarA/Handoff/Core/SectorExport.lean</li><li>• CNNA/PillarA/Handoff/Outputs/Closed.lean</li><li>• repo_inventory/raw_export/*.json</li><li>• generated_exports/consumption_classification_map.generated.tsv</li></ul> | No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean; CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean; CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.; Natural placement is before blue-object semantic correction and any new Lean content; do not patch later consumer phases.; Preserve or improve auditability; neutralize names and consolidate only with facades/regression evidence. | RD08 records the consolidation decision rule. Same-semantics wrappers may be unified; proof-stage modules must remain separated until a stronger canonical proof module preserves their role.; No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.; phase-semantic module names; unclassified declarations; duplicate wrapper modules; stale object/proof mappings. | Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.; Mandatory before any Lean merge: retain old modules as facades in pass 1; record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.; Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed; refactor-only changes must compile locally before closeout.; Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.; No mathlib complication in audit. Later code movement may expose import-minimization issues; resolve by preserving existing import surfaces first.; No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths. |
| 1.1.1.1.5 | O      | refactor   | Lean;<br>Governance;<br>Phase | —             | —                | <ul style="list-style-type: none"><li>• CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li><li>• CNNA/PillarA/Handoff/Core/SectorExport.lean</li><li>• CNNA/PillarA/Handoff/Outputs/Closed.lean</li><li>• repo_inventory/raw_export/*.json</li><li>• generated_exports/consumption_classification_map.generated.tsv</li></ul> | No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean; CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean; CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.; Natural placement is before blue-object semantic correction and any new Lean content; do not patch later consumer phases.; Preserve or improve auditability; neutralize names and consolidate only with facades/regression evidence. | RD08 records the consolidation decision rule. Same-semantics wrappers may be unified; proof-stage modules must remain separated until a stronger canonical proof module preserves their role.; No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.; phase-semantic module names; unclassified declarations; duplicate wrapper modules; stale object/proof mappings. | Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.; Mandatory before any Lean merge: retain old modules as facades in pass 1; record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.; Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed; refactor-only changes must compile locally before closeout.; Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.; No mathlib complication in audit. Later code movement may expose import-minimization issues; resolve by preserving existing import surfaces first.; No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths. |

| Phase     | Status | Phase type | Context class                 | Added modules | Modified modules | Reusable existing modules                                                                                                                                                                                                                                                                                                                      | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 1.1.1.1.5 | O      | refactor   | Lean;<br>Governance;<br>Phase | —             | —                | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li> <li>CNNA/PillarA/Handoff/Core/SectorExport.lean</li> <li>CNNA/PillarA/Handoff/Outputs/Closed.lean</li> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/consumption_classification_map.generated.tsv</li> </ul> | <p>No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean; CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean; CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.;</p> <p>Natural placement is before blue-object semantic correction and any new Lean content; do not patch later consumer phases.;</p> <p>Preserve or improve auditability; neutralize names and consolidate only with facades/regression evidence.</p> | <p>RD08 records the consolidation decision rule. Same-semantics wrappers may be unified;</p> <p>proof-stage modules must remain separated until a stronger canonical proof module preserves their role.;</p> <p>No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.;</p> <p>phase-semantic module names; unclassified declarations; duplicate wrapper modules; stale object/proof mappings.</p> | <p>Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.;</p> <p>Mandatory before any Lean merge: retain old modules as facades in pass 1;</p> <p>record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.;</p> <p>Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed;</p> <p>refactor-only changes must compile locally before closeout.;</p> <p>Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.;</p> <p>No mathlib complication in audit. Later code movement may expose import-minimization issues;</p> <p>resolve by preserving existing import surfaces first.;</p> <p>No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths.</p> |
| 1.1.1.1.5 | O      | refactor   | Lean;<br>Governance;<br>Phase | —             | —                | <ul style="list-style-type: none"> <li>CNNA/PillarA/Arithmetic/BoundarySource/GeneratedSubstrateRoute.lean</li> <li>CNNA/PillarA/Handoff/Core/SectorExport.lean</li> <li>CNNA/PillarA/Handoff/Outputs/Closed.lean</li> <li>repo_inventory/raw_export/*.json</li> <li>generated_exports/consumption_classification_map.generated.tsv</li> </ul> | <p>No Lean module added now. Potential future modules, if authorized: CNNA/PillarA/Handoff/Outputs/SectorExportAliases.lean; CNNA/PillarA/Handoff/Inputs/FeedbackInput.lean; CNNA/PillarA/Arithmetic/BoundarySource/CandidateLedgers.lean. Old modules remain as facades initially.;</p> <p>Natural placement is before blue-object semantic correction and any new Lean content; do not patch later consumer phases.;</p> <p>Preserve or improve auditability; neutralize names and consolidate only with facades/regression evidence.</p> | <p>RD08 records the consolidation decision rule. Same-semantics wrappers may be unified;</p> <p>proof-stage modules must remain separated until a stronger canonical proof module preserves their role.;</p> <p>No generator change. If consolidation touches generated Smoke files later, it must use the existing canonical generated-source/main-path seams and must not create a second generator path.;</p> <p>phase-semantic module names; unclassified declarations; duplicate wrapper modules; stale object/proof mappings.</p> | <p>Audit flows from code producers/import surfaces to consumers. If a consumer needs a consolidated API, the facade must be created at the producer/natural source layer, not patched in the consumer.;</p> <p>Mandatory before any Lean merge: retain old modules as facades in pass 1;</p> <p>record declarations_migrated_count, imports_changed, import_graph_delta, consumer_breakage_audit, legacy/coexistence status, and local full-build evidence.;</p> <p>Baseline remains Lean 4.28.0/mathlib snapshot. No new import may be assumed;</p> <p>refactor-only changes must compile locally before closeout.;</p> <p>Not a computation phase. Concrete evidence is the finite repository module/declaration graph plus direct code spot-checks of named files. Later executable refactors must preserve existing executable witnesses.;</p> <p>No mathlib complication in audit. Later code movement may expose import-minimization issues;</p> <p>resolve by preserving existing import surfaces first.;</p> <p>No noncomputable introduced. Later operational refactor cannot introduce noncomputable/classical pressure into executable paths.</p> |

| Phase | Status | Phase type     | Context class                 | Added modules | Modified modules | Reusable existing modules                                                                                                                                                                              | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                              | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                              | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 20.15 | W      | infrastructure | Database;<br>Mirror;<br>Phase | —             | —                | <ul style="list-style-type: none"><li>masters/<br/>cnnn_roadmap_master.yaml</li><li>generated_exports/*.tsv</li><li>repo_inventory/<br/>raw_export/*.json</li><li>scripts/generate_tables.py</li></ul> | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.;<br>Natural placement is DB-REVIEW/<br>generated-secondary publication infrastructure, not a CNNA mathematical pillar.;<br>Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content. | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.;<br>Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.;<br>second truth system risk;<br>manual DB table edits;<br>public write policy leak;<br>service_role key leak;<br>snapshot provenance loss | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.;<br>No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.;<br>None;<br>no Lean/mathlib code changes.;<br>No Lean noncomputable boundary involved. |
| 20.16 | W      | infrastructure | Database;<br>Mirror;<br>Phase | —             | —                | <ul style="list-style-type: none"><li>masters/<br/>cnnn_roadmap_master.yaml</li><li>generated_exports/*.tsv</li><li>repo_inventory/<br/>raw_export/*.json</li><li>scripts/generate_tables.py</li></ul> | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.;<br>Natural placement is DB-REVIEW/<br>generated-secondary publication infrastructure, not a CNNA mathematical pillar.;<br>Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content. | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.;<br>Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.;<br>second truth system risk;<br>manual DB table edits;<br>public write policy leak;<br>service_role key leak;<br>snapshot provenance loss | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.;<br>No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.;<br>None;<br>no Lean/mathlib code changes.;<br>No Lean noncomputable boundary involved. |
| 20.17 | W      | infrastructure | Database;<br>Mirror;<br>Phase | —             | —                | <ul style="list-style-type: none"><li>masters/<br/>cnnn_roadmap_master.yaml</li><li>generated_exports/*.tsv</li><li>repo_inventory/<br/>raw_export/*.json</li><li>scripts/generate_tables.py</li></ul> | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.;<br>Natural placement is DB-REVIEW/<br>generated-secondary publication infrastructure, not a CNNA mathematical pillar.;<br>Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content. | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.;<br>Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.;<br>second truth system risk;<br>manual DB table edits;<br>public write policy leak;<br>service_role key leak;<br>snapshot provenance loss | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.;<br>No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.;<br>None;<br>no Lean/mathlib code changes.;<br>No Lean noncomputable boundary involved. |
| 20.18 | W      | infrastructure | Database;<br>Mirror;<br>Phase | —             | —                | <ul style="list-style-type: none"><li>masters/<br/>cnnn_roadmap_master.yaml</li><li>generated_exports/*.tsv</li><li>repo_inventory/<br/>raw_export/*.json</li><li>scripts/generate_tables.py</li></ul> | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.;<br>Natural placement is DB-REVIEW/<br>generated-secondary publication infrastructure, not a CNNA mathematical pillar.;<br>Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content. | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.;<br>Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.;<br>second truth system risk;<br>manual DB table edits;<br>public write policy leak;<br>service_role key leak;<br>snapshot provenance loss | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.;<br>No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.;<br>None;<br>no Lean/mathlib code changes.;<br>No Lean noncomputable boundary involved. |



| Phase | Status | Phase type     | Context class                    | Added modules                                                                                                                                  | Modified modules                                                                                                                                                                                                                                                                                                                                                                                                                           | Reusable existing modules                                                                                                                                                                                                                            | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                               | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                                              | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| 20.19 | W      | infrastructure | Database;<br>Mirror;<br>Phase    | —                                                                                                                                              | —                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"><li>masters/<br/>cnna_roadmap_master.yaml</li><li>generated_exports/*.tsv</li><li>repo_inventory/<br/>raw_export/*.json</li><li>scripts/generate_tables.py</li></ul>                                               | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.;<br>Natural placement is DB-REVIEW/generated-secondary publication infrastructure, not a CNNA mathematical pillar.;<br>Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content.      | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.;<br>Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.;<br>second truth system risk;<br>manual DB table edits;<br>public write policy leak;<br>service_role key leak;<br>snapshot provenance loss | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.;<br>No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.;<br>None;<br>no Lean/mathlib code changes.;<br>No Lean noncomputable boundary involved.                                                                                |
| 20.20 | W      | infrastructure | Database;<br>Mirror;<br>Phase    | —                                                                                                                                              | —                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"><li>masters/<br/>cnna_roadmap_master.yaml</li><li>generated_exports/*.tsv</li><li>repo_inventory/<br/>raw_export/*.json</li><li>scripts/generate_tables.py</li></ul>                                               | No Lean modules added or modified. Future DB files may live under Repository/database/supabase or Repository/web once activated.;<br>Natural placement is DB-REVIEW/generated-secondary publication infrastructure, not a CNNA mathematical pillar.;<br>Keep DB schema normalized and queryable;<br>avoid large JSON blobs/PDFs/ZIPs as primary DB content.      | Reuse existing YAML/export/context-doc data;<br>do not create a second canonical planning model in DB without roundtrip contract.;<br>Supabase data must be generated/imported from Master YAML and generated exports with snapshot_id/hash evidence.;<br>second truth system risk;<br>manual DB table edits;<br>public write policy leak;<br>service_role key leak;<br>snapshot provenance loss | Consumers are public read-only frontend/API users and internal admin/import tooling only;<br>no pillar handoff or mathematical consumer rights.;<br>No Lean refactor. Future DB schema changes must preserve YAML roundtrip and public view compatibility.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>No live DB traversal yet. Future traversal is YAML/export rows -> normalized Postgres tables -> api.* views -> web frontend.;<br>None;<br>no Lean/mathlib code changes.;<br>No Lean noncomputable boundary involved.                                                                                |
| 20.21 | G      | documentation  | Generated;<br>Artifact;<br>Phase | <ul style="list-style-type: none"><li>Repository/<br/>archive/<br/>reports/<br/>V1_23_PDF_FQNT_SCALE_REPORT.md</li></ul>                       | <ul style="list-style-type: none"><li>Repository/<br/>scripts/<br/>generate_full_document.py</li><li>Repository/<br/>scripts/<br/>generate_context_doc.py</li><li>Repository/<br/>scripts/<br/>generate_context_doc.py</li><li>Repository/<br/>scripts/<br/>generate_tables.py</li><li>Repository/<br/>scripts/<br/>build_export_script_v1.8.py</li><li>README.md</li><li>Repository/<br/>masters/<br/>cnna_roadmap_master.yaml</li></ul>  | <ul style="list-style-type: none"><li>scripts/<br/>generate_full_document.py</li><li>scripts/<br/>generate_context_doc.py</li><li>scripts/<br/>generate_tables.py</li></ul>                                                                          | No Lean modules added/modified. Documentation generator scripts and YAML only.;<br>Natural placement is DB-REVIEW/documentation generator layer, not a mathematical pillar or CNNA generator.;<br>Do not split table generators solely for typography; use one central size-bump policy unless later profiles require explicit configuration.                    | Closed for typography: size profile is centralized in generators and not repeated by manual PDF edits.;<br>Typography behavior is generated from full/context/table generator source, not from edited PDF files.;<br>No Lean symbols affected. Avoid duplicate typography hacks outside the generators.                                                                                          | Human/tooling consumers only. No pillar handoff or mathematical consumer rights are granted.;<br>Regression checks are generate_tables.py, full PDF compile, context PDF compile, generated TeX size-profile audit and representative rendering.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>Concrete traversal is full-document PDF generation and Supabase context-document generation.;<br>No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.;<br>No Lean source edited;<br>no noncomputable boundary introduced.                                                    |
| 20.22 | G      | infrastructure | Database;<br>Mirror;<br>Phase    | <ul style="list-style-type: none"><li>Repository/<br/>archive/<br/>reports/<br/>V1_24_SUPABASE_IMPORT_ACTIVATION_PLAN_REPO_README.md</li></ul> | <ul style="list-style-type: none"><li>Repository/<br/>masters/<br/>cnna_roadmap_master.yaml</li><li>Repository/<br/>scripts/<br/>generate_context_doc.py</li><li>Repository/<br/>scripts/<br/>generate_full_document.py</li><li>Repository/<br/>scripts/<br/>generate_tables.py</li><li>Repository/<br/>scripts/<br/>build_export_script_v1.8.py</li><li>README.md</li><li>Repository/<br/>masters/<br/>cnna_roadmap_master.yaml</li></ul> | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>scripts/<br/>generate_context_doc.py</li><li>scripts/<br/>generate_full_document.py</li><li>generated_exports/*.tsv</li><li>repo_inventory/<br/>raw_export/*.json</li></ul> | No Lean modules. Future DB artifacts may be SQL migrations/import scripts/frontend contract files under Repository/database/supabase.;<br>Natural placement is DB-REVIEW/Supabase infrastructure sidechain, not a Lean derivation phase.;<br>Keep DB artifacts isolated under Repository/database/supabase and avoid leaking DB concerns into Lean module names. | Existing 20.15-20.20 were checked and kept; new phases are concrete activation subchain, not duplicates.;<br>Supabase rows must be generated from Master YAML/generated_exports/repo_inventory via canonical importer, not manual table edits.;<br>Existing broad DBR15-DBR20 contracts vs new concrete activation phases;<br>resolved by explicit broad-vs-activation split.                    | Consumers are public web/query/review tooling only. No pillar handoff or mathematical consumer rights are granted.;<br>No Lean refactor. Regression checks are generated tables/PDF, DB schema/import dry-run, live inventory/advisor output after authorized execution.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>Concrete traversal is Supabase project discovery, planned schema/import tables, and later live table/view verification.;<br>No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.;<br>No Lean source edited;<br>no noncomputable boundary introduced. |

| Phase | Status | Phase type     | Context class                 | Added modules                                                                                       | Modified modules                                                                            | Reusable existing modules                                                                                                                                                                                                             | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                               | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                                                                                                        | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| 20.23 | W      | infrastructure | Database;<br>Mirror;<br>Phase | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_24_SUPABASE_README.md</li></ul> | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li></ul> | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>scripts/generate_context_doc.py</li><li>scripts/generate_full_document.py</li><li>generated_exports/*.tsv</li><li>repo_inventory/raw_export/*.json</li></ul> | No Lean modules. Future DB artifacts may be SQL migrations/import scripts/frontend contract files under Repository/database/supabase.;<br>Natural placement is DB-REVIEW/Supabase infrastructure sidechain, not a Lean derivation phase.;<br>Keep DB artifacts isolated under Repository/database/supabase and avoid leaking DB concerns into Lean module names. | Existing 20.15-20.20 were checked and kept; new phases are concrete activation subchain, not duplicates.;<br>Supabase rows must be generated from Master YAML/generated_exports/repo_inventory via canonical importer, not manual table edits.;<br>Existing broad DBR15-DBR20 contracts vs new concrete activation phases; resolved by explicit broad-vs-activation split. | Consumers are public web/query/review tooling only. No pillar handoff or mathematical consumer rights are granted.;<br>No Lean refactor. Regression checks are generated tables/PDF, DB schema/import dry-run, live inventory/advisor output after authorized execution.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>Concrete traversal is Supabase project discovery, planned schema/import tables, and later live table/view verification.;<br>No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.;<br>No Lean source edited;<br>no noncomputable boundary introduced. |
| 20.24 | W      | infrastructure | Database;<br>Mirror;<br>Phase | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_24_SUPABASE_README.md</li></ul> | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li></ul> | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>scripts/generate_context_doc.py</li><li>scripts/generate_full_document.py</li><li>generated_exports/*.tsv</li><li>repo_inventory/raw_export/*.json</li></ul> | No Lean modules. Future DB artifacts may be SQL migrations/import scripts/frontend contract files under Repository/database/supabase.;<br>Natural placement is DB-REVIEW/Supabase infrastructure sidechain, not a Lean derivation phase.;<br>Keep DB artifacts isolated under Repository/database/supabase and avoid leaking DB concerns into Lean module names. | Existing 20.15-20.20 were checked and kept; new phases are concrete activation subchain, not duplicates.;<br>Supabase rows must be generated from Master YAML/generated_exports/repo_inventory via canonical importer, not manual table edits.;<br>Existing broad DBR15-DBR20 contracts vs new concrete activation phases; resolved by explicit broad-vs-activation split. | Consumers are public web/query/review tooling only. No pillar handoff or mathematical consumer rights are granted.;<br>No Lean refactor. Regression checks are generated tables/PDF, DB schema/import dry-run, live inventory/advisor output after authorized execution.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>Concrete traversal is Supabase project discovery, planned schema/import tables, and later live table/view verification.;<br>No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.;<br>No Lean source edited;<br>no noncomputable boundary introduced. |
| 20.25 | W      | infrastructure | Database;<br>Mirror;<br>Phase | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_24_SUPABASE_README.md</li></ul> | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li></ul> | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>scripts/generate_context_doc.py</li><li>scripts/generate_full_document.py</li><li>generated_exports/*.tsv</li><li>repo_inventory/raw_export/*.json</li></ul> | No Lean modules. Future DB artifacts may be SQL migrations/import scripts/frontend contract files under Repository/database/supabase.;<br>Natural placement is DB-REVIEW/Supabase infrastructure sidechain, not a Lean derivation phase.;<br>Keep DB artifacts isolated under Repository/database/supabase and avoid leaking DB concerns into Lean module names. | Existing 20.15-20.20 were checked and kept; new phases are concrete activation subchain, not duplicates.;<br>Supabase rows must be generated from Master YAML/generated_exports/repo_inventory via canonical importer, not manual table edits.;<br>Existing broad DBR15-DBR20 contracts vs new concrete activation phases; resolved by explicit broad-vs-activation split. | Consumers are public web/query/review tooling only. No pillar handoff or mathematical consumer rights are granted.;<br>No Lean refactor. Regression checks are generated tables/PDF, DB schema/import dry-run, live inventory/advisor output after authorized execution.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>Concrete traversal is Supabase project discovery, planned schema/import tables, and later live table/view verification.;<br>No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.;<br>No Lean source edited;<br>no noncomputable boundary introduced. |
| 20.26 | W      | infrastructure | Database;<br>Mirror;<br>Phase | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_24_SUPABASE_README.md</li></ul> | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_master.yaml</li></ul> | <ul style="list-style-type: none"><li>scripts/generate_tables.py</li><li>scripts/generate_context_doc.py</li><li>scripts/generate_full_document.py</li><li>generated_exports/*.tsv</li><li>repo_inventory/raw_export/*.json</li></ul> | No Lean modules. Future DB artifacts may be SQL migrations/import scripts/frontend contract files under Repository/database/supabase.;<br>Natural placement is DB-REVIEW/Supabase infrastructure sidechain, not a Lean derivation phase.;<br>Keep DB artifacts isolated under Repository/database/supabase and avoid leaking DB concerns into Lean module names. | Existing 20.15-20.20 were checked and kept; new phases are concrete activation subchain, not duplicates.;<br>Supabase rows must be generated from Master YAML/generated_exports/repo_inventory via canonical importer, not manual table edits.;<br>Existing broad DBR15-DBR20 contracts vs new concrete activation phases; resolved by explicit broad-vs-activation split. | Consumers are public web/query/review tooling only. No pillar handoff or mathematical consumer rights are granted.;<br>No Lean refactor. Regression checks are generated tables/PDF, DB schema/import dry-run, live inventory/advisor output after authorized execution.;<br>No Lean/mathlib version change. Lean data timestamp remains fixed.;<br>Concrete traversal is Supabase project discovery, planned schema/import tables, and later live table/view verification.;<br>No Lean/mathlib source complication introduced;<br>root manifest/toolchain unchanged.;<br>No Lean source edited;<br>no noncomputable boundary introduced. |

| Phase   | Status | Phase type | Context class                | Added modules                                                                                              | Modified modules                                                                        | Reusable existing modules                                                                                                                                                                                                                               | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                              | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                               | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 1.4.9   | W      | comparison | Reference;<br>Gate;<br>Phase | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC</li></ul> | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_masters</li></ul> | <ul style="list-style-type: none"><li>Structural/CayleyDickson/*.lean</li><li>Arithmetic/CayleyDickson/REFERENCE_GATE_REPORT</li><li>PillarB placeholder modules</li><li>scripts/generate_tables.py</li><li>scripts/generate_full_document.py</li></ul> | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.;<br>Natural placement is comparison/reference-gate layer, not a Foundational CNNA derivation layer.;<br>Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists. | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.;<br>The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.;<br>none identified | Comparison consumers may read CNNA-derived outputs;<br>CNNA derivation phases may not consume the comparison profile as a generator.;<br>No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.;<br>Baseline remains pinned Lean/mathlib snapshot;<br>live documentation is only a source-discovery hint until verified locally against the pin.;<br>Concrete traversal currently consists of source equation/profile extraction and target-object registration;<br>no CNNA runtime traversal yet.;<br>Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0;<br>implementation must recheck exact names and theorem availability.;<br>No noncomputable Lean code introduced.<br>Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed. |
| 1.4.9.1 | W      | comparison | Reference;<br>Gate;<br>Phase | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC</li></ul> | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_masters</li></ul> | <ul style="list-style-type: none"><li>Structural/CayleyDickson/*.lean</li><li>Arithmetic/CayleyDickson/REFERENCE_GATE_REPORT</li><li>PillarB placeholder modules</li><li>scripts/generate_tables.py</li><li>scripts/generate_full_document.py</li></ul> | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.;<br>Natural placement is comparison/reference-gate layer, not a Foundational CNNA derivation layer.;<br>Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists. | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.;<br>The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.;<br>none identified | Comparison consumers may read CNNA-derived outputs;<br>CNNA derivation phases may not consume the comparison profile as a generator.;<br>No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.;<br>Baseline remains pinned Lean/mathlib snapshot;<br>live documentation is only a source-discovery hint until verified locally against the pin.;<br>Concrete traversal currently consists of source equation/profile extraction and target-object registration;<br>no CNNA runtime traversal yet.;<br>Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0;<br>implementation must recheck exact names and theorem availability.;<br>No noncomputable Lean code introduced.<br>Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed. |



| Phase   | Status | Phase type | Context class                | Added modules                                                                                              | Modified modules                                                                        | Reusable existing modules                                                                                                                                                                                                                               | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                              | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                               | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 1.4.9.2 | W      | comparison | Reference;<br>Gate;<br>Phase | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC</li></ul> | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_masters</li></ul> | <ul style="list-style-type: none"><li>Structural/CayleyDickson/*.lean</li><li>Arithmetic/CayleyDickson/REFERENCE_GATE_REPORT</li><li>PillarB placeholder modules</li><li>scripts/generate_tables.py</li><li>scripts/generate_full_document.py</li></ul> | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.;<br>Natural placement is comparison/reference-gate layer, not a foundational CNNA derivation layer.;<br>Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists. | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.;<br>The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.;<br>none identified | Comparison consumers may read CNNA-derived outputs;<br>CNNA derivation phases may not consume the comparison profile as a generator.;<br>No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.;<br>Baseline remains pinned Lean/mathlib snapshot;<br>live documentation is only a source-discovery hint until verified locally against the pin.;<br>Concrete traversal currently consists of source equation/profile extraction and target-object registration;<br>no CNNA runtime traversal yet.;<br>Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0;<br>implementation must recheck exact names and theorem availability.;<br>No noncomputable Lean code introduced.<br>Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed. |
| 13.2.1  | W      | comparison | Reference;<br>Gate;<br>Phase | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC</li></ul> | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_masters</li></ul> | <ul style="list-style-type: none"><li>Structural/CayleyDickson/*.lean</li><li>Arithmetic/CayleyDickson/REFERENCE_GATE_REPORT</li><li>PillarB placeholder modules</li><li>scripts/generate_tables.py</li><li>scripts/generate_full_document.py</li></ul> | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.;<br>Natural placement is comparison/reference-gate layer, not a foundational CNNA derivation layer.;<br>Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists. | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.;<br>The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.;<br>none identified | Comparison consumers may read CNNA-derived outputs;<br>CNNA derivation phases may not consume the comparison profile as a generator.;<br>No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.;<br>Baseline remains pinned Lean/mathlib snapshot;<br>live documentation is only a source-discovery hint until verified locally against the pin.;<br>Concrete traversal currently consists of source equation/profile extraction and target-object registration;<br>no CNNA runtime traversal yet.;<br>Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0;<br>implementation must recheck exact names and theorem availability.;<br>No noncomputable Lean code introduced.<br>Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed. |

| Phase  | Status | Phase type     | Context class                     | Added modules                                                                                              | Modified modules                                                                        | Reusable existing modules                                                                                                                                                                                                              | Manifest / placement / compactness                                                                                                                                                                                                                                                                                                                                                              | Reuse / generator / duplicate risks                                                                                                                                                                                                                                                               | Implementation risk fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 13.6.1 | W      | comparison     | Reference;<br>Gate;<br>Phase      | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC</li></ul> | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_masters</li></ul> | <ul style="list-style-type: none"><li>Structural/CayleyDickson/*.lean</li><li>Arithmetic/CayleyDickson/Lean</li><li>PillarB placeholder modules</li><li>scripts/generate_tables.py</li><li>scripts/generate_full_document.py</li></ul> | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.;<br>Natural placement is comparison/reference-gate layer, not a Foundational CNNA derivation layer.;<br>Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists. | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.;<br>The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.;<br>none identified | Comparison consumers may read CNNA-derived outputs;<br>CNNA derivation phases may not consume the comparison profile as a generator.;<br>No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.;<br>Baseline remains pinned Lean/mathlib snapshot;<br>live documentation is only a source-discovery hint until verified locally against the pin.;<br>Concrete traversal currently consists of source equation/profile extraction and target-object registration;<br>no CNNA runtime traversal yet.;<br>Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0;<br>implementation must recheck exact names and theorem availability.;<br>No noncomputable Lean code introduced.<br>Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed. |
| 18.5   | W      | comparison     | Reference;<br>Gate;<br>Phase      | <ul style="list-style-type: none"><li>Repository/archive/reports/V1_25_HORN_QUATERNION_GEOMETRIC</li></ul> | <ul style="list-style-type: none"><li>Repository/masters/cnna_roadmap_masters</li></ul> | <ul style="list-style-type: none"><li>Structural/CayleyDickson/*.lean</li><li>Arithmetic/CayleyDickson/Lean</li><li>PillarB placeholder modules</li><li>scripts/generate_tables.py</li><li>scripts/generate_full_document.py</li></ul> | No Lean modules in V1.25. Future optional modules must be neutral ReferenceGates/ComparisonGate modules, not phase-named public APIs.;<br>Natural placement is comparison/reference-gate layer, not a Foundational CNNA derivation layer.;<br>Keep future comparison APIs small and isolated; no broad GA framework should be introduced before a concrete CNNA-derived comparison need exists. | Reuses existing CD, Pillar-B and side-comparison umbrella phases; new rows are target-gate refinements, not duplicate derivation phases.;<br>The only generator remains the CNNA Lean/provenance chain. External theory may only define target profiles for later comparison.;<br>none identified | Comparison consumers may read CNNA-derived outputs;<br>CNNA derivation phases may not consume the comparison profile as a generator.;<br>No Lean refactor in V1.25. Future module changes require import graph check, consumer discipline audit and full local Lean build evidence.;<br>Baseline remains pinned Lean/mathlib snapshot;<br>live documentation is only a source-discovery hint until verified locally against the pin.;<br>Concrete traversal currently consists of source equation/profile extraction and target-object registration;<br>no CNNA runtime traversal yet.;<br>Current web docs indicate available Quaternion and CliffordAlgebra APIs, but the project pin is Lean/mathlib 4.28.0;<br>implementation must recheck exact names and theorem availability.;<br>No noncomputable Lean code introduced.<br>Future use of analytic/normed quaternion APIs must be isolated from operational paths unless explicitly allowed. |
| 20.27  | G      | infrastructure | Tool;<br>Infrastructure;<br>Phase | —                                                                                                          | —                                                                                       | —                                                                                                                                                                                                                                      | Records table-renderer and YAML schema changes only;<br>no new Lean modules.;<br>Placed under 20.* because this is a planning/documentation tool schema change, not a Pillar-A mathematical object.;<br>No new script besides small patch helper; permanent changes are in the existing generator scripts.                                                                                      | Reuses existing status legend, phase-object-link and object register instead of creating a parallel object register.;<br>Context classes are type tags for review surfaces;<br>Lean code remains the only primary truth system.                                                                   | Consumers must read Context class before using status/link fields.;<br>Generated tables before/after are regenerated from the same YAML source;<br>old status columns remain intact.;<br>No mathlib dependency change.;<br>Complete generated-row traversal, not a mathematical instance traversal.;<br>none;<br>No Lean noncomputable boundary touched.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |